finite_volume58 frozen proof archive

Frozen proof-report archive: finite_volume58

This is a mechanical TeX translation of every bound Markdown proof report, plus prose reports explicitly named by the current proof map. It accompanies the edited manuscript, exact scripts and partial Lean development. It is an archival source collection, not a claim of end-to-end formal verification.

Reports retain their original dates, scopes and historical status wording. Later reports can supersede earlier bounds. The authoritative endpoint is results/fresh_a15_d15_dimension58_20261018.json. The current edited proof map is `docs/finite_volume58_lemma_map.md`.

The manifest records all report hashes and every recursively bound file. This export verifies source identity; it does not reprove the geometry.

Source 1

Source path: `docs/finite_volume58_reproduce.md`

SHA256: 25980f00c557a6c3d8ec0d2cf04a2a571b0504633035f3fcb238ed364c1fabe2

Reproducing the completed general finite-volume bound58

Run commands from the repository root with Python3. The completed endpoint is fresh_a15_d15_dimension58_20261018. It proves n≤58 for finite-sided, finite-volume real hyperbolic Coxeter polytopes. Later ambient58 frontiers concern a proposed57 bound and are not substitutes for this frozen certificate.

Levels of verification

1. **Identity and saved composition.** The final binder checks source/report hashes, complete saved finite/ideal domains, exact polynomial moment identities, local/count interfaces and final curvature arithmetic.
2. **Finite arithmetic replay.** The finite closure verifies all-last-item bounds and attaining predecessors of the complete saved tables, every item identity, all type comparisons and every contextual option. It does not merely trust a saved optimum or an optimizer success flag.
3. **Local geometry replay.** The two final local producers rederive their physical core calculations, native-map domains and current literal interfaces. Their written arbitrary-support and route-completeness arguments remain part of the mathematical proof.
4. **Historical dependency replay.** The transitive manifest identifies older geometric, classification and count packets. The endpoint does not execute all of them. Repeating the entire history is a distinct validation project; neither a successful hash audit nor216 literal faces replaces a universal geometric theorem.
5. **Proof-assistant verification.** The plain arithmetic export supports kernel checks. The internal geometric lemmas and higher tail must also be translated. The current Lean project is partial; this guide does not claim an end-to-end Lean proof of58.

Frozen endpoint and direct inputs

Artifact	JSON SHA256
dimension58 endpoint	60a6dbedc03b3c0690c80ab86f276e5767de786469b46468d0990e4a75d25985
finite closure	692e782f03d3dbc7ad230545819ccf37f6d9c69a8ed002bdeadbe8703c96c81e
simultaneous baseline	b192b2c6b9c27fa07a14d7cbf731a62c71c99da2de2460ea589e5d4a6d87a97d
completed59 higher tail	2b09dac21f742a216bd10229fe25fba3f1e4114b8f43fae1206dcb1840169bab

Each JSON's `sources` dictionary pins the source/data/report dependencies used at generation. Preserve those files unchanged. Use `--verify` on a frozen producer; do not regenerate over its certificate to hide a mismatch.

Commands

The smallest endpoint replay is:

```
python3 scripts/fresh_a15_d15_dimension58_20261018.py --verify
```

Its expected status starts with PASS_COMPLETE_A15_D15_DIMENSION58_COMPOSITION; the headline is dimension58, finite upper783/2, ideal surplus91/2 and constant4350.

The complete finite arithmetic replay is:

```
python3 scripts/fresh_a15_d15_ambient59_forest_closure_20261018.py --verify
```

The final two geometric replay commands are:

```
python3 scripts/fresh_finite_a15_three_coupon_zero_transfer_20261018.py --verify
python3 scripts/fresh_finite_d15_a4_a11_half_zero_transfer_20261018.py --verify
```

These are separate checks. A failure should be investigated at its first assertion and against the pinned inputs; do not change a frozen expected result merely to obtain `PASS`. They were already run successfully when the frozen proof was completed. The manuscript-translation task does not claim to have repeated all these historical computations.

Plain data for an independent arithmetic checker

The exporter produces finite_volume58_arithmetic_20261018.json. It reads the frozen count, computes only the exposed item/base/cross formulas, and copies the saved certificates. It does not optimize, reconvolve or prove the geometric interpretation. Its schema label is `finite-volume58-arithmetic-v1` and its status explicitly says `EXPORTED_FROZEN_DATA_NOT_A_FORMAL_PROOF`.

```
python3 scripts/export_finite_volume58_arithmetic_20261018.py --verify
```

For a clean checkout without the exported file, omit `--verify` once. Generation refuses to overwrite an existing export. JSON round-trip verification checks both the copied values and all source hashes. This packaging check is separate from a mathematical checker of the values.

The representation is deliberately simple:

- finite costs are integer units1/16; the cap is59 and upper is6264;
- rational ideal values are `[numerator, positive_denominator]`;
- `actual_types`, `representative_types` and `representative_map` encode all180 actual finite type representatives and their simultaneous map;
- base arrays and actual/upper pair matrices are explicit;
- each of5148 `tables` has a key, fixed rank5 count, excluded-rank2 flag, item ranks and weights,60 values and60 predecessor ranks;
- each of2574 `expanded_rows` records its fixed representative indices, fixed cost, residual rank and complete contextual options;
- each of1890 `ideal_profiles` records ranks,59 polynomial coefficients, V,G, normalized cost and both exact margins;
- all58 normalized affine envelopes and final constants are explicit.

A sound independent checker should establish these assertions, not just compare the final numbers:

1. Reconstruct every961 fixed vector and every2574 rank5 expansion and prove the residual split, including fixed A17 versus residual D17.
2. Verify simultaneous base/cross domination for every actual type and pair, and all item identities against the fixed components.
3. Check every last-item upper inequality. Induction then bounds arbitrary residual multisets. Predecessor equality supplies additive attainment but is not needed for an upper bound.
4. Prove coverage by the no-context, reserve-one and reserve-two branches, fixing all A2 multiplicities. Verify the two context formulas and every winning full-forest equality. Witness counts are presence guards.
5. Check every resulting upper$\le$6264. Do not assert actual attainment: the representative maximizer's actual cost is355.5.

6. Independently enumerate all rank58 terminal profiles with at most one positive rank below9. Recompute each polynomial product, not merely its symmetry, degree and coefficient sum.
7. Verify the exact curvature and cost margins, minimum normalized1/340, absolute minimum91/2, and the arithmetic783/2 and4350.
8. Connect these finite facts to the separately proved geometry, concentration, deficit-sum and higher-tail lemmas.

The export includes an `exclusions` list naming the geometric and all-rank statements not justified by the exported arithmetic. They are internal proof obligations, not permissions to add axioms for the final theorem.

Manuscript build

The standalone document is finite_volume58.tex. Its mathematical proof-atlas index links the internal local theorems and published inputs; the lemma map separates those categories in more detail.

```
pdflatex -interaction=nonstopmode -halt-on-error manuscripts/finite_volume58.tex
```

To keep build products out of the source tree, pass an existing directory with `-output-directory=/path/to/build`. Run a second pass to resolve cross-references. A successful TeX build is a formatting check, not a mathematical proof check. The translation is a manuscript draft backed by the frozen proof atlas; external refereeing and end-to-end formalization are not claimed.

Final arithmetic to keep visible

The upper and lower estimates concern the same deficit:

```
D5 ≤ (783/2) vf + sum C_actual(v)
D5 ≥ (783/2) vf + sum kappa(v) + 4350
kappa(v) − C_actual(v) ≥ 91/2.
```

Their difference is `0 ≥ 4350 + (91/2) vi > 0`. The finite equality is sufficient; a strict finite saving is unnecessary. The constant is added once, and a negative terminal ideal margin would not justify multiplying the concentration inequalities. The completed59 tail supplies dimensions at least60, completing the bound58.

Source 2

Source path: `results/dimension163_proof_status_20261017.md`

SHA256: 1d01da021d13aed9ebf4ee25580caff4e03509eb7732ded19e7d077eebebc46c

Status of the dimension-163 proof

The repository contains a **complete, internally checked computer-assisted proof** that every finite-sided, finite-volume real hyperbolic Coxeter polytope has dimension at most163. No unresolved mathematical step is currently identified in its final reviewed chain. Arithmeticity, compactness, simple cusps and a bound on the number of facets are not additional hypotheses.

The authoritative final case split binds the ordered-budget argument and independent arithmetic review, the all-rank estimates and the inherited geometric proof. The latter has an explicit internal geometric audit, including actual framing-wall incidence, common cusp-height ratios, endpoint Schur budgets and the compact-facet alternative. Standard published narrowness, classification and simple-polytope results are stated inputs.

The computational portions have exact certificates and separately implemented checks. The geometric portions have written internal reviews. These reviews were performed within the agent workflow; they are not external mathematician refereeing. The proof is not formalized in Lean or another proof assistant. External validation, publication priority and current-best literature status remain separate unestablished matters.

The word *candidate* in frozen historical reports expresses that validation status; it does not identify a remaining open case or unproved local inequality. It should not be read as reducing the result to numerical evidence. Those frozen reports and their hashes are preserved.

On2026-10-17 the final case/dependency checker was rerun successfully, checking 30 recursive manifests and83 source files. This rechecks that package's integrity, exact final inequalities and case split; it does not rerun every historical calculation or formally verify the geometric arguments.

```
python3 scripts/fresh_nikulin_two_budget_frontier_20261015_dimension163.py --verify
```

The gap between163 and the known examples motivates a stronger theorem. It is not a missing step in the existing argument.

Source 3

Source path: `results/dimension_bound_prokhorov_20261015_proof_review.md`

SHA256: 79d90fed91999743e372eeaf7f228738f87c54a392d3f79fadaed1a30774bf2b

Independent geometric and literature audit of the Prokhorov refinement

PASS, with a literature-status qualification. The proposed upper bound 990 follows from the cited published geometric lemmas and the new additive diagram estimate. It requires no arithmetic, compactness, ideal-vertex simplicity, finite-label cutoff, facet count, or source-chamber restriction. It is an explicit refinement of the published 995 argument. This review does **not** establish publication priority or a new strongest known bound: an older primary source announces a stronger bound below 300.

The small standalone checker and certificate bind the primary construction and independent arithmetic audit. They check the elementary identities below. The geometric audit is a source/proof review, not another computerized proof of the imported topological classification.

Why the edge estimate applies, including at cusps

Prokhorov's original paper, printed pp. 408 and 410, identifies the relevant edge diagram as elliptic of rank/order (n-1). Its unordered marked-node pairs determine the three-face angle incidences. Lemma 2 on p. 406 permits any positive edge constant (C); Proposition 5 supplies the good-face lower bound. The proposed argument changes only that constant, retaining every angle weight and all geometric lower bounds.

There is also a direct local explanation. An edge has points in hyperbolic space even when one or both endpoints are ideal. At a relative interior point, its incident reflections act on a positive-definite transverse space of dimension (n-1). Its diagram is spherical. The affine diagrams at ideal endpoints are different objects; they must not replace this edge diagram. Edge simplicity gives the local Boolean interval of containing faces, so deleting an unordered pair of incident facet constraints selects the corresponding three-face. In fact, the upper estimate only needs an injection into node pairs.

Consequently there is no omitted affine edge family, cusp multiplicity, or ordered-pair factor. Distances are ordinary numbers of graph edges, ignoring labels. Disconnected components contribute no pairs. Unbounded finite labels occur in (I_2(m)), whose unlabelled graph is always the two-node path. The complete finite real Coxeter classification has no exceptional connected family of unbounded rank beyond paths and (D_r).

An all-rank argument without a small-diagram census

Put (W=(2f_7+f_{15})/3), where (f_k) counts unordered node pairs at graph distance at most (k). For a path of rank (r≥ 17), (f_k=kr-k(k+1)/2) for (k=7,15). For (D_r), each of these counts is larger by one. Thus the deficits (29r-3W) are respectively 176 and 173.

For any component of rank (1≤r≤ 16), even the complete-graph bound suffices:

$$29r - 3W \geq 29r - 3\binom{r}{2} = \frac{r(61-3r)}{2} \geq \frac{13r}{2} > 0.$$

For a spherical forest of total rank (N≥ 27), a large component already supplies deficit at least 173, while all remaining deficits are positive. If every component is small, their combined deficit is at least (13N/2>173). Hence (W≤(29N-173)/3)

for every such forest. This weaker threshold than the primary's (N≥ 17) is sufficient for the dimension theorem, and avoids depending on any finite exceptional-graph calculation.

For a hypothetical dimension (n≥ 991), choose the single positive constant

$$C = C_n = \frac{29}{3} - \frac{173}{3(n-1)}.$$

It is uniform over all edges of this polytope. Dependence on the fixed dimension is permitted. The published lemma yields

$$n < 996 - \frac{5536}{n-1}, \qquad n^2 - 997n + 6532 < 0.$$

Writing (n=991+t) turns the last polynomial into (t^2+985t+586>0), a contradiction for every real (t≥ 0). At 990 it equals (-398), so this argument stops at 990. The primary's separate parity calculation agrees; the simpler substitution already proves the conclusion.

The older 300 announcement must remain visible

The official ICM 1986 proceedings contain Nikulin's *Discrete Reflection Groups in Lobachevsky Spaces and Algebraic Surfaces*, printed pp. 654–671. His definition on p. 654 makes CHRG mean finite-covolume reflection groups, including nonarithmetic groups. Printed p. 659 explicitly asserts dimension below 300. Pages 657–659 outline a narrow-face method and work out an affine-(A) cusp example, but do not supply the full numerical treatment of all diagram types needed to verify the final 300 claim from this text alone. The local audit read the original pages, not merely a search snippet.

Dolgachev's 2008 survey, printed p. 11, explicitly calls that result an announcement and cites Nikulin. More recent papers still quote 995, including the 2026 paper on strongly reversible geodesics. This disagreement in citation practice does not establish that the announcement is false, nor does the announcement alone constitute an independently checked complete proof here.

The accurate status is therefore: **990 is a fully explicit, independently audited consequence of the stated published Prokhorov inputs; a stronger historical primary announcement exists, and priority/current-best status is unresolved.** Recovering the full narrow-face constants is a substantive next task. No external announcement should erase that distinction.

For reproducible provenance, the source above was downloaded as the complete official 974-page first volume. Its PDF pages 756–773 contain printed pages 654–671; the local review excerpt was extracted from precisely those pages, without OCR substitutions. The source URL and page mapping, rather than a transient local path, are the durable reference.

The missing verification target is concrete: derive uniform explicit separation inequalities for every irreducible affine cusp type and every product of affine components, with the possible outgoing edge completions fully covered. Then compare those bounds with the narrow-face distance constraint and treat the alternative where the narrow facet has only finite vertices. The affine-(A) illustration alone does not establish those omitted cases. This is a proposed proof-reconstruction task, not an assertion that the announcement cannot be completed.

Separate conditional compact audit

The geometric argument in the compact-component report also passes source/proof review. At a finite vertex, inverse positive-definite Coxeter Gram blocks count nonright intrinsic two-face angles, giving the stated average component-pair inequality and the conditional rank bound (d≤4r). This counts all pairs within each component, not only graph edges.

Its stronger quadrilateral charge requires the full **flag/3-free** premise: every empty facet family contains a disjoint pair. Merely having no triangular two-faces is insufficient. Under that premise, opposite quadrilateral side facets are globally disjoint. If all adjacent pairs were right, the four normals would give two negative directions. Thus one direct spherical-diagram edge can be charged at a corner, and the corner plus released pair determines the two-face, making the charge injective. This justifies (q_4≤∑_v(d-c(v))) and the conditional bounds 11 or 10 stated there. The proof uses Alexandrov's definition and Lemma 4.3. These are precisely scoped consequences, with no unrestricted compact-dimension or novelty claim.

Scope of this review

The proof imports the published good/bad three-face and cusp analysis without attempting to re-prove every quasi-Lannér case. It checks that the unchanged construction meets the exact hypotheses needed for the improved edge constant. The separate arithmetic reviewer recomputes graph distances and scalar algebra independently. No floating-point evidence or finite search cutoff enters the dimension conclusion. The checker binds source/data only, allowing the primary report to add review links without invalidating the audit.

```
python scripts/dimension_bound_prokhorov_20261015_proof_review.py --verify
```

Source 4

Source path: `results/fresh_a11_cycle_coupon_domination_20261017.md`

SHA256: 946701ee5d8b324ff2ac67d0f87a84ee73d7a5fd1f016fd1beb4ea46f94162a7

The cycle12 A11 coupons fit the revised exact finite budget

Adding the two cycle12 coupons does not increase the revised finite-forest maxima. In particular the exact maximum at ambient rank69 remains544, equal to the finite curvature coefficient. The rule includes the whole-A17 endpoint deletion, the full A1/D11-short-leaf upgrade and every inherited coupon. This is a finite-budget theorem, not an ambient dimension bound.

The new pairs join whole A1 to the unique middle node of whole A11, at unit weight, or either node of whole A2 to that A11 middle, at weight one half. No old coupon uses A11 as its large component. Their total interaction costs with one A11 are therefore1 against A1,1 against A2, and0 against every other component.

Replace each A11 by D11. Both same-component costs are55. Under the existing upgraded D11 rule, interactions cost5/2 against A1,3 against A2, and3r/2 against any other component of rank r<=5. Interactions with larger components vanish. Each total is at least the corresponding new A11 total. Interactions between two large components vanish before and after replacement, so the replacement works successively throughout an arbitrary forest without double counting. All pair weights replace or supplement the stated distinct roles; an inherited half is never counted twice when upgraded to one.

The source and certificate check the interaction comparison against all228 representative finite types through rank75, including exceptional types and a symbolic I2 family. The arbitrary dihedral label is irrelevant to these cross-coupon roles. The rank bound75 is the domain of the inherited exact finite theorem.

The resulting forest is one counted in that exact theorem and has at least the enlarged-rule cost. Its maxima are therefore valid upper bounds. Conversely every saved maximizing forest contains no A11, so its cost is unchanged and still attains the same maximum. The checker directly recounts all76 typed witnesses, including rank0, using the separate graph-pair implementation from the finite-budget reviewer.

At rank69 the forests A1+4D17 and A1+4A17 still attain544. At rank68, 4D17 still attains544, exceeding the curvature coefficient527. The new coupons do not remove that lower-rank finite obstruction.

Generation and a fresh separate-process saved replay both terminated exit0/PASS on228 partner comparisons and76 attaining forests. A separate full source and mathematical read, including the imported direct graph pair-count routines, passed the complete role comparison, successive rank-preserving replacements and unchanged attainment argument. That read did not repeat the228/76 arithmetic checks or inherited Bellman calculations. Source, certificate and report are final/frozen. The finite-type classification and inherited exhaustive Bellman certificates retain their stated verification scopes.

Source 5

Source path: `results/fresh_a14_cd16_ambient59_frontier_20261018.md`

SHA256: 257219a7c1e324f44d4c7ed1c39d846a0994da807d2328f07b2387bad9ddb6cf

A14 and C/D16: finite787/2 with all ideal profiles positive

The simultaneous proved rule has exact formal finite maximum **787/2** at ambient59, attained by **D14 + 3A15**. This exceeds the finite curvature783/2 by2. All1,890 terminal ideal profiles are strictly positive: the minimum normalized margin is1/340 and the minimum terminal absolute surplus is91/2. Positive low9 concentration gives that same91/2 lower bound for every actual ideal-vertex surplus.

Thus any finite-sided, finite-volume real hyperbolic Coxeter polytope of dimension59 must satisfy

```
2 v_f >= 4350 + (91/2) v_i.
```

In particular it has at least2175 finite vertices, or at least2198 if it has an ideal vertex. The all-ideal subclass has dimension at most58, using the completed unrestricted59 theorem for higher dimensions. This is a consequence of this certificate, with no novelty or best-known claim for the all-ideal subclass. The unrestricted ceiling remains59: the finite obstruction prevents an unrestricted dimension58 conclusion.

The producer and certificate combine the frozen finite6/order13/order17 rule, the A14 local theorem and the C/D16 local theorem. The old394 certificate does not already contain these changes. The new rank1 interaction requires fresh residual tables and an enlarged key.

Local rules and simultaneous coverage

In a whole A14, delete the two reversal-related units(0,12),(1,13). Actual rank14 costs become A89/B87/D181/2, so the common representative allowance decreases91→181/2. The local theorem handles every changed selected map: the proper maps force a genuinely new endpoint whose projection is7/2>2, and the two whole cases seal all extra supporting whites away from the core. A distinct whole A15(1,13) unit pays1. There is no new finite coupon and no ideal-weight change.

In affine order16, delete C pairs(2,12),(3,13), and reduce the four D halves(1,14),(1,15),(10,12),(10,13) to7/16. Family totals become A80/B102/C101/D411/4, with all-family envelope411/64. The C maps force a new doubled terminal with projection8/3>2. The complete D contact and completion argument excludes proper factors or seals the four whole cases against arbitrary supporting whites. A distinct whole A4/A12 endpoint unit pays in two cases. In the other two cases the selected7/16, old whole A4-end/D12-short-leaf half and new whole A1/D15-node11 sixteenth are at three different corners and total1. No finite same-component D16 unit is spent.

The new ordinary A1/D15 coupon is at the unique long-arm neighbor of the fork center, canonical node11 in finite D15. Actual and representative component-pair costs both rise3/2→25/16. This increment is counted here; it is not treated as absorbed by the previous allowance. It is disjoint from all existing fork-center/short-leaf roles and the two contextual coupon families.

The simultaneous composition retains the preceding independent actual-corner priorities. A changed finite6 corner is handled first by its independent short5/cycle11 route, protected whole finite output or A1/A12 middle unit. None is an A14 distance12 pair or a new C/D16 long pair. On a face with no changed finite6 corner, apply the independent small-ideal/order13/contextual-order17 rules as before. Their short paths, smaller affine roles and specified whole-component coupons are unchanged by the present transfers.

Before any selected-core exclusion, the current short8 escape is paid independently. Old deep8 zeros give short7; deep7 zeros give short6 or their protected whole-factor alternatives; deep6 zeros give short5 or their explicit protected payments. These do not use the newly reduced long pairs. Both new local theorems patch their disjoint selected target families in the complete remaining overlap reduction: A14 outputs the retained whole A15 unit, while C/D16 outputs the distinct typed finite payments above. Neither theorem needs the other's reduced selected weight. The inherited low/disjoint/protected/adaptive branches retain their separate proofs. Absence from a finite map list is not used as an arbitrary-support argument.

Triangles retain their elementary short pairs, protected leaf/branch outputs and bounded exceptional whole components. The new distances are12 in finite A14 and10/11 in the order16 affine factors. Every local packet certifies triangle payment2 and quadrilateral payment1, and the listed retained outputs make their simultaneous use noncircular.

The producer aligns all216 saved current literal records and their actual marks with both local certificates. A14 has zero changes. Four selected D corners lose1/16; two distinct finite corners gain1/16. The affected face lower bounds are31/16,1,31/16,1, and all others are unchanged. This saved-interface check supplements the universal local proofs; it does not rerun native Grams or the historical corner census.

Actual and representative finite costs

All arithmetic uses integer sixteenths. Actual A14 decreases by32 units of this scale, whereas its rank14 representative decreases by8. The representative remains D14 with allowance181/2, attained by actual D14. Actual A14 costs89 and actual B14 costs87. The other bases retain the finite6 rule, including actual A15=101, D15=201/2, A16=441/4, D16=439/4, fixed A17=451/4 and residual D17=919/8.

Only the actual and upper A1/D15 cross costs increase, by one sixteenth. All32,400 ordered actual type-pair comparisons are checked after simultaneously replacing both ends by their representatives. This includes the preceding A1/A12 unit roles, A1/B12 half, A1/A15 and A1/A16 sparse roles, D5/D12 and D5/A12 charges. The old and new contextual costs remain respectively

```
a2*(a2-1)/4 if a10>0 and a1>=2,
a2*d5       if a9>0  and a1>=1.
```

Here a_i and d_i count whole components of their actual types. Their marked pairs are distinct. Whole unmarked witnesses certify presence only; their multiplicities do not multiply either cost. The upper rule allows any rank9/10 witness and replaces D5 by its rank5 A5 representative, preserving every actual triggered context.

Complete changed-key finite certificate

There are961 complete fixed multiplicity vectors at ranks9 through17. Fixed rank17 means A17; D17 remains a residual type. Expanding every permitted rank5 count gives2,574 vectors. Rank5 must be fixed because of its ordinary A1 interaction and its contextual A2 interaction. Residual ranks are1–4,6–8 and17–59. All ordinary residual–residual cross costs are zero, checked explicitly.

The old eight-coordinate interaction key is extended by the fixed rank15 multiplicity c15. This coordinate cannot be inferred by silently reusing a colliding old key. There are now961 distinct fixed keys and 2,574 distinct key/rank5 combinations. At scale16 the new item is

```
item_new(r, extended_key, k5)
  = item_previous(r, old_key, k5) + c15 if r=1,
```

and is unchanged for other residual ranks. The A14 base change is fixed, so is charged separately rather than subtracted from a residual item. The producer checks128,700 identities equating items with the new representative base and every interaction with the fixed forest.

Each key has an ordinary additive table and a table excluding rank2, giving5,148 fresh tables. Generation uses the item-first recurrence. The saved verifier checks every last-item upper inequality and one attaining predecessor at each positive residual rank. Induction proves both the bound and attainment. It checks6,723,288 inequalities and 303,732 equality predecessors; no numerical optimizer is used.

The complete9,191 context choices retain three exhaustive classes: ordinary no-context forests; reserve one A1 for the new rank9/rank5 context; reserve two A1 for the old rank10 context, also charging the new context when present. In the latter two classes every A2 count is explicit and the remainder uses a no-rank2 table. Reserving one A1 may allow extra A1 in the table, which is a safe upper relaxation for its intended class. If a winning candidate then omitted a positive old context cost, its same forest would be covered at a strictly greater value in the reserve-two branch. The ordinary branch has the same harmless undercount interpretation for reconstructed forests that happen to trigger a context. The source verifies equality of every winning value with its reconstructed full two-context forest.

The maximum is787/2. The actual formal witness is

```
D14 + 3A15: 181/2 + 3*101 = 787/2.
```

There are no ordinary cross or contextual costs in this witness. Formal attainment does not assert geometric realization by a Coxeter polytope. Exactly three fixed-multiplicity upper rows exceed783/2:

Representative forest	Upper cost	That literal forest's actual cost
D14 + 3D15	787/2	392
2D14 + D15 + D16	1569/4	1565/4
A4 + 2A5 + 3D15	392	781/2

The first upper value is attained by D14+3A15. The second is attained by2D14+A15+A16. The third mixes worst-type allowances and is only an upper representative row. These three rows are not a classification of every actual forest above curvature.

All ideal profiles and the positive-surplus extension

The producer independently enumerates the complete low9 terminal rank58 domain: arbitrary repeated ranks at least9 and at most one positive rank below9. Its1,890 profiles match the baseline and local certificates bijectively. The original saved h-polynomials are checked for positive integer coefficients, symmetry, degree58 and sum V=product(r+1). Their moment

```
G = sum_j min(j,58−j) h_j
```

gives curvature kappa=(5G−58V)/2; the equivalent absolute-deviation identity is also checked. This revalidates saved exact polynomial identities without recomputing the convolutions.

Only the order16 envelope changes, from103/16 to411/64. Hence each normalized margin increases by count(rank15)/64. Every row agrees exactly with the local C/D16 update and with the independently summed current all-family envelopes. All margins are positive, with normalized minimum1/340 at(9,9,9,15,16) and terminal absolute minimum91/2 at(58).

Let C_actual be the assigned cusp charge and C_upper the factor-envelope charge, so C_actual≤C_upper. The inherited low9 concentration reaches the complete terminal domain, decreases V and does not increase (kappa−C_upper)/V. Its low-rank ingredients are unchanged by the new order16 rule. Since all terminal normalized margins are positive, the two inequalities can be multiplied. Every original envelope surplus is at least its terminal surplus, hence at least91/2. The actual surplus is at least the envelope surplus. No product argument involving a negative terminal margin is used. The factor-fiber theorem already handles arbitrary products of affine factors; no simple-cusp hypothesis is added.

Global consequence and remaining obstruction

The producer independently computes and binds the finite curvature

```
(5*floor(58^2/4) − 59*58)/2 = 783/2
```

and the single extremum correction5*floor(59^2/4)=4350. Combining the proved local face inequalities, finite upper787/2 and ideal surplus91/2 with the inherited shape-curvature inequality gives

```
2 v_f >= 4350 + (91/2) v_i.
```

The correction4350 appears once. This excludes an all-ideal polytope in dimension59. The completed unrestricted59 theorem excludes higher dimensions, yielding the stated all-ideal ceiling58. For mixed or compact polytopes the positive finite deficit2 remains, so this is a necessary vertex inequality and not an unrestricted dimension58 bound. No arithmetic, compactness, facet-count or simple-cusp restriction is silently imposed. No comparison with the published best all-ideal dimension bound is asserted. No new example or external/formal verification is claimed.

Verification

Authoritative generation57727/chunkf8bf79 terminates exit0/PASS787/2 with three finite upper rows and no nonpositive ideal rows. The final frozen A14 and C/D16 reports are pinned by hash. Separate fresh saved replay56445/chunk7e95fe also terminates exit0/PASS with the same headline. The earlier changed-problem preflight12704/chunkf8b936 is not substituted for this source-bound final certificate. Direct final validation checks 382 source/data/report bindings, five local links and Python syntax.

The final saved replay checks all finite last-item inequalities and attainment, complete fixed/context domains, simultaneous typed costs, aligned216 literal changes, all1,890 ideal-domain/polynomial identities and the exact scalar update. It does not repeat historical geometric classification, native searches, link convolution or a global optimizer.

Root completed a full source/report mathematical composition read: PASS. It checked actual versus representative A14 deltas, the c15 key/item, all fixed/residual/context classes and winning full-cost equality, actual D14+3A15 attainment, three upper rows versus literal costs, complete1890 domain and saved moments, positive-only surplus extension, simultaneous local priorities and literal alignment, and the single4350 correction. A separate full producer/report mathematical composition read also passed, including exhaustive fixed-rank5/no2/reserve1/reserve2 coverage, last-item upper/attainment, simultaneous costs, actual cusp surplus direction and the limited all-ideal consequence. Neither reviewer executed the producer or replayed geometry, maps, DP, convolutions, scalar arithmetic or the historical census.

Source, data and report are final/frozen after direct validation. The completed unrestricted bound remains59; the positive ideal result and finite gap2 are the new simultaneous conclusions of this certificate.

```
python3 scripts/fresh_a14_cd16_ambient59_frontier_20261018.py --verify
```

Source 6

Source path: `results/fresh_a15_d15_ambient59_forest_closure_20261018.md`

SHA256: 021b4e474b3a15b805788d8736ff732bd102c0f93d8f5921c148a45720a2607c

Complete ambient59 finite upper391.5 with unchanged positive cusps

Under the simultaneous A15 and D15 zero-half transfers, every finite rank59 forest has charge at most **783/2=391.5**, equal to the finite curvature coefficient. Every actual ideal vertex retains surplus at least91/2. The global deficit comparison therefore requires

```
0 >=4350+(91/2)v_i,
```

which is impossible. This packet supplies the complete count and local composition for a separate dimension binder. It does not describe391.5 as an actual formal-forest maximum: it is the certified representative upper, and that distinction is sufficient for the contradiction.

The producer and certificate bind three frozen inputs:

- the current complete rule and finite/ideal certificate;
- the A15 three-coupon zero transfer;
- the D15 A4/A11 half transfer.

The source binds their final reports and every source/data dependency. No geometric realization of a formal forest is assumed.

One simultaneous actual-corner rule

The whole A15 pairs(0,12),(2,14) and whole D15 pair(0,11) change from one half to zero. Actual A15 and D15 both cost100; B15 remains97. Thus the rank15 representative allowance is100. All ideal weights and all other finite same-component weights remain unchanged.

Four ordinary whole-component increments are charged:

Small marked role	Large marked role	Increment per marked pair	Component-pair increment
A1	A12 endpoint	1/4	1/2
A1	A13 second node	1/4	1/2
A4 inner node	A13 second node	1/2	2
A4 endpoint	A11 endpoint	1/2	2

Different coupons have different component types or roles, so none is double counted. They apply to whole factors, not arbitrary subdiagrams. The old conditional A2/A2 and A2/D5 payments remain unchanged and are counted separately below.

Composition starts with the current independent finite6 and ideal actual-corner priorities. Their short5, protected leaf, whole small factor and typed-coupon outputs are retained. The current short8-to- short7-to-short6 routes remain independent of the new long rank15 zeros. In particular a short ideal path is handled by its established replacement theorem, including the cycle11 exception, rather than being presumed a unit.

If none of these prior branches applies, use the inherited full finite selected proof. An unchanged selected payment retains its old proof. A changed A15 selected payment is covered by one of its four complete whole-core contexts; a changed D15 selected payment is covered by one of its five complete contexts. Those local proofs exclude every proper completion or seal the entire core under arbitrary supporting whites. Their displayed alternatives are therefore whole actual factors.

The A15 proofs retain the A16(1,13)/(2,14) half and use either two new quarters or one new half at distinct other corners. The D15 proofs retain D16(1,12) at one half and add the distinct A4/A11 half. No A15 proof spends the newly zero D15 pair, and no D15 proof spends either newly zero A15 pair. The A14 theorem's A15(1,13) unit stays1, the finite D14 theorem's D15(1,12) half stays1/2, and the ideal15 theorem's finite D15(0,10) unit stays1. The current rank16/17 alternatives and ordinary A1/D15 node11 sixteenth are likewise distinct. The contextual coupons only gain optional additional ordinary payments; their conditions and amounts do not change.

The two local complete native domains and arbitrary-support proofs are bound inputs. Neither is replaced by the historical216 literal list. The present producer aligns the two independently reconstructed native literal audits with the same frozen baseline, compares every actual mark and corner delta, and verifies the combined lower payments. All216 rows have zero debit and zero credit here, so all face totals remain at least1. The elementary triangle payments stay at least2. Faces with at least five sides need no positive charge for the inherited deficit inequality.

Complete typed domination and finite domain

The charge unit in the source is1/16. Relative to the frozen baseline, the actual A15 base decreases16 units, the actual D15 base decreases8, and the common rank15 upper base decreases16. Actual and representative costs are checked separately.

Pair	New actual cost	Representative pair	New upper cost
A1/A12	5/2	A1/D12	5/2
A1/A13	33/16	A1/D13	33/16
A4/A13	8	A4/D13	8
A4/A11	2	A4/D11	8

The A1/D13 allowance rises by1/16 and A4/D13 by1. The A4/A11 addition fits its unchanged upper8. All32,400 ordered actual finite-type pairs through rank59 are checked after simultaneous type replacement, as is every same-component base. The inherited finite-type classification includes exceptional components and the arbitrary-label rank2 type. Contextual charges are dominated separately: their upper guards allow any rank9/rank10 representative and replace D5 by A5.

Every representative forest of total rank59 decomposes uniquely into the fixed rank9-through16 multiplicities, the separately fixed A17 multiplicity, and the remaining small or large components. The complete fixed domain has961 vectors. Every possible rank5 count is then fixed explicitly, giving2574 expanded vectors and2574 interaction keys. The key includes the rank15 count, required by the existing A1/D15 sixteenth. Fixed A17 and residual D17 are distinct representatives; residual ranks are1 through4,6 through8, and17 through59. Rank5 is not also permitted in the residual recurrence.

For each fixed vector and rank5 count, residual components have zero pairwise upper interaction. Their item charge is their upper base plus their full interaction with the fixed components. The producer checks all128,700 item identities. The only changes from the preceding item formula are

```
rank1: +c13,
rank4: +16*c13,
```

at scale16. Thus the existing interaction key is sufficient, but its tables must be recomputed. No old table is transported with changed item weights.

Context guards and complete Bellman certificates

For each key and rank5 multiplicity, the producer generates two additive tables: an ordinary table and a table excluding rank2. There are5148 tables. Generation uses item-first recurrence. Separate verification checks every possible last item as an upper inequality and at least one exact attaining predecessor at every positive rank. The6,723,288 inequalities and303,732

predecessors prove the additive upper by induction and certify its attainment in the representative item problem. This is distinct from actual typed-forest attainment.

The two nonadditive whole-factor payments require a complete split:

- With a whole A10 and at least two whole A1 factors, k whole A2 factors cost k(k−1)/4 in the old contextual rule. The upper guard allows any rank10 factor.
- With a whole A9 and at least one whole A1, k A2 factors and l whole D5 factors cost kl in the new contextual rule. The upper guard allows any rank9 and uses the fixed representative A5 count l.

Witness multiplicities do not multiply either charge. A forest with neither context is covered by the ordinary table. For the old context, reserve two A1 factors, fix every possible A2 count, and use the no-rank2 table for the rest. This branch includes both charges if both guards hold. For the new context without the old one, reserve one A1 and again fix every A2 count. Allowing additional A1 factors in that residual table is a safe relaxation. It cannot cause the winning reconstruction to omit a positive old-context payment: the two-A1 branch would then be strictly larger. The source checks exact equality between the winning complete representative forest cost and its calculated value.

All9191 context options are checked. This covers arbitrary numbers of both contextual factors and all remaining residual components; it does not sample a fixed set of suspect forests.

The largest representative upper is783/2. No expanded vector exceeds the curvature783/2. Its saved representative maximizer is

`8A1+3A5+3D12.`

That same literal typed forest has actual cost711/2=355.5. The upper uses conservative pair allowances. The certificate makes no claim that 391.5 is the exact maximum of actual forests or that this representative attainer is geometrically realizable. Only the complete upper is used in the global proof.

Unchanged positive ideal envelope

No ideal payment changes. The producer independently reconstructs the complete1890 terminal rank58 profiles, with arbitrary repetitions of ranks at least9 and at most one smaller positive rank. It rechecks the bound saved h-polynomials, degree58, symmetry, positive integer coefficients, product vertex counts and curvature moments. Every new row agrees exactly with the frozen current ideal row; no convolution is repeated.

For V=product(r+1), let C_upper be the saved all-family cusp-charge envelope and kappa its curvature. Every terminal normalized margin (kappa−C_upper)/V is positive. The minimum is1/340, at (9,9,9,15,16); the minimum absolute envelope surplus is91/2, at(58).

The inherited low9 concentration reduces V and does not increase the normalized margin. Since the terminal margins are positive, their product gives an absolute surplus lower bound for every original profile. The actual assigned cusp charge C is at most C_upper, so every actual ideal vertex has kappa−C at least91/2. The factor-fiber theorem handles products of affine factors; no simple-cusp hypothesis is added. No concentration of large factors is used.

The finite curvature783/2 and the strict global correction4350 are checked independently and against the frozen current certificate. The4350 is counted once. Combining the finite upper with the ideal surplus and simultaneous local payments gives the impossible necessary inequality stated at the start. A separate global binder will state the resulting dimension theorem and invoke the completed59 higher tail.

Verification and limits

Generation81617/chunke09a20 and separate fresh verification38621/chunk3c4d84 both terminated exit0/PASS. Two independent full source/report mathematical composition reads passed, including the fixed/residual/contextual domain, all actual and representative increments, disjoint whole-corner outputs, fresh item changes and Bellman certificate logic, representative-versus- actual upper distinction, positive-only ideal extension and single4350. The reviewers also read the current local reports and relevant preceding engine interfaces. Neither executed local maps, DP, convolution, literal or historical census checks. Final bindings, links and syntax metadata passed. The source, certificate and report are frozen. The producer rebuilds the complete changed finite tables and verifies the exact local interfaces and unchanged saved ideal

identities. It does not repeat the historical all-label census, local supporting-root geometry, original ideal convolutions or the shape-curvature theorem.

```
python3 scripts/fresh_a15_d15_ambient59_forest_closure_20261018.py --verify
```

Source 7

Source path: `results/fresh_a15_d15_dimension58_20261018.md`

SHA256: c153463b2b9b043eb3f076db1e6504548e0fc001072dd1455c7bd06f2014f845

Dimension at most58 from the simultaneous rank15 transfers

Every finite-sided, finite-volume real hyperbolic Coxeter polytope has dimension at most58. The proof composes completed local payment theorems, a complete finite upper certificate and the positive cusp surplus. It requires no arithmeticity, compactness, simple-cusp or numerical facet-count hypothesis.

The producer and composition certificate bind:

- the completed unrestricted bound59, excluding all dimensions at least60;
- the current simultaneous baseline, including the all-family ideal envelope and inherited global inequality;
- the A15 zero-half theorem and D15 zero-half theorem;
- the complete simultaneous finite and ideal certificate.

The bound59 proof and all inputs are frozen. This packet's generation and separate fresh replay passed, and three independent full source/report mathematical composition reads found no issue. Their precise scope is recorded below. No external refereeing or proof-assistant verification is asserted.

The local rule is simultaneous

The whole A15 pairs(0,12),(2,14) and whole D15 pair(0,11) change from weight1/2 to0. Their actual costs are both100; B15 stays97. Add these ordinary whole-component payments:

Small marked role	Large marked role	Increment
A1	A12 endpoint	1/4
A1	A13 second node from an end	1/4
A4 inner node	A13 second node from an end	1/2
A4 endpoint	A11 endpoint	1/2

All other finite weights and all ideal weights are unchanged. The two local theorems and the count bind the identical frozen baseline source, data and report. Their exact actual charges and representative upper allowances are compared, so the geometric rule and counting rule agree.

Consider an actual quadrilateral with arbitrary supporting roots. Apply the established finite6 and ideal actual-corner priorities first, including the independent short-path replacements. Their short5, protected leaf, whole small-factor and typed outputs remain available. A cycle11 half exception uses its own theorem; a short ideal path is not presumed a unit. The unchanged short8-to-short7-to-short6 arguments likewise terminate in retained independent alternatives.

In the remaining branch use the inherited complete finite selected proof. If its chosen payment is unchanged, that proof still applies. If it chooses one of the new A15 zeros, its complete native support is one of four whole contexts. If it chooses the new D15 zero, its complete support is one of five contexts. The local proofs cover every proper completion and arbitrary additional supporting white, not only a saved finite collection of faces.

For A15, the two Lorentz cores have all remaining contact projections at least4, while the two radical cores force either orthogonality or an independent short5 path. All displayed paying factors are therefore whole actual components. Each

case retains a distinct whole A16 half at(1,13)/(2,14). Two cases add two quarters at different corners; the other two add the A4/A13 half. Their total remains1.

For D15, proper cases1030/1613 require projection at least6>2. In1641 the forced fork center is saturated in the Lorentz core. Its product with a required new simple leaf would be positive, zero or at most−2, never−1; hence that completion is excluded. In the two whole cases, remaining inverse diagonals4,4,14 seal the entire core unless the independent short8 branch applies. A distinct D16(1,12) half and the new whole A4/A11 endpoint half then pay1 without the original D15 half.

These alternatives do not consume each other's new zeros. A15(1,13) remains a unit for the current A14 theorem. D15(1,12) remains a half for the current D14 theorem, and D15(0,10) remains a unit for the current ideal15 theorem. The current A16/D16/D17 and contextual coupons retain their stated marked roles. The A1/D15 node11 sixteenth is a cross-component role, distinct from the same-component zero.

The complete inherited low-core, high-disjoint, protected/adaptive and oversized-component routes remain valid. They cannot require a new long rank15 zero. The eligible native domains are536 patterns with 482 A15 maps and870 D15 maps, including failed searches; the eight and ten affected maps are all addressed above. This full routing and arbitrary-support argument is essential. The additional alignment of all216 current literal records, with zero new debit or credit, is an interface check rather than a universal face census.

Triangles retain lower payment2: their elementary short paths, protected features/leaves and exceptional whole factors are unchanged. Every quadrilateral retains lower payment1. Faces with at least five sides need no positive payment for the inherited five-minus-sides deficit inequality.

Complete finite upper783/2

The four new ordinary coupons cost1/2,1/2,2,2 per corresponding whole component pair. The A1/D13 representative allowance rises from2 to 33/16 and A4/D13 from7 to8. A1/A12 and A4/A11 fit their unchanged allowances5/2 and8. Actual A15 and D15 now both cost100, so their rank15 upper allowance is100. All32,400 ordered actual type-pair comparisons and every base comparison are verified simultaneously.

Every rank59 finite forest is covered by961 fixed large-component vectors. Rank5 multiplicities are fixed separately, giving2574 expanded vectors and interaction keys. Fixed rank17 means A17; residual D17 is explicitly included. The residual ranks are1 through4, 6 through8, and17 through59. Exceptional finite types and arbitrary rank2 labels are covered by the inherited representative classification.

The changed rank1 and rank4 interactions require fresh tables. The certificate supplies5148 ordinary/no-rank2 tables,128,700 exact item identities,6,723,288 all-last-item upper inequalities and303,732 attaining predecessor checks. These prove the additive upper by induction. All rank5 counts and all9191 contextual options are included.

The old contextual A2/A2 charge requires a whole A10 and two A1 factors; the new A2/D5 charge requires a whole A9 and one A1. Their upper guards permit any rank10/rank9 representative. The count fixes every possible A2 multiplicity and reserves one or two A1 factors as appropriate, using no-rank2 residual tables. The two-A1 branch includes both charges when both guards hold. The relaxed one-A1 branch is checked by reconstructing its full winning cost; it does not discard a contextual charge. Witness multiplicities never multiply a coupon.

The resulting upper for every actual forest is783/2. No expanded vector exceeds that value. This is a **representative upper**, not an asserted exact actual maximum. Its saved representative maximizer 8A1+3A5+3D12 has literal actual cost355.5. No geometric realization or actual attainment of391.5 is needed in the argument.

The global producer independently reconstructs the full fixed and expanded index domains and checks the saved upper results. The full Bellman proof and its separate replay are supplied by the frozen count; this binder does not repeat optimization or table generation.

Every ideal vertex retains surplus at least91/2

All ideal weights remain those of the current baseline. Its complete terminal rank58 domain has1890 profiles, allowing arbitrary repeated ranks at least9 and at most one smaller positive rank. The global producer independently reconstructs that domain and its bijection with the saved rows, then checks every saved h-polynomial identity:

```
V = product(r+1) = sum_j h_j,
G = sum_j min(j,58−j) h_j,
kappa = (5G−58V)/2.
```

The polynomials have degree58, positive integer coefficients and the required symmetry. Let C be the actual assigned cusp charge and C_upper its all-family envelope. Then C≤C_upper. Every terminal normalized margin(kappa−C_upper)/V is positive. Its minimum is1/340 at(9,9,9,15,16); the minimum absolute envelope surplus is91/2 at(58).

The inherited low9 concentration reduces V and does not increase the normalized margin. Positivity allows these two inequalities to be multiplied, giving absolute surplus at least91/2 for every original profile. Actual surplus kappa−C is at least the envelope surplus. No negative-margin multiplication or concentration of large factors is used. The established factor-fiber theorem covers affine products, so no simple-cusp assumption enters.

The dimension59 contradiction

Let v_f and v_i be the finite and ideal vertex counts of a proposed dimension59 polytope. The local payments and global shape-curvature inequality give upper and lower bounds on the same deficit quantity:

```
(783/2)v_f + sum_i C_i
    >= (783/2)v_f + sum_i kappa_i +4350.
```

The finite curvature is

```
(5 sum_{j=0}^{58} min(j,58−j) −58·59)/2
    = (5·841−3422)/2 =783/2.
```

The strict4350 correction is5 times the two-extrema difference sum_{j=30}^{58}(2j−58)=870. It is counted once. The producer checks these formulas independently and binds4350 to the preceding explicit necessary inequality2v_f≥4350+(91/2)v_i, rather than introducing an unbound constant.

Canceling the finite terms and using kappa_i−C_i≥91/2 yields

```
0 >=4350+(91/2)v_i >0,
```

a contradiction, including when there are no ideal vertices. Thus dimension59 is impossible. The completed unrestricted bound59 already excludes every dimension at least60; together these give dimension at most58. No assumption about face closure or construction of a lower-dimensional polytope is used.

Verification and scope

Generation chunk7463f3 and separate fresh replay chunk184de8 both terminated exit0/PASS. The binder checks the final frozen local/count/baseline reports, all source/data bindings, complete saved domains, ideal polynomial identities and exact curvature arithmetic. It does not rerun the historical all-label geometric census, supporting-root classification, convolutions, Bellman optimization or a polytope construction.

The root reviewer completed a full source/report mathematical composition read, including the completed bound59 report and both new local inherited proofs and the count. The construction reviewer completed full reads of this producer/report and the finite-closure producer/report. The exact-count reviewer completed a full read of this producer/report, having already read the finite closure and both local reports. Their checks cover the common frozen baseline, arbitrary-support routing and independent actual-corner priorities, retained alternatives, all coupon increments, the complete fixed/residual/contextual domain, representative upper versus actual cost, positive-only cusp-envelope extension, the single4350 correction and the completed59 higher tail. All three reviews passed without correction. These are mathematical source/report reviews; the reviewers did not execute this producer or repeat geometric maps, historical census, literal arithmetic, Bellman/DP computation or convolution/scalar replay. They do not constitute an independent reproof of every historical classification or shape-curvature theorem.

This is an internally checked computer-assisted mathematical proof. It is not an external refereeing or Lean/proof-assistant claim and produces no new finite-volume example. The separate compact arithmetic H13 optimization and bounded H9 construction experiments are not dependencies of this unrestricted dimension proof.

```
python3 scripts/fresh_a15_d15_dimension58_20261018.py --verify
```

Source 8

Source path: `results/fresh_a16_a17_full_payment_exchange_20261018.md`

SHA256: ced5478a43bd7245e29feb3006aa2cb36a3f684ff84c932435168b202f2669de

Finishing the A16/A17 exchange: costs110 and117

Lower the finite A16 halves(0,11) and(4,15) to zero, and restore the finite A17 halves(1,12) and(4,15) to units. The totals change from A16=111/A17=116 to **A16=110/A17=117**. Every ideal weight and every cross-component coupon is unchanged. Both diagram paths use consecutive node numbering starting at0; each pair set is invariant under reflection.

The source and certificate finish the two-corner exchange proved in the earlier rank16 theorem. The baseline is the current finite411/fractional rule. The additional A17 cost1 must be counted before any global conclusion; this packet performs no forest optimization and proves no dimension bound.

Complete selected maps and arbitrary-support replacement

All238 saved induced A16 maps in the complete394-pattern domain are validated from the native overlap Grams, including empty rows. Exactly four affected maps occur in contexts1657 and2067. In both, the marked component already has16 nodes and is the whole selected A16. There is no proper-completion case. The two changed pairs have distance11.

For each context the producer reruns the complete earlier `core_proof`, compares its exact result with the frozen certificate, and then changes only the two payments. It reconstructs the physical19-root norm2 Gram of signature(17,1,1), the radical, complete allowed contact domain, positive common-white and edge supports, and every finite corner.

The radical represents zero in the actual Lorentz space. Every allowed contact coordinate of an extra supporting white has nonzero radical coefficient. Nonzero acute contacts annihilating the radical must meet both signs; the complete mixed-sign checks give an independent short8 path. Otherwise every extra white is orthogonal to the entire core. A spectator bridge is excluded by applying the same argument to its first root meeting the core. Hence all displayed components used below are whole actual finite components, regardless of extra spectator rank.

The old proof supplies a selected A16 half and a distinct whole A17 half. Their literal marks are:

Context	Selected A16 corner	Distinct A17 corner
1657	1,15	2,15
2067	5,11	5,12

The complete canonical maps identify the A16 pair orbit(0,11)/(4,15) and the A17 pair orbit(1,12)/(4,15). Replace the selected half by0 and the distinct alternate half by1. Each face still receives1 without using an ideal corner, another same-component payment, or a new coupon.

Current rule and exact costs

The A16 weights are read from the current endpoint-transfer certificate and sum to111. The A17 weights are independently reconstructed from the 114-unit baseline, the earlier two restored halves, and the rank15 theorem's restored unit(1,15), giving116. Deleting two A16 halves saves1; raising two A17 halves to units costs1. The final full pair tables sum exactly to110 and117.

The rank15 proof's A16(1,13)/(2,14) halves and its A17(1,15) unit are explicitly preserved. D16 and D17 weights are unchanged by this exchange. In particular the independent new D16(1,11) zero theorem uses a distinct D5/A11 coupon and no A16 or A17 same-component payment; these two local changes have disjoint affected types and independent outputs.

All216 current literal configurations are reconstructed with their final cornerwise lower payments, including the order17 ideal quarters. None uses either changed A16 pair or restored A17 pair, so every inherited literal total is unchanged and remains at least1. This list records later local interfaces, not all possible quadrilaterals.

The complete selected/nonselected routing is the one in the earlier rank16 theorem. Low factors are too small; the protected leaf9/basic and adaptive bounded-component outputs do not use the changed distance11 pair. All other selected A16 images retain their old payment. Later rank15 alternatives are the retained half/unit sets just checked. Increasing A17 payments can only help other face inequalities.

Current independent ideal-corner priorities retain their short/basic paths or distinct typed coupons. The order15 and order17 transfers do not consume these long A16 pairs. If the radical alternative is a deep finite8 zero, the independent depth8 theorem gives short7; if that is a deep finite7 zero, the depth10 theorem gives short6. Their protected tree/cycle and small ideal outputs avoid the changed long pair. Thus the short-path alternative does not invoke the new exchange circularly. The finite7/ideal5 triangle theorem is likewise unchanged.

After these priorities, each changed selected context is paid by its distinct A17 unit. Faces with no changed actual A16 corner retain the preceding rule pointwise except for nonnegative A17 restorations. Every triangle still receives2 and every quadrilateral1. This theorem does not improve the completed global ceiling60 by itself.

Verification

Generation session1632, final chunkc25251, terminated exit0/PASS. Separate fresh replay session99017, final chunk8b8f22, also terminated exit0/PASS. Two peers completed full source/report mathematical reads, including the earlier support validation and whole-core proof: PASS. They checked arbitrary-contact radical sealing, the distinct whole A17 unit, exact current weight tables, retained rank15 roles, the complete saved-map and literal scopes, and independent short8-to-short7-to-short6 priorities. Neither repeated maps, inertia, literal or census arithmetic. The source validates the complete saved map domain, reconstructs both old radical proofs, both current pair tables and all216 literal records. It does not re-enumerate the historical all-label/contact census or perform a finite forest computation. No external referee or proof assistant is claimed. The source, certificate and report are frozen.

```
python3 scripts/fresh_a16_a17_full_payment_exchange_20261018.py --verify
```

Source 9

Source path: `results/fresh_a17_endpoint_d11_upgrade_forest_budget_20261017.md`

SHA256: 9472e4184f699c3c2383265da8ea6daec789bf1960cce72840f4e86a57a233e8

Exact finite budgets after the A17 endpoint deletion

The finite-corner rule with the whole-A17 endpoint unit removed has exact maximum544 in rank69. The same maximum holds after upgrading the two whole-A1/whole-D11-short-leaf halves to full units. Thus either explicit rule fits the rank69 finite curvature coefficient544 with zero slack. The D11 upgrade's geometric face coverage remains a separate obligation; this package proves only its finite-forest cost. No ambient dimension bound follows from this budget alone.

The source and certificate compute both variants separately in every rank1 through75. They retain the complete inherited coupons and use exactly the single proposed A17 endpoint deletion. The additional D11 rule is precise: for each marked pair in distinct whole components A1 and D11, upgrade weight1/2 to1 when the D11 mark is one of its two short fork leaves. Its fork-center weight remains1/2. All pairs are counted once.

Why rank17 requires two representatives

The previous exact budget proved componentwise type domination for the unchanged rule. Its rank17 representative A17 previously had the same base136 as B17/D17 and also two eligible cross-coupon positions. After removing one A17 pair, that argument no longer applies at rank17:

Type	Same-component cost	Broad cross-coupon positions
A17	135	2
D17	136	0

Keep **both** types. B17 is dominated by D17 with equal base cost and no coupons. All other rank representatives remain as before: A_r for rank1 through9, B10 at10, D_r at11 through16 and at least18. The D11 upgrade only strengthens D11's domination of other rank11 types. The small-type representatives A1/A2/A4 retain every special coupon. Noncrystallographic and exceptional finite types remain covered by the published finite-type classification and inherited same-rank domination.

For rank at most17 the unchanged base is binomial(r,2), except A17 now has135. For ranks at least18, the dominating D base is8r-32. There is no assertion that a maximizing abstract forest occurs at an actual finite vertex. Every actual finite vertex belongs to the counted class, which suffices for the upper bound.

Exact cost and complete maximization

Let a_i count A_i components for1<=i<=5, put `a=sum(i*a_i)`, let t count D11 through D16, and let u,b10,b11,b12,b13 count A17,B10,D11,D12,D13. The exact cost of a representative forest in the upgraded variant is

```
sum_components base(type)
  + a*(3t/2+u)
  + a1*b12 + a2*b10 + a4*b13
  + a1*b11.
```

Omit only the last term for the first variant. The broad three-mark D half-coupons and two-mark A17 half-coupons give the second term. The inherited D12 leaf upgrade, A2/B10 halves and A4/D13 neighbor halves give the next three terms. The new D11 addition has two leaves, each increased by1/2 against an A1, giving exactly `a1*b11`. This is an increment to the existing halves, so the same marked pair is never counted twice.

Enumerate every count vector of the eight coupon-bearing types `B10,D11,D12,D13,D14,D15,D16,A17` with total rank at most75. There are 1,608 vectors, as in the inherited complete enumeration. **D17 is not included in its A17 coordinate.** The residual additive components have ranks1 through9 or17 through75, represented by A_r in the first range and D_r in the second.

For each fixed large vector, double the costs. A residual rank r has base `r*(r-1)` at r<=17 and `16*r-64` at larger r. At r<=5 add `r*(3t+2u)`, and additionally `2*(b12+b11)` for r=1, `2*b10` for r=2, or `2*b13` for r=4. In the first variant set the upgrade coordinate b11 to zero while retaining D11 in t.

Only `(t,u,b10,b12,b13,upgraded_b11)` matters. There are377 tuples in the first variant and675 in the upgraded variant. For every tuple k, compute all76 Bellman values

```
F_k(0)=0,
F_k(s)=max_(allowed r<=s) (F_k(s-r)+weight_k(r)).
```

This is a complete induction over residual rank: delete any component to obtain a predecessor, and conversely attach the selected component to an attaining predecessor. Then maximize the large base plus the appropriate residual value over all feasible large vectors. Each saved maximum includes an attaining **typed** forest, checked by a separate direct evaluation of the full displayed cost formula. The type labels ensure an A17 coupon cannot be accidentally assigned to D17.

Results near the frontier

Both variants have identical maxima in every rank through75. In particular:

Rank	Maximum	One maximizing forest
66	513	D15+3D17
67	528	D16+3D17
68	544	4D17
69	544	A1+4D17
70	549	A2+4A17
71	555	A3+4A17
72	562	A4+4A17
73	570	A5+4A17
74	574	A1+A5+4A17
75	579	A2+A5+4A17

At rank69, A1+4A17 also attains544: its A17 base is540 and its retained cross-coupons total4. A1+4D17 attains544 without any cross-coupons. Keeping only the reduced A17 base at rank17 would have incorrectly underestimated the rank68 maximum, among other configurations.

At69 the finite curvature coefficient is544, so this local refinement removes the earlier finite deficit4. The existing ideal rank-profile inequalities remain an independent obligation. In a future global composition, zero finite slack can be sufficient together with positive ideal margins and the inherited strict positive global constant; this package does not establish those ideal margins.

Verification scope

Final generation and a separate fresh saved replay both terminated exit0/PASS after binding the frozen A17 local package. All22 direct bindings, syntax and5 local report links pass. The two runs exhaust the complete large-vector and Bellman domains, verify every attaining forest, compare the deletion-only result against the frozen old maximum, and compare the two new variants in all ranks. The finite-type domination proof is inherited except for the explicit two-representative

rank17 correction and the nonnegative D11 increment. There is no geometric realizability, new face coverage, external refereeing, formalization or new ambient dimension claim in this budget calculation. Root completed a full primary source and mathematical report read. It passed the A17/D17 distinction, unchanged domination at all other ranks, the precise `a1*b11` increment, all1,608-vector enumeration, complete residual rank domain and Bellman recurrence, typed attaining forests, and zero rank69 finite slack. The review did not infer any ideal or ambient dimension conclusion from the finite budget. Source, certificate and report are final/frozen.

```
python3 scripts/fresh_a17_endpoint_d11_upgrade_forest_budget_20261017.py
python3 scripts/fresh_a17_endpoint_d11_upgrade_forest_budget_20261017.py --verify
```

Source 10

Source path: `results/fresh_a17_endpoint_d11_upgrade_forest_budget_20261017_review.md`

SHA256: 72fe3b361fde7b834c3f1aff04086f337e689c3ee89d0f69c079173f831b330e

Separate certificate verification of the revised finite budgets

The reviewer independently verifies the two finite-budget certificates without importing producer arithmetic or any project module. Both rules have exact abstract finite-forest maximum544 in rank69. The second rule adds the whole-A1/whole-D11-short-leaf full upgrade; its geometric face coverage remains a separate obligation.

This is a separate algorithmic certificate reconstruction authored in the same agent lane. The producer source/report also received a separate parent-agent mathematical read. Neither activity is external refereeing or proof-assistant formalization.

The verifier constructs finite A/B/D component graphs directly. It counts every unordered marked pair by graph distance, whole-component rank and actual graph-leaf status; then it removes just the two A17 endpoints. Cross-component pairs are counted individually by the actual mark roles. An upgrade replaces a weight by the maximum required weight, so it cannot double-count an existing half. A17 has135 same-component units and D17 has136; their coupon eligibility is separately represented.

For each variant it checks221 classical type profiles and48,841 ordered partner comparisons against the claimed representatives. Exceptional finite types and arbitrary I2 occur only through rank8, where the base is complete and broad small eligibility is unchanged. Extra A1/A2/A4 coupons only favor the retained A representatives. The symbolic finite classification and all-rank component arguments remain mathematical inputs; this finite certificate review covers ranks through75.

The large-component domain is reconstructed by component number and nondecreasing multisets, independently of the producer's successive-rank count-vector recursion. It yields exactly the same1,608 vectors. For every resulting coupon tuple, the verifier creates an equivalent actual component environment and graph-counts every residual component's cost. It then checks every Bellman upper-bound alternative, and verifies the saved last-rank predecessor attains equality. Every candidate from every large vector is bounded by the saved global maximum. Finally, every attaining typed forest is counted directly from its marked pairs.

Check	A17 deletion only	With D11 upgrade
Independent large vectors	1,608	1,608
Residual tables	377	675
Bellman upper-bound alternatives	908,193	1,626,075
Attaining predecessors	28,275	50,625
Global upper bounds	22,504	22,504
Graph-counted typed witnesses	76	76

The76 witnesses include rank0. Both rank69 maxima are1088 in doubled units, and the finite curvature slack is exactly zero. The certificate does not claim a maximizing abstract forest occurs at a vertex of an actual finite-volume chamber.

Final generation0e43ab and separate fresh replaye771b5 against the frozen primary package both terminated exit0/PASS. All4 direct bindings, syntax and2 local links pass. Root completed a full independent-reviewer source and mathematical report read. It passed direct graph pair counting with max-based upgrades, exceptional/I2 domination scope, the independent multiset domain, equivalent coupon environments, every Bellman upper bound and attaining-predecessor interface, global upper bounds and typed witnesses. Both A17/D17 roles are retained, with no inference from sampled maxima. Root did not

duplicate the reviewer's arithmetic checks. Source, certificate and report are final/frozen. No new ideal cost, face coverage, ambient bound or finite-volume example is asserted here.

```
python3 scripts/fresh_a17_endpoint_d11_upgrade_forest_budget_20261017_review.py
python3 scripts/fresh_a17_endpoint_d11_upgrade_forest_budget_20261017_review.py --verify
```

Source 11

Source path: `results/fresh_a17_five_pair_alternate_transfer_20261018.md`

SHA256: 7a05ac84e36a33c738f816899828a4c45a39219f9fe27840bb7aab5f03d205ae

Five finite A17 pairs paid at alternate whole corners

Relative to the frozen D5/D12 rule, the whole finite A17 cost falls from467/4 to **451/4=112.75**. The saving is4 per component. Two explicit whole-component coupon families are added; their actual costs fit the existing representative upper costs. Triangles retain payment2 and quadrilaterals retain payment at least1. This is a local theorem, with no new finite forest maximum or dimension bound.

The producer and certificate bind and freshly replay the exact selected-corner pilot. That pilot is a diagnostic on its own. The complete current composition and arbitrary-support proof needed for the local theorem are given here.

Intrinsic rule and every new cost

Number A17 along0 through16. Make exactly these decreases:

Marked pairs	Old weight	New weight	Component saving
(2,12),(4,14)	1	0	2
(2,13),(3,14)	1	1/2	1
(2,14)	1	0	1

All sets and weights are invariant under path reversal. The distances are10,11 and12, so short8 and protected whole-leaf9 payments survive. All other A17 weights and every other finite same-component weight stay fixed. In particular the A17(0,13)/(3,16) units now used by the D17 leaf theorem remain units. The fractional pairs(1,12)/(4,15) and restored unit(1,15) also remain unchanged.

Add only these payments between distinct whole finite components:

- Either short fork leaf of D5 against either third node of A12: weight1. In canonical coordinates the roles are D5 nodes3/4 and A12 nodes2/9. Four marked pairs add4 per component pair.
- Whole A1 against either third or fourth node of A14: weight1/2. The A14 positions are2,3,10,11. Four marked pairs add2.

The D5 short leaves are distinct from its unique long endpoint0, used by every earlier D5/A12 coupon. The new actual D5/A12 total is27/4, including its existing11/4. A1/A14 previously had actual cost0 and now costs2. These are actual payments on whole components, not on arbitrary embedded subdiagrams. No new A4/A13 or A2/A14 cost is used.

Complete selected support

The complete finite A17 selected domain has140 eligible patterns in the inherited2,614-context high-overlap classification. The pilot freshly reconstructs every induced native embedding into A17, including failures: there are48 maps. Exactly five contexts have a changed marked image:

Contexts	Selected A17 pair	Other actual payment
1044,2057	(2,12) or(4,14), now0	New whole D5-short/A12-third unit
1040,2011	(2,13) or(3,14), now1/2	New whole A1/A14-fourth half
1009	(2,14), now0	Two distinct whole A1/A14-third halves

Every affected displayed selected component already has all17 nodes. There is no proper-completion case. Each corner's component types and all canonical marked images are computed from its native Gram matrix. The coupon corners are actual corners different from the selected one.

Context1009 is particularly useful: its two A1/A14 corners are distinct vertices of the quadrilateral, even though they have the same component types and marked orbit. Each contributes1/2, so the sum is1. No factor automorphism is counted as another corner. In the other rows either a distinct unit or a selected half plus distinct half gives1.

Whole-factor sealing for arbitrary extra roots

All five cases have the19-root L25/L25 physical norm-two core with signature(17,1,1). The pilot reconstructs its exact radical, the positive-definite15-root white support, all four strict edge principals and all four finite17-root corner principals. It also recomputes the selected corner and its complete necessary extra-white contact domain.

In any actual Lorentz realization, the Gram radical is a zero-vector relation. The displayed span contains a timelike vector; a vector both null and orthogonal to that span must vanish because the timelike orthogonal complement is positive definite. Every additional supporting white must annihilate this relation.

Every allowed contact position has a nonzero radical coefficient. Since actual root products are nonpositive, a nonzero extra contact must meet both radical signs. For every such pair the exact graph-distance check gives a path of length at most8 between adjacent black sides, with supporting whites inside. Truncation at consecutive black vertices gives an actual-corner path if needed. The independent current short8 route pays that branch.

Otherwise every additional supporting root is orthogonal to the entire core. A spectator chain could attach only through a first root touching the core, which is already excluded. Thus the displayed D5/A12 and A1/A14 factors at all paying corners remain whole with arbitrary supporting rank. This conclusion is stronger than merely knowing the selected A17 is whole and is proved separately by the radical argument. No upper bound on native contact magnitudes is imposed.

Complete current interfaces

The nonselected high-overlap domain is the inherited complete division: 2,391 cases with whole-component bound at most16,74 protected leaf9/basic ideal cases and nine adaptive cases. The remaining140 are exactly the selected domain just reconstructed. Low-core closures have bound12 and the high-disjoint branch is excluded. Smaller whole factors and protected distance9 roles cannot consume the changed A17 pairs.

The original finite A17 long-pair theorem's literal and later obligations remain part of the proof. The producer reconstructs every corner in all216 current literal records from the D5/D12 baseline. There is no debit and no new coupon credit in this list, so every recorded corner lower bound and every face total is unchanged. This does not assert that those216 records enumerate all quadrilaterals.

Later positive finite outputs are retained explicitly:

- The A16 exchanges and their fractional balance use A17(1,12)/(4,15), and the A15 endpoint exchange uses the restored A17(1,15) unit.
- The new D17 leaf theorem uses A17(0,13)/(3,16) units. Both are checked against the new A17 table and remain1.
- The D16 exchanges use their specified D17 node1 orbit, A16 halves, or whole-component coupons; the D15 and finite-rank14 refinements use their own smaller finite factors and D16 roles.
- The D17 node2 and node3 changes use A1/D5, A1/D14-center, A2/A14-third, A4/D13 and A1/A16 roles. None is changed here.

The producer directly checks stored positive whole-A17 corner arrays with physical Grams throughout its bound local inputs. None uses a changed pair. This is a supplementary interface check; the listed current output inventory and complete inherited routing give the universal composition, rather than an inference from a partial scan.

Apply the established independent actual-ideal-corner priorities first. Their universal tree/cycle, spectator and later order11–17 refinements retain their explicit short, protected whole-factor and typed-coupon outputs. The new order15 proof uses A1/D14 node9, D15(0,10), D5/D10, or the independent small ideal outputs; it spends no long finite A17 pair. The recent contextual rank13, order14 and D5/D12 arguments are likewise disjoint from these changes.

The short8 escape in the radical proof is independent: a finite8 deep zero invokes the established short7 replacement, and a finite7 deep zero invokes the depth6 short6/small whole-factor replacement. Its cycle11 half-weight exception has its own retained alternative. None uses a finite A17 marked distance10 or greater. Thus this argument does not appeal to the new theorem to pay its own short-path premise.

On the remaining selected finite branch, the five cases have the exact payments above. Other selected and nonselected routes retain their outputs. Faces with no changed actual A17 corner keep all old payments, with only nonnegative coupon additions. The elementary/protected triangle interfaces also remain unchanged. Consequently the simultaneous rule pays at least1 to every quadrilateral and2 to every triangle under the same finite-volume hypotheses as its predecessor.

Accounting, composition and scope

The source checks all32,400 ordered actual type pairs through rank59 under the existing simultaneous representative map. D5/A12 maps to A5/D12 with allowance21/2, which dominates the new27/4. A1/A14 maps to A1/D14 with allowance2, exactly its new cost. The separate ideal15 coupon raises the latter upper allowance further by1/16 when composed; the new A1/A14 payment does not use that extra room.

Actual and representative A17 bases both fall by64 in units1/16. A17 remains its own fixed species, separate from residual D17. Every representative base is checked again, as is symmetry of all actual cross costs. No representative cross cost increases. New payments are counted rather than declared absent because an upper allowance absorbs them. No complete finite optimization is performed in this packet.

The parallel D16 alternate-half rule, when included, consumes the unchanged A16(0,13)/(2,15) halves and adds A1/A15 third-node halves. These are disjoint from every A17 change and new coupon here. Its own proof and the complete simultaneous finite and literal accounting belong to the combined frontier certificate.

Final generation session10506/chunk877466 and separate fresh replay session44888/chunk3a2934 both completed with exit0 and `PASS_A17_FIVE_PAIR_ALTERNATE_TRANSFER`. The final producer explicitly checks the physical all-one gauge and norm-two diagonal of all five cores.

The arithmetic and ideal-lane reviewers each completed a full mathematical read of the promoted producer, its bound pilot and this report. Both passed the selected-domain, arbitrary-support seal, actual-corner, retained-output and simultaneous cost arguments. The ideal reviewer also checked the final physical-gauge guard. These reviews did not execute the historical census, embedding, inertia, literal or typed-comparison calculations. The author performed the final generation and separate fresh replay above.

The arithmetic uses exact rational Grams, complete native maps and current literal reconstructions. The historical all-label classification, necessary contact theorem, current short-path proofs and their universal scope remain bound inputs. No new ambient bound, geometric example, external refereeing or formal verification is claimed.

```
python3 scripts/fresh_a17_five_pair_alternate_transfer_20261018.py --verify
```

Source 12

Source path: `results/fresh_a2_d11_full_coupon_forest_budget_20261017.md`

SHA256: 7f83eb7152310eb29f0b0ca081ae666d744200cba2c4e8256ddf3b50aac63609

Conditional finite budget for full A2/D11 short-leaf coupons

Upgrading the existing A2/D11 short-fork-leaf half-payments to units leaves **every finite-forest maximum through rank75 unchanged** from the frozen D10/large-D budget. In particular, the maximum at rank66 remains495, attained by D66, with finite curvature slack0.

This is an exact conditional counting theorem. It does not prove the proposed coupon transfer or any ideal-pair reduction that uses it, and does not assert an ambient dimension bound.

Independent source · saved certificate

The precise change

A whole A2 has two possible marked roots, and a whole D11 has two short fork leaves. The four existing half-payments therefore have total cost2. Making each a unit raises that total to4, an increment of2 for each A2/D11 component pair. In doubled units this is exactly4a_2d_11.

All existing base costs and cross coupons remain. In particular the new D10 coupons still replace the dominated B10 contribution without adding it twice; the A1/D11 full upgrade is already present and receives no further increase. With the predecessor's notation the doubled representative cross cost becomes

$$a(2d_{10} + 3t + 2u) + 2a_1(d_{11} + d_{12}) + 2a_4d_{13} + 4a_2d_{11}.$$

The rank-preserving type-domination comparisons are repeated with this changed coupon function. Replacing a small rank2 component by A2 can add the new coupon and cannot lower the cost. Replacing an A11 or other rank11 component by D11 also remains valid. Thus the same representative types and complete1,608 large-component multisets apply.

Complete revised certificate

The source uses the predecessor's independently reviewed standard-library method and imports no project arithmetic. It generates the full large domain again, then compares each multiset, used rank, base cost and original three interaction coordinates with the frozen predecessor. The additional fourth coordinate d_11 distinguishes the new coupon; there are357 distinct interaction keys.

For each key, the residual costs of A1,…,A9 and D17,…,D75 are recounted from the original component-pair coupon function. Residual components still have no mutual coupons. Item-first unbounded-knapsack generation is verified by every last-item upper-bound inequality and a saved attaining predecessor. Every global candidate is bounded, and every reported maximum has a typed forest whose full component-pair cost attains it. Fresh replay checks the saved certificates without rerunning maximization.

Certificate part	Count
Large component multisets	1,608

Certificate part	Count
Revised interaction tables	357
Direct interaction-formula comparisons	109,344
Bellman upper-bound alternatives	860,013
Attaining Bellman predecessors	26,775
Global upper-bound alternatives	22,504
Typed global attainers	76

The checker additionally asserts equality of every global maximum with the predecessor, not merely that adding coupons cannot reduce it. Representative examples are:

Rank	Maximum	Typed attainer	Finite curvature slack
65	487	D65	−7
66	495	D66	0
67	503	D67	17/2
68	511	D68	16
69	520	A5+4D16	24
74	560	2A5+4D16	69
75	567	D75	161/2

The original same-component graph formulas and typed-cost exactness have not changed. The predecessor's all-rank D count, and the direct D18–75 graph checks repeated here, give D66 cost493+2=495.

Validation and scope

Final generation and a separate fresh saved-certificate replay both terminated directly with exit0/PASS. Logs are `/private/tmp/a2_d11_budget_gen.log` and `/private/tmp/a2_d11_budget_verify.log`. The geometric inputs to the predecessor are frozen and hash-bound. The proposed additional full A2/D11 transfer is deliberately an explicit hypothesis of this package; it is not silently treated as proved by the cost calculation.

The order16 reviewer read the entire successor source and report, including the final equality checks for all76 maxima. The precise four half-to-unit leaf increments, unchanged fork-center coupons, fourth interaction coordinate, rank2 residual cost, repeated type domination, Bellman/global upper bounds and attainment, predecessor comparisons and conditional scope all passed. This was a source/mathematical review, without numerical replay.

The parent reviewer read the complete successor delta and report against the already reviewed predecessor source. The doubled4a_2d_11 increment, fourth key, changed rank2 residual cost, repeated partner domination, complete global typed certificate and equality of all76 maxima passed. This review did not repeat the arithmetic. Twenty-two direct bindings, Python syntax and all three local report links pass. Source, data and report are final/frozen.

No ideal budget, new local face-payment theorem or global dimension conclusion is established here. A later composition may combine this exact budget with a separately completed coupon-transfer theorem and ideal-vertex comparison.

Source 13

Source path: `results/fresh_affine_d11_leaf_pair_transfer_20261017.md`

SHA256: c91b3c73c66507508dae9ff4fb3bbebdf3fea1dd22eaa72c9992cd1fc5a7d351

Four opposite-leaf transfers in affine D11

For an actual quadrilateral with a varying affine D11 factor of order12 at an ideal corner, each of the four opposite short-leaf pairs can be paid elsewhere after upgrading the whole-A1/whole-D11-short-leaf finite coupon from one half to one. The affine cost becomes62 instead of66. The other payments are retained paths of length at most5, whole finite components of order at most17, and inherited finite half-coupons. There is no arithmeticity, simple-cusp or spectator-rank hypothesis.

The source and certificate check the Schur identities and literal finite-corner configurations. This is a universal local transfer. Its new coupon must be included in any global finite budget; the A/B/C factors of order12 and any ambient dimension consequence remain separate obligations.

Actual star and end profiles

Use the inherited factor-fiber setup for W+a+b+c+d, with opposite pairs a,c and b,d. All roots have square2; the white and actual-edge diagrams are positive definite, corner diagrams are positive semidefinite, and the actual four-black Schur complement has determinant zero. At the chosen cusp, W+a+b splits as the varying affine factor F and a positive-definite spectator diagram Z orthogonal to F. Opposite walls have positive cusp height.

An adjacent black contact already pays at distance1. A shared spectator neighbor of c,d pays at distance2. Otherwise the native contact vectors x,y with Z have disjoint supports, and b.c=a.d=c.d=0. The opposite products a.c and b.d remain unrestricted native products.

Number the spine0–…–7, with leaves8,9 at0 and10,11 at7. Give the spine roots cusp coefficient1 and the leaves1/2. The effective resistance between opposite short leaves is7+2+2=11. Deleting a leaf leaves finite D11. The complete positive-semidefinite one-root extension list is isolated, simple attachment to its long endpoint (the old mate), doubled attachment there, or simple attachment to the next node (the old center). This follows from the finite/affine classification as in the preceding D12 transfer: exceptional types cannot contain D11, and multiple contacts would make a cycle together with a surviving fork.

For native opposite magnitude t, the end data are:

Profile	c0	Moment l	Height h	Cost g=2l/h+c0/h²
isolated	2	0	t/2	8/t²
mate3	1	1	(t+1)/2	4/(t+1)+4/(t+1)²
mate4	0	sqrt2	(t+sqrt2)/2	4sqrt2/(t+sqrt2)
center3	0	2	1+t/2	4/(1+t/2)

The checker directly eliminates the white part for all32 end profiles and64 ordered cross profiles. It verifies the resistance block, residual square c0+l²/11, far contact l/11, cross term lm/11 and determinant identity

```
det(S)/(h²k²)=1−(g_c+g_d)/11
             +(x/h−y/k)^T G_Z^(−1)(x/h−y/k)/11.
```

Every end has cost at most8. Equality occurs exactly when the new root is a simple leaf attached to either old short leaf, with height1/2: isolated t=1 or mate3 t=0. All other profiles have cost at most4. The separate diagrams Z+c and Z+d are positive definite as principal subdiagrams of actual edges.

A nonextreme end forces the extreme end's spectator contacts to vanish

Two nonextreme ends have total cost at most8<11 and cannot have zero determinant. If one end is extreme, the other needs cost at least3. Strict decrease of the displayed expressions and the native angle gaps leave precisely these five nonextreme choices:

Profile and opposite magnitude	Height	Cost	Required weighted spectator energy E
isolated, sqrt2	sqrt2/2	4	1
isolated, (1+sqrt5)/2	(1+sqrt5)/4	12-4sqrt5	9-4sqrt5
mate3, 1	1	3	0
mate4, 0	sqrt2/2	4	1
center3, 0	1	4	1

For isolated roots, positive height excludes t=0; labels at least6 give cost at most8/3. For nonextreme mate3, t>=1 and equality3 requires t=1. For mate4 or center3, any nonzero t>=1 gives cost below3. Thus this list covers arbitrary native finite labels, parallel and dotted opposite products.

Let x be the extreme end's contacts and y the nonextreme end's contacts. The scaled extreme contacts are2x. The nonextreme height is at most1, and E lies between0 and1. If x is nonempty and y is empty, inverse nonnegativity and diagonal lower bound1/2 give E>=4/2=2, impossible. If both supports are nonempty, take one node from each. Gram-metric Cauchy and disjoint supports give

```
E >= (2*x_i+y_j/h)^2/(4-2*Gij) > 9/8 > 1.
```

Here both native magnitudes are at least1, h<=1, and Gij>-2 in a finite diagram. This is again impossible. Therefore x=0. When E=0, positive definiteness also forces y=0.

Choose the adjacent corner containing the extreme new root and the far old mark, excluding the nonextreme new root. Every spectator is orthogonal to this chosen corner's displayed F-based components. If the extreme root attaches to the selected old leaf, that leaf is absent: the new root is a whole A1 and the far mark is a short leaf of whole D11. This gets the upgraded unit coupon. If it attaches to the old mate, the component is whole finite D12 and receives the inherited unit. In particular the center3 case never requires repayment at a second deleted affine pair.

Both ends extreme

Both heights are1/2 and both costs8. The determinant identity now gives

```
(x-y)^T G_Z^(-1)(x-y)=5/4.
```

If one support is empty, use the adjacent corner containing the spectator-free new root and the far old mark. The same whole A1/D11 unit or whole D12 unit pays, regardless of the other contact vector. The source checks all32 literal choices of old marks, leaf attachments and excluded black. These checks concern the selected finite corner; they do not claim that the displayed core without spectators is itself a realizable full face star.

If both supports are nonempty, the five-quarters spectator theorem gives either a path of length at most5 between c,d or exactly one whole A1 and one whole A3, contacted simply at leaves from opposite sides. The short path retains its finite8 or ideal5 unit.

For A1+A3, the checker constructs all32 literal configurations, including which new black contacts A1. Their18-root Grams have signature(16,1,1), their white diagrams and all four edge diagrams are positive definite, the old corner is affine, and all three new corners are finite. Every other spectator is orthogonal to this core. If a new root attaches to the selected old leaf, its adjacent corner has a whole A2 or A4 against a short leaf of whole D11, receiving an inherited half. An attachment to the mate instead joins the marks in a whole finite component of order at most17 and receives a unit. The saved corner totals are at least one in every case; the new full A1 coupon is not used here.

Payment interface and validation

The removed pairs are(8,10),(8,11),(9,10),(9,11), all at distance9. All other62 pairs remain at unit weight. The transfer applies to every actual quadrilateral using such a pair, including low-core, selected-core, protected and immediate-factor uses. The short paths and finite payments used in its conclusion do not call back to a removed ideal pair. Elementary triangle payments are not altered by this local theorem; their retained inclusion is a separate global-composition check.

Generation and fresh separate-process saved replay both terminated exit0/PASS on all32 end identities,64 cross identities, five nonextreme boundary profiles,32 arbitrary one-sided corner checks and 32 A1+A3 configurations with96 finite corners. A separate full internal source and mathematical review passed the complete native profile list, scaled Cauchy argument, whole finite-component choice without recursion, height-before-energy reduction and the sealed A1+A3 corner payments. The reviewer also checked compatibility with the finite A17 endpoint deletion: these new finite corners have at most16 roots. That review did not duplicate the32/64 arithmetic calculations. Source, certificate and report are final/frozen. No ambient dimension improvement or new finite-volume example is asserted by this package alone.

Source 14

Source path: `results/fresh_affine_d12_leaf_pair_transfer_20261017.md`

SHA256: dc8a3bfcfafdf3c951cb668e7c758237194fc0a3a83a55bdf638e728a82a0370

Four opposite-fork-leaf transfers in affine D12

Let an actual quadrilateral of a finite-sided finite-volume hyperbolic Coxeter polytope have a varying affine D12 factor F of order13 at an ideal corner. If its two marks are short leaves at opposite forks, its ideal unit may be removed after adding the full finite whole-A1/whole-D12-short-leaf coupon. The quadrilateral is then paid by a retained short path, a whole finite component of order at most17, this new full coupon, or two inherited halves. All four cross-fork-leaf pairs are covered. There is no arithmetic, simple cusp, facet-count or spectator-rank restriction.

The source and certificate verify the exact Schur identities and boundary configurations. The additional coupon fits the joint finite8 budget through ambient dimension75. This is a universal face-local transfer, so it also covers the immediate affine-D12 alternatives in earlier proofs, not only the saved preferred-core contexts. No new ambient bound is asserted here.

Actual face and complete profiles

Use the inherited factor-fiber setup for a face star W+a+b+c+d, with opposite pairs a,c and b,d. The white diagram and each white-plus-one-black edge diagram are strictly positive definite. Every adjacent corner is positive semidefinite. At the chosen ideal corner, W+a+b splits as F plus a positive-definite spectator diagram Z orthogonal to F. The actual four-black Schur complement has determinant zero. These properties hold also at a nonsimple ideal vertex.

Adjacent black contact already gives a path of length1 and a retained unit. Likewise a shared spectator contact of c and d gives a path of length2. Assume neither occurs. Thus b.c=a.d=c.d=0, and the nonnegative native contact vectors x,y of c,d with Z have disjoint supports. The opposite products a.c and b.d remain unrestricted native products, including parallel and dotted cases. Every nonzero native magnitude is at least1; if it exceeds1, it is at least sqrt2.

Number the spine of F as0–1–…–8, with short leaves9,10 at0 and11,12 at8. The marks are a in{9,10}, b in{11,12}. Give spine roots cusp coefficient1 and short leaves coefficient1/2. The scaled affine Gram is a Laplacian, and its resistance between a and b is8+2+2=12.

Deleting a short leaf leaves finite D12, with its long endpoint the mate of that leaf and its second node the old fork center. At this rank its complete one-root positive-semidefinite extensions are: an isolated root, a simple bond to the long endpoint, a doubled bond to that endpoint, or a simple bond to its second node. This is the same finite/affine classification used in the larger-D transfer. Exceptional finite or affine types cannot contain D12. Two contacts would create a cycle together with an existing fork and cannot give a finite or affine component. Thus this is an all-label classification, not a search with a maximum label or supporting rank.

Put t=-a.c, and call the mate L and the old center0. Eliminating F minus a,b gives the following complete left-end data; the right end is reflected.

Contacts in F minus a	Residual constant c0	Moment l	Height h	Cost g=2l/h+c0/h²
none	2	0	t/2	8/t²
L by3	1	1	(t+1)/2	4/(t+1)+4/(t+1)²
L by4	0	sqrt2	(t+sqrt2)/2	4sqrt2/(t+sqrt2)
0 by3	0	2	1+t/2	4/(1+t/2)

The residual square is c0+l²/12, the contact with the scaled far mark is -l/12, and the cross projection of the separated end

profiles is lm/12. The checker verifies these statements for all32 end profiles and64 ordered cross-profile pairs and verifies the formal determinant identity including spectators:

```
det(S)/(h²k²)=1−(g_c+g_d)/12
             +(x/h−y/k)^T G_Z^(−1)(x/h−y/k)/12.
```

Each end has g<=8. Equality occurs exactly when c is simply attached to one of the two old short leaves: either t=1 in the isolated profile or t=0 in mate3. These two extreme profiles both have h=1/2. All other profiles have g<=4. This follows from the native gap and the strictly decreasing expressions in the table, with isolated t>=sqrt2, nonextreme mate3 t>=1, and the last two profiles t>=0. In the isolated profile h>0 forces t>0: an opposite wall does not contain the chosen cusp. Thus the native lower bound t>=1 applies there.

A nonextreme end: the zero boundary seals the spectators

If at least one end is nonextreme, g_c+g_d<=12. If it is smaller, the normalized determinant is strictly positive even before the nonnegative spectator correction, which is impossible. Equality requires one extreme end of cost8 and one nonextreme end of cost4. The latter is precisely: isolated with t=sqrt2; mate4 with t=0; or center3 with t=0.

The spectator correction must vanish. Positive definiteness gives x/h=y/k; their disjoint supports then force x=y=0. Thus every spectator is orthogonal to F and both new roots, and cannot enlarge a displayed corner component.

Choose the adjacent corner containing the extreme new root and the old mark at the opposite end, excluding the nonextreme new root. If the extreme root was attached to the selected old leaf, that selected leaf is absent: the new root is a whole A1 and the other mark is a short leaf of whole D12. This receives the new full coupon. If it was attached to the mate instead, the long arm extends to whole finite D13, and both marks receive the old whole-component unit. Adding the orthogonal positive-definite Z leaves this actual corner finite.

There are48 literal cases: four old mark pairs, two choices of the extreme profile, three nonextreme equality profiles, and two orientations. The checker reconstructs every Gram, its signature(13,1,1), white and edge positivity, all corner semidefiniteness, and the chosen whole finite components. Doubled bonds are made rational by scaling only the relevant new root by sqrt2; its diagonal is4 and its single bond entry is-2. The chosen corner excludes that root, so its type and mark checks use the unscaled simple-bond Gram directly.

Both ends extreme: an arbitrary one-sided spectator needs no classification

Both new roots are simple leaves attached to old fork leaves and h=k=1/2. The determinant identity becomes

```
0=−1/3+D/3,  D=(x−y)^T G_Z^(−1)(x−y),
```

so D=1. This unscaled equality is used only after the heights have been proved equal to1/2.

If y=0, choose the corner containing d and a, which excludes c. All of Z is orthogonal to F and d, regardless of x, its support, or the ranks and types of its components. Exactly as above, this corner has whole A1 and D12 with the required short-leaf mark if d attaches to selected b; if it attaches to the mate, it has whole D13 containing both marks. It is finite and receives a unit. The case x=0 is reflected. No classification of the nonzero one-sided contact vector is needed. The32 literal choices in the checker verify this whole-component statement; their arbitrary spectator block is handled by the stated orthogonality argument.

Two nonempty supports: a short path or two sealed halves

The separate augmented diagrams Z+c and Z+d are strictly positive definite, as principal subdiagrams of the actual edge diagrams. Therefore the complete energy-one boundary theorem applies. With disjoint nonempty supports and D=1, the only possibilities are unit contacts on two distinct whole A1 components, the two endpoints of a whole A3, or the short fork leaves of a whole D4 through D7. Every unmentioned spectator component has zero contacts.

For A3 or D4 through D7, the two contacted nodes have distance2. Adding c and d gives a marked path of length4 at their adjacent actual corner. This is retained by both the finite8 and ideal5 rules, independently of any other supporting roots.

For two distinct A1 components, the literal core consists of F, c,d and one separate simple spectator leaf on each of c,d. Every other spectator is orthogonal to this17-root core. All16 choices of old marks and attachment leaves have signature(15,1,1); their three new corners are finite. If the new root attaches to the selected old leaf, the corresponding adjacent corner has a whole A2 against a short leaf of whole D12 and receives the inherited half. An attachment to the mate instead extends the long arm to D14 and receives the old whole-component unit. The checker gives four cases with two halves, eight with a unit and one half, and four with three units. Each receives at least one, with no use of the new full A1 coupon.

This exhausts all possible spectator behavior. Every use of a finite coupon has established whole actual components, rather than only a finite subdiagram inside an ideal corner.

Payment interface and verification

The four removed canonical pairs are(9,11),(9,12),(10,11),(10,12), each of graph distance10. The frozen order13 diagnostic identifies exactly these four as the difference between the78-pair D12 rule and its74 basic pairs. All basic and elementary-triangle payments are therefore retained. Since the transfer is universal, it covers every induced selected-core use and the `inner_next3` immediate alternative of the older affine-D14 proof. It does not need to enumerate those provenance cases again.

Generation and a fresh separate-process saved replay both terminated exit0/PASS on all32 end identities,64 cross identities,48 nonextreme configurations,32 arbitrary one-sided corner checks and16 two-A1 configurations with48 finite corners. A separate full internal mathematical and source review passed the nonextreme sealing, arbitrary one-sided shortcut, height reduction before the energy-one theorem, strict edge hypotheses and whole finite component arguments. It did not repeat the96 literal configurations/corner checks. Direct source bindings, syntax and report links pass. Source, data and report are final/frozen. No external refereeing, proof-assistant formalization or new finite-volume construction is claimed.

```
python3 scripts/fresh_affine_d12_leaf_pair_transfer_20261017.py
python3 scripts/fresh_affine_d12_leaf_pair_transfer_20261017.py --verify
```

Source 15

Source path: `results/fresh_affine_d13_long_pair_transfer_20261017.md`

SHA256: 0639ef2338507ee9454eaf135c7b1bb48bf4c7adce2b4a0bfd89a7b234ce6016

Universal long-pair transfer in the14-node affine D13 factor

Completed internally checked local theorem; verification scope below. Let an actual quadrilateral of a finite-sided finite-volume hyperbolic Coxeter polytope have an ideal corner whose varying affine factor is D13, of order14. Take one mark from each fork triple (the fork center and its two short leaves). There are nine pairs, at distances9,10 and11.

Each such quadrilateral is paid elsewhere by finite distance at most8 or a whole finite marked component of order at most17; ideal distance at most5 or a whole varying affine factor of order at most13; or two sealed finite half-coupons. The latter join a whole A1,A2,A4 or A5 component to a whole D11 or D13 component marked at its fork center or a short fork leaf. The finite-eight coupon budget includes these types, with the small side enlarged to rank at most5.

This permits removal of the nine specified ideal pairs, reducing an otherwise all-pair D13 factor cost from91 to82. It is a universal local theorem, not a claim that all other pairs in every small factor are paid, or that all finite-eight overlap cases have been covered. No new ambient bound or finite-volume construction is asserted.

The source and certificate recompute the14-node profile formulas, the newly non-strict cases and all72 extreme equality configurations. They preserve the frozen D14 predecessor.

Actual star and normalization

Write H=W+a+b+c+d, with opposite pairs {a,c},{b,d}. All roots have square2. W and each actual edge diagram W plus one black are positive definite. Each corner is positive semidefinite. The full actual face star has corank1 in its Lorentz realization.

At the selected ideal corner, the factor-fiber theorem gives the varying affine factor F containing a,b, plus a positive-definite spectator diagram Z orthogonal to F. The positive affine kernel represents the actual cusp, and the opposite sides have strictly positive heights. No simple-cusp or arithmetic assumption is needed. A nonzero adjacent black product already pays at distance1. A shared spectator neighbor of c,d gives a paid path of length2. These alternatives can be removed from the unpaid analysis.

Number the simple spine0,…,9, with left fork leaves10,11 and right leaves12,13. The affine kernel nu has coefficient1 on the spine and1/2 on the leaves. The marks range over a in{0,10,11} and b in{9,12,13}. The effective resistance is

```
R=9+2*[a is a leaf]+2*[b is a leaf].
```

The inherited all-rank Green identities apply at these ranks. The source also directly eliminates F minus the two marks, checking its normalized old2x2 block,96 individual end formulas and256 separated contact cross terms. For each end write its height h, moment l and norm constant c0. Its scalar cost is g=2*l/h+c0/h^2. The normalized determinant before spectators is

```
F0=1-(g_c+g_d)/R.
```

Writing x,y for the native nonnegative contacts to Z, spectator elimination adds

```
(x/h-y/k)^T G_Z^(-1)(x/h-y/k)/R >=0.
```

This is the same formally checked Schur identity as the predecessor; no small-affine exclusion from an earlier simple-cusp proof is imported.

Complete end profiles

Deleting a short marked leaf leaves a large finite D13 diagram. Its complete one-root positive-semidefinite extensions are isolated, simple attachment at its long endpoint (the old mate), doubled attachment there, and simple attachment at the next node (the old fork center). An exceptional finite/affine extension cannot contain this D13. A multi-contact cycle is incompatible with the surviving remote fork in any finite/affine classification. For native opposite magnitude t, the exact table is unchanged from the predecessor:

Profile	c0	l	h	Cost bound
isolated	2	0	t/2	8, equality only t=1
mate3	1	1	(t+1)/2	8, equality only t=0
mate4	0	sqrt(2)	(t+sqrt(2))/2	4
center3	0	2	1+t/2	4

The isolated height must be positive. Its non-extreme native magnitude is at least sqrt(2), giving cost at most4. The non-extreme mate3 has t>=1 and cost at most3. Thus the two extreme leaf profiles have height1/2 and are precisely a new simple leaf attached to either old fork leaf; every other leaf profile has cost at most4.

Deleting a marked fork center leaves D11 plus two isolated A1 roots. The large block has a long endpoint and its second node. A new root can be isolated from the large block, attach simply or doubly at the endpoint, or attach simply at the second node. The doubled-end and second-node alternatives produce a12-node affine component containing the new root and the far mark. The source verifies their positive kernels. Further acute contacts to that component vanish by its kernel, so the actual varying factor has order12 and is paid by the small13 rule.

If the new root attaches simply at the long endpoint, its further connections to the two old A1 roots are exactly: neither; one by3; one by4; or both by3. The large finite block excludes H/F/E alternatives, while an affine component's positive kernel forbids an extra acute contact. All these are the same complete finite/affine lists used in the D14 proof, with the large ranks now11 and13, still beyond every exceptional finite diagram.

For completeness, arbitrary contacts with only the two outer A1 roots are not replaced by a finite label search. If their magnitudes are u,v and the native opposite magnitude is t, then

`c0=2-(u^2+v^2)/2,  l=0,  h=t+(u+v)/2.`

The actual edge diagram gives $u^{2+v}2<4$. With no outer contact, positive height gives cost at most2. With one outer contact and t>0, the cost is at most2/3. With t=0 and a single magnitude u, the cost is8/u^2-2: its maximum6 occurs only at u=1, and otherwise the native gap gives cost at most2. With two nonzero outer contacts, h>=1 and c0<=1, so the cost is at most1. Thus the unique extreme center profile is a new simple leaf attached to either old outer leaf, native opposite contact zero, and height1/2.

All non-extreme center profiles have cost at most3. Equality3 occurs only for the simple inner-end attachment, with native opposite contact zero and height1. The other joined profiles have bounds14/9,4-2sqrt(2),1; an isolated center profile has bound2. These formulas cover every finite label and parallel or dotted native opposite contact. The source retains the14 exact scalar formula/derivative checks and recomputes the present pair margins.

The new non-extreme zero cases

The resistance and bounds are:

End types	R	Extreme sum	Maximum with a non-extreme end
center / center	9	12	9
center / leaf, either order	11	14	11
leaf / leaf	13	16	12

A strict inequality in the last column gives F0>0 and contradicts actual corank1 after the nonnegative spectator correction. Equality in the first two rows requires a center inner-end3 profile of cost3 at height1, and an extreme opposite end. Every other non-extreme combination has a strict gap.

For these zero cases, the spectator correction must vanish. Positive definiteness gives x/h=y/k. If either vector is nonzero they have a common support coordinate and the path c-e-d pays. Otherwise both vanish: every spectator is orthogonal to F,c,d.

At the adjacent corner obtained by deleting that marked center, the new simple inner-end root extends D11 to D12. Both marks lie in this whole finite D12 component. Orthogonality of every remaining spectator prevents enlargement or joining. It is paid by the small17 finite rule. The source checks all12 oriented configurations, their full signatures(14,1,1), positive-definite white blocks, all corners and the specified whole D12 unit payment. There is no assumption that a zero relaxed determinant alone is impossible.

Extreme heights and the three-quarter spectator equality

It remains that both ends are extreme. Their heights are **first proved to be h=k=1/2** by the native profile analysis. Their cost sum is R+3. Thus the zero-determinant equation becomes

`-3/R + 4D/R=0,    D=(x-y)^T G_Z^(-1)(x-y),`

and requires D=3/4. This is an unscaled native-contact quantity only after the height reduction.

A shared spectator neighbor is already paid. If both disjoint supports are nonempty, choose i in one and j in the other. If i,j are not adjacent, Gram-metric Cauchy gives D>=(x_i+y_j)^2/4>=1, impossible. If they are adjacent, the actual white-interior path c-i-j-d has length3 and pays under finite-eight or ideal-five. Consequently an unpaid extreme case has exactly one nonempty spectator support.

The frozen strict spectator spectrum applies: Z+c and Z+d are positive definite principal subdiagrams of actual edge diagrams. Its one-support D=3/4 equality forces one unit contact at an endpoint of a **whole simple A3 component**, with every other spectator contact zero. Hence all other supporting spectators are orthogonal to the entire equality core. The possible two-support A3 equality from the general spectrum was not silently discarded; it has the already paid length-three path just described.

All72 sealed equality configurations

The core now consists of the14-node affine factor, two simple leaves c,d attached to old fork leaves, and a simple A3 chain attached at one endpoint to c or d. There are nine mark pairs, four choices of the old attachment leaves and two choices of the spectator side, giving72 configurations.

Each19-root Gram has signature(17,1,1); its15-root white block is positive definite. The original corner is affine F with finite spectators, and the other three corners are finite. The source reconstructs all216 finite corner diagrams and classifies their whole marked components. Their half-coupon types are exactly:

Small component	Large component	Occurrences
A5	D11	24
A2	D11	24
A4	D13	24
A1	D13	24

Every large mark is a fork center or short fork leaf. The A5 case is why the safe budget must permit small rank5: deleting an old fork center retains an outer leaf, the new root and the three spectator-chain roots. When a marked fork leaf is deleted, the corresponding small component is A4 instead. These are whole finite components, not truncated fragments, because the spectator equality has sealed the entire core.

Exactly32 configurations have two halves and no unit;32 have a small17 unit and an additional half;8 have three small17 units. Thus every face receives at least one unit in total. The finite budget already admits all four types. This concludes the universal local transfer, independently of whether any particular core by itself defines a finite-volume chamber.

Verification scope

Generation and a fresh separate-process saved replay both terminated exit0/PASS for96 end identities,256 cross identities, two affine12 kernels,14 native-cost rows, all12 non-extreme zero cases and72 extreme cores with216 finite corners. Syntax, all14 direct source hashes and all6 report links passed. The source reuses the frozen predecessor's profile formulas and cost functions but recomputes the present14-node Grams and all new cases; it is a new producer, not a claim of source-independent arithmetic review. A separate full mathematical/source read found no gap in the complete D11/D13 profile reuse, all-label outer bound, normalized zero-case sealing, height-first three-quarter equality, short-path two-support alternative or whole-component coupon roles. That read did not independently repeat the96/256/72 calculations or review a global composition. The all-label classification and spectator equality remain explicit analytic proof steps. Source, data and report are final/frozen. No complete finite-eight overlap theorem, ambient dimension improvement, external refereeing or formalization is claimed.

Source 16

Source path: `results/fresh_affine_d14_long_pair_transfer_20261017.md`

SHA256: 09023a80827069078ebef7b32b7defaca0e8e9035ffa5e9e338558afe7aed8cc

Transferring all nine long pairs of an affine D14 factor

Completed local proof; verification status is recorded below. Let an actual quadrilateral in a finite-sided finite-volume hyperbolic Coxeter polytope have an ideal corner whose varying affine factor is D14, of order15. Choose one marked root from each fork triple, where a fork triple consists of a fork center and its two short leaves. There are nine such pairs, at distances10,11,12.

Each such quadrilateral has another unit-paid corner under the retained rules

- finite: marked distance at most9, or the marks in one whole connected component of order at most17;
- ideal: marked distance at most6, or the varying factor of order at most14;

or it has two sealed finite corners, each eligible for a half-payment between a whole small component of rank at most3 and a whole D12 or D14 component, with the latter mark its fork center or a short fork leaf. Consequently these half-payments permit removal of all nine specified D14 ideal pairs. Additional retained payments only strengthen this conclusion.

This is a universal face-local theorem, not an enumeration of the63 inherited preferred-core contexts. It permits arbitrary supporting white roots and nonsimple ideal vertices. No total finite-vertex budget or ambient dimension claim is made here. The checker and certificate verify the literal formulas and all equality configurations.

Actual face setup

Write the star as H=W+a+b+c+d, with opposite pairs {a,c} and {b,d}. Normalize roots to square2. The white diagram W and every actual edge diagram W plus one black root are positive definite. Each adjacent corner is positive semidefinite, and the full face star has corank1 after retaining its actual Lorentz span.

At the selected ideal corner W+a+b, the factor-fiber lemma gives the varying affine factor F containing a,b, plus a positive-definite spectator diagram Z orthogonal to F. The positive affine kernel of F represents that actual ideal point. The two opposite sides do not contain the point, so their heights h,k are strictly positive. This does not assume that the full ambient ideal vertex is simple.

If any adjacent black product is negative, that marked distance1 corner is already paid. Hence assume b.c=a.d=c.d=0. A common nonzero spectator contact of c,d similarly gives a paid path c-e-d of length2. No restriction is imposed on the opposite products a.c or b.d. Each is either zero or has magnitude at least1; if its magnitude is greater than1, it is at least sqrt(2). Parallel and dotted products are included.

Number F by the simple spine0–1–…–10, left leaves11,12 attached to0, and right leaves13,14 attached to10. Normalize the cusp coefficients to1 at all spine nodes and1/2 at the four leaves. The resulting scaled Gram is a graph Laplacian: spine conductance1 and fork-edge conductance1/2. For a in {0,11,12} and b in {10,13,14}, the resistance is

```
R=10+2*(a is a leaf)+2*(b is a leaf).
```

Eliminating F minus a,b leaves R^(-1)[[1,-1],[-1,1]] on the scaled marked roots. For a replacement near the a end, write its residual square as c0+l²/R and its cross contact with the scaled b as -l/R. The complete profiles below have separated left/right supports, so the projection cross term is lm/R. The weighted-tree Schur identity therefore gives

```
det(S)/(h² k²) = 1−(g_c+g_d)/R,
```

g_c=2l/h+c0/h².

Only its normalization, finite/affine classification and Green identities are used; none of that earlier report's global small-affine exclusions is imported. This source recomputes the end data for all nine present marked pairs and the separated projection term for every ordered profile pair, and rechecks the formal determinant identity.

Eliminating Z subtracts a positive-semidefinite matrix from the c,d block and adds

(1/R)*(x/h−y/k)^T G_Z^(−1)(x/h−y/k)

to the normalized determinant, where x,y are the nonnegative native contact vectors of c,d with Z. The heights are unchanged. Thus a strictly positive spectator-free determinant is already impossible in an actual quadrilateral.

Complete fork-leaf profiles

Suppose a is a short fork leaf, with mate L and center0. Deleting a leaves finite D14, with long endpoint L and second node0. Its complete one-root positive-semidefinite extensions at this rank are: an isolated root; a simple bond to L; a bond4 to L; or a simple bond to0. Finite D can extend only at its long end; the last two possibilities are the connected affine B/D extensions. A bond5 or an exceptional finite/affine extension cannot contain the old D14. Multiple contacts would create a cycle in a graph already containing the remote fork, which is not a finite/affine type. This is the same complete large-D extension list used in the weak-profile proof.

Put x=-a.c. The exact data are:

Contacts of c in F minus a	c0	1	h	Cost bound
none	2	0	x/2	8 at x=1; at most4 otherwise
L by3	1	1	(x+1)/2	8 at x=0; at most3 otherwise
L by4	0	sqrt(2)	(x+sqrt(2))/2	at most4
0 by3	0	2	1+x/2	at most4

In the isolated profile h>0 implies x>0. The native gap gives either x=1 or x>=sqrt(2). In the mate3 profile, either x=0 or x>=1. The costs decrease with x, and the displayed bounds follow at these exact endpoints. Thus

leaf end: g<=8, and every nonextreme profile has g<=4.

The two extreme profiles both say that c is a simple new leaf attached to one of the two old fork leaves: either the selected leaf a or its mate L. In both cases h=1/2. No small affine diagram is being forbidden to obtain these bounds.

Complete fork-center profiles, including all finite labels

Suppose a=0. Deleting a leaves a large finite D12 block B on nodes1,…,10,13,14, together with two isolated A1 leaves11,12. Its long endpoint is1 and its second node is2.

First consider the principal diagram B+c. By the complete large-D extension list, c is isolated from B, is joined to1 by3 or4, or is joined to2 by3. The latter two give connected affine components of order13. A positive affine kernel forces every other acute root of the actual corner to be orthogonal to that component. It is therefore the entire varying factor containing the marks b,c, and is already paid by the retained whole-factor cutoff14. The source checks both literal positive kernels.

If c is joined to1 by3, the component is finite D13. Adding its possible contacts with the two old A1 leaves gives precisely: neither leaf; one leaf by3; one leaf by4; or both leaves by3. This is again a direct finite/affine classification: the large component excludes H/F/E, while an extra doubled terminal gives an affine B component whose positive kernel forbids another contact. The both-simple case is affine D14 of order15 and is retained in the analysis. The contact x=-a.c remains arbitrary because a is absent from this adjacent corner.

The four unpaid large-block profiles have data

Additional outer contacts	c0	l	h	Maximum cost
none	1	1	x+1	3
one leaf by3	1/2	1	x+3/2	14/9
one leaf by4	0	1	x+1+sqrt(2)/2	4-2sqrt(2)
both leaves by3	0	1	x+2	1

It remains to allow arbitrary contacts with the two A1 leaves while c is orthogonal to B. Write their magnitudes as u,v>=0. Exact elimination gives

$c0=2-(u^2+v^2)/2,\ l=0,\ h=x+(u+v)/2.$

The actual edge W+c is positive definite, so $(u^2+v^2)/2<2$. This strict edge condition is needed; the corner condition alone would not suffice. The following bounds require no label cutoff:

- With no outer contact, h=x>0, hence x>=1 and g<=2.
- With one outer contact and x>=1, c0<=3/2 and h>=3/2, hence g<=2/3.
- With one outer contact and x=0, $g=8/u^2-2$. It is6 only at u=1; otherwise u>=sqrt(2) gives g<=2.
- With two nonzero contacts, u,v>=1, so h>=1 and c0<=1, giving g<=1.

Consequently

$\text{center end: } g<=6, \text{ and every nonextreme unpaid profile has } g<=3.$

The unique extreme is a simple new leaf c attached to either old outer fork leaf, with x=0. Again h=1/2. The case of an arbitrarily large opposite product and every admissible finite outer label has been covered.

Strict nonextreme gaps and the discrete equality

For center/center, center/leaf and leaf/leaf ends, the resistances are10,12,14. If at least one end is nonextreme, the cost sums are at most9,11,12, respectively. Thus the normalized determinant is at least1/10,1/12,1/7. Spectators cannot cancel these positive gaps.

Both ends must therefore be extreme. The old factor now has a simple new leaf c on one old left fork leaf and a simple new leaf d on one old right fork leaf. Both heights have been proved to equal1/2, and the total cost equals R+2. The determinant equation, including arbitrary spectators, is

$0=-2/R+(4/R)*D,$
$D=(x-y)^T\ G_Z^{-1}(x-y).$

Hence D=1/2. The height reduction precedes use of any unscaled contact gap.

If x,y have a common nonzero coordinate, the already-paid c-e-d path resolves the face. Otherwise their supports are disjoint. The positive-definite spectator gap states:

- If both supports are nonempty, Cauchy with e_i-e_j gives $D >= (x_i+y_j)^2/(4-2G_{ij})>1/2$.
- If exactly one is nonempty, D>=1/2, with equality only for one unit contact to an isolated A1 node of Z.
- If both vanish, D=0.

For completeness, in the second case equality in Cauchy with e_i forces x_i=1 and x=(1/2)G_Z e_i. Nonnegative x and nonpositive off-diagonal Gram entries force i to be isolated and every other coordinate zero. Thus the equality conclusion concerns the entire spectator diagram, including arbitrarily long indirect tails.

Exactly one extra isolated supporting node e therefore attaches simply to c or d. All other spectators are orthogonal to the whole18-root equality core. This is precisely the zero-residual extra wall allowed by the smaller17-root Lorentz core; it is not silently excluded by a strict inverse-diagonal test.

Whole finite components and the two half-payments

The equality core is completely specified: the15-node old F, the two simple leaves c,d attached to old fork leaves, and one simple e attached to c or d. The checker treats all nine mark pairs, all four choices of the two attachment leaves and both choices of the extra-spectator side, giving72 cases. In every case the full core has signature(16,1,1), its supporting white block is positive definite, its original corner is affine F plus an isolated A1, and the other three corners are finite.

Deleting a selected fork center leaves a D12 component containing the opposite mark, while the new local marked component is A2 or A3. Deleting a selected fork leaf leaves D14 at the other end and a local A1 or A2 if c attaches to that selected leaf. The large mark is always its fork center or a short fork leaf. These give the two half-coupon corners.

If a selected fork leaf instead has the new attachment on its mate, an adjacent finite corner has both marks in one connected component of order at most17 and already contributes a unit. The literal cases give32 configurations with exactly two halves and no old unit;32 with an old unit and an additional half; and8 with three old units. Every configuration receives at least1.

These are whole actual finite components. The spectator equality has proved orthogonality of every remaining supporting root to the equality core; no undisclosed supporting tail can enlarge or join the coupon components. At a finite ambient vertex the star already contains all incident facets. At the original nonsimple ideal vertex, the factor-fiber theorem is what identifies F as the full varying factor; no compact-style triangle transitivity is assumed.

The nine long pairs may therefore all be removed from the affine-D14 local rule after adding the stated half-coupons. This reduces that factor's all-pair cost from105 to96 when all other pairs are retained. The cost of the new finite coupons and the resulting global comparison are separate obligations.

Verification scope

Generation passes96 exact end-data calculations,256 separated-support cross identities, two positive affine13 kernels,14 scalar endpoint evaluations, four strict end-type margins and all72 equality configurations with216 finite corners. The determinant and spectator correction are checked as formal identities. The profile classification, continuous label bounds and spectator equality are analytic arguments; the finite evaluations are not a label cutoff or an enumeration of ambient chambers.

Final generation22597 and a fresh separate-process saved replay both terminated exit0/PASS. The affine component uses the standard rank label affineD14 for its15 nodes; finite D labels also denote rank. Syntax, all14 direct source hashes and all6 local links passed. A separate full mathematical/source read found no gap in the complete center/leaf extension lists, all-label outer-contact bounds, strict native gaps, the height1/2 reduction before the spectator gap, or whole-component sealing. That review did not independently repeat the96/256 formula calculations or72 equality configurations. Source, data and report are final/frozen. No external refereeing or proof-assistant formalization is claimed.

Source 17

Source path: `results/fresh_affine_distance8_feature_payment_20261017.md`

SHA256: 8eb322fa5de66b17dd880aca4e6f2c95185f42e021a95228c3578fbc7172b0d4

Affine distance8 plus feature-near distance9 pair budget

Let D be a connected affine Coxeter diagram with m nodes. Mark as features every vertex of degree at least3 and both endpoints of every nonsimple bond (label at least4, including the parallel bond). Give an unordered pair weight1 if its graph distance is at most8, or if its distance is9 and at least one endpoint is within14 graph steps of the feature set. Optionally give every pair weight1 when m≤17. Then the number P(D) of paid pairs satisfies

$$P(D) \le \min\{8m, \tbinom{m}{2}\}.$$

The rule is intrinsic under every weighted-diagram automorphism. The source and certificate give exact graph counts and verify the formulas below. This is an affine graph-cost lemma only. It does not prove that the weights pay every triangle or quadrilateral, and it does not assert dimension97 or any other new dimension bound.

Small diagrams and completeness

For m≤17, even paying every pair gives binom(m,2)≤8m. Thus the optional whole-component rule is safe without any further graph assumption. The certificate additionally checks all63 standard connected affine catalogue entries of order at most17: affine A1; the available A/B/C/D families; E6,E7,E8,F4,G2. It records each literal labelled graph and exact pair-distance histogram.

The standard affine Coxeter classification is an inherited input, as in the frozen classical graph-count package and affine endpoint catalogue. In orders m≥18 only the classical A/B/C/D families remain. The finite-rank checks corroborate the arbitrary-rank counting proof below; they are not a cutoff argument for completeness.

Arbitrary-rank counts

For m≥18 the exact counts are:

Affine family	Distance≤8	Extra feature-near distance9	Total
A cycle	8m	0	8m
C path	8m−36	min(m−9,32)	min(9m−45,8m−4)
B single fork and doubled far end	8m−35	2+min(m−11,31)	min(9m−44,8m−2)
D double fork	8m−34	4+min(m−13,30)	min(9m−43,8m)

Every total is at most8m. Since m≥18 implies8m≤binom(m,2), this proves the asserted budget in these ranks.

For an affine A cycle there are no features. Each distance k=1,…,8 contributes exactly m unordered pairs because2k<m. No distance9 pair is added.

For the tree families, number a spine with M vertices by0,…,M−1. Its short-pair count is sum(k=1,…,8)(M−k)=8M−36. A spine pair at distance9 is uniquely(i,i+9), with0≤i≤M−10, giving M−9 positions. The first min(a,L) positions and last

min(b,L) positions in a list of length L have union size min(L,a+b): if disjoint their sizes add, and otherwise their union is the entire list. This identity also covers a or b larger than L.

For C the whole graph is the spine, M=m. The doubled terminal bonds make0,1,M−2,M−1 features. The near-feature vertices satisfy i≤15 or i≥M−16. Therefore a distance9 pair(i,i+9) is added exactly when i≤15 or i≥M−25: the first16 and last16 positions. This gives min(m−9,32) extras.

For B, M=m−2 and two simple leaves are attached to the left spine endpoint0. The far spine bond(M−2,M−1) is doubled, so the features are0,M−2,M−1. Near-feature spine vertices satisfy i≤14 or i≥M−16. A distance9 spine pair is added when i≤14 or i≥M−25: the first15 and last16 positions, giving min(m−11,31). Each of the two left leaves adds its pair with spine vertex8, for two more distance9 pairs. At distances≤8 the leaves contribute16 leaf-spine pairs and one leaf-leaf pair, hence8M−36+16+1=8m−35.

For D, M=m−4 and two simple leaves attach to each spine endpoint. The features are0 and M−1. Near-feature spine vertices satisfy i≤14 or i≥M−15. A distance9 spine pair is added when i≤14 or i≥M−24: the first15 and last15 positions, giving min(m−13,30). The four leaves each have one distance9 pair along the spine, adding4. Opposite-end leaf pairs have distance M+1≥15 and contribute neither short nor extra payment. At distances≤8 the leaves contribute32 leaf-spine pairs and two same-end leaf-leaf pairs, hence8M−36+32+2=8m−34.

The deficits from8m are max(45−m,4), max(44−m,2) and max(43−m,0), respectively. Affine D therefore attains equality for every m≥43. The radius14 is sharp for this uniform rule: radius15 on affine D of order100 pays802 pairs, exceeding800. This diagnostic is also checked exactly.

Dividing the bound by m gives the per-factor form P(D)/m≤min(8,(m−1)/2). Applying it to actual cusp incidence multiplicities requires the separate geometric counting argument; no transfer is inferred merely from this graph estimate.

Exact checks and review scope

Generation and fresh saved replay both terminated exit0 with PASS_ALL_RANK_AFFINE_DISTANCE8_FEATURE _PAIR_BUDGET:444 breadth-first classical checks (orders18–128 in all four families),63 small affine catalogue checks and450 endpoint-interval checks. Every extra pair is counted in its literal spine-spine, leaf-spine or leaf-leaf category and compared with the formulas. Frozen labelled graph constructors are reused without alteration.

All six direct source hashes, Python syntax and four local report links pass. Frontier_literature independently read the mathematical proof and source and found no gap in the endpoint windows, leaf contributions, arbitrary-rank formulas, small-order reduction or sharpness diagnostic. Root separately read the complete source/report and inherited labelled constructors and confirmed the same counts and per-factor division. These are narrow mathematical/source reviews, not duplicate444/63/450 arithmetic replays. No optimizer is invoked. This is a complete affine-factor cost bound; face payment and ambient dimension conclusions remain outside its scope.

Source 18

Source path: `results/fresh_ambient59_payment_frontier_20261018.md`

SHA256: 25a8a07a9837b0430dd4f436a07786cd205b841b1bd8dc02dbeccbe89f887c37

Ambient59 frontier under the completed dimension59 rule

The unchanged rule has exact formal finite maximum **397.5**, versus curvature391.5, and **10 negative ideal terminal profiles**. The complete finite and ideal certificates therefore do not close ambient dimension59. The completed unrestricted bound remains59; this diagnostic asserts no bound58.

The producer and certificate bind the completed dimension59 proof, its depth-six local theorem, finite404 certificate, and sparse D16/ideal17 rule. No local weight, whole-component coupon or geometric hypothesis changes.

Complete finite rank59 comparison

The finite curvature coefficient is

$(5\cdot841 - 58\cdot59)/2 = 783/2 = 391.5.$

The independently enumerated fixed domain has961 multiplicity vectors. Fixed rank17 means A17; residual D17 remains included. Residual ranks are 1 through8 and17 through59. The interaction key has eight coordinates, including the fixed D16 multiplicity that records deletion of unused rank2 through5/D16 coupons. There are960 distinct keys.

All32,400 ordered actual type pairs are checked for simultaneous representative domination. Every actual same-component base is likewise dominated. Rank15 has actual A15=101, B15=97 and D15=201/2; its representative upper remains101. It is not identified with the actual D15 cost. The whole D5/D12 component-pair cost is45/4, and whole A2/D13 costs5. Every other current coupon is included through the frozen typed functions.

The producer generates960 new exact Bellman tables in eighth units. A separate checker proves all1,334,400 last-item upper inequalities and56,640 attaining predecessor equations. All49,011 item identities check the base plus every fixed-component interaction. Residual–residual and fixed–fixed interactions vanish. These checks prove the upper bound for every rank59 forest, rather than only those appearing among the named witnesses.

As a separate consistency check, each newly generated table agrees with the corresponding rank59 prefix of the frozen rank60 table. Generation does not import those old values. Fresh replay verifies the saved upper/attainment certificate and does not optimize again.

The exact maximum is795/2. Each of the following actual formal forests attains it:

D16 +D17 +D26,
D16 +A17 +D26,
2D16 +D27.

There are15 representative alternatives above curvature. The certificate retains all of them, including upper/actual distinctions. It also records these concrete actual obstructions:

Actual forest	Cost	Excess above391.5
D16+D17+D26, D16+A17+D26, 2D16+D27	397.5	6
2D17+D25, A17+D17+D25, 2A17+D25	396.5	5
3A1+4D5+3D12	395.5	4

Actual forest	Cost	Excess above391.5
10A2+3D13	394	2.5
3A15+D14	394	2.5

These are actual costs of formal forests. They are not claimed to occur at vertices of actual polytopes, and the displayed witnesses do not enumerate every actual type assignment. Actual attainment of the largest representative upper establishes the exact formal maximum397.5.

The singleton D59 costs only368. Further deletion of deeply interior distance-seven pairs in large components alone cannot resolve the whole finite comparison: the actual forests3A1+4D5+3D12 and10A2+3D13 have no component above rank13 and retain their stated costs under any change restricted to larger components. An improvement must address such smaller-component or coupon costs as well as the large-component maxima.

Complete ideal rank58 comparison

The current all-family total-cost envelopes at affine orders11 through17 are respectively

```
53, 62, 145/2, 82, 92, 103, 229/2.
```

Ranks below10 retain normalized envelope r/2. For order m at least18, the exact inherited family formulas are used individually before taking their maximum; no constant large-rank envelope replaces them. Every envelope is checked against all2,113 frozen rank59 rows before changing the link dimension.

The new complete terminal domain has1,890 sorted partitions of58, allowing arbitrary repeated ranks at least9 and at most one positive rank below9. No concentration of high ranks is used. For every partition, fresh integer convolution produces its link h-polynomial. The producer checks symmetry, vertex count V, the moment G, and both absolute-deviation and cumulative distribution identities, with the correct even link dimension58. It then computes

```
kappa=(5G-58V)/2,
margin=kappa/V-sum_r envelope(r).
```

There are10 negative rows and no zero rows:

Rank profile	Normalized margin	Envelope absolute margin
(9,9,9,9,9,13)	−1537/14000	−153700
(9,9,12,12,16)	−5241/57460	−26205
(9,9,9,9,10,12)	−5227/71500	−104540
(9,9,9,15,16)	−389/5440	−19450
(1,9,9,9,9,9,12)	−6893/130000	−137860
(8,9,9,16,16)	−1087/52020	−5435
(9,12,12,12,13)	−851/61516	−4255
(8,9,9,9,9,14)	−43/4500	−12900
(9,9,12,13,15)	−47/29120	−470
(6,9,9,9,9,16)	−19/59500	−380

These are upper-envelope deficits, not examples of actual cusps or polytopes. A more precise type-dependent theorem could also improve the comparison. The negative terminal absolute minimum is not extended to arbitrary profiles by multiplying inequalities: that positive-margin argument is unavailable here.

Quantified next targets, conditional only

If rank9 cost remains unchanged, the profile(9,9,9,9,9,13) requires an order14 total saving strictly greater than1537/1000 for positive margin. Its current maximum82 must become less than80.463. Current order14 families A/B/C/D cost70/82/81/82,

which the producer directly binds and asserts. Thus both B and D control the starting envelope, and C must also improve before that target is met.

The profile(1,9,9,9,9,9,12) independently requires an order13 saving greater than6893/10000 if lower-rank costs remain unchanged. Merely changing order14 does not repair it. Likewise the profile with one rank14 requires a saving in its own order15 envelope or in the other factors present.

One sufficient coordinated arithmetic target is:

Affine order	Current total maximum	Hypothetical saving
13	145/2	1
14	82	7/4
15	92	1/4
16	103	1/2
17	229/2	3/4

Under those simultaneous changes, every one of the1,890 terminal margins is positive. The normalized minimum would be21/5500 at(9,9,9,9,10,12), and the terminal absolute minimum would be91/2 at(58). Exact strict thresholds for each varied order with the other four changes fixed are saved in the certificate. Uniform savings1/4,1/2 or1 across every active order instead leave respectively5,3 or1 negative profiles.

The existing D14 fork-center pair(0,9) is already zero in the current canonical paid table, by the inherited D13 long-pair theorem. Reproving that deletion supplies no additional saving. The retained near-fork distance9 pairs and C14 basic distance9 pairs are separate research obligations.

These calculations do not supply local theorems. Each saving must hold for all families determining its envelope, preserve triangle and quadrilateral payments under the simultaneous rule, and include every new finite compensation cost. No such new geometry or finite compensation is assumed by this diagnostic.

Scope and verification

The ambient59 two-extrema constant is4350, counted once. Because the finite comparison has deficit6 and the ideal terminal domain has negative rows, this packet supplies no global contradiction or dimension58 claim. The completed dimension59 theorem remains unchanged.

Final source-bound generation session36632/chunk2d370a completed with exit0 and `PASS_AMBIENT59_PAYMENT_FRONTIER`. Separate fresh replay session91970/chunkd319b5 also completed with exit0/PASS. It checks the saved finite certificates and reconvolves the new ideal domain. Frozen source, data and report bindings are checked. The parent completed a full producer/report mathematical read: complete fixed/residual domains and typed domination, finite upper/attainment certificates, fresh even-link convolution identities, current family provenance, all ten deficits, conditional thresholds and the once-only 4350 constant passed. That review did not execute the DP, convolution, scalar arithmetic or historical geometric census. No native map search or global linear optimization is performed. Source, data and report are final/frozen; this remains a diagnostic, not a new dimension theorem.

```
python3 scripts/fresh_ambient59_payment_frontier_20261018.py --verify
```

Source 19

Source path: `results/fresh_ambient60_payment_frontier_20261018.md`

SHA256: 20d79d0469a2905b4cfd4e131a3d40f97823001713eb4f33f93a152a373c9a12

Exact ambient60 payment frontier under the completed dimension60 rule

The current rule does not close the ambient60 comparison. Its complete finite rank60 maximum is **413**, against curvature405. Among all2,113 terminal cusp profiles of rank59,77 have negative margins and none has zero margin. The smallest normalized margin is−1823/11560 at(9,9,9,16,16). The largest absolute terminal deficit is192250 at(9,9,9,9,9,14).

This is an exact obstruction diagnostic for the completed dimension60 rule, not a new local payment theorem or a dimension59 claim. Formal forests and cusp products in this calculation are not asserted to occur in an actual hyperbolic polytope. Negative terminal absolute margins are not extended to arbitrary nonterminal cusps.

Current finite rule and complete rank60 count

The finite420 certificate provides the exact current component and cross-component rules. The final whole A1/B10 doubled-terminal coupon is a unit, not the predecessor half: the source explicitly adds its doubled increment1 and rechecks simultaneous two-end domination for all33,489 typed component pairs. All four earlier coupon increments and the D17 restoration are retained. The current exceptional component totals are

Component	A15	B15	D15	A16	B16	D16	A17	B17	D17
Total	103	103	103	111	104	111	115	105	119

All other base weights, typed coupons, exceptional/I2 domination and the depth8 rule are inherited from the bound finite sources. The D16 upper representative now has its actual cost111. Some other representative coupon allowances remain conservative; the saved records distinguish the upper cost from the actual cost of each reconstructed forest.

Every finite forest is dominated component by component, on both ends of every coupon, by the following representatives. Rank6 uses E6, rank9 uses B9, ranks10..16 use D of the same rank, and A17 remains separate because of its cross-component roles. All other ranks at most8 use A; all other ranks at least17 use D. Whole A17 and D17 are thus different species throughout. The source checks base and simultaneous cross domination on the complete finite-type list through rank60, including the exceptional types and the arbitrary-dihedral representative.

There are1,044 possible multiplicity vectors for the nine fixed species B9,D10,…,D16,A17 with total rank at most60. They produce792 interaction keys. The remaining species have ranks1..8 or17..60. There are no cross-payments among these residual representatives, or among the fixed ones. For each residual rank the saved key gives its base cost plus every interaction with the fixed forest. The54,288 encoded item costs are compared directly with the current typed rule.

Generation computes new item-first unbounded knapsack tables through rank60 for all792 keys. The separate replay does not repeat that optimization: it checks every possible last item as an upper inequality and checks one attaining last item at every nonzero rank. These1,142,064 inequalities and 47,520 predecessors prove both the upper bound and attainment inductively. Every residual optimum is then combined with its entire fixed forest. The newly computed table values also agree with all792 appropriate prefixes of the frozen rank61 tables; they were not copied from those tables during generation.

The maximum is413. Actual formal forests attaining it include

- D60;

- D17+D43;
- 2D17+D26.

The finite curvature is

$$G_f = \sum_{j=0}^{59} \min(j, 59-j) = 870, \qquad \beta_{60} = \frac{5G_f - 59 \cdot 60}{2} = 405.$$

All17 fixed-vector alternatives whose upper cost exceeds405 also have an actual reconstructed witness above405. The certificate includes every such record. Particularly useful necessary obstructions are:

Actual forest	Cost	Deficit over405
D60 or2D17+D26	413	8
4D15	412	7
D16+D17+D27	412	7
D15+D17+D28	411	6
2D16+D28	411	6
D15+D16+D29	410	5
A2+2A5+3D16	408	3

The4D15 witness may use any four A15/B15/D15 components: their base costs agree and they have no mutual coupons. Thus a change only to D15 would not resolve that rank15 obstruction. If only the rank15 total changes, it needs at least7/4 saving per component. Other displayed witnesses require separate savings, and new coupon costs must be counted before drawing any sufficient finite conclusion.

One proposed additional rule would delete distance7 pairs whose two marks have feature depth at least11, with counts max(0,n−29), max(0,n−30), max(0,n−31) in A_n,B_n,D_n respectively. This packet does not prove that geometric rule. Even if it holds with all other costs unchanged, its exact rank60 maximum is still413: monotonicity gives the old upper413, while 2D17+D26 has no such pair and still attains413. This conditional statement requires no new optimization. In particular, removing the singleton D60 obstruction is insufficient; medium component ranks remain binding.

Complete ideal domain and exact moments

For a rank-r affine factor, its normalized cost is its total pair payment divided by r+1. The current costs are

$$c_r = r/2 \quad (1 \le r \le 9), \qquad (c_{10}, \ldots, c_{16}) = \left(\frac{53}{11}, \frac{63}{12}, \frac{73}{13}, \frac{83}{14}, \frac{94}{15}, \frac{105}{16}, \frac{116}{17}\right).$$

The source reads the completed proof's actual order11 and16 family totals, checks all2,365 prior normalized cost rows, and compares every tail family formula with the frozen ambient61 frontier. For order m at least18 the exact inherited family totals are

$$\begin{aligned} A_m &= 5m, \\ B_m &= 5m - 6 + \sum_{k=8}^{11} \min(m-k, 13), \\ C_m &= 5m - 15 + \sum_{k=6}^{9} \min(m-k, 14), \\ D_m &= 5m + 3 + \sum_{k=10}^{13} \min(m-k, 12). \end{aligned}$$

The maximum of these totals divided by m is used separately at each rank; there is no replacement by a constant tail envelope.

Only the inherited low9 concentration is used. Since all costs through rank9 are still r/2, it preserves total normalized cost and decreases the normalized curvature term until there is at most one factor of rank1..8; all other factors have rank at least9. No high-rank concentration or restriction on repeated large factors is made. Sorted recursion enumerates every partition of59 in this domain, giving2,113 profiles.

For each profile(r_i), generation and replay reconstruct

$$H(t) = \prod_i (1 + t + \cdots + t^{r_i}), \qquad V = \prod_i (r_i + 1), \qquad G = \sum_{j=0}^{59} [t^j] H(t) \min(j, 59 - j).$$

The normalized margin and terminal absolute margin are

$$M = \frac{5G - 59V}{2V} - \sum_i c_{r_i}, \qquad VM.$$

All coefficients are exact integers. Symmetry, length, total coefficient sum, the absolute-deviation identity and a cumulative-distribution identity are checked independently of the convolution accumulation. For this odd link dimension the CDF calculation centers at30; symmetry makes its mean absolute deviation equal to that centered at59/2. Thus it gives the same G/V=59/2−deviation. The odd-dimensional constant is handled explicitly.

There are77 negative profiles, no zero profiles. The full list is saved. Some leading normalized deficits are:

Affine ranks	Normalized margin
(9,9,9,16,16)	−1823/11560
(9,9,12,14,15)	−2311/15600
(9,9,11,15,15)	−1117/7680
(9,9,11,14,16)	−8251/61200
(9,9,9,9,9,14)	−769/6000

The singleton rank59 profile is positive. Its order60 family costs are A300,B346,C341,D351, so its normalized margin is9/10 and its absolute surplus is54. The largest terminal absolute deficit,−192250, occurs at (9,9,9,9,9,14), not at the normalized minimizer.

As in the inherited frontier, negative terminal absolute margins are not universal absolute lower bounds: the monotonic product-V argument requires positive margins before it can be applied in that direction.

Exact conditional local targets

These scenarios change only factor envelopes, leave ranks at most9 unchanged, and assert no new local theorem or finite coupon allowance. A total saving delta at affine rank r raises a profile's normalized margin by count(r)*delta/(r+1). Every saved scenario is checked on all2,113 profiles, including the currently positive ones.

Some individual orders need a reduction independently of all the others. While ranks at most9 are unchanged, the following profiles contain just one affected order and force the stated strict saving at that order regardless of improvements at other ranks at least10:

Affine order	Profile	Strict necessary total saving
13	(8,9,9,9,12,12)	>1559/15600
15	(9,9,9,9,9,14)	>769/400
16	(8,9,9,9,9,15)	>3361/3000
17	(9,9,9,16,16)	>1823/1360

Reducing every order11..17 total by1 still leaves three negative profiles. Reducing every such total by2 makes all profiles positive. A narrower sufficient rational scenario, leaving order11 unchanged, is:

Affine order	Current maximum	Conditional new maximum	Saving
12	63	125/2	1/2
13	73	145/2	1/2
14	83	331/4	1/4
15	94	92	2
16	105	415/4	5/4
17	116	229/2	3/2

This gives normalized minimum73/104000 at(9,9,9,9,11,12) and terminal absolute minimum54 at(59). The certificate includes the exact strict threshold for each coordinate while all other entries in this scenario are fixed. These are sufficient all-family envelope targets, not a claim that any particular Coxeter family can attain the savings for free. This scenario was selected by a small numerical linear-program scout followed by rational rounding; only the displayed exact row comparisons are used in the certificate, with no optimality claim about the scenario.

Changing order10 would invalidate the particular cost-preserving low9 reduction used here and would require a new terminal-domain argument or an expanded domain. This packet makes no such change.

Global status and verification

At ambient60 the strengthened two-extrema correction is4500, because each link contributes at least sum(2j−59 for2j>59)=900 and the two extremal coefficients total5. It is counted once. Current finite deficit8 and the negative ideal profiles keep the standard pointwise comparison open; they do not contradict the completed dimension60 proof, which excludes ambient61 and higher. No new finite-volume example is asserted.

- Producer.
- Exact certificate.

Final generation session22668, chunk5b0d55, and separate fresh replay session19716, chunkbc9b8f, both terminated exit0/PASS. The fresh replay checks saved finite Bellman certificates without reoptimizing, while reconstructing every ideal convolution and scalar comparison. It does not repeat any geometric census, local payment theorem or physical forest realization. Direct metadata checks pass13 source bindings, six local links and Python syntax.

The root agent completed a full source/report mathematical read: PASS. It checked the current A1/B10 increment and simultaneous typed domination, the complete fixed/residual partition and zero residual interactions, item identities and last-item induction, actual413 attainers and the conditional unchanged2D17+D26 witness, the odd-link CDF/symmetry identity, low9-only domain and exact tail formulas, conditional targets and the restriction on negative absolute margins, and the single constant4500. That review did not replay DP, convolution or any geometric census.

A second agent completed an independent full producer/report mathematical read, including the final residual-pair zero-interaction guard: PASS. It checked simultaneous typed domination and the actual A1/B10 upgrade, the complete1,044-vector/792-key decomposition and last-item certificates, the actual413 witnesses against405, the complete low9-only2,113-profile domain with exact family tails and odd-link moment identity, and the conditional-only targets and thresholds. Saved-summary inspection confirmed the17 offending finite alternatives,77 negative ideal profiles and unchanged 413 in the hypothetical deep7-only scenario. This review did not repeat DP, convolution, scalar arithmetic, graph-distance counts or any geometric census. Source, certificate and report are final/frozen.

```
python3 scripts/fresh_ambient60_payment_frontier_20261018.py
python3 scripts/fresh_ambient60_payment_frontier_20261018.py --verify
```

Source 20

Source path: `results/fresh_ambient61_payment_frontier_20261018.md`

SHA256: dffcb6dc7f4ab772c5b8e49a085dbd54a2be21277acc12528d3ed5cda8c55a8e

Ambient61 frontier under the completed dimension61 rule

The current proved simultaneous rule does not exclude ambient dimension61. Its complete finite rank61 maximum is **427**, against curvature420, and **21 of2,365** terminal ideal profiles have negative margins. The completed unrestricted ceiling remains61; this packet proves no bound60.

The producer and certificate bind the completed dimension61 proof, the current finite rule, its saved residual tables, and the current ideal closure. No frozen file is changed.

Finite rank61 maximum427

The same-component costs include A/B/D15=104, A16=113, B16=104, D16=225/2, A17=115, B17=105 and D17=120. All current coupons are retained, including the nine sparse increments, the D13 zero upgrades and actual E6/D11 total4. The D16 representative still uses conservative upper113, which covers actual A16=113 as well as actual D16=225/2.

There are1,132 complete fixed multiplicity vectors of B9,D10,…,D16,A17 with rank at most61 and854 interaction keys. The new source independently constructs this domain and identifies it with precisely the corresponding subset of the frozen rank62 domain. Residual components retain ranks1–8 or at least17, with D17 residual and A17 separately fixed. All59,996 residual item/vector comparisons have exactly their certified prior cost. Their required residual ranks are now at most61 and already lie in the saved tables through rank62. The source checks the table-list digest and reconstructs each attaining predecessor chain. No Bellman optimization or new table is needed.

The frozen cost functions are reused explicitly. The source verifies all34,596 simultaneous type-pair comparisons through rank61 and all base inequalities, preserving the distinction between actual costs and the conservative D16 representative. The complete upper maximum is427. Two actual forests attain it:

```
D17+D44: 120+307 =427;
A16+D45: 113+314 =427.
```

Neither has a cross coupon. Thus427 is an exact actual-rule maximum, not merely an upper allowance. Formal forest attainment makes no claim that a polytope realizing that forest exists.

The certificate saves every fixed-vector upper and the cost of its named actual forest. Eleven representative upper alternatives exceed420. Some conservative values are not attained by their named D16 forest; they must not all be presented as actual obstructions.

The following literal actual witnesses impose necessary savings on any new rule whose cross costs otherwise remain unchanged:

Actual formal forest	Current cost	Required total saving to reach420
D61	426	6
D17+D44	427	7
A16+D45	427	7
D15+D46	425	5
3D15+A16	425	5

Actual formal forest	Current cost	Required total saving to reach420
2A16+D29	425	5
A17+D44	422	2
A1+4D15	422	2

The singleton D61 is essential: improvements confined to finite ranks15, 16 or17 cannot eliminate this obstruction. For large finite D_r the current rule is

```
8*r-33-max(0,r-32),
```

so for r>=32 it equals7*r−1. At r=61 this is426. A new universal large-component argument is required, rather than just another isolated small-component saving. The witness list gives necessary conditions; it is not asserted to characterize every sufficient new rule.

Complete current ideal rank60 comparison

The exact factor envelope is linear r/2 for affine rank r<=9. At the remaining small ranks it is

Affine rank r	Current normalized maximum
10	54/11
11	63/12
12	73/13
13	83/14
14	94/15
15	106/16
16	116/17

For factor order m>=18, use the exact family costs, divided by m:

```
A: 5*m;
B: 5*m-6 +sum_(k=8)^11 min(m-k,13);
C: 5*m-15+sum_(k=6)^9  min(m-k,14);
D: 5*m+3 +sum_(k=10)^13 min(m-k,12).
```

The envelope is the maximum of those four ratios. No uniform163/23 tail is substituted. The source compares this complete cost function against all2,641 saved ambient62 cost rows before applying it at ambient61.

Only the proved low9 concentration is used. Hence the terminal domain consists of all rank60 partitions with at most one part1–8 and arbitrary repeated parts at least9. There is no high-rank concentration step. An optional-low-first recursion constructs all2,365 profiles.

For each profile the source multiplies the simplex H-polynomials by direct integer convolution, checks symmetry and total V, computes G, and compares the moment formula with both absolute-deviation and cumulative-distribution identities. The even link dimension60 is handled explicitly. All rows, including the complete polynomial, exact costs and both margins, are saved.

There are21 negative profiles and no zero profiles. The normalized minimum is

```
-6483/42592 at (10,10,10,15,15),
```

while the minimum terminal absolute margin is

```
-80800 at (9,9,9,9,9,15).
```

Other independent obstructions include (15,15,15,15), with normalized margin−469/4096, and (10,10,10,10,10,10), with margin−3189/161051. The certificate lists all21 negative rows. These are scalar obstructions to the current payment proof, not examples of hyperbolic polytopes.

The negative terminal absolute minimum is **not** promoted to a uniform absolute bound for every unreduced cusp. The positive-terminal argument uses monotonicity of V together with a positive margin; that implication cannot be reused with a negative terminal margin.

Conditional ideal targets

The following are exact scalar queries, not geometric payment theorems. Their hypothetical extra finite coupons, if any, are not counted.

An all-family affine-order11 maximum decrease of1/4, from54 to215/4, together with an all-family order16 decrease of1, from106 to105, makes all2,365 terminal profiles positive. The normalized minimum becomes 3/250 at(9,9,9,9,9,15), and the terminal absolute minimum becomes64 at the singleton(60). These changes affect ranks10 and15; the unchanged linear low-rank costs preserve the same complete concentration domain. If those simultaneous local rules were proved with their finite costs accounted for, the positive-terminal argument would extend that absolute 64 to all unreduced ideal links.

The exact threshold queries sharpen the target:

- With the order16 total decrease fixed at1, an order11 total decrease **strictly greater than1159/11616** is sufficient. Equality leaves the profile(10,10,10,15,15) at zero.
- With the order11 total decrease fixed at1/4, an order16 total decrease **strictly greater than3397/4160** is sufficient. Equality leaves (9,9,12,15,15) at zero.

Every row is checked, including zero-margin rows and nonpositive rows with no occurrence of the varied rank. An order11-only decrease cannot repair all profiles because eight negative rows have no rank10 factor. An order16-only decrease cannot repair all profiles because two negative rows have no rank15 factor. Two half-unit decreases also fail: they leave two negative profiles. These facts select two distinct geometric targets without pretending that a scalar adjustment proves a local rule.

Curvature, constant and validation

At ambient61 the link dimension is60. A finite simplex link has V=61, G=900 and

`kappa=(5*900-60*61)/2=420.`

The extremum difference is

`sum_(2j>60)(2j-60)=930.`

The two-extrema correction is therefore4650. It is recomputed at this dimension; the preceding ambient62 constant4805 does not carry over numerically. The current finite deficit7 and negative ideal profiles prevent a new contradiction with the present rule.

Generation811515 and separate fresh replaydffc45 both terminated exit0/PASS: finite427,2,365 ideal profiles,21 negative and constant4650. The computation performs no Bellman optimization, no finite-model linear program and no geometric census. A separate peer completed a full source/report mathematical read, including the restricted fixed-vector domain and unchanged residual items, conservative versus actual costs, complete low9-only profiles, exact tails, even-parity moment identities, conditional strict thresholds, and the negative-absolute-margin limit. The review passed without a scalar, DP or arithmetic replay. The finite functions and residual proof are explicitly inherited, while the changed-rank domain and ideal convolution are newly computed. All conditional targets remain separated from the proved dimension61 theorem and its current local rule.

Source 21

Source path: `results/fresh_ambient62_payment_frontier_20261018.md`

SHA256: 3c893e649b651c5dc9e556720bf7c8b2e44c8edf36d5f342d9935d388b4eaa7f

Ambient62: exact current deficits and conditional payment targets

The completed unrestricted ceiling remains **62**. At the next ambient dimension62, the current finite rule has exact rank62 maximum **441**, against curvature434. The current ideal comparison has genuine negative profiles as well. This diagnostic identifies those obstructions and checks a small collection of explicitly conditional cost targets. It proves no new local transfer, dimension61 theorem or example.

The source and saved certificate bind the completed62 proof, the complete nine-coupon finite budget, the current ideal envelopes and the all-rank family counts. Every predecessor remains unchanged. The implementation uses only the standard library; there is no global or numerical optimizer.

The finite obstruction is seven units

Retain all nine new typed component-pair increments, every old coupon, the deep11 finite8 zeros, and current A16/D16/D17 costs114/114/125. B9, E6 and A17 remain distinct representatives. The conservative E6/D11 allowance4 is kept separate from its actual cost2, and type domination compares both ends simultaneously, as in the completed budget.

At rank62 there are1,227 complete large-component multiplicity vectors and921 distinct seven-coordinate interaction keys. A separate rank-first dynamic program reconstructs every residual value, agreeing with every corresponding entry in the frozen rank63 item-first tables. Verification checks all1,426,629 last-item upper inequalities and57,102 attaining predecessors,66,258 literal interactions and all1,227 global alternatives. All35,721 ordered type pairs are checked for simultaneous domination.

The conservative maximum is441 and has an actual-cost attaining forest

```
D17+D17+D28: 125+125+191 =441.
```

Thus this is the exact maximum for the current counting rule, rather than merely a conservative upper estimate. Formal attainment does not assert that this diagram is realized at a vertex of a polytope. The finite curvature is434, leaving deficit7.

The following complete finite counts retain all coupons. A rank15 drop applies to every finite family of rank15; a rank16 drop applies to A16 and D16, with B16 left at104. A D17 drop affects only D17, preserving the separate A17 representative. The proposed D13-zero coupon upgrade adds1 per whole A2/D10-center pair and1/2 per whole A3-middle/D10-center pair; it is additional to, not a replacement for, the existing halves.

Conditional changes from the completed62 rule	Exact maximum	Attainer
None	441	D17+D17+D28
D13-zero coupon increments only	441	D17+D17+D28
D17 cost lower by4	438	2D15+2D16
Rank15 lower by1; A16/D16 lower by2; D17 lower by4; coupon increments	435	D17+D45
Same, with D17 lower by5	434	D17+D45

Conditional changes from the completed62 rule	Exact maximum	Attainer
D17 lower by4 and all D ranks at least33 lower by1, with the other drops and coupon increments	434	D17+D45

The last two rows are sufficient counting targets at unchanged listed coupon costs. They are **not proved geometric rules**. In particular, lowering only D17 or only the rank15/16 costs leaves other exact forests obstructing the target. Every scenario rechecks type domination and complete upper/attainment certificates; the actual cost equals the upper maximum on each displayed attainer. Any additional coupons required by future geometric proofs must be counted again.

A sharper existing ideal tail removes an artificial obstruction

With the uniform tail163/23 and the usual9/17 concentration, the complete rank61 terminal domain has2,004 profiles,64 negative and none zero. Its minimum is−3379/17408 at(15,15,15,16). The single rank61 factor has margin−2/23 under that uniform envelope.

That singleton failure is not a new geometric obstruction. The existing ideal5/R5 proof gives the following exact costs for every factor order m at least18:

```
A: 5m;
C: 5m−15 + sum(min(m−k,14), k=6,...,9);
B: 5m−6  + sum(min(m−k,13), k=8,...,11);
D: 5m+3  + sum(min(m−k,12), k=10,...,13).
```

At order62 the maximum is D361. The single-factor normalized margin is therefore7−361/62=73/62, with absolute surplus73. Every saved all-rank family count is compared with these literal formulas. Their geometric spine/counting proof remains an inherited input, not a consequence of the comparisons.

One cannot simply substitute these nonconstant costs in the old2,004-row terminal list: the high-rank concentration move need not preserve their sum. Instead retain **arbitrary repeated ranks at least9**, and concentrate only the ranks below9. Those moves preserve the linear cost r/2, decrease normalized curvature, and stop at rank9. There is at most one residual rank1 through8. This complete low9-only domain contains2,641 profiles; there is no high-rank concentration assumption.

Using the maximum of the four exact family formulas at each rank gives 62 negative profiles and none zero. Every negative profile has largest rank at most16. The normalized minimum remains−3379/17408 at (15,15,15,16). The most negative absolute value is−57009 at (1,10,10,10,15,15). These are exact failures of the current comparison, not constructions or nonexistence conclusions. In particular, reducing the large-rank uniform envelope is unnecessary to address these remaining negative profiles.

Conditional ideal targets use the complete larger domain

For every profile, a factor-cost decrease delta at affine rank r improves its normalized margin by count(r)*delta/(r+1). Here rank r means factor order r+1. All following costs are maximum totals across all four families of that order, not a reduction in only one family. The small concentration boundary remains9, since none of these scenarios changes ranks1 through9.

Conditional total-cost decreases by affine rank	Negative profiles
r12:1/2	45
r12:1/2, r15:1, r16:1/2	8
Also r13:1/2	3
Also r10:1/2	1
Same, but r16:3/4	0
Keep r16:1/2 and additionally lower r11 by1/2	0

The one remaining profile in the fourth row is(9,9,11,16,16), with margin−2/1445. Fixing drops1/2 at ranks10,12,13 and1 at rank15, the exact rank16 threshold for nonnegative margins is87/170. This is attained by that profile, and no profile lacking rank16 remains negative or zero. Both conditions are checked explicitly; all such zero rows would be saved separately in the threshold certificate. A strictly larger reduction therefore gives strict positivity. Choosing3/4 gives normalized minimum663/71680 at(9,9,13,15,15). The alternative final row gives minimum 23/4160 at(9,9,12,15,16). Both have absolute minimum73 at(61).

An order16 improvement alone cannot repair every current ideal deficit: even after the D13-zero decrease, nine negative profiles contain no rank15 factor. The saved threshold record lists them explicitly. This rules out that narrow scalar strategy, not more general geometric changes.

For the two positive scenarios, the low-rank concentration moves also decrease V, the product of factor orders. Since terminal normalized margins are positive, the terminal absolute bound73 consequently extends to every ideal profile under the stated hypothetical envelopes. This positive-product argument is not applied to the current negative rows. The exact scalar calculation uses sliding-window polynomial products, the doubled absolute-deviation identity and a parity-correct cumulative distribution identity for G; here the link rank61 is odd.

Status and limits

At ambient62 the strengthened extremum constant is5*floor(62²/4)=4805. A compatible complete future rule with finite cost at most434 and positive ideal surplus would suffice for a strict global contradiction. The conditional finite and ideal rows do not establish that rule: their geometric transfers, simultaneous payment and any further finite coupons remain separate obligations. The current established ceiling stays62, compact arithmetic21 and arbitrary compact29.

Final source-bound generation56911 (terminal chunk65d97d) and separate fresh saved-certificate replay24837 (terminal chunk93eaa3) both exited0/PASS, including the explicit zero-row threshold guard. The initial generation and replay also passed before that guard was added; no numerical conclusion changed. A separate full source/report mathematical read passed the simultaneous typed domination, B9/E6/A17 separation, all nine increments and the conservative allowance, complete finite split and Bellman proof, frozen table equality, larger exact-tail domain, odd-rank CDF identity, conditional gains and positive-terminal V argument. That reviewer did not rerun the finite or ideal arithmetic, or any geometric census. Root also completed a full source/report mathematical read of the independent rank-first reconstruction, all coupon increments and representative allowances, complete finite domain and actual attainment, parity-correct scalar identities, exact-tail domain and conditional scope. Root requested the explicit zero-row guard used in the final threshold certificate. This review did not repeat the dynamic programs or scalar arithmetic. The current finite reconstruction compares all frozen table entries; saved replay checks certificates without recomputing a numerical optimizer. The local geometric censuses and arbitrary-rank classification are inherited, with no new replay claimed here. All15 direct bindings, six report links and Python syntax pass. Source, data and report are final/frozen at this diagnostic scope.

Source 22

Source path: `results/fresh_ambient62_three_transfer_frontier_20261018.md`

SHA256: e3934e88e8664aed44f2a106cb4ae0a71dc1b01e6a2712fc7ce6472f712025e2

Three local transfers in the ambient62 payment frontier

The three completed local changes can be used simultaneously: the whole order13 affine D center pair becomes zero, two whole order16 affine B singleton units are deleted, and four whole order17 affine D halves become zero. Their all-family totals are respectively73,106,116. The enlarged finite rule has exact rank62 maximum441, against curvature coefficient434. Among the complete2,641 terminal cusp profiles of link rank61, exactly three still have negative margin. This improves the current local rule but supplies no dimension61 theorem; the completed unrestricted ceiling remains62.

The source and certificate bind the completed62 proof, the complete ambient62 frontier, and the three local theorems: D13 zero, B16 singleton deletion, and D17 zero with E6/D11 units. The exact arithmetic certificate reconstructs the scalar comparison and checks finite-cost absorption; arbitrary-support geometric coverage remains in those complete bound local proofs, not in a new census here.

Simultaneous actual-corner payment

All finite and ideal weights not specified by those three theorems stay as in the completed62 rule. The finite changes are precisely:

Whole components and marked roles	Old total per component pair	New total	Increment
A2 either endpoint / D10 fork center	1	2	1
A3 unique middle / D10 fork center	1/2	1	1/2
E6 endpoint of either length2 arm / D11 short fork leaf	2	4	2

For each actual quadrilateral, use the following exhaustive priority. These tests concern all actual corners; they do not depend on which corner an older proof selected.

1. A changed C11 endpoint corner gives a distinct path1 or2 unit by its independent zero theorem.
2. Otherwise, a changed B12 center/terminal corner gives its retained whole order12 D center/leaf unit, finite D12 leaf9 unit, or distinct typed A2/B9-terminal and A2/D11-center halves.
3. Otherwise, a D13 center corner gives a distinct full unit by the new zero theorem: path at most4, a retained order11 B/D center unit, or the upgraded whole A2/A3-middle versus D10-center coupon. It does not use its deleted original half.
4. Otherwise handle a cycle11 distance5 corner by its independent half theorem. Its original half and distinct retained half, or finite short unit, pay. Its ideal outputs have distance at most5 and its finite outputs distance at most6.
5. Otherwise handle a changed order11 B/C/D tree corner by the complete coupled half theorem. Its distinct immediate halves or its specified finite components and coupons pay without recursively requesting a unit from another changed

pair. Any C11 zero corner was already handled. None of these outputs is a deleted order16 B or order17 D distance10 pair.

6. If a changed order17 D corner now occurs, apply the complete D17 zero theorem. In its no-short8 branch, all64 proper completions are impossible and each of its four sealed whole-affine contexts has a distinct whole E6/D11 unit. Neither original ideal half nor a finite D16/D17 same-component unit is used.
7. If no changed order17 corner occurs, apply the complete B16-deletion theorem to the actual face. All its ideal weights agree pointwise with the present rule on this face: a changed D13 corner would have triggered step3, and a changed D17 corner step6. The extra finite coupons are nonnegative. Thus its whole-face payment transfers.

The short8 alternative in step6 is the complete current route bound by the D17 theorem, not an assumption that every short8 corner has weight1. The current63 proof's18 small tree interfaces retain all such pairs; the later changed D13 center was already handled in step3. Cycle11 and order11 tree half cases were handled in steps4–5. Small exceptional components retain their units. The universal cycle8 interface handles complementary arcs8 through11 by protected nondeep outputs, including the sealed cycle16 case; arcs at least12 have strict gap7/48. Larger radius-five tree alternatives use their retained paths, whole-leaf roles or typed coupons. A finite deep8 zero invokes its independent depth11 corridor and constructive short7 replacement. This shorter route uses finite paths at most7 or its protected ideal roles and does not recurse through another deep8 zero. The new B16 and D17 deletions are all distance10 and are not used by those routes. The unchanged sparse finite deletions are also distance at least10.

This also explains why step7 is not used to justify step6's short-path premise: the short-path payment was established by the independent earlier branches and corridor/cycle/tree theorems. The complete B16 proof itself patches its selected no-short8 domain by strict physical norm-two projection obstructions, with its low, disjoint, protected, later and finite interfaces retained.

Triangles keep payment at least2. Their inherited finite7/ideal5, protected ideal6/7 and whole-small-factor outputs are unchanged; the D13 distance8 and B16/D17 distance10 deletions are absent from those outputs. Every new finite coupon is nonnegative. Consequently the three-theorem rule still pays every quadrilateral at least1 and every triangle at least2, without a new global conclusion.

Exact finite obstruction with all three coupon upgrades

The sparse finite certificate already bounds the old actual E6/D11 total2 by a representative allowance4. It needs that allowance to dominate the separate actual E6/A11 total4 when A11 is replaced by D11. The new E6/D11 actual total4 fills that allowance exactly. It does not change any representative cost.

The complete ambient62 frontier's saved scenario1 already adds the D13 increments a2$d10+(a3$d10)/2 to the actual rule and its representative upper rule. We use that scenario's complete rank62 upper certificate, with its1,227 large-component vectors and921 interaction keys. No Bellman table or optimization is recomputed.

For each of35,721 ordered type pairs through rank62, the source checks the old actual doubled cost plus both applicable D13 increments and the new E6 increment against the saved upper doubled cost plus the D13 increments of the two representatives. Both types are replaced simultaneously. Only the two ordered E6/D11 rows absorb the new doubled increment4, and their new actual and upper values are both8. This is a pointwise proof for every forest, not a check only at selected attainers. Same-component costs and their domination are unchanged. B9, E6 and A17 retain their separate representative roles, and every old typed coupon remains included.

As an additional consistency check, all1,227 saved formal-attainer forests are recounted with doubled increment4$e6$d11 and remain below their unchanged representative values. The inherited complete upper bound is441. The actual forest

```
D17 + D17 + D28
```

still attains125+125+191=441. It has no interaction from any of the three new coupon upgrades. The D28 cost is8*28−33=191; its depth11 finite8 deletion count is zero. Thus441 is the exact actual-rule maximum, not merely an upper envelope. The finite curvature coefficient is434, so the finite margin remains−7.

Complete exact-tail cusp comparison

For affine rank r, the normalized factor costs that change are

```
r=12: 147/26 -> 73/13
r=15: 107/16 -> 106/16
r=16: 118/17 -> 116/17.
```

These follow from family totals65/73/73/73 at order13, 80/105/103/106 at order16, and85/111/108/116 at order17. The unchanged factor-fiber multiplicity is V/(r+1), where V is the link vertex count; the normalized savings are therefore1/26,1/16,2/17 per respective factor.

For orders m>=18 the inherited exact classical costs, rather than the coarser uniform163/23 envelope, are used:

```
A: 5m
B: 5m-6  + sum_{k=8}^{11} min(m-k,13)
C: 5m-15 + sum_{k=6}^{9}  min(m-k,14)
D: 5m+3  + sum_{k=10}^{13}min(m-k,12).
```

The normalized tail cost is their maximum divided by m. We do not apply high17 concentration to these nonconstant tail costs. Instead the complete terminal domain consists of at most one rank1 through8 part and arbitrary repeated ranks at least9, summing to link rank61. Its2,641 profiles are enumerated independently with the optional low part first, and compared bijectively to the saved frontier domain.

For each profile, direct bounded polynomial convolution independently reconstructs h, V and G; the doubled absolute-deviation identity checks G. Every original polynomial, curvature, cost and margin is compared exactly with the frozen row before applying the three savings. The curvature is(5G−61V)/2. All2,641 rows have nonzero current margin, with exactly these three negative rows:

Affine ranks	Normalized margin
10,10,13,13,15	−11779/379456
9,9,13,15,15	−271/10240
9,13,13,13,13	−113/5488

The first is the normalized minimum. These are exact obstructions to the present sufficient payment comparison; they do not assert the existence of a polytope with those links.

For targeting a next local theorem only, assume the whole order14 all-family total can additionally fall from84 to167/2, with no omitted finite-cost obligation. Subtracting1/28 per rank13 factor makes all 2,641 terminal margins strictly positive. The normalized minimum is 2741/371712 at(10,10,11,15,15); the absolute minimum is73 at(61). No geometric order14 reduction is proved by this packet, and no prospective extra finite coupon is included in that conditional query.

The inherited convex-order argument and cusp-link shape identity justify only the low9 concentration used here. It preserves the linear cost below9 and decreases normalized curvature and V. For the conditional strictly positive terminal margins, the absolute73 lower bound therefore extends to unreduced profiles. No such positive all-profile conclusion is inferred from the three negative current rows.

Even the conditional ideal improvement leaves the separate finite margin−7. The ambient62 global constant is4805. Neither the current comparison nor the conditional ideal-only calculation establishes a contradiction for arbitrary finite and ideal vertex counts. The unrestricted ceiling remains62.

Validation and scope

The source uses only the standard library. Its preparation checkf9ec21, final source-bound generation67777e and separate fresh replayd53025 all terminate exit0/PASS. The final generation binds the frozen D17 report. The runs check35,721 simultaneous type-pair inequalities,1,227 saved formal-attainer comparisons and all2,641 independently reconvolved original scalar rows, followed by both changed-cost comparisons.

Root and a separate reviewer completed full source/report mathematical reads. Both checked the independent actual-corner priority and the short8/deep11 route, B16 pointwise transfer on the remaining faces, D13-expanded scenario1 and simultaneous E6 absorption, actual441 attainment, complete optional-low domain with unrestricted high repeats, direct convolution and absolute-deviation identities, exact factor savings, and the conditional positive-terminal-only absolute73 bound. Neither

reviewer repeated scalar arithmetic, finite DP, geometric censuses or historical classification. Direct source bindings, local links and Python syntax are audited before freezing.

The computations here are exact saved-interface, type-pair absorption and scalar comparisons. There is no new DP, optimizer, physical-support census, arbitrary-label classification or proof-assistant formalization. Source, data and report are final/frozen.

```
python3 scripts/fresh_ambient62_three_transfer_frontier_20261018.py
python3 scripts/fresh_ambient62_three_transfer_frontier_20261018.py --verify
```

Source 23

Source path: `results/fresh_ambient63_ideal_margin_closure_20261018.md`

SHA256: 5c8b71019edd38ba3e2b3ff70e1a2cb7e5f0e3dad38c12bd2191014d9f9ace2f

Exact positive ideal margins at ambient dimension63

For the specified order11/12/13 total-cost envelopes54,63,147/2, every ideal rank profile at ambient dimension63 has positive margin. The unique normalized minimum is3505/201344 and the unique absolute minimum is139/46. This is an exact scalar conclusion. Geometric validity of the new local weights, their finite coupon costs and a simultaneous global theorem are separate obligations; this packet asserts no new dimension bound.

The source and certificate reconstruct every profile by a multiplicity recursion and every polynomial by sliding-window sums. They import no project arithmetic module and agree with all scalar rows of the independent ambient63 frontier, whose implementation uses sorted-rank recursion and direct convolution.

Specified envelopes and complete domain

Let r denote affine rank, so the factor has r+1 nodes. The normalized envelope used in the calculation is

r	c(r)
1 through9	r/2
10	54/11
11	21/4
12	147/26
13	6
14	94/15
15	107/16
16	118/17
17 and larger	163/23

Only the three middle values differ from the completed dimension64 ideal calculation. They require total pair-cost decreases1/2,1,1/2 at orders11,12,13 respectively. Finite coupons do not enter these ideal costs, but must be included separately at finite vertices.

The all-rank concentration theorem transfers rank from a smaller to a larger factor while decreasing normalized curvature. For two positive ranks below9, stop when one vanishes or the larger reaches9. For two ranks at least18, stop when the smaller reaches17. These transfers preserve the sum of c(r), since the respective ranges are linear and constant. They strictly increase the bounded sum of squared ranks, so they terminate.

Every terminal rank62 profile consequently has arbitrary multiplicities of9 through17, at most one rank1 through8, and at most one rank at least18. The source chooses the nine middle multiplicities, then the remaining low/high factors, generating each of the2,204 profiles once. The ranks always sum to62. Changing only the three listed middle costs does not alter either concentration boundary or the completeness of this domain.

Exact calculation and absolute bound

For a profile(r_i), define

```
H(t)=product_i(1+t+...+t^r_i), V=H(1),
G=sum_j [t^j]H(t)*min(j,62-j),
margin=(5G-62V)/(2V)-sum_i c(r_i).
```

The factor-fiber formula gives at most V times the cost sum at an actual cusp. All coefficients, V, G, curvature, cost and margins here are exact. Every polynomial and scalar field matches the frozen frontier, and each new margin matches its three-order conditional scenario. The gain from the old normalized margin is exactly

```
count(rank10)/22 + count(rank11)/12 + count(rank12)/26.
```

All2,204 margins are positive. The unique normalized minimum3505/201344 occurs at(10,10,12,15,15). The unique absolute minimum139/46 occurs at(62).

For any concentration transfer delta from a<=b,

```
(a+1)(b+1)-(a-delta+1)(b+delta+1)
  =delta*(b-a+delta)>0.
```

Thus V decreases together with normalized margin. Since the terminal margin is positive, their product decreases as well. The terminal absolute minimum therefore gives surplus at least139/46 for every actual cusp profile, including those with many small or large factors. No simple-cusp or arithmetic assumption is needed for this reduction.

The inherited shape-curvature estimate supplies the additive constant5*floor(63²/4)=4960. To use it in a new dimension theorem requires a complete compatible payment rule with all three specified ideal envelopes and finite maximum at most899/2. This scalar calculation does not supply those geometric or finite hypotheses.

Verification scope

Generationd7483c and separate fresh replay14112e both terminated exit0/PASS on every profile, all independently implemented field comparisons and both unique extrema. A separate full mathematical/source read passed the complete multiplicity domain, sliding-window implementation, concentration boundaries, cost preservation, V monotonicity, positive absolute surplus, constant4960 and explicit conditional geometric scope. The reviewer did not rerun the 2,204-row arithmetic. The actual factor-fiber, concentration and shape-curvature arguments are inherited written mathematical inputs, not consequences of this numerical table. No external refereeing or proof-assistant verification is claimed. Source, certificate and report are final/frozen after these checks.

Source 24

Source path: `results/fresh_ambient63_payment_frontier_20261018.md`

SHA256: f82a1ccafb23cb0b70d7b5e4538d4ce7e2692d2a58a57b3a3a6e72c9560b27ac

The next finite and ideal obstructions at ambient63

Under the completed dimension63 payment rule, the exact finite maximum at rank63 is **453**, exceeding the curvature899/2 by7/2. The complete ideal comparison at link rank62 has **2,204 profiles,21 negative margins and no zero margins**. Thus the current payment inequalities do not exclude dimension63. The completed unrestricted bound remains63.

The source and certificate compute these obstructions and a few explicitly conditional targets. They bind the completed proof, its enlarged finite coupon budget, and the completed ideal envelopes. No local weight reduction, new dimension bound or example is proved here.

Exact current finite maximum

All current finite payments are retained, including the depth11 zeros, D16=116 and D17=127, the whole E6/D11 half coupon, and the whole A1/B10 doubled-terminal half. Three representative forests attain453:

Forest	Cost
D15+3D16	105+3*116=453
2D15+D16+D17	2*105+116+127=453
2D17+D29	2*127+199=453

These are formal forest attainers, not assertions that corresponding vertices occur in a real chamber. The list is the set of canonical attaining witnesses returned by the complete component-vector comparison; it is not a classification of every typed maximizing forest. For example, an A16 has the same same-component cost116 as D16, so reducing D16 alone would leave analogous A16 obstructions.

The calculation covers all finite types. It keeps E6 at rank6 because of its coupon, and A17 separately because of its small-component interactions. Every other type is dominated by A_r for r<=9 and D_r for r>=10. The source checks same-component and every type/partner cross-cost dominance under each tested scenario, including arbitrary dihedral labels through the inherited uniform rank2 representative. There are192 types and36,864 type/partner comparisons per scenario.

All D10 through D16 and A17 multiplicities of total rank at most63 form 700 complete vectors, giving328 distinct coupon environments. The remaining components have ranks1 through9 or at least17. The exact interaction key and literal coupon formula are the same as in the frozen enlarged budget; in particular E6/D11 has doubled cross cost4, and A1/B10 has doubled cost1. The latter remains dominated once by D10, without double-counting B10.

For each environment, rank-first dynamic programming computes every residual value through rank63. The current tables agree entry-for-entry with the corresponding prefixes of the frozen item-first rank64 tables. An independent saved-table verification then checks every possible last-component inequality and an attaining predecessor for every rank. Each global vector reconstructs a typed attaining forest whose total same-component and cross-component cost is recounted directly.

Per scenario the certificate contains544,152 Bellman upper inequalities, 20,664 attaining predecessors,39,200 literal interaction comparisons and 700 global alternatives. Induction on residual rank proves both the upper bound for every forest and formal attainment; no sampled forest maximum is being extrapolated.

Finite targets worth distinguishing

Every scenario below preserves all current coupons and every unlisted same-component cost. A rank16 reduction is applied to both A16 and D16; B16 remains104. A rank15 reduction applies to A15,B15,D15 together.

Conditional costs	Exact rank63 maximum	Slack below899/2
Current rule	453	−7/2
A16/D16=114 only	453	−7/2
D17=125 only	453	−7/2
A/B/D15=104, A16/D16=115, D17=125	449	1/2
A16/D16=114, D17=125	449	1/2

The last row is the narrowest tested family target: two more units saved on both A16 and D16, and two on D17, would close the finite comparison with the enlarged coupons unchanged. Geometric feasibility is unproved. In particular the present D17 cost127 came from a deliberate restoration of two units, so deleting them again cannot be presumed compatible with the D16 saving.

The three current witnesses impose useful necessary inequalities. Write delta15,delta16,delta17,delta29 for same-component savings from the current costs, with delta16 also valid for the A16 substitute. To reach899/2 one must have

```
delta15+3delta16 >=7/2,
2delta15+delta16+delta17 >=7/2,
2delta17+delta29 >=7/2.
```

These witness inequalities are necessary, not sufficient. The complete DP proves sufficiency only for the explicit scenarios in the table. The D29 witness shows why work confined to ranks15/16 cannot settle the finite side. Conversely, changing D17 alone leaves D15+3D16 intact.

Complete ideal rank62 domain

The current normalized costs, with r the affine rank rather than factor order, are

```
c(r)=r/2 for1<=r<=9;
c(10)=109/22, c(11)=16/3, c(12)=74/13, c(13)=6,
c(14)=94/15, c(15)=107/16, c(16)=118/17;
c(r)=163/23 forr>=17.
```

The inherited all-rank convex-order argument concentrates two ranks below9 until one vanishes or the other reaches9; two ranks at least18 concentrate until the smaller reaches17. Both moves preserve this cost sum and decrease normalized curvature. Thus all actual rank62 cusp profiles reduce to arbitrary multiplicities of9 through17, at most one rank1 through8, and at most one rank at least18. The2,204 profiles are the complete domain at rank62, not the2,419 profiles appropriate to the previous rank63 link.

For each profile the source computes the entire product polynomial H=product(1+t+…+t^r), its value V=H(1), and

```
G=sum_j [t^j]H*min(j,62-j),
margin=(5G-62V)/(2V)-sum c(r).
```

Direct convolution is used; an exact cumulative-distribution identity checks G independently from the coefficient vector. All rows and all21 negative rows are saved. The largest normalized deficits begin as follows.

Affine ranks	Margin
(10,10,12,15,15)	−22543/201344
(10,11,11,15,15)	−5653/50688
(10,10,11,15,16)	−18281/197472
(10,10,10,16,16)	−27785/384659

Affine ranks	Margin
(11,12,12,12,15)	−863/13182
(9,11,12,15,15)	−2237/39936

Two further profiles show that improving only factor orders16 and17 can never finish the ideal comparison:

```
(10,10,10,10,10,12): -8607/190333,
(10,10,10,10,11,11): -4021/95832.
```

They contain only factors of orders11,12,13. Changes in larger factors cannot change their margins. A one-unit drop in the order16 envelope alone still leaves10 negative profiles.

Conditional ideal targets

The following three all-family total-cost targets make every one of the 2,204 margins positive, with minimum3505/201344 at(10,10,12,15,15):

Factor order	Current maximum	Conditional maximum	Present bottleneck
11	109/2	54	C has54.5; B already54
12	64	63	B has64; C already63
13	74	147/2	D has74; B and C already73

All other family maxima remain unchanged. These are total pair-cost drops of1/2,1,1/2, not normalized cost drops. The normalized gains are divided by factor orders11,12,13 respectively.

With the first two drops fixed, the exact additional order13 drop needed for nonnegative margins is4239/15488, attained at(10,10,12,15,15). A strict increase above this threshold makes every margin positive; the proposed half-unit drop is sufficient. There is no unaffected negative profile hidden by that one-parameter calculation.

An alternative two-order target lowers the order11 maximum by3/4 and the order12 maximum by1. It gives minimum157/8788 at(11,12,12,12,15). With the order12 drop fixed, the exact order11 threshold is22543/36608. This alternative requires reducing B as well as C at order11, since its target maximum is215/4=53.75. The three-order target may therefore be more convenient geometrically, although no such transfer is supplied here.

All tested changes leave the low linear range through rank9 and high constant range from rank17 unchanged. The same2,204-profile domain remains complete for every conditional scenario; no old concentration restriction is silently reused after changing a boundary cost.

If a complete compatible local theorem achieved a finite maximum at most 899/2 and positive ideal margins, the already proved global constant 5*floor(63²/4)=4960 would finish the ambient63 comparison. Both new local obligations remain open. The diagnostic does not promote that conditional observation to a dimension62 result.

Verification scope

The first generationc98670 passed all current and conditional exact checks. Final source-bound generation02d48e and separate fresh saved replay35f38f both terminated exit0/PASS. Direct audit6c2cc3 passed all12 source bindings, all5 report links and source syntax. The source imports no project arithmetic modules. Its current finite tables are independently reconstructed and compared with the frozen enlarged-coupon tables; each conditional table has its own complete upper/attainment certificate. Geometric classification, the existing payment theorem, and all-rank concentration remain inherited written inputs. No new coupon, pair deletion, half transfer, or global dimension theorem is proved by this computation.

A separate agent completed a full source/report mathematical and scope review, also reading the frozen finite-budget formulas. It passed the complete threshold9/17 domain, current-versus-conditional costs, rank-first Bellman upper and attainment proofs, typed coupon domination, simultaneous A16/D16 targets and exact threshold interpretation. A light independent standard-library read of the saved summaries confirmed the stated scenario results and absence of unaffected zero rows in

the two successful threshold claims. That reviewer did not rerun the profile arithmetic or DP optimization. No concrete issue was identified. Source, data and report are final/frozen; the completed dimension63 proof and its separate aggregate are unchanged.

```
python3 scripts/fresh_ambient63_payment_frontier_20261018.py
python3 scripts/fresh_ambient63_payment_frontier_20261018.py --verify
```

Source 25

Source path: `results/fresh_ambient64_finite_closure_20261018.md`

SHA256: 79307b74aee8e7d373c093d86c1b28fed1a81ab5c080bcd2e78f85257688ed7f

The finite-vertex budget at ambient64 is now464

There is a complete local face-payment rule with triangle totals at least2, quadrilateral totals at least1, and **exact finite-vertex maximum464 at rank64**. This equals the required finite curvature464. The previous finite maximum472 is no longer an obstruction to excluding dimension64. The ideal-vertex inequalities remain unresolved, so the completed general dimension bound remains64.

The binder and certificate combine three proved local transfers with the previously checked exact conditional budget. They do not turn the remaining ideal deficits into a contradiction.

The complete simultaneous rule

Start from the completed dimension64 payment theorem. Apply the depth11 zero rule: a finite distance8 pair in a whole A/B/D component has weight0 when both marks are at least11 from every endpoint, branch or nonsimple-bond endpoint. All other finite payments and coupons initially remain unchanged.

Next apply the two-unit exchange. In canonical finite D numbering, delete D16(0,11),(0,12) and restore D17(1,12),(1,13) to weight1. The resulting costs are D16=116 and D17=127. The other rank16/17 costs remain A16=116, B16=104, A17=114 and B17=105. Every coupon is unchanged.

These two rules have a complete noncircular compatibility proof in the exchange report. A changed deep8 corner supplies a short7 alternative whose constructive outputs avoid the two D16 long-pair deletions. An affected D16 selected context either supplies the same retained short-path branch or seals its19-root core, giving a whole restored D17 unit at a different corner. All literal, adaptive and universal interfaces are checked. The triangle proof uses finite7 and is unchanged.

Finally apply the affine cycle11 half rule to all11 distance5 pairs in that whole ideal factor. Its cost becomes99/2, with no finite payment or coupon added. This composition can be stated directly in terms of actual weights. If no cycle11 corner changes, the preceding complete rule retains its old total. If a cycle11 corner changes, the cycle theorem supplies either a finite unit at distance at most6 or a distinct terminal corner with a retained ideal short5 payment of at least1/2. The original cycle corner retains1/2. No output uses a deleted finite8 or finite D16 long pair. The cycle theorem separately excludes such a corner on any triangle. Thus all three rules preserve the required face totals simultaneously, without assuming the old cycle unit survives.

Discharging the exact finite-budget hypotheses

The last scenario in the exact ambient64 frontier has parameters feature_depth=11, deep_pair_weight=0, D16_drop=2 and D17_drop=−2. These are exactly the now-proved changes above, including the increase of the D17 cost. No new coupon has been introduced, and the cycle transfer changes only ideal costs.

That scenario's full finite-forest optimization is consequently applicable. It covers all finite Coxeter types and all rank64 forests, with every existing cross-component coupon. Its753 large-component count vectors give348 coupon environments;38,025 type/partner comparisons justify the dominating representatives. The exact Bellman certificates contain597,168 upper inequalities and22,272 attaining predecessors. The literal coupon formula is checked42,921 times,

and all753 global alternatives have a reconstructed attaining forest. Classical large-rank deep-pair counts are the proven path-start formulas; exceptional types and arbitrary dihedral labels retain their previously classified representatives.

The exact maximum is464, attained by the formal forest4D16. This is attainment in the finite-vertex optimization domain, not a claim that this forest occurs in an actual dimension64 polytope. Its finite slack against curvature464 is0.

The remaining ideal obstruction

The affine cycle11 cost is lower, but other order11 families still have cost55. Therefore the maximum over types, used in the complete ideal comparison, remains unchanged. The same1,513 terminal ideal profiles still have24 negative margins, with minimum−62665/419628 at ranks (10,10,11,16,16). The stated profiles are conservative envelope cases; no assertion of geometric realizability is made.

The inherited global additive constant is5120. Zero finite slack would be sufficient if the remaining ideal margins were nonnegative, but they are not yet so. The separate frontier gives sufficient conditional all-family targets and the required enlarged2,419-profile domain. Those ideal targets remain obligations; this packet proves no dimension63 bound.

Verification

Generation96a342 and separate fresh replayf3ee6d both terminated exit0/PASS. The binder checks the exact rule parameters, frozen dependency bindings, finite maximum/attainer, certificate counts and unchanged ideal deficits. It does not regenerate the inherited Bellman tables or geometric censuses. Their completed exact checks and written review scopes remain explicit inputs. The new contribution here is discharging the local geometric hypotheses of that exact budget and composing the three local rules.

A separate full binder/report mathematical read passed the exact scenario match, noncircular actual-cycle-corner split, unchanged finite coupons, maximum464 with formal attainer4D16, and remaining ideal deficits. The review included the conditional budget report and its interfaces, without rerunning the Bellman or scalar tables. Source, certificate and report are final/frozen. All review is internal; no external refereeing or proof-assistant verification is claimed.

Source 26

Source path: `results/fresh_ambient64_ideal_margin_closure_20261018.md`

SHA256: e467cee23830679ba7eaa7ca10126056f71737e08d99f71c4474829219330a54

Positive ideal-vertex margins at ambient dimension64

The new order11 and order17 factor envelopes give **positive margins for every possible ideal-vertex rank profile at ambient dimension64**. The minimum normalized margin is2385/146432; the minimum absolute surplus is 240/23. The complete finite coupon budget and simultaneous global payment composition are separate obligations. This scalar packet alone asserts no new dimension bound.

The source and certificate independently reconstruct the complete corrected profile domain. They use rank multiplicities and sliding-window polynomial multiplication, rather than the earlier diagnostic's sorted-rank recursion and direct convolution. No project arithmetic helper is imported.

Exact factor envelopes

The affine11 cycle transfer and coupled order11 tree transfers give total costs A=99/2, B=54, C=109/2, D=53. Hence the maximum divided by factor order is109/22 at affine rank10. The tree transfers add a finite half coupon between a whole A1 and the doubled terminal of a separate whole B10.

The order17 refinement gives A85/B111/C108/D118, with envelope118/17 at affine rank16. Its finite E6/D11 coupon increases from a quarter to a half on the specified two-by-two marked roles. These coupon costs must be included on the finite side; they do not enter an ideal factor's pair total.

All other ideal factors retain the previous completed envelopes. For affine rank r the complete normalized upper envelope is

Rank r	c(r)
1 through9	r/2
10	109/22
11	16/3
12	74/13
13	6
14	94/15
15	107/16
16	118/17
17 and larger	163/23

Exceptional affine factors have rank at most8 and retain their earlier envelopes. The factor-fiber multiplicity remains V/(r+1) for each marked pair, so the cost at an actual cusp is at most V times the sum of c(r).

Complete rank-profile reduction

Use the inherited all-rank convex-order concentration theorem. For two positive ranks below9, transfer rank from the smaller to the larger until the smaller vanishes or the larger reaches9. For two ranks at least18, transfer until the smaller reaches17.

Every transfer decreases normalized curvature and preserves the sum of costs: the respective cost ranges are linear and constant. It strictly increases the bounded sum of squared ranks, so the procedure terminates.

The total affine rank is63. A terminal profile therefore has arbitrary multiplicities of9 through17, at most one rank1 through8, and at most one rank at least18. There is no additional restriction on repeated middle ranks. The source enumerates each of the2,419 profiles exactly once by choosing the nine middle-rank multiplicities and then the two possible extreme factors.

The older1,513-profile domain, based on a linear envelope through rank10, would not be complete after changing the rank10 cost. The larger domain is essential here.

Exact curvature calculation and absolute surplus

For a profile(r_i), set

```
H(t)=product_i(1+t+...+t^r_i),  V=H(1),
G=sum_j [t^j]H(t)*min(j,63-j),
margin=(5G-63V)/(2V)-sum_i c(r_i).
```

Every polynomial, V, G, curvature, charge and margin is computed exactly. Every row agrees with the corresponding scalar data in the earlier conditional frontier. The new normalized margins agree with its simultaneous cost drops of 1/2 at factor order11 and1 at factor order17. This comparison uses all 2,419 rows, not only the extremal examples.

The unique minimum normalized margin is2385/146432, at ranks (10,11,12,15,15). The unique minimum absolute margin V times the normalized margin is240/23, at the single-factor profile(63). All rows are positive.

The absolute lower bound also holds before concentration. For a transfer delta from a<=b to the larger factor,

```
(a+1)(b+1)-(a-delta+1)(b+delta+1)
  =delta*(b-a+delta)>0.
```

Thus V decreases at every transfer, including disappearance of a factor, whose contribution then becomes1. Normalized margin also decreases and its terminal value is positive. Their product consequently decreases. The terminal absolute minimum240/23 therefore bounds every actual cusp, without a simple-cusp, arithmetic or number-of-factors hypothesis.

The shape-curvature extremum estimate, with the quantitative strengthening recorded in the previous ideal-margin proof, supplies constant5*floor(64²/4)=5120. The finite side must have maximum at most464 to combine this positive constant and ideal surplus into a dimension64 contradiction. This packet does not establish that finite hypothesis or the full simultaneous face-payment theorem.

Verification scope

Final source-bound generation621472 and separate fresh replay902f1b both terminated exit0/PASS on all2,419 profiles, both unique extrema and every field comparison with the independently implemented earlier conditional calculation. The certificate binds the completed local proof packets. Its rank-multiplicity recursion and sliding-window polynomial arithmetic are separate implementations of those scalar checks.

A full separate source/report mathematical read, including the inherited convex-order and shape-curvature arguments, passed the complete threshold9/17 domain, preserved costs, V-product monotonicity, positive absolute surplus, factor envelopes and constant5120. That reviewer did not rerun the2,419 rows. Source, data and report are final/frozen. All review is internal; external refereeing and proof-assistant verification remain absent. The geometric factor-fiber theorem and all-rank concentration theorem remain explicitly inherited inputs; scalar arithmetic does not prove those geometric statements.

Source 27

Source path: `results/fresh_ambient64_payment_frontier_20261018.md`

SHA256: 2c1a8526421925f5e1c798ec41019f3d67820b2660f1e250789e342df6c7778d

Exact ambient64 payment frontier

Under the completed dimension64 payment rule, the exact finite-vertex maximum at ambient64 is **472**, against curvature464. The complete current ideal comparison has **1,513 terminal profiles**, of which24 have negative margins. Its minimum is−62665/419628 at affine ranks (10,10,11,16,16). Thus the current rule does not exclude dimension64.

The source and certificate also give conditional targets. They prove scalar and finite-forest statements only. No proposed pair deletion, replacement coupon, changed local payment theorem or dimension63 bound is asserted here.

Exact current finite maximum

All components of rank at most17 retain their current weights. The frozen finite budget has exact rank64 maximum472 on that restricted domain, attained by four whole D16 components, each of cost118.

For a forest with a component of rank r>=18, the proof of the depth17 budget applies unchanged: its type is A/B/D, it participates in no current cross-component coupon, and D dominates its modified same-component cost. Remove that component and bound the remaining modified forest by the old unrestricted maximum at rank64−r. The47 exact upper bounds have maximum469, attained at r=64 by whole D64. The next bounds are467 at r=48 and933/2 at r=47. Hence this branch is strictly below the all-small maximum472.

This proves the sharp current maximum without assuming that the remainder has small components. A separate conditional optimizer below also returns 472 in its baseline calibration.

In particular, if coupons stay unchanged, the witness4D16 requires reducing the D16 same-component cost by at least2 to reach464. That change alone is insufficient: whole D64 still costs469. These are formal finite vertex forests; the calculation does not assert that any such vertex occurs in an actual dimension64 polytope.

Complete current ideal domain

For affine rank r the current normalized factor envelope is r/2 for r<=10; then16/3,74/13,6,94/15,107/16,7 for r=11,…,16; and163/23 for r>=17. These are the unchanged envelopes of the ambient65 ideal closure.

The inherited convex-order theorem allows concentration of two low ranks until one vanishes or the other reaches10, and of two ranks at least18 until one reaches17. Each move decreases normalized curvature and preserves cost, because the relevant cost ranges are linear and constant. Thus a complete terminal domain at link rank63 has arbitrary multiplicities of10,…,17, at most one rank1,…,9, and at most one rank>=18. The source enumerates all1,513 such partitions.

For each partition r, it constructs

```
H(t)=product_i(1+t+...+t^r_i), V=H(1),
G=sum_j [t^j]H(t)*min(j,63-j),
margin=(5G-63V)/(2V)-sum_i c(r_i).
```

Every polynomial, G, V, exact cost, normalized margin and absolute margin is saved. There are24 negative margins and no zeros. The main deficits are:

Affine ranks	Normalized margin
(10,10,11,16,16)	−62665/419628
(10,10,10,10,10,13)	−29765/204974
(1,10,10,10,16,16)	−50224/384659
(10,10,10,10,11,12)	−12997/103818
(10,10,12,15,16)	−48131/427856
(1,10,10,10,10,10,12)	−15835/190333
(10,10,10,11,11,11)	−5365/69696
(9,11,11,16,16)	−5/2601
(3,10,10,10,10,10,10)	−1927/644204

All24 negative rows are retained in the certificate, not just this table. The last profile has absolute deficit21197 and contains no affine rank above10. Therefore improvements confined to factor orders>=12 cannot settle the ideal comparison. A reduction at order11, a lower-order change, an exclusion of that actual cusp shape, or a new global compensation argument is necessary for this profile. The formal product profile is not claimed geometrically realizable.

Conditional ideal targets and the necessary expanded domain

Let a be a reduction in the maximum **total pair cost at factor order11** (affine rank10), and b a reduction at factor order17 (affine rank16). Their contributions to a profile's normalized improvement are

`a*count(rank10)/11 + b*count(rank16)/17.`

Reducing the rank10 cost destroys the old low-rank linear range through10. It would be invalid to test only the original1,513 profiles. Instead retain arbitrary multiplicities of9,…,17, at most one1,…,8 and at most one>=18. The low range through9 and the high constant range remain unchanged. The resulting **2,419 profiles** are a complete reduction domain for every scenario here. They include the1,513 original rows with identical scalar data; the current baseline has26 negatives on this larger domain.

Conditional total pair-cost changes	Negative profiles	Minimum margin
Order11:55→54	1	−5/2601
Order11:55→54.5; order17:119→118.5	0	167/419628
Order11:55→54.5; order17:119→118	0	2385/146432
Order11:55→54; order12:64→63	0	177/4624
Order11:55→54.5; order12:64→63	2	−9235/427856

With a=1/2, the exact necessary and sufficient b threshold for nonnegative margins on this domain is24517/49368, attained at(10,10,11,16,16). The choice b=1/2 exceeds this threshold and gives the displayed strictly positive minimum. With b=1, the corresponding a threshold is12997/37752, attained at(10,10,10,10,11,12). If instead order12 improves by1, the a threshold is48131/77792 at(10,10,12,15,16). No profile independent of the varied parameter remains negative in any of these three threshold checks.

These are **all-family envelope targets**, not changes to just one affine type. In particular affine A10 is an11-cycle: all55 pairs are already at distance at most5. Lowering the order11 maximum requires a new transfer for an elementary short5 pair in that cycle, in addition to any needed B/C/D changes. Deleting unused long pairs only in tree families cannot achieve this target. The hypothetical order17 reduction must lower its actual worst family, presently affine D16 of total cost119; the other families have smaller totals.

Any additional finite coupons needed for these ideal transfers must be counted separately. The scalar improvements here do not grant a free finite replacement.

Conditional finite targets with all existing coupons retained

For clarity, the diagnostic separately computes the complete rank64 finite maximum under the following proposed same-component changes. Nothing in this section proves their local geometric availability.

Deep distance8 rule	D16 cost	D17 cost	Exact rank64 maximum	Attaining forest
Current depth17 half	118	125	472	4D16
Depth17 half	116	125	469	D64
Depth17 zero	116	125	465	D17+D47
Depth17 zero	116	124	464	D17+D47
Depth11 zero	116	127	464	4D16

The last row is the parent's proposed two-unit exchange: remove two D16 units and restore two D17 units, combined with a prospective depth11 zero rule. Its finite arithmetic is sufficient even though its D17 cost rises. Its local proofs are not inputs to this diagnostic and remain separate. For every row, A16=116, B16=104, A17=114 and B17=105 stay unchanged.

The all-rank counts of affected distance8 pairs in finite A/B/D are max(0,r−2t−8), max(0,r−2t−9), max(0,r−2t−10), respectively, for feature depth t. They follow from the same path-start intervals as in the frozen depth17 budget. A zero weight subtracts two per pair from doubled old cost; a half subtracts one. No rank<=17 component is affected at either depth.

For each scenario the source independently regenerates all753 multisets of components D10,…,D16,A17 with total rank<=64, and checks equality with the frozen domain restricted to64. Their unchanged coupon interactions give348 five-coordinate environments. Residual types are A1,…,A9, with E6 replacing A6 when it dominates, and D17,…,D64. A17 remains a separate large type because of its distinct coupon. Finite type domination and all 38,025 type/partner coupon inequalities are checked in each scenario; the uniform I2 representative covers all dihedral labels.

An item-first integer Bellman calculation is followed by all597,168 last-item upper inequalities and22,272 attaining-predecessor checks per scenario. The literal component/coupon formula is checked42,921 times; all753 global alternatives have a reconstructed typed attaining forest. These checks account for arbitrary numbers of components and every unchanged coupon, including E6/D11 and A1/A2–D14 roles. They are not a sample of promising forests. The saved tables and the fresh replay are part of the certificate.

If a future compatible local theorem gives finite maximum<=464 and the positive ideal target above, the inherited global additive constant at ambient64 is5120. Zero finite slack would be sufficient: the ideal surplus and positive global constant would give a contradiction. This diagnostic does not establish those local hypotheses or a dimension63 theorem.

Verification scope

Final generation16c58d and separate fresh saved replay244610 both terminated exit0/PASS. A separate full source/report mathematical read passed the current strip bound, corrected2,419-profile concentration domain, all-family target distinction, complete753-vector/348-environment finite decomposition, type domination, explicit coupon formula and Bellman upper/attainment proof. The reviewer compared the coupon formula with its frozen provider definitions; it did not independently rerun the scalar or dynamic-programming arithmetic. No gap was identified in this diagnostic scope. The current finite proof binds the frozen sharp all-small table and the proved deep17 large-type/coupon interfaces. The ideal and conditional finite arithmetic here imports no project arithmetic module; it uses exact Fraction/integer operations and explicit formulas. The inherited convex-order theorem, finite type classification, current geometric payment theorem and coupon interpretations remain bound proof inputs. No historical geometric census is rerun. Direct source bindings, Python syntax and local report links pass. Source, data and report are final/frozen. Reviews are internal; external refereeing and proof-assistant formalization are absent.

```
python3 scripts/fresh_ambient64_payment_frontier_20261018.py
python3 scripts/fresh_ambient64_payment_frontier_20261018.py --verify
```

Source 28

Source path: `results/fresh_ambient65_ideal_margin_closure_20261017.md`

SHA256: e12fe1f2cd7aa36b8966e4310b765bd91b8adb1022ac31a1b9995bbca25d2541

All cusp margins at dimension65 are positive

The proved local improvements now resolve the entire ideal-vertex comparison at ambient dimension65. Every cusp has normalized surplus at least61/1936 and absolute surplus at least445/23. Consequently every finite-sided, finite-volume real hyperbolic Coxeter polytope whose vertices are all ideal has dimension at most64. The unrestricted ceiling remains65: its finite-vertex budget still exceeds curvature by7.

The source and certificate check all1,650 terminal profiles. This replaces the earlier conditional ideal targets by compatible proved local rules and an exact budget for the new coupon.

Compatible local rules and complete profile domain

The order15 B, C and D refinements give costs A75/B94/C91/D94 without added finite cost. The order17 D transfer gives A85/B111/C108/D119 after adding the precisely typed E6/D11 quarter coupon. Its complete finite budget leaves every maximum through75 unchanged. These changes affect distinct ideal orders, preserve their short and triangle interfaces, and introduce only that one new nonnegative finite coupon. The earlier order13 maximum74 and all other payments remain in force.

For affine rank r, the resulting normalized envelopes are:

Rank	Cost c(r)
1 through10	r/2
11	16/3
12	74/13
13	6
14	94/15
15	107/16
16	7
17 and larger	163/23

Use the inherited all-rank convex-order concentration theorem. Concentrate two ranks below10 until one disappears or the other reaches10. Concentrate two ranks at least18 until one reaches17. These moves decrease normalized curvature and preserve total cost, because the respective cost ranges are linear and constant. Each increases the bounded sum of squared ranks.

The complete terminal domain therefore has arbitrary multiplicities of 10 through17, at most one1 through9 and at most one rank at least18. There is no extra duplicate-rank reduction. All1,650 terminal profiles of total rank64 are enumerated.

For each profile define H(t)=product_i(1+t+…+t^r_i), V=H(1), and G=sum_j[t^j]H(t)*min(j,64-j). The exact normalized margin is

`m = (5G−64V)/(2V) − sum_i c(r_i).`

Its minimum is61/1936 at(9,10,10,10,10,15). Every margin is positive. All polynomials, V and G agree with the frozen complete-domain predecessor, and all new margins agree with its formerly conditional simultaneous target.

Absolute surplus and a stronger global constant

The absolute surplus V*m has minimum445/23, attained by the single rank64 profile. This lower bound also applies before concentration. Indeed, for 0<a<=b and a positive allowed transfer delta,

```
(a+1)(b+1)-(a-delta+1)(b+delta+1)
  = delta*(b-a+delta) > 0.
```

Thus V decreases under every concentration, with a vanished factor having value1. Normalized margin also decreases, and its terminal value is positive. Their product consequently decreases. The minimum over the terminal domain bounds the absolute surplus for every actual cusp.

The shape-curvature theorem has a quantitative extremum argument that admits a sharper constant here, and in fact for all the same finite-volume Coxeter polytopes. At every vertex figure, finite or ideal, H is a product of polynomials 1+t+…+t^r, so every coefficient h_j for0<=j<=d is at least1. Its proof gives

```
E-G = sum_(j>d/2) h_j*(2j-d)
    >= sum_(j>d/2)(2j-d)
    = floor((d+1)^2/4).
```

At each of the two distinct global extrema the cutting hyperplane misses the entire vertex figure, giving E rather than G uncut edges. Their combined extra contribution to D5 is therefore at least 5*floor(n^2/4), strengthening the earlier5(n-1). No new geometric hypothesis is needed. At n=65 this constant is5280.

The resulting theorem and remaining finite obstruction

Write v_f and v_i for the numbers of finite and ideal vertices. The compatible triangle/quadrilateral payment theorem gives

```
D5 <= 487*v_f + sum_(ideal v) V_v*sum_i c(r_vi).
```

The strengthened shape inequality gives

```
D5 >= 480*v_f + sum_(ideal v) (5G_v-64V_v)/2 + 5280.
```

Combining them with the absolute ideal surplus proves the necessary dimension65 inequality

```
7*v_f >= 5280 + (445/23)*v_i.
```

In particular, v_f=0 is impossible. The completed unrestricted ceiling65 already excludes dimensions66 and higher, so the all-ideal ceiling is64. The same inequality requires at least755 finite vertices in any remaining dimension65 example, or at least758 if it has an ideal vertex. These are necessary conditions, not claims that such examples exist.

There is no upper bound on v_f in this argument. In particular the inequality does not eliminate dimension65 in general. The formal finite forest D17+3D16 still costs487 against480. Arithmeticity, simplicity of cusps and a facet-count bound are not assumptions of the all-ideal result. No new finite-volume example or arithmetic dimension bound follows.

Verification status

The local input theorems and the complete finite budget have finished generation, fresh replay and separate internal mathematical reviews. Final generationa94cea and separate saved replay967ad6 terminated exit0/PASS on all1,650 rows. The independent scalar reconstruction uses a different enumeration and arithmetic implementation and agrees on every saved scalar field and both unique extrema. Its separate replay passes.

A separate full source/report mathematical composition review reread the shape-curvature and factor-fiber proofs, final finite budget and independent scalar scope. It checked the5280 extremum constant, product monotonicity, absolute445/23 surplus, simultaneous payment rules and the all-ideal64 conclusion, finding no unresolved step. It did not duplicate the scalar arithmetic, geometric censuses or finite-budget replay. The geometric census and all-rank convex-order theorem retain their explicitly stated inherited review scopes. No external review or proof-assistant formalization is claimed.

```
python3 scripts/fresh_ambient65_ideal_margin_closure_20261017.py
python3 scripts/fresh_ambient65_ideal_margin_closure_20261017.py --verify
```

Source 29

Source path: `results/fresh_ambient65_ideal_margin_closure_20261017_review.md`

SHA256: 1ff042db6cec0e79a7e107747f69372af5b9f93d87d23349dafbdaced9a635e1

Independent reconstruction of the ambient-65 cusp margins

All **1,650 terminal profiles** and every saved scalar field agree with an independent standard-library calculation. The unique minimum normalized margin is **61/1936**, at ranks (9,10,10,10,10,15). The unique minimum absolute margin is **445/23**, at the single rank64 factor.

The independent checker imports no producer or project arithmetic. Its certificate records compact scalar rows, the two complete extremal rows, and matching SHA-256 digests of all complete primary and independently reconstructed rows. It compares the scalar data in the primary certificate.

The enumeration differs from the producer's sorted-partition recursion: it chooses an optional low rank1..9 and an optional high rank18..64 first, then enumerates all nonnegative multiplicity vectors at ranks10..17 with the remaining total. The resulting1,650 rank64 profiles are distinct, and their set equals the complete saved row set.

For each profile, integer sliding-window convolution computes all65 coefficients of $H=\prod(1+x+\cdots+x^r)$. The checker verifies symmetry and $V=H(1)=\prod(r+1)$. It computes G using symmetric cumulative-distribution tails,

```
G = Σ[k=1..32] (V − 2 Σ[j=0..k−1] H_j),
```

and separately checks the absolute-deviation identity

```
2G = 64V − Σ[j=0..64] H_j |64−2j|.
```

The exact rational cost schedule is r/2 through10; then16/3,74/13,6,94/15,107/16,7 at ranks11..16, and163/23 at every rank$\geq$17. The curvature is $(5G-64V)/2$. Every H coefficient, V, G, cost, curvature, normalized margin and absolute margin is compared with its primary row; all margins are strictly positive. Both stated minima and their unique attaining profiles are independently recovered.

Generation and a separate fresh `--verify` invocation both terminated exit0/PASS. This is an independent scalar arithmetic/interface check. It does not repeat geometric censuses, the all-rank concentration proof, the finite-coupon budget, or the changing additive global-constant argument. The whole primary source/report is therefore not bound as an independently reviewed global theorem; the complete scalar rows are bound by their canonical content digest. This distinction permits the separate global-constant proof to be finalized without implying that it was audited here.

Source 30

Source path: `results/fresh_b14_b15_singleton_completion_transfer_20261017.md`

SHA256: 0ee2ce06396ff08c2d62bb081cc416b8d2f58f7d681f790b94a48aed3c65592e

Two redundant singleton payments in affine B factors

Remove the canonical pair `(0,10)` in affine B13 of diagram order14, and `(1,11)` in affine B14 of diagram order15. Their costs fall from87 to86 and from97 to96, respectively. No finite coupon or replacement half is needed: every selected-core occurrence of either pair would require a norm-two supporting root whose forced projection has square greater than2. The required actual affine factor is therefore impossible in an otherwise unpaid face.

The source and certificate check the complete saved support, all64 affected induced maps, and their exact necessary projection inequalities. This package makes no ambient dimension assertion. It changes only these two singleton weights; other possible reductions remain outside its scope.

The affected rules and complete map support

For order14 the B diagram has spine0–1–…–11, a doubled terminal bond10–11, and two short leaves12,13 at0. For order15 it has spine0–1–…–12, a doubled terminal bond11–12, and short leaves13,14 at0. The proposed singleton in each diagram has distance10 and is fixed by every diagram automorphism.

The order14 rule starts with its87 retained pairs in the frozen complete small-factor map certificate. The order15 rule starts with the97-pair previous B14 reduction, so its three previously deleted pairs keep their existing proofs. The new source checks all elementary triangle pairs, every pair at distance at most8, and every basic ideal5 pair remain paid.

Target diagram	Deleted pair	Affected selected contexts	Affected maps	New cost
Affine B13, order14	0,10	23	46	86
Affine B14, order15	1,11	11	18	96

Five contexts occur for both targets, so these are29 distinct old-core contexts and34 target/context occurrences. The complete context lists and every marked map are saved. The map counts include only maps with the changed marked image; maps of the same unmarked pattern to other pairs are unchanged. This avoids treating every map in an affected context as an affected payment.

A necessary projection bound with arbitrary additional contacts

The complete overlap theorem applies after the retained short8 alternatives have been handled. Its cross-core exclusions fix the literal old core H. In all contexts used here, every old core root has square2, every nonzero off-diagonal entry is-1, and H has signature `(order(H)-1,1,0)`. The checker reconstructs its exact inverse J and its actual preferred corner. In particular it checks the necessary core-contact domain for any additional supporting white root, using the corner's finite inverse diagonals or affine kernels.

The saved contact domain splits into two end sets. Both diagonal inverse blocks are entrywise strictly positive. A further white root meeting both ends would give an adjacent marked path of length at most8. The source recomputes the distance bound; its actual maximum in these cases is6. Such a path already resolves the face, so in an otherwise unpaid face all contacts of a new white e with H lie in one end set.

Let a be the nonnegative vector of contact magnitudes of e with H. Since H is nonsingular and Lorentzian, its orthogonal complement in the ambient Lorentz space is positive definite. Thus

`projection_H(e)^2 = a^T J a <= e^2 = 2`.

If the actual target factor forces a contact of magnitude q at old node i, entrywise positivity on that end gives

`a^T J a >= q^2 J_ii`.

This remains a necessary inequality when e has further contacts anywhere on the same end. Contacts on the opposite end are already resolved by the short-path alternative. No unspecified contact is set to zero, and there is no restriction on the number, labels or indirect chains of additional supporting whites.

All projection comparisons use the norm-two normalization before any diagonal congruence: a label4 bond has magnitude sqrt(2), hence square2. The new source does not introduce a rational gauge or change the squared norm of e. Consequently the obstruction for such a bond is `2 J_ii>2`.

Every map forces the offending doubled terminal

Let K be the displayed connected marked component of the preferred core corner, and suppose it lies in an actual affine B factor F of the indicated order. Its embedding into the fixed target is one of the saved induced maps. The target's doubled terminal is absent from K, while its adjacent node is in K and is one of the two marked nodes. This is checked for every one of the64 maps. The required terminal is therefore a supporting white e with a forced label4 contact to that old marked node i.

This e must be a genuinely new root outside H. An old root absent from the preferred corner is one of its two omitted black roots and cannot belong to F. Every other old corner root outside K belongs to a different connected component and is orthogonal to K. None can be the terminal, which must have a nonzero bond to i. The source verifies these old corner components and their zeros against K. This elementary adjacency argument identifies the witness even if other nodes of F were to lie in old components reached by indirect paths; no no-bridge lemma or completion of the whole factor is needed.

For every map the exact old full-core inverse diagonal satisfies J_ii>1. The complete obstruction values are:

Target order	J_ii	Forced projection lower bound2 J_ii	Number of maps
14	5/4	5/2	2
14	2	4	8
14	7/2	7	8
14	8	16	16
14	43/2	43	8
14	26	52	4
15	5/4	5/2	2
15	2	4	6
15	7/2	7	8
15	8	16	2

Every lower bound is strictly greater than the root's square2. Thus the extra terminal cannot exist unless a retained short payment already resolved the face. This excludes the selected affine-factor occurrence itself; it is not an assertion that a finite displayed subdiagram has become a whole finite component.

Coverage beyond the selected maps

The complete finite8/ideal5 theorem and its successors first cover triangles and marked paths of length at most8. All their required elementary B pairs retain unit weight here. Universal cycle rules are unchanged. The universal tree and D-transfer immediate whole-affine alternatives have order at most13, so they impose no new order14 or15 obligation. Their existing successors also remain unchanged.

After the short-path reduction, low-core configurations have whole marked component order at most12, and disjoint high cores are excluded by their all-label theorem. Finite whole-component, protected-leaf and adaptive finite payments keep their rules. Protected or basic ideal outcomes have marked distance at most9 and retain unit weight. The remaining high-core overlap cases are precisely those covered by the complete small-factor map certificate. It lists the23 and11 affected contexts here, and every affected marked map has the obstruction just proved.

Every triangle therefore still receives at least2 and every quadrilateral at least1. The finite budget is unchanged, including all existing halves and units. The intrinsic ideal pair costs become86 and96. Other types are unchanged, so the order14 maximum is86, and the order15 maximum is96 (shared with its unchanged D rule). A global dimension comparison is a separate task.

Verification scope

Primary generation86321 and the separate fresh saved replay (terminal command7d085b) both completed exit0/PASS. Syntax, all16 direct source bindings and all6 report links passed. A separate exact SymPy calculation, without importing producer arithmetic, independently reconstructed the inverses of all13 distinct old core Grams, checked their Lorentz signatures using characteristic-polynomial Sturm counts, and checked all64 forced projection inequalities across the34 target/context occurrences. The checker validates every direct input binding, reconstructs each literal Lorentzian core and preferred corner, verifies the exact inverse end blocks and contact-domain coverage, and checks every affected induced map and terminal projection inequality. The complete overlap and map censuses remain frozen inputs; they are not independently re-enumerated. No external refereeing, formalization or new dimension claim is made.

A separate full internal source/report mathematical read passed. It checked the genuinely new terminal witness, the norm-two doubled-contact bound, entrywise-positive inverse blocks with arbitrary additional contacts, the retained short-path alternative and the complete local-domain composition. It did not repeat the frozen overlap or induced-map censuses. Source, data and report are final/frozen.

Source 31

Source path: `results/fresh_balanced_rank16_order14_ambient60_frontier_20261018.md`

SHA256: 6444c9855bb509df1e14719da966d6e714bee6418f3af79774490bfd0578d9cb

Balanced finite costs409.5 and17 remaining ideal profiles

The combined ambient60 calculation has finite maximum **819/2=409.5**, against curvature405, and **17 negative terminal ideal profiles** among 2,113, with none zero. The preceding figures were411 and28. This is a further local and counting improvement; a dimension59 theorem remains unfinished. The completed unrestricted bound is still60.

The producer and certificate combine four local changes with the preceding fractional rule. The finite computation counts every new payment and restoration together.

One simultaneous rule

The D16 zero transfer first reduces D16 from111 to110. It adds two units per whole D5/A11 component pair, at the unique D5 long endpoint and either A11 second node. Their actual old cost was0. The unchanged representative A5/D11 allowance 19/2 dominates the new actual cost2 at both ends; this is checked for every finite type pair, rather than omitted from the count.

The A16/A17 exchange reduces A16 from111 to110 and increases A17 from116 to117. Its two sealed whole contexts replace a selected half and a distinct alternate half by zero and a full unit. The A17 increase is essential to the combined count.

The eighth balance increases the two D16 endpoint/fork-leaf halves to5/8 and decreases the two associated D17 halves to3/8. Each of the four sealed quadrilaterals still receives5/8+3/8=1. Its complete earlier selected D17 domain does not use either reduced pair. The resulting same-component costs are:

Finite type	Cost
A16	110
D16	441/4
A17	117
D17	469/4

The order14 C theorem removes the two remaining nonbasic distance10 units without a new finite charge. Their genuinely new doubled terminals would have projection square greater than2. All600 eligible native searches and28 affected maps are covered by its complete arbitrary-contact argument. The order14 family costs become A70/B82/C81/D82, so the maximum decreases83 to82.

These proofs preserve one simultaneous triangle/quadrilateral rule. Handle the completed independent B/C11 actual-corner priorities and the new C14 priority with their universal short/basic alternatives. The C14 contradictions need no replacement finite payment. The current order15/17 ideal transfers consume unchanged typed coupons, not the newly reduced long A16 or D17 pairs. Every ideal weight outside C14 is unchanged.

For finite changed contexts, the D16 zero proof uses its new D5/A11 unit; the A16 proof uses a distinct whole A17 unit; the four D16 fork contexts use their distinct5/8 and3/8 payments. None consumes another newly reduced selected pair. The rank15 A16/D16 halves, A17 unit and other later finite outputs remain retained. All independent short8-to-short7-to-short6

routes and their protected tree/cycle outputs are inherited, avoiding these long changed pairs. Thus the local arguments do not rely on the new inequality they are proving.

The binder reconstructs all216 inherited literal configurations from their native Grams, beginning with the D16 zero and eighth-balance payments. It checks every A16/A17 and C14 marked image anew. There is no additional debit or credit; the four earlier D16 eighth credits remain present. Each total is at least1. This finite list verifies later literal interfaces; the universal coverage belongs to the local proofs, not to this list.

After these priorities and their complete selected/nonselected routing, all remaining actual corners retain the preceding pointwise rule or an increased payment. Triangles receive at least2 and quadrilaterals at least1.

Complete finite count

All costs are scaled by8. Relative to twice the preceding quarter-unit rule, the base changes are A16:−8, D16:−6, A17:+8 and D17:−2. The actual D5/A11 cross cost increases16; the representative cross rule does not change. All earlier coupons, including D5/D12 cost47/4, and the deep finite7 deletions remain included.

The complete inherited rank60 domain has1,044 fixed-component vectors and792 interaction keys. Simultaneous domination checks cover33,489 type pairs. The residual D17 item decreases2 eighth-units; fixed D16 and A17 costs change explicitly. New tables are generated for every key. All54,288 item identities match the complete cross rule. Replay checks 1,142,064 Bellman upper inequalities and47,520 attaining predecessors, without invoking an optimizer. Fixed-fixed and residual-residual representative cross costs are zero in the declared split.

The maximum is3276 eighth-units, or409.5. All three formal forests

```
2D16+D28, D16+D17+D27, 2D17+D26
```

attain this cost under the actual rule. This proves equality for the formal forest optimization; it does not assert geometric realizability by a hyperbolic vertex. Fifteen representative alternatives still exceed405. The finite deficit is9/2, reduced from6.

Complete ideal count and remaining obstruction

The binder independently reconstructs the preceding exact integer convolutions and scalar rows on the complete2,113-profile rank59 domain: at most one part1 through8, with every other part at least9. It checks agreement with the frozen fractional frontier before applying the saving

```
count(rank13)/14.
```

This is the all-family order14 maximum change83→82, with the correct factor-fiber multiplicity. Eleven formerly negative profiles become positive, leaving17 negative and none zero. The normalized minimum remains−1117/7680. Every other current ideal envelope, including order17 maximum227/2, is unchanged.

Only the inherited low9 concentration is used. Negative terminal absolute margins are not extended by the positive-margin comparison to arbitrary unreduced cusps. No such universal negative estimate is needed or claimed here. At ambient60 the finite curvature is405 and the single two-extrema correction is4500, counted once. The remaining finite deficit and negative ideal profiles prevent a dimension59 conclusion from this calculation.

Verification status

Generation session98396, terminal chunk925c18, and separate fresh replay session73364, terminal chunk659b9f, both terminated exit0/PASS. Two peers completed full source/report mathematical composition reads: PASS. They reread the inherited finite cost and domain interfaces, independent local priorities and ideal convolution/concentration engine. They checked the joint eighth scaling, actual new coupon and representative domination, the fixed A17 versus residual D17 split, complete Bellman upper/attainment logic, three actual attainers, current216 literal interfaces, exact order14 saving and the incomplete ambient60 conclusion. Neither reviewer repeated DP, scalar convolutions, map, literal or historical census arithmetic. The producer checks all bound source hashes, exact finite cost interfaces, typed domination, new Bellman certificates, complete ideal convolutions and current literal configurations. It imports the universal geometric theorems as frozen inputs and does

not rerun their historical all-label censuses. No external mathematician review or proof-assistant verification is claimed. Source, certificate and report are final/frozen; the preceding packets remain frozen.

```
python3 scripts/fresh_balanced_rank16_order14_ambient60_frontier_20261018.py --verify
```

Source 32

Source path: `results/fresh_broad_finite_half_coupons_20261017.md`

SHA256: 53b13128bae00e5f771984adb639916f697f82ffd013606265cc0029b79d351c

Broad finite-component half-payments retain the9n budget

For every finite Coxeter vertex of rank n<=108, the following payments have total at most9n. Retain the previous unit payment when the two marked roots belong to the same finite component and either their distance is at most9 or that whole component has order at most17. Add a half-payment for a pair in different whole components if:

- one component has rank at most4, with its marked node arbitrary;
- the other is D12,D13,D14,D15,D16, with its mark the fork center or one of the two short-arm leaves; or is A13 or A15, marked at its unique middle.

Each eligible cross-component pair receives one half, once. This includes all the preceding A1/D16, A2/D14, A1/D15, A2/D13 and A1/A15-center coupons. It also allows A13-center and the D12/D14 finite corners arising in the new local analysis. Eligibility always concerns whole actual components.

The exact certificate and checker verify the following elementary budget. The inherited finite distance9 estimate bounds the base cost of every component of rank r by9r. For r<=17 the base cost is exactly binom(r,2), since every pair is paid.

Let a be the sum of the ranks of all components of rank at most4. Their base-cost slack is at least15a/2, because

```
9r-binom(r,2) >= 15r/2       for1<=r<=4.
```

Let t count the listed large components D12,…,D16,A13,A15. Each has rank at least12, base-cost slack at least24, and at most three eligible marked nodes. The total new cross-component cost is therefore at most 3at/2. All other components keep their original nonnegative slack. Hence

```
cost <= 9n + a*(3t/2-15/2)-24t,       a+12t<=n.
```

For n<=108, t<=9. If t<=5 the coefficient of a is nonpositive. At t=6,7,8,9 the respective maximum values of a are36,24,12,0, giving excess bounds-90,-96,-138,-216. Thus the displayed surplus is always nonpositive. This proves the claim for every finite vertex in that range, independently of its other components or finite labels.

The checker uses exact rational arithmetic for all ten count cases and the two endpoint choices of a in every dimension1,…,108. The endpoint checks corroborate the linear inequality; they are not a cutoff in the finite-diagram classification. Generation and a separate saved replay both terminated exit0/PASS. A separate agent read the complete source and proof and checked the component slack, cross-pair count and all count cases. That read does not repeat the inherited finite classification or prove any face payment. This source/data/report are final/frozen.

This is a budget theorem. A separate geometric argument must establish that a face receives these payments, including that the components remain whole after all supporting whites are added. No ambient dimension bound or new polytope is claimed by this package alone.

```
python3 scripts/fresh_broad_finite_half_coupons_20261017.py
python3 scripts/fresh_broad_finite_half_coupons_20261017.py --verify
```

Source 33

Source path: `results/fresh_center_reversion_rank16_balance_ambient60_frontier_20261018.md`

SHA256: 741fc285012fe3626dbe61a5aa2bce6adb230379a23140aec00173b0f9a987bc

Center reversion and finite corner balance: exact maximum408

The complete ambient60 rule now has exact formal finite maximum **408**, down from409, while all2,113 terminal ideal margins remain positive. The ideal normalized minimum stays31/6000 and the absolute minimum stays54. Every finite-sided, finite-volume real hyperbolic Coxeter polytope of dimension60 consequently satisfies

$$3\,v_f \geq 4500 + 54\,v_i.$$

This requires at least1500 finite vertices, or1518 if an ideal vertex exists. The all-ideal ceiling remains59 and the unrestricted ceiling remains60. The remaining finite deficit is3 against curvature405; no dimension59 theorem for general polytopes is asserted.

The producer and certificate start from the frozen order12/order16/depth8 rule. They certify two successive compatible changes, retaining every other weight and coupon:

Stage	Ideal order17 maximum	Finite maximum	Actual formal attainer
Frozen predecessor	227/2	409	4A1+4D5+3D12
Center reversion	114	817/2	2D17+D26
Reversion plus A/D balance	114	408	4D15

Each formal forest is an optimization witness, not a claimed geometric vertex or polytope. The current finite D15 cost remains102 throughout. Any further D15 change needs a separate theorem and budget.

Revert exactly one coupled ideal/finite transfer

The center-quarter theorem had replaced the affine D16/order17 halves(0,10),(2,12) by quarters, and upgraded the whole D5 unique long endpoint against the whole D12 fork center from1/2 to3/4. Restore the two ideal halves and return that single eligible finite marked pair to1/2. This is exactly the relevant part of the previously proved D114 rule. It increases the ideal family cost by1/2 and decreases the finite D5/D12 component-pair total by1/4, from47/4 to23/2.

This reversion does not reduce the separate D5-long-end/D12-short-leaf units or the finite D17 transfer's D5-short-leaf/D12-center-or-short-leaf units. Their canonical marked sets are disjoint:

Role	D5 mark	D12 mark	Current weight
Reverted center pair	0	9	1/2
D114 short-leaf pairs	0	10,11	1
Finite D17 compensation	3,4	9,10,11	1

The combined D5/D12 total is therefore 15/2+1+3=23/2. No other finite cross-component charge changes.

The producer reconstructs the complete old whole-core proofs in contexts1314 and2529 and compares them to the frozen center certificate. Both use the same19-root physical norm2 core of signature(17,1,1). The whole affine corner has a positive

kernel. Its actual corner Gram is positive semidefinite, so every extra supporting white is orthogonal to this kernel. Its individual products with the affine roots are nonpositive, forcing orthogonality to the entire affine component. The full-core radical represents zero in Lorentz space: its represented vector is null and orthogonal to a span containing a timelike vector. Its nonzero opposite coefficients at the two omitted black roots force an extra white to contact both or neither. Contacting both gives a distinct actual length2 unit; otherwise every extra white is orthogonal to the entire core. A spectator chain cannot attach indirectly, because its first attachment would be such an extra white.

Hence the displayed whole D5 and D12 at the paying corner remain whole for arbitrary supporting whites. Each of the two faces now receives its original ideal half and a distinct finite half, totaling1. The producer checks the distinct corner identities, full intrinsic marked roles and every entry of the restored136-pair ideal table. The table equals the D114 predecessor exactly; the family totals are A85/B111/C108/D114.

The original complete140-pattern/488-map selected-domain proof and selected/nonselected routing are inherited from the frozen D114 and center packets. Only these two full-factor contexts used the new center three-quarter coupon to compensate the selected quarter. Reverting them restores their original half-plus-half proof. Other preexisting routes already worked with the center coupon at1/2. All subsequent local outputs are also independent of its temporary extra quarter: the new order12 theorem uses A1/B11, finite B12 or retained D11 payments; order16 uses D5/D11; order13 uses D10/D12 roles or A4/B9; order15 uses A4/A5 against D11; other order17 transfers use their separate marked D5/D12 or E6/D11 roles. The finite D17 short-leaf compensation is the disjoint set in the table. No newly established route needs this particular D5-long-end/D12-center payment to exceed1/2.

Thus this is a compatible reversion of both sides of an existing local transfer, not an arbitrary deletion of finite charge from the present whole-face inequality.

A16/A17: one eighth plus seven eighths

The frozen A16/A17 exchange uses two distinct whole finite corners in contexts1657 and2067. Its present selected A16 weight is0 and alternate A17 weight1. Assign the A16 orbit(0,11)/(4,15) weight1/8 and the A17 orbit(1,12)/(4,15) weight7/8. Their sum stays1 at each affected face.

The producer validates all238 saved induced A16 maps in the complete 394-pattern domain, including the empty rows, and isolates exactly the four maps in those two contexts. It reconstructs both original radical seals and compares them to the frozen proofs before changing the two payments. These19-root cores have signature(17,1,1). Every possible extra-white contact has nonzero radical coefficient; both-sign contacts give the independently paid short8 alternative. Otherwise all extra whites are orthogonal to the core, including the first root of any putative spectator bridge. The displayed A16 and A17 are therefore whole components, and the two actual paying corners are distinct.

This also addresses use of the reduced A17 orbit elsewhere. It was originally absent from the retained A17 rule and was restored to halves for exactly these two A16 exchanges. The later full-payment version raised those halves to units while lowering the A16 halves to zero. Returning the A17 values only to7/8 leaves them above that original half level; all other interfaces of the original exchange retain at least their required payments. Only its two selected A16 obligations need the new numerical balance, and they receive1/8+7/8=1. The complete inherited selected/nonselected and later provenance is the one checked by that exchange, not just the four displayed maps.

The rank15 theorem's A16(1,13)/(2,14) halves and A17(1,15) unit are explicitly retained. The two A16 weights gain1/8 each and the two A17 weights lose1/8 each, giving A16=441/4 and A17=467/4. All other A16/A17 weights and all coupons remain unchanged.

D16/D17: seven eighths plus one eighth

The frozen D16/D17 balance has D16(0,14)/(0,15) at5/8 and D17(1,15)/(1,16) at3/8. Increase the D16 orbit to7/8 and lower the D17 orbit to1/8. The complete original selected D17 domain has140 patterns and168 induced maps; the producer validates each saved map and finds no occurrence of the reduced orbit.

Absence in this selected domain is not sufficient by itself. These D17 pairs were restored specifically for four later D16 exchanges: 1644,1664,1931,2324. The producer reconstructs each entire old radical seal and compares it with the bound certificate. In each, the selected whole D16 corner and distinct whole D17 corner have the precise canonical orbits above. The new total is7/8+1/8=1.

The arbitrary-support sealing is the same complete mixed-sign radical argument just described: it establishes whole components or an independent short8 alternative, without restricting contact magnitudes, spectator rank or indirect tails. The complete original nonselected routing and every later dependency were included in the prior balance. No other route uses the reduced D17 orbit. Raising the D16 weights cannot hurt other payments. The source preserves D16(1,11)=0 and the rank15 D16(1,14)/(1,15) halves explicitly. The final totals are D16=443/4 and D17=467/4. No coupon is added.

Simultaneous current coverage and exact literal accounting

The three changes act on disjoint marked sets. The center reversion uses an ideal order17 pair and a cross-component D5/D12 coupon; the two finite exchanges use only their specified whole A/D16–17 pairs. None uses another exchange's changed payment. Their eight reconstructed core arguments therefore compose directly, with each exceptional total still1.

The current independent ideal priorities are retained: order12 C then B, the earlier B/C11 and order13 routes, order16, and the separate order14, 15 and17 transfers use their protected short paths or named whole coupons. In particular the new order12 and16 alternatives do not use either reduced same-component A17/D17 orbit or the reverted center coupon. The complete selected/nonselected and later finite proof routes then receive either an unchanged payment or one of the exact new totals above. No inference is made that an old preferred witness necessarily remains paid merely because it was preferred.

All radical short8 escapes are established independently first. A deep finite distance8 zero supplies short7; the current depth8 finite7 theorem supplies short6 or a retained whole A8 unit. Its protected small-component, leaf and branch outputs are not any of the present long A17/D17 pairs. Likewise the triangle finite7/ideal5 proof and its fixed-leaf exceptions do not use these long orbits or the temporary center-coupon quarter. Triangle minimum2 and quadrilateral minimum1 persist.

For each of the two stages, the producer reconstructs all216 current literal configurations from their native Grams, starting from the final 409 packet's cornerwise lower payments. Center contexts1314 and2529 each gain an ideal quarter and lose the distinct finite coupon quarter; their lower totals remain1. In the final balance stage, four finite D16 corners gain1/4 in three other records:663 has two gains and total7/4; 676 and829 have one gain each and total11/8. There are no A16/A17 or D17 literal debits. All216 totals remain at least1 at each stage. This supplements the full universal/interface argument; it is not a census of all actual quadrilaterals.

Two complete finite certificates

Actual costs and representative upper costs are kept distinct. Both stages count the reduced D5/D12 total23/2 and the equal A5/D12 upper allowance23/2. All other actual and upper cross costs are inherited from the combined409 certificate, including A4/B9, D5/D11 and A1/B11. Every one of33,489 ordered type pairs is checked simultaneously in each stage, and every same-component representative still dominates its actual types.

The complete split retains all1,044 fixed vectors and792 interaction keys. Each stage generates new residual tables because the rank5 interaction with D12 changes; the final stage also changes residual D17. In eighths, the item delta is-2 times the D12 multiplicity at rank5, and in the final stage another-4 at rank17. Fixed A17 and representative D16 base changes are counted separately; they are not canceled by assuming equal component multiplicities.

For each stage the producer checks54,288 exact item identities, all fixed-fixed and residual-residual zero interactions,1,142,064 Bellman upper inequalities and47,520 attaining predecessors. Last-item induction proves upper bounds at every residual rank; attaining predecessors prove the representative table values. Replay checks these certificates without calling an optimizer.

After center reversion alone the exact maximum is817/2, attained by 2D17+D26. Eleven representative alternatives remain above405. The former 409 forest4A1+4D5+3D12 falls to406 because its twelve D5/D12 component pairs each lose1/4.

After both finite exchanges the exact maximum is408, attained by4D15, with ten representative alternatives above405. The previous large-type witnesses2D17+D26, D16+D17+D27 and2D16+D28 each cost815/2=407.5. Actual witnesses are evaluated using actual base and cross costs, not their potentially larger representative allowances. These statements are exact optimization results about formal forests only.

Ideal comparison and scope of the consequence

The producer reconstructs every integer convolution and scalar row of the frozen409 packet and checks exact equality before modification. The sole ideal cost change is the restored half-unit total at order17. For each rank59 terminal profile the normalized penalty is

`count(rank16)/34.`

All2,113 profiles in the complete low9-only concentration domain remain strictly positive. The normalized minimum is31/6000 at (9,9,9,9,9,14), and the absolute minimum is54 at(59). The finite exchanges change no ideal cost. Every multiplicity of rank9 and above remains in the domain; no extra concentration of middle or large factors is used.

Since all terminal normalized margins are positive, the inherited low9 concentration comparison, together with its non-increasing link-vertex product, extends the absolute surplus54 to every actual cusp. The factor-fiber multiplicities are unchanged. Combining the final finite upper408 with finite curvature405 and the single two-extrema correction 4500 gives3v_f≥4500+54v_i. The higher-dimensional tail is the completed unrestricted60 theorem. No geometric finite-volume example, unrestricted dimension59 theorem, external refereeing or proof-assistant verification is asserted.

Verification

Generation session37110/chunk6f653c and separate fresh replay 32042/chunk2f1286 both terminated exit0/PASS. Direct metadata checks pass89 bindings,7 local links and Python syntax. Root completed a full producer/report mathematical composition read and reread the underlying A/D exchange sources: PASS. The review checked the center reversion and disjoint coupon roles, eight whole-factor seals, original A17 half-level provenance, D17 selected absence and four later obligations, independent current priorities, both literal stages, exact residual-item changes, representative domination, actual408 witness and positive absolute54 comparison. It did not execute arithmetic, DP, maps or geometry. A second independent full producer/report mathematical composition read passed, including a reread of the frozen A/D exchange reports and interfaces. That review checked the same whole-factor and later-use coverage, independent short-path priority, exact base/item shifts, complete two-stage certificates and actual witnesses, both literal stages and positive-only ideal extension. It did not replay arithmetic, DP, maps or censuses. Source, data and report are final/frozen after both reviews and direct metadata checks. The eight old seals, complete saved native domains, two216-literal audits, two finite certificates and all2,113 ideal convolutions are checked. The historical all-label classification, universal contact domains and complete nonselected/interface geometry remain explicit bound inputs; no historical geometric census is rerun.

```
python3 scripts/fresh_center_reversion_rank16_balance_ambient60_frontier_20261018.py --verify
```

Source 34

Source path: `results/fresh_compact_dimension_bound_20261015.md`

SHA256: aec0f677cc583e5bf9188fa8040189797037eba93cb59ac8d14592d5193c5a3d

A finite local target for the compact dimension-29 problem

Any hypothetical compact hyperbolic Coxeter polytope in dimension 29 must have a vertex in an explicit list of **2,126 spherical Coxeter forests**, with only nine possible component-rank partitions. This is an unconditional necessary condition: no arithmetic, facet-count, or nerve assumption is imposed. It is **not an exclusion of dimension 29** and is not a classification of full polytopes.

The exact checker and saved certificate enumerate the complete local list. They check graph weights, integer partitions and the combinatorial identity below. A separate independent review reproduces the h-vector arithmetic and9/2,126/1,354 counts, and checks the occurrence argument. It inherits the connected graph weights after source review instead of repeating the66-graph census. The spherical-diagram classification and the global planar-face inequality remain published theorem inputs.

The unrestricted compact upper bound remains 29 in the checked primary survey, Lee–Marquis, Theorem 3.10 and Remark 3.12. Alexandrov's stronger bound 12 concerns the additional hypothesis that every missing face has size 2; it does not exclude general compact dimension 29. Alexandrov, Theorem C and §4.

The exact combinatorial threshold is 84

Use Vinberg's planar-angle weights. For a vertex v write σ(v) for its total weight, and for a k-gonal two-face F write σ(F) for the sum at its corners. The published construction has σ(v)≤3n and σ(F)≥5−k. Its componentwise weights and face proof are in the original Vinberg paper, §§1 and 6, printed pp.70–72 and 80–91.

For n=29, every vertex lies on 406 two-faces. The exact odd-dimensional average-side bound is 116/28=29/7. Consequently $f_2>98f_0$, and

$$\sum_v \sigma(v) \geq 5f_2 - 406f_0 > 84f_0.$$

Thus a universally valid weighting with vertex bound 84 would suffice. This is a less demanding numerical target than substituting c=23/8 into the coarser formula n<8c+6. Existing Vinberg weights give 87, so the outstanding gap is three units per vertex in a uniform bound, or an appropriate global averaged inequality.

There is an exact version of this comparison. Let $h_0,\ldots,h_{29}$ be the h-vector of the dual simplicial polytope; $h_i=h_{29-i}$, $h_i\geq 0$ and $h_0=1$. The standard face-count identities give

$$f_2 - 98f_0 = \sum_{i=0}^{14}(14-i)(15-i)h_i.$$

Define s(F)=σ(F)−(5−k_F)≥0. Double-counting planar angles gives the exact balance

$$\boxed{\sum_v(\sigma(v)-84) = 5\sum_{i=0}^{14}(14-i)(15-i)h_i + \sum_F s(F).}$$

Its right side is at least 1050 using h_0 alone. Because $\sigma(v) \le 87$, at least 350 vertices must have $\sigma(v) > 84$. This modest lower count uses no additional h-vector theorem. The certificate checks all 15 coefficients of the identity.

There is a necessary occurrence refinement. Set every angle weight on a face with five or more sides to zero, retaining the original weights on triangles and quadrilaterals. Denote the new vertex total by $\sigma_s(v)$. All face inequalities remain valid: on a larger face the new total is $0 \ge 5-k$. Also $\sigma_s(v) \le \sigma(v) \le 87$. The same balance therefore forces at least 350 vertices with $\sigma_s(v) > 84$. **At least one listed extremal corner has more than 84 units of weight on its incident triangles and quadrilaterals.** This supplies the small-face occurrence premise; extremal vertex existence alone would not supply it.

Complete enumeration of the extremal vertex types

Let $w(r)$ be the contribution of an irreducible spherical component of rank r, and $\delta(r) = 3r - w(r)$. The result of applying the published weighting is independent of the irreducible type at fixed rank:

Rank	$w(r)$	$\delta(r)$
1	0	3
2	2	4
3	6	3
4	12	0
5	10	5
6	15	3
7	21	0
8	24	0
9	27	0
$r \ge 10$	$r+19$	$2r-19$

For the classical A/B/D graphs the checker constructs every relevant end segment of order at most 6 and counts the union of its node pairs with adjacent pairs. It directly checks 66 graphs of ranks 8 through 29. At ranks 8 through 12 it also checks the end-segment list against a separate exhaustive connected-subset definition. For D, invariance under swapping the two fork leaves is enforced. The E_8 contribution $28-4=24$ is an explicit published input; the checker does not independently reconstruct the four pictured exceptional pairs or repeat Vinberg's face proof.

Since a rank 29 forest has $\sigma = 87 - \Sigma\delta$, the strict inequality $\sigma > 84$ requires $\Sigma\delta \le 2$. All components therefore have ranks 4,7,8,9 or 10; there can be at most two rank 10 components. Exhausting all 4,565 partitions of 29 leaves exactly the following list.

Component ranks	σ	Typed forests	Without H_4
4+4+4+4+4+9	87	378	168
4+4+4+7+10	86	420	240
4+4+4+8+9	87	420	240
4+4+7+7+7	87	300	200
4+7+8+10	86	240	192
4+7+9+9	87	120	96
4+8+8+9	87	150	120
7+7+7+8	87	80	80
9+10+10	85	18	18
Total		**2,126**	**1,354**

The available irreducible types are $A_4/B_4/D_4/F_4/H_4$; $A_7/B_7/D_7/E_7$; $A_8/B_8/D_8/E_8$; and A/B/D in ranks 9 and 10. Repeated equal-rank components are unordered. The certificate saves every typed forest, rather than only these counts. B and C are the same unoriented Coxeter type here.

No angle-label cutoff was used. Every finite $I_2(m)$, including arbitrarily large m, has deficit 4 and therefore cannot occur in an extremal forest. In fact all extremal vertices use only labels 3,4,5. Of the 2,126 types, 385 contain E_7, 245 contain E_8 and 772 contain H_4.

A known obstruction to the most obvious optimization

Vinberg explicitly warns in §6.12, printed p.91, that a uniform improvement $c<3$ is obstructed when nonnegative weights depend only on the planar-angle diagram and vanish across its components. E_7 supplies the obstruction through quadrilateral stars. This is a published limitation of that local architecture, not a new theorem proved by this checker. It must not be ignored when setting up a reweighting program. Original remark.

A rank 29-specific weighting depending on the full forest, weights across components, or a global charge argument requires separate analysis; the remark alone is not being promoted into an impossibility theorem for all such strategies. The exact balance above provides a test: any proposed global charge must control its positive vertex excess while retaining the nonnegative face slacks.

Our Lorentz-inertia ledger supplies valid forbidden configurations in arbitrary Lorentz dimension only under their literal spherical-face and contact premises. The strongest high-label configurations are not present inside the extremal vertices, whose components have no label ≥ 7. They cannot simply be declared unavoidable by increasing dimension.

Concrete next verification target

The 2,126 forests are finite entry points for an exhaustive local analysis. They are not by themselves a finite algorithm deciding every chamber completion. For each forest T and each pair of its nodes, delete those nodes to form a rank 27 white diagram W. A triangular or quadrilateral two-face through that angle introduces respectively one or two further black nodes.

The occurrence refinement gives a particularly small certificate target. If the original vertex weight is $W\in\{85,86,87\}$, a vertex with $\sigma_s>84$ can lose at most $W-85$ units to larger faces. Thus it suffices, for **each** listed type, to prove that some marked pairs of total original weight at least $W-84$ cannot occur on triangles or quadrilaterals. The required blocked weight is respectively 1, 2 or3. A pair of weight2 counts twice. This is an exhaustive marked-angle coverage requirement, rather than a claim that one arbitrary failed extension eliminates a corner type.

The necessary conditions can be stated exactly before any numerical search:

1. Each adjacent pair of black nodes, together with W, gives a positive-definite rank 29 vertex diagram. Each single black node together with W is also positive definite.
2. After taking the Schur complement of W, a triangular star has a 3×3 residual of Lorentz signature (2,1). A quadrilateral star has a 4×4 residual of rank 3 and signature (2,1,0); the opposite black pairs give indefinite residual pairs. All full-star principal subdiagrams must have negative index at most one, and compactness excludes parabolic subdiagrams.
3. Every discrete label and continuous dotted distance must remain covered. New rank-two spherical factors can arise after deletion, even though the original extremal forest has none; such $I_2(m)$ tails cannot be truncated without proof.

These conditions give the appropriate interface for the universal inertia obstructions. Rejecting all small-face extensions for every type would suffice; more economically, the blocked-weight coverage just described would exclude compact dimension 29. Alternatively, an exact charge or modified-weight certificate satisfying every retained triangle/quadrilateral constraint could contradict the balance identity. **Neither rejection nor that global certificate has yet been produced.** Merely verifying a local corner Gram, or rejecting a bounded selection of extension labels, would not achieve the goal.

Separate arithmetic refinement

For compact arithmetic groups in dimension at least 22, the relevant ground-field candidates are $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{5})$, and $\mathbb{Q}(\cos(2\pi/7))$; this is a separate hypothesis from the unconditional reduction. Vinberg, Theorem 2 and §7.

An H_4 component contains a label 5 edge. Its length-two cyclic product gives $4\cos^2(\pi/5)=(3+\sqrt{5})/2$, so its ground field must contain $\sqrt{5}$. $\mathbb{Q}(\sqrt{2})$ does not contain $\sqrt{5}$, and the degree-three field $\mathbb{Q}(\cos(2\pi/7))$ has no quadratic subfield. Therefore those two arithmetic cases retain only the 1,354 H_4-free extremal forests. $\mathbb{Q}(\sqrt{5})$ retains all 2,126 at this stage. Labels 4 in B/F components have rational squared cosine and are not removed by this argument. No arithmetic lattice or chamber classification is claimed.

Generation and saved replay pass:

```
python scripts/fresh_compact_dimension_bound_20261015.py --verify
```

The compact-star search remains frozen at its 60,015-visit cursor. This package does not change any prior construction, obstruction or search certificate.

Source 35

Source path: `results/fresh_compact_dimension_bound_20261015_review.md`

SHA256: 00266f108a248a1136be4e3ec3edd21f639f4ae477a75960ee37a7ea3625a58e

Narrow independent review of the compact-dimension-29 reduction

PASS for the necessary-condition reduction. The independent checker and certificate reproduce the h-vector coefficients and nine rank partitions, with 2,126 typed forests and 1,354 after removing (H_4). Generation and saved replay pass. No compact dimension is excluded.

This review inherits the connected-component weight table and global planar-face inequality from the primary package, which explicitly cites Vinberg's original proof. It does not independently repeat the 66 classical graph counts, exceptional (E_8) picture, or geometric face proof. The arithmetic implementation imports no producer code and enumerates bounded component multiplicities rather than recursively enumerating partitions.

For a simple 29-polytope, each vertex belongs to ($\binom{29}{2}$=406) two-faces. The strict odd-dimensional Nikulin bound gives (f_2>98f_0), hence total weight greater than (84f_0). Pairing the symmetric dual h-vector terms independently gives

$$f_2 - 98f_0 = \sum_{i=0}^{14}(14-i)(15-i)h_i.$$

Double-counting face-angle weights therefore proves the primary's exact balance identity. Its lower bound 1050 uses only (h_0=1) and nonnegativity. Since each vertex has weight at most 87, at least 350 vertices must have weight greater than 84; negatively weighted excesses at other vertices can only increase that requirement.

From the stated component table, the deficit (3r-w(r)) is nonnegative, is zero exactly at ranks 4,7,8,9, and is one at rank 10. Every other rank has deficit at least three. An extremal rank-29 forest consequently has only those five ranks and at most two rank-10 components. The independent bounded-multiplicity count reconstructs all nine possibilities and their unordered typed multiplicities. In particular the unbounded (I_2(m)) family cannot enter: its rank-two deficit is four, independent of (m). There is no label cutoff.

The separate arithmetic field reduction is valid: an (H_4) component has the cyclic product (4\cos^2(π/5)=(3+$\sqrt{5}$)/2), forcing ($\sqrt{5}$) into the cyclic-product field. Thus it is absent in either specified field not containing ($\sqrt{5}$). This does not exclude label 4 or B/F components.

The proposed next step needs an occurrence argument, which the primary author supplied during review and which passes independently. Set every weight at a face with at least five sides to zero, keeping the original triangle/quadrilateral weights. The face lower bound remains valid because (0≥ 5-k) there. Modified vertex totals only decrease, so they remain at most 87, and the same balance still forces at least 350 vertices with modified subtotal greater than 84. Each such vertex has original weight greater than 84, belongs to the listed types, and is incident to a triangle or quadrilateral. Consequently a complete rejection of all such small-face stars through all listed types would exclude compact29. No such rejection has been performed; the current result is the necessary-condition reduction.

```
python scripts/fresh_compact_dimension_bound_20261015_review.py --verify
```

Source 36

Source path: `results/fresh_cusp_factor_payment_framework_20261017.md`

SHA256: c65cfc5a4409a8cc83fcf0a083a6c1a7a29bb7bc7d649c11435773877e70ccab

Cusp-factor payments and a positive-semidefinite spectator lemma

This package proves a restricted factor-weighted dimension inequality and an exact Schur-complement monotonicity lemma. The restriction is essential: **every irreducible affine factor at every cusp has rank at least9**. Under that restriction, at most two factors per cusp imply dimension at most117. This does not improve the unrestricted bound163. The determinant lemma is intended to support a finer argument without discarding nonsimple cusps.

The checker and saved data certify the exact algebra below. Finite graph/model checks corroborate the all-rank proofs; they do not supply a cutoff argument.

Cusp factors and actual two-face corners

Let P be a finite-sided finite-volume hyperbolic Coxeter polytope of dimension n. Write an ideal vertex diagram as a disjoint union of irreducible affine diagrams with ranks r_vi, where sum_i r_vi=n−1. Its Euclidean link is the product L_v=product_i Delta^(r_vi). Put

```
s_v = number of factors,       V_v = product_i (r_vi+1).
```

Thus V_v is the number of edges at v and the link has (n−1)V_v/2 edges. These are the standard finite-volume cusp-link and simplicity-at-edges facts used by Khovanskii, §6 and Theorem10, also recorded in the frozen truncation note.

Every connected affine subdiagram is an entire irreducible factor of its actual cusp. Indeed its positive kernel gives a nonzero null vector: pairing its positive root combination with a strict chamber center is negative. Acuteness makes that vector feasible against every other wall. A feasible ideal point is extreme in the projective closure, by strict convexity of the closed Klein ball, hence is an ideal vertex. Its affine subdiagram lies in one irreducible cusp component, whose proper principal subdiagrams are positive definite. The published containment of every parabolic subdiagram in its unique ideal vertex diagram appears in Felikson–Tumarkin, §1.

An actual two-face has an elliptic supporting set W of n−2 walls. Each of its edges adds one side normal, giving a positive-definite set of n−1 walls because the edge has an interior hyperbolic point. At an ideal corner, the corresponding link edge varies in exactly one simplex factor i. W retains r_vi−1 nodes in that factor and r_vj nodes in every other factor. Adding its two marked sides restores the full affine factor i; all other factors have one node deleted and are elliptic. In particular the marked pair lies in the varying factor. Every pair of nodes of factor i occurs in exactly V_v/(r_vi+1) such link-edge fibers.

This description does not assume that the cusp itself is simple. An independent internal conceptual read checked this corner/fiber multiplicity and the closed-edge argument below; it did not repeat the scalar or graph checks.

A restricted distance-seven payment theorem

Assume from now on that every r_vi>=9. There is no affine subdiagram of order at most9. A minimal indefinite diagram is Lannér or quasi-Lannér of order at most10; the quasi-Lannér case contains an affine proper subdiagram of order at most9 and is therefore excluded. Every minimal indefinite diagram is consequently Lannér of order at most5. The finite classification and its use here are inherited explicitly from the affine-exclusion proof.

Assign a two-face corner weight1 when its two side nodes are at graph distance at most7, and weight0 otherwise. For a triangle, a minimal indefinite subdiagram of its star contains all three sides. Its order is at most5. A tree joining the three marked nodes has at least two marked-to-marked paths avoiding the third node, each of length at most3; these give two paid corners. Cyclic minimal diagrams only supply additional short paths. Thus every triangle receives at least2.

For a quadrilateral, W plus either opposite pair is indefinite. If it were positive definite, the usual acute projection of the strict center would give a point on both opposite closed sides. If it were positive semidefinite and singular, its positive affine kernel would give a nonzero feasible ideal point on both opposite closed sides. Both contradict their disjointness, including when the quadrilateral has ideal vertices. Let L and M be minimal indefinite subdiagrams for the two opposite pairs. Each contains both of its marked nodes, has order at most5, and has distance at most3 from a white node to a nearest marked node. The two subdiagrams cannot be orthogonal, because that would give two negative directions. Their overlap or a joining bond therefore supplies a path of length at most3+1+3=7 between adjacent sides, avoiding the other two marked nodes. Every quadrilateral receives at least1. This is the small-core proof of the frozen distance theorem, using only actual edge/corner supports rather than simple ideal vertices.

At a finite vertex the total weight is at most7n. At an ideal vertex, a factor of order m=r+1>=10 is classical, and its number of pairs at distance at most7 is at most7m. The all-rank path, cycle, single-fork and double-fork formulas are proved in the distance package. The current checker verifies the245 inherited literal graph rows and their source hashes; these checks are not the justification for arbitrary rank. Multiplying by the fiber count yields the exact budget

```
C(v) <= sum_i [V_v/(r_vi+1)] * 7(r_vi+1) = 7 s_v V_v.
```

The exact weighted inequality

Let v_f count finite vertices and define

```
beta_n = n(n-6)/8                (n even),
```

```
beta_n = (n-1)(n-5)/8            (n odd).
```

For the original polytope, not its truncation, Khovanskii's reduction yields a strict average-face bound for a bounded polytope simple at edges, as justified immediately below, and gives

```
D5 = sum_(two-faces F) (5-number_of_sides(F))
```

```
> beta_n v_f + (beta_n/n) sum_v V_v.
```

Here the total two-face/vertex incidence is binom(n,2)v_f+(n−1)sum_v V_v/2. Here is the strictness step rather than a tacit upgrade of the published weak bound. Choose a generic linear function on the bounded projective polytope. For each vertex w, let E_w be the number of edges in its simple vertex figure and E_w^0 the number of those edges not cut by the hyperplane through w parallel to a level hyperplane. Theorem9 gives the exact identities I=sum_w E_w and2f2=sum_w E_w^0. Theorem8, as applied in Theorem10, gives2E_w/E_w^0<=eta_n, where eta_n=4(n−1)/(n−2) for even n and4n/(n−1) for odd n. All denominators are positive. At a global minimum vertex the cutting hyperplane misses the vertex figure, so E_w^0=E_w and this ratio is2<eta_n. Thus I/f2, the positively weighted average of these local ratios, is strictly less than eta_n. This proves the displayed strict lower bound for D5=5f2−I. The argument applies to the bounded projective closure, including ideal vertices. The checker verifies the resulting coefficient identities and parity expressions. The corner payments give

```
D5 <= 7n v_f + 7 sum_v s_v V_v.
```

Consequently

```
beta_n/(7n) < [n v_f + sum_v s_v V_v] / [n v_f + sum_v V_v].
```

This is the proved factor-weighted inequality. If every s_v<=s, the right side is at most s, and hence n<=56s+5. At N=56s+6, beta_N=7sN, which the strict inequality excludes without any additional finite-vertex assumption. The all-rank threshold check is

```
8(beta_(N+2t)-7s(N+2t)) = 4t^2+2Nt,
```

```
8(beta_(N+1+2t)-7s(N+1+2t)) = 4t^2+(2N+4)t+N+6.
```

Thus the rank-at-least9, at-most-two-factor class has n<=117. These are restricted consequences, not claims about arbitrary finite-volume polytopes. The frozen all-simple bound29 is already stronger for s=1.

Orthogonal positive-definite spectators improve a scalar contradiction

The following identity has no factor-rank restriction. After eliminating the supporting whites of an affine marked corner, suppose alpha>0 and h,k are nonzero, and its four-root Schur matrix is

```
[ alpha  −alpha   B−h   −A  ]
[−alpha   alpha  −B     A−k ]
[ B−h     −B      C     −E  ]
[−A        A−k   −E      D  ].
```

Its determinant is

```
(hk−kB−hA)^2 − alpha(Ck^2+Dh^2+2Ehk).
```

Additional elliptic white spectators orthogonal to the old affine factor may couple to the two new roots c,d. Eliminating them subtracts a positive-semidefinite matrix [[p,q],[q,r]] from the c,d block, so C becomes C−p, D becomes D−r and E becomes E+q. With u=1/h and v=1/k, the normalized determinant changes by

```
alpha(pu^2−2quv+rv^2) = alpha ||u z_c−v z_d||^2 >= 0.
```

The equality uses any Gram realization of that positive-semidefinite correction; h,k are unchanged. Therefore every strict positive determinant contradiction of this form survives such spectator attachments. For an actual quadrilateral the complete supporting set has n−2 independent roots and the four residual normals lie in a Lorentz three-space, so the determinant must vanish. This explains why a strict scalar contradiction still excludes the configuration after the spectators are restored.

This identity transfers scalar portions of the affine-cycle proof and affine-tree proof. It does not itself transfer their entire payment theorem. A generic finite-component payment can change upon extension (for example, an abstract E7 diagram embeds in E8), while role-compatible spectator extensions in the actual proof require a separate literal check. This sentence does not assert that the specific D6-to-E7 configuration admits such an extension through its permitted new-root position. Also, a small affine diagram that was forbidden under simple cusps can now be a genuine cusp factor. Those separate cases require a new argument before a3s_v V_v budget can be asserted.

Two factors do not force absence of small affine diagrams

There is an exact local cusp model in every dimension m+1, m>=3, with link Delta^1 times Delta^(m−1). It shows why a factor-count hypothesis alone cannot replace the minimum-rank hypothesis above.

Use a Lorentz space with null nu,eta satisfying nu.eta=−1, an orthogonal unit e, and an orthogonal A_(m−1) root system alpha_1,…,alpha_(m−1). Put theta=sum alpha_i. The normals are

```
e, nu−e; alpha_1,...,alpha_(m−1), nu−theta.
```

The two components are affine A1 and affine A_(m−1), mutually orthogonal, and both positive null relations sum to nu. Their Gram rank is m and their normal-span rank is m+1. A strict timelike center is

```
H=m nu+eta−e/2−sum_(i=1)^(m−1) [i(m−i)/(2m)] alpha_i.
```

Its products with the first pair are−1/2 and with each remaining normal−1/m, while

```
H^2=−2m+1/4+(m^2−1)/(12m)<0.
```

The horospherical link is the bounded product of the two standard alcoves. These formulas prove the all-rank local realization;22 exact matrix checks corroborate them. No finite-volume completion in arbitrary dimension is claimed.

Verification and scope

Generation passes:245 inherited graph rows with their source manifest,171 scalar incidence checks, eight exact factor-count thresholds, the symbolic four-root determinant and positive-semidefinite correction, and22 rational cusp models. A fresh

saved replay also terminates with PASS; syntax, all eight direct source hashes and all seven local report links pass. The separate geometric review described above is limited to the factor-corner and closed-edge steps. A separate root mathematical/source review passes the strict Khovanskii reduction, factor geometry, distance-seven scope and positive-semidefinite identity; see the root review. That review does not duplicate the producer arithmetic. The source, data and this report are final/frozen. This package does not assert a full fine-weight extension, an unrestricted dimension improvement, or novelty of the restricted consequence.

Source 37

Source path: `results/fresh_cusp_finite_port_triangle_capacity_20261017.md`

SHA256: 6ca9cb26b25df0c06632ecc3d0f9cdc80aa29210c11d1a243e332dae582f3a8f

A sharp finite-ended cusp-port bound for one- and two-ideal triangles

Let P be a finite-sided, finite-volume acute hyperbolic Coxeter n-polytope. Write T_j for the number of triangular two-faces with exactly j ideal vertices. At an ideal vertex v write its combinatorial link as

$$L_v = \prod_{i=1}^{s_v} \Delta^{r_{vi}}, \qquad a_v = \#\{\text{edges from v to a finite vertex}\}, \qquad R_5(v) = \sum_i \min(r_{vi}, 5).$$

Let T_j(v) count those triangles through v. Then

$$\boxed{T_1(v) + T_2(v) \le a_v R_5(v)}, \qquad \boxed{T_1 + 2T_2 \le \sum_{v \text{ ideal}} a_v R_5(v)}. \tag{1}$$

In particular, if every ideal vertex is simple, then

$$\boxed{T_1 + 2T_2 \le 5E_{fi}}. \tag{2}$$

The constant5 in (2), and the uniform per-factor constant used in (1), are attained by the classical finite-volume H9 simplex L24. No new dimension bound or new hyperbolic example is claimed. For nonsimple cusps, the factor sum in (1) is essential to this proof and is not replaced by5.

Factor faces and the valid completion argument

A port at v is an old edge through v, hence a vertex of L_v. Color it finite or ideal according to its other endpoint. An old two-face through v gives an edge of L_v. Every edge of a product of simplices varies in one factor, so it lies in a unique maximal simplex factor fiber F. Fixing the other factor coordinates makes F a simplex Delta^{r_i}. The face correspondence for vertex links gives an actual (r_i+1)-face P_F of P with link F at v. Thus **v is simple in P_F**, even if it is nonsimple in P. Every finite vertex of P_F is also simple.

These local facts for general finite-volume acute chambers are recorded in Potyagailo–Vinberg, printed p.64. They do not say that arbitrary induced faces have Coxeter angles. The argument below is combinatorial; its later simplex-size restrictions come from the original-wall-star lemmas, not from treating P_F as a Coxeter polytope in its own right.

Fix a finite-ended port va within F. Suppose vab and vac are triangular faces. The three ports generate a three-face G of P_F. The vertices v and a are simple in G and each already has the three neighbors among v,a,b,c. If G had another vertex, deleting b,c would disconnect {v,a} from it. This contradicts Balinski's three-connectivity theorem, printed p.432. Thus G is a tetrahedron. In particular vbc is triangular, regardless of whether b,c are finite or nonsimple ideal vertices elsewhere in P_F.

All triangular neighbors of the chosen port consequently form a clique. Let that clique together with va have k ports and generate the k-face A. Only v and a need be simple: both already have degree k among the k+1 displayed vertices. If A contained any further vertex, deletion of the other k−1 displayed vertices would disconnect {v,a}. Balinski's k-connectivity

excludes this. Therefore A is the whole k-simplex. This also handles k=2 directly as the initial triangle; k=1 has no triangle contribution.

These maximal simplex blocks **within the fixed fiber** have disjoint sets of finite-ended ports. Indeed, if two blocks shared such a port, the preceding completion applied to all its triangular neighbors would merge them. They may share ideal-ended ports. Every triangle through v with at least one other finite vertex belongs to exactly one such block. The two types counted are T_1(v) and T_2(v).

No maximality in the ambient face poset is required when applying the frozen ideal-count classification: it bounds **every actual simplex face**, including these fiber-maximal blocks. No degree assertion is made about their undisplayed ideal neighbors.

The coefficient calculation

In one block let f>=1 be its number of finite-ended ports and i>=0 its number of ideal-ended ports. The block simplex has dimension k=f+i, with f finite vertices and i+1 ideal vertices, because v itself is ideal. Its contribution is exactly

$$T_1(v;A) + T_2(v;A) = \binom{f}{2} + fi = f\left(\frac{f-1}{2} + i\right). \tag{3}$$

The original-star theorem gives k<=9. The seven-simplex theorem allows at most three ideals or all eight vertices ideal; the latter case is excluded here by f>=1. The eight- and nine-simplex bounds give at most three ideals as well. Hence for k>=7, i<=2 and

$$\frac{f-1}{2} + i = \frac{k+i-1}{2} \le 5.$$

For k<=6 the same expression is at most k−1<=5. Thus the new coefficient does not require extending the six-simplex theorem to the unproved one-finite/six-ideal case. That case, with f=1,i=5, is explicitly retained in the exact checker. Also k<=r_i+1 inside this factor face, so the expression is at most r_i. Consequently each block contributes at most f*min(r_i,5).

Sum over the disjoint finite-port sets in a fiber. For any one factor i, each finite-ended port of the full link belongs to exactly one fiber of that factor. Summing over all fibers therefore contributes at most a_v*min(r_i,5), and then summing over factors proves the local bound in (1). A T_1 triangle is counted at one cusp and a T_2 triangle at two; this gives the global identity and inequality. Finally sum_v a_v=E_fi. A simple cusp has one factor, proving (2).

The standalone Fraction checker enumerates all28 allowed (k,f,i) types under the inherited safe envelopes, and verifies the coefficient formula. The checks through factor rank64 only confirm the explicit all-rank argument above; no rank cutoff is used in its proof. Formal maximizing types are not asserted realizable.

Sharpness and the previous counterexample

The literal L24 certificate already proves that this rational Gram defines a finite-volume H9 Coxeter simplex with seven finite and three ideal vertices. Its full geometry was also checked by the independent Fraction review. The current checker reconstructs its ten-by-ten Gram and inverse without project arithmetic, verifies all three positive affine kernels, and recounts the triangle incidences.

At each cusp, the link is Delta^8, with seven finite-ended and two ideal-ended ports. Thus T_1(v)=21, T_2(v)=14, and

$$T_1(v) + T_2(v) = 35 = 5a_v.$$

Globally T_1=63, T_2=21 and E_fi=21, so T_1+2T_2=105=5E_fi. This supplies actual sharpness, not only an abstract port distribution.

The finite-vertex T2 counterexample does not contradict the proof. Its two triangles share the finite–ideal edge A–Iinf, and Iinf has the square link Delta^1 x Delta^1. At Iinf, the two triangular link edges {A,I1} and {A,I2} belong to different factor fibers. They never enter one application of the simple-at-cusp completion argument. The checker reconstructs these fibers

from the saved full edge/face incidences and verifies (1) at all three cusps. We have not restored triangle-port transitivity at the finite vertex A.

Exact residual budget and limits

Combine (1) with the sharp finite-edge inequality and the original-face budget from the finite-volume report. With its notation b_n, alpha_n, v_f and E(L_v), one obtains

$$\boxed{\begin{aligned} 2T_3 + Q \;&\geq (b_n - 14n/3)v_f + \sum_v a_v\big(14/3 - R_5(v)\big) \\ &\quad + \alpha_n \sum_v E(L_v) + T_1. \end{aligned}} \tag{4}$$

Indeed the previous lower demand on 2(T_2+T_3)+Q loses at most sum_v a_v R_5(v)−T_1 when the T_2 contribution is subtracted. This is a new necessary tradeoff, not an upper payment bound for T_3 or Q. It is vacant for the all-ideal problem in the sense that a_v=T_1=T_2=0 there; the remaining all-ideal triangles and quadrilaterals still need control. For simple cusps R_5(v)<=5, while multiple small factors can make R_5(v) as large as n−1. No multiplicity-free generalization of (2) is claimed.

The two primary geometric sources above were opened on2026-10-17. The one-factor completion argument received an independent conceptual check from construction_audit. The parent independently read the full proof and source, including factor-face construction, finite-port disjointness, ideal-count scope, sharpness and residual algebra, and found no gap. Neither read is described as a second arithmetic implementation.

FINAL/FROZEN. Generation and a fresh saved-data replay pass. Syntax, all18 direct source hashes, eight local report links and the two primary source links were checked. This is an internally reviewed incidence lemma with exact counting and inherited geometric sharpness, not an ambient dimension improvement or external refereeing.

Source 38

Source path: `results/fresh_cusp_link_shape_curvature_20261017.md`

SHA256: 0946759e18de93d13ae4729b95026b6cb92c59e3c48c0f2ef33c61032b8ebd3e

Exact curvature from the complete cusp-link shape

This is a stronger global counting inequality for finite-volume hyperbolic Coxeter polytopes. It retains all ranks in each cusp's product of simplices, rather than replacing them by a single worst-case average. It supplies no ambient dimension bound on its own. An exact local example below shows why the coarsest distance payment still needs improvement.

Let P have dimension n>=3, let d=n−1, and let D5 be the sum of5 minus the number of sides over its two-faces. Write v_f for its number of finite vertices. At an ideal vertex v let the Euclidean link be

L_v = product_i Delta^(r_vi), sum_i r_vi=d.

Define

H_v(t)=product_i (1+t+...+t^(r_vi))=sum_(j=0)^d h_vj t^j,

V_v=H_v(1), G_v=sum_j h_vj min(j,d−j),

kappa(L_v)=(5/2)G_v−dV_v/2.

With beta_n=n(n−6)/8 for even n and beta_n=(n−1)(n−5)/8 for odd n, the proved inequality is

D5 >= beta_n v_f + sum_(ideal v) kappa(L_v) + 5(n−1).

No hypothesis on cusp simplicity, factor count, minimum factor rank, arithmeticity, or induced Coxeter faces is imposed. Finite volume supplies simple vertex figures (products of simplices at ideal vertices) and a bounded projective closure. The geometric facts are the same as in the factor-payment framework.

Proof retaining the local cut count

Choose a generic linear function on the bounded projective polytope. At every vertex w, form its d-dimensional simple vertex figure L_w and the hyperplane section induced by the level hyperplane through w. Let E_w count all edges of L_w and E_w^0 count edges not met by that section. The exact reduction identities in Khovanskii, Theorem9, printed p.49 are

I = sum_w E_w, 2 f2(P) = sum_w E_w^0,

where I is the total number of vertices of all two-faces. Therefore

D5=sum_w [(5/2)E_w^0−E_w].

Here is a direct lower bound for E_w^0 in terms of the complete H-polynomial, rather than a ratio bound. Orient the edges of the simple vertex figure by a generic linear function representing its cutting hyperplane. A vertex of index j has j downward edges and d−j upward edges. Assign each uncut edge below the section to its upper endpoint and each uncut edge above the section to its lower endpoint. Its contribution is j or d−j, respectively, and is at least min(j,d−j). The number of vertices of index j is the H-coefficient h_j, by the simple-polytope index formula (Khovanskii, Theorems2 and6). Consequently

E_w^0 >= G(L_w)=sum_j h_j min(j,d−j).

At a global minimum or maximum vertex of P, the section misses the entire vertex figure, so E_w^0=E_w. For any simple d-polytope, symmetry of H and the index formula give

E_w−G(L_w)=sum_(j>d/2) h_j(2j−d) >= d,

because h_d=1 and all H-coefficients are nonnegative. The distinct global minimum and maximum thus supply an additional total at least2d in the uncut-edge sum, or5d in D5. This proves the quantitative slack in the displayed theorem. In particular it also proves the strict version without tracking the additive constant.

For a finite vertex, L_w is a d-simplex: H=1+t+…+t^d, G=floor(d^2/4), and E=d(d+1)/2. Substitution gives exactly beta_n. At an ideal vertex, the product formula for face polynomials and the substitution H(t)=F(t−1) give the displayed product H_v; E=dV_v/2. This proves the entire inequality in every dimension. No finite partition census is used.

A probabilistic expression and a usable relaxation

Let independent X_i be uniformly distributed on the integers0,…,r_i. Since their sum has probability generating polynomial H(t)/V,

kappa(L)/V = 3d/4 − (5/2) E|sum_i X_i−d/2|.

Thus the complete link shape enters through a mean absolute deviation, with exact rational value. Its variance is

Var(sum_i X_i)=sum_i r_i(r_i+2)/12.

Cauchy–Schwarz gives the convenient rigorous relaxation

kappa(L)/V >= 3d/4 − (5/2)sqrt(sum_i r_i(r_i+2)/12).

The exact H-polynomial value is generally stronger. A product of many small factors can therefore have substantially more curvature available than the universal beta_n V/n bound. This is a combinatorial improvement; the corner-payment theorem must still be established for the proposed geometric weights.

An exact obstruction to one coarse local comparison

Consider the legitimate Euclidean cusp-link type

Delta^30 × Delta^30 × Delta^30 × Delta^30 × Delta^30 × Delta^13.

Its link dimension is163, so it is a local cusp model for H164. Choose each irreducible affine diagram to be an A cycle. On the five31-node cycles and the14-node cycle, every node pair has distance at most15. Thus the distance15 rule charges, per link vertex,

sum_i r_i/2 = 163/2.

The exact curvature margin is

kappa(L)/V − 163/2 = −69328961/200404057 < 0.

Therefore a proposed proof that pays every cusp separately by this distance15 budget cannot be justified by the exact link curvature alone. The global additive slack is positive, but is not a substitute for the failed per-cusp coefficient inequality when cusp counts are unrestricted. This is a scoped obstruction to that payment comparison, not a counterexample to a dimension theorem: no finite-volume H164 completion of this local cusp model is asserted. It is compatible with the independently proved unrestricted bound163.

The smaller distance7 budget is available only under the separate rank restriction of the factor framework. Fine-weight extensions could lower the large-factor cost further, and are a geometric obligation separate from this exact counting lemma.

Exact checks and status

The checker and data verify179 simplex-to-beta identities,17 product-link polynomials, symmetry and variance identities, the extremal slack, and the exact negative margin and cycle pair counts. The all-rank proof is the local index argument above; the finite lists corroborate its formulas. Generation and fresh saved replay both terminate with PASS. Syntax, all five direct source hashes and the local report links pass. The root independently read the all-profile index proof, including

the quantitative global-extremum slack, and found no gap; see the root review. That review does not duplicate the finite arithmetic checks. This source, data and report are final/frozen. No optimizer or new dimension claim is part of this package.

Source 39

Source path: `results/fresh_cusp_spectator_role_payments_20261017.md`

SHA256: 22a825bb48fc41b67891720c5caa2e332dd84311cb8dc6c338e769e226e7156e

Role-specific spectator extensions preserve two exceptional payments

Two finite-component payments in the affine-tree argument survive arbitrary acute positive-semidefinite spectator extensions, even without excluding small affine cusp factors. The relevant E7 component cannot extend to finite E8 through its allowed attachment position. Its only nontrivial extension is affine E7, where the marked pair is still short. The analogous D8 fork pair extends only to affine E8 and also remains short.

These are partial payment lemmas, not a complete nonsimple-cusp quadrilateral theorem or a new dimension bound. The checker and data independently calculate the inverse entries, affine kernels and small graph costs used below.

The precise E7 role

In the D6-to-E7 configurations of the frozen affine-tree proof, the new cap d attaches by a simple bond to one old fork leaf. The resulting E7 tree has arms of lengths1,2,3, with d at the end of the length-two arm. The other marked root is either the branching vertex or the leaf on the length-one arm. Their distances from d are2 and3.

Every spectator under consideration is orthogonal to every old affine-factor root. In this corner it can therefore meet this E7 only at d. With norm-two normalization, the E7 inverse entry is

```
(G_E7^-1)_(d,d)=2.
```

For any spectator with nonzero contact magnitude t, the Coxeter condition gives t>=1, while positive semidefiniteness of its principal extension gives

```
2-2t^2 >= 0.
```

Hence t=1. The extension is the affine E7 diagram with arms1,3,3. Its kernel is strictly positive. In a positive-semidefinite full corner Gram, a kernel vector of a principal block is also annihilated by the full matrix. Since all cross products are nonpositive, that positive kernel forces every further spectator to be orthogonal to the whole affine component. Thus the component is exactly E7 or this affine E7, with arbitrary orthogonal components elsewhere. This exhausts every number of spectators and every label; it is not a bounded attachment search.

In the finite E7 case the original finite weight pays the pair. In the affine case its unchanged distance2 or3 pays it under the ideal rule below. Extending the length-two arm does not produce finite E8, whose arms would have lengths1,2,4. A generic abstract E7-to-E8 embedding is therefore irrelevant to this attachment role.

The D8 sibling pair

In the fork-leaf P6 exception, the opposite finite component is D8. Its marked roots c,d are the two fork leaves, and every other node of this component is an old affine-factor root. Spectators may meet this D8 only at c and d. The relevant inverse block is

```
[[2,3/2],[3/2,2]].
```

If both contact magnitudes are nonzero, they are at least1 and their projection cost is at least7, exceeding the available norm2. If exactly one is nonzero, its cost is2t^2, so positive semidefiniteness again forces t=1. Adding that one simple-

bond spectator gives affine E8, with arms1,2,5. Its positive kernel then forces all other spectators to be orthogonal to the entire component. The complete possibilities are D8 itself or one such affine E8 extension.

The marked pair remains at distance2. In finite D8 it lies in the common fork-invariant end segment; in affine E8 it is paid by the short-distance ideal rule. No exclusion of affine E8 is used.

The construction agent independently checked the role geometry and both positive-semidefinite arguments conceptually. That read is separate from, and does not duplicate, the producer matrix calculations here.

A factor rule with cost at most three times its order

For an irreducible affine factor of order m, use weight1 on:

- every pair if m<=7;
- pairs at distance at most3 if m>=8;
- additionally, for a classical factor, pairs in a fork-invariant end segment of order at most6.

All other pairs have weight0. The total cost is at most3m for every irreducible affine diagram. For m<=7, binom(m,2)<=3m. The classical cycles, paths, single forks and double forks of orders8 through13 each have cost exactly3m; the checker counts all24 literal graphs. For m>=14 the all-rank short-pair and end-segment formulas are those proved in the frozen affine weight interface, giving exactly3m. The exceptional affine E6,E7,E8 diagrams have respective costs21,20,23 and respective orders7,8,9. The affine F4 and G2 diagrams have order at most5 and are already covered by the small-order bound. The rank-one affine factor is also covered.

Consequently, if this rule and the finite weights eventually satisfy every face-payment obligation, their ideal-vertex cost would be at most3s_v V_v by the proven factor-fiber multiplicity. This sentence is conditional: these two spectator lemmas do not establish all face payments. New affine components after replacing a central root, small-affine triangle cores, and general all-finite-corner transfers in the presence of small cusps remain separate cases.

Verification and status

Generation and a fresh saved replay pass. The checks include the literal E7 inverse diagonal and D8 inverse block, three affine extensions with positive kernels and every single-node deletion positive definite,24 small classical graphs, three exceptional affine costs, and the348 inherited graph rows. The all-spectator conclusion is the positive-kernel proof above. The all-rank cost statement uses explicit classical formulas and the finite affine classification, not extrapolation from the graph rows. Syntax, all five direct source hashes and local report links pass. The root independently reviewed the mathematical/source scope, including the two inverse-entry arguments, the positive-kernel isolation step and the all-rank cost accounting, and found no gap. That review did not duplicate the matrix calculations or graph census. This package is final and frozen; no global payment or ambient dimension conclusion is asserted.

Source 40

Source path: `results/fresh_cycle11_half_pair_transfer_20261018.md`

SHA256: ec117a499ddad7cf8e87426e467d54cde295c0956428b2a8bbda5140abcdecd5

Half payments on the affine11-cycle distance-five pairs

In the current payment rule, replace each of the11 distance5 unit pairs in a whole varying affine11-cycle by a half. Its total pair cost decreases from55 to99/2. Every affected quadrilateral still receives at least1, and no triangle can contain such a marked cycle pair. No finite payment or coupon is added or changed. Arbitrary supporting spectator diagrams and all native labels are allowed.

The source and certificate corroborate the proof below with exact matrices and scalar identities. This is a local theorem. Other order11 affine families still have cost55, so the all-family envelope does not yet improve. No new dimension bound or finite-volume example is asserted.

Complete actual-corner profiles

Write an actual quadrilateral star as W+a+b+c+d, with opposite pairs {a,c},{b,d}. At the chosen ideal corner its actual varying affine factor F is the11-cycle, marked at distance5; the remaining spectator diagram Z is positive definite and orthogonal to F. These are the inherited actual factor-fiber facts. All roots have squared norm2. Both actual edge diagrams W+c and W+d are strictly positive definite. Heights h=−c·nu and k=−d·nu at the positive kernel nu of F are strictly positive.

An adjacent black contact or a common nonzero spectator coordinate of c,d gives a marked path of length1 or2 in another actual corner. Such pairs retain their full units. Hence suppose these alternatives are absent, so the adjacent-black products vanish and the native nonnegative contact vectors x,y to Z have disjoint supports.

Deleting either mark from F leaves A10. The complete positive-semidefinite one-root extensions are isolated; one3 or4 bond at either endpoint; one3 bond at either second node; or3 bonds at both endpoints. This is the same all-label path-extension classification used in the cycle12 transfer and its weak-profile predecessor. No finite or affine exceptional diagram contains A10. The classical classification gives precisely the listed possibilities, with no imposed native-label cutoff. Adding spectators cannot repair an excluded principal extension.

Each of the six single-contact profiles gives an adjacent marked path of length at most6. The old endpoint opposite the attachment is a simple leaf with a degree-two neighbor. Both degrees are fixed after all spectators are restored because Z is orthogonal to F. No classical affine component has that configuration; exceptional affine components are too small. Thus the whole connected component containing the marks is finite, and all other components are positive-definite principal parts of Z. The adjacent corner is finite and its distance<=6 unit is unchanged. The source checks all six physical-norm2 core matrices, strict positive definiteness, marked distances and fixed leaf/neighbor roles.

If either profile has both endpoint3 contacts, its adjacent corner contains a full affine11-cycle, obtained by replacing the deleted old mark by the new black root. The positive kernel and actual corner positive semidefiniteness force every further root to be orthogonal to this cycle: its nonpositive products have zero positive-kernel sum. Hence it is the whole varying affine factor, with the two marks again at distance5. This different actual corner receives a half, as does the original corner. Their two halves already total1. This argument does not require the other opposite root's profile or height to be fixed.

Only two isolated profiles remain. They contact F only at their respective opposite old marks. Their positive heights are therefore native contact magnitudes:1, sqrt2, phi=(1+sqrt5)/2, or at least sqrt3. Parallel and dotted opposite products are included in the last range.

The three isolated height cases

The old arcs have lengths5 and6, so alpha=1/5+1/6=11/30. The inherited Schur identity, reverified symbolically in the source, is

```
0 = 1−2alpha/h²−2alpha/k²
      +alpha∗(x/h−y/k)^T G_Z^(−1)(x/h−y/k).
```

The final term is nonnegative. If both heights are at least sqrt2, the first three terms are at least4/15. If one height is1 and the other at least sqrt3, they are at least1/45. Thus the only unordered height pairs that remain are(1,1),(1,sqrt2),(1,phi).

For any nonzero x-coordinate i, disjoint supports and Gram-metric Cauchy give

```
(x/h−y/k)^T G_Z^(−1)(x/h−y/k) >= x_i²/(2h²) >=1/(2h²).
```

There is the corresponding bound for a y-coordinate. These bounds are used one coordinate at a time; they are not added.

At(1,sqrt2), the required normalized spectator energy is3/11<1/2. Consequently the height1-side support is empty. The other side would have unscaled one-sided energy6/11. The complete strict spectrum below one then forces a unit contact at a leaf of whole A_r with r/(r+1)=6/11, so r=6/5, impossible.

At(1,phi), the required energy is25/11−sqrt5. It is positive but strictly less than1/(2phi²), the smaller of the two possible nonempty-coordinate bounds. Both supports must therefore be empty, which would give energy0. This is again impossible. Reflection treats the reversed ordered heights.

At(1,1), the required unscaled energy is14/11. The next argument shows that this necessarily supplies a marked c–d path of length at most5.

Spectator energy14/11 forces a short marked path

Both separate augmented diagrams Z+c and Z+d are strictly positive definite. Every finite Coxeter diagram is a forest, so each new root can contact at most one node in any connected component of Z. If both roots contact one component, the all-rank inverse argument in the five-quarters spectator theorem applies to that component. We use its independently stated intermediate bound, not its conclusion specialized to energy5/4: contact nodes at distance>=4 contribute at least4/3 to the difference energy. The proof allows arbitrary native magnitudes, uses the strict inverse-diagonal restrictions, and covers all classical ranks and exceptional types. Every other component contributes nonnegative energy. Since14/11<4/3, a mixed component must instead give contact distance<=3, hence a c–d path of length<=5 with only spectator whites inside.

Suppose there is no mixed component. Every contacted component then contributes a one-sided positive energy. Each is at least1/2, so there are at most two active components. We need the following complete one-component spectrum up to14/11:

```
r/(r+1), r>=1 integral; 1; 6/5; 5/4.
```

Here is its all-rank derivation. There is only one contact within a component by the forest argument. Its magnitude is1 or sqrt2: a native magnitude at least phi already gives energy>=phi²/2>14/11. For magnitude sqrt2, the inverse diagonal must be<=7/11<1. The inherited below-one subtree argument forces a whole A_r leaf, and r/(r+1)<=7/11 forces r<=7/4. Only A1 remains, with energy1.

For a unit contact on A_r, write N=r+1 and the inverse diagonal as i(N−i)/N. A leaf gives r/(r+1). An interior position with both i,N−i>=3 has diagonal>=3/2, too large. The remaining positions have one of these integers equal2, giving2−4/N; the energy bound forces N<=11/2. Thus N=4 or5, giving1 or6/5.

For B_r, the all-rank inverse diagonals are the integers1,…,r−1 and r/2, so only1 fits. For D_r they are the spine integers and r/4 at the short leaves, leaving1 and5/4, the latter at D5. Exact exceptional Grams show every diagonal of E6,E7,E8,F4,H3,H4 exceeds14/11. Finally, for I2(m), m>=5, the diagonal2/(4−theta²) increases with the native bond theta and already exceeds14/11 at theta=phi. The cases m=3,4 are A2,B2. This proves the spectrum without a spectator-rank or label cutoff.

One active component cannot have energy14/11, since it is none of those values. With two active components, each is at most14/11−1/2=17/22<1. Each is therefore r/(r+1) with r<=3. The six unordered sums from {1/2,2/3,3/4} are checked exactly; none equals14/11. No unmixed-support case remains. The required short c–d path follows.

At its actual corner, a finite marked path<=5 retains a full unit. An ideal marked path<=5 also retains a full unit, except possibly another distance5 pair in an affine11-cycle; that one retains a half. In every case this distinct corner contributes at least1/2. Together with the original chosen half it gives the required total1. This completes the quadrilateral proof. It never assumes that the original changed pair still has a full unit.

No triangle has an affected corner

For an actual triangle, write its third black root as c and put the old affine11-cycle marks at a=0,b=5. Both other corner diagrams are positive semidefinite. The A10 extension classification applied to F\a allows c to contact only cycle nodes{1,2,9,10}; applied to F\b it allows only{3,4,6,7}. These sets are disjoint. The old marks themselves are interior positions in the opposite deleted paths, so contacts to a or b are also forbidden. Hence c is orthogonal to all of F.

Its height at the positive kernel of F would then be zero, whereas the third side of the actual triangle does not contain this ideal vertex and has strictly positive height. This contradiction excludes every triangle with the affected corner. Spectators cannot repair an excluded principal PSD extension, so the conclusion has arbitrary-support scope. Rotation/reflection treats all11 distance5 pairs. Triangle coverage is therefore unchanged without importing an extra triangle coupon.

Simultaneous-rule compatibility and verification

All alternative finite paths have length<=6 and all other terminal marked paths have length<=5. Therefore this proof is compatible both with the completed depth17 half rule and with a prospective depth11 zero rule on finite distance8 pairs. No such distance8 unit or half is consumed here. The other retained short5 ideal units and all coupons are unchanged. The new cycle weights are invariant under its full dihedral automorphism group. The11 half pairs and44 unit pairs have total99/2; the other order11 families and their current envelope55 are not altered.

Generation95bef3 and separate fresh saved replay97502e both terminated exit0/PASS. Two other agents completed full source/report mathematical reads, including the inherited strict below-one spectrum and the five-quarters proof's intermediate4/3 bound. They passed the complete A10 extension profiles, arbitrary-rank/magnitude spectator classification, native-height gates, whole-cycle positive-kernel sealing, distinct-half payments, fixed-leaf finite outcomes and triangle contact-set exclusion. Neither review duplicated the matrix/scalar arithmetic or any historical census. No gap was identified in the stated local scope. The source imports no project arithmetic helper. It checks six physical single-profile matrices, a positive-kernel whole cycle, the formal isolated determinant and all native-height gates, the exact finite exceptional inverse diagonals, all six two-component energy sums, the triangle contact-set exclusion and the complete weighted pair partition. Classical all-rank inverse formulas, the finite/affine classification, actual factor-fiber theorem and mixed-component4/3 bound remain explicit inherited proof inputs. The finite exceptional checks are complete cases; no finite numerical rank sample is used as an arbitrary-rank proof. All15 direct bindings, seven local report links and Python syntax pass. Source, data and report are final/frozen after the recorded reviews. All reviews are internal; external refereeing and proof-assistant formalization are absent.

```
python3 scripts/fresh_cycle11_half_pair_transfer_20261018.py
python3 scripts/fresh_cycle11_half_pair_transfer_20261018.py --verify
```

Source 41

Source path: `results/fresh_cycle12_ideal5_transfer_20261017.md`

SHA256: 53e69a52b1a72f0b62e37744ea315dab1642404f8c5b52fee9fd6df73877902f

Cycle12 antipodal transfer with sealed finite coupons

FINAL/FROZEN after generation, fresh saved replay and full source/proof review. An actual quadrilateral whose varying affine factor is a12-cycle, with old marks at antipodal distance6, has a retained finite8/ideal5 payment or receives one unit from these finite coupons:

- A marked whole A1 against the unique middle node of a whole A11: weight1.
- Either marked node of a whole A2 against the unique middle node of a whole A11: weight1/2.

Thus its six antipodal cycle pairs can be removed, retaining the60 pairs at distance at most5. This is a local quadrilateral transfer. A finite vertex budget, triangle-retention check and global dimension composition are separate obligations; none is claimed here.

The source and certificate adapt the complete cycle13 proof, using the frozen energy-one classification. The literal checks use rational congruences and Fraction congruence elimination; the formal determinant check uses SymPy. They corroborate the written arbitrary-rank argument rather than enumerating spectators.

Complete endpoint profiles

Use the actual factor-fiber face star W+a+b+c+d, with opposite pairs a,c and b,d. At the old ideal corner its varying factor F is the12-cycle and the remaining diagram Z is positive definite and orthogonal to F. The white-plus-one-black edge diagrams are strictly positive definite. A nonzero adjacent black contact or a shared spectator contact already gives the retained short-path payment, so assume these are absent. Write the nonnegative contact vectors to Z as x,y; their supports are disjoint.

Deleting a from F gives A11. The complete positive-semidefinite one-root extension list is isolated, one3 or4 bond at either endpoint, one3 bond at either second node, or3 bonds at both endpoints. This inherited all-label classification still applies: no exceptional finite or affine diagram contains A11. In each single-contact case the adjacent marked distance is at most6. The opposite fixed simple leaf and its fixed degree-two neighbor exclude every classical affine completion; spectators can attach only at the new root and cannot alter those fixed degrees. Thus the entire connected corner component is finite and pays finite8. All other components are principal parts of Z and are positive definite. The source verifies the six core positive-definite diagrams, distances and fixed leaf/neighbor roles.

Only isolated and both-end3 profiles remain. Put alpha=1/6+1/6=1/3. The normalized Schur determinant is separately concave in reciprocal heights u=1/h,v=1/k. Spectators add

$$\alpha\|x/h - y/k\|^2_{G_Z^{-1}} \geq 0.$$

The four ordered rectangle minima are$-1/3$ for two isolated profiles, zero for either mixed order, and1/3 for two both-end3 profiles. All16 rectangle vertices are checked exactly. Two both-end3 profiles are therefore impossible under the unpaid hypothesis.

For isolated/both-end3 the base determinant is

$$1 - \frac{2}{3}u^2 - \frac{2}{3}v, \qquad 0 < u \leq 1, \quad 0 < v \leq \frac{1}{2}.$$

It vanishes only at u=1,v=1/2. Corank forces equality in the nonnegative spectator correction. Positive definiteness of Z gives x/h=y/k; disjoint supports then give x=y=0. The adjacent corner containing the isolated new root has that root as a whole A1 and the other old mark as the middle of a whole A11. This corner is finite, so the unit coupon pays. Reversing the ordered profiles is identical.

The isolated/isolated boundary

The formally verified determinant identity is

$$0 = 1 - \frac{2}{3h^2} - \frac{2}{3k^2} + \frac{1}{3}\|x/h - y/k\|^2_{G_Z^{-1}}.$$

The positive native heights are1 or at least sqrt2. If both are at least sqrt2, the base term is at least1/3. If one exceeds sqrt2, it is strictly positive even when the other is1. If the heights are1 and sqrt2 in either order, the base term is zero; disjoint-support equality again forces x=y=0. Both adjacent corners have whole A1/A11 components, and one unit coupon suffices.

The remaining case has h=k=1 and

$$(x - y)^T G_Z^{-1}(x - y) = 1.$$

Both separate augmented diagrams Z+c,Z+d are strictly positive definite principal edge diagrams. Consequently the complete energy-one theorem applies, with no bound on the rank of Z.

If one support is empty, choose the adjacent corner containing that spectator-free new root. It is a whole A1; the old cycle with the other mark deleted is a whole A11, marked at its unique middle. The remaining spectators form finite components. This corner is finite and its unit coupon pays, regardless of which allowed one-sided energy-one profile occurs on the excluded new root. In particular no bounded list of B/D spectator ranks is needed for this argument.

If both supports are nonempty, the energy-one theorem gives unit contacts to two distinct whole A1 components, to the two endpoints of a whole A3, or to the two short leaves of a whole D_r with4<=r<=7. In the same-component cases there is a c–spectator–spectator–spectator–d path of length4. At the opposite actual corner its connected component is finite or affine; all other components are positive-definite principal parts of the old white diagram and Z. It therefore receives the retained finite8 or ideal5 unit. The source illustrates A3→A5, D4→E6 and D5→affine E6, and detects a negative direction for the proposed D6/D7 opposite corners. The short-path argument already suffices without this illustrative split.

For two distinct whole A1 spectator components, each adjacent corner has a whole A2 containing its new black root, against the whole A11 middle. The two half-coupons sum to one. The energy classification seals the two spectator components, and all other spectators are orthogonal to the core; there is no unaccounted finite extension that changes these coupon types.

Whole-component and verification scope

Whenever a coupon is used, spectators are right to all of F and the selected new root has either no spectator contact or exactly its sealed whole-A1 contact. Hence the displayed A11 and A1/A2 components are whole components of the selected corner. Every other component is finite, so the entire corner is finite. In A11's path order the marked node is index5 (zero-based), its unique middle. The source checks this role directly.

The checker constructs15 literal cases: two native-height equality cases, two mixed-profile equalities, the two-A1 split, five one-sided examples, and five same-component twin-leaf examples. Each white-plus-one-black edge Gram is positive definite. Every admissible full local star has one negative and one zero direction, its old corner is affine, and the saved corner coupons or length-four path supply a unit. The two impossible twin-leaf extensions are explicitly rejected by their corner inertia.

Generation and a fresh separate-process saved replay both terminated exit0/PASS on the six single-profile checks,16 rectangle values, formal determinant,15 literal cases and60/6 cycle-pair partition. Direct source bindings, Python AST and local report links passed.

Root read the full source and proof, including the inherited value and profile formulas: unique mixed equality, both-end positivity, the fixed leaf/degree-two exclusion of affine completion with arbitrary spectators, native-height reduction before energy one, disjoint-support equality, untouched-side whole A1/A11 sealing, the split A2/A11 halves and the same-component path-four alternative including nonsimple ideal corners. The review also checked the rational congruences and unique-middle coupon role, and found no dependence on an A17 endpoint payment. It did not duplicate the15 literal inertia or16 rectangle calculations. No new ambient bound, physical chamber, outside refereeing or proof-assistant formalization is asserted.

```
python3 scripts/fresh_cycle12_ideal5_transfer_20261017.py
python3 scripts/fresh_cycle12_ideal5_transfer_20261017.py --verify
```

Source 42

Source path: `results/fresh_cycle13_ideal5_transfer_20261017.md`

SHA256: 894d8568f4dd5d80838797fc95fb58c99aa63bfded057e8191b4a08e2ad44085

The missing cycle13 distance-six transfer

An actual quadrilateral with a varying affine cycle of order13 and marks at distance6 has a retained payment at another corner. This permits paying only the65 cycle pairs of distance at most5. Its13 distance-six pairs may be removed. No additional finite coupon is needed, and no restriction on arithmeticity, cusp simplicity or spectator rank is imposed.

The source and certificate adapt the complete cycle14 transfer to arc lengths6 and7. The decisive new input is the strict spectator spectrum below one. The earlier order13 support diagnostic explicitly left this universal short-path obligation open; absence from its preferred maps alone did not remove these cycle pairs.

Use the actual factor-fiber face star W+a+b+c+d, with opposite pairs a,c and b,d. At the chosen ideal corner, the cycle F is orthogonal to a positive-definite spectator diagram Z. White-plus-one-black edge diagrams are strictly positive definite. Opposite heights h,k at its positive cusp kernel are positive. Adjacent black contacts or a shared nonzero spectator contact already supply a retained short path, so assume neither occurs.

Deleting a from F leaves A12. Its complete one-root positive-semidefinite extensions are isolated, a bond3 or4 at either endpoint, a simple bond at either second node, or simple bonds at both endpoints. Exceptional finite and affine types cannot contain A12. This is the same all-label path extension classification used in the preceding cycle theorem.

Each of the six single-contact profiles gives a finite marked corner: the opposite fixed simple leaf and its fixed degree-two neighbor cannot be part of a classical affine completion. Additional spectators are orthogonal to the old path, so those degrees remain fixed. Affine cycles have no leaf, affine C has doubled terminal bonds, affine B has a doubled terminal at one end and a fork at the other, and affine D has forks at both ends. Exceptional affine diagrams are too small. Hence the complete connected corner component is finite, even if spectators attach at the new root. Its marked distance is at most7 and receives the finite8 unit. The checker reconstructs all six positive-definite cores, distances, and fixed leaf/neighbor degrees. Every other corner component is a principal part of the positive-definite spectator diagram, so the actual corner is finite as claimed.

Only isolated and both-end3 profiles remain. The inherited normalized Schur determinant is separately concave in the two reciprocal heights. On the full allowed rectangles its four ordered profile-pair minima are

Profiles	Minimum
isolated, isolated	-5/21
isolated, both-end3	1/14
both-end3, isolated	1/14
both-end3, both-end3	8/21

All16 rectangle values are checked exactly. Spectators add a nonnegative quadratic form. Thus any case containing both-end3 is impossible under the unpaid-face hypothesis.

For two isolated profiles, put alpha=1/6+1/7=13/42. The determinant identity, reverified formally in the source, is

```
det(S)/(h²k²)=1−2∗alpha/h²−2∗alpha/k²
             +alpha∗(x/h−y/k)^T G_Z^(−1)(x/h−y/k).
```

The isolated positive heights are native opposite-contact magnitudes: each equals1 or is at least sqrt2. If either is at least sqrt2, the first three terms are at least1-3*alpha=1/14>0, a contradiction. Consequently h=k=1 before using any unscaled contact energy. The determinant equation then forces

```
D=(x-y)^T G_Z^(-1)(x-y)=4-1/alpha=10/13.
```

The supports of x,y are disjoint, and Z+c,Z+d are strictly positive definite principal edge diagrams. The complete strict spectrum therefore applies: below one the possible energies are0, r/(r+1) for integral r>=1, 2-sqrt2, and1-sqrt5/5. The rational value10/13 is neither irrational exception and would require r=D/(1-D)=10/3, which is not integral. This contradiction exhausts the final profile. Rotation and reflection cover all13 distance-six pairs. The proof uses neither a finite cutoff on spectator rank nor a finite list of native angles.

Generation and a fresh separate-process saved replay terminated exit0/PASS on the six single-profile diagrams,16 exact rectangle values, formal isolated determinant, native-height gate, spectrum exclusion and65/13 pair partition. A separate full internal source and mathematical read passed the complete inherited profile scope, fixed-leaf exclusion of affine completion, strict edge hypotheses and height-before-energy argument. That read did not repeat the6/16 arithmetic checks. Direct bindings, syntax and report links pass. Source, data and report are final/frozen. This is a local transfer, with no new ambient dimension or chamber claim; external refereeing and proof-assistant formalization are not asserted.

```
python3 scripts/fresh_cycle13_ideal5_transfer_20261017.py
python3 scripts/fresh_cycle13_ideal5_transfer_20261017.py --verify
```

Source 43

Source path: `results/fresh_cycle14_ideal5_transfer_20261017.md`

SHA256: bbecb12165b73f2f6a8777c010c70b3ac306421f03b9042341d030161fa532a5

Cycle14 ideal-five transfer

Completed internally checked local theorem; verification scope below. An actual quadrilateral with an ideal corner in a14-node affine cycle, whose marked distance is6 or7, is paid elsewhere by finite marked distance at most8, an ideal short path of length at most2, or two sealed finite half-coupons. The coupons join a whole A1 or A2 component to a whole A13 component marked at its unique middle. No arithmetic or simple-cusp assumption is imposed.

Consequently the cycle14 pair rule may retain just its70 pairs of distance at most5; its other21 pairs are transferred as stated. This is not a claim about other14-node affine types, total vertex budgets, complete high-core overlaps or an ambient bound.

The source and certificate bind the frozen spectator-energy spectrum, finite-eight/ideal-five local theorem and the earlier cycle14 equality proof.

Write the actual star W+a+b+c+d, with opposite pairs {a,c},{b,d}. At the selected ideal corner the varying cycle F is orthogonal to a positive-definite spectator block Z. The actual edge diagrams W+c and W+d are positive definite. Heights h,k of the opposite sides relative to the positive affine cusp kernel are strictly positive. These are the inherited actual factor-fiber facts; no compact-only port closure is used.

The old marked arcs are(P,Q)=(6,8) or(7,7). The complete one-root extension list of the old A13 path consists of isolated, endpoint3/4, second-node3 and both-end3 profiles. Each single profile produces a finite marked corner, even with arbitrary spectators: the fixed opposite simple leaf and its fixed degree2 neighbor exclude a large affine completion. Its marked distance is L-j+1<=8 in all12 possibilities. The checker verifies their positive-definite diagrams, marked distances and fixed leaves. Thus unpaidness leaves only isolated and both-end3 profiles.

For these profiles the inherited normalized Schur determinant is nonincreasing in reciprocal arc lengths and concave in reciprocal heights. All32 exact rectangle values are recorded. Every ordered pair containing both-end3 has a strictly positive minimum. In the isolated pair, either native height is1 or at least sqrt(2). If either is at least sqrt(2), the lower bound1-3*alpha is positive, where alpha=1/P+1/Q. Hence h=k=1 before using any unscaled contact bound.

With x,y the nonnegative native contacts of c,d to Z, the zero determinant requires

$$D=(x-y)^T\ G_Z^{-1}(x-y)=4-1/\alpha.$$

The required value is4/7 for(6,8), and1/2 for(7,7). If x,y share a nonzero coordinate, the actual path c-e-d of length2 already pays. Otherwise the supports are disjoint. Moreover Z+c and Z+d are positive definite as principal subdiagrams of the actual edge diagrams. These are exactly the strict hypotheses of the frozen spectator spectrum; they are not inferred merely from semidefinite corner data.

That spectrum excludes D=4/7, so the distance-six case is impossible under unpaidness. For D=1/2, the inherited elementary gap says two disjoint nonempty supports give D>1/2, while one nonempty support gives D>=1/2 with equality only for a single unit contact at an isolated A1 spectator. The other contact vector vanishes. Every remaining spectator is orthogonal to F,c,d and that isolated spectator.

The resulting sealed core is the14-cycle0,…,13, with a=0,b=7, new simple leaves c=14 at0 and d=15 at7, and one simple spectator e=16 at c. Reflection exchanges the spectator side. The checker reconstructs its signature(15,1,1), positive-definite white block and all four corner diagrams. The two adjacent finite corners have whole A13/A2 and A13/A1 components. In both A13 paths the marked root has endpoint distances6,6, so each of the stated halves applies. The other finite corner has separate A2 and A1 marked components. Orthogonality of all remaining spectators proves these are whole actual components, so the two halves sum to one without an extension assumption.

The original equality-core realization and complete cone audit are frozen in the predecessor and are not repeated here. The local transfer does not require that the small core alone define a finite-volume chamber, and no new chamber is asserted.

Generation and a fresh separate-process saved replay both terminated exit0/PASS for12 single-profile checks,32 rectangle values, two height/spectator cases, three finite equality corners and the70/21 cycle pair partition. Syntax, all10 direct source hashes and all5 report links passed. A separate full mathematical/source read found no gap in the fixed-leaf finite corner, discrete-height ordering, strict edge hypotheses of the spectrum, isolated-A1 equality or whole A13-middle sealing. That read did not independently repeat the matrix/census calculations or review a global theorem. The all-label spectrum and inherited finite/affine profile classification remain explicit proof inputs. Source, data and report are final/frozen; no external refereeing or formalization is claimed.

Source 44

Source path: `results/fresh_cycle16_half_coupon_transfer_20261017.md`

SHA256: 5672775d7c0e2114df97fd8c7c15353e29643cc0e3351dffb71d739fe19a8290

Antipodal cycle16 payments from two finite half-coupons

Complete local theorem; verification metadata below. Let Q be an actual quadrilateral of a finite-sided finite-volume hyperbolic Coxeter polytope. Suppose one ideal corner has a varying affine A factor on16 nodes, with the two marked nodes antipodal. Then either some corner is paid by finite marked distance at most9 or ideal marked distance at most7, or Q has two finite corners whose marked roots lie in separate, whole components A1 and A15, with the A15 mark its unique middle node. Assigning one half to each of these two finite corner types pays Q.

Consequently the16-cycle factor can use just its112 pairs of distance at most7, dropping its8 antipodal pairs, provided this finite half-coupon is included. This is a universal local statement, including nonsimple cusps and arbitrarily many supporting whites. It supplies no global finite-vertex budget or ambient dimension bound by itself.

The source and exact certificate bind the frozen mixed finite-nine/ideal-seven lemma, its weak-profile and spectator argument, and the actual factor-fiber theorem. The existing infinite-volume order16 example is used only as the literal equality core, not as a finite-volume construction.

Complete antipodal profile reduction

Write the face star as W+a+b+c+d, with opposite pairs {a,c} and {b,d}. Assume no finite-distance9 or ideal-distance7 corner is already paid. Then every adjacent black product is zero, since a nonzero such product supplies a length-one marked path. At the selected ideal corner,

```
W+a+b = F orthogonal-sum Z,
```

where F is the16-node affine cycle, a,b are antipodal, and Z is positive definite. The two old arc lengths are P=Q=8. The factor-fiber theorem makes this decomposition valid even when the ambient cusp has several affine factors: the other factors contribute spherical one-node deletions to Z.

The complete one-root positive-semidefinite extensions of the old A15 path give the same eight profiles as in the frozen weak-profile proof: isolated; a simple endpoint attachment on either arc; a doubled endpoint attachment on either arc; a simple second-node attachment on either arc; and simple attachments at both endpoints. Completeness here applies directly at rank15: connected finite possibilities are A16, B16 and D16, and the connected affine possibility is the16-cycle. Exceptional finite or affine diagrams are too small. A root touching several old nodes cannot produce another finite or affine type containing this induced A15 path.

For each of the six single-attachment profiles, the adjacent corner containing the replacement root and the other old mark has a finite A16/B16/D16 principal component. The opposite endpoint of the old path remains a simple leaf whose neighbor has degree2. Supporting spectators are orthogonal to every old F root, so they cannot change those two degrees. No large connected affine A/B/C/D diagram has that configuration; exceptional affine diagrams are too small. Hence the full component stays finite. The actual factor-fiber theorem then excludes an ideal corner with these two marks in that finite component.

The marked distance is8 for an endpoint attachment and7 for a second-node attachment. Thus each single profile already gives a finite-distance9 payment, regardless of how many spectator roots enlarge the finite component. The checker verifies all six principal positive-definite diagrams, distances and fixed-leaf configurations explicitly. Under the unpaidness assumption, only the isolated and both-end3 profiles remain for c and d.

The unique determinant equality

Let h=-c.nu and k=-d.nu, where nu is the positive affine null vector with unit coefficients on F. These heights are strictly positive because the opposite sides do not contain the selected ideal point. Put u=1/h and v=1/k. An isolated profile has h>=1 and constant c0=2; a both-end3 profile has h>=2, c0=0 and equal moments1 on the two arcs. The bounds use only the least nonzero acute Coxeter contact magnitude1; they impose no upper cutoff on opposite labels or dotted products.

At P=Q=8 the inherited normalized determinant becomes

```
F = 1 - (l_c*u+l_d*v)/2 - (c0*u^2+d0*v^2)/4,
```

with (l,c0)=(0,2) for isolated and (1,0) for both-end3. Its four profile-pair minima on the allowed height rectangles are respectively

c profile	d profile	Minimum
isolated	isolated	0
isolated	both-end3	1/4
both-end3	isolated	1/4
both-end3	both-end3	1/2

All16 rectangle values are exact rational checks. The only possible zero is the isolated pair, since

```
F = ((1-u^2)+(1-v^2))/2,
```

and 0<u,v<=1. Therefore h=k=1. The native products are a.c=b.d=-1, and c,d have no other contact with the old cycle. This forces the literal simple16-cycle with a simple leaf at each antipodal mark.

Eliminating arbitrary spectator whites subtracts a positive-semidefinite matrix [[p,q],[q,r]] from the c,d block. The source independently verifies the formal determinant change, after dividing by $h^{2*k}2$:

```
Delta = (p/h^2 - 2q/(h*k) + r/k^2)/4 >= 0.
```

The full face star has corank1, so F+Delta=0. It follows that the isolated equality holds and Delta=0. If the spectator contact vectors of c,d are x,y and its Gram is G_Z, then at h=k=1 this equality is

```
(x-y)^T G_Z^(-1) (x-y) = 0.
```

Positive definiteness forces x=y. Any common nonzero coordinate gives a white-interior path c-e-d of length2, immediately paying that actual corner, whether finite or ideal. Under the unpaidness assumption x=y=0. Thus every supporting spectator is orthogonal to the complete18-root equality core. This is a statement about the entire spectator block, not a bounded-rank attachment test.

Sealed finite corners and the coupon

Number the cycle0,…,15 and its leaves16,17 attached at0,8. Supporting whites in the core are1,…,7 and9,…,15. The four adjacent marked pairs are (0,8), (8,16), (16,17), (17,0). The rational core has signature(16,1,1), with radical

```
(-4,-3,-2,-1,0,1,2,3,4,3,2,1,0,-1,-2,-3,-2,2).
```

The selected corner is affine A15. At (8,16) and (17,0), the marked components are A15 and A1, and the A15 mark is7 edges from either endpoint, hence its unique middle. The last corner has its two marks in different A1 components, with two unmarked A7 components. All three corners are finite. Orthogonality of every extra spectator seals the displayed components as whole actual vertex components; it cannot silently turn A15 or A1 into a larger diagram.

The two half-coupons therefore contribute exactly1. They are defined using whole finite components, with the A15 marked node its unique center. No coupon is assigned merely because an A15 subdiagram happens to occur inside a larger component. The source checks every literal corner, both middle-node identifications, and the total payment.

This proves the stated local transfer. It also explains why the earlier infinite-volume local countermodel is compatible with the new rule: its two previously unpaid finite corners now pay one half each. Neither that example nor this transfer asserts the existence of a finite-volume chamber containing the equality configuration.

Verification and review scope

Generation and fresh saved replay both terminated successfully. The certificate checks six finite single-profile diagrams,16 rational rectangle values, the formal spectator determinant correction, the literal18-root signature and radical, all four corner diagrams, two half-coupons, and the112 retained/8 removed ideal pairs. Syntax, all8 direct source hashes and all5 local report links passed. A separate internal mathematical/source review read the complete source and proof, finding no gap in the rank15 extension completeness, fixed-leaf finite-corner argument, unique rectangle equality, arbitrary-spectator equality and sealing, or whole-component half-coupon. That review did not independently rerun the rectangle or core arithmetic. The inherited complete extension classification and actual factor-fiber geometry are stated proof inputs; this package does not repeat the old full cone census or published classification tables. Source, data and report are final/frozen. No global dimension conclusion, external refereeing or proof-assistant formalization is claimed.

Source 45

Source path: `results/fresh_d10_coupon_large_d_forest_budget_20261017.md`

SHA256: 18e8b58d06f0a9437d817f2943fcefd12b0712fc1e5dd648d0d86ea90f614aef

Exact finite-forest budget with D10 coupons and large-D half payments

For the specified finite payment rule, the exact maximum at total rank66 is **495**, attained by the whole finite diagram D66. This equals the finite curvature coefficient at that rank. The computation covers every rank from0 through75 and retains all existing cross-component coupons.

This is a finite counting theorem conditional on the separately proved geometric payment rules. It does not recompute ideal-vertex costs or assert a new ambient dimension bound.

Independent source · exact certificate

Payment rule and type domination

The input large-D transfer changes the whole finite D_n cost to8n−33 for n≥20 and adds a half-payment between every mark of a whole finite component of rank at most5 and each short fork leaf of a separate whole D10. D18 and D19 keep8n−32. The rank17 theorem supplies A17=114 and D17=127; the weaker upper bound B17≤127 is enough here. Other same-component costs and existing coupons are unchanged.

The finite Coxeter classification and the unchanged base costs are inherited inputs. The checker separately compares every same-rank representative and every coupon partner among228 type representatives, including I2(*) for all labels at least4. Its51,984 comparisons establish the following domination:

- Ranks1 through9 are represented by A of that rank.
- Ranks10 through16 are represented by D of that rank.
- Rank17 retains A17, which has cross coupons, and D17, which has none; B17 is dominated by D17.
- Ranks18 through75 are represented by D of that rank.

Each replacement preserves rank and can only increase both the base cost and every coupon interaction. Applying the replacements one component at a time therefore gives an upper bound for every finite forest.

In particular, B10 has base45 and its old A2 coupon has total cost1. D10 also has base45, but the new A2/D10 coupons have total cost2. Thus D10 dominates B10. The representative formula includes the new D10 contribution and **does not add the old B10 coupon a second time**. The old A1/A11 unit and A2/A11 half coupons are explicitly included before type domination; D11 dominates them as well.

Let d_r count whole D_r components, u count A17 components, t=d_11+⋯+d_16, and let a_s count residual A_s components for1≤s≤5. Write a=Σs a_s. In doubled units the representative cross cost is exactly

$$a(2d_{10} + 3t + 2u) + 2a_1(d_{11} + d_{12}) + 2a_4d_{13}.$$

The same-component costs are doubled too, so the entire calculation uses integers.

Complete domain and exact certificates

The implementation imports no project or producer arithmetic. It enumerates large components by their number and a nondecreasing rank tuple, rather than using the earlier producers' consecutive-rank count recursion. There are exactly1,608 multisets of D10,D11,…,D16,A17 with total rank at most75. This is exhaustive because every such component has rank at least10, so there are at most7.

Their residual interaction depends only on

$$(L, b, c) = (2d_{10} + 3t + 2u,\ d_{11} + d_{12},\ d_{13}),$$

giving152 distinct tables. For each original multiset and every residual component, the checker compares this compressed formula against the explicit type-pair coupon function. No dummy component is used to represent the total rank.

The remaining components are A1,…,A9 and D17,…,D75. They have no mutual cross coupons. Once a large multiset is fixed, their contributions are additive. An item-first unbounded-knapsack calculation produces each residual table. A separate verifier checks, for every fitting rank r and total n,

$$V(n) \geq V(n - r) + c(r),$$

and checks a saved predecessor attaining equality. Induction from V(0)=0 proves both the upper bound for every residual forest and attainment. Fresh replay uses these saved certificates; it does not rerun the maximizing algorithm.

At each total rank the global certificate dominates every fitting large-multiset/residual-table combination. An explicit typed forest is independently recounted using the original component-pair coupon function and attains the reported maximum. Every attaining component is an actual representative with its stated exact cost, so using B17≤127 as an upper bound does not turn these maxima into merely unattained bounds.

Certificate part	Count
Large component multisets	1,608
Compressed interaction tables	152
Direct interaction-formula comparisons	109,344
Bellman upper-bound alternatives	366,168
Attaining Bellman predecessors	11,400
Global upper-bound alternatives	22,504
Typed global attainers, including rank0	76

Maxima and direct large-D count

The finite curvature coefficient is β_n=n(n−6)/8 for even n and (n−1)(n−5)/8 for odd n. The following values are exact.

Rank	Maximum	Typed attainer	β_n minus maximum
64	480	4D16	−16
65	487	D65	−7
66	495	D66	0
67	503	D67	17/2
68	511	D68	16
69	520	A5+4D16	24
70	527	D70	33
71	535	D71	85/2
72	543	D72	51
73	551	D73	61
74	560	2A5+4D16	69

Rank	Maximum	Typed attainer	β_n minus maximum
75	567	D75	161/2

The D_n formula also has a direct graph proof. Removing one short fork leaf leaves a path of order n−1. For n≥18 that path has8n−44 pairs at distance at most8; the restored leaf adds9, giving8n−35. Exactly three leaf-marked pairs have distance9. The long-end pair keeps weight1. For n≥20 the two short-fork-leaf pairs have weight1/2 each, giving8n−33 in total; for n=18,19 all three still have weight1, giving8n−32. The checker builds each D graph from18 through75 and independently counts these pair classes. In particular D66 has493 short pairs and total leaf-nine weight2, hence495. The argument, rather than the finite graph check alone, proves the stated all-rank formula.

Validation and scope

Generation and a separate fresh saved-certificate replay both terminated directly with exit0/PASS. The logs are `/private/tmp/d10_large_d_budget_gen.log` and /private/tmp/d10_large_d_budget_verify.log. The source binds the final local large-D theorem, the rank17 costs, the earlier finite-budget certificate and relevant coupon inputs. Verification does not invoke an optimizer or import their arithmetic functions.

The parent reviewer read the entire source and report. The inherited finite-classification scope, all228 typed partner comparisons, separate A17 role, preservation of A11 coupons, D10 domination without B10 double counting, complete1,608-vector domain and152-key compression, Bellman upper bounds and attainment, global typed witnesses and the direct D66 count493+2=495 all passed. This was a full source/mathematical read, without duplicating the certificate arithmetic.

Two other reviewers separately read the full source and report. Both passed the coupon accounting, distinct rank17 types, complete multiset/residual decomposition, interaction compression, all-last-item inequalities with attaining predecessors, typed global exactness and all-rank D count. They did not repeat the arithmetic, Bellman verification or geometric census.

Nineteen direct bindings, Python syntax and all four local report links pass. Source, data and report are final/frozen. The proof of geometric face coverage, the inherited finite classification and earlier coupon-transfer theorems remain separately bound inputs. No ideal scalar comparison or new dimension conclusion is asserted by this package.

Source 46

Source path: `results/fresh_d15_balanced_ambient60_frontier_20261018.md`

SHA256: 738c5ff6f3d2e3c28fb70c83179ac97e86d512059d60d4a71ed1b1cd6f38bb48

D15 counted with the balanced rule: finite maximum407.5

Combining the new whole finite D15 half transfers with the frozen center-reversion and A/D16–17 balance gives an exact formal rank60 finite maximum **815/2=407.5**. All2,113 ideal terminal margins remain positive, with normalized minimum31/6000 and absolute minimum54. The remaining finite deficit is5/2 against curvature405.

Every finite-sided, finite-volume real hyperbolic Coxeter polytope of dimension60 therefore satisfies

$$(5/2)\ v_f \geq 4500 + 54\ v_i.$$

This requires at least1800 finite vertices, or1822 if an ideal vertex exists. The all-ideal ceiling remains59, and the unrestricted ceiling remains60. No general dimension59 theorem or geometric realization of a formal attaining forest is asserted.

The producer and certificate combine the balanced408 rule with the D15 local theorem. All other pair weights and cross-component costs remain unchanged.

Local changes and compatibility

The whole finite D15 pairs(0,11),(0,12),(1,12) change from units to halves, lowering its actual total102 to201/2. Their distances are11, 12 and11. The complete local theorem reconstructs all536 eligible native patterns and870 maps, isolating24 changed maps in eleven contexts. It proves four proper-completion exclusions, four saturation maps in one further context, and six whole-component replacements. The latter use the selected half plus either a distinct existing whole D16 half or an existing whole A2/D13-fork-center half. No new coupon is introduced.

The delicate saturation proof is an actual norm2 argument. A forced unit contact with inverse diagonal2, within a positive inverse end, forces the sole old-core contact and zero residual. Two required new leaves would then have product2; alternatively a saturated new fork center's required simple leaf can have product only0, positive or at most−2. Its genuinely new status follows because no other old corner root has the required adjacency. Both-end contacts give the independent short4/8 branch. The theorem therefore handles arbitrary supporting contacts before imposing any zero, rather than merely testing a displayed completion matrix.

The two D16 compensation pairs are(1,12) and(1,13), both still halves in the final balanced table. Neither belongs to the rebalanced D16(0,14)/(0,15) orbit. The A2/D13 mark is fork center10, distinct from the previously upgraded short-leaf marks11/12. Its existing half and the whole A2/D13 total5 are unchanged. The producer asserts these specific retained interfaces against both frozen packets.

Conversely, the center reversion and A/D exchanges use their explicit whole D5/D12, A16/A17 or D16/D17 payments; they consume no newly reduced D15 pair. Their eight whole-factor proofs and all earlier ideal priorities remain valid. Current order12 uses its D11, B12 or A1/B11 outputs, order13 its smaller-factor roles or A4/B9, order16 D5/D11, and the remaining ideal transfers their stated protected paths or separate typed coupons. None requires a changed D15 long pair at a unit.

Apply the independent current ideal/short-path branches first and then the complete finite selected/nonselected reduction with these compatible local replacements. The D15 theorem retains the entire inherited2614 routing, low-core/high-disjoint, protected/adaptive and later positive interfaces. Only its24 changed selected maps require the new argument; other completions of the same proper displayed component retain their old proofs. The A/D exchange obligations retain their separate full totals. There is no circular use of a deleted unit in an alternative corner.

If a short8 finite alternative has zero weight, the established distance8 corridor supplies short7. The current depth8 finite7 theorem then supplies short6 or its retained whole A8 unit. These protected leaf/branch and small-component outputs avoid all three new long D15 pairs. Triangle payment2 and quadrilateral payment1 therefore continue under one simultaneous rule. This is the complete local proof-routing argument, not an inference from the saved literal records alone.

All current literal payments

The producer starts from the balanced408 packet's final216 cornerwise records, including the reverted ideal17 center halves and the extra D16 balance credits. It reconstructs every native Gram and every finite D15 marked component, maps its pair canonically and subtracts1/2 exactly when it is one of the three new changed pairs. Each debit is also compared with the independently bound D15 local certificate.

Exactly eight corners receive a debit, all for(0,12). Every corner lower bound stays nonnegative and every face total remains at least1. The center-reversion cancellations and A/D balancing payments are retained throughout. These checks supplement the complete arbitrary-support and selected/nonselected coverage described above.

Actual rank15 costs and a different upper allowance

The current actual same-component costs are

Type	Actual cost
A15	101
B15	97
D15	201/2

The retained rank15 representative is labelled D15 but receives the **upper allowance101**, not its actual cost100.5. This distinction is essential: using100.5 as the rank15 upper would fail to dominate A15. The source therefore defines separate actual and upper base functions. All other actual and representative base costs are inherited unchanged.

Every one of33,489 ordered type-pair comparisons is checked anew, using the actual base on the actual type and the upper base on its representative, with all current cross-component allowances. No new cross-component cost occurs. In particular the reverted actual D5/D12 and representative A5/D12 costs remain23/2, and the order12/order16/order13 compensation charges remain present.

Rank15 belongs only to the fixed ranks9 through17 and is not a residual item. Thus all792 residual tables are exactly the frozen balanced tables: no dynamic program or optimizer is run again. The certificate compares the complete saved tables by exact equality and checks their 1,142,064 last-item upper inequalities and47,520 attaining predecessors. The54,288 item identities still include every fixed interaction, and both fixed-fixed and residual-residual cross costs remain zero.

All1,044 fixed multiplicity vectors are enumerated. For each vector the new upper value is exactly the old upper minus one times its rank15 multiplicity, or-8 c15 in eighth-units. This is a complete fixed-vector update, not removal of only the former maximizing forest.

Exactly8 representative alternatives remain above405. The resulting maximum is815/2, attained under the actual rule by 2D17+D26. This witness uses no rank15 representative gap. The former four-D15 actual forest falls to402, while4A15 costs404; neither controls the new maximum. Some representative forests can still overestimate their displayed actual type assignments, which is harmless because the global upper is matched by an actual formal witness.

Unchanged positive ideal comparison

The D15 theorem changes no ideal weight. The producer nevertheless reconstructs all preceding integer convolutions and exact scalar rows and compares the complete ideal result with the frozen408 certificate. All2,113 terminal profiles of total rank59, with at most one part1 through8 and all others at least9, remain strictly positive. The minimum31/6000 is at(9,9,9,9,9,14); the absolute minimum54 is at(59).

The established low9 concentration decreases normalized margin and does not increase the link-vertex product. Positivity of every terminal margin therefore extends the absolute54 surplus to every actual cusp. All factor-fiber multiplicities and exact high-rank family envelopes remain unchanged.

The simultaneous local payments give

```
D5 ≤ (815/2) v_f + sum_i C(i),
```

while the inherited ambient60 shape-curvature inequality gives

```
D5 ≥ 405 v_f + sum_i kappa(i) + 4500.
```

The two-extrema constant4500 is counted once. Subtracting and using kappa(i)−C(i)≥54 gives the stated(5/2)v_f inequality. The completed unrestricted60 theorem supplies the higher-dimensional tail for the all-ideal class bound59. The positive finite deficit5/2 prevents an unrestricted dimension59 conclusion from this comparison.

Verification scope

Generation99169/chunk1e68e4 and separate fresh replay47475/chunk241405 both terminated exit0/PASS after both input reports were frozen. Direct metadata checks pass95 bindings,4 local links and Python syntax. Root completed a full producer/report mathematical composition read: PASS. That review checked actual versus upper rank15 costs, simultaneous typed domination, the complete fixed-rank update and unchanged residual certificates, actual attainment, current literal debits and retained D16 halves, unchanged ideal reconstruction and the positive54 consequence. It did not execute numerical calculations, DP, censuses or literal checks. A second independent full producer/report mathematical composition read passed: it checked the same distinct actual/upper costs, complete fixed-vector and residual certificate, actual witness, updated literal accounting, disjoint whole-component compensation and independent current short-path routes, and unchanged positive ideal extension. That review did not replay arithmetic, DP, literals or censuses. Source, data and report are final/frozen after these reviews and direct metadata checks. The producer checks the unchanged complete residual certificates, all fixed-vector and typed comparisons, current216 literal records and reconstructed2113 ideal rows. The new D15 local geometry and earlier arbitrary-support classifications, concentration, factor-fiber and shape-curvature proofs remain explicit bound inputs. No historical geometric census is repeated, and no external refereeing or proof-assistant validation is claimed.

```
python3 scripts/fresh_d15_balanced_ambient60_frontier_20261018.py --verify
```

Source 47

Source path: `results/fresh_d16_d17_d14_coupon_forest_budget_20261017.md`

SHA256: 2c109ca38bb38e69a9b28d971040e2029358a671bb1c961da35138ee321ad45d

Exact finite budgets after the rank16/17 and D14-role changes

With the proposed finite costs A16=116, B16=104, D16=118 and D17=125, and the added A1/A2–D14 role coupon, the exact rank65 forest maximum remains487. If every irreducible component has rank at most17, however, the exact maximum is479, attained by D17+3D16. The curvature allowance is480. The previous small-component obstruction is therefore removed by this local rule; large finite components still obstruct the unrestricted comparison.

The source and certificate prove a conditional counting theorem. The geometric justification is the separate local transfer. No ambient theorem is asserted by this budget package alone.

The exact rule counted

Retain every payment in the frozen E6/D11 budget, except for the four stated same-component reductions. The inherited A17 and B17 costs remain114 and105. Add a half between every mark in a whole A1 or A2 component and the unique long-arm neighbor of the fork center of a separate whole D14 component. That single D14 role gives total increments1/2 and1 per component pair. It is distinct from the existing fork-center and short-leaf half-coupons.

All finite types are checked for domination by the chosen representatives. In particular D16 has cost118, which dominates A16=116 and B16=104; retaining their old costs120 would invalidate this step. D17 cost125 dominates the inherited B17=105, while A17 remains separate because of its coupons. E6 continues to represent rank6 and preserves its D11 quarter-coupon interaction. Arbitrary I2 labels retain their uniform rank2 representative bound.

The complete1,608 large vectors consist of D10 through D16 and A17. The interaction key has five coordinates:

```
(2*d10+3*(d11+...+d16)+2*a17,
 d11+d12, d13, d11, d14).
```

Relative to the predecessor, the last coordinate adds d14 to the doubled rank1 residual cost and2*d14 to the doubled rank2 cost. Rank6 still gets the existing*2d11. Each D16 large component has doubled base236, and the residual D17 item has doubled base250. There are626 distinct keys.

Two exact domains are computed. The unrestricted domain allows residual ranks1 through9 and17 through75. The small-component domain allows only 1 through9 and17 as residual ranks. Together with the large-vector list, this is precisely the domain of all forests whose components have rank at most17. No new coupon joins two residual components.

Upper and attaining certificates

Each of the two domains uses item-first integer knapsack generation. Replay reads the saved tables and checks every last-item upper inequality, an equality predecessor at every table entry, and literal component-pair interaction formulas. It reconstructs an actual typed forest attaining every global maximum through75 and checks every large-vector alternative.

Combined check across both domains	Count
Literal residual interaction formulas	125424
Bellman upper inequalities	1944982

Combined check across both domains	Count
Attaining Bellman predecessors	93900
Global alternatives	45008
Typed global attainers	152

The unrestricted maxima never exceed the frozen predecessor's maxima. At65 the value487 is still attained by D65, so lowering only the small components cannot prove a general ceiling64. With component ranks at most17 the exact value is125+3*118=479. This is a complete upper-and-attainment calculation, not a search over selected partitions.

Verification

Generation70315 and separate saved replay16089 both terminated exit0/PASS. A separate full source mathematical review checked domination, the new role's exact cost, the five-coordinate compression, both complete domains, Bellman upper/equality verification and typed global attainment. It did not independently replay the arithmetic. The same reviewer completed the final report review and confirmed its conditional counting scope.

The local geometric transfer and composition with ideal margins remain separate proof obligations for any resulting global corollary. This package neither repeats the geometric censuses nor asserts a new chamber.

```
python3 scripts/fresh_d16_d17_d14_coupon_forest_budget_20261017.py
python3 scripts/fresh_d16_d17_d14_coupon_forest_budget_20261017.py --verify
```

Source 48

Source path: `results/fresh_d16_d17_eighth_balance_20261018.md`

SHA256: 3786fc72577499c9b1e8d8aea6d5d10f515906928d4de2b27f4147fab2b19e2d

D16/D17 balance by distinct five-eighth and three-eighth payments

Replace two D16 halves by5/8 and two D17 halves by3/8. The four affected later exchanges still receive exactly1 at two distinct actual corners. The component costs become **D16=441/4** and **D17=469/4**, formerly110 and235/2. All ideal weights and all cross-component coupons are unchanged. Triangles retain payment at least2 and quadrilaterals at least1. A finite forest maximum and any ambient consequence require a separate certificate.

The producer and certificate bind the new D16 zero and D5/A11 unit, the current finite/ideal rule, the D16 fork exchange and the complete selected D17 support.

In canonical D16 the spine is0 through13, with short leaves14 and15. Give pairs(0,14),(0,15) weight5/8 instead of1/2. In canonical D17 the spine is0 through14, with short leaves15 and16. Give pairs(1,15),(1,16) weight3/8 instead of1/2. Both orbits have marked distance14. The changes respect the diagram automorphisms. The D16 zero(1,11) remains zero, and the D16(1,14)/(1,15) halves used by the rank15 theorem remain halves.

Complete selected support and the later dependency

The complete saved D17 selected domain consists of140 eligible patterns and168 induced maps. The producer validates the saved images, injectivity, every edge and nonedge label, canonical marked pairs and diagram symmetry. It finds no occurrence of either reduced D17 pair. Domain completeness is inherited from the bound selected-support census; no new search over arbitrary supporting diagrams is claimed here.

Absence from that selected domain is not the whole proof. The two D17 pairs were originally zero and were restored to halves specifically to pay four later D16 fork exchanges. Those later obligations must be changed at the same time. The present source reconstructs the complete old seal in each of1644,1664,1931,2324 and compares it exactly with the frozen D16 exchange before changing its payments.

Context	Actual D16 corner marks	Distinct actual D17 corner marks	New sum
1644	(1,11)	(2,11)	5/8+3/8=1
1664	(1,12)	(2,12)	5/8+3/8=1
1931	(1,11)	(1,12)	5/8+3/8=1
2324	(2,11)	(2,12)	5/8+3/8=1

The marks in the table are the literal old-core coordinates. Complete canonical maps send the first corner to D16(0,14)/(0,15) and the second to D17(1,15)/(1,16). Each canonical orbit describes one actual marked pair at its corner; it is not counted twice because two automorphisms are possible. The two paying corners in every row are distinct.

Why the factors are whole with arbitrary supporting whites

Each of the four reconstructed proofs has a physical norm2 core with 19 roots and signature(17,1,1). Its common-white principal is positive definite, as are the common-white-plus-one-black edge principals and the finite corner principals used

by the exchange. The inherited actual-corner computation is repeated to recover the complete possible extra-white contact domain, rather than treating a displayed D16 or D17 subdiagram as automatically whole.

Let z be the exact Gram radical. In the actual ambient Lorentz space its represented vector is zero. Indeed, it is null and orthogonal to the old span, which contains a timelike vector; the orthogonal complement of that timelike vector is positive definite. Every extra supporting white e therefore satisfies

```
sum_i z_i (e · root_i) = 0.
```

All permitted contact positions have nonzero radical coefficients. Native acute products are nonpositive. If e has any nonzero contact, the relation forces contacts at positions of both radical signs. For every such pair, the reconstructed graph distances supply an opposite-black path of length at most8 through e. This tests the entire necessary contact domain, with no upper bound on contact magnitudes or the number of additional supporting roots.

In the remaining no-short8 branch, every extra supporting white is orthogonal to the entire old core. A chain of further spectator roots cannot attach to a displayed corner component: its first attachment would itself be an extra white contacting the core. Thus the finite D16 and D17 components in the table are whole actual components, even with arbitrary spectator rank. Their prescribed marked-pair weights are applicable. The source retains the old complete radical seal and changes only the two numerical payments in each proof.

The short8 alternative is paid independently under the current rule. If a finite8 output is a deep zero, the established depth8 theorem gives short7 in the full supporting star. If a finite7 output is a depth10 zero, the established depth10 theorem gives short6. The retained tree/cycle and small-factor routes give short, protected or typed-coupon payments, not either present distance14 D17 pair. The radical argument therefore does not appeal recursively to its own new balance.

All other actual-face interfaces

The complete D17 nonselected routing remains inherited: the low-core whole-component bound is12, the high-disjoint branch is excluded, and the2614 high-overlap cases split into2391 with bound at most16, 74 protected leaf/basic cases, nine adaptive finite cases and the140 selected patterns. Smaller whole components cannot be D17. The protected leaf9 and bounded distance9 roles cannot be the present distance14 pairs. No changed pair occurs in the complete selected D17 domain.

The current ideal-corner priorities are independent of these long finite pairs. The order11 cycle/B/C and earlier coupled routes use short paths, their small whole factors or typed coupons. B12 and D13 use their specified short/protected roles and separate whole-component coupons. D105 uses protected distance9 alternatives or its D5/D11 coupon. The B/C14 and B15 projection/residual routes use exclusions or the independent short8 route. The order15 B/D and fractional ideal order17 transfers use their distinct whole-component coupons, not a same-component D17 distance14 half. Every ideal weight is unchanged by this packet.

The later finite transfers likewise keep their positive alternatives. The rank15 D alternatives consume the unchanged D16(1,14)/(1,15) halves; the other A16/rank16 alternatives use the specified whole A17 payments or separate coupons. The D17 A4/D13 and D5/D12 transfers use their own distinct coupon corners. The new D16(1,11) deletion uses a distinct whole D5/A11 unit. None of those arguments newly consumes a reduced D17 pair. The four D16 fork contexts in the table are the explicit exception and have been rebalanced, rather than silently inherited with old halves.

This yields a complete proof-routing argument: apply the independent current ideal/short-path branches first, then the selected/nonselected finite reduction and its later replacement interfaces. Every old paying route either avoids the reduced D17 orbit or is one of the four reconstructed exchanges with total5/8+3/8. An increase in the D16 orbit cannot harm any other face. This reasoning does not assume that a previous preferred witness remains paid merely because it was preferred.

The producer also reconstructs all216 current literal obligations from their native Grams and the D16-zero packet's final cornerwise lower payments. No literal corner receives a D17 debit. Four D16 corners receive an extra1/8, in three configurations from the order16 finite cross-payment package:663 has two credits and total5/4;676 and829 each have one credit and total9/8. All other recorded lower totals stay at least1. These216 checks supplement the complete actual-face argument; they are not a census of every quadrilateral.

The elementary triangle and finite7/ideal5 triangle proofs use retained short/basic pairs or their fixed small-factor/leaf exceptions. None uses the reduced distance14 orbit. Consequently their minimum2 is unchanged, along with the quadrilateral minimum1 just proved.

Accounting and verification

Each of the two D16 pairs gains1/8, so110 becomes441/4. Each of the two D17 pairs loses1/8, so235/2 becomes469/4. No cross-component coupon is added or upgraded. The equal and opposite component-cost changes do not cancel automatically in an arbitrary forest: component multiplicities can differ. This packet supplies no forest optimization or ambient bound; the completed unrestricted bound remains60.

Generation session87634, terminal chunk943cb8, and separate fresh replay session47186, terminal chunk643765, both passed exit0. The producer checks the complete saved140/168 selected interface, reconstructs all four old whole-core seals and their exact new totals, and audits all216 current literal configurations. It preserves the current depth10 dependency. It does not rerun the historical overlap census or a finite forest count.

The construction agent completed a full producer and inherited-interface mathematical read, including the original D16 seal and current D17 selected/nonselected, literal and later-payment arguments, and wrote this report: PASS. No map, inertia, literal, census or optimizer replay was performed in that read. Root and Exacthub also completed full source/report mathematical reads: PASS. They checked the four distinct whole-factor payments, complete saved-domain and later-interface coverage, arbitrary-white sealing and independent short-path route, exact costs and literal changes. The review reads did not repeat maps, inertias, the historical census or a forest calculation; root's separately recorded fresh replay remains the execution evidence above. Final metadata verifies29 immutable bindings, six local links and the Python syntax tree. Source, data and report are final/frozen. No external referee or proof-assistant verification is claimed.

```
python3 scripts/fresh_d16_d17_eighth_balance_20261018.py --verify
```

Source 49

Source path: `results/fresh_d16_partial_weight_transfer_20261017.md`

SHA256: 72571673391c601f181faf6ecc3fe0803835841b69b0dad40ab718530f5bd64f

Ten half-weights on the order16 affine D factor

Starting from the completed finite8/ideal5 payment theorem and its successors, ten further pair weights in affine D15 fall from1 to1/2. Its diagram has16 nodes, and its total pair weight falls from111 to106. Every affected quadrilateral still receives at least1. No new finite coupon is needed. The order16 maximum consequently becomes107, attained by the unchanged B rule; the unchanged A and C costs are80 and103. This is a local weighted transfer, with no ambient dimension assertion.

The source and certificate cover every affected map from the frozen complete small-factor census. They do not assume that a displayed finite subdiagram is the whole affine factor. The proof below controls every possible completion to the actual order16 factor, including contacts with roots outside the displayed component.

The intrinsic weighted rule and its complete support

The canonical affine-D15 diagram has spine0–1–…–11, leaves12,13 at0, and leaves14,15 at11. Its previously deleted nine pairs remain at weight0. All still-paid pairs at distance greater than9 receive weight1/2:

Representative	Distance	Pairs in its orbit	Selected contexts
0,10	10	(0,10),(1,11)	8
1,14	11	(1,14),(1,15),(10,12),(10,13)	12
2,14	10	(2,14),(2,15),(9,12),(9,13)	36

The first and third context lists overlap in two cases, leaving54 distinct contexts. The complete328 affected induced maps produce66 distinct literal completions after identifying maps that give the same Gram with the old core labels fixed. The source checks every map, its marked image and all induced bonds, then records its completion index through the grouped output. All three pair orbits are invariant under every automorphism of the target.

The new weights sum to `111-10/2=106`. Every marked path of length at most8, and every elementary ideal triangle payment, retains weight1. The old nine zero weights retain their existing replacement proofs. In particular no half-weight is being treated as a deleted pair.

Residual control before completing a factor

The frozen complete overlap theorem applies when no retained short8 payment has already resolved the face. Its cross-bond exclusions determine each literal old core H. Of the54 affected contexts,46 use its ordinary Lorentz-residual argument and8 use its relevant-component argument. Every H here is simply laced, has norm-two roots and signature `(order(H)-1,1,0)`.

Write J for the inverse Gram of H. The actual preferred corner determines the necessary contact domain for an additional supporting white root. The source reconstructs that corner, including all its finite blocks and inverse diagonals or affine kernels, and checks equality with the saved domain. The two end sets cover this domain. A root touching both ends supplies a marked path of length at most8. Within either end, J is entrywise positive and every two distinct native contacts have projection square greater than2. Therefore every remaining extra root has at most one contact with H.

For a contact at i, let p=J_ii and rho=2-p. The projection and positive-space residual squares of a simple-contact root are p and rho. All relevant contact nodes have p greater than1, so the next possible Coxeter magnitude, sqrt(2), already has projection square `2p>2`. Thus these contacts must be simple; there is no label cutoff. The possible p=1 contacts outside the relevant component, including their doubled-contact alternatives, receive the separate exact tests inherited from the residual proof.

In each ordinary context all candidate nodes have p>1. For two candidate roots the exact test is

`J_ij^2 > (2-J_ii)(2-J_jj).`

If J_ij is positive, the two roots have positive actual product and cannot coexist. If it is negative, their actual product is negative and their new bond supplies a marked path of length at most8. The source recomputes every test and shortcut length. Hence at most one additional supporting white root can contact H in an otherwise unresolved face. Any such root has a single simple contact. All later whites are orthogonal to H. The inherited inverse bound controls entire attached components, so this statement is not restricted to the immediately adjoining nodes.

In each of the eight relevant-component contexts, the marked component has exactly one permitted entry node i. When its attaching root G is present, there can be at most one further core-contact root C. C contacts a node outside the marked component and is orthogonal to G. The only two residual possibilities needed here are:

p_G	p_C	Projection cross product	Residual determinant	Maximum attached tail rank if C exists
4/3	4/3	2/3	0	0
5/4	1	1/2	1/2	1

These entries, and every other excluded contact type and doubled alternative, are recomputed exactly. The complete residual/no-bridge proof applies with the now-checked shortcut threshold8. For determinant0 the two residuals have a positive zero relation, so no core-orthogonal white can attach to either. For determinant1/2 the inverse GG entry is2; a white touching G must be the unique equality root in their residual span, and acuteness seals it against every further core-orthogonal white. Thus the tail bounds0 and1 include indirect chains and prohibit bridges to C.

The marked base component has order13 in the first row and12 in the second. With C present, its whole component can therefore have order at most14 in either case. An actual order16 affine factor excludes C. Without C, G is the sole root contacting H, as in the ordinary case. This also shows that no other old component of H can enter the actual affine factor through an indirect bridge.

Every completion is either impossible or a sealed whole factor

Fix one of the328 maps of the marked base component into an actual affine D15 factor F. The no-bridge argument just given ensures that the remaining nodes of F are new roots outside H: they cannot be old roots from another component of the preferred core corner. Consequently each missing target node is a new supporting white, with its bonds to the base fixed by the map.

If two or more missing nodes must touch H, the one-contact-root conclusion excludes the map. If its forced entry node has J_ii>2, projection already excludes the norm-two new root. In every other proper completion exactly one new root G touches H. Its single simple contact is fixed by the target. Every other new node is orthogonal to all of H; the entire Gram on H and the missing F nodes is therefore determined, including its zeros against old components outside the marked base. This is where the arbitrary-support argument is needed: merely drawing the missing target edges would not justify these other zeros.

Exact rational inertia of every remaining proper completion is `(17,2,0)`. Two negative directions are impossible in the ambient Lorentz space. Thus no proper completion actually occurs. The exhaustive outcome is:

Outcome	Distinct completions	Affected maps
Requires multiple core-contact roots	20	64
Forced single contact has projection square greater than2	14	112

Outcome	Distinct completions	Affected maps
Forced full Gram has two negative directions	22	72
Displayed factor already whole; sealed	10	80
Total	66	328

The22 inertia exclusions arise in18 contexts; four contexts have two different literal completions. All completed matrices and their exact signatures are saved. The source also checks the positive affine kernel on the target F before using the full-matrix obstruction.

For the ten surviving cases, H consists of two L25 diagrams sharing a reversed edge, has18 roots and signature `(17,1,0)`, and its preferred component is already the entire16-node affine F. The necessary contact domain has no feasible projection candidate. Hence every additional supporting white is orthogonal to all of H. The remaining white Gram is positive definite, since it supports an actual hyperbolic face. Each other corner is an orthogonal sum of its displayed finite core corner and that positive-definite remaining diagram. Every displayed finite component used below is therefore a whole actual component. This does not assume a simple cusp or any upper bound on the number of spectators.

Actual finite halves in the ten surviving contexts

The checker reconstructs all40 literal corners, the white positive-definite principal and all four strict edge principals, the affine positive kernel, and every whole finite component and marked role. The needed finite halves are:

Contexts	Whole small component	Whole large component and marked role
1312,2495	A4, endpoint	D12, fork center
1975,2368,2505,2512	A4, endpoint	D12, short fork leaf
1979,2372,2536,2540	D5, marked node	D11, short fork leaf

These are all already permitted by the whole-small-component coupon budget. The D5 mark is allowed to be any node of that whole small component. Each finite half is at a different actual corner from the selected ideal half. Their sum is1. Four cases additionally have a whole finite D16 unit; the proof does not need to spend that extra payment.

Composition and cost

The complete finite8/ideal5 theorem first handles the short-path cases, then its low-core, disjoint-core and complete overlap domains. The present changes concern only distance10 or11 pairs in order16 D factors, so every short8 unit and elementary triangle unit is unchanged. The immediate whole-affine alternatives in the inherited local transfers have order at most13; their rules are unchanged here. The other domains are retained for specific reasons: the low-core theorem gives whole component order at most12; the all-label high-disjoint theorem excludes its entire domain; finite whole-component, protected-leaf and adaptive finite outcomes retain their finite rules; and protected/basic ideal outcomes have marked distance at most9, so retain their full weight. The complete1,153-pattern overlap map certificate contains all remaining possible order14–17 factors. Its54 contexts listed here are exactly those with an image among the ten changed pairs. Their rejected maps cannot represent actual factors, and each surviving one now receives an ideal half and an existing finite half. Thus a list of54 contexts is used only after the complete domain split, not as a census of all quadrilaterals.

Thus triangles still receive at least2 and quadrilaterals at least1. The finite budget is unchanged. At an ideal vertex with link-vertex count V, the contribution of this factor is now `106*V/16`, using the existing factor-fiber multiplicity, which is linear in the pair weights. The unchanged B cost107 is the largest order16 cost. A global scalar comparison must use107/16 and must check its own concentration moves; this package makes no dimension assertion.

Verification scope

Primary generation62629 and separate fresh saved replay45499 both terminated exit0/PASS. Syntax, all22 direct bindings and all6 local report links passed. An additional calculation using SymPy characteristic polynomials and exact Sturm

root counts independently checked the nine distinct full Grams underlying all22 Lorentz-index exclusions and all10 sealed cases; their signatures agree. It did not import the producer's inertia routine. The checker revalidates all direct input bindings, the54 affected support cases and328 maps, every residual inequality used here, all66 completed Grams or projection/contact obstructions, and the40 surviving corners. The frozen complete overlap and induced-map enumerations remain inputs; they are not independently enumerated again. No external refereeing or formal proof-assistant verification is claimed.

Two separate internal full source/report mathematical reads passed. The root review also reread the inherited residual/no-bridge proof and checked the new actual-corner reconstruction. The second review independently checked why the completion zeros are forced before the index-two test, why an exterior contact would bound the relevant whole component by14, and why the surviving finite components are whole. Both checked the full local-domain composition. Neither review independently repeated the frozen overlap or induced-map census. Source, certificate and report are final/frozen.

Source 50

Source path: `results/fresh_d17_existing_three_corner_transfer_20261018.md`

SHA256: 03606f81300acb76cac429d80045de042b560cf216001059d1352666e56b837a

Two finite D17 reductions using existing payments

Starting with the frozen D5/D12 alternate-corner rule, change whole finite D17 pair(2,12) from1 to5/8 and pair(2,14) from1/2 to0. Canonical D17 has spine0 through14 and short leaves15,16 at14. Its actual same-component cost decreases from467/4 to **927/8=115.875**, a saving7/8. No coupon, ideal weight or other finite weight changes.

The two newly reduced pairs are intrinsic singleton orbits, at distances10 and12. Every triangle still receives at least2 and every quadrilateral at least1. The complete formal rank59 maximum remains397.5: the previous upper remains valid under pointwise decreases, and2D16+D27 is an unchanged actual typed attainer. This is not a dimension58 proof or a new example.

The producer and certificate reconstruct the complete selected finite-D17 support, four exact full-core seals and all216 current literal interfaces. They use the actual coupons already present, rather than representative counting allowances.

Complete selected obligations and distinct payments

The inherited complete2,614-context high-overlap domain has140 eligible selected patterns for a whole finite D17 component. The producer freshly reconstructs all168 induced maps, including the patterns with no maps, and compares the whole result with the frozen D17 support certificate. Exactly eight maps in four contexts use either changed pair. Every affected displayed component already has17 roots; no proper completion or additional finite embedding is omitted.

Contexts	Changed selected corner	Other actual paying corners
1062,2525	D17(2,12), new5/8	Whole D5 long endpoint / whole A1, existing3/8
1017,1264	D17(2,14), new0	Whole A1 / D14 fork center, existing1/2; distinct whole A2 / A14 third node, existing1/2

The first row spends the new3/8 coupon proved in the frozen D5/D12 packet. There is one eligible long endpoint in D5 and one mark in A1. The selected5/8 and that distinct3/8 sum to1. Its other two corners are unused. In particular no new D5/A12 coupon is introduced.

The second row uses two corners other than the selected D17 corner. The A1/D14 fork-center half is part of the original broad whole-component rule; its canonical D14 mark is11. The A2/A14 half is the previously proved A16 third-node coupon: either node of A2 against canonical node2 or11 of A14. It is distinct from the A14 second-node coupon at1/12 and is already fully charged in the current actual A2/A14 total4. These two halves sum to1 with no payment at the reduced selected corner.

All types, component memberships and complete canonical mark images are extracted from the actual corner Gram matrices. The two paying corners in each row are distinct. The proof does not confuse a graph automorphism's multiple images with multiple actual corners.

Arbitrary supporting roots and whole components

Each affected context has the same19-root L25/L25 physical norm-two core, with signature(17,1,1). The exact radical is

```
(-6,-3,-4,-2,-5,-4,-3,-2,-1,0,
   6,3,4,2,5,4,3,2,1).
```

The source checks the positive-definite15-root white support, every white-plus-one-black edge support and all four finite17-root corners. It reconstructs the selected actual corner and its complete necessary extra-white contact domain using the inherited contact theorem.

In an actual Lorentz realization the radical represents zero. It is null and orthogonal to the displayed span, which contains a timelike vector, and that vector's orthogonal complement is positive definite. Every additional supporting white must therefore annihilate the radical. All allowed contact positions have nonzero radical coefficients. Since actual wall products are nonpositive, a nonzero contact vector must meet positions of both radical signs.

For every pair of opposite signs the producer checks an actual-corner white-interior shortcut of length at most8 through the extra white. The inherited independent current short8 route then pays the face. Otherwise every additional supporting white is orthogonal to the entire core. This covers arbitrary contact magnitudes and any number of extra roots. An indirect spectator bridge would have a first root touching the core and is excluded by the same argument.

In that remaining branch all displayed factors at every corner are whole actual factors. Thus the D5/A1, A1/D14 and A2/A14 coupons above apply without a spectator-rank bound. The first row does not assume that a whole selected D17 alone seals the other factors; the full-core radical argument supplies that additional conclusion.

Current whole-face routing and later interfaces

The full selected/nonselected division is inherited from the completed D17 theorem:2,391 high-overlap contexts have whole-component bound at most16,74 use protected leaf9 or basic ideal payments, nine use the independent adaptive finite outputs, and140 are the selected domain reconstructed here. The low-core whole bound is12 and the all-label high-disjoint domain is excluded. Neither protected distance9 outputs nor smaller whole components use either present D17 singleton.

The complete later positive-output inventory is unchanged except for the two selected routes explicitly replaced here:

- The finite A15/A16 exchanges restore or use whole A17 payments, or their specified A2/A13, A1/A13 and A2/A14 coupons.
- The D15 reductions and the later finite-rank14 refinements use distinct D16 halves, finite D15 halves or small-component coupons.
- The four D16 fork exchanges and their fractional rebalance use D17(1,15)/(1,16), a disjoint orbit checked directly by the producer.
- The D17 node3 and node4 transfers use their own selected pairs and whole A4/D13 or D5/D12 coupons; their other same-component D17 units are not consumed as alternatives.
- The original D16 unit exchange's restored D17(1,12)/(1,13) obligations were already replaced by their explicit sparse coupons. Neither is one of the new node2 reductions.
- The recent contextual rank13, order14 B/C/D and D5/D12 rules retain their stated small whole factors, short paths and typed coupons.

The producer also reconstructs all216 current literal obligations from the frozen D5/D12 packet. It maps every finite D17 marked component under all canonical embeddings. No literal corner uses either new singleton; every saved corner lower bound and face total stays unchanged. This confirms the literal interfaces rather than asserting that216 records enumerate all actual quadrilaterals.

Apply the established independent actual-ideal-corner priorities first. Their universal tree/cycle and spectator arguments use retained short paths, protected leaf roles, small whole components or explicit typed coupons, including all later refinements already bound by the D5/D12 baseline. None requires these two finite long node2 roles. The existing short8-to-short7 and short7-to-short6 replacements remain constructive under the present decreases: their protected features, whole small alternatives and current cycle11 replacement avoid these distance10/12 pairs. Thus the short8 escape in the radical argument is independent of the theorem being proved here.

On the remaining finite selected branch, unchanged selected outputs keep their old payment, and the four affected contexts have the exact alternatives above. Faces without a changed actual corner retain their whole previous payment. Triangles use the retained elementary short and protected outputs, so still receive2. The simultaneous quadrilateral minimum is1, without a circular use of the new selected reductions.

Counting consequence and verification scope

The total decrease is3/8+1/2=7/8 per whole D17. The actual and representative D17 bases both decrease by14 in units1/16. The source checks domination of every actual finite type through rank59; A17 remains its separate fixed representative and is unchanged. All actual and representative cross-component costs, including the contextual rule, remain unchanged.

The complete upper397.5 from the frozen D5/D12 certificate therefore continues to hold. Direct evaluation of the unchanged formal forest 2D16+D27 gives397.5, proving that the maximum itself remains exact. No new Bellman optimization is required for this conclusion. The number of fixed-vector alternatives above391.5 has not been recounted here. The ideal rule is unchanged and the completed unrestricted bound is59.

Generation session58007/chunk1144ae and separate fresh replay session88973/chunkf6514e terminated exit0/PASS, with eight affected maps and216 unchanged literal records. The ideal and arithmetic peers each completed a full source/report mathematical composition read: PASS. They checked the complete selected domain, four full-core seals and necessary contact domains, all whole-component marked roles, distinct payments, current literal and later-output interfaces, independent short8-to-short6 composition, exact cost decrease, and the unchanged attainer proving the monotone finite upper remains exact. The arithmetic peer also read the inherited selected/nonselected D17 routing and current D5/contextual interfaces. Neither review executed maps, inertias, literal arithmetic, the historical census or Bellman optimization. The source performs exact rational native-map, inertia, radical, corner-role and literal checks. Historical all-label geometry, the necessary contact theorem, current short-path composition and finite-count completeness remain explicit inherited mathematical inputs. No external refereeing or proof-assistant verification is asserted.

```
python3 scripts/fresh_d17_existing_three_corner_transfer_20261018.py --verify
```

Source 51

Source path: `results/fresh_d17_fractional_ambient60_frontier_20261018.md`

SHA256: 43f472cf7c004cec39dd7f3c540a097a3e42f3b1c3eca0b9ab8c870da02eade6

Finite maximum411 and28 negative ideal profiles at ambient dimension60

The simultaneous local changes lower the complete finite rank60 maximum from413 to **411** and reduce the negative terminal ideal profiles from49 to **28**. The required finite curvature is405. Thus this is a completed local and counting improvement, but it does not exclude dimension60 or prove dimension at most59. The completed unrestricted dimension bound remains 60.

The producer and certificate combine the preceding rank15/order15 rule with three completed local transfers:

Transfer	Same-component change	Additional whole D5/D12 cost
Finite D17	119 to235/2	3
Ideal order17 short-leaf transfer	116 to114	1
Ideal order17 center transfer	114 to227/2	1/4

All three finite increments are counted. In canonical finite numbering, their changed D5/D12 marked pairs are respectively `{3,4} × {9,10,11}`, `{0} × {10,11}` and `{(0,9)}`. These sets are disjoint. The actual whole-component total increases from15/2 by17/4 to **47/4**. The final ideal order17 family costs are A85, B111, C108, D227/2, with all-family envelope227/34.

Simultaneous local rule

The finite transfer replaces three D17 halves by zero. Its six selected configurations use distinct full D5-short-leaf/D12 units and no ideal payment. Arbitrary extra supporting roots either produce an independent short8 route or are orthogonal to the full radical core, sealing those whole components. The protected D17 alternatives of the D16 exchange remain halves.

The ideal short-leaf transfer removes four affine D16 halves using a distinct whole D5-long-end/D12-short-leaf unit. The center transfer reduces two other affine D16 halves to quarters, retaining an ideal quarter and adding a distinct whole D5-long-end/D12-center three-quarter payment. Their arbitrary-white arguments use the positive affine kernel and the full Lorentz-core radical: any nonorthogonal extra support gives a paid length2 path; otherwise the finite coupon components remain whole. Neither ideal theorem spends a same-component finite payment.

Consequently the new ideal routes can be used before the finite theorem. Their short-path escapes retain the current independent depth8-to-short7 and depth10-to-short6 alternatives. The earlier B/C11, cycle/tree, B12/D13, D105, B/C14, B/D15 and order17 routes retain their stated short, protected or whole-component outputs. Faces with no newly changed ideal corner retain the finite theorem's ideal weights pointwise and receive only increased finite coupons. The finite theorem's changed cases have their own independent units. This is a composition of the universal local arguments; the finite list below is an additional interface check.

All216 inherited literal configurations are reconstructed from their native Grams and the finite theorem's latest cornerwise lower payments. Positive-definite corners are decomposed into actual finite components; their marked types distinguish D5 long endpoints from short leaves and D12 centers from short leaves. Actual affine corners are independently mapped into the order17 diagram. Exactly six configurations change:

Contexts	Ideal debit	Distinct finite increment	New total
1978,2371,2535,2539	1/2	1/2	1
1314,2529	1/4	1/4	1

All other literal payments are unchanged. Each corner stays nonnegative and every total is at least1. These216 records are the inherited literal obligations, not an exhaustive census of actual quadrilaterals. Complete selected/nonselected coverage and arbitrary extra supports remain the local theorems' geometric inputs. Every triangle still receives at least2 and every quadrilateral at least1.

Complete finite maximum, including deep deletions

The finite calculation uses quarter units throughout. Its base cost is twice the preceding doubled cost, less four times the already proved depth10 deep7 deletions, and less6 for a whole D17. The numbers of deep7 deletions are `max(0,r−27)`, `max(0,r−28)` and `max(0,r−29)` for A, B and D factors. They are now included in the upper rule itself, not only subtracted when displaying actual witness costs.

The actual D5/D12 interaction gains17 quarter units. Its representative A5/D12 allowance gains the same amount. All33,489 simultaneous type-pair comparisons verify domination at both endpoints, including the changed base costs. An A5 representative allowance need not be an actual A5 payment, so each recovered forest's actual and upper costs are stored separately.

The producer independently generates792 residual tables for all1,044 fixed-component vectors at ranks9 through17. Residual ranks are1 through8 and17 through60. There are no residual/residual or fixed/fixed interactions; the item identities include all fixed/residual terms. For rank5, the new term is17 times the number of fixed D12 components, recovered exactly as the key's `c11+c12` entry minus its `c11` entry.

Fresh replay checks1,142,064 last-item upper inequalities,47,520 attaining predecessors and54,288 item identities. It does not repeat the optimizing recurrence. Every representative forest is covered. The maximum is1644 quarter units, or411. The actual formal forest2D16+D28 attains411, proving equality of the actual formal-forest maximum with the representative upper. Formal attainment makes no geometric-realizability assertion.

Sixteen representative alternatives exceed405; two have strict gaps between their upper allowances and actual recovered costs. The old 2D17+D26 witness now costs410. Four D15 factors still cost408, and 2D16+D28 supplies the current maximum deficit6. Thus further finite savings are required even if the ideal side is completely resolved.

Complete ideal comparison and remaining conditional targets

The binder independently reconstructs all2,113 terminal rank59 partitions with at most one part1 through8 and all other parts at least9. Every integer H-polynomial convolution, vertex product, weighted moment and scalar margin is recomputed. It first reproduces every preceding rank15/order15 row, then subtracts the exact order17 saving from its cost:

```
(116 − 227/2) ∗ count(rank16)/17 = 5∗count(rank16)/34.
```

The actual factor-fiber multiplicity V/17 is included in this normalized expression. Exactly21 old negative profiles become positive, leaving28 negative and none zero. No remaining negative profile contains rank16. The minimum normalized margin is−1117/7680 at(9,9,11,15,15); the minimum terminal absolute margin is−123905 at(9,9,9,9,11,12). A negative terminal absolute margin is not promoted to a universal bound for unreduced cusps. Only the proved low9 concentration is used; high factors are not concentrated.

A saved conditional calculation shows that further total factor-cost savings1/2,1/2,1/4,5/4 at affine ranks11,12,13,15 respectively would make all2,113 profiles positive. Those are orders12,13,14,16. The hypothetical normalized minimum would be73/104000 and absolute minimum54. These further geometric transfers are not proved, and any extra finite payments they require are not included. This conditional scalar target is not a theorem.

At ambient60 the finite curvature is

```
(5∗sum(min(j,59−j),j=0..59) − 59∗60)/2 = 405.
```

The single global two-extrema correction remains4500 and is counted once. The unresolved finite deficit6 and negative ideal profiles prevent the nonnegative vertex-surplus argument. No new dimension conclusion follows.

Verification scope

Generation session95482, final chunk3a59d5, terminated exit0/PASS. Separate fresh replay session76239, final chunkbe16ca, also terminated exit0/PASS. Two peer agents completed full producer/report mathematical composition reviews: PASS. They checked the arbitrary-support local priorities, disjoint role costs and retained finite alternatives; quarter-unit accounting and deep7 subtraction exactly once; complete typed/fixed/residual domains, item identities and Bellman upper/attainment logic; six net-zero literal transfers; and the complete ideal domain, factor-fiber saving, conditional-only targets and single global correction. Both also reread the inherited finite/scalar interfaces. Neither reran DP, convolution, literal inertia, geometric censuses or other arithmetic. One compared the saved summaries and six changed literal totals directly. The three local source/data/report packets were frozen before generation. The current binder reconstructs the complete finite and ideal arithmetic and all216 literal interfaces, and checks their immutable dependencies. It does not rerun the inherited all-label geometric censuses or formally verify their universal geometric arguments. This source, certificate and report are frozen. No external refereeing, proof-assistant verification or new finite-volume example is claimed.

```
python3 scripts/fresh_d17_fractional_ambient60_frontier_20261018.py --verify
```

Source 52

Source path: `results/fresh_d17_partial_weight_transfer_20261017.md`

SHA256: 46af0e7682a4fa9b6cda516679cb690abd05e42f5a9629b5a23bf76a3b244e48

Twelve half-weights on the order17 affine D factor

Completed local argument; verification is recorded below. Starting from the frozen complete finite8/ideal5 face-payment theorem and its D17 singleton successor, reduce twelve further ideal pair weights from1 to1/2. They form four automorphism orbits in affine D16, whose diagram has17 nodes. Its total pair weight falls from126 to120. No new finite coupon is needed.

This is a weighted rule. The twelve pairs are still paid by1/2; deleting them would not be justified by the present argument. Every affected selected ideal half has a different, actual finite corner carrying an existing half. Their sum remains1. All elementary triangle payments and all marked paths of length at most8 retain weight1. The source and certificate save the complete weight vector and the literal supporting cases.

The finite-vertex budget is unchanged. At a cusp with link vertex count V, this factor's contribution becomes `120*V/17`, by the existing factor-fiber multiplicity. This local result does not assert an ambient dimension bound or any reduction of the whole order13 cycle rule.

The complete affected support

The canonical affine-D16 diagram has spine0–1–…–12, leaves13,14 at0 and leaves15,16 at12. The previously removed singleton `(1,11)` remains at weight0. The new half-weight orbits are:

Representative	Distance	All pairs	Selected contexts
0,10	10	(0,10),(2,12)	1314,2529
0,11	11	(0,11),(1,12)	1311,2494
1,15	12	(1,15),(1,16),(11,13),(11,14)	1974,2367,2504,2511
2,15	11	(2,15),(2,16),(10,13),(10,14)	1978,2371,2535,2539

The source obtains the context support from every emitted marked image in the frozen complete1,153-pattern mapping certificate, and verifies equality with these lists. Each orbit is invariant under every diagram automorphism. All other weights remain as in the singleton successor. In particular the last four-pair nonbasic orbit is retained at weight1.

Every listed context has the same literal19-root core: two L25 diagrams sharing their far node9. The overlap certificate supplies absence of additional cross-core edges whenever no retained short payment has already resolved the face. The proof below does not assume that an abstract finite pattern is the entire ideal factor; in these twelve cases the literal preferred component is already the full affine D16 diagram.

Actual-kernel sealing applies uniformly

Number the first L25 by0 through9 and the second by10 through18 with shared9. The affine component F omits roots3 and13. The full19-root Gram has signature `(17,1,1)`, and its radical is an actual zero relation because its span contains a timelike vector. The radical coefficients at3 and13 are nonzero and have opposite signs.

For any additional actual supporting white root e, the positive affine kernel of F and positive semidefiniteness of the corresponding actual corner force e to be orthogonal to every F root. Pairing e with the full-core radical then forces its contacts

with3 and13 either both to vanish or both to be nonzero. In the latter case the adjacent marks3,13 have a path3–e–13 of length2 and receive a retained unit.

Otherwise every additional supporting white is orthogonal to the entire19-root core. Each remaining corner is consequently a direct orthogonal sum of the displayed finite core corner and the positive-definite remaining white diagram. All displayed components remain whole components. This proves both the finiteness of these corners and the absence of indirect tails or bridges that could invalidate a finite coupon. No simple-cusp hypothesis is used.

The checker verifies this setup separately for all twelve marked contexts: the full signature and radical; positive affine kernel; actual white positive definiteness; all four strict edge principals; exact corner connected components and marked roles. These are48 literal corner decompositions. They share a geometry proof, not an assumption that the marked roles are interchangeable.

The existing finite halves

In every affected context one adjacent finite corner has an existing half-payment between a whole small component of rank at most5 and a fork center or short fork leaf of a whole D11–D16 component. Representatives of the four groups are:

Representative context	Whole small component	Whole large component and mark	Finite half
1314	D5, arbitrary marked node	D12, fork center	1/2
1311	A4, endpoint	D13, fork center	1/2
1974 or2367	A4, endpoint	D13, short fork leaf	1/2
1978 or2371	D5, arbitrary marked node	D12, short fork leaf	1/2

The reversed contexts give the same component roles. These payments are already included in the whole-small-component finite budget. The D5 mark need not be a leaf: the inherited rule permits any mark in the whole small component. No D12 short-leaf upgrade is used here; its old half suffices, even when the small component is D5 rather than A1.

The selected ideal corner now receives1/2, and this distinct finite corner receives1/2. Hence the quadrilateral still receives at least1. Four of the twelve contexts have an additional whole finite D17 unit; retaining that extra payment is harmless. The other eight have exactly the required two halves under the recorded rules.

Transfer to the complete face-payment theorem

The original complete finite8 reduction first resolves short marked paths, then classifies the remaining low/disjoint/overlapping minimal-core domains. The newly reduced pairs all have distance greater than9. Thus every short8-first alternative remains a unit; the elementary ideal triangle rule also retains all its units. The source checks those inclusions explicitly.

In the remaining complete overlap domain, any selected marked image whose weight changed is one of the twelve contexts above, by the complete frozen map support. If an extra white yields the length-two path, the quadrilateral receives a unit. Otherwise the preferred ideal corner contributes its half and the sealed finite corner contributes the existing half. Selected pairs outside these four orbits retain their previous weight and proof. The prior zero-weight singleton keeps its own separately validated two-finite-half replacement.

This argument preserves the inequalities “triangle total at least2” and “quadrilateral total at least1.” It does not reinterpret a half-weight as a binary paid unit, and it adds no cost to any finite vertex. The ideal-factor sum is linear in the weights, so the twelve half-unit reductions subtract exactly6 from126. A new scalar comparison must use this weight sum, not the number of pairs with nonzero weight. In particular the old concentration move eliminating repeated rank15 parts can change direction when the rank16 coefficient falls, and is not imported here.

Verification scope

Generation25567 and a separate fresh saved replay16515 both terminated exit0/PASS for the complete saved-map support, all four symmetry orbits, all twelve sealed contexts and48 literal corner checks, and the exact120 total. Syntax, direct bindings and report links passed. Every input status and recorded direct binding is verified.

Two separate agents completed full source and written-mathematics reviews. Both passed the weighted composition: selected ideal halves combine with distinct sealed finite halves, short8 and triangle units remain unchanged, the previous zero singleton retains its own proof, and the actual factor-fiber cost is linear in the weights. The root review also checked that the inherited immediate whole-affine alternatives have order at most13, so no unlisted immediate D17 obligation is being changed. Neither review repeated the48-corner arithmetic or the1,153-pattern map census. The complete overlap and induced-map searches remain explicit frozen inputs; they are not repeated here. Source, data and report are final and frozen. This report makes no new scalar, dimension, external-review or formalization claim.

Source 53

Source path: `results/fresh_d17_singleton_pair_coupon_20261017.md`

SHA256: 7919a25ec9deeaf706e1e0439a864792790265ee3f1f97b6da71510737cc4867

One fewer paid pair in the order-17 affine D factor

The complete finite8/ideal5 payment rule can remove one additional pair from the 17-node affine D diagram if it adds a finite half-payment between a whole A4 component, marked at an endpoint, and a whole D13 component, marked at the unique long-arm neighbor of its fork center. The ideal factor cost decreases from 127 to 126. This is a local face-payment transfer; its finite-vertex budget and any resulting dimension bound are separate obligations.

Here the standard affine type is **affine D16**, with 17 nodes and affine rank 16. Labels such as "D17 target" in the inherited enumeration count diagram nodes. The new finite D13 coupon uses the ordinary finite type and therefore has 13 nodes.

The exact checker and saved certificate bind the complete 2,614-context overlap certificate, its 1,153-pattern induced-map census, and the complete finite8/ideal5 payment theorem. They do not replace those completeness proofs with the single core below.

Complete support diagnostic and the only affected context

Use the canonical affine D16 graph with spine 0–1–…–12, leaves 13,14 at 0 and leaves 15,16 at 12. The inherited paid set has 127 pairs, including 110 basic pairs. Its 17 remaining pairs split into six automorphism orbits:

Representative	Distance	Orbit size	Selected contexts requiring the orbit
0,10	10	2	2
0,11	11	2	2
1,11	10	1	1
1,15	12	4	4
2,15	11	4	4
3,15	10	4	16

The certificate saves the actual context keys for every row. The singleton pair 1,11 is fixed by every one of the eight graph automorphisms. Removing it leaves an invariant 126-pair set containing the whole basic set and every elementary triangle pair. In particular, the removed pair has distance 10: it is not a short8-first payment, a distance-at-most9 feature payment, or an immediate whole-factor alternative of order at most13. The earlier literal D16/D17 transfers end in finite coupons, finite units or length-two paths and do not use it.

Across every saved marked image in the complete current census, only context 2518 requires this pair. Its full key is

`left L25 with marks{3,4}; right L25 with marks{3,4}; identify left9=right9`.

Its preferred corner is the physical pair 4,14. This exact whole-core match matters: the inherited overlap certificate excludes every additional cross-core edge, including simultaneous or larger native contacts. The local argument below therefore has the literal 19-root Gram as a proved input in this context. It is not asserted for an arbitrary abstract embedding of an affine factor.

Literal core and all four corners

Number the left L25 by 0,…,9 and the right copy by 10,…,18, sharing 9. Every edge has label3. The left edges are

`0–1, 0–2, 2–3, 0–4, 4–5, 5–6, 6–7, 7–8, 8–9`,

with the corresponding right edges obtained by adding 10 to nodes0,…,8 and retaining the shared node 9. The opposite black pairs are{3,4} and{13,14}; all other displayed nodes are white. Arbitrary additional supporting whites E are allowed.

The checker verifies the full signature(17,1,1), positive definiteness of the 15-node white diagram and each of the four 16-node actual-edge diagrams. The four 17-node displayed corners are:

Marks	Displayed corner	New half-payment
3,13	finite A4 + A9 + A4, marks in different A4 components	0
3,14	finite A4 + D13; A4 endpoint 3, D13 long-arm neighbor 14	1/2
4,13	finite D13 + A4; D13 long-arm neighbor 4, A4 endpoint 13	1/2
4,14	affine D16 of order 17; marked distance 10	0

The finite D13 fork center is 10 in the second row, with short leaves 11,12 and long-arm neighbor 14. It is 0 in the third row, with short leaves 1,2 and long-arm neighbor 4. Thus these roles are distinct from the earlier fork-center/short-leaf coupons. The two new half-payments sum to 1.

For an independent realization check, the certificate matches the actual noncompact quadrilateral with sides 3,4,13,14 and four vertices in the frozen finite-volume H17 pyramid ledger. That example is corroboration; the following proof applies to any actual quadrilateral having this literal marked core.

Additional supporting whites either pay immediately or preserve both coupons

Let F be the 17-node affine diagram obtained by omitting 3 and 13. The actual corner 4,14 contains F and is ideal. Its positive affine kernel has coefficients 1 at its four leaves and 2 on the spine. Every extra supporting white e belongs to this corner. Its nonpositive contacts with F, together with the positive kernel, force e to be orthogonal to each root of F. Equivalently, the represented positive null vector is proportional to the ideal corner ray, to which every incident wall is orthogonal. This uses the actual corner and does not assume a simple ideal vertex.

The full 19-root Gram has radical

`(−6,−3,−4,−2,−5,−4,−3,−2,−1,0,6,3,4,2,5,4,3,2,1)`.

It is an actual linear relation among the normals. Its represented vector is null and orthogonal to the indefinite core, hence orthogonal to a timelike vector; that orthogonal complement is positive definite, so the represented vector vanishes. Pairing the relation with e and using e perpendicular to F gives

`−2(e·n3) + 2(e·n13) = 0`.

If the equal contacts are nonzero, the actual corner 3,13 contains the white-interior path 3–e–13. It has a unit payment: finite distance 2 is paid, and at an ideal corner the actual varying-factor lemma places the connected marked pair in its affine factor, where distance 2 is paid. This already supplies the quadrilateral's required unit.

Otherwise every extra supporting white is orthogonal to all 19 core roots. The extra-white diagram is positive definite as a principal subdiagram of the actual face support. At each of the three displayed finite corners the full corner diagram is therefore the orthogonal sum of its positive-definite displayed diagram and this extra-white diagram. These are actual finite corners, and the displayed A4 and D13 components cannot be enlarged or joined by extra whites. Both coupon components are whole finite components, with precisely the endpoint and long-arm-neighbor marks listed above. The two half-payments survive and give the required unit.

Consequently the local transfer is valid with arbitrary extra supporting whites and nonsimple ideal vertices. Additional ambient ideal walls do not enlarge the whole finite components used by the coupons.

Composition interface and verification scope

Replace only the order 17 affine D paid set by the saved 126-pair set and add only the stated A4-endpoint/D13-long-arm-neighbor half-coupon. The complete predecessor's triangle coverage and short8-first replacements remain valid. Every other selected image is retained or has its existing transfer; the sole new omission is the proved context 2518 above. Thus triangles still receive at least 2 and quadrilaterals at least 1, conditional on the already frozen complete predecessor theorem. The ideal pair cost for this target is 126. Establishing the total finite cost of the enlarged coupon rule is deliberately left to a separate budget proof.

Generation and separate fresh saved replay both terminated exit0/PASS. Syntax, all19 direct input hashes and all6 report links passed. Root completed a full source/mathematical read, including the imported literal Gram, inertia and component helpers, and found no gap in the affine-corner orthogonality, actual radical relation, length-two alternative, whole-component coupon roles or complete saved-map support diagnostic. That review did not duplicate the orbit/core arithmetic or the inherited overlap and embedding searches.

This checker reconstructs the six support orbits from the saved complete maps and directly verifies the one core's rational signature, positive kernel, four strict edge diagrams, four corners and coupon roles. It does not rerun the 82,197-identification or 1,153-pattern searches. Published diagram classifications and the predecessor's complete geometric arguments remain explicit inputs. Source, data and report are final/frozen. No new ambient dimension bound is asserted here, and no frozen predecessor was changed.

Source 54

Source path: `results/fresh_d5_d12_alternate_coupon_rebalance_20261018.md`

SHA256: 9688dd0de94f289f7d7289d3358c4146b84904dd1610124c474b58ff7425c9ff

Alternate whole corners for the D5/D12 coupon

Starting from the frozen rank13 contextual rule and the now proved order14 D-near/fork rule, make the following simultaneous changes. A component in this statement is a whole connected component of the actual finite corner diagram.

- Lower the payment between the unique long-arm endpoint of D5 and either short fork leaf of D12 from7/8 to1/2.
- Add3/8 between the unique long-arm endpoint of D5 and either endpoint of a whole A12.
- Add3/8 between that D5 endpoint and a whole A1.

The added amounts are increments, not replacements for other payments. All old broad halves, finite same-component weights, ideal weights and other sparse coupons remain unchanged. Every triangle still receives at least2 and every quadrilateral at least1, under the inherited finite-volume actual-corner hypotheses. The four affected sealed faces each receive1/8+1/2+3/8=1 at three distinct corners.

The complete formal rank59 finite maximum remains **795/2=397.5**, above the required783/2=391.5. The number of representative large-component alternatives exceeding391.5 falls from14 to12. This is a local payment and counting improvement; the completed unrestricted dimension bound remains59. No new chamber or new dimension bound is claimed.

Formal rank59 forest	Previous actual cost	New actual cost
3A1+4D5+3D12	791/2	391
8A1+3D5+3D12	1557/4	783/2

The second forest shows why the new A1/D5 cost must be counted. The statement concerns formal forest costs, not realizability of these forests by hyperbolic Coxeter polytopes.

The producer and exact certificate contain the reconstructed full-core proofs, marked alternatives, all216 current literal interfaces and the complete finite upper/attainment certificate. No historical overlap census or ideal convolution is rerun.

Four distinct whole-corner alternatives

The original order17 transfer has a complete140-pattern selected domain,488 successful native maps, and32 affected maps in contexts1978,2371,2535,2539. Each affected map already contains the entire17-root affine factor. Its current ideal weight is1/8 under the sparse-D16 and order17 rebalance. The latter used a distinct D5-long-end/D12-short-leaf payment7/8.

The new producer reconstructs the four full19-root Gram certificates using the inherited native routine, compares every reconstructed seal with the frozen rule, and independently extracts every marked finite component at all four corners. It checks the full signature(17,1,1), positive-definite supporting whites and all four strict edge principals. Here are the three paying corners; numbers are literal core labels.

Context	Ideal corner, weight1/8	D5/D12 corner, weight1/2	New alternate corner, increment3/8
1978	(1,15)	(1,13)	(3,13): A12 endpoint3 / D5 long endpoint13
2371	(2,15)	(2,13)	(3,13): A1 node3 / D5 long endpoint13
2535	(5,11)	(3,11)	(3,13): D5 long endpoint3 / A12 endpoint13
2539	(5,12)	(3,12)	(3,13): D5 long endpoint3 / A1 node13

Each row uses three different corners. The fourth corner contributes zero to this lower bound. No whole D17 or D16 same-component payment at that fourth corner is used. In particular, no previously deleted long finite pair is silently restored.

The component roles are intrinsic. D5 has fork-arm lengths1,1,2, so its unique long-arm endpoint is canonical node0. The two endpoints of A12 have canonical images0 and11. The two D12 short leaves have images10 and11. The producer checks all canonical images, rather than choosing one graph isomorphism. This makes every added and reduced marked role invariant under the corresponding component automorphisms.

Arbitrary supporting whites

The entire affine component consists of the core roots other than3 and13. Its positive kernel and nonpositive actual wall products force any additional supporting white root to be orthogonal to all seventeen affine roots. The full19-root core has a negative direction and a one-dimensional radical. Its radical is an actual zero vector relation: the orthogonal complement of a subspace containing a negative direction is positive definite. The radical coefficients at3 and13 have opposite nonzero signs.

Pairing an additional white with this zero relation forces its two remaining contacts either both to vanish or both to be nonzero. In the second case the path3–white–13 joins the adjacent black sides of the actual corner(3,13) and has an independently retained length2 payment. On the remaining branch every additional supporting white is orthogonal to the full core. A longer bridge is also impossible, because its first root touching the core would violate this conclusion.

Consequently all the displayed finite components, including the new A12 and A1 alternatives, remain whole in the actual corner after adjoining an arbitrary orthogonal positive-definite spectator forest. The old seal applies to every corner, not only the previously paying D5/D12 corner. This supplies the required arbitrary-support argument for the new three-corner total. It assumes neither a bound on spectator rank nor a simple ideal link.

Complete payment-use inventory and current composition

The decreased roles return to their original broad half. Therefore any earlier proof which used only that half remains valid. The complete output audit in the frozen sparse-D16 rule and its rank13 refinement is retained here. The following separates all possible uses by the actual marked role, rather than by a broad component-cost allowance.

Geometric interface	Payment retained here
Universal short cycle/tree, low/disjoint and protected supporting routes	Their original broad halves, specified A-path coupons, retained short paths or whole same-component alternatives
Original order17 partial-weight transfer	Any D5 mark against a D12 center/short leaf only requires the old half
Four order17 long-end/short-leaf upgrades,1978/2371/2535/2539	The three distinct whole corners just proved, or their independent length2 alternative
Order17 center transfer and its later reversal	D5 long endpoint0 / D12 center9 remains1/2
Finite-D17 D5/D12 unit transfer	All six roles $\{3,4\}\times\{9,10,11\}$ remain full units

Geometric interface	Payment retained here
Other later ideal/finite sparse transfers	Their separately specified A4/D13, A4/D12, A5/D11, E6/D11, D5/D11, D5/A11, small B/D or A-path roles remain unchanged
Current rank13 contextual proof	Its A2/A13-middle, A2/D13-center, A1/A13-middle and context-dependent A2/A2 payments remain unchanged
Current order14 D-near/fork theorem	A4/D10-short-leaf, A1/D13-neighbor9, whole finite D14, retained D11 half or independent short/BC11 outputs; no D5/D12 output

The original partial transfer explicitly permits an arbitrary marked node of a whole D5 against a D12 short leaf at weight1/2. The only later upgrade of the long-end/short-leaf roles was the four-seal theorem above. The finite-D17 unit theorem instead uses the two D5 short leaves, canonical3/4, and is disjoint. The long-end/center role0/9 is also disjoint and unchanged. Thus the current component-pair cost is the broad15/2 plus the six half-unit increments3, namely21/2; there is no residual charge from either long-end/short-leaf upgrade.

Use the same actual-corner priorities as the current completed proof. The independent small-ideal transfers and short8-to-short7-to-short6 routes retain all their protected outputs, including the whole-A8 alternative and the current depth6 finite7 exceptions. The current B/C/D14 basic/near-fork refinements are resolved by their own independent outputs. On the remaining selected order17 branch, the complete 140-pattern native map domain has exactly the four affected contexts above. Their full-core proof now gives1/8+1/2+3/8. Unchanged selected images retain their prior proof; low, disjoint, protected, immediate and later routes retain the outputs listed in the inventory.

This is a replacement of the actual paying outputs of the exhaustive face proof. It is not an assertion that every quadrilateral with a D5/D12 corner belongs to one of four saved cores. No new arbitrary supporting-white restriction is inferred from absence in a saved census. Triangles retain their elementary/same-component payments and total2; all old broad half payments survive, and the new coupons are positive.

As an exact simultaneous interface check, the producer starts with the current rank13 literal lower bounds and aligns all216 records and all marked corners with the frozen order14 D-near/fork interface. The latter has no literal debit or credit. The new rule has four D5/D12 debits, two A12 credits and four A1 credits. Four paired records stay at1. The two additional A1 credits occur in the order16 partial-transfer records[2372,0] and[2540,0], whose totals rise to11/8. All216 totals remain at least1. This corroborates the complete output proof; it does not replace it.

Exact finite accounting and the complete rank59 certificate

In units of1/16 the actual component-pair changes are

- D5/D12:−12, because two marked pairs each lose3/8;
- D5/A12:+12, because two endpoints each gain3/8;
- A1/D5:+6, because one marked pair gains3/8.

The resulting actual component costs are respectively21/2,11/4,3/8. The newly proved A1/D13-neighbor half is included: the actual total A1/D13 cost is2. It was the optional complete-count branch in the frozen rank13 certificate and now has its independent geometric proof.

The same representative type map is used as in that certificate. The only changed representative costs are A5/D12:−12 and A1/A5:+6. The D5/A12 increase is absorbed by the reduced A5/D12 allowance. The source verifies simultaneous base and cross-cost domination for all180 actual finite types through rank59, hence32,400 ordered type pairs, including every added and reduced coupon. The nonlinear context coupon is separately dominated by the same representative trigger: any actual whole A10 supplies a fixed rank10 component, and A1/A2 counts are preserved by the representative map.

Fix every count of ranks9 through17 exactly as in the inherited 961-vector domain. Also fix every possible count k5 of A5 in the representative forest, from0 to the remaining rank divided by5. There are2,574 such combined vectors and2,572 distinct interaction keys. The residual item ranks are1,2,3,4,6,7,8 and17 through59: rank5 is excluded because its entire multiplicity is already fixed. The fixed-rank17 A17 and residual-rank17 D17 roles remain distinct, as in the inherited typed count.

All costs involving a fixed A5 are included explicitly. In particular, each residual A1 gains6*k5 units. The source checks every residual item cost against the complete fixed-component cross sum, for128,700 item identities. Residual components have no mutual pair cost after rank5 has been fixed, apart from the already isolated contextual A2/A2 interaction.

For each key, the producer makes two item-first unbounded tables: one with all residual ranks and one excluding rank2 as well. The replay does not optimize. It checks every possible last-item upper inequality and one attaining predecessor for every rank0 through59. The5,144 tables give6,718,064 exact Bellman upper inequalities and303,496 nonzero-rank attaining predecessors.

The untriggered option uses the ordinary table. If a fixed rank10 component is present, the triggered option reserves two A1 components, fixes the total number j of A2 components, and uses the no-rank2 table for the remaining rank. Its contextual increment is4*j*(j−1) units. The no-rank2 table may add further A1 components, so the reservation does not constrain the actual A1 count to two. Every triggered forest has exactly one such total j and is covered; every untriggered forest is covered by the ordinary table. All6,455 options are checked. The reconstructed maximizing forest is evaluated with the full nonlinear rule, including its actual trigger, so an ordinary-table attainer that happens to trigger the bonus cannot be overlooked.

Each combined fixed-vector row has an exact upper value and an attaining representative forest. Taking the maximum over k5 gives the complete961 large-vector alternatives. Their maximum is6360/16=795/2. The actual typed forests D16+D17+D26, A17+D16+D26 and2D16+D27 all attain it. Twelve alternatives remain above6264/16=783/2. There is therefore no dimension58 conclusion from this certificate.

Verification and limits

Generation session15966 terminated exit0/PASS (terminal chunkb28479; saved log4503c1). The initial runs stopped at an overly narrow expected count for positive A1/D5 credits; the final guard records all four. No failure of a payment inequality or finite certificate occurred. Fresh saved replay session10149 terminated exit0/PASS (terminal chunk383bd1; saved loge69b1b), with exact saved-object equality and all upper/attainment checks. After the descriptive path-terminology precision, final source-bound generation4937/3a01df and fresh saved replay81754/2b1408 both terminated exit0/PASS (logs2872db andc64781).

The arithmetic peer completed a full producer/report mathematical read, including the inherited full-seal and actual/upper/context/item interfaces. The review passed the three distinct whole-corner payments, retention of broad halves and finite-D17 short-leaf units, complete current coupon inventory and literal composition, exhaustive rank5 fixing, the ordinary/no-rank2 contextual split, simultaneous type domination, all-last-item upper/attainment design and actual397.5 witnesses. It did not execute geometry, census, finite tables or scalar arithmetic.

The ideal/finite peer also completed a full producer/report mathematical read. It passed all four arbitrary-support seals and marked roles, disjoint short-leaf/center obligations, current D14 and contextual interfaces, exhaustive residual decomposition, type domination and attainment. Its requested prose precision now explicitly identifies 3–white–13 as the short path at the adjacent black sides of actual corner(3,13). It did not execute seals, maps, literal or Bellman checks.

Root completed a full current producer/report mathematical composition read, including that terminology precision. The three distinct payments, whole-corner sealing, original-half output inventory, actual and representative increments, complete rank5 fixation and ordinary/no-rank2 context split passed. This review did not execute arithmetic or rerun the complete inherited census. All three reviews are internal agent reviews. Final metadata validation checks317 direct bindings,10 local report links and source syntax. Source, certificate and report are final and frozen.

The calculation reconstructs four exact native full-core seals and all216 literal interfaces. Complete historical map enumeration, all-label supporting classifications, the original exhaustive low/disjoint/high-overlap reduction and the current triangle and short-path theorems remain explicitly inherited frozen inputs. No numerical normal search, geometric census, ideal scalar replay, external mathematical review or formal proof-assistant verification is claimed. The attained forest costs are formal counting statements, not finite-volume or compact hyperbolic examples.

Source 55

Source path: `results/fresh_deep_finite8_cycle_interface_20261017.md`

SHA256: 15087e6a21bd5acd1410848354df10682720e72f962c9772375aefd5538aa7ef

The cycle interface survives deep finite-distance-eight halves

The inherited ideal-five cycle and classical-tree replacements remain valid if unit payments at graph distance8 in whole finite A/B/D components are reduced to halves when **both marks have distance at least t from every endpoint, branch and endpoint of a nonsimple bond**, for either t=17 or t=20. All other current payments, including whole-leaf9 payments and separate-component coupons, retain their existing values.

This is a local interface theorem. It does not prove that the proposed finite rule pays every quadrilateral, give its finite-forest maximum, or improve the unrestricted dimension ceiling65. The source and certificate extend the finite8 cycle replacement. For each depth the additional calculation has36 ordered profile pairs and144 exact rectangle vertices. The strict minima are29/144 at depth17 and73/336 at depth20. This local statement is independent of whether a separate corridor theorem has been proved at either depth; it does not itself assert a depth17 high-overlap classification.

Exactly which old finite units can change

At an actual ideal cycle corner, write its varying affine factor as F and its orthogonal positive-definite spectator diagram as Z. Let the old marked arcs have lengths P,Q, with P<=Q. An adjacent black root c has one of the complete inherited profiles: isolated, endpoint3, endpoint4, second-node3, or both-end3. The classification and actual factor-fiber argument allow arbitrarily many spectators and arbitrary native labels; they are not a bounded list of full vertex diagrams.

For a single attachment at position j=1 or2 on an arc of length L, the adjacent finite marked distance is L-j+1. The endpoint of the old path opposite the attachment, and its degree-two neighbor, stay fixed because Z is orthogonal to F. This whole-component simple leaf excludes an affine completion in the relevant ranks. The adjacent corner is finite even after restoring all spectators.

An endpoint4 profile has a doubled bond at c. A second-node3 profile has an existing fork one step from c. Neither gives a changed deep pair, regardless of any additional contacts. Their stronger inverse-diagonal sealing, when used by the predecessor to pay a whole-leaf9 pair, is also unchanged. Thus only endpoint3 with L=8 can lose a unit.

In this endpoint3 case the fixed opposite old endpoint is a leaf of the whole actual component. Its distance from the other marked root is exactly **complementary_arc-1**. Spectators cannot remove the leaf or shorten this distance in a finite Coxeter tree. Therefore:

Cycle arcs	Endpoint3 attachment	Why the distance8 pair is unchanged
(6,8)	8-arc	Old mark is5 from a whole-component leaf
(7,8)	8-arc	Old mark is6 from a whole-component leaf
(8,Q), 8<=Q<=t	8-arc	Old mark is Q-1<t from that leaf
(8,Q), Q>=t+1	8-arc	The sole newly residual family

These are all possibilities in the current cycle replacements, whose marked distances are6 through9. The source records the15 finite endpoint cases at depth20 and12 at depth17 in the first three rows. At(8,8), reflection treats either8-arc. Cycle12 and cycle13 single-profile payments have distances at most6 and7; the cycle14 distance8 profile is the first row above.

Consequently a full-star path of length6 or7 at an ideal cycle corner still has its old replacement even when that ideal pair itself has weight zero. The small-cycle equality outputs are sealed small whole components or unchanged separate coupons. In the classical-tree branch, the immediate finite outputs have distances at most7; the ideal outputs have a persistent feature one step from a mark. Its sole additional P=6 equality is the sealed whole A7 pair. These outputs cannot be changed deep finite-distance8 pairs. This uses the complete ideal-five local transfer and its cycle12/13/14 successors, rather than treating every short ideal pair as a literal unit.

The new endpoint requires an actual spectator neighbor

It remains to treat P=8,Q>=t+1, where t is17 or20. The corner produced by endpoint3 on the8-arc starts with a finite A_(P+Q) path whose new endpoint is c. If c has no spectator neighbor, it is a leaf of the whole component and is not deep. A changed pair therefore forces at least one actual spectator neighbor of c. This is an assertion about the whole actual component, not its displayed A path. Indirect extension cannot begin without such a direct neighbor.

Normalize every root to squared norm2. Write h,k>0 for the native heights of the two opposite black roots c,d against the positive kernel of F, and x,y>=0 for their contact-magnitude vectors with Z. Every nonzero native magnitude is at least1. A shared nonzero coordinate gives an actual c-white-d path of length2, which retains a full payment whether that corner is finite or ideal. We can therefore suppose the supports are disjoint.

Any direct adjacent-black contact already supplies an unchanged distance1 payment. In the unpaid branch, the adjacent-black products that vanish in the inherited determinant formula therefore still vanish.

If c is the newly residual endpoint3 profile, choose one spectator neighbor i. Then x_i>=1,y_i=0. With G the positive-definite norm-two Gram matrix of Z and z=x/h-y/k, the Gram-metric Cauchy inequality gives

```
z^T G^(-1) z >= (z^T e_i)^2/(e_i^T G e_i)
                 = x_i^2/(2h^2) >= 1/(2h^2).
```

The corresponding bound for d is1/(2k^2). This does not require h=k, does not restrict other spectator contacts, and does not use a rank or label cutoff. If both profiles require neighbors, use the c-side bound throughout the entire rectangle. The two coordinate bounds are not added, and the chosen side does not change between rectangle vertices.

Complete constant-profile tail and exact strict gap

Put A=1/8, B=1/Q, u=1/h, v=1/k, and use the inherited nonnegative profile moments l1,l2,m1,m2 and norm constants c0,d0. Let L=u*l1*+*v*m1 and M=u*l2*+*v*m2. The inherited normalized determinant before spectator correction is

```
F=1-2(A*L+B*M)-(A+B)(c0*u^2+d0*v^2)-A*B*(L-M)^2.
```

Eliminating Z adds (A+B) z^T G^(-1) z. The required-neighbor bound therefore replaces c0=1 by1/2, or d0=1 by1/2 for the fixed d-side choice. All modified norm constants remain nonnegative. The source checks the corrected formula against the raw formula plus this explicit correction at every saved vertex.

For every Q>=t+1 the complete residual profile list is the same:

```
isolated, end3_0, end3_1, end4_1, next3_1, both_ends3.
```

The index0 denotes the8-arc. The old end4_0 and next3_0 profiles still pay through their nondeep distance8 and distance7 corners. The new end3_0 profile requires the spectator neighbor. On the Q-arc, the three single profiles have distances Q,Q,Q-1, so they remain in the old unrestricted list for all Q>=18. Isolated and both-end3 remain unrestricted. Their inherited minimum heights are respectively 1,1,1,sqrt(2),1,2, independent of Q. Thus this is a genuine constant-list tail, not an extrapolation from a changing finite sample.

With corrected constants nonnegative, F is concave in(u,v) on the closed reciprocal-height rectangle. Its minimum occurs at a vertex. It is also nonincreasing in B because the derivative is

```
-2*M-(c0*u^2+d0*v^2)-A*(L-M)^2 <= 0.
```

Since B<=1/(t+1), it suffices to take Q=t+1 and the four rectangle vertices for each of the36 ordered profile pairs. The source checks both tails:

Feature depth t	Tail Q>=	Ordered pairs	Exact vertices	Strict minimum
17	18	36	144	29/144
20	21	36	144	73/336

All288 values are strictly positive. Both minima are attained for end3_0/end3_0 with u=v=1. A strictly positive lower bound contradicts the exact zero determinant required by the actual quadrilateral star. No equality or additional spectator case remains in this tail.

All other cycle ranges and the classical-tree branch retain their old payments and proofs. Existing finite rank16/17 refinements preserve the short8 and whole-leaf9 interfaces used here. The large finite-D leaf9 half transfer also creates no new cycle dependency: a second-node3 distance9 output requires an arc of length10 and P<=9, hence a sealed whole D component of order at most19. Its two changed large-D leaf orbits start at order20. Existing separate coupons are unchanged.

Verification scope

Final two-depth generation672f44 and separate fresh replay0277d7 both terminated exit0/PASS on all27 protected endpoint cases,72 ordered pairs and288 exact rectangle vertices. The earlier depth20-only run is superseded. The producer deliberately shares the frozen exact arithmetic, profile and determinant implementations.

Two other agents completed full source/report mathematical reads, including the inherited cycle profile/classification and fixed-side Cauchy interfaces. Their final parameter-delta reads passed both depths, whole-component leaf/neighbor reasoning, complete constant-tail domains, concavity, reciprocal-Q monotonicity and unchanged tree/small-cycle/leaf9 outputs. They did not re-enumerate the inherited all-label geometry. One also checked the two minimum values algebraically at u=v=1, without repeating the288-value producer computation.

A separate saved scalar checker terminated exit0/PASS, chunk576e4b. It imports no project module and uses explicit normalized endpoint moments with rational Fraction arithmetic; the sqrt(2) endpoint4 contact cancels its reciprocal height. It reconstructs all288 raw values, corrections and lower bounds, all72 pair minima and corrected constants, the27 endpoint cases and both complete scalar domains. It independently obtains the minimum5/16-2/Q at Q=18 and21. This is an independent scalar reconstruction, not an independent proof of the geometric profile classification or the full payment system. The complete helper has also received a separate source/mathematical read.

All direct input bindings and report links, and both Python syntax checks, pass. No complete changed payment system, finite budget, external refereeing or formal verification is claimed. Source, data and report are final/frozen after the recorded reviews.

```
python3 scripts/fresh_deep_finite8_cycle_interface_20261017.py
python3 scripts/fresh_deep_finite8_cycle_interface_20261017.py --verify
```

Source 56

Source path: `results/fresh_deep_finite8_dimension64_20261018.md`

SHA256: 75a3177e5cbfff3b8e2967306bbea26b5419280865920cedbf77ac26acd62267

Unrestricted finite-volume dimension at most64

Every finite-sided, finite-volume real hyperbolic Coxeter polytope has dimension at most64. The new argument excludes dimension65; the completed ceiling65 excludes every higher dimension. There is no arithmeticity, compactness, simple-cusp or cusp-factor hypothesis, and no facet-count restriction beyond finite sidedness.

The source and certificate bind the local and counting theorems and check the current small-factor interfaces. The written local-to-global composition is given here. Generation, fresh replay and two separate full internal composition reviews have passed. This is a complete internally checked computer-assisted proof; the exact verification scopes are recorded below.

The single change to the current rule

Retain the complete current face-payment rule, including all ideal weights and all finite cross-component coupons. Its latest finite refinement is the rank16/17 transfer, and its ideal envelope is the completed dimension65 cusp-margin theorem.

Make exactly one further change. A unit at marked distance8 in a whole finite A/B/D component becomes a half if both marks have distance at least17 from every endpoint, branch and endpoint of a nonsimple bond of that whole component. All other weights retain their values. This condition is intrinsic to the labelled whole component; it makes no choice of an embedding or core. Only components of rank at least43 can be affected. In particular every finite short7 unit, every whole small-component payment and every existing cross-component coupon is retained.

Triangles and quadrilaterals

The finite7 triangle theorem requires only finite distances at most7, ideal distances at most5 or feature-protected distances6/7, and whole ideal factors of order at most9. All these payments remain literal units in the current rule and under the new change. Thus every triangle still receives at least2.

For a quadrilateral, divide into two cases according to its actual corners. If none has a changed finite pair, every corner weight is exactly its old value. The completed old local theorem therefore supplies total weight at least1. This inherits the actual numerical weights, without requiring that each intermediate branch of its original proof be re-run under new rules.

If at least one actual corner has a changed pair, apply the universal deep-corridor theorem at that corner. It supplies a path of length at most7 joining the two opposite black pairs, with only supporting white roots inside the path. Such a path joins adjacent sides of the actual quadrilateral. Consequently it is a marked path in another actual corner, with all its supporting roots present. The statement includes low and high minimal cores, disjoint and overlapping cores, all native labels and arbitrary spectators. In particular it does not require the old no-short8 assumption.

If this corner is finite, its marked distance is at most7 and its unchanged unit pays. If it is ideal, the actual factor-fiber theorem puts the two marks and the path in the same varying affine factor. The following complete division treats the possible ideal factors.

- In a factor of order at most11, every pair is still a unit. This includes every exceptional affine factor.
- For classical tree factors of orders12 through17, every pair at distance at most7 is a literal unit in the current rule. The binder reconstructs all18 labelled target graphs and their exact current pair weights, checks their total costs against the current envelope, and checks this inclusion.

- For a larger classical tree factor, distance at most5 or distance6/7 within5 of a feature is a retained elementary ideal unit. In the remaining feature-free case the completed radius-five tree transfer supplies a finite path of length at most7, an ideal path with a persistent feature one step from a mark, or the sealed whole finite A7 equality. Each output is unchanged. None is a deep finite-distance8 pair.
- For a cycle, distances at most5 are units. Missing short7 pairs have distance6 or7. The completed cycle12/13/14 transfers and the universal larger-cycle transfer cover these pairs. Their compatibility with the present finite halves is precisely the deep finite8 cycle interface. The small-cycle distance8 outputs retain a whole-component leaf at distance5 or6 from a mark and cannot be deep. Sealed small-component and separate-coupon outputs also retain their values. In the sole newly residual universal family, an8-edge arc and complementary arc at least18, a forced actual spectator neighbor gives the strict determinant gap29/144. Thus that family cannot lose the required total payment.

The binder checks all six cycle targets of orders12 through17 as well as the18 tree targets. For the order13 C factor it uses the73-pair set explicitly adopted in the completed dimension71 proof, whose five omitted pairs have distance at least10. The earlier support diagnostic alone is not treated as a local theorem. All other current small-factor sets are taken from their completed local successors. Their exact costs are:

Factor order	A	B	C	D
12	60	64	63	62
13	65	73	73	74
14	70	84	84	82
15	75	94	91	94
16	80	107	103	106
17	85	111	108	119

These cases exhaust the affine classification. Each either supplies an unchanged full unit or a complete retained transfer totaling at least1. The original changed half need not be added to this payment; no unit is counted twice. Thus every quadrilateral still receives at least1 under the simultaneous current ideal rule and new finite rule. There is no obligation to repeat the later selected-map censuses: faces with unchanged actual weights inherit their old coverage, and every face with a changed actual weight has just been handled by the universal short7 alternative.

The dimension65 contradiction

The exact conditional finite budget now applies, because its local hypothesis has been proved above. Its maximum over all rank65 finite Coxeter forests is479, attained by D17+3D16. If a forest has a component of rank at least18, that component participates in no current cross-coupon. Strip it off and use the old unrestricted maximum for the remainder; all48 resulting bounds are at most953/2. The all-small branch attains479. This argument covers arbitrarily many components and does not assume the remainder has only small components.

All ideal weights are unchanged, so the complete dimension65 cusp-margin theorem applies without a new optimization or partition census. It includes arbitrary products of Euclidean simplex factors in a nonsimple cusp. For each actual cusp let C_v be its current corner-payment cost and kappa_v its shape-curvature contribution. Its exact all-profile consequence is

```
kappa_v − C_v >=445/23.
```

Let v_f and v_i count finite and ideal vertices, and let D5 be the sum of 5 minus the number of sides over all two-faces. Nonnegative payments and the proved triangle/quadrilateral coverage give

```
D5 <=2T+Q <=479*v_f + sum_(ideal v) C_v.
```

The inherited strengthened shape inequality at dimension65 gives

```
D5 >=480*v_f + sum_(ideal v) kappa_v +5280.
```

Combining them yields the impossible inequality

```
0 >= v_f +(445/23)*v_i +5280.
```

Hence dimension65 is impossible. The completed ceiling65 already excludes all dimensions at least66, proving the ceiling64. In particular the result is not a dimension65-only obstruction with an unproved higher-dimensional tail. The separate compact arithmetic ceiling21 and arbitrary compact ceiling29 are unchanged. No new finite-volume example is claimed.

Verification scope

The universal corridor and cycle-interface packets have completed exact generation, fresh replay and separate internal mathematical reads. Their precise scopes, classification inputs and independent-checker limitations are recorded in their reports. The new finite budget has likewise completed generation, replay and a separate full mathematical read. The1,650 cusp comparisons and their complete reduction domain are inherited from the frozen ideal-margin theorem and its independent reconstruction. The existing ceiling65 and its independent reconstruction supply the higher tail.

The new binder checks direct source bindings, all24 current small classical factor interfaces, the local scalar interfaces and the final strict inequality. It imports no project arithmetic module. It does not re-enumerate the historical geometric censuses or re-run their finite-budget optimization. The present report supplies the mathematical case split connecting these checked inputs. Final generation eda92d and separate fresh replay102fa2 both terminated exit0/PASS on all24 interfaces. Metadata check5b609a passed all88 direct bindings, Python syntax and all11 local report links; the conditional budget's10 bindings, syntax and three local links also passed.

Two other agents completed separate full source/report composition reads. They checked the actual-corner split, full-star short-path interpretation, all small-factor provenance including the completed C13 adoption, unchanged triangle inputs, larger-tree outputs, complete small/universal-cycle interfaces and whole-component coupon protection. They also read the conditional finite budget, checked its applicability to the simultaneous rule, and checked the unchanged absolute ideal surplus, extremum constant, strict contradiction and higher-dimensional tail. Neither found an unresolved step. Neither repeated the24-row arithmetic, historical geometric censuses, Bellman calculation or1,650 ideal scalar comparisons. These are mathematical/source reviews with the inherited computational checks explicitly retained, not independent implementations of the whole proof. Source, certificate and report are final/frozen.

All reviews are internal. External mathematical refereeing and proof-assistant formalization are absent.

```
python3 scripts/fresh_deep_finite8_dimension64_20261018.py
python3 scripts/fresh_deep_finite8_dimension64_20261018.py --verify
```

Source 57

Source path: `results/fresh_depth10_finite7_zero_transfer_20261018.md`

SHA256: 51e5c1686d51a07afbca2387d2bc9f05bb21c893d50cdc8b00105a56c275916c

Feature depth10 suffices for the finite7 zero transfer

The depth11 theorem can be strengthened to feature depth10. In the completed dimension60 rule, delete finite distance7 payments in whole A/B/D components when both marks are at least ten steps from every endpoint, branch and endpoint of a nonsimple bond. Every triangle still receives at least2 and every quadrilateral at least1. All other payments remain unchanged.

The source and certificate check the new equality argument below. Final generation, fresh replay and two full internal mathematical source/report reviews pass.

The shorter retained path

Use the predecessor's actual quadrilateral star W+a+b+c+d, with opposite pairs a,c and b,d. Retain eight outward whites on each side of the distance7 pair a,b. They form A24, with a at8 and b at15. The full outward arms have ten whites. Their A10 inverse diagonals at distances 3 through8 are

```
24/11, 28/11, 30/11, 30/11, 28/11, 24/11.
```

All exceed2. Exactly the predecessor's strict actual-edge and no-short6 arguments restrict c's retained contacts to6,7,8,9 and d's to14,15,16,17. Both vectors are nonzero: a minimal indefinite core has order at most10, so its component through a after deleting c has radius at most8 and fits in the retained path. Feature depth10 keeps that component simple. Disconnected deletion is still allowed.

Put J=J24 and alpha=1/25, with factors u_i=i+1 on the left and u_i=24-i on the right. Every residual entry

```
C[i,j]=J[i,j]-u_i*u_j/25
```

is strictly positive. Its left diagonal is

```
77/25, 72/25, 63/25, 2,
```

and its right diagonal is the reflection. Thus, for nonzero native contact vectors x,y, with U=sum x_i*u_i and V=sum y_i*u_i,

```
q-2 >= U^2/25 >0,
r-2 >= V^2/25 >0,
s=U*V/25.
```

The Schur determinant is at least zero. A strictly positive determinant again gives two negative directions and is impossible. Unlike the depth11 argument, equality must now be resolved explicitly.

The unique equality contradicts a genuine supporting white

Equality requires x^T C x=y^T C y=2. Every nonzero native coordinate is at least1; all residual entries are positive, and only one diagonal on each side equals2. Hence

```
x=e9, y=e14,
q=r=6, s=4.
```

The Schur block is [[-4,-4],[-4,-4]]. The full principal Gram on A24+c+d has inertia(24,1,1), and its radical is

`k=(J(e9−e14), 1, −1).`

In particular k's old endpoint0 coefficient is1/5. The source verifies every entry of the full Gram-kernel identity, not just its two-dimensional Schur counterpart.

This radical is an actual linear relation among the roots. Indeed the principal span contains a timelike vector. Any vector in its radical is orthogonal to that timelike vector and has norm zero; the timelike orthogonal complement in the ambient Lorentz space is positive definite. The radical vector therefore maps to zero. It pairs to zero with every other actual root, including roots outside this principal diagram.

Let z be the next outward supporting white on the left, at distance9 from a. It exists because the feature depth is at least10. Inside the retained old path it has precisely the simple contact with endpoint0. Let x_z=-c.z and y_z=-d.z. Pairing z with k forces

`−1/5 − x_z + y_z = 0.`

Both x_z and y_z must be0 or1. To see this, use the full ten-white arm in either actual edge. Its inverse diagonal at z is18/11. Any nonzero nonsimple native contact has squared magnitude at least2, giving projection norm at least36/11>2, contrary to strict edge positivity. Additional arm contacts only increase this lower bound. The only remaining finite native labels are2 and3, with magnitudes0 and1.

The required difference y_z-x_z=1/5 is therefore impossible. All four allowed contact pairs are checked exactly. This excludes the unique zero-determinant case without limiting spectator rank, other contacts or the ambient Coxeter labels.

The full-star short6 alternative consequently holds at feature depth10.

Retained payments and finite counting consequence

The predecessor's actual-corner case split applies unchanged: a newly zero finite7 corner forces a short6 actual corner; its current ideal cycle/tree alternatives supply a full payment. If no newly zero corner occurs, the completed dimension60 rule applies pointwise.

All finite7 outputs used in that short6 argument have a fixed leaf five steps from a mark or a branch one step from it. The existing finite8 outputs have their separately established depth8 protection. Small sealed components and separate-component coupons are unchanged. Thus reducing the new feature threshold from11 to10 loses no such alternative.

For triangles, all14 elementary distance7 payments have a persistent feature within five steps of a mark. The sole exceptional distance7 alternate has a marked whole-component leaf. Their protections remain strictly below10. The source checks these interfaces in the completed predecessor certificate; it does not repeat the1,025-triple enumeration.

The deleted pair counts in A_n,B_n,D_n are respectively max(0,n-27), max(0,n-28), max(0,n-29). This follows by placing the first mark at least ten steps from the left endpoint and the second at least ten from the right endpoint, doubled-bond endpoint, or fork center. Both marks lie on the undecorated spine, seven steps apart.

This still leaves the complete ambient60 finite maximum at413: the current frontier gives upper413, deleting nonnegative payments cannot increase it, and the actual formal forest2D17+D26 loses no pair and still attains413. This conclusion needs no new DP. It is an obstruction of the payment rule, not a claimed geometric realization. Curvature remains405; a dimension59 theorem still needs further finite and ideal improvements.

Verification scope

Generation `2c69dd` and separate fresh replay `039503` both terminated exit0/PASS. Two peer agents completed full source/report mathematical delta reads after their earlier conceptual reviews. Both checked the unique equality, full radical and its actual Lorentz interpretation, the genuine ninth white, complete zero/simple contact alternatives, disconnected minimal-core interface, retained short6/triangle outputs, all-rank pair counts and unchanged413 counting conclusion. Neither repeated matrix, scalar, triangle-census, DP or classification arithmetic. These are internal agent reviews, not external mathematician refereeing. The producer checks A10/A24 inverses, strict arm bounds, every residual block entry, unique equality positions, exact Schur data, the full principal radical and all allowed contacts with the ninth white. It shares

the predecessor's simple rational path-inverse routine. Minimal-core, actual-face and current short6/triangle arguments remain explicit inherited theorem inputs. There is no new geometric census, finite optimization, ideal convolution, external refereeing or formal proof-assistant validation. No new finite-volume example is established.

```
python3 scripts/fresh_depth10_finite7_zero_transfer_20261018.py
python3 scripts/fresh_depth10_finite7_zero_transfer_20261018.py --verify
```

Source 58

Source path: `results/fresh_depth11_finite7_zero_transfer_20261018.md`

SHA256: f97e719172fefd235824950e3aaa6a63a39ca7aca257afe77b5349fc1cd9e66d

Deep finite distance-seven payments can be deleted

Start with the complete simultaneous rule of the dimension60 proof. Set to zero precisely those finite distance7 pairs in whole A/B/D components for which both marks have graph distance at least11 from every endpoint, branch and endpoint of a nonsimple bond. Every triangle still receives at least2 and every quadrilateral at least1. All other finite weights, all ideal weights and all separate-component coupons remain as in the completed rule, including the existing deep finite8 zeros.

The source and certificate check the matrix and triangle interfaces below. Generation, fresh replay and two full internal mathematical reviews pass. This is a local theorem; it supplies no new dimension bound by itself.

A retained A26 path forces a shorter actual corner

Use the inherited actual quadrilateral star W+a+b+c+d, with opposite pairs a,c and b,d. All roots have squared norm2. W and W plus any one black root are strictly positive definite; W+a+c and W+b+d are indefinite. Every nonzero native contact magnitude is at least1. The full actual Gram has at most one negative direction. These are the arbitrary-support inputs of the universal corridor theorem.

Suppose a,b form a newly zero finite pair. Their distance is7, and feature depth11 supplies two outward simple arms of eleven whites. Keep nine on each side. The retained principal diagram is A26, with a at index9 and b at index16. A feature exactly eleven steps from a mark is allowed: the eleven-white path up to it still has simple bonds. The entire retained path lies inside the actual finite component.

Assume there is no white-interior path of length at most6 between an adjacent black pair. Then c.d=c.b=d.a=0. The opposite products c.a and d.b remain unrestricted. Write

```
J_n[i,j]=(min(i,j)+1)*(n-max(i,j))/(n+1)
```

for the inverse norm-two A_n Gram. In an eleven-white outward arm, the diagonal entries at outward distances3 through9 are

```
9/4, 8/3, 35/12, 3, 35/12, 8/3, 9/4.
```

Each exceeds2. A nonzero c- or d-contact at one of these positions would already project with squared norm greater than2 onto that white arm, contradicting the positive-definite actual edge. Additional contacts cannot reduce the bound because the path inverse is positive.

The no-short-path assumption excludes every remaining retained c-contact except positions7,8,9,10: an inward contact at least two steps from a gives a path to b of length at most6; contacts within two outward steps of b do so as well. The reflected d-support is 15,16,17,18. Contacts outside the retained path remain unrestricted.

Both retained contact vectors are nonzero. Indeed choose a minimal indefinite principal K inside W+a+c. It contains a,c, is connected, and has at most10 roots by the inherited all-label minimal-core theorem, including its disconnected-semidefinite-deletion bridge. The component B through a in K minus c has at most9 roots and lies in the original whole finite tree. Feature depth11 makes B a simple path, entirely within eight steps of a; it cannot pass through the absent b. Hence B lies in the retained A26. Connectedness forces c to contact B. This argument does not assume K minus c is connected. Reflection gives the same conclusion for d.

A uniform negative-definite Schur block

Let x,y be the nonnegative retained contact vectors, J=J26, and put q=x^T J x, r=y^T J y, s=x^T J y. On the left support set u_i=i+1; on the right set u_i=26-i. Define

```
alpha=122/3267,
U=sum_left x_i*u_i, V=sum_right y_i*u_i.
```

On either four-position block the matrix

```
C[i,j]=J[i,j]-alpha*u_i*u_j
```

is entrywise nonnegative and every diagonal is at least2. Each contact vector has a native coordinate at least1. Thus x^T C x>=2 and y^T C y>=2. The cross block factors exactly, giving

```
q-2 >= alpha*U^2 >0,
r-2 >= alpha*V^2 >0,
s = U*V/27.
```

Since c.d=0, elimination of A26 leaves the Schur block

```
[2-q, -s; -s, 2-r].
```

Its diagonals are negative and its determinant satisfies

```
(q-2)*(r-2)-s^2
  >= (alpha^2-1/27^2)*U^2*V^2
   = U^2*V^2/43923
  >= 4096/43923 >0,
```

because U,V>=8. It is negative definite, producing two negative directions in an actual Lorentz Gram matrix. This contradiction proves the full-star short6 alternative. There is no spectator-rank, contact label, contact-magnitude or supporting-extension cutoff.

The source checks the A11/A26 inverses and positive leading determinants, all seven arm bounds,32 residual entries, eight diagonal bounds and16 cross factorizations. These finite checks verify formulas used for every allowed nonnegative contact vector; they do not enumerate that infinite domain.

Short6 ideal corners still supply a complete payment

At a finite corner, the resulting path of length at most6 retains a unit. At an ideal corner, factor-fiber geometry places its marks in one varying affine factor. The current rule retains every distance-at-most5 unit except the distance5 halves on the11-cycle. Handle that exception first using its independent theorem: its alternative finite paths have length at most6, or a distinct retained half combines with the original half. No distance7 finite payment is used.

Small exceptional affine factors retain their whole-factor units. Every current classical-tree change has distance at least8, so the small-tree short6 units and radius-five protected-feature units are unchanged. For a remaining unpaid distance6 corner, the only universal cases are a cycle of order at least12 or a feature-free classical tree.

For cycles, the inherited profile classification gives adjacent finite single-contact corners. A single endpoint3 attachment on an arc of length L has marked distance L, and the opposite old endpoint remains a whole-component leaf at distance other_arc minus1 from the other mark. An endpoint4 attachment has a nonsimple bond at its new mark; a second-node3 attachment has a branch one step from it. The latter two can never produce a newly deep pair.

The cycle12 proof has only finite single-contact paths of length at most6. The cycle13 proof can give distance7 on the longer arc, but its old mark is five steps from the fixed leaf. The cycle14 proof at arcs6/8 can give distance7 only through a second-node attachment, which has the nearby branch. Its distance8 endpoint output is protected by a fixed leaf five steps from the old mark, as already required by the current deep8 rule. Thus every one of these payments is retained.

For cycles of order at least15 with shorter arc6, the other arc is at least9. A single-contact distance7 output cannot occur: the short-arc outputs have distances5 or6, while the long-arc outputs have distance at least8. The possible distance8 output is a second-node profile on the9-arc and has a fixed branch. The inherited P=6 local proof also uses retained leaf9 outputs, small

sealed components, and whole A1/A17 coupons. These all keep their current weights. Its height and spectator arguments are therefore unchanged; no new cycle tail is assumed from sampled ranks.

In the same report's radius-five classical-tree proof at P=6, an inner second-node3 profile supplies a path of length5. Every other non-equality case has a strict determinant gap. The unique equality seals a whole A7 component with marked distance6. These outputs are also unchanged.

Now consider any actual quadrilateral. If a newly zero finite7 corner occurs, the corridor and the short6 alternatives just established supply a complete payment under the new rule. If no such corner occurs, every actual corner weight agrees pointwise with the completed dimension60 rule. Its existing coverage applies. This actual-corner case split avoids recursively asking a deleted finite7 pair to pay.

Triangle coverage

The source freshly reconstructs all1,025 high-core marked triples of the finite7 triangle theorem and checks exact agreement with its saved record. Every one of its14 elementary distance7 payments has a persistent branch or nonsimple-bond feature within five steps of a marked endpoint. Restoring supporting roots cannot erase that feature. At a finite corner such a pair cannot be newly deep; if the actual distance shortens, it is also retained.

The four exceptional alternate lengths are6,6,6,7. In the last case, marked triple(1,2,9), the alternate pair(1,9) has mark1 protected as a leaf of the whole actual component by the inherited affine E8 kernel argument. Its depth is zero, so its finite unit remains. The ideal exceptional-factor payment is unchanged. Low cores already give two paths of length at most5 by the unchanged all-label Steiner argument. Thus every triangle retains two distinct units.

Verification and remaining scope

Generation `2d1eb8` and separate fresh saved replay `a2aecd` both terminated exit0/PASS. Two peer agents completed full mathematical producer/report reads. They also reread the arbitrary-support minimal-core bridge, including disconnected deletion, the current rule and relevant short6 cycle/tree proofs, and the inherited triangle/E8 interfaces. Both passed the support restrictions, nonzero-contact argument, shifted Schur bound, noncircular actual-corner case split and retained triangle payments. Neither repeated matrix, marked-triple, scalar, DP or classification arithmetic. These are internal agent reviews. The producer uses its own rational path formulas and reuses the inherited triangle graph/kernel routines. The minimal-core order bound, actual-corner/factor-fiber geometry, complete affine extension classification, and old short6 cycle/tree arguments remain explicit written theorem inputs. No historical core classification, finite DP, ideal convolution, external refereeing or proof-assistant check is repeated.

This changes the generic finite rule in sufficiently large components. It does not remove the ambient60 obstruction2D17+D26, whose ranks are too small for a depth11 distance7 pair. The completed dimension bound therefore remains60 pending further local improvements and counting. No new finite-volume polytope is constructed.

```
python3 scripts/fresh_depth11_finite7_zero_transfer_20261018.py
python3 scripts/fresh_depth11_finite7_zero_transfer_20261018.py --verify
```

Source 59

Source path: `results/fresh_depth11_finite8_zero_transfer_20261018.md`

SHA256: 227854073503670aa463389504902c51e63907c33f6c25b8f93a259fb17b051d

Deep finite distance-eight payments can be deleted at feature depth11

Retain the current complete finite and ideal payment rule, except assign weight zero to a marked distance8 pair in a whole finite A/B/D component when both marks are at distance at least11 from every endpoint, branch and endpoint of a nonsimple bond. Every other weight and coupon remains. Then **every triangle receives at least2 and every quadrilateral at least1**.

This improves the completed depth17 half rule in two ways: the required feature distance falls to11, and the affected pair receives zero rather than one half. The source and certificate check the new exact matrix and cycle inequalities. A separate finite budget is needed for subsequent counting improvements. No bound below64 is claimed here. Generation, fresh replay and a separate full source/report mathematical review have passed. The verification scope is recorded below.

A shorter forced corridor

Use the actual quadrilateral star W+a+b+c+d, with opposite pairs {a,c} and {b,d}, and roots of squared norm2. The white diagram and W plus any one black are positive definite, while W+a+c and W+b+d are indefinite. These are the same actual-face inputs as in the completed universal depth17 theorem.

Suppose W+a+b is a finite corner with the stated distance8/depth11 pair. Its whole finite tree contains two simple outward arms of11 whites. Keep nine whites on either side of the eight-edge a-to-b path. These27 roots form an induced A27, with a at position9 and b at17, indexed from zero. The other25 roots are supporting whites. The two extra whites on each outer arm remain available for the actual-edge estimate.

Assume that the full star has no path of length at most7 joining the two opposite black pairs with only whites inside. In particular c.d, c.b and d.a vanish. The own opposite contacts c.a and d.b are unrestricted. For an A_n path its norm-two inverse is

$$J_n[i,j]=(\min(i,j)+1)*(n-\max(i,j))/(n+1).$$

Any retained outer contact at distance j=3,…,9 from its old mark gives, in an actual edge, projection square at least

$$J_{11}[j-1,j-1]=j*(12-j)/12 >=9/4>2.$$

Every nonzero native contact has magnitude at least1, and all inverse entries are positive, so further contacts cannot lower this estimate. It contradicts strict edge positivity. Contacts farther along the middle path, or within two steps beyond the opposite old mark, supply a path of length at most7 and are excluded by the current assumption. Thus c and d can contact the retained A27 only at

$$c: 7,8,9,10; \qquad d: 16,17,18,19.$$

Both contact vectors are nonzero. Indeed choose a minimal indefinite principal K inside W+a+c. It contains both a,c. The published all-label order bound is10, with the exact Chein/Vinberg classification bridge recorded in the depth17 theorem. The connected component B through a in K minus c has at most9 roots and radius at most8 from a. Since B lies in the actual finite tree, feature depth11 makes it a simple subpath. It cannot pass through b, which is absent from K. Nine outward whites and the seven inner whites therefore contain B. Connectedness of K forces c to contact B. The reflected argument applies to d. No core-overlap or finite-label cutoff has entered.

Exact uniform Schur inequality

Let x,y be the two nonnegative contact-magnitude vectors, J=J_27, and q=x^T J x, r=y^T J y, s=x^T J y. Both vectors have a coordinate at least1, so positive inverse entries give

```
q,r >=40/7.
```

Put U=sum x_i*(i+1) on the left block and V=sum y_j*(27-j) on the right. The cross block has s=U*V/28. On either same-side block,

```
J[i,k] >=(17/308)*u_i*u_k,
```

where u_i=i+1 on the left and u_i=27-i on the right. Consequently

```
s^2 <=(121/289)*q*r.
```

Eliminating A27 from its Gram with c,d leaves

```
[2-q, -s; -s, 2-r].
```

Both diagonals are negative. Writing q=40/7+u and r=40/7+v gives the following lower bound for its determinant:

```
36/289 +(382/289)*(u+v) +(168/289)*u*v >0.
```

The Schur matrix is negative definite, contradicting the single negative direction available in the actual Lorentz space. Hence the full-star short7 alternative exists. The source checks the A11/A27 inverse identities, all positive leading determinants, seven edge gates,32 same-block bounds, 16 cross factorizations and the exact positive determinant coefficients. The estimate applies to arbitrary nonnegative native magnitudes and arbitrary supporting spectators. It is not inferred from a sampled contact census.

The cycle interface at the shorter depth

The completed cycle interface identifies every finite distance8 output that could change. Endpoint4 has a nonsimple bond at its new mark, and second-node3 has a nearby branch; neither is deep. An endpoint3 output has a fixed whole-component leaf at distance complementary_arc minus1 from the old mark. Thus cycles with arcs (6,8), (7,8), or (8,Q) for8<=Q<=11 have unchanged outputs. There are six such endpoint cases.

Only the universal family with arcs P=8,Q>=12 remains. A changed endpoint3 output forces an actual spectator neighbor of its new root. Otherwise that root is itself a whole-component leaf. The predecessor's fixed-side unequal-height Cauchy inequality then subtracts1/2 from that profile's norm constant in the determinant lower bound. If both profiles require neighbors, use one fixed side throughout; no two coordinate bounds are added.

For every Q>=12 the residual list consists of the same six profiles: isolated, endpoint3 on the8-arc, endpoint3 or endpoint4 or second-node3 on the Q-arc, and both-end3. The long-arc distances are Q,Q,Q-1, all at least11, so no change of classification occurs in this tail. The corrected norm constants and moments are nonnegative. The determinant lower bound is concave in reciprocal heights and nonincreasing in1/Q. It suffices to check Q=12 and the four height-rectangle vertices for every ordered profile pair.

All36 pairs and144 exact rational endpoint values are strictly positive. The minimum is

```
5/16-2/12=7/48.
```

It occurs only for the two endpoint3 profiles on the8-arc. This contradicts the zero determinant of an actual quadrilateral star in the unpaid branch. There is no newly surviving equality case. The source uses normalized rational moments; for the endpoint4 profile the square root cancels its reciprocal height exactly, as in the frozen independent scalar checker.

The argument never uses the weight of the changed distance8 pair. It supplies retained alternatives or excludes the remaining configuration. It therefore permits weight zero. The earlier half was not an indispensable term in a two-corner payment sum. All other cycle ranges retain their proofs, including whole-leaf9 and separate-component coupons. Small-cycle outputs have a nearby fixed leaf or sealed small components. The classical-tree transfer outputs finite paths of length at most7, persistent ideal features, or the sealed whole A7 equality. These are all unchanged.

Complete simultaneous local coverage

Use the actual-corner case split of the completed dimension64 composition. If a quadrilateral has no newly changed finite corner, all its actual weights equal the proved current values and its total is at least1. Otherwise the new corridor supplies an actual adjacent-corner path of length at most7. Finite short7 pairs remain units. In ideal corners the current small-tree short7 units, elementary ideal units and complete tree and cycle transfers remain valid by the preceding interface argument. This supplies total at least1 without adding any part of the deleted corner's old weight. No cyclic reuse of a changed finite payment occurs.

The existing24-factor inclusion certificate is unchanged because all ideal weights remain unchanged. The completed finite7 triangle theorem uses no finite distance8 unit, so every triangle still receives at least2. All current cross-component coupons and all remaining same-component weights keep their values. This proves the stated complete local rule.

Verification status

Primary generation35f9a0 and separate fresh replay29c166 terminated exit0/PASS. A separate full source/report mathematical review passed the forced corridor, complete contact supports, nonzero-vector argument, uniform Schur obstruction, constant-profile cycle tail and fixed-side Cauchy correction. It checked that the actual-corner split never consumes the deleted half and that all triangle, ideal and coupon interfaces remain. No unresolved step was identified. The reviewer did not repeat the144 endpoint values or any inherited geometric census. The source imports no project arithmetic module; its finite identities do not replace the stated inherited classification and actual-face proofs. The short7 small-factor inclusion check is inherited from the frozen dimension64 binder. No new finite-forest optimization, ideal-envelope improvement, dimension theorem, polytope, external refereeing or proof-assistant formalization is asserted. All direct source bindings, syntax and local report links pass. Source, certificate and report are final/frozen.

```
python3 scripts/fresh_depth11_finite8_zero_transfer_20261018.py
python3 scripts/fresh_depth11_finite8_zero_transfer_20261018.py --verify
```

Source 60

Source path: `results/fresh_depth17_rank65_conditional_budget_20261018.md`

SHA256: 2469e228710c5aebf6cd9a3120deab1cdfadadc6a90850e90ddb3c580caaf976

The depth17 half rule has exact finite maximum479 at rank65

Suppose the current finite payment rule is changed only as follows: a unit at marked distance8 in a whole finite A/B/D component becomes a half when both marks are at distance at least17 from every endpoint, branch and endpoint of a nonsimple bond. Every other payment and coupon stays fixed. Then the **exact maximum over all finite Coxeter forests of rank65 is479**, attained by D17+3D16. The finite curvature allowance is480.

This is a conditional counting theorem. It does not prove face coverage for the modified weights or assert a new ambient dimension bound. The source and certificate use the completed current finite budget.

Strip one large component

If every component has rank at most17, no weight changes. The frozen budget gives maximum479, with the stated attaining forest.

Otherwise select a component of rank r>=18. Finite Coxeter classification leaves only A_r, B_r or D_r. None participates in a current cross-component coupon: every designated large coupon component has rank at most17 and every small partner has rank at most6. The source checks the literal inherited coupon function against all finite types through rank65, including exceptional and uniform dihedral representatives.

For A_r, B_r and D_r, respectively, the numbers of changed pairs are

```
max(0,r-42), max(0,r-43), max(0,r-44).
```

These follow by counting starting positions of an8-edge segment at the specified distance from the path endpoints, doubled bond or fork. The source independently constructs all144 relevant finite graphs for 18<=r<=65 and counts every eligible pair by breadth-first distances. It checks the formulas and saves every counted pair.

The inherited doubled same-component bounds are16r-68 for A/B, 16r-64 for D18/D19, and16r-66 for D_r with r>=20. Halving one unit subtracts one from the doubled cost. The new D bound dominates the new A/B bounds at every relevant rank, as the source checks explicitly.

Let M(s) be the completed old unrestricted maximum at rank s, taken from the frozen certificate. The remaining modified forest costs no more than M(65-r), since the modification only decreases weights. Thus a forest with a component of rank at least18 has doubled cost at most

```
new_doubled_D_cost(r) + 2*M(65-r).
```

All48 values are computed exactly. Their maximum is953, uniquely at r=65. Therefore this branch costs at most953/2=476.5. The other branch has exact maximum479, and D17+3D16 still attains it. This proves the complete rank65 result without another knapsack optimization or a claim that the remaining forest contains only small components.

Scope and checks

The argument retains all existing coupons, including E6/D11 and the A1/A2–D14 role addition. It introduces none. The rank65 result would combine with the already positive dimension65 ideal margins only after a complete local payment theorem supplies its geometric hypothesis.

Generation52c6ab passed144 graph counts,48 large-rank bounds and28,512 zero-cross-coupon comparisons. Fresh saved replay104c66 terminated exit0/PASS. An initial graph-construction run detected a missing D-fork edge before any certificate was written; the final graph has both fork leaves and asserts connectivity. The old finite-budget certificates are bound inputs; this package does not rerun their Bellman tables or geometric payment proofs. A separate full source/report mathematical read passed the literal pair-count formulas, absence of large-component cross-coupons, domination and one-component stripping argument. That reviewer did not repeat the graph counts or the inherited Bellman calculation. No gap was identified within this conditional counting scope. Source, certificate and report are final/frozen.

```
python3 scripts/fresh_depth17_rank65_conditional_budget_20261018.py
python3 scripts/fresh_depth17_rank65_conditional_budget_20261018.py --verify
```

Source 61

Source path: `results/fresh_depth6_dimension59_20261018.md`

SHA256: 9bc6ca854bc7bceb8e8f5a50915f391c0cec345f901be09e251d09917069cdd8

Dimension at most59 from the depth-six transfer

Every finite-sided, finite-volume real hyperbolic Coxeter polytope has dimension at most59. The argument below composes completed local payment theorems and exact counting certificates. It imposes no arithmeticity, compactness, simple-cusp or numerical facet-count hypothesis.

The producer and composition certificate bind four completed inputs:

- the unrestricted dimension60 proof, which excludes every dimension at least61;
- the sparse D16 and fractional ideal17 rule, including all current finite and ideal payment interfaces;
- the universal depth-six local theorem, which preserves triangle payment2 and quadrilateral payment1;
- the complete conditional404 count, whose geometric hypothesis is discharged here by that local theorem.

Generation and separate fresh replay have passed, followed by three full internal mathematical composition reviews. This packet is final and frozen. No external refereeing or proof-assistant verification is asserted.

A single simultaneous payment rule

Start with the frozen sparse-D16 rule. Its ideal order17 total is229/2, its precise whole D5-long-end/D12-short-leaf coupon is7/8, and its unused rank2 through5/D16 broad coupons have been removed. Whole A1/D16 coupons and every required universal and later local output remain. All other finite exchanges, whole-component coupons and ideal refinements are as specified in that completed predecessor.

The sole additional change deletes a finite distance-seven unit whenever both marks have distance at least six from every endpoint, branch vertex and endpoint of a nonsimple bond in their whole classical component. The newly affected same-component pairs have weight one before deletion; pairs already zero under the depth-eight rule remain zero. There is no new cross-component cost and no ideal change.

The local proof and conditional count bind the identical source, data and report of the sparse predecessor. Their all-rank deletion formulas agree exactly. Thus the counting rule is the rule proved geometrically, rather than a separate hypothetical envelope.

Consider an actual quadrilateral, with all its supporting roots. If none of its actual corners has a newly deleted pair, its corner weights agree pointwise with the frozen rule. Its complete old payment theorem applies. This branch does not assume that an earlier preferred witness survives a change elsewhere.

If a changed finite corner occurs, apply the depth-six theorem directly to that corner. It gives either an adjacent-black white-interior path of length at most six, or a distinct paying corner with a retained whole finite A8/D9 or affine D9/E8 marked pair. The independent current short-six theorems supply total payment at least one in the former case. Their shorter paths, fixed leaves, features at distance at most five, whole small components and typed coupons all survive the new deletion. In particular, the cycle11 half-weight exception has its own independent replacement theorem; a short path is not automatically treated as a unit. The latter case pays a unit in a finite component of rank at most9 or an affine factor of order at most10. Its marked whole-component leaves prevent the new deep-pair deletion from applying.

For clarity, the geometric input covers arbitrary additional supporting whites. Its complete eighteen retained profiles and two classified E10 patches lead to766 unbounded Schur branches. The765 strict branches contradict Lorentz signature. The

remaining branch is not discarded: the arbitrary-contact endpoint projection has lower bound5/2 for a square-two root, so the two replacement marks must be whole-component leaves. The finite and affine classifications give exactly the retained small alternatives just listed. Disconnected minimal-core deletions and possible contacts of the additional patch whites are included in that proof. This composition imports the completed argument, not just its saved finite profile list.

Triangle payments remain at least two. The inherited elementary distance-seven payments have a persistent feature within five steps, and the exceptional alternate has a fixed whole-component marked leaf. The other triangle routes retain their old weights. Consequently all triangles and quadrilaterals satisfy the required lower payments under one simultaneous rule. Faces with at least five sides require no positive payment for the inherited five-minus-sides deficit inequality.

The finite upper bound is404

For a classical component of rank n, the total deleted distance-seven pair counts are max(0,n−19), max(0,n−20), max(0,n−21) for A, B and D. The preceding depth-eight counts were respectively max(0,n−23), max(0,n−24), max(0,n−25). The finite certificate subtracts precisely these differences and keeps every other actual and representative cost.

The complete typed decomposition has1,044 fixed vectors and1,042 interaction keys. Fixed components use the inherited representatives for ranks9 through16, together with separately fixed A17 components. Residual components have ranks1 through8 or17 through60, including D17. The original conditional report misstated the residual range as starting at18. Its frozen clarification records the correct domain. The source, generated tables and every certificate check always included residual D17. No bound or computation changes because of this prose correction.

The certificate checks simultaneous domination for all33,489 ordered actual type pairs and every same-component base. It proves all54,288 residual item identities, including interactions with the fixed forest. Residual–residual and fixed–fixed interactions vanish. New exact integer Bellman tables have1,502,564 last-item upper inequalities and62,520 attaining predecessors. These certificates cover every forest of total rank60. Actual and representative costs remain distinct: A15 costs101, B15 costs97 and D15 costs201/2, while the rank15 representative upper cost is101.

The maximum is404. The actual formal forest4A15 attains4×101=404; geometric realization of that forest is not required for the upper bound. All six preceding large-component maxima decrease from407.5 to403.5. The complete enumeration includes all other forests and all current coupons, rather than checking only those six cases.

The global producer checks the frozen certificate bindings, complete fixed-vector and key counts, saved maximum and actual witness. It does not repeat Bellman optimization; the preceding certificate already provides the complete induction and attainment proof.

Every ideal vertex has surplus at least54

Ideal weights are unchanged from the sparse predecessor. The sorted terminal rank59 domain allows arbitrary repetitions of ranks at least9 and at most one positive rank below9. It has exactly2,113 profiles. The producer independently reconstructs that domain and checks its bijection with the saved rows.

For each saved link h-polynomial, it verifies symmetry, positive integer coefficients, degree59, and

```
V = product(r+1) = sum_j h_j,
G = sum_j min(j,59−j) h_j,
kappa = (5G−59V)/2.
```

Let C be the actual assigned cusp cost and C_upper its saved all-family envelope, so C≤C_upper. The checked normalized margin is (kappa−C_upper)/V. Every terminal margin is positive; the minimum is31/6000 at(9,9,9,9,9,14). The envelope surplus kappa−C_upper has minimum54 at(59). The original convolutions and all-family factor envelopes remain explicit completed inputs; the global producer checks their saved polynomial and scalar identities without regenerating them.

The inherited low-nine concentration reduces the link vertex count V and does not increase the normalized margin. It takes every actual profile to this complete terminal domain. Since all terminal margins are positive, both inequalities can be multiplied: each original envelope surplus is at least the terminal surplus, hence at least54. The actual surplus kappa−C is at least this envelope surplus. This step uses no concentration of large factors and no product argument with a negative margin. The factor-fiber payment theorem already handles products of affine factors, so no simple-cusp assumption is introduced.

The dimension60 contradiction

Let v_f and v_i be the numbers of finite and ideal vertices of a proposed dimension60 polytope. For each ideal vertex i write its curvature as kappa_i and its assigned cost as C_i. The complete finite count and the local face payments give the inherited upper and lower bounds on the same global face-deficit quantity:

```
404 v_f + sum_i C_i
    >= 405 v_f + sum_i kappa_i +4500.
```

Here405 is the finite curvature coefficient:

```
(5 sum_{j=0}^{59} min(j,59−j) −59·60)/2
    = (5·870−3540)/2 =405.
```

The4500 is the inherited two-extrema correction, counted once. Its dimension arithmetic is5×sum_{j=30}^{59}(2j−59)=5×900=4500. The producer checks this against the sparse predecessor's explicit saved necessary ambient60 inequality; it is not an unbound new constant.

Since every kappa_i−C_i is at least54, rearrangement yields

```
0 >= v_f +54 v_i +4500 >0,
```

a contradiction. Thus dimension60 is impossible. The completed bound60 already excludes dimensions at least61, so the resulting bound is59. No inheritance of a face-closure property or construction of a lower dimensional polytope is used.

Verification and scope

Generation chunk14a314 and separate fresh replay chunk4910e1 each completed with exit0 and PASS_COMPLETE_DEPTH6_DIMENSION59_COMPOSITION. All theorem source/data/report bindings, including the residual-domain clarification, are checked. The local classification, arbitrary-support geometry, finite Bellman proof, ideal factor envelopes, concentration theorem and global shape-curvature inequality are explicit completed inputs. No historical geometric census, convolution, DP optimization or construction is rerun by this global producer.

The root, construction and arithmetic-lane reviewers each completed a full internal source/report mathematical composition read. Their scopes cover the common frozen baseline, actual-corner priority, protected alternatives, typed finite404 certificate, residual D17 clarification, unchanged ideal envelopes, positive-only concentration, the single4500 correction and the completed60 higher tail. The construction and arithmetic reviewers also confirmed the final wording distinguishing newly affected units from old zeros and actual cusp cost from its saved upper envelope. The arithmetic review additionally read the depth-six and depth-eight proofs, the classical catalogue report and E10 metadata, the endpoint certificate and inherited short-six/triangle interfaces. These reviews did not execute profile, polynomial, matrix, census, DP or convolution computations, and do not claim a fresh review of every historical dependency.

This is an internally checked computer-assisted mathematical proof. It is not a Lean proof or an external peer-review claim, and it produces no new finite-volume example.

```
python3 scripts/fresh_depth6_dimension59_20261018.py −−verify
```

Source 62

Source path: `results/fresh_depth6_finite6_order13_order17_ambient59_frontier_20261018.md`

SHA256: 0dab0a9fbbbc9cf9eed55e9006a444bd723cd626d719d60a8f41eb90122451ea

Deep finite6 with the order13 and contextual order17 transfers

The simultaneous proved rule has exact formal finite maximum **394** at ambient59, attained by **A14 + 3A15**. This is 5/2 above the finite curvature 783/2. Three fixed-multiplicity representative alternatives remain above that curvature. The complete 1,890 terminal ideal profiles have one negative row, `(9,9,9,15,16)`, with normalized margin `-69/5440`. The completed unrestricted dimension bound remains59; this packet does not prove dimension58.

The producer and exact certificate combine three frozen local theorems with the complete preceding finite and ideal rule. The finite residual costs change, so the certificate contains freshly generated tables. The separate verifier checks every last-item inequality and an attaining predecessor without running the generation recurrence. Future rank14 or order16 proposals are excluded.

Local inputs and simultaneous face payments

The baseline is the frozen order15/D16/D17 composition. It includes the D5 alternate-corner payments, the contextual A2/A2 rule, the A17 exchange, both finite D17 transfers and the D16 transfer. All their actual coupons, representative allowances and retained roles remain present.

The finite6 theorem sets a finite distance6 unit to zero when both marked roots have depth at least6 from every endpoint, branch and multiple-edge feature. In a whole classical factor its additional deleted-pair counts are

Whole type	Additional deleted units
A_n	max(0,n−18)
B_n	max(0,n−19)
D_n	max(0,n−20)

Other finite types have no change. These are additional deletions from the current distance7/8 and long-pair rule; they are subtracted once. The local packet supplies the all-rank feature-distance argument and its complete adaptive exceptions, rather than inferring a universal rule from this finite count.

Its new payment is a unit between a whole A1 and either middle root 5 or6 of a whole A12. The actual component-pair increment is2. The whole A1/D12 representative allowance is already5/2, so no upper cross cost increases. The theorem's escapes use the independent short5 and triangle routes or its explicit protected whole-factor alternatives. In particular its exceptional whole A13 distance6 unit is retained by the endpoint/leaf condition.

The order13 theorem gives ideal family totals A65, B143/2, C71 and D71. The all-family envelope is143/26, down from145/26. Its only new finite payment is a whole A1 against the third root from the doubled terminal in a whole B12, of weight1/2. This adds1/2 per component pair, also within the unchanged A1/D12 allowance. Its whole B13 alternative is a protected finite unit; the other outputs are the independent small-ideal/short routes and the explicitly retained whole B11/D11 payments.

The contextual order17 theorem deletes the two affine D16 center halves `(0,10)` and `(2,12)`. The order17 totals become A85, B111, C108 and D227/2, with envelope 227/34 in place of229/34. Its two complete whole-core seals provide two

distinct finite halves: the existing D5-long/D12-center half and a new whole A2-endpoint/D5-long-endpoint half, conditional on whole unmarked A1 and A9 components at that same finite corner. The new component-pair cost is1 when the witnesses exist. Multiple witnesses do not multiply that cost.

The arbitrary-support and complete selected/nonselected routing proofs remain inputs from those bound local reports. The following actual-face priority makes their simultaneous use noncircular. If a face contains any newly changed finite6 corner, apply the universal finite6 theorem first. Its short5/triangle/protected outputs and A1/A12 unit survive both new ideal transfers. The short branches use their independent current payment rules, not either newly deleted long ideal pair.

If there is no changed finite6 corner on the face, its finite weights agree pointwise with the preceding rule, with only positive new coupons. Apply the order13 C-first/B-second and the contextual order17 priorities from their reports. Their short/whole-factor outputs and two distinct contextual halves remain available. This transfers the preceding whole-face inequalities without assuming that an arbitrarily chosen old witness survives a deletion. Triangles retain total at least2 and quadrilaterals total at least1.

The producer additionally aligns all216 saved current literal records with the baseline and all three local certificates. All corner arrays, marks and lower bounds agree before the explicit local changes. Finite6 and order13 have zero deltas on this domain. Contextual order17 has two half debits and two distinct half credits, in cases1314 and2529; both totals remain1. There are2,496 local-corner comparisons and every face total is at least1. This is an exact saved-interface check. It does not reconstruct native Grams or repeat the historical corner census.

Actual costs, representative costs and contextual witnesses

All finite arithmetic uses integer sixteenths. For a whole type `T`, the new actual base is the preceding actual base minus sixteen times the deleted count above. The same formula is applied to its upper representative. The source compares the count to the frozen finite6 formula for every actual type through rank59.

The only new ordinary component-pair increments are A1/A12:+2 and A1/B12:+1/2. They are distinct types, not two additions to one actual pair. All inherited actual increments remain included. The resulting selected totals are

Actual component pair	Total
A1/A12	2
A1/B12	1/2
A1/D12	5/2
D5/D12	21/2
D5/A12	27/4

The complete32,400 ordered type-pair checks verify simultaneous two-end replacement: each actual base is bounded by its representative base, and each actual pair cost is bounded after replacing both ends. This is not an unproved sequence of one-end dominations. B9, E6 and A17 retain their inherited distinct representatives. Actual D16 costs 439/4 while its representative allowance441/4 is attained by A16. Fixed rank17 means A17 of cost451/4; residual rank17 means D17 of cost919/8. Neither receives a finite6 deletion.

Two separate contextual costs must also be counted. Write `a_i,d_i` for whole-component counts of the actual types. The old cost is

```
a2*(a2-1)/4, if a10>0 and a1>=2; otherwise0.
```

It charges two marked A2 components and needs an unmarked A10 and two unmarked A1 witnesses. The new cost is

```
a2*d5, if a9>0 and a1>=1; otherwise0.
```

It charges an A2/D5 pair and needs an unmarked A9 and A1. The charged component pairs and the roles are distinct. Witness components certify presence only; their multiplicities are never factors in either cost.

The upper rule relaxes an A9 or A10 witness to any component of that rank, and replaces D5 by the A5 representative. These replacements preserve every actual triggered context and can create additional upper contexts. They therefore give a valid simultaneous upper bound.

Complete fresh finite certificate

The inherited complete fixed domain has961 multiplicity vectors at ranks9 through17. Expanding every possible rank5 multiplicity gives 2,574 vectors and2,572 distinct interaction-key/rank5 combinations. Rank5 is fixed because the D5/A1 coupon creates a rank1/rank5 interaction. Residual ranks are1–4,6–8 and17–59, with their inherited representatives. Every ordinary residual–residual cross cost is zero; the source checks this explicitly. The two contextual costs are handled outside the additive residual tables.

For each key and fixed rank5 count, the new item is exactly

```
item_new(r,key,k5) = item_previous(r,key,k5)
                     -16*deleted(residual_representative(r)).
```

There are128,700 direct checks that this equals the new representative base plus every interaction with the fixed components. Finite6 changes large residual items, so the preceding tables cannot simply be reused. For each key the producer generates an ordinary table and a table excluding rank2, giving5,144 fresh tables.

The context enumeration uses three branches. The ordinary table bounds the class with no contextual cost. If a rank9 witness and rank5 factor are present, reserve one A1, enumerate every A2 count `j`, and use a no-rank2 table for the remaining rank. Add the new cost `j*k5`. If a rank10 witness is present, separately reserve two A1, enumerate every `j`, and add the old cost `j*(j-1)/4`, together with `j*k5` whenever the rank9/rank5 condition holds. These branches cover all forests, including those with both contexts. Rank2 is excluded from the remainder so no charged A2 is hidden in a table.

The one-A1 branch may allow extra A1 in its residual table. It is an upper relaxation for forests having only the new context. Conversely, each candidate constructs a representative forest whose cost is at least that candidate: a missing old contextual bonus is nonnegative. If a maximizing one-A1 candidate had a positive omitted old bonus, its same forest would be covered at a strictly greater value by the two-A1 branch. Thus the maximum is exact for the representative rule. The certificate reconstructs each winning forest and explicitly checks equality with its full two-context cost. The ordinary branch has the same harmless undercount interpretation for reconstructed forests which happen to trigger a contextual payment.

The saved-table verifier checks every last-item inequality `V(n)>=V(n-r)+w(r)` and one equality predecessor at each positive rank, starting at `V(0)=0`. Induction proves both an upper bound for every residual forest and an attaining residual forest. There are6,718,064 upper inequalities and303,496 equality predecessors. The finite context enumeration has9,191 choices. No numerical optimizer is used.

The complete maximum is394. The actual formal forest

```
A14 + 3A15: 91 + 3*101 = 394
```

has no cross or contextual cost and attains it. Thus394 is sharp for the actual stated formal payment rule, even though some other upper rows use incompatible worst-type allowances. Formal attainment does not assert that this forest occurs in a geometric Coxeter polytope.

Exactly three fixed-multiplicity alternatives exceed783/2:

Representative forest	Upper cost	Literal forest's actual cost
D14 + 3D15	394	392
2D14 + D15 + D16	1573/4	1565/4
A4 + 2A5 + 3D15	392	781/2

The first upper cost is attained by A14+3A15. The second is attained by2A14+A15+A16. The third mixes the rank15 maximum base with D15 cross allowances and is only a representative upper row. The table is not a classification of all actual forests above curvature. It identifies the remaining binding component ranks without promoting future unproved cost changes.

Ideal comparison and limits

The bound baseline supplies the complete1,890 rank58 terminal profiles, obtained by the low9 concentration argument with at most one factor below9 and arbitrary repeated larger ranks. Its saved link polynomials, curvatures, vertices and rank-

specific all-family envelopes are inputs. The source compares the entire profile set and link-vertex counts with both final local packets and independently recomputes the new cost sum. It does not recompute the convolutions or reclassify affine factors.

The only new normalized savings relative to the baseline are

`count(rank12)/13 + count(rank16)/17.`

They are the order13 and order17 total savings of one divided by their respective factor orders. The complete comparison has one negative row and no zero rows. Its remaining row is

Factor ranks	Normalized margin	Terminal absolute margin
9,9,9,15,16	−69/5440	−3450

No positive-terminal/product argument is applied to that negative absolute margin. It is a terminal obstruction, not a uniform lower bound over arbitrary cusp products.

As conditional terminal arithmetic only, an additional all-family order16 total saving strictly greater than69/340 would make all rows positive. A saving1/4 gives normalized minimum1/340 and terminal absolute minimum91/2. The threshold check finds no unaffected nonpositive row. These prospective changes affect ranks above the low9 concentration threshold, but their geometry and any new finite compensation are not assumed or counted here.

Ambient59 has finite curvature783/2 and strict extremum constant4350, counted once. The remaining finite deficit5/2 and ideal negative row prevent a dimension58 conclusion from this packet. No new example, external review or formal proof-assistant verification is claimed.

Verification

Authoritative generation session80994/chunk1f585d and separate fresh saved-certificate replay session21070/chunk56c5d1 both exited0. Their captured logs confirmed `PASS_DEEP_FINITE6_ORDER13_ORDER17_AMBIENT59_FRONTIER 394 3 1`. The earlier arithmetic preflight is not substituted for this final source-bound generation. All three local reports are frozen and pinned by final hashes before authoritative generation.

Root completed a full producer/report mathematical composition read: PASS. The scope includes simultaneous typed domination, fresh additive certificates, the fixed-rank5/A17 versus residual-D17 split, both context guards and the winning-forest equality argument, actual394 attainment, finite6-first actual-corner composition, the216 saved interfaces and complete saved ideal comparisons with their conditional-only scope. No research computation was rerun by that reviewer.

An independent full producer/report mathematical composition read by construction_audit also passed. It checked preservation of the short5/cycle11/triangle and protected whole-factor outputs, pointwise composition on the other faces, ordinary coupon absorption, the full context split and relaxed-candidate proof, Bellman upper/attainment, actual-versus-representative costs and the ideal threshold scope. It did not execute geometry, censuses, DP, table checks, convolutions or a solver.

The author's fresh replay checks finite table inequalities/attainment, the complete finite domain and actual costs, saved literal alignment and exact saved ideal-row updates. It does not repeat native geometry, historical classifications, support censuses, link convolution or a global model optimization. Final integrity validation checks349 direct source bindings, six local links and the producer AST. The packet is final/frozen.

Source 63

Source path: `results/fresh_depth6_finite6_zero_transfer_20261018.md`

SHA256: cc2deec8ac4391dd9889b9d1591b25eb2074ce1ac904d0a99456d6232743f8a6

Finite distance six at feature depth six

Delete the finite distance-six unit whenever both marked nodes have graph distance at least six from every endpoint, branch vertex and endpoint of a nonsimple bond in their whole classical component. Add a unit between a whole A1 and either middle node of a whole A12. Relative to the current combined rule, every triangle still receives at least two and every quadrilateral at least one. All ideal weights and all representative cross-component allowances remain unchanged. This is a local theorem; no new dimension bound or complete finite-forest maximum is asserted here.

The producer and certificate promote the complete adaptive diagnostic by resolving its universal supporting-white obligations and all seven remaining branches. The diagnostic itself remains frozen with its more limited scope. In particular, its unsuccessful single-mark projection test is not treated as a seal.

Actual star and the supporting-white restriction

Use the actual quadrilateral star W+a+b+c+d, with opposite pairs a,c and b,d and all roots of square two. W plus any one black node is positive definite; actual corners are positive semidefinite; W+a+c and W+b+d are indefinite. The full Gram has at most one negative direction. These are the inherited actual-star hypotheses, with arbitrary additional supporting whites allowed.

The changed finite corner contains a simple A19 path, numbered 0 through18, with a=6 and b=12. Work first on the branch where no adjacent black pair has a white-interior path of length at most five. Then c.b=d.a=c.d=0. An additional supporting white can contact the retained path only at 0 or18: every interior position already has two simple path neighbors and is within five steps of a or b, so another contact would create a forbidden branch feature. No additional white contacts both endpoints, since that would create a cycle in the selected finite forest. These restrictions hold for all supporting whites, regardless of their labels or the ambient rank.

The retained W is A6+A5+A6. Its inverse entries are nonnegative and the smallest diagonal is5/6. Three nonzero contacts of c give projection at least5/2>2. A label at least five has squared magnitude greater than5/2, giving projection greater than25/12>2. Thus at most two retained white contacts occur, each of magnitude one or sqrt(2).

The no-short-five condition permits positions0,1,2,3,4,5,7,17,18. All163 contact vectors with at most two nonzero entries are checked. Strict W+c positivity and semidefiniteness of (A19 minus a)+c leave exactly19 profiles, in certificate order:

```
0 empty;
1 {18}; 2 {18:sqrt(2)}; 3 {17};
4 {7}; 5 {7,18}; 6 {7:sqrt(2)};
7 {5}; 8 {5,18}; 9 {5,7}; 10 {5:sqrt(2)};
11 {4}; 12 {3}; 13 {2}; 14 {1};
15 {0}; 16 {0,18}; 17 {0,7}; 18 {0:sqrt(2)}.
```

Unwritten magnitudes are one. Reflect i to18-i for d. The opposite products t=-a.c and u=-b.d are independently zero or at least one; the calculation includes arbitrarily large dotted products.

Low projections force one of two actual extra whites

Let J be the A19 inverse and q the projection of c onto that path, including its a-contact t. If q is at most two, t is zero. The only such profiles have indices0,1,2,3,14,15,16,18. In each, c has no retained contact within four white steps of a.

Take a minimal indefinite principal diagram K in W+a+c. It contains both a and c; deleting either is positive definite. K is connected, has order at most ten, and its component through a after deleting c inherits every branch or nonsimple-edge feature from the selected finite component. Such a feature is at distance at least six from a. Also the distance from a to c in K is at least six: a shorter contact would lie in the retained path where the profile excludes it, and reaching an additional white through an endpoint already takes at least seven steps. Thus K cannot have order at most five.

The inherited complete higher-order catalogue has26 diagrams and768 ordered distinct finite-deletion pairs. The exact screen allows disconnected K minus c. Under the two distance conditions only diagram25, with (a,c)=(8,1) or(8,2), survives. It is the E10 tree with simple edges

`0–1, 0–2, 2–3, 0–4, 4–5, 5–6, 6–7, 7–8, 8–9.`

Its c-contact is five steps from a. It cannot point toward b, since that would give a short adjacent c-b path. Therefore the complete retained profile is {1}. There is an additional actual supporting white z of one of two types:

- z extends the path simply at0 and is orthogonal to c and the rest of the retained path;
- z is joined simply to c and is orthogonal to the retained path.

The explicit E10 maps in the preceding adaptive theorem relabel to the following maps from its nodes0 through9, with c=19 and z=21:

```
extension: 1,19,0,21,2,3,4,5,6,7;
separate:  1,0,19,21,2,3,4,5,6,7.
```

For the extension, a second retained contact of z would make a selected finite cycle. For the separate white, the map fixes orthogonality to0 through7 and the feature restriction excludes every other interior position. A possible contact at18 is excluded by strict W+c positivity: the left A6 contributes10/7 to the c-projection, and the far A6 together with z contributes at least7/8. Their sum129/56 exceeds two. For a native far-end magnitude p at least one, the latter diagonal is 1/(2-p^2*6/7), at least7/8 whenever the denominator is positive; a nonpositive denominator already contradicts the white-block positivity.

In both cases d.z=0, since otherwise c and d have an adjacent path of length at most four. If reflected witnesses are needed on both sides, they are distinct and orthogonal. Equality or a mutual contact gives a c-d path of length at most five whenever a separate white occurs; two extensions would instead make a cycle in the selected finite forest. Thus the positive block is the retained path with zero, one or two simple extensions and zero, one or two separate A1 roots. No unspecified supporting contact is assigned zero without these arguments.

Unbounded Schur branches and native gap

After the forced enlargement, both black projections q,r exceed two. Write s for their cross projection. Split each opposite magnitude into zero or1+T, respectively1+U, with T,U nonnegative. The Schur block is

`[[2-q,-s],[-s,2-r]].`

The complete829 adaptive branches have822 strictly positive determinant polynomials with nonnegative exact Q(sqrt(2)) coefficients. Both eigenvalues are negative, contradicting the Lorentz index. Branches with a low projection other than the forced singleton have already been excluded by the complete minimal-core argument. A profile pair with a short c-d path is handled independently below.

Exactly seven branches remain, given as (c-profile,d-profile,t,u):

```
(0,4,1,0), (0,5,1,0), (4,0,0,1),
(4,4,0,0), (4,5,0,0), (5,0,0,1), (5,4,0,0).
```

In a remaining branch with a nonzero opposite product, all nonconstant determinant coefficients are nonnegative and substitution sqrt(2) for that product gives a strictly positive determinant. The native gap between one and sqrt(2) therefore forces that product to be exactly one. Up to reflection the seven branches give four exact21-root Grams. Each has inertia(20,1,0), and every retained W plus one black is positive definite. The following arguments address arbitrary additional supporting whites in all four cases.

Four residual configurations

Write c=19,d=20. All unspecified core contacts vanish.

Mixed case: c joins6 and d joins11, both simply. An additional white touching d can contact only d,0,18: a c-contact gives a short path, and interior contacts violate the selected feature restriction. With H=G^-1, the H-block on0,18 is positive definite. Minimizing the projection quadratic form over both endpoint coordinates without sign restrictions gives at least(9/4)*v_d^2, hence greater than two. Since the core already contains a timelike direction, its orthogonal complement is positive definite; a square-two root cannot have this projection. Thus d is a whole-component leaf. Its neighbor11 has fixed degree two at the actual corner (a,d), whose marked distance is six. A finite classical completion pays a retained unit because a mark is a whole leaf. Finite exceptional completions have rank at most eight and also retain their basic distance-six units. An affine classical leaf adjacent to a degree-two node along a simple edge is impossible; the exceptional affine alternatives have order at most nine and retain their units. This pays at a distinct actual corner.

Mixed cycle: c joins6; d joins11 and0. The corner (a,d) contains the affine13-cycle on0 through11 and d. Its strictly positive kernel vector is all ones. In a positive-semidefinite actual corner, every additional white is orthogonal to this whole cycle: its nonpositive contacts have zero sum against the kernel. A white touching c can therefore contact only c and18 in the core. Free minimization over18 gives projection at least(78/7)*v_c^2>2. Thus c is a whole A1 at corner (c,d). The component through d is the whole A12 path

```
5,4,3,2,1,0,d,11,10,9,8,7.
```

It is sealed by the affine kernel and the c-seal. Here d is middle node6, or5 in reverse order. The new whole A1/A12 middle-node unit pays this distinct corner. This is the only new coupon.

Central case: c joins7 and d joins11. It suffices to prove that at least one of c,d is a whole leaf. Suppose an additional white z touches c. It cannot touch d, and its only possible retained contacts are0,18, at most one of them. In the positive-definite W+c principal, the three relevant blocks are A6: the left path0 through5, the middle path c,7 through11, and the right path13 through18. Strict positivity and native magnitudes leave exactly

```
{c:1}, {c:sqrt(2)}, {c:1,0:1}, {c:1,18:1}.
```

Indeed a label at least five already exceeds the single-contact budget; with two contacts, any sqrt(2) exceeds the budget18/7. Exact projection onto G excludes {c:sqrt(2)} and {c,0}, whose projections are7/3 and29/12. The survivors are {c} and {c,18}, with projections7/6 and17/12. Reflection leaves {d} and {d,0} for a white w touching d.

The whites z,w must be distinct and orthogonal, since otherwise they give a c-d path of length two or three. The two-white Schur determinants for the four surviving contact pairs are, in order,

```
−13/12, −7/8, −7/8, −5/6.
```

Each gives an additional negative direction beyond the one in G. Therefore both marks cannot have additional neighbors. At least one is a whole simple leaf with fixed degree-two neighbor7 or11. The actual corner (c,d) has marked distance six and pays a retained unit by the same finite/affine classification as the mixed case. The diagnostic's failed one-mark lower bound13/18 is superseded by this two-neighbor argument, not silently strengthened.

Central cycle: c joins7; d joins11 and0. The same affine13-cycle seals all additional whites against0 through11 and d. A possible white touching c contacts only c and18; its minimized projection is at least (78/31)*v_c^2>2. Thus c is sealed. Corner (c,d) contains the whole A13 path obtained by appending c to the mixed-cycle A12 path. Its marked distance is six and c is a whole leaf, so its unit remains. Reflection handles the other three residual branches.

Independent short paths and triangles

Every current finite marked path of length at most five retains its payment. Every current ideal short-five unit is also unchanged. The exception is an affine11-cycle pair at distance five, whose half is handled by its independent replacement theorem. That proof uses shorter retained paths, a distinct cycle11 half, or finite paths of length six. Its only two length-six single profiles, end3_1 and end4_1, have a fixed whole leaf1 at distance four from the old marked node5. They therefore cannot trigger the present deletion. Consequently the short-five branch still gives a full quadrilateral payment without invoking the present theorem recursively.

The producer reconstructs all1,025 high-core triangle triples. Every one of their54 elementary distance-six payments has a persistent branch or nonsimple-edge feature within five steps. The three exceptional alternate paths of length six have a protected whole leaf within two, one and zero steps of a mark, respectively. The fourth exceptional alternate has length seven and is unchanged. Lower-order cores retain the two paths of length at most five supplied by the inherited all-label Steiner argument. Every triangle still receives two.

For simultaneous composition, inspect actual corners. If a newly changed finite distance-six corner occurs, use this universal proof directly. If none occurs, all actual corner weights equal the predecessor weights plus the nonnegative new coupon, so the completed predecessor whole-face payment applies. This priority avoids requiring every historical preferred witness to survive in a different hypothetical face. The independent short-five interface above is the only replacement theorem invoked within the new proof.

Actual costs and verification

For A_n,B_n,D_n the new deleted-pair counts are respectively max(0,n-18), max(0,n-19), max(0,n-20). The admissible marks form a simple spine interval; counting distance-six starting positions gives these formulas for every rank. Direct graph enumeration checks236 diagrams through rank80. These are additional distance-six deletions, not replacements for the preceding distance-seven deletion counts.

The whole A1/A12 coupon adds exactly two to that actual component-pair cost, previously zero. The unchanged representative A1/D12 allowance is 5/2. All32,400 ordered actual type pairs with each type's rank at most59 satisfy simultaneous domination, including every inherited coupon; every actual same-component cost is dominated as well. Actual A12 and D12 are different types and their costs are not added to one another. All216 current literal replacement configurations are reconstructed; none of their corner payments changes and every total remains at least one. This finite check supplements the universal geometry above.

Generation session22306/chunk9f27a2 passed with `PASS_DEPTH6_FINITE6_ZERO_LOCAL_TRANSFER`. Separate fresh replay session29495/chunkea8eac also passed with that status. The construction peer completed a full producer, adaptive-diagnostic and report mathematical composition read: PASS. Its scope covered the complete contact domain, disconnected minimal-core screen and relabelled E10 witnesses, mutual and opposite witness orthogonality, unbounded/native branch reduction, arbitrary supporting-white seals, central two-neighbor argument, independent short-five routes, triangle protections, actual-corner priority and full actual versus representative cost accounting. That review did not execute the profile, Schur, inertia, literal, type-pair or catalogue calculations. The root also read the inherited short-five cycle11 and triangle proofs in full.

The source, data and report are final and frozen. These are internal mathematical reviews, not external refereeing or formal verification. The producer freshly reconstructs the adaptive profiles, catalogue screen and Schur branches, the four exact residual arguments, triangle interfaces, graph counts, literal configurations and actual type-pair domination. It does not rerun the historical catalogue census, prove the written arbitrary-support reasoning by a proof assistant, or perform a finite-forest dynamic program. The explicit relabelled E10 maps above are inherited from the preceding exact map check.

```
python3 scripts/fresh_depth6_finite6_zero_transfer_20261018.py --verify
```

Source 64

Source path: `results/fresh_depth6_finite7_adaptive_transfer_20261018.md`

SHA256: 65e3c3dbae70ae1db09293710ff888e1ee8f18b4f30f468e14e7f0cafc4441bb

Finite distance seven at feature depth six

In the current simultaneous face-payment rule, delete the unit for a finite marked pair at distance seven whenever both marks are at least six steps from every endpoint, branch vertex and endpoint of a nonsimple bond in their whole A, B or D component. Every triangle still receives at least two and every quadrilateral at least one. No new coupon is required. This strengthens the depth-eight theorem.

The current baseline is the sparse D16 and fractional order17 rule. The producer and exact certificate address local geometry and all-rank deleted-pair formulas. A complete finite-forest count and global dimension conclusion are separate obligations. Generation, separate fresh replay and two full internal mathematical composition reviews pass. This local packet is frozen.

Actual star and complete retained profiles

Use the inherited actual quadrilateral star W+a+b+c+d, with opposite pairs a,c and b,d, and normalize each root to square two. Each W plus one black is positive definite; adjacent corners are positive semidefinite and opposite pairs with W are indefinite. The full Gram has at most one negative direction. Work on the branch with no adjacent black pair joined by a white-interior path of length at most six. Consequently c.b=d.a=c.d=0.

The changed corner supplies a simple path A20, numbered 0 through19, with a=6 and b=13. Features can occur at its endpoints but not in its interior. Additional supporting whites remain arbitrary. The retained W splits as A6+A6+A6. Its inverse entries are nonnegative and its smallest inverse diagonal is6/7. Three nonzero contacts from c already give projection at least18/7>2. A contact label at least five has square greater than5/2, likewise giving projection greater than15/7>2. Thus c has at most two retained white contacts, each1 or sqrt(2).

The no-short-six assumption leaves positions0,1,2,3,4,5,7,19. Exhaust all129 contact vectors on these positions. Strict positivity of W+c and semidefiniteness of the adjacent retained corner (A20 minus a)+c leave exactly18 profiles:

```
empty;
{19}, {19:sqrt(2)};
{7}, {7,19}, {7:sqrt(2)};
{5}, {5,19}, {5,7}, {5:sqrt(2)};
{4}, {3}, {2}, {1};
{0}, {0,19}, {0,7}, {0:sqrt(2)}.
```

Unwritten coefficients are one. The d profiles are reflected by i→19−i. The opposite magnitudes t=−a.c and u=−b.d are independently zero or at least one; dotted and parallel values are included.

The two forced supporting-white alternatives

Let J be the A20 inverse, x the c profile including x_a=t, and q=x^T J x. If q>2 no enlargement is necessary. Suppose q≤2. Then t=0, since J_aa=14/3. Every retained contact of c is at distance at least five from a: the only profiles with q≤2 are empty, {19}, {19:sqrt(2)}, {1}, {0}, {0,19}, {0:sqrt(2)}.

Choose a minimal indefinite principal diagram K in W+a+c. Both a and c belong to K; deleting either leaves a positive-definite diagram. K is connected and has order at most ten. The order bound and the compact Lannér bound of five are explicit published classification inputs, already used by the preceding theorem. The complete classical higher-order catalogue reproduces all26 minimal diagrams of orders six through ten.

Let B0 be the component through a of K minus c. It is an induced subtree of the actual selected finite component. Any branch or endpoint of a nonsimple edge in B0 is also such a feature in that actual component; it must be at distance at least six from a. Leaves of B0 are deliberately not treated as features, since the actual component may extend them. The distance from a to c in K is at least six. Indeed, a contact at distance at most four in B0 would be inside the retained path, where the profile has none. Reaching outside that path needs at least seven white steps. A diagram with at most five nodes is therefore impossible.

Enumerate all768 ordered pairs of distinct finite-deletion nodes (a,c) in all26 higher-order diagrams, allowing K minus c to be disconnected. Require the two necessary distance conditions just stated. Exactly two ordered pairs survive: diagram25, with (a,c)=(8,1) or(8,2). Its edges are

```
0–1, 0–2, 2–3, 0–4, 4–5, 5–6, 6–7, 7–8, 8–9,
```

all simple. These are the two relevant positions in the classical E10 tree. The c contact in B0 is five steps from a. It cannot be on the b side, since that would give an adjacent c-to-b path of length three. It is therefore retained position1, and the complete profile is {1}. The exact labelled maps give one of the following necessary witnesses:

- **Path extension:** an additional supporting white z is joined simply to retained position0. Its products with c and positions1 through7 vanish. As a member of the selected finite tree it cannot contact any other retained path node: that would make a cycle. Thus the retained positive block extends to A21.
- **Separate white:** an additional supporting white z has c.z=−1 and is orthogonal to retained positions0 through7. It is in a different component of K minus c. Any contact to an interior position8 through18 would create a feature within six steps of a or b in the selected actual finite component. The only other possible retained contact is position19. If present, strict W+c positivity fails: the left A6 contributes10/7 to the projection of c; the right A6 together with z contributes at least7/8. Their sum129/56 exceeds two. Hence z is orthogonal to the entire A20 and supplies an independent A1 block.

In the second argument, the native contact to19 is at least one. The inverse diagonal at z in A6+z is $1/(2-p^2\cdot 6/7)\geq 7/8$; a nonpositive denominator already contradicts the positive-definite supporting-white block. Extra contacts cannot restore positivity. The c contacts in the displayed principals are specified by the exact E10 map and the retained profile.

In either alternative d.z=0: otherwise there is an adjacent c-to-d white path of length four or two, respectively. Reflect the entire argument for d when its original projection $r\leq 2$. If both sides need witnesses, the two extra whites are distinct and orthogonal. Equality or a nonzero mutual contact gives a short adjacent path of length at most five whenever a separate white is involved. If both are path extensions, equality or a mutual contact makes a cycle in the selected finite component. These arguments allow every unspecified supporting root; there is no assumption that K minus c is connected or that all whites lie in the displayed core.

Complete adaptive Schur certificate

Use the positive block consisting of A20, any forced simple endpoint extensions, and any forced separate A1 roots. Both c and d now have projection greater than two. A separate A1 contributes exactly1/2 to its own projection and zero to the cross projection. Path extensions use the exact A21 or A22 inverse. All new contacts have just been forced, rather than assigned for convenience.

For every pair of retained profiles, first discard a pair already giving a c-to-d white path of length at most six. Split t into0 or1+T and u into0 or1+U, with $T,U\geq 0$. If a projection is at most two, exclude the branch unless its profile is the forced singleton above; in that case test both possible witness types. There are766 adaptive branches.

Writing q,r,s for the two new projections and their cross projection, the Schur block is

```
[[2–q, –s], [–s, 2–r]].
```

In765 branches the polynomial $(q-2)(r-2)-s^2$ has nonnegative exact Q(sqrt(2)) coefficients and strictly positive constant, with minimum5/84. This makes both eigenvalues negative, contradicting Lorentz signature. The certificate handles arbitrary nonnegative T,U, not a finite angle grid. The sole remaining branch is t=u=0 with c joined simply to7 and d joined simply to12; it needs no witness. Its determinant is−4/7. This branch is Lorentz compatible and is not discarded by the sign test.

The remaining branch forces both replacement marks to be leaves

Let G be the complete22-root Gram A20+c+d in that remaining branch. Its signature is(21,1,0). Suppose an additional actual supporting white z touches c. No-short-six forces its contacts with b,d and retained positions9 through17 to vanish. Its contacts with any middle white 7 through12 also vanish: W+c already contains the path c−7−…−12, and an extra z touching c and that path would create a cycle in the positive-definite actual edge diagram W+c.

Consequently the magnitude vector v=−(z.root) can be supported only on

```
L={0,1,2,3,4,5,6,c},        R={18,19},
```

with every entry nonnegative and v_c≥1. Put H=G^−1. The block H_RR is positive definite. Minimize v^T H v over the R coordinates with no sign restriction; this only weakens the lower bound. The remaining matrix

```
H_LL − H_LR (H_RR)^−1 H_RL
```

is entrywise positive and has c,c entry5/2. Thus the squared norm of the projection of z onto the core span is at least5/2. The core span is nondegenerate and contains a timelike direction, so its orthogonal complement in actual Lorentz space is positive definite. A square-two root cannot have that projection. Therefore no additional supporting white touches c. Reflection proves the same for d.

In the actual opposite corner containing c,d, both are whole-component leaves joined by the simple length-seven path c−7−…−12−d. The finite classification permits only A8 or D9. The affine classification permits only affine D9 of order ten or affine E8 of order nine. Finite exceptional diagrams have smaller diameter; doubled endpoint bonds and affine cycles cannot contain this whole-leaf simple path. Each named marked pair retains its unit under the current rule. Additional supporting roots may attach inside the path, but the classification includes precisely those allowed completions. No claim that the entire A20 is sealed is needed.

The paying corner is distinct from the deleted selected corner. This completes the no-short-six branch without any new coupon.

Current-rule composition and all-rank counts

On the complementary short-six branch use the already independent short-path routes from the depth-eleven theorem and their current composition in the depth-eight theorem. Their finite distance-seven/eight outputs have a feature within five steps or a fixed whole-component marked leaf; their remaining outputs are retained shorter paths, separate coupons or small ideal factors. Threshold six preserves all those outputs. Every elementary triangle distance-seven payment likewise has a feature within five steps, and the exceptional triangle alternate has a fixed whole-component leaf. The later finite rank15–17 changes concern distances at least ten; the changed ideal order17 roles have distance eleven. None is used recursively as a short-six output. The current sparse-D16 removal keeps the entire existing output inventory. Triangle total two and quadrilateral total one are consequently preserved simultaneously.

The numbers of newly deleted distance-seven pairs, relative to paying all such pairs, are max(0,n−19), max(0,n−20), max(0,n−21) in A_n,B_n,D_n. Both marks must lie in the unique simple spine interval at feature depth at least six. Counting its possible starting positions gives these formulas for every rank. Direct graph enumeration through rank80 checks them in236 finite diagrams. Relative to the old depth-eight rule, subtract the corresponding old counts n−23,n−24,n−25, each truncated at zero. All other payments remain unchanged.

Verification status

Generation session89253, terminal chunk965e1a, and separate fresh replay session27237, terminal chunk80313a, both pass exit0. Each reconstructs all129 candidate profiles,768 ordered classical finite-deletion pairs, the two exact E10 maps, all766 adaptive polynomial branches, the central inverse/Schur seal and236 graph counts. They do not rerun the historical 26-diagram attachment census or the preceding geometric theorems.

The construction and order16 peers each completed a full mathematical source/report composition read, including the inherited short-six and triangle interfaces. Both checked disconnected minimal-core deletion, genuine new supporting whites, unspecified contacts and their exclusion, mutual patch orthogonality, unbounded polynomial substitutions, the Lorentz projection inequality, whole-leaf completion classification and all-rank counts. Neither executed the package or repeated the

catalogue, profile, Schur, inertia or historical convolution computations. Neither found a correction. These are internal agent reviews, not external mathematician refereeing.

Metadata check9cd315 verifies16 direct source bindings, source syntax and local report links. It is an integrity check, not another mathematical replay. The classical catalogue, current short-six and triangle geometry, and finite/affine classification are explicit inherited inputs. No proof-assistant verification is claimed.

```
python3 scripts/fresh_depth6_finite7_adaptive_transfer_20261018.py --verify
```

Source 65

Source path: `results/fresh_depth6_sparse_ambient60_forest_frontier_20261018.md`

SHA256: 75f2d0a56f759815644f298f060697a8d3be6afe12f38cc3fe1dff359ecca399

Conditional depth-six finite forest certificate

Under the stated depth-six payment hypothesis, the complete finite rank60 maximum is **404**, attained by the formal forest4A15. The finite curvature threshold is405. All2,113 terminal ideal profiles retain positive margins, with normalized minimum31/6000 and absolute minimum54. This packet proves the counting implication; it does not prove the geometric hypothesis or claim a new unconditional dimension bound.

The producer and certificate bind the frozen sparse D16 and ideal17 rule. That predecessor proves the simultaneous local rule, complete finite maximum 407.5 and positive ideal comparison. The only proposed change here deletes finite distance-seven units when both marks have feature depth at least six, instead of eight. Every other same-component weight and every typed cross-component coupon stays unchanged.

The separate geometric obligation is the adaptive depth-six transfer. This certificate intentionally does not bind a draft version of that theorem. A subsequent composition must bind its completed proof and verify the local triangle and quadrilateral hypotheses before using the conditional count.

Exact change and complete finite domain

For a whole classical component of rank n, the proposed total numbers of deleted distance-seven units are

Type	Frozen depth-eight rule	Proposed depth-six rule
A	max(0,n−23)	max(0,n−19)
B	max(0,n−24)	max(0,n−20)
D	max(0,n−25)	max(0,n−21)

The producer subtracts the difference between these columns from the actual and representative base costs. The formula applies at every rank; the finite enumeration needs ranks only through60. No rank9 through17 base changes, and all cross-component costs remain exactly those of the frozen predecessor. In particular, actual A15/B15/D15 costs remain101,97,201/2. The representative rank15 upper cost is101; it is not asserted to equal the actual D15 cost.

The unchanged representative map is checked simultaneously at both ends of every cross-component pair. All33,489 ordered actual type pairs are covered, and every actual same-component cost is at most its representative cost. The frozen sparse-D16 interaction key includes the number of fixed D16 components. Thus the removal of rank2 through5/D16 coupons remains counted, while whole A1/D16 coupons are retained.

Every representative forest has a unique fixed multiplicity vector for ranks9 through17 and a residual forest with ranks1 through8 or18 through60. The producer independently enumerates all1,044 fixed vectors, giving1,042 interaction keys. It asserts that fixed–fixed and residual–residual cross-component interactions vanish. For each fixed vector and every residual rank, it verifies that the item weight is exactly the representative base plus all interactions with the fixed components. These are54,288 explicit item identities, so the decomposition omits no interaction.

For each key, a new item-outer integer Bellman table is generated in eighth units. The fresh checker does not rerun the optimization: it checks every last-item inequality and one attaining predecessor at every positive rank. The1,502,564 inequalities prove the upper bounds by induction; the62,520 predecessors prove attainment for each residual table. Every fixed-vector total is then checked against its reconstructed representative forest and against the predecessor's upper bound. All1,044 alternatives are retained.

The largest upper value is3232 eighths, or404. The actual forest4A15 costs 4×101=404 and attains this upper bound. This is formal cost attainment; no assertion that the forest occurs at a vertex of an actual polytope is needed or made. There is no representative alternative above405.

The six preceding large-component obstructions each lose exactly four units: 2D17+D26, A17+D17+D26, 2A17+D26, D16+D17+D27, A17+D16+D27, and2D16+D28 all decrease from407.5 to403.5. The previously removed small coupon obstructions remain removed. The count includes every other forest, not just these six witnesses.

Unchanged ideal comparison

The ideal rule is unchanged, including order17 total229/2. Independently enumerate the sorted rank59 terminal domain: arbitrary repeated parts at least9 and at most one positive part below9. Its2,113 rows agree exactly with the frozen predecessor. For each saved h-polynomial, the checker verifies positive coefficients, symmetry, degree59, vertex count V, the exact moment

```
G = sum_j min(j,59−j) h_j,
```

and curvature (5G−59V)/2. It checks the saved normalized cost, positive margin and absolute margin identities against the inherited exact data. This is a saved-polynomial check; the original convolutions and classified factor envelopes are inherited, not recomputed here.

The normalized minimum remains31/6000 at(9,9,9,9,9,14). The absolute minimum remains54 at the singleton(59). The inherited low-nine concentration decreases both V and the normalized margin. Because every terminal margin is positive, the product comparison extends the absolute lower bound54 to every actual cusp. No concentration of large factors, and no multiplication of a negative margin by a smaller V, is used.

Conditional global implication

For ambient dimension60, the finite curvature coefficient is405 and the two-extrema correction is4500, counted once. If the proposed rule preserves triangle payment2 and quadrilateral payment1 for every actual face, the inherited global inequalities give

```
404 v_f + sum_i C_i >= 405 v_f + sum_i kappa_i + 4500,
```

where kappa_i−C_i≥54 at every ideal vertex. Consequently

```
0 >= v_f +54 v_i +4500 >0.
```

This contradiction is conditional here on the separate complete local theorem and its simultaneous composition. The completed unrestricted bound60 is unchanged by this counting-only packet. No new polytope, external referee validation or proof-assistant validation is asserted.

Verification

Generation session67497/chunk039233 completed with exit0 and PASS_CONDITIONAL_DEPTH6_SPARSE_AMBIENT60_FOREST_FRONTIER. A separate fresh replay, session15892/chunk933837, also completed with exit0 and the same PASS status. It checks saved Bellman upper/attainment certificates without optimizing again. Frozen input hashes are checked before either run.

The root agent and construction reviewer each completed a full internal source/report mathematical review: all-rank cost differences, simultaneous type domination, complete fixed/residual decomposition, Bellman proof, actual witness, unchanged positive ideal comparison and conditional scope. Neither reviewer reran the DP, convolutions or geometry. These are internal reviews, not external refereeing or proof-assistant validation.

```
python3 scripts/fresh_depth6_sparse_ambient60_forest_frontier_20261018.py --verify
```

Source 66

Source path: `results/fresh_depth6_sparse_ambient60_forest_frontier_20261018_domain_clarification.md`

SHA256: 2371e53c45bda179729402d3f52ec9559ecc9c03e6b21feed7963275ec03a9bd

Clarification of the residual rank domain

The frozen conditional404 report contains a prose typo: its residual domain says ranks1–8 or18–60. The correct domain is **ranks1–8 or17–60**, with A17 separately present in the fixed domain and D17 retained among the residual items.

The producer, generated tables, fresh certificate checks and reconstructed forests already use this correct domain. No source, data, bound or proof inequality changes. The arithmetic-lane peer identified the typo in a third mathematical source/report review after the report was frozen. The original report and its hash are preserved.

Source 67

Source path: `results/fresh_depth8_d17_rank61_forest_budget_20261018.md`

SHA256: 79e2c6f4bc2b8a085e3e0e5e180a6e613b4b0374fde1171dbfbbb42f8d70718a

Exact finite rank61 maximum425 after depth8 and D17=118

Under the simultaneous completed dimension61 payment rule, the proved depth8 distance-eight zeros and the new finite D17 cost118 rule give **exact finite rank61 maximum425**. The formal forest

```
A16 + 3D15: 113 + 3*104 =425
```

attains it. Ambient dimension61 has finite curvature420, so a deficit5 remains. The singleton D61 now costs420 instead of426. No ambient dimension60 theorem follows from this finite counting packet.

The source and certificate bind the completed dimension61 rule, its finite costs, the depth8 theorem, the D17 existing-half transfer and the prior ambient61 frontier. All other finite weights, ideal weights and typed coupons retain their proved current values. Further rank15/16 decreases are not assumed.

One simultaneous local rule

The depth8 theorem deletes only distance8 pairs deep in whole classical components; the D17 theorem halves only four distance10–12 pairs in a whole D17. Their changed sets are disjoint, but composition also requires checking their replacement outputs.

The D17 theorem's eight sealed cases use existing whole A4/D13 neighbor, center or short-leaf halves. These coupons are unchanged by depth8. Its other escape is the independent generic short8 route. Whenever that route reaches a newly deleted deep finite8 pair, the new corridor supplies its constructive short7 alternative. The depth8 theorem proves the required Q=9, Q=10 and Q>=11 cycle interfaces separately, together with the retained tree/protected-feature interfaces. They consume no finite D17 distance-at-least10 payment.

Apply the completed independent ideal-corner priorities first, as in both local reports. Their outputs are short paths, protected whole factors or typed coupons and remain unchanged. In the remaining branch the complete finite reduction uses the D17 sealed replacements and the depth8 short7 route. The216 latest literal configurations involve components too small to contain a deep8 pair and already pass the simultaneous D17 debit. More explicitly, the bound `T.literal_inputs` records have at most19 roots, and all832 corner supports `white union marks` have order16 (460 supports) or17 (372 supports). A whole finite component there has rank at most17. A distance8 pair with both feature depths at least8 requires at least25 roots in A,26 in B or27 in D. A separate saved-input size check verified these216/832 counts without repeating a geometric classification or the budget calculation. Triangles use unchanged finite short7 and retained ideal interfaces. Consequently one payment rule simultaneously gives at least2 per triangle and at least1 per quadrilateral. No new geometric census is needed here; the local theorems and their recorded whole-face arguments are inputs.

Actual base costs and exact deep-pair counts

The retained small-rank costs are:

Type	Actual cost
A15, B15, D15	104
A16	113

Type	Actual cost
B16	104
D16	225/2
A17	115
B17	105
D17	118

All finite types of rank at most14 retain every same-component pair. Exceptional and arbitrary-label dihedral types are included at their actual ranks. For each classical type of rank r>=18, the number of distance8 pairs at feature depth at least8 is respectively

`A: max(0,r-24), B: max(0,r-25), D: max(0,r-26).`

For A, orient its path from0 to r-1. A distance8 pair is (i,i+8), and both marks have depth at least8 exactly when 8<=i<=r-17. The B terminal bond contributes its two endpoint features, shifting the upper limit down by1. In D, the fork center and two short leaves shift it down by2. The selected pair lies on the long simple spine in both cases. This proves the formulas at every rank. The source independently constructs each labelled A/B/D tree and uses graph distances to check both the old depth11 and new depth8 counts for all132 types of rank18 through61.

Before deep zeros, the established finite8/leaf9 rule has base8r-34 in A/B, base8r-32 in D18/D19, and base8r-33 in D_r for r>=20. Subtract the new counts above. In particular the new D_r tail is7r-7 for r>=26, so D61 has cost420. The old residual tables cannot be reused: their large-item weights still used the depth11 subtraction. This packet generates new tables and checks their certificates afresh.

All current coupons and simultaneous type domination

The source independently writes the current doubled coupon function and checks the nine sparse increments against the bound predecessor source. It retains the whole-small/D10 and D11–D16 rules, A11/A13/A15/A17 roles, the exact A1/A2 D11 upgrades, A4/D13 neighbor, A1/A2 D14 neighbor, B9/B10 terminal and sparse A/D-path coupons. In particular the D13-zero A2/D10 and A3/D10 increments and actual E6/D11 total4 remain present. There is no additional cross-component charge in either new local theorem.

Replace finite types by the established representatives: paths at ranks at most8, with E6 kept at rank6; B9 at rank9; D_r at ranks at least10; and a separate A17. This preserves total rank. The D16 representative is conservatively allowed113, so it still bounds A16=113 even though the actual D16 cost is112.5. Every other representative has its actual cost.

For every ordered pair of the186 finite types through rank61, including the exceptional and arbitrary-label dihedral cases, the source verifies

```
representative_base(a) >= actual_base(a),
cross(rep(a),rep(b)) >= cross(a,b).
```

There are34,596 pair checks. The second inequality replaces both ends simultaneously. This matters for sparse coupons whose small or large type changes; sequential one-end domination is not assumed. Summing these inequalities bounds every actual forest. The finite-type classification and the geometric validity of the coupons are inherited theorem inputs, not conclusions of these finite comparisons.

Complete new residual certificates

Separate fixed B9, D10 through D16 and A17 components from residual components of ranks1 through8 or D-rank at least17. Enumerating all fixed multiplicities of total rank at most61 gives1,132 vectors. These are independently reconstructed and compared with the prior frontier's complete domain. They produce854 interaction keys.

There is no coupon between two fixed components or between two residual components. Every cross interaction is between a residual component and a fixed one. The seven-coordinate key records all such interactions, including the distinct E6 and A17 roles. All59,996 encoded residual item costs are checked against a literal recount of the actual coupon function.

Generation computes an unbounded item-first exact dynamic program for each key. Its entries are doubled costs. A separate last-item certificate then checks, for every residual rank s and every allowed final item r,

`value[s] >= value[s−r] + item_cost[r].`

It also saves an allowed r attaining equality at each positive s. Induction from value[0]=0 proves both the upper bound for every residual forest and an attaining decomposition. Saved replay checks all these inequalities and predecessors without rerunning the optimizing recurrence. There are1,276,730 upper inequalities and52,094 attaining predecessors.

For every one of the1,132 fixed vectors, the source adds its exact fixed upper cost to the certified residual value at the remaining rank. It reconstructs and literally recounts an attaining representative forest. The largest upper is425. Its displayed representative3D15+D16 has actual cost424.5, so that alone would not prove actual attainment. Replacing its D16 by A16 gives the formal actual forest3D15+A16, cost425 with no cross coupon. Hence425 is the exact maximum of the actual payment rule, not only a representative upper bound. A formal attaining forest need not be realized by a hyperbolic polytope.

The six representative alternatives above curvature420 are:

Representative forest	Certified upper	Cost of those named actual types
3D15+D16	425	849/2
A1+4D15	422	422
2D16+D29	422	421
D14+D15+2D16	421	420
D16+D45	421	841/2
A3+2A5+3D16	841/2	419

The conservative upper and literal actual columns are intentionally separate. Some A16 substitutions attain the larger upper; others can lose cross coupons and need a fresh literal recount. This table selects future geometric targets and does not authorize any unproved decrease.

Curvature and verification scope

At ambient61 a finite link has dimension60, V=61 and

```
G=sum_(j=0)^60 min(j,60−j)=900,
curvature=(5∗900−60∗61)/2=420.
```

The new maximum425 therefore remains5 too large for the desired finite comparison. The ideal deficit recorded in the separate ambient61 frontier also remains unchanged here. The completed unrestricted ceiling is61.

Final generation `8cf16d` and separate fresh replay `dad836` both exited successfully with PASS. The source uses only the standard library and reconstructs the complete fixed-vector domain, all typed domination checks, graph-distance counts, new residual tables and exact upper and attaining certificates. It does not repeat a geometric classification, native contact census, ideal convolution or global model optimization. The root and construction reviewer completed full source/report mathematical reads, including the independent type formulas, exact counting domain, residual-table certificates and simultaneous local interfaces. Neither repeated the DP, graph-distance arithmetic or geometric census. The separate216-interface size check above terminated with PASS (`8f806e`, session73717).

```
python3 scripts/fresh_depth8_d17_rank61_forest_budget_20261018.py
python3 scripts/fresh_depth8_d17_rank61_forest_budget_20261018.py −−verify
```

Source 68

Source path: `results/fresh_depth8_finite7_profile_transfer_20261018.md`

SHA256: b7b6d23d2f1e705ecb1a62c6af1c0f69fcdf0fdb44b2dec11de9d412b7fd32ff

Finite distance7 at feature depth8: complete profiles and a sealed equality

The local rule deletes finite distance7 payments in whole A/B/D components when both marks are at least eight steps from every endpoint, branch and endpoint of a nonsimple bond. All other weights and coupons are unchanged. The argument below strengthens the completed depth10 rule. Exact generation, a separate fresh replay and two independent internal mathematical composition reviews pass; no new finite forest maximum or dimension bound is asserted by this local packet.

The producer binds the current order13 rule. The proof uses a complete finite contact classification with arbitrary opposite-wall products; it does not extrapolate sampled labels.

Actual star and retained path

Use the inherited actual quadrilateral star W+a+b+c+d, with opposite pairs a,c and b,d. Roots have square2. W and each W plus one black are strictly positive definite; adjacent corners are positive semidefinite, and opposite pairs with W are indefinite. The full Gram has at most one negative direction. Assume no adjacent black pair has a white- interior path of length at most6. Then c.b=d.a=c.d=0.

The newly changed finite corner supplies the retained path A24, numbered 0 through23, with a=8 and b=15. There are eight outward supporting whites on either side. A feature exactly eight steps away may be at an old endpoint; all bonds inside this path are simple. Extra supporting whites and their contacts outside the retained path remain arbitrary.

On an eight-white arm the inverse diagonals at outward distances3 through6 are2,20/9,20/9,2. A nonzero contact at any of these sites violates strict actual-edge positivity. Together with the no-short6 assumption this leaves c's retained white contacts only at

`0,1,6,7,9,22,23.`

Its contact at the opposite black a is independent, of magnitude t. The d profile is the reflection i→23−i, with opposite product u at b. Each of t,u is0 or at least1, including arbitrary parallel/dotted products.

Complete seventeen white-contact profiles

The retained W principal splits as A8+A6+A8. Its smallest inverse diagonal is6/7, and its inverse entries are nonnegative. Thus three nonzero white contacts would already have projection at least18/7>2. There are at most two. A label at least5 has contact square at least (3+sqrt5)/2>5/2, giving projection greater than15/7>2. Each nonzero white contact is consequently1 or sqrt2.

Exhaust all99 such vectors on the seven positions. Impose exact strict positivity on W+c and semidefiniteness on the adjacent retained corner (A24 minus a)+c. Both are single-root Schur tests against positive path blocks. Exactly these seventeen profiles remain (unwritten coefficients are1):

```
empty;
{23}, {23:sqrt2}, {22};
{9}, {9,23}, {9:sqrt2};
{7}, {7,23}, {7,9}, {7:sqrt2};
```

```
{6}, {1};
{0}, {0,23}, {0,9}, {0:sqrt2}.
```

These are necessary profiles, not claimed realizable chambers. The strict and weak inequalities are kept distinct, and no unknown Gram entry is set to zero. Outside contacts can only impose further conditions.

A central contact is forced, including disconnected deletion

At least one c contact at6,7,a=8,9 is nonzero. To prove this, choose a minimal indefinite principal K in W+a+c. The inherited complete theorem bounds its order by10, and it contains a,c and is connected. The component B through a in K minus c has at most9 roots. Feature depth8 makes this component a simple retained path: reaching an old endpoint requires all nine roots, leaving no room for a branch beyond that endpoint. B cannot pass through the absent b. Connectedness forces c to contact B.

Suppose t=0 and no contact6,7,9 occurs. Rightmost contacts22/23 cannot reach B. If c contacts0, the whole nine-root path0..8 is required, so K has exactly those roots and c. The seventeen-profile list makes its only contact in K a simple or doubled bond at0: K is finite A10 or B10.

The only other possibility is c's single unit contact at1. The path1..8 and c already use nine roots. With no extra root K is A9. At most one extra root is possible. If it joins B in the selected finite component, feature depth8 permits only extension to0 or9; K is D10 or A10. If it is in a separate component of K minus c, connectedness requires a contact to c. The strict edge contains the seven whites1..7 and c, an A8 with c its endpoint. Its inverse diagonal8/9 forces the new bond to be3 or4, so K is again A10 or B10. No further extra component fits the order bound. Every alternative contradicts indefiniteness. This argument does not assume K minus c connected. Reflection proves the d central-contact statement at14,b=15,16,17.

Complete Schur certificate and its unique equality

Let J be the exact A24 inverse. Extend each white profile by x_a=t and y_b=u. Put q=x^T J x, r=y^T J y and s=x^T J y. The central-contact statement and positive inverse entries imply q>2 and r>2. The full principal Schur block is

```
[[2-q, -s], [-s, 2-r]].
```

For every pair of the seventeen profiles, discard pairs already giving a white-interior c-to-d path of length at most6. Split each opposite product into t=0 or t=1+T, and u=0 or u=1+U, with T,U≥0. In the zero branches require the central white contact just proved. Exactly581 branches remain. The determinant polynomial

```
(q-2)(r-2)-s²
```

has nonnegative exact Q(sqrt2) coefficients in every branch. In580 its constant coefficient is strictly positive, with exact minimum47/25. Thus it gives two negative directions, impossible in Lorentz space. The only remaining branch has t=u=0 and x=e9,y=e14. Its determinant is zero, with q=r=6 and s=4. This is an exhaustive coefficient certificate for unbounded T,U, not a numerical parameter grid. The candidate finite profiles were themselves exhaustively obtained from the native gap and strict edge bound.

Equality seals the finite path and supplies a retained unit

The full A24+c+d Gram has inertia(24,1,1) and radical

```
k=(J(e9-e14),1,-1),   k_0=1/5, k_23=-1/5.
```

Its represented vector is zero in actual Lorentz space: the span contains a timelike vector, whose orthogonal complement is positive definite. Every extra white therefore annihilates this radical.

If the actual selected finite component extends the old path, take its first extra root z. It can attach only at endpoint0 or23. An interior attachment creates a branch within seven steps of a or b; a second attachment creates a cycle, both forbidden by the finite component and the feature-depth assumption. By reflection take endpoint0, with bond magnitude p. Projection onto A24 gives $p^2 \cdot 24/25 < 2$, hence p=1 or sqrt2.

For p=1, the eight outward whites and z form A9, with inverse diagonal 9/10 at z. Strict actual-edge positivity restricts x_z=−c.z and y_z=−d.z to0,1,sqrt2. Radical annihilation requires

```
y_z-x_z=1/5,
```

which none of the nine pairs satisfies. For p=sqrt2 the same white principal is B9 with inverse diagonal9/2 at z; both contacts must vanish, contradicting radical annihilation with nonzero endpoint contribution. Thus there is no first attachment: the selected component is whole A24.

Every other supporting white is orthogonal to A24. Its radical relation now says x_z=y_z; a nonzero value gives the adjacent c-z-d path of length2. In the no-short6 branch both vanish. All extra whites are therefore orthogonal to the entire core. In the corner with marks c,d, the middle path c−9−…−14−d is a whole A8, with both marks leaves. Its distance7 unit is retained: its feature depth is zero. It is distinct from the deleted selected corner and pays the face without any new coupon.

Current payments, triangles and counting scope

Outside equality, the proof forces a full-star short6 corner. The completed depth11/depth10 arguments show that every current ideal short6 alternative uses retained short paths, finite7/8 paths protected by a feature within five steps, a fixed marked leaf, or separate coupons. Reducing the threshold to8 preserves these outputs. The later order13/14 changes have ideal distances at least9; rank15–17 finite changes have distances at least10. None changes these short6 replacements.

Every one of the14 elementary triangle distance7 payments has a feature within five steps; the exceptional distance7 alternate has a fixed whole-component marked leaf. These retain their units. Thus the new actual-corner priority preserves triangle payment2 and quadrilateral payment1 under the current simultaneous rule.

The new deleted pair counts are max(0,n−23), max(0,n−24) and max(0,n−25) for A_n,B_n,D_n. All finite compensation is zero. A complete new forest count, including every existing coupon and restoration, is a separate obligation. No dimension59 conclusion is asserted here.

Verification status

Generation session6477, terminal chunk93f980, and separate fresh replay session88473, terminal chunk51cfa0, pass exit0. Both reconstruct all99 candidate white vectors,17 admitted profiles and581 coefficient branches. All coefficient signs are determined in Q(sqrt2) by rational square comparisons, without floating-point positivity tests. The producer also checks central-contact finite boundaries, the full radical, every endpoint-extension contact and the whole A8 alternative. Direct graph counts corroborate the all-rank spine formulas in236 A/B/D graphs through rank80. No finite-forest optimization or ambient comparison is performed.

The construction peer completed a full producer/report mathematical read, including the disconnected minimal-core cases, exact polynomial domain, Lorentz radical relation, equality sealing, current short6 and triangle interfaces, and all-rank counting formulas. This was a source and proof review, not an execution or arithmetic replay. The ideal-lane peer also completed a full producer/report mathematical composition read, checking the complete native reduction, disconnected-core argument, exact sign and monotonicity logic, radical sealing and the distinct whole A8 unit. That review additionally checked compatibility with the new order12 and order16 transfers. It did not execute or repeat profile, polynomial, matrix, graph or historical-census computations. The minimal-core classification and current short6/triangle geometry remain explicit inherited inputs. No external referee or proof-assistant validation is claimed.

```
python3 scripts/fresh_depth8_finite7_profile_transfer_20261018.py --verify
```

Source 69

Source path: `results/fresh_depth8_finite8_zero_transfer_20261018.md`

SHA256: 8eddb81ade5943e69c453029cc865d74ef4b2640855ec1f9e174f5d901f8b8e3

Deep finite distance-eight payments can be deleted at feature depth8

Start with the complete simultaneous payment rule of the dimension61 proof. Set to zero every remaining payment at a distance8 marked pair in a whole finite A/B/D component when both marks have distance at least8 from every endpoint, branch and endpoint of a nonsimple bond. Retain every other finite weight, every ideal weight and every component-pair coupon. Then **every triangle receives at least2 and every quadrilateral at least1**.

The change lowers the feature threshold of the completed depth11 zero theorem. Its main new step is an estimate for the Schur diagonal after subtracting the new root's squared norm, rather than separate bounds on the projection and correlation. A second argument excludes the minimal cores that could escape the shorter retained path. The two small cycle boundary cases are handled separately; their profile domains differ from the long tail.

The source and certificate check the exact matrix and scalar inequalities. This is a local payment theorem. It does not itself give a complete finite-forest maximum, improve the ideal envelopes or prove a dimension bound below61.

The actual-face inputs and the shorter path

Use an actual quadrilateral star W+a+b+c+d with opposite black pairs {a,c} and {b,d}, normalized to squared root norm2. W and W plus any one black are positive definite. W plus either opposite pair is indefinite. These are the established actual-face inputs of the universal corridor theorem. Every distinct-root product is nonpositive, and every nonzero native contact magnitude is at least1. Arbitrarily large finite labels, parallel products and dotted products are included.

Suppose W+a+b has the stated whole finite component and marked pair. Feature depth8 supplies two simple outward arms of eight whites, in addition to the eight-edge a-to-b path. A feature exactly at distance8 is allowed: all bonds before it are still simple. Keep six outward whites on either side. The resulting21 roots form an induced A21 with

```
a at6, b at14, indexed from0.
```

Its other19 roots are supporting whites. The two additional whites on each outward arm remain available for actual-edge estimates.

Assume there is no path of length at most7 between the opposite black pairs whose internal roots are white. Then c.d=c.b=d.a=0. The own-opposite products c.a and d.b remain unrestricted. For a norm-two A_n path write

```
J_n[i,j]=(min(i,j)+1)*(n-max(i,j))/(n+1).
```

In either full eight-white outward arm, the inverse diagonals at distances j=3,4,5,6 from the old mark are respectively

```
2, 20/9, 20/9, 2.
```

A contact of magnitude at least1 already has projection square at least2. The actual edge containing that white arm and c is strictly positive definite, so equality is also excluded. Positivity of the path inverse means that extra contacts cannot reduce this obstruction. This excludes those four outer positions on either arm, and likewise for d.

A c-contact at an inner position at least two steps from a supplies a path to b of length at most7. A c-contact within two steps beyond b does the same. Hence the only possible retained contact positions are

```
c: 4,5,6,7;       d: 13,14,15,16.
```

No bound on their contact magnitudes is imposed. Contacts outside A21 remain allowed throughout.

Nonzero retained contact, including a disconnected deletion

Choose a minimal indefinite principal K inside W+a+c. It contains both a and c, since deleting either leaves a principal part of a strictly positive-definite actual edge. K is connected and has order at most10. The published Chein/Vinberg order bound, including the bridge for dotted order2 diagrams and disconnected positive-semidefinite deletions, is explicitly recorded and reviewed in the bound universal corridor report. This proof uses that all-label order ceiling, not a bounded search.

Let B be the connected component through a in K minus c. It has at most nine roots and lies in the same whole finite tree as a. Feature depth8 forces it to be a simple path: reaching a possible feature at distance8 already uses all nine available nodes, leaving no node for branching or a bond beyond the feature. It cannot pass through b, which is absent from K. Connectedness of K forces c to contact B.

Suppose c has no retained A21 contact. All of B toward b lies inside the retained path. On the outward side, the only possible contacts missed by that path are distances7 and8. The full eight-white edge arm gives

```
J8[6,6]=14/9, J8[7,7]=8/9, J8[6,7]=7/9.
```

At distance7 a nonsimple contact has projection at least28/9>2, so only a simple bond is possible. At distance8 a label at least5 has squared native magnitude at least(3+sqrt5)/2>5/2, giving projection greater than 20/9>2. Thus only labels3 and4 are possible there. Contacts at both positions have projection at least(14+8+14)/9=4 and are impossible.

There are now five exhaustive cases.

- If the contact is only at distance8, B already has nine roots. There is no further node in K beyond c. Appending c with bond3 or4 gives finite A10 or B10, contradicting indefiniteness.
- If the contact is only at distance7 and B has nine roots, the ninth B-node is either one inward step from a or outward distance8. The simple c-contact then gives finite A10 or D10, respectively.
- If the contact is only at distance7 and B has eight roots, it is exactly a and the seven outward whites. With no further white in K, appending c gives finite A9.
- In the same eight-root case, K can instead contain one further white z outside B. The definition of B makes z orthogonal to B, and connectedness forces z to contact c. The strictly positive-definite diagram K minus a is an A8 path (the seven outward whites and c) with z attached to its end. Its inverse endpoint8/9 excludes a c-z label at least5. A bond3 or4 makes the full K finite A10 or B10.

The second bullet has two alternatives, giving five cases in the certificate. In particular **K minus c need not be connected**; the extra z case is essential and is not silently discarded. The source checks positive pivots for every finite outcome. For B10 it uses the explicit rational congruence that scales its terminal root by sqrt2.

Every case contradicts minimal indefiniteness. Therefore c has a nonzero retained contact. Reflection proves the same for d. The argument permits arbitrary supporting spectators and does not assume that the two minimal indefinite cores intersect.

A sharper uniform Schur inequality

Let x,y be the nonnegative contact-magnitude vectors against A21 and put J=J21, q=x^T J x, r=y^T J y and s=x^T J y. Each vector has a coordinate at least1. On the left set u_i=i+1, on the right set u_i=21-i, and put

```
alpha=17/352, U=sum_left x_i*u_i, V=sum_right y_i*u_i.
```

On either four-position block the matrix

```
C[i,j]=J[i,j]-alpha*u_i*u_j
```

is entrywise nonnegative and every diagonal is at least2. Thus one nonzero native coordinate gives x^T C x>=2, and similarly for y. The cross block of the path inverse factors exactly. Consequently

```
q-2 >= alpha*U^2 >0,
r-2 >= alpha*V^2 >0,
s = U*V/22.
```

Since c.d=0, eliminating A21 leaves the Schur matrix

```
[2-q, -s; -s, 2-r].
```

Its diagonal entries are negative and its determinant satisfies

```
(q-2)*(r-2)-s^2
  >= (alpha^2-1/22^2)*U^2*V^2
   = (3/11264)*U^2*V^2
  >= 1875/11264 >0,
```

where U,V>=5. It is negative definite, giving two negative directions in an actual Lorentz Gram matrix. This is impossible. The full-star short7 alternative therefore exists at feature depth8.

The source checks the A8/A21 inverses, all positive leading determinants, 32 residual block entries, eight diagonal lower bounds and16 cross factorizations. The bound applies to all nonnegative native magnitudes; it is not inferred from a finite contact sample.

The cycle interface has two distinct boundary domains

The inherited cycle interface and finite8 replacement give the complete single-root profile classification. Endpoint4 has a nonsimple bond at its new mark. Second-node3 has a fixed branch next to its new mark. Neither can give a changed deep finite8 pair. Endpoint3 on an arc of length8 is the only affected single-contact profile.

Its opposite old endpoint stays a whole-component leaf, at distance complementary_arc minus1 from the old mark. Thus the outputs for arcs (6,8),(7,8),(8,8) remain nondeep and unchanged. Only P=8,Q>=9 remains. If the endpoint3 root on the8-arc has no spectator neighbor, that new root is itself a leaf and its pair is also unchanged. An unpaid changed pair therefore forces an actual spectator neighbor.

At Q=9 a second neighbor requirement is inherited: endpoint3 on the 9-arc otherwise gives a whole-leaf9 payment. The complete residual lists are as follows. Subscript0 denotes the8-arc, and subscript1 the Q-arc.

Q	Residual profiles	Profiles requiring a spectator neighbor
9	isolated, end3_0, end3_1, both_ends3	end3_0, end3_1
10	isolated, end3_0, end3_1, end4_1, both_ends3	end3_0
>=11	isolated, end3_0, end3_1, end4_1, next3_1, both_ends3	end3_0

At Q=9, end4_1 is a sealed whole-leaf9 unit and next3_1 a nondeep distance8 unit. At Q=10, next3_1 is a sealed whole-leaf9 unit. Their whole finite components have ranks17 or18, so the separate large-D leaf9 reductions starting at rank20 do not affect them. These profiles must be discharged before taking the residual domain; retaining all six tail profiles at Q=9 or10 would give false nonpositive lower bounds. For Q>=11 all three long-arc distances Q,Q,Q-1 are at least10, so the six-profile list is constant.

The spectator argument is the inherited unequal-height one. Adjacent black contacts or a common spectator contact already pay through paths of length1 or2. Otherwise the spectator supports x,y are disjoint. For one required neighbor i of c, the norm-two spectator Gram G gives

```
(x/h-y/k)^T G^(-1)(x/h-y/k) >= x_i^2/(2h^2) >=1/(2h^2).
```

Use the c-side if it requires a neighbor, otherwise the d-side. The side is fixed for the entire reciprocal-height rectangle, even if both require neighbors. Two coordinate bounds are never added.

With A=1/8, B=1/Q, u=1/h, v=1/k, L=u*l1*+*v*m1 and M=u*l2*+*v*m2, the inherited normalized determinant is

```
1-2*(A*L+B*M)-(A+B)*(c0*u^2+d0*v^2)-A*B*(L-M)^2,
```

plus the nonnegative spectator term. The chosen Cauchy bound reduces that endpoint3 norm constant from1 to1/2. Corrected constants and moments are nonnegative. The lower bound is concave on the reciprocal-height rectangle and nonincreasing in B. Four rectangle vertices suffice for each ordered profile pair; Q=11 bounds the entire constant-list tail.

Region	Ordered pairs	Exact rectangle values	Strict minimum
Q=9	16	64	1/18
Q=10	25	100	1/10
Q>=11	36	144	1/11

All308 values are positive, contradicting the exact zero determinant of an unpaid actual star. No equality case remains. The source independently normalizes the eight inherited profile records in Q(sqrt2); the endpoint4 square roots cancel against their reciprocal heights. This scalar check does not purport to reprove the inherited geometric classification.

The other cycle regions are unchanged. The classical-tree short7 route uses finite paths at most7, persistent ideal features or sealed whole A7; the small-cycle routes have fixed nearby leaves or sealed small components. They never consume a newly deleted deep distance8 payment.

Composition with the current dimension61 rule

The precise baseline is the completed dimension61 rule, including all rank15/rank16 changes, the A17 restoration, D17 cost120, D13-zero coupon increments and actual E6/D11 coupon cost4. No historical cheaper ideal rule is silently substituted for it.

First handle the baseline's independent actual-corner priorities: C11 endpoint zero, B12 center/terminal zero, D13 center zero, cycle11 distance5 half, and the coupled order11 B/C/D halves. Their proofs are bound in the sparse composition, order11 composition and current ideal closure. Their outputs are paths of length at most7, retained whole affine center units, typed small whole-component coupons, whole finite components of rank at most14, whole A9/A18 endpoint8 pairs, or whole D18 leaf9 pairs. Endpoint8 pairs are nondeep, and the other outputs miss the new deletion. Distinct retained halves still add to1. These priorities remain independent of the generic short8 route.

In the remaining branch, consider the actual corners of the face. If none has a newly deleted finite pair, all its weights agree pointwise with the proved dimension61 rule and its coverage is unchanged. If a newly deleted corner occurs, the new corridor gives a full-star path of length at most7. At a finite corner it is an unchanged unit. At an ideal corner the established elementary, small-tree and protected-feature units or the complete cycle/tree replacements apply. The priority types above are absent on the whole face, so no alternative is asked to spend one of their removed units. The cycle and tree interfaces just proved supply the same retained payments under the depth8 rule.

The later order17 D zeros and order14 B/C or order16 B deletions have distance10 or11. They cannot remove a short7 output. The predecessor's 18 small classical tree targets have their short8 roles explicitly checked, with the later C11/B12/D13 exceptions already discharged by priority. For uses of the generic short8 route inside those later proofs, the new corridor replaces a deep finite8 output by short7, and the new Q=9/Q=10/tail cycle analysis replaces the old depth11 cycle interface. No invocation asks the deleted distance8 pair itself to pay.

Every recent finite rank15/rank16/D17 decrease has distance at least10; these and all existing coupons keep their exact current values. The new deep pairs require components larger than those small ranks in any event. The existing finite7/ideal5 triangle theorem, including protected ideal 6/7 and small whole-factor interfaces and the cycle11 triangle exclusion, uses no finite distance8 payment. Hence triangles still receive2 and quadrilaterals1 under the single simultaneous rule stated above.

Verification scope

Final generation a91fd9 and separate fresh replay9f838f both terminated exit0/PASS, obtaining corridor margin1875/11264 and the three cycle minima1/18,1/10,1/11. Two peer agents completed full source/report mathematical reads. Both checked the five missed-retained-contact cases, the disconnected K-minus-c extra white, the strict A8 equality exclusion, the shifted Schur bound, both small cycle domains, the fixed-side Cauchy correction and the constant-profile tail. They also reread the current ideal priorities and protected finite outputs. Both passed the simultaneous current61 composition and triangle interface without a mathematical correction. A prose typo was corrected. Neither reviewer reran the path matrices,308 scalar values or inherited classification/census. The source imports no project arithmetic module. Its exact checks certify the finite matrix/profile inequalities and bind the completed inputs; the all-label minimal-core order bound, profile classification, factor fibers and actual-corner geometry remain explicit written theorem inputs. No historical geometric census,

forest optimization or ideal convolution is repeated. No new dimension theorem, finite-volume example, external refereeing or proof-assistant formalization is asserted. Direct source bindings, Python syntax and local report links pass. Source, certificate and report are final/frozen after these completed checks and reviews.

```
python3 scripts/fresh_depth8_finite8_zero_transfer_20261018.py
python3 scripts/fresh_depth8_finite8_zero_transfer_20261018.py --verify
```

Source 70

Source path: `results/fresh_dimension225_hostile_20261015.md`

SHA256: 45b5115b8157a342015abb3338dcc609fa48f85cfd1d2b4b71161bfd01426d7d

Hostile audit of the narrow-facet dimension argument

The repaired argument passes this proof audit. The product-cusp branch gives $n \le 225$ when the chosen narrow facet contains an ideal vertex. The separate coarse compact-facet lemma gives $n \le 143$ if any facet is compact. Together with Nikulin's narrow-facet theorem, these establish $n \le 225$ for a finite-sided, finite-volume real hyperbolic Coxeter polytope, under the standard published affine and quasi-Lannér classification inputs stated below. No arithmeticity, simplicity at ideal vertices, or finite-label cutoff is assumed. This is an independent proof review and small exact certificate, not a novelty or current-best literature claim.

The initially suggested compact-facet bound $n \le 31$ did **not** follow from its proposed import. Vinberg's finer compact-chamber weights use global absence of parabolic subdiagrams. We found an exact counterexample to that localization premise. The replacement proof uses graph distance at most17 and permits affine diagrams; it does not invoke the unsupported finer transfer.

The reviewed producers are the product-cusp proof and compact-facet proof. The independent checker and saved data import no producer arithmetic. They independently reconstruct six exceptional affine energies by grounding a potential, verify the scale and rank identities, check the parity thresholds and distance counts, and certify the rational counterexample below. The geometric audit is given explicitly here; these finite calculations are not presented as a machine verification of every geometric statement or of the classification inputs.

Published hypotheses checked

Nikulin, ICM1986, printed pp.654–655 defines the relevant reflection groups using finite-volume chambers, including nonarithmetic groups. Theorem A supplies a facet whose framing hyperplanes have pairwise distance strictly below arccosh7. His subsequent discussion explicitly treats ideal vertices on that facet. Thus compactness of the ambient chamber is not an input to this narrow-facet theorem. For outward normals of square2, the strict hyperplane-distance bound contradicts a pairing ≤ -14.

Prokhorov, printed pp.407–408, Propositions1–2 supplies minimal indefinite/quasi-Lannér extraction and maximal diagram order10. His Proposition3 also records the spherical graph-distance pair count. We read these primary statements directly, rather than inferring them from a secondary dimension bound.

Vinberg, printed pp.70–71,76–77,81–88 supplies the simple-polytope counting input and spherical pair bound. His original fine-weight face proof invokes Proposition9, whose no-parabolic assertion assumes a compact ambient chamber. The new compact-facet proof replaces that step.

Product-cusp audit

The following points address the possible rank, scale and endpoint gaps.

1. **Full cusp rank and scales.** A finite-volume ideal link is a product of Euclidean Coxeter simplices. If the factor ranks are r_a, their sum is $N=n-1$. The positive affine combinations c_a are positive multiples $k_a c$ of one future null vector. For terminating walls x,y set $\sigma_a=-x\cdot c_a$ and $\tau_a=-y\cdot c_a$. The ratio $q=\sigma_a/\tau_a$ is independent of a. In the

quotient, dividing the factor difference by k_a and multiplying its energy by σ_aτ_a cancels k_a². Consequently the identity

$$-x\cdot y = (1/2)\Sigma_a\ \sigma_a\tau_a\ E_a - q - 1/q$$

uses no equal-scale or arithmetic assumption. The checker verifies this cancellation in independent null coordinates with separate symbolic factor scales.

2. **Actual outgoing edges and endpoints.** Omitting one node per affine factor gives a positive-definite support S of N walls and a vertex of the horospherical link. It determines an actual edge. Finite volume gives a finite or ideal second endpoint. An additional terminating wall is not another original cusp wall: a deleted original cusp wall is strict along the corresponding outgoing edge away from the original cusp. Acuteness and the full cusp span then give $\sigma_a,\tau_a>0$. Indeed, vanishing contact with one factor would make the common cusp pairing zero; nonnegative contact rows force all factor rows to vanish, leaving a norm-two vector in the null orthogonal complement of the full cusp span, which is impossible.

3. **The framing-wall choice covers ideal endpoints.** The retained N-node support lies in the full affine diagram at a second ideal endpoint. Each component of rank r contains at most r nodes of this positive-definite support. Equality of the totals forces exactly one missing node in each component. If the narrow-facet root lies in a component of rank at least2, its missing node has a finite angle with it. If its component is affine A1, choose a missing node of another component; the resulting pair is orthogonal. A second component exists in the relevant $n\geq 226$ range. Thus the chosen terminating wall has a positive-definite pair with the narrow-facet root. Projection of a strict interior point proves an actual ridge, so this is precisely a wall controlled by Nikulin's theorem.

4. **The four-contact budget is valid at both endpoint types.** The Schur complement of S in S+x is nonnegative, so $p_S^{\top} G_S^{-1} p_S \leq 2$. A positive-definite diagonal-two Z-matrix has a nonnegative inverse and inverse diagonal at least1/2. Every nonzero normalized Coxeter contact has magnitude at least1, including any parallel or dotted contact. Hence at most four retained entries are nonzero. The two walls spoil at most eight factors; marking the narrow-facet factor separately gives at most nine. No upper bound on the deleted-node pairing is needed.

5. **Omissions are selected before the contact rows.** In a non-narrow factor the fixed point-contact choices are retained even if the factor later proves spoiled. Low ranks $r\leq 9$ use only energy nonnegativity. In high ranks, only affine A/B/C/D occur. The closed inverse formulas give the asserted localization. For example, in modified B_r the left row is supported in nodes0,1,2, while the right row is in $r-3,r-2,r-1,r$; normalized coordinates3 through $r-3$ have fixed difference1/2, giving $r-5$ coordinates. The analogous C modification leaves coordinates4 through $r-2$ with fixed difference1. D has at least $r-4$ such coordinates, and the weaker $r-5$ estimate suffices. Positive normalized contact rows are convex combinations of the localized point rows, so the coordinates stay fixed for arbitrary allowed contact magnitudes.

6. **The cycle bound is uniform.** For A_r, write $m=r+1$. The two separating arcs have lengths $\mathrm{floor}(m/2)-4$ and $\mathrm{ceil}(m/2)-4$. Minimizing their constant-flow energy gives their product divided by their sum. For $r\geq 10$ this exceeds $(r-8)/4$: the surplus is1/4 when m is even and $(k-4)/(2(2k-7))$ when $m=2k+1$. The checker verifies the identities. Thus no sampled-rank inference is needed.

7. **Final normalization.** Unspoiled energy is at least $M(2r/9)$, where $M=\max(q,1/q)$. Each marked factor loses at most2M. The independently reconstructed exceptional scale-invariant costs are A1:1/2, E6:4/3, E7:2, E8:2, F4:1, G2:2/3. Summing and using $q+1/q\leq 2M$ gives $-x\cdot y\geq M(N/9-11)$. For $n\geq 226$, $N\geq 225$ and this is at least14. It contradicts the strict narrow-facet bound. This proves exactly the conditional product-cusp result.

A concrete obstruction to the original fine-weight import

Use the following Gram matrix, with each normal of square2:

```
G = [[ 2,-1, 0, 0],
     [-1, 2,-1,-1],
     [ 0,-1, 2,-1],
     [ 0,-1,-1, 2]].
```

It has determinant−3 and signature(3,1). Its inverse has diagonal $(0,-2,-4/3,-4/3)$, and every entry is nonpositive. The four vectors $v_i=-G^{-1}e_i$ are future causal, satisfy $v_i\cdot n_j=-\delta_{ij}$, and give one ideal and three finite simplex vertices. Their

sum is a strict center of square−22. Facet0 is the compact triangle with vertices v_1,v_2,v_3. Nevertheless, its three black nodes1,2,3 form affine A2. All proper principal diagrams are positive semidefinite; the whole four-node diagram is indefinite but not Lannér. Hence this compact triangle's star contains **no Lannér subdiagram**.

This explicitly disproves the attempted no-affine/Lannér-extraction localization. It does not disprove the numerical face-weight inequality: in this example the original three corner weights are each2. A new argument is necessary, and the following coarse one supplies it.

Hostile audit of the coarse compact-facet repair

Minimal indefinite diagrams have the required classification scope. Such a diagram is connected and every proper principal subdiagram is positive semidefinite. If a proper connected affine block A occurs, choose an adjacent outside node v. Its pairing with A's positive kernel is strictly negative. Restricting to that kernel and v gives an indefinite two-dimensional form. Minimality forces the whole diagram to be A+v; its coupling to the radical makes the full form nonsingular of signature(|L|−1,1). Any other proper affine block must likewise have codimension one. Thus neither zero eigenvalues nor an affine block with disconnected additional nodes evade the Lannér/quasi-Lannér classification. If all proper blocks are positive definite, the usual one-node Schur complement gives the same nonsingularity. The published order bound10 therefore applies.

The opposite-pair block really is indefinite. Let W be the n−2 white normals of a compact quadrilateral Q, and U a pair of opposite black sides. If W+U were positive definite, acute projection would give an actual finite common point of the opposite sides in Q. If it were singular positive semidefinite, a positive kernel of an affine component gives a nonzero future null combination c. Positivity of the whole block implies c is orthogonal to all of W+U; acuteness gives c·n≤0 for every other chamber wall. Thus c is an ideal point on Q, contradicting its compactness. Both alternatives are excluded. This reasoning uses global acuteness and compactness of Q, and permits cusps elsewhere in P.

The graph paths belong to actual corner diagrams. For a compact triangle, a minimal indefinite diagram contains all three black nodes and at most seven white nodes. Two pairs of black nodes have connecting paths avoiding the third; each path has at most eight edges. For a compact quadrilateral, minimal indefinite diagrams L⊂W+U and M⊂W+V each contain their opposite black pair and at most eight white nodes. They cannot be orthogonal in an index-one space. An overlap or connecting edge supplies a path between the two black sets; a shortest such path has only white internal nodes and length at most8+1+8=17. Its endpoints are consecutive sides of Q. These paths lie in the full positive-definite ambient vertex diagrams where the weights are counted.

Weight1 on ambient node pairs at distance at most17 therefore gives triangle sum≥2 and quadrilateral sum≥1. For k≥5, nonnegativity gives the required sum≥5−k. The spherical pair bound is17n at an ambient finite vertex. It also follows elementarily: components of order≤35 have at most binom(r,2)≤17r pairs; higher-rank spherical diagrams are paths or D forks, with at most r pairs at any specified distance1,…,17.

A compact facet F has dimension d=n−1 and is simple, since its vertices are finite vertices of P. Its two-faces are compact two-faces of P. Nonnegative weights on its angles count only a subset of all ambient vertex pairs, so the upper bound remains17n=17(d+1), rather than17d. The symmetric nonnegative h-vector gives average lower bounds d(d−6)/8 for even d and (d−1)(d−5)/8 for odd d, strictly. For d=143+2t and144+2t, subtracting17(d+1) gives respectively

$$(1/2)t^2+36t+3/2,$$
$$(1/2)t^2+(73/2)t+19.$$

Both are positive for t≥0. Thus a compact facet forces n≤143. All constants and the rank shift have been independently checked.

Exhaustiveness of the dichotomy and conclusion

A finite-volume polytope is the convex hull of finitely many finite or ideal vertices. A facet without ideal vertices is compact. Apply Nikulin's theorem to choose a narrow facet. If it has an ideal vertex, the audited product-cusp argument gives n≤225. Otherwise the audited coarse compact-facet argument gives n≤143. The cases exhaust the chosen facet. Therefore their combination yields n≤225 under the stated finite-sided, finite-volume Coxeter hypotheses.

The audit does not promote a fixed-lattice or fixed-source obstruction into a dimension theorem: neither proof fixes a field, lattice, source polytope, facet count, or cusp factor count. It does rely on the published narrow-facet theorem, affine Coxeter classification, minimal-indefinite order bound, and simple-polytope h-vector facts. Broader literature priority and publication status are separate from correctness of this derivation.

```
python scripts/fresh_dimension225_hostile_20261015.py
python scripts/fresh_dimension225_hostile_20261015.py --verify
```

Source 71

Source path: `results/fresh_distance10_extension_20261017.md`

SHA256: ca25a1a09f0400f94a1a83dc8dfcf6f379e7700f29a1d3b67beb8b94242e0084

Bounded supporting extensions for two distance-10 overlap cores

For the two overlap families below, every actual supporting-white extension leaves a preferred marked corner component of order at most **17**. Both families permit nonzero Euclidean residuals; their supporting diagrams need not be isolated unchanged. The proof controls arbitrarily many additional whites through a residual norm budget, not a finite search cutoff.

This is a local result under the assumption that no path of length at most 10 joins the two marked opposite pairs with white internal nodes. It is not a complete distance-10 classification, an ambient dimension bound or a finite-volume realization claim. The exact checker and certificate use the literal L22 and L25 diagrams from the frozen census.

Common geometric setup and residual budget

All normals have square 2. Let G be the Gram matrix of one of the stated nonsingular Lorentzian cores. The actual supporting white diagram W is positive definite; W plus any one black root is positive definite; W plus an adjacent black pair is positive semidefinite. These are the actual face-star hypotheses used in the completed distance-11 proof.

For an additional supporting white e with core contacts b, orthogonal projection onto the core span has square `b^T G^-1 b`. The core contains a timelike vector, so its orthogonal complement is positive definite. Write

$$e = e_{\mathrm{core}} + w, \qquad w^2 = 2 - b^T G^{-1} b \geq 0.$$

At a preferred corner, the nonnegative inverse of each finite component restricts a new nonzero contact to inverse diagonal at most 2. The magnitude of such a norm-two Coxeter contact is at least 1. The positive kernel of an affine component forces every extra contact there to vanish. The omitted black nodes are allowed separately. These tests give the explicit end domains below.

We will use the following elementary fact. Suppose at most one supporting white e contacts the core, its residual has square ρ, and all remaining supporting whites are core-orthogonal. Their Gram matrix E is a positive-definite principal submatrix of W. If c is the contact vector of e with those roots, Euclidean projection gives

$$c^T E^{-1} c \leq \rho.$$

Each contacted connected component costs at least 1/2: projecting onto any one contacted square-two root already costs at least 1/2. Hence $\rho \leq 2/3$ permits at most one contacted component.

If that connected component has at least three nodes, choose a connected three-node principal subdiagram through a contacted node. When the node is an end of a path with bond magnitudes a,b≥1 and possible extra contact c≥0, its inverse diagonal is

$$\frac{4 - b^2}{8 - 2(a^2 + b^2 + c^2) - 2abc} \geq \frac{3}{4}.$$

The denominator is positive. After clearing it, the inequality becomes `6a²+2b²+6c²+6abc−8>=0`, immediate from a,b≥1. If the contacted node is the middle of the path, the corresponding difference is `6a²+6b²+2c²+6abc−8>=0`. Projection onto

a larger positive-definite component cannot cost less than projection onto this three-node subspace. Thus a contacted component of rank at least three costs at least 3/4.

For a two-node connected component with bond magnitude $t \geq 1$, the endpoint inverse diagonal is `2/(4-t²)>=2/3`. Equality requires t=1. If both nodes are contacted, positivity of the inverse gives a larger cost; at the minimal simple diagram the cost is already 2. Therefore:

- if $\rho = 2/3$, the only nonempty attached component is A1 or A2, all contacts simple, and e meets only one end of A2;
- if $\rho = 1/2$, only one A1 with a simple contact is possible.

This bounds the **whole** attached component, including roots connected indirectly to e. Every other component is orthogonal to both e and the core. No rank cutoff on the ambient supporting diagram was used.

Four-node reversed L25 overlap

Use the norm-two L25 edges

`0-1, 0-2, 2-3, 0-4, 4-5, 5-6, 6-7, 7-8, 8-9,`

all labelled 3. The second copy has map

`(0,1,2,3,4,5,6,7,8,9) -> (10,11,12,13,14,15,9,8,7,6).`

The resulting 16-root Gram has determinant -3 and signature (15,1). Deleting centers 0 and 10 leaves a positive-definite block and Schur matrix

$$\begin{pmatrix} -1/18 & -1/9 \\ -1/9 & -1/18 \end{pmatrix}.$$

Each black pair is one of {1,2}, {1,3}, {2,3} in its own copy. Choose preferred root 2, 1, 2 respectively. In all nine contexts the preferred marked component has order 14. Its finite inverse or affine kernel restricts contacts to the end sets {1,2,3} and {11,12,13}. A root contacting both ends would give a marked path of length at most 4. The exact checker also confirms that any unspecified direct cross edge between the exclusive core parts would give a path of length at most 10. The core Gram is consequently fixed under the hypothesis.

The full inverse block on either end is

$$\begin{pmatrix} 2 & 2 & 1 \\ 2 & 10/3 & 5/3 \\ 1 & 5/3 & 4/3 \end{pmatrix}.$$

All entries are positive. The norm budget and the Coxeter contact alphabet therefore leave only two one-end types:

- **A:** a sole contact -1 at node 1 (or 11), projection square 2 and zero residual;
- **B:** a sole contact -1 at node 3 (or 13), projection square 4/3 and residual square 2/3.

Node 2 is impossible because its inverse diagonal exceeds 2. Two nonzero contacts at one end already cost more than 2. The next possible single-contact magnitude after 1 is sqrt(2), which also exceeds the budget at either remaining node. Some contexts eliminate type A already at the preferred-corner test; retaining it in the following analysis is conservative.

The cross and same-end inverse entries give complete control of multiple core-contacting whites. The A root is uniquely in the core span. Its product with a same-end B root is 1, contradicting acuteness; with an opposite-end B root it is -2, contradicting positive definiteness of W. Opposite A roots have product -3, also impossible in W. A second copy of the same A root cannot occur in a positive-definite supporting diagram. Thus an A root excludes every other core-contacting white. It is automatically orthogonal to all core-orthogonal whites and adds at most one node.

For two B roots at the same end, Cauchy in the positive residual space gives product at least `4/3-2/3=2/3`, violating acuteness. For B roots at opposite ends, the product is at most `-4/3+2/3=-2/3`, so there is a nonright edge between them. That edge gives a marked white path of length at most 5, forbidden by hypothesis. Hence there is at most one B root in total.

Apply the residual lemma with $\rho = 2/3$. Only an A1 or A2 tail can attach to that B root, with all bonds simple. At most three new connected roots occur: B plus at most two core-orthogonal whites. The marked component therefore has order at most `14+3=17`. Other supporting whites cannot connect to it.

The checker verifies all allowed local variants: six context-permitted A choices and 54 B choices with tail lengths 0, 1 or 2. Every supporting edge star remains positive definite and all four adjacent corners remain positive semidefinite. Six maximal-tail variants attain marked component order 17 at this local Gram level; no finite-volume chamber realization is inferred.

Reversed L22/L25 terminal edge

Number L22 by 0 through 8 and map the L25 nodes by

`(0,1,2,3,4,5,6,7,8,9) -> (9,10,11,12,13,14,15,16,8,7).`

The 17-root Gram has determinant -2 and signature (16,1). Its two-center Schur matrix is

$$\begin{pmatrix} -5/36 & -1/9 \\ -1/9 & -1/18 \end{pmatrix}.$$

Use the same three black-pair choices and preferred roots. The nine contexts cover both orientations after exchanging the cores. Preferred marked components have order at most 15; a detached node 4 in some contexts is retained in the contact analysis. The allowable contact domains are contained in {1,3,4} on the L22 end and {10,11,12} on the L25 end. Contacts at both ends produce a marked path of length at most 5. Every extra direct cross edge gives a path of length at most 10.

The inverse block on the L22 end is

$$\begin{pmatrix} 7/2 & 3 & 3/2 \\ 3 & 4 & 2 \\ 3/2 & 2 & 3/2 \end{pmatrix},$$

and the L25 end has strictly positive entries and diagonal minimum 4. Thus no L25-end contact is possible. On the L22 end the only possibility is a sole contact -1 at node 4: projection square 3/2, residual square 1/2. Every other node, multiple contact or larger single Coxeter contact exceeds the norm budget.

Two roots of this type would have product at least `3/2-1/2=1`, violating acuteness. There is therefore at most one. The residual lemma with $\rho = 1/2$ permits only one attached A1. At most two new connected whites occur, and the marked component has order at most `15+2=17`.

All 18 local variants, with tail length 0 or 1, pass the exact supporting-edge and adjacent-corner checks. Three attain marked component order 17. In contexts where node 4 is detached from the marked component, its extension does not enlarge that marked component.

Actual ideal corners and validation scope

At an actual ideal corner, the factor-fiber incidence lemma places the varying affine factor inside W plus the two side roots. The foregoing bounded connected component is therefore the component relevant to the actual face payment; additional cusp walls outside that star cannot enlarge it. The supporting-diagram argument uses W positive definite throughout and does not assume the entire ambient cusp is simple.

Exact generation passed: 18 preferred contexts and 78 permitted extension variants, with exact inverse blocks, signatures, residual endpoint Grams, all supporting-edge positive-definiteness checks and all adjacent-corner semidefinite checks. The general residual bound is a written proof using every indirect supporting root, not a claim extrapolated from those 78 variants. Generation and a fresh saved replay both passed. Seven direct source hashes, Python syntax and all local report links passed. The parent independently read the full mathematical argument and found no gap; that review did not duplicate the finite matrix calculations. Source, data and report are final/frozen.

Source 72

Source path: `results/fresh_distance11_extension_review_20261017.md`

SHA256: d98a23239035c55f533da84328d0f43d4254d80218345b00b6aae2e759584b18

Supporting-white extensions of the three-node L25 overlap

This is an exact local lemma for one remaining distance-11 configuration. It does not establish a complete distance-11 payment rule or a new dimension bound. All roots below have square 2 and all diagrams are principal diagrams of an actual acute Coxeter face star.

Let two copies of the frozen census diagram L25 share their three terminal nodes in reverse order. Its edges are

`0–1, 0–2, 2–3, 0–4, 4–5, 5–6, 6–7, 7–8, 8–9,`

all labelled 3. The second copy has the node map

`(0,1,2,3,4,5,6,7,8,9) -> (10,11,12,13,14,15,16,9,8,7).`

Assume the two pairs of opposite black roots have no connecting path of length at most 11 whose internal nodes are white. Then every additional supporting white root is orthogonal to this 17-root core, with at most one exception. The exceptional root has contact -1 with node 3 alone, or with node 13 alone. In either case it lies in the core span. A preferred adjacent corner therefore has an isolated marked component of order at most **16**, even after all supporting whites are included.

Exact core and the nine marked contexts

The checker reads only the literal L25 diagram from the frozen exact census; it imports no producer arithmetic. Deleting nodes 0 and 10 gives a positive definite 15-root block. Its Schur complement is

$$\begin{pmatrix} -1/15 & -1/10 \\ -1/10 & -1/15 \end{pmatrix}.$$

Thus the full Gram matrix G is nonsingular of signature $(16, 1)$, with determinant -2.

The checker examines all $\binom{7}{2}^2 = 441$ choices of marked pairs outside the common path. Exactly nine have no marked path of length at most 11: each pair is one of $\{1,2\},\{1,3\},\{2,3\}$, with the second pair translated by 10. All four adjacent corner diagrams in each case are positive semidefinite. An unspecified extra edge between the two exclusive core parts always creates a path of length at most 11, so the displayed core has no unexamined free cross contacts under this hypothesis.

Choose the preferred marked node 2 for the pair $\{1,2\}$, node 1 for $\{1,3\}$, and node 2 for $\{2,3\}$; use the translated choice on the right. Delete the other black node on each side. The resulting preferred corner has a connected marked component of order 15 and marked distance 12.

For each finite component of this corner, the checker computes its inverse diagonal. A new square-two root cannot have a nonzero Coxeter contact at a node whose inverse diagonal exceeds 2: the squared projection already exceeds its norm, since the inverse of a spherical Coxeter Gram matrix is entrywise nonnegative. For an affine component, its positive kernel forces every contact of a further root to vanish whenever the extended corner is positive semidefinite. After also allowing contacts at the two omitted black roots, the possible contact domain is contained in

$$E_L = \{1, 2, 3\}, \qquad E_R = \{11, 12, 13\}.$$

Every allowed node is within one edge of its own marked pair. A root contacting both ends would therefore produce a marked white path of length at most 4. Every nonorthogonal supporting white is consequently supported at just one end.

The only two one-end directions

On either end the corresponding block of G^{-1} is

$$J_E = \begin{pmatrix} 7/2 & 4 & 2 \\ 4 & 6 & 3 \\ 2 & 3 & 2 \end{pmatrix}.$$

Let b be the nonpositive vector of native contacts of a proposed root with the core. Every nonzero magnitude is at least 1. The core span contains a timelike vector, so its orthogonal complement is positive definite. Projection therefore gives the necessary inequality

$$b^T G^{-1} b \le 2.$$

All entries of J_E are strictly positive. Its first two diagonal entries exceed 2, and its last is 2. Hence the only nonzero one-end possibilities are exactly $b = -e_3$ and $b = -e_{13}$. They attain equality, so their orthogonal residual is zero: the actual roots are uniquely

$$r = -G^{-1} e_3, \qquad s = -G^{-1} e_{13}.$$

Their coordinates in the ordered core basis are

```
r = (-4,-2,-3,-2,-3,-2,-1,0,1,2,6,3,4,2,5,4,3),
s = (6,3,4,2,5,4,3,2,1,0,-4,-2,-3,-2,-3,-2,-1).
```

Direct exact multiplication gives $r^2 = s^2 = 2$, the prescribed sole contacts, and

$$(r, s) = -2.$$

Both cannot be supporting roots of the quadrilateral: the supporting diagram W is positive definite, whereas their two-root Gram matrix is singular. A second root in the same direction is also impossible for distinct normalized facet normals. Every remaining supporting white that is orthogonal to the core is automatically orthogonal to the exceptional root, because that root lies in the core span. Thus there is at most one extra node in the marked component, and no indirect extension through other whites.

Corner and orientation checks

For all nine contexts and either one-root choice, the checker verifies positive definiteness of the supporting white diagram and positive semidefiniteness of all four adjacent corners. Among the 18 preferred-corner variants the marked component is finite of order 16 in six cases, finite of order 15 in four, and affine of order 15 in eight. An extra node disconnected from that component does not enlarge it.

These are feasible local acute contact sets. For example $x = -G^{-1}\mathbf{1}$ has square $-1001/2$, contact -1 with every core root and contact -13 with each of r, s. This establishes coherent orientation for the local calculation; it is not a finite-volume realization claim.

For an actual ideal corner, the factor-fiber lemma identifies its varying affine factor inside W plus the two side roots. Additional incident walls outside that corner star do not enlarge the marked component relevant to the face payment. This is the same geometric scope used in the frozen distance-12 proof and its independent review.

Verification scope

Exact generation and a fresh saved replay both passed. The saved certificate records all nine marked contexts, exact preferred-corner contact restrictions, both unique roots, and all 18 extension variants. The arithmetic uses only Python integers and `Fraction`. The six direct source hashes, Python syntax and all local report links passed. The parent independently read the mathematical argument and found no gap; that review did not duplicate the exact corner census. No claim is made here about the other distance-11 core overlaps or an ambient dimension bound.

Source 73

Source path: `results/fresh_distance11_extension_review_20261017_mixed.md`

SHA256: b0ba8a1f46eedcb45240cef78058600f40305c07e5f413c752a5463a9216eeeb

Radical isolation of the mixed L22/L25 overlap

In each of the nine marked contexts below, the mixed L22/L25 core is orthogonal to every additional supporting white root whenever there is no marked white path of length at most 11. A preferred adjacent corner consequently has an isolated marked component of order at most **16**. This is a local lemma; the classification establishing that these are the surviving mixed contexts is outside this certificate, as is any new ambient dimension bound.

Use the literal diagrams from the frozen quasi-Lannér census. Number L22 by 0 through 8, and map the L25 nodes by

```
(0,1,2,3,4,5,6,7,8,9) -> (9,10,11,12,13,14,15,16,17,8).
```

Thus L22 node 8 and L25 node 9 are shared. The left black pair is one of $\{1,2\},\{1,3\},\{2,3\}$; the right pair is the image of one of those three L25 pairs. These nine cases also cover the nine reverse orientations after exchanging the copies. Every additional cross bond between their exclusive parts creates a marked path of length at most 11, checked independently for all nine contexts. Hence, under the no-short-path hypothesis, there is no free cross contact in the following core Gram matrix.

The actual linear relation

The exact checker deletes the two fork centers 0 and 9. The other 16 roots have a positive definite Gram matrix; the two-center Schur complement is

$$\begin{pmatrix} -3/20 & -1/10 \\ -1/10 & -1/15 \end{pmatrix}.$$

It is negative semidefinite of rank one. The 18-root Gram matrix therefore has inertia $(16,1,1)$. Its radical is spanned by

```
z = (-4,-2,-3,-2,-1,-3,-2,-1,0,6,3,4,2,5,4,3,2,1).
```

This Gram relation is an actual relation among the ambient normals. Indeed, the corresponding linear combination is orthogonal to the core span, which contains a timelike vector, and has square zero. The orthogonal complement of that timelike vector is positive definite, so the combination vanishes.

Preferred corners and supporting contacts

Choose node 2 from $\{1,2\}$, node 1 from $\{1,3\}$, and node 2 from $\{2,3\}$, on each copy. Delete the other marked root on each side. All nine preferred corners are positive semidefinite, and their marked roots lie in one connected component. There are three finite components of order 16, two finite components of order 15, and four affine components of order 15.

Every proposed extra supporting white belongs to the full adjacent-corner star. For each finite component of the preferred corner, positivity and the entrywise nonnegative inverse restrict a nonzero contact to inverse-diagonal entries at most 2, since every nonzero norm-two Coxeter contact has magnitude at least 1. For each affine component, its strictly positive kernel forces all extra contacts to that component to be zero. The omitted black roots are included separately in the allowed contact domain.

The certificate gives the exact inverse diagonals or positive kernel for every component and records the resulting domain. In every case:

- the zero-radical node 8 is excluded from that domain;
- each permitted negative-radical node is within distance at most 2 of the left black pair;
- each permitted positive-radical node is within distance at most 1 of the right black pair.

Let e be an additional supporting white, with nonpositive core contacts $b_i = (e, r_i)$. The actual relation gives $\sum_i z_i b_i = 0$. If any contact is nonzero, the contacts must include both a negative-z node and a positive-z node. The resulting marked path through e has length at most

$$2 + 1 + 1 + 1 = 5,$$

contrary to the hypothesis. Thus every additional supporting white is orthogonal to the entire core. There can be no indirect extension of the preferred marked component either, because the first connecting root would already have a nonzero core contact.

The actual corner-factor statement supplies the same interpretation at ideal corners: the varying affine factor is contained in the supporting white diagram plus the two side roots. Hidden incident walls elsewhere in the cusp do not enlarge this marked component.

Verification and scope

Generation and a fresh saved replay both passed. The certificate records the exact radical and Schur matrix, all nine preferred-corner contact domains, positive definiteness of each supporting white diagram with one black root, and positive semidefiniteness of all 36 adjacent corners. It also checks every possible additional exclusive-part cross bond for a short path.

The checker uses the frozen standalone `Fraction` helpers from the three-node-overlap certificate, without numerical linear algebra or producer imports. Six direct source hashes, Python syntax and five report links passed. The exact-hub agent independently read the written proof and source and found no gap; this was a conceptual/source review without a duplicate arithmetic replay. This package makes no finite-volume realization or full distance-11 classification claim.

Source 74

Source path: `results/fresh_distance14_unrestricted_dimension156_20261017.md`

SHA256: 72f0eeb03af05848ba50ec00a201f4163000b43a182514865ac17fcfd4444aa3

An unrestricted dimension156 bound from distance14 payments

Complete internally checked computer-assisted proof. Every finite-sided, finite-volume real hyperbolic Coxeter polytope has dimension at most156. There is no arithmetic, compactness, cusp-simplicity, minimum-factor-rank, factor-count or facet-count hypothesis. The proof combines the inherited bound163 with the new exclusion of dimensions 157 through163. No new finite-volume example is constructed.

Two ingredients are new: a distance14 quadrilateral payment and a convex-order reduction of arbitrary cusp-rank partitions to a small, explicit family. The argument uses no unproved fine-payment extension at small cusp factors.

The radius-seven obstruction cannot occur on both sides

Use the actual two-face star W plus its side roots. As in the distance15 proof, W plus one side is positive definite, and W plus an adjacent pair is positive semidefinite, including ideal corners with nonsimple ambient cusps. Opposite closed sides are disjoint, so W plus either opposite pair is indefinite. These are actual-face facts, not assumptions about an arbitrary abstract diagram.

Let U={u1,u2}, V={v1,v2} be opposite pairs. Choose inclusion-minimal indefinite diagrams L in W+U and M in W+V. Both marked deletions of each core are positive definite because they lie in actual edge supports. The frozen complete minimal-diagram classification and its 384 finite-marked-pair calculations show that every core node has distance at most7 from its marked pair. Orders at most5 have the stronger elementary radius3 bound. At radius7 the only possibilities are the following, with labels from the frozen catalogue:

Core	Black pair	Unique farthest node
L23	0,1	8
L25	1,2 or1,3 or2,3	9

Suppose no adjacent black pair in the quadrilateral can be joined by a white path of length at most14. Then L and M are disjoint: a shared white node would give a path of length at most7+7. They cannot be orthogonal because both are indefinite and the ambient negative index is one. Any bond between them must join nodes of radius7 on both sides; otherwise it gives a white path of length at most14. There is only one radius7 node in each listed case, so the unique possible cross bond joins these two farthest nodes. All other cross products vanish.

The L23 case is impossible even before considering the other core. In its finite deletion of black0, the inverse Gram diagonal at node8 is9/2 (and after deleting black1 it is4). An exterior root of square2 with a nonright contact there has magnitude at least1. Entrywise nonnegativity of the finite inverse makes its Schur cost at least9/2, contradicting positive semidefiniteness. The exterior farthest root is white, so this extension is in fact contained in an actual finite edge diagram. Additional nonpositive contacts can only increase this cost.

It remains to join two literal L25 diagrams at their respective node9. Each has a simple A3 tail7–8–9. These two tails belong to an actual corner and have no other cross bonds. Eliminating the two interior nodes on each side leaves diagonal4/3 at each proposed bridge endpoint. The bridge magnitude must therefore be at most4/3. A nonright Coxeter product is1 or at

least sqrt2, so the bridge has label3, magnitude1. This includes all finite labels, parallel and dotted possibilities; there is no angle cutoff.

The resulting20-node diagram has negative index two. Each L25 has a fork center0 with short arms of lengths1 and2 and a long arm of length6. Eliminating the short arms leaves effective diagonal5/6 at each fork center. The two centers are joined by a13-edge simple path. Eliminating its12 internal nodes leaves

```
[ -7/78  -1/13 ]
[ -1/13  -7/78 ],
```

whose eigenvalues are−1/6 and−1/78. Every eliminated block is positive definite. Thus the actual star would have two negative directions, which is impossible. The checker also derives this Schur complement directly from the literal20-root Gram.

Consequently every quadrilateral has an adjacent black pair connected by a white path of length at most14. The triangle argument is unchanged: a minimal indefinite core has order at most10 and contains all three black roots, giving two marked paths of length at most8 in the corresponding actual corners.

Unrestricted vertex costs

Give weight1 to a corner whose marked roots have graph distance at most14 in its full finite or affine vertex diagram; give weight0 otherwise. The paths just proved occur inside the actual corner diagrams, so triangles receive at least2 and quadrilaterals at least1. Larger polygons require only nonnegative payment. Extra factors at an ideal vertex do not change the marked pair's varying factor.

The all-rank classical graph formulas bound pairs at each distance by the component order; small exceptional types satisfy the bound directly. Thus a finite vertex costs at most14n. For an affine factor of rank r and order r+1, both the distance bound 14(r+1) and the total-pair bound r(r+1)/2 apply. The actual cusp-fiber multiplicity therefore gives, for a cusp link with V=product(r_i+1),

```
cost/V <= sum_i min(14,r_i/2).
```

This holds for every irreducible affine type and any number of factors.

Concentrating rank partitions without changing this cost

For centered discrete uniforms X_r, the shape identity is

```
kappa/V = 3d/4−(5/2)E|sum_i X_(r_i)|,    d=sum_i r_i.
```

For integers1<=a<=b, replacing (a,b) by(a−1,b+1) increases their sum in convex order. Here is a direct all-rank proof. Before centering, the probability at j of U_a+U_b, with each U_r uniform on0,…,r, is

```
min(j+1,a+1,b+1,a+b−j+1)/[(a+1)(b+1)].
```

After the replacement the denominator is a(b+2), smaller by b−a+1. The new probability is strictly larger on the tails jb, and strictly smaller on the central interval a<=j<=b. On that interval the probabilities are respectively1/(b+2) and1/(b+1). Both distributions are symmetric about the same mean(a+b)/2 and have total mass one. Symmetrizing any convex test function makes it nondecreasing with distance from that mean. Subtracting its value at the crossing radius proves that the change in expectation is nonnegative. This proves convex order, including a=1, where X_0 is the constant zero.

Adding the other independent uniforms preserves the comparison, by applying it to the convex function x -> E|x+Y|. Consequently the replacement decreases normalized curvature at fixed d.

Apply this transfer only to two ranks strictly below28, stopping when one becomes0 or the other28, or to two ranks strictly above28, stopping when the smaller becomes28. In each case sum min(14,r_i/2) is unchanged. Ranks0 can be discarded. Repeating terminates, for example because the integer sum of squared ranks strictly increases while d stays fixed. Every initial rank partition therefore has curvature-minus-cost at least that of a profile

```
(28,...,28), optionally one a with1<=a<=27,
optionally one b>=29.
```

There is at most one rank of each optional kind. This is a complete comparison for all partitions, not a census of sampled cusp shapes.

Exact exclusion of dimensions157 through163

For each d=156,…,162 the checker enumerates every profile of the displayed reduced form with sum d. It computes the exact curvature from polynomial convolution. A separate inclusion-exclusion formula for the truncated first moment independently verifies each result. Every reduced profile satisfies

```
kappa/V > sum_i min(14,r_i/2).
```

The smallest margin over the entire checked range is 1476616/20511149, attained at d=156 with ranks(28,28,28,28,28,16). Every other rank partition inherits the inequality from concentration. Also beta_n>14n throughout n=157,…,163.

The actual face payments give

```
D5 <= 14n v_f + sum_ideal_v V_v sum_i min(14,r_vi/2).
```

The unrestricted shape theorem gives

```
D5 >= beta_n v_f + sum_ideal_v kappa(L_v) + 5(n−1).
```

These inequalities contradict each other for every n in that range. The already completed unrestricted163 proof excludes all higher dimensions. Hence n<=156.

The distance14 comparison alone does not exclude n=156: at d=155, the local link profile(28,28,28,28,28,15) has margin−4172075/82044596. This is a limitation of the payment comparison, not an ambient example or a claim that the dimension156 bound is sharp.

Verification scope

The checker and certificate bind the frozen catalogue, the radius proof, actual factor geometry, unrestricted shape inequality and inherited163 proof. New arithmetic checks the forbidden Gram configurations, the all-rank probability identities and every concentrated profile in the necessary seven dimensions. No previous large catalogue is re-enumerated.

Generation and a fresh saved replay pass all924 concentrated profiles, the physical Gram obstructions and direct dependency hashes. Separate construction-agent and frontier-agent mathematical/source reads pass the actual-face geometry, unique farthest-node bridge, all-label bridge gate, convex-order transfer and complete partition reduction. Neither read claims to repeat the inherited classification census.

The independent review records a separately implemented rational-arithmetic check of the literal Gram obstructions and all924 reduced profiles, together with its saved replay and written mathematical review. Its arithmetic does not import this producer's implementation. The written geometric steps and convex-order proof are not formally verified by source-integrity checks. No unresolved step is currently identified. External refereeing, proof-assistant formalization and publication-priority claims remain unestablished.

Source 75

Source path: `results/fresh_distance14_unrestricted_dimension156_20261017_review.md`

SHA256: 36dc27a4ab25624aba51284d37cd81184332a89390b6d98789a2d22457a38e95

Independent review of the unrestricted dimension156 proof

Mathematical and source review PASS: no gap identified. The primary proof excludes dimensions157 through163 for every finite-sided finite-volume real hyperbolic Coxeter polytope. Together with the inherited unrestricted163 theorem, this proves dimension at most156. No arithmetic, compactness, simple-cusp, minimum-rank, factor-count or facet-count hypothesis is used.

This review independently checks the new physical Gram obstructions, the graph-radius cases, the convex-order reduction and all924 reduced cusp profiles using a separate Fraction-only checker. Its certificate binds the final producer source, data and report. It imports no producer arithmetic and does not use SymPy. The complete high-rank minimal-diagram classification, actual cusp-fiber interpretation and unrestricted shape theorem remain explicitly inherited proof inputs; their large censuses are not repeated here.

Distance14 at actual polygon corners

The actual-face incidence argument extends to ideal corners without assuming that the ambient cusp is simple. An edge interior is hyperbolic, so W plus one side is positive definite. Opposite closed sides of a genuine quadrilateral have no common finite or ideal point. The acute projection and positive affine-kernel arguments therefore exclude PSD for either opposite block. Each inclusion-minimal indefinite core in that block contains both black roots, and both black deletions are PD. This is exactly the marked domain of the inherited catalogue.

The reviewer separately recomputes384 marked graph-distance arrays from the catalogue edges, finding the same four radius-seven cases and their unique farthest nodes. If no white path of length at most14 joins the two opposite black sets, the minimal cores are disjoint. A shared white node would give a path of length at most14. Two disjoint orthogonal indefinite cores would already contradict ambient negative index one. Every cross bond must join radius-seven nodes, since either smaller distance yields a path of length at most14. Thus the only possible cross bond is between the two unique farthest nodes; all other cross products are zero. Choosing a shortest marked path removes any internal black node, so a surviving path lies in an actual adjacent corner diagram.

For L23 the reviewer uses a rational congruent Gram: scale node8 by sqrt2, changing its squared norm to4 and its last bond to-2. The inverse diagonal in the original norm2 normalization is twice the rational inverse diagonal. Deleting black0 or1 gives respectively9/2 and4. The exterior farthest node is white and would extend a finite edge-support subdiagram with a nonright product of magnitude at least1. Its Schur residual is negative. This excludes L23 without a field or label restriction and without relying on a common-center choice for the hypothetical new node.

For two L25 cores, their A3 tails7–8–9 consist entirely of white nodes. The union of the tails is therefore PD. Eliminating the two inner nodes on each side leaves diagonal4/3 at the far endpoints. A bridge of magnitude at least sqrt2 is impossible; any nonright Coxeter bridge must consequently have label3. Dotted and parallel bridges are covered by the same inequality.

The literal joined20-root Gram is reconstructed separately. Removing the two fork centers leaves the positive18-root block A12+2A2+2A1. Rational elimination gives the Schur complement

```
[[-7/78, -1/13], [-1/13, -7/78]],
```

with two negative eigenvalues$-1/6$ and$-1/78$. Thus this last possible distance15 configuration also cannot occur. The triangle argument continues to give two paths of length at most8; no new claim about maximal-simplex incidence at ideal

vertices is involved.

Costs at finite and arbitrary ideal vertices

The inherited all-rank finite and affine graph formulas bound the distance-at-most14 pair count by14 times component order. The small exceptional diagrams have at most4 times component order pairs in total, which is enough here. Finite forests therefore cost at most14n.

At an arbitrary cusp, a paid marked pair lies in one affine factor. For a factor of rank r and order r+1, its number of paid pairs is at most min(14(r+1),r(r+1)/2). The actual factor-fiber multiplicity divides this by r+1 when normalized by the link vertex count V. This gives exactly sum min(14,r/2). It does not substitute total incident wall count n for a nonsimple cusp, or charge pairs from different affine factors as though they were adjacent.

Concentration and complete partition coverage

The transfer(a,b) to(a−1,b+1), 1<=a<=b, preserves the total rank and mean. On the common integer support0,…,a+b, the new-minus-old mass is positive for jb and negative for a<=j<=b. On the central interval the difference is−1/((b+1)(b+2)); on either tail it is the old rising or falling numerator times (b−a+1)/[(a+1)(b+1)a(b+2)]. These formulas are checked independently for all6,561 positive pairs with a+b<=162, which includes every transfer needed in the target dimensions.

The same formulas prove the all-rank convex-order statement. Center the support and take the chord of an arbitrary convex test function through the points ±(b−a+1)/2. The function lies below that chord on the negative-mass central interval and above its extension on the positive-mass tails. Mass and first moment cancel the chord, so the expectation change is nonnegative. Convolution with the other independent uniforms preserves this comparison. Hence normalized curvature cannot increase under the transfer.

The cost-preserving endpoints matter. For low factors the receiving rank must start below28: use1<=a<=b<=27 and stop at0 or28. For high factors both start above28 and the smaller stops at28. Thus the cost sum min(14,r/2) is unchanged at every step. Strict increase of the integer sum of squared ranks proves termination. Only repeated28s, at most one rank1..27, and at most one rank>=29 can remain. Zero ranks contribute a constant uniform and zero cost and may be dropped. This proves domination for every initial rank partition, regardless of its number of factors.

The independent enumeration chooses the optional large rank first; Euclidean division of the remaining total by28 uniquely supplies the repeated blocks and small remainder. Counts for link dimensions 156 through162 are129,130,131,132,133,134,135, totaling924. Integer sliding-window convolution is checked coefficient by coefficient against an independent bounded-composition inclusion-exclusion formula. All curvatures, costs and margins agree with the final producer certificate. The smallest margin is

```
1476616/20511149 > 0,
```

at ranks(16,28,28,28,28,28). The finite budgets beta_n>14n are separately checked throughout n=157,…,163. At n=156 the displayed local profile (15,28,28,28,28,28) instead has margin−4172075/82044596; this correctly limits this payment comparison without asserting an ambient example.

Composition and validation status

All actual two-faces receive the required5−k payment, and the complete partition reduction bounds every ideal-vertex contribution. The unrestricted shape inequality has finite coefficient beta_n, ideal curvature kappa and the additional positive5(n−1) term. It therefore contradicts the distance14 upper payment throughout dimensions157–163. The already complete unrestricted163 proof covers every dimension above163. There is no finite upper-rank sampling gap or conditional cusp hypothesis in this composition.

Independent generation52415 and fresh saved replay87174 both exited0 with the identical saved certificate. Source syntax, direct manifests, the producer's direct source bindings and local report links pass. This review's source, data and report are final and frozen. Source-integrity checks are separate from the mathematical proof reads. The completed result is an internally checked computer-assisted proof; external refereeing, publication priority and proof-assistant formalization are not claimed.

Source 76

Source path: `results/fresh_distance8_adaptive_leaf_overlap_20261017.md`

SHA256: 5a809dd0846a9aca65b16793f7244f67457dbce141a243aca24aa84a715b5ceb

Nine adaptive finite-leaf payments for distance-eight overlap cores

This local lemma closes nine explicitly listed high-core overlap contexts. It does **not** constitute a complete distance-eight overlap classification, and gives no new dimension bound. Generation and a separate fresh saved replay both passed. Root and `exact_hub_counts` completed separate full source/mathematical reads with no gap identified; neither review claims a duplicate nine-core arithmetic run or full overlap census.

The input is an actual quadrilateral face star, with opposite marked pairs, no marked path of length at most eight, and one of the nine literal core identifications in the source and data. The core has exactly its displayed edges: exclusion of further cross-core bonds is a separate obligation of the full overlap classification. Any additional supporting wall is a white root. Every edge principal diagram is positive definite, and every corner principal diagram is positive semidefinite. Normalize roots to square two.

The conclusion in every case is a **finite** corner paid by distance at most eight, or by a marked leaf at distance nine, or by a whole connected component of order at most sixteen. Thus these cases use the finite rule already budgeted in the finite8 coupon package. No new ideal pair or coupon is introduced.

Complete one-root contact domains

All nine cores are nonsingular Lorentzian with rational, simply-laced Grams. Their four literal corner Grams are positive definite. Let `J` be the inverse core Gram and let `x` contain the nonnegative contact magnitudes of an additional white root. Its orthogonal residual is spacelike or zero, hence

`x^T J x <= 2`.

For each finite corner principal Gram `K`, the corresponding necessary bound is `x_K^T K^-1 x_K <= 2`. Every inverse `K^-1` is entrywise nonnegative. Therefore setting every nonzero contact to one gives a necessary test for each support, with no upper-label cutoff assumption. The source enumerates **every subset** of the union of eligible core nodes, discarding either a length-at-most-eight one-root shortcut or one of these monotone corner violations.

For all remaining supports, the inverse-core submatrix is entrywise positive except for one two-coordinate pattern and its renumberings. In the positive cases its unit quadratic value is a lower bound. A value above two excludes the support for all labels; a two-contact value exactly two forces both contacts to be one. In the exceptional negative-entry two-coordinate case, completing the square gives a lower bound of four times the square of one nonzero coordinate, hence at least four. This excludes it for **all real** contact magnitudes at least one; it is not a unit-label inference.

Every surviving singleton has inverse diagonal at least `4/5`. Its contact square is at most `5/2`, while a Coxeter label at least five has square at least `(3+sqrt(5))/2 > 5/2`. Thus only labels three and four remain, and their squares one and two are checked exactly. The saved domain records include all support rejection counts and the complete-square certificates. The nine graphs are constructed afresh from the frozen high-rank minimal-diagram catalogue, rather than loaded from an exploratory overlap JSON.

Two-root exclusion and protected leaves

For additional roots with core-projection product `a < 0` and residual squares `rho,sigma >= 0`, the orthogonal complement of the Lorentz core is positive definite. If `a^2 > rho*sigma`, residual Cauchy implies their actual inner product is negative. They therefore have a native bond. Whenever the two contact paths and this bond give a marked path of length at most eight, the star is already paid. The saved exact tests use squared quantities, so contacts of magnitude `sqrt(2)` cause no approximation. Shortest paths to opposite marked sets avoid other marks internally and survive in the corresponding actual corner.

A simple marked leaf whose neighbor remains of degree two cannot belong to a classical affine component: affine A has no leaves, affine C has doubled terminal bonds, and affine B/D simple leaves meet a branch. Every chosen base component has at least ten nodes, excluding exceptional affine diagrams. The actual factor-fiber lemma says the marked component at an ideal corner is the whole varying affine factor, so unlisted ideal walls cannot repair this obstruction. Consequently such a protected leaf produces a finite corner, and its distance-nine pair receives the finite leaf payment. Core-orthogonal whites cannot alter a core leaf or its neighbor.

Four of the nine contexts have only two possible contact locations on the remote tail. Their opposite marked simple leaf and degree-two neighbor are untouched in every extension. This proves their finite distance-nine payment directly.

Three more contexts have a corner whose two marked nodes are simple leaves with already protected degree-two neighbors. The literal two-root inequalities exclude simultaneous contacts at the two leaves in an unpaid star. A single root meeting both leaves was already excluded by the one-root shortcut test. At least one marked leaf therefore stays untouched. This is an **adaptive** choice of protected leaf; neither end is asserted to be untouched in every realization.

The two reversed five-node contexts

These are opposite orientations of the same remaining configuration. In the first numbering the marked corner is `(0,11)`, its finite base component has twelve nodes, and the remote support set is `{1,2,3}`. If any supporting root contacts that set, the residual inequalities forbid every root contacting marked node11, unless a short paid path appears. Node11 and its degree-two neighbor are then protected, giving the finite leaf-nine payment.

Suppose instead that the remote support set is untouched. The opposite endpoint3 and its neighbor2 remain a simple leaf and degree-two neighbor, so this entire marked component is finite. Its only relevant contacting root `G` has a unit contact at node11, projection square `5/4`, and residual square `3/4`. The only other possible profile that can coexist with `G` is a unit contact `B` at node13, outside the chosen base component. Its projection square is one and the projection product is `1/2`. All other profiles are excluded by positive projection and residual Cauchy. The native product `G.B` must be zero: any value at most minus one exceeds the residual Cauchy bound. Their residual Gram and inverse are

```
S     = [[3/4, -1/2], [-1/2, 1]]
S^-1  = [[2, 1], [1, 3/2]].
```

There is at most one `G`. If `G` is present there is at most one `B`: two distinct B roots would have opposite unit residuals, while G would need product `-1/2` with both. A doubled contact at the outside node has zero residual and is incompatible with G.

If B is absent, any whole connected component of core-orthogonal whites contacting G has total rank at most three. This is the established endpoint-inverse bound `m/(m+1)`: the available residual norm is `3/4`. It controls indirect attached nodes, not just G's immediate neighbors. If B is present, any core-orthogonal root contacting G costs at least the inverse entry two; equality forces a unique unit G-contact and zero B-contact. Such a root lies in the span of the two residuals. Every further root contacting B would then have positive product with it, and every root right to both is also right to it. Thus no indirect bridge reaches an arbitrary B-tail. This is the same two-residual no-bridge argument, applied to the displayed literal matrix.

The whole finite marked component therefore has order at most `12+1+3=16`. It receives the existing whole-component payment. The reversed orientation is checked separately with its literal renumbering.

Scope and verification

The exact source reconstructs nine cores and all36 corner Grams, enumerates the complete support domain before deriving its label bound, and verifies each adaptive pair inequality and both residual matrices. The methods are four universally

protected marked leaves, three adaptive leaf choices, and two adaptive leaf-or-bounded-tail choices. The proof retains the actual-edge/corner hypotheses, the absence of short marked paths, and the separate no-extra-cross-edge premise. It does not promote the exploratory2614 count to a complete theorem, nor does it address the ideal costs of the other overlap contexts.

The final verification scope is nine reconstructed cores,36 rational corner checks, complete powerset support filtering, all surviving label3/4 profiles, and the stated adaptive/residual certificates. Root additionally inspected the inherited positive-semidefinite/corner helpers and two-residual proof. The source/data are stable; only this verification metadata was finalized after replay.

Source 77

Source path: `results/fresh_distance8_high_disjoint_20261017.md`

SHA256: 6884fc0d740e37c9fba00f328027fb332c4b14f60d75b6d36bee31a9489d2aac

All-label exclusion of disjoint high cores at distance greater than eight

Complete local computer-assisted proof; primary generation and fresh replay PASS. Under the actual quadrilateral positivity conditions and absence of an adjacent marked path of length at most8, its two minimal indefinite cores cannot be disjoint if both have orders6 through10. This lemma does not assert a dimension bound or control overlapping cores and their extensions. It is supplementary to the new ideal-corner replacement route.

The source and certificate use exact Fraction arithmetic, positive diagonal congruences and explicit nonnegative witnesses. The independent checker reconstructs the native catalogue and arithmetic separately.

Complete domain and inverse gates

Let W be the supporting white roots of an actual quadrilateral, with opposite black pairs U,V. Normalize each root to square2. The required conditions are W positive definite, W plus any one black positive definite, W plus an adjacent black pair positive semidefinite, and the full star having at most one negative direction. All off-diagonal entries are nonpositive. The frozen high-order catalogue supplies every minimal indefinite diagram of orders6 through10.

Its384 marked contexts have maximum white-to-nearest-mark distance7. Assuming no adjacent marked path of length at most8, a joining bond must have white endpoints of distances summing to at least8, hence each distance is at least1. Every external contact must pass both positive-definite black-deletion inverse tests. Of2404 ports,754 have both normalized inverse diagonals strictly below2 and can support a nonzero contact.

These ports yield12135 ordered candidate joins. For each, let x,y be the inverse diagonals at its two endpoints in the separate white blocks. A bond of squared magnitude b^2 requires $b^2xy<1$ for white positive definiteness. At a black deletion it also requires b^2 times each endpoint inverse diagonal to be less than2. Nonnegative inverse entries make these necessary conditions valid even when other contacts are present.

The simple-bond white test excludes3037 joins, each with a saved nonnegative vector of square at most0. The other9098 joins admit label3. Exactly178 also survive the necessary label4 tests. Every remaining label at least4 is excluded by one of the two displayed inequalities. In those 178 cases, label at least5 has squared magnitude at least $(3+\sqrt{5})/2>5/2$, and the exact endpoint inverse test excludes it. Parallel and dotted contacts have still larger magnitude. No numerical upper bound on labels is imposed.

Single joins and genuine corner obstructions

The resulting9276 single-join label cases have266 underlying labelled endpoint types. For262 types, one endpoint has a zero diagonal in the inverse of its full indefinite core: its deleted principal diagram is affine, so the cofactor is zero. Minimal indefiniteness plus a finite black deletion gives full signature$(\text{order}-1,1,0)$. Eliminating this full core across a single rank-one joining bond changes the other core by zero. The joined matrix therefore has two negative directions, for every bond magnitude. This argument also handles full cores with an odd label4 cycle, where no rational square-root gauge exists.

Four remaining types have explicit rational congruences and exact inertia checks. Altogether9256 single-join label cases have two negative directions. The remaining20 cases have an adjacent corner with a saved nonnegative vector of strictly negative square, contradicting positive semidefiniteness. Strict negativity matters here: a null vector alone would not exclude

an ideal corner. The implementation promotes a zero-pivot witness to a negative witness only when its product with another coordinate is negative; the resulting quadratic value is checked exactly.

The earlier exploratory count22 included two joins already excluded by white positivity. The final20 count is over the surviving single-join domain.

Every multiple-cross-edge subset

There can be up to seven individually admissible cross edges for a fixed ordered context pair. The source enumerates every subset of size at least2, giving10990 subsets. For10987, already replacing every cross bond by a simple bond produces a nonnegative nonpositive-square witness in an actual edge principal matrix. Increasing any cross magnitude only decreases its quadratic value, so that witness excludes every native label assignment.

The three remaining subsets have two edges each. Their individual label gates leave six assignments in total. For L25 with black pair{0,1}, and for black pair{1,4}, only(3,3) is possible; both have negative index2. For black pair{1,2}, the assignments are(3,3),(3,4),(4,3),(4,4). The first has negative index2; the other three have actual-edge witnesses. Thus no multiple-edge assignment survives. With no cross edges the two cores themselves give two negative directions. This completes the local disjoint-core proof.

Verification scope

Primary generation65276 and fresh saved replay6403 both terminated exit0 with PASS_DISTANCE8_ALL_LABEL _HIGH_DISJOINT_EXCLUSION. The replay recomputes the full domain and compares it with the saved certificate. The separate independent generation reconstructs26 native catalogue Grams,142 finite deletions and49 ideal deletions, and checks all ports, label gates, edge subsets, witnesses and inertia records without importing producer arithmetic. Its final binding and replay are recorded in its linked report.

The catalogue classification and actual-face positivity are explicit inherited mathematical inputs. This local result has no external refereeing or proof-assistant formalization and makes no construction claim.

Source 78

Source path: `results/fresh_distance8_high_overlap_20261017.md`

SHA256: 84e0c3cd3dbd16545038ccdf0161e4561eaeecd241532e877d38c294643c41d1

Complete distance-eight high-core overlap domain and supporting extensions

This is a complete local classification of the high-core **overlap** branch for an actual quadrilateral with no marked path of length at most eight. It is not a dimension theorem. Disjoint high cores and minimal cores of order at most five are covered by the separate frozen high-disjoint and low-core packages. Compression of the small ideal pair sets is checked separately before a full local payment composition is asserted.

The source reconstructs this domain directly from the frozen catalogue of all26 minimal indefinite diagrams of orders6 through10. It reads no private exploratory JSON. The earlier exploratory count2614 is therefore an expected count checked by the new enumeration, not an input certificate.

Geometric and combinatorial domain

Write the quadrilateral's opposite black pairs as U,V and its supporting white roots as W. The actual relative interior of the face makes W elliptic; the interior of each actual edge makes W plus its single black root elliptic. Opposite closed sides are disjoint even for ideal quadrilaterals, so W plus either opposite pair is indefinite. Consequently its minimal indefinite core contains both black roots and both black deletions are elliptic. These are the same closed-face and strict-edge hypotheses established in the earlier complete quadrilateral proof.

The inherited complete high catalogue gives384 black-pair contexts and maximum distance seven from either black set. If two cores meet in a white root, absence of a marked path of length at most eight forces the two distances to sum to at least nine. Each is therefore at least two. The new source enumerates every injective identification of eligible white nodes, checking every induced edge and nonedge, including all possible overlap sizes. There is no connected-overlap assumption. The displayed candidate count6,390,952 is a combinatorial superset count;82,197 identifications pass the injectivity and induced-label consistency tests and are the actual consistent domain.

At a shared node, an unshared neighboring root of the other core is white. If the corresponding diagonal of either black-deleted inverse is at least two, its nonzero contact would contradict positive definiteness of the actual edge principal diagram. This all-label inverse test rejects75,947 consistent identifications. The remaining actual-edge diagrams are checked by nonnegative quadratic witnesses: a square at most zero contradicts their strict positive definiteness. Rational positive-diagonal congruence is used where available. Otherwise the all-unit-bond comparison is an upper bound on the quadratic form of a nonnegative vector, so its nonpositive witness also applies to every larger native bond. These edge witnesses reject3,636 additional identifications, leaving exactly2,614 oriented marked contexts on154 underlying labelled cores.

No unknown edge between the two copies is silently set to zero. For every surviving context, every absent cross-copy edge is inserted at its smallest nonzero magnitude. Either the new edge creates a marked path of length at most eight, or an actual-edge nonnegative witness excludes it. There are154,502 shortcut tests and5,528 saved edge witnesses. Increasing that new bond, or adding any further nonpositive cross-products, can only decrease the witness square. Thus these tests exclude arbitrary simultaneous cross-bond additions and every label, not merely one simple extra edge in isolation.

Supporting-wall extension certificates

For every surviving core the source reconstructs all four corner Grams and their finite/affine components. At any actual ideal corner the marked component is precisely the full varying affine cusp factor. This is the established factor-fiber statement,

and does not assume a simple cusp. The remaining corner components are elliptic spectator diagrams.

The certificate uses the same rational Lorentz projection and residual arguments as the distance9 independent extension proof, with the shortcut threshold explicitly reduced to eight. Every numerical assertion needed by a chosen case is regenerated. The extension methods are:

- **1,329 radical cases.** The indefinite core makes each Gram radical vector an actual linear relation. Allowed contacts on its two signs would create a marked path of length at most eight. Thus every extra supporting white is orthogonal to the marked core component, whose full order is bounded by seventeen.
- **823 positive-residual cases.** Corner inverses first restrict the contact domain. Every remaining root has one native contact; all two-contact supports are excluded by entrywise-positive projection bounds. Cauchy either excludes two such roots or forces a short paid bond. The one possible root has residual norm below one. The whole rank of core-orthogonal attached components is bounded using the connected inverse lower bound `m/(m+1)`, so indirect chains are included. The resulting whole marked component has order at most seventeen.
- **259 relevant-component cases.** Only roots that can reach the marked component need have residual norm below one. The exact positive-projection, forced-orthogonality, residual-Gram inverse and no-bridge tests control a possible exterior residual component. The saved matrices and inequalities establish the whole-component bound, including indirect attachments; this does not discard a low-projection root merely because it is outside the marked base component.
- **120 inverse-isolated components.** Every potential contact to the marked component has projection norm above two. Contacts at both ends would give a short path, and each same-end inverse block is positive. Thus all additional whites are right to the marked component and its original order is its whole order.
- **74 protected-leaf cases.** A marked leaf and appropriate fixed core nodes cannot acquire additional contacts. Eighteen cases have a fixed simple leaf with a fixed degree-two neighbor in a component of order at least ten, forcing the actual corner to be finite. The other56 have a persistent core feature within five of a mark. If their corner is finite, the protected marked leaf gives the finite distance-nine payment; if ideal, the distance-nine pair with a feature within five is already in the basic ideal5 rule. This argument does not bound an arbitrary growing ideal component by seventeen.
- **9 adaptive cases.** The separately replayed adaptive leaf lemma now has its previously conditional no-extra-cross-edge premise discharged by this complete cross-edge census. Four cases have a universally protected marked leaf, three choose an untouched end adaptively, and two use a finite leaf or a finite component of order at most sixteen.

In the first four methods, a finite corner is paid by the existing whole-component cutoff seventeen. An ideal corner has an actual full affine marked component of order at most seventeen. Orders at most thirteen are already entirely paid; orders14 through17 require the separate complete induced-map check. In the last two methods, the payment is already finite or basic ideal5, with no new ideal pair.

What is and is not established

Every surviving high-core overlap has one of the six stated local certificates. The result is complete for this marked-core domain and for arbitrary additional supporting whites, subject to the actual face/edge/corner hypotheses. It uses published/inherited finite and affine Coxeter classifications and the frozen minimal-indefinite catalogue; it is not a formal proof-system verification of those classifications. The exact program is not a search over ambient dimensions or a finite cutoff on attached-chain lengths.

The new small-affine mapping package must still verify the precise selected pair sets and any removed-pair replacements. The complete triangle/short-path reduction, small-core and disjoint branches must then be explicitly composed. No stronger global bound follows from this overlap certificate alone.

Final generation49951 and separate fresh saved replay38628 both terminated exit0/PASS. Root completed a full source/mathematical read, also inspecting the inherited rational-congruence, positive-semidefinite obstruction, corner, context and union helpers and the residual/no-bridge proof. No gap was identified. This review did not duplicate the82,197-context,2,614-extension or1,153-map arithmetic censuses. Source, data and report are final/frozen; exact small-factor inclusion and the full local binder are separate successor files.

Source 79

Source path: `results/fresh_distance8_ideal5_small_factor_20261017.md`

SHA256: 2cc07e884396e686a7b106f9f4fb962c97337dcb8732fd7db3a610f6e2d156ba

Retaining the ideal5 pair costs after the complete finite8 overlap classification

This is a complete induced-map census for the selected small-factor patterns from the new distance-eight overlap certificate. It retains the exact frozen ideal5 pair sets and costs; no ideal cost is increased. The full finite8/ideal5 face-payment composition is a separate statement. No new dimension bound is asserted.

The checker reads the new authoritative2614-context JSON. It does not use private scout data or import the old949 domain as coverage of the new cases. Among the selected whole-component certificates, exactly1153 permit an affine component of order at least14:313 have bound14,354 bound15,346 bound16 and140 bound17. The other methods either bound the whole component by13, pay a finite corner, or directly give a basic ideal5 pair.

For each of those1153 patterns and every classical affine target A/B/C/D of order14,15,16,17, the checker enumerates every induced label-preserving map when the target order does not exceed the certified whole-component bound. The same exhaustive backtracking checks both edges and nonedges, while preserving both marked-root roles. The pattern is only the marked connected base component; its actual completion may contain arbitrary additional supporting whites up to the certified bound. All resulting marked images are saved. This is a complete finite completion test, not a claim that every abstract embedding is geometrically realizable.

Every marked image belongs to the retained paid pair set or to a separately justified deletion set. The retained A/B/C/D costs remain:

Factor order	A	B	C	D
14	70	87	84	82
15	75	100	95	96
16	80	107	103	111
17	85	111	108	127

For cycle targets the omitted pairs are transferred by the universal cycle14 or order-at-least15 local theorems, with the existing A13/A15/A17 finite coupons. For D targets of orders14 and15, the omitted nine fork pairs are exactly the universal D13 and D14 transfer domains. Their immediate whole-affine alternatives have orders12 and13, so the retained whole-factor cutoff13 suffices.

For D targets of orders16 and17, the omitted nine fork pairs use the earlier literal sealed-core replacements. This step cannot be inferred merely from an abstract small-factor embedding. The new source therefore matches the **whole marked-core key**, including both opposite black pairs and every identified root, to the frozen old949 records. Exactly nine cases are affected at each order:

- order16: old indices384,394,401,648,663,676,814,829,842;
- order17: old indices383,393,400,647,662,675,813,828,841.

These are exactly the old sealed replacement domains, without any additional context. The new full overlap certificate excludes every extra cross-core bond, and the existing sealing lemmas control arbitrary supporting whites. The replacements pay finite whole components or finite half-coupons, or a marked path of length two; they do not require a removed small ideal pair to remain paid.

The all-rank ideal cost proof is unchanged because the paid sets themselves are unchanged. Orders at most13 cost at most half the affine rank per link-vertex incidence. Above the four explicit small orders, the retained ideal5/R5 all-rank formulas give common normalized factor cost `163/23`. The exact rank envelope is therefore still

```
r <= 12: r/2
r = 13:  87/14
r = 14:  20/3
r = 15:  111/16
r = 16:  127/17
r >= 17: 163/23.
```

The certificate concerns actual cusp factors through the factor-fiber theorem and the new whole-component bounds; no all-simple cusp assumption is introduced. The complete new overlap domain must be combined with the frozen low-core, disjoint-core, triangle and short-path transfer lemmas before asserting a local face-payment theorem.

Final bound generation84160 and separate fresh saved replay48284 both terminated exit0/PASS. Root completed a full source/mathematical read of the1,153-pattern selection, complete induced mapping and the literal order16/17 marked-core correspondence, with no gap identified. That review did not duplicate the embedding search or published finite/affine classification. Source, data and report are final/frozen.

Source 80

Source path: `results/fresh_distance8_ideal_corner_replacement_20261017.md`

SHA256: 83b6c9eb437a0eb39ac3caf1fbd482fba313cae120824b311006100273d1417a

Replacing an ideal distance-nine quadrilateral payment

Final, internally checked local theorem. Let Q be an actual quadrilateral face of a finite-sided finite-volume hyperbolic Coxeter polytope, with no simplicity assumption at ideal vertices. Suppose an ideal corner has its two marked roots at graph distance9 in the varying irreducible affine factor F, of order greater than17. Then either a mark lies within14 steps of a feature of F, or an adjacent black pair of Q is joined by a path of length at most8 whose internal vertices are supporting white roots. A feature means a vertex of degree at least3 or an endpoint of a bond with label at least4.

There is no global exclusion of small affine diagrams. The theorem is compatible with arbitrarily many spectator factors and supporting whites. It is a local replacement lemma, not a new ambient dimension bound.

The checker and certificate verify the exact weak-profile inequalities. Their all-rank classification and geometric interpretation are proved below. The corresponding affine graph rule has the separately proved cost at most8m on a factor of order m.

Actual corner and spectator reduction

Write the face star as H=W+a+b+c+d, with opposite pairs {a,c} and {b,d}. All roots have square2. The white diagram W is positive definite; W plus any one black supports an actual edge and is also positive definite. Each adjacent corner is positive semidefinite. The full star has rank n+1 and corank1: its four side normals project to a spanning three-dimensional Lorentz space perpendicular to W.

At the chosen ideal corner W+a+b, the factor-fiber lemma gives the varying affine component F containing a,b, plus a positive-definite spectator forest Z orthogonal to F. This statement concerns the actual face star, not the entire set of facets at a nonsimple cusp.

Assume for contradiction that there is no path of length at most8 from {a,c} to {b,d} with white interior. In particular b.c=a.d=c.d=0. Let nu be the positive affine kernel combination of F. It is a nonzero future null vector representing this actual ideal corner. Neither opposite side c nor d contains that corner, so

```
h=−c.nu>0,  k=−d.nu>0.
```

The opposite contacts x=-a.c and y=-b.d may be zero, any finite Coxeter magnitude, parallel2 or dotted. Every nonzero magnitude is at least1. No upper label or distance cutoff is imposed on x,y.

Eliminate the whites of F first. The frozen four-root determinant identity applies. Eliminating Z afterwards subtracts a positive-semidefinite matrix [[p,q],[q,r]] from the c,d block. By the spectator identity, this changes the normalized determinant by

```
alpha∗(p/h² − 2q/(h∗k) + r/k²) >= 0,
```

where alpha>0. The heights are unchanged because Z is orthogonal to F. Thus a strictly positive determinant before the spectator subtraction contradicts the required corank1 afterwards as well.

Affine cycles: complete weak profiles

Suppose F is affine A. Its two a-to-b arcs have edge lengths P,Q≥9, since a,b have graph distance9. Deleting a leaves A_(P+Q−1), of rank at least17. Its complete one-root positive-semidefinite extensions give the following c profiles: isolated; one simple bond at either endpoint; one doubled bond at either endpoint; one simple bond at the second node from either endpoint; or simple bonds at both endpoints. In ranks at least17, finite exceptional diagrams cannot occur. The connected finite extensions are A, B and D, and the only connected affine extension of the entire path is its simple cycle. Disconnected spectators do not enlarge this list, because the principal old path plus c is itself positive semidefinite. This is the same full finite/affine classification used in the previous cycle proof; here none of its small-affine exclusions is used.

For an attachment to an arc, let j=1 at an endpoint and j=2 at its next node. Let l=(l1,l2) be attachment magnitude times j on the two arcs. The norm constants, heights and weak bounds are:

Profile	c0	l	h	Minimum h
isolated	2	(0,0)	x	1
endpoint3 on arc i	1	unit vector i	x+1	1
endpoint4 on arc i	0	sqrt(2) times unit vector i	x+sqrt(2)	sqrt(2)
second-node3 on arc i	0	2 times unit vector i	x+1	1
both endpoints3	0	(1,1)	x+2	2

The isolated minimum follows from h>0 and the native Coxeter gap. All other minima use only x≥0. Define analogous d0,m,k for d, and u=1/h, v=1/k.

There is one additional necessary length inequality. If c and d both attach on the same arc, at positions j_c,j_d from its opposite ends, their path through that arc has length P−j_c−j_d+2. It has white interior because j_c,j_d≤2 and P≥9. The no-short-path assumption therefore implies

```
P >= max(9,7+j_c+j_d).
```

Apply the corresponding inequality to Q. If one attachment is absent, the original bound9 suffices. This refinement is essential for two second-node attachments: their common arc must have length at least11.

Put A=1/P, B=1/Q, L=u*l1*+*v*m1 and M=u*l2*+*v*m2. The normalized four-root determinant is exactly

```
F=1−2∗(A∗L+B∗M)
    −(A+B)∗(c0∗u²+d0∗v²)−A∗B∗(L−M)².
```

For fixed u,v≥0, F is nonincreasing in A and B. For fixed A,B≥0 it is jointly concave in u,v: its quadratic terms are negative diagonal squares and one negative square. It therefore attains its minimum on the permitted height rectangle at a rectangle vertex. For each of the64 ordered profile pairs, the checker uses the largest permitted reciprocal arc lengths, then checks all four rectangle vertices in Q(sqrt(2)). All256 values are at least5/81. Hence F≥5/81>0 for every actual arc length and every allowed height. This is an arbitrary-rank, continuous-height inequality, not extrapolation from finitely many lengths.

Consequently an ideal affine-A corner of marked distance9 cannot occur in an unpaid quadrilateral. Additional spectator contacts only increase this positive determinant.

Feature-free classical tree corridor

In order greater than17, the other affine factors are the classical B/C/D trees. Suppose both marks are farther than14 from every feature. They lie on the undecorated simple-bond spine, at distance P=9. Beyond each mark, the outer component is a finite B or D diagram attached at its long endpoint and has rank at least9. Indeed a fork feature is at least15 simple steps away, and a doubled bond's nearer endpoint is also at least15 steps away; retaining the entire end decoration only increases the outer order. Deleting either mark therefore leaves two large finite B/D components.

A complete one-root positive-semidefinite extension of these two old components has eight possible profiles: isolated; an endpoint3, endpoint4 or second-node3 attachment to either component; or simple attachments to both long endpoints. For a single component of rank at least9, its finite one-root extension only prolongs the long endpoint by3. Its affine one-root extensions add4 at that endpoint or3 at its second node, whenever the old type permits that profile. We safely allow both affine

alternatives for either old type. If both old components are joined, their two remote decorations force the corresponding affine C/B/D diagram, joined by simple endpoint bonds. No other connection fits the finite/affine classification; an additional branch or internal doubled bond would violate it. In particular this argument explicitly retains the one-component affine extensions that were excluded in the earlier simple-cusp corridor proof.

Normalize nu to coefficient1 along the spine and at all relevant ports. For c, let the outer component be the one beyond a, and the inner component the one containing b. After eliminating the white roots of F, its scalar data and weak cost are:

Profile	l	c0	Minimum h	Maximum 2*l/h+c0/h²
isolated	0	2	1	2
outer endpoint3	0	1	1	1
outer endpoint4	0	0	sqrt(2)	0
outer second-node3	0	0	1	0
inner endpoint3	1	1	1	3
inner endpoint4	sqrt(2)	0	sqrt(2)	2
inner second-node3	2	0	1	4
both endpoints3	1	0	2	1

These constants follow from exact all-rank inverse entries. The long endpoint and next node of either finite B_r or D_r have inverse diagonals1 and2. The corridor inverse-column argument gives the endpoint column. For the second-node column, put1 at the long endpoint and2 on every subsequent spine node, with value sqrt(2) at the doubled B terminal or1 at each D fork leaf. Direct multiplication by the native norm-two Gram gives the unit vector at the second node, proving its inverse diagonal2 in every rank. On the middle A_(P−1) path, the corresponding entries are (P−1)/P and2(P−2)/P. The cross term for the separated inner contacts is l*m/P; their positions are at most2 from opposite ends and P=9, so no overlap exception occurs.

Thus B=l/P, C=c0+l²/P, A=m/P, D=d0+m²/P and E=l*m/P in the four-root identity. Cancellation gives

```
det(S)/(h²*k²)
  = 1−[2*l/h+c0/h²+2*m/k+d0/k²]/P.
```

Each single-root cost is at most4. With P=9, the determinant is at least1/9>0. The checker verifies all eight cost bounds and64 ordered pairs exactly. The same estimate holds for every P≥9 under the same outer-rank hypothesis. Once again spectator elimination preserves the contradiction.

Consequence and verification scope

The two branches prove the stated local theorem. In the tree case, failure of the feature alternative forces the corridor contradiction. In the cycle case there are no features, and the weak-profile contradiction applies directly. Every resulting short path selects an actual adjacent corner of Q; if that corner is ideal, the factor-fiber lemma places both marked roots and the path in its varying affine factor.

Generation and a fresh separate-process saved replay both terminated exit0 with PASS_UNIVERSAL_IDEAL_DISTANCE9_CORNER_REPLACEMENT. All16 direct source bindings and source syntax passed. The parent agent read the complete source and proof without finding a gap; that was a mathematical/source review rather than an independent arithmetic replay.

The separate independent review reconstructed the algebra without importing producer arithmetic. It matched all256 cycle values, eight tree costs and64 pair gaps, and checked the full formal determinant, PSD correction and four symbolic all-rank inverse identities. Its full mathematical/source read also checked the actual factor-fiber geometry, positive heights, complete enlarged extension lists, arc-length refinement and long-tail reduction. That report records its own final bound replay and hashes.

The source rechecks the formal determinant identity and binds the actual-corner, spectator and finite/affine classification inputs. It does not enumerate new high-core diagrams, assert that a finite numerical rank range exhausts them, or provide an ambient dimension theorem. External refereeing and formalization remain separate.

Source 81

Source path: `results/fresh_distance8_ideal_corner_replacement_20261017_review.md`

SHA256: 1c5c03daeacc1e7336cca5fb1a16af8b2733518f8d1a2bf07d18602a78a93a98

Independent review of the ideal distance-nine replacement

PASS: mathematical/source review and independent exact reconstruction. The producer report, final SHA256 83b6c9eb437a0eb39ac3caf1fbd482fba313cae120824b311006100273d1417a, proves the stated local quadrilateral replacement without assuming simple cusps or excluding small affine diagrams. No mathematical gap was identified. The review checker and saved reconstruction import no producer or other project arithmetic code.

The actual ideal corner consists of its varying affine factor and a positive definite spectator forest. The opposite sides do not contain that ideal point, so their heights against the positive null combination are strictly positive even when the full cusp is nonsimple. The four residual side normals span a Lorentz three-space. Consequently their four-by-four determinant must vanish. Eliminating the orthogonal spectator forest changes the normalized determinant by a positive semidefinite quadratic form and cannot invalidate the strictly positive contradiction.

The one-root extension lists are complete by the stated finite/affine classification. A long A path admits only the eight displayed oriented profiles. In the classical tree case, feature distance greater than14 puts both marks on the simple spine with long B/D outer components. Their retained forks or doubled bonds prohibit other classical extensions; exceptional types are too small. Joining both long components gives the corresponding affine tree through simple long-end bonds. Including the single-component affine profiles, even where the actual edge condition could exclude them, only enlarges the necessary domain.

The height bounds use only nonnegative native contacts and the Coxeter gap between zero and one. In the cycle case, attachments at positions `j_c,j_d` on a common arc give a white-interior path of length `P−j_c−j_d+2`; excluding paths of length at most8 gives exactly `P >= max(9,7+j_c+j_d)`. The determinant expression is nonincreasing in the reciprocal arc lengths and concave in the two reciprocal heights. Its rectangle checks therefore cover all real heights and all ranks. For the feature-free tree, the all-rank endpoint/second-node inverse diagonals1 and2 give the stated profile table. The separated contacts on the middle path give the cross term `l∗m/P`, and each single-root cost is at most4, giving the strict gap at `P=9`.

The independent SymPy checker reconstructs all eight cycle profiles and all256 rectangle values, matching the uniform minimum `5/81`. It separately reconstructs all eight tree costs and all64 ordered pair gaps, matching the minimum `1/9`. It computes the full symbolic four-root determinant directly, derives the spectator correction by substitution, and verifies the four path inverse identities symbolically with an indeterminate length. These are independent arithmetic checks, not calls to the producer's field class or determinant routine.

Generation and a fresh saved replay pass. The review retains the published finite/affine classification and the inherited actual factor-fiber geometry as explicit inputs; it does not claim an independent classification census. The conclusion is the local replacement lemma. An ambient dimension bound requires a separately checked payment and scalar composition. External refereeing and proof-assistant formalization are not claimed.

Source 82

Source path: `results/fresh_distance8_ideal_dimension97_20261017.md`

SHA256: 7922b4dce4448026057c072540f12f47ab582d9fa055ece647c80d35e13df28c

Finite distance nine and ideal feature payments: dimension97

Complete internally checked computer-assisted proof. Primary generation and fresh saved replay pass. Separate mathematical/source review and independent exact scalar reconstruction pass, with no identified gap. The argument below gives dimension at most **97** for every finite-sided, finite-volume real hyperbolic Coxeter polytope. There is no arithmetic, compactness, simple-cusp, cusp-factor-rank or facet-count restriction. The previously completed compact-facet ambient ceiling **79** is unchanged.

This reduction inherits the complete dimension108 proof and its independent internal review. It changes only the ideal-vertex payments. The source and certificate bind the new local geometric and all-rank counting inputs, then check the scalar transfer. The separate distance-eight core explorations are not prerequisites.

Corner rule and payment of every face

At finite vertices retain the completed rule: a pair receives weight1 when its graph distance is at most9 or its connected component has order at most17. The total finite-vertex cost remains at most9n.

At an ideal vertex, use its actual irreducible affine factors. A feature is a vertex of degree at least3 or either endpoint of a nonsimple bond. Give a pair in one factor weight1 in any of these cases:

- its distance is at most8;
- the factor has at most17 vertices;
- its distance is9 and at least one mark is within14 graph steps of a feature.

Every other pair has weight0. This is intrinsic under diagram automorphisms. The factor-fiber theorem identifies each actual ideal face corner with a pair in its varying factor, including at nonsimple cusps.

The inherited triangle proof supplies two corner paths of length at most8. Both remain paid. For a quadrilateral, the dimension108 proof supplies a corner of distance at most9 or with marked component of order at most17. A finite corner, a small component, or a distance-at-most8 corner remains paid immediately. In the sole remaining case it is an ideal distance9 corner in a factor of order greater than17. If it is near a feature, the new rule pays it. Otherwise the universal ideal-corner replacement supplies an adjacent corner path of length at most8, which is paid. That lemma has a separate independent exact and mathematical review.

Thus every triangle receives at least2 and every quadrilateral at least1. Nonnegative weights suffice for all larger polygons. The total assigned weight is at least the sum D5 of5 minus the number of sides over all two-faces.

The replacement lemma uses complete one-root finite/affine extension lists, weak positive cusp heights and an exact four-root determinant. It needs no exclusion of small affine diagrams. Its cycle branch has uniform normalized determinant gap5/81; its feature-free tree branch has gap1/9. Positive-semidefinite spectator elimination preserves these contradictions. Its geometric scope is essential; a numerical height or rank cutoff would not justify this transfer.

Ideal cost and all-rank shape reduction

The affine feature-budget theorem proves that a connected factor with m vertices has at most

```
min(8m, m(m−1)/2)
```

paid unordered pairs. Its proof covers every rank by explicit path, cycle and fork counts. In particular affine D reaches8m in all orders m>=43; the estimate does not rely on a finite rank sample.

At a cusp with affine ranks r_i, write d=n−1 and V=product(r_i+1). Each pair in factor i occurs in exactly V/(r_i+1) face-corner fibers. The total cost divided by V is therefore at most

```
sum_i min(8,r_i/2).
```

For the link H-polynomial h(t)=product_i(1+t+…+t^(r_i)), put G=sum_j h_j min(j,d−j). The inherited shape inequality gives normalized ideal curvature (5G−dV)/(2V), together with a strictly positive global slack. The finite curvature coefficient is

```
beta_n=n(n−6)/8                  if n is even,
beta_n=(n−1)(n−5)/8              if n is odd.
```

The inherited all-rank convex-order transfer applies to cost sum min(8,r_i/2), with cap16. It reduces all integer rank partitions of d to copies of16, at most one smaller positive part and at most one larger part. The cost is preserved and the curvature can only decrease under this reduction. Therefore it suffices to check those shapes.

Exact dimension conclusion

For link dimensions d=97 through107, the checker enumerates all957 reduced shapes. Each has strictly positive curvature minus cost. The minimum is112101/1419857, at ranks(1,16,16,16,16,16,16) and ambient dimension98. It also checks beta_n>9n in every ambient dimension98 through108. The resulting lower bound for D5 exceeds the upper bound from corner payments, a contradiction.

The completed108 theorem excludes every larger ambient dimension. Consequently the new ceiling is97. At ambient97, the profile consisting of six ranks16 has margin−185352/1419857. This proves that this particular per-cusp scalar comparison cannot lower the ceiling to96; it is not a realized finite-volume polytope.

Finite weights were retained, so the completed compact-facet ceiling79, compact-arithmetic ceiling21 and unrestricted compact ceiling29 are unchanged. No new finite-volume construction is asserted.

Verification scope

The producer computes the link polynomial by exact integer convolution, then checks the curvature again by a separate inclusion-exclusion moment formula. It verifies all957 shapes, the finite coefficients and the97 boundary profile, and binds the completed108 proof and both new local lemmas. Final generation and fresh saved replay both terminated exit0 with `PASS_FINITE9_IDEAL8_FEATURE_DIMENSION97`. The separate independent checker reconstructs all957 shapes, minima and the boundary using a different shape enumeration, sliding-window convolution and cumulative-tail inclusion-exclusion identity. Its exact values agree throughout. Two separate internal mathematical/source reads confirm the face-payment transfer, finite9n cost, ideal factor multiplicity, cap16 reduction and inherited108 tail. The linked independent review records its final bindings and saved replay; it does not rerun the inherited108 high-core census. External mathematician refereeing, publication priority and proof-assistant formalization are separate unestablished matters.

Source 83

Source path: `results/fresh_distance8_ideal_dimension97_20261017_review.md`

SHA256: 22ab6b461b28bb5a7500c87b9cbcbba3ed436cbaf7aafe91d2e95519fea80f7b

Independent review of the finite-nine / ideal-eight dimension97 proof

PASS: independent exact scalar reconstruction and mathematical/source composition review. The final primary proof has SHA256 7922b4dce4448026057c072540f12f47ab582d9fa055ece647c80d35e13df28c. No missing mathematical case was identified in its deduction of dimension at most97 for finite-sided finite-volume real hyperbolic Coxeter polytopes. The compact-facet ambient ceiling79 remains unchanged.

The independent checker and saved data import no producer or other project arithmetic code. They enumerate the reduced shapes by first choosing the exceptional smaller/larger ranks and then forcing the multiplicity of16. This reproduces exactly957 shapes for link dimensions97 through107, with no missing or repeated profile.

For each profile, a sliding-window integer convolution reconstructs every coefficient of `H(t)=product_i(1+...+t^r_i)`. Independently, write

```
product_i(1-t^(r_i+1)) = sum_b a_b t^b,
s = number of ranks, D = floor(d/2), V = product_i(r_i+1).
```

The cumulative-tail identity gives

```
G = D*V - 2*sum_b a_b binom(D-b+s,s+1).
```

This follows by summing `G=sum_(t=1)^D [V-2 Pr_count(S<t)]` and the hockey-stick identity for bounded compositions. It is distinct from the producer's truncated first-moment formula. Both calculations agree with `G=sum_j h_j min(j,d-j)` on every profile and the boundary profile. The checker additionally verifies the centered absolute-deviation expression for curvature. All operations are exact integers and Fractions.

Every scalar row and per-dimension minimum agrees with the primary data. The minimum curvature-minus-cost is `112101/1419857`, and the six-rank16 boundary at ambient97 has margin `-185352/1419857`. The finite slack `beta_n-9n` is positive for every `n=98,...,108`. Generation and a fresh saved replay pass all957 profiles and the boundary comparison.

The written composition was checked separately from these scalars:

- The inherited108 proof pays triangles by two paths of length at most8, and quadrilaterals by a distance-at-most9 corner or a component of order at most17. Finite corner weights remain unchanged.
- At an ideal corner the varying affine factor contains the marked pair and any connecting corner path. Its pair multiplicity is exactly `V/(r_i+1)`, including nonsimple cusps. The all-rank affine graph bound therefore gives normalized cost `sum_i min(8,r_i/2)`.
- The only lost old payment could be a distant ideal distance9 pair in a large factor. The independently reviewed universal replacement supplies another paid corner. Neither a simple-cusp assumption nor a global exclusion of small affine diagrams is inserted.
- The inherited convex-order rank transfer is valid for an arbitrary cap. With cap16, concentrating two ranks below16 or two above16 preserves the stated cost and cannot increase curvature. The sum of squared ranks strictly increases, so reduction terminates with only copies of16 and at most one positive rank on either side. This proves coverage of all cusp partitions, not merely the957 displayed shapes.
- The unrestricted shape inequality has positive global slack. The scalar comparisons exclude dimensions98 through108, and the completed inherited108 theorem excludes the entire higher tail.

The published finite/affine classifications, actual face/cusp geometry, all-rank convex-order argument and inherited108 proof remain explicit mathematical inputs. This review does not rerun a high-core census or claim formal verification. A separate frontier-agent mathematical/source read also passes the payment, cap16 and dimension-tail composition, without duplicating the957-profile arithmetic. External refereeing and publication priority are not asserted.

Source 84

Source path: `results/fresh_distance8_low_core_20261017.md`

SHA256: 634e921bd21f0d5c39c1da343f21389368a522a9e49489b493d6226e11bcf8b0

Complete small-core domain at distance eight

Final, internally checked. Exact generation and a fresh saved replay passed, and a separate narrow mathematical/source review found no gap in the stated scope. This is a local face-payment lemma, with no new ambient dimension claim.

Let H=W+U+V be the star of an actual quadrilateral in a finite-volume hyperbolic Coxeter polytope. W is its set of supporting white roots; U,V are the opposite black pairs. Normalize every root to square two. Suppose there is no path of length at most eight from U to V with white internal nodes. If a minimal indefinite core in W+U or W+V has order at most five, the configuration is impossible or has an adjacent marked corner whose **entire connected component has order at most12**, even after all additional supporting whites are restored.

The source and certificate use the unchanged complete order-four/five graphs in the distance-nine low-core certificate, and the high-core catalogue and actual-face/cusp conventions of the completed distance-nine proof. The new ports, joins, identifications, corner domains and extension conclusions are recomputed. There is no assumption of arithmeticity or simple ideal vertices.

Domain and the unbounded order-three family

A connected core of order k containing two black roots has marked radius at most k−2. The complete high-order catalogue has maximum radius seven. If both cores have order at most five, every possible cross edge gives a path of length at most seven, and every overlap gives a still shorter path. With no cross edge their two indefinite blocks already give two negative directions. Thus only a small/high pairing needs attention.

A minimal order-two core has only its two black nodes. Every contact with a high core gives a path of length at most eight; otherwise the indefinite blocks are orthogonal. It cannot occur in an unpaid configuration.

An order-three core has a single white w, at distance one from its black pair. An overlap is impossible, because1+7=8. Any disjoint cross edge must meet a high port at distance seven. The strict actual-edge inverse test leaves precisely L25 node9, with high black pairs {1,2}, {1,3}, {2,3}. Its full inverse diagonal is exactly

```
(G_L25 inverse)99 = 0.
```

Write the small Gram as

```
[[2,-x,-a],[-x,2,-b],[-a,-b,2]].
```

The black-black magnitude x can have an unbounded finite label or be parallel. A truly minimal order-three core has x≤2: a dotted black-black pair would already be an indefinite proper order-two subdiagram. Its two black deletions are positive definite. If one also applies the low port inverse test, $a^2,b^2<3$, so the white-incident labels are right,3,4 or5. No finite bound on the black-black label follows or is assumed.

There is nevertheless a uniform exclusion. Every cross contact lands at high node9. Eliminating the nonsingular Lorentzian high core leaves the small Gram unchanged, since the Schur correction is identically zero. The high core supplies one negative direction and the small indefinite core supplies another. This holds for **every real x,a,b and cross magnitude for which the small core is indefinite**, including the physical unbounded-label family. It also covers zero cross contact. The proof is a symbolic identity, not a sampled angle limit.

The same zero-correction identity permits an arbitrary column of low contacts when all high endpoints are node9; the finite computations below retain every such contact pattern rather than rely on a single-edge approximation.

New finite order-four/five domain

The frozen low-core census proved, before enumeration, that order-four finite labels are at most six and order-five labels at most five. It enumerated all weighted graphs, not just trees, and found32 and14 canonical minimal-indefinite types. The current checker recomputes every black pair with positive-definite deletions and every white port of distance at least one.

For disjoint joins, low radius one requires high radius seven; radius two requires high radius at least six; radius three requires high radius at least five. The high inverse filter gives18 relevant ports, belonging to L18,L22,L25. Every such port has maximum normalized black-deletion inverse diagonal at least6/7. Actual-edge positive definiteness therefore forces any cross magnitude beta to satisfy beta²<7/3. Since the label-five squared magnitude is (3+sqrt(5))/2>7/3, only labels3 and4 remain. This includes parallel and dotted exclusions and is not an imposed label cutoff.

The checker applies both endpoint inverse tests and the supporting-white Schur condition

```
beta² (G_Wlow inverse)xx (G_Whigh inverse)yy < 1.
```

It then eliminates the full high core and computes the exact inertia of the remaining small Schur matrix. The726 low-first joining-label cases split as follows:

Low order	Simple two-negative joins	Simple white-PD rejections	Label-four two-negative joins	Label-four white-PD rejections
4	294	11	17	6
5	342	22	15	19

Thus668 joins have two negative directions and58 violate white positive definiteness. The131 underlying small Schur types are recorded exactly. Unlike the distance-nine case, some doubled joins pass the initial white tests; their full two-negative inertia is essential.

There are417 marked context pairs with at least one surviving edge choice:249 have one possible cross-edge position,117 have two and51 have three. Every one of the321 subsets containing at least two positions already violates actual-edge positive definiteness when all selected bonds are simple. Explicit nonnegative vectors of nonpositive square certify these rejections. Larger native magnitudes only decrease the squares, so all mixtures of simple and doubled edges are covered.

Complete overlaps, including two shared nodes

An overlap needs marked-distance sum at least nine. Order four therefore contributes radius two against high radius seven. Order five contributes radius two against seven, or radius three against six or seven. All raw eligible high ports are retained before the inverse test: with two shared nodes an endpoint need not have an exclusive new neighbor, so filtering every shared high node in advance by its port inverse would be invalid.

The checker tests all injective partial identifications and every contact on their shared roots. Among630 raw candidates,523 are consistent. Conditional inverse tests reject422; actual-edge tests reject35. The66 remaining low-first contexts have14 literal core types, all involving L25. They comprise12 order-four and54 order-five contexts. Reversing the cores covers the other orientation.

All four adjacent corner Grams are positive semidefinite in every survivor. An additional bond between exclusive parts of a core either creates a path of length at most eight (2,184 tests) or has an actual-edge obstruction (12 tests). Those nonnegative witness certificates remain valid after any extra acute contacts. Thus each full surviving core Gram is fixed before its inverse or radical is used.

Complete supporting-white extensions

At a selected corner, a further supporting white is right to every affine component by its positive kernel. On a finite component a possible contact must have normalized inverse diagonal at most two. The omitted black roots are retained in the possible-contact set. These tests cover contacts anywhere on the core.

The66 cases divide into three complete mechanisms.

- **39 radical cases.** The core has one negative direction and a one-dimensional radical. In an actual Lorentzian realization the radical is a linear root relation: its null root combination lies in the positive-definite perpendicular space to a timelike vector and hence is zero. The selected corner contact domain avoids every zero radical coordinate. Acuteness and the relation force any nonzero contact vector to meet both signs, giving a path of length at most eight. Every extra supporting white is therefore orthogonal to the core.
- **24 cases with no relevant contact.** The nonsingular core is Lorentzian. The contact domain splits into two positive inverse end blocks. A root meeting both ends gives a short path; in either one end its projection cost is at least any one contacted inverse diagonal. Every possible contact in the marked component has full-core inverse diagonal greater than two. Thus no outside root can touch that component. Projection-one contacts elsewhere may allow arbitrarily long outside chains; no bound on those irrelevant chains is claimed or needed. There is no route into the marked component without a first contact to it.
- **Three sealed simple extensions.** The only potentially relevant core-contact node has inverse diagonal p=(5+sqrt(5))/4. All other contacts are excluded by the same end-domain tests. A doubled bond already costs2p>2, so the one allowed contact is simple. Its positive-space residual has square rho=(3−sqrt(5))/4<1/2. Two roots with that same contact would have actual product at least2p−2>0, so at most one occurs. Every remaining core-orthogonal norm-two white has product magnitude at most sqrt(2rho)<1 with it. The normalized acute Coxeter gap forces that product to be zero. The new root therefore has no indirect tail or bridge, and adds at most one node.

The source retains every corner decomposition and then chooses a certified corner. The first two mechanisms give component order at most12; the third gives at most12 after its one possible new root. Arbitrarily many additional supporting whites are allowed in these arguments. The conclusion is not a bounded one-root search.

At an ideal corner, the inherited actual factor-fiber lemma places the whole varying affine factor inside W plus the marked pair. Other incident cusp walls cannot enlarge the component outside this calculation. Hence the component bound has the same meaning for finite and ideal corners, without simple-cusp assumptions.

Exact certificates and scope

Finite/affine positive-semidefinite recognition uses the frozen exact tree/cycle criterion. Every failed strict actual-edge test now includes an explicit nonnegative nonpositive-square vector. A shortest cycle uses an all-ones vector; for a forest, positive leaf Schur eliminations construct the witness by positive coefficients until the first nonpositive pivot. The source verifies each stored vector in the native normalized Gram. Squared tree bonds lie in Q(sqrt(5)); the remaining small matrices and inverse/radical calculations use exact SymPy arithmetic. No floating-point test is used.

Generation15769 and fresh saved replay59083 both terminated with exit0 and PASS_COMPLETE_DISTANCE8_SMALL_CORE_DOMAIN. The replay recomputed the exact certificate and verified its input bindings. The parent agent independently read the complete new source and proof, checking the radius domain, analytic triangle Schur argument, doubled-join and multiple-edge coverage, conditional overlap inverse tests, nonnegative obstruction certificates, all three extension mechanisms and ideal-factor transfer; this was a narrow mathematical/source review, not a second independent enumeration.

The result completes the small-core portion of a distance-eight argument only. The remaining high-core disjoint/overlap and extension geometry, and any resulting dimension theorem, are separate obligations. No external refereeing, formalization, construction or publication-priority claim is made.

Source 85

Source path: `results/fresh_distance9_low_core_20261017.md`

SHA256: 85daf5c370134b41ed6b39daf7650187494ece212f984e9381a34540bce3514a

Complete small-core domain for distance-nine quadrilateral payments

FINAL / FROZEN. Generation and fresh saved replay PASS; narrow internal mathematical/source review PASS. This is a local classification and isolation lemma. It does not establish a new ambient dimension bound or complete the remaining high-rank distance-nine analysis.

Consider an actual quadrilateral face star in a finite-volume hyperbolic Coxeter polytope, with supporting white roots W and opposite black pairs U,V. Normalize roots to square 2. Assume that no U-to-V path of length at most nine has white internal nodes. If a minimal indefinite core of W+U or W+V has order at most five, the configuration is either impossible or has an adjacent marked corner whose whole connected component is isolated and has **order at most 12**.

The source and certificate cover every finite label and every additional supporting white under these hypotheses. They bind the high-rank catalogue and the actual-face, positive-kernel, and cusp-factor arguments of the completed distance-ten proof. No frozen predecessor is changed.

Why smaller cores enter here

Each minimal indefinite core contains both of its black roots: omitting either black leaves a positive-definite actual-edge support. Its proper principal subdiagrams are positive semidefinite. The inherited complete classification bounds its order by ten, and the high-order marked radius by seven.

A connected core of order k with two marked roots has radius at most k−2. Thus cores of order at most three cannot participate in an unpaid disjoint join: their radius plus seven plus the joining edge is at most nine. Two cores of order at most five are excluded similarly. An overlap with a small core requires a shared node with distance sum at least ten. Consequently only order-five radius-three nodes can overlap a high-order radius-seven node.

Disjoint joins are different. Order-four radius-two ports can meet radius-seven ports. Order-five radius-two or radius-three ports can meet the appropriate radius-six or radius-seven ports. The order-five radius-three tree list alone would miss this domain.

Complete all-label enumeration of orders four and five

Every edge of a connected four-node graph belongs to a connected proper three-node subdiagram. If its norm-two magnitude exceeds sqrt(3), adjoining any nonright edge already gives negative determinant, even when the third possible bond is removed. Hence every bond in an order-four minimal indefinite core has label at most six. Equality [6,3] can be affine, so label six must be retained. Parallel and dotted contacts are excluded by the same proper-principal test.

For order five, extend an edge to a connected proper four-node subdiagram. A label at least six gives an indefinite connected triple, or the affine [6,3] equality. In the latter case the connected fourth node has a negative product with its positive kernel, so the four-node diagram is indefinite. Thus order-five labels are at most five. These bounds are proved before enumeration; neither is a numerical cutoff.

The checker enumerates every weighted graph on four nodes with edge alphabet {right,3,4,5,6}, and every graph on five nodes with alphabet {right,3,4,5}. It retains exactly the connected indefinite graphs whose one-node deletions are all positive

semidefinite. Every smaller proper principal subdiagram lies in such a deletion. Black pairs are selected only from strictly positive-definite deletions.

The positive-semidefinite recognizer is elementary and exact. For a tree component, leaf Schur elimination uses squared bond magnitudes 1,2,(3+sqrt(5))/2,3 in Q(sqrt(5)); a zero pivot with a remaining nonzero incident bond is indefinite. A component containing a cycle can be positive semidefinite only when it is precisely a simple affine A cycle. Indeed, the all-ones vector on the cycle has nonpositive square. Equality requires all its bonds simple, and its positive kernel excludes every chord or attachment. This handles disconnected finite/affine deletions without assuming a selected diagram list.

Canonicalization tests every vertex permutation, preserving all bond labels. The resulting census is:

Order	Labelled minimal-indefinite graphs	Canonical types	Relevant marked ports	Inverse-admissible ports
4	392	32	17	11
5	1,050	14	68	18

All marked pairs and ports are retained within each canonical graph. A relabelling therefore cannot omit a marked case. The exact far-port inverse diagonal is computed from spherical-deletion determinants. A native simple contact requires both inverse diagonals strictly below two; a doubled contact requires both strictly below one. The strict inequalities come from actual-edge positive definiteness, not an ideal-corner semidefinite test.

The high-order inverse filters leave seven relevant ports, all in L25: three radius-seven cases at node 9, three radius-six cases at node 9, and one radius-six case at node 8. Their literal contexts and inverse values are saved.

Disjoint joins and every possible joining label

Each low port has a two-node white path; each relevant high port has a three-node white path. Coefficients (1,1/2) give square at most 3/2 on the first path, and (1,2/3,1/3) give at most 4/3 on the second. With native cross magnitude beta, their combined square is at most

```
17/6 − 2 beta.
```

Supporting-white positive definiteness forces beta<17/12. This permits precisely labels 3 and 4; label 5, parallel contacts, and dotted contacts are excluded. Any other acute bonds only strengthen the nonpositive-vector test.

For a proposed cross label, the inverse tests at both endpoints are applied. Eliminating the two spherical white blocks gives the additional strict condition

```
beta² (G_Wlow inverse)xx (G_Whigh inverse)yy < 1.
```

The complete low-first count after the endpoint tests is:

Low core order	Simple joins with two negative directions	Simple white-PD rejections	Label-four white-PD rejections
4	33	0	2
5	60	2	7

Thus all nine possible doubled joins fail the white test. The 93 surviving simple joins have 23 underlying types, with inertia (12,2,0) for order-four low cores and (13,2,0) for order-five low cores. Tree parts are eliminated exactly using squared bonds. At most five cyclic nodes remain, and their exact symbolic congruences are recorded. No floating-point signature is used.

One must also consider multiple cross edges. Six low-first marked-context pairs admit two candidate simple edges. In three, using both produces a wholly white simple four-cycle; in the other three it produces a wholly white simple triangle. The certificate stores the vertex set and its all-ones affine kernel. These contradict positive definiteness even after extra acute contacts are restored. Every nonempty subset of allowed cross edges is therefore excluded. Reversing low/high roles gives the corresponding ordered cases without changing any obstruction.

The two low/high overlap cores

For an overlap, only an order-five radius-three node can be shared. Every such low marked context has at most one eligible node, so larger or different identifications are impossible. After the high inverse gate, there are 30 low-first candidates. Twenty-four fail the low inverse test. The other six pass actual-edge positive definiteness and must be handled geometrically; they are not discarded as numerical near-misses.

There are two literal 14-root cores, each with three high marked pairs. Use low nodes 0 through 4. The right L25 copy maps its node 9 to the shared low node, and its nodes 0 through 8 to 5 through 13.

- The low fork has edges 0–4, 1–4, 2–3, 3–4, with labels 3,4,3,3. The low black pair is {0,1}; node 2 is shared. Its Gram has signature (12,1,1).
- The low path has edges 0–4, 1–3, 1–4, 2–3, with labels 3,3,3,5. The low black pair is {2,3}; node 0 is shared. Its Gram has signature (13,1,0).

The high black pairs are {1,2}, {1,3}, and {2,3}, transported into the right copy. Reversing the two cores gives 12 ordered surviving contexts.

Every unspecified contact between exclusive core nodes would be a marked path of length at most nine. There are 36 such pairs in each low-first context, all checked exactly. Thus no additional negative cross bond can change either core Gram while the no-short-path hypothesis holds.

For every adjacent corner, the checker decomposes the principal Gram into finite and affine components. A further supporting white is orthogonal to any affine component by its positive kernel. On a finite component, a nonzero norm-two contact can occur only at inverse diagonal at most two. Omitted black nodes are included separately in the possible contact domain. These necessary domains include contacts to every core node; they do not assume that new whites meet only marked endpoints.

Isolation of the fork core

In the literal normalization above, the fork core has radical

`(-1,-sqrt(2),0,-1,-2,6,3,4,2,5,4,3,2,1).`

This is an actual root relation in any Lorentzian realization. Its root combination is orthogonal to a timelike vector in the indefinite core span and has square zero; the timelike perpendicular space is positive definite, so the combination vanishes.

Each chosen adjacent-corner contact domain omits the zero coordinate. Its negative coordinates are on the low end and its positive coordinates on the high end. Acuteness and the relation force any nonzero external contact vector to touch both signs. The saved distances then give a marked path of length at most four through that new white. Therefore every additional supporting white is orthogonal to the core. The chosen marked component has order 12.

Isolation of the label-five path core

This core is nonsingular Lorentzian. The selected corner permits contacts only in two end sets. Touching both ends gives a marked path of length at most four. Within either end, the full inverse-Gram block is entrywise positive, and every diagonal is strictly larger than two. The smallest relevant diagonal is 8+4sqrt(5).

For a norm-two external root with contact vector b, projection onto the Lorentzian core gives

`b^T G^{-1} b <= 2,`

because the perpendicular complement is positive definite. A nonzero one-end contact has magnitude at least one and therefore already exceeds two; extra contacts at that end only increase its cost. A two-end contact gives the forbidden short path. Thus every extra supporting white is orthogonal to this core as well. The selected connected marked component has order at most 12.

This is complete isolation, not merely a bounded search for single-node extensions. Arbitrarily many indirectly connected supporting whites cannot enlarge a component to which none of them is incident. At an ideal corner the inherited factor-fiber

lemma puts the entire varying affine factor in W plus the marked pair, so additional cusp walls outside the face star cannot enlarge the paid component either. No simple-cusp assumption is made.

Validation and scope

Generation 59182 and fresh saved replay 5697 both terminated successfully. The parent independently read the all-label bounds, forest/cycle PSD recognizer, complete permutation/port domain, joining-label exhaustion, white-cycle obstructions, and all six overlap contact-domain and isolation arguments, and found no gap. That review did not duplicate the million-graph census or its exact arithmetic. The certificate contains all canonical low graphs, all relevant marked ports, 104 low-first joining-label candidates, the six multiple-edge obstructions, all 30 low-first overlap candidates, all adjacent-corner decompositions for the six survivors, and 216 extra-cross shortcut checks.

The conclusion is restricted to the complete small-core portion of a distance-nine face-payment argument. It supplies neither the remaining high-order extension classification nor an ambient ceiling. The source uses the frozen independent distance-ten review's exact quadratic-field class, and the small cyclic remainders and extension matrices use exact SymPy arithmetic. A fresh replay is a recomputation by this source, not an independent implementation. No external refereeing, proof-assistant formalization, construction, or publication-priority claim is made.

Source 86

Source path: `results/fresh_distance9_supporting_extensions_review_20261017.md`

SHA256: 4a074a5c70634d64581c90c1884c142bd6b560c05c1760c416db712f03062677

Independent review of the exceptional distance9 supporting extensions

The proposed residual-space argument closes the supporting-white extension problem on the supplied69 exceptional contexts. Independent exact reconstruction gives nine full-core types,24 relevant-root cases and46 contexts with no relevant core-contact node. The resulting marked-component order is at most16. No gap was found in the indirect-tail or no-bridge argument under the stated hypotheses.

The independent source imports no producer arithmetic. Its certificate reconstructs normalized Grams directly from the frozen quasi-Lanner census, using SymPy rational matrix arithmetic. The69 literal contexts and selected corners are explicit inputs in the source. This review checks those contexts, rather than independently proving that they exhaust all high-core overlaps. The whole distance9 classification, unknown core cross-edge exclusions, low-order cases and dimension composition remain separate obligations.

Each reconstructed core H has inertia(order(H)−1,1,0). Consequently its orthogonal complement in the ambient Lorentz space is positive definite. For every selected corner, the source recomputes its connected finite or affine blocks, their exact inverse diagonals or positive kernels, the complete necessary core-contact set, and the marked component. Affine blocks permit no external contact: their positive null relation and acuteness force all such products to vanish. Finite blocks give the usual inverse-diagonal bound2. The selected corner places every feasible projection-one contact outside the marked component.

Write J for the inverse normalized full-core Gram. Suppose the short-path alternative has not already paid the face. The saved end partition has maximum left distance plus maximum right distance plus2 at most9. A supporting white root contacting both ends therefore supplies a short path. Within either end, all entries of J are positive, and the exact minimum projection for any two distinct contacts is greater than2. Since every nonzero acute Coxeter contact has magnitude at least1, no multicontact root remains. Thus every core-contact root has a single contact. The candidate nodes satisfy1≤J_ii≤2. If J_ii>1, only a simple bond is possible; if J_ii=1, a simple or doubled bond is possible. Larger labels have projection square greater than2.

For a simple-contact root at node i, write its core projection as v_i and its positive-space residual as g_i. Then

$$v_i^2 = J_{ii} = p_i, \qquad g_i^2 = 2 - p_i = \rho_i, \qquad r_i \cdot r_j = J_{ij} + g_i \cdot g_j.$$

Cauchy bounds the residual product. The exact pair tests either force a positive actual product, force a nonzero native edge giving a path of length at most9, or place the actual product strictly above−1. In the last case the discrete Coxeter gap and acuteness force the product to be0. All doubled-contact candidates at projection-one nodes are separately checked: their residual vanishes, so the sign of the exact core cross product either excludes coexistence or forces a short path. These checks also exclude simultaneous relevant roots, apart from the single relevant root being analysed.

Let G be that relevant root, with0≤rho<1. There is at most one other allowable core-contact type C when G is present, and C contacts a node outside the marked component. The actual roots G,C must be orthogonal. If its simple-contact projection is p_C>1, two distinct roots at the same contact node would have actual product at least2p_C−2>0. If p_C=1, two such roots could be acute only by being orthogonal with opposite unit residuals. Both cannot have the same strictly negative residual product with G. Hence at most one C occurs. Identical complete roots are a single actual wall, not two occurrences.

The four exact residual possibilities are:

p_G	p_C	projection cross product P	residual determinant	inverse GG entry
4/3	4/3	2/3	0	—
3/2	1	1/2	1/4	4
7/4	1	1/2	0	—
5/4	1	1/2	1/2	2

In each case the residual Gram is S=[[rho,−P],[−P,rho_C]]. If det(S)=0, its kernel has strictly positive coordinates, so a positive combination of the two residual vectors is zero. Any white root orthogonal to the full core has nonpositive products with both G and C; the positive relation forces both products to be zero. No indirect attachment can meet either root.

If det(S)>0, S is a positive definite Stieltjes matrix and S^{-1} has nonnegative entries. A norm2 white root E orthogonal to the core, but joined to G, has contact magnitude at least1. Its projection on span(g,c) therefore has squared norm at least(S^{-1})_GG. If that entry is greater than2, E is impossible. If it equals2, every inequality must be equality: E has contacts(−1,0), no component outside span(g,c), and

$$E = -2g - (S^{-1})_{CG}c.$$

For any other core-orthogonal white Z, acuteness gives g·Z≤0 and c·Z≤0, hence E·Z≥0. Acuteness again forces E·Z=0. Also E·C=0. The equality determines E uniquely. Thus the one permitted branch is sealed; it cannot bridge to another outside component or continue through further core-orthogonal whites.

If C is absent, all other supporting whites coexisting with G are orthogonal to H. Let E be a whole connected component among them, of rank m, and let b be its nonnegative contact magnitudes with G. Positive-space projection gives b^T G_E^{-1}b≤rho. For every connected positive definite norm2 Coxeter Gram of order m,

$$(G_E^{-1})_{ii} \geq m/(m+1).$$

This bound does not assume a bounded rank classification. Choose a simple spanning tree. Its nonnegative adjacency matrix is entrywise at most the original one, so positive definiteness and the Neumann-series inverse comparison reduce to that tree. Root the tree at i. The Schur complement gives inverse diagonal1/(2−sum of rooted child inverse diagonals). By induction, child subtrees of sizes m_j contribute at least m_j/(m_j+1), whose sum is at least(m−1)/m. This yields m/(m+1). The base case is1/2. The same inverse comparison handles larger bond magnitudes.

A contacted component consequently satisfies m≤floor(rho/(1−rho)). Two distinct contacted components would each cost at least1/2, contradicting rho<1. The bound controls the entire indirect component, including roots having no direct contact with G. It is not merely a bound on the first attached node. In the present cases the maximum alone-tail rank is3.

The no-bridge conclusion now follows in both alternatives. With C present, no core-orthogonal route can continue from G except the sealed single root just described; G and C themselves are orthogonal. With C absent, at most one complete finite component of the stated rank can attach to G, and all other core-contact root types have already been excluded or would force a short path. If there is no relevant core-contact node, nothing can enter the marked core component at all. The source combines these bounds with each actual preferred component and obtains order≤16 in all69 contexts.

Generation and a fresh saved replay both pass PASS_INDEPENDENT_DISTANCE9_SUPPORTING_EXTENSION_CONTEXTS. The replay repeats the exact69-context construction and all residual tests. This is an independent local arithmetic and mathematical review; it makes no independent complete-overlap enumeration or new dimension claim.

Source 87

Source path: `results/fresh_e6_b10_half_coupon_forest_budget_20261018.md`

SHA256: 0db9f24b0868bf964efe8e9e2e7544eb7bdbf0f9064229abd7ecc913c0087dc7

Enlarged E6/D11 and A1/B10 coupons retain the finite maximum464

At rank64 the exact maximum finite cost remains **464**, attained by the forest of four D16 components, after both of these additions:

- Upgrade the typed whole E6/D11 coupon from 1/4 to 1/2 per eligible marked pair, increasing its component-pair cost from 1 to 2.
- Add a half coupon between a whole A1 and the doubled-terminal mark of a whole B10, adding 1/2 per component pair.

The first supports the order17 D118 refinement. The second is included explicitly in this counting rule; this report does not assert the separate ideal order11 theorem that may use it. All other coupons and the proved depth11 zero rule and finite D16/D17 exchange are retained from the finite closure. The source and saved certificate use standard-library exact arithmetic. This is a finite counting theorem, not an ambient dimension bound.

Complete finite-type reduction

Work in doubled costs. Same-component costs are unchanged: A16/D16 have 232, B16 has 208, A17 has 228, B17 has 210 and D17 has 254. For ranks at most 15 the conservative same-component bound is r(r−1). For A/B/D of rank r≥18, start from 16r−68 for A/B, 16r−64 for D18/D19 and 16r−66 for D≥20. The new whole-component deep distance8 zero rule subtracts twice the number of eligible pairs, respectively

`max(0,r-30), max(0,r-31), max(0,r-32).`

These are the same proved all-rank formulas used in the frozen finite closure; their geometry is not reproved here.

Keep A17 separately because of its cross-component coupons. Every other finite component is dominated by A_r for r≤9, except that E6 is retained at rank6, and by D_r for r≥10. The inherited finite-type classification includes A/B/D, E6/E7/E8/F4/H3/H4 and arbitrary dihedral labels. The source checks same-component dominance and every one of 38,025 type/partner comparisons at the enlarged coupon weights.

The changed B10 interface is explicit. With A1, doubled cross cost for D10 is 2 and for B10 is 1; with A2 the costs are 4 and 2. Thus replacing B10 by D10 still dominates both its new A1 half and old A2 coupon. Other partners give no additional B10 cost. E6 continues to dominate other rank6 components, now with doubled E6/D11 cross cost 4. No old B10 coupon is added again after replacing it by D10.

Complete forest domain and exact certificates

Separate all D10 through D16 components and all A17 components. Their multiplicities form 753 complete count vectors of total rank at most 64. All remaining components have ranks1–9 or at least17, with E6 as the rank6 representative and D_r as the representative for r≥17.

For counts d10,…,d16,a17, the five-coordinate interaction key is

`(2d10+3(d11+...+d16)+2a17, d11+d12, d13, d11, d14).`

There are 348 keys. Let the coordinates be (b,u,a,d,e). The residual item of rank r has its doubled same-component cost plus these cross terms:

- rb when r≤5;
- an additional 2u+e for r=1;
- an additional 4d+2e for r=2;
- an additional 2a for r=4;
- an additional 4d for r=6.

The last term is the upgraded E6/D11 cost. The new B10 coupon disappears only after its explicit D10 domination check. These formulas also retain the A1/D11 and A1/D12 unit upgrades, A4/D13 neighbor coupon, A1/A2–D14 neighbor halves, A17 coupons and all broad small-component coupons. The source compares the compressed formula with literal typed component interactions 42,921 times.

For each key, generation uses item-first unbounded dynamic programming. The saved verifier does not rerun that optimization. For every residual rank n it checks all last-item upper inequalities

```
V[n] ≥ V[n−r]+w[r],
```

and one saved attaining predecessor with equality. Induction on n proves both the upper bound for every residual forest and attainment by a typed forest. There are 597,168 inequalities and 22,272 predecessors.

Each of the 753 large-component vectors is then combined with the exact residual value at its remaining rank. Its attaining typed forest is reconstructed and its full same-component plus cross-component cost is recounted literally. This covers every rank64 forest after domination. The largest doubled value is 928, hence cost 464, and four D16 components attain it. They carry no cross coupons, so this attainment is unchanged by either addition. It is attainment in the formal finite-forest domain, not an assertion that such a vertex occurs in an actual chamber.

Geometric and global scope

The baseline finite rules are already proved. Enlarging a finite coupon cannot invalidate any old face-payment lower bound. The new E6/D11 half is exactly the coupon required by the order17 refinement. Thus the finite budget 464 remains available when that refinement is composed with the baseline rule, including the optional A1/B10 half.

The new order11 ideal application and the complete ideal curvature comparison are separate obligations. No conclusion about dimension63, physical realizability, or a new example follows from this budget alone. The certificate does not repeat the inherited geometric classification, complete overlap census or all-rank deep-corridor proof.

Validation

Final source-bound generation dfea64 and separate saved verification abf19c both terminated exit0/PASS. They passed all complete count-domain, type-domination, Bellman and typed-attainment checks. The saved verification uses only integer/Fraction arithmetic and saved Bellman tables; no global or local LP is invoked, and it does not regenerate the optimizing tables.

The parent and two other reviewers each completed a full mathematical source/report read. They checked typed coupon accounting and domination, E6/A17 distinctions, complete large/residual decomposition, interaction compression, Bellman induction and typed attainment. Those source reviews do not duplicate the arithmetic enumeration, the inherited finite-type classification or geometric payment censuses.

All ten direct source bindings, the Python syntax and all four local report links pass. Source, certificate and report are final and frozen; there are no pending computations.

Source 88

Source path: `results/fresh_e6_d11_quarter_coupon_forest_budget_20261017.md`

SHA256: 197c8ddba86965b285ed62d617eb31c9837fa871b7fbcd8a6204240d08e12ad8

The E6/D11 quarter coupon leaves every finite maximum through75 unchanged

Adding a quarter payment between either long-arm endpoint of a whole E6 component and either short fork leaf of a separate whole D11 component leaves every finite-forest maximum through rank75 unchanged. In particular, the maximum at ambient65 remains487 against curvature480, and at66 it remains495 against495. The new local ideal transfer is therefore affordable within the existing finite budget, but it does not remove the finite obstruction at65.

The source and certificate prove this conditional counting statement. The geometric justification of the coupon is the separate order17 D transfer. There are two eligible roots of each type, so the added total per component pair is22(1/4)=1. No old coupon has E6 as its small component.

Complete finite counting

Start with the frozen full A2/D11 finite budget. Retain its same-component payments and every existing cross-component coupon. The only change is the additional E6/D11 cost just stated.

For type domination, represent every rank6 component by E6, whose same-component upper cost is unchanged and whose new interaction is nonnegative. All other representatives stay the same. The source checks every actual-type/partner comparison, including exceptional types and the full I2(m) family represented by its inherited uniform cost.

The1,608 complete large-component vectors and357 interaction keys remain valid. Large components are D10 through D16 and coupon-bearing A17. Residual components have ranks1 through9 and17 onward. There is no coupon joining two residual components; the new E6/D11 interaction has its D11 member in the large vector. For each fixed vector, add exactly2*d11 to the doubled residual cost of rank6. Other residual costs remain unchanged.

Generation uses item-first unbounded knapsack tables. The verifier checks every last-item upper inequality and an attaining predecessor for every entry. It also checks the compressed interaction formula against literal component-pair costs, reconstructs actual typed forests, and checks every global alternative at every rank through75. The resulting certificate has:

Check	Count
Large component vectors	1608
Residual interaction tables	357
Literal interaction formula checks	109344
Bellman upper inequalities	860013
Attaining Bellman predecessors	26775
Global forest alternatives	22504
Typed global attainers	76

Every resulting maximum agrees exactly with the frozen predecessor. This is an upper-and-attainment proof, not a search over selected forests. It reuses the predecessor's complete large-vector domain and payment formulas; it does not independently re-prove the geometric payment rules.

Verification

Generation79676 and separate saved replay40871 both terminated exit0/PASS. A separate internal full source/report mathematical review, including the imported base, cross-cost, representative, domain and residual-cost helpers, found no unresolved step. It did not independently replay the arithmetic. The replay reads the saved tables and checks all upper and attaining relations rather than rerunning their item-first generator. No new ambient theorem or example is asserted here.

```
python3 scripts/fresh_e6_d11_quarter_coupon_forest_budget_20261017.py
python3 scripts/fresh_e6_d11_quarter_coupon_forest_budget_20261017.py --verify
```

Source 89

Source path: `results/fresh_exact_finite8_forest_budget_20261017.md`

SHA256: db27fff369213805c024f134c0d384b42547376f41a5c55fdb8c182b20db4148

Exact maxima of the current finite-corner rule

The unchanged finite-corner payment rule has maximum cost **548 in rank69** and **553 in rank70**, over all abstract finite Coxeter forests of those ranks. The first is attained by `A1+A17+A17+A17+A17`; the second by `A2+A17+A17+A17+A17`. These are universal upper bounds at actual finite vertices, with equality witnesses in the larger class of abstract vertex diagrams. No ambient realization of a maximizing diagram is asserted.

The source and certificate give the exact maximum in every rank1 through75. This is a refinement of the existing8n budget. It changes no face payment, adds no finite coupon, and proves no new ambient dimension bound by itself.

The complete unchanged rule

The inherited base rule pays one on a same-component pair if its distance is at most8, or its distance is9 with a whole-component leaf mark, or its whole component has rank at most17. Its exact component cost is binom(r,2) for r<=17, 8r-34 for A/B at r>=18, and8r-32 for D at r>=18. These are all-rank graph identities using the finite Coxeter classification, not sampled formulas.

Retain all broad half-payments: between distinct whole components, one of rank at most5 with arbitrary mark and the other D11 through D16 at its three fork marks, A13/A15 at its middle, or A17 at its two specified path positions. Also retain the A4/D13 neighbor halves, and the D12 short-leaf upgrade and A2/B10 halves. Every unordered marked pair is counted once.

Domination by one representative per rank

For the maximization problem each connected finite type may be replaced by the following type of the same rank without decreasing its total cost:

Rank	Representative	Reason
1 through5	A_r	Same complete base cost and broad small-component eligibility; A1/A2/A4 retain every additional small-type coupon
6 through9	A_r	Same complete base cost; no cross-component coupon
10	B10	Same complete base cost, plus its A2 coupon
11 through16	D_r	Same complete base cost; three broad large marks dominate the one A13/A15 middle mark; D12/D13 retain their additions
17	A17	Same complete base cost, plus its two broad large marks
18 and above	D_r	Base8r-32 dominates the A/B base8r-34; no cross-component coupon

The exceptional finite types and arbitrary I2(m) occur only among the small ranks already covered. The domination argument uses counts of all eligible marked pairs, not a graph embedding or a geometric deformation. Every coupon coefficient is nonnegative. It is therefore valid to replace one component at a time: no other component's contribution decreases. Conversely, every listed representative is itself a finite Coxeter type, so a maximum attained among these representatives is also attained in the full abstract finite-forest class.

Let a_i count representative A_i components for1<=i<=5, let `a=sum_i i*a_i`, let t count D11 through D16, and let u,b10,b12,b13 count A17,B10,D12,D13 respectively. For a representative forest the cost is exactly

```
sum_components base(rank)
  + a*(3t/2+u) + a1*b12 + a2*b10 + a4*b13.
```

There are three broad eligible marks on each D and two on each A17, each paid at one half against all a small marked nodes. The D12 upgrade increases each of its two short-leaf weights by one half against each A1, giving a1*b12. A2/B10 has two pairs of weight one half, giving a2*b10. The two A4 endpoints against the distinct D13 neighbor give a4*b13. These additions involve disjoint pair roles and count no pair twice.

Complete exact maximization

Enumerate all count vectors of the eight large coupon ranks10 through17 whose total rank is at most75. There are1,608 such vectors. For fixed large counts the cost of each remaining connected component is additive: its rank is1 through9 or at least18, and its entire cross-payment contribution is fixed by the large counts.

Only the tuple `(t,u,b10,b12,b13)` matters for this residual additive problem. There are377 distinct tuples. Double every cost to use integers. For each tuple k let w_k(r) be the doubled residual component cost, including all its cross-payments, and define

```
F_k(0)=0,
F_k(s)=max_(allowed r<=s) [F_k(s-r)+w_k(r)].
```

Allowed ranks are1,…,9 and18,…,75. A rank greater than s cannot occur. This Bellman identity is an induction proof of completeness: removing any one component of a residual forest leaves a lower-rank residual forest; conversely every maximizing predecessor may be extended by that component. The positive component ranks make the recursion finite. The certificate stores all76 values and an attaining last rank for every tuple.

For each ambient rank n, maximize the doubled large-component base cost plus F_k(n minus their rank) over every large vector of rank at most n. This covers every representative forest. The source also reconstructs an attaining forest for each n and checks its full cost directly using the displayed coupon formula.

Maxima near the current frontier

Rank n	Exact maximum	Slack below8n	One maximizing forest
66	513	15	D15+3 A17
67	528	8	D16+3 A17
68	544	0	4 A17
69	548	4	A1+4 A17
70	553	7	A2+4 A17
71	559	9	A3+4 A17
72	566	10	A4+4 A17
73	574	10	A5+4 A17
74	578	14	A1+A5+4 A17
75	583	17	A2+A5+4 A17

At ambient69 the finite curvature coefficient is544. The unchanged payment rule can cost548, so even this exact abstract-forest budget exceeds the available coefficient by4. Its obstruction is explicit: the four A17 components already cost544, and their half-coupons against A1 add4. Thus simply sharpening the estimate of the same rule cannot make every finite-vertex

coefficient fit at69. A smaller geometric payment rule, restrictions on actual vertex diagrams, or a global compensation argument would be needed. This does not prove that the abstract obstruction occurs in an actual finite-volume chamber.

At ambient70 the exact553 cost is7 below the finite curvature coefficient560, whereas the older8n bound only gives equality. The existing dimension argument may retain that older bound and its positive global additive constant; this refinement is a separate theorem with no new ideal costs.

Verification status

Primary generation and a fresh separate-process saved replay both terminated exit0/PASS on all1,608 large vectors,377 residual tables and75 positive-rank maxima. The independent certificate review uses no producer arithmetic. It verifies all885,950 Bellman upper-bound alternatives and28,275 attaining predecessors,22,504 global upper bounds, and all maximizing forests by direct marked-pair counting. It reconstructs the large-vector domain by component number and nondecreasing multisets, and separately checks type domination on228 finite-type profiles and51,984 ordered partner comparisons, including the symbolic arbitrary I2 family. Its report records final generation and fresh replay.

A separate full internal source/report mathematical read passed the type domination, exact coupon counts, complete Bellman and large-vector domains, and the distinction between abstract attainment and actual realizability. Finite-type classification and the unchanged geometric payment rule are inherited inputs; the present computation does not establish any new face coverage. No external refereeing, formalization, ambient realization or new dimension theorem is claimed. Source, certificate and report are final/frozen.

Source 90

Source path: `results/fresh_exact_finite8_forest_budget_20261017_review.md`

SHA256: c025c803742581d1e0b3c051789c55a252c0e8da86e201f379a5599f1561e23d

Independent review of the exact finite-corner budget

The exact finite-forest budget passes a complete mathematical/source read and a separate certificate check with no producer arithmetic imports. The current unchanged rule has exact abstract-forest maximum **548 at rank69** and **553 at rank70**. These are universal upper bounds for actual finite vertices; attaining abstract diagrams do not assert ambient realizability. No new face coverage or dimension bound follows from this budget alone.

The review source independently checks the type domination, complete large-component domain, every Bellman alternative and an attaining predecessor, and every global maximum and attaining forest. Final generation and a separate fresh saved replay both terminated with exit0/PASS; the saved certificate binds the frozen primary source, data and report.

Complete type domination and coupon count

The finite Coxeter classification is inherited: the only unbounded-rank connected families are A, B and D; the exceptional types have rank at most8, and arbitrary I2(m) has rank2. The whole-component rule pays every internal pair through rank17. At larger rank its inherited exact base costs are$8r-34$ for A/B and$8r-32$ for D. These all-rank path/fork identities were read as proofs, not inferred from a finite sample.

The reviewer independently counts twice the cross-payment between any two component types. A small component of rank $r\le5$ contributes$3r$ against D11–D16, r against A13/A15, and$2r$ against A17. Add2 for each whole-component pair A1/D12, A2/B10 or A4/D13. The first addition upgrades two short-leaf halves, the second adds two A2-mark halves, and the third adds two A4-endpoint halves at the distinct D13 long-arm neighbor. These roles are disjoint from the appropriate previously counted pairs.

This verifies the claimed exact representative-forest formula. It also proves componentwise domination: replace ranks1–9 by A, rank10 by B10, ranks11–16 by D, rank17 by A17 and larger ranks by D. Internal cost cannot decrease, nor can the cross-payment against any other component. Because all interactions are pairwise and nonnegative, replacements one at a time preserve the inequality for an entire forest. Every representative is itself a finite Coxeter type, so domination gives both a valid upper search space and attainable abstract maximizers.

As an additional independent check, the reviewer constructs228 type profiles through rank75 and verifies all51,984 ordered partner comparisons. One symbolic rank-two profile covers every I2(m),$m\ge4$; I2(3)=A2 is handled separately because of the special A2/B10 coupon. The symbolic profile's cost and broad small-component eligibility do not depend on m. This check is not a new proof of finite Coxeter classification; that classification and the inherited same-component rule remain mathematical inputs.

Bellman certificates prove every residual optimum

The producer enumerates large-rank counts successively. The reviewer instead enumerates the number of large components and all nondecreasing rank tuples from10 through17 with total at most75. Exact set equality verifies all1,608 count vectors, their ranks, base costs and377 distinct coupon keys.

For each coupon key, the remaining ranks are1–9 or18–75. Their costs are additive once the large counts are fixed. The reviewer derives each residual cost by summing its independently counted interactions with a representative large-component

multiset, rather than calling the producer's residual formula.

For every saved state s and every allowed component rank r≤s, it checks

`saved_F(s) >= saved_F(s−r) + independently_counted_cost(r).`

It also checks that the saved last rank is permitted and attains equality. There are **885,950 upper-bound alternatives** and **28,275 attaining predecessors** across377 tables. Induction from F(0)=0 proves both directions: no residual forest exceeds the saved value, and following positive-rank predecessors constructs a forest attaining it. This is certificate verification, not a heuristic optimizer or a sample of forests.

The reviewer then checks **22,504 global alternatives**: for every total rank and every feasible large count vector, the reported optimum dominates its exact large cost plus the residual-table value. It independently counts the full pair cost of each reported maximizing forest and verifies equality with the selected alternative. All75 positive-rank maxima are thus certified; the zero-rank base case is also checked.

Consequences and limits

At rank69 the witness is `A1+4 A17`. Four A17 components cost544 internally; their existing half-coupons against A1 add4, giving548. The finite curvature coefficient is544, so this exact abstract-forest budget still exceeds it by4. Improving only the estimate of this same rule cannot eliminate that abstract obstruction. It does not follow that this vertex diagram occurs in an actual finite-volume polytope.

At rank70 the witness is `A2+4 A17`, with cost553. The finite curvature coefficient is560, leaving slack7. This refinement may be combined with a separately proved ideal-vertex and face-payment argument; this package does not supply those arguments.

The unchanged four coupon packages are bound directly: the base leaf-nine rule, broad A17/D11 rule, A4/D13-neighbor addition, and D12/B10 additions. No geometric payment is added or removed here.

Verification status

The independent complete comparison, final hash-bound generation and separate fresh saved replay all terminated with exit0/PASS. The full producer source and proof, together with the inherited coupon definitions and all-rank base-count arguments, received a mathematical read with no identified gap. A separate root review read the complete independent checker and its mathematical argument, including the alternate component-count enumeration, coupon type domination, Bellman upper bounds and attaining predecessors, and all global bounds and witnesses; that review did not duplicate this certificate arithmetic. The 16 direct source bindings, report links and Python syntax were checked. No full raw-forest enumeration, physical vertex realization, external refereeing or proof-assistant formalization is claimed.

Source 91

Source path: `results/fresh_finite7_ideal5_triangle_payments_20261017.md`

SHA256: 00d3f334ffefc3b926edfdbe7b319a7f6792d83e23fe94511ede7dc60b71bf75

Finite distance seven suffices for the triangle payments

Every triangle of a finite-sided finite-volume real hyperbolic Coxeter polytope receives at least two units under the following rule:

- At a finite corner, pay a marked graph distance at most7.
- At an ideal corner, pay distance at most5, or distance6 or7 when a mark is within5 steps of a branch vertex or an endpoint of a nonsimple bond in the actual varying affine factor.
- Retain the whole-factor ideal payment for factors of order at most9.

This sharpens the finite-distance requirement in the completed finite8 triangle theorem. It does not yet allow removing finite distance8 units from the global rule: quadrilateral coverage and the full changed finite budget remain separate obligations. The current ambient ceiling stays65.

The source and certificate re-enumerate all1,025 high-core marked triples and verify the inherited four exceptional interfaces.

Proof and arbitrary supporting diagrams

Choose an inherited minimal indefinite core containing all three black side roots. Every other root of the core supports the triangle. A corner path avoiding the third black persists after all supporting roots are restored. Its displayed branch or nonsimple bond also persists. At an ideal corner, the actual factor-fiber theorem places the two marks and their path in the same varying affine factor.

For core order at most6, the unchanged all-label Steiner-tree argument gives at least two distinct marked-pair paths avoiding the third mark, each of length at most5. They satisfy the new rule.

For orders7 through10, the complete inherited core catalogue gives 105,224,336,360 marked triples, respectively. The source repeats every triple and every actual corner path/feature test, now discarding every elementary payment whose path has length8. No new exception appears. Exactly the same four triples of the simply-laced10-node L25 core need a non-elementary payment:

Marked triple	Alternate marked pair	Path length	Protection
(0,1,9)	(0,9)	6	Affine E8 positive kernel
(0,2,9)	(0,9)	6	Affine E8 positive kernel
(1,2,8)	(1,8)	6	Finite E8 inverse
(1,2,9)	(1,9)	7	Affine E8 positive kernel

Each has one elementary payment, and its alternate pair is distinct from that payment. The predecessor's exact kernel and inverse argument protects a simple leaf and its degree-two neighbour in the entire alternate marked component, allowing arbitrary additional supporting roots. No classical irreducible affine Coxeter graph has this protected simple-leaf configuration. Consequently the component is finite or exceptional affine of order at most9. The displayed finite path has length at most7; the exceptional affine case has its whole-factor unit. Both cases retain the second payment.

The proof does not require that a protected leaf be a mark. It does not infer whole-component finiteness from a truncated diagram. It uses the same arbitrary-support E8 argument as the predecessor, with the sharper explicit path lengths checked here.

Verification scope

Generation61e45a and a separate fresh saved replay332ec3 both terminated exit0/PASS. The source repeats all1,025 triples, checks exactly the four old exceptions and reconstructs the E8 kernel/inverse interfaces. The graph and matrix routines are shared with the predecessor; this is not an independent arithmetic implementation or a new core classification.

A separate full source and inherited-proof read passed the low-core all-label argument, new path threshold, four distinct protected alternatives and arbitrary-support scope. That reviewer did not rerun the numerical census. The complete new report also passed that review. All11 direct source bindings, three local report links and Python syntax pass. Source, certificate and report are final/frozen. No quadrilateral theorem, changed finite maximum, new ambient bound, new polytope, external refereeing or formal verification is asserted.

Source 92

Source path: `results/fresh_finite8_a17_d11_coupon_budget_20261017.md`

SHA256: fdbb7ff1ba9bb15a88622628174b262155d82790922dd0cbdc9b6a7a5a15d6be

Finite-eight cost with whole A17 and D11 half-coupons

FINAL/FROZEN: generation, fresh saved replay and separate full mathematical/source review PASS.

For a finite Coxeter vertex of dimension n<=80, the following rule has total cost at most **8n**. Retain the unit payments of the finite-eight budget: two marks in the same whole component pay one if their distance is at most8, their distance is9 with an actual whole-component leaf mark, or that component has rank at most17. For marks in distinct whole components, add one half when one component has rank at most5 and the other is one of:

- D11,…,D16, marked at the fork center or either short leaf;
- A13 or A15, marked at its unique middle;
- A17, marked at path position5 or11, with positions numbered0,…,16.

Count each unordered marked pair once. All earlier half-coupons are retained. The A17 rule covers both sealed cycle18 coupons from the ideal-five local theorem. That theorem needs only an A1 on the small side. Allowing all ranks at most5, and also allowing D11, is a proved counting enlargement; this statement does not claim a new geometric equality configuration uses either enlargement.

Counting proof

The predecessor proves the all-rank base costs: at rank r>=18, types A and B cost8r-34 and type D costs8r-32. Every component of rank r<=17 costs at most binom(r,2)<=8r. These bounds include all finite labels through the finite-type classification; their validity does not come from extrapolating a graph sample.

Let a be the total rank of small components of rank at most5. Their base slack is at least6a. Let t count whole components D11,…,D16,A13,A15. Each uses at least11 ranks, has at most three eligible marks, and has base slack at least8. Let u count whole A17 components. Each uses17 ranks and has two eligible marks. Its base slack is zero, so it cannot be included in t. All remaining components have nonnegative base slack.

There are at most a*(3t+2u) eligible cross-component pairs, at half a unit each. Consequently the excess above8n is bounded by

```
E = a*(3t/2+u-6)-8t,
a+11t+17u<=n<=80.
```

If the coefficient of a is nonpositive, E<=0. Otherwise maximize a at 80-11t-17u. The complete possible (t,u) pairs and maxima are:

t	u	Maximum a	Coefficient of a	Maximum E
3	2	13	1/2	-35/2
4	1	19	1	-13
4	2	2	2	-28
5	0	25	3/2	-5/2
5	1	8	5/2	-20
6	0	14	3	-6

t	u	Maximum a	Coefficient of a	Maximum E
7	0	3	9/2	-85/2

Completeness is elementary: t<=7; for t<=2, the allowed u makes the coefficient nonpositive, while the rank inequality bounds u by2 for t=3,4, by1 for t=5 and by0 for t=6,7. When t=3 only u=2 has positive coefficient. All displayed excesses are negative. This proves the counting theorem in every dimension through80 without a rank-partition enumeration.

The source and certificate additionally check both endpoints of the affine function E for every allowable integer n,t,u. The whole-component and actual-leaf hypotheses are essential. The geometric assertion that every triangle and quadrilateral receives enough payment is a separate obligation; **no ambient dimension improvement follows from this counting theorem alone**.

Verification scope

Generation and a separate fresh saved replay both terminated exit0/PASS for all1,487 endpoint checks. A second agent read the complete proof and source, including every component slack, the cross-component pair count and all seven positive-coefficient cases, and found no gap. That review did not duplicate the1,487-case arithmetic run. This package reuses the predecessor's all-rank finite base proof and verifies the new cross-component allowance. No external refereeing or proof-assistant formalization is claimed.

Source 93

Source path: `results/fresh_finite8_a4_d13_neighbor_coupon_budget_20261017.md`

SHA256: 61d396f2384e4a5a0badb81930324c3c72317e3353420f99ea6cd5f76dcc26b5

Finite-eight cost with the whole A4/D13-neighbor half coupon

For every finite Coxeter vertex of dimension n<=80, the complete frozen finite8/A17/D11 payment rule remains within **8n** after adding the following half payment:

> For marks in distinct whole components, pay one half when one mark is an endpoint of a whole A4 and the other is the unique non-leaf neighbor of the fork center of a whole D13.

Count every unordered marked pair once. The new D13 root is the first root on the long arm adjacent to the fork center. It is neither the center nor a short leaf, so these pairs do not duplicate the inherited D13 coupon. The whole-component hypotheses are essential. Applying this counting allowance to a particular quadrilateral requires a separate local transfer proof. This package changes no ideal payment and proves no new ambient bound.

The source and certificate are separate successors to the frozen spectrum/C3 checkpoint. That checkpoint is not modified or extended by this package.

Counting proof

Retain the notation of the frozen budget. Let a be the total rank of whole components of rank at most5. Let t count whole components D11 through D16, A13 and A15; let u count whole A17 components. Let b count whole D13 components, so 0<=b<=t.

All same-component unit payments, actual leaf-nine payments and inherited half-coupons remain unchanged. The predecessor proves that small components have base slack at least 6a, every component counted by t has base slack at least8 and at most three old eligible marks, and each A17 has two eligible marks and zero base slack. All other components have nonnegative base slack. These estimates use its all-rank finite-type base proof, not an enumeration of forests up to dimension80.

A whole D13 has all binom(13,2)=78 same-component pairs paid and hence slack 8*13-78=26. Relative to the old lower bound8, each D13 supplies18 additional units of slack and uses two extra ranks relative to the old minimum11. Thus

```
a+11t+2b+17u <= n <= 80.
```

Each whole A4 has exactly two endpoints and the D13 has one new eligible root. The two new pairs each cost one half, giving one unit per A4/D13 component pair. The number of whole A4 components is at most a/4. Consequently the added payment is at most a*b/4, while the refined slack is at least 6a+8t+18b. The excess above8n is therefore at most

```
E_new = a*(3t/2+u-6+b/4)-8t-18b
      = E_old + b*(a/4-18),
E_old = a*(3t/2+u-6)-8t.
```

The old rank inequality a+11t+17u<=80 still holds, so the frozen theorem gives E_old<=0. If b=0, there is no new term. If b>=1, then t>=1 and the refined rank inequality gives a<=80-11-2=67. Hence

```
b*(a/4-18) <= -5b/4 < 0.
```

This proves the enlarged rule in every dimension through80. No new table of partitions or geometric existence assumption is needed. In particular, the estimate deliberately ignores the additional slack available from a whole A4, so it is a conservative allowance.

Exact certificate and scope

The independent-of-optimizer Fraction checker verifies the component costs and the analytic dominance identity. It also checks both endpoints of the affine function in `a` for every allowed integer `n,t,b,u`, for every `1<=n<=80`. There are3,465 distinct endpoints in total and138 at dimension80. Counting both endpoints separately when they coincide at `a=0` gives140 at dimension80; this explains the difference from the exploratory count. All38 positive-coefficient triples at dimension80 have strictly negative maximum excess. The1,487 inherited budget endpoint records are checked as well, and their frozen source/data/report hashes are bound.

FINAL/FROZEN. Primary generation and a separate fresh saved replay both terminated exit0/PASS. Root read the complete source and mathematical proof, including the distinct new root role, component-pair count, refined rank/slack bounds and analytic dominance; no gap was identified. That review did not duplicate the3,465 endpoint arithmetic run. All4 direct source bindings,4 local report links and source syntax pass. These checks establish the finite budget only. They do not assert that a proposed D17 pair can be removed, that its replacement corners contain the required whole A4 and D13, or that a new global curvature comparison succeeds.

```
python3 scripts/fresh_finite8_a4_d13_neighbor_coupon_budget_20261017.py
python3 scripts/fresh_finite8_a4_d13_neighbor_coupon_budget_20261017.py --verify
```

Source 94

Source path: `results/fresh_finite8_cycle_replacement_20261017.md`

SHA256: 33a5320dacc54b69623dcde3eccc717f28bcf24dbe37e42c6ce0e91ab3dd0ee3

Finite-eight and marked-leaf-nine replacements

Completed internally checked local theorem; verification scope is recorded below. Let Q be an actual quadrilateral of a finite-sided finite-volume hyperbolic Coxeter polytope. No arithmetic or simple-cusp hypothesis is imposed. Use the following corner payments:

- A finite corner pays one if its marked roots are connected at distance at most8, or at distance9 with one mark a leaf of its **whole actual connected component**.
- An ideal corner pays one at marked distance at most6, or at distance7,8,9 with a mark within6 steps of a feature of its varying affine factor. Features are vertices of degree at least3 and endpoints of bonds of label at least4.
- The already proved half-coupon between a whole A1 component and the unique middle of a whole A15 component is retained.

Then an ideal corner in an affine cycle of order at least15 at marked distance7,8 or9 is replaceable by these payments at the other corners. The only equality case uses two of the stated half-coupons. The same replacement holds for a classical affine tree at those distances if both marks are farther than6 from its features.

Adding the existing unit payments on whole finite components of order at most17 and whole affine factors of order at most14 preserves this statement. This package repairs a specific finite-distance-nine dependency of the mixed ideal-six proof. It does **not** establish the complete distance-eight high-core overlap theorem, cover every preferred small-factor pattern, prove a total finite budget, or improve the ambient dimension bound.

The source and certificate contain740 exact cycle rectangle values,66 single-profile finite diagrams, three isolated exceptional analyses and192 tree cases.

Actual star, profiles and spectators

Write the actual face star as H=W+a+b+c+d, with opposite black pairs {a,c},{b,d}. The white diagram W and each actual edge diagram W+one black are positive definite. At the chosen ideal corner W+a+b, the factor-fiber lemma gives the varying irreducible affine factor F containing a,b and a positive-definite diagram Z orthogonal to F. This remains true at nonsimple cusps.

Normalize roots to square2. A direct adjacent black bond gives a unit payment, so in the unpaid analysis b.c=a.d=c.d=0. Let nu be the positive affine kernel representing the actual chosen cusp. The opposite sides do not contain this cusp, so h=-c.nu and k=-d.nu are strictly positive. Every nonzero native contact magnitude is at least1.

For a cycle of order m, let the two old marked arcs have lengths P,Q, with P the marked distance and Q>=P. Here P is7,8 or9 and m>=15. Deleting a leaves the long A_(m-1) path. The complete one-root positive-semidefinite extension list is isolated; endpoint3 or endpoint4 on either end; second-node3 on either end; or both-end3. Finite exceptional extensions are too small. The weak-profile theorem supplies this classification, exact heights, Schur identity and its all-rank interpretation. Nothing here assumes absence of small affine diagrams elsewhere in the polytope.

For a single attachment at position j=1 or2 on an arc of length L, the adjacent marked distance is L-j+1. The old endpoint opposite this attachment is a simple leaf whose neighbor has degree2. Both remain fixed under all spectator contacts, because Z is orthogonal to the old cycle. The adjacent connected component has at least m>=15 nodes. No classical affine A/B/C/D graph has such a fixed simple leaf with a degree2 neighbor; exceptional affine graphs are too small. By the actual

factor-fiber theorem, a connected component containing both marks of an ideal corner would have to be affine. Therefore this adjacent corner is finite, including after all supporting whites have been restored.

Distances at most8 now pay immediately. The new issue is precisely distance9.

Which distance-nine marks remain leaves?

An endpoint4 profile is a finite B_m component with the new mark c at its doubled terminal. A second-node3 profile is a finite D_m component with c a short fork leaf. Every spectator is orthogonal to every old path root, so a spectator touching this component can touch only c.

The exact inverse diagonal at c is

```
B_m doubled terminal:  m/2,
D_m short fork leaf:   m/4.
```

These follow by one Schur complement of A_(m-1): its endpoint inverse diagonal is (m-1)/m and its second-node diagonal is 2(m-2)/m. The native attachment squares are2 and1 respectively. Since m>=15, both displayed values exceed2. Any spectator with nonzero contact magnitude at least1 would give negative Schur norm in the principal extension by that root, contradicting positive semidefiniteness of the actual corner. Hence every spectator is orthogonal to c. The new mark remains a leaf of the whole actual component, and the distance-nine leaf rule pays.

An endpoint3 profile instead gives A_m, with inverse diagonal m/(m+1). It may extend. If c has no spectator neighbor it remains a whole-component leaf and again pays. Thus an unpaid endpoint3 profile at distance9 **must** have a spectator neighbor. This necessary contact supplies a useful determinant correction; no finite list of spectator forests is needed.

An unequal-height spectator bound

Let G be the positive-definite norm-two Gram matrix of Z, and let x,y>=0 be the native contact-magnitude vectors from c,d to Z. If they share a nonzero coordinate, c-e-d is an actual white-interior path of length2, which pays at that adjacent corner. Otherwise their supports are disjoint.

Suppose c is the unpaid endpoint3 profile just described, and choose one of its spectator neighbors i. Then x_i>=1 and y_i=0. For z=x/h-y/k, Cauchy in the positive-definite Gram metric gives

```
z^T G^(-1) z >= (z^T e_i)^2/(e_i^T G e_i)
                = x_i^2/(2h^2) >= 1/(2h^2).
```

This uses no equality between h and k and no lower bound on other coordinates of z. The analogous bound holds for d. If both sides require neighbors, use **one fixed side only** throughout the height rectangle; two coordinate bounds are not added.

Write A=1/P, B=1/Q, u=1/h, v=1/k and use the inherited profile moments l,m and norm constants c0,d0. Put L=u*l1*+*v*m1 and M=u*l2*+*v*m2. Before the spectators, the normalized determinant is

```
F=1-2(A*L+B*M)-(A+B)(c0*u^2+d0*v^2)-A*B*(L-M)^2.
```

Spectator elimination adds (A+B) z^T G^(-1)z. The new bound therefore replaces c0=1 by c0=1/2 for one required-contact endpoint3 profile, or replaces d0 analogously. The modified constants remain nonnegative. Consequently this lower bound is still nonincreasing in A,B and concave in u,v. Its minimum over the permitted height rectangle occurs at a vertex, exactly as in the inherited determinant proof. A strictly positive lower bound contradicts the actual star's corank1.

Complete cycle partition

For each P, split Q into its permitted values up to10 and the single monotone range Q>=11. No endpoint-only sample is being extrapolated: once Q>=11 the surviving profile list is constant and the reciprocal-arc monotonicity applies. The retained lists follow only from the finite-eight and leaf-nine deductions above. In particular, an endpoint3 profile on an arc of length9 uses the fixed-side spectator correction.

The source evaluates all740 rectangle vertices for185 ordered profile pairs across the11 regions. Of these pairs,182 have a strict gap; the only three others are isolated/isolated, treated in the next section. The strict-pair minima are:

P	Q or lower bound	Strict-pair minimum
7	8	11/56
7	9	1/9
7	10	1/35
7	>=11	5/77
8	8	1/4
8	9	1/18
8	10	1/10
8	>=11	1/11
9	9	1/9
9	10	2/15
9	>=11	1/9

The checks use exact Q(sqrt(2)) arithmetic and also verify that the modified-profile formula equals the original formula plus its claimed correction. The66 explicit single-profile diagrams verify positive definiteness, marked distances, the fixed opposite leaf and the inverse diagonals. Their role is corroboration; arbitrary rank is covered by the classification, inverse formulas and monotone tails.

The three isolated cases

For isolated profiles, h,k are native opposite-contact magnitudes. They are1 or at least sqrt(2). With alpha=1/P+1/Q, if either is at least sqrt(2), the determinant is at least1-3*alpha>0 in all three exceptional regions. Hence h=k=1 before applying the unscaled spectator-gap argument.

The zero determinant then requires

`D=(x-y)^T G^(-1)(x-y)=4-1/alpha.`

The three required values are4/15 at(P,Q)=(7,8),1/16 at(7,9), and0 at(8,8). The unscaled gap proof gives D>1/2 for two nonempty disjoint supports, D>=1/2 for exactly one, and D=0 if both vanish. Thus the first two are impossible without an already paid common-neighbor path.

For(8,8), D=0 implies x=y. A nonzero common coordinate pays by a length-two path; otherwise every spectator is orthogonal to c,d as well as F. This is exactly the sealed cycle16 equality core of the half-coupon theorem: two opposite simple leaves on the16-cycle, with no further supporting contacts. Its two adjacent finite corners have whole A1/A15 components, with the A15 mark its unique middle. Their two half-payments sum to one. The geometric equality-core and sealing proof is inherited; this package makes no new finite-volume construction claim.

The tree branch already uses only finite paths of length at most7

The classical tree argument of the mixed ideal-six theorem needs no finite-distance-nine payment. Outside feature radius6, the outer finite components have type B of rank at least8 or D of rank at least9, so its complete eight-profile list remains valid.

At P=7, the only cost-four profile, inner second-node3, gives a path of length6 and pays. At P=8 it gives a path of length7. If that corner is finite it pays under finite-eight. If ideal, the old attachment node remains a degree3 feature and the new mark is one step away, so the ideal feature rule pays. Spectators cannot remove its three existing edges. All other profiles give positive gaps at least1/7 and1/4 respectively. At P=9 all ordered profiles already give gap at least1/9. The192 exact cases here recheck those bounds and record the finite-eight reason explicitly.

This proves the stated tree transfer with the cheaper finite rule. Together with the cycle result it removes that specific local obstruction to a future finite-eight theorem. It does not turn the old distance-nine preferred-corner census into a complete distance-eight census: the latter has additional overlap configurations and separate small-factor obligations.

Verification and review scope

Generation and a fresh separate-process saved replay both terminated exit0/PASS for740 cycle rectangle values,66 finite profile checks, three exact isolated cases and192 tree cases. Syntax, all11 direct source hashes and all7 local report links passed. A separate full mathematical/source review found no gap in the fixed-leaf finite-corner argument, inverse-diagonal sealing, unequal-height fixed-side Cauchy correction, constant-profile monotone tails, isolated cases or tree payment transfer. That review did not independently reproduce the740 rectangle values. An additional conceptual review accepted the unequal-height coordinate Cauchy bound and the decision to use only one fixed side. Source, data and report are final/frozen. External refereeing and formalization are not claimed.

Source 95

Source path: `results/fresh_finite8_d12_b10_coupon_budget_20261017.md`

SHA256: 0717c93d8739fc351eeff827809bd57f5f19f3e7e1eadf9ec63c54c4c91d00c6

Finite cost eight with two order-13 transfer coupons

At every finite vertex in ambient dimension at most75, the following enlarged corner rule has total cost at most8n. Retain the entire previous rule, including the A4/D13-neighbor half-payments, and make exactly these changes:

- Upgrade the existing half-payment to one for a pair in distinct whole A1 and D12 components, when the D12 mark is a short fork leaf. Its fork center is not upgraded.
- Add a half-payment for a pair in distinct whole A2 and B10 components, when the B10 mark is its terminal node incident to the doubled bond. Either node of A2 is eligible.

Each unordered marked pair is counted once. These two additions have disjoint component types, and neither duplicates the A4/D13-neighbor rule. The source and certificate verify the exact finite integer inequality below. Local geometric transfers are separate obligations; this counting statement alone removes no ideal payment.

Let a be the total rank of components of rank at most5. Let t count the inherited large types D11 through D16, A13 and A15; let b and c count the D12 and D13 components among them. Let u count A17 and v count B10. Then

`b+c<=t,  a+11t+b+2c+17u+10v<=n<=75.`

The inherited same-component cost is at most8 times component rank. Small components supply at least6a of slack. Each of the t large components supplies at least8; D12 supplies30, hence22 extra, and D13 supplies26, hence18 extra. Whole B10 has all45 unordered pairs paid and supplies35 slack. A17 supplies no slack. All other components have nonnegative slack.

The old small/large half-payments cost at most a(3t/2+u). The retained A4/D13-neighbor addition costs at most ac/4. For the D12 upgrade, each A1 has two eligible D12 pairs whose weights each rise by1/2. Its added cost is thus one per A1/D12 component pair, at most ab. Each A2/B10 component pair has two eligible pairs of weight1/2, giving cost one; the number of A2 components is at most a/2, so this costs at most av/2.

Consequently the surplus above8n is at most

`E=a*(3t/2+u-6+b+c/4+v/2)-8t-22b-18c-35v.`

For every fixed nonnegative integer t,b,c,u,v satisfying the rank constraint, E is affine in a. Its maximum on the permitted interval occurs at a=0 or a=n-11t-b-2c-17u-10v. The checker exhausts all those integer tuples for every n from1 through75 and both distinct endpoints. All13,034 values are nonpositive exact rational numbers. This proves E<=0 throughout the whole permitted domain. It is a finite proof with an explicit rank bound, not a sampling of finite Coxeter forests.

The restriction to75 is intentional. A previously completed dimension bound may handle higher ambient dimensions in a subsequent global argument; this report makes no assertion that these additional coupons fit through80.

Verification status

Generation and a fresh separate-process saved replay both terminated exit0/PASS on all13,034 exact endpoints. A separate full internal source and mathematical review passed the component slacks, disjoint coupon increments, rank relaxation, affine endpoint reduction and complete integer loop ranges. That review did not duplicate the13,034-row calculation. No

local geometric transfer was inferred from the budget check. Source, data and report are final/frozen. No new ambient bound, external refereeing or proof-assistant formalization is claimed here.

```
python3 scripts/fresh_finite8_d12_b10_coupon_budget_20261017.py
python3 scripts/fresh_finite8_d12_b10_coupon_budget_20261017.py --verify
```

Source 96

Source path: `results/fresh_finite8_ideal5_combined_payments_20261017.md`

SHA256: bb75329fd65933fe5a784fe21cecc64b5464523db6ffa8693b94c88392fd19d1

Complete finite8 / ideal5 face payments with the existing finite coupons

This package completes the finite8 quadrilateral coverage that was explicitly left open in the77 proof. It proves a local payment theorem and its vertex budgets; it does **not** lower the current unrestricted dimension bound77. The exact unchanged-envelope obstruction is recorded below.

Let P be a finite-sided finite-volume hyperbolic Coxeter polytope of dimension at most80. Finite vertices are simple; ideal vertices may have arbitrarily many simplex factors. At a finite corner give a unit when its two marks have graph distance at most8, lie in one whole connected component of order at most17, or have distance9 with at least one mark a leaf of that whole component. Add the frozen half-coupons between a whole small component of rank at most5 and a whole D11–D16 component at a fork center/short leaf, a whole A13/A15 component at its unique middle node, or a whole A17 component at positions5/11. These are exactly the finite8 budgeted rules.

At an ideal corner, use the actual varying affine cusp factor. Pay all pairs in factors of order at most13. For orders14 through17 use the exact frozen ideal5 pair sets, as retained by the new complete map census. At larger orders pay distance at most5, or distance6 through9 when an endpoint is within5 of a feature (a branch vertex or endpoint of a nonsimple bond).

Every triangle receives at least2 and every quadrilateral receives at least1. The finite-vertex sum is at most8n. The normalized ideal-factor envelope, for affine rank r, is unchanged:

r	factor cost per link-vertex incidence
1–12	r/2
13	87/14
14	20/3
15	111/16
16	127/17
at least17	163/23

The finite and ideal bounds use their established actual-corner multiplicities. An ideal vertex with link factors of ranks r_i and V link vertices contributes the sum of its factor pair costs multiplied by `V/(r_i+1)`. No port-transitivity statement at a nonsimple cusp is assumed.

Complete face coverage

The ideal5/finite8 triangle theorem already gives two payments per triangle. The new checker verifies that every elementary ideal triangle pair in each of the16 small target diagrams remains paid. The whole-factor alternatives in that proof have order at most9 and are retained. Thus changing finite9 to finite8 creates no triangle gap.

For a quadrilateral, first handle an existing adjacent marked path of length at most8. If the corner is finite it is paid directly. If ideal and not already paid, the universal finite8 cycle, ideal5 local and cycle14 replacements give another retained payment. Factors of order at most13 are wholly paid. For classical tree factors of orders14 through17 the exact retained pair sets contain every pair of distance at most8; the checker verifies this explicitly. The only missing small short pairs are cyclic and lie in the universal cycle domains. For larger tree factors, an unpaid path is outside feature radius5, exactly the hypothesis

of the tree replacement. These replacements already use finite8, protected-leaf9 or the retained coupons; finite distance9 with two nonleaf marks is never imported here.

We may therefore assume the star has **no** marked path of length at most8. Choose minimal indefinite opposite-side cores. The actual edge diagrams make both marked deletions positive definite; the closed opposite-side argument applies even when the quadrilateral has ideal vertices. Minimal cores of order at most5, including the unbounded opposite-black product in order3, are fully handled by the frozen distance8 low-core package. Its66 surviving low/high overlaps have full marked-component bound at most12, which is retained on both finite and ideal sides.

For two high cores, disjoint cores are excluded by the complete all-label disjoint certificate. Overlapping cores are exactly the new 82,197-consistent-identification domain: strict edge tests leave2,614 contexts, and every possible additional cross-core edge is either a short path or excluded by a nonnegative quadratic witness. All2,614 have a complete supporting-white certificate. The finite outcomes pay by whole component at most17, a protected marked leaf at distance9, or the new adaptive finite lemma. Other outcomes are basic ideal5 payments or full varying affine components of order at most17.

Every selected pattern with possible factor order14–17 is included in the new complete1,153-pattern induced-map census. Its images require no new ideal pair beyond the retained sets and their validated transfer domains. The omitted cycle pairs have universal replacements. The nine omitted D13/D14 fork pairs have universal transfers whose immediate whole-affine alternatives have orders12/13. These alternatives remain wholly paid.

The removed D pairs at orders16 and17 are different: their transfer is literal, not an arbitrary embedding argument. The new census matches all18 affected whole marked-core keys exactly to the frozen sealed cases, and the new full overlap certificate supplies the absence of extra cross-core edges. The local replacements then give finite whole-component units or finite half-coupons, or a length-two marked path. Their old global no-short9 reduction is **not** imported as coverage of the new domain. The source checks the saved finite outcomes and confirms that every old finite unit is still paid by distance at most8 or whole component at most17. It likewise rechecks all144 D13/D14 equality outcomes under the finite whole-component rule. This closes the change from the older finite9 statement without treating its weaker finite hypothesis as interchangeable.

Vertex costs and exact remaining scalar obstruction

The finite8 coupon budget proves the sum at each finite vertex is at most8n for n at most80. There is no longer an extra n charge for general finite distance-nine pairs. The exact ideal pair sets and all-rank ideal5/R5 spine formulas are unchanged, so the envelope in the table is the same as in the77 theorem.

The fixed-envelope shape comparison still fails at dimension77 on the formal rank profile `(12,16,16,16,16)`. Its normalized curvature-minus-budget is

`-16664/1085773`.

The link has1,085,773 vertices, so its unnormalized cusp deficit is16,664. The improved finite term has positive slack68 at n77, but no universal finite-vertex multiplicity relation has been proved that compensates this cusp deficit. The global strict shape constant is380; it does not by itself cancel an unbounded number of such formal cusp contributions. This profile is a scalar obstruction, not a claimed geometric realization.

For that single profile, eliminating its negative margin would require lowering the order13 factor's total pair cost by `16664/83521`, or lowering each of its four order17 factor costs by `4166/63869`. One whole pair would exceed either threshold, but such a reduction must be geometrically proved for the corresponding worst factor family and checked against all profiles. The source records the exact reduced-profile minima at dimensions70 through77; all remain negative with the unchanged envelope. It therefore makes no new dimension claim.

Verification scope

The binder verifies all frozen input digests, the complete domain and selected-pattern interfaces, exact small pair-set equality, triangle and short8 retention, immediate ideal alternatives, and finite-only literal replacement payments. It also recomputes the unchanged-envelope diagnostic profiles. It does not repeat the82,197 overlap construction or1,153-pattern embedding search; those are separate freshly generated and replayed certificates.

Final generation83039 and separate fresh saved replay85271 both terminated exit0/PASS. All51 direct source hashes, syntax and report links passed. Root completed a full source/mathematical composition read, including the preceding binder

and triangle proof, and found no gap in the short8-first reduction, complete domain split,1,153 maps, literal finite8-safe replacements,144 equality checks or the distinction between the improved finite budget and unchanged ideal envelope. This review did not duplicate the reduced-profile arithmetic or the separate overlap/embedding censuses. Source, data and report are final/frozen. Published finite/affine and minimal-indefinite classifications remain explicit mathematical inputs. No external refereeing or proof-assistant formalization is claimed.

Source 97

Source path: `results/fresh_finite8_ideal5_local_replacement_20261017.md`

SHA256: 3ea61aed17df260b3dcf2f480abbd1b00069477b2958d06eac7719d0659f3fe9

Local finite-eight and ideal-five quadrilateral transfers

Completed internally checked local theorem; verification scope is below. Consider an actual quadrilateral of a finite-sided finite-volume hyperbolic Coxeter polytope. Keep the following payments:

- Finite unit: marked distance at most8, one whole marked connected component of order at most17, or distance9 with a mark a leaf of its whole connected component.
- Ideal unit: marked distance at most5, or distance6,7,8,9 with a mark within5 steps of a feature in the varying affine factor. A feature is a vertex of degree at least3 or an endpoint of a bond of label at least4.
- Finite half-coupons: whole A1 against whole A15 marked at its unique middle; and whole A1 against whole A17 marked at distances5 and11 from its two endpoints.

An ideal cycle corner of order at least15 at marked distance6,7,8 or9 is replaceable using these payments at the other corners. The same holds for an ideal classical-tree corner at these distances when both marks are farther than5 from every feature. The half-coupons occur only in explicitly sealed cycle equality cores. There is no simplicity or arithmetic assumption, and supporting spectator diagrams are arbitrary.

This is a local theorem. The source and certificate do not establish a complete high-core overlap classification, remove all small-factor obligations, prove the total finite budget or assert a new ambient dimension bound. In particular this statement alone does not replace the frozen dimension80 proof.

Inherited actual-star facts and the new issue

Use H=W+a+b+c+d with opposite pairs {a,c},{b,d}, all roots of square2. The white block and W plus any one black are positive definite. At the chosen ideal corner the actual factor-fiber lemma gives a varying irreducible affine factor F containing a,b and a positive-definite spectator block Z orthogonal to F. The positive affine kernel is the actual cusp ray. The opposite sides have strictly positive heights h,k because they do not contain that ray.

The frozen finite-eight/leaf-nine theorem proves the cycle statement at marked distances7,8,9. In its cycle proof the only alternative ideal payments are direct adjacent bonds or shared spectator paths of length2. All its longer single-profile alternatives are finite, and its sole equality uses the sealed A15-middle coupons. Thus its **cycle** conclusion already survives the present reduction from ideal-six to ideal-five. The tree branch requires a new radius-five argument and is handled separately below.

It remains to analyze a cycle with shorter marked arc P=6 and other arc Q>=9, since its order is at least15. We retain the complete eight one-root profiles and exact normalized determinant from the weak-profile theorem. A single profile is finite even after spectators are restored: the fixed opposite simple leaf and its fixed degree2 neighbor exclude all sufficiently large affine completions, as proved in the predecessor.

Its adjacent marked distance is L-j+1. Distances at most8 pay. At distance9, endpoint4 and second-node3 have inverse diagonals m/2 and m/4 at their new marked leaf, both greater than2; no spectator can contact that root, so the whole-component leaf-nine rule pays. Endpoint3 either remains a leaf or has a spectator neighbor. In the latter case, absent a shared neighbor of c,d, the predecessor's unequal-height Cauchy estimate adds at least alpha/(2h^2), or alpha/(2k^2) for one fixed selected side, where alpha=1/6+1/Q. This replaces that endpoint profile's norm constant1 by1/2 and preserves the concavity and reciprocal-arc monotonicity. No two coordinate lower bounds are added.

The complete P=6 partition

The retained profiles are constant on each of Q=9,10,11,12 and Q>=13. For the last range the determinant lower bound is nonincreasing in reciprocal Q, so Q=13 controls the entire infinite tail. The source evaluates400 exact height-rectangle vertices for100 ordered profile pairs. Ninety-three pairs have strictly positive minima. Seven pairs need the discrete-height argument below.

The only exceptional profiles are isolated and endpoint3 on the long arc. For an isolated profile h is1 or at least sqrt(2); for endpoint3, h=x+1 is1 or at least2. This includes every finite Coxeter label and parallel or dotted opposite products. For each of the seven pairs, the source evaluates both smaller height rectangles obtained by imposing either higher-height alternative. All14 resulting minima are strictly positive. Consequently every exceptional case has h=k=1 **before** the unscaled spectator gap is used.

The normalized zero-determinant equation now requires D=(x-y)^T G_Z^(-1)(x-y) to have the following value:

Q	Ordered profiles	Required D	Resolution
9	isolated / isolated	2/5	excluded by gap
10	isolated / isolated	1/4	excluded by gap
10	isolated / endpoint3, either order	1/16	excluded by gap
10	endpoint3 / endpoint3	0	sealed finite A11
11	isolated / isolated	2/17	excluded by gap
12	isolated / isolated	0	two A1/A17 half-coupons

If x,y share a nonzero spectator coordinate, their length-two marked path already pays. Otherwise their supports are disjoint. The unscaled spectator-gap theorem gives D>1/2 for two nonempty disjoint supports and D>=1/2 for exactly one nonempty support. Thus all five positive required values, each strictly below1/2, are impossible. If D=0, positive definiteness gives x=y; after the shared-neighbor alternative has been removed, both vanish. Every remaining supporting spectator is therefore orthogonal to F,c,d, sealing the complete equality core.

This is an all-label argument. The seven exceptional rectangles are not treated by inferring a positive gap from a negative relaxed minimum; their native height possibilities and exact spectator equality are resolved explicitly.

Two sealed cycle cores

For Q=10, number the16-cycle0,…,15, with a=0,b=6. At the endpoint3/endpoint3 equality, c=16 attaches simply to15 and d=17 attaches simply to7; their native opposite contacts are zero. All other new contacts vanish. At the opposite corner W+c+d, the marked component is the path on7,…,15,c,d, of order11. Every other spectator is orthogonal to it. Hence this is a whole finite A11 component, paid by the retained small17 rule. In fact the two adjacent finite marked components also have order16 and pay. The source checks the core signature(16,1,1), positive-definite white block and all four corners.

For Q=12, number the18-cycle0,…,17 with a=0,b=6. The equality attaches c=18 simply to0 and d=19 simply to6, with no other contacts. Its two adjacent finite corners have whole components A1 and A17. In the A17 path the old mark is five steps from one endpoint and eleven from the other. Both half-coupons are therefore of exactly the new stated type and sum to one. The third finite corner has its marks in separate A1 components. The source checks signature(18,1,1), the white block, all four corner signatures, both full component decompositions and the two endpoint distances.

These literal cores describe a necessary equality inside an actual quadrilateral. No claim that either complete small chamber has finite volume is needed or made. The whole-component coupon interpretation follows from the proved spectator equality, not merely from a truncated diagram.

The radius-five tree corridor

Outside distance5 from all features, both marks lie on the undecorated simple spine. The outer components left after deleting a mark have type B of rank at least7 or D of rank at least8. The inner component containing the other mark is larger by at least the six-step marked separation. The classical eight-profile list of the weak-profile theorem therefore still applies, apart from one potential finite/affine classification addition at the smallest outer rank.

The outer D8 component can acquire an affine E8 extension by prolonging a short fork leaf by a simple bond. This entire affine subdiagram consists of the outer whites and c: it lies in the actual **edge** diagram W+c, which is positive definite. It is therefore impossible. The source checks its positive kernel explicitly, with D8 chain0–1–2–3–4–5, fork leaves6,7 and new root8 attached to6: the coefficients are(1,2,3,4,5,6,4,3,2). Finite exceptional extensions cannot have more than8 nodes, and the doubled B component cannot sit inside a simply-laced E diagram. Larger inner components and profiles joining both old components have no newly available exceptional extension. Thus no arbitrary rank cutoff or omitted D8 case enters the profile reduction.

Use the same eight scalar profiles, with maximum single-side cost g=2*l/h+c0/h^2. Its maximum4 occurs only at inner second-node3; all other costs are at most3, with equality only at inner endpoint3 with h=1. The normalized determinant before spectators is

`F=1-(g_c+g_d)/P.`

At P=6, inner second-node3 gives an actual adjacent path of length5 and pays. At P=7 or8 it gives a path of length6 or7. A finite such corner pays under finite-eight. At an ideal such corner, the old attachment node retains three incident edges and is a feature one step from the new marked root, so the radius-five ideal rule pays. At P=9, every profile pair already has positive gap at least1/9.

After these immediate payments, all P=7,8 cases have positive gaps. At P=6 the sole zero case is two inner endpoint3 profiles. Put u=1/h and v=1/k. Here

`F=[(3-2u-u^2)+(3-2v-v^2)]/6,`

with 0<u,v<=1. Equality forces u=v=1 and zero spectator correction. A shared spectator contact pays by a path of length2; otherwise all spectators are orthogonal to F,c,d. The middle white path A5 with the two new endpoints c,d is then a whole finite A7 marked component. Its marked distance is6, so it pays even by finite-eight, without using the small17 allowance. The two outer finite blocks remain separate.

The source records256 ordered tree cases:45 immediate payments,210 strictly positive gaps, and this one sealed A7 equality. This completes the stated local tree transfer at P=6,7,8,9 in every rank satisfying the feature-distance condition.

Verification scope

Generation and a fresh separate-process saved replay both terminated exit0/PASS for400 cycle rectangle values, all14 high-height gates for the seven exceptional ordered pairs, two literal cycle equality cores,256 tree cases and the affine E8 positive-kernel exclusion. Syntax, all14 direct input/self hashes and all6 report links passed. Two separate full mathematical/source reads found no gap in the height-first cycle analysis, fixed-side correction, whole-component A11/A17 sealing, radius-five D8 edge exclusion, persistent features or sealed A7 equality. Neither review independently reran the400/14/256 arithmetic, re-enumerated the inherited profile catalogue, or asserted a global bound. The finite/affine classification, actual factor-fiber theorem and formal determinant remain explicitly inherited proof inputs. Source, data and report are final/frozen; external refereeing and formalization are not claimed.

Source 98

Source path: `results/fresh_finite8_leaf9_half_coupon_budget_20261017.md`

SHA256: 00451680f07672df335c4b8107a5e537e7350e6c742bb83727c328b7514ab8ea

Finite distance8 and marked-leaf distance9 payments cost at most8n

For a finite Coxeter vertex of rank n<=80, give a unit to a marked pair in the same whole connected component when at least one condition holds:

- the marked distance is at most8;
- the distance is9 and at least one mark is a leaf of the whole component;
- the whole component has rank at most17.

Add the previous broad finite half-payments: one half for a pair in different whole components, one of rank at most4 with arbitrary mark, and the other D12,…,D16 marked at its fork center or short leaf, or A13/A15 marked at its middle. Each eligible pair is counted once. **The total cost is at most8n.**

The source and certificate check the identities and inequalities below. This is a vertex-budget theorem only. Whether these payments cover every required face remains a separate geometric obligation, including the actual-leaf and whole-component premises. No new ambient dimension bound follows from this budget alone.

All-rank finite-component cost

For rank r<=17 the base cost is binom(r,2)<=8r. For r>=18 the finite Coxeter classification leaves only A_r, B_r and D_r. The A/B graph is a path; its number of pairs at distance at most8 is

`sum_{k=1}^8(r-k)=8r-36.`

The two endpoints each contribute one further distance9 pair. These are distinct at r>=18, giving total8r-34.

Construct D_r from an A_(r-1) path by duplicating one terminal leaf at its neighbor. The added leaf has eight spine vertices and its mate within distance8, contributing nine pairs. Hence the short-pair count is

`8(r-1)-36+9=8r-35.`

Each of its three leaves has exactly one vertex at distance9. No two leaves are at distance9 in this rank range, so the three additional pairs are distinct. The resulting base cost is8r-32. These formulas hold for every r>=18; checking ranks18,…,80 merely corroborates the identities in the range used below. No graph sample is used to extrapolate the classification.

Half-payments fit the remaining slack

Let a be the sum of ranks of components of rank at most4. Their total base slack below8 times rank is at least13a/2, since

`8r-binom(r,2)>=13r/2 for1<=r<=4.`

Let t count the listed large coupon components. They have rank at least12, base slack at least8 (the minimum is D16) and at most three eligible marked nodes each. The new cross-component cost is at most3at/2. Every other component has nonnegative slack by the preceding all-rank formulas. Thus

```
total cost<=8n+a*(3t/2-13/2)-8t,
a+12t<=n<=80.
```

Here t<=6. For t<=4 the coefficient of a is nonpositive. For t=5 and6, the largest possible a is20 and8, respectively, giving maximum excesses -20 and-28. This proves the8n budget throughout the stated dimension range. The checker also verifies both possible endpoint values of a for every admissible t and n=1,…,80.

Verification scope

Generation and a separate fresh saved replay both terminated exit0/PASS, including126 graph-formula checks and622 budget endpoints. A separate agent read the complete source and mathematical argument, checking the all-rank path/fork counts, component slacks, cross-payment count and seven cases for t; that review does not repeat the622-case computation. This package is final/frozen. The finite-type classification is an inherited mathematical input. No triangle or quadrilateral coverage, new construction, external refereeing or formalization is asserted.

```
python3 scripts/fresh_finite8_leaf9_half_coupon_budget_20261017.py
python3 scripts/fresh_finite8_leaf9_half_coupon_budget_20261017.py --verify
```

Source 99

Source path: `results/fresh_finite_a14_near_endpoint_zero_transfer_20261018.md`

SHA256: 9a3315bd2a960ae6202d8f2ad9b7c7b4e9d0d47ee94c971a25350d134cf85c58

A14 costs89 using a distinct whole A15 unit

The current whole finite costs become **A14=89, B14=87 and D14=181/2**. The common rank14 representative allowance therefore falls91 to181/2. Triangle payment at least2 and quadrilateral payment at least1 are preserved for arbitrary finite-sided, finite-volume real hyperbolic Coxeter polytopes. No new coupon is needed; every other finite and ideal weight is unchanged. This local theorem does not assert a new ambient dimension bound or a complete updated finite-forest maximum.

The producer and certificate start from the frozen combined current rule and the rank14 theorem. Number A14 along0 through13. Set the two units(0,12),(1,13) to zero. These are the full reversal orbit, with distance12. All other A14 pairs remain units, including the longest pair(0,13). The unchanged whole A15 pair(1,13) is the replacement used here.

Complete native support

The producer freshly searches all600 eligible selected patterns for A14, including patterns with no induced embedding. It checks exact agreement with all552 maps in the frozen complete rank14 inventory. The changed orbit occurs in exactly eight maps: two in context1157, two in1617, and four in1640. The first two contexts already display the whole A14; the third displays A13 and needs one genuinely new white root.

Every affected core is checked in the physical norm-two gauge. Its white principal and each white-plus-one-black principal are positive definite. The actual selected corner, omitted black roots, complete necessary-contact domain, Lorentz signature and full inverse Gram matrix are reconstructed. The proof below permits arbitrary additional supporting whites and arbitrary native contact magnitudes.

The two whole cases force a distinct whole A15

In contexts1157 and1617 the old core has18 roots and signature(17,1,0). Let J be its inverse physical Gram matrix. The inherited necessary-contact lemma partitions all permitted contacts of a further supporting white into two ends. Each same-end block of J is entrywise positive. A root meeting both ends gives an adjacent-black path of length at most8, and the independent current short-path theorem already pays that branch.

Consider the remaining branch. Because the selected A14 is stipulated to be a whole actual component, any further white is orthogonal to its14 roots. The only remaining allowed core-contact positions are 2,3,12,13, with inverse diagonals14,4,14,4. A nonzero native contact between norm-two roots has magnitude at least1. For a contact vector x supported in one end, same-end positivity gives

```
x^T J x >= 4 > 2.
```

But the old core spans a timelike subspace, whose orthogonal complement is positive definite. The projection of a square-two root must therefore have square at most2. No such contact is possible. This argument only bounds the forced contact from below; it does not set unspecified products to zero or impose an upper bound on their labels.

Thus every extra supporting white is orthogonal to the entire old core. An indirect spectator chain cannot attach to one of its components: the first extra root on any such chain would itself meet the core. All displayed finite components at all four corners are whole actual components, even if additional mutually connected spectators are present.

The exact finite-component and marked-role reconstructions then give:

Context	Changed selected corner	Distinct paying corner	Whole marked pair	Payment
1157	0,11	0,12	A15(1,13)	1
1617	1,10	2,10	A15(1,13)	1

The canonical A15 pair is fixed by reversal. Its weight is an existing unit in the rank15 exchange. The current A15 total remains101; the subsequent finite-distance-six and distance-seven deletions do not reach rank15, and the later sparse finite reductions change other types. This unit is retained in the current rule. The other displayed components are unmarked spectators and do not multiply the payment. In each case the selected A14 zero and the distinct whole A15 unit give total at least1.

Every proper completion is impossible in the unpaid branch

Context1640 has a17-root nonsingular Lorentz core and a displayed A13. Each of its four changed A14 maps requires a missing endpoint with a simple contact to core node1 or11. At either coordinate, J[i,i]=7/2.

The endpoint is genuinely a new supporting white. It belongs to the actual selected corner, so cannot be an omitted black root. Nor can it be an old white from a different component: all such old whites are orthogonal to the displayed component, whereas the endpoint must have the specified simple contact to it.

If the new root meets both permitted ends, the independent short8 route applies. Otherwise all additional contacts are in one positive end block, and its projection square is at least7/2>2. This contradicts its norm2. The producer checks each target embedding, forced endpoint, old target neighbor and core coordinate. There is no unexamined saturation case and no arbitrary zero-filled completion matrix.

Simultaneous current payments and full routing

The complete inherited high-overlap domain has2614 contexts, partitioned as1378 with whole-component bound at most13,83 protected/adaptive outputs, 553 displayed components larger than14, and600 selected patterns. The first and third classes cannot have a changed whole A14 selected pair. Protected finite outputs use short paths or whole leaves at distance9; the protected ideal outputs are unchanged. The low-core bound12 and the complete high-disjoint exclusion are unchanged. Thus the selected domain is used after the full geometric reduction, rather than treated as a sample of all actual faces.

The existing independent ideal-corner priorities retain their smaller affine factors, typed coupons and protected short-path outputs. The later finite rank15–17 transfers retain their specified alternatives and typed roles. The finite-distance-six proof uses short5, marked leaves, whole small finite components or its A1/A12 middle coupon; none is a newly deleted distance12 pair in A14. The order13 and contextual order17 outputs are also unchanged. All216 current literal configurations are reconstructed with their native matrices and actual marked components; no corner payment in them changes. This supports the interface claim and does not replace the arbitrary-support arguments.

The short8 escape above is independent of this theorem. A previously deleted deep finite8 pair has its existing short7 replacement. A deep finite7 zero has its short6 or retained whole-small-component replacement. The newer finite6 zero has its independent short5 or protected whole finite output. Leaf and feature restrictions and the separate cycle11 replacement preserve those payments. These routes do not spend the new A14 distance12 zeros or appeal to the new A14 theorem to justify itself.

Follow the existing complete whole-face reduction and these independent priorities. Unchanged selected images keep their previous proofs. A changed selected A14 image has one of the two whole replacements or four impossible proper maps proved above. Hence every quadrilateral still pays at least1. All triangle requirements are also preserved: the elementary pairs have distance at most7, and their exceptional whole-component leaf/branch outputs are unchanged. A14 short9 and all basic pairs are explicitly checked to remain units.

The source checks every current actual base against its representative base and all32,400 ordered type pairs through rank59. Only actual A14 decreases by2 and representative rank14 by1/2; every component-pair allowance and both contextual costs

remain unchanged. The current finite394 tables are not rerun here. A subsequent combined certificate must also count the separate C/D16 theorem's new A1/D15 sixteenth.

The earlier rejected shortcut remains rejected

The frozen rank14 report rejected a proposed unit at a different A2/A13 corner: its A13 mark is second from an end, while the existing unit coupon requires an endpoint. That exact role mismatch is rechecked and preserved in this certificate. The present proof instead uses the whole A15 corner in the table above, with its independently retained unit. No A2/A13 coupon is reinterpreted or increased.

The longest A14 pair(0,13), including context1639, is not deleted. Its alternative corners have different A15 roles whose weights need not be units. The selected orbit and its complete support are essential to the statement proved here.

Verification

Generation68801/chunk6f039a and separate fresh replay90485/chunk336604 both completed exit0/PASS. An earlier run stopped before writing data at a tuple-versus-JSON-list comparison of the preserved historical rejection; encoding that comparison corrected the serialization issue without changing any geometry or weight. The producer checks600 complete selected patterns and552 native maps, two whole Lorentz seals, all four proper-completion projections, distinct actual A15 unit corners, the current216 native literal interfaces and32,400 typed inequalities. The full historical core census, arbitrary-support necessary-contact theorem, independent short-path and triangle theorems remain explicitly inherited inputs. No forest optimizer or global dimension proof is run.

The ideal-lane and construction peers each completed a full mathematical source/report read, including the inherited basic, proper-completion and necessary-contact helpers. Both checked the arbitrary-white seals, genuinely new forced endpoints, distinct retained A15 unit, complete selected routing, independent short-path escape, current interface composition, actual versus representative cost, and the preserved rejected A2/A13 shortcut. Both returned PASS with no mathematical correction. Neither review executed native maps, inertia, literal checks, a historical census or a forest computation. Metadata check 6edbec passed371 source bindings, five local links and Python syntax. This packet is final and frozen. External refereeing and proof-assistant formalization are not asserted.

```
python3 scripts/fresh_finite_a14_near_endpoint_zero_transfer_20261018.py --verify
```

Source 100

Source path: `results/fresh_finite_a15_endpoint_coupon_zero_20261018.md`

SHA256: 7a137085243e41636980c7ce00c56d16029a129c161d5c7b7532c9854f0b15b7

Finite A15 cost103 with an explicitly charged coupon increase

The whole finite A15 cost can fall from104 to103 by deleting its two current endpoint-near-endpoint halves and upgrading one existing finite coupon. The upgrade **costs two additional units per whole A2/A13 component pair**. This is a complete internally checked local transfer; the extra cost must be included in any subsequent forest bound. It is not a free finite-budget saving or a new dimension theorem.

Number A15 along its path from0 to14. Make precisely these changes:

- A15 pairs(0,13),(1,14): weight1/2 becomes0.
- Between two separate whole finite components A2 and A13, a mark at either A2 node and either A13 endpoint: the existing weight1/2 becomes1.

All four eligible cross-component marked pairs are charged. Their cost per A2/A13 component pair rises from2 to4; other A2/A13 roles retain their weights. All other finite and ideal weights remain unchanged. Every triangle still receives at least2 and every quadrilateral at least1.

The source and certificate use the completed rank15 half theorem, the latest D17 cost118 theorem and the independently completed depth8 theorem. The local conclusion is compatible with their simultaneous rule.

The two actual selected contexts are already sealed

The rank15 theorem validates all536 eligible selected patterns and482 labelled A15 maps. Exactly contexts1156 and1616 reach the two changed marked pairs. The displayed marked component is already the whole A15, so no unexamined proper completion is introduced by this further change. The source revalidates that entire saved map domain and reconstructs both old selected proofs. Completeness of the underlying all-label high-overlap domain remains the bound predecessor's theorem input.

Both contexts have the inherited19-root radical core of signature(17,1,1). The white support and each actual-edge support are positive definite. In an actual Lorentz realization the radical represents zero: it is null and orthogonal to a span already containing a timelike vector. Every added supporting white must therefore annihilate the radical.

The complete inherited allowed-contact list has no zero radical entry. A nonzero nonnegative contact row annihilating it must meet both signs. Every possible opposite-sign pair gives a full-star path of length at most8 between adjacent black sides through that new white. Thus either the independent short8 route pays, or every added supporting white is orthogonal to the entire core. A spectator chain cannot evade this: its first root touching the core would already have such a contact row. The source reconstructs the radical and every mixed-sign path bound; the argument covers arbitrary numbers of added whites and native labels.

In the latter branch a distinct actual corner has whole A2 and A13 components, with the marks at the stated endpoints. Previously this corner supplied half and the selected A15 corner supplied half. Now the distinct whole-component coupon supplies the full unit by itself. The selected A15 corner can receive zero. The two cases have exactly the same geometry and sealing proof as before; the change is the explicit reassignment of payment between distinct actual corners.

The new unit is one coupon with a changed weight. It is not added on top of the old half. At a finite vertex each eligible A2/A13 component pair has two choices of A2 mark and two choices of A13 endpoint. This explains both the cost4 and

the genuine increase2 stated above.

Simultaneous current interfaces

The baseline literal records are the latest216 after D17=118, including the preceding rank15/rank16 changes and restored A17 halves. Each actual finite shared A15 component is mapped to its canonical path before any debit. None of these records uses either newly zero pair. All216 totals therefore remain at least1 even without counting the coupon increase. They are the complete inherited literal interface, not an enumeration of every quadrilateral.

The other recent finite selected replacements use whole D16 halves, whole-small/D13 roles, restored A17 halves or A4/D13 halves. The source checks their positive contributions and confirms that none spends a same-component A15 pair. The two changed A15 contexts above are patched explicitly. The new coupon can only increase every other contribution.

The predecessor's complete finite-rank15 routing still applies. Low cores and nonselected high-overlap contexts have smaller component bounds, protected/basic or adaptive whole-leaf outputs, or the unchanged selected alternatives. Both new zeros have distance13. They miss the short8 and whole-leaf9 routes. The two complete selected proofs above replace every affected A15 image, rather than extrapolating from a successful finite sample.

Apply the current independent actual ideal-corner priorities before the selected reduction, exactly as in the completed61 and depth8 compositions. C11/B12/D13, cycle11 and coupled order11 outputs use short paths, small whole components, protected endpoints or typed coupons. The later ideal order17/order16/order14 proofs use the independently established short8 escape or their sealed whole-component alternatives. None needs an A15 distance13 pair. If a short8 output is a deep finite8 zero, the new depth8 corridor gives a short7 path and its completed cycle/tree interface pays independently. This argument never invokes the present A15 theorem to justify that escape.

In a face with no newly changed A15 corner, the previous complete rule already pays and the coupon increase only helps. Otherwise the complete selected/nonselected reduction and the two patched cases supply a full unit. The finite7/ideal5 triangle route and protected6/7 interfaces use no finite distance13 payment; their coverage is unchanged.

Consequently the simultaneous local rule has A15=103, D17=118 and the depth8 zeros, with B15/D15 and every other same-component cost unchanged at this stage. Its additional2 per A2/A13 pair is an outstanding counting obligation for any ambient-dimension claim. No existing frozen budget is silently reused after adding that cost.

Verification scope

Generation82786/fe1988 and fresh replay37161/ada911 both terminated exit0/PASS. Two peer agents completed full producer/report mathematical reads, including the inherited support validator and radical sealing routine. They passed the complete saved-domain scope, distinct whole coupon unit, explicit increase2, latest216 zero-debit interface, recent selected-output noninterference and independent short8/depth8 route. Neither reviewer reran embedding, inertia, radical or literal arithmetic for this packet. No mathematical correction was requested.

The source revalidates the complete saved A15 embedding domain, reconstructs both exact radical seals, checks the four coupon roles and their increase, reconstructs all216 latest literal interfaces and checks recent finite-transfer contributions. It does not repeat the historical geometric census or compute a new forest bound. No new polytope, ambient theorem, external refereeing or proof-assistant verification is asserted. Direct bindings, syntax and local links pass. Source, certificate and report are final/frozen.

```
python3 scripts/fresh_finite_a15_endpoint_coupon_zero_20261018.py
python3 scripts/fresh_finite_a15_endpoint_coupon_zero_20261018.py --verify
```

Source 101

Source path: `results/fresh_finite_a15_three_coupon_zero_transfer_20261018.md`

SHA256: 90b45e20ba0835656a32481f8d42eb47bf2e96d82c38da1c986e1f0fd30efaa2

A15 costs100 with three explicitly charged finite coupons

Set the two remaining whole A15 halves(0,12),(2,14) to zero, numbering its path0 through14. Its total becomes **100**. Add three ordinary whole-component payments:

Small component and marked role	Large component and marked role	Increment per marked pair	Total increment per component pair
A1	A12 endpoint	1/4	1/2
A1	A13 second node from an end	1/4	1/2
A4 inner node	A13 second node from an end	1/2	2

These are additive increments; the proof does not infer that any old payment at a different role is available. The resulting simultaneous rule preserves triangle payment2 and quadrilateral payment1 for arbitrary finite-sided, finite-volume real hyperbolic Coxeter polytopes. Every ideal weight and every other finite base remains unchanged.

The producer and certificate start from the frozen current combined rule and reconstruct the complete rank15 exchange. They assume no further D15 reduction. At this stage D15 still costs100.5, so the rank15 representative allowance becomes100.5, not100. The report does not itself assert a new ambient dimension bound.

Complete selected domain and arbitrary supporting whites

The source freshly reconstructs all536 eligible selected A15 patterns, including failed embeddings, and compares all482 native maps with the complete frozen inventory. The changed reversal orbit has exactly eight maps: two each in contexts1029,1612,1671,2416. All four already display a whole15-node A component. There are no proper-completion cases for this orbit.

For each context the producer reruns the inherited exact physical Gram, white-principal and four edge-principal checks, canonical marked roles and whole-core sealing argument, and compares the reconstructed proof with its frozen record. All roots use norm2. The subsequent coupon check recognizes actual marked roles by every native labelled diagram map, rather than by a single favorable labeling.

In1029 and1612 the core has18 roots and nonsingular Lorentz signature (17,1,0). The actual whole A15 premise forbids further supporting whites from touching its component. The complete remaining allowed contacts have inverse-Gram diagonals4,8,14. Within each end the inverse is entrywise positive. A nonzero native contact has magnitude at least1, so any one-end contact has projection square at least4, contradicting the square-two norm. Extra same-end contacts cannot reduce the bound. Meeting both ends gives an independently paid path of length at most5. Consequently, in the remaining branch, every extra white is orthogonal to the entire core.

In1671 and2416 the core has19 roots and signature(17,1,1). Its exact radical represents zero in any ambient Lorentz realization, because the core contains a timelike vector and its orthogonal complement is positive definite. Every additional white must annihilate that radical. All permitted contact coordinates have nonzero radical coefficients. A nonzero contact

vector with nonnegative magnitudes must therefore meet both signs. Each opposite-sign pair gives a path of length at most5 between adjacent black sides; every such pair is checked. This is again the independent short-path branch. Otherwise no additional white meets the core.

These statements permit arbitrary contact magnitudes, numbers of spectators and indirect spectator chains. A first extra white on a chain reaching an old component would already meet the core, which has just been excluded. Thus all displayed components at the other corners are whole actual components. No bounded spectator enumeration or arbitrary assignment of unspecified products to zero is used.

Distinct corner payments

All four cases retain a distinct whole A16 half at canonical pair (1,13)/(2,14). These are two canonical images of one actual marked pair, not two payments. That half is retained by the current A16 rules. The new payments occur at different actual corners:

Contexts	Retained payment	New payment at another corner	Further new payment	Total
1029,1612	whole A16 half	whole A1/A12 endpoint quarter	whole A1/A13 second-node quarter	1
1671,2416	whole A16 half	whole A4 inner/A13 second-node half	none	1

The selected A15 weight is now zero. All roles in this table are reconstructed exactly. The whole A2 or A1 spectators seen in the first pair of configurations are not conditions on the new coupon; the rule is ordinary and applies whenever its two marked whole components have the specified types and roles. Similarly, no witness condition is imposed on the A4/A13 half. This makes the required finite accounting explicit and avoids silently using a contextual payment as an unrestricted one.

The A15 unit(1,13), needed by the recent A14 theorem, remains1. The longest A15 pair(0,14) retains its previously proved zero and restored A17 alternative; it is not part of this new exchange.

Full current composition

The fresh finite-map domain is used only after the inherited complete low-core, high-disjoint and high-overlap reductions. The rank15 inventory's full routing is bound by the producer. Cases with a smaller whole-component bound or more than15 displayed component nodes cannot require this selected whole A15 orbit. Protected/basic outputs retain their short paths and whole-leaf roles. In the eligible domain, only the four complete contexts above need a replacement; unchanged selected images keep their old proofs.

The ideal weights are unchanged, including the order16 improvement and the contextual order17 rule. Their independent priorities retain their smaller affine alternatives, protected short paths and whole typed coupons. The existing finite6 theorem retains its short5, cycle11, whole-leaf/small-component and A1/A12 middle outputs. Its middle coupon and the new A1/A12 endpoint quarters are both counted. The A14 exchange retains its explicit A15(1,13) unit. The required A16 half pair in the table above is unchanged by all later A16/D16 and D17 rules. The new ordinary coupons are nonnegative and alter none of these previously retained weights.

The short-path escapes are independent of this new conclusion. The two Lorentz end tests and the radical sign tests already produce short5, which cannot itself be a deleted finite distance6/7/8 pair. The established ideal short-path rules and their protected alternatives are unchanged. More generally the current short8-to-short7-to-short6 routes retain their separately proved short5 and protected outputs; they do not require a newly deleted long A15 pair. Triangles likewise retain all elementary short pairs, feature/leaf alternatives and bounded exceptional whole factors. All basic and short9 A15 pairs are explicitly checked to remain units.

The historical current216 literal configurations are reconstructed with native matrices and aligned actual marks. They have no A15 debit and no new coupon credit. The four freshly checked whole configurations above are additional local obligations,

not inferred from that historical literal list. The universal contact proofs and full routing are essential; unchanged216 saved totals alone would not establish the new theorem.

Every added finite cost is included

Each coupon counts actual marked pairs, once per pair. A1 has one mark, A12 has two endpoints, A13 has two second-from-end nodes, and A4 has two inner nodes. Thus the total increments are1/2,1/2,2 as listed above, with no multiplier for diagram automorphisms or witnesses.

Actual component pair	Old cost	New cost	Old representative allowance	New representative allowance
A1/A12	2	5/2	5/2	5/2
A1/A13	25/16	33/16	2	33/16
A4/A13	6	8	7	8

The last two increases are not claimed to fit the old representative allowances. The producer explicitly raises A1/D13 by1/16 and A4/D13 by1, and verifies all32400 ordered actual type pairs through rank59 after simultaneous replacement of both components. It also checks every actual base against its new representative base. Both old contextual costs are unchanged.

At integer scale16, a future residual item of rank1 must gain the fixed rank13 multiplicity, and a rank4 item must gain16 times that multiplicity. The existing interaction key already records that count. No forest count is supplied by this local packet. In particular, a conditional count using an additional D15 saving does not discharge that separate local theorem.

Verification

Generation43660/chunk29da32 and separate fresh replay73801/chunk9de566 terminated exit0/PASS. Two independent internal agent mathematical reads of the full producer/report and inherited whole-support proof passed: order16_fractional and construction_audit (the latter source/report chunks7b1567/cf615c, inherited proof524450 and saved seal inspection80777d). These were mathematical source/report reviews, not extra executions or independent reruns of the native-map, inertia, literal, typed or census calculations. They are not external mathematician refereeing. The producer checks all536 selected patterns and482 native maps, four arbitrary-white whole seals, exact distinct-corner payments, all current216 native literals and32400 typed inequalities. Historical core classification and the independent short-path/triangle theorems are explicit inherited inputs. No ideal convolution or forest optimization is run.

```
python3 scripts/fresh_finite_a15_three_coupon_zero_transfer_20261018.py --verify
```

Source 102

Source path: `results/fresh_finite_a16_d16_d17_sparse_coupon_transfer_20261018.md`

SHA256: 018d14cbdb206aa05ab6933a48d4ba6e48761ec92f57d3743401129c9d47e071

Sparse finite coupons lower A16/D16/D17 costs to114/114/125

Relative to the completed dimension63 payment rule, finite same-component costs can be changed to **A16=114, B16=104, D16=114 and D17=125**, with the six whole-component coupons below. Every triangle still receives at least2 and every quadrilateral at least1. Ideal weights are unchanged. A separate complete finite-forest count is required; this local theorem does not assert an ambient dimension bound.

The source and certificate bind the completed63 rule, the preceding finite exchange, and its complete selected and literal interfaces. Every frozen predecessor is preserved.

The new intrinsic finite rule

Use path numbering0 through n−1 for A_n. For D_n use the spine0 through n−3, with its two short leaves n−2,n−1 at n−3. Delete these unit pairs:

Whole component	Deleted pairs	Old cost	New cost
A16	(0,12),(3,15)	116	114
D16	(0,10),(0,13)	116	114
D17	(1,12),(1,13)	127	125

These sets are invariant under the complete labelled automorphism groups. Every marked distance is at least10. All finite short8 pairs and protected whole-leaf9 pairs remain units. Existing halves, other deletions, deep11 distance8 zeros, and all old coupons retain their values.

Add the following payments between distinct **whole** finite components. An endpoint or second node is counted at either end of an A path; the D5 long endpoint is the leaf on its length2 arm. The two E6 long endpoints are the leaves on its two length2 arms.

First marked component	Second marked component	Weight	Total per component pair
A4 endpoint	A12 endpoint	1	4
A4 endpoint	A13 second node	1	4
D5 long endpoint	A12 second node	1	2
either A2 node	A13 endpoint	1/2	2
either A2 node	A14 second node	1/2	2
E6 long endpoint	A11 second node	1	4

These roles are disjoint from the old coupons on the same type pairs. For example the old A2/A13 cycle coupon marks the middle of A13, whereas the new half marks an endpoint. No cost is assigned merely because a larger component contains one of these diagrams as a subdiagram.

Complete selected support and proper D16 completions

The complete finite induced-embedding routine is rerun on every eligible selected component in the frozen2,614-context high-overlap domain. The two A16 deletions occur only in contexts1651,2026. The D16 deletions occur in501,504,1031,1282,1534,1537,1614,1621,1688,2547. No selected D17 image uses either removed restored pair; those units were introduced solely for the previous exchange's four contexts1681,1685,2500,2532, which are handled explicitly below. Thus map absence is not used to discard their actual predecessor obligation.

Six of the D16 contexts are proper completions with a nonsingular Lorentzian core. All roots are normalized to square2. The inherited contact theorem partitions possible contacts into two ends. A root touching both ends gives an actual short8 path; within each end the inverse block is entrywise positive. Hence unspecified additional contacts can only increase a projection estimate. A first target root adjacent to the old marked component is genuinely new: an old other corner component is orthogonal to that component, and an omitted black root does not occur at this corner.

For501 and1534, a missing short fork leaf must have a unit contact at an old node with inverse diagonal6. Its projection has square at least6, contradicting its norm2.

For504 and1537, two distinct missing short fork leaves are orthogonal in the required D16 diagram. Both contact the same old node with inverse diagonal3/2. Each residual square is at most1/2, but their residual inner product has magnitude at least3/2. Indeed both contact vectors lie in the same positive inverse end block, while their actual inner product is0. The residual two-by-two determinant is at most-2. The orthogonal complement of the Lorentz core is positive definite, so this is impossible.

For1031 and1614, the first missing root is the fork center. Its forced unit contact has projection square4/3. A nonsimple contact would already give at least8/3. Any additional contact in the same end raises the projection beyond2; the other end gives a short8 path. Thus this center has exactly the one core contact and residual square2/3.

Its two prospective short leaves cannot be old other-corner roots, since the center is orthogonal to all those roots. A new leaf has no contact with the old marked component, as prescribed by the actual D16 diagram. For every remaining permitted core position, the exact test gives one of three obstructions: projection greater than2; projection2 and a strictly positive product with the forced center; or a short path of length at most8 through the required center–leaf bond. Same-end multiple contacts also have projection greater than2. Therefore both new leaves are orthogonal to the entire core. Their actual mutual product is0 and each has product-1 with the center. The residual principal is

```
[ 2/3  -1  -1 ]
[  -1   2   0 ]
[  -1   0   2 ],
```

with signature(2,1,0), contradicting positive definiteness of the residual space. These arguments allow arbitrary extra supporting roots and all native contact magnitudes; unspecified products are never set to zero without the stated necessity.

Ten sealed radical alternatives

All remaining selected cases, and all four old exchange cases, have the same19-root L25/L25 core identified at node9. Its signature is(17,1,1), with radical

```
(-6,-3,-4,-2,-5,-4,-3,-2,-1,0,
  6, 3, 4, 2, 5, 4, 3, 2, 1).
```

The radical represents the zero vector in an actual Lorentz realization: it is null and orthogonal to a span containing a timelike vector. Every additional supporting white must annihilate it. All permitted contact positions have nonzero radical coefficients; every opposite-sign pair gives a checked short8 path. Therefore, unless such a path pays, every additional white is orthogonal to the whole core. Its displayed finite corner components are whole and cannot acquire an indirect tail or bridge.

The source reconstructs all four finite corners, their full components, and the marked roles, obtaining these alternatives:

Contexts	New finite payment replacing the deleted unit
1651,2026	A4-endpoint/A12-endpoint unit

Contexts	New finite payment replacing the deleted unit
1282,1621	A2/A13-endpoint half plus distinct A2/A14-second-node half
1688,2547	E6-long-endpoint/A11-second-node unit
1681,2500	A4-endpoint/A13-second-node unit
1685,2532	D5-long-endpoint/A12-second-node unit

These ten alternatives use no deleted same-component unit. The latter four discharge exactly the obligations previously paid by restoring D17(1,12),(1,13); the restoration is not simply undone without replacement.

Literal and universal compatibility

All216 literal configurations of the preceding finite theorem are reconstructed and debited simultaneously. Two require the new A2/A13 endpoint half: cases394 and648 of the order16 finite-cross package lose their D16(0,13) unit but retain an old A2/D13 short-leaf half at another corner. The new half restores total1. The old affine-kernel/inverse argument in that package seals every finite component at these corners, so the new coupon is applied to whole factors. Two other configurations gain the A4/A12 unit. The remaining212 totals are unchanged, and the minimum over all216 remains1. Literal compatibility is not a claim that these216 configurations enumerate every quadrilateral.

The complete low-core bound is12; the all-label high-disjoint domain is excluded. The protected/adaptive branches use basic ideal or whole-leaf9 payments. Their two bound16 trees have distance9 and miss every deletion. Universal cycle/tree alternatives retain finite short paths, protected whole-leaf9 pairs, small components, or separate coupons. The later A16 long-pair uses are among the literal configurations just checked. The later baseline63 order11 outputs are whole finite components of order at most14, A9/A18 endpoint8, D18 leaf9, E8 path6, or typed coupons. The order17 E6 transfer uses no finite same-component D16/D17 unit.

If a required short8 payment is a deep finite8 zero, the independent depth11 corridor supplies its constructive short7 alternative. The completed63 small-tree, cycle-tail and protected-feature interfaces pay that alternative without using the present distance10-or-longer deletions. Every finite short7 unit and every ideal weight is unchanged. Thus this argument does not invoke the unchanged-corner branch of the new theorem to justify a payment used by its changed-corner branch.

Triangles retain the completed finite7/ideal5 coverage. The new finite coupons only add payments. The selected-component classification, proper-completion exclusions, radical alternatives, complete literal audit and universal interfaces therefore prove simultaneous face payment under the baseline63 ideal rule. Future ideal refinements must provide their own compatible changed-ideal-first composition; none is assumed here.

Verification and counting scope

Generation42537 and separate fresh saved replay28025 both terminated exit0/PASS. The complete selected-map domains contain394 eligible A16 patterns/238 maps,394 D16 patterns/578 maps, and140 D17 patterns/168 maps. The new checks reconstruct eight proper-map exclusions in six contexts, ten radical transfers, complete canonical roles and all216 literal totals. All143 direct bindings, Python syntax and four local links pass.

Root and a separate reviewer completed full producer/report mathematical reads. Root also read the inherited complete selected/literal routines and the whole-component sealing proof for old cases394/648; the separate reviewer read the inherited endpoint-constraint helper. Both checked the first-entry argument, all extra/native contact alternatives, two-leaf Cauchy and fork residual exclusions, radical whole-factor control, exact new coupon roles, restored-unit obligations, simultaneous literal debits and baseline63 short8/deep11 interfaces. Neither reviewer re-enumerated selected maps, inverses, historical classifications or literal arithmetic. Historical all-label classifications remain inherited theorem inputs. Source, data and report are final/frozen. All reviews are internal; no external refereeing or proof-assistant verification is claimed.

The new actual coupons must be counted, rather than declared free. Some can be dominated by existing representative-type budgets; the E6/A11 unit may require a larger conservative E6/D11 representative allowance. Such an allowance is an upper bound, not an extra actual payment rule. A separate complete forest certificate must include all interactions before these local costs are used in a dimension comparison. No global optimizer, new ambient bound, or example is supplied by this packet.

Source 103

Source path: `results/fresh_finite_a16_endpoint_sparse_half_transfer_20261018.md`

SHA256: ec7cf23d62899ead9c63dc332b5c79d496675fa5a621c57ef6c33469caaa7a9f

Finite A16 cost111 with two explicit sparse half coupons

The current whole finite A16 cost can decrease from113 to111 by replacing four unit weights with halves. Two new whole-component coupon roles pay the missing halves. They must be included in a subsequent forest budget; this packet establishes the local payment rule only.

Number A16 consecutively from0 to15. Halve(0,13),(2,15),(1,13),(2,14). Keep every other same-component weight, including the earlier A16 halves and the restored A17 halves, unchanged. Add precisely these coupons:

- Whole A1 versus either endpoint of whole A13: half per marked pair, total1 per component pair.
- Either node of whole A2 versus either third node of whole A14 (indices 2 and11): half per marked pair, total2 per component pair.

These are additional roles. The first does not coincide with the old A13-middle coupon; the second does not coincide with the old A14-second coupon at indices1 and12. No other finite or ideal weight changes. Triangles retain total at least2 and quadrilaterals total at least1.

The source and certificate bind the completed rank16 exchange, D17 cost118 rule, current rank15 rule and depth8 theorem.

Complete selected domain and four seals

The inherited finite A16 selected domain has394 eligible patterns and238 labelled maps. Every saved map is validated against the original native Gram and target path. Exactly four contexts reach a new half:

Contexts	Changed A16 orbit	Distinct whole-component half
1028,1611	(0,13),(2,15)	A1 / A13 endpoint
1013,1138	(1,13),(2,14)	A2 / A14 third node

Each displayed marked component already has16 roots and is the whole actual A16. Thus there is no omitted proper completion in these affected maps. Completeness of the original all-label domain and embeddings is an explicit frozen predecessor input; validation here is not a new classification.

Each context has the19-root norm2 radical core of inertia(17,1,1). The white support and every actual-edge support are positive definite. The exact radical is null and orthogonal to a span containing a timelike vector, so it represents zero in an actual Lorentz realization. Every additional supporting white must annihilate it.

The complete inherited allowed-contact list has nonzero radical coefficient at every allowed node. An extra white with a nonzero nonnegative native-contact vector must therefore meet both radical signs. Every opposite-sign pair has a saved full-star path of length at most8 between adjacent black sides through this white. Either the independent short8 route pays, or every additional white is orthogonal to the whole core. This also excludes an arbitrary chain growing from a displayed component: its first attachment to the core would already contradict this contact argument. No bound on the number of extra whites and no zero assignment to unspecified native contacts is used.

In the no-short8 branch all displayed factors at the four actual corners are whole factors. The source reconstructs those corners, identifies the two marked components and checks every canonical mark. The selected A16 corner supplies half;

a distinct corner has exactly the coupon in the table and supplies half. The two contributions sum to1. No other corner is needed. The first coupon has1*2 marked choices and the second has2*2, proving the respective component-pair charges1 and2.

Complete simultaneous interfaces

The source starts from the latest216 literal obligations after D17=118, including all earlier rank15/rank16 changes and A17 restorations. It reconstructs each actual finite A16 factor and debits the four pairs simultaneously. No literal corner uses one of the new halves; every total remains at least1 even before adding the coupons. Positive alternative contributions in the earlier rank15, rank16 and latest D17 transfers also avoid these pairs. In particular the existing A17 restoration remains present, and the D16 halves used by rank15 are untouched.

The inherited complete finite-rank16 routing handles all nonselected high-overlap contexts, low cores and disjoint cases through bounded whole factors, protected/basic marked pairs or their established alternatives. The394 eligible patterns above cover every possible selected long whole-A16 image. All four changed pairs have distance at least10 and every retained basic pair remains a unit. They cannot remove a short8 or protected whole-leaf9 payment. The later literal interfaces are checked separately, rather than inferred from selected absence.

As in the current61 composition, independent actual ideal-corner priorities are used first. Their short-path, protected-component and typed-coupon outputs do not consume these A16 long pairs. The later order14 proper-completion arguments use the independent short8 route. If such a short8 output is a deep finite8 zero, the completed depth8 corridor supplies a short7 path, and the separate cycle/tree interfaces pay. This escape does not invoke the present A16 transfer. The four radical seals therefore apply in the remaining branch without a circular payment. The unchanged finite7/ideal5 triangle route uses none of these new long-pair halves.

The new B15/D15 existing-payment proposal, when composed, uses only its selected D15 half and a whole A2/D13 short-leaf half. The A15 endpoint-zero proposal uses a whole A2/A13 endpoint unit. Those particular alternatives do not spend an A16 long pair; conversely the present four seals require neither a B15/D15/A15 same-component payment nor a D16 change. Their final simultaneous counting remains a separate obligation.

Counting scope

The two sparse roles have actual costs1 and2, even if a subsequent representative upper rule already allows enough to absorb them. They must not be silently omitted. In the current rule the total A1/A13 cost would become3/2, and total A2/A14 cost would become4. The corresponding current A1/D13 and A2/D14 representative allowances are3/2 and4. These equalities suggest a compatible upper budget, but a complete simultaneous typed domination and forest certificate is outside this local packet. No ambient dimension improvement is asserted.

Verification scope

Generation `48506/36d81b` and separate fresh replay `68451/7f59e3` both terminated successfully with PASS. The producer validates all saved selected maps, reconstructs four radical/whole-factor proofs and all216 current literal interfaces. It does not repeat the original all-label census or compute any forest optimum. The root and a separate local-theorem reviewer completed full source/report mathematical reads, including the inherited complete-map validation interface. They checked radical sealing, actual whole-component roles and costs, old exchanges, literal guards and the independent current/depth8 short-path composition. Neither repeated map, radical, inertia, literal arithmetic or the historical census.

```
python3 scripts/fresh_finite_a16_endpoint_sparse_half_transfer_20261018.py
python3 scripts/fresh_finite_a16_endpoint_sparse_half_transfer_20261018.py --verify
```

Source 104

Source path: `results/fresh_finite_a17_endpoint_payment_support_20261017.md`

SHA256: ac946608877ce9bd54d5d1a565ac2bc8fde3f90eba63b783d4408a8a19525b3e

The whole finite A17 endpoint unit is redundant

This package audits the complete support of one proposed change to the frozen dimension69 face-payment rule: at a finite corner whose two marks are the opposite endpoints of a **whole finite A17 component**, remove its unit. Canonical A17 is the path0–1–…–16, so the removed pair is(0,16). The same-component cost is136 before and135 after this change. The pair is fixed by path reflection. All102 short8 and actual-leaf9 pairs are retained, and every cross-component coupon is unchanged, including A17 positions5/11.

The complete local provenance audit below finds no lost triangle or quadrilateral payment: triangles still receive at least2 and quadrilaterals at least1. This is a local refinement conditional on the frozen complete payment theorem and its subsequent transfers. It asserts no ambient dimension improvement. In particular, it does not recompute the optimum over finite forests, and the remaining ideal scalar obstructions have not been discharged by this change.

The checker and certificate bind the complete finite8/ideal5 theorem, its complete all-label overlap and disjoint domains, and every later local transfer used in the dimension69 composition. The distinction between actual whole components and displayed subdiagrams is retained throughout.

The first reductions do not need the removed pair

The triangle theorem uses finite marked distance at most8, or retained ideal elementary pairs and whole affine factors of order at most9. A pair at distance16 in a finite A17 cannot supply any of these payments.

For a quadrilateral with an adjacent marked path of length at most8, a finite corner pays directly. For an ideal corner, the universal cycle and tree replacements use finite short8, distance9 at an actual whole-component leaf, or the existing coupons between separate components. The finite8 cycle theorem covers the long cycle arcs and radius6 tree branch; the ideal5 local replacement covers arc6 and radius5 trees. Their finite single-profile conclusions are stated without blanket small-component units. The separate cycle13 and cycle14 proofs likewise use finite paths of length at most7/8 or retained cross-component coupons.

The sealed cycle18 equality is an especially relevant interface: its two half-payments are between whole A1 and whole A17 at positions5/11. They do not have two marks in A17, and they do not depend on a same-component endpoint unit. The checker records both literal corners and confirms their halves remain unchanged. The cycle16 and cycle14 coupon components are A15 and A13 respectively. No triangle, short-path or universal cycle branch loses a payment.

We may therefore assume no marked path of length at most8. The low-core theorem's66 surviving overlaps have a whole marked-component bound12. The all-label high-core disjoint theorem excludes that entire branch. These results are independent of the removed A17 pair.

All2,614 high-overlap contexts

The frozen82,197-identification census and its no-extra-cross-edge proof leave exactly2,614 supported contexts. The new checker routes **every** one as follows.

Route	Contexts	Why the endpoint unit is unnecessary
Whole marked component at most16	2,391	Cannot be whole A17
Protected finite leaf9 or basic ideal feature payment	74	The distance9 unit is retained
Adaptive finite payment	9	Actual leaf9 or whole component at most16
Bound17, no induced finite-A17 embedding	125	The selected connected component cannot lie in A17
Bound17, induced finite-A17 maps	15	All48 maps avoid the endpoint pair

For the140 bound17 cases, use the saved actual core Gram and its selected connected marked component. In a positively gauged Gram, a native simple bond is characterized by `4*g_ij^2 = g_ii*g_jj`; this avoids confusing a rational congruence with norm-two normalization. An induced connected subdiagram of A17 must be a simple path. A connected path embeds into A17 precisely as a consecutive interval, in either orientation. The checker enumerates all such maps and checks every edge and nonedge. This over-approximates actual completions: excluding the endpoint image from this larger set requires no additional extension assumptions.

Exactly15 selected components are paths admitting maps, with48 maps and 10 distinct canonical marked pairs:

`(2,11),(2,12),(2,13),(2,14),(3,12),(3,13),(3,14),(4,13),(4,14),(5,14).`

None is(0,16). The other125 selected components fail the connected simple-path test and therefore cannot lie in finite A17. This also has a direct graph interpretation: any connected subdiagram of A17 containing both endpoints already contains all17 nodes. A proper extension cannot turn the marks of one of the listed path patterns into both endpoints.

The9 adaptive contexts were not folded into this embedding count. Their independently proved alternatives are4 protected leaves,3 adaptive choices of a protected leaf, and2 leaf-or-tail cases with whole bound16. Their distance and whole-order interfaces are checked separately. The same is true of the74 protected contexts. Thus the48 maps are not presented as the entire quadrilateral domain.

The old literal D16/D17 replacements

The complete small-factor map theorem matches18 literal old replacement contexts,9 each for affine orders16 and17. Their proof says either an extra supporting root gives a retained length-two marked path, or every extra supporting white is orthogonal to the full core. In the latter case the displayed finite components are whole actual components.

The new checker reconstructs every finite corner from the saved literal Gram and black marks, verifies its components and positive definiteness, and subtracts the proposed endpoint unit when appropriate. Seventeen cases lose no payment. The only affected case is old order17 case662. It originally had three finite units. Its marks(3,13) are indeed the endpoints of a whole finite A17, so that unit is removed. The other two finite marked components are both D17, each retaining its whole-component unit. The new finite total is therefore2, still above the required1.

This check is necessary: absence from the preferred overlap maps alone would not have covered this nonpreferred finite corner. The sealing and length-two alternatives also explain why arbitrary additional supporting whites cannot change this conclusion.

Universal D13/D14 transfers and their immediate alternatives

The two universal opposite-fork transfers each have72 sealed equality configurations. The checker reconstructs all144 complete Grams and all corners, verifies the saved signatures and components, and recomputes the retained unit-plus-half total. Every total is at least1.

The D14 equality cores have18 roots and16-root corners, so no A17 component can occur. The D13 equality cores have19 roots. Exactly8 of their finite corners have whole A17; the marked positions are(0,13) in the deterministic path orientation,

equivalent to(3,16) after reflection. None has marks at both endpoints. All8 units survive. The finite nonextreme zero cases have whole D12 components and also survive.

The universal proofs' immediate ideal alternatives have orders12 and13, not finite A17. Their refinements in the dimension71 composition are the universal affine-D12 transfer, the B12 singleton strict gap and the cycle13 transfer. These are included below. Consequently this argument covers the immediate and universal uses as well as selected-core occurrences.

Later local transfers through dimension69

The new checker visits the explicitly typed finite-corner interfaces of all subsequent transfers, rather than inferring safety from their names.

Transfer	Finite units/coupons that can supply its conclusion
Order17 D singleton	Separate A4/D13 halves; length-two alternative
Order17 D partial weights	Whole D17 units or separate A4/D13 and A5/D12 halves
Order16 D partial weights	Whole D16 units or separate A4/D12 and A5/D11 halves
Order15 B three pairs	Whole D14 or whole A12/A13 finite alternatives
Order13 B singleton and promoted long pair	Whole B13 or separate A1/D12, A2/B10 coupons
Universal affine D12 opposite leaves	Whole D13/D14/A15, full A1/D12, or separate A2/D12 halves
Order14/order15 B singleton zeros	Strict projection exclusions; no finite alternative needed

The D16 partial-weight proof's proper-completion exclusions remain valid before any finite payment is used. In its10 sealed cases the finite corners have16 roots. The weighted D17 source supplies12 sealed19-root cores; its only same-component units have typeD17. The A4/D13 new singleton coupon remains between different components. The order13 B long-pair promotion is read in its actual dimension71 composition: two contexts give whole B13 and two give the full whole-A1/D12-short-leaf coupon.

For the universal D12 transfer, arbitrary one-sided spectators leave a whole A1/D12 or D13 corner; two-sided energy-one cases give a short path or two sealed A1 spectators. Its literal same-component units are D13, D14 or A15. The whole-component arguments and arbitrary-support scope are inherited from that universal theorem, not extrapolated from the finite table. None of these routes needs the finite A17 endpoint unit.

Status and verification scope

This establishes the local refinement by exhaustion of the predecessor's coverage routes and all later transfer interfaces. It preserves the triangle minimum2, quadrilateral minimum1 and every ideal pair weight. Only the one whole finite A17 endpoint pair is changed. It does not claim that any other unsupported long pair can be deleted.

The checker independently enumerates the140 A17-eligible overlap interfaces and48 path maps, reconstructs18 literal old replacements and 144 universal D equality configurations, checks all relevant successor finite-unit types and cycle18 halves, and verifies all direct frozen bindings. It inherits the full all-label census, the arbitrary spectator completion/sealing arguments, and the published finite/affine diagram classifications. It does not rerun those censuses or assert external refereeing or proof-assistant formalization.

Final generation27989 and separate fresh replay86207 both terminated exit0/PASS. All78 direct bindings, syntax and4 local report links pass. Root completed a full source and mathematical report read. That review passed complete2,614-context routing; the native-simple path test in positive gauges and all48 induced maps; the sole nonpreferred legacy662 debit retaining two D17 units; all144 sealed D equality configurations including eight nonendpoint A17 units; cycle18 separate-component halves; and the successor typed-unit interfaces. The full geometric census and arbitrary-support sealing proofs remain inherited inputs and were not rerun by that review. Source, certificate and report are final/frozen. No global dimension or exact finite-forest maximum is asserted here.

```
python3 scripts/fresh_finite_a17_endpoint_payment_support_20261017.py
python3 scripts/fresh_finite_a17_endpoint_payment_support_20261017.py --verify
```

Source 105

Source path: `results/fresh_finite_b15_d15_existing_payment_transfer_20261018.md`

SHA256: ddabd354250b2b7915e3662549136966e2093224c1e5fc22b3d30c5f49c40205

Finite B15 and D15 cost103 with existing payments

The simultaneous finite rule can reduce B15 and D15 from104 to103 without adding a coupon. In the canonical finite diagrams, delete B15 pair(0,13) and replace the two D15 unit weights at(1,13),(1,14) by halves. All other weights stay unchanged. Triangles still receive at least2 and quadrilaterals at least1. No new forest maximum or ambient bound is claimed here.

The source and certificate bind the completed rank15 rule, the D17 cost118 theorem and depth8 theorem. The new D15 alternative is an already charged half between a whole A2 and a short fork leaf of a separate whole D13. It introduces no new marked role or cost.

Complete selected support

The frozen rank15 inventory supplies all536 eligible selected high-overlap patterns. The source validates each saved labelled map against its native Gram and the canonical target:240 B15 maps and870 D15 maps. It also checks every marked image, automorphism and retained basic pair. Original all-label high-overlap classification and completeness of that inventory remain explicit predecessor inputs; this packet does not classify the diagrams again.

No B15 image uses(0,13). Exactly eight D15 contexts use(1,13)/(1,14). Contexts1646,1666,1933,2326 display a proper D14 component. Contexts1162, 1175,1910,2303 already display the whole D15 component. The source handles both kinds rather than treating selected absence alone as a whole-face proof.

Four impossible proper completions

All four proper cases have a17-root norm2 Lorentz core with inertia (16,1,0). Its white support and each actual-edge support are positive definite. In every saved map the unique missing D15 node is node0, joined by a simple bond to mapped node1. Therefore an actual D15 completion requires a new supporting white with a contact of magnitude1 at the corresponding old core root. That root has full-core inverse diagonal7/2.

This white is genuinely outside the old core. The old roots absent from the source corner are its other black facets, not incident supporting whites. The remaining old corner components are orthogonal to the marked component and cannot supply the required simple bond. These sets are explicit in the saved map and end-projection records.

The inherited complete allowed-contact theorem applies to arbitrary native contacts of this white. If contacts reach both end sets, the saved distance bound gives a full-star path of length at most8 between adjacent black sides. Otherwise its contact vector is nonnegative and supported in one end. Each inverse end block is entrywise positive, so its projection onto the old core has square at least7/2, even with additional or larger contacts. Since the core span contains a timelike direction, its orthogonal complement in an actual Lorentz realization is positive definite. A norm2 white cannot have projection square greater than2. Thus the no-short8 branch admits no such completion. Unspecified contacts have not been set to zero.

Four whole-component seals

In each remaining context the marked D15 is already the entire actual finite component. Any additional supporting white is orthogonal to every root of that component. Subtracting those nodes from the complete allowed-contact set leaves three roots with inverse diagonals4,4,14. Contacts at both ends again give a short8 payment. A nonzero single-end contact has magnitude at least1 and, by the same positive inverse-block argument, projection square at least4. Hence every additional white is orthogonal to the entire core in the no-short8 branch. A chain cannot extend a displayed factor: its first contact with the core would violate the same statement.

All four actual finite corners are reconstructed. At the selected one the D15 pair now receives half. A distinct corner has whole A2 and whole D13, with the D13 mark at either short fork leaf. Its existing half supplies the other half. The canonical images are A2 nodes0/1 and D13 nodes11/12. The source checks those whole factors and actual marks, rather than assigning a coupon to a proper subdiagram. These two distinct half payments sum to1. Other available payments are not needed, including any A15 payment in these corners.

Remaining and simultaneous interfaces

The inherited2614-case routing partitions into1691 cases with marked component bound at most14,83 protected or adaptive cases,304 with a displayed component larger than15, and the536 eligible patterns above. The first three groups use smaller whole factors, unchanged protected or basic marked pairs, or their established alternatives. The complete low-core and disjoint domains likewise retain their established smaller component or protected outputs. Universal cycle/tree routes use short8 or protected whole-leaf9 pairs. The new changed pairs all have distance at least10, and every retained basic pair remains a unit.

The source reconstructs the latest216 literal interfaces after all current rank15/rank16 changes and D17=118. It debits both new B15 and D15 changes simultaneously, rather than composing separate optimistic trials. Every total remains at least1. It also checks the positive replacement contributions in the earlier rank15, rank16 and latest D17 proofs: none consumes a newly changed B15/D15 pair. The new whole A2/D13 half is already included in the current coupon rule. Increasing the separate A2/A13 endpoint coupon or deleting an A15 long pair cannot invalidate the two payments used here.

Use the completed independent actual ideal-corner priorities first. Their short paths, whole small components and typed coupons are unchanged. The later order14 B/C proper-completion obstructions use the independent short8 route, not these finite distance-at-least10 weights. If a generic short8 output reaches a deep finite8 zero, the completed depth8 corridor produces a short7 alternative and its cycle and tree interfaces pay independently. This escape does not invoke the present selected transfer. In the branch without such an ideal corner or short8 escape, the complete finite selected/nonselected routing applies, and the eight contexts above are its only changed selected images. The finite7/ideal5 triangle route is untouched.

Consequently this is a simultaneous local payment theorem. A separate complete finite-forest count is still required before an ambient claim.

Verification scope

Generation `82637/0dda0b` and separate fresh replay `15534/b335e4` both terminated successfully with PASS. The source reconstructs the complete saved map domain, eight exact geometric certificates and all216 latest literal interfaces. It does not rerun the historical all-label classification, optimize a forest budget or prove an ambient dimension bound. The root and a separate local-theorem reviewer completed full producer/report mathematical reads, including the inherited `X.constraints` end-projection interface and the earlier rank15 proof. They checked forced new-root status, arbitrary-contact bounds, whole-component sealing, coupon roles and simultaneous routing. Neither repeated the map, projection, literal arithmetic or geometric census. All216 latest literal totals are unchanged.

```
python3 scripts/fresh_finite_b15_d15_existing_payment_transfer_20261018.py
python3 scripts/fresh_finite_b15_d15_existing_payment_transfer_20261018.py --verify
```

Source 106

Source path: `results/fresh_finite_d15_a4_a11_half_zero_transfer_20261018.md`

SHA256: 5fc4e6622636fbce7dcc3f349028b657d2f868d2593ebfadaf51488c97adfe61

Finite D15 cost100 using a whole A4/A11 endpoint half

The whole finite D15 marked pair(0,11), currently of weight1/2, can be assigned weight0. Add weight1/2 between an endpoint of a **whole A4** component and an endpoint of a distinct **whole A11** component. The actual D15 cost falls201/2 to100. The new coupon costs2 per A4/A11 component pair and fits the unchanged A4/D11 representative allowance8. All triangles retain payment at least2 and all quadrilaterals at least1. This is a local payment theorem with a pointwise counting comparison, not a new complete forest optimization or dimension bound.

The producer and certificate bind the current simultaneous rule and the earlier three-singleton D15 theorem. The latter supplies the complete inherited arbitrary-support contact theorem and its exact five relevant proofs, which are reconstructed here. The current ideal rule, its positive cusp surplus, and all other finite payments remain unchanged. In particular A15 still costs101 in this packet, so the rank15 representative remains101.

Exact change and complete selected domain

Number finite D15 by its spine0 through12 and short leaves13/14 at12. The changed pair has distance11 and is fixed by every labelled automorphism. Its zero does not change any short8 pair, protected leaf9 pair or elementary triangle payment.

The full inherited high-overlap domain has2614 contexts. Their partition is1691 with whole selected-component bound at most14,83 protected or adaptive cases,304 with displayed selected component larger than15, and 536 eligible selected patterns. The producer re-enumerates every native induced map of these536 patterns into D15, including empty searches, and compares all870 maps to the frozen complete domain. Exactly ten maps in five contexts use(0,11):

Contexts	New obligation
1030,1613	Proper A14 completion, excluded by projection6
1641	Proper completion, excluded by a saturated fork center
1682,2501	Whole D15, replaced by two distinct halves

The other maps in1641 use the retained pair(1,12). They are included in the full inherited proof comparison but are not claimed to be changed by this theorem. Selected-map completeness alone is not the universal argument: the support, sealing and inherited nonselected routes below are also required.

Arbitrary supporting whites and whole factors

Every relevant core uses physical norm2 roots and gauge1. Its full Gram is nonsingular Lorentzian, while the white subsystem and white plus any one black framing root are positive definite. The producer explicitly reconstructs these checks through the inherited proof routine. Let J be the inverse full-core Gram and x the nonnegative magnitudes of an additional white root's products with the core. Its projection square is x^t J x, at most2 because the orthogonal complement of a subspace containing a timelike vector is positive definite.

The inherited contact theorem restricts contacts to two displayed ends. If a white meets both ends, there is an actual path between adjacent black sides of length at most8. Its independent current payment is handled first. In the remaining case all

contacts lie in one end and the corresponding inverse block is entrywise positive. Extra contacts can therefore only increase the relevant lower bound. Nonzero native contact magnitudes are at least1; there is no bound on the number of outside whites or an artificial assumption that their unspecified contacts vanish.

For1682 and2501, the displayed selected component already has15 roots. When the actual selected component is D15, it is the whole component. Every extra supporting white is orthogonal to all its roots. The only remaining possible core contacts have inverse diagonals4,4,14. Any nonzero contact on one end gives projection at least4>2, impossible. Thus, outside the independently paid short8 case, all extra whites are orthogonal to the entire18-root core. A spectator bridge would have a first root meeting the core and is excluded by the same argument.

Consequently every displayed factor at the other corners is whole in the actual corner. The certificate reconstructs all four corners, not only the old selected witness. Two distinct paying corners are:

Context	Retained half	New half
1682	whole D16 pair(1,12)	whole A4 endpoint / whole A11 endpoint
2501	whole D16 pair(1,12)	whole A4 endpoint / whole A11 endpoint

The A4 mark has canonical images0 and3; the A11 mark has images0 and10. The other A11-related displayed corner has different factors and is not used as an endpoint coupon by assumption. The selected D15 zero, the retained whole D16 half and the new distinct half total1. Both paying corners are distinct from the zero corner and from each other. The current D16 weighted table explicitly retains(1,12) at1/2.

Proper completions

In1030 and1613 the displayed selected component is A14. Completing it to the changed D15 pair requires a new short fork leaf simply joined to the mapped fork center. The forced old-core coordinate has inverse diagonal6. It is a genuinely new supporting white: another old component at the selected corner is orthogonal to the displayed component, and the omitted black roots are not roots of that corner. The positive inverse end block gives projection at least6>2 under arbitrary other same-end contacts. Meeting both ends already supplies the independent short8 payment. No coupon is required in these cases.

In1641, the changed maps require a new fork center12 and leaves13/14. The center's simple contact to mapped target11 is at an inverse diagonal2. Positivity forces exactly this one unit contact and no others; saturation of norm2 forces zero orthogonal residual. Its only old-core nonzero product is−1 with the already mapped target11, so neither required new leaf can be another old root.

For either orientation, the inverse row on the opposite contact end is

`(-6,-3,-4,-2).`

The product of the saturated center with another new root is therefore positive if that root contacts the same end, zero if it contacts neither, and at most−2 if it contacts the opposite end. These alternatives exhaust the case without an independent both-end short4 route. None equals the required simple bond−1. Thus the changed D15 completion is impossible. The old proof, including all inverse entries and physical edge checks, is reconstructed and compared with its frozen certificate.

Current whole-face, triangle and later interfaces

Use the same independent actual-corner priorities as the current rule. A newly affected deep finite6 corner invokes its independent short5 or whole small-factor payment. Earlier deep7/deep8 and ideal priorities likewise give their established short8-to-short6 routes, protected leaf outputs or explicitly typed factors. None needs D15(0,11). When these earlier branches do not apply, the inherited finite selected proof is used. If its selected payment changes here, one of the five complete contexts just proved applies; if not, its old payment remains. This avoids inferring universal coverage from the finite literal list.

The1691 low-bound and304 oversized displayed cases cannot choose a whole D15 of the changed kind. The83 protected/adaptive cases keep their short and leaf9 payments. The inherited low-core component bound12 and high-disjoint

exclusion are unchanged. Triangles use finite paths of length at most7 and ideal order at most5; their payments are unchanged.

The exact later roles that might otherwise be confused with this zero remain separate:

- The finite D14 theorem and its saturated extra-white cases use D15(1,12), still of weight1/2. Its complete positive list is checked.
- The universal order15 fractional theorem uses whole D15(0,10), still a unit; its saved whole finite alternative is checked explicitly.
- The current ideal C/D16 coupon uses a mark at node11 of a whole D15 opposite a distinct A1. It is a cross-component sixteenth, not the same-component pair(0,11), and is unchanged.
- The current A14 theorem uses whole A15(1,13). The finite A15/A16/D16/ D17 exchanges use their distinct named A-type, rank16/17, or typed cross-component payments. The D16(1,12) half used here survives all of those changes. The existing D15(0,12) and(1,12) halves also remain.

The earlier D15 theorem already checked all its preceding positive finite D15 interfaces. The later current local reports bound by this packet supply the named independent priority interfaces above. No general subdiagram occurrence is promoted to a whole-component coupon.

The producer also reconstructs all216 current literal configurations from their native Grams, aligning all current cornerwise lower payments. No corner debits D15(0,11), and none of these216 records receives the new coupon. All corner lower bounds stay nonnegative and all face totals stay at least1. These literal checks supplement the arbitrary-support proof; they are not a census of all actual faces.

Finite accounting and verification

Each whole A4/A11 component pair has two endpoint choices in each factor. Four halves therefore add2. The previous actual A4/A11 charge is0. Its unchanged representative A4/D11 allowance is8. The producer checks all32,400 ordered current type pairs, together with all base comparisons, against the unchanged representative rule. Actual D15 falls100.5 to100 while its representative stays101 because actual A15 still costs101 in this packet. All other actual costs are unchanged.

This proves the old complete forest upper remains valid by pointwise domination. It does not compute a new sharp maximum, assert a particular representative forest is an actual attainer, or prove dimension58. Any simultaneous A15 successor requires its own complete combined count.

Generation12992/chunkfe9580 and separate fresh replay1801/chunkc3e7c4 both terminated exit0/PASS. Two independent full source/report mathematical reads also passed. Both included the complete selected domain, inherited physical edge/white checks and L.proof, genuine new-root and saturated completion exclusions, whole-component seals, distinct payments, current positive-role separation and exact representative accounting. Neither reviewer executed the map, inverse, literal, type-pair or historical census calculations. Final source bindings, local links and Python syntax were checked; the source, certificate and report are frozen. The producer reconstructs the exact selected maps, inherited local proofs, current literal corners and type comparisons. The historical all-label census, universal contact theorem and established short-path replacement proofs remain explicit bound inputs. No proof-assistant or external refereeing claim is made.

```
python3 scripts/fresh_finite_d15_a4_a11_half_zero_transfer_20261018.py --verify
```

Source 107

Source path: `results/fresh_finite_d15_three_singleton_half_transfer_20261018.md`

SHA256: 7dcfbd0056b95d4b53a04c34eddce21d15dc0895e89b89d724e3e0d918ebe70b

Whole finite D15 cost201/2 using three existing half transfers

The three whole D15 marked pairs(0,11),(0,12),(1,12) can each be reduced from1 to1/2. The actual D15 cost falls102 to **201/2=100.5**. There is no new coupon or increase in another payment. Triangles retain minimum2 and quadrilaterals minimum1 under the current simultaneous rule. This is a local payment theorem, not a new dimension bound or finite forest optimization. A15 still costs101, so any later rank15 representative allowance must be at least101; using100.5 for all rank15 types would be incorrect.

The producer and certificate start from the rank15 rule and reconstruct the216 literal interfaces of the frozen combined409 rule. They supersede the earlier unexecuted shortlist. The subsequent diagnostic supplied a useful list of obligations but is not a proof input to this theorem.

Complete native support and geometric setting

Use D15 spine0 through12 and short leaves13/14. The three changed pairs are invariant under its labelled automorphisms and have distances 11,12,11. Every short8 and protected leaf9 pair is retained.

The producer freshly runs the complete injective induced labelled-map search on all536 eligible inherited high-overlap patterns, including empty rows, and compares all870 resulting maps to the frozen scout. Exactly24 maps in eleven contexts use the newly reduced pairs. A proper displayed component can also have other, unaffected completions; those maps retain their previous proofs and payments. The affected union is

`1030,1147,1269,1283,1284,1613,1622,1623,1641,1682,2501.`

For each context the original norm2 Gram, physical gauge, marked corner, allowed contacts, omitted black roots and both contact ends are checked exactly. Each full core is nonsingular Lorentzian; the white subsystem and white plus either single framing root are positive definite. Write J for the full-core inverse and x for the nonnegative magnitudes of an additional supporting white's products with the core. The projection square is x^t J x, bounded above by the root's square2, because the orthogonal complement of a timelike-containing span is positive definite. Every nonzero native contact has magnitude at least1.

The inherited actual-contact theorem places all possible contacts in two ends. Meeting both supplies an independently paid full-star path of length at most8. In the remaining branch, contacts lie in one end, and each inverse end block is strictly entrywise positive. Consequently additional contacts cannot cancel a forced positive projection. None of the following arguments assigns unspecified outside contacts zero by assumption. Spectator rank and contact magnitudes are unrestricted.

Six whole-component seals and distinct existing halves

In1147,1269,1283,1622,1682,2501 the displayed selected component already has15 nodes. Whenever its actual type is D15, it is therefore the whole actual component. An additional supporting white is orthogonal to it. The only remaining possible core contacts have inverse diagonals4,4,14. Any nonzero single-end contact has projection square at least4>2. Thus, unless the independent short8 route applies, every additional white is orthogonal to the entire core. An indirect spectator bridge would have a first white meeting the core and is excluded by the same argument.

All displayed finite factors at the other corners are therefore whole actual factors. Each face has the selected half and a distinct existing half, whose exact marked roles are reconstructed:

Contexts	Selected D15 half	Distinct retained payment
1147,1269	(1,12)	A2 versus D13 fork-center half
1283,1622	(0,12)	D16(1,13) half
1682,2501	(0,11)	D16(1,12) half

The A2 mark may be either node. The D13 mark is its unique fork center, canonical node10. The prior rank15 exchange increased only the D13 short-leaf marks11/12 to units; this center half is unchanged. The total actual A2/D13 component-pair cost remains5. The two D16 singletons are both halves in the current bound weight table; neither is part of the later D16/D17 balancing orbit(0,14)/(0,15). The two paying corners are distinct in all six cases, and their total is1.

Four strict proper-completion exclusions

Contexts1030 and1613 display A14, requiring one new short fork leaf to complete D15 with changed pair(0,11). Its simple bond to the mapped fork center forces a unit contact at a core coordinate of inverse diagonal6. Contexts1284 and1623 display D14 and require the missing long endpoint0 to complete the changed pair(1,12). Its simple bond to target1 forces a unit contact at a coordinate of inverse diagonal7/2.

These missing roots are genuinely new supporting whites. The other old roots at the selected corner are orthogonal to the entire displayed component, so none can supply the required adjacent node; the omitted black roots do not belong to that corner. The producer checks these old sets explicitly. The same-end positive inverse block gives projection at least6 or7/2, exceeding2, while meeting the other end supplies the independent short8 route. This excludes all eight changed maps in these four contexts, without a replacement coupon.

The four saturation maps in context1641

This16-root core has signature(15,1), with12 nodes in the displayed component. The allowed ends are(0,1,2,3) and(10,11,12,13); meeting both already gives a path of length at most4. At the relevant coordinates i=1 or11 one has J_ii=2. A forced simple contact there saturates the entire norm2 projection budget. Positivity of the inverse end block then forces exactly that single unit contact, with every other old-core contact zero. The positive definite orthogonal complement forces the residual vector to vanish. These zeros are consequences of saturation.

For changed pair(1,12), the missing target nodes are0,13,14. The two required short leaves13/14 both have a simple bond to the mapped fork center at the same relevant core coordinate i. Both leaves are new: their required adjacency excludes other old corner components, and the omitted black roots are absent. Saturation gives them identical projection and zero residual. Their mutual product is therefore2, impossible for two distinct acute-angled supporting walls (and also incompatible with the required D15 orthogonality). This excludes both orientations of this changed pair.

For changed pair(0,11), the missing target nodes are12,13,14. The new fork center12 has a simple bond to mapped target11, at core coordinate i=1 or11. It is saturated in exactly the same way. Its only nonzero product with an old core root is−1 at this already mapped target11. Neither missing leaf13/14 can therefore be another old root. Each would be a new supporting white with contact magnitudes y, and would need product−1 with this saturated center.

The exact inverse row on the opposite end is

`(-6,-3,-4,-2).`

Since the center has zero residual, its product with that new leaf is sum_j J_ij y_j. A contact vector on the same end gives a positive product; the zero vector gives0; any nonzero contact vector on the opposite end gives a product at most−2, because nonzero native magnitudes are at least1. None gives the required−1. Contacts on both ends already invoke the independent short4 route. Thus even one required simple leaf is impossible, excluding both remaining maps.

The certificate checks the actual inverse entries, all target bonds, all four missing-node sets and both orientations. No residual Cauchy estimate, numerical sign test or bound on extra white roots is needed.

Simultaneous whole-face coverage

The inherited all-label routing has2614 high-overlap contexts:1691 have whole marked-component bound at most14;83 use protected/adaptive outputs; 304 display more than15 nodes;536 form the selected domain re-enumerated here. A new whole D15 payment cannot occur in the first or third class. The protected83 retain their short and leaf9 outputs. The low-core whole-component bound is12 and the high-disjoint exclusion is unchanged. Thus the eleven proofs above cover every newly decreased selected map.

The source also checks positive D15 corners in all earlier rank15, rank16, D17, A15, A16 and D16 transfer proofs. None consumes a newly decreased pair. The subsequent A16/A17 and D16/D17 balances use only their named rank16/17 corners, preserving the two D16 halves used here. The current independent ideal routes are applied first: the order12 alternatives use A1/B11, whole B12 or D11; order13 uses its D10/D12 roles or A4/B9; order16 uses D5/D11; and the remaining order14,15,17 routes use their earlier protected paths and separate typed coupons. These do not require the new distance11/12 D15 payments to exceed1/2.

The short8 alternatives are established independently of this theorem. A deep finite8 zero has its short7 route. The current depth8 finite7 theorem then gives short6 or its whole A8 unit. Its protected cycle/tree, leaf and small-component alternatives use none of the changed long D15 pairs. Triangle finite7/ideal5 payments are unchanged. If no changed whole D15 pair occurs, the previous actual payments remain unchanged.

Finally the producer reconstructs all216 current literal configurations from their native Grams, starting from the combined409 cornerwise lower payments. Exactly eight D15(0,12) corners lose1/2; neither of the other changed singletons occurs. Every corner lower bound remains nonnegative and every face total remains at least1. These finite interface checks supplement the universal argument; they are not asserted to enumerate all actual quadrilaterals.

Verification and limits

Final generation session35654/chunk3b2a0c and separate fresh replay 58562/chunk7fefb9 both terminated exit0/PASS. An earlier replay exposed integer-versus-string dictionary keys at the JSON boundary; the encoder now normalizes keys before comparison. The correction changes no mathematical calculation. The successful final runs re-enumerate the complete native map domain and reconstruct the eleven exact cores and216 literal interfaces. The historical all-label narrowness census, universal contact theorem and nonselected routing remain explicit bound inputs. Two independent full source/report mathematical reads have passed, including the arbitrary-contact saturation, whole-component seals, genuine-newness arguments, inherited contact and nonselected routing, distinct existing halves and current simultaneous interfaces. Neither reviewer executed the maps, inverse, literal or historical census checks. These are internal agent reviews, not external mathematician refereeing or proof-assistant verification. Source, certificate and report are frozen. This local certificate does not rerun a finite forest count or claim a new dimension theorem. Its actual rank15 costs are A15=101, B15=97, D15=201/2.

```
python3 scripts/fresh_finite_d15_three_singleton_half_transfer_20261018.py --verify
```

Source 108

Source path: `results/fresh_finite_d16_alternate_half_zero_transfer_20261018.md`

SHA256: dd940e8d23642709a1af782dcca9457286abb9e8b7477875a24045ac5242e11c

Two finite D16 halves paid at distinct whole corners

Starting from the frozen D5/D12 alternate-corner rule, set the whole finite D16 weights at canonical pairs(2,14),(2,15) from1/2 to0. Add1/2 between a whole A1 and node2 or12 of a whole A15, the third node from either endpoint. This is an increment to the old rule. All other finite weights, ideal weights and coupons remain unchanged.

Every triangle still receives at least2 and every quadrilateral at least1 under the inherited finite-volume actual-corner hypotheses. The actual D16 component cost falls443/4 to439/4. The unchanged A16 cost441/4 limits the representative rank16 upper cost to441/4: the actual saving is1, but the representative saving is only1/2. The new actual A1/A15 total3/2 fits its unchanged A1/D15 representative allowance3/2. No forest optimization or new dimension bound is claimed here.

The producer and certificate validate the complete saved D16 support, reconstruct eight full-core seals and all four finite corners of each, and check all216 current literal obligations and32,400 simultaneous ordered type pairs.

Complete selected domain and two distinct payments

Number D16 by spine0 through13, with short leaves14,15 at13. The changed pairs form one automorphism orbit and have distance12. All finite short8 and protected whole-leaf9 payments remain intact. The producer verifies the orbit under every native automorphism.

The original rank16 exchange contains the complete394-pattern/578-map selected D16 domain. Every saved map is revalidated for injectivity, all native edges and nonedges, and the marked images. Exactly16 maps in eight contexts use the two changed pairs. Every selected component already has16 nodes, so it is the whole stipulated D16; there are no proper displayed-component extensions to classify in this affected domain. Completeness of the saved domain remains an inherited theorem, rather than a new search.

The source reconstructs the old full-core certificate in each affected context and compares it exactly with its frozen predecessor. It then extracts every finite corner and all canonical marked images again. The following table gives the paying corners in literal core labels. The selected D16 corner now contributes zero.

Context	Selected D16, now zero	Existing short-leaf half	Distinct alternate half
1035	(0,11)	(1,11): A1/D13	(0,13): A16
1049	(0,12)	(1,12): A1/D13	(0,13): new A1/A15
1905	(1,10)	(1,11): A1/D13	(3,10): A16
1964	(1,14)	(1,12): A4/D13	(3,14): A16
2298	(2,10)	(2,11): A1/D13	(3,10): new A1/A15
2357	(2,14)	(2,12): A4/D13	(3,14): new A1/A15
2421	(4,11)	(2,11): A4/D13	(4,13): A16
2429	(4,12)	(2,12): A4/D13	(4,13): new A1/A15

Every row uses two distinct corners, each paying1/2. Additional nonnegative payments are unnecessary for this lower bound. Unmarked finite components displayed at a corner, such as A2 or A1, are retained in the saved decompositions and do not change the marked roles.

The D13 mark in every row is a short leaf, with canonical images11,12; it is never its fork center. Thus the recent contextual reduction of an A2/D13-center coupon is irrelevant here. The A4 mark uses the old broad whole-small-component half, which permits any A4 marked node; no endpoint condition is silently added. The four whole A16 alternatives have canonical pairs(0,13),(2,15), both still1/2 in the current balanced table. In the other four cases the marked components are whole A1 and whole A15, and the latter mark has exactly images2,12. The added half is therefore intrinsic under reflection and has cost1 per A1/A15 pair.

Arbitrary supporting whites and whole factors

Contexts1035,1049,1905,2298 have18-root physical norm-two cores of signature(17,1,0). The selected finite corner is exactly the whole displayed D16. Any additional supporting white is orthogonal to that whole component: a nonzero contact would enlarge the marked connected component and contradict its actual type.

The full necessary contact domain leaves only the two omitted black roots, one in each contact end set. Their inverse-Gram diagonal entries are4 and8, and each inverse end block is entrywise positive. These facts are recomputed by the inherited full contact-constraint routine, not obtained by setting unspecified products to zero.

If an extra white touches both ends, the checked graph distances give an actual white-interior short path of length at most8, paid by the independent current short-path route below. Otherwise every nonzero contact lies in one end. A native nonzero norm-two contact has magnitude at least1, so its projection to the old Lorentz span has square at least4, exceeding its own square2. The orthogonal complement of a nonsingular Lorentz span is positive definite. Such a contact is therefore impossible; larger labels and additional same-end contacts only increase the projection bound. In the no-short8 branch every extra white is orthogonal to the entire core.

Contexts1964,2357,2421,2429 instead have19-root cores of signature (17,1,1). Their exact radical is a zero-vector relation in any actual Lorentz realization: it is orthogonal to a span containing a timelike vector. Every permitted contact position has nonzero radical coefficient. Nonpositive wall products and annihilation of the radical force a nonzero extra-white contact vector to meet both coefficient signs. Every such opposite-sign pair gives a checked white-interior path of length at most8 between adjacent black sides. Thus either the independent short8 route pays or every extra white is core-orthogonal.

These arguments cover arbitrary contact magnitudes, any number of extra whites and indirect spectator bridges: a bridge has a first white touching the core. All supporting-white and white-plus-one-black principals and all four finite corners are checked positive definite. Consequently the displayed factors at every corner remain whole after adjoining the arbitrary orthogonal positive-definite spectator forest. This seals both paying corners, including the new A1/A15 alternative; positive definiteness of a displayed subdiagram alone is not used as a substitute for that assertion.

Current routing and later positive obligations

The original complete finite16/17 reduction supplies all remaining routes. Low cores have marked-component bound at most12. The protected and adaptive branches use retained short paths, actual whole-leaf9 pairs, basic ideal outputs or their specified whole-order16 distance9 pairs. The all-label high-disjoint exclusion is unchanged. Every other selected supporting component that can lie in whole D16 is in the complete394-pattern domain. The eight affected contexts are exactly those replaced above; all other selected outputs keep their payments.

The later finite exchanges explicitly retain their D16 alternatives:

Later proof	D16 half roles retained
Rank15 existing-half transfer	(1,14),(1,15)
D15 three-singleton transfer	(1,12),(1,13)
Finite rank14 transfer	(2,13)

The source binds these exact role lists and checks all five weights in the final D16 table. The D16/D17 fractional balance uses D16(0,14), (0,15), a disjoint orbit. The A16/A17 balance uses A16(0,11),(4,15), also disjoint from the existing A16 halves used here. Other sparse transfers use their separately specified whole-small/D13, D5/D12, A-path, E6/D11 or other typed coupon roles, all unchanged.

A supplementary read-only saved-array query examined30 positive whole-D16 corners in the bound local packets' direct stored-Gram proof arrays. Its only hits on the new orbit were these eight original rank16 proofs (query9785d3). This corroborates the named later-output inventory; it is not claimed as an exhaustive geometric census or as a substitute for the universal routing argument.

Resolve the current independent actual-ideal-corner priorities first. The universal cycle/tree and later order11–17 rules use their retained short/protected outputs, specified small whole factors or typed coupons. The current order12 near-terminal outputs use B11/B12 or D11; order13 uses its D10/D12/D13 or B9 outputs and contextual compensation; order14 uses retained D11, finite D14, A4/D10 and A1/D13 roles. None uses the changed whole-D16 distance12 pair. The D5/D12 alternate rule uses its three already distinct whole corners and remains unchanged.

The short8 branch used above is independent of this new selected theorem. A deleted deep finite8 pair gives a constructive short7 replacement; the current depth6 finite7 theorem then gives short6 or its explicit whole-A8 alternative. The protected cycle/tree and whole-leaf9 outputs survive. These branches do not consume the new long D16 zeros. Triangle elementary, protected and whole-small-factor routes likewise remain intact and still give2.

If no actual changed D16 corner occurs, the entire old face proof retains its payments, with only a possible positive coupon increment. If a changed corner occurs after the independent priorities, the complete selected/nonselected reduction and the replacements above provide the same quadrilateral lower bound1. This is a whole-face composition, not a claim that every face with any D16 corner is one of the eight displayed configurations.

The parallel new local packets are compatible without being silently included in this baseline cost calculation. The ideal15 refinement outputs A1/D14-node9, finite D15(0,10), D5/D10 or small ideal payments. The finite-D17 leaf theorem uses A17(0,13)/(3,16) units or A4/D13-shortleaf plus A1/A16-node3/12. The root D17-node2 exchange uses A1/D5, A1/D14-center and A2/A14-third coupons. The new A17 exchange uses its stated D5-short/A12-third and A1/A14-third/fourth coupons. None uses D16(2,14)/(2,15), the A16 alternatives here or the new A1/A15-third roles. Their own changes and coupon costs must be included separately by any combined certificate.

Literal audit and exact type accounting

The source starts from the frozen D5/D12 lower bounds and reconstructs all216 literal configurations with exact actual marked components. Neither the deleted D16 orbit nor the new A1/A15-third role occurs in these stored obligations. There are zero debits and zero credits; every lower bound and total is unchanged, and the minimum total is1. This exact simultaneous check includes earlier replacement obligations, but does not itself establish universal supporting-diagram completeness.

In units of1/16, the actual D16 base decreases by16 and the A1/A15 cross cost increases by16. Its old total8 becomes24. The unchanged A1/D15 representative allowance is24. The representative D16 base decreases by8, to1764, equal to the unchanged actual A16 base441/4. The new actual D16 base is1756, or439/4.

For all180 actual finite types through rank59, the producer checks base domination and all32,400 ordered actual type pairs under the same simultaneous representative map. Every new actual cost is nonnegative and bounded by its representative cost. No representative cross allowance or contextual trigger changes. In particular, the contextual rank13 A2/A2 rule is retained. The saving can enter a later complete count without a new interaction key, but this local packet does not recompute a forest maximum or claim that any old maximum has already fallen.

Verification and limits

The complete current-literal/type preflight session40114 terminated exit0/PASS (terminal4a2dce; summarye32fd2). An earlier preflight stopped at a list-shape guard for the saved rank14 role; the guard was corrected from a nested pair list to the actual single pair[2,13]. No payment or domination inequality failed.

Source-bound generation99536/544528 and separate fresh replay 94172/4b3425 both terminated exit0/PASS, with complete saved-object equality. Metadata checkd64328 passed326 source bindings, seven local links and the producer syntax tree. The source revalidates578 saved maps, reconstructs eight exact seals and16 affected maps, and checks all216 literal rows and32,400 type pairs. It does not rerun the historical overlap census, native map search, forest optimization, ideal scalar convolution or any geometric chamber construction.

The arithmetic peer completed a full producer/report mathematical read, including the inherited full-core and map-validation routines. The review passed the four Lorentz inverse4/8 seals, four radical seals, intrinsic whole-corner roles and distinct payments, complete saved-map interface, current216 zero deltas, retained later positive outputs, independent short8/depth6 route and actual/representative accounting. The parallel new local outputs were also checked for disjointness. The review did not execute maps, inertia, literal arithmetic, type-pair checks or finite optimization.

The ideal/finite peer separately completed a full producer/report and inherited full-core routine read. It passed arbitrary-support sealing before whole-factor payment, the four existing A16 and four new A1/A15 alternatives, all five retained later D16 roles, the independent short8-to-short7-to-short6/whole-A8 route, literal no-hit scope and the actual439/4 versus representative441/4 distinction. It did not execute maps, inertia, literal arithmetic, a census or type-pair checks.

Root completed a full final producer/report mathematical read (cde47b/db4401), together with its previously completed reads of the inherited full-core and map-validation routines. The review passed all eight seals and old/new corner routes, actual versus representative savings, retained obligations and the unchanged216 literal interface. It did not execute the producer or repeat the arithmetic. No correction was requested. Source, data and report are final and frozen; no owned process remains active. No external referee or proof-assistant validation is claimed.

```
python3 scripts/fresh_finite_d16_alternate_half_zero_transfer_20261018.py --verify
```

Source 109

Source path: `results/fresh_finite_d16_d17_long_pair_half_transfer_20261017.md`

SHA256: 7c40f186263891499f92a8faef3fd04cd26af68d664eb94cf89acced2c414f54

Finite rank16 and17 savings with one D14 half-coupon

The same-component finite costs can simultaneously be reduced to **A16=116, B16=104, D16=118 and D17=125**, preserving every triangle's payment of at least2 and every quadrilateral's payment of at least1. The only added payment is a half-coupon between any marked node of a whole A1 or A2 component and the unique long-arm neighbor of the fork center of a separate whole D14 component. Its added cost is1/2 per A1/D14 component pair and1 per A2/D14 pair. Every ideal weight and all other finite payments remain unchanged.

The source and certificate prove this local transfer. A finite-forest maximum and any dimension consequence require a separate budget. In particular no unrestricted ceiling64 is asserted here.

Canonical simultaneous rule

Number finite D_n by the spine0–1–…–(n-3), with its two short leaves n-2,n-1 attached to n-3. Change these unit weights to halves:

Whole component	Changed marked pairs	Old total	New total
D16	(1,12),(1,13),(1,14),(1,15)	120	118
D17	(2,13),(2,14),(2,15),(2,16)	127	125

Each set is invariant under the full labelled-diagram automorphism group. The distances are11,12,13,13; hence no short8 or protected whole-leaf9 payment changes. All nine old finite D17 deletions remain deleted.

Number A16 as path0–…–15, and B16 by that path with the bond14–15 doubled. Delete the following unit payments:

- A16: (0,14),(0,15),(1,14),(1,15).
- B16: (0,11),(0,12),(0,13),(0,14),(0,15), (1,12),(1,13),(1,14),(1,15), (2,13),(2,14),(2,15), (3,14),(3,15),(4,15),(5,15).

These sets are also invariant and disjoint from the short8 and whole-leaf9 sets. Finite C16 has the same labelled graph as B16. The old rank17 A/B costs114/105 remain unchanged.

In D14 the new coupon's distinguished node is10 in the same numbering: the fork center is11 and its short leaves are12,13. Node10 is uniquely specified intrinsically. The old coupon already assigns halves at 11,12,13 to every mark in a separate whole component of rank at most5. The present addition concerns node10 only and small components A1/A2 only; it neither duplicates the old coupon nor enlarges its rank cutoff.

Complete selected support

Use the completed finite8/ideal5 face reduction and its current compatible successors. In the complete2,614-context high-overlap domain, a whole finite target of rank16 needs only394 patterns:2,045 contexts have whole marked component bound below16,92 already have a selected base larger than16, and74 protected plus9 adaptive contexts have their separate interfaces. A rank17 target needs140 patterns, with2,391 smaller-bound contexts and the same83 protected/adaptive contexts handled separately.

Every remaining pattern's connected selected component is mapped injectively and inducedly into each target, retaining every native label and nonedge. The native label is determined from 4*g_ij^2/(g_ii*g_jj), so a positive diagonal gauge never changes an angle. The backtracking embedding routine tries every unused target node consistent with the assigned labels and nonedges; its degree filter is only necessary. It thus includes every actual whole-component map, even if some emitted maps need not be geometrically realizable.

Target	Eligible patterns	Patterns with maps	Complete maps
A16	394	70	238
B16	394	28	100
D16	394	240	578
D17	140	75	168

None of the selected A16/B16 images is deleted. In contrast, every proposed D half has actual selected support. Exactly eight contexts and16 maps per D target are affected:

```
D16: 1146,1161,1174,1185,1268,1909,2302,2491
D17: 1017,1034,1048,1059,1264,1904,2297,2488
```

All16 selected bases are already full finite D16 or D17 components. There is no proper-completion case to discard. The proof therefore uses alternate actual corners, rather than absence from a map census.

All-label radical sealing and the alternate halves

Every affected case has the same19-root norm-two core, constructed from two copies of catalogue diagram25 identified at their node9. Its exact Gram has signature(17,1,1), with radical coefficients

```
(−6,−3,−4,−2,−5,−4,−3,−2,−1,0,
   6, 3, 4, 2, 5, 4, 3, 2, 1).
```

The checker reconstructs the white15-root positive-definite principal, all four strict edge principals, and all four finite corner principals. It recomputes the selected corner's complete necessary contact domain from its actual finite blocks and compares every field with the frozen high-overlap certificate.

In an actual Lorentz realization a radical combination of this core represents the zero vector: it is null and orthogonal to a span containing a timelike vector, whose orthogonal complement is positive definite. Consequently every additional supporting white's contact vector must annihilate the displayed radical. Every allowed contact position has a nonzero coefficient. Contacts of only one sign cannot annihilate it, because acute contact magnitudes are nonnegative. For every opposite-sign pair of allowed positions, the checker exhibits a marked path of length at most8 through that white. This uses no upper-label cutoff and allows arbitrarily many simultaneous contacts. Shortest paths can be truncated at consecutive black vertices to obtain the actual adjacent-corner path.

Unless the already-retained short8 branch pays, every additional white is therefore orthogonal to the entire19-root core. This excludes all direct attachments, indirect tails and bridges to its displayed components. Every displayed finite component is a whole actual vertex component. Arbitrary extra supporting whites form separate positive- definite blocks, so the alternate corners remain finite; no simple-cusp assumption has entered.

Each D16 context has a separate whole A2/D14 corner, and each D17 context a separate whole A1/D14 corner. The D14 mark is precisely the counterpart of the changed large mark:

Large mark in D16/D17	Alternate D14 mark	Payment
12 /13	10, long-arm neighbor of fork	New half
13 /14	11, fork center	Existing half
14 or15 /15 or16	12 or13, short fork leaf	Existing half

The new role is needed in only two contexts of each rank, namely 1185,2491 and1059,2488. Each configuration receives its retained selected D half plus the alternate half, totaling1. No other finite same-component unit is spent in this argument. All component types, canonical marked images and coupon roles are checked from the full corner Gram.

Simultaneous nonselected and universal interfaces

Selected support alone would not justify these changes. In particular, the A16 endpoint pair is used by an older literal ideal replacement. The checker reconstructs216 complete literal configurations and debits all four new finite rules simultaneously, after first applying the completed A17/B17/D17 deletion:

Source	Configurations
Old affine-order16 and order17 replacements	9+9
Universal affine D13 and D14 equalities	72+72
Order17 partial-weight replacements	12
Order16 partial-weight replacements	10
Universal affine D11 A1+A3 spectator equalities	32

Each finite shared component is reconstructed and mapped to its whole finite target. Every resulting literal total is at least1. Exactly25 configurations lose payment:

- Old order16 case663 loses its A16 endpoint unit but keeps two D16 units, at pairs(0,14)/(0,15), which are unchanged. Its total stays2.
- Twenty-four D14 equality configurations lose one D16 half. Eight of these also lose an A16 unit. The remaining totals are at least1.

The old A17 endpoint case662 still keeps its two D17 units. No new D17 half affects any literal unit in these216 configurations. The D11 equalities' A16 images are(1,12)/(3,14), both retained. The optional D16 units displayed in the later partial-weight proof are unchanged and are not needed by that proof, which uses an ideal half and a finite cross-component half.

The other branches have explicit retained interfaces:

- Finite triangles use distance at most8. The short-path quadrilateral branch is unchanged. Its universal ideal-cycle/tree replacements use finite paths of length at most8, actual whole-leaf9 pairs, or separate-component coupons. The small P6 equalities are whole A11 or A7; the cycle18 equality uses separate A1/A17 coupons. The cycle16 one-root finite alternatives have distances7 or8.
- The complete low-core closure has bound12. The all-label high-disjoint domain is excluded geometrically. The74 protected cases retain their actual leaf9 or basic ideal payment.
- The nine adaptive cases retain their leaf9 payments. In the two bound16 branches the selected base is a tree with marked distance9. Its complete D16 induced maps avoid every changed pair; A16/B16 completions also retain that distance9 and no deleted pair has that distance. A larger actual target is excluded by the bound16 proof.
- Universal D12 nonextreme alternatives are whole D13; its sealed equality alternatives are at most15. Universal D11 one-sided alternatives are whole D12 or a separate A1/D11 coupon. Its32 remaining equalities were included above. Their finite D alternatives are D13/D15, with the only rank16 path being A16.
- The B15 older finite alternatives have whole order at most14. The later B13 and B12 alternatives have order at most13 or separate coupons. The newer B/C missing-terminal proofs introduce no finite alternative. Cycle12/13/14 outputs have short finite paths or separate coupons. Immediate D13/D14 ideal outputs are whole affine factors of orders12/13, whose own completed refinements remain.
- The new affine D15 and D17 transfers explicitly consume no finite D16/D17 same-component units. Their existing A4/A5–D11 halves and E6–D11 quarter remain unchanged. The large-finite-D half transfer concerns ranks at least20 and uses its separate small/D10 coupon.

The frozen all-label geometric census, universal supporting-root classification and residual sealing arguments remain inherited inputs. This package re-enumerates the affected finite induced maps and checks the new radical argument and all simultaneous literal totals. It does not equate216 literal configurations with all quadrilaterals.

Thus the new rules preserve complete face coverage. They leave all current ideal envelopes intact, including the proved positive dimension65 ideal margins. Counting their finite cost is separate: large finite components can still dominate the unrestricted maximum. The existing short8 interface is retained explicitly; this report does not assume the still-separate finite7 quadrilateral route.

Verification status

Final generation79090 and separate fresh saved replay45872 both terminated exit0/PASS on the complete finite-target maps,16 radical cores,32 changed D maps,216 simultaneous literal totals and the stated later interfaces. Python AST, all125 direct bindings and all3 local report links pass. The final source explicitly checks that the two newer literal domains have no whole rank17 finite component, before using their direct old weights without an earlier rank17 debit.

Root completed a separate full source/report mathematical read, including the imported native-label and complete induced-embedding helpers. The review passed canonical role invariance, all-label radical sealing, the alternate whole A1/A2–D14 coupon, A16/B16 unsupported-pair deletion, and all216 simultaneous literal debits and universal interfaces. A separate saved query confirmed the absence of a whole rank17 unit in the order16 partial-weight package. This review did not independently re-enumerate the selected maps or repeat the upstream geometric census. No second independent mathematical read or independent arithmetic reconstruction is claimed. Source, data and report are final/frozen. No external refereeing or proof-assistant verification is claimed.

```
python3 scripts/fresh_finite_d16_d17_long_pair_half_transfer_20261017.py
python3 scripts/fresh_finite_d16_d17_long_pair_half_transfer_20261017.py --verify
```

Source 110

Source path: `results/fresh_finite_d16_d17_two_unit_exchange_20261018.md`

SHA256: fb09f5ccb40eda524fc095cab00820b437c30356729b36bbea4765352848e297

Two finite D16 units exchanged for two D17 units

The current finite same-component costs can be changed from D16=118, D17=125 to **D16=116, D17=127**, with every ideal weight and every coupon unchanged. Triangles still receive at least2 and quadrilaterals at least1. This is compatible with the completed depth11 zero rule for finite distance8 pairs. No ambient dimension bound follows from this local theorem alone.

The source and certificate check the complete affected finite maps, all-label radical argument and simultaneous literal interfaces. The baseline is the completed D16/D17 half transfer as inherited by the dimension64 proof. The depth11 zero theorem is used only for the simultaneous-rule compatibility below.

The rule and its exact selected support

Number finite D_n by the spine0–1–…–(n-3), with short leaves n-2,n-1 at n-3. Delete the D16 unit pairs(0,11),(0,12). Restore the previously deleted D17 pairs(1,12),(1,13), giving each weight1. Both sets are invariant under all labelled-diagram automorphisms. Their marked distances are11,12, so every finite short8 and protected whole-leaf9 pair is retained. The earlier D16/D17 halves and all other earlier deletions are unchanged. In particular A16=116, B16=104, A17=114 and B17=105 remain unchanged.

The complete2,614-context high-overlap census has394 eligible patterns for a whole finite component of rank16. The same complete induced native embedding routine as the baseline gives578 D16 maps. Exactly four contexts map a selected pair to one of the two new deletions:

```
(0,11):1685,2532
(0,12):1681,2500.
```

All four selected bases are already whole D16, with component-order bound16. There is no proper-completion case here. Each context has the same19-root norm-two core, catalogue pair(25,25) identified at node9, with signature(17,1,1). Every other eligible selected D16 map is retained. Restoring D17 weights cannot invalidate an earlier coverage inequality.

Radical sealing supplies a whole restored D17 corner

The exact radical vector is

```
(−6,−3,−4,−2,−5,−4,−3,−2,−1,0,
   6,3,4,2,5,4,3,2,1).
```

For each of the four contexts the checker reconstructs the15 white roots, their positive-definite Gram, all four strictly positive-definite edge principals and all four finite corner principals. The selected corner's finite blocks, omitted black roots, preferred marks and complete possible contact domain are recomputed from the actual Gram and compared with the frozen certificate.

In an actual Lorentz realization the displayed radical combination represents the zero vector. Indeed it is null and orthogonal to a span containing a timelike vector, whose orthogonal complement is positive definite. Hence the contact vector of any further supporting white must annihilate this radical. Every possible contact position has nonzero radical coefficient. Nonnegative native contact magnitudes confined to one sign cannot have zero radical pairing.

For every opposite-sign pair of possible contact positions, the source checks a path of length at most8 between the opposite black sets through that added white. Truncate at consecutive black vertices if necessary; this gives a path in an actual corner, with only supporting whites inside. Thus either the complete short8 replacement branch supplies the face payment, or every extra white is orthogonal to the entire19-root core. This argument allows all native magnitudes and any number of extra whites.

In the latter branch every displayed component is whole: there is no direct attachment, tail or bridge from the other supporting roots. Those other roots form separate positive-definite blocks. In each of the four contexts a distinct corner has a whole finite D17 component marked at (1,12) or(1,13), up to its complete canonical embeddings. Its restored unit pays1. The deleted D16 corner is assigned0 and is not used. The source reconstructs these whole-component types and marked images from each full corner Gram; it does not merely infer them from a diagram sketch.

Nonselected interfaces and simultaneous depth11 zeros

The baseline's216 complete literal configurations are reconstructed and all new debits and credits applied simultaneously to its already retained weights. Only four configurations change, all in the order16 partial-weight package, at cases(1975,0),(1979,0),(2505,0),(2536,0). Each loses one of the new D16 units and still has total1 from an ideal half and a separate finite cross-component half. The other212 totals remain unchanged. The minimum over all216 configurations is1. These216 cases are the literal interfaces of the universal proof, not a substitute for all quadrilaterals.

The remaining inherited branches retain their explicit outputs:

- Finite triangles use the completed finite7 theorem. No changed pair has distance at most7.
- The low-core whole-component bound is12. The all-label high-disjoint domain is geometrically excluded. The74 protected and nine adaptive contexts retain their basic ideal or whole-leaf9 payments. Both adaptive bound16 marked trees have distance9, and all their D16 maps avoid the two new deletions.
- Universal ideal-tree short-path replacements use finite short paths, protected whole-leaf9 pairs, small whole components or separate-component coupons. The universal D12 and D11 exceptional whole finite outputs have order at most15, apart from the A16 outputs included in the literal audit. Later B alternatives have order at most14 or use coupons. The newer B/C terminal proofs consume no finite same-component unit.
- The cycle replacements use short finite paths, small whole A11/A7 components or separate coupons. The newer affine D15/D17 transfers explicitly consume no finite D16/D17 same-component unit. The large finite D half transfer uses a separate small-component/D10 coupon.

For compatibility with the depth11 zero rule, the only apparent new dependency is a short8 alternative extracted during radical sealing. If it is finite and its distance8 pair is assigned0, the depth11 universal corridor supplies an actual short7 alternative. Its constructive changed- corner branch uses finite short7 payments and the explicitly retained ideal tree/cycle replacements; none consumes the deleted D16 distance11 or12 pair. It does not appeal to the unchanged-corner branch of the new exchange theorem. Ideal short8 alternatives use the completed replacements with the same depth11 cycle interface. Their finite outputs are retained short7, protected finite8/whole-leaf9, small whole factors or coupons. The four newly used whole D17 units also have distance11 or12, so are unaffected by the depth11 rule. The216 literal configurations have too few roots to contain a deep11 finite distance8 pair.

Consequently the two new rules compose without a circular payment assumption. The original depth11 argument remains valid after the D16/D17 exchange because its constructive outputs are precisely those unchanged short-path, protected and coupon payments. Ideal weights and their envelopes have not changed. A separate finite-forest budget is needed to use the new costs globally.

Verification status

Generation c4c221 and separate fresh saved replay7c4f9b both terminated exit0/PASS. The source checks all394 selected patterns/578 complete D16 maps, four radical transfers,216 simultaneous literal configurations and the stated saved adaptive/later interfaces. It binds the complete preceding proof chain and the depth11 theorem. The arbitrary-support justification is the written radical and short-path composition argument above. Historical geometric censuses are inherited with their recorded review scopes; this package does not independently regenerate them. All134 direct bindings, Python syntax and five local report links pass.

A separate full source/report mathematical read, including the baseline selected-map and universal-interface account, passed. It checked the whole-D16 selected bases, radical annihilation and possible-contact domain, the distinct restored whole-D17 alternative, literal versus universal interfaces, and noncircular composition with depth11 zeros. That review did not repeat the map or literal arithmetic. No unresolved mathematical step is identified in this local theorem. Source, certificate and report are final/frozen. All review is internal; no external refereeing or proof-assistant verification is claimed.

```
python3 scripts/fresh_finite_d16_d17_two_unit_exchange_20261018.py
python3 scripts/fresh_finite_d16_d17_two_unit_exchange_20261018.py --verify
```

Source 111

Source path: `results/fresh_finite_d16_fork_exchange_20261018.md`

SHA256: 8ac18b4643f4d3f81dd937f0a3ee03a05859250c603112d8f2d40490cc47c0f9

Finite D16 cost111 with a D17 restoration and one sparse half-coupon

This local exchange reduces finite D16 from225/2 to111, restores finite D17 from118 to119, and adds a half-payment between an endpoint of a whole A4 component and a third node from an end of a separate whole A12 component. Its component-pair cost is2. These changes must be counted together; this report does not assert a new finite-forest maximum or dimension bound.

In canonical D16, the spine is0–1–…–13 and the short leaves are14,15. Change the unit pairs(0,14),(0,15),(2,12) to halves. In canonical D17, restore the currently zero pairs(1,15),(1,16) to halves. All remaining same-component finite weights and every ideal weight stay unchanged. The new A4/A12 coupon has canonical roles A4 nodes0,3 and A12 nodes2,9, each of the four marked pairs receiving1/2.

The producer and exact certificate consume the complete saved394-pattern/578-map D16 selected domain. They validate each saved induced map, reconstruct the affected contact and whole-component arguments, and audit all216 current literal configurations. They do not repeat the historical minimal-core or overlap census.

Proper completions of the singleton(2,12)

Three selected contexts,1011,1031,1614, have proper displayed subcomponents of the proposed actual D16 factor. Let G be the old physical norm-two Lorentz Gram and J=G inverse. An additional supporting root has nonnegative native contact magnitudes x; the positive orthogonal complement gives x-transpose J x$\leq$2. The inherited complete contact theorem splits all possible contacts into two ends. Contact with both ends already supplies a short8 path. Each one-end inverse block is strictly positive, so extra same-end contacts only increase a forced projection bound.

Every first-entry root adjacent to the old marked component is outside the entire old core. Other old corner components are orthogonal to that component, and the omitted black roots are not walls at this corner. Thus the projection argument does not mistake an old spectator for a new root.

In1011, a forced simple contact has inverse diagonal6. Four of the1031/1614 maps force an additional contact with inverse diagonal10/3. Each exceeds norm2, excluding that completion in the no-short8 branch.

The remaining two maps, one in1031 and one in1614, require the three missing target nodes13,14,15. Node13 is the fork center; its forced simple contact is at a node k with Jkk=2. One-end positivity and the no-short8 branch force its entire contact vector to be exactly the unit vector at k. Its residual in the positive complement is zero.

Consider either required simple leaf f of that fork. An old other-corner root is orthogonal to the center, so f must also be outside the old core. If y is its contact vector, its product with the center is exactly the sum of Jkj yj: there is no center residual term. Every possible negative coefficient Jkj gives a full-star path between the opposite black roots through k, the center, f and j, of length3,4 or5. In the no-short8 branch all such yj therefore vanish. The remaining sum is nonnegative, contradicting the required simple product-1. Arbitrary extra contacts of f are covered by this sign argument; no unproved zero-contact completion is imposed.

The proper exclusions use all-label inequalities and the complete inherited necessary contact domains. They are not a finite sample of unit-contact extensions.

Whole factors and distinct alternate halves

The remaining eight affected selected contexts have the whole marked D16 already displayed.

In1060 and2489 the18-root core is nonsingular Lorentzian. A supporting white cannot contact the actual whole D16 factor. The only remaining allowed old-core contact nodes are the two omitted black roots, in opposite ends, with inverse diagonals4 and8. A single-end contact exceeds norm2; contacts at both ends give short8. Hence every further white is orthogonal to the entire core in the no-short8 branch.

The six other contexts1644,1664,1931,2324,2436,2515 have19 roots with signature(17,1,1). The saved radical is an actual linear relation: the core has rank18 and contains a Lorentz span, whose orthogonal complement in the ambient Lorentz space is positive definite. Every allowed additional-white contact coordinate has nonzero radical coefficient. Orthogonality to the relation forces contacts at both radical signs; every such mixed-sign pair gives a short8 path. In the no-short8 branch all additional whites are therefore orthogonal to the entire core.

This seals all finite corner components used below as whole actual factors, even with arbitrary further spectators. The source also verifies strict positive definiteness of the common-white and white-plus-one-black principals and reconstructs all four finite corners.

Selected contexts	Changed selected half	Distinct alternate half
1060,2489,2436,2515	D16(2,12)	Whole A4 endpoint / whole A12 third node
1644,1664,1931,2324	D16(0,14) or(0,15)	Restored whole D17(1,15) or(1,16)

Each row supplies1/2+1/2 at two distinct corners. All four latter contexts contain the stated whole D17 alternative. Only two have a whole A17 alternate, so an A17-only restoration would not prove this exchange. No additional A17 payment is introduced.

Cost and current-rule compatibility

The D16 saving is three halves, hence225/2−3/2=111. The two restored D17 halves add1, hence118+1=119. The new coupon has two eligible A4 endpoints and two eligible A12 third nodes, giving four halves and total2 per component pair.

This coupon is a real new payment. The older whole A4-end/A12-end units cost4 per component pair; the two role sets are disjoint, so their combined cost becomes6. That equals the existing broad A4/D12 representative allowance. This observation explains why a separate finite budget may absorb the coupon; it is not a substitute for that budget's complete domination and forest certificate.

All changes to D16 are at distance at least10. Every short8 pair, protected finite leaf9 pair and triangle pair is retained. The existing D16(1,14),(1,15) halves used by the rank15 theorem are explicitly checked unchanged. Increasing D17 payments cannot invalidate any earlier face inequality.

The simultaneous rule includes the completed dimension61 proof, the proved depth8 finite-zero transfer, D17 cost118, A15 endpoint zero with its upgraded A2/A13 coupon, B15/D15 cost103, and A16 cost111 with its two new coupons. The latter three finite packets have zero debits on all216 inherited literal configurations; their saved cornerwise lower payments agree with the D17=118 baseline. The source checks those equalities before applying all new D16 debits, D17 restorations and A4/A12 additions together. Every resulting literal total is at least1. It also checks that the positive selected alternatives of the intervening rank15/rank16/D17/A15/A16/B15/D15 transfers do not consume a changed D16 unit.

The complete actual-face proof uses the inherited priority, not only this selected map list. Apply the independent changed-ideal-corner routes C11, B12, D13, cycle11 and coupled B/C/D11 first. Their finite outputs are small components or typed coupons; their immediate ideal outputs are retained short paths. The order17 E6 zero theorem and B16/B14/C14 projection transfers use their distinct coupons or the independent short8 route. None requires the new long D16 units. Later ideal D16 distance9 halves are a separate local packet and are not imported here.

For any short8 finite output that is newly deep-zero, the proved depth8 corridor theorem supplies a short7 path in the full supporting star. The complete short7 ideal tree/cycle route then gives retained finite payments or short ideal alternatives, without returning to an affected long D16 pair. This ordering prevents a circular appeal to the changed payment itself.

After those priorities, the complete inherited selected/nonselected reduction applies. Low cores have whole marked factors too small to be D16; high-disjoint cases are excluded; protected leaf9 and adaptive finite outputs remain paid. Every eligible actual D16 selected image is in the saved complete394-pattern domain. Only the eleven contexts analyzed above meet the changed pairs. An unchanged selected image keeps its value; a changed image is impossible or receives the selected half plus a distinct alternate half. If no changed actual finite corner occurs, all old finite weights agree pointwise except for nonnegative restorations and additions. The literal216 audit handles the inherited later local interfaces in addition to this complete reduction; it is not treated as a census of all quadrilaterals.

Triangles retain the completed minimum2, and quadrilaterals retain minimum1 under the simultaneous rule. A separate finite-forest certificate is required before using the changed costs for an ambient bound.

Verification

Generation24949 (`ba04f0`) and separate fresh saved replay90546 (`f4242e`) both terminated exit0/PASS. The ten proper maps give eight projection exclusions and two zero-residual fork exclusions; the eight whole-factor cases close as stated. Four D16 half-debits occur among the216 current literal configurations, all of whose new totals remain at least1. The source performs bounded exact Fraction arithmetic with inherited core and component helpers. Complete saved-map coverage and the universal actual-face reduction remain explicitly inherited inputs. Root and Exacthub each completed a full source/report mathematical read, including the inherited X.constraints interface. They checked the arbitrary-contact zero-residual fork proof, whole-factor seals, distinct restored D17 or new A4/A12 halves, the111/119/+2 costs, current depth8 priority and simultaneous literal accounting. Neither review repeated the map, inertia, literal or historical census arithmetic. Final source/data bindings, local links and Python syntax passed metadata validation; no census, finite-forest optimizer, external refereeing or proof-assistant verification is claimed.

```
python3 scripts/fresh_finite_d16_fork_exchange_20261018.py
python3 scripts/fresh_finite_d16_fork_exchange_20261018.py --verify
```

Source 112

Source path: `results/fresh_finite_d16_second_pair_zero_transfer_20261018.md`

SHA256: 9acfd1695071a51d127049026f66789114d73e2999604155fbc0c2e0a01e5c4a

Finite D16 cost110 with a whole D5/A11 unit

Delete the finite D16 unit at canonical pair(1,11). Its cost decreases 111 to **110**. Add a unit between the unique long-arm endpoint of a whole D5 and either second-from-end node of a separate whole A11. There are two such pairs, so the new actual D5/A11 cost is2, formerly0. The existing A5/D11 representative allowance19/2 dominates it without increasing. This local theorem preserves triangle payment2 and quadrilateral payment1; a complete forest count is separate.

The producer and certificate start from the simultaneous finite411/fractional ideal rule. Every ideal weight, the finite D17 total235/2, all other finite pairs and all previous coupons are unchanged. The finite D16 numbering has spine 0 through13 and leaves14,15 at13. Pair(1,11) has distance10 and is fixed by its diagram automorphisms. All basic pairs, short8 and protected leaf9 payments remain, as do the D16(1,14)/(1,15) halves used by the rank15 proof.

Complete selected support

The producer validates all578 saved induced D16 maps in the complete 394-pattern selected domain, including rows with no map. It checks their native labels and marked images against the inherited overlap Grams. Exactly ten maps in six contexts meet(1,11):

Contexts	Displayed component order	Affected maps	Argument
504,1537	14	4	Forced projection above norm2
1031,1614	13	2	Two orthogonal fork leaves violate Cauchy–Schwarz
1188,2527	16	4	Whole-core radical seal and distinct D5/A11 unit

Some proper displayed patterns also map to other marked pairs. Only their maps to(1,11) are excluded here; their other images retain the old rule. Completeness of the saved domain remains the inherited census. This packet validates every saved map without claiming a new exhaustive search.

Proper completions with arbitrary extra contacts

For each proper case the checker reconstructs the physical norm2 Gram G, its nonsingular Lorentz inverse J, the actual marked corner, and the full necessary contact domain. As in the earlier D16 exchange, the allowed contacts split into two ends. A root meeting both ends gives an independent short8 path. Every entry of each one-end inverse block is strictly positive.

In the remaining branch a new supporting root with nonnegative contact magnitudes x has projection square $x^{T}Jx\leq 2$. In504 and1537, the missing endpoint0 and missing short fork leaf force simple contacts with inverse diagonals8 and4. Either is already greater than2. Extra contacts in the same end only increase the projection square; contacts in both ends already have the paid short-path alternative.

In1031 and1614 the missing target nodes are0,14,15. The two leaves14,15 are both simply joined to the same displayed fork center k, are mutually orthogonal, and have Jkk=4/3. Each leaf must be genuinely outside the old core: other old corner components are orthogonal to the marked component, whereas these leaves have the prescribed nonzero contact with k; omitted black roots are not walls of this corner.

In the no-short8 branch both contact vectors x,y are supported in the same end containing k, with xk=yk=1. Strict positivity gives

$x^{T}Jx \geq 4/3, \quad y^{T}Jy \geq 4/3, \quad x^{T}Jy \geq 4/3.$

The core's orthogonal complement is positive definite. The residual vectors therefore have squares at most2/3, whereas their mutual product is$-x^{T}Jy \leq -4/3$ because the actual leaves are orthogonal. Their residual Gram determinant is at most4/9−16/9=−4/3, contradicting positivity. Arbitrary further contacts are included in these inequalities; no unspecified contact has been set to zero. The other missing endpoint, whose forced projection square is2, is not needed for the contradiction.

Two whole radical cores and the new unit

Contexts1188 and2527 have the whole selected D16 displayed inside a 19-root norm2 core of signature(17,1,1). The producer reconstructs its radical, all possible contacts, the positive-definite15-root common-white support, all16-root edge supports and all17-root corner supports.

The represented radical is zero: it is null and orthogonal to a span containing a timelike vector, whose orthogonal complement is positive definite. Every further supporting white annihilates the radical. All allowed contact coordinates have nonzero radical coefficients, so any nonzero nonpositive contact vector must meet both signs. Every such pair gives an independently paid short8 path, as checked on the complete allowed domain. Otherwise all extra whites are orthogonal to the core. A spectator bridge would have a first root contacting it, already excluded.

In that orthogonal branch all displayed corner components are whole actual finite factors. The distinct coupon corners are literal marks (0,13) in1188 and(3,10) in2527. At each, the marked components are D5 and A11; their complete canonical maps give D5 mark0 and A11 marks1/9. The selected D16 corner is assigned0 and the distinct new coupon gives1. No ideal or other same-component finite payment is spent.

Current interfaces and actual cost

All216 current literal configurations are reconstructed from their native Grams and the finite411 packet's final cornerwise lower payments, already including the ideal order17 quarter transfer. No corner uses the new D16 zero and no new coupon occurs in this list. Thus all216 lower totals remain at least1. The literal list supplements, rather than replaces, the complete actual-face argument.

The low-core bound12, high-disjoint exclusion,74 protected and nine adaptive high-overlap routes remain inherited. Their protected short/basic or whole-order outputs do not use this distance10 pair. Every selected D16 image is in the complete394-pattern domain above. Later rank15/rank16 alternatives retain their explicitly protected D16 halves and D17 payments. The new finite D17 and ideal order17 theorems pay using their own distinct whole-component coupons, not D16(1,11).

Use the current independent changed-ideal-corner priorities first. Their small-factor, protected-path and typed-coupon outputs remain paid. In a short8 escape from either new argument, a deep finite8 zero invokes the independent depth8 short7 theorem; a deep finite7 zero invokes the independent depth10 short6 theorem. The retained tree/cycle and small ideal interfaces supply their payments without this long D16 pair. The current order15 B/D and fractional order17 routes do not consume a same-component finite unit. Hence no short-path branch appeals circularly to the new theorem. The finite7/ideal5 triangle proof is unaffected.

On the remaining selected/nonselected branch, an unchanged selected pair retains its old payment; a changed image has an impossible completion or the distinct unit above. Faces with no changed actual D16 corner have their old weights plus a nonnegative new coupon. Thus every quadrilateral still receives1 and every triangle2 under one simultaneous rule.

The two new whole D5/A11 pairs each receive1, for actual cost2. Their representatives A5/D11 already have allowance19/2 in the current rule. This pointwise cost domination uses both component replacements. It does not mean the new actual payment is free, or prove a forest maximum by itself. The completed dimension bound remains60.

Verification

Generation session81712, final chunkb05c68, terminated exit0/PASS. Separate fresh replay session89318, final chunk60ce45, also terminated exit0/PASS. Two peers completed full source/report mathematical reads, including the inherited contact and map-validation interfaces: PASS. They checked genuine newness of forced roots, arbitrary same-end contacts, the orthogonal-leaf Cauchy obstruction, whole-core radical sealing and intrinsic coupon roles, complete saved-map scope, current literal and later interfaces, independent short-path priorities, and the actual cost versus representative allowance. Neither repeated map, inverse, inertia, literal, census or finite-budget arithmetic. An earlier draft assertion incorrectly required every marked image of an affected proper pattern to be changed; it was removed before any certificate was written. Complete maps are still validated, and only the ten actual changed images enter the new exclusion proofs.

The source, certificate and report are final/frozen. No historical all-label census, finite forest optimization, external refereeing, formal proof-assistant verification or new example is supplied.

```
python3 scripts/fresh_finite_d16_second_pair_zero_transfer_20261018.py --verify
```

Source 113

Source path: `results/fresh_finite_d17_a4_d13_half_transfer_20261018.md`

SHA256: 6fed0b3f037b7ed5afd1e2a7043c07e027d61468f61454b810104fd30b76dce3

Finite D17 cost118 using existing A4/D13 halves

The current whole finite D17 payment decreases from120 to118 without adding a coupon or changing any other finite or ideal weight. Four long pairs become halves. Every triangle still receives at least2 and every quadrilateral at least1 under the simultaneous rule used in the completed dimension61 proof. This is a local payment theorem. No finite-forest maximum or new ambient dimension bound is asserted here.

Number D17 by spine0 through14, with short leaves15 and16 attached to14. The changed pairs are

`(3,13), (3,14), (3,15), (3,16):  1 -> 1/2.`

Their distances are10,11,12,12. The changes are invariant under the full labelled diagram automorphism group. All short8 and whole-leaf9 units remain unchanged, as do every earlier zero and half. The total is118.

The producer and certificate bind the cost120 predecessor, the latest rank15 theorem and its simultaneous rank16/A17 exchange. They reconstruct the complete selected maps and debit the latest216 literal obligations, rather than an earlier rule with larger payments.

Eight complete selected contexts

Reconstruct all140 eligible selected patterns into finite D17, including unsuccessful embeddings. The resulting168 maps agree exactly with the frozen complete domain. The four changed pairs occur only in these eight contexts, with two native maps each:

Contexts	Selected D17 role	Distinct existing half
2049,2508	(3,13)	whole A4 endpoint / whole D13 long-arm neighbor of fork center
1293,2019	(3,14)	whole A4 endpoint / whole D13 fork center
1944,2032,2041,2337	(3,15) or(3,16)	whole A4 endpoint / whole D13 short fork leaf

Every displayed marked component already has17 nodes and is the entire selected D17. No proper completion case is omitted. The shared core is the same19-root L25/L25 core used by the predecessor, with physical diagonal2 and signature(17,1,1).

The source reconstructs all four finite corners of each context from the actual Gram matrix. Exactly one corner other than the selected one has the stated A4 and D13 factors. Their canonical marks are A4 indices0 or3, and D13 indices9,10,or11/12 respectively. In each case the retained selected half and that distinct coupon half sum to1. No other payment is needed for these eight selected contexts.

The neighbor half is the existing whole A4/D13-neighbor coupon. The center and short-leaf halves belong to the existing whole-small/D11–D16 rule, which permits every marked node in any whole finite component of rank at most5. These allowances remain in the current complete finite rule. They are not added again or counted twice by this theorem.

Arbitrary extra supporting roots and whole factors

The source rechecks the physical Gram, its radical, the complete allowed contact positions, the selected corner, the positive-definite15-root white support, all four16-root edge supports, and all four17-root corner supports. The core radical is

```
(-6,-3,-4,-2,-5,-4,-3,-2,-1,0,
   6,3,4,2,5,4,3,2,1).
```

In any actual Lorentz realization this radical represents the zero vector: it is null and orthogonal to a span already containing a timelike vector. Every further supporting white therefore annihilates it. The inherited complete contact theorem restricts such a white to the saved allowed positions. Every allowed position has a nonzero radical entry.

A nonzero acute contact vector annihilating this radical must have a contact at positions of each sign. For every opposite-sign pair of allowed positions, the literal graph-distance check supplies a path of length at most8 between the two marked sides through that extra white. Thus either the independently established short8 route pays, or every extra supporting white is orthogonal to the entire core. This argument does not bound the number of extra whites or assume unit contacts.

In the latter branch both alternative factors are whole actual finite components. A spectator chain joining the core would need a first new root with a core contact, which has already been excluded. The existing whole-component coupon therefore applies. The selected and coupon corners are distinct actual corners of the face, so their halves can be added. The argument is the predecessor's complete zero-radical sealing proof evaluated at these eight new marked contexts; it is not an inference from finitely many sampled extra roots.

Simultaneous latest literal obligations

The producer reconstructs every one of the216 inherited literal configurations from their saved native Grams. Its baseline is the final rank15 packet after all rank15 decreases and the rank16/A17 exchange. Every displayed finite shared D17 component is mapped to the canonical diagram before the four new debits are applied simultaneously.

Exactly24 configurations change, all in the frozen D13 long-pair transfer. Eight use the center role(3,14), and16 use one of the leaf roles(3,15)/(3,16). The neighbor role(3,13) affects none. Every new individual contribution is nonnegative and every full literal total remains at least1. The other192 configurations are unchanged.

These records include the three-corner replacements as well as the four-corner records; no uniform four-corner schema is assumed. They are the complete inherited literal interface, not an enumeration of every quadrilateral. The full selected/nonselected theorem supplies that larger scope.

The intervening rank15 and rank16 transfers use whole A2/A13 endpoint coupons, retained finite D16 halves, whole-small/D13 coupons and restored whole A17 halves. The producer checks that none of their selected replacement contributions is a finite D17 same-component pair. In particular this further D17 decrease does not spend the half needed by the latest rank15 transfer.

Complete face coverage and current priorities

The cost120 predecessor supplies the exhaustive finite17 routing. In the complete2,614 high-overlap domain,2,391 contexts have whole-component bound at most16,74 have protected leaf9/basic ideal outputs, nine have adaptive leaf9 or whole-order16 outputs, and140 form the selected domain reconstructed here. The low-core bound and all-label high-disjoint theorems are unchanged. None of those other finite outputs can use one of the present D17 distance-at-least10 roles.

Apply the current independent actual ideal-corner priorities first, as in the completed61 proof. C11, B12, D13, cycle11, the coupled order11 B/C/D cases and ideal D17 zero produce retained short paths, protected small whole factors or the already counted typed coupons. B16 and B14/C14 projection exclusions have only the independent short8 escape. None consumes the finite D17 long pairs changed here.

If none of those branches applies, use the complete current selected reduction with the eight replacements above. Its other selected finite entries retain their values; all later literal obligations pass the simultaneous216-record check. The new finite replacements do not alter an ideal weight or a later replacement premise.

The universal tree/cycle and protected-feature routes retain their finite short paths, whole-leaf9 roles, prescribed small factors and coupons. If short8 reaches a deep finite8 zero, the completed depth11 corridor supplies its independent constructive short7 alternative. No part of the new radical sealing proof invokes the new local payment theorem to justify that escape. The inherited triangle route uses only unchanged finite short7 and ideal interfaces and therefore still pays2.

These arguments establish one simultaneous local payment rule. The prospective depth8 corridor strengthening and any future finite15/16 decreases are separate, unbound work and are not premises here.

Verification scope

Final source-bound generation89918/5bc07e and separate fresh replay 51409/27374f terminated exit0/PASS. The producer reconstructs140 patterns/168 native maps, the eight19-root radical cases/16 affected maps, exact whole-component roles and all216 current literal totals. It does not repeat the historical all-label overlap/contact classification or perform a finite-forest optimization.

A separate peer completed a full producer/report mathematical read and reread the predecessor's radical sealing proof, the allowed-contact construction and the latest literal/current61 interfaces. That review passed the mixed-sign radical argument, whole A4/D13 roles, distinct existing halves, complete selected/literal routing, intervening rank15/16 compatibility and independent depth11 short7 escape. It did not repeat maps, inertias, literal arithmetic or the historical geometric census. Root separately completed a full producer/report mathematical read, checking the140/168 selected domain, eight sealed cases/16 maps, arbitrary supporting-white argument, actual existing coupon roles, all216 latest literal debits and intervening rank15/16 compatibility. Root did not repeat a geometric census or arithmetic. Neither review found a gap.

Direct source bindings, local report links and Python syntax pass. Source, data and report are final/frozen. The current cost118 theorem does not depend on any prospective depth8 or further small-rank change.

```
python3 scripts/fresh_finite_d17_a4_d13_half_transfer_20261018.py
python3 scripts/fresh_finite_d17_a4_d13_half_transfer_20261018.py --verify
```

Source 114

Source path: `results/fresh_finite_d17_cost120_rank62_budget_20261018.md`

SHA256: abe7764c8853887aef9ea14fc0bcd9bc0b2ae1dacc35026e8f84b1013fd1bebd

Finite D17 cost120 lowers the complete rank62 maximum to438

The complete current finite payment rule, with only the newly proved finite D17 reduction125→120, has exact rank62 maximum **438**. The formal forest2D15+2D16 attains it. The finite curvature is434, leaving a deficit4. This is an exact counting result for the proved local rule; it does not establish an ambient dimension61 bound.

The producer and saved certificate bind the frozen D17 five-unit reduction and the current three-transfer finite rule. The earlier complete ambient62 frontier supplies the exact certificate engine, and the sparse rank63 budget supplies the inherited finite classification, marked coupon rules and complete multiplicity domain. Frozen parent files are unchanged.

Precisely one new component reduction

All current coupons remain. In particular, all nine sparse increments, the D13 zero-transfer increments of1 per whole A2/D10 pair and1/2 per whole A3-middle/D10 pair, and the E6/D11 unit-coupon total4 are included. The new D17 theorem adds no coupon.

The relevant same-component costs are

Component	Current cost
A15, B15, D15	105
A16, D16	114
B16	104
A17	114
B17	105
D17	120

Every other same-component weight, including deep11 finite8 zeros and the large-D fork-leaf halves, is inherited unchanged. No proposed rank15 reduction, A16/D16 reduction, or further large-D reduction is used. The local D17 proof separately checked current literal obligations and the newer B14/C14 short8 escape interface; the counting computation here does not repeat that geometric proof or its embedding census.

The reused engine calls its input delta `conditional_changes` and marks `geometric_change_proved_here` false. These are inherited engine field names: its exact delta here is D17−5 and the already proved D13 coupon upgrade, both bound to frozen geometric packets. The wrapper identifies this proved rule explicitly. The engine does not itself prove geometry.

Complete domain and current actual costs

Replace finite types simultaneously at both ends of each component pair. E6 remains distinct at rank6, B9 at rank9, and A17 is retained separately from D17. Otherwise the representatives are A at ranks at most8 and D at ranks at least10. The source checks all35,721 ordered type-pair inequalities and all component-base inequalities through rank62. This simultaneous comparison preserves the D5/A12 sparse coupon, which would not be justified by a one-end-only replacement.

The complete large multiplicity domain consists of B9,D10,…,D16,A17 with total rank at most62:1,227 vectors. The residual components have ranks1–8 or at least17. There are921 distinct seven-coordinate interaction keys. Large components have no mutual coupon; residual components have no mutual coupon. All remaining interactions are counted in their exact key-dependent additive costs. The decomposition and each literal interaction identity are inherited and checked by the bound engine.

The old engine's actual-cost field predates the E6/D11 upgrade and still uses total2, while its conservative representative upper already uses4. The wrapper therefore adds doubled4 for each E6/D11 component pair and recounts **every** reconstructed formal forest directly under the true current actual rule. It also repeats every simultaneous type-pair inequality with that actual total4. Thus no stale actual-cost field is used to claim attainment or the present maximum. The corrected values are stored in `current_actual_comparisons`.

Upper certificates and attainment

Generation computes the genuinely changed residual tables once. Fresh verification reads those saved tables and checks every possible last-item upper inequality and an attaining predecessor at each rank. Induction therefore proves each table's complete upper bound and attainment in the representative problem. The checks comprise1,426,629 upper inequalities,57,102 attaining predecessors,66,258 literal interaction identities and1,227 global alternatives. No numerical optimizer or finite-model linear program is used.

The largest representative upper value is438. The actual forest

```
D15 + D15 + D16 + D16
```

has no cross coupon and has cost105+105+114+114=438. This proves the exact actual-rule maximum, not merely an upper allowance. Formal attainment does not assert that this vertex forest is realized in a polytope. Actual A16/D16 substitutions in this attainer are explicitly recounted in the certificate. A separate cheap literal cost check also confirms all18 choices of two unordered A/B/D15 components and two unordered A/D16 components have the same cost438.

Precisely five fixed large-multiplicity alternatives have actual cost above434. Each listed forest attains its corresponding representative upper bound:

Representative formal forest	Exact current cost
2D15+2D16	438
A1+3D15+D16	435
3D15+D17	435
2D16+D30	435
D16+D46	435

This table is a complete list of offending representative multiplicity alternatives, not a claim that these are the only typed actual forests. For example the first has18 type ties. The A1 row must not be expanded by base-cost ties alone: changing D15 or D16 can remove its cross coupons. These witnesses identify finite ranks15 and16 as the remaining binding families after the D17 improvement. The separate ideal comparison and any new geometric reductions remain distinct tasks.

Verification and scope

Generation session63562 completed with terminal chunk5fe3ab, exit0/PASS. Separate fresh saved-certificate verification session29562 completed with terminal chunkaf30d9, exit0/PASS. Both report maximum438 and finite slack−4. The literal type-tie and saved-alternative diagnostic was a separate bounded exact query (terminal chunk00066e); it did not generate new tables. No frozen geometric census or global optimization was rerun.

The root reviewer and a separate peer completed full source/report mathematical reads, including the inherited component-base, representative, cross-cost, domain and exact-certificate engine. These reads passed for the single new D17 reduction, current D13/E6 coupons, simultaneous type comparisons, complete domain, saved upper/attaining proof, actual438 witness and representative/type-tie distinction. Neither review reran the arithmetic or dynamic programming.

The arithmetic is exact, but the count intentionally reuses the frozen certificate engine; it is not an independent reimplementation of its Bellman verifier or finite-type classification. This packet supplies a complete finite count for the stated current rule, without a new global dimension claim.

Source 115

Source path: `results/fresh_finite_d17_d5_d12_unit_transfer_20261018.md`

SHA256: c1567b083f18af6ed7b520d9a1bf987c7a5403303cdf0e2cfc78d1e51755fd18

Finite D17 cost235/2 using a whole D5/D12 unit

The whole finite D17 cost decreases from119 to **235/2**. Three existing halves become zero. In exchange, six whole-component D5/D12 half payments become units, adding **3 per component pair**. Every triangle remains paid by at least2 and every quadrilateral by at least1 under the simultaneous current rule. The additional finite cost must be counted before drawing a dimension conclusion; this packet supplies no new forest bound.

Number D17 by spine0 through14 and short leaves15/16 at14. The changed pairs are(4,14),(4,15),(4,16), at distances10,11,11. The set is invariant under every labelled automorphism. All finite short8 and protected leaf9 units remain, as do the D17(1,15)/(1,16) halves used by the D16 exchange.

The coupon is intrinsically defined: the mark in a whole D5 is either short fork leaf, canonical3/4; the mark in a separate whole D12 is its fork center or either short fork leaf, canonical9/10/11. Each such pair changes1/2 to1. There are2×3=6 pairs, so the increment is3 and the total actual D5/D12 cost becomes21/2 from15/2. The D5 long-arm endpoint is not in this changed set.

The producer and certificate bind the current frontier, rank15 exchange, D16 exchange and D17 restoration, and the six original D17 radical transfers. The coupon is an upgrade within the already proved and counted whole-small/D11–D16 half rule.

Complete selected domain and six whole-factor transfers

The saved complete selected domain has140 eligible patterns and168 induced maps into D17. The producer validates all maps, all target labels and all marked images, including patterns with no map. Completeness of that domain remains the bound predecessor's census; no new exhaustive map search is claimed. Only six contexts and12 maps meet the changed pairs:

Contexts	Selected D17 pair	Distinct whole-component unit
1297,2063	(4,14)	D5 short fork leaf / D12 fork center
1950,2072,2078,2343	(4,15) or(4,16)	D5 short fork leaf / D12 short fork leaf

Every selected component already has17 nodes. Thus all six are whole D17 cases; no proper completion is discarded. The producer reconstructs each old radical proof and compares it with its frozen record before replacing the two half payments by the single distinct coupon unit.

All six use the19-root norm2 core of signature(17,1,1), with radical

```
(-6,-3,-4,-2,-5,-4,-3,-2,-1,0,
   6,3,4,2,5,4,3,2,1).
```

Its represented radical is zero in the actual Lorentz space: it is null and orthogonal to a span containing a timelike vector. Hence every extra supporting white annihilates it. The complete inherited contact theorem restricts contacts to positions with nonzero radical coefficients. Any nonzero acute contact vector annihilating the relation meets both signs. Every such

opposite-sign pair gives a marked full-star path of length at most8 through that white, as checked on the complete allowed domain.

If that short-path alternative does not occur, every extra white is orthogonal to the whole core. A bridge through further spectators would have a first root meeting the core and is excluded by the same argument. The displayed D5 and D12 are therefore whole actual components. The producer also rechecks positive definiteness of the15-root common-white support, each16-root edge support and all four17-root corner supports. It recovers the exact short-leaf and center roles from those finite components. There is precisely one coupon corner distinct from the selected zero corner; its upgraded unit supplies the required total1.

This argument allows arbitrarily many extra supporting roots and arbitrary native contact magnitudes. It is not based on a sample of extensions.

Current literal interfaces and universal coverage

The producer reconstructs all216 current literal configurations from their native Grams, using the rank15 exchange's final cornerwise lower payments at D16=111 and D17=119. No corner uses one of the three changed D17 pairs. No literal coupon increment is needed either. Every old lower total remains at least1, with nonnegative individual payments. Three-corner and four-corner schemas are both retained. These216 records are the inherited literal interface, not a census of all quadrilaterals.

The positive same-component alternatives of the later rank15, rank16, A15, A16 and D16 exchanges are checked separately. None uses a changed D17 pair. In particular the restored D17 pairs(1,15)/(1,16) remain halves. The six selected halves in the table above are explicitly replaced by their new units; they are not assumed to remain as old alternatives.

The original finite17 theorem supplies the complete nonselected routing: among2,614 high-overlap contexts,2,391 have marked whole-component bound at most16,74 have protected leaf9/basic ideal outputs, nine have adaptive leaf9 or whole-order16 outputs, and140 form the selected domain. The low-core bound is12 and the high-disjoint branch is excluded. None of the protected or smaller-factor outputs can use these distance-at-least10 pairs in a whole D17.

Apply the current changed-ideal-corner priorities first. The completed B11/C11 routes use short paths, whole B11/B12, small factors or typed coupons. D105 uses protected distance9 outputs or its distinct whole D5/D11 coupon. The new order15 B/D theorem uses projection exclusions, short paths and whole A4/D11 or A5/D11 units, expressly without spending a same-component finite unit. The earlier order17 E6 transfer uses its distinct E6/D11 coupon. These outputs remain paid. The cycle/tree and other established ideal routes retain their stated short-path, small-factor and coupon interfaces.

When a short8 finite output is a deep finite8 zero, its independent depth8 theorem supplies short7. When that output is a deep finite7 zero, the independent depth10 theorem supplies its short6 alternative. Their protected outputs have distance at most9 or separate coupons and avoid the three new D17 zeros. Thus the radical argument does not invoke its own new payment theorem to justify the short8 escape. The finite7/ideal5 triangle theorem likewise uses no affected pair and retains total2.

After those priorities, the complete selected/nonselected reduction and the literal interfaces above give the simultaneous face theorem. Every ideal weight is unchanged. The additional whole-component payments are nonnegative, so cases with no changed finite corner retain their prior inequalities. The completed unrestricted bound60 remains the global result.

Verification scope

Generation session57629, final chunk895600, and separate fresh replay session86552, final chunk926a28, both terminated exit0/PASS. Two peer agents completed full source/report mathematical reads, including the inherited radical seal, selected-map and literal interfaces: PASS. They checked the arbitrary-contact argument, exact whole-component roles and cost, unchanged literal payments, retained later alternatives and independent current short-path priorities. Neither repeated map searches, inertia, arithmetic, the historical census or a forest optimization.

The producer validates the saved selected maps, reconstructs the six exact radical arguments and all216 current literal configurations, and checks later positive finite interfaces. It does not repeat the historical all-label/contact census, optimize a forest budget or formalize the proof in a proof assistant. The universal supporting-root argument and complete inherited

routing are explicit mathematical inputs and are explained above. The source, certificate and report are frozen. No external refereeing is claimed.

```
python3 scripts/fresh_finite_d17_d5_d12_unit_transfer_20261018.py
python3 scripts/fresh_finite_d17_d5_d12_unit_transfer_20261018.py --verify
```

Source 116

Source path: `results/fresh_finite_d17_five_unit_reduction_20261018.md`

SHA256: 640ca5a286a75d5f7c229f3373b81a022dae1e480b5a1feadb96e256d90e4565

Finite D17 cost120 with no new coupon

The current whole finite D17 payment can be reduced from125 to120 while retaining every other finite weight and every coupon. Two long pairs become zero and six become halves. Every quadrilateral still receives at least1 and every triangle at least2. The result is compatible with the completed62 rule and the three subsequent D13/B16/D17 ideal transfers. It supplies a local payment theorem, not a new finite-forest maximum or ambient dimension bound.

Use D17 numbering with spine0 through14 and short leaves15,16 attached to14. The changed weights are:

Pairs	Old weight	New weight	Total saving
(0,12),(0,13)	1	0	2
(0,14),(0,15),(0,16)	1	1/2	3/2
(4,14),(4,15),(4,16)	1	1/2	3/2

The sets are invariant under both labelled diagram automorphisms. All changed distances are at least10. Every short8 and whole-leaf9 pair retains its unit. The old four halves beginning at node2 are unchanged.

The producer and certificate bind the current sparse finite theorem, its complete selected and literal interfaces, and the simultaneous three-transfer rule. The only cross-component payment used below is an already budgeted whole-small-component half: any marked node in a whole component of rank at most5, paired with the fork center or short fork leaf of a separate whole D11–D16 component. In particular a D5 mark is allowed; the rule is not restricted to small paths or rank at most4.

Complete selected support

The complete induced-embedding routine is rerun on all140 eligible selected patterns in the bound high-overlap certificate, including failed embeddings. It reconstructs168 maps into finite D17. None uses any of the five changed pairs beginning at0.

The remaining three changed pairs occur in exactly six contexts and12 maps. Every affected marked component is already the full17-node finite D17, in the same19-root L25/L25 core. Thus no proper finite completion requires a new projection argument.

Contexts	Selected D17 pair	Distinct existing half
1297,2063	(4,14)	whole D5 short leaf / whole D12 fork center
1950,2072,2078,2343	(4,15) or(4,16)	whole D5 short leaf / whole D12 short leaf

The displayed D5 short leaves have canonical images3,4. The D12 fork center is9 and its short leaves10,11. The source reconstructs the complete finite components and all marked images directly from the Gram matrix. In each context there is exactly one such coupon at a corner distinct from the selected D17 corner. Its half plus the retained selected half is1. No other finite or ideal contribution is needed for these six cases.

Whole-component sealing for arbitrary supporting roots

The common core has physical diagonal2, signature(17,1,1), and radical

```
(−6,−3,−4,−2,−5,−4,−3,−2,−1,0,
   6, 3, 4, 2, 5, 4, 3, 2, 1).
```

The radical is the zero vector in an actual Lorentz realization: its represented vector is null and orthogonal to a span containing a timelike vector. Every additional actual supporting white therefore annihilates it. The inherited complete contact theorem supplies the possible contact positions; every such position has a nonzero radical coefficient. The source rechecks that domain, the actual selected corner, the positive-definite15-root white support, all four16-root edge supports, and all four finite17-root corner supports.

Any nonzero acute contact vector annihilating the radical must meet positions of both signs. For every opposite-sign pair of allowed positions the complete distance check gives a marked path of length at most8 through that supporting white. Thus either the retained short-path route pays, or every additional white is orthogonal to the entire old core. This argument permits arbitrary extra supporting roots and arbitrary native contact magnitudes; it never assigns an unproved zero to a possible contact.

In the latter branch the displayed D5 and D12 are whole actual components. An indirect spectator bridge would still need a first new root contacting the core, which has just been excluded. The existing half-coupon is therefore legitimate. This is the same complete radical argument as in the bound sparse finite theorem, now evaluated at six different marked contexts with different already-paid roles.

Nine literal debits and all216 interfaces

Absence from selected maps alone would not justify removing the five pairs beginning at0. Those pairs have actual obligations in older literal transfers. The source reconstructs all216 configurations of the current sparse theorem, preserving every current sparse coupon, and debits all eight changed pairs simultaneously.

Nine configurations are affected:

Source configuration	Debited D17 roles	Old recorded total	New recorded total
old finite-cross393,647	(0,14) unit to half	3/2	1
old finite-cross662	two(0,15)/(0,16) units to halves	2	1
old finite-cross675,828	one(0,15)/(0,16) unit to half	3/2	1
D17 partial1974,2504	(0,13) unit to zero	2	1
D17 partial1978,2535	(0,12) unit to zero	2	1

The four partial-weight configurations retain a distinct finite whole-small/D12-or-D13 half and an ideal D17 half. Their ideal marks are explicitly mapped into the latest order17 zero rule: all their canonical images still have weight1/2. They are not in its newly zeroed E6-coupon orbit. The old A17 endpoint deletion in configuration662 is also preserved; its two surviving D17 halves alone give1.

All other207 recorded totals are unchanged. Every individual payment is nonnegative and every configuration has total at least1. The literal domain includes184 four-corner records and32 inherited three-corner replacement records; the checker preserves each full record instead of assuming a uniform four-corner schema. These216 interfaces are not asserted to enumerate all quadrilaterals. They are the complete affected literal obligations inherited by the current finite theorem.

Full face coverage and current ideal compatibility

The original complete finite17 theorem supplies the nonselected coverage. The low-core marked bound is12. Among all2,614 high-overlap contexts,2,391 have whole-component bound at most16,74 use protected leaf9/basic ideal payments, nine use adaptive leaf9 or whole-order16 payments, and the remaining140 form precisely the selected domain reconstructed

here. The all-label high-disjoint domain is excluded by its bound theorem. None of the protected or adaptive finite outputs has one of the changed distance-at-least10 roles.

The universal tree/cycle replacements retain finite short paths, whole-leaf9 roles, their prescribed small whole factors or separate coupons. Later order11 outputs are whole finite components of order at most14, whole A9/A18 endpoint8, D18 leaf9, E8 path6, or typed coupons. The D13 center zero theorem uses whole D10-center coupons, retained order11 center units or paths of length at most4. The B12 zero theorem uses its whole D12 leaf9 or typed coupons. The D17 zero theorem uses a distinct E6/D11 unit and expressly uses no finite D16/D17 same-component unit. The current B16 deletion supplies only necessary projection exclusions or its retained short8 route. All these outputs avoid the present finite D17 long pairs.

Apply the independent current ideal-corner priorities first, as in the bound three-transfer theorem. Their paying branches remain valid under this finite change for the reasons just given. In the remaining branch repeat the complete current no-short8 finite/ideal reduction: the selected finite D17 entries now use the six sealed transfers, unaffected entries keep their weights, and the literal obligations use the simultaneous216-record audit. The B16 selected affine obstructions remain independent of these finite entries; no new whole-face fall-through is used to prove another branch's premise.

If a short8 route reaches a finite deep8 zero, the independent depth11 corridor supplies its constructive short7 alternative. Its small-tree, cycle-tail and protected-feature outputs avoid finite D17 distance10 or longer. Thus the radical proof's short8 alternative is already established without invoking the new finite theorem recursively. Every finite short7 and every ideal weight is unchanged. The inherited triangle theorem therefore still supplies total2.

This proves simultaneous face payment under the current three-transfer ideal rule with no added coupon. A subsequent finite-forest certificate must use D17=120 and the other actually proved component costs; the earlier conditional rank15 or A16/D16 reductions are not imported. No claim about ambient61 follows from this local theorem alone.

The subsequent B14 and C14 projection exclusions are compatible as well. Their only paying escape is the independent retained short8 route just described; neither consumes a finite D17 distance-at-least10 unit. Their distinct selected affine entries and the present selected finite entries can therefore be patched in the same complete reduction. Their scalar consequence and this finite-forest recount remain separate certificates.

Verification scope

Final source-bound generation7afe75 and separate fresh replayaf8415 both terminate exit0/PASS. The source reconstructs140 selected patterns/168 maps, six full radical contexts/12 affected maps, all216 current literal records and the four retained current ideal-half interfaces. It binds the existing whole-small coupon explicitly. No historical high-overlap, arbitrary-label or complete contact classification is repeated.

Root and a separate peer completed full producer/report mathematical reads. Root checked all selected, radical, literal and current-priority interfaces; the peer additionally reread the inherited marked-role, literal-audit, selected-map and literal-input helpers. Both passed the zero-radical/complete-contact sealing, distinct existing whole D5/D12 half, all current simultaneous debits and retained ideal halves, and the exhaustive nonselected and short8/deep11 composition. The peer also confirmed compatibility with its B14/C14 exclusions. Neither reviewer repeated maps, inertias, literal arithmetic, historical censuses or a forest budget. Direct bindings, local links and Python syntax are audited before freezing. Source, data and report are final/frozen.

```
python3 scripts/fresh_finite_d17_five_unit_reduction_20261018.py
python3 scripts/fresh_finite_d17_five_unit_reduction_20261018.py --verify
```

Source 117

Source path: `results/fresh_finite_d17_leaf_a16_half_transfer_20261018.md`

SHA256: 56ccbce32eb877f3de0d3b1b4b474bd850d0cf48b1046ee67051eb4b17769a76

Two finite D17 leaf halves paid at other whole corners

Starting from the current D5/D12 alternate-corner rule, set the whole finite D17 weights at(3,15),(3,16) from1/2 to0. Add1/2 between a whole A1 and canonical node3 or12 of a whole A16. The actual D17 same-component total falls467/4 to463/4, saving1. The only new component-pair cost is A1/A16:+1; its new actual total1 is below the existing A1/D16 representative allowance3/2.

Every triangle still receives2 and every quadrilateral at least1 under the inherited finite-volume actual-corner hypotheses. No ideal weight changes. This local theorem gives no new dimension bound or finite forest optimization. The completed unrestricted ceiling remains59.

The producer and certificate validate the complete saved selected support, reconstruct four full-core seals, and debit the latest216 literal interfaces. They also check the new actual coupon against every current representative type pair through rank59. The larger, unbudgeted proposals in the preceding node3 support scout are not used.

Complete selected domain and exact alternate payments

Number D17 by spine0 through14, with short leaves15,16 at14. The two changed pairs form one intrinsic reflection orbit and have distance12. The source checks invariance under every diagram automorphism. All other whole-D17 pair weights, short/protected units and the separate whole-A17 weights remain unchanged.

The earlier D17 A4/D13 theorem reconstructed every eligible selected pattern and native map. The new source verifies the entire140-pattern/168-map domain against the bound overlap classification, including empty-map patterns, distinct induced maps, all labels and the actual marked images. Exactly eight maps in four contexts use the changed leaf orbit. All four displayed selected components already have17 roots, so no proper completion is omitted. The historical map search itself is inherited, not rerun.

Contexts	Payment after deleting the selected half
1944,2032	A distinct whole A17 unit at canonical(0,13), equivalently(3,16)
2041,2337	The existing whole A4 endpoint/D13 short-fork-leaf half, plus the new distinct A1/A16 node3-or12 half

The A17 unit already exists in the current balanced weight table. It is not one of the restored or fractionally rebalanced A17 pairs. The producer checks both canonical images against that exact table and retains them explicitly. In these two cases it uses only the unit; the existing A4/D13 half is unnecessary additional payment.

In the other two cases, the full finite corner has a whole A16 and a separate whole A1. The marked A16 node has exactly canonical images3 and12 under all path automorphisms. These are fourth-from-end nodes, not the previously used A13 middle or A14 second/third nodes. The new half and the existing whole A4/D13-shortleaf half occur at two distinct corners; neither is the deleted selected D17 corner. Their sum is1. The new rule is invariant under A16 reflection and counts its two marked positions once each, giving component-pair cost1.

Arbitrary supporting whites and whole factors

Each affected case is the19-root L25/L25 core with physical norm2 scaling and signature(17,1,1). The producer recomputes its rational radical, positive-definite15-root white support, all four strict edge supports, the selected corner's complete necessary contact domain, and every finite corner decomposition. It compares this result with the frozen original seal.

The radical is an actual zero-vector relation in any Lorentz realization: the span contains a timelike vector, and its orthogonal complement is positive definite. Every allowed contact position for an additional supporting white has a nonzero radical coefficient. Annihilation of the relation, together with nonpositive actual wall products, forces any nonzero contact vector to touch positions of both radical signs. For every opposite-sign pair, the saved exact distance calculation gives an actual white-interior path between adjacent black sides of length at most8. The independent current short8 route pays that branch.

If no such shortcut occurs, every additional supporting white is orthogonal to the entire core. This treats arbitrary contact magnitudes and any number of additional roots. An indirect spectator bridge would have a first root touching the core and is excluded by the same argument. Hence all displayed factors at every corner stay whole: the A17 in the first two cases, and the A4/D13 and A1/A16 components in the other two. The proof does not infer whole-factor sealing from positive definiteness of the selected D17 alone.

Current whole-face composition and later uses

The complete finite-D17 routing has2,614 high-overlap contexts: 2,391 with whole-component bound at most16,74 with protected leaf9 or basic ideal outputs, nine with independent adaptive outputs, and 140 in the selected domain just validated. The low-core and all-label high-disjoint reductions are unchanged. The proper small-component, short and protected outputs cannot use the new distance12 leaf pair. The four replacements above handle precisely its selected uses.

All current independent actual-ideal-corner priorities retain their stated outputs. The B/C11, C11/B12/D13, cycle and later order12–17 refinements use short paths, protected whole factors or separately specified coupons. The current order14 near-fork proof uses retained D11 halves, A4/D10-shortleaf or A1/D13-neighbor halves, a whole finite D14 unit or an independent short path. The recent contextual rank13 and D5/D12 exchanges also use distinct marked roles. None consumes this finite D17 node3/short-leaf half.

The short8 escape remains independent of this theorem. The proved finite8 and finite7 replacements reduce to the current constructive short7/short6 routes or their explicit protected whole-factor alternatives. None requires a distance12 pair in whole D17. The triangle proof uses the same retained elementary/protected outputs and receives2. No deleted long pair is silently restored to pay a short-path branch.

Later finite positive outputs are separately respected. A15/A16 path exchanges use their specified whole A17 roles or sparse path coupons; D15 and rank14 exchanges use their own D16/finite smaller-factor roles. The D16 fork exchange and fractional balance use D17(1,15)/(1,16), which are disjoint from(3,15)/(3,16). The D17 node4 transfer uses its D5/D12 units. In the directly stored positive-corner arrays of the bound current local packets, the only positive whole-D17 corners using the newly removed orbit are exactly the four original contexts replaced here. The source checks these arrays and their canonical images. This direct scan supplements the complete written routing argument; it is not presented as a universal geometric census.

The producer also reconstructs all216 current literal obligations from the frozen D5/D12 packet, matching each source, case and actual marked corner. There are16 D17 half-debits and eight A1/A16 half-credits. Every individual lower bound remains nonnegative and every total stays at least1. No strengthening from the unused A17 units is needed for these literal totals. These checks include the old affine-D13 replacement obligations; they are not discarded because they belong to an earlier packet.

Thus the whole previous face proof can be replayed with the same actual-corner priorities and the four replacements above. Faces whose proofs use no changed output retain their old payment. The new positive coupon only adds payment at other corners. There is no circular use of the selected D17 theorem in its own short8 escape.

Exact accounting and separation from other proposals

The source binds the actual current A1/A16 component-pair cost0. There are two new marked pairs, each1/2, so the actual total becomes1. It verifies all32,400 ordered actual-type pairs through rank59 after this increment, using the same simulta-

neous two-end representative map. A1/A16 maps to A1/D16 with allowance3/2. No representative cross cost increases. It also checks every representative base after lowering D17 by1, with A17 retained as a separate type.

The rule leaves the contextual A2/A2 trigger and every other actual coupon unchanged. Consequently it can be used in a subsequent complete finite count without a new cross-cost allowance, but this packet does not reconstruct the full forest tables or claim a reduced maximum.

The distinct node2 D17 proposal does not enter the hypotheses here. Its changed roles(2,12),(2,14) are absent from these new replacements; the replacements use A17 units, A4/D13 and A1/A16. Conversely the node2 proposal's A1/D5, A2/A14-third and A1/D14-center payments are unchanged here. Their simultaneous accounting and any added ideal coupons remain the responsibility of the combined certificate. The retained A17(0,13)/(3,16) units are now explicit obligations for future weight changes.

Verification scope

Final source-bound generation49744/a5b164 and separate fresh replay 35615/c0b025 completed exit0/PASS with complete saved-object equality. Root and the order16 agent each completed a full producer/report mathematical read, including the inherited full radical seal. Their reviews covered the complete saved domain, actual whole-component payments, simultaneous literal changes, bounded later-interface audit, arbitrary-support routing and unchanged representative-cost absorption. Neither review executed the code or repeated maps, inertia, literal arithmetic, type-pair enumeration or a historical census.

The source validates the complete saved140/168 maps, four exact full seals, whole-component mark roles, current216 totals, direct positive-corner interfaces and type cost domination. It does not rerun the historical overlap search, all-label classifications, full finite optimization, ideal convolution or any global arithmetic model. Those broader proof inputs are bound frozen artifacts, with their stated mathematical scope preserved.

```
python3 scripts/fresh_finite_d17_leaf_a16_half_transfer_20261018.py --verify
```

Source 118

Source path: `results/fresh_finite_d17_long_pair_support_20261017.md`

SHA256: b968047ccea17a4993ff2d42e486595cc35fd98df593b7c25f295afa631af619

Simultaneous long-pair deletions in whole finite rank17 components

Starting from the completed dimension68 face-payment rule, the same-component costs of whole finite A17, B17 and D17 can be reduced to114,105 and127 respectively. Every triangle still receives at least2 and every quadrilateral at least1. All ideal weights and all finite cross-component coupons remain unchanged. This is a local refinement; an exact finite-forest budget and any dimension consequence are separate obligations.

The source and certificate reconstruct all affected selected maps and literal replacement interfaces. The complete all-label core classification, extension bounds and arbitrary-support sealing arguments are explicitly inherited from the frozen predecessors.

Canonical rules

Number A17 as the path0–1–…–16. Number B17 by the same path with the bond15–16 doubled. Number D17 by the spine0–1–…–14 with short leaves15 and16 attached to14. Every rule is invariant under the full automorphism group of the corresponding labelled diagram.

Write B for all pairs at distance at most8, together with distance9 pairs having an endpoint that is a leaf of the whole component. These basic sets have sizes102,102,104 respectively. They are retained in their entirety. The new paid sets, all at unit weight, are specified exactly as follows.

For A17, retain B, the following ten selected pairs, and(0,13),(3,16):

```
(2,11),(2,12),(2,13),(2,14),(3,12),
(3,13),(3,14),(4,13),(4,14),(5,14).
```

This gives114 pairs. The previously deleted endpoint pair(0,16) remains deleted. Relative to the completed rule of cost135,21 further pairs are removed. They are the reflection orbits represented by

```
(0,10),(0,11),(0,12),(0,14),(0,15),
(1,10),(1,11),(1,12),(1,13),(1,14),(1,15).
```

The first ten orbits have size2; the last is fixed by reflection.

For B17, retain B and(2,11),(3,12),(4,13), giving105 pairs. The labelled B17 diagram has no nontrivial automorphism, so this is intrinsic after choosing the doubled end as specified. Thirty-one old units are removed.

For D17, retain every old pair except

```
(0,10),(0,11),(1,10),(1,11),(1,12),
(1,13),(1,14),(1,15),(1,16).
```

Only the last two are exchanged by its nontrivial automorphism. The nine deletions give cost127. They include the internal distance9 pair(1,10), whose endpoints are not whole-component leaves. No protected leaf9 pair is removed. The certificate also lists every retained and removed pair and every automorphism, avoiding any dependence on a drawing convention.

Complete routing before the finite rank17 question

The predecessor's finite triangle payments use marked distance at most8, which is retained. Its ideal triangle payments do not change.

For quadrilaterals, the complete reduction first resolves a marked path of length at most8. A finite such corner retains its unit. If that corner is ideal, the universal finite8/ideal5 replacements are still available:

- For long cycles, single-contact finite alternatives have distance at most8 or distance9 at an actual whole-component leaf. The inverse-diagonal sealing and fixed-side spectator arguments in the finite8 replacement establish these roles with arbitrary supporting spectators.
- The distance6 extension of that argument has the same finite-path alternatives. Its only small-component equality is whole A11, and its cycle18 equality uses separate A1/A17 halves at A17 positions5/11. The radius-five tree equality uses a whole A7 path of distance6. These are the actual conclusions of the local replacement, rather than an inference from its more permissive small17 statement.
- The cycle13/14 proofs use finite paths of length at most7/8, or unchanged separate-component coupons. The new cycle12 proof uses finite paths of length at most6, paths of length4, or separate A1/A11 and A2/A11 coupons. In particular no one-sided arbitrary-spectator branch depends on a long same-component rank17 unit.

After this branch, the complete low-core closure has66 chosen contexts, each with whole marked-component bound at most12. The all-label disjoint high-core branch is excluded geometrically. The complete high-overlap census leaves2,614 contexts, routed as follows:

Route	Contexts	Effect of the deletions
Whole marked component at most16	2,391	No rank17 component can occur
Protected actual leaf9 or retained basic ideal payment	74	Payment is retained
Adaptive leaf9 or whole-component bound16	9	Payment is retained
Whole-component bound17	140	All induced finite A/B/D17 maps are enumerated

The bound17 cases do not stand for the entire quadrilateral domain. The other2,474 contexts, the low and disjoint cases, and the universal short-path branch have their separate reasons above. The frozen A17 provenance audit already verifies these routing counts against the full82,197-identification core census; this source binds that audit and its underlying certificates.

All finite A17/B17/D17 selected maps

For every one of the140 bound17 cases, take its connected selected marked component in the saved actual core Gram. A positive diagonal congruence does not change its native angle. For distinct nonorthogonal roots define

```
rho_ij = 4*g_ij^2/(g_ii*g_jj).
```

The values1 and2 mean native labels3 and4. Any other value cannot occur in an induced subdiagram of finite A17, B17 or D17. Signs are checked negative; orthogonality is kept as an explicit nonedge. Thus a rational gauge is never mistaken for the norm-two root normalization.

The induced-map search tries every unused target vertex consistent with every already assigned edge and nonedge. Its degree filter is only a necessary condition. It therefore enumerates every injective labelled induced embedding, including every possible embedding coming from an actual whole-component extension. It need not prove that every emitted embedding is geometrically realizable: the over-approximation is safe because all its marked images are retained.

Target	Eligible patterns with maps	Complete induced maps	Distinct marked pairs
A17	15	48	10
B17	5	18	3

Target	Eligible patterns with maps	Complete induced maps	Distinct marked pairs
D17	75	168	20

Every selected image is in the retained set. Patterns with no map cannot be subdiagrams of that actual target. Finite exceptional irreducible types have rank at most8, so these three types exhaust finite irreducible rank17. The finite type called C17 has the same labelled Coxeter graph as B17. Components of every other rank keep their entire preceding rule.

Nonpreferred finite corners in literal replacements

Absence from selected maps does not suffice: an old ideal transfer may pay at another finite corner. The checker therefore reconstructs174 complete literal configurations:

Source	Configurations
Old affine-order16 replacements	9
Old affine-order17 replacements	9
Universal affine D13 equality	72
Universal affine D14 equality	72
Later affine-order17 partial weights	12

It obtains each actual corner by adding its two black marks to the saved white set, recomputes its connected components, and enumerates full rank17 isomorphisms into each canonical finite target. Every rank17 match has a positive-definite full corner Gram. These are whole components: the old literal and partial-weight cores use the actual affine kernel and a full-core radical to show that any additional supporting contact either gives a retained length-two marked path or vanishes on the whole core. The universal D13/D14 equalities force all remaining spectator contacts to vanish. This excludes both direct attachments and indirect tails or bridges; it is not inferred merely from a displayed subdiagram.

The literal unit images are:

Type	Canonical marked pairs
A17	(0,13),(3,16),(0,16)
B17	None
D17	(0,12),(0,13),(0,14),(0,15),(0,16),(3,14),(3,15),(3,16)

Every listed unit is retained except the already-deleted A17 endpoint pair. The checker simultaneously applies all three new finite rules and recomputes all174 totals, retaining their inherited half-payments and ideal half-weights. Every total is at least1. The sole debit is the old affine-order17 case662: its A17 endpoint unit was already absent, and its two D17 units both remain. Its retained total is2. In particular the new D17 deletion does not consume the payments that justified the old A17 endpoint deletion.

The universal D13 nonextreme zero cases use whole D12, not rank17. The immediate D13/D14 alternatives are whole affine factors of orders12/13, whose current refinements have separate interfaces below. Thus neither the literal equality census nor the selected-map census is being used as a substitute for the universal nonextreme or immediate branches.

Every later interface in the completed rule

The checker traverses the explicitly typed finite-unit corners of the successor packages and records their actual whole-component types. The only rank17 units are the four D17 corners of the partial-weight package, already included among the174 configurations above.

Later transfer	Relevant finite alternatives
Affine-order17 singleton	Separate A4/D13 halves; short path
Affine-order17 partial weights	Preserved D17 units; existing finite halves plus an ideal half
Affine-order16 partial weights	Whole D16 or separate-component halves
Order15 affine B reduction	Whole D14 or a finite component of order at most14 extending A12
Order13 affine B transfers	Whole B13 or separate A1/D12 and A2/B10 coupons
Universal affine D12 transfer	Whole D13/D14/A15 or separate A1/A2 versus D12 coupons
Order14/15 affine B zero transfers	Projection exclusions; no finite alternative
Cycle12 transfer	Finite paths of length at most6 or separate A11 coupons
Universal affine D11 transfer	Whole D12, or literal D13/D15/A16, or separate D11 coupons
Order12 affine B half-transfer	Whole B11/B12 or retained separate D11 halves
Order12 affine C deletion	No new finite alternative

The B15 and B11 whole finite alternatives include their arbitrary-support sealing/exclusion statements; they are not only types observed in sampled corners. For D12 and D11, the one-sided arbitrary-spectator arguments select the spectator-free new root's whole finite component. Their two-sided alternatives are either retained short paths or sealed finite corners of order at most15/16. Those conclusions cover unbounded spectator rank without extrapolating from the saved finite tables.

All cross-component coupons remain unchanged, including the A17 positions5/11 coupons, both full A1/D11 and A1/D12 upgrades, the B10/D13 special coupons, and the two new A11 coupons. No new finite payment is introduced. No ideal pair is altered, and none of these preserved routes calls back to one of the newly deleted finite units.

Verification status

Generation9167 and fresh separate-process replay5320 both terminated exit0/PASS on the complete selected maps, all174 simultaneous literal totals, the retained basic sets and automorphism invariance, and every recorded later interface. All94 direct bindings, Python AST and six local report links pass. The full geometric core census and arbitrary-support classification/sealing remain frozen inherited inputs.

A separate full source/report and universal-interface review passed. That reviewer reread the finite8 cycle/local proofs to check their actual finite-path-at-most8 or protected whole-leaf9 conclusions, the A11/A7 equalities and separate A17 coupons, the cycle13/14 finite paths and coupons, and the later arbitrary-spectator D12/D11/B11 whole components of order at most16 or separate-component alternatives. A separate small standard-library check validated the234 saved induced maps, basic-pair inclusions and all174 rational totals, including the sole inherited case662 debit retaining2. This check did not re-enumerate the maps, recompute corner inertias, repeat the geometric census, or calculate a finite budget.

Root completed a full source and mathematical report read, including the imported induced-map routine's completeness, normalization in positive gauges, and the universal finite8 cycle/local proofs. That review passed the retained whole-leaf9/basic pairs and automorphism invariance, complete 2,614-context routing, all140 eligible finite-target interfaces, the simultaneous174 literal debit logic with case662 retaining2, and every newer finite output of order at most16 or between separate components. Root did not repeat the high-core census or corner inertia calculations. Source, certificate and report are final/frozen. No exact finite-forest maximum, ambient dimension theorem, new polytope, external refereeing or proof-assistant formalization is claimed here.

```
python3 scripts/fresh_finite_d17_long_pair_support_20261017.py
python3 scripts/fresh_finite_d17_long_pair_support_20261017.py --verify
```

Source 119

Source path: `results/fresh_finite_rank14_existing_payment_transfer_20261018.md`

SHA256: 0e9fbeec07722b00198ef84ea3f749b24af46aca7076387402e7be17ed0c1bd7

Finite B14 and D14 reductions with every saturated extension covered

The current whole finite costs become **A14=91, B14=87 and D14=181/2**. No other finite or ideal weight changes, and no coupon is added or increased. Every triangle retains payment at least2 and every quadrilateral at least1 under the completed depth-six rule. This is a local theorem; the common rank14 upper envelope remains91 because A14 is unchanged. It does not lower the completed unrestricted dimension bound59.

The producer and certificate bind the complete rank14 inventory, the current sparse-D16/ideal17 baseline, and the depth-six theorem.

Number B14 along0 through13, with double bond12–13. Set its four pairs (0,13),(1,13),(2,13),(3,13) from unit to zero. Number D14 with spine0 through11 and two short leaves12/13 attached to11. Set its singleton pair(0,11) from unit to half. These sets are invariant under all labelled diagram automorphisms. Their distances are at least10, so the basic short8 and whole-leaf9 pairs remain unchanged.

Complete selected support

The producer freshly reconstructs all600 eligible selected patterns for each finite rank14 type, including failed maps. It agrees exactly with the saved552 A maps,260 B maps and1,104 D maps. No changed B pair occurs in this domain. The changed D pair meets exactly13 contexts:

- Six whole radical contexts:554,616,1240,1276,1342,1543.
- Two whole nonsingular Lorentz contexts:1284,1623.
- Four strictly impossible proper completions:624,1158,1555,1618.
- One saturated proper completion:1641.

The whole contexts already display all14 nodes of the selected D14. The actual-component premise prevents additional supporting whites from touching that component. The proof below addresses all contacts to the remaining core, including contacts of arbitrary native magnitude and arbitrary numbers of added whites.

Radical cases and distinct retained halves

For each radical core the exact signature is(n−2,1,1). White and white-plus-one-black supports are positive definite. The core radical represents zero in any Lorentz realization, since the core contains a timelike vector and its orthogonal complement is positive definite. Every further supporting white must annihilate this radical.

Every position in the inherited complete allowed-contact set has a nonzero radical coefficient. A nonzero nonnegative contact vector annihilating the radical must meet both signs. Each opposite-sign pair gives a white-interior path of length at most8 between adjacent black sides; the producer checks every such pair. Therefore either the independent current short8 route pays or every additional white is orthogonal to the entire core. An indirect spectator chain cannot evade this: its first root meeting the core would have a forbidden contact.

In the latter branch the displayed factors are whole actual factors. The selected D14 half has a distinct retained half as follows:

Contexts	Retained alternative
554,1543	Whole D15 pair(1,12)
616,1240	Whole D16 pair(2,13)
1276,1342	Whole A4 second node versus whole D13 fork center

The first half is part of the completed D15 theorem. The second remains in the current D16 weights, checked directly against the bound eighth-balance table. The third is the existing broad whole-small/D13 fork-center half; that rule allows every mark in A4. It does not require the A4 mark to be an endpoint. The later A2/D13 short-leaf upgrade does not change this center payment. Each selected half plus its distinct alternative sums to1.

Nonsingular cases: the extra white must be handled, not discarded

Contexts1284 and1623 have17-root signature(16,1,0). Write J for the inverse physical norm-two Gram matrix. The inherited contact theorem splits all allowed contacts into two ends. Meeting both ends already gives a short8 path. Each same-end inverse block is entrywise positive, so a unit contact at i contributes at least J[i,i] to the projection square of a new norm-two white. Additional same-end contacts increase it.

After excluding contacts to the stipulated whole selected D14, every remaining inverse diagonal is at least2. Precisely positions3 and13 have diagonal2; the other remaining diagonal is6. Thus a nonzero extra white can only have one simple contact at3 or at13. Its projection already has square2, so its orthogonal residual is zero. The vector is uniquely determined, not a free extra normal.

Two whites with the same contact position would coincide and have product2, contradicting distinct acute walls. Whites at the two different positions have product $J[3,13]=-2$. Their two-node Gram is semidefinite, contradicting the positive-definite supporting-white Gram. Consequently there is at most one extra white touching the core. Every core-orthogonal white is also orthogonal to this extra white, because the latter lies entirely in the core span. This excludes indirect extensions as well.

The producer reconstructs all three possibilities in each context: no extra white, one at3, or one at13. It checks white and edge positivity and all four actual finite corners in each branch. In1284 the no-extra and position3 branches give the distinct D15(1,12) half; position13 gives D16(2,13). In1623 the no-extra and position13 branches give D15(1,12), while position3 gives D16(2,13). All these factors are whole by the argument above. Together with the selected D14 half they always pay1.

Every proper completion is excluded

For contexts624,1158,1555,1618, each affected induced map requires a genuinely new white with a specified simple contact. The exact inverse diagonal at that forced contact exceeds2; the smallest is7/2. Contacting the opposite end supplies the independent short8 route, and additional same-end contacts only increase the projection. Thus no such actual completion exists in the no-short8 branch. Unspecified contacts are not silently set to zero.

Context1641 displays a12-node path. A changed D14 completion requires both missing short leaves12/13 attached to its existing fork center11. Each new leaf has forced unit contact at an inverse diagonal exactly2. Same-end positivity forces every other core contact to vanish, and the residual is zero. The two leaves then have identical projections and are the same vector, with product2 rather than the required0. They cannot be distinct actual supporting walls. Every affected map is checked, including path reversal.

Simultaneous interfaces and universal coverage

The rank14 inventory reconstructs the216 current literal configurations against the sparse-D16/ideal17 baseline. This producer applies all five decreases together. Exactly eight D14 corners lose half; all216 full totals remain at least1. No changed B pair occurs in these configurations. It also checks positive same-component rank14 outputs in the inherited recent finite-transfer interfaces against the new pair sets. These saved checks supplement the complete actual-face proof, rather than standing in for it.

The complete high-overlap domain has2,614 contexts:1,378 have component bound at most13;83 have protected/adaptive outputs;553 display more than14 nodes; and600 are the selected domain above. The first and third groups cannot use a changed whole rank14 pair. The protected/adaptive outputs retain their short paths, finite whole-leaf9 roles, basic ideal roles and typed coupons. The low-core bound is12, and the inherited high-disjoint exclusion is unchanged. Thus the absence of the four B pairs and the13 complete D proofs discharge every affected selected obligation; no unsupported inference from selected absence alone is used.

Apply the existing independent changed-ideal-corner priorities first. Their universal tree/cycle and later factor routes use retained short paths, protected distance9 roles, whole small components, or the current typed coupons. The long finite15–17 transfers have their own checked replacement interfaces; none spends the newly decreased long rank14 pairs. In a short8 escape, a deep finite8 zero is replaced by the completed short7 route, and a deep finite7 zero by the completed depth-six short6 route. Their protected alternatives have small whole components or retained features and do not invoke this new theorem. This establishes the escape independently of the conclusion being proved.

Triangles use unchanged short7/ideal5 and protected-feature interfaces, so their payment2 is preserved. If no actual corner has a newly changed weight, the old whole-face theorem applies unchanged. Otherwise use the complete routing and replacements above. Hence the simultaneous quadrilateral payment remains1.

Rejected A14 shortcut and limits of the saving

An initial proposal to delete A14 pairs(0,12)/(1,13) failed an exact marked-role assertion before producing a certificate. Contexts1157 and 1617 have a distinct whole A2/A13 corner, but its A13 mark is at the second node (canonical1/11), whereas the existing unit coupon applies at the endpoint (0/12). The producer records both exact mismatches. The proposed free transfer is withdrawn; A14 and all coupons remain unchanged. This is a mathematical scope correction, not a relaxed check.

The three actual rank14 costs therefore become91,87,90.5. A14 still requires representative upper91. No forest optimization is run, and the completed59 proof remains valid because all changes decrease costs while preserving face payments. These local improvements alone do not close the ambient59 comparison or produce a new polytope.

Verification

Final generation session78607, terminal chunk71fa94, completed exit0 with PASS_FINITE_RANK14_EXISTING_PAYMENT_TRANSFER. Separate fresh replay session23142, terminal chunk3970d7, also completed exit0/PASS. Two internal peers completed full producer/report mathematical reads. Both reread the inherited contact helper; the arithmetic peer additionally read the corner, finite-component, marked-map and literal helpers, the D15 proof and the current sparse/depth-six interfaces. They checked all arbitrary-support saturation cases, distinct retained halves, proper exclusions, the complete selected/nonselected partition, current literal and short-path priorities, and the rejected A14 role identification. No correction was identified. Neither reviewer executed map, inertia, literal or census calculations. These are internal agent reviews, not external refereeing or formalization. Source, data and report are final after these checks. Earlier failed preflight assertions did not produce data. The source checks exact native maps, physical Grams, radicals, projection identities, marked roles, saturated extensions and simultaneous literal arithmetic. It inherits the historical all-label catalogue and actual-face routing; it does not rerun that geometric census or formalize the proof.

```
python3 scripts/fresh_finite_rank14_existing_payment_transfer_20261018.py --verify
```

Source 120

Source path: `results/fresh_finite_rank14_long_pair_support_scout_20261018.md`

SHA256: 88e635946247b175d2c2fe0be15d149d3ee6e96874522372216976197d24af9c

Complete selected and literal rank14 inventory

This diagnostic enumerates the finite A14, B14 and D14 long-pair support against the current sparse-D16/ideal17 rule. It proves no new payment rule or dimension bound. Each of these finite components currently costs91; the later depth-six distance-seven deletion does not affect rank14.

The producer and data bind the complete inherited high-overlap catalogue and reconstruct every eligible native induced map, including failed searches. There are600 eligible selected patterns per type: the stated component bound is at least14 and the displayed connected marked component has at most14 nodes. The numbers of labelled maps are552,260 and1,104 for A, B and D respectively.

All long-pair automorphism orbits are recorded: six for A, ten for B and six for D. Every affected selected context and map is retained, including proper displayed components that would require additional actual walls. B14 pairs(0,13),(1,13),(2,13),(3,13), with double bond12–13, have no selected support. Every A14 and D14 long pair has selected support. Absence from this selected domain alone does not prove that a weight can be removed from a universal face-payment rule.

The producer also reconstructs all216 literal configurations against the current baseline, with its current per-corner lower payments. Each orbit is separately tested at half and zero weight; these trials are not combined. All half trials retain total payment at least1. The D14 zero trials at (0,11) and at the orbit{(0,12),(0,13)} each have eight violating literal records, whose remaining payment is1/2. All A and B trials have zero literal debit. Passing these finite tests is only a necessary interface check; it does not replace arbitrary-support geometry or later-route review.

The next useful A14 case is the orbit{(0,12),(1,13)}. Its complete selected obligations are whole contexts1157/1617 and proper context1640. The longest pair(0,13) instead reaches context1639. These are different obligations; no transfer is inferred here for either orbit.

Generation session19371, terminal chunk972232, completed with exit0 and PASS_FINITE_RANK14_SELECTED_LITERAL_SUPPORT_SCOUT. Separate fresh replay session50311, terminal chunke03226, also completed exit0/PASS. An internal peer completed a full source/report scope read and a bounded saved-summary comparison; no issue was identified. That reviewer did not repeat the native maps, literal arithmetic or historical geometric census, and did not reread the inherited helpers. The source performs exact map, orbit and literal arithmetic, using inherited geometry and classification helpers. It does not repeat the original all-label geometric census, optimize a finite forest or establish a new polytope.

```
python3 scripts/fresh_finite_rank14_long_pair_support_scout_20261018.py --verify
```

Source 121

Source path: `results/fresh_finite_rank15_existing_half_transfer_20261018.md`

SHA256: 4fd9e9fe75061bed5c437da345322ebdf3f1dccb5417eb35e52a075f18e4a868

Finite A15, B15 and D15 each cost104

Each whole finite rank15 family can lose one unit of same-component payment while keeping every triangle paid by at least2 and every quadrilateral by at least1. The changes use existing payments only. They compose with the frozen rank16/A17 exchange and current order14 ideal closure. No new finite coupon or dimension bound is asserted here.

The producer and certificate bind the complete rank15 support inventory, the current D17 cost120 theorem, the rank16 exchange, and the current ideal closure. The inherited complete all-label classification and contact domains remain inputs; the new arbitrary-support consequences are proved below.

Intrinsic changed pairs

Use the path0 through14 for A15 and B15, with B15's doubled bond13–14. For D15 use spine0 through12 and short leaves13,14 at12.

Whole component	Changed pairs	New weight	Saving
A15	(0,13),(1,14)	1/2	1
B15	(0,14)	0	1
D15	(0,13),(0,14)	1/2	1

All other pair weights remain unchanged. Each changed set is invariant under every labelled diagram automorphism, and every changed distance is at least10. All finite short8 and protected whole-leaf9 units are retained. Each new total is104, down from105. Finite C15 has the same Coxeter graph as B15; exceptional irreducible finite types have rank at most8.

The new A15 halves will use the existing whole-A2/whole-A13 endpoint coupon of weight1/2. The new D15 halves will use the already retained whole-D16 pairs(1,14),(1,15), each of weight1/2. These are distinct actual corners, so the two halves give a full unit. No original unit is counted after its weight has been reduced.

Complete selected support and the two A15 cases

The frozen scout reconstructs all536 eligible selected patterns for each finite rank15 target, including failed embeddings. There are482 A15, 240 B15 and870 D15 induced maps. The new producer validates every saved map against the native labelled Gram, every marked image, and the complete affected lists. It consumes the frozen exhaustive enumeration rather than repeating that search.

The A15 orbit occurs only in contexts1156 and1616. In both, the displayed marked component already has all15 nodes. The full19-root core has physical diagonal2, signature(17,1,1), and the saved radical is checked exactly. The white Gram and each actual edge Gram are strictly positive definite; every displayed corner is finite.

Because the core contains a timelike vector, its radical represents the zero vector in the actual Lorentz space. Every additional supporting root annihilates this vector. Every possible old-core contact position has a nonzero radical coefficient. The checker verifies that any two allowed positions of opposite sign give a marked path of length at most8 through the extra white. Therefore, in the branch without a full-star short8 alternative, every additional supporting white is orthogonal to the entire core. This permits arbitrary native magnitudes and arbitrarily many extra roots. An indirect bridge would require a first new root meeting the core, which is likewise excluded.

The resulting whole finite components at a distinct corner are:

Context	Selected A15 marks	Distinct coupon corner	Exact marked roles
1156	(0,11)	(2,11)	whole A2 endpoint / whole A13 endpoint
1616	(1,10)	(1,12)	whole A13 endpoint / whole A2 endpoint

The source reconstructs every finite component and its canonical images: A2 marks map to0,1 and A13 marks to0,12. These are exactly the existing half-coupon roles in the frozen sparse finite rule. Their whole-component status follows from the preceding sealing, not merely the displayed positive-definite Gram. The selected new A15 half plus this distinct half is1. The proof does not use either alternative A16 corner, so it is independent of the simultaneous A16 payment changes.

Four D15 cases: whole-component projection seals all spectators

The D15 orbit occurs only in contexts1645,1665,1932,2325. Each displayed marked component is already the full15-node finite D15. Thus, when the actual selected whole component is D15, every extra supporting white is orthogonal to every node of that displayed component.

The full18-root core is nonsingular Lorentzian, with physical root norm2. The complete necessary contact domain and its two end sets are recomputed using the bound contact theorem. Each inverse end block is entrywise positive; a root contacting both ends gives an independently paid full-star short8 path. In the one-end branch the squared projection of a root with contact magnitudes q_i is q^T G^{-1}q. Its residual lies in the positive orthogonal complement, so this projection is at most its actual norm2.

After imposing orthogonality to the whole selected D15, the only remaining possible contact positions and inverse diagonals are:

Contexts	Remaining nodes	Inverse diagonals
1645,1665	2,3,13	14,4,4
1932,2325	3,12,13	4,14,4

Every nonzero native norm-two contact has magnitude at least1. Entrywise positivity makes its full one-end projection at least the corresponding diagonal, which is strictly greater than2. Extra contacts cannot lower this bound. Thus no extra white contacts the core in the no-short8 branch. This is an all-contact projection exclusion, not a completion obtained by assigning unknown contacts zero. It also excludes a later indirect tail or bridge. Consequently every component in every displayed corner is a whole actual component.

Each case has a distinct whole D16 corner:

Context	Selected D15 marks	Distinct D16 marks
1645	(1,11)	(2,11)
1665	(1,12)	(2,12)
1932	(1,11)	(1,12)
2325	(2,11)	(2,12)

All four D16 marked pairs have canonical images(1,14),(1,15). Their current weight is1/2. The rank16 exchange changes only D16 pairs(2,13),(2,14),(2,15), so this half is unchanged; the producer checks that disjointness and the actual bound weights. In every case, the new selected D15 half and the distinct retained D16 half give1. No new D16 unit is assumed, and no cross-component payment is added.

Unsupported B15 endpoint and simultaneous literal audit

No complete selected B15 map uses(0,14). This observation alone would not prove deletion. The source also reconstructs all216 current literal replacement configurations after D17=120, applies all three rank15 changes simultaneously, and applies the frozen rank16/A17 exchange. Every whole-rank15 corner to be debited has a recorded baseline unit.

Exactly16 D15 corners in16 inherited affine-D14 equality records lose1/2. Each record retains total at least1. No A15 or B15 literal corner is affected; the rank16/A17 exchange has no changed corner in these216 records. All payments are nonnegative, and every complete record still totals at least1. The saved domain includes both three-corner replacements and four-corner configurations; no uniform four-corner schema is assumed. This audit also retains all prior D17 and sparse finite debits.

Coverage outside the selected cases

The complete inherited reduction supplies the following exhaustive routes. They are not inferred from the six examples above.

- The low-core marked whole-component bound is12, so none of the changed whole-rank15 roles can occur. The all-label high-disjoint branch is geometrically excluded by its frozen theorem.
- Of the2614 high-overlap contexts,1691 have marked whole-component bound at most14. The83 protected/adaptive contexts use basic ideal payments, actual whole-leaf9, or the adaptive finite distance9 interface. These are all retained.
- The remaining840 bounded contexts have bound at least15. In304 their displayed connected marked component already has more than15 nodes, so it cannot lie in a whole rank15 component. The other536 are exactly the complete native-map domains used here. A15 and D15 affected maps use the proved transfers; B15 has none. All other selected images retain their prior weight.
- Universal finite8/ideal5 cycle and tree replacements give finite paths of length at most8, protected actual whole-leaf9, sealed small factors, or their separate typed coupons. The deeper finite8 zero route supplies its independently established short7 replacement. None needs these finite distance-at-least10 roles.
- The older affine-D13/D14 equality cases and later D11 spectator equalities are included in the216 literal records. Their arbitrary one-sided branches instead give whole D12/D13 or separate small/D coupons. The remaining earlier affine-B finite alternatives have order at most14. The order15 and order17 affine-D coupon transfers use whole A4/A5/E6 versus D11 roles and no finite D15 long unit. The rank16/A17 exchange uses restored A17 halves or separate small/D13 halves.

Apply the independently established current ideal-corner priorities before the selected no-short8 reduction: C11 zero, B12 zero, D13 zero, cycle11, BCD11, and order17 zero have their proved distinct short paths, center roles, finite leaf9, small whole components or coupons. In particular the newer BCD11 mixed-component alternatives are protected A9/A18 endpoint8, D18 leaf9 or E8 path6; their unmixed alternatives are typed small-component coupons. These outputs survive the rank15 changes.

The later B16 and B14/C14 exclusions use the independently available short8 route or exact impossible completions. Their changed affine pairs have distance at least10 and are not paying outputs of the current short route. The ideal closure's current-corner order therefore applies without calling the present finite theorem to justify its own short-path premise. In the remaining no-short8 branch, patch the complete selected finite domain with the six transfers, retain the unaffected selected weights, and use the simultaneous216 literal audit. This is the same full-domain reduction as the bound current finite theorem, with the newly affected rank15 obligations handled explicitly.

The current triangle theorem uses finite distance at most7 and unchanged ideal elementary interfaces. Hence it retains total2. The complete quadrilateral routes above retain total1. The B15 unsupported endpoint is therefore legitimately removed, and both half-orbit transfers hold with arbitrary supporting roots. The whole face theorem is simultaneous with the frozen rank16/A17 exchange. A finite-forest comparison and any ambient conclusion remain separate obligations.

Verification scope

Final generationdcb9e6 and separate fresh replay1182bb both terminated exit0/PASS. Root and a separate peer completed full producer/report mathematical and interface reads. Both checked arbitrary-support sealing, the existing whole A2/A13 coupon, the retained D16 half despite the rank16 exchange, simultaneous literal debits, and nonselected/current

ideal/short8/deepzero coverage. Root additionally read the final explicit 1691/83/304/536 routing guards; the peer read the complete frozen scout and reread the inherited positive-end contact helper. Neither reviewer repeated arithmetic, map search, inertia calculations or historical censuses. No mathematical gap was identified in this stated scope. Direct bindings, local links and Python syntax pass. Source, data and report are final/frozen. The producer validates the complete frozen536-pattern map domains, recomputes the six radical/Lorentz core proofs and exact finite marked roles, and reconstructs all216 simultaneous literal records. It checks the existing coupon and retained D16 half against their frozen rules. It does not repeat the original all-label core census, native-map search, arbitrary-support contact classification or any finite-forest optimization.

```
python3 scripts/fresh_finite_rank15_existing_half_transfer_20261018.py
python3 scripts/fresh_finite_rank15_existing_half_transfer_20261018.py --verify
```

Source 122

Source path: `results/fresh_finite_rank15_long_pair_support_scout_20261018.md`

SHA256: ae84e2b833c5ad42dbc2557bc35bee17b4bb93d8e0bbf93c865be6db94b8366a

Finite rank15 long-pair support inventory

All three current rank15 classical finite types A15, B15 and D15 still have cost105, with every same-component pair paid1. This diagnostic reconstructs their complete selected and literal long-pair obligations against the now completed finite D17=120 rule. It proves no new deletion, half transfer, finite maximum or ambient bound.

The source and certificate bind the finite D17 reduction and the current ideal closure. No global dynamic program is run. Proposed orbit debits below are tested one orbit at a time; separate passing trials are not asserted to compose.

Complete saved domains

For each finite target, reconstruct every labelled induced embedding from the536 high-overlap selected patterns with whole-component bound at least15 and displayed order at most15. Failed pattern searches remain in the saved domain. Physical Gram entries are converted to native labels using the diagonal-invariant squared contact ratio, rather than assuming every historical core is in the same root gauge.

Whole finite target	Eligible patterns	Induced maps	Invariant distance-at-least10 orbits
A15	536	482	9
B15	536	240	15
D15	536	870	10

Every long pair in A15 or D15 occurs in this selected domain. B15 has eight unsupported long singletons:

`(0,12),(0,13),(0,14),(1,13),(1,14),(2,14),(3,14),(4,14).`

Here A/B use spine0 through14, with B's doubled bond13–14. D uses spine0 through12 and leaves13/14 at12. All tested orbits are invariant under the full labelled automorphism group. All short8 and whole-leaf9 units are excluded from the candidate list.

The source also reconstructs all216 current literal configurations, including their three- and four-corner records. It begins with the finite D17=120 retained totals, so it does not accidentally spend a unit already removed by that theorem. Every finite shared rank15 component is identified by its entire actual displayed corner component and all native maps to the appropriate finite target. The source records every affected corner and tests both unit-to-half and unit-to-zero debits for each candidate orbit.

These literal tests are exact for the saved configurations. They do not seal a displayed component against arbitrary extra supporting roots. That geometric whole-component step remains necessary for a local theorem.

Narrow candidates for one unit of saving in every family

The following shortlist avoids every protected basic pair.

Type and proposed change	Selected obligations	Literal trial under current D17=120
A15:(0,13),(1,14), unit to half	Full rank15 radical contexts1156,1616	No affected literal records; minimum remains1
B15:one of the eight unsupported long singletons, unit to zero	No selected map	No affected literal records; minimum remains1
D15:(0,13),(0,14), unit to half	Full rank15 Lorentz contexts1645,1665,1932,2325	16 affected records; every new total remains at least1

Each proposed row would save1 in its own type. None is proved by this inventory. The A cases need an arbitrary-support radical sealing proof and a distinct retained half at another corner. The D cases need control of all further roots orthogonal to the stipulated whole finite D15, then a legitimate alternate whole-component payment. For B, absence in the selected and literal lists must still be combined with the complete nonselected, universal and later interfaces; absence alone is insufficient.

Some stronger deletions encounter explicit saved obligations. Deleting the D15 center pair(0,12) entirely would lower eight old D14 equality records to1/2; halving it preserves their total1 but saves only1/2 in the factor. Deleting the D15 endpoint/short-leaf orbit(0,13),(0,14) entirely would lower16 such records to1/2, whereas halving that orbit preserves1. These are concrete obstructions to a direct debit-only deletion argument, not nonexistence statements about alternative transfers.

Other small-support A candidates are the endpoint pair(0,14), supported only by full-rank radical context1638, and pair(1,13), supported only by full-rank radical context1142. Each singleton has no affected current literal record. A zero transfer would require an independent full unit; a half transfer would save only1/2. The certificate retains the complete support for these and all other long orbits, rather than choosing one solely by its small context count.

Remaining geometric and global obligations

The inherited low-core bound, high-disjoint exclusion, protected/adaptive branches and universal tree/cycle routes are not rerun by this scout. A proposed theorem must explicitly verify that all their possible rank15 outputs survive. Likewise all later ideal and finite transfers must be composed under the current actual-corner priority and complete short8 route. Passing216 literal records does not replace that written domain coverage.

The current ideal closure changes no finite payment. The rank15 A/B/D type ties therefore matter in any prospective forest improvement; reducing only the representative D15 cost cannot silently lower actual A15/B15. The separate finite-budget lane reports its actual maximizing forests. This packet does not bind an unfinished budget, optimize a new one, or claim that a rank15-only saving resolves every remaining finite obstruction.

Verification scope

Final generation63d147 and separate fresh replayd8cd33 both terminated exit0/PASS. Root completed a full source/report read at this bounded scope, including the complete536 native-map domains, all216 literal records after D17=120, the baseline-unit guard, invariant long orbits and separate single-orbit debit trials. That review did not rerun maps, geometry or arithmetic. Source, data and report are final/frozen. The producer validates frozen bindings, reconstructs the three536-pattern domains and all216 literal interfaces, identifies every invariant long orbit, and performs the exact hypothetical debits. It imports the inherited native-map and literal-domain helpers. It does not repeat the original all-label geometric census, infer whole-component sealing from sampled Grams, or promote an unsupported-pair list into a payment rule.

```
python3 scripts/fresh_finite_rank15_long_pair_support_scout_20261018.py
python3 scripts/fresh_finite_rank15_long_pair_support_scout_20261018.py --verify
```

Source 123

Source path: `results/fresh_finite_rank15_rank16_rank62_budget_20261018.md`

SHA256: 60909ae838465dc15043050dd8845d3137628b894826787b220e9ff78995a4ed

Rank15/rank16 exchange gives exact finite rank62 maximum434

The complete finite payment rule after the two frozen local exchanges has exact rank62 maximum **434**, attained by D17+D45. The finite curvature is434, so the comparison has slack0. Both geometric exchange packets are final and source-bound. No ambient dimension61 conclusion is asserted by this finite counting packet alone.

The source and certificate use the frozen complete438 budget and its saved exact residual certificates. The new local inputs are the rank15 existing-half transfer and the rank16 payment exchange. The latter includes a necessary A17 restoration; omitting it would misstate the simultaneous rule.

Actual rule and a conservative rank16 representative

The proved changes, all in whole finite components, are:

Component	Old cost	New actual cost	Representative upper cost
A15, B15, D15	105	104	104
A16	114	113	113
D16	114	225/2	113
A17	114	115	115

B16 remains104. D17 remains120 under its frozen five-unit reduction. Every other same-component cost and every current cross-component coupon is unchanged, including all sparse coupons, both D13 zero-transfer increments, and actual E6/D11 total4. Both local inputs explicitly state that they add no cross-component coupon.

The D16 representative deliberately uses the upper allowance113 although its actual cost is225/2. This is needed to dominate A16, which has actual cost113. A17 remains a separate large type, with its increased cost115. Neither its same-component restoration nor its existing small-component cross coupons are folded into D17.

The source reconstructs the current doubled base and cross-cost formulas using only the standard library. For every finite type a through rank62 and every pair a,b, it verifies

```
upper_base(R(a)) >= actual_base(a),
cross(R(a),R(b)) >= cross(a,b).
```

There are189 type representatives and35,721 ordered pair comparisons. The replacement is simultaneous at both ends. It retains B9 and E6 explicitly and does not assume that changing one endpoint alone preserves all sparse coupons. Arbitrary dihedral labels use the inherited uniform rank2 representative; the finite classification and original coupon theorems are inherited, not independently reclassified here.

The rank15 transfer uses the old D16 pairs(1,14),(1,15) at half. The source checks their weights directly in the frozen sparse rule and checks that neither belongs to the rank16 exchange's changed orbit (2,13),(2,14),(2,15). Thus the rank15 witness does not spend a D16 half decreased by the rank16 change. Full geometric compatibility remains the content of the two local theorems and the eventual global composition.

Why the residual tables do not change

The complete fixed large list is B9,D10,…,D16,A17. Its multiplicity vectors have total rank at most62. The source independently enumerates all1,227 vectors and matches them bijectively to the frozen alternatives. They have921 distinct interaction keys. Residual components have ranks 1–8 or at least17, with D17 among the residual types and A17 only in the fixed large list.

All changed component bases occur in the fixed list: ranks15 and16, and the separately retained A17. Cross costs do not change. Consequently each residual item has exactly the same base and exactly the same interaction with a given fixed vector as before. The source verifies this directly for all66,258 literal item/vector comparisons and checks each result against the seven-coordinate key formula. No residual rank15 or rank16 component is hidden in this assertion.

The frozen438 packet already proved every residual table's upper bound and attaining-predecessor certificate. Since the allowed item domain, all item costs and each residual rank are unchanged, that same theorem applies without recomputing a table. The new certificate binds the entire predecessor packet and records a digest of its exact saved table list. It follows saved predecessor chains only to reconstruct the formal attaining forests; it does not rerun a Bellman optimizer or repeat the earlier1,426,629 Bellman inequalities.

For fixed counts c15,c16,a17, the doubled fixed cost changes by exactly

```
−2∗c15 −2∗c16 +2∗a17.
```

Here c16 uses the conservative representative allowance113. The source checks this identity literally, adds it to each of the1,227 complete old alternatives, and recounts the resulting typed forest with both the new actual and upper costs. The new actual cost never exceeds the shifted upper cost.

Maximum and actual attainment

The largest shifted upper value is868 in doubled units, hence434. Three saved representative alternatives reach that upper value:

Representative forest	Upper cost	Actual cost of those named types
D17+D45	434	434
D16+D46	434	867/2
2D15+2D16	434	433

The last two illustrate why a conservative representative value need not be an actual cost for the named D16 forest. They are not used as actual attainment witnesses. The first forest is sufficient:

```
cost(D17+D45) =120+314 =434.
```

There is no cross coupon between those two components. The D45 value is the inherited large-D formula8*45−33 minus its13 deep11 finite8 zeros. Both components are untouched by the new rank15/rank16 exchange, so this is an actual-rule witness, not a conservative allowance. Formal forest attainment does not claim a polytope realization.

Together with the complete upper certificate and frozen local rules, this proves the exact finite counting maximum434. The finite curvature in ambient62 is also434. Thus the finite comparison has zero slack, in contrast to the preceding438 rule's deficit4. The separate ideal comparison and complete global face-payment composition are not conclusions of this arithmetic packet.

Validation and scope

Final source-bound generation (terminal chunkc2ed7d) and separate fresh verification (terminal chunk746c65) both exited0/PASS, reporting434. The final source checks the exact frozen source/data/report hash triples for both local exchanges, their actual cost records, their absence of new coupons and their simultaneous retained-D16-half interface. No conditional local input remains in this certificate.

The root reviewer and two separate peers completed full source/report mathematical reads. They checked current coupon totals, simultaneous type domination with the conservative D16 upper, the A17 restoration, the complete fixed-vector/residual split, unchanged residual costs, the exact shifts, actual attainment and the precise inherited Bellman scope. None independently reran the1,227-vector arithmetic, residual table proof or geometric census. Final hash additions introduced no arithmetic change after those reads.

This is an independent standard-library recount of the new component formulas, simultaneous domination and fixed-vector arithmetic. It inherits the frozen residual Bellman certificates and geometric classification; it is not a new geometric census, a new global model optimization, or an independent replay of the old table proof.

Source 124

Source path: `results/fresh_finite_volume_simplex_truncation_20261017.md`

SHA256: ccb06da5e00d3b8075e96d743dda73836b9cd0a00a789848fdc72693c59b9727

Compact simplex ports and exact cusp truncation in finite volume

The disjoint-port theorem extends to **compact simplex faces through finite vertices** of a finite-volume acute Coxeter polytope, even when its ideal vertices are nonsimple. It does not extend to triangles containing ideal vertices. The exact square pyramid below demonstrates the distinction.

Truncating the ideal vertices produces a simple projective polytope and gives explicit face-count identities. This is not, by itself, a route to a better dimension bound: the artificial cusp faces already exceed the usual simple polytope corner-payment budget. A more useful starting point is Khovanskii's inequality on the original polytope, with the cusp contributions retained. It leaves a precise missing noncompact-face incidence inequality.

The standalone Fraction checker and exact certificate verify the identities against 914 product links, all-rank polynomial coefficients, and the literal pyramid. The proofs for arbitrary ranks are given here; bounded checks are not used as extrapolation. No new dimension bound, external review, or novelty claim is made.

Geometric scope and primary inputs

Let P be a finite-sided, finite-volume acute Coxeter polytope in hyperbolic n-space, n >= 3, and use its closed projective polytope in the Klein model. Write v_f for its number of finite vertices. Finite vertices are simple. At an ideal vertex v the link is combinatorially

$$L_v = \prod_{i=1}^{s_v} \Delta^{r_{vi}}, \qquad r_{vi} \geq 1, \quad \sum_i r_{vi} = n - 1.$$

The projective polytope is simple at edges. These local facts and the later average-face formula are stated in Potyagailo–Vinberg, printed pp.64–65. The graph-connectivity input is Balinski's original theorem, printed p.432. Khovanskii's original Theorem 10, printed p.49, and §6 apply the average-face estimate to polytopes simple at edges, explicitly including finite-volume reflection chambers. All three primary sources were opened on 2026-10-17.

Transfer of the compact-simplex port theorem

At a finite vertex a, the incident edges are its ports. Simplicity at a gives a Boolean face interval: any d ports generate a unique d-face. Relate two ports only when their two-face is a **compact** triangle.

Suppose the triangles abc and abd share the edge ab. They lie in the three-face generated by their three ports at a. Their vertices a,b,c,d are finite. In that three-face a and b each have degree three, with neighbors exactly the other three displayed vertices. If there were another vertex, deleting c and d would disconnect {a,b} from it, contrary to Balinski's three-connectivity. The three-face is therefore a tetrahedron, and it is compact. This argument does not require any undisplayed vertex to be simple. Compact-triangle adjacency of ports is consequently transitive.

Let a class have d >= 2 ports, and let F be their generated d-face. The neighbors a1,…,ad of a are finite because they belong to compact triangles. Every pair ai,aj is an edge of F. Thus a,a1,…,ad form a complete graph, and each already has degree d

in F. All these vertices are finite and hence simple in F. There can be no edge to another vertex. Graph connectivity implies that F has exactly these d+1 vertices, so F is a compact simplex.

Conversely, a compact simplex through a gives a clique, hence lies in one class. It follows that distinct maximal compact simplex faces of dimension at least two through a use disjoint ports. The frozen compact disjoint-port proof therefore supplies the same one-port capacities at every finite vertex. An isolated port ending at an ideal vertex is not a compact simplex edge; it must not be silently included in this conclusion.

The theorem gives no port capacity at a nonsimple ideal vertex, and no partition of all triangles into compact simplex aggregates. Those are separate obligations in any finite-volume charge argument.

Exact counterexample to including noncompact triangles

In the Lorentz form diag(-1,1,1,1), take outward normals

$$b = (0,0,0,-1), \quad s_1 = (1,2,0,1),\ s_2 = (1,0,2,1),\ s_3 = (1,-2,0,1),\ s_4 = (1,0,-2,1).$$

Their squares are 1,4,4,4,4. Base/side products are -1, successive side products are zero, and opposite side products are -4. Thus the chamber has base/side dihedral angle pi/3, right successive sides, and parallel opposite sides. H=(2,0,0,1) has square -3 and product -1 with every normal, so it is a strict common future center.

In the affine chart t=1 the closed polytope is exactly

$$0 \le z \le 1, \qquad |x|, |y| \le (1-z)/2.$$

It has four finite base vertices (1,+/-1/2,+/-1/2,0) and the ideal apex nu=(1,0,0,1). Its cusp link is the bounded square Delta1 x Delta1, so this is a finite-volume Coxeter square pyramid. At the finite vertex (1,1/2,1/2,0), the two triangular side faces share the edge to nu, whereas the third two-face through its ports is the square base. Triangle adjacency is not transitive if these noncompact triangles are allowed.

The checker reconstructs all vertices, active sets, edges and faces from the inequalities, not from an assumed incidence table. Cutting by z <= 1/2 gives a combinatorial cube. Its new outward normal (1,0,0,2) has product +1 with every old side normal: even this elementary truncation is not acute. Accordingly no Coxeter-angle or acute local-face theorem is imported to the truncated polytope.

Exact truncation counts

Take sufficiently small disjoint projective vertex cuts at all ideal vertices, and call the result Q. Each new vertex belongs to a truncation facet and corresponds to a vertex of L_v. Since L_v is simple, this new vertex is simple. The old finite vertices are unchanged, so Q is a simple Euclidean/projective polytope.

For one cusp suppress v and put d=n-1, V=f0(L), E=f1(L). Let T and R be the numbers of triangular and square two-faces of L. Then

$$\begin{aligned} V &= \prod_i (r_i+1), & E &= \tfrac{1}{2}dV, \\ T &= \tfrac{1}{6}V\sum_i r_i(r_i-1), & R &= \tfrac{1}{4}V\sum_{i<j} r_i r_j, \\ f_2(L) &= T+R, & 3T+4R &= \binom{d}{2}V. \end{aligned} \tag{1}$$

Indeed a two-face of a product is either a triangular two-face in one simplex factor or an edge in each of two factors. The other factors supply vertices. This proves (1) for all ranks, including a two-dimensional link whose full polytope itself is counted as its one two-face.

Every old positive-dimensional face persists after truncation. The new positive-dimensional k-faces are precisely the k-faces of the links, while an old ideal vertex is replaced by V vertices. Therefore

$$\begin{aligned} f_0(Q) &= f_0(P) + \sum_v (V_v - 1) = v_f + \sum_v V_v, \\ f_1(Q) &= f_1(P) + \sum_v E_v, \\ f_2(Q) &= f_2(P) + \sum_v (T_v + R_v). \end{aligned} \tag{2}$$

Let N_{k,s} count old k-gonal two-faces having s ideal vertices. An ideal corner contributes one additional side after truncation, so

$$f_{2,\ell}(Q) = \sum_{k+s=\ell} N_{k,s} + \mathbf{1}_{\ell=3} \sum_v T_v + \mathbf{1}_{\ell=4} \sum_v R_v. \tag{3}$$

In particular triangles(Q)=N_{3,0}+sum T_v, while quadrilaterals(Q)=N_{4,0}+N_{3,1}+sum R_v. A noncompact triangle with two or three ideal corners becomes a pentagon or hexagon, respectively.

Counting old incidences also gives

$$2f_1(P) = nv_f + \sum_v V_v, \qquad S(P) := \sum_{F^2 \subset P} |F^2| = \binom{n}{2} v_f + \sum_v E_v. \tag{4}$$

Here E_v counts old two-faces through v, because they correspond to edges of its link. The polygon side count equals its vertex count, including ideal vertices. These formulas do not replace E_v by binom(n,2), which would incorrectly assume ideal vertices simple.

For D_j(P)=sum_F(j-|F|), (2)–(4) yield

$$D_4(Q) = D_4(P) + \sum_v (T_v - E_v), \qquad D_5(Q) = D_5(P) + \sum_v (2T_v + R_v - E_v). \tag{5}$$

The demand on the surviving old faces, without the new link faces, is

$$D_5^{\mathrm{old}}(Q) = 5f_2(P) - \binom{n}{2} v_f - 2 \sum_v E_v. \tag{6}$$

Each old ideal corner is counted twice here: once as its old vertex and once as the extra side created by truncation.

Why simple truncation does not close the charge argument

For n >= 4 write

```
\eta_n=\begin{cases}4(n-1)/(n-2)&n\text{  even},\\4n/(n-1)&n\text{  odd},\end{cases}  \quad b_n=\begin{cases}n(n-6)/8&n\text{ even},\\(n-1)(n-5)/8&n\text{ odd}.  \end{cases}
```

The strict simple-polytope inequality S(Q)/f2(Q)<eta_n gives D5(Q)>b_n f0(Q), since S(Q)=binom(n,2)f0(Q).

Consider the usual nonnegative corner-payment scheme in which every triangle receives at least 2 and every quadrilateral at least 1; this suffices to dominate 5-|F| on all two-faces. The artificial faces of one cusp alone require at least 2T+R, all supported on its V new vertices. Formula (1) gives the exact all-rank lower bound

$$\frac{2T+R}{V} = \frac{5\sum_i r_i^2 + 3d^2 - 8d}{24} \geq \frac{d(d-1)}{8}, \tag{7}$$

because sum r_i^2 >= sum r_i=d. Its excess over b_n is at least

$$\frac{(n-1)(n-2)}{8} - b_n = \begin{cases} (3n+2)/8 & n \text{ even}, \\ 3(n-1)/8 & n \text{ odd}. \end{cases} \tag{8}$$

It is strictly positive for every cusp partition. Thus a nonnegative corner-payment model on Q with total budget at most b_n at **every** new vertex is impossible already on the new link faces. This is an algebraic obstruction to this particular uniform-budget method, not an existence claim about high-dimensional reflection polytopes.

Subtracting those known demands instead leaves

$$D_5^{\mathrm{old}}(Q) > b_n v_f - \sum_v \bigl(2T_v + R_v - b_n V_v\bigr). \tag{9}$$

Every displayed cusp deficit is positive. Finite-vertex compact-simplex port savings alone provide no bound on their sum; finite-volume chambers can even have no finite vertices. A proof using truncation would therefore need an additional aggregate bound relating cusp mass to genuine old-face incidences. It cannot discard the new faces while retaining the original simple-polytope lower bound.

A usable original-polytope budget, and the missing theorem

Khovanskii's Theorem 10 avoids the artificial faces altogether. Using its estimate in the conservative weak form S(P)/f2(P) <= eta_n, (4) gives

$$D_5(P) \geq \alpha_n S(P) = b_n v_f + \alpha_n \sum_v E_v, \qquad \alpha_n = 5/\eta_n - 1. \tag{10}$$

No strictness for this nonsimple-at-vertices application is needed or claimed here. For a prospective nonnegative original-face payment rule, let B(a) bound the total payment at a finite vertex a, and C(v) the total payment over original two-face corners at the ideal vertex v. A necessary condition for all face lower bounds to hold is

$$\sum_{a\ mfinite} (B(a) - b_n) + \sum_{v\ mideal} (C(v) - \alpha_n E_v) \geq 0. \tag{11}$$

Consequently a strictly negative right-hand upper bound would give a dimension contradiction. This retains the exact cusp factors and is a concrete target for a future incidence theorem. It is not yet an inequality bounding C(v) from above.

Ambient n	Finite-vertex benchmark b_n	Cusp benchmark per old incident two-face alpha_n
23	99/2	9/46
24	54	9/46
25	60	1/5
26	65	1/5

The transferred simplex capacities apply only to compact triangles and their compact maximal simplex aggregates. A complete application of (11) still needs payments for noncompact triangles and quadrilaterals, their interaction with the affine product at each cusp, and simultaneous global incidence restrictions. Local affine classification or the individual port theorem alone supplies none of these missing inequalities.

There is a published model of the stronger kind of argument needed: Potyagailo–Vinberg's §§4–5 couple cusp counts across neighboring faces to obtain a lower bound on the number of facets of a five-dimensional right-angled polyhedron, then apply the higher-face average inequality. The right-angle assumptions are essential there; that result is not an unrestricted estimate. This package changes neither the repository's dimension163 proof package nor any arithmetic bound.

FINAL/FROZEN: generation and a fresh saved replay passed. The checker is one standalone exact implementation. The geometric transfer and truncation counts, and formulas (7)–(11), received a narrow internal conceptual read from exact_hub_counts. That read did not independently recheck Khovanskii's source hypotheses and is distinct from an independent arithmetic replay.

Source 125

Source path: `results/fresh_full_a39_corridor_high_domain_20261017.md`

SHA256: 98d016172938d90373991852a39347b821d158ac2b97f10a2354841a1ae67166

Depth17 suffices for the full high-domain corridor alternative

For every corner of the complete frozen 6,703-context high-overlap domain, the following statement holds. If the actual whole finite A/B/D component has its two marked roots at distance 8, and both marks have distance at least 17 from every endpoint, branch and endpoint of a nonsimple bond, then the full actual supporting star has an opposite-black path of length at most 7.

This is a **short-path alternative**. Only in the branch with no such path anywhere in the full supporting star does it exclude the deep corridor. The absence of a path in the saved old core alone is insufficient. There is no new pair weight, complete payment system or dimension bound here.

The source and exact certificate cover all 26,812 corners of the inherited high domain, including all four corners of the 251 formerly unresolved contexts and all earlier structural outputs. No old principal matrix is required to be nonsingular or Lorentzian. The proof instead uses a forced positive definite path in the proposed actual completion.

The forced path and every possible old embedding

Normalize all actual wall roots to squared norm 2. A feature of a finite A/B/D diagram is a vertex of degree other than 2, or either endpoint of a bond whose label exceeds 3. Depth at least 17 at each mark, together with marked distance 8, gives two outward simple arms of length 17 and the simple path joining the marks. All vertices other than the two marks are supporting whites. The two omitted black roots do not belong to this actual corner.

Keep the first 15 outward whites on each arm. Together with the marks and the seven whites between them, these form an induced A39 diagram. Number its vertices 0 through 38; the left and right marks are at 15 and 23. The unused two outward whites on each side are retained as geometric inputs for the A17 contact estimates below, but are not needed in the final matrix. A feature at outward distance exactly 17 is harmless: all bonds inside the displayed A17 arm are simple.

The 342 old native cores are reconstructed from their frozen diagrams and positive diagonal gauges. They have order at most 19, so any displayed corner has at most 17 roots. Their gauges are 1 or 2; dividing by the square roots of the gauges puts every relevant product in Q(sqrt(2)). The source checks all four left/right choices in every old context and independently reconstructs the displayed component partitions.

A displayed component containing a mark must be a simple path. An existing branch or nonsimple bond remains a feature in the actual component and is at distance at most $16 < 17$ from the mark. An existing cycle cannot occur inside a finite A/B/D tree. The checker saves one such witness for each rejected nonsimple marked block.

If the two marks are in the same displayed path, that path must already give distance 8: a tree cannot acquire a second, shorter route. Its embedding in A39 is then fixed once the two marks are placed at 15 and 23. If the marks lie in different displayed paths, orient each path from its outward endpoint toward the eventual joining segment. Let t_L,t_R be the inward distances from the marks to the chosen endpoints. The connecting segment must contain exactly

```
k = 7 - t_L - t_R
```

additional whites, with k at least 1. The two displayed components are orthogonal, so no zero-white connection is possible. Every orientation of both paths is enumerated, with singleton orientations counted once. A connected displayed path has at most 17 roots and already spans 8 edges between the marks, so its two outward tails together use at most 8 roots. For

disconnected marked paths, their combined order is at most 17 and each is nonempty; either path has at most 16 roots and outward extent at most 15. Thus every old embedding fits inside the 15 retained outward whites on each side. The source asserts these distinct bounds before checking every old pair product against the corresponding A39 entries. This also covers paths joined through old spectator roots: those roots may occupy positions not selected in the reduced old embedding. Their identities and extra contacts are not assumed away. The Gram tests use a necessary subset of actual roots only.

All-label restrictions leave three white slots per omitted black

Write b for the omitted left black root and d for the omitted right black root. The full-star no-short7 hypothesis implies b·d=0 and forbids any contact from b to the right mark, or from d to the left mark. Their products with their own marks are fixed native products and remain in the calculation.

For a norm-2 A17 arm, the inverse is positive and its diagonal at outward distance j is j(18-j)/18. The omitted black and all these whites belong to an actual edge diagram, which is strictly positive definite. Hence:

- At outward distance 1, only nonzero contacts of labels 3 or 4 are possible. A label at least 5 has squared magnitude greater than 5/2, and (17/18)(5/2)>2.
- At distance 2, only label 3 is possible, since (16/9)·2>2.
- At distances 3 through 15, even a unit contact gives projection greater than 2, so no contact is possible.

Additional nonpositive products cannot lower these projection bounds, because the inverse is positive. Strictness excludes equality as well. The retained path stops at outward distance 15. Contacts to the unused outer whites at distances 16 and 17 are not claimed to vanish: those whites supply the A17 inverse bound and are omitted from the final principal matrix. These statements have no finite label cutoff.

In the inner A7 segment, b contacting a white at distance j at least 2 from the left mark gives a route to the right mark of length 9-j, at most 7. Only the first inner white can therefore meet b in the no-short7 branch. Its A7 inverse diagonal is 7/8, which excludes label 5 and every larger label by the same squared-magnitude comparison. Contacts to the first two outward whites beyond the opposite mark also give short paths; more distant contacts are already excluded by the other A17 arm. The argument for d is the reflection of this argument.

Thus b can contact, apart from its own mark at 15, only positions 14,13,16, with allowed nonzero labels respectively (3 or 4), 3, (3 or 4). Root d has the reflected slots 24,25,22. The source first checks every fixed old contact. A forbidden one supplies either a recorded short path, an A17 projection obstruction, or a strict edge label obstruction. Four orientations are rejected by the last reason; the remaining fixed-contact exclusions use the former two reasons.

One omitted black cannot contact both consecutive outer whites. The positive A2 inverse is (1/3)[[2,1],[1,2]], and two unit contacts already have projection square 2, contrary to strict edge positivity. The remaining slot products are enumerated exactly as 0,1,sqrt(2), respecting every old fixed contact. They are an overestimate of realizable completions, which suffices for an obstruction.

Two necessary Gram tests

For a norm-2 simple path of order n, the inverse entries, using indices from 0, are

```
J_n[i,j] = (min(i,j)+1)(n−max(i,j))/(n+1).
```

The source verifies this inverse formula by multiplication for n=2,7,15,17,39. Deleting the two marked roots from A39 gives three independent white blocks A15+A7+A15. Let J_W be their inverse. For the positive contact vectors x,y of b,d with the A39 path, strict positive definiteness of each actual edge requires

```
2 − x_W^T J_W x_W > 0,     2 − y_W^T J_W y_W > 0.
```

Every profile violating either strict inequality is impossible. Otherwise eliminate the full positive definite A39 block. Because b·d=0, the remaining two-by-two Schur matrix is

```
[ 2 − x^T J_39 x       −x^T J_39 y     ]
[   −x^T J_39 y      2 − y^T J_39 y    ].
```

In every surviving enumerated profile its first diagonal is strictly negative and its determinant strictly positive. It therefore has two negative directions. By congruence the full Gram matrix would have two negative directions, impossible for actual vectors in a real Lorentz space. This argument requires no assumption on the old core's signature. Adding further actual roots or spectator contacts cannot repair a principal Gram matrix with two negative directions.

All signs and products use the exact Q(sqrt(2)) helper. No floating-point tolerance or optimizer is used.

Complete counts and scope

Corner outcome	Corners
Nonsimple marked block incompatible with depth 17	16,627
Connected marks at a fixed distance other than 8	2,187
No admissible distance-8 joining orientation	14
Complete corridor/contact calculation	7,984
Total	26,812

The last row of cases has 13,644 complete path orientations. Fixed contacts exclude 8,337 orientations. The remaining 5,307 use 379 distinct contact templates, with every template's multiplicity saved. There are 4,983 distinct joint contact profiles. With orientation multiplicities restored, 57,582 profiles fail strict actual-edge positivity and 50,137 give two negative directions. None remains unresolved within this deep-corridor hypothesis.

This settles the proposed deep-pair alternative throughout the **high-overlap domain**, including the formerly unresolved 251 contexts. It does not settle those contexts for arbitrary pair shapes, and does not complete the finite-7 payment theorem. Low/disjoint cases, the actual retained pair maps and all affected universal payment interfaces must still be checked before changing a weight or importing a finite budget. No global conclusion changes. No new finite-volume polytope is constructed.

Conditional rank65 graph counts

A separate literal BFS count in whole finite A65/B65/D65 diagrams gives 23/22/21 pairs at distance 8 whose two endpoints both have feature distance at least 17. The source saves all eligible unordered pairs; the independent helper reconstructs the graphs and all distances without producer imports. If an additional complete payment argument later authorized halving exactly those pairs, the inherited same-component costs 486/486/487 would become 949/2,475,953/2, respectively. These are hypothetical arithmetic values, not an authorized finite payment system or a finite-budget theorem. In particular, this local high-domain alternative alone does not justify those changes.

Verification and inherited scope

The uncompressed root pilot covered all 26,812 corners and found the same orientation-weighted outcomes. The authoritative producer independently reconstructs the old native matrices and all corner/path embeddings, compresses identical fixed-slot domains, and saves every exact profile. Authoritative generation 93617 and separate fresh saved replay 11768 both completed terminal exit0/PASS. No floating-point tolerance, local optimizer or global solver is used in this packet.

The separate saved-template checker imports no project modules. It computes generic rational Gauss–Jordan inverses for A39 and its 37 white roots, checks the A17/A7 all-label inequalities directly, and decides Q(sqrt2) signs by rational isolating intervals rather than the producer's squared-norm sign test. It reconstructs every saved 379-slot domain and 4,983 joint profile, reconciles all 13,644 orientation multiplicities, and checks the optional rank65 graph-count table. Independent saved checker 44786 completed terminal exit0/PASS; all 12 direct source bindings were checked. It does not independently reconstruct old native cores, fixed-contact exclusion witnesses, old-block embeddings, all-label geometry or the inherited overlap census.

The frozen A43 predecessor remains unchanged and is directly bound. The inherited all-label high-overlap census is a frozen proof input. This successor reconstructs its native matrices, corner partitions and complete new corridor/contact domain, but does not repeat the original overlap census. Root and construction peer each completed a full producer, report and independent-helper mathematical/source read. Both checked depth 17 feature persistence, the connected/disconnected

embedding bounds, unused outer whites, all-label slots, physical gauges, strict edge positivity, the index-two contradiction, independent arithmetic implementation and conditional rank65 counts. Neither review duplicated the 4,983 profile arithmetic or the inherited native/embedding/high-overlap census.

```
python3 scripts/fresh_full_a39_corridor_high_domain_20261017.py
python3 scripts/fresh_full_a39_corridor_high_domain_20261017.py --verify
python3 scripts/check_full_a39_corridor_saved_20261017.py
```

Source 126

Source path: `results/fresh_full_a43_corridor_high_domain_20261017.md`

SHA256: fc6489456e9d959798607ffc702ef4c52a48aca856ef085760d41a4b53fa5074

A full corridor settles the deep-pair question throughout the high domain

For every corner of the complete frozen 6,703-context high-overlap domain, the following statement holds. If the actual whole finite A/B/D component has its two marked roots at distance 8, and both marks have distance at least 20 from every endpoint, branch and endpoint of a nonsimple bond, then the full actual supporting star has an opposite-black path of length at most 7.

This is a **short-path alternative**. Only in the branch with no such path anywhere in the full supporting star does it exclude the deep corridor. The absence of a path in the saved old core alone is insufficient. There is no new pair weight, complete payment system or dimension bound here.

The source and exact certificate cover all 26,812 corners of the inherited high domain, including all four corners of the 251 formerly unresolved contexts and all earlier structural outputs. No old principal matrix is required to be nonsingular or Lorentzian. The proof instead uses a forced positive definite path in the proposed actual completion.

The forced path and every possible old embedding

Normalize all actual wall roots to squared norm 2. A feature of a finite A/B/D diagram is a vertex of degree other than 2, or either endpoint of a bond whose label exceeds 3. Depth at least 20 at each mark, together with marked distance 8, gives two outward simple arms of length 20 and the simple path joining the marks. All vertices other than the two marks are supporting whites. The two omitted black roots do not belong to this actual corner.

Keep the first 17 outward whites on each arm. Together with the marks and the seven whites between them, these form an induced A43 diagram. Number its vertices 0 through 42; the left and right marks are at 17 and 25. The unused three outward whites on each side are retained as geometric inputs for the A20 contact estimates below, but are not needed in the final matrix. A feature at outward distance exactly 20 is harmless: all bonds inside the displayed A20 arm are simple.

The 342 old native cores are reconstructed from their frozen diagrams and positive diagonal gauges. They have order at most 19, so any displayed corner has at most 17 roots. Their gauges are 1 or 2; dividing by the square roots of the gauges puts every relevant product in Q(sqrt(2)). The source checks all four left/right choices in every old context and independently reconstructs the displayed component partitions.

A displayed component containing a mark must be a simple path. An existing branch or nonsimple bond remains a feature in the actual component and is at distance at most 16 from the mark. An existing cycle cannot occur inside a finite A/B/D tree. The checker saves one such witness for each rejected nonsimple marked block.

If the two marks are in the same displayed path, that path must already give distance 8: a tree cannot acquire a second, shorter route. Its embedding in A43 is then fixed once the two marks are placed at 17 and 25. If the marks lie in different displayed paths, orient each path from its outward endpoint toward the eventual joining segment. Let t_L,t_R be the inward distances from the marks to the chosen endpoints. The connecting segment must contain exactly

```
k = 7 - t_L - t_R
```

additional whites, with k at least 1. The two displayed components are orthogonal, so no zero-white connection is possible. Every orientation of both paths is enumerated, with singleton orientations counted once. All retained embeddings fit inside A43; all old pair products are checked against the corresponding A43 entries. This also covers paths joined through old

spectator roots: those roots may occupy positions not selected in the reduced old embedding. Their identities and extra contacts are not assumed away. The Gram tests use a necessary subset of actual roots only.

All-label restrictions leave three white slots per omitted black

Write b for the omitted left black root and d for the omitted right black root. The full-star no-short7 hypothesis implies b·d=0 and forbids any contact from b to the right mark, or from d to the left mark. Their products with their own marks are fixed native products and remain in the calculation.

For a norm-2 A20 arm, the inverse is positive and its diagonal at outward distance j is j(21-j)/21. The omitted black and all these whites belong to an actual edge diagram, which is strictly positive definite. Hence:

- At outward distance 1, only nonzero contacts of labels 3 or 4 are possible. A label at least 5 has squared magnitude greater than 5/2, and (20/21)(5/2)>2.
- At distance 2, only label 3 is possible, since (38/21)·2>2.
- At distances 3 through 18, even a unit contact gives projection greater than 2, so no contact is possible.

Additional nonpositive products cannot lower these projection bounds, because the inverse is positive. Strictness excludes equality as well. These statements have no finite label cutoff.

In the inner A7 segment, b contacting a white at distance j at least 2 from the left mark gives a route to the right mark of length 9-j, at most 7. Only the first inner white can therefore meet b in the no-short7 branch. Its A7 inverse diagonal is 7/8, which excludes label 5 and every larger label by the same squared-magnitude comparison. Contacts to the first two outward whites beyond the opposite mark also give short paths; more distant contacts are already excluded by the other A20 arm. The argument for d is the reflection of this argument.

Thus b can contact, apart from its own mark at 17, only positions 16,15,18, with allowed nonzero labels respectively (3 or 4), 3, (3 or 4). Root d has the reflected slots 26,27,24. The source first checks every fixed old contact. A forbidden one supplies either a recorded short path, an A20 projection obstruction, or a strict edge label obstruction. Four orientations are rejected by the last reason; the remaining fixed-contact exclusions use the former two reasons.

One omitted black cannot contact both consecutive outer whites. The positive A2 inverse is (1/3)[[2,1],[1,2]], and two unit contacts already have projection square 2, contrary to strict edge positivity. The remaining slot products are enumerated exactly as 0,1,sqrt(2), respecting every old fixed contact. They are an overestimate of realizable completions, which suffices for an obstruction.

Two necessary Gram tests

For a norm-2 simple path of order n, the inverse entries, using indices from 0, are

```
J_n[i,j] = (min(i,j)+1)(n−max(i,j))/(n+1).
```

The source verifies this inverse formula by multiplication for n=2,7,17,20,43. Deleting the two marked roots from A43 gives three independent white blocks A17+A7+A17. Let J_W be their inverse. For the positive contact vectors x,y of b,d with the A43 path, strict positive definiteness of each actual edge requires

```
2 − x_W^T J_W x_W > 0,     2 − y_W^T J_W y_W > 0.
```

Every profile violating either strict inequality is impossible. Otherwise eliminate the full positive definite A43 block. Because b·d=0, the remaining two-by-two Schur matrix is

```
[ 2 − x^T J_43 x       −x^T J_43 y     ]
[   −x^T J_43 y      2 − y^T J_43 y    ].
```

In every surviving enumerated profile its first diagonal is strictly negative and its determinant strictly positive. It therefore has two negative directions. By congruence the full Gram matrix would have two negative directions, impossible for actual vectors in a real Lorentz space. This argument requires no assumption on the old core's signature. Adding further actual roots or spectator contacts cannot repair a principal Gram matrix with two negative directions.

All signs and products use the exact Q(sqrt(2)) helper. No floating-point tolerance or optimizer is used.

Complete counts and scope

Corner outcome	Corners
Nonsimple marked block incompatible with depth 20	16,627
Connected marks at a fixed distance other than 8	2,187
No admissible distance-8 joining orientation	14
Complete corridor/contact calculation	7,984
Total	26,812

The last row of cases has 13,644 complete path orientations. Fixed contacts exclude 8,337 orientations. The remaining 5,307 use 379 distinct contact templates, with every template's multiplicity saved. There are 4,983 distinct joint contact profiles. With orientation multiplicities restored, 57,582 profiles fail strict actual-edge positivity and 50,137 give two negative directions. None remains unresolved within this deep-corridor hypothesis.

This settles the proposed deep-pair alternative throughout the **high-overlap domain**, including the formerly unresolved 251 contexts. It does not settle those contexts for arbitrary pair shapes, and does not complete the finite-7 payment theorem. Low/disjoint cases, the actual retained pair maps and all affected universal payment interfaces must still be checked before changing a weight or importing a finite budget. The unrestricted dimension ceiling remains 65. No new finite-volume polytope is constructed.

Verification

The first full-domain pilot used direct, uncompressed contact enumeration and found the same orientation-weighted outcomes. The authoritative producer compresses identical fixed-slot domains and saves every one of the 4,983 distinct exact profile checks. Generation 81249 terminated exit0/PASS (terminal chunk c22b97). Separate fresh saved replay 96652 terminated exit0/PASS (terminal chunk 8076b4). An earlier producer run exposed a missing explicit fixed-label rejection for four orientations; it was corrected before this successful generation. The final calculation records these rejections.

Two other agents completed full producer/report mathematical reads. They checked complete connected/disconnected embeddings, the old size bound and persistent-feature argument, physical gauge normalization, all-label slot completeness, arbitrary spectator contacts, strict white-edge positivity and the Lorentz index-two contradiction. Neither repeated the 4,983-profile arithmetic or the inherited overlap census.

A separate saved-template checker completed exit0/PASS (terminal chunk 8cba59). It imports no project module, uses generic rational Gauss-Jordan inversion instead of the path-inverse formula, and decides Q(sqrt(2)) signs with rational isolating intervals instead of the producer's squared-norm sign comparison. It reconstructs all 379 saved slot domains, checks all 4,983 edge/Schur profiles exactly, and reconciles the 13,644 orientation multiplicities. A second agent read the complete helper and confirmed this scope. The helper does not independently reconstruct the old native cores, fixed-contact exclusion witnesses, path embeddings, all-label geometry or original high-overlap census.

The inherited all-label high-overlap census is a frozen proof input; this packet reconstructs its native matrices, corner partitions and complete new corridor/contact domain, but does not repeat the original overlap census.

```
python3 scripts/fresh_full_a43_corridor_high_domain_20261017.py
python3 scripts/fresh_full_a43_corridor_high_domain_20261017.py --verify
```

Source 127

Source path: `results/fresh_h17_d16_payment_bottleneck_20261017.md`

SHA256: 809170d0134ea11ab63a54e2cbaf573f92b063256902202bc5a09ebf2902d79e

The D16 cusp payment bottleneck in the known H17 pyramid

The known finite-volume H17 pyramid realizes the difficult17-node affine D16 factor. With the retained finite rule (unit payment for distance at most9 **or connected component order at most17**) and unit payments at its other E8 ideal corners fixed, its actual faces force all136 D16 pair weights to be at least1 under any rule invariant under the intrinsic D16 diagram automorphisms. The resulting factor cost is136=8·17. This is a local obstruction to reducing that particular diagram-based budget, not a new dimension theorem.

The exact checker and saved certificate reconstruct every two-face through this cusp from the already frozen complete H17 cone and face ledger. The chamber is the two-L25 diagram sharing node9. Its82 cone rays comprise80 finite vertices and two proper ideal vertices: the apex supported on9 has factors affine E8×E8; the vertex supported on{3,13} has the17-node affine D16 factor. The finite-volume, actual-facet and complete-cone assertions are inherited from that frozen chamber package, not inferred from a partial list of proposed corners.

Exact face deficits

The D16 cusp is simple. Every pair among its17 incident facets determines one actual two-face, giving136 faces:72 triangles and64 quadrilaterals. For each pair the checker independently reconstructs the face vertices and actual side facets from the saved cone rays, then checks agreement with the frozen exhaustive face ledger. It computes each marked distance in the actual corner star, including the correct varying E8 factor at the nonsimple apex.

The fixed comparison rule is:

- at finite corners, weight1 if the marked distance is at most9 or the connected marked component has order at most17, otherwise0;
- at the other ideal cusp, weight1 on the actual nine-node E8 varying factor;
- at the D16 cusp, leave the weight as an arbitrary nonnegative real variable for its marked pair.

A triangle requires total payment2 and a quadrilateral requires1. The exact required D16 payment is the positive part of that demand minus the fixed payments at all other corners. Every resulting deficit is0 or1.

Face type	Faces	Pairs with deficit1	Pairs with deficit0
Triangle	72	60	12
Quadrilateral	64	49	15
Total	136	109	27

The49 mandatory quadrilateral pairs have distance histogram

`2:1, 3:2, 4:3, 5:4, 6:5, 7:6, 8:7, 9:6, 10:5, 11:4, 12:3, 13:2, 14:1`.

In particular **15 pairs at distance greater than9 are required by actual quadrilaterals under the retained finite rule**. All other corners of these quadrilaterals are finite, so this conclusion is independent of any payment at the E8 cusp. It is not an artifact of a catalogue containing unrealizable stars.

For comparison, omitting the retained small-component finite payments and using finite distance9 alone gives62 mandatory quadrilateral pairs, including all26 pairs at distance greater than9. There are still60 mandatory triangle pairs, hence122 total. The small-component rule removes13 of these quad deficits (11 long pairs and two at distance9). The certificate saves both rules' corner payments and face deficits separately. The26-long-pair statement must not be transferred to the retained finite rule.

Fourteen of the60 mandatory triangles also contain the E8×E8 apex. Their deficits do depend on fixing the other ideal payment at1. Allowing a different weight there may change the D16 demands; this report makes no lower-bound claim for arbitrary simultaneous changes of both cusp rules. Every face, corner payment and deficit is saved explicitly, not only as an aggregate count.

Diagram invariance forces all136 pairs

The D16 diagram has two fork-leaf interchanges and the reversal of its unique double-fork spine. These generate its full group of8 automorphisms. The checker verifies every generator against the literal graph, constructs the group, and saves for each of the136 pairs an automorphism taking it to one of the109 primary mandatory pairs. Since those109 are also among the122 comparison demands, the same witnesses establish closure for both baselines.

Consequently any intrinsic-diagram-invariant nonnegative real weighting satisfying the stated fixed-baseline face demands must give every pair weight at least1. Its cost is therefore at least136. Conversely unit weight on every pair satisfies these demands, so this local minimum is exactly136.

Without diagram invariance, the109 primary mandatory pairs may receive weight1 and the27 optional pairs weight0 in this particular chamber. Under the distance9-only comparison the corresponding counts are122 and14. These are statements about this complete fixed chamber and the specified fixed baselines, not a universal context-dependent weighting theorem. Additional ambient incidence information is therefore a possible way around the diagram-only obstruction.

Relation to the nonincident E8 carriers

Every one of the64 quadrilaterals through the D16 cusp has all19 roots in its original face star. Both E8 factors of the other cusp occur in that star, but that cusp is not on the quadrilateral. Thus the required long-pair faces have explicit nonincident small-factor carriers at the same other cusp. This supplies concrete data for a future cross-cusp charging argument. It does not prove a bounded multiplicity or a global curvature inequality.

The mixed finite9/ideal7 replacement lemma and its improved triangle paths remain valid. This audit explains why those local improvements alone do not remove the order17 whole-factor payment cost in a diagram-invariant scheme.

Verification scope

Generation and fresh saved replay both terminated successfully:136 reconstructed faces,109 primary and122 comparison unit deficits,8 automorphisms and136 saved orbit demands. Syntax, all7 direct source bindings and all4 local links pass. Independent internal source/proof review found no gap, and a separate read-only inspection recomputed every saved primary/comparison corner payment and face deficit and all136 automorphism witnesses. Those reviews did not repeat the original H17 cone or volume proof. This audit uses elementary exact graph and rational cone-incidence checks; it makes no new construction or dimension claim. No frozen predecessor was edited. Source, data and report are final/frozen.

Source 128

Source path: `results/fresh_h17_half_coupon_transfer_20261017_review.md`

SHA256: 69af5283c7ee248b4beba0d66c4cbad5c3a42247315f2e71027d3a96f44e8844

Independent local transfer for the nine dropped D16 pairs

The nine pairs between the two fork triples of the17-node affine D16 diagram can be removed in a literal shared-L25 H17 quadrilateral star if the stated finite half-payments are added. The transfer survives arbitrary additional supporting whites: either they give an immediate length-two payment, or they are orthogonal to the entire19-root core and preserve its finite component types.

The independent exact checker and certificate verify the core signature, radical, affine positive kernel and all27 finite corners in the nine cases. They use Fraction elimination and literal graphs, without importing the proposed global producer's arithmetic. The actual small chamber is the frozen known finite-volume H17 pyramid; the transfer below concerns any actual quadrilateral with this same literal core and role assignment.

Local hypotheses and finite coupons

Let the19-node core be two copies of L25 sharing node9, with the numbering of the D16 bottleneck audit. Its opposite black pairs are

`{a,3}` and `{b,13}`, with a in{0,1,2} and b in{10,11,12}.

All other core roots are supporting whites. Further supporting whites E are allowed, and their Gram is positive definite because the face has an interior hyperbolic point. The four displayed blacks are the actual four sides. The Gram on the displayed19 roots is assumed to be the literal core, including its zero contacts; obtaining that literal core from a general classification is a separate obligation.

Retain a unit finite payment when the two marks have distance at most9 or lie in a connected component of order at most17. Add weight1/2 at either of the following kinds of finite corner:

- the marks lie in distinct A1 and D16 components, with the D16 mark at its fork center or either short fork leaf;
- the marks lie in distinct A2 and D14 components, with either A2 mark allowed and the D14 mark at its fork center or either short fork leaf.

The component names here are finite types: D16 has16 roots and D14 has14 roots. These half-payments are disjoint from the retained finite rule, since their marks lie in different components. This note proves the required local transfer, not a universal bound on the sum of all such coupons.

Extra supporting whites cannot change the required components

Write F for the17-node affine D16 subdiagram obtained by omitting3 and13 from the core. The actual corner retaining a,b contains F. It is necessarily ideal, since a finite corner cannot contain a singular principal diagram. Its positive affine kernel has coefficients1 at the four leaves and2 elsewhere. Every supporting white e in E lies at that corner. The positive kernel and nonpositive Coxeter contacts force e to be orthogonal to every root of F.

The exact19-root Gram has signature(17,1,1) and radical

`(-6,-3,-4,-2,-5,-4,-3,-2,-1,0,6,3,4,2,5,4,3,2,1)`.

This radical is an actual linear relation in the ambient Lorentz realization. Indeed, its represented vector is null and orthogonal to the indefinite core, hence to a timelike vector; a timelike vector's orthogonal complement is positive definite, so the represented vector must be zero.

Since e is already orthogonal to F, pairing that relation with e leaves

$-2(e\cdot n3)+2(e\cdot n13)=0.$

If these equal contacts are nonzero, the actual corner with marks3,13 contains the white-interior path3–e–13. That corner has a unit payment under the finite distance9 or ideal distance7 rules. At an ideal corner the actual factor-fiber lemma places this path in the varying affine factor. Thus this case already pays the quadrilateral.

Otherwise every e in E is orthogonal to all19 core roots. The finite corner stars then split as the displayed finite core corner plus the positive-definite diagram E. They remain finite, and the marked core components cannot be enlarged or joined by E. Consequently their literal A1/D16 or A2/D14 coupon types, and their existing connected-component payments, persist unchanged.

This argument concerns the supporting whites in the actual face star. It does not assume a simple ideal vertex or claim that all ambient facets incident to the ideal corner are among them. Those extra ideal facets are irrelevant to the finite coupon corners.

All nine literal cases

In F, the fork triples are{0,1,2} and{10,11,12}. Their nine cross-pairs are exactly the eight pairs at distance at least13 plus the pair of fork centers at distance12. The checker verifies that F has corank1 and the other three corner stars are positive definite of order17, for every case.

Dropped ideal pair	Old finite payment total	Added coupon total
0,10	0	1
0,11	1	1/2
0,12	0	1
1,10	1	1/2
1,11	3	0
1,12	1	1/2
2,10	0	1
2,11	1	1/2
2,12	0	1

Only four cases need coupons. Their two half-paid marked pairs are:

Dropped pair	First finite corner	Second finite corner
0,10	0 in D14 fork center;13 in A2	3 in A2;10 in D14 fork center
0,12	0 in D16 fork center;13 in A1	3 in A2;12 in D14 short leaf
2,10	2 in D14 short leaf;13 in A2	3 in A1;10 in D16 fork center
2,12	2 in D16 short leaf;13 in A1	3 in A1;12 in D16 short leaf

Each required pair of half-payments sums to1. Together with the sealing argument, this proves the local transfer for every actual quadrilateral satisfying the explicit core hypotheses, with arbitrary additional supporting whites.

Verification and limits

Generation and fresh saved replay both terminated successfully. Syntax, all7 direct source hashes and all4 local report links pass. The independent finite arithmetic covers one19-root signature/radical, one17-node affine positive kernel, nine15-node white supports and all27 finite corner matrices. It also checks the literal marked component roles and each face payment. A

separate internal written-proof review passed the actual-corner affine-kernel argument, actual radical relation, length-two alternative and finite-component sealing; that review did not duplicate this arithmetic or any global embedding census.

This package does not reproduce the full old108 embedding census, prove that these are the only affected cases in that census, establish the global finite coupon budget or the all-rank ideal budget, or compose a new dimension theorem. Those are distinct obligations in the proposed parent argument. No frozen predecessor was edited. Source, data and report are final/frozen.

Source 129

Source path: `results/fresh_high_rank_quasilanner_census_20261017.md`

SHA256: deee1eee4ab3e01e508f6d23c995d3d66557d0724b80d533fabdcf6fca065b56

Exact reproduction of the order-six-to-ten quasi-Lannér census

The exact source examines **146,262 attachments to 24 affine source diagrams**, retains 84 labelled attachments, and identifies **26 diagrams up to labelled graph isomorphism**. The saved census records every retained attachment, its explicit isomorphism, every vertex deletion and the relevant diagonal-decrease thresholds.

Number of nodes	Nonisomorphic diagrams
6	12
7	3
8	4
9	4
10	3
Total	26

These are **classical diagrams, not new examples or a new classification**. The result reproduces the higher-order part of M. Chein's 1969 classification, printed pp.15–16. Chein's printed p.4 states the order-ten ceiling, attributing it to Vinberg, and describes the finite/affine principal-subdiagram criterion. Those classification facts are published mathematical inputs here; a search restricted to ten nodes would not itself establish the ceiling. A repository copy of the primary PDF is also available through the separately frozen finite-edge package.

Why the finite search covers these orders

Use the usual norm-two Gram convention: diagonal entries are 2, a missing edge denotes label 2, label 3 has product -1, and label 4 has product `-sqrt(2)`. A minimal indefinite diagram has indefinite full Gram matrix and positive-semidefinite every proper principal submatrix. It is connected, since otherwise a smaller connected component would already be indefinite.

For orders six through ten, the compact Lannér branch is absent: the published compact classification has at most five nodes. Consequently at least one proper principal submatrix is singular positive semidefinite and contains a connected affine component A. Its kernel has strictly positive coordinates. If another node is joined to A, the enlarged block cannot be positive semidefinite: its affine kernel has a nonzero pairing with the new node. Indeed a vector of zero quadratic form in a positive-semidefinite matrix must lie in its full kernel, while the positive coefficients and nonpositive Coxeter contacts make this pairing strictly negative.

Connectivity supplies such an adjacent outside node. Minimal indefiniteness therefore forces A together with that node to be the whole diagram. Thus every diagram under consideration is an irreducible affine diagram of order `n-1` with one additional, nonorthogonal node. This also explains why an arbitrary disconnected affine forest is not needed as a source.

There are no edge labels 5 or larger. Any edge in a connected diagram with at least six nodes can be extended to a connected induced subdiagram on five nodes, by adjoining adjacent vertices. This is a proper principal subdiagram and must be finite or affine. The standard finite/affine classification has no connected five-node diagram containing an edge label at least 5. The smaller exceptions, such as the noncrystallographic finite diagrams or the rank-two families, cannot survive this extension.

Dotted pairs are already indefinite two-node subdiagrams, so are excluded immediately. A parallel pair is a connected affine two-node subdiagram; adjoining any adjacent node makes a proper indefinite subdiagram by the same positive-kernel argument. Parallel bonds are therefore also excluded at these orders. The attachment alphabet is exactly `2,3,4`, including orthogonality; there is no numerical cutoff on a hidden label or dotted distance.

The complete affine source list has ranks four through eight:

- rank 4: affine A4, B4, C4, D4 and F4;
- rank 5: affine A5, B5, C5 and D5;
- ranks 6, 7 and 8: affine A, B, C, D and the corresponding E type.

These are the 24 source models. For a source on `n-1` nodes the program tests all `3^(n-1)-1` nonzero attachment vectors. Their total is 146,262.

Exact acceptance, signature and isomorphism checks

Every one-node deletion is decomposed into connected components. Each component must be a recognized finite or affine Coxeter diagram. Finite and affine trees are recognized by their complete labelled tree codes; the affine A cycles are recognized separately. This supplies exact positive semidefiniteness of every one-node deletion, and hence of every proper principal submatrix. Rejection depends on this classification, not on approximate eigenvalues.

For every newly encountered isomorphism type the source reconstructs the full Gram matrix over exact rational numbers and `sqrt(2)` and proves its determinant negative. Interlacing with the positive-semidefinite one-node deletions then gives precisely one negative eigenvalue. The nonzero determinant excludes a radical, so the full signature is `(n-1,1,0)`.

Refinement is used only to select possible isomorphism matches; the actual merges use a complete labelled graph-isomorphism test. All 84 retained attachments have an explicit saved map to one of the 26 final diagrams. The classical order counts are asserted only after enumeration, as a comparison check; they are not used to prune or accept candidates.

The export distinguishes finite from affine node deletions. For a diagram with a unique affine deletion j, it also records, for every other deleted node b, the exact threshold

$$\tau_b = \frac{1}{(G_{\widehat{b}}^{-1})_{jj}}.$$

All these thresholds are checked positive and rational. They describe positive-semidefinite rank-one diagonal decreases at j. Applying them to an actual simplex face with additional white supporting roots still requires the appropriate geometric and Schur-complement argument.

Scope and frozen validation

The finite/affine classifications, compact Lannér order bound and general order-ten ceiling are explicit published inputs. The computation exhausts the stated attachment domain and reproduces the classical higher-order diagrams. It does not identify an induced geometric simplex face with a Coxeter simplex, settle arbitrary extra supporting roots or black/white partitions, or establish a new ambient dimension bound.

FINAL/FROZEN: generation 76418 and fresh saved replay 93878 passed; the root agent confirmed the latter terminated with exit code zero. The saved replay log ends with the complete `146262 / 84 / 26 / 24` PASS result. No additional full replay was started to prepare this report. Python syntax, all four direct source-manifest hashes and local report links were checked. The frozen script and JSON were not edited.

Artifact	SHA-256
Source	ab08dbff4912545541bba7117ce66848269e527e4f53270d cde138ef27812399
Saved JSON	6fe2aa3a6b5cd9ef70e25c2efea1d484f2f2c4a17219a9e1ea 1966c8bc0e2729

The saved complete rejection-trace digest is 04bbd355d85dc1f1ba411e7e179b12af09fb89fb5f6d5e66dcf4ab565a6c8ce3.
This is a reproducible classical census and exact data package, not a novelty claim or a formal proof-assistant verification.

Source 130

Source path: `results/fresh_ideal5_combined_payments_20261017.md`

SHA256: b3548ef283e20478205cde5924f0a20e839ef1f26ca1fb148920022b7580e4c0

Complete finite9 and ideal5 face payments

For every finite-sided finite-volume hyperbolic Coxeter polytope of dimension n≤80, the rules below give each triangular two-face total corner payment at least2 and each quadrilateral at least1. Finite-vertex payment is at most9n. At an ideal vertex the exact factor costs have the piecewise envelope stated below. This is a complete local-payment and counting composition, including nonsimple cusps and arbitrary supporting white roots. The ambient dimension comparison is separate.

The checker and certificate bind the frozen proofs and check their interfaces. The complete core classification used here is the finite9 classification. No claim of complete finite8 overlap coverage is needed or made.

The rules

At a finite vertex, pay1 when the marks lie in the same whole irreducible component and either their graph distance is at most9 or that component has order at most17. In addition, pay one half between distinct whole components when one has rank at most5 and the other is one of:

- D11 through D16, marked at its fork center or either short fork leaf;
- A13 or A15, marked at its unique middle;
- A17, marked five steps from one endpoint and eleven from the other.

Each qualifying pair receives one half, with no duplicate coupon for multiple descriptions. The unit and coupon conditions use the full actual finite vertex diagram, not a selected subdiagram.

At an ideal vertex, use the full varying affine factor containing the two marks, as supplied by the actual factor-fiber theorem. Pay every pair when its order is at most13. At orders14–17 use the exact automorphism-invariant pair sets in the 571-pattern census. At larger orders pay distance at most5, and distances6–9 if an endpoint is within5 of a branch or a nonsimple-bond endpoint. All these ideal payments are zero or one.

Triangle coverage is retained

The complete finite8/ideal5 triangle theorem already pays every triangle twice by finite distance at most8, ideal factors of order at most9, or ideal distance at most5 and feature-near distances at most8. The present finite rule contains that finite rule. The source checks the inclusion of every elementary triangle pair in every order14–17 target. Conceptually, cycles have no features, so their removed distances greater than5 were never elementary triangle payments. The nine removed D pairs have minimum distance9 at order14 and larger minimum distances in higher orders, whereas the triangle theorem's feature-near paths have length at most8. Factors of order at most13 are wholly retained. Thus no replacement of one triangle corner by only one total quadrilateral payment is used.

Quadrilateral coverage from the complete finite9 domain

Start with an actual quadrilateral. If it has a finite marked corner path of length at most9, its unit payment remains. Suppose an ideal corner has such a path. It pays immediately at distance at most5 or within the radius5 feature neighborhood, unless it is one of the explicitly removed pairs.

For a classical tree factor outside that neighborhood, the complete finite8/ideal5 local theorem replaces marked distances6–9 by retained units or sealed finite coupons. The same theorem treats cycles of order at least15, including the sealed A1/A15

and A1/A17 equalities. The separate cycle14 theorem treats its remaining distances6 and7 using the strict spectator spectrum and the sealed A13-middle coupons. Smaller factors are wholly paid.

All D pairs deliberately deleted from the basic rule are handled by the specific transfers below. Therefore, when no corner has yet paid and no transfer has already supplied a total unit, there can be no adjacent marked path of length at most9. This is exactly the original hypothesis of the complete dimension108 core classification, not a silently weakened no-short8 hypothesis.

That classification includes its complete all-label low-core supplement and949 high-core overlap contexts. The low-core selected component bounds are at most12. At a finite selected corner, the retained small17 rule pays. At an ideal selected corner, the whole varying factor has the proved bound at most17. Orders at most13 pay. For orders14–17 the actual marked principal component gives an induced labelled embedding into the full factor; all such maps are in the571-pattern census, with its extension-bound filter. Thus its pair is in the union before the certified deletions. No arbitrary provenance, favorable map, or minimum across incompatible realizations is selected.

Every deleted pair has a retained alternative

The cycle transfers above directly use the present short/feature rule and the listed whole finite coupons. They do not leave an unexamined ideal distance6 or7 obligation behind.

For order14 affine D13, the universal long-pair transfer replaces all nine cross-fork pairs. Its immediate ideal alternatives are paths of length at most3 or the two literal affine components of order12; these all pay. Its other conclusions are sealed whole finite components or small components of ranks1,2,4,5 paired with D11 or D13 fork marks, all in the finite rule.

For order15 affine D14, the preceding universal transfer has only two immediate whole-small ideal alternatives, both of order13. They arise from the center deletion D12 plus the new root, with either an endpoint4 or a second-node3 attachment. They are wholly paid. A possible order14 affine component after another outer attachment is **not** an immediate paid alternative in that proof: it remains among the bounded nonextreme profiles and is eliminated by the strict determinant gap. The only short ideal alternatives used are the adjacent-product or common spectator paths of length1 or2, not an arbitrary distance6 pair. Its sealed small-rank$\le$3 against D12/D14 coupons are included.

For orders16 and17, the nine affected preferred-core contexts at each order are matched exactly to the saved order16 and order17 replacements. Those are literal sealed-core arguments, not a new universal center-profile extension. A nonorthogonal supporting white gives a retained length2 path; otherwise the saved finite unit/coupon corners sum to at least1. The source checks that every nonzero terminal payment in all18 cases is finite. There is consequently no hidden order14 ideal payment inherited from their older global rules. Their core matching is applied only after the original no-short9 reduction has been restored.

These implications complete quadrilateral coverage. They are direct contradictions to a face whose total new payment is less than1: a direct new ideal unit, a direct finite unit, or two sealed halves resolves it. No cyclic chain of ideal transfers is assumed to terminate without proof.

Finite and ideal budgets

The finite8 coupon budget already covers exactly the stated coupons together with distance$\le$8, distance9 at a whole-component leaf, and whole finite components of order at most17, for n$\le$80. Its total is at most8n.

To restore all finite distance9 pairs, nothing is added in ranks at most17. At rank r$\ge$18 the only irreducible finite diagrams are A, B and D. Each has r$-$9 pairs at distance9. Exactly two of these pairs contain a leaf for A/B, and exactly three for D; there are no leaf-to-leaf distance9 exceptions at these ranks. The additional costs are r$-$11 and r$-$12, respectively, each at most r. Summing over components adds at most n. The new finite total is therefore at most9n with all enlarged coupons included. The certificate checks the literal pair sets at every relevant rank18–80, alongside this rank-independent path count.

The small-factor costs and the all-rank ideal5/R5 spine proof give the factor rank envelope

Rank r	Normalized factor cost bound b(r)
1–12	r/2
13	87/14

Rank r	Normalized factor cost bound b(r)
14	20/3
15	111/16
16	127/17
at least17	163/23

For an actual cusp whose link is a product of simplices of ranks r_i, let V be its number of incident edges. Each pair in a varying factor of order r_i+1 repeats V/(r_i+1) times. Hence the ideal-vertex payment is at most V times the sum of b(r_i). The actual link and factor-fiber lemmas justify this multiplicity even at nonsimple ideal vertices; no global simple-cusp or arithmetic assumption has entered.

Verification scope

The checker binds the complete inherited core and low-core proofs, the1025 triangle cases, the571-pattern target sets, both universal D transfers, both nine-case sealed replacements, the cycle/weak-profile theorems and the finite coupon budget. It checks all16 triangle-pair inclusions, both literal order12/order13 immediate alternatives, terminal finite payment totals, the small costs and every finite9 incremental pair set. The all-label classification, local geometric lemmas and arbitrary-rank counting proofs remain explicit inherited mathematical inputs. Generation and a fresh separate-process saved replay both terminated exit0/PASS. A separate full source/mathematical read found no gap in the short-path reduction before the949-case classification, triangle-pair retention, immediate order12/13 alternatives, literal finite replacements, finite9 incremental cost or actual cusp multiplicities. That review did not rerun the571-map census, the949 core census or the saved row arithmetic. Syntax, all38 direct bindings and report links pass. Source, data and report are final/frozen. No external refereeing or proof-assistant formalization is claimed.

Source 131

Source path: `results/fresh_ideal5_finite8_triangle_payments_20261017.md`

SHA256: 559194b3706685d049df14cb816530394aa35c797a1c6b13dc3f58e9658b050d

Triangle payments with finite distance8 and ideal distance5

Every triangular face of a finite-sided finite-volume hyperbolic Coxeter polytope receives total payment at least2 under the following rule:

- a finite corner pays1 when its marked roots have distance at most8;
- an ideal corner pays1 when its varying affine factor has order at most9, or its marked distance is at most5, or its marked distance is at most8 and a mark is within5 of a branch or a nonsimple-bond endpoint.

This includes nonsimple ideal vertices and arbitrary additional supporting white roots. The checker and certificate enumerate the finite cases and their exact E8 data. This is a triangle theorem, not a quadrilateral theorem or a dimension bound.

Complete core domain

The actual triangle setup and minimal-indefinite-core reduction are inherited from the preceding triangle proof. Such a core contains the three black roots, with all other roots supporting the face. Deleting any black root gives an actual corner subdiagram and therefore a positive-semidefinite Gram. Graph paths avoiding that third black root remain corner paths when all supporting whites are restored. Likewise a branch or nonsimple bond already present in the marked component persists. The actual cusp-factor theorem places those paths and features in the full varying affine factor at an ideal corner.

For core order at most6, a Steiner tree through the three marks has at least two marked-pair paths avoiding the third mark, each with at most5 edges. This argument allows all labels in the small-core families.

For orders7,…,10, the complete inherited high-core catalogue gives 1,025 marked triples:105 at order7,224 at order8,336 at order9 and360 at order10. The checker computes each marked path and every feature distance in the actual corner subdiagram. All but four triples have at least two elementary payments. Each exception is in the simply-laced10-node L25 tree, whose edges are

`0–1, 0–2, 2–3, 0–4, 4–5, 5–6, 6–7, 7–8, 8–9.`

The exceptional triples are(0,1,9),(0,2,9),(1,2,8),(1,2,9). Each already has one elementary payment; the next argument supplies a different corner payment. No unexamined supporting-white extension is discarded.

Three positive-kernel resolutions

For the triples containing9, the corner formed by the other two marks contains the full affine E8 diagram on nodes0,…,8. Its positive kernel is

`(6,3,4,2,5,4,3,2,1).`

The source verifies this kernel and a positive-definite principal8-node matrix, proving positive semidefiniteness. Any additional supporting white belongs to this actual corner. Its nonpositive products with a positive affine kernel must all vanish. Thus every extra supporting white is orthogonal to all nine displayed nodes.

At the indicated alternate corner a simple leaf and its degree-two neighbor are consequently fixed, including after every supporting white is restored:

Triple	Alternate corner	Fixed simple leaf	Fixed neighbor	Marked distance
(0,1,9)	(0,9)	3	2	6
(0,2,9)	(0,9)	1	0	6
(1,2,9)	(1,9)	1	0	7

The fixed leaf belongs to the marked connected component; it need not itself be a mark. None of the classical irreducible affine A/B/C/D diagrams has a simple leaf whose neighbor has degree two. Hence this actual corner is finite, in which case the displayed distance is at most8, or the varying factor is exceptional affine, of order at most9. Either case pays1.

The finite-E8 resolution

For the remaining triple(1,2,8), the corner(1,2) contains finite E8 on nodes0,…,7, together with node9. Write its normalized Gram as E. The exact inverse is entrywise nonnegative, with diagonal

`(30,8,14,4,20,12,6,2).`

An added supporting root has square2. Its nonnegative contact magnitudes x with E must satisfy x^T E^(-1)x<=2, because the augmented principal diagram belongs to a positive-semidefinite actual corner. Every nonzero native contact has magnitude at least1. Nonnegative inverse entries imply that any contact on nodes0,…,6 alone already costs more than2. Thus all additional supporting whites are orthogonal to those seven nodes. Contacts at7 are allowed; the argument does not assume they vanish.

At the alternate corner(1,8), node1 remains a simple leaf and node0 remains its degree-two neighbor. The marked path1-0-4-5-6-7-8 has length6. As above, the entire marked component is finite or exceptional affine of order at most9, so this corner pays. Arbitrary attachments farther along the path cannot change its protected leaf pattern.

Verification scope

The source verifies all1,025 triples, the exact four exceptional domains, their distinct alternate payments, positive-definite corner pivots, the affine E8 kernel, and every entry of the finite E8 inverse. The all-label small-core argument, positive-kernel orthogonality, projection inequality and affine-type exclusion are analytic parts of the proof. The catalogue classification and actual-face/cusp reduction are explicit inherited inputs.

Generation and a separate fresh saved replay both terminated exit0/PASS. A separate agent read the complete source and mathematical proof, including the small-core Steiner argument, all four E8 mechanisms, native contact gap, protection of the actual leaf pattern and actual-cusp transfer. That review does not duplicate the1,025-case census or the inherited classification. The parent also independently obtained the E8 inverse diagonal and affine kernel using symbolic matrix arithmetic. This package is final/frozen. No quadrilateral coverage, ideal-factor cost, new ambient bound, external refereeing or formalization is asserted.

```
python3 scripts/fresh_ideal5_finite8_triangle_payments_20261017.py
python3 scripts/fresh_ideal5_finite8_triangle_payments_20261017.py --verify
```

Source 132

Source path: `results/fresh_ideal5_piecewise_dimension74_20261017.md`

SHA256: b8639db0608c4d18e04e0e52b6a2ccc05c702a2e47c49975c480e241376d2678

Unrestricted finite-volume dimension at most 74

The completed local three-pair replacement and the exact global comparison give the following computer-assisted conclusion:

Every finite-sided, finite-volume real hyperbolic Coxeter polytope has dimension at most 74.

There is no arithmetic, compactness, simple-cusp, cusp-factor or facet-count restriction. This strengthens the completed dimension-75 proof. The source and certificate check the new local interfaces and all 999 reduced profiles at dimension 75. The independent review reconstructs every profile and agrees on every exact row.

What changes locally

The order-15 affine B reduction removes exactly three invariant paid pairs, (0,11), (11,13) and (11,14), from the canonical affine B14 diagram. Its order is 15 and rank is 14. The cost falls from 100 to 97. The three pairs have distances 11,12,12, so no basic short-path or elementary-triangle payment is lost.

The complete saved induced-map census assigns these pairs to exactly seven overlap contexts. Six have a preferred whole finite D14 that no supporting white can touch. In the seventh, the marked A12 can only remain whole or acquire a single residual-zero endpoint extension to whole A13. That added root lies in the Lorentzian core span and cannot attach to core-orthogonal whites; its exact products exclude every second core-contact white. In all seven cases the rest of the corner is an orthogonal positive-definite white block. Thus these are actual finite corners, paid by the existing whole component cutoff 17. An abstract finite subdiagram at an ideal corner would not suffice; the local proof establishes the complete finite outcome.

No finite coupon or finite cost is added. Every other selected pair and local replacement stays as in the completed finite8/ideal5 theorem and its dimension-75 refinement. All triangle payments remain at least two and quadrilateral payments at least one. The finite cost remains at most 8n through dimension 80. The prior D17 singleton replacement and its finite A4/D13 coupons are retained without change.

At order 15 the other costs are A75, C95 and D96, so the largest cost is now 97. The normalized ideal-factor envelope at affine rank r becomes:

Rank r	Normalized cost
1 through 12	r/2
13	87/14
14	97/15
15	111/16
16	126/17
17 and larger	163/23

The previous arbitrary-rank estimates are unchanged. Actual nonsimple-cusp incidence multiplicities still multiply each factor cost by its link vertex count divided by that factor's order, as in the inherited factor-fiber theorem. The only new envelope entry is at rank 14.

Complete concentration and exact comparison

For a cusp of total rank d=n-1 with factor ranks r_i, put

```
H(t)=product_i(1+t+...+t^r_i), V=H(1),
G=sum_j [t^j]H(t)*min(j,d-j).
```

The normalized curvature term is (5G-dV)/(2V). Subtracting the sum of the costs in the table gives the ideal margin. The inherited convex-order lemma permits concentration (a,b) to (a-1,b+1), with 1<=a<=b, without increasing normalized curvature. The cost must simultaneously be preserved or increased; this is checked again for the altered envelope.

Low parts concentrate until one vanishes or the other reaches 12. Duplicate 13 parts may be replaced by 12 and 14, raising cost by exactly 4/105. Duplicate 15 parts may be replaced by 14 and 16, raising it by exactly 7/2040. Both values are strictly positive, although small. Two parts at least 18 concentrate until one reaches 17, preserving their constant cost. Each step strictly increases the bounded integer sum of squared parts at fixed total rank, so moves that recreate a middle rank cannot cycle.

Every partition therefore reduces to repeated 12,14,16,17, at most one 13 and one 15, at most one part from 1 through 11, and at most one part at least 18. The source enumerates every such partition of 74.

Ambient dimension	Reduced profiles	Minimum ideal margin	Finite slack
75	999	63/74698	95/2

The minimum is attained at (10,12,12,12,12,16). Every margin is an exact positive rational number; no floating tolerance is used. The finite coefficient is beta_75=74*70/8, so beta_75-8*75=95/2>0. The inherited global shape inequality also has strictly positive constant 5*(75-1)=370. Summing the retained local face payments and comparing them with these finite and ideal budgets gives a contradiction at dimension 75. The completed dimension-75 theorem excludes every dimension at least 76. Together they prove the stated ceiling 74.

At ambient dimension 74, the formal rank profile (1,12,12,12,12,12,12) has margin -168143/742586. Thus this envelope comparison does not give a dimension-73 bound. This profile is a scalar obstruction to the present argument; no polytope realizing it is asserted.

The separate compact arithmetic ceiling 21 and arbitrary compact ceiling 29 are unchanged. The unrestricted result makes the effective ambient ceiling for the compact-facet alternative 74; its older auxiliary proof still gives 79. No new finite-volume construction is claimed.

Verification status

Primary generation and a fresh separate-process saved replay both terminated exit0/PASS for the local interfaces and all 999 exact profiles. The local three-pair proof has passed generation, fresh replay and two internal mathematical/source reviews. The complete inherited overlap and map censuses retain their frozen validation records and are not rerun here. The separate standard-library scalar reconstruction agrees on every one of the 999 rows, the minimum, concentration increments and boundary. It uses a different enumeration and both polynomial and cumulative curvature calculations. Two complete internal source/mathematical reads of the assembled argument found no identified gap in the local interfaces, cost envelope, concentration, global inequality or inherited higher tail. Those reads did not repeat the inherited geometric censuses. Direct source bindings and Python syntax pass. Source, data and report are final/frozen.

All reviews are internal to the agent workflow. External mathematician refereeing and Lean or other proof-assistant formalization are not claimed.

```
python3 scripts/fresh_ideal5_piecewise_dimension74_20261017.py
python3 scripts/fresh_ideal5_piecewise_dimension74_20261017.py --verify
```

Source 133

Source path: `results/fresh_ideal5_piecewise_dimension74_20261017_review.md`

SHA256: 49f02d8ef688011275a029cb3dcb6c1e6548cc568e2fd0869308a9e96e638051

Independent review of the dimension74 comparison

The independent exact reconstruction and full written composition pass. The primary proof establishes dimension at most74 for every finite-sided, finite-volume real hyperbolic Coxeter polytope, without arithmetic, compactness or cusp-shape restrictions. The completed dimension75 theorem supplies all higher dimensions; the new comparison excludes dimension75.

The reviewer imports only standard-library arithmetic. Its certificate independently enumerates999 reduced profiles, compares every primary row and reconstructs the seven new literal rational cores. The complete inherited overlap and all-label supporting-white censuses remain explicit frozen inputs.

Independent scalar reconstruction

Only the affine-rank14 cost changes, from100/15 to97/15. The envelope is

```
r/2 for1<=r<=12;
87/14 at13;97/15 at14;111/16 at15;126/17 at16;
163/23 forr>=17.
```

The reviewer chooses optional low, middle and high ranks first, then recursively enumerates all multiplicities of12,14,16,17 at the remaining total rank. It imports no producer profile list. Sliding-window convolution forms H(t)=product_i(1+t+…+t^r_i), with V=H(1) and G=sum_j h_j min(j,d-j). A separate inclusion-exclusion calculation expands product_i(1-t^(r_i+1))=sum_l a_l t^l and obtains

```
G=floor(d/2)*V
  -2*sum_l a_l*binom(floor(d/2)-l+s,s+1).
```

The two values agree for every profile. Polynomial symmetry, coefficient sum and the absolute-deviation identity

```
(5G-dV)/(2V)=3d/4-(5/2)*E|sum U_i-d/2|
```

are also checked. All999 rows at total rank74 match the primary certificate, including every coefficient, curvature, cost and margin. The minimum ideal margin is63/74698 at(10,12,12,12,12,16). The finite slack is95/2 and the positive global term is370. At ambient74, the boundary profile(1,12,12,12,12,12,12) has margin-168143/742586; it is a limitation of this comparison, not an asserted polytope.

The inherited convex-order move(a,b) to(a-1,b+1) cannot increase normalized curvature. Low ranks concentrate to12 or zero at unchanged linear cost. Duplicate13 can become12 and14 with cost increase4/105; duplicate15 can become14 and16 with increase7/2040. Both increments remain strictly positive under the changed envelope. Two ranks at least18 concentrate until one reaches17 at unchanged constant cost. Every move increases the bounded integer sum of squared ranks, so the process terminates even if middle ranks are recreated. The terminal domain is exactly the one independently enumerated. The unchanged15 all-rank affine cost branches and their maxima are independently recomputed and matched to the frozen branch certificate.

Three removed pairs and seven complete finite outcomes

The full source and mathematical proof of the local affine-B reduction were read without an identified gap. It removes only(0,11),(11,13),(11,14) from the order15 affine B14 target. Independent breadth-first search gives distances11,12,12. The leaf swap preserves their union. Their complete saved marked-image support consists exactly of1284,1623,1641,1646,1666,1933,2326. All16 elementary-triangle inclusions remain valid, including the prior dimension75 singleton deletion in order17 affine D.

The reviewer reconstructs every local Gram from the literal L25 tree and the saved identifications, rather than importing a producer matrix helper. Rational congruence elimination and Gauss-Jordan inversion independently check the signatures, white and displayed-corner positive definiteness, marked components, contact projections and endpoint extensions. These small checks verify the seven listed outcomes; they do not reconstruct the complete all-label contact classification that supplies their candidate lists.

In six contexts the core has signature(16,1,0), and the preferred component is D14. Every allowed nonzero direct contact is at node3 or13, outside that component. No extra-white path can reach it without a first direct contact. The complement in the actual corner consists of supporting whites and is positive definite. Thus the entire corner is finite and D14 is whole, so the already paid whole-component rule supplies a unit.

In context1641 the core has signature(15,1,0), and the preferred component is A12. Only its two endpoints can receive relevant direct contacts. Each endpoint-contact root has projection square2. Its residual lies in the positive-definite perpendicular to the Lorentz core and has square zero, so the residual is exactly zero. It is therefore orthogonal to every core-orthogonal white. Against another allowed core-contact white, its actual product is a verified projection product: positive, or at most-2. The former violates acuteness for distinct actual walls; the latter violates strict positive definiteness of two supporting whites. Identical contact and zero residual fix the same normal, rather than another occurrence.

If no endpoint-contact root exists, A12 remains whole. If one exists, it gives a whole A13. Both exact extension matrices are checked. The remaining corner components are again positive-definite white blocks. This establishes actual finite corners, not merely finite subdiagrams inside an ideal corner. No additional finite coupon is required.

The argument uses the complete frozen no-extra-cross and all-label support domain for each literal core. The removal is not justified by an arbitrary abstract affine embedding. Basic, triangle and short8-first payments remain unchanged because the three omitted pairs have distance greater than9.

Global composition and validation

The complete finite8/ideal5 theorem and the dimension75 refinement retain all other local payments and the whole A4/D13 finite coupons. Since the new seven-case replacements use existing finite units, the finite budget remains 8n through80. At order15 the other type costs75,95,96 are below97, so the new rank14 envelope is valid. Actual nonsimple-cusp factor costs retain the link-vertex multiplicity divided by factor order; no simple-cusp hypothesis is introduced.

Every ideal profile at dimension75 has positive margin, the finite margin is95/2 and the inherited global shape inequality retains370. This contradicts the summed local face payments in dimension75. The completed dimension75 theorem excludes dimensions at least76, proving74. The compact arithmetic21 and arbitrary compact29 ceilings are unchanged. No new construction or dimension73 result is asserted.

Independent bound generation and separate fresh saved replay both terminated exit0/PASS. Every999-row scalar field and both comparison interfaces match the frozen primary. All25 direct bindings, syntax and4 local links pass. Root completed a full source and mathematical read of the independent scalar algorithms, all-rank branches, seven literal-core reconstructions, local interfaces, bindings and this entire report, finding no gap. That peer read did not duplicate the999-row arithmetic, seven rational checks or inherited censuses. Source, data and report are final/frozen; all frozen predecessors are unchanged.

This is an internal computer-assisted proof review. External refereeing, publication priority and Lean or other proof-assistant formalization are not claimed. The inherited2,614-context overlap and1,153-pattern embedding censuses are not rerun by this package.

Source 134

Source path: `results/fresh_ideal5_piecewise_dimension75_20261017.md`

SHA256: 760cad1955be1ad85680a5f294aa89a0a055d13aa25cc877fd2ca7892f267a16

Dimension at most75 from the D17 singleton transfer

Complete internally checked computer-assisted proof; validation scope is below.

Every finite-sided, finite-volume real hyperbolic Coxeter polytope has dimension at most75. No arithmetic, compactness, cusp-shape or facet-count restriction is imposed. The completed dimension77 proof supplies the higher-dimensional tail; the new comparison excludes76 and77.

The change is one pair in the17-node affine D factor. The singleton-pair transfer replaces it by two finite half-payments, while the enlarged finite budget retains cost8n through dimension80. The complete finite8/ideal5 theorem provides the rest of the face coverage. This composition uses the completed 2,614-context classification, not the older no-short-nine hypothesis.

Local change and its coverage

In the standard17-node affine D diagram, the pair(1,11) on the simple spine has distance10. It is fixed as an unordered pair by every diagram automorphism. Removing it changes the D cost from127 to126. The other order17 types retain costs85,111,108, so the worst normalized factor cost changes from127/17 to126/17. Every other rank cost is unchanged.

The complete1,153-pattern map census requires this pair in exactly one whole marked-core context, index2518 of the new2,614-context domain. Its positive affine kernel forces every extra supporting white to be right to the17-node affine component. The core radical then makes its contacts with the two remaining black roots equal. Nonzero contacts supply a paid path of length2. If they vanish, the entire core is sealed. Its two adjacent finite corners have whole A4 and D13 components, with the A4 mark an endpoint and the D13 mark the unique long-arm neighbor of the fork center. Each receives the new half-payment, giving total1.

This literal whole-core argument is necessary: an abstract affine embedding alone would not seal the finite components. The complete overlap theorem excludes every extra cross-core bond, and the local lemma handles all extra supporting whites. The new pair has distance10, so all elementary triangle payments and short-path reductions remain unchanged. The pre-existing removed D pairs retain their own sealed or universal transfers.

Finite counting and the ideal envelope

Each pair of whole A4 and D13 components contributes at most one additional unit: two A4 endpoints, one specified D13 node, half a unit each. If a is the total rank of small components and b is the number of D13 components, the extra cost is at most ab/4. Every D13 has base slack26 rather than the generic8 already charged in the finite8 proof. The resulting excess bound is the old bound plus b(a/4-18). When b>0 the rank constraint gives a<=67, making this correction negative; when b=0 it vanishes. Thus the full finite-vertex budget remains8n, including all old coupons.

The new normalized ideal-factor envelope, for affine rank r, is

```
r/2 for1<=r<=12;
87/14 at13;20/3 at14;111/16 at15;126/17 at16;
163/23 forr>=17.
```

Actual cusp-factor multiplicities are those of the inherited factor-fiber theorem: factor pair costs multiply by the link-vertex count divided by factor order. No simple-cusp assumption is introduced.

Exact global comparison

Use the inherited convex-order theorem: transferring(a,b) to(a-1,b+1), with1<=a<=b, cannot increase normalized curvature. Concentrate two ranks at most11 until one vanishes or the other reaches12; their linear budget is preserved. Replace duplicate13 by12 and14, increasing cost by5/21. Replace duplicate15 by14 and16, increasing cost by83/408. Concentrate two ranks at least18 until one reaches17; their constant cost is unchanged. Every move increases the bounded integer sum of squared ranks, so it terminates. Terminal profiles have arbitrary multiplicities of12,14,16,17 and at most one each of13,15,a rank1..11 and a rank at least18.

The checker enumerates every terminal profile in the two remaining dimensions. For `H(t)=product(1+t+...+t^r)`, put V=H(1) and `G=sum_j [t^j]H(t) min(j,d-j)`, where d is total affine rank. The inherited normalized curvature is `(5G-dV)/(2V)`. Subtracting the new envelope gives:

Ambient dimension	Profiles	Minimum ideal margin	Finite margin
76	1053	11615/448188	57
77	1111	41558/485537	68

Both ideal and finite margins are strictly positive, and the inherited global shape inequality retains the positive constants375 and380. Hence both dimensions are impossible. The77 theorem excludes every larger dimension, giving75.

At ambient75 the formal profile(12,12,12,12,12,14) has negative margin `-16873/171366`. This is a limitation of the comparison, not a realized hyperbolic polytope. This package makes no dimension74 claim.

Verification status

Primary generation and a fresh separate-process saved replay terminate exit0/PASS, including all2164 exact scalar rows and the local certificate interfaces. Two separate full source/mathematical reads pass the complete composition, including the new whole-component coupon, the short8-first reduction and the inherited higher-dimensional tail.

The separate standard-library-only independent reconstruction matches every primary scalar row, both minima and the boundary profile. It independently enumerates the reduced partitions and computes curvature using sliding convolution, cumulative inclusion-exclusion and an absolute-deviation identity. These checks do not duplicate the upstream 2,614-context classification,1,153-pattern embedding census or every inherited geometric proof; those retain their frozen validation records. All reviews are internal. External refereeing and proof-assistant formalization are not claimed. Source, data and report are final/frozen.

Compact arithmetic21 and arbitrary compact29 are unchanged. The unrestricted result also gives an effective compact-facet ambient ceiling75, while the older auxiliary compact-facet proof remains79.

```
python3 scripts/fresh_ideal5_piecewise_dimension75_20261017.py
python3 scripts/fresh_ideal5_piecewise_dimension75_20261017.py --verify
```

Source 135

Source path: `results/fresh_ideal5_piecewise_dimension75_20261017_review.md`

SHA256: 0437bf7ab88f1a50a115a5109630e479828b49caeb728f58955a49a5e0bd3ff4

Independent review of the dimension75 comparison

The independent exact comparison and complete written composition pass. The primary argument proves that every finite-sided, finite-volume real hyperbolic Coxeter polytope has dimension at most75, without an arithmetic, compactness or cusp-shape restriction. This review uses the completed dimension77 theorem for the higher-dimensional tail. It finds no open mathematical dependency in the new composition.

The checker imports only Python's standard library. Its certificate independently reconstructs all2,164 reduced scalar profiles, compares every primary row and checks the new local interfaces. It does not rerun the inherited2,614-context overlap or1,153-pattern embedding searches.

Independent arithmetic and complete reduction

The normalized ideal-factor budget at affine rank r is

```
r/2 for1<=r<=12;
87/14 at13;20/3 at14;111/16 at15;126/17 at16;
163/23 forr>=17.
```

The checker chooses the optional low rank1,…,11, optional13 and15 and optional rank at least18 first, then recursively enumerates all multiplicities of 12,14,16,17 at the remaining total rank. This differs from the primary's anchor-first loops and imports no producer list.

Sliding-window convolution forms H(t)=product_i(1+t+…+t^r_i). For d=sum r_i, V=H(1) and G=sum_j h_j min(j,d-j), a separate inclusion-exclusion calculation expands product_i(1-t^(r_i+1))=sum_l a_l t^l and obtains

```
G=floor(d/2)*V
  -2*sum_l a_l*binom(floor(d/2)-l+s,s+1).
```

Both exact values agree for every profile. The checker also verifies polynomial symmetry, H(1)=V and the absolute-deviation identity

```
(5G-dV)/(2V) = 3d/4-(5/2)*E|sum U_i-d/2|.
```

Subtracting the new ideal budget gives the following exact minima.

Ambient dimension	Profiles	Minimum ideal margin	Finite slack beta_n-8n
76	1053	11615/448188	57
77	1111	41558/485537	68

The first minimum occurs at(11,12,12,12,12,16), the second at(12,12,12,12,12,16). The ambient75 boundary profile(12,12,12,12,12,14) has margin-16873/171366. It is a formal obstruction to this comparison, not a realized polytope or a dimension74 theorem.

The inherited convex-order theorem permits transfer from(a,b) to(a-1,b+1) for1<=a<=b without increasing normalized curvature. Concentrating two ranks at most11 preserves their linear cost until one vanishes or the other reaches12. Replacing

duplicate13 by12 and14 raises cost by5/21. Replacing duplicate15 by14 and16 raises it by83/408. Concentrating two ranks at least18 until one reaches17 preserves their constant cost. Each move strictly increases the bounded integer sum of squared ranks, so even moves that recreate a middle rank cannot cycle. The terminal domain is exactly the independently enumerated one. Positivity there therefore covers every partition of the same total rank.

The checker additionally reconstructs the15 all-rank B/C/D affine cost branches and their exact ratio maxima160/23,75/11,163/23 from spine-window formulas. Each branch has cost/m=a+b/m, so its endpoints and limiting value5 exhaust its extrema. This is an all-rank calculation, not extrapolation from sampled ranks. All16 small-factor cost and elementary-triangle inclusion interfaces are checked.

The new local transfer

The full source and written proof of the D17 singleton transfer were read. Here D17 counts nodes: the standard affine type is D16. Removing only the automorphism-fixed pair(1,11) reduces its cost127 to126. Independent breadth-first search gives its distance10. The checker validates all eight saved automorphisms, the invariance of the new paid set and retention of all elementary triangle pairs; it does not claim an independent automorphism enumeration.

The complete saved induced-map data contain precisely one affected context, 2518, the two L25 copies marked{3,4} and identified at node9. The reviewer reconstructs its19-root Gram from the literal edge list using rational arithmetic. Independent congruence elimination gives signature(17,1,1), a positive-definite 15-root white diagram, four positive-definite16-root edge diagrams and the four stated corner signatures. Direct multiplication verifies the saved affine positive kernel and full-core radical. The two finite coupon corners have whole A4 and D13 components with the correct endpoint and long-arm-neighbor roles. These small checks are independent of the local producer's helpers.

The geometric handling of additional whites is essential. At the actual ideal corner containing the17-root affine diagram, its positive kernel and nonpositive contacts force every extra supporting white to be orthogonal to every affine root. No simplicity assumption is needed. The full-core Gram radical represents a vector orthogonal to a timelike core vector; its zero square in that positive orthogonal complement makes it an actual zero relation. Pairing it with an extra white equates that white's contacts with the two omitted roots.

If those equal contacts are nonzero, the opposite corner has a paid path of length2. Otherwise all additional whites are orthogonal to the entire core. Their Gram is positive definite as part of the actual face support, so the two displayed finite corners stay finite and their A4/D13 components stay whole. The two new half-payments then supply the required unit. This proves the transfer for arbitrary additional whites and nonsimple ideal vertices.

The literal whole-core hypothesis is supplied by the completed overlap theorem, which excludes extra cross-core bonds. An arbitrary affine embedding alone would not justify sealing the finite components. The proof and its saved-map interface retain this distinction. The distance10 omission changes no short8 reduction, elementary triangle payment or immediate small-factor payment.

Finite budget and global composition

The full source and proof of the new finite budget also pass the narrow review. A whole A4/D13 component pair adds one unit: two A4 endpoints, one specified D13 node and half-weight each. If a is total small-component rank and b is the number of D13 components, this is at most ab/4. Each D13 has unit-pair cost78 and slack26 against8r, supplying18 more than the previously reserved8 and consuming two additional ranks beyond the generic large-component bound11. Thus

```
E_new = E_old+b(a/4−18).
```

For b>0, the rank bound in dimensions at most80 gives a<=67, so the added term is at most-5b/4. For b=0 it vanishes. The old finite8 budget therefore continues to cover every old coupon and the new one. This argument was read in full; the3,465 endpoint enumeration was not independently rerun here.

The completed finite8/ideal5 theorem supplies all remaining triangle and quadrilateral payments, using the full 2,614-context classification rather than the old no-short9 domain. Only the one saved D17 pair changes. Its new transfer, all16 elementary triangle inclusions and existing removed-pair transfer interfaces preserve the required face payments. Actual nonsimple-cusp multiplicities multiply a factor cost by the link-vertex count divided by its order; no simple-cusp restriction enters.

The finite coefficient exceeds8n by57 and68 in dimensions76 and77. Every ideal profile has the positive margin above, and the inherited shape inequality retains positive global terms375 and380. This yields the contradiction in both dimensions. The completed dimension77 proof excludes the higher tail. The unrestricted conclusion also gives an effective compact-facet ambient ceiling75; the separate compact arithmetic21 and arbitrary compact29 results are unchanged.

Validation scope

Independent generation and a separate fresh saved replay both terminated exit0/PASS. All28 direct source/data/report bindings, syntax and all6 local report links pass. The checker independently recomputes every scalar row, the boundary, all-rank cost branches and the stated small local/interface checks. The full primary and local mathematical composition was read without an identified gap. A separate complete source and mathematical peer read passed the alternate enumeration, sliding convolution, cumulative identity, absolute-deviation check, concentration argument, independent inertia and literal core, saved-map and automorphism interfaces, all-rank branches and final bindings. It also read this full report. That peer read did not duplicate the2,164-row arithmetic or inherited censuses.

The inherited diagram classifications, full overlap and map searches retain their frozen validation records. They are not independently reconstructed by this package. All reviews are internal; external refereeing, publication priority and proof-assistant formalization are not claimed.

Source, data and report are final/frozen. No frozen predecessor was modified.

Source 136

Source path: `results/fresh_ideal5_piecewise_dimension77_20261017.md`

SHA256: ca1d068067d050d369a25f5a56aeaf9e3a7ba6224c0cc30b6124f66d2844a105

Dimension77 from the complete finite9/ideal5 payment rule

Completed internally checked computer-assisted proof. Every finite-sided, finite-volume real hyperbolic Coxeter polytope has dimension at most77, without arithmetic, simple-cusp or facet-count assumptions. The argument below combines complete local geometric proofs with an exact finite scalar comparison; it does not assume the uncompleted finite8 overlap classification.

The source and certificate bind the complete local theorem, its 571-pattern census and all-rank counts, and the completed dimension80 proof with its independent review. The local theorem gives finite cost at most9n for n≤80 and ideal cost at most V times the sum of the factor-rank envelope

```
b(r)=r/2 for1≤r≤12;
b(13)=87/14; b(14)=20/3; b(15)=111/16; b(16)=127/17;
b(r)=163/23 forr≥17.
```

The key small-factor change is the selective order14 rule. It retains87 pairs in affine B,84 in C,82 in D and70 in the cycle. These are complete pair sets with every actual preferred-component embedding included, except where a separate universal or sealed finite-coupon transfer replaces it. The combined proof checks that no elementary triangle payment or immediate small-affine payment is lost by these removals. The context-specific D removals at orders16 and17 have distances greater than9, so they are used only after the complete no-short9 classification; they do not create a circular short-path reduction.

Complete reduction of cusp-link shapes

For the actual link L=product_i Delta^(r_i), put d=sum_i r_i=n−1 and

```
H(t)=product_i(1+t+...+t^r_i)=sum_j h_j*t^j,
V=product_i(r_i+1),       G=sum_j h_j*min(j,d-j).
```

Its normalized curvature is (5G−dV)/(2V). The inherited convex-order theorem shows that a transfer(a,b)→(a−1,b+1), for1≤a≤b, can only decrease this curvature while all other ranks remain fixed.

The new envelope drops below the former linear cost at rank13, so it would be incorrect to concentrate all low parts up to13. Instead concentrate two parts strictly below12 until one vanishes or the other reaches12; their total cost is unchanged. Replace two13 parts by12 and14, increasing cost by5/21. Replace two15 parts by14 and16, increasing cost by107/408. Finally concentrate two parts strictly above17 until one reaches17; their constant cost is unchanged. Every operation can only decrease curvature minus cost.

Each operation strictly increases the integer sum of squared ranks at fixed total d, and this sum is bounded by d^2. Thus the process terminates even though several kinds of moves may be interleaved. Its terminal shapes have arbitrary copies of12,14,16,17; at most one13 and at most one15; at most one low part1–11; and at most one high part at least18. The source enumerates every such shape, so its finite comparison covers every integer partition of each relevant link dimension.

Exact comparison and the zero finite margin

The primary checker computes H,G,V by integer convolution and checks the same curvature through a separate cumulative inclusion-exclusion moment identity. The exact reduced-shape counts and minima are:

Ambient dimension	Profiles	Minimum normalized ideal margin	Finite slack beta_n−9n
78	1173	409611/2171546	0
79	1239	63437/167042	21/2
80	1310	3585185/7619196	20

Here beta_n=n(n−6)/8 for even n and (n−1)(n−5)/8 for odd n. The overall minimum occurs at ambient78 and ranks(1,12,16,16,16,16). All3,722 ideal margins are strictly positive.

The inherited link-shape inequality is

```
D5 ≥ beta_n*v_f + sum_cusps kappa(L) + 5(n-1).
```

The face-payment theorem gives an upper bound9n*v_f plus the ideal budgets. In dimension78 the finite coefficient is exactly equal to9n, so it must not be described as a strictly positive finite margin. The global term5(n−1)=385 is strictly positive even when there are no cusps; when cusps are present their margins are positive too. Dimensions79 and80 also have positive finite margins. The inequalities therefore contradict the existence of a polytope in every dimension78,79,80. The completed80 theorem excludes all larger dimensions. Together they give n≤77.

At ambient77, the formal profile(12,16,16,16,16) has margin −16664/1085773. This comparison does not exclude77. The profile is not a constructed hyperbolic Coxeter polytope or an assertion of sharpness.

The former auxiliary compact-facet argument still has its proved bound79; the unrestricted77 theorem now gives the stronger effective bound77 for that subclass. The separate compact dimension29 and compact-arithmetic dimension21 results are unchanged. No new finite-volume example is claimed.

Verification scope

The separate independent review records its reconstruction and mathematical scope. Primary full generation and a fresh separate-process saved replay both terminated exit0/PASS; syntax and all19 direct bindings also pass. The separate standard-library checker independently reconstructed and compared all3,722 rows, the three minima including their global slack, the boundary example, all15 affine branches and all16 small target pair sets with their triangle inclusions. Its final bound generation and saved replay are recorded in that separate review. A full source/mathematical read found no gap in the complete local composition, cost directions, terminating concentration, actual cusp multiplicities, zero finite margin at78 or inherited80 tail. The complete local binder also passed generation, fresh replay and separate full mathematical/source reads. Source, data and this report are final/frozen.

The proof imports the published finite/affine classification, the complete old minimal-core and overlap proofs, and the explicitly linked local geometry. Their arithmetic and source-review scopes are recorded in their own frozen packages. A scalar pass alone would not prove the theorem; the complete local composition is an essential input. All reviews here are internal. External mathematical refereeing, publication priority and proof-assistant formalization remain unestablished.

Source 137

Source path: `results/fresh_ideal5_piecewise_dimension77_20261017_review.md`

SHA256: fb8027c4e5a40431fa4cad5cb10e82dd2c56fa4aa0c4a34e70664297cd26003a

Independent review of the ideal-five dimension comparison

The full written composition and independent exact comparison pass: the argument proves dimension at most77 for every finite-sided, finite-volume real hyperbolic Coxeter polytope. No arithmetic or cusp-shape restriction is imposed. The checker imports only standard-library arithmetic. It independently enumerates3,722 reduced profiles in ambient dimensions78,79,80, computes curvature in two different ways, and derives15 all-rank affine cost branches. The certificate compares these results with the primary proof.

Scalar calculations

The envelope under examination is

```
b(r)=r/2 for1<=r<=12;
b(13)=87/14; b(14)=20/3; b(15)=111/16; b(16)=127/17;
b(r)=163/23 forr>=17.
```

The checker chooses the optional ranks1,…,11,13,15 and at least18, then recursively enumerates all multiplicities of12,14,16,17 at the remaining total rank. Sorted shapes are deduplicated; no producer list is imported. Sliding-window convolution computes

```
H(t)=product_i(1+t+...+t^r_i),
V=product_i(r_i+1),  G=sum_j h_j min(j,d-j).
```

A separate inclusion-exclusion calculation expands the numerator product_i(1-t^(r_i+1))=sum_l a_l t^l and uses

```
G=floor(d/2)*V-2*sum_l a_l*binom(floor(d/2)-l+s,s+1).
```

Both exact values agree. Polynomial symmetry, H(1)=V and the independent absolute-deviation identity for normalized curvature are checked for every shape. Curvature minus the proposed cost gives:

Ambient dimension	Profiles	Minimum margin	Finite slack beta_n-9n
78	1173	409611/2171546	0
79	1239	63437/167042	21/2
80	1310	3585185/7619196	20

The least margin occurs at(1,12,16,16,16,16). The ambient77 boundary shape (12,16,16,16,16) has margin-16664/1085773, and its finite slack is-9. This is a limitation of the proposed comparison, not an actual chamber.

Complete shape reduction

Use the inherited convex-order theorem: transferring(a,b) to(a-1,b+1) for 1<=a<=b cannot increase normalized curvature, with all other factors fixed. Two ranks at most11 can be concentrated until one vanishes or the other reaches12; their linear budget is preserved. The new rank13 cost is less than13/2, so the older reduction that concentrated through13 is not valid.

Instead replace duplicate13 by12 and14. This increases the budget by5/21. Replace duplicate15 by14 and16, increasing it by107/408. Concentrate two ranks at least18 until one becomes17; their constant budget is unchanged. Each move therefore cannot increase curvature minus cost. It strictly increases the bounded integer sum of squared ranks, proving termination.

Terminal partitions have arbitrary multiplicities of12,14,16,17, at most one rank1,…,11, at most one13, at most one15 and at most one rank at least18. The independent enumeration above exhausts that domain. Positivity for all terminal profiles proves positivity for every partition of the same rank.

Independent all-rank ideal counts

For factor order m>=18, the proposed ideal rule pays distance at most5 and distance6,…,9 when a mark is within5 of a branch or nonsimple-bond feature. The independent spine-interval formulas are

```
A: 5m;
B: 5m−6  +sum_{s=8,9,10,11} min(m−s,13);
C: 5m−15 +sum_{s=6,7,8,9} min(m−s,14);
D: 5m+3  +sum_{s=10,11,12,13} min(m−s,12).
```

For B the short-distance contribution is5m-14, and each of the four longer distances adds two fork-leaf pairs plus the spine window of width13. For D the short contribution is5m-13; four leaf pairs at each longer distance give the constant16 adjustment, and the spine window has width12. C has a path with two end feature windows of combined width14. Opposite fork ends are too distant to produce an extra leaf pair at m>=18.

Each min term has one breakpoint. The five branches per type have cost/m of the form a+b/m, whose extrema occur at endpoints or the limiting value5. Independent exact maxima are160/23 for B,75/11 for C and163/23 for D. This establishes the all-rank envelope from the stated rule, without extrapolating a finite graph sample. Small-order preferred-pattern coverage must separately justify the exceptional rank13,…,16 values.

Finite budget and composition obligations

The enlarged finite-eight budget is at most8n through n=80, including whole-small rank5, D11 and A17 coupons. Changing from finite-eight/leaf-nine to full finite-nine adds at most n: for a component of rank r>=18, the additional nonleaf distance-nine pairs number r-11 in types A/B and r-12 in type D. Smaller components were already fully paid. Thus the enlarged finite-nine rule costs at most9n through80.

At n=78 the finite coefficient equals9n, rather than strictly exceeding it. The inherited shape inequality nevertheless retains its strictly positive constant5(n-1)=385. Thus nonnegative finite slack and strictly positive ideal margins give the contradiction, even when the polytope has no ideal vertices. The positive global slacks at79 and80 are390 and395. The completed dimension80 theorem supplies the higher tail.

Complete local composition review

The full source and mathematical argument of the combined local theorem, the small-factor census and the primary comparison were read. The following interfaces close the geometric obligations; the scalar enumeration alone would not suffice.

The finite-nine rule contains the prior finite-eight triangle rule. At ideal factors of order14 through17, a separate breadth-first distance reconstruction checks that every elementary triangle pair remains in the saved paid set. The16 target costs agree with the stated rank envelope. No replacement paying a quadrilateral only once is mistaken for a triangle payment of two.

For a quadrilateral with a marked path of length at most9, the retained finite rule or the complete ideal-five local transfers supply a unit or sealed finite half-coupons. The order14 cycle uses its separate strict spectator-spectrum transfer. The universal affine-D13/D14 transfers have immediate small affine alternatives of orders12/13, both wholly paid. All other terminal alternatives are short paths or the listed whole finite components. Their source and proof reviews are recorded in the local packages.

After this reduction, the original no-short-nine hypothesis holds for the complete949-case classification. Its low-core supplement has component bound at most12, and the571 remaining large-bound patterns are exhaustively embedded with labels and nonedges respected. Orders16/17 D deletions have minimum distances11/12, checked independently here; thus they

never require their context-specific transfer before the no-short-nine reduction. After classification their affected context sets match the saved sealed-core proofs. Every nonzero terminal payment in those18 cases is finite, and a nonorthogonal supporting root supplies a retained length-two path. No unpaid whole affine order14 conclusion or cyclic ideal transfer is imported.

The all-rank spine counts and actual cusp-factor multiplicity V/(r+1) justify the ideal costs also at nonsimple cusps. The inherited shape inequality retains its positive global-extremum contribution, and the concentration moves preserve or increase costs while decreasing curvature. The old reduction through rank13 would fail with the new budget; the independently reconstructed12/14/16/17 reduction avoids that error.

No gap was found in this complete composition. It uses the finite-nine classification; complete finite-eight overlap coverage remains a separate open problem and is not needed for77.

Verification status

The complete read-only run, generation and fresh separate-process saved replay all terminated exit0/PASS, comparing every primary profile field, all three minima and the boundary profile. The checker independently reconstructed all15 infinite-order affine branches and checked the16 small-factor pair sets and126 finite-cost increment interfaces. A separate full source/mathematical read of this reviewer checked the enumeration, cumulative-tail identity, branch extrema, graph inclusions, no-short-nine reduction and positive global slack, finding no gap. That peer read did not independently rerun the3,722-row arithmetic or the571/949 censuses.

The independent checker verifies20 direct source/data/report bindings. It does not rerun the571-pattern map census,949-core classification, historical contact arithmetic or all preceding geometric proofs; those retain their explicit frozen validation records. The new composition has a separate full source and mathematical read. Source, data and report are final/frozen after direct binding, syntax and local-link checks. All reviews are internal; external mathematician refereeing and proof-assistant formalization are not claimed.

Source 138

Source path: `results/fresh_ideal5_small_factor_payments_20261017.md`

SHA256: 3782a3e81034e940c09ee23533304560ff27b58d3d535ea4681960852252b818

Complete small-factor census for the ideal5 rule

This package reconstructs every inherited preferred-component pattern that can occur in an affine factor of order14 or larger, and records every labelled embedding into the four classical targets of orders14–17. It also supplies the all-rank cost of the basic ideal5 rule. Local face coverage and the global dimension comparison are separate composition steps; no dimension claim is made by this census alone.

The source and certificate preserve the complete frozen distance9 classification. They do not substitute the still separate distance8 overlap domain.

Exact domain and pair sets

At a cusp, a feature means a vertex of degree at least3 or an endpoint of a bond with label at least4. The basic rule pays marked distance at most5, and distances6–9 when a mark is within5 steps of a feature. This rule is intrinsic under weighted-diagram automorphisms.

The complete949 high-core overlap cases have preferred marked components with proved extension bounds at most17. Exactly571 have bound at least14: 134 of bound14,177 of bound15,180 of bound16, and80 of bound17. Each pattern is reconstructed from its literal two cores, overlap map and preferred corner. For a target of order m, only patterns with bound at least m and order at most m are considered. The inherited explicit backtracking tries every unused target vertex consistent with all previous edges **and nonedges**. It therefore enumerates all induced labelled injections, not only one graph isomorphism or one orbit representative.

The paid set is the union of the basic rule and every marked image, followed by precisely these validated removals:

- In cycles, remove distances greater than5. Order14 uses the new cycle14 transfer; orders15–17 use the ideal5 local theorem, including its sealed cycle16 coupon.
- In affine D, remove the nine pairs across the two fork triples. Orders14 and15 use the universal D13 and D14 transfers. Orders16 and17 use the previously proved literal sealed-core replacements in order16 and order17. The checker matches the complete affected case-index sets to those saved proofs, not merely the number of removed pairs.

The exact retained costs are:

Factor order	A cycle	B	C	D
14	70	87	84	82
15	75	100	95	96
16	80	107	103	111
17	85	111	108	127

All emitted maps, marked images, basic pairs, additional pairs, removed pairs and final pairs are saved. Thus the table is accompanied by literal pair sets rather than only scalar totals. Exceptional affine types have order at most9 and remain wholly paid in the intended composition.

All-rank basic cost

For every classical affine factor of order m≥18, the exact basic costs are

Family	Cost
A	5m
C	5m−15 + sum of min(m−k,14), k=6,7,8,9
B	5m−6 + sum of min(m−k,13), k=8,9,10,11
D	5m+3 + sum of min(m−k,12), k=10,11,12,13

Here is an arbitrary-rank derivation. A cycle has no features and exactly m pairs at each distance1–5. For a tree, use a simple spine with M vertices. At distance at most5 the spine contributes5M−15. A spine pair at distance k has M−k possible starting positions. Paying the first a and last b positions of a list of length L selects exactly min(L,a+b) positions, including when the endpoint intervals overlap.

For C, M=m and the doubled terminal bonds make the first two and last two vertices features. The radius5 neighborhoods reach the first seven and last seven spine vertices. At each distance k=6–9, the paid starting positions are the first seven and last seven, giving min(m−k,14).

For B, M=m−2, with two short leaves at the left end and a doubled bond at the right end. The left branch neighborhood contributes the first six starting positions; the right doubled end contributes the last seven. Thus the spine extras are min(m−k−2,13), k=6–9. The two leaves contribute ten pairs at distances1–5 and their mutual pair, changing the short count to5m−14. They also contribute two pairs at each of distances6–9, all within one step of the branch. Adding eight gives the displayed5m−6 constant.

For D, M=m−4, with two leaves at each end. Each branch gives six starting positions, so the spine extras are min(m−k−4,12), k=6–9. The four leaves contribute twenty short leaf-spine pairs and two same-end leaf pairs, giving5m−13. Their sixteen extra leaf-spine pairs change this to5m+3. Opposite-end leaf pairs have distance M+1≥15 and do not contribute. These arguments cover every m≥18 and do not extrapolate from a finite sample.

The four minima change one at a time, so each tree family has five linear branches, of slopes9,8,7,6,5. C changes at orders20–23; B at21–24; D at22–25. The certificate saves all15 branches and their exact endpoint comparison with163m/23. The final slope5 tails have increasing slack. The respective normalized maxima are75/11 for C,160/23 for B, and163/23 for D; the common bound163/23 is attained by D at order23. The cycle bound5 is smaller.

Consequently the intended factor-rank envelope, writing r=m−1, is

```
r/2 for 1≤r≤12;
87/14 for r=13;
20/3 for r=14;
111/16 for r=15;
127/17 for r=16;
163/23 for r≥17.
```

This is a counting envelope. Its use for actual cusp incidences additionally requires the factor multiplicity theorem and complete face-payment transfer.

Verification scope

Generation passed the complete571-pattern embedding census, all16 small target pair sets, exact correspondence of the order16/17 removed contexts, 248 breadth-first graph checks and all15 infinite-tail branch comparisons. A fresh separate-process saved replay also terminated exit0/PASS. The finite graph checks corroborate the written spine proof. A separate complete source/mathematical read found no gap in the inherited domain filter, all-edge/nonedge backtracking, pair unions, exact deletion-case correspondence or all-rank spine counts. That review independently derived the15 branches and their maxima, but did not duplicate the571-pattern map census. Syntax, direct source bindings and report links pass. Source, data and report are final/frozen; no claim of external refereeing or formalization is made.

Source 139

Source path: `results/fresh_ideal5_weighted_dimension65_20261017.md`

SHA256: f9f2e6c5ae9baf977d37d296978a1ca523b2c63ef8e2519b5d420c7788edcabf

Unrestricted finite-volume dimension at most65

Every finite-sided, finite-volume real hyperbolic Coxeter polytope has dimension at most65. The new argument excludes dimension66; the completed ceiling66 theorem excludes all higher dimensions. Arithmeticity, compactness, simple cusps, a bound on the number of cusp factors and a bound on the number of facets are not additional hypotheses.

The source and certificate assemble the local payment theorems, complete exact finite budget and all1,659 ideal comparisons. Final verification status is recorded below.

Simultaneous local payment rules

Start with the completed ceiling66 rule, including the finite rank17 costs A114/B105/D127 and all its existing cross-component coupons. Apply these three refinements:

1. The large finite D theorem halves the two short-fork-leaf distance9 payments in a whole D_n, n>=20, saving1. It adds a half-coupon between any mark of a whole finite component of order at most5 and a short fork leaf of a separate whole D10.
2. The order14 B theorem deletes two further redundant ideal pairs, lowering B86 to84. Every affected actual factor would require a new doubled-terminal root whose norm-two projection exceeds2. No finite payment changes.
3. The order12 B/C theorem removes B's two remaining halves and C's two remaining nonbasic units, giving B64 and C63. Upgrade the existing whole A2/D11 short-fork-leaf halves to units. This adds2 per A2/D11 component pair; the already full A1/D11 payments remain full.

These refinements have compatible complete provenance. All finite short8 units and all protected ideal units at distances at most9 remain. The large-D theorem's affected high-overlap routes are fully partitioned: 2,531 whole components of order at most17,18 protected simple leaves, nine adaptive simple-leaf cases,40 cases with two protected halves, and16 cases with the new small-component/D10 coupon. Exact residual bounds control whole supporting components, including indirect bridges. Its universal cycle D leaf-nine alternatives have order at most19 and are unchanged.

The order12 B/C proof reconstructs all758 eligible high patterns, all28 affected B maps and four C maps,14 literal B geometric interfaces and16 finite-payment variants. Radical or zero-residual sealing makes the required finite components whole. The strengthened A2/D11 and existing A1/D11 payments give finite totals1 or2 even after the ideal B half is removed. Both affected C contexts already have whole finite B11 components; their missing-terminal projections also give the direct contradiction16>2. The universal immediate B alternative is excluded by the inherited strict determinant gaps. Low, short-path, protected and all later interfaces are checked at their actual outputs.

Neither B/C proof consumes the new D10 coupon or a large-D half. Conversely, the large-D proof never needs a removed distance10 B/C ideal pair. The order14 projection proof introduces no replacement payment. Thus the three theorems apply simultaneously: every triangle receives at least2, every quadrilateral at least1, and all weights remain nonnegative.

Exact finite maximum495, including both coupon changes

The D10/large-D budget and its full A2/D11 successor count the actual simultaneous rule. The successor is an abstract counting theorem with an explicit geometric hypothesis; the completed B/C theorem above supplies that hypothesis here.

Finite type domination retains D10 through D16 and coupon-bearing A17 as large components, with A1 through A9 and D17 onward as residual types. D10 replaces B10: its new rank2 interaction is already larger than the old A2/B10 coupon, which is not added again. A17 remains separate from D17. Existing A11 coupons and the A1/D11, A1/D12 and A4/D13 upgrades are preserved.

Write a=sum_(s=1..5) s*a_s, let d_r count D_r components, let u count A17 components, and put t=d_11+…+d_16. The exact doubled cross cost is

```
a*(2*d_10+3*t+2*u)
  +2*a_1*(d_11+d_12)+2*a_4*d_13+4*a_2*d_11.
```

The last term is precisely the new A2/D11 increment. All1,608 large multisets through rank75 are covered. Their357 distinct residual tables have860,013 Bellman upper alternatives and26,775 attaining predecessors. All22,504 global alternatives and76 typed attaining forests are checked. Generation uses an item-first algorithm; replay checks every last-item upper inequality and a predecessor attaining equality. This proves the exact maxima without importing the geometric producers' arithmetic.

Every maximum through75 remains unchanged after the A2/D11 upgrade. At66 the maximum is495, attained by D66. Its direct graph count is493 short8 pairs plus total leaf-nine weight2. The finite curvature coefficient is66*(66−6)/8=495. Zero finite slack suffices because the global inequality has a strictly positive additive constant.

Actual cusp weights and corrected complete concentration

For an affine factor of rank r, hence order r+1, the normalized cost is bounded above by c(r):

Rank r	c(r)
1 through10	r/2
11	16/3
12	75/13
13	6
14	32/5
15	107/16
16	120/17
17 and larger	163/23

Only the order12 and order14 maxima change: their A/B/C/D costs are 60/64/63/62 and70/84/84/82, respectively. Other envelopes are inherited. For a cusp of link dimension d and factor ranks r_i, set

```
H(t)=product_i(1+t+...+t^r_i), V=H(1),
G=sum_j [t^j]H(t)*min(j,d-j).
```

The actual factor-fiber theorem gives V/(r_i+1) copies of each factor pair, including nonsimple cusps. The normalized ideal margin is

```
(5G-dV)/(2V) - sum_i c(r_i).
```

Use the inherited all-rank convex-order theorem: concentrating two ranks decreases normalized curvature. For a margin reduction, the total cost must also stay the same or increase.

Low ranks concentrate only up to10, preserving their linear cost. Two ranks at least18 concentrate until one reaches17, preserving their constant cost. Two rank13 parts may become12 and14: the cost increase is11/65. Every move increases the bounded integer sum of squared ranks, so the process terminates.

The old duplicate12 move is invalid: its cost increment is now−8/39. Consequently the complete terminal domain retains arbitrary copies of 10,11,12,14,15,16,17, with at most one13, one part1 through9, and one part at least18. No old terminal list is reused after its proof ceases to apply.

At ambient66, d=65. All1,659 terminal profiles have strictly positive margin. The exact minimum is

13367/495924,

attained at(10,11,12,16,16). The primary computes every integer polynomial, V and G and every rational margin. It also compares each row with the previous conditional scalar diagnostic; the new B/C theorem and finite budget now supply that diagnostic's missing geometric and finite-cost obligations.

Strict contradiction and scope of the bound

Let D5=sum_(two-faces)(5−number of sides), and let v_f count finite vertices. Face coverage gives

D5 <= 495∗v_f + sum_(ideal v) V_v∗sum_i c(r_vi).

The shape inequality gives

D5 >= 495∗v_f + sum_(ideal v) (5G_v−65V_v)/2 + 325.

Every ideal difference is positive, the finite difference is zero, and the global constant325 is positive. These inequalities contradict each other, including the all-finite and all-ideal cases. Dimension66 is therefore impossible. The completed ceiling66 excludes every dimension at least67, proving the ceiling65.

At ambient65 the new comparison still has two obstructions. Its finite maximum is487 at D65, against curvature480. Among1,528 terminal ideal profiles,16 have negative margin; the minimum is−78444/485537 at (12,12,12,12,16). These are formal comparison obstructions, not realized polytopes or an optimality claim. No ceiling64 is asserted.

Compact arithmetic remains at most21 and arbitrary compact Coxeter at most29. The effective compact-facet ambient ceiling becomes65, while the older auxiliary argument remains79. No new finite-volume example or construction dimension follows from the proof.

Verification status

The local theorems and both finite-budget packages have completed generation, fresh replay and separate full internal mathematical/source reviews. The large-D local proof also has an independent16-core rational arithmetic/interface check. Their inherited all-label classifications and arbitrary-support geometry retain their explicitly recorded scopes.

Primary generation and a separate fresh saved replay both pass. Full internal source and mathematical composition reviews find no unresolved step. An independently implemented verifier agrees with all1,659 primary scalar rows and all1,528 lower-boundary profiles. It also reconstructs the full1,608-vector/357-table finite upper-and-attainment certificate and checks290 supplied B/C maps, 16 finite-payment variants and four C projection bounds. It imports no project arithmetic; original geometric-search completeness and all-rank structural theorems retain their separate recorded reviews. Its saved certificate and replay status are recorded in that report.

This is a complete, internally checked computer-assisted proof. The internal reviews were performed within the agent workflow. No external refereeing, proof-assistant formalization or publication-priority claim is made.

```
python3 scripts/fresh_ideal5_weighted_dimension65_20261017.py
python3 scripts/fresh_ideal5_weighted_dimension65_20261017.py --verify
```

Source 140

Source path: `results/fresh_ideal5_weighted_dimension65_20261017_review.md`

SHA256: df9b7dc16a598a49ba2461a67c26ec91a1e62ee47a3381b6a2160ee636714818

Independent scalar and composition review for ceiling65

The independent rational reconstruction agrees with all1,659 required ideal profiles: every margin is positive and the minimum is 13367/495924 at(10,11,12,16,16). The independent finite-certificate check gives the exact maximum495 at rank66, attained by D66. The complete source and written composition read has found no gap. Final input bindings, generation and a separate fresh saved replay all pass. This independent review is complete and frozen.

Reviewer source, saved certificate and the primary proof use different scalar enumeration and polynomial algorithms. The reviewer imports no project or producer arithmetic. The original geometric censuses and all-rank structural theorems remain separately bound inputs.

Independent scalar arithmetic and concentration

The new cost function is r/2 through rank10, then

```
c(11)=16/3, c(12)=75/13, c(13)=6,
c(14)=32/5, c(15)=107/16, c(16)=120/17,
c(r)=163/23 for r>=17.
```

The order12 A/B/C/D costs60/64/63/62 and order14 costs70/84/84/82 give the two changed entries. All other entries are checked against the preceding completed rule. The unchanged constant high-rank envelope remains an inherited all-rank result.

The concentration argument has the correct direction. The inherited convex-order theorem says that spreading two rank summands decreases normalized curvature. Low ranks are spread only until one vanishes or the other reaches10, preserving their linear cost. Two ranks at least18 are spread until one reaches17, preserving constant cost. Two rank13 parts become12 and14, increasing cost by11/65. Each operation increases the bounded integer sum of squared ranks, so the reductions terminate. The old duplicate12 move would decrease cost by8/39 and is not used.

The reviewer enumerates the optional low part, optional13 and optional high part first, then independently recurses over multiplicities of 10,11,12,14,15,16,17. The primary instead recurses directly over sorted rank summands. The two domains must agree exactly, not just in size or minimum. At link dimension65 there are1,659 profiles.

For each profile the reviewer builds the integer h-polynomial by a sliding-window convolution. It checks its symmetry, degree and value V at1, and calculates G by summing coefficients against min(j,d−j). A separate inclusion-exclusion cumulative-composition identity computes the same G:

```
G=floor(d/2)*V
  -2*sum_shift a_shift*binom(floor(d/2)-shift+s,s+1).
```

Here the a_shift are coefficients of product(1−t^(r_i+1)), and s is the number of factors. The absolute-deviation formula

```
kappa/V=3d/4-(5/2)*E|sum U_i-d/2|
```

provides a third check. Every polynomial, vertex count, G, curvature, cost and margin is compared with the final primary data.

At the lower boundary n=65 the independent domain has1,528 profiles; 16 have negative margin. Its minimum is−78444/485537 at (12,12,12,12,16). The finite maximum487 also exceeds curvature480. These are comparison obstructions, not claimed polytopes or an optimality theorem. No ceiling64 follows here.

Local interfaces and simultaneous payments

The full mathematical/source read covers the order12 B/C transfer, order14 B projection transfer, and large finite D transfer, together with their application in the primary composition.

The new B/C rule removes only distance10 ideal pairs; the B14 rule removes distance11 pairs. Entire elementary/basic units and all short8 units remain. Independently constructed B12/C12 graphs reproduce the 64/63 retained basic pair sets. The reviewer checks every edge and nonedge of all290 supplied successful B/C induced maps; it does not re-enumerate the758 eligible-pattern search or claim a new proof that no other maps exist. B14's77 saved elementary triangle pairs remain inside its84 unit pairs.

All16 B finite-payment variants are independently recounted from their actual saved whole components and marked roles. Eight total1 and eight total2 after the preferred ideal half is removed. Twelve use the new A2/D11 upgrade and four the existing A1/D11 upgrade. The four C missing-terminal entries independently give the native norm-two bound 2*s_i*J_ii=16>2. These checks retain the distinction between a root's physical normalization and its rational gauge. Original corner inertia, whole-component sealing and arbitrary supporting-root completeness are inherited geometric proofs, not inferred merely from these totals.

The large-D theorem changes short-fork-leaf distance9 payments only in whole finite D_n with n>=20. Its selected high-overlap routes partition all2,614 cases; the low and disjoint routes are separately covered. Universal cycle D leaf-nine alternatives have order at most19. The remaining protected alternatives supply two halves or a half plus the new whole small-component/D10 coupon. A separate earlier review in this session independently reconstructed all16 full-core inverses, end powersets, residual tables and D10 cuts; that arithmetic is not repeated by this scalar reviewer.

The B/C proof uses whole finite components of order at most13 and no new D10 coupon or large-D half. The large-D proof uses retained short8 or protected ideal units, not either deleted distance10 B/C pair. The B14 proof excludes an actual completion and creates no replacement payment. The earlier rank17 finite deletions do not remove any of the small finite alternatives used here. Thus no replacement chain loses its payment or becomes circular. Triangle coverage remains at least2 and quadrilateral coverage at least1 under the simultaneous rule.

Independent finite upper bound and attainment

The full A2/D11 finite budget is conditional as a counting theorem. The completed B/C local theorem supplies exactly its extra geometric hypothesis: four marked A2/D11-short-leaf weights rise from one half to one, adding2 per component pair. The doubled increment is4*a2*d11.

The finite-budget source/report read checks all old coupons, including A11, A1/D11, A1/D12 and A4/D13. D10 replaces B10 and already dominates its A2 interaction; the old A2/B10 cost is not added to D10 again. Coupon-bearing A17 remains separate from D17. The finite-type classification and the successive same-rank domination argument remain reviewed inputs.

The independent verifier reconstructs all1,608 large-component count vectors by consecutive-rank count recursion, distinct from the producer's component-number multiset enumeration. It computes each residual cost from explicit component-pair coupons and compares every compressed interaction table. For all357 tables it checks860,013 last-item upper inequalities and26,775 attaining predecessors. It checks all22,504 global combinations and independently recounts all76 typed attaining forests. No optimizing routine is called.

The saved inequalities imply an upper bound by induction on residual rank; each attaining predecessor proves the reverse inequality. The complete global alternatives and typed attainers establish exact maxima, not only upper estimates. The looser B17 bound is harmless because its representative D17 is an actual attaining type. A direct D66 graph count finds493 short8 pairs and three leaf-nine pairs, two of them halved, giving495.

Global conclusion and verification limits

At ambient66 the finite curvature coefficient and maximum both equal495. Every ideal profile has positive margin and the global shape inequality has additive constant325. Consequently zero finite slack still gives a strict contradiction, including the all-finite case. The actual cusp-fiber multiplicity V/(r+1) applies also to nonsimple cusps. The completed ceiling66 excludes dimensions at least67. The local theorems, finite budget, ideal comparison and higher-dimensional tail therefore compose to the stated ceiling65, with their final bindings and replays complete.

This review is independent scalar and certificate arithmetic plus a complete source/mathematical composition read. It is not a duplicate all-label geometric census, a new proof of the convex-order or cusp-fiber theorems, external refereeing or proof-assistant formalization. It gives no new finite-volume construction.

Final primary-bound generation and a separate fresh saved replay both terminated with exit0 and PASS_INDEPENDENT _DIMENSION65_SCALARS_AND_INTERFACES. All recorded source bindings, Python syntax and local report links pass. The parent independently executed the comparison-only mode against the final primary data, including the full finite upper/attainment checks; that run also terminated with exit0/PASS.

The parent and the order12-transfer author separately read the complete reviewer source and report and found no mathematical or scope issue. Those peer reads did not duplicate the arithmetic. Their review covers the optional-first scalar domain, local component/payment interfaces, finite certificate induction and attainment, and the noncircular global composition. No jobs remain live for this package.

Source 141

Source path: `results/fresh_ideal5_weighted_dimension66_20261017.md`

SHA256: cc2aa3c33e4acaa47acd5c8086a74de912fa7ffcf9389fd18c09851cc0baf497

Unrestricted finite-volume dimension at most66

Every finite-sided, finite-volume real hyperbolic Coxeter polytope has dimension at most66. The new proof excludes dimensions67 and68 by reducing unnecessary long-pair payments in whole finite rank17 components. The completed dimension68 theorem excludes all higher dimensions. There is no arithmetic, compactness, simple-cusp, cusp-factor or facet-count restriction.

The source and certificate assemble the local theorem, exact finite budget and both complete ideal comparisons. The final verification scope is recorded below.

Reducing finite payments without changing ideal weights

The complete local theorem changes only same-component unit payments on whole finite A17,B17,D17. Their costs become114,105,127, down from135,136,136. The old A17 endpoint deletion remains in force. All finite pairs at distance at most8, and all distance9 pairs with a marked leaf of the whole component, remain. Every cross-component coupon and every ideal pair weight is unchanged.

These deletions have complete face provenance, not merely an absence in a selected list. The entire2,614 high-core overlap domain splits into 2,391 cases with whole component bound16,74 protected cases,9 adaptive cases and140 bound17 cases. Every induced finite rank17 map is enumerated for the latter:48 A maps,18 B maps and168 D maps. Their marked pairs are retained. Native labels are recovered from the scale-invariant quantity 4*g_ij*^2*/(g_ii*g_jj), so a rational root gauge cannot change the type. Low-core, disjoint-core and universal short-path branches are separately covered by their complete inherited classifications and proofs.

The local checker also reconstructs174 complete old and later literal configurations and applies all three new rules simultaneously. Their totals remain at least1. The sole debit is the already-removed A17 endpoint in old case662; both finite D17 alternatives remain, totaling2. Thus the D17 change preserves the very payments used to justify the earlier A17 deletion.

Universal transfer proofs are inspected at their actual conclusions. Their unbounded-spectator alternatives use finite short paths, distance9 at a whole-component leaf, explicitly sealed small components, or unchanged cross-component coupons. The long-cycle B/D leaf claims follow from inverse-diagonal sealing; a possible extension of an A endpoint instead forces the positive spectator correction already in that proof. Later rank17 literal units are explicitly retained. A former permissive "whole finite component at most17" phrase is never silently imported after deleting one of its units.

Consequently triangles retain at least2 and quadrilaterals at least1. There is no new coupon cost and no change in actual cusp multiplicity.

Exact finite maxima for the actual new rule

The full variant of the rank17 finite-forest calculation assumes A17 base114, D17 base127 and B17 base at most127. The local theorem now supplies those hypotheses, with B17 actually105. The other finite types and all coupons are unchanged. The conditional budget is therefore applicable to this actual face-payment rule.

The type reduction retains A17 separately because it has coupon roles; non-coupon B17 is dominated by D17. The two A11 cycle coupons remain included through the unchanged A11-to-D11 domination theorem. There is no double counting of a

half that was upgraded to one.

Enumerating every1,608 large-component count vector through rank75 leaves 675 additive Bellman tables. Each table gives exact upper bounds with attaining predecessors, and every global combination is checked. The separate standard-library verifier imports no producer arithmetic. For the full variant it independently checks the type/coupon domination, every count vector,1,626,075 Bellman upper alternatives,50,625 attaining predecessors,22,504 global upper alternatives and all76 attaining forests including rank0.

At ambient67 the exact finite maximum is504, attained by the single component D67. At68 it is512, attained by D68. These equal8n−32 and are strictly below the corresponding curvature coefficients. The old4D17 obstruction has disappeared; its new cost is508 at rank68.

The ideal rule is unchanged

For affine rank r the normalized factor cost remains

r	c(r)
1 through10	r/2
11	65/12
12	75/13
13	43/7
14	32/5
15	107/16
16	120/17
17 and larger	163/23

At a cusp with link vertex count V, the actual factor-fiber theorem multiplies a factor's pair cost by V/(r+1), even at nonsimple cusps. For d=n−1 and factor ranks r_i, put

```
H(t)=product_i(1+t+...+t^r_i), V=H(1),
G=sum_j [t^j]H(t)*min(j,d-j).
```

The ideal margin is(5G−dV)/(2V) minus the sum of c(r_i). The all-rank convex-order lemma and the completed68 concentration argument remain valid without change: keep arbitrary copies of10,11,13,14,15,16,17; at most one12; at most one part between1 and9; and at most one part at least18. The duplicate12 move raises cost by23/1092. Low-rank concentration stops at10, not11. Every move increases a bounded integer squared-rank sum and decreases the margin.

Both complete terminal domains are freshly enumerated and evaluated:

Ambient dimension	Profiles	Minimum ideal margin	Minimizing ranks
67	1,761	851/165888	(11,11,11,11,11,11)
68	1,906	411499/3234816	(11,11,11,11,11,12)

All3,667 margins are positive. The primary uses the frozen exact68 polynomial implementation; a separate global reviewer reconstructs the domains and arithmetic independently.

Two strict contradictions and the higher-dimensional tail

Let D5 sum5 minus the number of sides over all two-faces, and let v_f count finite vertices. Face coverage gives

```
D5 <= M_n*v_f + sum_(ideal v) V_v*sum_i c(r_vi).
```

The inherited shape inequality gives

```
D5 >= beta_n*v_f + sum_(ideal v) (5G_v-dV_v)/2 + 5(n-1).
```

The exact coefficients are:

n	M_n	beta_n	Finite slack	Global additive constant
67	504	1023/2	15/2	330
68	512	527	15	335

Every finite and ideal difference is positive, and the global constant is positive. The two inequalities contradict one another in each dimension, including when all vertices are finite or all are ideal. The completed ceiling68 excludes n>=69, so no dimension at least67 is possible. This proves the ceiling66.

At ambient66 this same comparison still fails in two separate ways: D66 has finite cost496 against curvature495, and the formal cusp profile (13,13,13,13,13) has ideal margin−1565/10976. These are comparison obstructions, not realized polytopes or an optimality claim. No ceiling65 is asserted.

Compact arithmetic remains at most21 and arbitrary compact Coxeter polytopes at most29. The effective compact-facet ambient ceiling is66; the older auxiliary proof remains79. No finite-volume construction or new example dimension follows from this proof.

Verification scope

The local deletion theorem has completed generation, fresh replay and two full internal mathematical/source reviews, including the universal finite17 interfaces. Its complete inherited all-label censuses and supporting-root sealing proofs retain their distinct verification scope. The finite-budget calculation has a separate exhaustive certificate checker. Primary global generation and a fresh separate-process replay both terminated exit0/PASS. The independent global reconstruction agrees with every one of the3,667 polynomial, vertex-count, curvature, cost and margin rows, using its own optional-first enumeration, sliding convolution, cumulative-composition and absolute-deviation identities. It independently re-enumerates all234 finite rank17 selected induced maps, reconstructs the canonical paid sets, recomputes174 saved simultaneous literal totals and directly counts the attaining D67/D68 pair costs. The reviewer report records final bound generation and replay.

Separate full internal source/mathematical reviews passed the conditional budget's application to the actual local rule, the distinct A17 coupon role, unchanged A11 domination, cusp multiplicities and concentration, both strict global contradictions, and the higher-dimensional tail. The negative boundary66 profiles remain correctly distinguished from actual realizations. No identified proof obligation remains in this composition. Source, data and report are final/frozen.

No external refereeing, proof-assistant formalization or publication priority is claimed.

Source 142

Source path: `results/fresh_ideal5_weighted_dimension66_20261017_review.md`

SHA256: d4eed0fca8bf200a819c520f5bf02847e6860209d0541272dfc3f2049da12f1f

Independent reconstruction of the dimension66 comparison

The dimension66 primary proof passes independent reconstruction of its complete ideal arithmetic and the newly changed finite rank17 interfaces. The reviewer imports only the Python standard library; it imports no primary or project arithmetic.

Independent source and saved certificate.

Independent complete ideal domains

The primary enumerates sorted partitions by adjoining one part at a time. This reviewer instead chooses the optional low part0 through9, the optional rank12 part, and the optional high part at least18 first. It then enumerates multiplicities of the repeated ranks10,11,13,14,15,16,17 to fill the remaining rank. The categories are disjoint, and duplicate profiles are explicitly forbidden. Thus this is a separate enumeration of every terminal profile in the proved concentration domain.

For each profile, the reviewer multiplies the uniform polynomials using a sliding window. The primary uses direct nested integer sums. Every coefficient is compared, along with the total link vertex count V and symmetry. The curvature is reconstructed in three mutually checked forms:

```
G = sum_j h_j min(j,d-j),
G = floor(d/2)*V - 2*sum_t a_t binom(floor(d/2)-t+s,s+1),
kappa/V = 3d/4 - (5/2)*E|sum U_i-d/2|.
```

Here s is the number of factors and a_t are the coefficients of product_i(1-z^(r_i+1)). The second identity sums the cumulative bounded composition formula by the hockey-stick binomial identity. Binomial terms outside their natural domain are zero. All arithmetic uses integers and exact fractions, with no floating-point comparisons.

Every primary profile polynomial, G, normalized curvature, factor cost and margin agrees with the independent reconstruction:

Ambient dimension	Profiles	Minimum ideal margin	Minimizing ranks
67	1,761	851/165888	(11,11,11,11,11,11)
68	1,906	411499/3234816	(11,11,11,11,11,12)

All3,667 margins are strictly positive. The independent source also checks the duplicate12 cost increment23/1092, the linear low-rank cost through10 and the constant high-rank cost from17. The all-rank convex-order proof is inherited, as is the actual cusp-factor multiplicity. The changed finite rule leaves both that ideal envelope and the complete terminal domain unchanged.

Finite rank17 interfaces reconstructed separately

The reviewer constructs the canonical finite A17, B17 and D17 labelled graphs directly. It independently counts their short8 and actual-leaf9 basic pairs and reconstructs the prescribed retained rules of costs 114,105,127. It computes all diagram automorphisms and checks invariance of the removed sets.

For all140 eligible contexts from the complete2,614-context overlap certificate, it recovers native labels from the original positively gauged Gram matrices using4*g_ij^2/(g_ii*g_jj). It then independently enumerates induced labelled maps using a fixed breadth-first pattern ordering. This differs from the producer's adaptive next-vertex choice. Every permitted target image is tried, testing all earlier edges and nonedges. Necessary degree pruning does not discard a valid embedding.

The complete independently enumerated map sets agree exactly:

Target	Contexts with maps	Maps	Distinct marked images
A17	15	48	10
B17	5	18	3
D17	75	168	20

Every selected marked image is retained. The reviewer also recomputes each debit from the saved whole-component marked images under the three independently reconstructed rules, then recomputes all174 literal totals. Every total remains at least1. This part checks exact payment arithmetic; the literal core inertias and the whole-component sealing proofs are inherited from the frozen local theorem and its separate full reviews. The complete original geometric census is not rerun here.

Finite budgets and strict composition

The exact finite-forest upper bounds are supplied by the separately reviewed finite-budget certificate. This reviewer checks that its full variant uses A17=114, D17=127 and B17<=127, satisfied by the actual B17 cost105, and that all76 saved maxima agree between the finite producer and its independent verifier.

The attaining forests at the two excluded dimensions are checked directly: construct finite D67 and D68, count every basic paid pair, and obtain 504 and512, respectively. Neither witness contains a rank17 component or a cross-component coupon. The Bellman upper-bound and type-domination certificate is not duplicated by this global reviewer.

The independently reconstructed global coefficients are:

Ambient dimension	Finite maximum	Finite curvature	Finite slack	Global constant
67	504	1023/2	15/2	330
68	512	527	15	335

Together with the positive ideal margins, these give the two strict contradictions in the primary proof. The completed ceiling68 supplies the tail in every dimension at least69. This composition covers arbitrary finite-volume Coxeter polytopes, including nonsimple cusps and either absence of finite vertices or absence of ideal vertices.

The next comparison obstruction is also independently checked. At ambient66, the formal profile(13,13,13,13,13) has margin-1565/10976. Direct counting in D66 gives496, against finite curvature495, a deficit1. The entire primary boundary record agrees. Neither obstacle is claimed to be a realized hyperbolic polytope or an optimality statement.

Review and verification scope

The reviewer read the complete primary source and mathematical report. The simultaneous local deletions, unchanged coupons, separation of A17 from D17 in the finite budget, complete concentration, actual cusp multiplicity, two strict global contradictions and completed higher tail all pass. The arithmetic comparison has terminated exit0/PASS on every profile and all stated interfaces.

Generationbd9375 and fresh separate-process saved replay58927 both terminated exit0/PASS after binding the final primary source, certificate and report. All25 direct bindings, Python AST and four local report links pass.

Root read the entire independent source and its final interface changes. The independent optional-first domain, sliding-window/CDF/absolute-deviation identities, fixed breadth-first induced-map enumeration, exact retained rules,174 saved

literal debits/totals, direct D67/D68 counts, witness fields and full ambient66 boundary comparison passed that mathematical/source review. Root did not independently rerun this reviewer's arithmetic. Source, certificate and report are final/frozen. No external refereeing, publication-priority claim, proof-assistant formalization or new construction is asserted.

```
python3 scripts/fresh_ideal5_weighted_dimension66_20261017_review.py
python3 scripts/fresh_ideal5_weighted_dimension66_20261017_review.py --verify
```

Source 143

Source path: `results/fresh_ideal5_weighted_dimension68_20261017.md`

SHA256: 421ebd84ae64e9f837213236723583bf96b7354209948c95ffa022e8bfc96c48

Unrestricted finite-volume dimension at most68

Every finite-sided, finite-volume real hyperbolic Coxeter polytope has dimension at most68. The proof excludes dimension69 by combining the complete order12 affine transfers with an exact finite-corner budget. The completed dimension69 theorem excludes every dimension at least70. There is no arithmetic, compactness, simple-cusp, cusp-factor or facet-count hypothesis.

The source and certificate bind the local proofs and calculate the complete new domain of2,060 cusp profiles. Verification scope is recorded below.

One simultaneous rule for all faces

Start with the complete finite8/ideal5 rule of the preceding theorem. The following are the only changes to its weights. Finite type names denote rank; affine order denotes the number of nodes.

Factor	Change	New same-factor cost
Whole finite A17	Delete its endpoint pair	135
Affine A11, order12	Delete the six antipodal pairs	60
Affine B11, order12	Give the two distance10 pairs weight1/2	65
Affine C11, order12	Delete the endpoint pair at distance11	65
Affine D11, order12	Delete the four opposite short-leaf pairs	62

The finite cross-component rule has three specified changes: upgrade the existing A1/D11-short-leaf half to one; add a unit joining whole A1 to the middle node of whole A11; and add halves joining either node of whole A2 to that A11 middle. All other weights and coupons remain those of the completed predecessor. An upgrade replaces the previous half rather than adding a second copy.

The A17 deletion audits every selected and indirect use of its endpoint unit. Its sole debited old literal configuration, case662, still has two finite D17 units. All short finite paths and the protected leaf9 pairs remain, as do the old cross-component coupons. Its later-transfer audit includes the D13/D14 equality cases and the order16/order17 half-weight rules.

The cycle12 theorem is universal: it covers every quadrilateral with an antipodal marked pair in a whole affine A11 factor, including short-path and selected-core uses. Its exact end-profile inequalities and energy-one spectator classification replace the removed pair by a short ideal path, a finite unit, or the stated A11 coupons. All its finite unit outputs have rank below17 or lie on a path of length at most6, so the A17 endpoint deletion cannot remove them.

The D11 theorem likewise covers selected, low-core and immediate-factor occurrences. Its all-rank spectator lemma resolves the energy5/4 boundary. The surviving alternatives are a short path of length at most5, whole finite units, the full A1/D11-short-leaf coupon, or two inherited finite halves. The literal A1+A3 spectator cases have finite corners of rank16; the arbitrary one-sided corners use whole A1/D11 or D12. None requires the deleted finite A17 endpoint.

The B11 half-transfer covers all14 affected high contexts and28 induced maps. Radical cores seal the finite alternative; nonsingular cores admit only a uniquely forced zero-residual simple extension, or already have a sealed finite B11 component.

A retained ideal half plus a finite half supplies one. The complete69 low-core variants do not use either changed pair. Every immediate D13 occurrence of these B pairs is excluded by a strict Schur gap of1/9 or1/11, including arbitrary spectators. No finite coupon is added.

The C11 endpoint deletion independently reconstructs all induced weighted embeddings of the758 eligible high patterns. Its15 successful contexts and30 maps avoid the endpoint pair. The complete low domain and immediate D13 alternatives are B or D, with no C occurrence. The report checks each later transfer's actual terminal outputs, including the new cycle12/D11 interfaces and A17 case662. No finite coupon is added.

These changes compose without a circular replacement. First discharge any path of length at most8 by the complete short-path branch, using the new universal cycle12 theorem if it is an antipodal pair. This need not leave a unit at that particular corner: the cycle theorem pays the whole face, returning finite coupons or ideal paths of length at most4. It therefore does not recurse into another changed pair. New D alternatives use finite payments or ideal paths of length at most5, retained in every family. After this short-path branch, new B alternatives use retained finite halves/units and the same retained B half. The C and A17 deletions preserve those outputs. The older transfers have only the finite, short-path, same-order D half or explicit small B/D outputs checked in the local provenance reports. The latter are covered by the universal D transfer and the complete B immediate analysis. The complete2,614 high overlaps,66 chosen low closures with three optional extensions, and disjoint-core exclusions remain the classification inputs; no sampling replaces them.

Triangles retain both required payments. The finite elementary rule uses distance at most8, whereas the only deleted finite pair has distance16. For ideal factors the elementary rule uses distance at most5, or distance at most8 with a nearby feature; exceptional whole factors have order at most9. The cycle's deleted antipodal distance6 pairs have no feature. The B/C/D changed pairs have distances10/11/9. The source reconstructs the distances and feature sets and checks every elementary triangle pair at weight one. Thus triangles receive at least2 and quadrilaterals at least1 under this single rule, including at nonsimple cusps.

The exact finite budget includes every new coupon

The revised forest theorem and its independent checker prove the exact finite maximum544 at rank69 after the A17 deletion and D11 upgrade. They keep A17 and D17 separate: the former has base135 and coupon roles, while the latter has base136 without those roles. The complete finite-type domination and1,608 large-component vectors reduce the remaining problem to exact Bellman bounds with attaining witnesses. The independent implementation counts pairs directly and verifies every upper-bound and attainment certificate.

The two cycle12 coupons also fit544, by the A11 domination theorem. Replace each whole A11 component by D11. Both have same-component cost55. The new A11 total cross-cost against A1 or A2 is1; upgraded D11 has totals 5/2 and3 respectively. Against every other component D11 costs at least as much. Large/large interactions vanish, so successive substitutions preserve rank and never decrease the total cost. The old maximizing forests omit A11 and remain attaining witnesses. In particular A1+4D17 attains544. This establishes the budget for the actual simultaneous rule, not a smaller rule omitting its new coupons.

Ideal envelope and the changed concentration domain

All affine factors of order12 are classical A/B/C/D, so their maximum cost is65. The older factors are unchanged. A cusp whose Euclidean link has V vertices and factor ranks r_i therefore has total weight at most V times the sum of c(r_i), with

Affine rank r	c(r)
1 through10	r/2
11	65/12
12	75/13
13	43/7
14	32/5
15	107/16
16	120/17
17 and larger	163/23

The factor multiplier is V/(r+1), from the inherited actual factor-fiber count; it does not require a simple cusp. For total rank d=n-1 set

```
H(t)=product_i(1+t+...+t^r_i), V=H(1),
G=sum_j [t^j]H(t)*min(j,d-j).
```

The normalized ideal curvature is(5G-dV)/(2V). Subtracting the sum of c(r_i) defines the ideal margin.

The inherited all-rank convex-order lemma decreases this curvature when two ranks are concentrated. The old low-rank cap11 cannot be reused: rank11 now has a cheaper cost. Instead concentrate pairs of ranks below10 only until one disappears or reaches10. Their total linear cost is preserved. Keep arbitrary rank11 parts. Two rank12 parts can be replaced by11 and13, increasing cost by

```
65/12 + 43/7 - 2*(75/13) = 23/1092 > 0.
```

Two ranks at least18 concentrate until one reaches17, preserving the constant high-rank cost. Every nontrivial move increases the bounded integer sum of squared ranks. Hence it suffices to retain arbitrary copies of10,11,13,14,15,16,17; at most one12; at most one rank1 through9; and at most one rank at least18. Curvature decreases and cost does not, so the terminal margin is no larger than the original margin.

At d=68 this complete terminal domain has2,060 profiles. Direct exact polynomial multiplication gives positive margin for every profile. The minimum is3641/13608, attained at(11,11,11,11,11,13). No old terminal profile list is reused.

The strict contradiction in dimension69

Let D5 be the sum of5 minus the number of sides over all two-faces, and let v_f count finite vertices. Face coverage and the finite/ideal budgets give

```
D5 <= 544*v_f + sum_(ideal v) V_v*sum_i c(r_vi).
```

The inherited shape inequality gives at n=69, d=68

```
D5 >= 544*v_f + sum_(ideal v) (5G_v-dV_v)/2 + 340.
```

Here beta69=(69-1)(69-5)/8=544. Every ideal difference is strictly positive, the finite difference is zero, and the additional constant340 is positive even when there are no ideal vertices. The two inequalities contradict each other. This excludes dimension69; the completed ceiling69 supplies the higher-dimensional tail, proving ceiling68.

The present finite rule does not prove a further reduction: at ambient68 its maximum is still544, attained by4D17, while beta68=527. The resulting finite deficit17 remains even though all1,906 terminal ideal profiles there were positive in the conditional comparison. No ceiling67 is asserted.

Compact arithmetic remains at most21 and arbitrary compact Coxeter polytopes at most29. The effective compact-facet ambient ceiling is68; the older auxiliary proof remains79. There is no new finite-volume example or new construction dimension.

Verification status

Primary generation and a fresh separate-process replay both terminated exit0/PASS on the complete2,060 profile domain and all direct local interfaces. The independent standard-library reconstruction agrees with every profile polynomial, vertex count, curvature, cost and margin using a separately enumerated optional-first domain, sliding convolution, cumulative-composition and absolute-deviation identities. It independently reconstructs all264 canonical order12 pair weights and elementary triangle sets,228 A11 partner comparisons and76 finite attaining witness costs. Its own report records the final bound generation and replay.

Separate full internal source/mathematical reads passed the corrected concentration domain, actual cusp multiplicities, simultaneous finite budget, zero finite slack, global additive strictness and inherited tail. A further composition review checked the short8-first ordering and every new local interface; small independent saved-set checks found elementary triangle-set sizes60/61/60/62 for A/B/C/D, all retained at unit weight. No lost payment or circular replacement was identified.

These global checks do not repeat the inherited all-label geometric censuses, supporting-root closure proofs or complete Bellman upper-bound calculation. Those frozen proofs and their separate checkers retain their stated scopes. Source, certificate and report are final/frozen. All reviews are internal; no external refereeing or proof-assistant formalization is claimed.

Source 144

Source path: `results/fresh_ideal5_weighted_dimension68_20261017_review.md`

SHA256: 39db4afa45150bbde505d19fcbd959d224462ea8cf0f0cd445f39f2bfe2d93c0

Independent scalar and composition review of the dimension68 proof

The independent reconstruction agrees with all2,060 exact ideal-profile rows in the dimension68 proof. The complete mathematical composition read finds no remaining logical obligation in that proof. It excludes dimension69, and the completed dimension69 theorem excludes every higher dimension. There is no arithmetic, compactness, simple-cusp or cusp-factor restriction.

The reviewer source imports only the Python standard library. Its certificate records the independent scalar data, pair weights, finite domination checks and exact input bindings. Final generation/replay status is given below. This is internal verification, not external refereeing or a proof-assistant formalization.

Complete profile reconstruction

The reviewer enumerates optional exceptional parts first, then recurses over multiplicities of10,11,13,14,15,16,17. This differs from the primary's nondecreasing-part recursion. The exceptions are at most one low part between1 and9, one12, and one part at least18. At total rank68 the two independent domains agree exactly on2,060 profiles, with no duplicates.

For each profile the reviewer multiplies the uniform-factor polynomials by a sliding window, instead of the primary's nested coefficient loops. It checks symmetry and total mass V. It then calculates G in two ways:

```
G = sum_j h_j min(j,d-j),
G = floor(d/2)*V - 2*sum_shift a_shift
                       * binom(floor(d/2)-shift+s,s+1).
```

The second formula is the cumulative bounded-composition identity, with the coefficients a_shift obtained from the numerator product `product(1-z^(r+1))`. It also checks the independent absolute-deviation formula

```
kappa/V = 3d/4 - (5/2)*E(abs(sum_i U_i - d/2)).
```

Every profile's h-polynomial, V, G, curvature, budget and margin agrees with the primary. All margins are positive; the minimum is3641/13608 at `(11,11,11,11,11,13)`.

The changed concentration argument is essential. Ranks at most10 have linear cost and concentrate only up to10; rank11 is retained. Reusing the old cap11 argument would be invalid. The exact gain for `12+12 -> 11+13` is23/1092, positive. High-rank concentration preserves the constant cost. Every move increases the bounded integer squared-rank sum, so iteration terminates. Curvature decreases and cost is preserved or increased; positivity in the complete terminal domain therefore implies positivity for every cusp profile of the given total rank.

The unchanged large-affine envelope is checked again from its15 exact piecewise-linear spine branches and endpoint ratios, yielding163/23. The all-rank graph-count proof and published affine classification remain inherited inputs.

One simultaneous local rule

The reviewer independently constructs the four canonical12-node affine graphs and their graph distances. It reconstructs all264 pair weights:

Family	Cost	Changed pairs
A	60	Six antipodal distance6 pairs removed
B	65	Two terminal/fork-leaf distance10 pairs receive one half
C	65	The distance11 endpoint pair removed
D	62	Four opposite-fork-leaf distance9 pairs removed

Every elementary ideal triangle pair still has unit weight. The cycle has no feature, so its antipodal pairs were not elementary triangle payments. Exceptional whole-factor triangle alternatives have order at most9. The sole finite deletion has distance16, outside finite short8.

The complete source and proof reads cover the following interfaces. The universal cycle12 transfer returns finite units/coupons or ideal paths of length at most4. The universal D11 transfer returns finite payments or ideal paths of length at most5. Neither can recurse into another changed ideal pair. Their finite outputs avoid the deleted finite A17 endpoint.

For the B rule, the14 affected cases split into eight radical sealed cores, two already-whole finite B11 components, and four nonsingular cores admitting zero or one forced supporting root. The reviewer recounts all16 saved sealed payment variants: an ideal half plus at least a finite half gives one. It independently recomputes the three immediate-D13 strict-gap arithmetic identities,1/9 and1/11. The core projection/contact classification, actual radical argument and whole component sealing are mathematically reviewed inherited local proofs; their28 maps and exact core inertias are not re-enumerated here.

The C theorem checks all758 eligible high patterns,66 low closures with three optional variants, immediate whole B/D factors and all downstream outputs. Its fifteen successful C contexts/30 maps avoid the endpoint. Its proof also checks that the finite A17 deletion's only debited legacy case662 still has two finite D17 units. The complete inherited high/low classifications and closure proofs remain explicit inputs. This review does not describe those earlier computations as a new independent census.

The short8-first reduction, universal transfers, complete no-short8 core split, protected payments and literal later half-weights thus form one noncircular rule. In particular, an old whole-small-factor phrase is not silently reused after deleting its pair. The local reports inspect the actual alternate factors, short paths or finite coupon outcomes.

The finite budget includes the A11 coupons

The reviewer reconstructs the new coupon interaction totals directly, without importing the finite-budget producer. Each A11 has internal cost55, equal to D11. Its new total interaction is1 with A1 or A2 and0 otherwise. Upgraded D11 has respective totals5/2 and3 and dominates every other partner. All228 finite-type partner rows agree with the saved domination certificate, including the symbolic dihedral family.

Interactions between two large components vanish. Replacing A11 components successively by D11 therefore preserves rank and never decreases total charge. The frozen exhaustive Bellman bound remains an upper bound for the enlarged rule. Its separate no-producer review is bound here; its full Bellman enumeration is not repeated.

The reviewer independently recounts all76 typed attaining forests with the new coupons. Their costs agree exactly; none contains A11. Hence they still attain the inherited upper bounds. In rank69 both `A1+4D17` and `A1+4A17` have cost544. The A17/D17 distinction is retained: their internal costs are135 and136, respectively, and their coupon eligibility differs.

At n69 the finite curvature coefficient is544. All ideal margins are positive and the global shape inequality has the extra constant340. Subtracting the payment upper bound from that shape lower bound is strictly positive even with zero cusps. This excludes69; the frozen ceiling69 supplies n>=70, proving68.

At ambient68 the checked profile `11^5+12` has positive ideal margin 411499/3234816, but the finite maximum remains544 while its curvature coefficient is527. The finite deficit17 prevents inferring ceiling67 from this comparison. The reviewer makes no such further claim.

Verification status and limits

Independent scalar preparation and the full primary-row/interface comparison both terminated exit0/PASS. After the primary report froze, final hash-bound generation and a separate fresh saved replay also terminated exit0/PASS. All45 direct

bindings, report links and Python syntax passed. Source, certificate and report are final/frozen. The new arithmetic consists of2,060 profile reconstructions,264 pair weights,228 partner comparisons,76 attaining forests and the explicitly stated saved local-payment sums. Full mathematical/source review covers the local composition, concentration and strict global contradiction.

Root separately read the complete reviewer source and mathematical algorithms, passing the optional-first domain, sliding convolution, cumulative and absolute-deviation identities, canonical BFS/pair weights, independent partner comparisons and typed witnesses, and the explicit inherited-upper-bound scope. That source review did not separately rerun the2,060-row arithmetic.

Complete all-label core/overlap classifications, arbitrary spectator geometry, physical supporting-root closures and the exhaustive Bellman upper-bound certificates retain their frozen validation scopes. They are not replaced by finite samples or claimed as newly reimplemented here. The result is an internally checked computer-assisted proof, with no claim of outside refereeing or Lean-style formalization.

Source 145

Source path: `results/fresh_ideal5_weighted_dimension69_20261017.md`

SHA256: 600a9acb16136c1e30a1d591a1ef6375b7c841d1378968d3aa25b7bf72be5c4e

Unrestricted finite-volume dimension at most69

Every finite-sided, finite-volume real hyperbolic Coxeter polytope has dimension at most69. The argument excludes dimension70 using two new local projection obstructions and an exact cusp-profile comparison. The completed dimension70 theorem supplies all higher dimensions. Arithmeticity, compactness, simple cusps, cusp-factor restrictions and a fixed number of facets are not required.

The source and certificate bind the local theorem and all1,339 reduced cusp profiles. Verification status appears below.

The two changed payments

Retain the entire preceding finite8/ideal5 rule, including all order13 transfers, order16/order17 half-weights and finite coupons. In affine B13 of order14 remove the singleton pair(0,10); in affine B14 of order15 remove (1,11). The complete local theorem reduces their costs87 to86 and97 to96, with no new finite payment.

Every affected selected-core occurrence forces a missing terminal root with a doubled bond to an old marked node. The terminal cannot be another old corner root: the other old components are orthogonal to that marked component, and the two omitted black roots are absent from the selected corner. It is therefore a genuinely new supporting white root of square2.

The old core H is nonsingular and Lorentzian. Its complete necessary contact domain splits into two ends with entrywise-positive inverse blocks. Touching both ends supplies an already-paid short path. Otherwise the forced doubled contact alone gives a projection lower bound2(H^-1)_ii. All64 affected maps have this bound at least5/2, strictly greater than2. Additional contacts on the same end only increase the projection. Thus the actual affine occurrence is impossible in an otherwise unpaid face. No unspecified full-matrix contacts are assigned zero in this argument.

There are23 affected order14 contexts with46 maps and11 order15 contexts with18 maps; five contexts occur in both lists. Every elementary triangle, short8 and basic ideal payment remains at unit weight. The inherited low-core, disjoint-core, protected, immediate and finite alternatives are unchanged for the precise domain reasons in the local theorem. Its complete selected-map support exhausts the remaining changes. Consequently all triangles still receive at least2 and all quadrilaterals at least1.

The finite budget stays8n through dimension75. The independently reconstructed13,034 budget endpoints are inherited through the completed70 theorem. No new finite budget is needed for the present dimension conclusion.

Ideal costs and complete concentration

The order14 costs are now A70,B86,C84,D82, with maximum86. The order15 costs are A75,B96,C95,D96, with maximum96. Every other cost is unchanged.

Affine rank r	Normalized factor cost c(r)
1 through11	r/2
12	75/13
13	43/7
14	32/5

Affine rank r	Normalized factor cost c(r)
15	107/16
16	120/17
17 and larger	163/23

At a cusp with V vertices in its Euclidean link, a factor of order r+1 multiplies its pair-weight sum by V/(r+1). Hence the total cusp cost is at most V times the sum of c(r_i), including at nonsimple cusps.

For total cusp rank d=n-1, set

```
H(t)=product_i(1+t+...+t^r_i), V=H(1),
G=sum_j [t^j]H(t)*min(j,d-j).
```

The normalized curvature is(5G-dV)/(2V), and the ideal margin subtracts the sum of c(r_i). The inherited convex-order lemma decreases this curvature under concentration of ranks. The complete preceding reduction domain remains valid. Low ranks concentrate up to11 with unchanged linear cost. Two ranks12 become11 and13; the cost gain is now19/182, still positive. Two ranks at least18 concentrate until one is17, preserving their constant cost. Every nontrivial move increases the bounded integer sum of squared parts, proving termination.

It suffices to keep arbitrary copies of11,13,14,15,16,17, at most one12, at most one low rank1 through10 and at most one rank at least18. No further rank-elimination rule is assumed. At total rank69 this domain contains 1,339 profiles, all with strictly positive exact ideal margin. The minimum is1819/51840, attained at(11,11,11,11,11,14).

Zero finite slack and the strict global contradiction

Let D5 sum5 minus the number of sides over all two-faces. Triangle and quadrilateral coverage gives the upper bound

```
D5 <= 8n*v_f + sum_(ideal v) V_v*sum_i c(r_vi).
```

The inherited shape inequality gives

```
D5 >= beta_n*v_f + sum_(ideal v) (5G_v-dV_v)/2 + 5(n-1).
```

At n=70, beta_n=70*64/8=560=8n. Thus the finite coefficient has **zero slack**, not negative slack. Each ideal coefficient has strictly positive slack, and the additional global constant is345. Subtracting the upper bound from the lower bound is therefore strictly positive even if there are no ideal vertices. This contradiction excludes dimension70. The completed ceiling70 excludes n>=71, proving the ceiling69.

At ambient69 the formal profile(11,11,11,11,11,13) still has negative margin -2029/13608. This is a limitation of the present comparison, not a realized cusp or a claim of optimality. The finite coefficient also needs stronger control for a further reduction. No bound68 is asserted.

Compact arithmetic remains at most21 and arbitrary compact Coxeter polytopes at most29. The effective compact-facet ambient ceiling is69; the older auxiliary proof remains79. No new finite-volume construction or new example dimension follows from this proof.

Verification status

Primary generation and a fresh separate-process saved replay both terminated exit0/PASS on all1,339 profiles and both changed local interfaces. The independent standard-library reconstruction agrees with every profile polynomial, curvature, cost and margin, the minimum and the negative boundary profile. It also reconstructs both canonical B targets, their entire saved affected-map support,13 distinct old core inverses and signatures, actual preferred-corner contact domains, positive end blocks and all64 forced terminal inequalities. Its own report records its final generation and fresh replay.

A separate full internal source/mathematical review passed the primary local composition, concentration moves, finite-budget equality, global additive strictness and higher-dimensional tail. The frozen complete core/map censuses and unchanged finite-budget reconstruction remain inherited, with their stated scopes. All reviews are internal; no external refereeing or proof-assistant formalization is claimed. Source, certificate and report are final/frozen.

Source 146

Source path: `results/fresh_ideal5_weighted_dimension69_20261017_review.md`

SHA256: c8eb5953bb48b6f679cb8274c4ffb33b6cd5bc6a9ab799f9d7a76aaadf550ec4

Independent scalar and composition review of dimension69

The dimension69 proof passes the independent exact reconstruction and mathematical review described here. No remaining gap was identified in the reviewed scope. The review source uses only the Python standard library and imports no project arithmetic. Its saved review certificate binds the frozen primary and its direct inputs. Final generation and a separate fresh saved replay both passed; their scope is recorded below.

Independent scalar calculation

At ambient dimension70 the cusp rank is69. The reviewer enumerates profiles by choosing an optional high rank, an optional low rank and an optional12 first, then filling the remaining rank with copies of11,13,14,15,16,17. The primary uses its inherited repeated-ranks-first enumeration. Exact set equality gives the same1,339 terminal profiles, without duplicates.

For each profile the reviewer computes the polynomial `H(t)=product_i(1+t+...+t^r_i)` by sliding-window convolution. It checks symmetry, the vertex count V, and independently recomputes G by the cumulative bounded-composition identity

`G=floor(d/2)*V-2*sum_shift a_shift*binom(floor(d/2)-shift+s,s+1),`

where `sum a_shift*t^shift=product_i(1-t^(r_i+1))` and s is the number of factors. It also checks the absolute-deviation identity

`curvature/V=3d/4-(5/2)*E|sum U_i-d/2|.`

All coefficients, vertex counts, G values, normalized curvature, costs and margins agree exactly with every primary row. The minimum margin is **1819/51840**, attained at `(11,11,11,11,11,14)`. The finite slack is **0**, and the strict global additive constant is345. The ambient69 boundary profile `(11,11,11,11,11,13)` has margin **−2029/13608**; it is a formal obstruction to this comparison, not a claimed realized cusp.

Only the rank13 and14 coefficients change, becoming43/7 and32/5. The complete concentration domain remains valid: low-rank concentration has unchanged linear cost, two12 ranks become11 and13 with exact cost gain 19/182, and the constant-cost high ranks concentrate to at most one rank above17. Every nontrivial move increases the bounded integer squared-rank sum. No further concentration or elimination of ranks13 through17 is assumed. The reviewer also reconstructs the unchanged fifteen all-rank affine cost branches and their endpoint/limit maxima, rather than inferring an all-rank maximum from a finite sample.

Independent affected-map and projection checks

The new local theorem deletes one pair from each of the order14 and15 affine B rules. The reviewer independently builds both canonical target graphs, their distances, feature sets and two automorphisms. It verifies that each changed pair is an invariant distance10 singleton, and reconstructs the paid sets of sizes86 and96. All elementary triangle, short8 and basic ideal pairs remain units.

To determine the complete affected support, the reviewer derives marked images directly from all506 saved order14 maps and548 saved order15 maps in the frozen complete small-factor certificate. These1,054 maps occupy422 target/context rows. The derived marked-image sets agree with the saved sets, and select precisely46 and18 affected maps in23 and11 contexts.

Five contexts occur for both targets: there are29 distinct literal contexts and34 target/context occurrences. Every affected map is checked injectively against the labelled old core Gram and appears exactly once in the new local certificate.

For the thirteen distinct old core Grams, independent Fraction Gaussian elimination reconstructs the full inverses. Exact congruence elimination, including a two-by-two hyperbolic pivot when necessary, checks their Lorentz signatures. For each target/context occurrence the reviewer checks the supporting-white positive principal and all four strict edge principals, reconstructs the preferred finite corner and its connected components, and derives the complete necessary contact domain from the finite-block inverse diagonals and the two omitted blacks. The result agrees with the local certificate and the original overlap record.

The reviewer then checks that the two end sets partition this domain, their full inverse blocks are entrywise positive, and touching both ends supplies a retained short path. It derives the shortcut length from the actual old core, rather than accepting the saved bound. The other old corner components are orthogonal to the marked component, and the two omitted blacks are absent from the preferred corner.

Every affected map forces the target's missing doubled terminal to touch an old marked node i. Because the terminal is outside the displayed component but adjacent to it, it cannot be another old corner component; an omitted black also cannot belong to the actual corner. Thus it is a new supporting white. In the norm-two normalization its required contact has square2. All64 exact calculations give `2(H^-1)_ii>2`, with minimum 5/2, and agree with the local certificate's witnesses.

The mathematical argument permits arbitrary further contacts. Contacts on both ends give the already-covered short path. If all contacts lie in one end, its entrywise-positive inverse block makes the forced term a lower bound for the full projection square. The core is Lorentzian, so the perpendicular residual has nonnegative square and a norm-two root's projection cannot exceed2. This contradiction excludes the affine occurrence. No unspecified Gram entries are assigned zero, no indirect tail cutoff is imposed, and no displayed finite subdiagram is silently treated as a whole component.

The complete old core and induced-map enumerations remain frozen inputs. This review checks all saved target-map images and every affected literal projection; it does not independently repeat those original exhaustive censuses.

Global composition and zero finite slack

The full local domain split is unchanged. Triangles and short8 paths retain their required units. Low-core alternatives have whole component order at most12; the high-disjoint branch is excluded; protected/basic ideal payments have distance at most9; finite and universal cycle alternatives retain their rules; and the universal immediate whole-affine alternatives have order at most13. The complete remaining overlap support is the one checked above. Thus triangle and quadrilateral totals remain at least2 and1, and no finite coupon or extra finite cost is introduced.

The finite budget remains8n through dimension75. Its previously independently checked13,034 endpoints are inherited through the dimension70 review without repeating that unchanged enumeration. The new order14 maximum is86; the order15 maximum is96, shared by B and D. Every other small-order cost and the all-rank tail are unchanged. The actual cusp factor-fiber count multiplies a factor's pair weight by V/(r+1), including at nonsimple cusps.

At n=70 the shape inequality has finite coefficient `beta_70=70*64/8=560=8n`. Subtracting the face-payment upper bound from the shape lower bound leaves zero times the finite-vertex count, strictly positive coefficients at all ideal vertices, and the unconditional additive term345. The difference is therefore positive even with no ideal vertices. Positive finite slack is not required. This excludes dimension70, and the completed dimension70 theorem excludes every higher dimension, establishing the unrestricted ceiling69.

No arithmeticity, simple-cusp, factor-count or facet-count hypothesis is added. The construction problem and compact-arithmetic models are separate; this review supplies no new example, external refereeing or formalization.

Verification status

Independent scalar preparation, complete primary/local comparison, final hash-bound generation and separate fresh saved replay all terminated exit0/PASS. Syntax, all27 direct source bindings and all8 local report links pass. The full primary source and written proof, the new local source and written argument, and the global composition received a complete mathematical read; no correction was required. The parent independently read this reviewer's full source and mathematical reconstruction, including scalar enumeration and all three curvature identities, inverse/inertia and actual-corner routines,

contact-domain and map coverage, and the zero-finite-slack composition. No gap was identified; that read did not repeat the reviewer's1,339-profile calculation.

The bound primary hashes are source 013692566433013fa44efad68c9abb21c7bdfa666f56a3e51e8da4697cdacf0d, data 70a9bc7d0d5e13252b6e9e7743fb989759bc5d4521901d548b5130bda39ecb8c, and report 600a9acb16136c1e30a1d591a1ef6375b7c841d1378968d3aa25b7bf72be5c4e. All validation is internal. The primary and original censuses are frozen dependencies with the distinct verification scopes described above.

Source 147

Source path: `results/fresh_ideal5_weighted_dimension70_20261017.md`

SHA256: ebf0d1b19f7f1ef4a0a4486c64cf93cb3fc00895964b99543b142617d4431257

Unrestricted finite-volume dimension at most70

Every finite-sided, finite-volume real hyperbolic Coxeter polytope has dimension at most70. The proof uses the new weighted order16 local theorem and an exact comparison in dimension71. The completed dimension71 theorem supplies all higher dimensions. There is no arithmeticity, compactness, simple-cusp, cusp-factor or facet-count restriction beyond finite sidedness.

The source and certificate bind the complete local transfer and compare all1,432 reduced cusp profiles. Verification and review status are recorded below.

The changed local rule

Retain the entire preceding finite8/ideal5 face rule, including the order13 transfers, finite coupons and twelve half-weights in order17. The only change is in affine D15, a factor with16 nodes. Its ten still-paid pairs at distance greater than9 now have weight1/2. The complete local proof reduces its total weight from111 to106 without adding any finite coupon.

Every elementary triangle payment and every short8 unit remains unchanged. The low-core, disjoint-core, protected-feature, immediate-factor and adaptive finite alternatives retain their old proofs, for the explicit domain reasons given in the local report. The complete induced-map census contains54 affected selected contexts and328 affected maps. The local proof controls actual affine-factor extensions, including arbitrary contacts and indirect tails, before fixing their completed Gram matrices. Of66 distinct literal completions,20 require multiple core-contact roots,14 require projection square greater than2, and22 have two negative directions. These cannot occur. The ten remaining factors are already whole and their cores are sealed against all further supporting white roots.

Each surviving configuration has a different actual finite corner carrying an existing half-payment: whole A4 against a whole D12 fork center or short leaf, or whole D5 against a whole D11 short leaf. These are whole actual components, including in the presence of arbitrarily many spectators. The finite half and the remaining ideal half total1. Thus all triangles still receive at least2 and all quadrilaterals at least1.

The unchanged finite budget is8n through dimension75, including dimension71. Its13,034 exact endpoint cases and mathematical composition were independently reconstructed in the preceding review. No new finite expense is added here.

The ideal-factor envelope

The order16 costs are now A80, B107, C103 and D106. Hence the maximum is107, and the rank15 coefficient falls from111/16 to107/16. All other ranks retain their preceding costs:

Affine rank r	Normalized cost c(r)
1 through11	r/2
12	75/13
13	87/14
14	97/15
15	107/16

Affine rank r	Normalized cost c(r)
16	120/17
17 and larger	163/23

For an actual cusp with factor ranks r_i and V vertices in its Euclidean link, each pair in a factor of order r_i+1 occurs V/(r_i+1) times. Fractional weights respect this exact counting formula. Therefore the cusp cost is at most V times the sum of c(r_i). No simple-cusp assumption is used.

Complete reduction and exact comparison

For a cusp of total rank d=n-1, define

```
H(t)=product_i(1+t+...+t^r_i), V=H(1),
G=sum_j [t^j]H(t)*min(j,d-j),
normalized curvature=(5G-dV)/(2V).
```

The ideal margin is this curvature minus the sum of c(r_i). The preceding convex-order lemma decreases curvature under concentration of two ranks. We retain exactly its complete reduction domain. Low ranks concentrate up to11 without changing their linear cost, leaving at most one rank1 through10. Two12 ranks become11 and13, increasing cost by16/91. Two ranks at least18 concentrate until one is17, preserving their common constant cost. These moves do not involve the changed rank15 cost. Each move strictly increases the bounded integer sum of squared parts, so the procedure terminates.

Every partition consequently reduces to repeated11,13,14,15,16,17, at most one12, at most one rank1 through10, and at most one rank at least18. This complete domain need not be minimal. In particular repeated15 parts are retained and no additional elimination rule is assumed.

At ambient dimension71 the total rank is70. The complete reduced domain has 1,432 profiles. Every exact ideal margin is positive; the minimum is

```
23641/338688,
```

attained at(11,11,11,11,13,13). The finite-vertex slack is

```
beta_71 - 8*71 = (70*66)/8 - 568 = 19/2.
```

To finish, let D5 be the sum of5 minus the number of sides over all two-faces. If T and Q count triangular and quadrilateral faces, then

```
D5 <= 2T+Q <= total corner payments
   <= 8n*v_f + sum_(ideal v) V_v*sum_i c(r_vi).
```

The inherited shape inequality gives

```
D5 >= beta_n*v_f + sum_(ideal v) (5G_v-dV_v)/2 + 5(n-1).
```

At n=71, every finite and ideal coefficient on the lower-bound side exceeds the respective upper-bound coefficient, and the additive constant is350. This is impossible. The completed ceiling71 excludes every n>=72; together these arguments establish the ceiling70.

At ambient70, the formal profile(11,11,11,11,11,14) still has negative margin -1637/51840. It contains no rank15 factor, so further improvement of the order16 rule alone does not remove this obstruction. This is a limitation of the present comparison, not a realized cusp, a polytope or an optimality claim for70. No bound69 is asserted.

Compact arithmetic remains at most21, and arbitrary compact Coxeter polytopes remain at most29. The effective compact-facet ambient ceiling becomes70; its older auxiliary proof remains79. No new finite-volume example or new construction dimension has been established.

Verification status

Primary generation and a fresh separate-process saved replay both terminated exit0/PASS on all1,432 comparisons and the saved local interfaces. The independent standard-library reconstruction agrees with every profile, its full polynomial and curvature, the minimum, the negative boundary profile and the changed local interface. It also independently checks all328 saved affected maps, the nine distinct full Grams underlying the nontrivial completions, and all40 sealed corners and their whole finite half-payments. Its own report records its generation and replay.

A separate full internal source/mathematical read passed the new local composition, the unchanged complete concentration domain, the finite budget, the higher-dimensional tail and the global contradiction. The local geometric census is an explicit frozen input, not independently enumerated by this global checker or the new reviewer. The unchanged13,034 finite-budget endpoint reconstruction remains inherited from the frozen71 review. All reviews are internal; external mathematical refereeing and proof-assistant formalization are absent. Source, certificate and report are final/frozen.

Source 148

Source path: `results/fresh_ideal5_weighted_dimension70_20261017_review.md`

SHA256: 841ec43914d42c528419f8a72dddb2e3a33f31d372acc4946289231ad0da715d

Independent scalar and composition review of the dimension70 proof

The dimension70 proof passes the independent exact scalar comparison and the mathematical composition review described here. No remaining logical gap was identified in the reviewed scope. The review source uses only the Python standard library and imports no producer arithmetic. Final hash-bound generation and a separate fresh saved replay both terminated with exit0/PASS; the saved review certificate binds the frozen primary and all direct inputs.

Independent exact scalar calculation

At ambient dimension71, the cusp rank is70. The reviewer selects the optional high rank, optional low rank and optional rank12 first, then fills the remaining rank with arbitrary copies of11,13,14,15,16,17. The producer uses its inherited repeated-parts-first enumeration. Exact set equality confirms that both enumerate the same1,432 profiles without duplicates.

For each profile the reviewer multiplies the factors `1+t+...+t^r` by sliding-window convolution. It independently checks the central coefficient sum using the cumulative bounded-composition identity

`G = floor(d/2)*V - 2*sum_shift a_shift*binom(floor(d/2)-shift+s,s+1),`

where `sum a_shift*t^shift = product(1-t^(r+1))`. It also verifies the absolute-deviation formula

`curvature/V = 3d/4 - (5/2)*E|sum U_i-d/2|.`

All polynomial coefficients, vertex counts, G values, curvature, costs and margins agree exactly with all1,432 primary rows. The minimum margin is **23641/338688**, attained at `(11,11,11,11,13,13)`. The finite-vertex slack is **19/2**, and the inherited global additive slack is350. At ambient70, the boundary profile `(11,11,11,11,11,14)` has margin **−1637/51840**. This negative formal profile establishes neither realizability nor optimality of the theorem.

Only the rank15 coefficient changes, from111/16 to107/16. The unchanged concentration moves preserve or increase cost: low ranks concentrate to11; two12 ranks become11 and13, increasing cost by16/91; and two ranks at least18 concentrate until one is17, preserving the constant tail cost. Integer squared-part sums strictly increase and are bounded. Thus the inherited complete terminal domain remains valid. Repeated rank15 parts are retained. The reviewer also reconstructs the unchanged fifteen piecewise affine cost branches and their endpoint/limit maxima; it does not infer an all-rank maximum from a finite sample.

Changed local interface and literal checks

The order16 local transfer changes ten still-paid affine-D15 pairs from1 to1/2. The reviewer reconstructs the canonical16-node graph, its distances and eight automorphisms. The ten pairs are exactly the previously paid pairs at distance greater than9 and form an invariant set. Its complete120-entry weight vector sums to106. The nine inherited zero pairs stay zero, while all94 elementary triangle/short8 pairs retain weight1. No finite coupon is added.

From the frozen small-factor map certificate, the reviewer independently selects all54 affected cases and compares the complete328 saved maps. Each literal map and marked image is checked against the actual core's labelled Gram. Their66 grouped completion outcomes are:

Outcome	Literal completions	Maps
Multiple required core-contact roots	20	64
Required contact projection square exceeds2	14	112
Completed Gram has two negative directions	22	72
Already-whole sealed factor	10	80

Independent Fraction Gaussian elimination computes the saved core inverses. Exact congruence elimination, including two-by-two hyperbolic pivots when needed, recomputes the signatures of the nine distinct full Grams underlying the22 index exclusions and ten sealed completions. It also checks the14 projection exclusions from inverse diagonals. The20 multiple-root outcomes use the independently checked required-contact counts and the local theorem's reviewed at-most-one-contact-root lemma; this review does not claim a new exhaustive classification of supporting-root profiles.

For all ten sealed completions, the reviewer independently verifies the positive affine kernel, the white and four edge principals, all40 corner signatures, their entire finite components and the marked coupon roles. The finite halves are whole A4 against whole D12 at a fork center or short leaf, or whole D5 against whole D11 at a short leaf. Each occurs at a different corner from the selected ideal half. Every total is at least1; four cases have additional finite unit payment. These checks use actual corner occurrences rather than arbitrary finite subdiagrams.

The mathematical/source read checks why the completed Gram's unspecified entries really are zero. The inherited contact and no-bridge lemmas give at most one relevant core-contact root. In the relevant-component cases, the possible extra exterior root would bound the whole marked component by14, so it cannot coexist with the actual order16 factor. Once the forced factor is already whole, no additional supporting white can contact its core. The displayed finite components are therefore whole even with arbitrarily many spectators. This geometric justification is necessary before an exact inertia calculation can exclude a completion.

The original complete core/overlap and induced-map censuses remain explicit frozen inputs. This review compares their saved affected support and the literal completions; it does not rerun those original censuses.

Complete proof composition

The local modification concerns only distance10 or11 pairs. It leaves short8 units, elementary triangle units and all inherited nine zero-pair replacement proofs intact. The complete prior split into low, disjoint, overlapping, protected/basic and immediate-factor alternatives is retained. Only the54 affected saved contexts change, and each possible completion is either excluded or receives the ideal half plus an existing finite half.

The finite budget remains8n through dimension75. Its13,034 endpoint checks and proof were independently reconstructed in the dimension71 review; they are bound and inherited here without repeating that unchanged enumeration. The order16 maximum is now107, attained by affine B, while D has cost106. All other small-order costs and the all-rank tail are unchanged.

For an ideal vertex with V link vertices, a factor of order r+1 contributes its weighted pair sum multiplied by V/(r+1). Fractional weights preserve this counting formula, including at nonsimple cusps. The shape inequality, the positive ideal margins and finite slack then contradict the total face-payment upper bound at dimension71. The completed dimension71 theorem excludes all higher dimensions. Together these prove the unrestricted ceiling70 under the stated finite-sided, finite-volume real-hyperbolic Coxeter hypotheses.

This argument adds no arithmeticity, simple-cusp, cusp-factor or facet-count restriction. It is separate from the compact-arithmetic partial models. No new construction, external mathematical refereeing or proof-assistant formalization is asserted.

Verification status

The independent scalar preparation, complete primary-row/local-interface comparison, final hash-bound generation and separate fresh saved replay all terminated with exit0/PASS. The final commands completed directly, without outstanding process handles. All21 direct source bindings, Python syntax and nine local report links pass. The saved primary hashes are source 83d1d3a33f86a25153f381497071e96d8a885a694306d6eb1bcf0045496e6beb, data 7f683b9d1a9ce11cec0393

35d9008b2da50cda380a2f0e0a57a27f6b192795d7, and report ebf0d1b19f7f1ef4a0a4486c64cf93cb3fc00895964b99543b142617d4431257.

The local source and written proof, and the full primary source and composition, received a complete mathematical read in the scopes above. The parent also independently read this reviewer's entire source and its mathematical reconstruction, including scalar enumeration, inverse/inertia routines, all saved maps and corner roles, and the global interface; no gap was identified. That source review does not claim to repeat this reviewer's1,432-profile calculation. No upstream exhaustive census is claimed to have been independently repeated by this reviewer. The source, data and report are final/frozen; all validation is internal.

Source 149

Source path: `results/fresh_ideal5_weighted_dimension71_20261017.md`

SHA256: 7c45782e06c76f303a4d3c6f9571f10475d5c8f28bfa5a87fbafd06949d365dc

Unrestricted finite-volume dimension at most71

The order13 transfers and the weighted order17 rule, together with the exact global comparison, give the following computer-assisted conclusion:

Every finite-sided, finite-volume real hyperbolic Coxeter polytope has dimension at most71.

There is no arithmetic, compactness, simple-cusp, cusp-factor or facet-count restriction. The source and certificate bind the new local interfaces and verify all4,913 reduced cusp profiles in dimensions72, 73 and74. The completed dimension74 theorem handles every higher dimension. Verification status is recorded below.

The complete modified face rule

Retain the complete finite8/ideal5 face theorem and its validated successors, with these changes. At finite vertices use the joint coupon budget: the old half between whole A1 and a short leaf of whole D12 becomes one, and a new half joins whole A2 to the doubled terminal of whole B10. All earlier rules, including A4/D13-neighbor halves, remain. The total finite cost is at most8n through dimension75. This dimension range covers all new comparisons here.

At ideal factors of order at most12, all pairs remain paid. At order13 the four classical affine costs are now A65, B75, C73 and D74. At orders14,15,16, the prior rules and costs are unchanged. At order17, twelve affine-D16 pair weights fall from one to one half, reducing its cost126 to120. Other order17 types are unchanged. The larger-order ideal5/R5 rule is unchanged.

The essential coverage statement remains: every triangle receives at least2 and every quadrilateral at least1. Here is the full interface argument, including the short-path and immediate-factor obligations.

For order13 cycles, the universal cycle13 transfer removes exactly the13 distance-six pairs. Its remaining65 pairs have distance at most5. The missing pairs occur in the short8-first stage of the original proof, so absence from selected overlap maps would not suffice. The new universal transfer supplies a finite short-path payment or an ideal short path. Its final isolated profile requires energy10/13, excluded by the complete strict spectator spectrum. It needs no new finite coupon.

For affine B12 of order13, remove the singleton(0,10) and the two distance11 pairs(10,11),(10,12), keeping the other75 pairs. The B12 singleton proof seals the two singleton cores and supplies the two finite halves, one old and one new A2/B10 half. Its four saved conditional distance11 contexts are promoted here:166 and1744 have an existing whole finite B13 unit; 170 and2137 have whole A1 against a short leaf of whole D12 and receive the newly budgeted full coupon. The source checks the actual whole types and marked roles in those saved finite corners. Their sealing is the same positive-affine-kernel and full-core radical argument as in the singleton proof. The B12 proof also supplies a strict positive determinant gap for all universal D14 `inner4` alternatives, so none of those formerly immediate payments is silently lost.

For affine D12 of order13, the universal opposite-fork-leaf transfer removes all four distance10 leaf pairs, leaving74. This covers all127 affected selected contexts and all universal D14 `inner_next3` leaf-pair alternatives. It uses the new full A1/D12 coupon where needed; its two-sided boundary instead gives a short path or two existing halves.

For affine C12, retain the73-pair basic-plus-selected set in the complete order13 support certificate. Its five omitted pairs have distance at least10. Every elementary triangle pair and every pair of distance at most8 remains a unit. Every selected marked image is retained, and there is no C-type immediate order13 demand. Thus this change needs no new geometric

local transfer. The primary checker checks all these inclusions for the other three types, the complete801 eligible-pattern domain including larger component bounds, and all six literal D14 immediate records.

These exhaust the new order13 obligations in the complete face theorem. Its low-core alternatives have whole component order at most12. The D13 universal transfer has immediate factor order12. Its D14 transfer has the order13 B/D alternatives just discharged. The other sealed order16/17 transfers end in finite payments or retained short paths. The general feature-free tree replacement ends in finite or retained feature-path payments. There is no additional whole-order13 assumption being imported.

At order17 use the twelve half-weight transfers. All affected pairs have distance greater than9, so short8 and triangle units are unchanged. The complete saved marked-image support consists of twelve sealed shared-L25 contexts. In each, the selected ideal corner keeps one half and a different actual finite corner has an existing half. Thus the face still receives at least1. This is a sum of weights, not a claim that a half is a full payment. The prior zero-weight D17 singleton retains its two-finite-half replacement. No finite cost is added by these twelve changes.

The original complete low/disjoint/overlapping core classification and induced-map censuses remain explicit frozen inputs. The modifications above check every place their previous ideal payments change. All finite coupons apply to whole actual finite components, not merely finite subdiagrams inside ideal corners.

The new ideal envelope and concentration domain

The resulting normalized ideal-factor costs are bounded as follows:

Affine rank r	Normalized cost
1 through11	r/2
12	75/13
13	87/14
14	97/15
15	111/16
16	120/17
17 and larger	163/23

The full small-order maxima are taken over all four types. In particular order17 has A85, B111, C108 and D120. The all-rank tail is unchanged. For an actual nonsimple cusp with factor ranks r_i and V link vertices, each factor pair-weight sum is multiplied by V/(r_i+1), exactly as in the inherited factor-fiber theorem. Fractional weights preserve this linear counting formula.

For total cusp rank d=n-1 write

```
H(t)=product_i(1+t+...+t^r_i), V=H(1),
G=sum_j [t^j]H(t)*min(j,d-j).
```

The normalized curvature term is (5G-dV)/(2V). Subtracting the sum of the costs in the table gives the ideal margin.

The inherited convex-order lemma decreases normalized curvature when (a,b) with1<=a<=b is concentrated to(a-1,b+1). We use only moves that also preserve or increase total cost. Low ranks concentrate up to11, leaving at most one rank from1 through10. Two ranks12 may become11 and13: their cost increases by exactly16/91. Two ranks at least18 concentrate until one is17, preserving the constant tail cost. Each move strictly increases the bounded integer sum of squared parts at fixed total rank, so the process terminates.

Every partition therefore reduces to repeated11,13,14,15,16,17; at most one12; at most one low rank1 through10; and at most one rank at least18. The older move eliminating repeated15 is not used: its cost difference has become negative under the new rank16 coefficient. Arbitrarily many rank15 factors are explicitly included in the new enumeration.

Exact global comparison

Every reduced profile in each of the three dimensions has positive ideal margin. The checker evaluates exact rational values throughout:

Ambient dimension	Reduced profiles	Minimum ideal margin	Finite slack beta_n-8n
72	1530	6875/49152	18
73	1635	41401/145152	28
74	1748	135455/304128	37

The respective minimizing profiles are(11,15,15,15,15), (11,11,11,11,13,15), and(10,11,11,11,15,15). Here beta_n is n(n-6)/8 for even n and(n-1)(n-5)/8 for odd n. The inherited global shape inequality also has the strictly positive constant5(n-1). Summing the retained face-payment inequalities and comparing with the finite and ideal vertex budgets gives a contradiction in each of dimensions72,73,74. The completed74 theorem excludes all dimensions at least75. Together they prove the ceiling71.

At ambient dimension71, the profile(11,11,11,11,11,15) has margin -5597/248832. Thus this envelope does not prove a bound70. This is a scalar obstruction, not a polytope construction or a claim that such a cusp is realized.

The separate compact arithmetic bound21 and arbitrary compact bound29 are unchanged. The effective ambient ceiling for the compact-facet alternative becomes71; its older auxiliary proof remains79. No new finite-volume polytope or new example dimension is claimed.

Verification status

Primary generation and a fresh separate-process saved replay both terminated exit0/PASS on all4,913 profiles and the new local interfaces. The separate standard-library reconstruction agrees with every exact profile row, minimum and boundary. It uses a different partition enumeration, sliding polynomial convolution, cumulative-G formula and absolute-deviation identity. It also independently reconstructs all13,034 finite-budget endpoints and the order13/weighted17 interfaces.

Two full internal source and mathematical reads of the assembled argument found no identified gap in the complete local provenance, whole-component coupons, weighted face sums, actual cusp multiplicities, new concentration domain, finite budget or inherited higher-dimensional tail. Every local input has generation, fresh replay and internal mathematical/source reviews. All31 direct primary source bindings, Python syntax and local links pass. The complete inherited overlap and induced-map censuses retain their frozen validation records and are not rerun here. Source, data and report are final/frozen. All reviews are internal to the agent workflow. External mathematician refereeing and Lean or other proof-assistant formalization are not claimed.

```
python3 scripts/fresh_ideal5_weighted_dimension71_20261017.py
python3 scripts/fresh_ideal5_weighted_dimension71_20261017.py --verify
```

Source 150

Source path: `results/fresh_ideal5_weighted_dimension71_20261017_review.md`

SHA256: 9f51ba699a84c54bc9528e796c8ac70283cdef2657991f507a118e115e451993

Independent scalar and composition review of dimension71

Completed independent exact reconstruction; fresh replay passed. This review supports the unrestricted dimension71 theorem. It imports no producer or project arithmetic code. Its standard-library source and saved certificate independently reconstruct all4,913 scalar profiles, all13,034 finite-budget endpoint cases, and the changed order13/order17 composition interfaces.

The primary's local geometric theorems have also received a full source and mathematical read in the scopes described below. The inherited complete core, profile and induced-embedding censuses are frozen dependencies; this review does not claim to independently repeat those censuses or every local Gram calculation.

Independent scalar calculation

The normalized factor-cost envelope is

Rank	Cost
1 through11	r/2
12	75/13
13	87/14
14	97/15
15	111/16
16	120/17
17 and above	163/23

The independent enumeration chooses the optional high rank, optional low rank and optional rank12 first, then recursively fills the remainder with11,13,14,15,16,17. The producer enumerates the repeated parts first and obtains the optional parts from its remainder. Exact profile-set equality verifies both completeness and absence of duplicates.

Low parts concentrate toward11 without changing their linear cost. Two12 parts become11 and13, increasing cost by16/91. Two parts at least18 concentrate until one reaches17, preserving their common constant cost. These moves decrease normalized curvature by the inherited convex-order result. Each nontrivial move strictly increases the integer sum of squared parts, bounded by the square of their fixed total. Thus termination and the stated reduced domain follow. Repeated15 parts are retained: their old elimination has the wrong cost direction after the rank16 reduction.

For each profile, sliding-window convolution constructs the coefficients of `H(t)=Π(1+...+t^r)`. Independently, the bounded-composition numerator `Π(1−t^(r+1))` gives a cumulative-count identity

`G=floor(d/2)V−2 Σ_s a_s binom(floor(d/2)−s+k,k+1).`

This agrees exactly with `Σ_j h_j min(j,d−j)`. The curvature `(5G−dV)/(2V)` also agrees with the absolute-deviation formula `3d/4−(5/2) E|Σ U_i−d/2|`. Integer convolution and rational comparisons therefore have two distinct exact reconstructions in addition to comparison with the primary output.

Ambient dimension	Profiles	Minimum ideal margin	Minimizing ranks	Finite slack
72	1530	6875/49152	11,15,15,15,15	18
73	1635	41401/145152	11,11,11,11,13,15	28
74	1748	135455/304128	10,11,11,11,15,15	37

Every primary profile row matches, including its full h-polynomial, vertex count, G, normalized curvature, cost and margin. The boundary profile `(11,11,11,11,11,15)` at ambient71 has margin `-5597/248832`, also matching independently. It is a scalar limitation of this envelope, not a realized cusp or polytope.

The unchanged all-rank B/C/D feature bounds are independently evaluated at every piecewise-linear breakpoint and their limits. On each branch the normalized cost has form `a+b/m`, so endpoints and the limiting value suffice. Their maxima are160/23,75/11 and163/23, respectively. This is an all-rank branch argument, not a finite numerical cutoff.

Finite budget and local interfaces

The new finite8 budget applies through ambient75, covering all three new dimensions. The review independently enumerates counts of D12,D13,other inherited large components,A17 andB10 first, and only then all admissible dimensions and the two small-rank endpoints. The resulting13,034 exact rows equal the producer's rows. Every surplus is nonpositive. This domain ordering differs from the producer's dimension-first loop splitting a total large-component count into its D12/D13 subcounts.

The component slacks and disjoint coupon increments also pass source/mathematical review: the D12 short-leaf upgrade costs at most `a*b`, the A2/B10 halves cost at most `a*v/2`, and the inherited A4/D13-neighbor charge remains `a*c/4`. Their rank constraint and affine endpoint reduction are valid. No geometric transfer follows merely from that numerical budget; those transfers are checked separately.

The review reconstructs the four canonical order13 affine diagrams and their distances without project graph helpers. It checks all elementary triangle pairs, changed short8 pairs, selected marked-image supports and immediate-factor interfaces against the primary. The resulting costs are A65,B75,C73,D74.

- The cycle13 theorem replaces all13 distance-six pairs, including their use in the original short8-first stage. Its fixed simple leaf and degree-two neighbor force the single-contact corner component to be finite even with arbitrary spectators. The remaining profiles force both isolated heights to1 before the unscaled energy10/13 is used. The strict spectrum excludes that value. The full source/proof was read; the six/16 arithmetic cases were not separately rerun here.
- The B12 singleton theorem seals the selected radical cores and supplies whole finite coupon components. The four saved conditional distance11 cases are checked separately: two have a finite B13 unit, and two receive the now-budgeted whole A1/D12-short-leaf unit. Its strict universal inner4 gaps1/5 and1/6 discharge the separate immediate-D14 provenance. Full source/proof review found no missing contact or indirect-tail assumption; the underlying801-pattern census remains inherited.
- The universal D12 leaf theorem covers every use of its four removed pairs. At its zero boundary, the positive spectator quadratic form and disjoint supports force both contact vectors to vanish. With one-sided contact, choosing the corner that excludes the contacting root avoids any spectator classification. With two-sided contact, equal heights are established before the strict energy-one theorem is applied; its short-path or two-whole-A1 outcomes give the stated payments. Whole actual components and strict edge hypotheses were checked in the source/proof read; the96 literal configurations were not rerun here.
- The D16 weighted theorem changes twelve order17 pairs to weight1/2. Their complete saved support consists of twelve sealed cores, each with an existing half at a distinct finite corner. The review checks the full136-entry weight vector, its exact sum120, unchanged triangle/short8 unit weights, and all twelve saved payment totals. The former zero-weight singleton retains its prior proof. A half-weight is never treated as a binary unit.

The canonical graph checks and exact saved interfaces are independent computations. The associated full source/mathematical reads are separate from the geometric censuses they bind.

Global composition and scope

The finite cost8n, ideal factor-fiber multiplicity `V/(r+1)`, and fractional pair weights compose linearly. All triangle totals remain at least2 and quadrilateral totals at least1. The new local replacements cover the formerly paid order13 short, selected and immediate cases; unchanged cases retain the earlier complete face theorem.

Every reduced ideal profile at ambient72–74 has positive margin, and the finite slacks are18,28,37. The inherited global shape inequality has the positive constant `5(n−1)`. Summing the face payments therefore contradicts the vertex budgets in these three dimensions. The completed dimension74 theorem excludes every dimension at least75. This proves the unrestricted ceiling71 under the stated finite-sided, finite-volume real-hyperbolic Coxeter hypotheses, without arithmeticity, simple-cusp or factor-count restrictions.

The negative boundary margin does not prove optimality of71. No new example, compact-arithmetic dimension result, external refereeing or proof-assistant formalization is asserted. The compact arithmetic partial models remain separate from this geometric proof.

Verification status

All4,913 scalar rows,13,034 budget rows and composition interfaces passed comparison with the primary. The final generated certificate binds the frozen primary source, data and report. A fresh separate-process saved replay, run directly by the root agent as session15598, terminated exit0/PASS on the complete reconstruction. All33 direct source bindings, Python syntax and local report links pass.

A separate full internal source/mathematical read passed the independent partition enumeration, three curvature calculations, all-rank branch bounds, alternative finite-budget enumeration, canonical graph reconstruction and weighted local composition. That read did not duplicate the4,913 scalar calculations or inherited geometric censuses. Source, data and report are final/frozen. All reviews are internal; no external refereeing or proof-assistant formalization is claimed.

Source 151

Source path: `results/fresh_ideal6_combined_payments_20261017.md`

SHA256: 842135eb5e1043a1ebeb823d81a5d3f0dfca8de51a29fd8535b8b0bb26f2a5b4

Complete finite9/ideal6 payments and piecewise cusp budgets

This package combines the local arguments into a single payment rule for arbitrary finite-sided finite-volume hyperbolic Coxeter polytopes. Every triangle receives at least2 and every quadrilateral at least1. For n<=108 every finite vertex costs at most9n. At an ideal vertex with simplex-factor ranks r_i and V link vertices, the cost is at most V*sum_i b(r_i), where

```
b(r)=r/2 for1<=r<=13;
b(14)=20/3; b(15)=111/16; b(16)=127/17;
b(r)=95/13 forr>=17.
```

The binder and certificate retain the complete source dependencies. A global dimension comparison is separate.

The rule

At finite vertices retain unit payments for same-component marked pairs of distance at most9 or whole-component order at most17, together with the broad finite half-coupons. Those coupons apply to whole components, with one of rank at most4 and the other a specified fork-marked D12,…,D16 or middle-marked A13,A15.

At ideal vertices pay every pair in a varying factor of order at most14. For larger factors the basic unit rule is distance at most6, or distance7–9 with an endpoint within6 of a branch or nonsimple bond endpoint. At orders 15,16,17 add all induced marked images of the complete inherited preferred components. Remove the nine D pairs at each of those orders: the two fork centers and the eight pairs spanning the two fork triples. Their finite replacements are proved separately, including all supporting whites.

The exact resulting A/B/C/D costs are

Factor order	A	B	C	D
15	90	100	95	96
16	96	107	103	111
17	102	111	108	127

The small-factor census enumerates every relevant induced labelled map of437 preferred contexts. The new direct affine-D14 transfer handles all nine order15 pairs universally, so it handles all63 inherited contexts affected by their deletion without assuming that their original small cores are whole actual stars. The frozen order16 and order17 arguments provide the other two nine-pair transfers. Every finite coupon they use is included in the broader rule.

Every face is paid

The triangle theorem applies directly: its short paths and feature witnesses remain paid, and its sole affine-E8 exception uses a whole factor of order at most9. None of the nine removed D pairs at orders15–17 belongs to these short/feature payments.

For a quadrilateral, suppose first that some actual corner has a path of length at most9. A finite corner pays immediately. An ideal factor of order at most14 pays in full. An unpaid pair in a larger classical factor has distance7,8 or9 and lies outside

the relevant feature neighborhood. The mixed6 replacement handles distance7 cycles and tree distances7–9. For cycles of distance8 or9, the preceding mixed7 theorem gives another finite-distance9 or ideal-distance7 payment when the factor order is at least17. If that ideal-distance7 corner is still unpaid, the new mixed6 theorem applies there. Since the whole face was assumed unpaid, each such transfer directly contradicts that assumption; no iteration or termination premise is needed. The antipodal order16 cycle case is handled by its finite A15-center coupons. No cycle of order15 has distance8 or9. These cases cover every short path.

Otherwise there is no short9 path at any corner, precisely the hypothesis of the inherited preferred-component theorem. Its complete small-core classification gives a preferred component of order at most12; the complete high-core classification gives one of order at most17. A finite preferred corner pays by its whole-component rule. At an ideal corner, order at most14 pays in full, and every possible image at orders15–17 was included above. Only the nine removed D pairs can remain; their universal finite transfer pays the face. Thus every quadrilateral receives at least1. Polygons with at least5 sides need no positive lower payment for the D5 comparison.

Vertex budgets and scope

The broad finite-coupon theorem proves the9n bound for every n<=108, using only whole-component ranks, slack and at most three eligible large marks. At factor order m<=14, binom(m,2)/m=(m-1)/2. The table gives the exceptional normalized maxima20/3,111/16,127/17. For order at least18 the preferred component bound allows no extra pair. The mixed6 all-rank spine formulas and their12 linear branches give a common normalized bound95/13; cycles cost6 per node. These are all-rank arguments, not extrapolation from a finite graph sample.

The actual cusp-factor theorem counts each pair in a rank-r factor with multiplicity V/(r+1). This gives the stated ideal cost V*sum_i b(r_i), without a simple-cusp, arithmetic or minimum-factor-rank assumption.

The direct-D14 proof and this composition have received separate internal mathematical reads. These check the complete large-D extension profiles, arbitrary supporting spectators, whole-component sealing, the short-path case split, and retention of every triangle and preferred-component payment. The binder checks the exact nine removed D14 pairs against the local certificate, all72 equality payments, all12 small-factor costs and all12 large-factor linear branches. Generation and saved replay are recorded in the dimension80 verification package and its archive. No external refereeing or proof-assistant formalization is claimed.

Source 152

Source path: `results/fresh_ideal6_piecewise_dimension80_20261017.md`

SHA256: 918c37a7f8bcf1eae66871dcd0602698fd8095b2958b3931fa2a4037ee98e95f

Dimension80 from the finite9/ideal6 payment rule

Completed internally checked computer-assisted proof. Every finite-sided, finite-volume real hyperbolic Coxeter polytope has dimension at most80, without an arithmetic, simple-cusp or facet-count hypothesis. The local geometric arguments, their complete composition and the scalar comparison have passed the internal checks specified below. External refereeing and proof-assistant formalization remain separate, outstanding checks.

The source and certificate bind the combined local payment theorem, the completed86 theorem and its independent review. The new local input has finite cost at most9n for n<=108 and ideal cost V*sum_i b(r_i), with

```
b(r)=r/2 for1<=r<=13;
b(14)=20/3; b(15)=111/16; b(16)=127/17;
b(r)=95/13 forr>=17.
```

The order15 improvement rests on a universal transfer of nine long affine D14 pairs to finite half-payments. It is not an assumption that the63 saved small-core contexts have no further supporting whites. The broad finite budget covers every finite coupon in the combined proof.

Reduction of all cusp-link shapes

For a link L=product_i Delta^(r_i), put d=sum_i r_i=n-1,

```
H(t)=product_i(1+t+...+t^r_i)=sum_j h_j*t^j,
V=product_i(r_i+1),       G=sum_j h_j*min(j,d-j).
```

The normalized curvature is kappa(L)/V=(5G-dV)/(2V). The inherited convex-order argument proves that transferring(a,b) to(a-1,b+1) for1<=a<=b can only decrease this curvature, with other factors fixed.

Concentrate two ranks below13 until one vanishes or the other reaches13. Their linear cost is preserved. Replace two14 parts by13 and15; this increases cost by5/48. Replace two15 parts by14 and16; this increases cost by107/408. Concentrate two ranks above17 until one reaches17; their constant cost is preserved. In each case curvature minus cost cannot increase.

The middle two moves can create each other's exceptional ranks, so a single pass is not a termination proof. Instead every move strictly increases the integer sum of squared ranks at fixed total d, which is bounded by d^2. After all available moves stop, there are arbitrary copies of13,16,17, at most one low part1,…,12, at most one14, at most one15, and at most one high part at least18. Checking all these reduced shapes therefore covers every integer partition of d.

Exact comparison

The primary checker computes H,G,V by exact integer convolution and checks the equivalent absolute-deviation identity. The independent checker uses a different enumeration, sliding-window convolution and a cumulative inclusion-exclusion identity. All5,008 reduced shapes in dimensions81–86 agree and have positive curvature minus the ideal cost.

Ambient dimension	Profiles	Minimum normalized margin	Finite slack beta_n-9n
81	742	940675/11294304	31
82	776	1132393/9143008	41
83	813	3418145/9143008	105/2
84	852	15382621/27429024	63
85	892	221295/326536	75
86	933	12970945/15059072	86

Here beta_n=n(n-6)/8 for even n and(n-1)(n-5)/8 for odd n. The overall minimum occurs in ambient81 at ranks(2,13,13,13,13,13,13).

The inherited shape inequality gives

`D5 >= beta_n*v_f + sum_cusps kappa(L) + 5(n-1).`

The actual face payments give the upper bound9n*v_f plus the ideal budgets. The strictly positive finite and ideal margins contradict these inequalities for every n=81,…,86. The completed86 theorem excludes all n>=87. Together these two ranges give n<=80. The compact-facet ambient ceiling79 and compact/arithmetic bounds are unchanged.

At ambient80, the formal profile(1,13,13,13,13,13,13) has margin -1131729/7529536. This particular comparison therefore does not exclude80. It is a formal link-shape obstruction, not a constructed polytope or a claim that the bound is sharp. Lowering order14 costs or a different global allocation is the next required improvement.

Verification scope

The independent review states its arithmetic and composition scope. Primary generation and fresh saved replay both terminated exit0/PASS, as did generation and replay of the combined local-payment binder. A separate agent independently compared every field of all5,008 scalar profiles, the six dimension minima and the boundary profile, and independently derived the12 all-rank affine branches. The complete source and written composition received a separate mathematical read, including the D14 profiles and sealing, the short-path/no-short-path split, termination of concentration, actual cusp multiplicities and the86 tail. These checks identify no remaining gap in the stated argument.

The independent scalar checker does not duplicate the inherited minimal-core classification or local geometric censuses. Those remain explicitly linked proof inputs with their own certificates and review scopes. Final version bindings and saved replays are recorded in the independent review and the dimension80 checkpoint archive. All reviews here are internal; no external mathematical refereeing, publication priority or proof-assistant formalization is asserted.

```
python3 scripts/fresh_ideal6_piecewise_dimension80_20261017.py
python3 scripts/fresh_ideal6_piecewise_dimension80_20261017.py --verify
```

Source 153

Source path: `results/fresh_ideal6_piecewise_dimension80_20261017_review.md`

SHA256: c82c5c75bfe22a627dd69b221a84ae9d223173c8cea99f1e96d9c0be27b37571

Independent review of the dimension80 comparison

FINAL/FROZEN: independent arithmetic, written/source composition review, final hash-bound generation and fresh saved replay PASS. The independent checker imports no producer or project arithmetic. It reconstructs all5,008 reduced profiles using standard-library integer/Fraction arithmetic and compares every saved scalar field with the primary certificate. The complete primary proof and local composition have been read; no mathematical gap was identified within their stated inherited inputs.

Independent arithmetic

The proposed normalized cusp-factor budget is

```
b(r)=r/2 for1<=r<=13;
b(14)=20/3; b(15)=111/16; b(16)=127/17;
b(r)=95/13 forr>=17.
```

For each link dimension d=80,…,85, the independent enumeration chooses the optional high part in{0,18,…,d}, optional low part in{0,…,12}, and the presence or absence of14 and15. It then chooses the multiplicities of17 and16. The remaining nonnegative rank must be divisible by13, forcing its multiplicity. Zero parts are omitted and sorted shapes are deduplicated. This exhausts the stated reduced domain without using a producer list.

Sliding-window convolution independently computes

```
H(t)=product_i(1+t+...+t^r_i)=sum_j h_j t^j,
V=product_i(r_i+1),  G=sum_j h_j min(j,d-j).
```

The code checks polynomial length, symmetry and H(1)=V. It separately expands product_i(1−t^(r_i+1))=sum_l a_l t^l, combining repeated shifts. For s factors and q=floor(d/2), the cumulative-tail identity gives

```
G=qV-2 sum_l a_l binom(q-l+s,s+1).
```

This agrees exactly with the coefficient computation for every profile and the boundary profile. The absolute-deviation identity is also checked:

```
kappa/V=(5G-dV)/(2V)
       =3d/4-(5/2)E|sum_i U_i-d/2|.
```

Ambient dimension	Profiles	Minimum curvature minus cost	Finite slack beta_n−9n
81	742	940675/11294304	31
82	776	1132393/9143008	41
83	813	3418145/9143008	105/2
84	852	15382621/27429024	63
85	892	221295/326536	75
86	933	12970945/15059072	86

The minimum is940675/11294304 at ranks(2,13,13,13,13,13,13). The ambient80 boundary profile(1,13,13,13,13,13,13) has margin−1131729/7529536. This boundary is a limitation of the particular scalar comparison, not a realized chamber or a sharpness claim.

Completeness of the shape reduction

The inherited convex-order theorem says that transferring(a,b) to(a−1,b+1) for1<=a<=b can only decrease normalized curvature, with the other uniform factors fixed.

Two ranks at most12 can be concentrated until one vanishes or the other becomes13; their linear budget is preserved. Two ranks14 can be replaced by13 and15, increasing the budget by5/48. Two ranks15 can be replaced by14 and16, increasing it by107/408. Two ranks at least18 can be concentrated until one becomes17, preserving the constant budget. In every case curvature minus budget cannot increase.

The two middle-rank moves can create each other's exceptional rank, so they must be allowed to repeat rather than applied only once in a fixed order. Termination still follows: every elementary transfer strictly increases the integer sum of squared ranks at fixed total d. The process is bounded and ends with at most one14 and one15. Low transfers stop at13, high transfers stop at17, and the middle moves produce only13,14,15,16. The final partition therefore consists of arbitrary copies of13,16,17, at most one low rank1,…,12, at most one14, at most one15, and at most one high rank≥18. Every possible terminal shape is enumerated above.

The helpful cost increases are checked exactly, not treated as cost preservation. Positivity at a reduced shape implies positivity at the original shape.

Local finite-budget review

A separate narrow source/mathematical read of the broad finite half-coupon budget passes. Let a be the total rank of whole finite components of rank at most4, and let t count whole components D12,…,D16,A13,A15. The small-component base slack is at least15a/2. Each listed large component has rank at least12, slack at least24 and at most three eligible marked nodes. The total cross-component half-payment is at most3at/2, counted once per eligible pair. Thus

```
cost<=9n+a*(3t/2-15/2)-24t,    a+12t<=n.
```

For n<=108, t<=9. The coefficient of a is nonpositive for t<=5. For t=6,7,8,9 the maximal excesses are−90,−96,−138,−216. This proves the finite budget independently of the small components' particular finite types. It is a counting theorem only: the separate local geometry must ensure that a proposed coupon refers to whole actual components.

Independent all-rank ideal cost calculation

For factor order m>=18, a direct spine-interval count gives the following costs for the ideal distance6/feature6 rule:

```
A: 6m;
B: 6m-14 + sum_{s=9,10,11} min(m-s,15);
C: 6m-21 + sum_{s=7,8,9} min(m-s,16);
D: 6m-7 + sum_{s=11,12,13} min(m-s,14).
```

For D, the distance-at-most6 contribution is6m−19; each distance7,8,9 adds four leaf-to-main pairs and a main-spine window of length min(m−4−distance,14). For B, the corresponding short contribution is6m−20, with two leaf-to-main pairs per added distance and main windows of combined width15. For C the short contribution is6m−21 and the two end windows have combined width16. Opposite-end leaf pairs do not enter these distances at m>=18.

The independent checker constructs the twelve linear branches from their window breakpoints and compares all branch coefficients with the local certificate. On each branch cost/m is a+b/m, so its maximum occurs at an endpoint or at the limiting value6. The exact normalized maxima are36/5 for B at m=25,85/12 for C at m=24, and95/13 for D at m=26. This proves the all-rank95/13 envelope without extrapolating a finite graph sample. The exceptional orders15,16,17 use the explicitly bound small-factor census, with normalized maxima20/3,111/16,127/17. Whole factors of order at most14 cost binom(m,2)/m=(m−1)/2.

Local geometry and composition review

The complete source and written proof of the universal D14 long-pair transfer were read separately. The large finite-D extension profiles are exhaustive in the stated endpoint domains. At a fork center, the arbitrary outer contacts obey the strict edge-PD inequality u²+v²<4. Nonextreme end profiles have strictly positive determinant margin. In the remaining cases both heights are1/2 before the spectator-gap argument is applied. Its equality case permits exactly one unit contact to an isolated spectator; all other supporting whites are orthogonal to the resulting core. Thus the finite coupons refer to whole actual components. The72 literal equality cases and their corner/coupon reconstruction remain producer arithmetic inputs, not a second census by this reviewer.

The combined payment theorem retains the complete triangle payment: removed D pairs are too distant to remove the short or feature witnesses. For a quadrilateral with some corner path of length at most9, finite corners and small ideal factors pay directly. The mixed6 tree and cycle replacements cover the remaining distance7 cases and tree distances7–9. For a large cycle of distance8 or9, the preceding mixed7 transfer gives finite9 or ideal7 payment, and the latter is covered by mixed6 if necessary. The order16 antipodal cycle has its separate finite A15-center coupons. These implications are applied under the assumption that the entire face is unpaid, so no iterative termination assumption is needed.

If no corner has a path of length at most9, the hypothesis of the inherited complete preferred-component theorem is exactly satisfied. Its low and high cases give components of order at most12 and17 respectively. Finite components pay directly; ideal components of order at most14 pay in full. Every induced marked image at orders15–17 is included in the small-factor census, except the explicitly removed nine D pairs at each order. The universal D14 transfer and the frozen order16/order17 transfers pay those exceptions. The combined binder matches the nine removed order15 pairs to the D14 marked-pair domain. The broad finite budget includes all finite coupons used in these transfers and is valid throughout the needed dimensions81–86.

At an actual nonsimple cusp, a pair in a rank-r simplex factor has multiplicity V/(r+1). The inherited actual-factor theorem therefore gives the ideal budget V*sum b(r), with no simplicity, arithmetic or minimum-factor-rank restriction. The inherited shape inequality has lower bound beta_n*v_f+sum kappa(L)+5(n−1). Every finite and ideal coefficient exceeds its payment cost in dimensions81–86 by the exact positive margins above. The completed dimension86 theorem supplies the tail n>=87. The resulting composition proves the stated unrestricted bound80; the compact-facet ambient bound79 is unchanged.

Verification scope

All5,008 primary scalar rows, six minima and the boundary row agree exactly. Polynomial coefficients, cumulative-tail curvature and absolute-deviation curvature agree for all5,009 tested shapes. All twelve independently reconstructed affine branches agree with the local binder. The complete shape reduction and primary/local mathematical composition have received the source reads described above. The parent reviewer separately checked the D14 local proof and this independent scalar source.

The local geometric classifications, the complete inherited preferred-component census, and the actual cusp-factor/shape theorems remain explicit dependencies. This review does not duplicate their full geometric or embedding censuses. Final generation and a separate fresh saved replay both terminated with exit0/PASS after the primary report was frozen. All31 direct source hashes, source syntax and local report links were checked. No external refereeing, publication-priority claim or proof-assistant formalization is asserted.

The saved review binds primary source SHA256 7c160770b5c8503eda48ae723d83ef0b7c6c540f37acf9f554d7249152d96567, primary JSON 0779c84a01fd5d9a4179db3ffcbad0dcd3b9744b6f8c2f5c13487d122bf5c133, and primary report 918c37a7f8bcf1eae66871dcd0602698fd8095b2958b3931fa2a4037ee98e95f.

```
python3 scripts/fresh_ideal6_piecewise_dimension80_20261017_review.py
python3 scripts/fresh_ideal6_piecewise_dimension80_20261017_review.py --verify
```

Source 154

Source path: `results/fresh_ideal6_small_factor_frontier_20261017.md`

SHA256: 5eabdcbf0a6cf18549f44ef79b9a0d632d0aae136a71622707fd12605c1cad72

The small-factor obstruction to the ideal-distance6 improvement

The complete inherited preferred-component census shows why the proposed distance6 ideal rule does not yet lower the completed dimension86 bound. Order15 affine D still receives all105 pairs, normalized cost7. The old boundary profile with ranks(1,14,14,14,14,14,14) therefore retains its negative margin, regardless of improvements at larger ranks.

The checker and data retain all437 preferred contexts with component extension bound at least15:177 with bound15,180 with bound16 and80 with bound17. For each target order, contexts with a smaller proven extension bound are omitted. Every injective induced labelled map is enumerated by the inherited explicit backtracking routine; edges and nonedges are both checked. No graph-library matcher is used.

The proposed basic ideal rule pays graph distance at most6 and distances7,8,9 when an endpoint is within6 of a branch or a nonsimple bond endpoint. The table adds all preferred-pattern images. It then removes only the nine D pairs at each of orders16 and17 whose replacement by finite payments has already been proved in the frozen preceding chain.

Factor order	A cost	B cost	C cost	D cost
15	90	100	95	105
16	96	107	103	111
17	102	111	108	127

The triangle part of the new rule has its own complete local proof. Universal quadrilateral replacement, small-cycle transfers, complete finite budgets and global comparison are separate obligations. This embedding census is not by itself a complete payment theorem.

The next concrete local problem

At order15, the analogous proposed D deletion consists of the nine pairs at distance at least11 or joining the two fork centers. It affects63 distinct preferred contexts. They involve mixtures of catalogue diagrams18,22,25 and several overlap types, so treating only the nine longest-overlap L25/L25 stars would not close the domain. All63 indices and their labelled maps are saved. No order15 pair is removed from the reported costs.

There is an additional reason not to copy the preceding sealing proof. For the two L25 cores overlapping in three consecutive terminal nodes, the inverse contact block is2[[1,-1],[-1,1]], whereas the order16 proof used4 on the diagonal. A single unit contact now has projected norm2, so the old strict norm contradiction does not apply. Possible equality extensions need their own analysis. This observation is a direction for the next calculation, not a universal exclusion or a proposed finite-volume example.

Verification scope

The census inherits the complete distance9 preferred-core classification; it does not silently broaden that classification's no-short9 hypothesis. The all-label geometry and extension bounds remain explicit inputs from the dimension108 package.

Old order16/17 deletions retain their original mathematical prerequisites. No new ambient dimension bound or construction follows from this package.

Generation and a separate saved replay both terminated exit0/PASS. A separate agent read the complete new source and report and found no identified issue in the domain filter, induced-map search, pair unions or stated limitation. That review did not repeat the embedding enumeration or the inverse contact calculation. The archive separately checks every emitted labelled map and the pair unions. This source, data and report are final/frozen.

```
python3 scripts/fresh_ideal6_small_factor_frontier_20261017.py
python3 scripts/fresh_ideal6_small_factor_frontier_20261017.py --verify
```

Source 155

Source path: `results/fresh_ideal6_triangle_payments_20261017.md`

SHA256: 3f54ea4f448f5ea94f771506205848098383b433d93b5f6f0956b9e39f845131

Triangle payments with finite distance9 and ideal distance6

This local theorem gives every actual triangular two-face total corner payment at least2 under the following rule. A finite corner pays1 if the marked sides have graph distance at most9 in its full vertex diagram. An ideal corner pays1 if its varying affine factor has at most9 nodes, or the marks have distance at most6, or their distance is at most9 and an endpoint is within6 of a branch vertex or an endpoint of a nonsimple bond. A corner pays at most1 under this rule. There are no cross-component coupons in this triangle theorem.

The exact checker and data enumerate all920 marked triples in the inherited minimal indefinite diagrams of orders8,9,10. The complete classification is an explicit input from the high-rank census. The actual supporting-white and cusp-factor geometry is inherited from the factor framework. This package alone gives no new ambient dimension bound.

Reduction to the finite census

Let W be the elliptic supporting diagram of an actual triangular two-face, and let its sides be a,b,c. Its full star is indefinite. Every W together with two sides is a finite or ideal corner and hence has positive-semidefinite Gram matrix. Consequently any minimal indefinite subdiagram contains all three sides. Such a diagram has order at most10 by the stated classification. For order at most7, a Steiner tree on the three marked nodes has at most6 edges. At least two of its marked-to-marked paths avoid the third mark. Each gives a paid corner, including at an ideal vertex.

For orders8,9,10, remove the third mark for each corner and find its marked path and branch/bond features. There are224,336,360 triples respectively. Only nine triples fail the pure distance6 test. The feature rule pays at least two corners in all920 triples except diagram25 with marks(1,2,9). Every saved feature witness persists under additional supporting whites: an existing branch or nonsimple bond cannot disappear, and an existing path cannot become longer. At an ideal corner the connected marked core lies in its varying affine factor. At a finite corner every saved path has length at most9. Thus these are witnesses in the full corner, not assumptions that a small finite subdiagram remains a whole finite component.

The one exceptional triple

Diagram25 is simply laced, with edges

`0–1, 0–2, 0–4, 2–3, 4–5, 5–6, 6–7, 7–8, 8–9.`

Deleting9 leaves affine E8 on nodes0,…,8. This is present at corner(1,2). The exact checker verifies its positive kernel, rank8 and a positive-definite eight-node principal matrix. Every extra supporting white is orthogonal to this entire affine diagram: positive semidefiniteness at corner(1,2) forces its pairing with the positive kernel to vanish, and all individual pairings are nonpositive. Corner(1,2) already pays through its path1-0-2.

At corner(1,9), deleting2 leaves A8+A1, with marked path

`1–0–4–5–6–7–8–9`

and isolated node3. The checker verifies this labelled path and positive definiteness exactly. The preceding orthogonality forces node1 to remain a simple-bond leaf and its neighbor0 to retain degree2 in the full corner. No classical irreducible

affine diagram has a simple-bond leaf whose neighbor has degree2: affine A has no such leaf, affine C has nonsimple end bonds, and the simple leaves of affine B and D meet branch vertices. The remaining irreducible affine diagrams have at most9 nodes. This uses the standard finite/affine classification already assumed by the census.

Therefore, if corner(1,9) is finite, its length7 path pays1. If it is ideal, its varying affine factor must be exceptional and has at most9 nodes, so the whole-factor rule pays1. Either way the triangle receives its second payment, with arbitrary extra supporting whites included.

Verification scope

The checker uses standard-library rational arithmetic and no project arithmetic imports. It verifies all920 triples, every path/feature witness, and the literal exceptional Gram calculations. It does not redo the inherited classification, formalize the geometric argument, or prove a quadrilateral payment rule. Generation and a separate saved replay both terminated exit0/PASS. A separate agent read the complete source and written argument and found no identified gap in the kernel, fixed-leaf, feature-persistence or lower-order steps. That review did not rerun the920 arithmetic cases or the inherited classification. No external refereeing or proof-assistant validation is claimed. This source, data and report are final/frozen.

```
python3 scripts/fresh_ideal6_triangle_payments_20261017.py
python3 scripts/fresh_ideal6_triangle_payments_20261017.py --verify
```

Source 156

Source path: `results/fresh_ideal7_cross_dimension91_20261017.md`

SHA256: 892a2689396aad5f772ca6c7fd9f901bdca3069acddbc6b6bf6184e6c4e4326b

Finite cross-component payments and ideal cost seven and a half: dimension91

Complete internally checked computer-assisted proof. This argument gives dimension at most **91** for every finite-sided, finite-volume real hyperbolic Coxeter polytope. There is no arithmetic, compactness, simple-cusp, cusp-factor-rank or facet-count hypothesis. Verification and its limits are recorded below. The compact-facet ambient ceiling79, compact-arithmetic ceiling21 and arbitrary compact ceiling29 are unchanged.

The source and certificate combine the new local payment theorem with the completed dimension97 proof. The independent review reconstructs the scalar arithmetic separately. No construction or sharpness assertion is involved.

Payment of actual faces

Retain finite weight1 for connected marked pairs at distance at most9 or in a whole irreducible component of order at most17. Add weight1/2 between A1 and D16, or between A2 and D14, when the D mark is its fork center or a short fork leaf. All component types refer to whole components of the actual finite vertex diagram.

At ideal vertices use the intrinsic factor rules of the local theorem: whole factors of order at most16; selected order17 pairs; and for larger orders distance at most7 together with distances8/9 near a diagram feature. The theorem proves that every triangle receives at least2 and every quadrilateral at least1. Its complete preferred-component embedding census handles the order17 exception. A Lorentzian radical relation seals the exceptional19-root stars against outside support, so the two finite half-payments remain valid in every actual ambient face star. This step is not an inference from the known H17 example alone.

Thus the sum of all assigned corner weights is at least

```
D5 = sum over two-faces F of (5 - number of sides of F).
```

Larger polygons cause no difficulty because weights are nonnegative.

Finite and ideal cost bounds

The local theorem proves finite-vertex cost at most9n for every n<=108. If a,b,c,d count A1,A2,D14,D16 components, its explicit bound is

```
9n - 9a - 17b - 35c - 24d + (3/2)ad + 3bc <= 9n,
a + 2b + 14c + 16d <= n.
```

The proof allows arbitrary other finite components. The extra finite payments therefore cost no additional total budget in the dimensions needed here.

Every irreducible ideal factor of order m costs at most

```
min(15m/2, m(m-1)/2).
```

The proof covers every rank by path, cycle and fork counts; the saved finite BFS range is a corroboration, not the justification of the tail. For a cusp with affine ranks r_i, let d=n-1 and V=product_i(r_i+1). The actual factor-fiber theorem gives

multiplicity V/(r_i+1) for each pair in factor i, including at nonsimple cusps. Its total normalized cost is consequently at most

```
sum_i min(15/2, r_i/2).
```

For h(t)=product_i(1+t+…+t^(r_i)) and G=sum_j h_j min(j,d-j), the inherited shape curvature inequality has normalized ideal coefficient (5G-dV)/(2V). Its finite coefficient is

```
beta_n = n(n-6)/8              for even n,
beta_n = (n-1)(n-5)/8          for odd n.
```

It gives a lower bound for D5 with a strictly positive global slack. Therefore coefficients at least the corresponding payment budgets would already contradict the payment upper bound. The comparisons below are strict at every vertex type.

All-rank reduction and exact comparison

The inherited convex-order transfer applies to an arbitrary integer cap, here15. It preserves the cost sum_i min(15/2,r_i/2) and can only decrease the curvature when a partition is concentrated to copies of15, at most one smaller positive part and at most one larger part. Thus those shapes suffice for all rank partitions.

The source enumerates all477 reduced shapes for link dimensions91 through96:

Ambient dimension	Reduced shapes	Minimum curvature minus cost	Finite slack beta_n-9n
92	77	10121/4194304	161
93	78	1497773/6291456	175
94	79	3493651/8388608	188
95	80	1307621/2097152	405/2
96	81	3244935/4194304	216
97	82	13980081/14680064	231

The overall minimum is10121/4194304 at ranks(1,15,15,15,15,15,15), ambient dimension92. Every listed ideal margin and finite slack is positive, contradicting the two bounds for D5. These dimensions cannot occur. The completed97 theorem excludes all larger dimensions, so the new ceiling is91.

At ambient91, the six-rank15 profile has margin-428955/2097152. This specific scalar comparison therefore cannot prove a ceiling90. The profile is a formal possible link shape, not a realized finite-volume polytope or an existence result.

Verification scope

The producer uses exact integer convolution and checks the curvature a second way by an inclusion-exclusion moment formula. It binds the local payment theorem, its mixed replacement input, the completed97 proof and review, and the factor/shape/convex-order arguments. Its full generation and fresh saved replay both terminate exit0 with `PASS_FINITE_CROSS_IDEAL7_DIMENSION91` for all477 profiles.

The separate checker uses no project arithmetic: it enumerates the reduced shapes differently, reconstructs h by sliding-window convolution, and compares a cumulative-tail inclusion-exclusion formula for G and an absolute-deviation curvature identity. Its independent exact comparison passes every polynomial, G, curvature, cost and margin for all477 profiles, all six minima, the finite coefficients and the boundary. Separate internal mathematical/source reviews confirm the local payments and budgets, actual factor multiplicities, cap15 concentration and completed97 tail, with no identified gap. The linked review records its saved certificate and final replay status. These checks do not rerun the inherited complete core classifications. External refereeing and proof-assistant formalization are not asserted.

Source 157

Source path: `results/fresh_ideal7_cross_dimension91_20261017_review.md`

SHA256: 82402efd009bea861aceb7a1a1426c38845dc98912b8e7473a44060ea5d43daf

Independent scalar and composition review of the dimension91 argument

FINAL: independent exact generation and fresh saved replay both PASS. The checker uses only Python's standard library and exact integers/Fractions. It imports neither the producer nor any project arithmetic module. It independently reconstructs all477 cap15 profiles and the dimension91 boundary. A separate written/source read finds no identified gap in the composition, with the inherited geometric classifications and the new selective-order17 payment theorem retained as explicit inputs.

The primary result under review is the dimension91 proof. The review does not claim a new independent80-pattern graph-embedding census or a second derivation of the complete dimension108 classification.

Independent exact scalar reconstruction

For each link dimension d=91,…,96, the checker chooses the exceptional large part b in{0,16,…,d} and small part a in{0,…,14} first. The number of15-parts is then forced by d−a−b, which must be nonnegative and divisible by15. Removing zero parts and sorting gives every reduced profile exactly once after deduplication. This differs from the producer's loops over the number of15-parts and the small part.

For a rank profile, sliding-window multiplication independently reconstructs

```
H(t)=product_i(1+t+...+t^r_i)=sum_j h_j t^j,
V=product_i(r_i+1),  G=sum_j h_j min(j,d-j).
```

Polynomial length, symmetry and H(1)=V are checked. A second calculation expands only the numerator product_i(1−t^(r_i+1))=sum_l a_l t^l. With s the number of factors and q=floor(d/2), the cumulative-tail/hockey-stick identity gives

```
G=qV-2 sum_l a_l binom(q-l+s,s+1),
```

where out-of-range binomial terms are zero. It agrees with the coefficient calculation in all477 profiles and the boundary profile. The checker also verifies

```
kappa/V=(5G-dV)/(2V)
       =3d/4-(5/2)E|sum_i U_i-d/2|.
```

The budget is sum_i min(15/2,r_i/2). Every checked reduced shape has strictly positive curvature-minus-budget. The minimum is10121/4194304 at ranks(1,15,15,15,15,15,15), ambient dimension92. The six-rank15 profile at ambient91 has margin−428955/2097152. This negative boundary is a limitation of this particular comparison, not a realized polytope or a sharpness claim.

The finite curvature coefficients beta_n=n(n−6)/8 for even n and (n−1)(n−5)/8 for odd n exceed9n in all ambient dimensions92 through97. Final comparison checks every reconstructed polynomial, G, curvature, cost, margin, per-dimension minimum and boundary against the producer's saved rows.

Mathematical composition reviewed

The new local payment package retains finite connected distance9/whole-component17 payments and adds half-payments only between whole A1/D16 or A2/D14 components at the specified fork positions. Every triangular face retains two path7 unit payments. A purported quadrilateral with total payment below1 cannot have an adjacent path of length at most9: at a finite corner it pays directly, and at an ideal corner it is either paid or invokes the mixed replacement theorem. The completed no-short-path classification therefore remains applicable.

The order17 exception treatment explicitly uses all induced labelled embeddings of the80 inherited preferred components with upper bound17. Its retained and added pairs cost119,111,108,127 on affine A,B,C,D respectively. The nine D deletions are restricted to literal two-L25 stars sharing node9. In each listed case roots3 and13 are adjacent black sides, and the complementary17-root diagram is the full affine D factor at an actual ideal corner. A supporting white is orthogonal to that factor by its positive kernel and acuteness. The actual Lorentz-space radical relation then forces its two remaining contacts to be both nonzero or both zero. Both nonzero give a paid length2 path. Thus an unpaid star is sealed from all further supporting whites. The saved finite corner components survive unchanged and the listed unit or two half-payments complete its demand. This is the key universal extension argument; inspecting a known H17 chamber alone would not suffice.

The finite coupon cost is bounded by(3/2)ad+3bc for counts a=#A1,b=#A2,c=#D14,d=#D16. Their inherited budget savings relative to9n are9a,17b,35c,24d. For n<=108, d<=6 makes a((3/2)d−9)<=0. If c<=5, the b,c contribution is nonpositive. The only remaining cases c=6 or7 satisfy b<=12 or5 respectively and are strictly negative. This proves finite cost<=9n throughout the dimensions used in the conclusion, independently of all other finite component types.

The all-rank affine cost formulas use explicit spine/fork counts, not extrapolation from the444 BFS checks. Their three linear branches have slopes9,8,7; checking the finite branch endpoints and the increasing slack of the final slope7 branch proves cost<=15m/2 at every order m>=18. The selective17 table and whole smaller factors also satisfy this bound. Total pairs additionally give cost<=m(m−1)/2. Thus a rank r factor has cost at most(r+1) min(15/2,r/2).

At a nonsimple cusp, the inherited actual factor-fiber theorem gives multiplicity V/(r_i+1) for each pair of factor i. The entire ideal cost is therefore at most V sum_i min(15/2,r_i/2), with no minimum-rank, factor-count, arithmeticity or simple-cusp restriction.

Why finitely many profiles suffice

The all-rank convex-order transfer replaces integer ranks1<=a<=b by(a−1,b+1), increasing the centered sum in convex order. Applied only to two ranks below15 or two above15, and stopped at0 or15, this preserves sum min(15/2,r_i/2) and can only decrease normalized curvature. The integer sum of squares strictly increases at fixed d, so repetition terminates with copies of15, at most one smaller positive rank and at most one larger rank. Hence the477 checks cover every cusp-rank partition in the six relevant dimensions, rather than a sample of shapes.

The inherited shape inequality gives

```
D5 >= beta_n*v_f + sum_cusps kappa(L) + 5(n-1).
```

Actual corner payments give the corresponding upper bound with finite cost9n and the ideal budgets above. The independently positive margins and beta_n>9n contradict these inequalities for n=92,…,97. The completed dimension97 theorem excludes every larger dimension. The final local and primary certificates are bound; the resulting ceiling is91 for all finite-sided finite-volume real hyperbolic Coxeter polytopes. The compact-facet ambient ceiling79 is inherited unchanged.

Verification scope

Independent reconstruction and comparison pass all477 profiles and the boundary. Final hash-bound generation and a separate fresh saved replay both terminated successfully. Every reconstructed primary row and all six dimension minima agree exactly; all bound source hashes were rechecked. The written/source review covers the payment composition, finite coupon budget, all-rank cost, factor multiplicity, cap15 reduction and inherited tail. It does not rerun the inherited high-core classification or the new80-pattern embedding census. External human refereeing, publication priority and proof-assistant formalization are not established here.

Source 158

Source path: `results/fresh_ideal7_finite_cross_payments_20261017.md`

SHA256: 7c9d08979421838d5620593c74f428977727b55798941b0c57f588f085c41d4a

Ideal cost seven and a half with finite cross-component half-payments

This local argument replaces whole order17 ideal payments selectively and preserves the total finite-vertex cost at most9n for n<=108. Every triangle receives at least2 and every quadrilateral at least1. Every irreducible ideal factor of order m costs at most min(15m/2,m(m-1)/2). An ambient dimension conclusion requires the separate curvature comparison.

The source and certificate inherit the complete distance9/order17 classification and its independent review. They do not presume that the known H17 chamber is the only ambient chamber containing one of its face stars.

The new weights

At finite vertices retain weight1 for connected marked pairs at distance at most9 or in a component of order at most17. Add weight1/2 for a pair in different irreducible components in either of these cases:

- one component is A1 and the other is D16;
- one component is A2 and the other is D14.

In either case the mark in D must be its unique degree3 fork center or one of the two short fork leaves. Either A2 node is allowed. All these definitions refer to whole irreducible components of the actual finite vertex diagram. Other disconnected pairs retain weight0.

At ideal vertices use the actual varying affine factor, as justified by the factor-fiber theorem. For order at most16 pay every pair. For order at least18 pay distance<=7, and distances8 or9 if either mark lies within7 steps of a feature. A feature has degree>=3 or meets a bond of label>=4.

At order17 the rules are:

Affine type	Paid pairs	Count
A16 cycle	distance<=7	119
B16	distance<=9, plus a fork leaf paired with the spine node nine steps from the fork	111
C16	distance<=9	108
D16	all pairs except distance>=13 and the pair of fork centers	127

The B exception consists of two pairs. The D deletion consists of nine pairs: four at distance14, four at distance13, and the two fork centers at distance12. These rules are intrinsic under diagram automorphisms. Every ideal weight is0 or1.

Why all faces remain paid

The mixed finite9/ideal7 theorem gives two length<=7 triangle paths, so their two unit payments survive. It also handles an ideal quadrilateral corner at distance8 or9 in any factor of order>=17 outside the feature neighborhood. Its replacement

has finite distance<=9 or ideal distance<=7 and hence a unit payment.

Consequently a quadrilateral of total weight<1 can have no adjacent white-interior path of length<=9. At a finite corner such a path would already pay1. At an ideal corner it either pays directly or invokes the mixed replacement. Thus the complete inherited no-short-path classification applies unchanged, including all its cross-bond exclusions.

It produces a preferred connected marked component of order at most17 after every supporting-white extension. A finite preferred corner or an ideal factor of order<=16 still pays1. The sole issue is an ideal factor of order17. The preferred core component must embed as an induced labelled subdiagram of that actual factor, with its two marks retained.

Only80 of the949 inherited high-order contexts have upper bound17; all remaining contexts and the complete low-core supplement have smaller bounds. The new checker reconstructs each of these80 preferred components and enumerates every injective induced labelled embedding into all four classical affine order17 diagrams. It uses explicit backtracking: every unused target vertex consistent with every previously mapped edge and nonedge is tried. Necessary degree pruning does not exclude an embedding. The domain is complete by induction on the number of mapped vertices.

Taking all preferred marked images together with the basic short/feature rule gives costs119,111,108,136 on A,B,C,D. In D, the nine deleted pairs can arise only in context indices383,393,400,647,662,675,813,828,841. Every one is two L25 diagrams sharing node9, with the exact19-root Gram already fixed by the inherited cross-bond tests. No other core, label, embedding or indirect extension is discarded.

Sealing the nine exceptional stars

In each of those stars the actual corner obtained by omitting black roots3 and13 contains the entire17-node affine D16 factor F. An additional supporting white e is orthogonal to every root of F: its products are nonpositive and the positive affine kernel pairs to zero with e.

The19-root core is indefinite, so a Gram-radical vector is an actual linear relation of its roots in Lorentz space. Its coefficients at3 and13 are nonzero with opposite signs. Since e is orthogonal to all other core roots, the relation forces e to contact both3 and13 or neither. Contacting both gives the adjacent white-interior path3-e-13 of length2, a unit payment at that actual corner. Thus in an otherwise unpaid quadrilateral every extra supporting white is orthogonal to the entire 19-root core. This seals all its finite components; no outside tail can enlarge them or turn a finite corner into an ideal one.

The checker inspects all four corner decompositions in each of the nine literal stars. Five contexts already have a finite unit payment in the retained connected-component rule. The other four, indices383,400,813,841, each have two finite half-payments of the stated A1/D16 or A2/D14 kind. The pair of fork-center marks uses two A2/D14 payments; two opposite fork-leaf marks use two A1/D16 payments; the mixed case uses one of each. Every exceptional quadrilateral therefore receives at least1.

The separate half-payment certificate and review reconstructs the nine literal stars, their Gram signatures and kernels, and every finite corner over the rational numbers. It also gives a separate written proof of the support-sealing argument. It does not replace the complete80-pattern embedding enumeration above.

This is a universal local argument about actual face stars. Its support sealing is essential: merely observing the same finite component types in the known H17 chamber would not prove the transfer.

Every-rank ideal-factor cost

Whole factors of order<=16 have m(m-1)/2<=15m/2 pairs. The explicit order17 costs above are also<=15m/2. In order m>=18, direct spine counts give:

Type	Distance<=7 pairs	Extra distance8/9 pairs
A	7m	0
C	7m-28	min(m-8,18)+min(m-9,18)
B	7m-27	4+min(m-10,17)+min(m-11,17)
D	7m-26	8+min(m-12,16)+min(m-13,16)

For C, the simple path has m vertices. The feature neighborhoods at its two ends have nine positions each. At distance k=8 or9, the starting positions paid from the two ends form intervals whose union has size min(m-k,18). For B the spine has m-2 vertices and end neighborhoods of sizes8 and9. For D it has m-4 vertices and both neighborhoods have size8. The union count is always min(number of positions,sum of end lengths). Each fork leaf contributes one extra spine pair at each distance; this gives4 in B and8 in D. Opposite fork leaves are farther than9 in these orders. Their same-fork pairs were already included in the short count.

These are all-rank formulas. The checker also corroborates them by444 literal BFS counts through order128; that finite range is not the proof of the infinite tail. Each sum of minima has three linear branches. For C they are9m-45,8m-19,7m+8, breaking at26 and27. For B they are 9m-44,8m-17,7m+11, breaking at27 and28. For D they are 9m-43,8m-15,7m+14, breaking at28 and29. Comparison with15m/2 at the finite interval endpoints, and the increasing slack on each final branch, proves the desired bound for every m. The smallest D slack is1/2 at m=29. The finite exceptional affine types all have order<=9.

Total finite cost remains9n in the relevant dimensions

Let a,b,c,d be the numbers of whole components A1,A2,D14,D16 at a finite vertex. Their retained costs are0,1,91,120 respectively, and the inherited cost on every other component of rank r is at most9r. The new coupon cost is at most(3/2)ad+3bc. Thus

```
cost <=9n-9a-17b-35c-24d+(3/2)ad+3bc,
a+2b+14c+16d <= n.
```

For n<=108, d<=6 makes a((3/2)d-9)<=0. If c<=5 then b(3c-17)-35c<=0. For c=6, b<=12 gives at most12-210<0; for c=7, b<=5 gives at most20-245<0. The remaining term-24d is nonpositive. Hence total finite cost is at most9n. This analytic argument permits arbitrary other finite components and all their Coxeter labels. The source additionally enumerates all admissible a,b,c,d in dimensions 92 through108 as an exact finite corroboration.

Verification status

Primary generation passes the complete80-pattern embeddings, all nine sealed replacements,444 large-factor graph counts, the small affine catalogue, nine unbounded linear branches and the finite-budget checks. Fresh saved replay also terminates exit0 with the identical certificate. A separate full mathematical/source read checks the embedding domain, the reduction to the inherited no-short-path classification, sealing, finite9n budget and every-rank ideal counts. That review independently compares the saved order17 paid sets with the intrinsic rules above; it does not duplicate the80-pattern embedding enumeration or444 BFS runs. The separate rational half-payment certificate independently reconstructs the exceptional stars and their finite payments. The new package does not rerun the inherited complete high-core census; it binds that completed proof and uses its explicit order and extension bounds. External refereeing and formalization are not asserted.

Source 159

Source path: `results/fresh_ideal_distance6_mixed_replacement_20261017.md`

SHA256: d00644e0e39e2cfc8f2ead5a1aa382026ae080f91a62bdfa5f2e1eb3980c0cee

Mixed finite-nine and ideal-six quadrilateral replacements

Completed local proof; final verification status below. Use unit finite payments for marked distance at most9. At ideal corners use unit payments for marked distance at most6, or for distance7,8,9 when an endpoint is within6 graph steps of a feature. A feature is a vertex of degree at least3 or an endpoint of a bond of label at least4.

The following statements hold for actual quadrilaterals in finite-sided finite-volume hyperbolic Coxeter polytopes, without an arithmetic or simple-cusp assumption.

- An ideal marked pair at distance7 in an affine cycle of order at least15 has another corner paid by finite distance9 or ideal distance6.
- At cycle order14, the same conclusion holds unless the face has two sealed finite corners of types A1/A13 and A2/A13, with the A13 mark its unique middle. A half-payment at each of these whole-component pairs pays the face.
- For a classical affine tree, an ideal marked pair at distance7,8 or9 whose endpoints are farther than6 from all features has another corner paid by the rule above.

The source and certificate also give the all-rank costs of this basic ideal rule. They do **not** establish a new dimension bound or remove the separate preferred-component obligations in small factors. In particular the remaining order15 D-factor payment obstruction is not resolved here. Triangle payments and the total finite budget are separate tasks.

Actual face setup and inherited classification

Write the face star as H=W+a+b+c+d, with opposite pairs {a,c},{b,d}. If a corner has a nonzero adjacent black product it already pays, so in the unpaid analysis all adjacent black products are zero. At a chosen ideal corner, W+a+b is a varying affine factor F plus a positive-definite spectator diagram Z orthogonal to F. The actual factor-fiber theorem gives this description even at nonsimple cusps.

The positive affine kernel represents the actual ideal point. The opposite sides c,d do not contain it, so their heights h,k are strictly positive. We retain the complete finite/affine one-root profile lists and normalized Schur determinant from the frozen weak-profile proof and mixed finite-nine/ideal-seven proof. These classification inputs do not assume a global exclusion of small affine subdiagrams.

Eliminating the spectator block changes the normalized determinant by

```
alpha * (x/h−y/k)^T G_Z^(−1) (x/h−y/k) >= 0,
```

where alpha>0 and x,y are nonnegative native contact-magnitude vectors of c,d with Z. Thus a strictly positive spectator-free determinant contradicts the actual face star's corank1. A negative relaxed determinant alone proves nothing; the small-cycle analysis below handles its equality condition exactly.

Cycle factors of order at least17

Choose the shorter old marked arc as P=7. The other arc has Q>=10. The complete long-A profiles are isolated, endpoint3 on either arc, endpoint4 on either arc, second-node3 on either arc, and both-end3.

A single-attachment profile gives an adjacent component containing the old A_(m-1) path and the new root. The old endpoint opposite the attachment stays a simple leaf with a degree2 neighbor, since spectators are orthogonal to the old factor. No large affine A/B/C/D diagram has this configuration, and exceptional affine diagrams are too small. Therefore this adjacent marked component and its actual corner are finite. This is the fixed-leaf argument already proved in the cycle16 transfer; it applies here to old paths of rank at least13.

For an attachment at position j=1 or2 on an arc of length L, the finite marked distance is L-j+1. Under unpaidness it must exceed9, forcing L>=9+j. Thus all single profiles on the short arc7 are excluded. On the long arc an endpoint profile requires Q>=10 and a second-node profile Q>=11.

Only five profiles remain. The normalized determinant is nonincreasing in the reciprocal arc lengths and concave in the reciprocal heights, as in the frozen formal identity. For every ordered profile pair, use Q>=11 if a second-node profile occurs and Q>=10 otherwise. The checker evaluates all100 height-rectangle vertices, with exact minimum1/35. Positive heights and arbitrary positive-definite spectators therefore cannot produce a zero determinant. This proves the cycle statement for every order at least17, not just for tested orders.

The exact spectator gap at orders14,15,16

Here P=7 and Q=7,8,9. Every single profile already yields a finite marked distance at most9. The checker verifies the18 literal positive-definite single-profile diagrams, their marked distances and their fixed opposite leaf. Only isolated and both-end3 profiles remain. The12 ordered profile pairs give48 exact rectangle values. Every pair involving both-end3 has a strictly positive minimum. Hence only isolated/isolated can remain.

For isolated profiles the heights are exactly the native opposite contact magnitudes. They are1 or at least sqrt(2). If either height is at least sqrt(2), the normalized determinant has lower bound

```
1-3*alpha > 0,       alpha=1/7+1/Q <=2/7.
```

Consequently h=k=1. This step precedes the contact gap: no lower bound on coordinates of a divided contact vector is being assumed. The normalized determinant equation now requires

```
D = (x-y)^T G_Z^(-1)(x-y) = 4-1/alpha.
```

Its required values are

Cycle order	Q	Required D
14	7	1/2
15	8	4/15
16	9	1/16

Any common nonzero coordinate of x,y gives a path c-e-d of length2, already paid. Otherwise their supports are disjoint. Each nonzero native contact magnitude is at least1.

If both vectors are nonzero, choose i in the support of x and j in the support of y. Cauchy in the positive-definite Gram metric gives

```
D >= (x_i+y_j)^2 / (4-2G_ij) > 1/2.
```

Indeed x_i+y_j>=2, while positive definiteness of the two-node principal block gives -G_ij<2. This argument allows any finite labels and arbitrary other spectator components.

If just one vector is nonzero, say x, Cauchy with the coordinate vector e_i gives D>=x_i^2/2>=1/2. Equality forces x_i=1 and x=(1/2)G_Z e_i. Nonnegativity of x and nonpositivity of the off-diagonal Gram entries then force i to be isolated in Z and x to have its single unit contact there. The other vector is zero. If both vectors vanish, D=0, which does not equal any required value.

Thus orders15 and16 are impossible in the unpaid analysis. Order14 forces precisely one simple contact to one isolated A1 spectator, on one side only. This proves the first statement and reduces the second to a completely specified equality core. The strict bound and its equality case are universal positive-definite Gram arguments, not finite enumerations of spectator forests.

The sealed order14 equality core

Number the cycle0,…,13, take a=0,b=7, and attach c=14 to0, d=15 to7, and the single spectator e=16 to14, all by simple bonds. The reflected configuration is equivalent. Every other contact among these17 roots is zero. Equality in the spectator gap has already shown that all remaining supporting spectators are orthogonal to every root of this core.

The two adjacent finite corners have whole marked components A13/A2 and A13/A1. In both A13 paths the old marked root is6 edges from either endpoint, so it is the unique middle. The third finite corner has the marks in A2 and A1, with unmarked old-path components. The two specified half-coupons contribute1. They refer to whole actual finite components; orthogonality of all remaining spectators prevents enlargement or hidden joining of these components.

The source checks the literal core signature(15,1,1), its radical, the supporting positive-definite white block, every corner and both coupons. It also verifies that the equality configuration is an actual local quadrilateral: a strict common center has norm -204, each wall has a strict timelike facet witness, and the complete face cone has three finite rays and one ideal ray.

For scope, the full chamber is infinite volume. Its complete nonnegative dual cone, subject to one radical balance equation, has58 rays:27 finite,5 ideal and26 exterior. A positive displacement of an exterior ray toward the strict center gives an explicitly checked strictly feasible future null vector. This local equality example is not promoted as a finite-volume construction or a novelty claim. The transfer theorem is conditional on an actual quadrilateral and does not require that this entire small chamber be finite volume.

Classical trees outside feature radius6

Both marks are on the simple spine. At a doubled end, distance greater than6 from both endpoints of the doubled bond leaves an outer B component of rank at least8. At a fork end, distance greater than6 from the fork center leaves an outer D component of rank at least9. These are literal node counts, not distances that count a doubled bond as an ordinary extra edge.

The same eight complete classical extension profiles as in the preceding corridor proof apply at these thresholds. A B8 component contains a doubled bond, so it cannot acquire an exceptional E extension; F4 and its affine analogue are too small. A D9 component already has more nodes than any finite E diagram or any proper subdiagram of affine E8. A replacement meeting both components produces an even larger diagram. Consequently the large B/D extension classification has no newly available exceptional profile at the reduced radius.

Each profile has cost g=2*l/h+c0/h^2 at most4. The unique cost4 profile is an inner second-node simple attachment; every other profile has cost at most3. With marked spine distance P, the normalized determinant is

```
F = 1-(g_c+g_d)/P.
```

For P=7, an inner second-node attachment produces an adjacent path of length6, so is already paid. Removing it leaves F>=1/7. For P=8, it produces a path of length7. In the core the new marked root is a simple fork leaf, at distance1 from the degree3 attachment node; these three old/new incident edges remain in that actual corner. Spectators may add neighbors to the new root, but the old attachment node remains a feature. Therefore this corner is finite-distance9 paid or ideal-distance7 within the new feature neighborhood. Removing the profile leaves F>=1/4. For P=9, no removal is needed: F>=1/9.

The source checks192 ordered profile/distance cases:30 immediate payments and162 positive gaps. Spectators only increase the determinant. This proves the tree statement in every rank satisfying the actual distance-from-features condition.

All-rank costs of the basic ideal rule

For order m>=18, the exact counts are

```
A: 6m,
C: 6m-21+min(m-7,16)+min(m-8,16)+min(m-9,16),
B: 6m-14+min(m-9,15)+min(m-10,15)+min(m-11,15),
D: 6m-7+min(m-11,14)+min(m-12,14)+min(m-13,14).
```

The A cycle has exactly m pairs at each distance1,…,6 and no features. For C, the path contributes6m-21 short pairs. At each distance7,8,9, the allowed start positions lie in two endpoint intervals of length8, so their union has min(m-k,16) positions.

For B, the simple spine has m-2 nodes. Short pairs total6m-20, including the one extra pair between the two fork leaves. At each of the three long distances there are two fork-leaf pairs, plus two spine start intervals of lengths7 and8. This gives the constant6m-14 and the three minima with cap15. For D, short pairs total6m-19, including the two fork-leaf sibling pairs. At each long distance there are four leaf-spine pairs, and two spine intervals of length7. For m>=18 no opposite-fork leaf pair has distance at most9. This gives6m-7 and the cap14 minima. The interval-union formulas cover their overlap exactly.

Each expression has successive linear slopes9,8,7,6. Its normalized maximum is85/12 at C order24,36/5 at B order25, and95/13 at D order26. The checker records all12 finite/infinite linear branches and proves the bound on each from its endpoint or positive infinite-tail slack. The332 literal graph counts at orders18,…,100 corroborate the formulas; the proof of the infinite tail is the spine count and linear-branch analysis, not extrapolation.

These are costs of the basic rule only. The inherited preferred-component images may require additional pairs at small orders, and the order14 coupon needs a total finite-budget proof if used globally. This package makes neither inference automatically.

Verification scope

Generation and fresh saved replay both passed100 large-cycle rectangle values,48 small-cycle rectangle values,18 single-profile finite diagrams,192 tree cases,58 complete equality-core rays and332 affine graph counts. The12 all-rank linear branches and exact spectator-gap equality are also checked or proved as described. Syntax, all11 direct source hashes and all6 local report links passed. A separate full mathematical/source review found no gap in the height-one gate, unscaled spectator gap and equality, sealed finite components, fixed-leaf argument, persistent P8 feature, or all-rank count formulas. It did not independently rerun the100/48 rectangle values or192 tree-profile cases. An additional narrow conceptual review checked the unscaled spectator gap and the required ordering of the height reduction, without reproducing the cycle determinant substitutions. Source, data and report are final/frozen. External refereeing and proof-assistant formalization are not claimed.

Source 160

Source path: `results/fresh_ideal_distance7_mixed_replacement_20261017.md`

SHA256: 4cc49d377ed8d3e3aeafe57c1d2afd823c98445526b8c7aa62f90041c43e35e0

Mixed finite-nine and ideal-seven local payments

Completed internally reviewed local results. The following statements concern actual faces of finite-sided finite-volume hyperbolic Coxeter polytopes, with no arithmeticity or simple-cusp assumption.

1. Every triangle has two distinct adjacent black pairs joined by white-interior paths of length at most7.
2. Suppose an ideal corner of an actual quadrilateral has marked distance8 or9 in a varying affine factor of order at least17. Either one mark is within7 graph steps of a feature, or some corner is paid by finite distance at most9 or ideal distance at most7. A feature is a vertex of degree at least3 or an endpoint of a bond of label at least4.

The source and certificate prove the finite exact parts below. These are local payment statements, not an ambient dimension bound. In particular they do not replace the old whole-component payments on factors of order16 or17 without an additional argument.

There is also an exact order16 obstruction to the naive local rule: an actual quadrilateral in an infinite-volume H16 Coxeter chamber receives no payment from finite distance9 and ideal distance7 with whole components only through order15. Its complete cone and strict escape are certified below. It is **not** a finite-volume counterexample to statement2 or to a proposed global theorem.

Two length-seven paths at every triangle

The triangle star has the supporting positive-definite white diagram W and three black roots. It is indefinite, while deleting any black gives the positive-semidefinite star at an actual corner. Consequently a minimal indefinite principal subdiagram L contains all three black roots. The inherited Lanner/quasi-Lanner classification bounds its order by10, as in the completed dimension108 proof.

If L has order at most9, take a minimal spanning tree connecting its three marks. Its three arms have total length at most8. If the branch point is one of the marks, the two nonempty arms each have length at most7. Otherwise all three arms are positive; adjoining the shortest arm to either other arm gives two paths of length at most7. These paths contain no third mark internally, so their interiors consist of whites. This argument also covers every lower-order minimal core, independently of labels.

Only the three order10 diagrams in the complete frozen catalogue remain. The checker tests all360 black triples in these three literal graphs. For each pair it deletes the third black before finding a shortest path. Every triple has at least two such paths of length at most7. The largest second-smallest length is exactly7. These paths lie in the corresponding actual corner diagrams. At an ideal corner the factor-fiber theorem puts both marks and the path in the varying affine factor. Thus two triangle payments survive lowering ideal short distance to7.

Quadrilateral setup and cycle factors

Use H=W+a+b+c+d with opposite pairs {a,c},{b,d}, and square-two normals. Assume that no corner receives finite-distance9 or ideal-distance7 payment. In particular there is no adjacent black bond: such a bond would have distance1 at its actual corner. Hence b.c=a.d=c.d=0.

At the selected ideal corner, W+a+b is its varying affine factor F plus a positive-definite spectator forest Z orthogonal to F. The positive affine kernel gives the actual future ideal point nu. The opposite sides c,d do not contain it, so h=-c.nu and k=-d.nu are positive. These facts hold at nonsimple cusps by the actual factor-fiber theorem.

We use the complete eight long-A profiles, weak height bounds and four-root determinant from the frozen ideal distance-nine replacement. No small-affine exclusion is restored here. For a cycle, write its two old marked arcs as P,Q. Their minimum is8 or9, and P+Q≥17. It is safe to relax this to P,Q≥8 and P+Q≥17.

A single endpoint3/endpoint4/second-node3 profile for c has an important extra consequence. In the adjacent corner W+b+c, the old cycle minus a is a path of rank at least16. The old endpoint opposite the attachment remains a simple leaf, and its neighbor remains degree2: both are old F roots, spectators are orthogonal to them, and the single attachment is at the other end. This fixed leaf and neighbor cannot occur in a connected affine A/B/C/D diagram. Exceptional affine diagrams are too small to contain the old path. Thus the full component containing b,c is finite, even after arbitrary spectators are restored. The actual factor-fiber theorem then prevents this corner from being ideal: at an ideal face corner its two marks would have to lie in the varying affine component.

Let the attachment be on an arc of length P, at position j=1 for an endpoint or j=2 for a second node. The finite marked distance from c to b is P−j+1. If it were at most9 this corner would already pay. Therefore an unpaid single profile forces

```
P >= 9+j:  P>=10 at an endpoint,  P>=11 at the second node.
```

This handles attachments on a long arc as well as on a short one; it does not incorrectly assume that every single-profile corner has distance at most9. The same reasoning applies to d. Isolated and both-end3 profiles impose no such additional lower bound.

Use the weak heights h>=1 for isolated, endpoint3 and second-node3; h>=sqrt(2) for endpoint4; and h>=2 for both-end3. For each ordered profile pair, impose its arc lower bounds. If both lower bounds remain8, the integer condition P+Q>=17 splits the domain into P>=8,Q>=9 or P>=9,Q>=8. Otherwise the lower bounds already sum to at least17.

With A=1/P, B=1/Q, u=1/h, v=1/k, and the inherited moment data, the normalized determinant is

```
F=1−2∗(A∗L+B∗M)
    −(A+B)∗(c0∗u²+d0∗v²)−A∗B∗(L−M)².
```

It is nonincreasing in A,B and jointly concave in u,v. The checker evaluates the four height-rectangle vertices for every resulting reciprocal-arc box. All272 exact values are at least1/18. Therefore F>0 throughout the domain, contradicting the full face star's corank1.

As before, eliminating arbitrary spectator whites changes this normalized determinant by

```
alpha∗(p/h² − 2q/(h∗k) + r/k²) >= 0
```

for a positive-semidefinite [[p,q],[q,r]] and alpha>0. It cannot remove the contradiction. Thus the cycle part of statement2 is complete for every factor order at least17.

Classical trees away from features

In these orders only affine A/B/C/D types occur. For a B/C/D factor suppose both marks lie farther than7 from every feature. They are on its simple spine. At a doubled end, the nearer doubled-bond endpoint is a feature, so the outer finite B component has rank at least9. At a fork end, the fork center is a feature, so the outer finite D component has rank at least10. These are literal node counts, not distances that accidentally count a doubled edge as a simple extension.

The eight complete large-B/D extension profiles and their all-rank Schur constants are those proved in the preceding replacement lemma. They include both one-component affine alternatives, endpoint4 and second-node3. Each root's normalized scalar cost g=2*l/h+c0/h² is at most4; the only profile attaining this bound is inner second-node3. The next-largest maximum is3 for inner endpoint3. The determinant after eliminating the factor whites is

```
F=1−(g_c+g_d)/P,
```

where P is the old marked spine distance.

For P=9, F>=1/9 directly. For P=8, an inner second-node3 attachment gives a path from the new root to the opposite old mark of length P−1=7. That actual corner is already paid, whether finite or ideal. Excluding that profile for either

root leaves g_c,g_d<=3, so F>=1/4. The checker records all128 ordered profile/distance cases:15 immediate length-seven paths and113 strictly positive gaps. Spectators again only increase the determinant. This proves the tree part, and therefore statement2 with feature radius7.

Exact local obstruction at cycle order16

Take a simple cycle on0,…,15, add a simple leaf16 at0 and a simple leaf17 at8, and set every other contact to zero. Let G be its norm-two Gram. Its exact signature is(16,1,1), so the quotient by its radical is a Lorentz space for H16. The checker gives a vector h satisfying G h=-1 and h²=-204. Thus there is a strict common interior. For every root i, h+e_i/2 is a timelike witness on its wall and strictly inside every other wall, proving that all18 walls are actual facets.

Set W={1,…,7,9,…,15}, with black order(0,8,16,17). W is positive definite. Its four projected side normals have Gram

```
[[ 1/4,-1/4,-1, 0],
 [-1/4, 1/4, 0,-1],
 [-1,    0,  2, 0],
 [ 0,   -1,  0, 2]].
```

The complete face cone has four rays: three finite vertices and one ideal vertex. Hence this is an actual quadrilateral. At the ideal corner the marked roots0,8 are antipodal at distance8 in the16-node affine A cycle, which has no features. At each other corner the marked roots lie in different finite components: A15+A1, A15+A1, and A7+A7+A1+A1. Thus finite-distance9 does not pay any of them. The ideal-short7/whole-factor15 rule pays no corner of this actual quadrilateral.

The full chamber is nevertheless **infinite volume**. In dual contact coordinates y=-Gx, the cone is y>=0 subject to one radical relation. Its extreme rays are the two zero-coordinate axes and the64 balanced pairs of positive and negative radical coordinates. The exact census gives35 finite,7 ideal and24 exterior rays. This is a complete cone census because a nonnegative vector in one homogeneous balance equation decomposes into those axes and balanced pairs.

For a particularly explicit escape, let v be the ray with y=e4. Then v²=1 and h.v=-9. With

```
t=(sqrt(285)-9)/204>0,
```

the vector v+t*h is null, pairs negatively with h and has strictly negative product with all18 wall normals. It gives an open ideal escape region. No finite-volume realization or completion is claimed. This example shows exactly why the local mixed replacement theorem cannot simply include factor order16 without additional payment or global geometry.

Verification scope

Generation and fresh saved replay both terminated successfully, checking the272 cycle rectangles,128 tree cases,360 marked triples and66 cone rays. Syntax, all11 direct source hashes and all7 local report links also passed. A separate internal mathematical/source review read this complete source/report and the inherited replacement proof, finding no gap in the triangle argument, cycle finite-corner and arc restrictions, rectangle argument, radius-seven tree reduction, spectator correction, or order16 cone and escape. That review did not independently reimplement the arithmetic or rederive the inherited classification tables. The source binds the frozen complete high-core classification and the existing determinant/factor-fiber proofs; it makes no new all-label enumeration or ambient dimension claim. External refereeing and proof-assistant formalization remain separate. Source, data and report are final/frozen.

Source 161

Source path: `results/fresh_large_affine_factor_quad_payment_20261017.md`

SHA256: e83368c329af16bf100f33f20f0422d66849f3c99dd81aca6430fea75c1d9904

Quadrilateral payment at a large affine factor with spectators

Final internally reviewed ideal-corner payment lemma. Let P be a finite-sided finite-volume hyperbolic Coxeter polytope. Suppose its diagram has no connected affine subdiagram of order at most9. If an actual quadrilateral has an ideal corner whose varying irreducible affine factor has order m>=18, that quadrilateral receives payment at least1 under the finite Vinberg rule and the affine rule described below. Other irreducible factors at the cusp may be present. This statement does not assert payment for quadrilaterals with four finite corners, or an ambient dimension bound.

The affine rule pays a marked pair at graph distance at most3 or in a fork-invariant end segment of order at most6. Its cost is at most three times the factor order in every rank, by the role-payment package. The finite rule is the original published Vinberg rule. The new checker and data verify the additional scalar and inverse-entry calculations below. Complete extension lists and unbounded parameters are handled in the proof, not by a rank or label cutoff.

The original face star and the spectator correction

Write the four corners as W+a+b, W+b+c, W+a+d and W+c+d. The supporting white diagram W has order n-2 and is positive definite. Each W+one black supports an actual edge and is positive definite, even if an endpoint is ideal. The full star has rank n+1 and corank1, by projecting its four side normals to the three-dimensional Lorentz space perpendicular to W.

At the chosen ideal corner, W+a+b consists of the varying affine factor F, containing both a and b, together with a positive-definite spectator forest Z orthogonal to F. This is the actual factor-fiber description; it does not assume the ideal vertex is simple. The full cusp may have other incident normals not in the face star. Every normal has square2.

The positive kernel of F gives a nonzero vector nu representing this actual ideal vertex. Since c and d are opposite sides of the quadrilateral, neither contains that vertex. Therefore h=-c.nu and k=-d.nu are strictly positive. This argument replaces orthogonality to the entire simple-cusp normal span; a single factor need not span that null hyperplane.

First eliminate only the white roots of F, leaving the spectators. The frozen cycle and tree formulas give a four-root Schur determinant. Eliminating Z subsequently subtracts a positive-semidefinite matrix [[p,q],[q,r]] from the c,d block. The normalized determinant changes by

```
alpha*(p/h^2 - 2q/(h*k) + r/k^2) >= 0,
```

where alpha>0 is the old marked-pair residual diagonal. The heights h,k are unchanged. This is the exact PSD spectator identity. Consequently every strictly positive scalar contradiction remains valid with any number of spectator roots and arbitrary allowed contacts to them. A finite-component payment requires a separate check that its marked roles remain paid; those checks appear below.

If any of b.c, a.d or c.d is negative, the corresponding adjacent marked pair is paid. In the remaining analysis all three products are zero. The opposite contacts x=-a.c and y=-b.d are arbitrary Coxeter contacts: zero, a finite cosine magnitude, parallel2, or dotted. No field restriction is imposed.

Cycle factors

For an affine A factor, deleting a leaves the connected A_(m-1) path. Its complete one-root finite/affine extensions are exactly the five classes in the frozen cycle proof. Spectators do not enlarge this principal-diagram list: every connected component of a positive-semidefinite Coxeter diagram is finite or affine.

All height exclusions in that proof use only affine diagrams of orders3,4 and9. They therefore hold under the present hypothesis. The old cycle has m>=18, so its unused arc supplies the same induced affine-E8 leaf witness. Both arc lengths are at least4 at an unpaid corner. The all-rank reciprocal-arc argument gives a strictly positive determinant before spectator elimination, and the PSD identity preserves that contradiction. Thus the complete cycle branch transfers.

A bounded affine witness for every excluded simple leaf

Consider a classical affine tree with main-path marks a,b, distance P>=4, and outer ends B_r or D_r. D2 denotes the two fork leaves. The factor order is m=r+s+P+1. A new simple leaf at a, its first inner neighbor, or its second inner neighbor has available opposite all-3 path lengths

`L=r-1+j,  R=m-3-L,  j=0,1,2.`

These lengths stop before a doubled terminal bond and may end at one chosen fork leaf. If L,R>=2, their sum m-3>=15 ensures that one is at least5. The leaf together with path arms of lengths2 and5 induces affine E8, of order9. Extra fork leaves are omitted and introduce no additional edge among the selected vertices.

Only the following near-end exceptions have L<2. At j=0 they are B1, B2 and D2; the whole short end, a, the new leaf and the first inner neighbor give respectively an affine four-node doubled star, affine B4, or affine D4. At j=1 the only exception is B1, giving affine B4 on five nodes. There is no exception at j=2. The right path has length at least4-j. Thus every simple-leaf exclusion needed here uses an explicit induced affine diagram of order at most9, even when the outer end is arbitrarily long. This replaces the older proof's use of a possibly long affine B/D witness.

Complete main-path replacements and their bounds

Deleting a separates its outer B/D end from the inner B/D component containing b. A connected one-root extension contains whole old components. The original finite/affine classification in the short-end proof still gives every join touching both components. The one addition is a D2 join using one outer leaf by a bond4 and the inner long port by a bond3, leaving the second A1 separate. It has outer projection data q=1,t=1/sqrt(2), hence g<=2/(1+1/sqrt(2))<14/9. Other extra doubled joins would put a doubled bond in the interior of a large classical component and are excluded by the finite/affine classification.

An outer-only extension is still strictly spherical: outer+c is a subdiagram of the actual edge W+c. Thus q<2, and the arbitrary-contact Cauchy–Schwarz argument from the short-end proof applies unchanged. Its small F4/E8 exclusions have order at most9. The D2 arbitrary-contact bound is unchanged. These give g<=14/9, with no contact-label cutoff.

The inner component may now acquire its own affine extension while the outer end remains separate. For a large B/D inner component, the complete additional profiles are a bond4 at its long endpoint and a bond3 at its next node. Together with isolated and ordinary endpoint3, their residual data are:

Profile	c0	l	h	Bound on g=2l/h+c0/h^2
isolated	2	0	x	1
inner endpoint3	1	1	x+1	1
inner endpoint4	0	sqrt(2)	x+sqrt(2)	2sqrt(2)/(1+sqrt(2))
inner next-node3	0	2	x+1	4/(1+sqrt(2))

For isolated x=1 and endpoint3 x=0, the preceding leaf argument applies. Isolated x=sqrt(2) gives a doubled star (or affine C2 at the doubled end). Endpoint3 x=1 gives affine A2. Endpoint4 x=0 gives the four-node affine doubled star. Next-node3 x=0 is the interior leaf just treated, and x=1 gives an affine four-cycle. Thus the stated lower height bounds hold in every rank. The case x=0 for an isolated root would give h=0, already excluded by the actual quadrilateral geometry.

For all ordinary profiles, therefore,

```
g <= g_* = 4(sqrt(2)-1).
```

The weighted-tree Green identity is unchanged. The contact supports lie at the first or second surviving path node; at P=4 the two second nodes can coincide, and the same Green entry l*m/P remains valid. Hence

```
det(S)/(h^2*k^2) = 1-(g_c+g_d)/P
                 >= 3-2sqrt(2) > 0.
```

The complete small-inner exceptions remain D6→E7 and D7→E8. This is the same finite/affine deletion classification: exceptional affine extensions have order at most9 and are excluded, while the newly allowed classical affine profiles are the two rows just added. Because m>=18, at most one inner component is small; the other has rank at least9.

Finite exceptional roles and the extra E8 scalar rows

For D6→E7, the added d is the leaf at the end of the length-two arm. Its inverse diagonal is2. Spectators can touch that E7 only at d; the role lemma makes every nonzero extension affine E7, which is forbidden here. Thus no spectator touches it, and the original finite E7 payment persists.

For D7→E8, d is again the end of the length-two arm, now with inverse diagonal4. A nonzero spectator contact costs at least4, exceeding square2. Thus every spectator is orthogonal to d and to this entire E8 component. In the D2/P5 case the marked branching vertex is paid by Vinberg's published special-vertex lemma. This argument uses the actual d position, not an arbitrary embedding of E7 or E8.

For the main-path D3/P4 exception, retain the old notation with outer A3 roots L,0,L', a the next main node, and d attached to L'. The frozen formulas give alpha=1/4, A=1/2, D=5/4 and v=1/k<=2/3. All old scalar contradictions survive the PSD correction. There are two new inner profiles and the formerly directly paid outer-only profiles to check.

The complete outer-only finite extensions of A3 are an endpoint bond3,4 or5, or a middle bond3. An endpoint5 cannot coexist with d in the positive-semidefinite opposite corner: its connected five-node diagram contains a bond5 and is neither finite nor affine. Endpoint4 at the same leaf as d gives the forbidden affine doubled star. The remaining cases have the following exact data and height bounds:

c profile	C	E	upper bound for u=1/h
endpoint3, same leaf as d	5/4	3/4	1/(sqrt(2)+1/2)
endpoint3, other leaf	5/4	1/4	1/(sqrt(2)+1/2)
middle3	1	1/2	1/(1+sqrt(2))
endpoint4, other leaf	1/2	sqrt(2)/4	1/(1+sqrt(2)/2)

For endpoint3, x=0 gives the leaf affine E8 and x=1 an affine four-cycle. For middle3, x=0 gives affine D4 and x=1 affine A2. For endpoint4, x=0 gives affine B4 with one doubled terminal arm. All these witnesses have order at most9. Substitution into

```
F=(1-Bu-Av)^2-alpha*(C*u^2+D*v^2+2E*u*v)
```

with B=0 gives a strictly positive minimum at the displayed positive upper bounds and v=2/3. The checker calculates all four minima exactly.

For the two new inner profiles E=0 and (B,C)=(sqrt(2)/4,1/2) or (1/2,1). Both have u<=1/(1+sqrt(2)). Their polynomials are respectively

```
1-u/sqrt(2)-v-v^2/16+sqrt(2)*u*v/4,
1-u-v-v^2/16+u*v/2.
```

Each is affine in u and concave in v, so the four rectangle corners suffice; all eight values are strictly positive. Spectator elimination only increases F. This exhausts the additions in the main-path E8 exception, rather than assuming a small finite component remains small after attaching spectators.

In the fork-leaf P6/E8 exception the old scalar cases are unchanged. The opposite D8 sibling pair permits only an affine-E8 spectator extension by the role lemma, so here it permits none. Its finite end-segment payment persists. The doubled-mate profile still contains a forbidden four-node affine doubled star. No new terminal profile is available beyond the four complete finite/affine profiles already in the frozen tree proof.

Terminal marks and conclusion

At a doubled terminal the old bound is g<=5/2; at a fork leaf it is g<=4. Their complete four-profile lists and all exclusions use only affine diagrams of order at most9. Replacing the opposite ordinary main-path bound by g_* still gives

```
1−(5/2+g_∗)/6 > 0,
1−(4+g_∗)/6 > 0.
```

The exceptional opposite E7/E8 roles were checked above. For two terminal marks, R=m-1>=17 and the old total bound is at most8. The supports are separated and the same Green identity applies. All positive inequalities persist with arbitrary spectators.

Every unpaid large classical ideal-corner case is therefore either paid by a separately preserved finite component or contradicts the actual corank-one determinant. This proves the stated ideal-corner branch.

The no-small-affine hypothesis follows, in particular, if every irreducible cusp factor has rank at least9. Requiring every factor to have rank at least17 would also ensure that every ideal corner falls in the present large-factor branch. That does **not** yet establish a dimension theorem: the preliminary U,V decomposition in the frozen all-finite-corner transfer excluded potentially large affine private diagrams by simplicity, beyond its later local carriers of order at most17. That remaining use is not removed by this report.

Verification status

Generation passes the generic margins, six new E8 profile calculations, the E8 inverse entry and two explicit affine-star kernels. The all-rank Green identity and old exceptional rows are inherited from frozen proofs, with the changed hypotheses checked above. A fresh saved replay also passes. Syntax, all seven direct source hashes and local links pass. Separate narrow root and construction mathematical/source reviews pass the actual factor geometry, bounded leaf witnesses, expanded profile accounting, E7/E8 roles, six new scalar rows and spectator correction. They do not claim an independent re-enumeration of every inherited finite/affine extension list or a duplicate arithmetic replay. The source, data and report are final and frozen. No previously frozen file is changed.

Source 162

Source path: `results/fresh_large_finite_d_leaf_half_transfer_20261017.md`

SHA256: 2774ceef3099cd23df609156d8c5eec5e18985380e670cd64bbae0cf433d3d7a

Half payments at short fork leaves in large finite D components

In the completed dimension66 payment rule, replace the two unit payments at distance9 from the short fork leaves of a **whole finite D_n, n>=20** by halves. Add a half-payment between any marked root of a whole finite component of order at most5 and a short fork leaf of a separate whole finite D10. All other payments, including ideal weights, remain unchanged. Then triangles still receive at least2 and quadrilaterals at least1. The same-component cost of D_n becomes8n−33 for n>=20.

The source and certificate check the affected exact local interfaces. A finite-forest maximum and any dimension consequence are separate obligations. The review status is recorded at the end; no new ambient dimension bound is claimed here.

The complete affected domain

All finite marked paths of length at most8 remain paid, so triangles and the initial short-path branch retain their finite payments. In the universal cycle transfer, a second-node attachment gives a whole D_m and marked distance L−1 on an arc of length L. To need a distance9 payment, L=10. The shorter cycle arc P is6,7,8 or9, hence m=P+10<=19. These D payments are unchanged. An endpoint4 gives whole B, also unchanged. An endpoint3 gives whole A if its new marked leaf has no spectator contact; if it has a contact, that mark is no longer a leaf and the existing spectator correction applies. The other mark has two fixed path neighbours. Thus this branch cannot need a changed short-fork-leaf payment in D_n with n>=20. The radius-five tree branch and the cycle12/13/14 successors use shorter finite paths, small whole components or unchanged separate coupons.

After the short-path branch, the complete low-core closure has whole component bound12 and the disjoint high-core branch is excluded. In the complete high-overlap census, 2,531 of2,614 selected contexts have whole marked-component bound at most17. Their payments are unchanged. The nine adaptive contexts protect a simple marked leaf with its fixed degree-two neighbour, or give a whole component of order at most16. A short fork leaf of D_n has a degree-three neighbour, so none of their chosen payments changes.

The remaining74 contexts originally use a protected marked leaf at distance9. Eighteen protect a simple leaf and its degree-two neighbour; again they cannot use a changed short fork leaf. Reconstructing every corner of each of the other56 contexts gives:

Outcome	Contexts	Retained payment
Two distinct protected leaf-nine corners	40	At least one half at each
One protected corner plus the new coupon	16	One half plus one half

Every protected ideal alternative retains its old unit payment. A finite alternative either keeps its unit or is precisely one of the new halves. The first40 contexts therefore need no further geometric inference.

All later finite alternatives in the rank17 interface audit have whole order at most17, shorter finite paths or separate coupons. The change introduces no dependence on a subsequently removed ideal pair.

The sixteen remaining contexts

Use the norm-two full old core H, with exact inverse J. Each of these cores is nonsingular Lorentzian. Write a,b for the selected marks, with b its short fork leaf, and a',b' for the respective opposite black roots. The selected base component K is already the complete labelled finite D_s, with13<=s<=17. It contains every old core root except a',b'. The old root a' is a simple leaf attached only to a in H.

If the actual selected component is not D_n with n>=20, its original unit is unchanged and the face is covered. Suppose it is such a D_n. An induced copy of D_s with both short leaves and its fork present can extend inside D_n only along its existing long arm. Thus all new nodes of that component lie beyond its old long endpoint j. The checker reconstructs both canonical D20 maps, while this graph argument covers every n>=20 without extrapolating a finite rank sample.

Delete the mark a and insert a'. The fork-side part left by deleting a is exactly D10, because a was at distance9 from the short leaf b. All ten of these roots already lie in K, and a' is orthogonal to them. This D10 is a **whole** component of the new actual corner: all other roots of the selected D_n are separated from it by deleting a, and every supporting root outside that whole D_n was orthogonal to it. No unexamined new node or indirect bridge can enlarge this D10.

It remains to bound the whole component containing a'. This requires the residual argument below; being isolated in the displayed corner alone would not suffice.

Complete contacts and residual bounds

Let j be the old long endpoint of K, and p its preceding node. The complete necessary contact domain of any additional supporting white is exactly {a',p,j,b'}. Its two end sets are {a',p,j} and {b'}. The inverse on each end is entrywise positive. A root meeting both ends already gives a retained marked path of length at most8.

The exact inverse entries, reconstructed separately for all16 contexts, are

```
J[a',a'] = r in {5/4,3/2,7/4,9/4},
J[p,p] = 2,  J[j,j] = 1,
J[a',p] = 1,  J[a',j] = 1/2.
```

Every multi-node support on one end has unit quadratic value greater than2. Entrywise positivity excludes it for arbitrary native contact magnitudes, each at least1. Every singleton has inverse diagonal at least1; labels at least5 have squared contact magnitude greater than2 and are excluded. The complete remaining singleton profiles have labels 3 or4 and are explicitly enumerated. These are all-label necessary tests, not a finite-label search assumption.

If r=9/4, no extra root can meet a', so its whole component is A1. Otherwise the only possible root meeting a' is a unit-contact root G, with residual square rho=2−r<=3/4 in the positive-definite orthogonal complement of H. Two such roots have a necessarily positive actual product, so G is unique.

For each other surviving core-contact profile the exact residual Cauchy test gives one of three outcomes. A root at p, a double-contact root at j, or a second G would have positive actual product with G and is impossible. A root meeting b' either is already excluded by its inverse diagonal or has forced negative product with G; its native bond gives the marked path a'–G–root–b' of length3 and the face is already paid. The only profile that can coexist with G in an otherwise unpaid face is a unit-contact root B at j.

The native G·B must be zero: a product at most−1 would require residual magnitude at least3/2, larger than the available Cauchy bound. At most one B can coexist with G. Two B residuals would have to be opposites, while each has product−1/2 with the G residual, a contradiction.

If B is absent, every additional root reachable from G is orthogonal to H. The established connected inverse lower bound m/(m+1) controls the entire attached finite forest. Two nonempty components cost at least1>rho, and a component of rank4 costs at least4/5>rho. Therefore there are at most three such roots and the whole a' component has order at most1+1+3=5. This controls indirect chains, not merely neighbours.

If B is present, their residual Gram is

```
S = [[2−r,−1/2],[−1/2,1]].
```

For r=5/4 its inverse is [[2,1],[1,3/2]]. A core-orthogonal white meeting G costs at least2 in the inverse form. Equality forces its sole contact to be the unit G contact and the root to have residual E=−2g−b, with no additional orthogonal component. A further root meeting B but right to G would then have positive product with E, impossible; roots right to G and B are also right to E. This prevents an indirect bridge into an unbounded B-tail. There is at most one such E.

For r=3/2 the inverse G diagonal is4, so no core-orthogonal root can meet G. For r=7/4, S has positive kernel(2,1); every core-orthogonal white must be orthogonal to both G and B. These are the same exact no-bridge mechanisms as the inherited residual theorem, with the literal matrices checked here. In every case the whole a' component has order at most5 and stays disjoint from the whole D10.

The alternate actual corner is finite. If it were ideal, the actual factor-fiber theorem would require both varying marks in the same irreducible affine factor, whereas b lies in the whole finite D10. Its new half-coupon and the original corner's retained half therefore sum to one. This completes all16 remaining configurations.

Counting and verification scope

There are two short fork leaves in D_n, each with one root at distance9 along the long arm. The two changed pairs are distinct and exchanged by the diagram automorphism. Replacing their units by halves saves1, giving8n−33. Between a component of order s<=5 and one D10 the new coupons have total cost at most s: there are two short leaves and s possible marks, each with half weight. Existing coupons are unchanged.

Generation85418 and fresh separate-process replay74699 both terminated exit0/PASS. The checker reconstructs all74 protected contexts, every alternate protected corner, the16 literal Lorentz inverses, complete necessary contact supports and native label domains, all residual inequalities and both D20 interface maps in each coupon case. It records the nine adaptive and universal cycle interfaces. The geometric completeness of the original census and the inverse-rank lemma are inherited inputs.

A separate full source/report mathematical review passed. It reread the complete contact/end-domain helper, protected-leaf helper, connected inverse-rank and no-bridge proof, universal finite8 cycle/local proofs, adaptive nine-case proof and later rank17 interface audit. The whole attached-forest bound, exclusion of other core-contact profiles, residual equality and positive-kernel sealing, all-rank D_n long-arm restriction, whole exposed D10, and finite alternate corner all passed. That review did not duplicate the16 inverse calculations,74-corner enumeration or original geometric census.

A second reviewer independently inverted all16 saved Grams using rational arithmetic and checked their Lorentz signatures without producer imports. All160 end-block entries,128 nonempty end subsets,64 multisupport exclusions,72 singleton profiles,60 G-interaction rows and12 residual matrices agree. The check reconstructs all16 exposed D10 components, their short fork marks and orthogonality to a', and every edge and nonedge of all32 D20 embeddings. Its complete source/report read also passed the all-rank extension and no-bridge arguments. That check did not reproduce the original census, protected-leaf classification, inverse-rank lemma or a global finite budget. Source, certificate and report are final/frozen. No finite-forest maximum or ambient bound follows from this local package alone.

```
python3 scripts/fresh_large_finite_d_leaf_half_transfer_20261017.py
python3 scripts/fresh_large_finite_d_leaf_half_transfer_20261017.py --verify
```

Source 163

Source path: `results/fresh_multi_ideal_triangle_scope_20261017.md`

SHA256: d4789b2c2e606db746acb14c5e39add6af4d795e29572235e7dd0ba5040801a9

Multi-ideal triangle obstructions and a bound on simplex-face size

Two exact finite-volume Coxeter square pyramids show that triangle-port transitivity can fail for triangles with exactly two ideal vertices, and also for triangles with three ideal vertices. In both examples the chosen cusp is simple and its whole link is a single simplex factor. Thus choosing a simplex-factor face at that cusp does not repair the argument: the other endpoint of the shared edge can still be a nonsimple ideal vertex.

A separate original-wall-star argument does give a justified bound on aggregate size: **any actual simplex face of a finite-volume hyperbolic Coxeter polytope has dimension at most9**. This does not assume that the induced face angles are Coxeter angles. It sharpens the one-ideal cusp block budget, without bounding the ambient dimension or covering the remaining multi-ideal triangles.

The standalone checker and certificate use exact Fraction pairs in Q(sqrt2). They reconstruct both pyramids, all vertices, active sets, face incidences, Gram gates, and ideal-support semidefiniteness. They also check the block maximum below by dynamic programming through 256 ports. Its proof for all port counts is given separately here.

Two literal Coxeter counterexamples

Use the Lorentz form diag(-1,1,1,1). Put

$$b = (0,0,0,-1), \quad s_1 = (a_1,1,0,a_1), \quad s_2 = (a_2,0,1,a_2), \quad s_3 = (a_3,-1,0,a_3), \quad s_4 = (a_4,0,-1,a_4).$$

All five normals have square1. Their products are

$$b \cdot s_i = -a_i, \qquad s_i \cdot s_{i+1} = 0, \qquad s_1 \cdot s_3 = s_2 \cdot s_4 = -1.$$

Indices on the successive sides are cyclic. Choose a_i=cos(pi/m_i) with m_i in {3,4}. Then all native contacts are right, pi/3, pi/4 or parallel. The common future center H=(2,0,0,1) has square-3 and pairs strictly negatively with every normal.

In the chart t=1 these inequalities give the rectangular pyramid

$$0 \le z \le 1, \qquad -a_3(1-z) \le x \le a_1(1-z), \qquad -a_4(1-z) \le y \le a_2(1-z).$$

Its apex is nu=(1,0,0,1). Its base vertices are the four combinations of x in {a_1,-a_3} and y in {a_2,-a_4}, with z=0. Both choices below put all vertices in the closed ball, with bounded Euclidean links at the ideal vertices, hence give finite-volume Coxeter chambers.

Two-ideal triangles. Set (m_1,m_2,m_3,m_4)=(4,4,3,3). The base vertex

$$v = (1, 1/\sqrt{2}, 1/\sqrt{2}, 0)$$

is ideal; the other three base vertices are finite. The faces on s_1 and s_2 are triangles, each with precisely the ideal vertices v and nu and one finite vertex. They share edge v–nu. The third two-face through the three ports at v is the quadrilateral base. Thus the two relations at v do not complete to a third triangular relation.

Three-ideal triangles. Set every m_i=4. All five vertices are ideal. The same two side triangles and base square give the same failure, now with all three vertices of each triangle ideal.

In both cases v is incident with exactly three facets. Its link is the affine (4,4,2) triangle and is combinatorially Delta2, so the selected factor fiber is the entire link. The apex nu is incident with four facets; its link is Delta1 x Delta1. Positive null relations include

$$\sqrt{2}\,b + s_1 + s_2 = \sqrt{2}\,v, \qquad s_1 + s_3 = (a_1 + a_3)v, \qquad s_2 + s_4 = (a_2 + a_4)v.$$

The shared ideal edge therefore has a nonsimple endpoint nu. This is exactly where the degree-three/Balinski argument fails. No inference from a numerical near-ideal point or an unverified Gram signature is involved.

These examples rule out unrestricted transitivity and a disjoint-port capacity1 for multi-ideal simplex blocks even within a simplex cusp fiber. They do not rule out a different incidence inequality using additional endpoint data.

A simplex-face bound from the original Coxeter star

Let A be an actual k-dimensional simplex face of an n-dimensional finite-volume Coxeter polytope P, with k >= 2. Every vertex of A is finite or ideal. Let W be the n-k original ambient facet normals supporting A. They are independent and span a positive-definite subspace.

Each facet of A is positive-dimensional, of dimension k-1. Simplicity at edges gives exactly n-k+1 ambient facets containing it. Hence each facet of A supplies **one** additional original normal b_i beyond W. These k+1 black normals are distinct. This produces the original star

$$\Sigma = W \cup \{b_0, \dots, b_k\}, \qquad |\Sigma| = n + 1.$$

This does not say that an ideal vertex of P has only n incident facets. The n normals obtained by deleting b_i are merely a selected subset of the walls through the corresponding vertex of A.

Project the black normals orthogonally off W. The projected normals are positive multiples of the geometric facet normals of A. Since A is a nondegenerate projective k-simplex, its k+1 vertex rays are independent; its facet normals form the corresponding dual basis. Their span is the full Lorentzian (k+1)-space. Thus the Gram matrix of Sigma has signature (n,1) and in particular is indefinite. No assertion about the projected angles being Coxeter angles is used.

For every i, the star Sigma minus b_i consists of normals through an actual vertex of A. It is positive definite at a finite vertex and positive semidefinite at an ideal vertex, since the latter normals lie in the orthogonal complement of a null vector. Therefore any indefinite principal subdiagram of Sigma must contain **all k+1 black nodes**.

Choose an inclusion-minimal indefinite principal subdiagram L. It is connected and every proper principal submatrix is positive semidefinite. If every proper submatrix is positive definite, L is Lanner. Otherwise a proper connected affine component C has a positive null vector. An outside node adjacent to C makes C plus that node indefinite by its nonzero coupling to that null vector. Minimality forces these to be all of L. This also excludes a proper affine component accompanied by extra orthogonal spherical or affine components. Thus L has precisely the Lanner/quasi-Lanner scope of the standard classification.

The classification input is Prokhorov's §2, Proposition1, printed p.407: the largest quasi-Lanner diagrams have order10. The primary article record identifies the original paper and translation. Its indexed primary PDF text was retrieved during this check; a fresh full-PDF fetch timed out. The same exact input and the minimal-indefinite reduction were already explicitly reviewed in the frozen compact-facet proof. This package does not independently reclassify those diagrams.

Since L contains all black nodes,

$$k + 1 \le |L| \le 10, \qquad \boxed{k \le 9.} \tag{1}$$

For k=1 the same conclusion is trivial. The local face-incidence facts used to obtain W and the black normals are stated in Potyagailo–Vinberg, printed p.64.

If the **ambient P is compact**, it has no affine subdiagram, so L is Lanner and the familiar order-five bound instead gives k<=4. Compactness of A alone does not justify that improvement: deleting a white node from L can still leave an affine subdiagram containing all black nodes, even when all black deletions are positive definite. The report makes no order-five claim in that case.

Sharper one-ideal cusp budget

The one-ideal factor-fiber theorem gives disjoint blocks of k ports in a fiber with m=r_i+1 ports. By (1), every nontrivial block has 2 <= k <= 9. Define

$$F_9(m) = 36\lfloor m/9 \rfloor + \binom{m \bmod 9}{2}. \tag{2}$$

Then sum binom(k,2) <= F9(m). To prove this for every m, move a port from a smaller nonempty block to a largest block of size below9. The objective does not decrease, and increases when both blocks have size at least2. A leftover block of size1 is an unused port and contributes zero. Repeat until all blocks except the last have size9; the remainder has size m mod9. This proves (2) without a rank cutoff. The exact dynamic-programming check is an independent finite confirmation of this elementary optimization.

For a cusp with factor ranks r_i and V=product(r_i+1), every triangle with exactly one ideal vertex belongs to one such block. Hence

$$\boxed{2N_{3,1}(v) \le V \sum_i \frac{2F_9(r_i+1)}{r_i+1}.} \tag{3}$$

For r_i<=8 the summand equals r_i. For larger factors it is at most8, so in particular

$$2N_{3,1}(v) \le V \sum_i \min(r_i, 8) \le \min((n-1)V, 8s_v V). \tag{4}$$

This improves the previous worst-case bound 2E_v=(n-1)V when the cusp has large factors. It does not pay for triangles with two or three ideal vertices, or by itself provide the alpha_n E_v budget required by the original-face inequality. The exact counterexamples above show why that missing scope cannot be filled by simply repeating the transitivity proof.

The complementary finite-ended-port theorem groups triangles having at least two finite vertices into maximal simplex faces, which may have several ideal vertices. Those blocks are not the same as the maximal one-ideal blocks in (3), so their incidence variables must not be identified without an additional allocation argument.

FINAL/FROZEN. Generation and a fresh saved replay passed, as did source hash, syntax and report-link checks. Exact_hub_counts independently read the original-star/minimal-indefinite argument and the block-filling proof; the primary classification was not independently re-enumerated. This is an internally checked face-size lemma and two exact local counterexamples, not a new ambient dimension ceiling, external review or novelty claim.

Source 164

Source path: `results/fresh_nikulin_avoid_node_20261015.md`

SHA256: 9573209316bc3069fa14289a6b1143f3f5edfe445c2348f5ead22426cfa7810d

Avoiding the narrow-facet node at a cost of 2/9

Uniform local lemma. In every irreducible affine Coxeter factor of rank r, and for every forbidden node s, one can prescribe two omissions i,j different from s, before either terminating contact row is known, such that

$$\sigma\,\tau\,E / M + 2C \geq 2(r-1)/9. \qquad (1)$$

Here the roots have square two, C is the sum of the two retained-support Schur costs, σ,τ are the positive affine null heights, $M=\max(\sigma/\tau,\tau/\sigma)$, and E is the normalized-row energy. The rows satisfy the actual endpoint hypotheses from the frozen product proof: they are nonnegative, every nonzero entry is at least one, and each endpoint's total Schur cost is at most two. No upper contact or Coxeter-label bound is imposed.

Together with the reviewed non-narrow estimate, this gives a **research proof candidate $n\leq 181$** for finite-sided finite-volume real hyperbolic Coxeter polytopes. The new lemma is not yet independently reviewed. The 189 checkpoint stays frozen and remains the preceding reviewed result. No novelty, publication-priority, or current-best claim is made.

The checker and data rebuild all 34 exceptional forbidden-node cases, the 16 modified classical cases in ranks 2 through 9, the three E8 inexpensive-contact identities, and the all-rank scalar comparisons. The unbounded-rank localization and geometric framing inputs are explicitly inherited from the frozen proofs and their reviews; finite computations are not substituted for those arguments.

Definitions and fixed choices

For an affine Gram matrix G with positive kernel δ, use the normalization $\min \delta=1$. For the two prescribed omissions i,j let S_i,S_j be the retained supports and put

$$C = p_{Si}^{T}\, G(S_i)^{-1}\, p_{Si} + q_{Sj}^{T}\, G(S_j)^{-1}\, q_{Sj},$$
$$\sigma = \delta\cdot p, \quad \tau = \delta\cdot q, \quad M=\max(\sigma/\tau,\tau/\sigma).$$

For normalized point rows define

$$R_{ab} = (e_a/\delta_a - e_b/\delta_b)^{T}\, G^{+}\, (e_a/\delta_a - e_b/\delta_b).$$

The normalized rows p/σ and q/τ have difference of energy E. In particular,

$$\sigma\,\tau\,E/M = \min(\sigma,\tau)^{2}\,E. \qquad (2)$$

The inverse retained Grams are entrywise nonnegative. A retained nonzero entry therefore costs at least the corresponding inverse diagonal. A terminating row cannot vanish on a whole affine factor, since all affine null sums are positive multiples of the initial cusp ray and its null height is positive.

Use these fixed rules in the single factor containing the forbidden narrow-facet node:

- A_r, $r\geq 2$: rotate an opposite pair of cycle nodes to avoid s. The existing estimate is rotation invariant.
- D_r, $r\geq 4$: use leaves from opposite fork ends, switching to the other leaf if necessary to avoid s.
- B_r: use a left fork leaf and the right short end. Switch left leaves if necessary. If the short end is forbidden, replace that omission by its adjacent interior node.
- C_r: use both ends, replacing a forbidden end by its adjacent interior node. The two endpoint cases are diagram reversals.
- A_1: both omissions are the only node other than s; coincident omissions are allowed in this factor.
- E6: choose the first two of leaves 2,4,6 not equal to s.

- E7: choose 4,7, except when s is 4 or 7, when the pair is 2,5.
- E8: choose 3,7, except when s is 3 or 7, when the pair is 1,6.
- F4: choose 1,3, except when s is 1 or 3, when the pair is 0,4.
- G2: choose the two nodes other than s.

The exact exceptional node numbering and complete Gram matrices are serialized. For E6/E7/E8 they are stars with arm lengths respectively (2,2,2), (1,3,3), and (1,2,5), numbered consecutively from the center along each arm. The rules do not depend on the subsequent contact rows, their zero sets, labels, or heights.

Unmodified classical cases

The reviewed contact-cost lemma already gives the stronger inequality

$\sigma\ \tau\ E/M + 2C \geq 2r/9$

for the ordinary A/B/C/D choices. A-cycle rotation and D-fork switching preserve those diagrams and estimates. Thus only the modified B and C endpoint choices require new work. This includes the separately checked A9 localization in the earlier review; no unstated large-rank threshold is being applied to rank nine.

Modified B and C choices

Consider B_r when the right short node is forbidden, and C_r when the left end is forbidden. The other C case is obtained by reversing its entire diagram. In the modified support the forbidden end becomes an isolated A1. Its inverse diagonal is 1/2.

When C=0 the rows are point contacts at their omitted nodes. The weighted point energy is $(r-1)/4$ for modified B_r with $r\geq 3$, and $(r-1)/2$ for modified C_r. Modified B2 instead has point energy 1. All these values are at least $2(r-1)/9$.

For ranks 2 through 5, every retained inverse diagonal is at least 1/2. Consequently $C>0$ implies $2C\geq 1\geq 2(r-1)/9$, proving (1) without using energy.

For ranks 6 through 9, the isolated A1 is the only inverse diagonal below one in either deletion support. If $C\geq(r-1)/9$, its charge already pays (1). Otherwise $0<C<(r-1)/9<1$, and exactly one retained entry can be nonzero across the two rows. It is the isolated A1 entry, of magnitude $b\geq 1$ and cost $b^2/2$. The other row is supported only at its omitted node. With i the modified omission, l the isolated node and j the opposite omission, write

$p=a\ e_i+b\ e_l,\quad q=c\ e_j,\quad u=\delta_l\ b/(\delta_i\ a+\delta_l\ b).$

Here $a\geq 0$, $c\geq 1$ and $0\leq u\leq 1$. The complete inverse-diagonal lists and the following three resistances are checked exactly for all eight cases:

Type	R_{ij}	R_{lj}	R_{il}	Resulting energy
B_r	$(r-1)/4$	$r/4$	1/4	$(r-1+u^2)/4$
C_r	$(r-1)/2$	$r/2$	1/2	$(r-1+u^2)/2$

The energy formula is the squared-distance interpolation identity

$E=(1-u)R_{ij}+uR_{lj}-u(1-u)R_{il}.$

Since $\min\delta=1$, every nonzero row has height at least one. Equation (2) therefore bounds the weighted energy below by $(r-1)/4$ for B, or $(r-1)/2$ for C. Either is at least $2(r-1)/9$, even without adding its positive contact cost.

For every $r\geq 10$, the inherited fixed-choice localization gives weighted energy at least $(r-5)/4$ in modified B and $(r-5)/2$ in modified C. The closed finite B/D inverse formulas show that the isolated A1 still has the smallest diagonal, 1/2; all other retained inverse diagonals are at least one. Thus $C>0$ implies $2C\geq 1$. At $r=10+t$, $t\geq 0$, the exact margins are

$(r-5)/4 +1-2(r-1)/9 = (t+9)/36 >0,$
$(r-5)/2 +1-2(r-1)/9 = (5t+27)/18 >0.$

The C=0 point-contact calculation already covers every rank. This proves (1) for all modified classical cases, without enumerating contact magnitudes or imposing a rank cutoff.

Direct exceptional cases

For A1,E6,E7,F4,G2 the prescribed pair has both point energy at least the target $2(r-1)/9$ and twice the minimum retained inverse diagonal at least that target. The C=0 and C>0 cases are therefore immediate. The records cover every forbidden node, not only diagram-orbit representatives.

The E7 replacement is useful here: pair 4,7 has point energy 3/2 and minimum inverse diagonal 3/2; pair 2,5 has the same point energy and minimum diagonal 2/3. Both meet the new target 4/3. No cheap-contact subcase is needed in E7.

For A1 the target is zero. Its two omissions coincide; two rows supported at this common node have E=C=0. Hence the old target $2r/9$ cannot hold uniformly under an avoid-node constraint, even at the level of admissible local contact rows. The additive loss 2/9 in (1) is therefore sharp for this uniform local accounting. This observation does not preclude a different global argument from improving the eventual dimension bound.

E8 and its replacement omissions

Here the target is 14/9. If $C\geq 7/9$ the charge alone suffices. In the remaining case $0<C<7/9$, the exact inverse diagonals force a single cheap retained contact.

For pair 3,7 this is node 8 after deleting 7, with cost $b^2/2$. Its null coefficients are $\delta_i=\delta_j=2$, $\delta_l=1$ and its resistances are (1/2,1,1/2). This is the previously reviewed E8 argument: if a=0, $\sigma\,\tau\,E/M\geq 1$ and $2C\geq 1$; if $a\geq 1$, both heights are at least two and $E=(1+u^2)/2\geq 1/2$, so $\sigma\,\tau\,E/M\geq 2$. Both cases exceed 14/9.

When s is 3 or 7 the new omissions are 1,6. Their null coefficients both equal three. The only inverse diagonals below 7/9 are nodes 7 and 8 after deleting 6, each equal to 2/3. Every other retained diagonal is at least 8/9. Two nonzero retained contacts would cost at least 4/3, so only one can occur. Its row is $p=a\,e_6+b\,e_l$, while the other is $q=c\,e_1$, with $c\geq 1$.

Cheap node l	δ_l	R_{6l}	R_{ll}	R_{6l}	E
7	2	2/9	7/18	1/6	$(4+3u^2)/18$
8	1	2/9	8/9	2/3	$2(1+3u^2)/9$

If $a\geq 1$, both heights are at least three and $E\geq 2/9$. Equation (2) gives $\sigma\,\tau\,E/M\geq 2$.

If a=0 and l=7, the heights are at least two and three. The weighted energy is at least $4\cdot 7/18=14/9$, as required. If a=0 and l=8, the energy is at least 8/9 and the cost contributes $2C\geq 4/3$, giving at least 20/9. These are all cases. The only discreteness used is that a is either zero or at least one.

Consequence for the narrow-facet dichotomy

Use the unchanged point/contact-cost omissions in every non-narrow factor and the above avoid-node choices in the single narrow factor. The frozen product argument then supplies actual terminating framing walls x,y. In particular, both outgoing edges retain the narrow node; at an ideal terminal point its positive-definite edge support omits one node per affine component, and a suitable missing wall has a finite angle with the narrow wall. The strict-center projection argument makes their intersection an actual ridge. No merely ideal-point incidence is substituted for Nikulin's framing condition.

Writing N=n−1, summing (1) and the stronger non-narrow estimate gives

$$\Sigma_a\,\sigma_a\,\tau_a\,E_a \geq M(2N/9-2\Sigma_a\,C_a-2/9)$$
$$\geq M(2N/9-8-2/9).$$

The two endpoint Schur budgets still give $\Sigma_a\,C_a\leq 4$, now including the narrow factor's own cost. The common height ratio and the Lorentz identity give

```
−(x,y) = (1/2)Σ_a σ_a τ_a E_a−q−1/q
        ≥ M(N−55)/9.
```

For n≥182, N≥181 and this is at least 14, contradicting the strict narrow-facet distance bound arccosh 7 for square-two normals. Thus the ideal-vertex branch gives n≤181. If the narrow facet has no ideal vertex, it is compact and the independently reviewed compact-facet bound n≤143 applies. The resulting candidate bound is n≤181.

Only the avoid-node accounting is new. The published narrow-facet theorem, affine classification, common-height identity, endpoint existence and actual-ridge bridge, and compact-facet proof remain the explicitly bound dependencies of the reviewed 189 checkpoint. The new lemma requires its own independent review before 181 replaces that checkpoint.

```
python scripts/fresh_nikulin_avoid_node_20261015.py
python scripts/fresh_nikulin_avoid_node_20261015.py --verify
```

Source 165

Source path: `results/fresh_nikulin_avoid_node_20261015_review.md`

SHA256: 166f4a37e37704290875c4aeddf3000e94ad6e04826aba30033979485bde3ec7

Independent review of the avoid-node estimate

The new local estimate and its dimension-181 consequence pass an independent mathematical and exact arithmetic review. This is an internal research review; external validation and novelty are not established.

The review checker and certificate import no arithmetic from the new producer. They reuse elementary Fraction elimination from the earlier independent review and reconstruct rational congruences of each physical Gram matrix. Energies come from grounded linear equations, with every equation checked, rather than the producer's regularized inverse. Every retained inverse diagonal is compared with the saved primary data.

All 34 exceptional forbidden-node choices pass. For E8, the full cheap-contact list is rebuilt, the mixture's constant, linear and quadratic coefficients are computed independently, and both zero and positive omitted-contact cases meet the target. All 16 modified B/C cases in ranks 2–9 pass; ranks 10–12 separately check the boundary formulas. These finite checks supplement the all-rank argument, not replace it.

The mathematical review checks the following quantifiers and boundaries:

- Omissions are prescribed before the terminating walls are selected. A-cycle rotation and D-fork interchange retain the earlier lemma; B/C end avoidance is precisely the modification treated in the frozen product proof's localization section.
- At ranks 6–9, cost below the target's half allows only the isolated A1 entry. Nonnegative inverse entries prevent cancellation. The row's omitted entry is either zero or at least one; every nonzero row has height at least one after the stated kernel normalization.
- For ranks at least ten, the inherited localization applies to the modified finite D or B component, including rank-nine D at the boundary. A positive retained cost contributes at least one after multiplication by two. The remaining margins are `(t+9)/36` for B and `(5t+27)/18` for C, with `t=r−10≥0`. The zero-cost case instead uses the point energy for all ranks.
- In E8's replacement omission pair, either retained diagonal is 2/3 or at least 8/9. Thus cost below 7/9 leaves one cheap entry. The reconstructed positive omitted-contact energy is at least two. With that contact absent, the node-7 energy is at least 14/9; node 8 has energy at least 8/9 and charge at least 4/3.
- Coincident omissions in A1 are permitted. They produce zero energy and zero cost, explaining the unavoidable 2/9 loss for this uniform local formulation. They do not invalidate the global argument: the other factors supply separation in the excluded dimension range.
- The summed endpoint Schur budget includes the narrow factor and remains at most four. The common positive height ratio is unchanged. Retaining the narrow node, followed by the finite/ideal terminal-wall choice already reviewed, makes both selected walls actual ridge neighbors of the narrow facet.

Hence the total energy is at least `M(2N/9−8−2/9)`, where `N=n−1` and `M≥1`. The Lorentz identity gives `−(x,y)≥M(N−55)/9`; at `n=182` this reaches 14. The separately reviewed compact-facet alternative still gives 143. The full combined research candidate is therefore **$n \le 181$**.

The inherited actual-ridge geometry, common-height scaling, classification inputs and compact-facet argument remain bound through the 189 package. No arithmetic assumption, bound on finite labels, simple-cusp assumption or fixed facet count was added.

```
python scripts/fresh_nikulin_avoid_node_20261015_review.py --verify
```

Source 166

Source path: `results/fresh_nikulin_coupled_charge_20261015.md`

SHA256: 054f3bc9cf40659a0a66e4ecdcc2e628bb911ad12db4f7500b92a63ca5029774

Coupling the contact charge to the common height ratio

Research proof candidate, pending independent review: every finite-sided finite-volume real hyperbolic Coxeter polytope has dimension at most **166**. The new step is a local charge estimate of 7/6 when the common affine height ratio is at most 21/20. For larger ratios the frozen reviewed 4/3 estimate is already sufficient. No arithmetic hypothesis is used.

The checker and exact data have passed generation and saved replay. This is producer verification, separate from independent proof review. All earlier 169 artifacts and the balanced unit-charge obstruction remain unchanged. There is no claim of publication priority, optimality, or formal proof-assistant verification.

Local statement and common choices

Use the normalization and geometric hypotheses of the frozen product-cusp proof. Each affine factor has rank r, Gram matrix G with root squares two, and positive kernel δ normalized by $\min \delta_k = 1$. The prescribed omitted nodes are i, j. The two nonnegative contact rows p, q are nonzero, and every nonzero entry is at least one. Each endpoint's total retained-support Schur cost is at most two. Set

$$\sigma = \delta \cdot p, \quad \tau = \delta \cdot q, \quad M = \max(\sigma/\tau, \tau/\sigma), \qquad C = p_{S_i}^T G(S_i)^{-1} p_{S_i} + q_{S_j}^T G(S_j)^{-1} q_{S_j}.$$

The normalized-row energy is $E \geq 0$, and

$$W = \sigma\tau E/M = \min(\sigma, \tau)^2 E.$$

For the same preset omissions as in the reviewed 4/3 lemma, we prove

$$\boxed{M \leq 21/20 \quad \Longrightarrow \quad W + \tfrac{7}{6} C \geq T}, \qquad T = \begin{cases} 2r/9, & \text{ordinary factor,} \\ 2(r-1)/9, & \text{factor containing the forbidden narrow-facet node.} \end{cases} \tag{1}$$

The choices are made before the terminating rows or their common ratio are known. In particular, ordinary E8 uses omissions `(1,6)`; the avoid-node E8 rules still use `(3,7)` or `(1,6)` as prescribed in the frozen proof. Both branches of the global argument use these identical choices.

This is compatible with the balanced counterexamples: at $M = 1$, the E8 common unit row has $C = 14/9$ and $W = 0$. A unit charge fails, while $(7/6)(14/9) = 49/27 > 16/9$.

The elementary height improvement

If the two rows have lower height bounds $\sigma \geq h_1 > 0$, $\tau \geq h_2 > 0$, then $M \leq B = 21/20$ implies

$$\min(\sigma, \tau)^2 \geq H(h_1, h_2) := \max\left\{\min(h_1, h_2)^2, \frac{\max(h_1, h_2)^2}{B^2}\right\}. \tag{2}$$

Indeed, the smaller height is at least the smaller individual lower bound, and is also at least the larger actual height divided by B. No contact-label cutoff or approximation enters this estimate.

The case $C \geq T/(7/6)$ proves (1) immediately. We therefore restrict to the strict small-cost range

$$C < 6T/7. \tag{3}$$

Every inverse retained Gram is entrywise nonnegative. A retained nonzero entry at node k costs at least its inverse diagonal c_{ik}, and two entries in the same row also contribute their nonnegative cross term. These facts give the complete finite exceptional support classification below.

Exceptional factors, including every forbidden node

The checker reconstructs the literal Grams, positive kernels, principal inverses and point resistances for A1, E6, E7, E8, F4 and G2. The six ordinary records and all 34 forbidden-node records exactly match the preset choices in the frozen proof.

For a single retained contact, orient the rows as

$$p = ae_i + be_l, \qquad q = ce_j, \qquad b, c \geq 1, \quad a = 0 \text{ or } a \geq 1.$$

Write R_{uv} for the energy of the normalized point-row difference $e_u/\delta_u - e_v/\delta_v$. If $a = 0$, then $E = R_{lj}$, while $\sigma \geq \delta_l$, $\tau \geq \delta_j$. Therefore

$$W + \tfrac{7}{6}C \geq H(\delta_l, \delta_j)R_{lj} + \tfrac{7}{6}c_{il}. \tag{4}$$

If $a \geq 1$, the heights are bounded below by $\delta_i + \delta_l$ and δ_j. With

$$u = \frac{\delta_l b}{\delta_i a + \delta_l b} \in [0, 1], \qquad E(u) = (1-u)R_{ij} + uR_{lj} - u(1-u)R_{il},$$

we obtain

$$W + \tfrac{7}{6}C \geq H(\delta_i + \delta_l, \delta_j) \min_{0 \leq u \leq 1} E(u) + \tfrac{7}{6}c_{il}. \tag{5}$$

The energy is a convex quadratic. Its minimum is checked exactly at both endpoints and, if it lies in the interval, at its stationary point. All **54 single-contact occurrences** in the full 40-record list satisfy (4) and (5). The smallest recorded excess over the target is $2363/1764 > 0$. Repeated forbidden-node choices are retained explicitly; the count is not a count of distinct diagrams.

The inverse-diagonal and cross-term census leaves exactly one two-contact pattern in (3): ordinary E7 with omissions `(3,6)`, one isolated A1 contact at node 4 in the first row and one at node 7 in the second. Its exact normalized energy is

$$E(u, v) = \tfrac{1}{2}(1 + u^2 + v^2) \geq \tfrac{1}{2}.$$

Both heights are at least one and the total cost is at least one, so

$$W + \tfrac{7}{6}C \geq \tfrac{1}{2} + \tfrac{7}{6} = \tfrac{5}{3} > \tfrac{14}{9}.$$

Every other two-contact support is excluded by its exact minimum cost, and three retained contacts exceed the threshold in every record. Thus this is complete support coverage with arbitrary real contact magnitudes satisfying the zero-or-at-least-one gap, not a scan of selected labels or unit rows. Zero retained contacts give the frozen point-row bound. The coincident-omission A1 avoid-node case has target zero and is immediate.

Classical A: point surplus pays the excess linear loss

The following part is unconditional in M. For affine A_r put $m = r + 1$, and choose opposite cycle nodes 0 and $h = \lfloor m/2 \rfloor$. Let

$$R(k) = \frac{k(m-k)}{m} \quad (0 \le k < m), \qquad R_{ij} = R(h).$$

For a retained node k, its inverse diagonal is $c_k = R(k)$, and its linear energy loss from the first omission is

$$\ell_k = R_{ij} + R(k) - R(k-h),$$

with arguments read modulo m. Reflection of the cycle interchanges the omissions, so the same complete list covers the other row.

The frozen linear-loss identity, with nonnegative inverse entries and the contact gap, gives

$$E \ge R_{ij} - \lambda C$$

whenever $\ell_k \le \lambda c_k$ at every possible retained contact. Since both heights are at least one, $W \ge E$. In the range (3), only nodes with $c_k < 6T/7$ and $c_k \le 2$ can occur. Thus for $\lambda \ge 7/6$,

$$W + \tfrac{7}{6}C \ge R_{ij} - (\lambda - 7/6)\frac{6T}{7}. \tag{6}$$

For each of the 18 cycle sizes $m = 2, \dots, 19$, the checker enumerates every node, takes the exact maximum of $7/6$ and ℓ_k/c_k over the allowed nodes, and verifies that (6) is at least $T = 2r/9$. This finite check covers all small ranks; it is not a cutoff on the theorem.

For larger ranks, the endpoint budget two implies that a retained contact is at cyclic distance at most two from its omission: for $m \ge 10$,

$$\frac{3(m-3)}{m} > 2,$$

and the inverse diagonal increases up to the antipode. Write $\varepsilon = m \bmod 2$. At distance $d \le 2$,

$$c = \frac{d(m-d)}{m}, \qquad \ell \le \frac{(m+\varepsilon)d}{m}, \qquad 6m\left(\tfrac{7}{6}c - \ell\right) \ge d(m - 7d - 6\varepsilon).$$

The last expression is nonnegative for every even $m \ge 20$ and odd $m \ge 21$; all four distance/parity classes are checked symbolically. Finally the point surplus is positive:

$$R_{ij} - 2r/9 = \begin{cases} (m+8)/36, & m \text{ even}, \\ (m-1)(m+9)/(36m), & m \text{ odd}. \end{cases}$$

This proves the ordinary estimate in every rank. Rotating the same omission pair to avoid a forbidden node preserves it and proves the weaker avoid-node target. A1 with coincident omissions is already handled separately.

Classical B, C and D

The complete prefix-coordinate identities and their index-class coverage are inherited from the frozen independent 4/3 review. The new checker explicitly re-evaluates all **24** classes at threshold $6T/7$.

For every non-distant contact the previously proved identity gives $\ell \le c$, which is stronger than needed. The only other classes have costs $r/4$, $r/2$, $(r-1)/4$, or $(r-1)/2$, and remain strictly above the new small-cost threshold. For example,

$$\frac{r}{4} - \frac{6}{7}\frac{2r}{9} = \frac{5r}{84} > 0,$$

and the modified B case is the same expression with $r-1$. Each all-rank substitution is checked as an exact polynomial with nonnegative coefficients in independent nonnegative rank/index parameters. This is an inherited all-rank coordinate proof with new threshold comparisons, not an inference from a finite rank sample.

The point resistances $r/4, r/2, (r-1)/4, (r-1)/2$ exceed their respective targets. Modified B2 is separate: its point energy is one, while a positive contact costs at least $1/2$, whose $7/6$ charge already exceeds the target $2/9$. Consequently all B/C/D cases satisfy (1), without using the height-ratio condition.

The global split at a common ratio

The frozen narrow-facet and actual-ridge arguments supply two terminating walls with a **single common ratio** M across all affine factors. The total endpoint costs satisfy $\sum C_a \le 4$. The sole avoid-node penalty remains $2/9$, and $N = \sum r_a = n-1$.

If $M \le 21/20$, summing (1) and applying the unchanged Lorentz identity gives

$$-(x,y) \ge M\left(\frac{N}{9} - 2\cdot\frac{7}{6} - \frac{1}{9} - 2\right) = M\frac{N-40}{9}.$$

If $M \ge 21/20$, use the frozen 4/3 estimate with the same omissions:

$$-(x,y) \ge M\frac{N-43}{9}.$$

For $N = 166 + t$, $t \ge 0$, the first lower bound is at least $14 + t/9$. The second is at least

$$\frac{21}{20}\frac{123+t}{9} = \frac{287}{20} + \frac{7t}{60} > 14.$$

Both contradict the strict narrow-facet bound $-(x,y) < 14$ for its norm-two framing normals. The ideal-vertex branch therefore excludes $n \ge 167$. The separately reviewed compact-facet branch gives $n \le 143$, so the combined candidate is $n \le 166$.

All geometric coverage conditions—finite-sided finite volume, cusp products, endpoint existence, common height ratio, actual framing ridges and the compact-facet alternative—are inherited from the reviewed 169 chain. The new proof is the smaller charge under a common-ratio condition and the two-branch optimization. Independent scrutiny of this new local step remains required.

```
python scripts/fresh_nikulin_coupled_charge_20261015.py
python scripts/fresh_nikulin_coupled_charge_20261015.py --verify
```

Source 167

Source path: `results/fresh_nikulin_coupled_charge_20261015_dimension166.md`

SHA256: bdb7d35d7ccde9c05dd9b61c01bdea78180634d687cf55230d5fff76b5ef55a7

Reviewed research proof candidate: dimension at most 166

Candidate theorem. Every finite-sided, finite-volume real hyperbolic Coxeter polytope has dimension **$n \le 166$**, without an arithmeticity, compactness, simple-cusp or facet-count assumption.

The common-height estimate and independent review sharpen the frozen 169 candidate. The case binder and certificate bind both branches and recursively verify the cited source hashes. This is an internally reviewed research proof candidate; external validation, novelty and current-best literature status remain unestablished.

Nikulin's narrow-facet theorem supplies a facet F whose framing hyperplanes have pairwise distance strictly below `arccosh 7`. If F contains an ideal vertex, its Euclidean link is a product of affine Coxeter simplices with total rank `N=n−1`. The reviewed endpoint and actual-ridge arguments supply two framing walls x,y with total retained-support cost at most four and one common height ratio `M≥1`. Nikulin, official ICM report, printed pp.654–655 and 657–659.

For `M≤21/20`, the local charge improves to `7/6`, giving `−(x,y)≥M(N−40)/9`. For `M≥21/20`, the earlier `4/3` charge gives `−(x,y)≥M(N−43)/9`. Both use identical preset omissions, including the node-avoidance rule for F. At `N≥166`, the first lower bound is at least 14 and the second at least `287/20>14`. Each contradicts narrowness. If F has only finite vertices, it is compact, and the separately reviewed compact-facet argument gives `n≤143`. The complete split yields 166.

The new estimate covers every affine type and every rank, with arbitrary contact magnitudes satisfying the Coxeter zero-or-at-least-one gap. Exact exceptional support enumeration is combined with symbolic classical-family inequalities. The published geometric and classification inputs remain mathematical dependencies, not formally certified software axioms.

For literature comparison, Prokhorov's published bound is 995 and Nikulin's ICM report announces the stronger bound below 300. These are distinct from the repository's candidate. The present improvement concerns unrestricted finite-volume polytopes; the published arithmetic ceiling 29 and noncompact arithmetic ceiling 21 are already much smaller. Prokhorov, printed p.402, Nikulin, printed p.659, Belolipetsky's survey.

```
python scripts/fresh_nikulin_coupled_charge_20261015_dimension166.py --verify
```

Source 168

Source path: `results/fresh_nikulin_coupled_charge_20261015_review.md`

SHA256: b0ecdc2d995fb9bde2e5d4aac542ea18a7f63849ea9c5c89c96af956a0be18cb

Independent review of the common-height charge and dimension 166

The independent checker passes for the new local estimate. It reconstructs exceptional energies using grounded rational linear algebra and rechecks the complete support census, rather than importing the producer's arithmetic. This supports an **internally reviewed research proof candidate**, not external validation or a priority claim.

For the same prescribed omissions as the frozen four-thirds proof, let `M` be the common height ratio and `C` the retained-support cost. The new statement is

```
M ≤ 21/20  ⇒  W + (7/6) C ≥ 2r/9,
```

with target `2(r−1)/9` in the single factor containing the narrow-facet normal. If the individual height lower bounds have squares `a,b`, the smaller actual height has square at least `max(min(a,b),max(a,b)/(21/20)²)`. This elementary improvement is valid for every nonnegative contact magnitude and is the only additional height hypothesis.

The independent reconstruction covers all 40 exceptional omission records and all 54 single-contact occurrences. It minimizes each normalized energy quadratic exactly. It also enumerates every subset of cheap contacts, including the nonnegative inverse-Gram cross costs. Only one multiple-contact support survives: ordinary E7, whose independently established energy is `(1+u²+v²)/2`; its cost is at least one, so its lower bound `5/3` exceeds the target `14/9`. No larger support survives. The review preserves every forbidden-node choice and verifies the literal normalized Grams.

For affine A, inverses of independently grounded path matrices recover the cycle resistances for all sizes 2 through 19. Each possible retained contact is checked against both its endpoint budget and the new strict small-cost threshold. Four symbolic distance/parity classes cover every larger cycle. For B, C and D, the review reuses the explicitly identified, independently proved coordinate identities in the frozen four-thirds review, and checks all 24 classes against the new threshold. These inherited identities and the published affine classification remain mathematical dependencies; this is not a new formalization of them.

The global arguments concerning actual framing ridges, terminating endpoints, one common height ratio, and the compact-facet alternative are inherited from the reviewed dimension-169 binder. They are not inferred from these local arithmetic checks. Writing `N=n−1`, the two bounds are

```
M ≤ 21/20:  −(x,y) ≥ M(N−40)/9,
M ≥ 21/20:  −(x,y) ≥ M(N−43)/9.
```

For `N=166+t`, `t≥0`, these are at least `14+t/9` and `287/20+7t/60`, respectively. Both contradict the strict narrow-facet threshold 14. The compact-facet branch still gives 143. Thus the combined candidate is **n≤166**, without an arithmetic, compactness or simple-cusp assumption. At `N=165,M=1`, the new scalar lower bound is below 14; this particular estimate does not establish 165.

The saved certificate binds the exact producer files and inherited review dependencies. The producer's frozen pending-review wording records its earlier status.

```
python scripts/fresh_nikulin_coupled_charge_20261015_review.py --verify
```

Source 169

Source path: `results/fresh_nikulin_defect_frontier_20261015.md`

SHA256: de3727604c2a1f6a89a51ed777d449d0480ca27a85053415056fa2d1b3a2b40b

A subunit residual-cost envelope: dimension-164 candidate

Research proof candidate, pending independent review: the reviewed dimension-165 argument sharpens to dimension at most **164**. The new point is that two large E8 defects leave less than one unit of cost for every other factor together. That residual budget allows only one retained contact, whose total energy loss is at most 2/3. This improves the aggregate loss from 40/9 to 38/9. There is no new uniform unit-charge claim.

The exact checker and certificate bind the frozen 165 source and its independent review. They use the latter's standalone rational grounded-matrix helper read-only. The new thresholds, complete low-cost singleton lists, classical tail inequalities, and aggregate cases are recomputed. No earlier artifact is modified. This package makes no publication-priority, current-best, or formal verification claim.

Unchanged local framework, with a larger height window

Use the same preset omissions, contact normalization, and geometric hypotheses as the reviewed 165 proof. For each affine factor of rank r, put

$$T = 2r/9 \quad \text{or} \quad 2(r-1)/9$$

in the unique factor avoiding the narrow-facet node. The energy is $W = \min(\sigma, \tau)^2 E$, and C is the sum of the two retained-support quadratic costs. The ratio $M = \max(\sigma/\tau, \tau/\sigma)$ is common across all factors. Each endpoint's total cost is at most 2, hence $\sum C \le 4$.

The 165 exceptional support lists and small-cycle convex domains are independent of the height-window endpoint. Re-evaluating all 71 singleton energy bounds, the inexpensive two-row E7 case, and all 580 certified small-cycle convex minima at

$$\boxed{M \le B = 21/20}$$

still proves the same unit-charge classification. The A large-rank and B/C/D proofs did not use height balance and remain unchanged. Thus every factor has $W + C \ge T$, except possibly the same 15 E7/E8 support occurrences:

Type of actual failing factor	Target	Minimum cost	Defect upper bound
Ordinary E8	16/9	14/9	2/9
Ordinary E7 or avoid-node E8	14/9	3/2	1/18

An actual failing factor means $T - W - C > 0$. The table describes necessary conditions, not an assertion that all listed support possibilities attain the stated defects. At most two failing factors are possible because three would consume at least $9/2 > 4$ cost.

The residual-product lemma

Under $M \le 21/20$, consider any collection of the prescribed affine factors with total retained cost **strictly less than 1**. Then

$$\boxed{\sum (T - W) \le 2/3.} \tag{1}$$

This includes ordinary and avoid-node factors of every rank.

First, for any positive definite retained Gram with root squares 2, its inverse diagonal is

$$J_{kk} = \frac{1}{2 - v^T B^{-1} v} \ge \frac{1}{2}.$$

Its inverse is entrywise nonnegative. Every nonzero contact is at least 1, so two retained coordinates, anywhere among the two rows of all the factors, already cost at least 1. The strict residual budget therefore allows **at most one retained coordinate globally**. A zero-cost factor has both rows supported at their omitted nodes and satisfies the reviewed point-energy bound $W \ge T$. It remains to bound the loss of a single-contact factor.

Exceptional factors

The checker reconstructs every retained inverse diagonal below 1 in all 40 ordinary/forbidden-node records. If $T \le 2/3$, inequality (1) for that factor follows from $W \ge 0$. Otherwise the complete singleton energy/height calculation gives the stronger $W \ge T$ at $B = 21/20$. Ordinary F4, whose target is 8/9, has no retained diagonal below 1. There is no hidden multi-contact case because the total cost is strictly below 1.

All A cycles

For affine A_r, set $m = r + 1$, choose omissions 0 and $h = \lfloor m/2 \rfloor$, and write $R(d) = d(m - d)/m$. A retained node at distance d has inverse diagonal $R(d)$. Diagonal below 1 forces $d = 1$: for $m \ge 4$, every $2 \le d \le m/2$ has $R(d) \ge 2(m - 2)/m \ge 1$; m=2,3 already have only distance 1.

Write the single-contact row as $p = ae_i + be_k$, where k is a neighbor of i, $b \ge 1$, and $a = 0$ or $a \ge 1$. The other row is supported at its omitted node j.

If a=0, both heights are at least 1, and the worst neighbor is the one toward the opposite omission. Hence

$$W \ge R(h - 1) = \begin{cases} m/4 - 1/m, & m \text{ even}, \\ (m^2 - 9)/(4m), & m \text{ odd}. \end{cases}$$

Subtracting the ordinary target $2(m - 1)/9$ and adding 4/9 gives

$$\frac{m^2 + 24m - 36}{36m}, \qquad \frac{m^2 + 24m - 81}{36m},$$

respectively. These are nonnegative for all even m≥2 and odd m≥3, as verified by their exact nonnegative-coefficient substitutions. Thus the loss is at most 4/9, already smaller than the required 2/3.

If a≥1, one height is at least 2, so the smaller height is at least $2/B$. For m=2,3 the target is at most 4/9, so there is nothing further to prove. For m≥4 the normalized-row energy is the squared distance from a point on the segment between the two adjacent point potentials to the other omission's potential. Its minimum, using the worse of the two adjacent directions, is

$$E_{\min} = \begin{cases} \dfrac{m(m-2)}{4(m-1)}, & m \text{ even}, \\ \dfrac{(m+1)(m-3)}{4(m-1)}, & m \text{ odd}. \end{cases}$$

These follow by minimizing the exact convex quadratic $R(h) - \ell u + R(1)u^2$, with $\ell = (m + \varepsilon)/m$ and $\varepsilon = m \bmod 2$. Both minima are at least $(m - 3)/4$. Therefore

$$W - T \ge \frac{m - 3}{B^2} - \frac{2(m - 1)}{9} = \frac{302m - 1102}{441} > 0 \qquad (m \ge 4).$$

Rotation proves the same ordinary estimate while avoiding a prescribed node when m≥3. The coincident-omission avoid-A1 case has target zero. This covers every A rank and every permitted omission case.

All B/C/D ranks

The complete 24 prefix-coordinate classes are recomputed from the frozen independent coordinate realization. Imposing inverse diagonal below 1 leaves only the following possibilities:

Family and position	Diagonal	Consequence
Ordinary B, right omission, node 0 or 1	$r/4$	r≤3, so T≤2/3
Modified B, omission r−1, node 0 or 1	$(r-1)/4$	r≤4, so T≤2/3
Modified B, omission r−1, node r	1/2	linear energy loss 0
Separate modified B2	at least 1/2	T=2/9
Modified C, omission 1, node 0	1/2	linear energy loss 0
Modified C, omission 1, node r	$(r-1)/2$	r=2, so T=2/9

All other classes have diagonal at least 1. In each zero-linear-loss case the nonnegative quadratic-remainder identity gives $E \geq R_{ij}$, and both heights are at least 1, so $W \geq R_{ij} \geq T$. In the small-target cases, $W \geq 0$ suffices. All modified choices are included via the previously reviewed fork/path symmetries and the explicit modified B/C classes.

The constant 2/3 in (1) is sharp for this local domain. In ordinary affine B3 with omissions (0,3), put both contact rows at node 0 with coefficient 1. Their normalized potentials agree, giving $M = 1$, $E = W = 0$, while the sole retained cost is $3/4 < 1$ and T=2/3. Its literal rationally scaled Gram and inverse diagonal are checked exactly. This is a local witness, not an assertion that an arbitrary product of extremal factors is realized by an ambient polytope.

Complete defect accounting

Let the total loss be $L = \sum(T - W)$. If at most one factor fails unit charge, then

$$L \leq \sum C + 2/9 \leq 38/9.$$

If two fail, classify them by the number of ordinary E8 factors:

Ordinary E8 among the two	Minimum combined cost	Remaining cost	Combined target	Total loss bound
0	3	≤1	28/9	$28/9 + 1 = 37/9$
1	55/18	≤17/18	30/9	$30/9 + 2/3 = 4$
2	28/9	≤8/9	32/9	$32/9 + 2/3 = 38/9$

The first row uses unit charge for all remaining factors, which do not fail by definition. The other two use the strict-subunit residual-product lemma; the individual failing factors themselves are bounded only by $W \geq 0$. This proves the full envelope

$$\boxed{L \leq 38/9 \quad \text{whenever } M \leq 21/20.} \tag{2}$$

It accounts for actual costs, not just a count of possible exceptional support records.

Dimension conclusion and scope

With $N = n - 1$, the sum of targets is $(2N - 2)/9$, including the sole avoid-node penalty. Equation (2) gives

$$\sum W \geq \frac{2N - 40}{9}, \qquad -(x, y) \geq M\frac{N - 38}{9} \quad (M \leq 21/20).$$

For $M \geq 21/20$, the unchanged four-thirds estimate with the same preset omissions gives $-(x, y) \geq M(N - 43)/9$. At $N = 164 + t$, these are at least

$$14 + t/9, \qquad 14 + 7/60 + 7t/60,$$

respectively. Both contradict the strict Nikulin narrow-facet pairing bound $-(x, y) < 14$. Thus the ideal-vertex branch excludes n≥165; the inherited compact-facet branch gives n≤143. Their combination is the candidate $\boxed{n \leq 164}$.

All geometric coverage—finite-sided finite volume, the narrow facet, compact/ideal alternative, actual framing and terminating walls, common height ratio, and endpoint Schur budgets—is inherited explicitly from the reviewed 165/169/166 chain. The new result is the widened local check and the complete strict-subunit residual envelope. No arithmetic restriction or finite-label cutoff is imposed.

For later work, a rank-residue argument alone does not rule out the relaxed extremal combination: two E8 factors, one B3, one avoid-A1 and 24 zero-cost ordinary E6 factors have total affine rank 164. Their individual endpoint cost budgets may obstruct this combination, which is stronger information than the summed cost used here. That is a separate future question, not part of this proof or an achieved bound below 164.

```
python3 scripts/fresh_nikulin_defect_frontier_20261015.py
python3 scripts/fresh_nikulin_defect_frontier_20261015.py --verify
```

Generation and saved replay pass, as do source manifests, syntax, and local links. Independent review of this new argument remains pending.

Source 170

Source path: `results/fresh_nikulin_defect_frontier_20261015_dimension164.md`

SHA256: 0dc49f46a76f8ead2ad1c45f66df58dc1cc9a9a6aa545eddfe084367e23aa8f1

Reviewed research proof candidate: dimension at most 164

Candidate theorem. Every finite-sided, finite-volume real hyperbolic Coxeter polytope has dimension **$n \le 164$**, without arithmeticity, compactness, a simple-cusp hypothesis or a facet-count restriction.

The residual-cost proof and independent review sharpen the frozen 165 candidate. The case binder and certificate verify the branch split and recursive source hashes. This is an internally reviewed research proof candidate; external validation, novelty and current-best literature status remain unestablished. Historical pending-review wording in the frozen primary package is superseded by the separate review.

Use the inherited narrow-facet geometry, actual framing walls, common null height ratio $M \ge 1$ and total retained-support Schur budget four. If the narrow facet has an ideal vertex, its affine factors have total rank $N = n-1$. For $M \le 21/20$, the same fifteen possible E7/E8 exceptional support occurrences are the only factors that can fail unit charging. Three actual failures would cost at least9/2, so at most two are possible.

Two ordinary E8 failures consume cost at least28/9, leaving strictly less than one everywhere else. Each retained inverse diagonal is at least1/2, so this residual product has at most one nonzero retained coordinate. The new all-rank lemma bounds its energy loss by2/3. The other defect cases are also checked, giving total loss at most38/9. Including the unique avoid-node penalty, the Lorentz pairing obeys $-(x,y) \ge M(N-38)/9$. For $M \ge 21/20$ retain the reviewed four-thirds estimate $-(x,y) \ge M(N-43)/9$, using identical preset omissions.

At $N = 164$ the two lower bounds are respectively at least14 and$14+7/60$, and both increase with N. They contradict the strict narrow-facet bound $-(x,y) < 14$ for squared-length-two normals. Thus the ideal-vertex branch excludes $n \ge 165$. If the narrow facet has no ideal vertex it is compact, and the frozen compact-facet proof gives $n \le 143$. These alternatives complete the candidate bound164.

The estimate covers every affine rank and every prescribed avoid-node case, with arbitrary real nonzero Coxeter contacts at least one. It does not use a finite-label cutoff. The independent review checks the widened height window, the exceptional singleton domains, symbolic classical families and aggregate arithmetic. The geometric dependencies are inherited from the reviewed 165 case split; the local certificate does not itself prove the narrow-facet theorem.

The near-range scalar estimate at $N = 163$ is below14, so this package claims no bound163. Keeping the two separate endpoint budgets is an ongoing investigation with its own files. The arithmetic ceiling29 and noncompact arithmetic ceiling21 are separate published results, unaffected by164.

Replay: `python3 scripts/fresh_nikulin_defect_frontier_20261015_dimension164.py --verify`.

Source 171

Source path: `results/fresh_nikulin_defect_frontier_20261015_review.md`

SHA256: b644fdc5d9ed26a8c6eb3917dc211957e980f56d09fc003022b5cd3c8d7116c8

Independent review of the strict-subunit envelope and dimension-164 candidate

PASS. The independent checker and certificate verify the new arithmetic and audit the proof in the primary report. Generation and saved replay pass. No correction to the primary proof was required. Its pending-review line is historical; this separate artifact records the completed review without changing that frozen report.

This is an internally reviewed research proof candidate for finite-sided finite-volume hyperbolic Coxeter polytopes, without an arithmetic restriction. It is not a publication-priority, current-best, or formal-verification claim. The narrow-facet and compact-facet theorems, actual framing-wall coverage, common height ratio, prescribed omission choices, and endpoint Schur budgets are inherited from the frozen reviewed 169/166/165 chain. The present review concerns the widened height window, the new all-rank strict-subunit lemma, and their aggregate consequence.

The checker imports no producer or previous-review arithmetic. It uses physical norm-two Gram matrices with exact SymPy radicals, rather than the primary's rationally scaled grounded helper. It reconstructs the six affine models, positive null vectors, retained inverses and potentials, and all forty ordinary/avoid-node records. Every inverse is positive definite with nonnegative entries; every physical inverse diagonal is at least one half. All complete low-cost support subsets are regenerated. This reproduces 71 singleton and 16 multiple-support occurrences, with exactly fifteen possible defect occurrences: four ordinary E8 and eleven ordinary-E7/avoid-E8 cases. The one additional E7 double-support case is verified directly from its two-variable energy polynomial and height bounds.

At the enlarged height ratio M <= 21/20, all singleton unit-charge inequalities still hold. A separate Fraction implementation reconstructs every cycle support domain and checks all 580 convex dual certificates. Their Gram matrices are explicit Euclidean prefix-potential Grams, so positivity does not depend on an optimizer. Equality feasibility, nonnegative barycentric weights, dual slacks, complementary slackness and objective equality are checked. The widened height lower bound is then applied afresh. Thus these are exact continuum bounds, not sampled contact values.

The strict-subunit reduction is valid. If the sum C of all retained quadratic costs is less than one, two nonzero retained coordinates are impossible: each has contact magnitude at least one and inverse diagonal at least one half, and all same-row cross terms are nonnegative. Coordinates in different factors or at the two different endpoints contribute separately. Consequently at most one retained coordinate survives across the entire residual product. Zero-cost factors have point energy W >= T, including the zero-target coincident-omission avoid-A1 case.

The unique possible contact is completely covered:

- The physical exceptional calculation checks all 65 retained positions with diagonal below one in the forty prescribed records. Every case with target T > 2/3 satisfies the stronger W >= T at M <= 21/20. The other cases use W >= 0.
- For every affine A cycle, inverse diagonal below one forces a neighboring retained node. The point-row energy gives loss at most 4/9. With a positive omitted contact, the energy is minimized on an explicit segment; the even/odd formulas and minimizers are checked symbolically. For cycle size at least four, the energy margin is at least (106 + 302t)/441 at size 4+t. Sizes two and three have target at most 4/9. Rotation preserves the ordinary estimate for all permitted avoid-node choices. The unchanged large-rank unit-charge tails are also rechecked as positive rational-polynomial identities.
- For B/C/D, the complete 24 prefix-coordinate classes and their index-domain coverage are inherited from the earlier independent all-rank review. This checker independently verifies their cost/resistance/loss identities and the new implication of diagonal below one. Each class either has diagonal at least one, has zero linear energy loss, or has an

exact bounded integer-rank domain with T <= 2/3. The separate modified B2 case is checked directly. This is an all-rank argument; the finite integer checks occur only where the strict diagonal inequality bounds the parameter.

It follows that any residual product of total cost strictly below one has total loss sum(T-W) <= 2/3. The bound is sharp in the stated local domain: the independently reconstructed norm-two affine B3 matrix has right-omission inverse entry 3/4 at node zero. Both rows at that same node give energy zero and loss 2/3. This is a local domain witness, not an asserted global polytope realization.

The defect accounting is exhaustive. An actual unit-charge failure consumes at least 3/2, so there are at most two. With at most one failure the loss is at most 4+2/9 = 38/9. With two failures, zero ordinary-E8 factors give at most 37/9 using unit charge on the remainder. One ordinary E8 leaves at most 17/18 cost and gives total loss at most 4; two leave at most 8/9 and give at most 38/9. The latter two budgets are strictly below one, exactly as required by the new lemma. No assumption that possible defect patterns actually attain their worst bounds is made.

For N=n-1, the prescribed targets sum to (2N-2)/9. Therefore the near-window pairing lower bound is M(N-38)/9. Outside the window the inherited four-thirds estimate gives M(N-43)/9. At N=164+t these are respectively at least 14+t/9 and 14+7/60+7t/60. Both contradict the inherited strict narrow-facet upper bound when n >= 165. The separate compact-facet branch remains at most 143. The resulting candidate bound is therefore n <= 164.

All primary input hashes and recursive source manifests are checked. The review does not rerun the full older global geometric certificates, but it binds their frozen checked chain and explicitly preserves their hypotheses. There is no finite-label cutoff and no extrapolation from a finite rank sample.

```
python3 scripts/fresh_nikulin_defect_frontier_20261015_review.py --verify
```

Source 172

Source path: `results/fresh_nikulin_dimension_181_candidate_20261015.md`

SHA256: fb5938e5c9a0132a6c8a548d46d9178d99666b041d9a56e62f25917538662731

Reviewed research proof candidate: dimension at most 181

Candidate theorem. Every finite-sided finite-volume real hyperbolic Coxeter polytope in H^n satisfies **n≤181**, without arithmeticity, compactness, simple ideal vertices, or a facet-count restriction.

The avoid-node estimate and its independent arithmetic and proof review sharpen the frozen 189 proof. The dependency binder and certificate bind the complete chain. This is an internally reviewed research proof candidate. External validation, priority and current-best status are unestablished.

Nikulin's narrow-facet theorem gives a facet F whose framing hyperplanes have pairwise distance strictly below arccosh 7. The proof uses actual ridge neighbors of F. With outward normals of square two, two such walls cannot have negative pairing at least 14. Official ICM report, printed p.655.

If F has an ideal vertex, its horospherical link is a product of irreducible affine Coxeter simplices, with ranks summing to N=n−1. Preset omissions select two outgoing edges in F. At their other endpoints, finite volume supplies terminating walls x,y. At an ideal endpoint, one chooses a missing node in the appropriate affine component, or an orthogonal component if F belongs to an A1 factor. The frozen product proof verifies that these walls meet F along actual ridges.

For affine factor a, let C_a be the sum of the two retained-support Schur costs, and σ_a,τ_a its positive null heights. Their ratio q is common to all factors. Put M=max(q,1/q). The endpoint Gram matrices give ΣC_a≤4. The unchanged non-narrow estimate and new avoid-node estimate give, respectively,

```
σ_a τ_a E_a / M + 2C_a ≥ 2r_a/9,
σ_a τ_a E_a / M + 2C_a ≥ 2(r_a−1)/9.
```

Only the single factor containing F's normal uses the second inequality. It holds for every forbidden node, with choices made before the contact rows are known. The proof covers all classical ranks analytically and checks every exceptional omission exactly, including cheap retained-contact cases; it imposes no label cutoff.

Summing and using the Lorentz identity yields

```
Σ_a σ_a τ_a E_a ≥ M(2N/9−8−2/9),
−(x,y) = (1/2)Σ_a σ_a τ_a E_a − q − 1/q
       ≥ M(N−55)/9.
```

At n≥182, N≥181, so the pairing is at least 14, contradicting narrowness. This branch gives n≤181.

If F has no ideal vertex, it is compact. The separately reviewed compact-facet argument gives n≤143. It allows affine subdiagrams in ambient stars and does not assume F is itself Coxeter. The two alternatives exhaust F and give the stated candidate.

The new local loss of 2/9 is sharp within the uniform avoid-node lemma: in A1, both omissions must be the only node other than the forbidden node, and equal point-contact rows have zero cost and energy. This does not prove that 181 is optimal for the global method or for hyperbolic Coxeter polytopes.

For historical comparison, Prokhorov's published Theorem C gives n≤995. Nikulin already announced the unrestricted n<300 in his ICM report, whose definition of the class assumes finite volume but not arithmeticity. Thus 181, if validated, improves both numerical bounds; a bounded literature check cannot establish novelty or rule out a later stronger theorem.

Prokhorov, printed p.402, Nikulin, printed pp.654 and 657–659. Khovanskii's 2006 author account still states 995 as Theorem 0.1. Author text.

The new primary report's pending-review sentence records its earlier frozen state. The completed review and this final binder supersede that status without rewriting the frozen report. Computational checks certify the exact arithmetic and source closure; the geometry and published classifications remain mathematical proof dependencies, rather than a formal proof-assistant development.

```
python scripts/fresh_nikulin_dimension_181_candidate_20261015.py --verify
```

Source 173

Source path: `results/fresh_nikulin_dimension_189_candidate_20261015.md`

SHA256: af414d020971f879f07c8c29cd65efe5fbb00e234348e941b81ed8cc6c7343e6

Reviewed research proof candidate: dimension at most 189

Candidate theorem. Every finite-sided finite-volume real hyperbolic Coxeter polytope in H^n satisfies **n≤189**. There is no arithmeticity assumption, no simple-ideal-vertex restriction, and no facet-count or finite-label cutoff.

This package combines the frozen 225 baseline, the new contact-cost estimate, and its independent proof and arithmetic review. It preserves every earlier artifact. The binder and certificate record their source, data, and report hashes and validate the recursive dependency manifests.

The conclusion is presented as a reviewed research proof candidate, not as an externally peer-reviewed theorem, a publication-priority assertion, or a determination of the current strongest known bound. Nikulin's ICM report already announces the unrestricted inequality n<300. That historical claim remains distinct from the explicit estimates reconstructed here. Primary report, printed pp.657–659.

Exhaustive case split

Nikulin's narrow-facet theorem supplies a facet F whose framing hyperplanes have pairwise distance strictly below arccosh 7. For outward normals of square two, the selected framing walls x,y therefore cannot satisfy $-(x,y)\geq 14$. The construction uses walls sharing actual ridges with F, so they belong to the precise framing set in the theorem. Theorem A, printed p.655.

If F contains an ideal vertex, its horospherical link is a product of affine Coxeter simplices with total rank N=n−1. The frozen product argument selects two outgoing edges, and terminating walls x,y that meet F in actual ridges. Their normalized affine null-height ratio q is common across all factors. With M=max(q,1/q)≥1, their Lorentz pairing is

$$-(x,y) = (1/2)\,\Sigma_a\,\sigma_a\,\tau_a\,E_a - q - 1/q.$$

At each terminating endpoint the total retained Schur cost is at most two. If C_a is the sum of the two endpoint costs in factor a, then $\Sigma_a C_a\leq 4$. The new estimate proves

$$\sigma_a\,\tau_a\,E_a\,/\,M \geq 2r_a/9 - 2C_a$$

for every factor except the one containing F's normal. That one factor retains the previous uniform loss allowance of two. Consequently

$$\Sigma_a\,\sigma_a\,\tau_a\,E_a \geq M(2N/9-10),$$
$$-(x,y) \geq M(N/9-7).$$

For n≥190, N≥189 and the final lower bound is at least 14, contradicting narrowness. This branch therefore gives n≤189.

If F has no ideal vertex, finite-sidedness and finite volume make F compact: its vertices are among the finitely many finite and ideal vertices of the ambient projective polytope. The already reviewed compact-facet lemma gives n≤143. It uses ambient vertex diagrams and quasi-Lannér extraction; it does not assume that F is itself a Coxeter polytope or that its ambient face stars have no parabolic subdiagrams.

These cases exhaust the chosen narrow facet. Thus n≤max(189,143)=189.

What the sharper estimate adds

The 225 proof charged a fixed loss to each of at most eight spoiled non-narrow affine factors. The new proof charges their loss to the actual total Schur cost, bounded by four independently of how many factors occur.

For classical factors and affine A_1,E_6,F_4,G_2, the cost of one retained nonright contact, together with the inherited localization estimate, directly pays the required loss. The two exceptional issues are E_7 and E_8, whose chosen deletions can leave an inexpensive A_1 contact. Exact inverse-diagonal checks show that cost below one permits only a single such contact. Three exact resistance values, 1/2,1,1/2, then give the interpolation identity $E=(1+u^2)/2$. Splitting according to whether the omitted-node contact is zero or at least one proves the same charge inequality. This is an all-label argument, not a search with a maximum Coxeter label.

The independent review reconstructs the exceptional arithmetic and checks the geometric and inequality cases. The earlier independent reviews continue to cover common null-height scaling, the actual-ridge endpoint selection, the prescribed omission choices, and the compact-facet branch. The primary energy-charge report's final pending-review sentence is historical: that report was frozen before the separate review finished; its completed review is bound in this package.

The binder checks source closure, the exhaustive case combination, and the exact scalar threshold. It imports no producer arithmetic. Published classifications and the geometric proof arguments remain explicitly identified mathematical inputs; these computational checks are not a formal proof-assistant development.

```
python scripts/fresh_nikulin_dimension_189_candidate_20261015.py
python scripts/fresh_nikulin_dimension_189_candidate_20261015.py --verify
```

Source 174

Source path: `results/fresh_nikulin_dimension_225_candidate_20261015.md`

SHA256: 170ccf0ff78482971ed07658b12766cf6c9cd6ab1ca886bf82ed784b4825904d

A research proof candidate for the finite-volume bound 225

Candidate theorem. Every finite-sided finite-volume real hyperbolic Coxeter polytope in H^n satisfies **n≤225**. No arithmeticity, compactness, simple-ideal-vertex, facet-count, or prescribed-diagram hypothesis is imposed.

This is a research proof candidate assembled from two explicit arguments and their internal independent checks. It is not a claim of publication priority, an externally peer-reviewed result, or the strongest currently known bound. Nikulin's 1986 ICM report already announces the unrestricted inequality n<300; the present work reconstructs explicit constants rather than treating 995 as the only historical assertion.

The dependency checker and certificate bind the exact source, data, and proof reports. They verify the case partition and scalar comparisons, and recursively validate the saved source manifests. They do not purport to turn the geometric arguments or the cited classifications into a formal proof-assistant development.

The two cases

Choose a narrow facet F supplied by Nikulin's narrow-facet theorem. Hyperplanes of any two framing facets whose hyperplanes meet H(F) have distance strictly less than arccosh 7. The proof below uses framing facets sharing actual ridges with F, a subset of that published framing set. With outward normals normalized to square two, such framing normals x,y cannot satisfy $-(x,y)\geq 14$. Nikulin, printed p.655, Theorem A.

If F has an ideal vertex, use the product-cusp argument. Its horospherical link is a product of irreducible affine Coxeter simplices with total rank $N=n-1$. Select two outgoing edges by deleting one node in each affine component, avoiding the node of F. The terminating walls can be chosen to meet F in actual ridges, including when the second endpoints are ideal and nonsimple.

At either endpoint the positive-semidefinite Schur bound is at most two. Inverse diagonals are at least one half, inverse entries are nonnegative, and every nonright norm-two Coxeter contact has magnitude at least one. Thus at most four retained contacts occur per endpoint. At most eight affine factors are affected; allowing one additional factor for the avoided narrow-facet node gives at most nine marked factors.

The full product proof proves a uniform affine energy bound $2r/9$ per factor of rank r, with loss at most two per marked factor. The two null-height ratios agree across all affine factors, without any arithmetic or equal-scale assumption. If M is the larger of this ratio and its reciprocal, then $M\geq 1$ and the exact Lorentz identity yields

$$-(x,y) \geq M\,((n-1)/9 - 11).$$

For $n\geq 226$ the right side is at least 14, contradicting narrowness. Hence this case has $n\leq 225$. The classical formulas, product arithmetic, and separate independent arithmetic review make the all-rank and exceptional constants explicit.

If F has no ideal vertex, it is compact. This uses finite-sidedness and finite volume: the projective closure of the polytope is the convex hull of finitely many finite and ideal vertices, and the vertices of a facet are among these. A facet with only finite vertices is therefore compact. It is simple because its vertices are finite vertices of an acute Coxeter chamber. It need not be a Coxeter polytope itself.

Apply the separate compact-facet lemma, which gives $n\leq 143$. Its weights are defined using the ambient spherical vertex diagrams. Minimal indefinite subdiagrams are allowed to be quasi-Lannér, of order at most ten. Compact triangles supply

two positive-weight corners; compact quadrilaterals supply one by a white path of length at most 17. At each ambient finite vertex, at most 17n pairs have that graph distance. The simple-polytope h-vector inequality on the facet contradicts $n \geq 144$.

Thus the two exhaustive cases give $n \leq \max(225,143) = 225$.

What was repaired during the audit

The first proposed treatment of the compact-facet case attempted to transfer Vinberg's finer compact-chamber weights directly. That would have used a false intermediate assertion: a compact marked face need not have an ambient star free of parabolic subdiagrams. A Coxeter tetrahedron with path labels 3–3–6 supplies an explicit counterexample.

The final compact-facet proof does not use that assertion. For a compact quadrilateral, the full block consisting of its white support and an opposite pair of black nodes is indefinite: a positive-definite block would give an actual intersection of opposite sides, while a singular positive-semidefinite block would give an ideal point on the compact quadrilateral. Minimal-indefinite extraction then permits quasi-Lannér diagrams. The original primary sources state the needed order-ten classification and extraction, and the proof separately explains why disconnected affine degeneracies cause no exception. Prokhorov, §2, printed pp.407–408.

The older frozen product report describes its result as conditional because its all-finite alternative was unresolved at that checkpoint. That report remains unchanged. The separate compact-facet proof supplies the missing branch here; it does not retroactively import an unproved compact-facet Coxeter property.

Evidence and literature status

The product review reconstructs the exceptional affine matrices and their energy constants with separate arithmetic, checks independent classical coordinate/localization examples, and verifies the all-rank scalar inequalities. The separate hostile mathematical review checks common null-height scaling, the contact budget, predetermined omission choices, and the actual-ridge endpoint bridge. It also audits the compact-facet source and geometry, with independent scalar replay. The combined certificate records the specific review files and their hashes.

The primary literature inputs are the narrow-facet theorem, spherical and affine Coxeter classification, quasi-Lannér order at most ten, the spherical graph-distance pair bound, and the simple-polytope h-vector relations. The new geometric reductions and estimates are written out in the bound reports; the checker does not silently substitute a finite numerical search for any all-rank statement.

Nikulin's printed pp.657–659 outline the narrow-facet method, work out an affine-A example, and announce $n < 300$. The earlier literature audit distinguishes that announcement from the fully explicit 995 proof and its independently audited 990 refinement. This combined candidate is presented for further mathematical scrutiny with that historical distinction preserved. No current-best or novelty assertion is made.

```
python scripts/fresh_nikulin_dimension_225_candidate_20261015.py --verify
```

Source 175

Source path: `results/fresh_nikulin_energy_charge_20261015.md`

SHA256: cbbd06781e531000418398ca6c4515e4bee59599d38042be316ad2d0ab2bb66c

Charging affine-factor losses to contact cost: a bound of 189

Research proof candidate. Every finite-sided finite-volume real hyperbolic Coxeter polytope in H^n satisfies **n≤189**.

This improves the separately frozen 225 candidate by one change: the loss in a spoiled affine factor is charged to its actual Schur-complement contact cost. The narrow-facet component retains the earlier loss allowance of two. No further choice optimization or finite-label enumeration enters this report. The published inputs, actual-ridge endpoint selection, affine product scaling, and compact-facet branch are unchanged. Publication priority and current-best status are not asserted.

The checker and certificate reconstruct the exceptional inverse diagonals and three small resistance identities exactly, and check the all-rank inequalities below. They bind the earlier geometric proofs rather than importing producer arithmetic. Independent review is separate from this primary package.

Setup retained from the 225 proof

Choose a narrow facet F. If it is compact, the frozen compact-facet lemma gives n≤143. Otherwise F contains an ideal vertex. Its link is a product of affine Coxeter simplices, of ranks r_a summing to N=n−1.

Use exactly the fixed omission choices from the product-cusp proof: opposite cycle nodes for A, the designated end nodes for B/C/D, and the recorded pairs for the exceptional types. Only the component containing F's normal uses the old avoid-node modification. These choices precede the selection of terminating contact rows.

Write x,y for the resulting two norm-two terminating walls, chosen to share actual ridges with F. In a component let p,q be their nonnegative negated contact rows, δ its positive affine kernel, and σ=δ·p, τ=δ·q. The ratio σ/τ is common to all components. Put M=max(σ/τ,τ/σ)≥1. Let E_a be the corresponding normalized-row energy in the Euclidean cusp quotient. The exact identity is

$$-(x,y) = (1/2)\ \Sigma_a\ \sigma_a\ \tau_a\ E_a - \sigma_a/\tau_a - \tau_a/\sigma_a.$$

For the two deletion supports S_i,S_j in this component define its **contact cost**

$$C_a = p_{Si}^{T}\ G(S_i)^{-1}\ p_{Si} + q_{Sj}^{T}\ G(S_j)^{-1}\ q_{Sj}.$$

Each endpoint's total Schur cost is at most two, so Σ_a C_a≤4. The inverses are entrywise nonnegative. Every nonzero contact is at least one, including contacts with an omitted node; no upper bound on its label or magnitude is used.

We prove, for every component other than the one containing F's normal,

$$\sigma_a\ \tau_a\ E_a\ /\ M \geq 2r_a/9 - 2\ C_a. \qquad (1)$$

If C_a=0, both rows are supported only on their omitted nodes. The frozen point-contact bound already gives σ_aτ_aE_a/M≥2r_a/9. It remains to consider C_a>0.

Classical and direct exceptional factors

A positive contact in a retained row costs at least the corresponding inverse diagonal, because its magnitude is at least one and all cross terms are nonnegative. Thus C_a is at least the smallest inverse diagonal among the two retained supports.

For affine A_r, r≥2, this minimum is r/(r+1). For r≤8, nonnegativity of energy bounds the deficit by 2r/9, and

$$2r/(r+1) - 2r/9 = 2r(8-r)/(9(r+1)) \geq 0.$$

For r≥9 the unchanged localization argument gives energy at least (r−8)/4 after division by M, so the deficit is at most 2−r/36. Here

$$2r/(r+1) - (2-r/36) = (r-8)(r+9)/(36(r+1)) \geq 0.$$

Both ranges prove (1). Affine A_1 is separate: its retained inverse diagonal is 1/2 and its baseline is only 2/9, so the zero-energy bound suffices.

For the unmodified B/D end deletions, the retained inverse minimum is at least one in ranks r≥4. The B_2 and B_3 presentations have minimum respectively 1/2 and 3/4; their zero-energy baseline is already smaller than twice that value. For r≤9 the bound 2r/9≤2 suffices. For r≥10, the old localization gives energy at least (r−5)/4 after division by M, so the deficit is at most 5/4−r/36<2. This proves (1) for B and D.

For C_r, r≥2, either end deletion leaves finite B_r, with inverse minimum one. The zero-energy argument covers r≤9. In larger rank the retained bound (r−5)/2 already has deficit at most 5/2−5r/18≤2. Thus (1) holds here as well. These claims use the closed inverse formulas in the frozen classical proof; the checker separately verifies the low B ranks where the generic minimum-one statement would be false.

The direct exceptional cases are also immediate from their exact inverse diagonals:

Affine type	Rank	Fixed omitted nodes	Point energy	Minimum retained inverse diagonal
A_1	1	0,1	1/2	1/2
E_6	6	2,4	4/3	4/3
F_4	4	0,4	1	1
G_2	2	0,2	2/3	2/3

In each row the point energy is at least 2r/9, and twice the last entry is also at least 2r/9. Zero energy therefore suffices whenever a retained contact occurs. The node numbering is fixed by the full matrices in the certificate.

The inexpensive E_7 and E_8 contacts

A minimum-diagonal argument alone would fail in these two types. The fixed E_7 omissions 3,6 each isolate an A_1 node, respectively 4,7. The fixed E_8 omissions 3,7 have just one inverse diagonal below one: node 8 after omitting 7. Each of these inexpensive diagonals is 1/2; every other retained inverse diagonal is at least one. The checker verifies the entire two inverse-diagonal lists, rather than only these three selected entries.

If C_a≥1, inequality (1) follows from nonnegative energy since 2r/9<2 for r=7,8. Suppose 0<C_a<1. Nonnegativity of inverse entries implies that there is exactly **one** retained nonzero contact across the two endpoint rows: it is one of the inexpensive A_1 contacts. Two such contacts would already cost at least one, and any other retained contact would cost at least one by itself.

Orient the two rows so that this contact, of magnitude b≥1, belongs to x. Let i be x's omitted node, l the inexpensive retained node, and j the other endpoint's omitted node. The kernel is normalized so that

$$\delta_i=\delta_j=2, \quad \delta_l=1.$$

The contact rows consequently have the exact forms

$$p = a\,e_i + b\,e_l, \qquad q = c\,e_j,$$
$$\sigma = 2a+b, \qquad \tau = 2c,$$
$$C_a = b^2/2.$$

Here c≥1 because the second row cannot vanish, and a is either zero or at least one. This last alternative is the only discreteness used: a nonright Coxeter contact cannot have magnitude strictly between zero and one.

For nodes u,v let R_uv be the squared energy of the normalized point-row difference e_u/δ_u−e_v/δ_v. In each of the three inexpensive-contact cases, exact inversion gives

```
R_ij=1/2,   R_lj=1,   R_il=1/2.                    (2)
```

If a=0, the normalized first row is the point row at l. Hence E_a=1. Since both σ,τ≥1,

```
σ τ E_a/M = min(σ,τ)² E_a ≥ 1,
2C_a=b²≥1.
```

Their sum is at least two, which exceeds the required baseline 2r/9.

If a≥1, both σ and τ are at least two. Put u=b/(2a+b). The first normalized row is the convex combination (1−u)e_i/δ_i+u e_l/δ_l. The standard squared-distance interpolation identity and (2) give

```
E_a = (1−u)R_ij+uR_lj−u(1−u)R_il
    = (1+u²)/2 ≥ 1/2.
```

Therefore σ τ E_a/M = min(σ,τ)² E_a≥2, so (1) follows even without adding its contact cost. The same argument applies when the inexpensive contact belongs to y. This closes all E_7/E_8 possibilities, including arbitrary large finite labels and dotted omitted contacts.

Summing the costs and completing the dichotomy

The one component containing F's normal keeps the frozen estimate

```
σ_a τ_a E_a/M ≥ 2r_a/9 − 2.
```

There is no need to charge that component's cost, or to optimize its omission choices. Summing (1) over the other components and using Σ_a C_a≤4 gives

```
Σ_a σ_a τ_a E_a ≥ M(2N/9 − 8 − 2).
```

Since σ_a/τ_a+τ_a/σ_a≤2M, the Lorentz identity implies

```
−(x,y) ≥ M(N/9−7).
```

For n≥190, N=n−1≥189 and the right side is at least 14. Both x,y are actual framing walls of the narrow facet, so this contradicts its strict distance bound arccosh 7. Thus the ideal-vertex branch now gives n≤189.

If the chosen narrow facet has only finite vertices, finite-sidedness and finite volume make it compact, and the unchanged compact-facet proof gives n≤143. The two cases therefore yield the stated candidate bound n≤189.

This argument changes the energy accounting only. It does not change the 225 baseline artifacts, the endpoint-existence argument, cusp rank decomposition, positive common null-height ratio, or the proof that the selected ending walls meet the exact narrow-facet framing condition. Its classification and primary-source scope are those documented in the frozen combined candidate and audit. Independent scrutiny of the new contact-cost step is required before using this sharper value as a checked checkpoint.

```
python scripts/fresh_nikulin_energy_charge_20261015.py
python scripts/fresh_nikulin_energy_charge_20261015.py --verify
```

Source 176

Source path: `results/fresh_nikulin_energy_charge_20261015_review.md`

SHA256: 29aa444e864c3158f53011d6045348af9e44b48ba5e1d392750707634a4a08f3

Independent hostile review of the contact-cost bound

PASS. No gap was found in the new charging argument giving the research candidate **$n \leq 189$** for finite-sided finite-volume real hyperbolic Coxeter polytopes, using the explicitly inherited geometric and published inputs. The review makes no publication-priority or current-best claim. The previously frozen 225 proof and review are unchanged.

This note reviews the new argument. Its independent checker and saved certificate use only standard-library `Fraction` arithmetic. They import no producer code or algebra engine. Exceptional energies are calculated by grounding a coordinate in rationally congruent affine Gram matrices, rather than using the producer's regularized inverse. All complete physical Gram entries, retained inverse diagonals, and relevant point energies agree.

Scope of the new proof

The proposed improvement changes only the loss assigned to a non-narrow affine factor. The prior hostile 225 audit supplies the finite/ideal narrow-facet dichotomy, actual-ridge terminating-wall construction, positive common null-height ratio, relative factor scales, and compact-facet bound $n \leq 143$. The classical localization and fixed omission choices are inherited from their bound primary proofs. No geometric hypothesis is silently strengthened, and the new calculation does not re-prove these inputs through a finite matrix census.

Let C be the sum of the two endpoint Schur costs in one factor. Their global sum is at most four, because each norm-two endpoint contributes at most two. Each retained inverse is entrywise nonnegative, and every nonzero norm-two Coxeter contact has magnitude at least one. Thus a retained contact costs at least its inverse diagonal, regardless of how large its label or dotted product is. An omitted contact is either zero or at least one as well; it has not been assumed finite.

The quotient energy is nonnegative. Writing M for the common maximum of the two reciprocal null-height ratios gives the exact identity

```
sigma*tau/M = min(sigma,tau)^2.
```

This identity is unchanged by a separate positive normalization of each affine kernel. Therefore the scale conventions in the exceptional calculations do not introduce a hidden common-scale assumption.

Exceptional factors and the low-cost cases

The independent checker reconstructs A1, E6, E7, E8, F4 and G2, with the fixed omissions from the earlier product proof. For F4 and G2 it clears radical edges by positive diagonal congruence and tracks the diagonal scaling exactly. The actual inverse diagonals, rather than the diagonals of the rescaled matrices, are compared. Positive definiteness of every retained block and nonnegativity of every inverse entry are checked.

The direct A1/E6/F4/G2 factors satisfy both needed inequalities: the zero-contact point energy is at least $2r/9$, and twice the minimum retained inverse diagonal is at least $2r/9$. Nonnegative energy then handles every positive cost. The low B2 and B3 presentations are independently reconstructed; their minimum retained inverse diagonals are 1/2 and 3/4, respectively. They are not incorrectly assigned the generic lower bound one.

For E7 and E8, costs at least one are harmless because $2r/9<2$. If $0<C<1$, the complete inverse lists leave precisely one nonzero retained contact across both rows. It is an isolated A1 entry of inverse diagonal 1/2. Two inexpensive contacts would already cost at least one; any other retained contact would do so alone. Thus, after possibly exchanging endpoints, the rows really are

```
p = a*e_i + b*e_l,    q = c*e_j,
sigma = 2a+b,         tau = 2c,       C=b^2/2,
```

with $b,c\geq 1$ and a either zero or at least one. The zero row is excluded by the inherited positive null-height argument. The independent matrices give delta_i=delta_j=2 and delta_l=1 in all three cheap cases: E7 has two, E8 one.

The grounded calculations independently yield R_ij=1/2, R_lj=1 and R_il=1/2. Hence the interpolation energy is exactly $(1+u^2)/2$, where $u=b/(2a+b)$. If $a=0$, the energy is one and $\min(\text{sigma},\text{tau})\geq 1$, so weighted energy plus twice the cost is at least $1+1=2$. If $a\geq 1$, $\min(\text{sigma},\text{tau})\geq 2$ and energy$\geq 1/2$, so weighted energy alone is at least two. Both exceed the required baseline $2r/9$. The inequalities remain valid for arbitrary allowed contact magnitudes; no angle truncation is present.

Classical factors and the A9 boundary

The all-rank charging comparisons have positive, explicit expressions. For A_r with $2\leq r\leq 8$ the zero-energy margin is $2r(8-r)/(9(r+1))$. For $r=9+t$, $t\geq 0$, the large-rank margin is

```
(t+1)(t+18) / (36(t+10)) > 0.
```

The corresponding B/D and C margins at $r=10+t$ are $37/36+t/36$ and $41/18+5t/18$. The inverse-minimum and energy assumptions here are exactly those in the bound classical proof; no rank sampling is substituted for their closed formulas.

There is one range extension requiring an explicit check: the old product report stated its localization estimates for $r\geq 10$, while the new A argument starts at $r=9$. In A9 the inverse diagonal at distance three from an end is $21/10>2$; all five middle entries exceed two. Thus only the first two and last two retained nodes can be contacted. Opposite deletions in the affine 10-cycle leave two intervening arcs of length one. Their effective conductance is 1/2, which is stronger than the needed $(9-8)/4=1/4$. The new boundary rank is therefore covered directly, rather than inferred from the old stated range.

Global conclusion and limitations

Every non-narrow factor now satisfies weighted energy divided by M at least $2r/9-2C$. Summing its cost losses gives at most eight. The one narrow factor retains the old loss allowance two, even if it itself also uses contact cost. This double allowance is conservative; no cost is omitted from the global upper bound.

Consequently total energy is at least $M(2N/9-10)$. In the Lorentz identity the reciprocal-height term is at most 2M, giving $-(x,y)\geq M(N/9-7)$. At $n\geq 190$, $N=n-1\geq 189$, so this is at least 14. The inherited actual-ridge result puts both selected walls in the framing set of the strict arccosh(7) narrow facet, contradicting that bound for square-two normals. The compact-facet branch remains $n\leq 143$. The two branches give $n\leq 189$ with the same generality as the earlier candidate.

This conclusion does not rely on arithmeticity, simplicity at ideal vertices, a single affine cusp factor, a finite label bound, or a common relative scale for distinct cusp factors. Its published-source and geometric dependencies are explicit. This is an independent research proof audit, not a claim of external peer review or a literature-priority determination.

Generation and saved replay, syntax, source manifests and local links pass.

```
python scripts/fresh_nikulin_energy_charge_20261015_review.py
python scripts/fresh_nikulin_energy_charge_20261015_review.py --verify
```

Source 177

Source path: `results/fresh_nikulin_height_charge_20261015.md`

SHA256: 9265376bded7bedea8573ab2363fdbc54dd62078bac520d83bc5eb170f2475bc

Near-balanced unit charge with a shared defect budget

Research proof candidate, pending independent review: every finite-sided finite-volume real hyperbolic Coxeter polytope has dimension at most **165**. The new step counts the few factors where unit charging can fail. Each such factor consumes at least 3/2 of a shared budget of four, so at most two can occur. This improves the accounting in the frozen, reviewed 166 candidate; it does not alter that package or assert a new uniform local charge coefficient.

The checker and certificate pass generation and saved replay. They import no producer arithmetic. This is producer verification; independent review remains necessary. No publication-priority, current-best, or formal-verification claim is made.

Local variables and the aggregate statement

Use exactly the preset omissions and geometric hypotheses of the reviewed four-thirds proof and common-height proof. An affine factor has rank r, Gram G with root squares two, positive kernel δ normalized by min δ=1, and omissions i,j. Its two contact rows p,q are nonnegative and nonzero; each nonzero entry is at least one. Each endpoint's retained-support cost, summed across all factors, is at most two. Put

$$\sigma = \delta \cdot p, \quad \tau = \delta \cdot q, \quad M = \max(\sigma/\tau, \tau/\sigma), \qquad C = p_{S_i}^T G(S_i)^{-1} p_{S_i} + q_{S_j}^T G(S_j)^{-1} q_{S_j},$$
$$W = \min(\sigma, \tau)^2 E, \qquad T = 2r/9 \quad \text{or} \quad 2(r-1)/9$$

in the unique factor avoiding the narrow-facet node. The ratio M is common to all factors. The same omission choices are used in both height ranges below.

For **M≤31/30**, every factor has W+C≥T, with the following controlled exception: a failing factor must be one of explicitly listed E7/E8 support patterns, must have C≥3/2, and always satisfies

$$W + C \geq T - 2/9. \tag{1}$$

Because the shared total cost is at most four, three failing factors are impossible. Thus, writing N=Σr=n−1 and summing all targets,

$$\sum W \geq \frac{2N}{9} - \frac{2}{9} - 4 - 2\frac{2}{9} = \frac{2N-42}{9}. \tag{2}$$

This does not contradict the exact balanced E8 counterexample to unit charge. That example is one of the expensive exceptions and remains admissible in the local domain.

The height estimate

If σ≥h and τ≥k, then the ratio bound B=31/30 gives

$$\min(\sigma,\tau)^2 \geq H(h,k) := \max\{\min(h,k)^2, \max(h,k)^2/B^2\}. \tag{3}$$

All arguments below permit arbitrary real contact magnitudes with the zero-or-at-least-one gap. No finite-label cutoff is used.

Complete exceptional support coverage

The checker reconstructs the literal affine A1, E6, E7, E8, F4 and G2 Grams, their positive kernels, every retained inverse and all 40 ordinary/avoid-node records. These records exactly match the frozen preset choices.

If C≥T, unit charging is immediate from W≥0. Otherwise every retained nonzero coordinate has inverse diagonal below T. All inverse entries are nonnegative, so setting each retained nonzero coordinate to one gives a rigorous support-cost lower bound, including every same-row cross term. The checker enumerates **every subset** of the cheap-coordinate list. No support-size cutoff is imposed. Precisely 71 singleton and 16 multiple-contact occurrences survive C<T; none has more than two retained coordinates.

For a singleton, orient the rows as p=a e_i+b e_l and q=c e_j, with b,c≥1 and a=0 or a≥1. Let R_uv be the normalized point-potential resistance and put

$$E(u) = (1-u)R_{ij} + uR_{lj} - u(1-u)R_{il}, \qquad 0 \leq u \leq 1.$$

When a=0, the lower bound is H(δ_l,δ_j)R_lj+c_il. When a≥1, it is H(δ_i+δ_l,δ_j) min E+c_il. The convex quadratic minimum is evaluated exactly at its endpoints and any interior stationary point. Both lower bounds meet T in **all 71 occurrences**, proving unit charging for every singleton.

One multiple-contact pattern is also harmless: ordinary E7 with omissions (3,6), retained nodes 4 and 7 on the respective rows. The cost is at least one and

$$E(u,v) = \tfrac{1}{2}(1+u^2+v^2).$$

If both omitted coordinates vanish, u=v=1, so W+C≥5/2. If either omitted coordinate is nonzero, one height is at least three, and (3) gives W+C≥9/(2B²)+1. Both bounds exceed T=14/9.

The remaining 15 occurrences need no energy estimate beyond W≥0:

Record	Support occurrences	Minimum cost	Maximum possible defect T−W−C
Ordinary E7	4	3/2	1/18
Ordinary E8	4	14/9	2/9
Avoid-node E8 with omissions (3,7)	7	3/2	1/18

The seven avoid-node occurrences are the separate prescribed forbidden-node records, not seven simultaneously used narrow factors. The larger uniform allowance 2/9 is safe for all records. Every possible defect consumes at least 3/2 of actual cost, proving the assertion used in (1) and (2). Zero retained contacts satisfy the inherited point-energy bound, checked again in every record. The coincident-omission A1 case has target zero.

All classical A ranks

For affine A_r set m=r+1 and choose opposite cycle nodes 0 and h=⌊m/2⌋. The kernel entries are one. The point resistance and retained inverse entries are

$$R(k) = \frac{k(m-k)}{m}, \qquad J_{kl}^{(i)} = \frac{R(k-i) + R(l-i) - R(k-l)}{2},$$

with arguments reduced modulo m. These are the frozen all-rank cycle identities. An explicit Euclidean realization is v_k=e_1+⋯+e_k−(k/m)(e_1+⋯+e_m), with v_0=0. Its inner products are min(k,l)−kl/m=(R(k)+R(l)−R(k−l))/2; the checker verifies this identity for every used point pair.

For m=2,…,17 the checker lists every retained coordinate with diagonal at most two, enumerates every subset, and retains precisely the supports whose unit-coordinate cost is at most two. Entrywise nonnegative inverses make this a complete necessary list for actual endpoint rows. For each pair with total minimum cost below T, it checks both possibilities—zero or nonzero—for each omitted coordinate.

Let U,V be the resulting full nonempty supports. The normalized endpoint potentials lie in conv{v_k:k∈U} and conv{v_k:k∈V}. Their heights are at least |U| and |V|. For each of **580 cases**, the certificate supplies an exact rational closest-pair dual witness proving E≥e. Combining it with (3) and the complete minimum support cost gives

$$W + C \geq H(|U|, |V|)e + C_{\min} \geq T. \tag{4}$$

The dual witnesses are independently checkable without trusting an optimizer. For the signed Gram H of the two point lists and the two simplex-sum constraints A, the saved nonnegative barycentric vector x and multipliers λ satisfy Ax=(1,1), Hx+A^{T}λ≥0, and complementary slackness. Positive semidefiniteness of H is supplied by the explicit Euclidean point-potential realization. These conditions prove the global quadratic minimum. Singular minimizer sets do not cause a completeness assumption: one checked dual witness suffices for each complete support case. No floating-point solver is used.

For all larger cycles a single analytic inequality replaces the finite calculation. The individual cost budget two excludes every retained node at cyclic distance d≥3, because 3(m−3)/m>2 for m≥10 and the inverse diagonal increases to the antipode. For d≤2, with ε=m mod 2, the frozen linear-loss identity gives

$$c = \frac{d(m-d)}{m}, \quad \ell \leq \frac{(m+\varepsilon)d}{m}, \quad \frac{\ell}{c} \leq \frac{m+\varepsilon}{m-2}.$$

Since C≤4 and W≥E, it follows that

$$W + C \geq R(h) - \frac{4(2+\varepsilon)}{m-2}. \tag{5}$$

For even m≥18 the excess of the right side over T is

$$\frac{m^2 + 6m - 304}{36(m-2)} > 0.$$

For odd m≥19 it is

$$\frac{m^3 + 6m^2 - 457m + 18}{36m(m-2)} > 0.$$

Substitution m=18+2t or 19+2t yields numerators with strictly positive constant terms and nonnegative remaining coefficients for every t≥0. Both symbolic substitutions are checked exactly. This proves all ranks; it is not an extrapolation from the 580 cases. Rotating the same opposite omission pair avoids a prescribed node and proves the stronger ordinary target whenever m≥3. The remaining affine A1 avoid-node case is already immediate.

Classical B, C and D

Use the complete 24 prefix-coordinate classes from the independently reviewed four-thirds proof. Re-evaluate their thresholds at C<T, rather than T/(4/3). All nondistant contacts have $\ell \le c$, so the linear-loss identity directly gives unit charging. The other contacts have cost r/4, r/2, (r−1)/4 or (r−1)/2, which exceeds T=2r/9 or 2(r−1)/9 and is impossible in this range. Every index-class substitution is checked symbolically with independent nonnegative rank and index parameters.

The same point resistances exceed the respective targets. Modified B2 has minimum positive retained cost 1/2>T=2/9, and zero-cost point energy one. Thus every B/C/D rank satisfies W+C≥T without using height balance. The proof inherits the previously reviewed completeness of these coordinate classes; the new strict threshold comparisons are recomputed.

Global split and dimension 165

The reviewed geometric chain supplies actual framing walls for a Nikulin narrow facet, one common M across the affine product, total cost at most four, the unique avoid-node loss 2/9, and the strict norm-two pairing bound −(x,y)<14. These are inherited hypotheses, not consequences of the local support calculations.

In the range M≤31/30, equation (2) and the unchanged Lorentz identity give

$$-(x, y) \ge M\frac{N-39}{9}. \tag{6}$$

For M≥31/30, retain the frozen four-thirds estimate with exactly the same preset omissions:

$$-(x, y) \ge M\frac{N-43}{9}. \tag{7}$$

At N=165+t, t≥0, (6) is at least 14+t/9, while (7) is at least

$$\frac{31}{30}\frac{122+t}{9} = 14 + \frac{1}{135} + \frac{31t}{270} > 14.$$

Both contradict the strict narrow-facet bound. Therefore the ideal-vertex branch excludes n≥166 and gives n≤165. The separately reviewed compact-facet branch still gives n≤143. Their combination is the stated candidate n≤165, with no arithmetic assumption and no restriction to simple cusps.

The improvement comes from an integer count of expensive exceptional factors. A smaller uniform local coefficient alone could not give it: the exact balanced E8 obstruction requires at least 8/7, whose scalar threshold still rounds to 166. The present argument neither assumes that 8/7 is a valid uniform upper coefficient nor changes any frozen omission rule.

```
python scripts/fresh_nikulin_height_charge_20261015.py
python scripts/fresh_nikulin_height_charge_20261015.py --verify
```

Generation, saved replay, direct source manifests, syntax and local links pass. The full unbounded geometry remains the explicitly inherited reviewed chain. This new aggregate proof and its complete finite exceptional checks are ready for independent scrutiny; prior 166 and earlier artifacts remain unchanged.

Source 178

Source path: `results/fresh_nikulin_height_charge_20261015_dimension165.md`

SHA256: a6e5217a3b6ea2fe4c0a75e62193a79f6b4c553e4d5df8bbfe9899c224538793

Reviewed research proof candidate: dimension at most 165

Candidate theorem. Every finite-sided, finite-volume real hyperbolic Coxeter polytope has dimension **$n \le 165$**, without arithmeticity, compactness, a simple-cusp hypothesis or a facet-count restriction.

The aggregate proof and its independent review improve the frozen 166 candidate. The case binder and certificate check the complete branch split, numerical threshold and recursive source hashes. This is an internally reviewed research proof candidate. External validation, novelty and current-best literature status remain unestablished. The primary report's historical pending-review wording remains unchanged.

Nikulin's narrow-facet theorem supplies a facet F whose framing hyperplanes have pairwise distance strictly below arccosh(7). If F has an ideal vertex, the reviewed geometric chain supplies affine factors of total rank $N=n-1$, actual framing walls x,y, one common height ratio $M \ge 1$, and total retained Schur cost at most four. These are inherited geometric hypotheses; the new finite calculations do not establish them. See the frozen geometric case binder and Nikulin's official ICM report, printed pp.654–655 and 657–659.

For $M \le 31/30$, unit charging fails only in listed E7/E8 support patterns. Each failure consumes cost at least 3/2 and loses at most 2/9. Since the total cost is at most four, at most two factors can fail. The aggregate loss is therefore at most 40/9. Including the unchanged avoid-node loss and Lorentz identity gives $-(x,y) \ge M(N-39)/9$. For $M \ge 31/30$ the inherited four-thirds estimate gives $-(x,y) \ge M(N-43)/9$, with the same preset omissions. At $N=165$ these lower bounds are respectively at least 14 and $14+1/135$; both increase thereafter. They contradict the strict narrow-facet bound $-(x,y)<14$ for normals of squared length two. The ideal-vertex branch thus gives $n \le 165$. The scalar estimate at $N=164$ is insufficient to exclude that case; no bound164 follows from this package.

If F has only finite vertices it is compact. The independently reviewed compact-facet argument gives $n \le 143$. These alternatives exhaust F, completing the bound165 case split. The arithmetic ceiling29 and the noncompact arithmetic ceiling21 are separate, smaller published bounds; the new unrestricted result does not lower them.

The independent checker uses standalone Fraction arithmetic, with no producer imports or numerical optimizer. It reconstructs all40 exceptional records,71 single-contact and16 multiple-contact occurrences, all580 complete-support convex-dual certificates,24 classical B/C/D classes and two symbolic A-family tails. The proof covers arbitrary real contact magnitudes satisfying the Coxeter gap and every affine rank. Published classification and geometry inputs remain mathematical dependencies.

Replay: `python3 scripts/fresh_nikulin_height_charge_20261015_dimension165.py --verify`.

Source 179

Source path: `results/fresh_nikulin_height_charge_20261015_review.md`

SHA256: 05d9dd9e050ac0bb6f5896bac1c47c00d796cd35ea3b82ff16ca9e0c9ba590a3

Independent review of the height-charge dimension-165 candidate

The new local estimates and their global accounting pass this independent review. No mathematical discrepancy was found. The resulting research proof candidate gives dimension at most 165 for finite-sided, finite-volume real hyperbolic Coxeter polytopes, conditional only on the explicitly inherited geometric chain already reviewed in the preceding packages. This is an internal mathematical and exact-arithmetic review, not a formal verification, publication-priority claim, or independent reproduction of all historical geometric inputs.

The primary proof, independent checker, and review certificate are separate artifacts. The checker uses standalone `Fraction` arithmetic, grounded matrix inverses, rational convex-dual verification, and polynomial coefficient arithmetic. It imports **no project or producer code, SymPy, optimizer, or floating-point solver**. It binds the final primary PY/JSON, not its report.

Exceptional factors: exhaustive support coverage

The checker independently builds the literal affine A1/E6/E7/E8/F4/G2 matrices. For the last two it uses rationally scaled root bases; the physical norm-two Gram is checked entry by entry through signs and squares. Grounding one node gives a rational principal inverse, from which the positive kernel, normalized kernel squares, and point resistances are recovered. Every principal deletion is positive definite, and every retained inverse is entrywise nonnegative. The ordinary and forbidden-node omission rules are reconstructed explicitly, rather than copied from the new record list.

All 40 records agree with the frozen preset choices. The zero-retained-contact case is checked in each record, including the coincident-omission A1 case, whose target is zero.

Write $T = 2r/9$, or $2(r-1)/9$ in the single forbidden-node factor. If actual total cost $C \geq T$, the unit-charge conclusion follows from $W \geq 0$. Otherwise each retained nonzero coordinate has inverse diagonal strictly below T. Because every nonzero contact is at least 1 and all inverse entries are nonnegative, replacing the contacts in a fixed support by 1 is a valid cost lower bound, including all same-row cross terms. The independent checker enumerates **every subset** of that full cheap-coordinate set. There is no support-size or finite-angle cutoff.

This reconstructs exactly 71 singleton and 16 multiple-contact occurrences; none of the latter has more than two retained coordinates. For each singleton, the resistance quadratic is reconstructed independently and minimized at the endpoints and its stationary point. With $B = 31/30$, the bound

$$\min(\sigma,\tau)^2 \geq \max\left\{\min(h,k)^2, \frac{\max(h,k)^2}{B^2}\right\}$$

is valid regardless of which endpoint has the larger actual height. Separate zero/nonzero omitted-coordinate cases give the claimed unit charge in all 71 occurrences. Multiplying a separately valid energy lower bound by a separately valid nonnegative height lower bound is safe; no unproved correlation is required.

The sole inexpensive multiple support is the ordinary E7 pattern with one contact at node 4 of the first row and node 7 of the second. Polarization of the grounded resistance data gives exactly

$$E(u,v) = \tfrac{1}{2}(1+u^2+v^2), \qquad C \geq 1.$$

If both omitted coordinates vanish, $u = v = 1$, so $W + C \geq 5/2$. Otherwise one height is at least 3; the common ratio bound gives $W + C \geq 9/(2B^2) + 1$. Both exceed 14/9.

The other 15 occurrences have cost at least 3/2 and possible deficit at most 2/9, just using $W \geq 0$. More precisely, the four ordinary E7 occurrences have cost 3/2 and deficit at most 1/18; the four ordinary E8 occurrences have cost 14/9 and deficit at most 2/9; the seven forbidden-node E8 records have cost 3/2 and deficit at most 1/18. The seven records represent alternative forbidden-node choices, not seven factors simultaneously present. The review does not assume every possible defect is attained; these are safe upper bounds on actual failure.

Small A cycles: complete domains and exact dual certificates

For each cycle size $m = 2, \ldots, 17$, a separately computed inverse of the grounded A-type path reconstructs all point inner products:

$$K_{ij} = \min(i, j) - ij/m.$$

These agree with the explicit Euclidean vectors $v_k = e_1 + \cdots + e_k - (k/m)\sum_{a=1}^{m} e_a$. Their squared distances are $R(i - j)$. Thus the signed Gram used for the two point lists is positive semidefinite, including singular and coincident-potential cases.

The independent checker lists all retained nodes whose individual cost is at most 2, enumerates all their subsets, and computes every same-row cross term. It then rebuilds both endpoint domains, all domain pairs with total minimum cost below T, and both zero/nonzero possibilities for each omitted coordinate. The complete list has **580 cases**, agreeing exactly with the primary. Actual contact coefficients may be arbitrary real numbers at least 1; the support enumeration therefore covers more than a finite label set.

For every case it verifies the saved rational barycentric vector x and multipliers λ:

$$x \geq 0, \quad Ax = (1, 1), \quad s = Hx + A^T\lambda \geq 0, \quad x_i s_i = 0.$$

It also checks $x^T Hx = -\sum \lambda_i$. For any other feasible z,

$$z^T Hz - x^T Hx = (z - x)^T H(z - x) + 2z^T s \geq 0.$$

This independently establishes the global convex minimum; no trust in the producer's active-set optimizer or its treatment of degenerate minimizers is needed. One verified dual witness suffices for each completely enumerated support case. Allowing zero barycentric coordinates relaxes the actual positive-contact domain, which is valid for a lower bound.

The actual heights are at least the two full support sizes. Combining the height estimate with the verified convex energy and support-cost lower bounds proves unit charge in every case.

Unbounded classical ranks

For a large A cycle, a retained coordinate at cyclic distance at least 3 costs at least $3(m - 3)/m > 2$, and is impossible under the individual endpoint budget. The inverse diagonal increases up to the antipode, so this excludes all larger distances, not merely distance 3. At distance $d \leq 2$, the two directions have linear losses $d(m \pm \varepsilon)/m$, where $\varepsilon = m \bmod 2$, while the cost is $d(m - d)/m$. Therefore the loss-to-cost ratio is at most $(m + \varepsilon)/(m - 2)$.

The frozen nonnegative-quadratic-remainder identity gives the energy lower bound after these linear losses. Since both heights are at least 1, $W \geq E$; since $C \leq 4$,

$$W + C \geq R(\lfloor m/2 \rfloor) - \frac{4(2 + \varepsilon)}{m - 2}.$$

The independent checker reconstructs the cleared excess numerators

$$m^2 + 6m - 304 \quad (m \text{ even}), \qquad m^3 + 6m^2 - 457m + 18 \quad (m \text{ odd}).$$

Substitution $m = 18 + 2t$ or $19 + 2t$ gives strictly positive constant terms and nonnegative remaining coefficients, with positive denominators. This proves every larger rank. It also verifies the cross-multiplied identities with the point resistance and target, independently of symbolic parsing. Rotation preserves the ordinary opposite-omission estimate and avoids any prescribed node when $m \geq 3$. The remaining A1 avoid-node case has target zero.

For B/C/D the review independently re-derives all 24 prefix-coordinate classes using rational interval segments and affine polynomials in nonnegative rank and node-index parameters. It verifies the point-energy surplus, and in each class either the linear loss is at most its inverse diagonal or that diagonal strictly exceeds T. The latter excludes the contact in the relevant range $C < T$. Modified B2 is checked separately by an exact grounded matrix. The geometric realization, complete class partition, and omission symmetries are the same explicitly inherited ones as in the independently reviewed four-thirds proof. No new rank cutoff or omitted classical family is introduced here.

Shared budget and common-height split

Define an actual failing factor by $T - W - C > 0$. The complete local analysis shows that it consumes actual cost at least $3/2$, and loses at most $2/9$. Since the two endpoint budgets give $\sum C \le 4$, three failing factors would cost at least $9/2 > 4$. Thus at most two factors can fail. This counts factors, not the number of candidate support records.

With $N = n - 1 = \sum r$ and the single avoid-node target penalty $2/9$,

$$\sum W \ge \frac{2N}{9} - \frac{2}{9} - 4 - \frac{4}{9} = \frac{2N - 42}{9}.$$

The unchanged Lorentz identity consequently gives $-(x, y) \ge M(N - 39)/9$ for $1 \le M \le 31/30$. In the other range the reviewed four-thirds estimate, with the **same preset omissions**, gives $-(x, y) \ge M(N-43)/9$. At $N = 165+t$, the respective lower bounds are at least

$$14 + t/9, \qquad 14 + 1/135 + 31t/270.$$

Both contradict the strict narrow-facet pairing bound $-(x, y) < 14$. The equality endpoint of the near range is handled by that strict bound. The ideal-vertex branch therefore gives $n \le 165$; the inherited compact facet branch gives $n \le 143$.

This step genuinely uses the shared cost budget and common height ratio. It does not assert a uniform unit-charge lemma, which is false, or assume that the balanced counterexamples disappear. The same omission choices in both ranges avoid any circular dependence on the subsequently found walls.

Dependencies and final status

The new checker and this review establish the changed local inequalities, complete support coverage, exact dual certificates, all-rank algebra, and global numerical accounting. The following remain explicit previously reviewed geometric inputs: Nikulin's narrow facet and its strict constant; the compact/ideal facet dichotomy; actual terminating walls and framing ridges; nonzero contact rows with the Coxeter gap; a common height ratio across all affine factors; the two endpoint Schur budgets; and the compact-facet 143 branch. They are bound through the frozen 169 and 166 chains. The present review does not silently replace those geometric premises by a finite computation.

Independent generation and saved replay pass. Source hashes, syntax, and local links pass. No primary changes were requested.

```
python3 scripts/fresh_nikulin_height_charge_20261015_review.py
python3 scripts/fresh_nikulin_height_charge_20261015_review.py --verify
```

Source 180

Source path: `results/fresh_nikulin_narrow_20261015.md`

SHA256: 52ad5cca34bee1f87bdc61483feaa49d208bcd8d798a4a9fb442a4b04f800751

Explicit separation at classical irreducible cusps of a narrow facet

Conditional geometric bounds, with all rank parameters explicit. Let (P⊂\mathbb H^n) be a finite-sided finite-volume Coxeter polytope. Suppose a Nikulin narrow facet (F) contains an ideal vertex whose **full** cusp diagram is one irreducible affine classical diagram. Then the following bounds hold:

Full cusp diagram	Bound on ambient dimension (n)
(\widetilde A_{n-1})	(n≤ 135)
(\widetilde B_{n-1})	(n≤ 51)
(\widetilde C_{n-1})	(n≤ 47)
(\widetilde D_{n-1})	(n≤ 55)

The proof supplies explicit parts of the missing narrow-facet argument. It does **not** prove a new unrestricted dimension bound, the full historical assertion (n<300), or a bound for every simple finite-volume polytope. Products of affine components and a chosen narrow facet having only finite vertices remain untreated. No novelty or strongest-known-bound claim is made.

The exact checker and certificate verify closed inverse formulas, Lorentz-coordinate identities, energy inequalities and the boundary indexing with `Fraction` arithmetic. The proofs below, rather than a finite rank cutoff, establish the all-rank statements.

Published input and geometric conventions

Nikulin's ICM report, printed p.655, gives a facet (F) such that the hyperplanes of any two facets whose hyperplanes meet (H(F)) have distance less than (arccosh7). Printed pp.657–659 describe terminating walls on cusp edges and illustrate affine (A), obtaining (n<140). We use the narrow-facet theorem and reconstruct the separation calculation independently. The frozen source audit records the exact primary provenance and the incomplete status of the full 300 reconstruction.

Use Lorentz signature ((n,1)), outward facet normals of square 2, and chamber inequalities ((h,α)<0). Acuteness gives ((α,β)≤ 0) for different walls. Every nonzero negative pairing has magnitude at least 1: intersecting walls give (2\cos(π/m)), (m≥ 3), while parallel or disjoint walls give at least 2. Thus no finite-label cutoff is needed.

At the assumed cusp there are (n) incident walls, with normals (α0,\ldots,α{n-1}). Deleting node (i) gives a positive-definite rank-((n-1)) edge support (S_i). Its geometric edge has the original cusp as one endpoint. Choose any additional facet (x_i) at its other endpoint. Finite volume ensures that endpoint is finite or ideal, not beyond the hyperbolic closure. Consequently the Gram matrix on (S_i∪{x_i}) is positive semidefinite. This remains true at a nonsimple ideal endpoint; no simplicity assumption is imposed there.

1. Schur localization, without endpoint-diagram enumeration

Write (J=G(S_i)^{-1}) and (p_j=-(x_i,α_j)≥ 0). Positive semidefiniteness gives

$$p_{S_i}^T J p_{S_i} \le 2.$$

All entries of the inverses below are nonnegative. Thus (J_{jj}>2) forces (p_j=0), because a nonzero Coxeter contact has (p_j≥ 1).

The needed inverse formulas are elementary. For (A_r), in path order (a,b=0,\ldots,r-1),

$$J_{ab} = \frac{(\min(a,b)+1)(r-\max(a,b))}{r+1}.$$

For (B_r), number the short end 0 and the remaining path (1,\ldots,r-1), normalizing **all** roots to square 2:

$$J_{00} = r/2, \quad J_{0j} = (r-j)/\sqrt{2}, \quad J_{ij} = r - \max(i,j) \quad (i,j \ge 1).$$

For (D_r), use fork leaves 0,1, attached to 2, followed by the path to (r-1):

$$J_{00} = J_{11} = r/4, \quad J_{01} = (r-2)/4, \quad J_{0j} = J_{1j} = (r-j)/2 \ (j \ge 2), \quad J_{ij} = r - \max(i,j) \ (i,j \ge 2).$$

Multiplication by their sparse Gram matrices proves these identities for every rank; all entries are positive. In (A_{n-1}), every diagonal except those within two positions of an end exceeds 2 when (n≥ 10). In (B_r), the same assertion holds for (r≥ 5); in (D_r), it holds for (r≥ 9). The two allowed end diagonals in (B,D) are 2 and 1.

It follows that, when the retained components meet those rank thresholds, the full contact row of (x_i) is supported only on

$$\{i-2, i-1, i, i+1, i+2\}.$$

For affine (A) the indices are cyclic. For the other families they are interior path indices. The contact at the deleted node (i) itself is unrestricted except by acuteness and the Coxeter gap; it has **not** been assumed finite.

2. A Lorentz row identity and its uniform lower bound

Let (c=∑δ_jα_j) be a future null cusp vector, with positive affine coefficients. Put

$$p_j = -(x,\alpha_j), \quad q_j = -(y,\alpha_j), \quad \sigma = \sum \delta_j p_j = -(x,c), \quad \tau = \sum \delta_j q_j = -(y,c).$$

Both (σ,τ>0). A zero contact row would make a norm-two vector orthogonal to the full cusp span, whose orthogonal complement is the null line (\mathbb Rc), an impossibility. Let (E) be the squared Euclidean norm of the image of (x/σ-y/τ) in (c^\perp/\mathbb Rc). Directly expanding its square gives the exact identity

$$\boxed{-(x,y) = \frac{\sigma\tau}{2}E - \frac{\sigma}{\tau} - \frac{\tau}{\sigma}.}$$

If all possible contact positions have the same affine coefficient (d), then the Coxeter gap gives (σ,τ≥d). When (E≥ 4/d^2), the identity implies

$$-(x,y) \ge \frac{d^2 E}{2} - 2.$$

Indeed, (1/σ^{2+1/τ}2≤ 2/d^2) and (στ≥d^2). This removes any need for an upper bound on the unknown deleted-node contacts, or a limiting argument involving those contacts.

3. The path energy for affine (B,C,D)

Write (m=n-1). In a null basis (c,d_0) with ((c,d_0)=-1), together with orthonormal spatial coordinates (e_1,\ldots,e_m), one may use:

- (\widetilde C_m): (α_0=$\sqrt{2}$(c-e_1)), (α*j=e_j-e*{j+1}) for (1≤j<m), (α_m=$\sqrt{2}$ e_m). The affine coefficients are (1/$\sqrt{2}$,1,\ldots,1,1/$\sqrt{2}$).
- (\widetilde B_m): (α_0=c-e_1-e_2), (α_1=e_1-e_2), (α*j=e_j-e*{j+1}) for (2≤j<m), (α_m=$\sqrt{2}$ e_m). The coefficients are (1,1,2,\ldots,2,$\sqrt{2}$).
- (\widetilde D_m): the preceding first roots and interior chain, with final fork roots (α{m-1}=e*{m-1}-e_m), (αm=e*{m-1}+e_m). The coefficients are (1,1,2,\ldots,2,1,1).

Deleting an interior node (i) leaves respectively

$$B_i + B_{m-i}, \qquad D_i + B_{m-i}, \qquad D_i + D_{m-i}.$$

After localization, all nonzero contacts are at interior nodes with common coefficient (d=1,2,2), respectively. The spatial coordinates of a normal with row (p) are, along the separating chain,

$$z_k = -\sum_{\ell=k}^{m} p_\ell.$$

For two rows supported within two nodes of (i<j), every coordinate indexed from (i+3) through (j-2) has normalized difference exactly (1/d) in absolute value. One row has no remaining mass there and the other has total contact mass (τ/d). Therefore

$$E \geq \frac{j-i-4}{d^2}, \qquad -(x_i, x_j) \geq \frac{j-i-4}{2} - 2 \quad \text{if } j-i-4 \geq 4.$$

Let (s) be the node of the narrow facet. Initially choose the deleted nodes as follows:

Family	Initial (i,j)	Component rank requirements
(C_m)	(5,m-5)	both (B) ranks at least 5
(B_m)	(9,m-5)	(D) rank at least 9, (B) rank at least 5
(D_m)	(9,m-9)	both (D) ranks at least 9

If (s=i), increase (i) by one; if (s=j), decrease (j) by one. At the excluded ranks these alternatives are disjoint, so at most one move occurs. All retained-component thresholds remain true. The resulting negative pairings are at least

$$\frac{m-19}{2}, \qquad \frac{m-23}{2}, \qquad \frac{m-27}{2},$$

respectively. They reach 14 at (m=47,51,55).

4. The cycle energy for affine (A)

The affine (A_{n-1}) Gram matrix is the cycle Laplacian; all null coefficients equal 1. The normalized row difference has total charge zero, with positive mass 1 in the five-node interval about (i) and negative mass 1 in the interval about (j).

The Euclidean energy (E) is the electrical flow energy on this cycle. This follows either by solving the root-coordinate differences or from the Laplacian pseudoinverse. On the two long arcs separating the contact intervals, conservation makes the flow constant, with respective values (u) and (u-1). If their lengths are (L_1,L_2), all remaining edge energies are nonnegative, and

$$E \geq L_1 u^2 + L_2 (u-1)^2 \geq \frac{L_1 L_2}{L_1 + L_2}.$$

The last inequality is the identity obtained by subtracting the right side and leaving (((L_1+L_2)u-L_2)^2/(L_1+L_2)). It covers arbitrary nonnegative contact distributions, not just single-node rows.

Choose opposite deleted nodes, avoiding (s) by rotating both one step if needed. Then

$$L_1 = \lfloor n/2 \rfloor - 4, \qquad L_2 = \lceil n/2 \rceil - 4.$$

For (n≥ 136), both lengths are at least 64, so (E≥ 32) and (-(x_i,x_j)≥ 14). This recovers the linear separation mechanism of Nikulin's worked affine-(A) example with an explicit boundary.

5. Why these are genuinely framing walls of the narrow facet

This incidence bridge is essential. Avoiding (s) means that the narrow-facet root belongs to both retained edge supports. Its inverse diagonal is at least 1 in the path cases, and at least ((n-1)/n>1/2) in the cycle case. The Schur bound therefore gives

$$|(x_i, \alpha_s)| < 2.$$

Thus their rank-two Gram is positive definite; the two hyperplanes meet **inside** hyperbolic space. More strongly, they share an actual ridge of (P). To see this, project a strict interior point (h) onto their common orthogonal complement. The inverse Gram is entrywise nonnegative, while the two pairings with (h) are negative. Consequently the projection has negative square and its pairing with every other wall remains strictly negative by acuteness. It is a point of the relative interior of that ridge. This argument does not rely on the terminating edge endpoint being finite.

Both terminating walls therefore belong to exactly the set controlled by the narrow-facet theorem. If their pairing is at most (-14), their hyperplane distance is at least (arccosh7), contradicting the published strict upper bound. The four dimension bounds follow.

Scope and the remaining problem

This proof closes the full-rank irreducible classical cusp calculation. The exceptional irreducible affine types have bounded rank at most 9; thus, using the standard affine classification, any **irreducible** cusp on the chosen narrow facet has ambient dimension at most 135. This is a conditional statement about that particular narrow facet. A noncompact polytope may have a narrow facet with only finite vertices, and a nonsimple ideal vertex has a product cusp diagram. Neither case is eliminated here.

For context, Felikson–Tumarkin's primary theorem already gives (n≤ 8) when the entire polytope is both simple and ideal. Their ideal hypothesis means every vertex is ideal, and is substantially stronger than the local hypothesis used here. The present bounds are proof-reconstruction lemmas, not a claim to improve that theorem or a verified new global bound for simple finite-volume polytopes.

For a product cusp the affine null vectors of the different factors are proportional but have independent relative scales. The row identity survives after choosing one common null vector. The localization and energy arguments above cannot simply be summed: an outgoing edge omits a node in **each** affine factor, and a terminating wall may distribute its contacts across those factors. Controlling that distribution and the narrow-facet incidence uniformly is the next unresolved problem. No claim that terminating walls contact only one component is used or justified here.

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python scripts/fresh_nikulin_narrow_20261015.py --verify
```

Source 181

Source path: `results/fresh_nikulin_narrow_20261015_compact_facet.md`

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A coarse dimension bound from a compact facet

Theorem. Let P be a finite-sided, finite-volume hyperbolic Coxeter polytope in H^n. If P has a compact facet, then **n≤143**. The facet need not itself be a Coxeter polytope. No narrowness assumption is required for this lemma.

The proof uses a coarser version of Vinberg's graph-distance weighting. It allows parabolic subdiagrams in the ambient face stars. This report establishes the compact-facet branch separately; any combination with the conditional narrow-facet product-cusp bound requires review of both proofs and their common hypotheses.

The small exact checker and certificate verify the final scalar inequalities, the h-vector coefficient minima, classical distance-count examples, and an explicit counterexample to the stronger localization premise discussed below. The geometric argument and the stated classification inputs are not replaced by a numerical test.

Published inputs and the localization issue

Prokhorov defines quasi-Lannér diagrams by minimal indefiniteness and records both the maximal order ten and extraction from any hyperbolic diagram in §2, Propositions 1–2, printed pp.407–408. These statements supply the classification input here. Primary paper.

Vinberg's §4 Proposition 10, printed p.77, bounds the number of unordered node pairs at graph distance at most c in a spherical diagram of order r by cr. His §1, pp.70–71, provides the symmetric nonnegative h-vector estimate for a simple polytope. Primary paper.

The finer compact-chamber weights cannot simply be imported: Vinberg's §3.2 and Proposition 9 use absence of parabolic subdiagrams, and §6.4 invokes that fact. Compactness of one marked face does not imply this absence. The Coxeter tetrahedron with path labels 3–3–6 has a compact triangular facet at node 0 but an affine G2 subdiagram on nodes 1,2,3. The checker verifies determinant $-1/16$, dual-ray squares $(0,-4,-12,-8)$, a strict timelike center, and all three spherical corners of facet 0. This diagnoses a gap in the finer transplant; it does not contradict Vinberg's compact theorem or the coarser proof below.

Minimal indefinite diagrams

All Gram matrices below have diagonal one and nonpositive off-diagonal entries. A minimal indefinite principal subdiagram L is connected: otherwise its indefinite component is a smaller such subdiagram. Every proper principal submatrix is positive semidefinite.

If every proper subdiagram is positive definite, L is Lannér. Otherwise choose a connected affine component A of a proper positive-semidefinite subdiagram. Its null vector has strictly positive coefficients. Connectedness of L supplies an outside node v adjacent to A. The Gram of A+v is already indefinite: evaluating on the positive null vector plus a sufficiently small positive multiple of v gives a negative value. Minimality therefore forces L=A+v. In particular the affine component has codimension one; there can be no additional disconnected affine or spherical part. Its nonzero coupling to the affine null vector makes L nonsingular with inertia $(|L|-1,1)$.

Consequently the general minimal-indefinite situation has exactly the Lannér/quasi-Lannér scope of the cited classification. This also follows directly from Prokhorov's Proposition 2. It gives $|L|\le 10$; the Lannér case has the stronger bound five,

which is not needed. A strict dotted edge, if present in a minimal indefinite subdiagram, already constitutes a two-node indefinite diagram and causes no exception to the size bound.

Compact two-face stars

Let Q be a compact two-dimensional face of P. Let W be the n−2 facet normals whose facets contain Q, and mark the other facets that cut its polygonal sides black. All corners of Q are finite. Therefore W together with each pair of consecutive black nodes is a positive-definite diagram of order n.

The whole star has a negative direction. One way to see this is to project its black normals off the positive subspace spanned by W: their hyperplanes bound the compact polygon Q in a Lorentzian three-dimensional space. The polygon side normals span that space. Every star subdiagram consequently has negative index at most one, as does the ambient Gram matrix.

Two elementary incidence facts will be used. A positive-definite set of wall normals of an acute chamber has an actual common face: project a strict interior timelike point orthogonally off their span. The inverse of a positive-definite Z-matrix is entrywise nonnegative, so all other wall inequalities remain strict and the projected vector stays timelike. If instead a principal block is positive semidefinite and singular, one connected affine component supplies a positive null combination c. This combination is nonzero since its pairing with a strict interior point is negative. It is future null, orthogonal to every normal in that positive-semidefinite block, and nonpositive against every other chamber normal. It is therefore an ideal point of the chamber. These facts do not require the entire chamber to be compact.

For a triangle, any indefinite subdiagram of the star must contain all three black nodes: omitting a black node puts it inside a positive-definite corner diagram. Choose a minimal indefinite subdiagram L. It has at most ten nodes. In a connected graph with three marked nodes, at least two marked nodes can each be deleted while the other two remain joined by a path. For example, use the leaves of the minimal subtree joining the marks in a spanning tree. The resulting two paths lie in L with one black node removed, hence have length at most eight. They lie in two actual corner diagrams of Q.

For a quadrilateral, split the black nodes into its two pairs U,V of opposite sides. The full principal block W+U is indefinite. If it were positive definite, the actual-face projection argument would make the opposite sides intersect in Q. If it were singular positive semidefinite, the null-vector argument would produce an ideal point on every white wall, hence in Q, contradicting compactness. The same reasoning applies to W+V. This is the required replacement for the global no-parabolic assertion.

Choose minimal indefinite subdiagrams L⊆W+U and M⊆W+V. Each must contain its two black nodes, since W plus either single black node is positive definite. Both have at most ten nodes and at most eight white nodes. They cannot be orthogonal: their two negative directions would give ambient negative index at least two. Thus they overlap in a white node or are joined by an edge. A shortest path between the black sets U and V in this connected union has only white internal nodes. It has length at most 8+1+8=17. Its two black endpoints are consecutive sides of Q, so the path is contained in their actual corner diagram.

Assign a planar angle of a compact two-face weight one if its two marked nodes have graph distance at most 17 in the full ambient finite-vertex diagram, and zero otherwise. The triangle argument gives total weight at least two; the quadrilateral argument gives at least one. Polygons with k≥5 need only nonnegative total weight. In all cases the total is at least 5−k.

Counting on a compact facet

Let F be a compact facet, d=n−1. It is a compact simple d-polytope because all of its vertices are finite vertices of the acute chamber. Its two-faces are also two-faces of P, so the preceding weights apply. At each vertex of F these weights count a subset of the unordered pairs in the full ambient spherical diagram, whose order is n. The published spherical distance bound therefore gives a vertex subtotal at most 17n=17(d+1). No reflection structure on F is being asserted.

Write f_i for the face numbers of F. Summing the two-face lower bounds gives

```
total weight ≥ 5 f_2 − binom(d,2) f_0.
```

For a simple polytope its nonnegative h-vector is symmetric with h_0=h_d=1. Pairing the coefficients binom(k,2) and binom(d−k,2) gives the strict bounds

```
f_2/f_0 > d(d−2)/8          if d is even,
```

```
f_2/f_0 > (d-1)^2/8        if d is odd.
```

Strictness follows from the nonzero endpoint coefficients, which exceed the middle coefficient minimum. It follows that the average vertex subtotal is strictly greater than d(d−6)/8 in even dimension, and (d−1)(d−5)/8 in odd dimension. Comparing with 17(d+1) gives a contradiction whenever d≥143. Explicitly, writing the odd dimensions as d=143+2t and the even ones as d=144+2t, eight times the difference of lower and upper bounds is respectively

```
4t²+288t+12 > 0,
4t²+292t+152 > 0.
```

Thus d≤142 and n≤143, as claimed.

The proof is deliberately coarse and makes no current-best or novelty claim. Its useful feature is that it permits the affine subdiagrams that obstruct the direct use of the finer compact-chamber weighting. The checker imports no earlier producer arithmetic. Generation and saved replay are run with:

```
python scripts/fresh_nikulin_narrow_20261015_compact_facet.py
python scripts/fresh_nikulin_narrow_20261015_compact_facet.py --verify
```

Source 182

Source path: `results/fresh_nikulin_narrow_20261015_products.md`

SHA256: 51769466dc2913e43c21cf16908d9906e4c1d3a396189bbeff8446e724edcc9d

A product-cusp separation bound for a narrow facet

Conditional theorem. If a Nikulin narrow facet of a finite-sided finite-volume real hyperbolic Coxeter polytope (P⊂\mathbb H^n) contains an ideal vertex, then (n≤ 225).

This treats products of affine cusp components, including arbitrarily many factors of small rank. It is a further explicit step in reconstructing the historical narrow-facet argument. It does **not** yet give an unrestricted bound of 225: a chosen narrow facet may have only finite vertices. The latter case, which Nikulin's outline proposes treating combinatorially, is still open in this reconstruction. No novelty or current-best claim is made.

The checker and certificate verify the finite exceptional constants and the all-rank scalar identities. The argument uses the inverse and coordinate formulas proved in the separate classical-cusp package, whose files remain frozen. The published narrow-facet input is Nikulin, ICM 1986, printed p.655.

Setup and the common height ratio

Use outward normals of square 2 in Lorentz signature ((n,1)). At an ideal vertex (v), a horospherical link is a compact Euclidean Coxeter polytope, hence a product of affine Coxeter simplices. Let their ranks be (r_a≥ 1), with

$$N := n-1 = \sum_a r_a.$$

Every factor has (r_a+1) nodes. Deleting one node in each factor specifies a vertex of the horospherical link, hence an outgoing edge of (P). Its retained support (S) has (N) roots and positive-definite block-diagonal Gram matrix. The standard Euclidean reflection classification, including the parallel two-wall affine (A_1) case, is the only diagram classification imported here.

Choose two such deletion tuples as described below, and let (x,y) be terminating walls at the other endpoints of their edges. The next section ensures these can be chosen to meet the narrow facet in actual ridges. Their contact rows against the initial cusp roots are nonnegative after negating pairings. In a factor (a), write these rows as (p_a,q_a), its positive null coefficients as (δ_a), and

$$c_a = \sum_j \delta_{a,j}\alpha_{a,j}, \quad \sigma_a = \delta_a \cdot p_a = -(x,c_a), \quad \tau_a = \delta_a \cdot q_a = -(y,c_a).$$

All (c_a) are positive multiples of the same future null vector. Therefore the ratio

$$q := \sigma_a/\tau_a > 0$$

is **the same in every factor**, although their individual scales need not agree. Put (M=\max(q,1/q)≥ 1).

The cusp quotient is an orthogonal direct sum of the Euclidean factor spaces. Let (E_a) be the squared norm of the factor component of (x/σ_a-y/τ_a). The Lorentz identity from the classical package becomes

$$\boxed{-(x,y) = \frac{1}{2}\sum_a \sigma_a \tau_a E_a - q - 1/q.}$$

The relative null scales cancel in each summand. This is why the proof does not require commensurable cusp scales, arithmeticity, or a prescribed horosphere.

Choosing terminating walls that really meet the narrow facet

Let the narrow facet have normal (α_s). In its affine factor, both deletion tuples avoid (s), so both outgoing edges lie in the narrow facet. At a finite second endpoint, every additional wall has finite angle with (α_s).

At an ideal second endpoint, let (T_b) be its affine components. Its full rank is again (N). The retained edge support (S) is positive definite with exactly (N) nodes, so it contains at most (rankT_b) nodes from each component. Equality of the total counts forces equality in every component. Thus exactly one node in each (T_b) is missing from (S), and these are additional terminating walls.

If the component containing (α_s) has rank at least 2, choose its missing node. Every two nodes in such an irreducible affine diagram have finite angle, so this wall meets (α_s). If that component has rank 1, its missing node is parallel to (α_s); instead choose a missing node of another component. That wall is right to (α_s). Another component exists for (n$\geq$ 3), the only range relevant to the claimed contradiction.

In either case the chosen terminating wall and (α_s) have positive-definite rank-two Gram. Projecting a strict interior point of (P) onto their common orthogonal complement gives a timelike point still strict against every other wall, because the inverse Gram is nonnegative and all distinct wall pairings are nonpositive. Hence they share an **actual ridge**, not merely a point at infinity. This proves that (x,y) belong to the exact framing-wall set controlled by the narrow-facet theorem.

At most eight factors can carry retained nonright contacts

At either second endpoint, the Gram matrix of (S$\cup${x}) is positive semidefinite. Thus

$$p_S^T G(S)^{-1} p_S \leq 2.$$

A positive-definite matrix with diagonal 2 has inverse diagonal at least (1/2), by its one-node Schur complement. A positive-definite Coxeter Gram is a symmetric (Z)-matrix, so its inverse is entrywise nonnegative. Every nonzero retained contact has magnitude at least 1. Consequently at most four entries of (p_S) can be nonzero. The same holds for (y).

Call a factor **spoiled** if either terminating wall has a nonright contact with any retained node in that factor. There are at most eight spoiled factors. Also mark the factor containing (s), regardless of its contacts. Thus at most nine factors are marked. In all unmarked factors each contact row is supported only on the deleted node. A terminating wall cannot have zero contact with an entire factor, since its null sum is a positive multiple of the cusp vector and its height is positive.

Point-contact energy in every affine type

For a factor with Gram (G), null vector (δ), and two chosen omitted nodes (i,j), define

$$R_{ij} = \left(e_i/\delta_i - e_j/\delta_j\right)^T G^+ \left(e_i/\delta_i - e_j/\delta_j\right), \quad L_{ij} = \min(\delta_i^2, \delta_j^2) R_{ij}.$$

The definition is unchanged by rescaling (δ). In an unspoiled factor the two contact magnitudes are at least 1. Since their null-height ratio is (q),

$$\sigma_a \tau_a \geq \max(q\delta_j^2, \delta_i^2/q) \geq M \min(\delta_i^2, \delta_j^2).$$

The omitted nodes can be chosen so that (L_{ij}$\geq$2r/9) for **every** irreducible affine type of rank (r):

Type	Chosen nodes	Exact or lower point-contact energy (L_{ij})
(\widetilde A_r, r≥ 2)	opposite cycle nodes	(⌊(r+1)^2/4⌋/(r+1)≥r/4)
(\widetilde B_r)	a fork leaf and the short end	(r/4)
(\widetilde C_r)	the two short ends	(r/2)
(\widetilde D_r)	fork leaves at opposite ends	(r/4)
(\widetilde A_1)	its two nodes	(1/2)
(\widetilde E_6)	two outer leaves	(4/3=2r/9)
(\widetilde E_7)	equal-weight nodes specified in JSON	(2)
(\widetilde E_8)	equal-weight nodes specified in JSON	(2)
(\widetilde F_4)	its two path ends	(1)
(\widetilde G_2)	its two path ends	(2/3)

The classical values follow from the explicit coordinates in the first package. For (B,D), the normalized point difference has (r) coordinates of magnitude (1/2); for (C), it has (r) coordinates of magnitude 1 and the minimum squared null coefficient is (1/2). The finite exceptional calculation is exact: the checker uses ((G+$\delta\delta^{T}$){-1}) on the orthogonal complement of (δ), where it agrees with (G^+). The certificate records each full matrix and node choice, rather than referring only to diagram names.

For every unmarked factor we use these choices. They do not need to avoid a narrow-facet node, since that factor was separately marked.

Every marked factor loses at most 2

We claim that in each marked factor of rank (r), the omission choices can be made so that

$$\sigma_a \tau_a E_a \geq M(2r/9 - 2).$$

For (r≤ 9), simply use nonnegativity of energy. If it is the narrow-facet factor, choose any node other than (s) for both omissions; they may coincide. Since (2r/9-2≤ 0), this proves the claim without a low-rank search.

For (r≥ 10), only the classical families occur. Use the following choices and Schur-localization estimates. Importantly, the choices are made **before** seeing the terminating contact rows.

- For (A_r), take opposite cycle nodes, rotating both to avoid (s) when necessary. Deleting a node leaves (A_r); the inverse-diagonal test localizes either row within two cyclic nodes of its deletion. The two long-arc energy estimate gives (E_a≥(r-8)/4). All null coefficients equal 1, so (σ_aτ_a≥M).
- For (B_r), take a left fork leaf and the right short end. Deletion leaves respectively (B_r) and (D_r). In the first row only nodes (0,1,2) can be contacted; the other row is supported within the last three nodes. If the short end is (s), delete the adjacent interior node instead; the retained diagram is (D_{r-1}+A_1), and the second row uses at most the last four nodes. If the chosen left leaf is (s), use the other left leaf. Thus at least (r-5) intervening spatial coordinates have normalized difference (1/2). Every allowed null coefficient is at least 1, giving (σ_aτ_aE_a≥M(r-5)/4).
- For (D_r), take fork leaves at opposite ends, switching to the other leaf if one is (s). Each deletion leaves (D_r); inverse-diagonal localization allows only the adjacent end cluster. At least (r-4) coordinates have normalized difference (1/2). The weaker bound (M(r-5)/4) suffices.
- For (C_r), take its two ends. If one is (s), move that deletion inward by one node; the retained diagram there is (A_1+B_{r-1}). Localization leaves at least (r-5) coordinates with normalized difference 1. The allowed null coefficients are at least ($1/\sqrt{2}$), giving (σ_aτ_aE_a≥M(r-5)/2).

These are applications of the same closed (A,B,D) inverse formulas and cumulative-coordinate calculations already proved in the first package. In all cases (r≥ 10) meets the needed rank thresholds, including (D_{r-1}) in the modified (B) choice. The elementary identities

$$\frac{r-8}{4} - (2r/9 - 2) = r/36, \qquad \frac{r-5}{4} - (2r/9 - 2) = r/36 + 3/4,$$

and

$$\frac{r-5}{2} - (2r/9 - 2) = 5r/18 - 1/2 > 0$$

prove the claimed uniform loss bound. No assumption about the contacts' maximum labels or magnitudes is used.

There is no circular choice here: in every non-narrow factor use the point-contact omissions from the preceding table. If that factor subsequently turns out spoiled, the same classical end choices satisfy these localization estimates; the low-rank estimate needs no change. Only the narrow factor uses the stated modified choices, and it is already included in the loss allowance whether or not it is spoiled.

Summing the energies

There are at most nine marked factors, so

$$\sum_a \sigma_a \tau_a E_a \geq M\left(\frac{2N}{9} - 18\right).$$

Since (q+1/q≤2M), the Lorentz pairing identity yields

$$-(x, y) \geq M\left(\frac{N}{9} - 11\right).$$

If (n≥ 226), then (N≥ 225) and the right side is at least 14. But both (x,y) are actual framing walls of the narrow facet, whose pairwise hyperplane distance is strictly below (arccosh7). For square-two normals this contradicts (-(x,y)≥ 14). Therefore (n≤ 225) under the stated ideal-vertex hypothesis.

The bound remains conditional. It proves that in dimension at least 226 a Nikulin narrow facet, if one existed in a finite-volume Coxeter polytope, would have only finite vertices. Turning that remaining case into an unconditional dimension bound requires a separate argument; it cannot be obtained by silently declaring the facet itself a Coxeter polytope.

```
python scripts/fresh_nikulin_narrow_20261015_products.py --verify
```

Source 183

Source path: `results/fresh_nikulin_product_independent_20261015.md`

SHA256: fff8673db85e25bfaa987628175eba7760c5dfb8384764661e84f003d35a60bc

Independent arithmetic for the product-cusp bound

The standalone checker and certificate independently verify the affine constants and scalar estimates in the conditional bound of 225. They import no producer arithmetic. Generation and saved replay pass.

The exceptional $\widetilde{A}_1, \widetilde{E}_6, \widetilde{E}_7, \widetilde{E}_8, \widetilde{F}_4, \widetilde{G}_2$ energies are reconstructed by inverting a proper finite subdiagram and pinning a null coefficient. This differs from the producer's modified pseudoinverse. Their weighted energies are respectively $1/2, 4/3, 2, 2, 1, 2/3$.

Independent Euclidean coordinates check all standard and modified deletion choices for the classical B, C and D factors at ranks 10, 12 and 17. Contact-simplex extreme points verify the claimed constant-coordinate intervals; these finite checks supplement the all-rank coordinate proof, rather than replace it. Symbolic parity and rank tails verify the scalar inequalities for every larger rank. At total cusp rank $N = 225 + t$, the final lower bound is exactly $14 + t/9$.

The geometric common-height identity, endpoint wall selection and narrow-facet hypothesis require the accompanying mathematical proof and its hostile review. This checker alone is not an unconditional dimension theorem.

```
python scripts/fresh_nikulin_product_independent_20261015.py --verify
```

Source 184

Source path: `results/fresh_nikulin_two_budget_frontier_20261015.md`

SHA256: e2a54cd23b9467b5ee6bd3ce04f2db4f1cc2f2a82521102f42c44568c67c07fa

Ordered endpoint budgets: dimension-163 candidate

Internal research proof candidate, pending independent review: the frozen, independently reviewed 164 argument sharpens to dimension at most **163**. The two endpoint Schur budgets are each at most two. Keeping their order, rather than replacing them by their sum, gives the complete near-balanced energy-loss envelope

$$\boxed{\sum_a (T_a - W_a) \le 4 \quad \text{if } M \le 21/20.} \tag{1}$$

The proof does not assert a uniform unit-charge estimate. Such estimates have known balanced counterexamples. Nor does it extrapolate from a finite rank or finite dihedral-label search.

The checker and certificate preserve all previous files. They use the frozen standalone Fraction grounded-matrix review helper read-only. New support lists, convex minima, radical bounds, tail inequalities, and ordered packing decisions are recomputed. Published geometric inputs and the compact-facet branch remain the explicitly inherited reviewed chain. No publication-priority, current-best, or formal-verification claim is made.

Framework and the residual lemma

Use exactly the preset omissions of the reviewed 164/165/169 packages. For a rank-r affine factor, let

$$T = 2r/9, \qquad T = 2(r-1)/9 \text{ in the sole avoid-node factor.}$$

Its endpoint contact rows p,q have nonzero entries at least one. The positive affine kernel δ is normalized by $\min \delta_i = 1$. Put

$$\sigma = \delta \cdot p, \quad \tau = \delta \cdot q, \quad M = \max(\sigma/\tau, \tau/\sigma), \quad W = \min(\sigma, \tau)^2 E.$$

Here E is the squared distance of the normalized Euclidean point-potential means. The same M applies to every factor. Write C_0, C_1 for its two retained-support quadratic costs. The actual endpoint geometry gives

$$\sum_a C_{a,0} \le 2, \qquad \sum_a C_{a,1} \le 2. \tag{2}$$

The new residual-product lemma is:

Residual endpoint budgets	Maximum total loss $\sum(T - W)$
$(10/9, 4/3)$	$20/9$
$(1, 3/2)$	$22/9$
$(3/2, 1)$	$22/9$

It covers arbitrary numbers and ranks of prescribed factors, with at most one avoid-node factor, for every $M \le 21/20$. All three maxima in the relaxed local domain are attained by the exact table records below. The proof first establishes a complete continuous envelope for every factor with $C_0, C_1 \le 3/2$, then performs finite exact ordered-budget packing.

Complete continuous domains at finite ranks

For an omitted node i and a retained node k, the inverse diagonal is $\delta_k^2 R(i,k)$, where R is the squared distance of the normalized point potentials. The retained inverse Gram is entrywise nonnegative. Therefore every actual support lies among all subsets of nodes with diagonal at most 3/2, and its cost is bounded below by evaluating every nonzero coordinate at one, including all same-row cross terms. No support-size cutoff is used.

For every pair of retained supports, the checker includes both possibilities for each omitted entry: zero, or at least one. Empty full rows are discarded because their positive affine height must be nonzero. If both retained supports are empty, the previously reviewed point-energy inequality gives $W \geq T$, so they contribute no positive loss.

For full supports U,V, the normalized means lie in the two finite convex hulls. The checker constructs an exact rational closest-pair certificate. For their signed Euclidean Gram K, barycentric vector x and the two simplex constraints A, it verifies

$$x \geq 0, \quad Ax = (1,1), \quad Kx + A^t\lambda \geq 0, \quad x_i(Kx + A^t\lambda)_i = 0.$$

The explicit Euclidean models make K positive semidefinite. These identities prove the global minimum $E \geq e = x^tKx = -\lambda_0 - \lambda_1$, including singular minimizer sets. There is no floating-point optimization in the certificate.

The full supports also give height lower bounds $s = \sum_{k\in U} \delta_k$, $t = \sum_{k\in V} \delta_k$. Thus

$$W \geq H_B(s,t)e, \qquad H_B(s,t) = \max\{\min(s,t)^2, \max(s,t)^2/B^2\}, \quad B = 21/20. \tag{3}$$

Square roots in s,t are either exact rational numbers or replaced by certified 64-bit rational lower bounds. H is increasing in each argument, so this is a valid downward energy bound. Surviving support-cost floors are checked to be exactly rational; radical lower bounds used while excluding other supports cannot discard a feasible support.

The finite coverage is A cycles of sizes 2 through 17, B/C ranks 2 through 7, D ranks 4 through 17, all ordinary/avoid E6/E7/E8/F4/G2 records, and the separate modified B2. This gives **91 models, 1,116 retained-support pairs and 3,630 continuous full-support domains**. Exactly 1,395 computed loss upper bounds are positive. Each is dominated, with the same modified/ordinary flag, by one of the following 28 rational records: the dominating record has no larger cost floor in either endpoint and no smaller loss upper bound.

Kind	Cost floors (C_0, C_1)	Loss upper bound
ordinary A1	$(1/2, 0)$	2/9
ordinary A2	$(2/3, 0)$	4/9
ordinary A4	$(4/5, 4/5)$	8/9
ordinary B2, B3, B4, B5	$(0, r/4)$, r=2,3,4,5	$2r/9$
ordinary D4, D5	$(r/4, 0)$, r=4,5	$2r/9$
ordinary E6	$(0, 4/3)$, $(4/3, 0)$	4/3
ordinary E7	$(1/2, 1)$, $(1, 1/2)$	14/9
ordinary E8	$(8/9, 2/3)$	16/9
modified B3 through B7	$((r-1)/4, 0)$, r=3,…,7	$2(r-1)/9$
modified E6	$(0, 4/3)$	10/9
modified E7	$(0, 3/2)$, $(2/3, 5/6)$, $(5/6, 2/3)$	4/3
modified E8	$(1, 1/2)$, $(8/9, 2/3)$	14/9
modified F4	$(0, 1)$, $(1/2, 3/4)$	2/3
modified G2	$(0, 1/2)$	2/9

The surviving table happens to consist of identical point rows with W=0. That observation is **not** the completeness premise: every continuous support domain is separately bounded by (3) and its saved convex certificate. The JSON records the dominating table index for every positive-loss domain.

Classical tails, with no rank cutoff

The following models are the same verified prefix-coordinate realizations as in the frozen unit-charge proofs. Let $s_k = e_1 + \cdots + e_k$. For B, use $v_0 = 0, v_1 = e_1, v_k = s_k/2$ for k≥2; the kernel squares are $(1, 1, 4, \ldots, 4, 2)$. For C, use $v_k = s_k$,

metric one half the usual Euclidean metric, and kernel squares $(1, 2, \dots, 2, 1)$. For D use the B prefix vectors through r−2 and $v_{r-1} = (s_{r-1} - e_r)/2, v_r = s_r/2$, with kernel squares $(1, 1, 4, \dots, 4, 1, 1)$.

Inserting these formulas into $\delta_k^2 R(i,k) \le 3/2$ gives exactly the retained-node lists below. Nodes not listed have larger diagonal, so positive cross terms cannot restore them. The omission itself is optionally present in each full support.

Type and ranks	Omissions	Possible retained nodes, in order	Fixed-coordinate bound for E	Height-square bound
B, r≥8	(0,r)	{1}; {r−1}	$(r-2)/4$	$2/B^2$
modified B, r≥8	(r−1,0)	{r−2,r}; {1}	$(r-3)/4$	$2/B^2$
C, r≥8	(0,r)	{1}; {r−1}	$(r-2)/2$	1
modified C, r≥8	(1,r)	{0,2}; {r−1}	$(r-3)/2$	1
D, r≥18	(0,r)	{1}; {r−1}	$(r-2)/4$	1

For example, every convex mean in the left B support has zero coordinates 2 through r−1, while every right mean has those coordinates equal to one half. This gives $(r-2)/4$. Its right full support has minimum kernel entry $\sqrt{2}$, so the common ratio gives the stated height bound. The other rows follow from the same fixed coordinates. Exact affine polynomials in r−8 or r−18 show that W−T is nonnegative in every row; only the D18 starting value is zero. These are all-rank identities, not tests at a large numerical cutoff. Literal boundary cases are also checked in the JSON.

For the remaining A cycles set m=r+1, $h = \lfloor m/2 \rfloor$, $\varepsilon = m \bmod 2$. For m≥18 even or m≥19 odd, a diagonal $d(m-d)/m \le 3/2$ forces d=1. Two retained coordinates in one row already cost more than 3/2. Thus each full support consists of its omission and at most one adjacent node.

If both full supports are singletons, their cyclic distance is at least h−2, and $W \ge R(h-2)$. Its excess over the ordinary target has numerator

$$m^2 + 8m - 9(4+\varepsilon)^2$$

over the positive denominator 36m. If either full support has two nodes, one height is at least two. The frozen exact linear-loss identity gives $E \ge R(h) - 2(m+\varepsilon)/m$, hence

$$W - T \ge \frac{(9 - 2B^2)m^2 + (2B^2 - 72)m - 81\varepsilon}{9B^2 m}.$$

Substituting m=18+2t or 19+2t gives positive constant terms and nonnegative remaining coefficients in both numerators. This proves **W≥T**, a stronger statement than the old tail's W+C≥T; the two are not conflated.

Rotating opposite A omissions avoids any prescribed node when m≥3. Fork swaps do the same in D. Ordinary B/C omissions already avoid all other nodes; their exceptional choices are the modified rows above. In these cases using the stronger ordinary target and treating the record as ordinary only enlarges the residual envelope. Modified A1 has target zero and needs no positive-loss record. All small boundary cases are in the finite census.

Exact packing and the actual failing-factor split

The table has minimum positive total cost at least one half. An exact finite recursion takes nondecreasing multisets of its records, respecting both ordered budgets and allowing at most one modified record. It recomputes the three residual maxima in the first table in 110, 151 and 163 memoized states. The witnesses are respectively an ordinary E8 plus B2, an ordinary E7 plus two B2, and an ordinary E7 plus ordinary B2 and modified B3. This is a complete finite packing calculation, not a linear relaxation or sampled set.

Now call a factor actually failing if $T - W - C_0 - C_1 > 0$. The frozen 164 proof, already valid at B=21/20, restricts such factors to its 15 listed E7/E8 support occurrences. Two are E7 same-row supports at nodes4 and7. The other endpoint row must then be supported on its omission. Exact convex energy and height bounds, with and without the same-row omitted entry, give W+3/2>14/9 in all four cases. These two occurrences cannot actually fail.

The checker reconstructs all remaining 13 rows and their ordered floors:

A selected actually failing factor	Ordered cost floors	Target	Residual budgets	Residual loss
ordinary E8	$(8/9, 2/3)$	$16/9$	$(10/9, 4/3)$	$20/9$
ordinary E7 or modified E8	$(1/2, 1)$ or $(1, 1/2)$	$14/9$	$(3/2, 1)$ or $(1, 3/2)$	$22/9$

If there is no actually failing factor, summing unit charge gives total loss at most four. Otherwise select **any one** such factor. Bound its own loss by its target, using W≥0, and apply the complete residual lemma to every other factor. Each row gives total loss at most four. This argument also includes additional actually failing factors in the residual product. If the selected factor was modified, still allowing one modified residual record is a harmless enlargement. Equation (1) follows.

Dimension consequence and sharp local accounting

The targets sum to $(2N - 2)/9$, where N=n−1 and the subtraction is the sole avoid-node penalty. Therefore (1) gives

$$\sum W \geq (2N - 38)/9, \qquad -(x, y) \geq M(N - 37)/9 \quad (M \leq 21/20).$$

For M≥21/20, the unchanged four-thirds estimate with the **same preset omissions** gives $-(x, y) \geq M(N - 43)/9$. At N=163+t the two lower bounds are respectively at least

$$14 + t/9, \qquad 14 + 7t/60.$$

They contradict the strict narrow-facet inequality $-(x, y) < 14$ for every t≥0. The ideal-vertex branch therefore gives n≤163; the inherited compact facet branch gives n≤143. Combined, the research candidate is $\boxed{n \leq 163}$, without arithmetic hypotheses.

The local loss constant four is sharp. Take two ordinary E7 factors with both rows at node7, and two ordinary B2 factors with both rows at node0. Their common M is one, W=0, the endpoint cost totals are exactly (2,2), and the total target is four. Adding a zero-target modified A1 and 24 zero-cost E6 factors with W=T gives total rank163. This is a checked configuration of the local charge domain, **not** an ambient polytope. Further improvement may need rank-sensitive accounting or additional geometric restrictions; no claim of an optimal dimension bound follows from this local witness.

Generation and saved replay both **PASS**; syntax, source manifests and local links also pass. The primary source, certificate and this report are frozen. The published narrowness, actual framing/terminating-wall existence, common height ratio, endpoint budgets and compact-facet coverage remain inherited from the frozen reviewed chain; this certificate addresses the new residual envelope and global accounting.

Source 185

Source path: `results/fresh_nikulin_two_budget_frontier_20261015_dimension163.md`

SHA256: a0b39b448d543ade48327fd5ab78e8c698834f008ced600aaa5f5f80071643f2

Reviewed research proof candidate: dimension at most 163

Candidate theorem. Every finite-sided, finite-volume real hyperbolic Coxeter polytope has dimension **$n \le 163$**, without arithmeticity, compactness, a simple-cusp hypothesis, or a facet-count restriction.

The new ordered-budget proof and independent review sharpen the frozen 164 candidate. This case binder and its certificate check the local outputs, complete geometric case split and recursive source hashes. They preserve every preceding artifact. Historical pending-review wording in the frozen primary report is superseded by the independent review bound here. External validation, novelty, and current-best literature status remain unestablished.

If the chosen narrow facet has an ideal vertex, its affine factors have total rank $N=n-1$. Use the inherited actual framing walls, common height ratio $M \ge 1$, prescribed omissions, and separate Schur budgets $C_0 \le 2$ and $C_1 \le 2$. For $M \le 21/20$ the new all-rank continuous envelope proves total target loss at most four. When unit charge fails, select any failing factor. An ordinary E8 factor leaves endpoint budgets at most (10/9,4/3), whose residual loss is at most20/9; its own target is16/9. An E7-type factor leaves (1,3/2) or (3/2,1), whose residual loss is at most22/9; its own target is14/9. Both sums are four, and additional failures are included in the residual product. With no failing factor the two total budgets already give four.

The independently checked envelope covers all 91 finite models and all 3,630 continuous support domains, plus analytic A/B/C/D tails at every remaining rank. It uses no finite-angle cutoff. The two same-row E7 possibilities are excluded by separate exact convex certificates, and the 28-record envelope is packed with both endpoint budgets in their stated order. The review uses independently reconstructed physical Grams, Fraction convex-certificate checks and a separate packing algorithm.

The targets sum to $(2N-2)/9$, including the sole prescribed avoid-node loss. Thus the near-range Lorentz pairing is at least $M(N-37)/9$. For $M \ge 21/20$ the unchanged reviewed four-thirds estimate, with the same omissions, gives at least $M(N-43)/9$. At $N=163+t$ these lower bounds are respectively at least $14+t/9$ and$14+7t/60$. They contradict the strict narrow-facet inequality $-(x,y)<14$ for norm-two walls for every $t \ge 0$. This branch excludes $n \ge 164$.

If the chosen facet has no ideal vertex, finite-sidedness and finite volume make it compact. The frozen compact-facet proof, with its hostile review, gives $n \le 143$. These exhaustive cases yield the candidate $n \le 163$. The published narrowness theorem and the reviewed incidence, common-height and compact-facet arguments remain explicit inherited inputs; the local arithmetic certificate alone is not a proof of those geometric statements.

Both scalar estimates at $N=162$ are below14, so this package claims no dimension162 bound. Its loss-four local witness concerns the relaxed charge domain only, not existence of an ambient hyperbolic polytope.

Generation and saved replay **PASS**. The binder checks all recursively referenced manifests, the reviewed threshold at $N=163$, and the preserved compact-facet143 branch.

```
python3 scripts/fresh_nikulin_two_budget_frontier_20261015_dimension163.py
python3 scripts/fresh_nikulin_two_budget_frontier_20261015_dimension163.py --verify
```

Source 186

Source path: `results/fresh_nikulin_two_budget_frontier_20261015_review.md`

SHA256: 27ad96416ddd7e531845560c0d43e7b7850c6316706264af31beb008daa034ec

Independent review of the ordered-budget dimension-163 candidate

PASS. The new ordered endpoint-budget argument is valid within the explicitly inherited geometric framework. The review checker and certificate independently reconstruct the new arithmetic, without producer, project, or prior-review arithmetic imports. The checked primary source and data remain unchanged.

This is an internally reviewed research proof candidate, not a claim of publication priority, current-best literature status, or formal verification. The primary report's pending-review statement is historical; this separate artifact supplies that review without changing the frozen report.

Independent arithmetic and domain coverage

The checker builds literal norm-two affine Grams for all relevant types, computes their positive kernels, and grounds the rational weighted Laplacian $\mathrm{diag}(\delta)G\,\mathrm{diag}(\delta)$. A separate Fraction Gaussian elimination gives the Euclidean point-potential Gram. For every omitted node in the finite census, its inverse physical spherical Gram is independently checked against

$$J_{kl} = \frac{\delta_k \delta_l}{2}\big(R(i,k) + R(i,l) - R(k,l)\big).$$

All inverse entries are nonnegative and every diagonal is at least one half. Since every nonzero contact is at least one, a support with endpoint cost at most 3/2 must be a subset of the nodes whose inverse diagonal is at most 3/2. The review enumerates every such subset and tests its **exact physical** cost, including all cross terms. This separately confirms the primary's radical lower-bound exclusion logic. No feasible support is lost by an interval test, and the retained costs are exactly rational.

The finite domain contains 91 models, 1,116 retained-support pairs, and 3,630 continuous full-support domains. Both omitted-entry alternatives, zero and at least one, are included. Empty full rows are excluded only because the inherited geometry gives positive affine height. The zero-retained-cost cases satisfy the independently checked point-energy bound $W \geq T$.

For each continuous domain, the saved barycentric vector and equality multipliers are checked against a separately reconstructed Gram. The nonnegativity, simplex constraints, gradient slack and complementary slackness prove the global convex minimum. This verifies all 3,630 exact KKT certificates, including singular minimizer sets. It does not depend on trusting a numerical optimizer or finding a new minimizer independently.

If the full support forces heights at least s and t and the actual common height ratio is at most $B=21/20$, then

$$\min(\sigma,\tau)^2 \geq \max\{\min(s,t)^2, \max(s,t)^2/B^2\}.$$

This bound is increasing in both support-height lower bounds. The exact rational enclosures for their square roots therefore have the correct direction. Multiplying by the nonnegative convex energy gives a valid lower bound for W over the entire contact domain, not only at its unit-contact point.

The review derives the 28-record Pareto frontier independently from all 1,395 positive-loss domains and verifies every saved domination map. A separate forward reachability calculation, which permits all record sequences rather than using the producer's ordered-multiset recursion, proves:

Ordered remaining budgets	Maximum relaxed loss
(10/9, 4/3)	20/9
(1, 3/2)	22/9
(3/2, 1)	22/9

The recursion terminates because each positive-loss record costs at least one half in total. It permits at most one modified factor. Negative-loss factors can be dropped when seeking an upper bound. Replacing avoid-node A/D cases, or already-avoiding B/C cases, by the larger ordinary target enlarges the domain and does not violate the one-modified-factor restriction.

All-rank tails

The finite census is supplemented by an analytic, not sampled, tail proof. The literal prefix coordinates give the following complete inverse-diagonal reductions. Interior distances below mean integers at least two.

- For ordinary B, omission 0 leaves node1 of cost1; the other node classes have costs k≥2 or r/2. Omission r leaves r−1 of cost1; the other classes have costs r/4 or r−k≥2. For modified B, omission r−1 leaves r−2 and r of costs1 and1/2, while its other classes have costs (r−1)/4 or r−1−k≥2. The opposite omission0 again leaves only node1.
- For ordinary C the only nodes left at the two ends are1 and r−1, each of cost1. The other costs are at least two or r/2. Modified omission1 leaves0 and2 with costs1/2 and1; the remaining end cost is (r−1)/2.
- For D, omission0 leaves only1, and omission r leaves only r−1, each of cost1. Other classes have cost at least two or r/4.

These exclusions hold for all B/C ranks r≥8 and D ranks r≥18. Inserting the coordinate formulas shows that arbitrary convex means in the surviving full supports differ in fixed coordinates. Their energies are at least (r−2)/4, (r−3)/4, (r−2)/2, (r−3)/2 and (r−2)/4, respectively. In both B cases one full support has all kernel coefficients at least $\sqrt{2}$, so the height-square factor is at least 2/B². The resulting W−T expressions are affine in r−8 or r−18 with nonnegative constant and positive slope. The checker verifies these symbolic expressions and separately checks their physical coordinate identities at the boundary ranks.

For A cycles of size m≥18 even or m≥19 odd, the resistance is $R(d) = d(m-d)/m$. Its concavity shows that any retained distance at least two exceeds the cost cap. Two distinct retained entries also exceed the cap, so each full support is its omission plus at most one adjacent node. Two singleton full supports have separation at least floor(m/2)−2. If a full support has two nodes, its height is at least two. Expanding the difference of the two means leaves a nonnegative square and a linear loss at most $2(m + (m \bmod 2))/m$. The independent symbolic calculation reproduces the primary's two positive parity polynomials and proves W≥T at every remaining rank. Rotations and fork swaps implement the stated avoid-node choices; avoid-A1 has target zero. Thus no unexamined large rank or dihedral-label cutoff remains.

Failing-factor split and dimension threshold

Completeness of the old 15 possible unit-charge defects is inherited from the frozen independent 164 review. The new review independently reconstructs their ordered cost floors. The two same-row E7 cases are eliminated by four fresh convex certificates; these give W+3/2>14/9 even when the omitted entry is allowed to vary. The remaining 13 support occurrences have exactly the stated E8 or E7-type floors.

If no factor actually fails unit charge, the total loss is at most the sum of the two endpoint budgets, namely four. Otherwise choose any actually failing factor. Bounding its own loss by T and applying the full residual envelope to all other factors yields either 16/9+20/9=4 or 14/9+22/9=4. This includes additional failing factors in the residual product; it does not require uniqueness or assume that every possible defect is actually realized. Allowing a modified residual factor even when the selected factor is modified only weakens the bound.

Consequently the near-balanced pairing bound is M(N−37)/9. The unchanged four-thirds estimate, using the same omissions, gives M(N−43)/9 outside the window. At N=163+t these are at least 14+t/9 and 14+7t/60. They contradict the

inherited strict narrow-facet bound at every $t \geq 0$. With the independently reviewed compact-facet branch at143, the combined candidate is $n \leq 163$.

The loss-four local witness and its rank163 padding are independently rechecked. They show sharpness of this relaxed loss constant, not existence of a hyperbolic chamber or optimality of the dimension bound.

The narrow-facet theorem, actual terminating-wall incidence, common height ratio, geometric Schur budgets, and compact-facet branch are inherited with their frozen review chain. The new review does not repeat or weaken those hypotheses. Generation and saved replay pass, as do recursive source manifests, syntax and report links.

```
python3 scripts/fresh_nikulin_two_budget_frontier_20261015_review.py
python3 scripts/fresh_nikulin_two_budget_frontier_20261015_review.py --verify
```

Source 187

Source path: `results/fresh_nikulin_unit_charge_20261015.md`

SHA256: e8acc1c075861e24ae07f436f13b56add4220e40155b42fb85b09ab69109461f

Unit charge fails; a uniform charge of 4/3 gives a 169 candidate

For the affine-factor energy estimate, coefficient one is impossible even after changing the prescribed omission pair. A coefficient of **4/3** does work uniformly, with a different non-narrow E8 pair and the existing avoid-node choices. It is optimal for this local lemma with slope 2/9 and omissions prescribed before the contact rows. This is a statement about that particular accounting scheme, not an optimal dimension bound.

The resulting **research proof candidate is n≤169** for finite-sided finite-volume real hyperbolic Coxeter polytopes. The reviewed 181 candidate remains frozen. The new argument requires independent review before it becomes the reviewed baseline. No priority or current-best claim is made.

The checker and certificate check all exceptional cases, complete E6/E7/E8 omission-pair obstructions, and symbolic all-rank classical formulas. They use no finite-label cutoff and import no producer arithmetic.

Precise local statement

Retain the notation and geometric hypotheses of the product proof and avoid-node proof. The affine roots have square two; the positive kernel δ has minimum entry one. For prescribed omissions i,j, contact rows p,q are nonnegative and nonzero, and every nonzero entry is at least one. Each endpoint's full Schur cost is at most two. Write

$$\sigma=\delta\cdot p,\quad \tau=\delta\cdot q,\quad M=\max(\sigma/\tau,\tau/\sigma),$$
$$C=p_{Si}^{T}G(S_i)^{-1}p_{Si}+q_{Sj}^{T}G(S_j)^{-1}q_{Sj},$$
$$W=\sigma\,\tau\,E/M=\min(\sigma,\tau)^2\,E.$$

The new inequalities are

$$W+(4/3)C\ge 2r/9 \qquad \text{(ordinary factor)},$$
$$W+(4/3)C\ge 2(r-1)/9 \qquad \text{(factor avoiding a fixed node)}.$$

Use the frozen ordinary choices except that E8 now uses nodes **1,6**. Use exactly the frozen avoid-node choices in the factor containing the narrow-facet normal. Thus every choice remains independent of the eventual contact rows. In particular, the new ordinary E8 choice does not replace the separate rule needed when node 1 or 6 must be avoided.

Exact obstruction to coefficient one

With the old ordinary E8 omissions 3,7, take the row at omission 7 to be p=e8 and the other row q=e3. Their heights are 1,2, the total cost is 1/2, and W=1. Consequently W+C=3/2<16/9; this pair would require a charge at least 14/9.

The certificate checks every unordered omission pair, not just the old choice. A pair whose zero-cost point rows have energy below 2r/9 is ineligible for every coefficient. Among the remaining pairs it tests all unit point-contact rows whose two individual costs are at most two. These are legal rows under the stated local hypotheses and use only contact magnitudes zero and one.

Type	Eligible omission pairs	Least necessary coefficient among them
E6	6 of 21	1
E7	1 of 28	10/9
E8	4 of 36	4/3

The four E8 pairs are (1,6), (2,6), (2,5) and (3,7), with respective necessary coefficients 4/3, 65/48, 14/9 and 14/9. For the best pair (1,6), choose p=e1 and q=e8. The heights are 3,1, the costs are 0 and 2/3, and W=8/9. Hence (16/9−W)/C=4/3 exactly. The uniform upper bound below therefore attains the best possible coefficient within this fixed-slope, preset-two-omission local statement. No claim that these local obstructions alone impose a global dimension lower bound is made.

A linear loss identity

Choose Euclidean point potentials v_k for the normalized point rows e_k/δ_k, defined up to a common translation. Put R_ab=|v_a−v_b|². The contact-row potentials are convex combinations

$$z_p=\Sigma_k\,(\delta_k\,p_k/\sigma)v_k,\qquad z_q=\Sigma_k\,(\delta_k\,q_k/\tau)v_k.$$

Set a=z_p−v_i, b=z_q−v_j and

$$\ell_{ik}=R_{ij}+R_{ik}-R_{kj}.$$

Direct expansion gives

$$E=R_{ij}-\Sigma_k\,(\delta_k\,p_k/\sigma)\ell_{ik}\\ \qquad -\Sigma_k\,(\delta_k\,q_k/\tau)\ell_{jk}+|a-b|^2. \tag{1}$$

The inverse diagonal for a retained node k after omission i is

$$c_{ik}=\delta_k^2\,R_{ik}.$$

All inverse entries are nonnegative, so the row cost is at least Σ_k c_ik p_k². Every convex weight δ_k p_k/σ is at most one; when it is nonzero, p_k²≥1. Therefore, if every possible retained contact satisfies $\ell_{ik}\le\alpha\,c_{ik}$ (and similarly for the other row), each linear loss in (1) is paid by α times its diagonal cost. Negative ℓ terms only improve the bound. It follows that

$$E+\alpha C\ge R_{ij},\quad\text{hence } W+\alpha C\ge R_{ij}, \tag{2}$$

because σ,τ≥1 and E≥0. This elementary identity uses the energy and cost together, instead of discarding all energy when a contact occurs.

For each factor with target T, the case C≥T/α is immediate. In the remaining case every retained contact has diagonal $c_{ik}<T/\alpha$ and $c_{ik}\le 2$. These restrictions eliminate the exceptional distant contacts in the following coordinate tables.

All classical ranks

For A_r let m=r+1, take opposite cycle nodes, and let ε=m mod 2. A node at cyclic distance d from an omission has

$$c=d(m-d)/m,\qquad \ell\le(m+\varepsilon)d/m.$$

For r≤5 the minimum retained diagonal r/(r+1) is already at least the small-cost threshold r/6, so a nonzero retained contact is impossible below that threshold. For r=6,7,8, the threshold excludes distance two and farther; only d=1 remains. For r≥9, the individual endpoint budget two excludes d≥3, since $3(m-3)/m>2$ for m≥10 and the diagonal increases up to the antipode. Thus d≤2.

In all these remaining cases, multiplying $(4/3)c-\ell$ by 3m gives the lower bound

$$d(m-4d-3\varepsilon)\ge 0.$$

For odd m≥11 the smallest case is m=11,d=2, where equality holds. The cases m=7,8,9,d=1 and even m≥10 are also nonnegative. Opposite-cycle point energy is at least 2r/9. Equation (2) proves the ordinary and rotated avoid-node estimates for every rank. A1 is handled separately below.

For B,C,D use orthonormal coordinates e1,…,er. The following point potentials give all required resistances directly by squared distances; their adjacent differences solve the corresponding normalized affine-root equations.

- B_r: `v0=0`, `v1=e1`, and `vk=(e1+…+ek)/2` for $2\le k\le r$. Kernel entries are 1 at nodes 0,1; 2 at nodes 2,…,r−1; and $\sqrt{2}$ at r.
- C_r: `v0=0`, `vk=(e1+…+ek)/√2` for $1\le k\le r$. Kernel entries are 1 at both ends and $\sqrt{2}$ at the interior nodes.
- D_r: `v0=0`, `v1=e1`, `vk=(e1+…+ek)/2` for $2\le k\le r-2$, `v_(r−1)=(e1+…+e_(r−1)−er)/2`, and `vr=(e1+…+er)/2`. Kernel entries are 1 at the four fork leaves and 2 elsewhere.

The ordinary omissions are 0,r. The modified B omission pair is r−1,0; the modified C pair is 1,r. Leaf switches and reversal cover the other forbidden-node cases. In all of these cases every retained contact has $\ell\le c$, except the following distant contacts, whose cost is already strictly above the threshold T/(4/3):

Case	Distant contacts	Their inverse diagonal	Threshold
Ordinary B	opposite left fork leaves, from omission r	r/4	r/6
Ordinary D	opposite fork leaves	r/4	r/6
Ordinary C	opposite end	r/2	r/6
Modified B, r≥3	left fork leaves, from omission r−1	(r−1)/4	(r−1)/6
Modified C	opposite end from omission 1; node 0 from omission r	(r−1)/2; r/2	(r−1)/6

For example, in ordinary B the interior-node costs and losses are `c=k, ℓ=k/2` from omission 0, and `c=r−k, ℓ=(r−k)/2` from omission r. A same-end fork leaf has $c=\ell=1$. In modified B the isolated short end has $c=1/2$ and $\ell=0$, so it causes no linear loss at all. In C the interior costs equal their linear losses. The certificate records the full 24-row index-class table and verifies its symbolic margins with independent nonnegative parameters for the node index and the excess rank; this is not an inference from a finite rank scan.

The point resistances are r/4 for ordinary B/D, r/2 for ordinary C, (r−1)/4 for modified B, and (r−1)/2 for modified C. Each is at least its required target. Equation (2), even with coefficient one on the remaining contacts, proves all these cases. Modified B2 is separate: its point energy is 1 and its minimum positive cost is 1/2, already sufficient for target 2/9. Ordinary B2/B3 are covered by the displayed coordinate formulas without assuming their inverse minima equal one.

Exceptional factors and every forbidden node

The checker reconstructs the six ordinary exceptional cases and all 34 forbidden-node cases from their literal Gram matrices. Let T be the target and discard the immediate case C≥3T/4. Nonnegative inverse entries then restrict retained contacts to nodes with diagonal strictly below 3T/4.

There are only 46 single-contact occurrences across these records, and one possible two-contact pattern. Every other support is excluded by its diagonal costs, including cross terms when two contacts lie in the same row. There is no restriction on the magnitude of any remaining contact beyond the endpoint cost and the gap between zero and one.

For one retained contact the rows are

```
p=a e_i+b e_l,  q=c e_j,
b,c≥1,  a=0 or a≥1.
```

Writing `u=δ_l b/(δ_i a+δ_l b)`, the normalized-row energy is the quadratic

```
E(u)=(1−u)R_ij+uR_lj−u(1−u)R_il.
```

If a=0, a valid lower bound for W+(4/3)C is

```
min(δ_l,δ_j)² R_lj +(4/3)c_il.
```

If a≥1, a valid lower bound is

```
min(δ_i,δ_j)² min_[0,1] E(u) +(4/3)c_il.
```

Both bounds meet T in every serialized occurrence. The quadratic minima are evaluated exactly at the two endpoints and, when applicable, the interior stationary point. The critical E8 ordinary case is omission 6, cheap node 8, opposite omission 1, with a=0: it gives equality 8/9+(4/3)(2/3)=16/9.

The only surviving two-contact pattern is ordinary E7: one isolated A1 contact at node 4 after deleting 3 and one at node 7 after deleting 6. Its exact energy is $(1+u^2+v^2)/2 \geq 1/2$; its total cost is at least one. Thus $W+(4/3)C \geq 11/6 > 14/9$. Three contacts already exceed the cost threshold. This accounts for all exceptional rows, not merely single-contact proposals.

The avoid-node rules remain those of the frozen 181 proof, including E7 pairs 4,7 or 2,5, and E8 pairs 3,7 or 1,6. A1 has ordinary point energy 1/2 and positive cost at least 1/2; its avoid-node target is zero. It remains the exact obstruction to eliminating the additive avoid-node loss 2/9 in this local scheme.

Global consequence and scope

The same geometric argument supplies actual terminating walls framing the chosen narrow facet, with a common height ratio q and total cost $\Sigma C \leq 4$. Only the single factor containing the facet normal uses the avoid-node target. Consequently

```
Σ σ_a τ_a E_a ≥ M(2N/9−(4/3)·4−2/9),
−(x,y) ≥ M(N−43)/9,  N=n−1.
```

At $n \geq 170$, $N \geq 169$, so the lower bound is at least 14, contradicting the strict narrow-facet distance bound arccosh 7 for square-two normals. The ideal-vertex branch gives $n \leq 169$. The unchanged reviewed compact-facet branch gives $n \leq 143$. Their combination is the stated 169 research proof candidate.

For clarity about constants, coefficient one with the same total cost four and avoid-node penalty 2/9 would give 163, not 145. The value 145 would correspond to eliminating the Schur-cost loss entirely. The exact E7/E8 examples explain why that unit-charge route does not work.

The published narrow-facet theorem, endpoint existence, common-height product identity, actual-ridge incidence bridge and compact-facet proof are inherited mathematical dependencies, bound through the frozen 181 checkpoint. The new results are the local obstruction, linear loss accounting and coefficient 4/3. Computation checks the exact identities and finite exceptional case coverage; the all-rank and geometric arguments remain mathematical proofs requiring independent scrutiny, not a formal proof-assistant verification.

```
python scripts/fresh_nikulin_unit_charge_20261015.py
python scripts/fresh_nikulin_unit_charge_20261015.py --verify
```

Source 188

Source path: `results/fresh_nikulin_unit_charge_20261015_dimension169.md`

SHA256: fccd442849ba2baa9489041f82c11f6be647110a5a8643a41c421474531d02fc

Reviewed research proof candidate: dimension at most 169

Candidate theorem. Every finite-sided, finite-volume real hyperbolic Coxeter polytope has dimension **n≤169**, without an arithmeticity, compactness, simple-cusp or facet-count assumption.

The four-thirds estimate and its independent review sharpen the frozen 181 proof candidate. The case binder and certificate bind the complete dependency chain. This is an internally reviewed research proof candidate; external validation, novelty and current-best literature status remain unestablished.

Nikulin's narrow-facet theorem supplies a facet F whose framing hyperplanes have pairwise distance strictly below `arccosh 7`. If F contains an ideal vertex, its Euclidean link is a product of irreducible affine Coxeter simplices of total rank `N=n−1`. Preset omissions select two edges in F, and actual terminating walls x,y can be chosen to meet F in ridges. Their two total Schur costs sum to at most four. Nikulin, official ICM report, printed pp.654–655 and 657–659.

For each affine factor, the new estimate is

```
σ_a τ_a E_a / M + (4/3)C_a ≥ 2r_a/9,
```

with target `2(r_a−1)/9` in the single factor containing F's normal. The height ratio is common across factors and `M=max(q,1/q)≥1`. The proof covers every affine type and all ranks, with no label cutoff. It combines a linear energy-loss identity with exact exceptional-case analysis; the ordinary E8 omission changes to nodes 1 and 6.

Summation and the Lorentz identity yield

```
−(x,y) = (1/2)Σ σ_a τ_a E_a − q − 1/q
       ≥ M(N−43)/9.
```

At `n≥170`, this is at least 14, contradicting narrowness for square-two normals. If F has only finite vertices, it is compact; the separately reviewed compact-facet argument gives `n≤143` without assuming that F itself is Coxeter. The two cases establish the candidate 169 bound.

For comparison, Prokhorov's published Theorem C gives 995, and Nikulin's ICM report already announces the stronger unrestricted bound below 300. The repository's candidate is separate from those literature statements and is not presented as an externally validated theorem. Prokhorov, printed p.402, Nikulin, printed p.659.

The compact and arithmetic dimension ceilings remain 29; the noncompact arithmetic range remains `n≤19` or `n=21`. The 169 candidate addresses the general finite-volume class and does not improve those stronger restricted bounds.

The primary report's pending-review sentence records its earlier frozen state. This binder records the completed independent review without rewriting that artifact. Arithmetic and source integrity checks do not replace the stated mathematical and published dependencies.

```
python scripts/fresh_nikulin_unit_charge_20261015_dimension169.py --verify
```

Source 189

Source path: `results/fresh_nikulin_unit_charge_20261015_review.md`

SHA256: 999a3694e1889ecdb7932c97125e7f4be8b2b55b45a14a215a9b88b188e8d4fc

Independent review of the four-thirds contact-cost estimate

The new local estimate and its dimension-169 consequence pass independent arithmetic and mathematical review. This is an internal research review. External validation, publication priority and current-best status remain unestablished.

The review checker and certificate import no arithmetic from the primary proof. They reuse only elementary Fraction matrix operations from an earlier independent checker. Exceptional energies are recovered from grounded rational linear systems, with every equation checked, rather than the producer's regularized inverse. The nonsimply-laced matrices are treated by rational diagonal congruences; the literal physical Gram entries and the normalized kernel squares are compared with the primary data.

Coverage of the local lemma

For each of the 40 ordinary and forbidden-node exceptional cases, the review rebuilds every retained inverse diagonal. It enumerates **every subset** of contacts whose diagonal costs are below the trivial-charge threshold, adding the inverse cross terms for contacts in the same row. All inverse entries are nonnegative, so setting every nonzero magnitude to one gives a valid support-cost lower bound. This establishes support coverage for arbitrary real contact magnitudes at least one, with no finite-label cutoff.

There are 46 single-contact occurrences and one remaining two-contact support. In the single-contact case, the energy is a convex quadratic in the normalized mixture weight. The review computes its three coefficients and exact minimum separately, then checks both possibilities for the omitted-node contact: zero, or at least one. The latter distinction supplies the correct lower bound on the two null heights. The sole two-contact support is ordinary E7; polarization of the independently reconstructed resistance matrix gives `(1+u²+v²)/2`. Its cost is at least one. Both cases meet their stated targets.

The complete E6/E7/E8 omission-pair census is independently reconstructed. Zero-cost rows eliminate the ineligible pairs; all unit point-contact rows satisfying each endpoint budget at most two give the remaining necessary coefficients. Their best values are 1, 10/9 and 4/3. In E8, the omissions `(1,6)` and contact rows at nodes 1 and 8 attain the 4/3 obstruction. This proves optimality for the **specified local lemma**, whose slope is fixed at 2/9 and whose omissions must precede the contact rows. These local rows need not be realizable together in a whole hyperbolic chamber, and no global optimality follows.

The all-rank step

The linear identity used by the producer has the correct sign. For point potentials `v_i`, write `z_p=v_i+a` and `z_q=v_j+b`. Expanding their squared distance gives

```
E=R_ij−Σw_k(R_ij+R_ik−R_kj)
       −Σz_k(R_ij+R_jk−R_ki)+|a−b|².
```

Each coefficient is a convex weight. When its contact is nonzero, that weight is at most one and the squared contact magnitude is at least one. Since the retained inverse has nonnegative entries, its cost dominates the sum of diagonal contributions. Thus a linear loss bounded by `(4/3)c_ik` is paid by the corresponding cost. Nonpositive losses require no payment. Finally `W=min(σ,τ)²E≥E`, since the normalized affine kernel has minimum entry one and both rows are nonzero. These steps use neither arithmeticity nor a bound on the contact labels.

The classical point potentials have an independent direct verification. For B and D use roots `α0=−e1−e2`, `αk=e_k−e_(k+1)`, with final root `√2 e_r` in B and final two roots `e_(r−1)−e_r`, `e_(r−1)+e_r` in D. In C use `α0=−√2 e1`, the same difference roots, and `αr=√2 e_r`. The prefix vectors in the primary proof satisfy, for every index,

`α_l·v_k = 1_(l=k)/δ_k − 1_(l=0)/δ_0.`

This follows by subtracting adjacent coordinates; the two fork/end equations check the boundary cases. It identifies the potentials with normalized point contacts, proves their squared-distance formula, and gives `c_ik=δ_k²R_ik` by deleting node i. It is an all-rank identity, not an extrapolation from small diagrams.

The reviewer computes distances from these coordinate vectors by symbolic interval intersections. It checks 24 index classes partitioning the required B/C/D rows, with independent nonnegative parameters for the node index and excess rank. Every class either satisfies the four-thirds loss inequality or has cost strictly above the trivial-charge threshold. Path reversal and fork automorphisms cover the stated symmetric rows and forbidden-node choices. The exceptional modified B2 pair is checked directly. No producer resistance table is imported.

For A, direct substitution in `R(d)=d(m−d)/m` gives linear losses `d(m±ε)/m` for the two directions from an opposite omission, where `m=r+1` and `ε=m mod 2`. Below the trivial-charge threshold, ranks 2–5 allow no retained contact, and ranks 6–8 allow only distance one. For `m≥10`, the endpoint budget excludes distances at least three because `3(m−3)/m>2`; monotonicity up to the antipode completes this restriction. For the remaining distances, the cleared margin is `d(m−4d−3ε)`, nonnegative in both parity classes. A1 and its coincident avoid-node omissions are included among the exact exceptional cases. Cycle rotation avoids any prescribed node in all other ranks.

Global hypothesis and threshold check

The frozen 181 binder supplies the geometric dependencies: actual ridge neighbors of the narrow facet, terminating walls at finite or ideal endpoints, a common null-height ratio across affine factors, the Lorentz pairing identity, and the compact-facet branch. The new omission in ordinary E8 is chosen before the contact rows and does not alter the separate avoid-node rule. Thus the framing argument still applies.

There are two endpoint budgets, each at most two. Exactly one affine factor contains the narrow-facet normal and incurs the avoid-node deficit 2/9. With `N=n−1`, the new local estimate therefore gives

```
Σ σ_a τ_a E_a ≥ M(2N/9−16/3−2/9),
−(x,y) ≥ M(N−43)/9,    M≥1.
```

At `n=170`, this is at least 14. It contradicts the strict narrow-facet distance bound `arccosh 7` for square-two normals. At `n=169`, the scalar lower bound alone is smaller than 14, so no further one-unit improvement has been silently inferred. The ideal-vertex branch gives 169; the inherited compact-facet branch gives 143. These two alternatives exhaust the chosen narrow facet.

The checker certifies the exact arithmetic and finite coverage supporting these arguments. The coordinate identities, geometric bridge and published classifications remain mathematical proof dependencies, not a formal proof-assistant development.

```
python scripts/fresh_nikulin_unit_charge_20261015_review.py
python scripts/fresh_nikulin_unit_charge_20261015_review.py --verify
```

Source 190

Source path: `results/fresh_nonincident_carrier_incidence_20261017.md`

SHA256: 3732043a118aa72a0b6e10c2ab6f0c21198dc21df5960eded4d5e82b991aac4c

Nonincident cusp carriers: exact multiplicity obstructions

The known finite-volume H17 pyramid has **1,048 unpaid quadrilateral faces** under the fine corner rule, of which987 are compact. Every one has the two affine E8 factors of the same nonincident cusp in its original face star. This disproves two natural proposed bounds: charging at most one unpaid face per edge of the carrier cusp link, or at most six per link vertex per small factor. It does not disprove a stronger bound with a larger constant, or one that uses additional geometry.

Separately, an explicit family of finite-volume H3 Coxeter chambers has arbitrarily many compact quadrilateral faces carried by a fixed-combinatorial-type cusp. Those faces are fully paid. Thus a useful global estimate must depend on the **payment deficit**, not merely on the presence of a nonincident carrier.

The checker and data preserve the distinction. No dimension bound is asserted.

Exact chamber and complete two-face ledger

Use the19-root shared-L25 Gram from the frozen selective-distance13 proof and its independent reconstruction. Its nodes split into

- L={0,…,8};
- R={10,…,18};
- the shared node9.

The Gram radical is

`(-6,-3,-4,-2,-5,-4,-3,-2,-1,0,6,3,4,2,5,4,3,2,1)`.

In covector coordinates p_i=x·n_i the chamber cone is p_i<=0 with z·p=0. Its complete extreme rays are the single-coordinate ray supported on9 and the81 two-coordinate rays supported on one node of L and one of R. A rational18-by18 inverse rechecks all squares and the strict common timelike center. There are80 finite vertices and two ideal vertices: the apex supported on9, and the base vertex supported on{3,13}. The frozen chamber proof supplies the proper cusp links and actual-facet checks; the new calculation independently recovers all82 rays and their signs.

The projective combinatorics is a pyramid over Delta8×Delta8. A two-face allows four covector coordinates, at least one from each of L,R; its supporting whites are the other15 roots. This is exhaustive: the restricted cone has dimension3 exactly in these cases, since its positive and negative z coordinates impose one independent relation. The actual sides are the coordinate inequalities whose zero sets contain an edge, not every allowed coordinate. The checker also recognizes every15-root support as a finite forest.

There are3,456 two-faces:

Family	Triangles	Quadrilaterals
In the base Delta8×Delta8	1,512	1,296
Containing the apex	648	0

The two affine E8 factors at the apex are exactly L and R. Their positive kernel coefficients are the absolute radical coordinates on the corresponding factor. Each positive root combination has covector−e_9, so both determine the same actual cusp. Its Euclidean link is Delta8×Delta8, with81 vertices and648 edges. The other cusp has one affine D16 factor and contributes no small factor of rank at most8.

For each E8 factor separately, the complete incidence ledger is:

Two-face relation to the carrier apex	Count
Incident triangles	324
Nonincident triangles	756
Nonincident quadrilaterals	1,296
Unpaid nonincident quadrilaterals	1,048
Compact unpaid nonincident quadrilaterals	987

Every quadrilateral has all19 normals in its original wall star. Both E8 carriers are therefore present, and neither is incident to that quadrilateral.

Literal quadrilateral payments

At a finite corner the checker uses Vinberg's actual rule: weight2 in a connected component of order at most4; weight1 in orders5 through7; in a larger classical component, weight1 for adjacency or membership in a connected end segment of order at most6 invariant under interchange of fork leaves. Otherwise the weight is0. The definition is the one in Vinberg, §6.1–2, printed pp.80–81, also bound in the frozen finite-component checker. It is not replaced by a short-distance finite rule.

Every connected marked component at a finite quadrilateral corner is A or D. Thus no literal E8 excluded-pair list is needed. At the other ideal vertex, the varying factor is affine D16 and the checker uses the frozen distance-three and invariant-end-segment rule. The small-factor weight2 modification from the nonincident-defect lemma does not alter any quadrilateral corner here: the small-factor cusp is not on these faces.

Total quadrilateral payment	All quads	Compact quads
0	1,048	987
1	243	240
2	5	5

The requested compact quadrilateral has opposite marked pairs{1,2} and{11,12}. Its four omitted pairs and connected marked components are

Omitted roots	Marked roots	Component	Payment
1,11	2,12	A17	0
1,12	2,11	A16	0
2,11	1,12	A16	0
2,12	1,11	A15	0

These are actual corners of the exact compact quadrilateral, not merely spherical subsets of a proposed star.

The complete triangle incidence ledger is included, but **individual triangle fine weights are not claimed**. Some triangle corners use E8; the inherited finite checker records the number of its four omitted pairs without labeling them. The primary text was reopened, but fresh image retrieval did not yield a readable diagram. No excluded pair is guessed from distance. This limitation has no effect on any quadrilateral count or counterexample above.

What the unpaid example rules out

Write Q_bad for the number of unpaid quadrilaterals and sum only over cusps which carry a small factor. This chamber gives

Q_bad=1048 > 648 = sum f1(link_v).

Also, with s_small(v) the number of small factors and V_v the link-vertex count,

Q_bad=1048 > 6·2·81=972,

and even the compact unpaid count987 exceeds972. Consequently an inequality

Q_bad <= c sum s_small(v)V_v

would require c>=524/81 in general, and c>=329/54 if restricted to compact unpaid quadrilaterals. This remains true if each defective face is assigned to just one of its carriers: all available small carriers lie at the same cusp, and the right side already includes both factors. These are necessary tests on a proposed multiplicity theorem, not an upper bound established by this example.

Unbounded carrier incidence with all faces paid

For every integer N>=1, take the upper-halfspace region

−1/2 <= x <= 1/2,

−1/sqrt2 <= y <= (2N−1)/sqrt2,

outside the N unit hemispheres centered at(0,sqrt2·k), k=0,…,N−1. In the Lorentz metric diag(−1,1,1,1), norm-one outward normals are

X±=(1/2,±1,0,1/2),

Y−=(1/sqrt2,0,−1,1/sqrt2),

Y+=((2N−1)/sqrt2,0,1,(2N−1)/sqrt2),

B_k=(k²,0,sqrt2·k,k²−1).

The pairings give:

- X±·B_k=−1/2, hence angle pi/3;
- B_j·B_k=1−(j−k)², hence adjacent caps meet at pi/2 and all others are dotted;
- the end cap meets its end y-wall at pi/4; other cap/end-wall pairs are dotted;
- the vertical x/y pairs are right, and each opposite vertical pair is parallel.

Every cap is an actual facet. Its visible patch lies over the rectangle x∈[−1/2,1/2], y∈[(2k−1)/sqrt2,(2k+1)/sqrt2], clipped only by the two global end walls when appropriate. It is a compact quadrilateral. The bottom is the upper envelope of these hemispheres. The nearest center has horizontal squared distance at most1/4+1/2=3/4, so the bottom height is at least1/2 everywhere. The finite horizontal rectangle and the integral of z^(−3) from this positive lower height to infinity prove finite volume. Only the point at infinity is ideal; its link is a bounded rectangle.

All finite vertices are

(x,y,z)=(±1/2,(2j−1)/sqrt2,1/2), j=0,…,N.

Thus there are2N+2 finite vertices and one proper ideal vertex. Endpoint finite corners are B3, interior finite corners A3. Each cap's four planar angles therefore has fine weight2, total8. The x-wall pair is the same affine A1 factor in every cap's star and its cusp is not on any cap. The other cusp factor is also A1, independent of N, and the link has four vertices.

This is an all-N symbolic argument; the six exact samples N=1,2,3,7,16,64 only check the displayed identities. It proves that no bound on **all** such carrier incidences can depend solely on cusp factor ranks or the combinatorics of its link. It does not refute a bound on unpaid incidences: these N faces are fully paid.

Verification and scope

Generation and a fresh saved replay pass:82 exact rays,3,456 two-face incidences, all1,296 quadrilateral payments, and six exact strip checks. Syntax, ten direct source hashes and all local report links pass. A separate internal symbolic-geometry review of the strip confirms its angles, complete vertex description, positive bottom height and sole proper rectangular cusp; it does not duplicate the H17 payment census. The known H17 chamber is not a new construction, and the strip family is used only as an elementary multiplicity obstruction; no novelty claim is made. The earlier dimension proofs and their frozen files are unchanged. Source, data and report are final/frozen.

Source 191

Source path: `results/fresh_order11_bc_zero_transfer_20261018.md`

SHA256: e7e96ef4276187ffcc55ad8877e37f02735130076930df827961ab8149198460

Universal order11 B/C zero transfers

The order11 family costs can become A99/2, B53, C52 and D53. In affine B10, delete the two remaining terminal/fork halves. In affine C10, delete the two distance9 pairs next to its already deleted terminal pair. Add one explicit finite upgrade: a whole A1 against the doubled terminal of a separate whole B10 now pays1 instead of1/2. Its actual added cost is1/2 per such component pair.

Every affected quadrilateral still receives at least1 and triangles retain at least2. The new rank10 ideal envelope is53/11. This is a local payment theorem; a simultaneous finite count and global scalar comparison remain separate. No new dimension bound is asserted.

The source and certificate build on the coupled B/C/D half theorem, the independent C endpoint-zero theorem, and the completed dimension61 rule. They also bind the new order16 D105 theorem. All root norms are2, with native simple magnitude1 and doubled magnitude sqrt2. No arbitrary rank or label cutoff is used.

Actual-star identities and direct alternatives

At an affected ideal corner, let a,b be its marks and c,d their opposite black roots. The actual varying affine factor F has positive kernel nu; the remaining supporting whites form a positive-definite block Z, orthogonal to F. Every actual edge is strictly positive definite and every actual corner positive semidefinite. These are the inherited actual factor-fiber facts. Write x,y for the nonnegative contacts of c,d to Z and h,k for their positive heights against nu.

An adjacent black contact gives a distinct marked path1; a common spectator contact gives a distinct marked path2. Both remain full units under every rule in this packet. Remove those branches. Adjacent black products now vanish and x,y have disjoint supports.

After eliminating F minus its marks, the scaled old-mark Gram is R^(-1)[[1,-1],[-1,1]]. For each opposite root let l be its moment and c0 its residual constant. Put g=2l/h+c0/h². Eliminating Z from the actual three-dimensional face quotient gives the determinant identity

```
0 = 1−(g_c+g_d)/R
      +(x/h−y/k)^T G_Z^(−1)(x/h−y/k)/R.              (1)
```

The final form is nonnegative. The source checks this symbolic identity at R9 and R10, as well as the physical profile and cross identities. It rejects floating-point constants in every saved symbolic value.

C: the near-terminal distance9 orbit

Number the affine C10 path0,…,10, with doubled bonds0–1 and9–10. The already deleted pair is(0,10). The new zero orbit is(0,9),(1,10), invariant under reflection. Consider a=0,b=9; the reflected case is identical. Kernel coefficients are1 on internal nodes and1/sqrt2 at the two terminal nodes. The exact resistance in(1) isR=9.

Deleting a leaves the finite B10 path1,…,10. Its complete PSD one-root extensions have no contact, a simple or doubled contact at its near endpoint1, or a simple contact at node2. The four heavy-end cost bounds are respectively4,3,2,4, including every possible contact to the omitted opposite mark. These are the same complete four profiles as in the endpoint-zero theorem; the source rechecks their physical identities at the new resistance9.

Deleting b leaves finite B9 on0,…,8 and an isolated white10. For the B9 block the norm-two inverse diagonal is1 at8,2 at7, and greater than2 everywhere else. An extension having two contacts inside B9 would contain a cycle. A connected finite diagram is a tree; the only affine cycle type is A, which cannot contain the old doubled bond. Exceptional finite/affine types are too small to contain this ten-node extension. Thus at most one B9 contact is possible. The Schur condition allows only simple/doubled at8 or simple at7, including equality.

Let q be the magnitude of the contact with old white10, and t>=0 the opposite product with the omitted mark b. Strict positivity of the actual edge containing10,d gives q<2; a nonzero native q is at least1. The PSD corner containing the complete B9+A1 block further bounds the sum of its projection costs. The complete necessary profiles are:

B9 contact	Additional q	Height k	Moment	c0	Maximum g
none	0, t>=1	t	0	2	2
none	1<=q<2	t+q/sqrt2	0	2−q²/2	3
simple at8	0<=q<=sqrt2	t+1+q/sqrt2	1	1−q²/2	3
doubled at8	q=0	t+sqrt2	sqrt2	0	2
simple at7	q=0	t+1	2	0	4

For example, a doubled contact at8 or a simple one at7 already has projection cost2 onto B9, forcing q=0. A simple contact at8 has cost1, forcing q²/2<=1. The continuous q intervals in the table are relaxed necessary domains containing all native possibilities. The first row has t>0 by positive height and hence t>=1 by the native product rule.

The source rederives all nine profile identities and all20 cross identities from the physical C Gram. For each upper bound it multiplies maximum−g by the positive height squared and checks a polynomial with nonnegative coefficients after shifting t to its lower endpoint and, in the tail-only row, q to1+u. Thus the inequalities cover every t and q in the stated infinite domains, not merely sampled native labels.

Both sides have g<=4. Equation(1) would therefore have right side at least1−8/9=1/9>0. A direct path1 or2 must occur after all, and supplies a distinct full unit. Neither the original near-terminal payment nor another changed affine payment is needed. The two new weights may be0, so C has55−3=52 after retaining the earlier endpoint zero.

B: retain all profiles instead of stopping at another affine half

Use the canonical affine B10 diagram with spine0,…,8, doubled bond 7–8 and short leaves9,10 at0. Set a=8,b=9. The new zero orbit consists of(8,9),(8,10); each formerly had weight1/2. The resistance isR=10.

Deleting either marked leaf gives B10 or D10. The complete four-profile classification in the coupled theorem applies: isolated, end3, end4, next3. It follows from the endpoint inverse diagonals1,2 and larger remaining diagonals, together with the same cycle/exceptional-type exclusion. We retain all four profiles, including those that formerly supplied another affine half.

At the heavy end every profile has cost at most4. At the fork end, end4 and next3 also have cost at most4. Isolated and end3 reach8 only at t=1 and t=0 respectively; each such extreme has height1/2 and one simple contact to the old selected fork leaf or its mate. Every nonextreme isolated profile has t>=sqrt2 and cost at most4; every nonextreme end3 has t>=1 and cost at most3. The exact source checks all8 physical profiles and their complete native-ray polynomial bounds.

If the fork is nonextreme, the total cost is at most8 and(1) is impossible. Hence it is extreme. Let x be its spectator vector and let y,k belong to the heavy root. Equation(1) requires

```
E=(2x−y/k)^T G_Z^(−1)(2x−y/k)=g_heavy−2<=2.          (2)
```

If x is nonzero at i, disjoint supports and Gram-metric Cauchy give E>=4x_i²/2>=2. Equality is necessary, forcing x_i=1 and

```
2x−y/k=G_Z e_i.
```

The nonpositive off-diagonal signs make i the only x-coordinate. If i has a neighbor j, then y_j>0 and the actual c–j–i–d path3 gives a retained full unit. Otherwise i is a whole A1 component, all other x/y contacts vanish, and the heavy cost is exactly4. The only heavy profiles with that cost are isolated t=1 and next3 t=0. No rank bound on Z was used in this argument.

If x is empty, choose the actual corner containing the extreme root and the far old heavy mark. A contact to the omitted old selected leaf leaves a whole A1 against a separate whole B10 at its doubled terminal. The upgraded unit pays the entire face. A contact to the old mate instead gives a whole finite B11 component and an unchanged unit. The heavy root is absent from this corner, so arbitrary contacts y cannot extend either selected component.

In the remaining equality case x is a single contact to a whole A1, y=0. If the extreme root attaches to the old mate, its selected finite component is whole B12, again paid fully. Otherwise its corner has whole A2/B10 and contributes the existing half. A second distinct corner supplies another existing half:

- heavy isolated t=1 leaves whole A1 against whole D10 at a short leaf;
- heavy next3 t=0 gives the whole affine D10 opposite-fork pair, which retains1/2.

The source constructs all four heavy-profile/old-fork-attachment combinations, verifies every strict white/edge principal, and identifies the two actual corner components, roles and positive affine kernel. No omitted spectator can extend them: Z is orthogonal to F, and the proved equality leaves exactly the one whole A1 and no other new contact. These two halves are distinct from the original B corner. Their sum is1, so the original B half is unnecessary. Both B weights may become0, leaving55−2=53.

Coupon cost and simultaneous priorities

The only finite change is whole A1/B10 doubled-terminal weight1/2→1. There is one marked choice in A1 and one eligible terminal in B10, so the actual increment is exactly1/2 per component pair. It must be written into any subsequent actual-cost rule.

The inherited finite representative maps B10 to D10 and A1 to itself. The old actual A1/B10 doubled cost is1; its new doubled cost is2. The unchanged A1/D10 representative doubled allowance is also2, for the two short-fork-leaf halves. The source checks these exact values against the frozen current coupon function. Every other type pair is unchanged. Thus every existing simultaneous representative upper comparison survives this upgrade; no residual optimization is needed for that pointwise statement. A subsequent combined forest certificate must still record the actual increment and all other new local changes.

For simultaneous quadrilateral coverage, first handle any actual new C near-terminal corner by its independent path1/2 unit. The earlier C endpoint-zero theorem has the same independent output. Next handle any new B zero corner by the argument above: its outputs are finite B11/B12 units, path3 units, the upgraded A1/B10 unit, or two distinct retained A2/B10 and A1/D10 or affineD10 halves. None is a changed C or B pair, a long finite rank15–17 payment, or a changed affineD16 pair.

If neither new C nor new B corner exists, weights on all actual ideal corners agree with the previous rule, and finite weights have only increased. The complete old whole-face proof therefore applies pointwise. This handles the old coupled D11 proof's occasional B-half output without recursion: an actual such B corner is in the earlier independent branch. The new order16 D105 theorem uses a whole D5/D11 finite half and retained short8; if that route reaches a B zero, its independent proof here supplies the complete payment.

All new changed pairs have distance9. Finite7/ideal5 triangle payments, protected ideal6/7 roles, whole affine order<=9 outputs and the cycle11 triangle exclusion remain unchanged. The whole finite outputs above have rank<=12, or are separate-component coupons, and therefore avoid all deep finite8 zeros and recent finite long-pair reductions. This proves one simultaneous local rule with triangle minimum2 and quadrilateral minimum1.

Verification scope

Final source-bound generation `287a9e` and fresh saved replay `b625d5` both exited0/PASS. A separate metadata audit passed all22 direct input/self bindings, all6 report links and source syntax. The parent and a second agent each completed a full source/report mathematical read, including the inherited physical diagram/profile and marked affine-type helpers. The parent also reread the old C endpoint theorem. Both checked the complete B9+A1 contact domains, C strict gap, all B profiles and native gaps, metric-Cauchy equality and whole-component sealing, distinct corner alternatives, explicit coupon increment, noncircular actual-corner priorities, and final full weighted-table and native-involution guards. Neither reviewer repeated profile, determinant, literal-corner, scalar or historical census arithmetic. No concrete gap was found in the stated local theorem.

The source performs only the exact small profile, determinant and literal-corner computations described above. Published finite/affine classification, actual factor-fiber geometry and the completed whole-face rule remain explicit inherited inputs. No native catalogue, arbitrary-support census, ideal convolution or finite DP has been rerun or claimed here. Source, certificate and report are final/frozen.

```
python3 scripts/fresh_order11_bc_zero_transfer_20261018.py
python3 scripts/fresh_order11_bc_zero_transfer_20261018.py --verify
```

Source 192

Source path: `results/fresh_order11_bcd_half_pair_transfer_20261018.md`

SHA256: 7e59403fa47e5195b089b558554c000448ad803d805bd60095244d1ac62be3b7

Coupled half payments for the order11 affine B, C and D factors

For whole varying affine factors of order11, halve the two terminal/fork pairs in B10, the one terminal pair in C10, and the four opposite-fork leaf pairs in D10. Their total pair costs become54,109/2,53 respectively. Every affected quadrilateral still receives at least1, provided the finite rule includes a half between a whole A1 and the doubled terminal of a separate whole B10. This adds exactly1/2 per such component pair. No other finite coupon is added. The proof allows arbitrary supporting spectators and all native labels.

Together with the frozen cycle11 theorem, the four order11 costs are

Family	Total pair cost
A cycle	99/2
B	54
C	109/2
D	53

The all-family normalized cost at affine rank10 is therefore109/22. The source and certificate check the exact identities and the complete finite boundary cases below. The finite coupon budget and global dimension comparison are separate obligations. No new dimension bound is asserted by this packet.

Intrinsic marked pairs and actual-star setup

All roots have physical squared norm2. A simple bond has contact magnitude1 and a doubled bond magnitude sqrt2. The following canonical descriptions specify automorphism-invariant pair sets; numbering only records the checks.

- C has path0,…,10, with doubled bonds0–1 and9–10. Halve(0,10), the two doubled terminal leaves, at distance10.
- B has spine0,…,8, doubled bond7–8, and fork leaves9,10 at0. Halve(8,9),(8,10), each joining the doubled terminal to a fork leaf, at distance9.
- D has spine0,…,6, fork leaves7,8 at0 and9,10 at6. Halve all four pairs between opposite forks, at distance8.

Write an actual quadrilateral star as W+a+b+c+d, with opposite pairs {a,c},{b,d}. At the chosen affected ideal corner F is the whole varying affine factor containing a,b. Its remaining supporting diagram Z is positive definite and orthogonal to F. These are the inherited actual factor-fiber facts. Every actual edge W+one black is strictly positive definite; every actual corner is positive semidefinite. The positive kernel nu of F gives strictly positive heights h=−c·nu and k=−d·nu.

Any adjacent black contact, or a shared nonzero spectator coordinate of c and d, gives a marked path of length1 or2 at another actual corner. Those pairs retain units in the simultaneous rule, so they already finish the payment. Henceforth adjacent black products vanish, and the native nonnegative contact vectors x,y to Z have disjoint supports. This is a branch of the proof, not an assumption about all quadrilaterals.

Complete all-label end profiles

Deleting a marked leaf from any of these three factors leaves a connected finite B10 or D10. Its simple long endpoint lies next to the omitted leaf (or is the old mate at a fork). The only possible positive-semidefinite one-root extensions, when restricted to this old ten-root component, are

1. no contact;
2. a simple contact at that long endpoint;
3. a doubled contact at that endpoint;
4. a simple contact at the next node.

Here is the complete classification argument. An extension with two old contacts would contain a cycle. A connected finite Coxeter diagram is a tree; the only connected affine diagrams with a cycle are affine A, which cannot contain the old doubled bond or fork. Exceptional finite or affine diagrams are too small to contain B10 or D10. For one contact, the native inverse diagonals of either old component are1 at the long endpoint,2 at the next node, and greater than2 everywhere else. The Schur condition q² Jii<=2 therefore permits only q=1 or sqrt2 at the endpoint, and only q=1 at the next node. This includes equality and imposes no label cutoff: the next native magnitude after sqrt2 is phi, whose square exceeds2. Adding spectator roots cannot repair a non-PSD principal extension.

The opposite product with the omitted old mark has magnitude t>=0 and is not restricted by that corner extension. It can be zero, an ordinary native magnitude, parallel or dotted. A positive nonzero native magnitude is at least1; if it exceeds1 it is at least sqrt2. The isolated profile requires t>0 because its height is positive. Call the four profiles isolated, end3, end4, next3. At a fork end, end3/end4 attach to the mate and next3 attaches to the fork center.

Choose kernel coefficients1 on the spine,1/sqrt2 at doubled terminal leaves, and1/2 at short fork leaves. Eliminating the nine supporting roots in F leaves the scaled old-mark matrix

```
(1/10) [[1,-1],[-1,1]].
```

For each new root write its height h, moment l, and residual constant c0 as follows; its end cost is g=2l/h+c0/h².

End	Profile	h	l	c0
Heavy	isolated	t/sqrt2	0	2
Heavy	end3	t/sqrt2+1	1	1
Heavy	end4	t/sqrt2+sqrt2	sqrt2	0
Heavy	next3	t/sqrt2+1	2	0
Fork	isolated	t/2	0	2
Fork	end3	(t+1)/2	1	1
Fork	end4	(t+sqrt2)/2	sqrt2	0
Fork	next3	1+t/2	2	0

The exact face Schur determinant, which vanishes because the actual quadrilateral quotient has dimension3, gives

```
0 = 1-(g_c+g_d)/10
      +(x/h-y/k)^T G_Z^(-1)(x/h-y/k)/10.              (1)
```

The source checks every physical Gram, the24 end rows,48 cross rows and the symbolic determinant identity. The spectator quadratic form is nonnegative; no limit on its rank or number of components was used.

Affine alternatives are distinct second halves

Every end4 or next3 profile gives a whole affine order11 factor at the adjacent corner replacing the omitted old mark by the new root. Its positive kernel, together with corner PSD and nonpositive products, forces every additional supporting root orthogonal to that factor. Thus its type and marked roles remain valid in the whole actual corner.

Original end	end4 target	next3 target
C heavy	C terminal pair	B terminal/fork pair
B heavy	B terminal/fork pair	D opposite-fork pair
B fork	C terminal pair	B terminal/fork pair
D fork	B terminal/fork pair	D opposite-fork pair

All targets are exactly pairs retained at1/2 by this theorem. The original corner retains1/2 and this different actual corner contributes1/2. This is a simultaneous two-half payment, with no recursive appeal to another transfer theorem. The source checks all12 oriented instances, positive kernels, whole factor types, marked leaf roles and distances.

Only isolated and end3 profiles remain. A heavy isolated cost is at most4; a heavy end3 cost is at most3. At a fork the maximum8 occurs exactly for isolated t=1 or end3 t=0. In either case the height is1/2, and the root has exactly one unit contact to the selected old leaf or its mate. Call these two profiles extreme. Every other remaining fork profile has cost at most4: an isolated nonextreme t is at least sqrt2, while an end3 nonextreme t is at least1 and has cost at most3. These bounds follow directly from the positive-denominator formulas, which decrease with t.

For C the total cost is at most8, contradicting(1). For B an extreme fork is necessary. For D at least one end must be extreme. This leaves only one extreme plus a nonextreme, or D with two extremes.

One extreme end: coordinate Cauchy and whole finite roles

Let the extreme end have height1/2 and contact vector x, and let the other end be nonextreme. Equation(1) requires

```
E=(2x−y/k)^T G_Z^(−1)(2x−y/k)=g_other−2,  0<=E<=2.
```

For a nonzero x-coordinate i, disjoint supports and Gram-metric Cauchy give E>=4x_i²/2>=2. If E<2, x is empty. At equality E=2, necessarily x_i=1 and equality in Cauchy forces

```
2x−y/k = G_Z e_i.                                    (2)
```

The sign pattern in(2) makes i the only x-coordinate. If i has a neighbor j in Z, then y_j>0, giving the c–i–j–d marked path of length3. That unchanged full unit finishes the proof. Otherwise i is a whole isolated A1 component, and(2) forces all remaining contacts of both roots to vanish. Consequently the selected extreme root has either no spectator contact, or exactly a unit contact to a whole A1, with every other supporting component orthogonal to the selected corner components.

Choose the adjacent corner containing that extreme root and the far old mark. If the extreme root attached to the omitted selected old leaf, it lies in a separate whole A1 or A2, while the far old mark lies in the whole B10 doubled terminal (B case), or a whole D10 short leaf (D case). If it attached to the old mate, the marks are in one whole finite B11/B12 or D11/D12 component. Thus the selected corner supplies at least1/2:

- A1/B10 doubled terminal uses the new half in this theorem.
- A2/B10 doubled terminal uses the existing typed coupon.
- A1/A2 versus D10 short leaf uses the existing small-component half.
- The joined finite component has a retained full unit.

The original ideal half plus this distinct corner's payment totals at least1. This argument explicitly determines whole components, so it does not mistake an induced A1/A2 or B10/D10 subdiagram for a coupon role.

Two D extremes: the complete spectator energy3/2 split

Both heights now equal1/2 and both end costs equal8. Equation(1) becomes

```
(x−y)^T G_Z^(−1)(x−y)=3/2.                           (3)
```

If either support is empty, select that extreme root and use the spectator-free corner just proved. Otherwise both supports are nonempty. Both separate augmented diagrams Z+c and Z+d are strict PD principal subdiagrams of actual edges. Therefore each root contacts at most one node in any connected spectator component, since a second contact would create a cycle in a finite diagram.

If a component has contacts from both sides at distance<=3, it gives a c–d path of length<=5. Such an ideal or finite corner contributes at least 1/2 in the simultaneous rule: only the cycle11 distance5 orbit can have half rather than a full unit. The original half then finishes the payment.

Suppose a mixed component has contact nodes at distance>=4. Apply the all-rank inverse calculation and positive quadratic remainder in the five-quarters spectator proof. We use its intermediate inequalities, not its conclusion for energy5/4. Writing the inverse block as [[a,c],[c,b]], the contribution is a p²+b q²−2cpq, with p,q>=1 and a,b<2 by strict separate extension. The complete possibilities with energy<=3/2 are as follows.

Whole component and roles	a	b	c	Unit energy
A5 endpoints	5/6	5/6	1/6	4/3
A6 endpoints	6/7	6/7	1/7	10/7
A7 endpoints	7/8	7/8	1/8	3/2
D6 long/short leaves	1	3/2	1/2	3/2
E6 eligible endpoints	4/3	4/3	2/3	4/3

For completeness, in A_r put N=r+1 and let d be the contact distance. The unit energy is d(N−d)/N>=2d/(d+2). Thus d>=7 is impossible. For d=4,5,6, the condition energy<=3/2 respectively gives N<=32/5,50/7,8; together with N>=d+2 these force N=6,7,8 and the endpoint pairs above. B has no far pair of allowed contact positions. In D only a long/short pair in D6 or D7 is possible, with energy r/4, so D7 is too large. E6 gives the displayed block; E7 has at most one allowable node and E8 none. F,H and I2 have diameter<=3. These are all ranks and native labels.

In each displayed block, either nonunit contact is at least sqrt2. The source checks that the gradients are nonnegative at(sqrt2,1) and (1,sqrt2), and the quadratic remainder is positive semidefinite. The minimum on either corresponding quadrant is therefore attained there, and both values exceed3/2. All surviving contacts are unit.

Every active component contributes at least1/2. For a one-sided component this follows from Jii>=1/2 and a native magnitude>=1, or from the complete spectrum below one. For a mixed component with disjoint contact nodes i,j, Gram-metric Cauchy applied to e_i−e_j gives

```
E >= (p+q)²/(4−2Gij) > 1/2,
```

because the positive-definite two-node block has Gij>−2. This lower bound is used to count additional active components, not added coordinatewise inside one component.

The unit A5,A6,E6 energies leave less than1/2 before the total3/2, so no other component can supplement them; alone they do not satisfy(3). Only a whole A7 endpoint pair or whole D6 long/short pair remains, with all other spectator contacts zero. These are whole components of Z by construction, not smaller subdiagrams chosen inside a larger spectator.

For A7 there are four choices of the two extreme attachments (selected old leaf or mate). If at least one attaches to its selected leaf, the c–d corner is finite with marked distance8 in whole A9 or A18 and has an endpoint mark. If both attach to mates, an adjacent finite whole D18 corner has a marked leaf at distance9. Each is a retained full unit. For D6 the two orientations and four old attachment choices give eight literal cases. The strict edge/corner tests exclude six; each remaining case has a finite whole E8 marked path of length6. The source checks all12 configurations with exact rational inertia, every actual edge, all four corners, component roles and finite marked distances. Other untouched spectator components are orthogonal direct summands and change none of these checks. The D18 leaf9 unit is retained: the large-D leaf halves begin at order20. The endpoint distance8 pairs cannot be depth11 pairs.

It remains to treat(3) with no mixed component. Every active component is one-sided with energy>=1/2, so at most three are active. With three, all have energy1/2; the below-one classification makes them whole A1 with unit contacts. Both roots have support, so one of them contacts exactly one A1. With two active components, each side has exactly one; the smaller energy is at most3/4. The same complete classification makes that side a whole A_r with a unit endpoint contact, r<=3. Select this extreme root.

If it attaches to its omitted old leaf, the adjacent corner has whole A_(r+1), of rank<=4, versus a separate whole D10 short leaf, earning the existing half. If it attaches to the mate, the adjacent marked component is whole finite D_(11+r), of rank<=14, earning a unit. The source checks all these cases for r=0,1,2,3, as well as the four B cases above:12 exact finite corners with their positive pivots and whole-component roles. This completes every support pattern in(3).

Triangles and simultaneous later interfaces

All changed B/C/D pairs have distance at least8. The inherited finite7/ideal5 triangle theorem uses a finite marked path at most7, an ideal path at most5, a protected ideal path6/7 near a feature, or a whole affine factor of order at most9. None of those units is changed here. The distinct cycle11 distance5 halves do not affect triangles, by the frozen cycle11 triangle exclusion. The new finite coupon only adds weight. Thus triangle payments remain valid without a new triangle census or coupon.

For quadrilaterals, if no actual corner has a changed B/C/D pair, its old coverage is unchanged (with any separate cycle11 change handled by that theorem). If a changed B/C/D corner exists, apply the complete proof above at that actual corner. Its immediate affine alternatives contribute a distinct half directly. No step asks a second changed B/C/D pair to supply a full unit, and no branch uses a refined order17 ideal orbit.

The finite outputs are whole components of rank<=14, whole A9/A18 endpoint distance8, whole D18 leaf9, whole E8 path6, or the precisely typed small A/D10 and A1/A2/B10 halves. They survive the depth11 zero rule on deep finite distance8 pairs and the D16/D17 exchanges. In particular the proof does not consume a generic whole-D16 or whole-D17 long-pair unit. The cycle11 proof uses finite paths<=6 or ideal paths<=5; the latter still contribute at least half after every change here. Conversely, the short<=5 branch above also requires only half from a cycle11 corner.

The separate order17 half-coupon successor can safely use a short8 alternative: even if it is one of the newly changed D10 pairs, the alternative corner retains1/2 and the original changed order17 corner retains1/2. This observation is an interface statement, not a new proof of that successor's other geometry. All new weighted sets are invariant under the automorphism groups of their whole native factors.

Verification scope

Generation35489d and separate fresh saved replay5544c5 both terminated exit0/PASS. The exact source checks24 end identities,48 cross identities,12 whole affine alternatives, 12 small finite corners, the complete five far inverse blocks and12 sealed far spectator configurations. The all-rank arguments are the classification, inverse formulas, positivity and support reasoning written above; bounded matrix checks are not being used as a rank or label cutoff. No previous selected-map census is needed for this universal local theorem.

The parent completed a full source/report mathematical read, passing the complete B10/D10 one-root classification, physical R10 identities, nonrecursive distinct affine halves, equality-Cauchy whole A1 sealing, complete energy3/2 far blocks and nonunit exclusions, literal A7/D6 boundary cases, unmixed whole A_r selection, and triangle/depth11/finite exchange/cycle11 interfaces. A second agent independently completed a full source/report mathematical and interface read of the same arguments, including the positive-quadratic remainder on each nonunit quadrant and the order17 half interface. Neither review reran the matrix, profile, scalar or inherited census arithmetic. Both found no concrete gap in the stated universal local theorem.

Direct audit60a6fd passed all25 source bindings, all9 report links and source syntax. Source and data were unchanged between generation and fresh replay; only this verification account was finalized afterward. Source, certificate and report are final/frozen. External refereeing, formal verification, a finite budget or a new ambient bound are not claimed by this local packet.

```
python3 scripts/fresh_order11_bcd_half_pair_transfer_20261018.py
python3 scripts/fresh_order11_bcd_half_pair_transfer_20261018.py --verify
```

Source 193

Source path: `results/fresh_order11_c_endpoint_zero_transfer_20261018.md`

SHA256: a94f1c701ecbb22e3af650c81439cfb386e91bbb79c146fa1fe5eb3f13c67363

The order11 affine C endpoint half can be deleted

In the whole varying affine C10 factor of order11, assign weight0 to its two doubled terminal leaves instead of the previous1/2. Every affected quadrilateral still receives at least1 at another corner. The C total decreases from109/2 to54, with no new finite coupon and no change to any other weight. Combined with the completed order11 rules, the family totals are A99/2, B54, C54, D53, so the normalized rank10 envelope is54/11.

The source and certificate check the full native profile calculation. This is a universal local theorem, with arbitrary supporting spectators and all native labels. It is compatible with the completed dimension63 rule, including cycle11 halves, B/D11 halves, the order17 refinement, finite depth11 zeros and the D16/D17 exchange. No new ambient bound is claimed.

Complete physical end profiles

Use an actual quadrilateral star W+a+b+c+d with opposite pairs{a,c},{b,d}. At the affected ideal corner let F be the whole varying C10 factor, with path0,…,10 and doubled bonds0–1 and9–10. Its marks are a=0,b=10. The remaining supporting diagram Z is positive definite and orthogonal to F, by the actual factor-fiber theorem. All physical roots have squared norm2. The positive kernel nu has coefficients1/sqrt2 at0,10 and1 at1,…,9. The opposite heights h=−c·nu and k=−d·nu are strictly positive.

If either c.b, c.d or d.a is nonzero, the corresponding actual adjacent corner has a marked path of length1. If c,d have a common nonzero contact coordinate in Z, their actual corner has a marked path of length2. Every such pair retains a full unit in the completed rule and the new rule. These are corners different from the chosen C endpoint corner. Thus suppose these alternatives are absent. The three adjacent-black products then vanish, and the nonnegative native contact vectors x,y to Z have disjoint supports.

The complete one-root extension classification from the coupled order11 theorem applies without discarding any affine alternatives. Deleting a or b leaves a whole displayed B10 principal diagram. Its norm-two inverse diagonals are1 at the simple endpoint near the omitted mark,2 at its next node, and greater than2 everywhere else. A one-contact PSD extension therefore allows a label3 or4 at that endpoint, or a label3 at the next node. Two old contacts create a cycle, impossible in a finite diagram or in a classical affine diagram containing the old doubled bond. Exceptional finite and affine diagrams are too small to contain B10. The no-contact profile is also allowed. These are all possibilities, not a label-limited census. Spectators cannot repair a non-PSD principal extension.

The magnitude t of the contact with the omitted opposite old mark is unrestricted except for its native Coxeter values: t=0, or t>=1, including parallel and dotted possibilities. The four profiles have the following exact physical quantities, with end cost g=2l/h+c0/h².

Profile on the old B10	h	l	c0	Native range	Maximum g
isolated	t/sqrt2	0	2	t>=1	4
endpoint3	t/sqrt2+1	1	1	t>=0	3
endpoint4	t/sqrt2+sqrt2	sqrt2	0	t>=0	2
next-node3	t/sqrt2+1	2	0	t>=0	4

For the isolated profile t>0 follows from positive height, hence t>=1. In every other profile positive height is automatic at t=0. All four cost functions decrease with t. The exact source also substitutes t=u+t_min and checks that maximum−g has a polynomial numerator with nonnegative coefficients and a strictly positive polynomial denominator on u>=0. Thus the stated maxima hold on the entire continuous rays, which contain every native label and every dotted opposite product.

The source independently constructs the norm-two C10 Gram, its positive kernel, the two B10 inverses and their only admissible contact positions. It rederives every height, moment and norm constant, and all16 cross identities. No project arithmetic helper is imported.

A strict contradiction without spending an affine half

Eliminating the nine white roots in F and scaling the old marks by their kernel coefficients leaves

```
(1/10) [[1,-1],[-1,1]].
```

The exact quadrilateral Schur determinant is the inherited identity

```
0 = 1-(g_c+g_d)/10
      +(x/h-y/k)^T G_Z^(-1)(x/h-y/k)/10.          (1)
```

The source rechecks its symbolic determinant directly. As in the actual factor-fiber framework, elimination of all supporting whites leaves the four side roots of a three-dimensional face quotient, so the determinant vanishes. The last term is nonnegative because Z is positive definite.

Every profile in the complete table has g<=4. Therefore the right side of(1) is at least

```
1-8/10 = 1/5 >0,
```

a contradiction. The source checks the16 ordered profile-pair bounds; their uniform minimum is exactly1/5. The spectator form was only bounded below by0, so this proof neither bounds spectator rank nor assumes the spectators empty or sealed.

In particular, endpoint4 and next-node3 are kept inside the strict calculation. The previous half theorem could use their other whole affine corners as second halves. That step is not used here: their costs are at most2 and4 respectively, so they cannot evade the contradiction even if their new affine endpoint payment has also been deleted.

Consequently at least one of the earlier direct alternatives must occur: an adjacent black contact or a shared spectator coordinate. A distinct actual corner then contributes a retained full unit at distance1 or2. The original endpoint weight is unnecessary and may be0. This proves quadrilateral coverage for every affected actual C corner, without a selected-core census or a completion assumption about other finite factors.

Simultaneous B/D, triangle and later interfaces

The new pair set consists of the unique terminal pair of the whole native C factor and is invariant under its reflection automorphism. It has distance10. All54 other pairs remain units, giving total54.

The simultaneous proof uses actual corners, rather than attempting to reuse the old two-half argument with a missing half:

1. If a quadrilateral has any newly changed C endpoint corner, the proof above supplies a full distance1 or2 unit at a different actual corner. This unit is unchanged by every completed finite or ideal refinement.
2. Otherwise every actual corner has exactly its completed baseline weight. The baseline face-payment theorem supplies its required total.

This directly handles the apparent coupling with the B/D order11 theorem. That proof sometimes outputs an actual C endpoint half. If such a corner occurs after its deletion, it lies in the first case and the new independent unit alternative pays the face. If no such actual corner exists, the old coupled proof's actual total is unchanged. There is no recursion between a missing C half and a B/D half, and no inference that the old half theorem alone justified weight0.

The finite7/ideal5 triangle theorem uses finite paths at most7, ideal paths at most5 or protected ideal paths6/7, and whole affine factors of order at most9. The changed distance10 pair in an order11 factor is none of those obligations. Triangle coverage therefore remains at least2. The cycle11 triangle exclusion is unchanged.

The only new terminal payments used here have distance1 or2. They survive the finite depth11 zero rule, all finite D16/D17 reductions, the order17 ideal half refinement, and every cycle11 or B/D11 half. No whole finite component or separate-component coupon is consumed. Every finite cost, including both enlarged coupons in the completed dimension63 budget, stays exactly as before. The order11 all-family envelope improves to54/11; the other still-open ambient63 ideal and finite obligations remain separate.

Verification scope

Generation08ec1b and separate fresh saved replay555738 both terminated exit0/PASS. Direct audit7e7778 passed all9 source bindings, all6 report links and source syntax. The parent completed a full source/report mathematical read, passing the complete B10 principal extension domain, the exact polynomial native-ray bounds for all four profiles, reflected cross identities, nonnegative spectator correction and changed-C-first actual-corner priority. It also passed the triangle and unchanged distance1/2 interfaces and absence of a new finite coupon. That review did not rerun the matrix/scalar computations or independently reprove the inherited finite/affine classification. No concrete gap was found. The checker covers both deleted B10 inverses, all four exact physical profiles, every native-ray upper bound, all16 cross/profile-pair bounds, the symbolic Schur identity, the changed orbit and exact weight total. The all-label completeness and arbitrary-support interpretation are the written finite/affine classification and actual-face argument above. No finite forest optimization, global scalar comparison, external refereeing, formal verification or new polytope is asserted here. Source, certificate and report are final/frozen. A requested additional peer read is not counted as completed in this report.

```
python3 scripts/fresh_order11_c_endpoint_zero_transfer_20261018.py
python3 scripts/fresh_order11_c_endpoint_zero_transfer_20261018.py --verify
```

Source 194

Source path: `results/fresh_order11_order16_dimension60_20261018.md`

SHA256: 9434e6c00eeb5bb52023aaf5c6411cffecd1d711bad1d99044f1b1a08c8ddeeb

Unrestricted finite-volume dimension bound60

The simultaneous local rule below excludes ambient dimension61. Together with the completed higher-dimensional exclusion, it gives dimension at most60 for every finite-sided, finite-volume real hyperbolic Coxeter polytope. Arithmeticity, compactness, simple cusps and a bound on the number of facets are not additional hypotheses.

The resulting contradiction is

```
0 >= 0*v_f + 64*v_i + 4650 > 0,
```

where v_f and v_i count finite and ideal vertices. The finite comparison is equality; the positive global constant still gives a contradiction when there are no ideal vertices. No vertex-count bound is used.

The source and certificate bind the completed local and counting inputs below. Final generation, fresh replay and two full internal mathematical composition reviews pass. This is a complete, internally checked computer-assisted proof; no unresolved mathematical step is identified in the reviewed chain.

One simultaneous local rule

Start with the completed dimension61 proof. Use the depth8 finite theorem and the complete finite transfers bound by the finite420 theorem. Their current finite costs are A15/B15/D15=103, A16/D16=111, B16=104, A17=115, B17=105 and D17=119. The last value includes the restored D17 halves needed for the new D16 exchange.

All four added finite coupon increments are present: A2/A13 endpoint upgrade2, A1/A13 endpoint half1, A2/A14 third-node half2 and A4-end/A12-third half2 per eligible component pair. All other costs are those of the complete finite420 rule.

Now apply the order16 D105 theorem and order11 B/C theorem. Their ideal family totals are:

Factor order	A	B	C	D	Maximum
11	99/2	53	52	53	53
16	80	105	103	105	105

The only additional finite cost is the whole A1/B10 doubled-terminal half upgraded to a unit, adding1/2 per component pair. This actual cost is counted explicitly below.

The whole-face argument uses actual-corner priorities. If an actual changed C11 corner occurs, its independent universal proof supplies a distinct full path1/2 unit. If an actual new B11 zero occurs, its universal proof supplies a path3 unit, a whole B11/B12 finite unit, the upgraded A1/B10 unit, or two distinct retained halves. The latter are A2/B10 together with A1/D10 or a whole affine D10 opposite-fork pair. None uses a newly changed ideal pair or a finite rank15–17 long pair. Thus these first two branches are independent of the old coupled B/C/D argument.

Next, a changed actual affine D15 pair of order16 is covered by its complete local theorem. Proper completions are excluded with arbitrary supporting contacts. Surviving whole factors use a distinct whole D5/D11 fork-center half. That coupon and its full component-pair cost15/2 are unchanged by every new finite transfer. Its independent short8 escape retains the

depth8-to-short7 replacement and the completed tree/cycle payment interfaces. Its nine adaptive finite outputs remain paid: seven protect a simple leaf at distance9 with a degree-two neighbor. Such a leaf cannot be a short-fork leaf affected by the large-D half rule. The other two either give the same protected leaf or retain the marked distance9 path inside a whole finite component of order at most16. All later finite rank15–17 reductions concern distances at least10; the depth8 change concerns distance8. Neither changes these outputs.

The D105 report originally checked literal records at the completed61 baseline, where finite D17 cost120. The present binder does not reuse those old totals. It matches all216 source/case identifiers and corner marks against the latest D16 exchange's literal audit at D17=119, and verifies that every D105 new debit is zero there. The latest finite totals remain at least1. This finite list covers inherited literal obligations; it is not a substitute for the full actual-face domain.

If none of the newly changed actual ideal corners occurs, all actual ideal weights agree pointwise with the complete finite420 local rule, and its proof applies. Finite weights have only increased by the last A1/B10 upgrade. The new ideal changes are at distance9; the retained finite7/ideal5 triangle argument and its protected6/7 interfaces are unaffected. Hence every triangle receives at least2 and every quadrilateral at least1 under one simultaneous rule.

Complete finite maximum remains420

The complete finite certificate includes all1,132 fixed-component vectors,854 residual tables and34,596 simultaneous type comparisons. It already counts every rank15/rank16 change, all four new coupons and the D17 restoration. Its fresh replay checks all1,276,730 last-item upper inequalities and52,094 attaining predecessors.

Only the new A1/B10 half-to-unit increment remains to be added. The previous actual doubled cost is1; the new doubled cost is2. The same-rank representative is A1/D10, whose unchanged doubled allowance is2. Every other actual and representative type-pair cost is identical. Thus the same complete representative upper rule dominates the new actual costs; no DP table changes or optimization is needed. This is a pointwise domination proof, not omission of the additional coupon.

The actual formal forests D61,2D17+D27,3D15+A16 and3D15+D16 still cost420 and contain no A1/B10 pair. The complete finite maximum is therefore exactly420. Formal forest attainment does not assert geometric realization by a hyperbolic polytope.

At ambient dimension61 the finite link has dimension60,61 vertices, and G=sum(min(j,60-j),j=0..60)=900. Its curvature is

```
(5*900 - 60*61)/2 = 420.
```

Every finite vertex consequently has nonnegative surplus, with lower bound0.

Every ideal vertex has surplus at least64

The ambient61 frontier provides complete exact factor envelopes and saved H-polynomials for every terminal rank60 partition with at most one part1 through8 and all other parts at least9. The binder independently reconstructs this entire2,365-profile domain and checks exact agreement.

For each saved polynomial it checks symmetry, positive coefficients, the sum V=product(r+1), its weighted moment G, curvature, the original factor cost and the saved margin relation. The polynomial convolution itself remains a completed predecessor computation.

Relative to the original frontier, the new normalized factor-cost saving is

```
count(rank10)/11 + count(rank15)/16.
```

These are the actual all-family maximum changes,54→53 and106→105. Every other rank-specific envelope is unchanged. The binder computes every new normalized and absolute margin exactly; all2,365 are positive. The normalized minimum is3/250 at(9,9,9,9,9,15), and the absolute minimum is64 at the single rank60 factor.

The inherited low9 concentration theorem only moves low-rank factors. Their normalized cost is still the linear function r/2 for r<=9. The move preserves that cost, decreases normalized curvature, and decreases V. It does not concentrate high-rank factors. Consequently an unreduced actual cusp has normalized margin at least its positive terminal margin and vertex count at least the terminal V. Its absolute surplus is therefore at least64. The actual factor-fiber multiplicity V/(r+1) is already included in these normalized costs. No simple-cusp or arithmetic hypothesis enters this step.

Global contradiction and higher dimensions

Let D5 be the sum of5 minus the number of sides over all two-faces. The simultaneous local rule and complete finite bound give

```
D5 <= 420*v_f + sum_(ideal v) C_v.
```

The inherited shape-curvature theorem gives the lower bound with one global two-extrema correction. Every coefficient of the simplex or product-of-simplices link H-polynomial is at least1. At link dimension60, each extreme vertex has

```
E-G >= sum_(2j>60)(2j-60) = 930.
```

The coefficient5/2 at each of the two extremes gives the single strict constant4650. This is the correction already present in the ambient61 frontier, counted once. Hence

```
D5 >= 420*v_f + sum_(ideal v) kappa(L_v) + 4650.
```

Subtracting the bounds and using kappa(L_v)−C_v>=64 yields the contradiction at the start. Dimension61 is excluded. The completed ceiling61 already excludes every dimension at least62, so the combined ceiling is60.

Compact arithmetic21 and arbitrary compact29 are unchanged. No new finite-volume example, external mathematician refereeing, publication-priority claim or proof-assistant formalization is supplied.

Verification scope

Generation `29dc73` and separate fresh replay `913e99` both exited0/PASS. The author and two peer agents completed full producer/report mathematical reads. The peers checked the independent C/B actual-corner priorities, D105's unchanged whole-component half and adaptive leaf/distance outputs, the latest216 literal interfaces, complete finite420 with the additional A1/B10 charge, the complete terminal partition domain and exact saving formula, positive-margin absolute extension, factor fibers and the single global correction. One peer additionally reread the full shape-curvature and factor-fiber proofs, adaptive-nine proof, and recent finite-transfer reports. Neither peer reran geometric censuses, DP, convolution or the 2,365-profile scalar arithmetic. These are internal agent reviews, not external mathematician refereeing or proof-assistant verification.

The final binder checks source integrity, current local cost interfaces, the extra coupon's representative domination, joint literal records, the entire terminal partition domain and saved-row scalar relations, finite curvature and the single strict constant. It does not rerun a historical geometric census, finite DP, scalar convolution or global optimization. Those completed inputs retain their own explicit verification scopes.

```
python3 scripts/fresh_order11_order16_dimension60_20261018.py
python3 scripts/fresh_order11_order16_dimension60_20261018.py --verify
```

Source 195

Source path: `results/fresh_order11_order17_dimension63_20261018.md`

SHA256: 1aaa4f9c0578b4c7d1b2dedd48fbadd7f723095ded0fe28dfdfdd95626e2784d

Dimension at most63 for finite-volume hyperbolic Coxeter polytopes

Every finite-sided, finite-volume real hyperbolic Coxeter polytope has **dimension at most63**. Arithmeticity, compactness, simple cusps and an extra bound on the number of facets are not hypotheses.

The source and certificate combine the new local rules and exact finite/ideal bounds below. The completed ceiling64 already excludes all dimensions at least65. It remains to exclude dimension64. This is a complete internally checked computer-assisted proof, with no identified unresolved mathematical step. It has not been externally refereed or formalized in a proof assistant.

A simultaneous complete face-payment theorem

Start from the complete finite464 payment rule. It includes the depth11 zero rule for deep finite distance8 pairs, the finite D16/D17 exchange giving costs116/127, and the affine cycle11 distance5 halves. Its triangles receive at least2 and quadrilaterals at least1. Add the following compatible changes.

The order11 tree theorem halves the two terminal/fork pairs of affine B10, the endpoint pair of affine C10, and the four opposite-fork pairs of affine D10. Add a finite half between a whole A1 and the doubled terminal of a separate whole B10. The ideal family totals are now A99/2, B54, C109/2, D53. The largest is109/2.

The order17 theorem lowers four distance10 pairs in the whole affine D16 factor from3/4 to1/2. Upgrade the finite coupon between an E6 length2-arm endpoint and a D11 short leaf from1/4 to1/2, requiring both components whole. The factor's cost falls from119 to118. Its companion family totals remain A85/B111/C108. There are four eligible finite marked pairs per E6/D11 component pair, so the finite component-pair cost rises from1 to2.

All other finite and ideal weights stay as in the baseline. The new finite coupons only increase payments. The following actual-corner case split proves simultaneous quadrilateral coverage and makes the dependency order explicit.

1. If a cycle11 distance5 corner is present, the independent cycle11 theorem gives its original half plus a distinct retained half, or a finite short unit. Its finite outputs have distance at most6; its other ideal outputs have distance at most5. None is changed by the new tree or order17 rules.
2. Otherwise, if a changed order11 tree corner is present, apply the complete coupled tree theorem at that corner. Its immediate affine alternatives contribute a distinct half directly, without recursively demanding a full unit from another changed pair. The other alternatives are finite whole components of order at most14, whole A9/A18 endpoint pairs at distance8, whole D18 leaf9, whole E8 path6, or the stated typed finite half coupons. These all survive the depth11 zero and finite D16/D17 exchange rules. No alternative uses an order17 distance10 pair.
3. Otherwise, if a changed order17 D corner is present, use its complete inherited geometric classification. The64 proper completions remain impossible unless the short8 alternative pays. A finite short8 pair retains its unit unless it is a deep distance8 zero; in that event the constructive depth11 corridor gives a short7 alternative. Its finite and ideal replacements avoid the order17 distance10 orbit. The old cycle6/7/8 and radius-five tree replacements retain their checked depth11 interfaces. In particular every pair at distance at most8 in the18 current tree targets of orders12 through17 is a literal unit. For cycle arc length8, complementary lengths8 through11 have protected nondeep finite outputs, including the sealed cycle16 case; lengths at least12 have the strict depth11 cycle-interface gap7/48. Larger tree alternatives use retained finite short paths, whole-leaf9 roles or separate coupons. If any resulting finite8 output is deep, its independent corridor supplies the protected short7 route just described. No new order11 corner changes

in this branch. Thus the short-path route is still sufficient. In each of the four whole-affine cases, the19-root radical seals a distinct whole E6/D11 corner; its new half plus the original ideal half equals1. No finite same-component D16/D17 unit is used.

4. If none of these new ideal corner types occurs, every ideal weight agrees with the complete baseline rule, and the only finite changes are nonnegative coupon increments. Its old quadrilateral coverage therefore remains valid.

The first case also handles a cycle11 corner that was already halved in the finite464 baseline; applying its standalone theorem again is harmless and avoids hiding a changed short-path payment inside another proof. The cases are exhaustive and do not assume that a removed unit survives.

Triangle coverage is inherited from the finite7/ideal5 theorem and the cycle11 triangle exclusion. The changed tree distances are8,9,10, and the changed order17 distance is10. Finite paths at most7, ideal paths at most5, protected ideal6/7 pairs and the small whole affine triangle factors retain the required payments. The cycle11 theorem excludes its affected corner on any triangle. Thus every triangle receives at least2 under the full simultaneous rule.

The enlarged finite budget is still464

The complete finite coupon certificate includes both additions: total E6/D11 component-pair cost2, and A1/B10 component-pair cost1/2. It retains all previous coupons, deep11 zeros, and D16/D17 costs116/127. The new B10 coupon remains dominated by replacing that finite component with D10; the E6 representative is retained separately.

All finite types and all rank64 forests are covered by753 large-component vectors,348 coupon environments,38,025 type/partner domination comparisons, 597,168 Bellman upper inequalities and22,272 attaining predecessors. Every global alternative has a reconstructed attaining forest and literal coupon recount. The exact maximum is464, attained formally by4D16. This formal attainment is not an existence assertion about an actual vertex.

The finite curvature coefficient is also464. A finite vertex figure has H(t)=1+t+…+t^63, V=64 and G=992, giving

```
(5G−63V)/2=464.
```

No positive finite slack is claimed or required.

Every ideal vertex has positive surplus

The independent ideal-margin calculation uses the proved order11 envelope109/22 and order17 envelope118/17, with every other current factor envelope unchanged. The inherited all-rank concentration theorem reduces total affine rank63 to2,419 profiles: arbitrary multiplicities of9 through17, at most one1 through8, and at most one rank at least18. The lower concentration boundary is9 after the rank10 cost change; the older1,513-profile domain would be insufficient.

For H(t)=product_i(1+t+…+t^r_i), V=H(1) and G=sum_j[t^j]H(t)*min(j,63-j), every normalized margin

```
(5G−63V)/(2V)−sum_i c(r_i)
```

is positive. Its unique minimum is2385/146432 at(10,11,12,15,15). The minimum absolute margin is240/23, attained by(63). Both normalized margin and V decrease under the permitted concentration moves, so the positive terminal absolute minimum bounds every actual cusp. This covers all cusp factor counts and ranks, without an arithmetic or simplicity assumption.

The strict global contradiction

The completed corner-payment theorem bounds the global defect D5 above by

```
D5 <= 464∗v_f + sum_(ideal v) V_v∗sum_i c(r_vi).
```

The inherited shape-curvature inequality, with its quantitative two-extrema strengthening, gives

```
D5 >= 464∗v_f + sum_(ideal v) (5G_v−63V_v)/2 +5120.
```

Here5120=5*floor(64²/4). The two actual global extrema and the positive coefficients of every product-of-simplices vertex figure supply this constant; there is no new geometric hypothesis. Subtracting the two bounds and using the absolute ideal surplus gives

```
0 >= (240/23)*v_i +5120,
```

which is impossible even when v_i=0. Dimension64 is therefore excluded. The completed ceiling64 supplies the entire higher-dimensional tail, so the resulting unrestricted ceiling is63.

Compact arithmetic remains at most21 and arbitrary compact at most29; their separate sharper arguments are unchanged. No new finite-volume example follows from this nonexistence theorem.

Verification scope

Final generation753c2e and separate fresh replay0bdd94 both terminated exit0/PASS. All34 direct bindings, Python syntax and nine local report links pass. The binder reconstructs the five newly changed family interfaces, checks all short7/short8 roles, including all18 inherited small-tree short8 targets, and finite coupon amounts, and verifies the finite464, ideal240/23, constant5120 and inherited higher-tail inputs. The new ideal source independently recomputes all2,419 scalar rows with a different enumeration and polynomial implementation. The enlarged finite budget has exact upper and attainment certificates.

Two separate full source/report mathematical composition reviews passed. They reread the relevant actual-corner, triangle, deep11 cycle/tree, finite-budget, factor-fiber, shape and concentration interfaces. The final review explicitly strengthened the distance8 branch to check all18 current small tree targets and to state the complete cycle8 and radius-five tree alternatives; final generation and replay include that change. A separate saved-only check also confirmed those18 tree pair sets. Neither reviewer repeated the2,419-row scalar computation, Bellman arithmetic or historical geometric censuses. The second full ideal-scalar review arrived after its local report froze and is recorded here without modifying that report.

Geometric classification, whole-component sealing, actual factor fibers, shape curvature and all-rank concentration remain written theorem inputs, with their stated internal review scopes. Neither source hashes nor a numerical solver substitute for those arguments. Historical geometric censuses are not independently re-enumerated by this composition binder. No external mathematical refereeing or proof-assistant verification is claimed. Source, certificate and report are final/frozen.

Source 196

Source path: `results/fresh_order12_affine_b_half_transfer_20261017.md`

SHA256: 583e9762be6bfc7c6dd63f3d00f0467b41c9ffe2a52e4486c82a3e9864112cca

Two half-weights in the order12 affine B factor

The complete inherited face-payment rule may assign weight1/2 to the two pairs joining the doubled terminal of affine B11 to its short fork leaves. All other pairs retain weight1. The order12 factor cost is therefore65 instead of66. No finite coupon is added, and this change is compatible with the whole finite A17 endpoint deletion.

Number the affine B11 diagram by the spine0–1–…–9, with the doubled bond8–9 and simple leaves10,11 at0. The changed pairs are(9,10),(9,11), both at distance10. They form the two-element orbit under the leaf swap. Every basic, short8 and elementary-triangle pair retains its full unit.

The source and certificate bind the complete order12 support diagnostic, the frozen2,614-context overlap proof, and the complete predecessor face-payment theorem. This is a refinement of that coverage theorem; it does not assert a universal transfer for an arbitrary quadrilateral containing a B11 pair without the stated coverage routes. No A/C order12 cost or ambient dimension consequence is asserted.

Complete provenance before the local calculation

The initial triangle and short-path reductions are unchanged because the two pairs have distance10. Protected finite-leaf9 and basic ideal feature payments are unchanged as well. The high-disjoint branch is excluded by the inherited all-label theorem independently of these weights.

The order12 diagnostic reconstructs all66 actual low-core chosen corners and their3 optional sealed-root extensions. Its actual whole affine B11 outputs use only canonical pairs(8,10),(8,11), retained at weight1. It does not infer low-core safety by embedding a finite pattern in an arbitrary affine completion.

The complete high-overlap support has exactly14 affected contexts:

`98,100,119,121,164,167,171,1452,1728,1734,1745,2121,2127,2138`.

All28 induced maps are recomputed from their saved native labelled patterns. They map the preferred marks to the changed orbit. Every other selected high pattern is unaffected. The only remaining immediate provenance is the `inner4` alternative of the universal affine-D13 transfer, treated separately below.

Actual roots and normalization

Use an actual quadrilateral star W+a+b+c+d, with W and every actual edge diagram positive definite. Each saved core has exactly its displayed edges: the inherited overlap theorem excludes every extra cross-core edge or supplies a retained marked path of length at most8. All further supporting walls are white roots, with arbitrary native labels and arbitrary total supporting rank.

Physical roots have norm2. The saved rational Gram uses a positive diagonal congruence with squared scale s_i, so its diagonal is2s_i. For a new physical norm2 root contacting old root i by native magnitude q, the inverse-core singleton projection square is `q²*s_i*(G^-1)_ii`. All calculations below use this expression. They do not regard a congruently scaled root of diagonal4 as a norm2 root.

The checker reconstructs the white and four strict edge Grams and the complete chosen-corner contact domain. It then uses the following three exhaustive classes.

Eight radical cores seal, or give a short path

Contexts98,100,119,121,1728,1734,2121,2127 have15-root cores of signature(13,1,1). Their preferred corner contains the whole12-root affine B11 factor and one additional old white root. The possible contacts of an extra supporting white are exactly the two omitted black roots and this old white root. The checker verifies the core radical and its nonzero coefficient at each allowed contact location.

Because the core is indefinite, its Gram radical is an actual linear relation among the normals: the represented vector is null and orthogonal to a timelike vector, whose orthogonal complement is positive definite, hence it is zero. Nonpositive root products then force any nonzero contact vector to meet both signs of this relation. The saved graph distances give a marked path of length at most3 through the new white in every such case. That path receives an unchanged unit.

If no such short path exists, every additional white is orthogonal to the entire core. The displayed finite corners remain finite after restoring the orthogonal positive-definite extra-white diagram, and their components are whole actual components. The finite alternatives include a whole B12 unit or an inherited whole-A2/whole-D11 half at a short fork leaf. In all8 cases the total retained finite payment is at least1/2. The preferred ideal B11 corner now contributes1/2, giving at least1 in total. This includes contexts with only a single finite half; the source does not silently replace that half by a unit.

Six nonsingular cores: complete contact-domain reduction

Contexts164,167,171,1452,1745,2138 have14-root nonsingular Lorentz cores. The complete allowed contact domain splits into two end sets. A root meeting both sets gives a retained marked path of length at most8. Within each end, the inverse-core submatrix is entrywise positive. For two distinct contacted nodes i,j the source checks the all-label lower bound

```
s_i*Jii + s_j*Jjj + 2*min(s_i,s_j)*Jij > 2.
```

This is valid because each native nonzero contact is at least1, and `sqrt(s_i*s_j)>=min(s_i,s_j)`. Thus any surviving root has a singleton contact. Exact singleton tests leave just one node, an omitted black; its normalized inverse diagonal is2. A native label at least4 would give projection square at least4 and is impossible. Its only allowed contact has label3, projection square2 and residual norm0.

For contexts164 and1452 the selected connected component is finite B11 of order11. The sole contact location lies outside it. No extra white can touch it, so it is a whole actual finite component, and its two marks already receive a full unit. Other actual corner components lie inside the positive-definite white diagram. These contexts therefore cannot be proper affine-B11 completions despite their abstract induced maps into the12-node target.

The other four contexts have the complete affine B11 factor at the preferred corner. There can be zero or one nonorthogonal extra white. Indeed its zero residual lies in the span of the nonsingular Lorentz core. Two roots with the same forced singleton projection would be the same normal, contradicting strict white positive definiteness. Every core-orthogonal white is also orthogonal to that forced root. This excludes indirect tails and bridges, not merely a second direct contact.

The checker reconstructs both possibilities. For the one-root case, scale the new root by sqrt2: its diagonal is4 and its sole Gram entry to the old sqrt2-scaled node is-2, representing a physical unit contact. The enlarged15-root Gram has signature(13,1,1). Every finite corner is reclassified with the enlarged gauge. Its relevant coupon components are whole A1 or A2 against whole D11 at a short fork leaf, and retain the inherited half. A finite B12 unit can also occur. In every variant the finite total is at least1/2 and the preferred ideal half brings the quadrilateral total to at least1.

The full A1/D11 upgrade of the separate affine-D11 theorem is not needed for any of these B cases. The pre-existing A1/D11 and A2/D11 halves suffice. All same-component units here have order at most13, so the finite A17 endpoint deletion cannot affect them.

The universal immediate-D13 alternative has a strict gap

The affine-D13 factor has spine0–…–9 and leaves10,11 at0 and12,13 at9. Its `inner4` profile attaches the new opposite root to node1 by a doubled bond, with arbitrary nonnegative native opposite contact t at0. For the far mark b in{9,12,13}, its resistance is

```
R=9 when b=9, and R=11 when b is a short leaf.
```

Direct elimination gives height `h=t+sqrt2`, moment `l=sqrt2` and residual constant `c0=0`. Hence the end cost is

`g=2sqrt2/(t+sqrt2) <= 2.`

The complete opposite-end cost is at most6 for the center or8 for a short leaf, including profiles that formerly paid immediately. The normalized four-black determinant is therefore at least1/9 or1/11. Arbitrary spectators add the inherited nonnegative positive-definite quadratic correction, so they cannot cancel this strict gap. The actual star has zero determinant, giving a contradiction. The right-end case is the labelled reflection.

The two affected literal immediate pairs are(12,14),(13,14), mapping to the changed B orbit. The far-center pair maps to the retained(0,9), but its same strict inequality is also checked. Thus no universal immediate occurrence relies on keeping the B pair at weight1.

Composition and verification scope

The complete domain split covers all selected, low, protected, short-path and immediate uses of the two changed pairs. Every later transfer in the dimension69 chain ends in finite payments, paths of length at most8, or the separately handled whole factors; none adds a new long B11 terminal/fork provenance. Triangles keep their minimum2 and quadrilaterals their minimum1 with these two half-weights.

The source recomputes all28 affected induced maps, verifies14 literal core/edge/contact interfaces, reconstructs8 radical sealed cores and both variants of4 nonsingular whole-affine cores, proves the2 proper patterns already finite, and checks all3 immediate strict gaps exactly. The inherited all-label overlap construction, finite/affine classification, positive-cusp factor argument and full low-core closure remain explicit inputs; their original censuses are not rerun.

Final generationc00256 and separate fresh replay78846 both terminated exit0/PASS. All19 direct bindings, syntax and3 local report links pass. Root completed a full source, report and helper mathematical read. It passed the normalized projection lower bounds under positive gauges, radical sealing, singleton residual-zero extension and exclusion of indirect tails, the already finite B11 cases, the universal D13 strict gaps, and the retained actual half/unit coupon roles. A separate construction-lane full source/report read, including the finite-block and graph-distance helpers, also passed the same local and complete provenance mechanisms. Neither read duplicated the14-core or28-map arithmetic. The all-label census and low-core closure retain their inherited verification scopes. Source, certificate and report are final/frozen. No external refereeing, formalization or new ambient dimension bound is asserted by this local package.

```
python3 scripts/fresh_order12_affine_b_half_transfer_20261017.py
python3 scripts/fresh_order12_affine_b_half_transfer_20261017.py --verify
```

Source 197

Source path: `results/fresh_order12_affine_c_endpoint_reduction_20261017.md`

SHA256: 2496929ede336da13b8426c5564ef308e53fc9c764e4b3a28da6912b534176c5

Removing the endpoint pair in the12-node affine C factor

The finite8/ideal5 face-payment rule of the completed dimension69 proof remains valid after removing just the two-terminal pair of an affine C11 factor of order12. In the canonical numbering this is `(0,11)`, at distance11. Every other one of its65 pairs retains weight1. Triangles still receive at least2 and quadrilaterals at least1. The change introduces no finite coupon and leaves the existing finite budget unchanged.

This is a local composition theorem. It does not by itself lower the maximum over all order12 affine families or the ambient dimension bound. In particular, the A/B order12 obligations and any coupons introduced by their separate transfers require their own proofs and joint budget.

The source and certificate make the complete provenance check below. They use the frozen order12 support diagnostic and reconstruct its C-target embeddings with a separate standard-library path algorithm. No producer arithmetic module is imported.

The invariant pair and the elementary payments

The canonical diagram is the path0–1–…–11 with label4 at0–1 and 10–11, and label3 elsewhere. Its two automorphisms are identity and reversal, so the unordered endpoint pair is intrinsically defined. Scaling the norm2 Gram by the positive vector with endpoint entries 1/sqrt2 and all other entries1 gives the path Laplacian. This verifies the stated affine type, including its one-dimensional positive kernel.

All65 pairs other than the endpoints are retained. The basic ideal5/R5 rule requires63 pairs; its only additional retained pairs are `(0,10)` and `(1,11)`, both at distance10. Every pair of distance at most8 is basic and therefore remains paid.

The complete triangle theorem uses ideal distance at most5, or distance at most8 with a mark within5 of a feature. Its whole exceptional factors have order at most9. Thus it never needs the removed distance11 pair. All other factor rules and finite corner payments are unchanged, so its two triangle payments remain valid, including at nonsimple cusps.

The short8-first quadrilateral reduction

Use the complete finite8/ideal5 composition. It first treats any adjacent marked path of length at most8. A finite corner pays by the unchanged finite rule. An ideal C11 corner of order12 pays directly, since all its short8 pairs remain. The other affine families retain their old short-path rules and universal transfers.

Those universal transfers also remain valid when their *conclusions* are examined. The cycle replacements at orders13,14 and at least15 use finite whole-component, finite short-path or finite coupon payments; their ideal alternatives are an adjacent black bond or a shared spectator path of length2. Their classical-tree alternatives have marked distance at most9 with a persistent nearby feature. None can call for the C12 endpoint pair of distance11. This includes the protected finite-leaf9 argument: its whole-component sealing is a finite outcome, not an implicit all-small-affine payment.

It is therefore legitimate to pass to a star having no marked path of length at most8, just as in the complete predecessor. The choice of minimal indefinite opposite cores, closed-opposite-edge argument and actual-edge positive definiteness are inherited unchanged. There is no new simplicity, compactness or arithmetic hypothesis.

Complete low and high core provenance

The complete low-core domain has66 chosen-corner closures and three optional sealed simple-root extensions, giving69 actual component variants. These are actual closed components, rather than finite patterns allowed to sit in an arbitrary larger affine diagram. The frozen exact signatures and positive kernels give12 affine variants of order12: two B and ten D. There is no C variant. All finite outcomes remain paid by the whole-component rule. Consequently no low-core branch loses a payment.

The complete high-domain predecessor excludes disjoint cores and treats all2,614 surviving overlaps, with every additional cross-core edge either a short path or excluded by its saved quadratic certificate. For a possible C factor of order12, every relevant selected marked component must have size at most12 and a complete component bound at least12. The diagnostic includes **all758** such patterns. A smaller bound rules out the proposed whole factor; a larger selected component cannot embed in it. Finite, protected and basic outcomes are unchanged.

The new source reconstructs all induced weighted embeddings of these 758 connected patterns into the canonical C path. A graph embedded in a path must itself be a path, and its image is a contiguous segment; the algorithm tries every segment and both orientations and checks every edge and nonedge. Hence this is a complete embedding search, not a sampling of the saved successful maps.

Exactly15 contexts have embeddings, with30 maps in total:

`148,159,164,575,602,620,1081,1328,1352,1450,1452,1549,1630,1695,2573.`

Their marked images are precisely

`(0,9), (0,10), (1,10), (1,11), (2,11).`

All are retained. Only164 and1452 need the nonbasic distance10 orbit. The source compares every reconstructed map and marked image with the frozen diagnostic, including the absence of a map in every other pattern. Thus the removal is supported by the complete actual-star classification and closure arguments, rather than mere absence from a partial map list.

Immediate alternatives and every later transfer

The current rule also reaches some small factors indirectly. These must be checked separately from the selected high/low patterns.

The D13 long-pair transfer has six marked immediate alternatives on whole order12 factors. Its `inner4` alternative is B and its `inner_next3` alternative is D. All six induced maps and their positive-kernel provenance are bound by the diagnostic; none is C. The corresponding D14 alternatives have whole factor order13. The positive affine kernel makes these whole factors, so they cannot conceal a C factor of order12.

The later reductions used in the dimension69 rule have these outputs:

Transfer	Payments other than its original selected ideal pair
D13/D14 universal long pairs	The immediate B/D factors just checked; short paths; sealed finite units or halves
D12 opposite-leaf transfer	Finite units/full A1–D12 coupons/finite halves, or ideal paths of length at most4
B12 singleton and its later short-leaf upgrade	Finite A1–D12 or A2–B10 coupons/units; short paths; the old immediate B13 branch is excluded by a strict Schur gap
B14 three-pair reduction	Whole finite D14, A12 or A13 components
D16 singleton removal	Finite A4–D13 neighbor halves, or a shared-neighbor path of length2
Order17 half-weight reduction	Finite halves plus the retained half at the same order17 D factor
Order16 half-weight reduction	An impossible completion or finite halves plus the retained half at the same order16 D factor
Latest B13/B14 singleton completions	Projection contradictions, with no new payment

Here finite type names use rank and affine orders count nodes. The source binds each precise local theorem and checks the literal terminal payment interfaces:144 D13/D14 equality configurations,96 D12 configurations, six B12 singleton/conditional-long cases, seven B14 cases, the D16 singleton's four corners, all saved order16/order17 half-weight completions, and the latest64 impossible B completion maps. Positive finite payments are checked against finite corner signatures; positive ideal halves remain on the stated D factor. The analytic short-path branches and whole-component sealing are inherited from the cited proofs. The older 18 finite-only D replacements and their all-label selected support are already checked by the complete predecessor and are unchanged here.

This list covers the successive changes from the complete finite8/ideal5 rule through the frozen dimension69 rule. No branch uses the removed C12 endpoint as an alternate unit or half-payment. In particular, an old theorem phrased with a whole-small-factor cutoff is not imported without inspecting its actual output factors or paths.

Two newer independent local interfaces are also checked for compatibility: the cycle12 transfer has only finite coupons or ideal paths of length at most4, and the D11 leaf transfer has only finite payments or ideal paths of length at most5. Their conclusions are safe for this C deletion. Their new finite coupons are **not** charged to the unchanged budget claimed here; composing either newer rule still requires its separate finite-budget proof. No unfinished B12 reduction is assumed by this package.

The separate finite A17 endpoint deletion is compatible as well. It introduces no alternative ideal pair. Its only debited old literal case662 retains two whole finite D17 units, checked again here; every later typed terminal unit avoids that finite A17 pair. Its other selected, short and low routes are the same routes audited above. Thus combining these two deletions does not compensate one removed pair by the other. The exact revised finite maximum remains the separate finite-forest budget's conclusion.

Conclusion and verification scope

The modified rule pays every triangle and quadrilateral with the same minimum as the completed predecessor. Its C11 factor cost is65, so that factor contributes `65V/12` at a cusp with V link vertices, by the unchanged actual factor-fiber multiplicity. Other affine costs, all finite payments and their established finite budget are unchanged. There is no new ambient bound or construction claim.

Final generation and a fresh separate-process saved replay both terminated exit0/PASS. All61 direct bindings,11 report links and Python syntax passed. Root completed a full source and mathematical report read, passing the contiguous-path/reversal search including all rejected patterns, the69 actual low variants, whole B/D immediate factors, all later finite, short-path and same-factor-half interfaces, elementary payment retention and explicit A17 case662 compatibility. That review did not independently repeat the758-pattern embedding arithmetic or the inherited classifications. Source, data and report are final/frozen.

The new computation reconstructs the C-target embedding search and checks saved interface arithmetic. It does not repeat the all-label minimal-core classifications, the2,614 supporting-white closures, low-core exact inertia calculations or the local transfers' continuous spectator proofs. Those frozen proofs and certificates are explicit inputs. No external refereeing or proof-assistant formalization is claimed.

Source 198

Source path: `results/fresh_order12_b_center_terminal_transfer_20261018.md`

SHA256: 9cbca44be2e9b74c558e616237dbf2d0aedd30696fdfff31ddc08a125d903559

Removing the order12 B fork-center / doubled-terminal unit

For a whole varying affine B11 factor of order12, remove the pair joining its fork center to its doubled terminal. Every affected quadrilateral is paid elsewhere after adding a half-payment between either marked endpoint of a whole finite A2 and the doubled terminal of a separate whole finite B9. The order12 B cost falls from64 to63. The new finite component-pair cost is1. A joint finite-forest bound remains a separate obligation.

This is a universal actual-face argument, with arbitrary supporting roots and native labels. It does not restrict to the eight preferred-core contexts in which the singleton occurs. The source and certificate check its exact Schur identities, finite inverse diagonals and boundary corners. No new ambient bound or finite-volume example is asserted.

Complete endpoint profiles

Write an actual quadrilateral star as W+a+b+c+d, with opposite pairs {a,c},{b,d}. The original ideal corner has whole varying factor F and positive-definite spectator diagram Z, orthogonal to F. Actual edges are strictly positive definite and actual corners positive semidefinite, as in the inherited factor-fiber framework.

Number F by the spine0–…–9, with doubled bond8–9, and short leaves 10,11 at0. The removed marks are a=0,b=9. Normalize physical root squares to2 and cusp coefficients to1 on0,…,8, to1/sqrt2 at9, and to1/2 at10,11. The marked resistance is9. The singleton is invariant under the fork-leaf swap and has graph distance9.

An adjacent black contact already supplies a retained unit at a different corner. A common spectator contact of c and d similarly supplies a distance2 unit. Otherwise their native nonnegative spectator contact vectors x,y have disjoint supports, and all adjacent black contacts vanish. Opposite products remain unrestricted: each magnitude is0 or at least1, and a value greater than1 is at least sqrt2, including parallel/dotted cases.

Deleting a leaves finite B9 on1,…,9 and two isolated A1 leaves. The inverse diagonals of B9 are1,2,3,…,8,9/2. Its complete one-root PSD extension therefore has no contact, a simple or doubled contact at1, or a simple contact at2. Two contacts would create a cycle containing the old doubled bond, which no finite or affine Coxeter diagram permits. Exceptional types are too small. The inverse diagonal test covers every native label, including equality; it is not a finite label search.

If there is a doubled contact at1 or a simple contact at2, the resulting affine component has a positive kernel. All additional contacts to the two old A1 leaves and to spectators must vanish. If the contact at1 is simple, Schur elimination of the whole B9 block requires $u^2+v^2<=2$ for the two outer contact magnitudes u,v. Thus the possible outer contacts are none, one simple, one doubled, or both simple. If there is no inner contact, the actual edge requires $u^2+v^2<4$; no label enumeration is needed below.

Deleting b leaves finite D11. Its long endpoint8 has inverse diagonal1, its next node7 has diagonal2, and every other inverse diagonal exceeds2. Its four complete profiles are no contact, simple or doubled contact at8, and simple contact at7. The same tree/classification and Schur arguments exclude every other profile.

Put t=-a.c on the center side, or t=-b.d on the terminal side. The exact endpoint data are:

End and profile	c0	Moment 1	Height h
Center, outer contacts only	$2-(u^2+v^2)/2$	0	t+(u+v)/2

End and profile	c0	Moment l	Height h
Center, simple contact at1	1−(u²+v²)/2	1	t+1+(u+v)/2
Center, doubled contact at1	0	sqrt2	t+sqrt2
Center, simple contact at2	0	2	t+1
Terminal, isolated	2	0	t/sqrt2
Terminal, simple contact at8	1	1	t/sqrt2+1
Terminal, doubled contact at8	0	sqrt2	t/sqrt2+sqrt2
Terminal, simple contact at7	0	2	t/sqrt2+1

Each endpoint cost is g=2l/h+c0/h². Eliminating the ten old supporting roots and then Z gives the exact identity

```
0 = 1−(g_c+g_d)/9 + E/9,
E = (x/h−y/k)^T G_Z^(−1)(x/h−y/k) >=0.
```

The source checks all eight endpoint rows, all sixteen separated cross products, the marked Schur matrix and the formal determinant identity.

A unique center extreme and three terminal possibilities

The center cost is6 only for t=0 and a single unit outer contact. This means c is simply attached to one of the old fork leaves10,11 and has height1/2. Every other center profile costs at most4:

- Without inner or outer contacts, positive height gives t>=1 and g<=2.
- For one outer contact, t=0 gives g=8/u²−2, at most2 unless u=1. For t>=1, c0<=3/2 and h>=3/2 give g<=2/3.
- For two nonzero outer contacts, c0<=1 and h>=1 give g<=1.
- Inner simple profiles cost at most3 because 0<=c0<=1 and h>=1. Inner doubled and next-node profiles cost at most2 and4 respectively.

Every terminal profile costs at most4. A nonextreme center would therefore give determinant at least1/9, even before E, which is impossible. The center must be the stated extreme. The determinant then requires g_d>=3. The native gap and strictly decreasing terminal formulas leave only:

Terminal profile	Opposite magnitude	Cost	Height	E
isolated	1	4	1/sqrt2	1
simple at8	0	3	1	0
simple at7	0	4	1	1

For the next-node profile, the actual corner W+a+d contains the whole affine D11 factor of order12. Its positive kernel seals the factor. The marks are its fork center and an opposite short leaf at distance8, an unchanged unit under the completed payment rule. This directly pays the face. The literal next-node template checks only that affine corner; it is not asserted to realize an entire face star without further spectators.

The remaining spectator block is forced exactly

In the other two cases h=1/2 and E<=1. If x has a nonzero coordinate i, disjoint support gives (x/h−y/k)_i=2x_i. Gram-metric Cauchy gives

```
E >= (2x_i)²/G_Z(ii) >=2,
```

a contradiction. Thus x=0. For the simple-at8 case E=0 forces y=0 too. All spectators are orthogonal to the entire displayed core. The actual corner W+a+d has a whole finite D12 with marks at the fork center and long endpoint, distance9. Its retained unit pays the face.

For the isolated case k=1/sqrt2 and E=1 give y^T G_Z^(-1)y=1/2. A nonzero coordinate has native magnitude at least1. Equality in Cauchy forces y_i=1 and y=(1/2)G_Z e_i. Nonnegative contacts and nonpositive off-diagonal Gram entries

force i to be an isolated whole A1 in Z, with every other contact zero. Thus exactly one spectator e attaches simply to d, while every remaining supporting root is orthogonal to F,c,d,e. Indirect tails and bridges are excluded by this equality, not merely omitted from a search.

The two relevant finite corners are consequently whole components:

Corner marks	Marked components and roles	Payment
b,c	B9 doubled terminal; A2 endpoint	new1/2
a,d	D11 fork center; A2 endpoint	existing1/2

The existing half is part of the D11/D13 coupon theorem. These two distinct corners total1 without the removed ideal unit. The third finite corner has separate A2 marked components and is not needed. Both choices of the old fork leaf give the same roles.

The source reconstructs both isolated-spectator cases and both simple-at8 cases, their full signatures, strict white and edge diagrams, all corners and actual finite component roles. Physical doubled bonds are handled by a recorded rational congruence; scaled diagonal4 roots are not treated as physical norm4 roots. The two next-node affine corners are checked separately.

Simultaneous rule and finite cost

The order12 B/C theorem already leaves exactly64 B pairs. Deleting only(0,9) leaves63. All short8 and elementary triangle pairs remain. No triangle payment changes.

For quadrilaterals, give priority to any actual corner with the newly removed B singleton and apply this universal theorem. Its outputs are distance1/2 units, the whole affine D11 center/leaf distance8 unit, the whole finite D12 center/endpoint distance9 unit, or the two stated whole finite halves. None is changed by the order11 C endpoint-zero proposal, the existing deep finite zeros or the finite D16/D17 exchange. If there is no affected actual B corner, all its old payments agree with the completed rule. This avoids an unproved provenance assumption for earlier short-path or immediate alternatives that happened to use the singleton.

The new half has two eligible A2 marks and one B9 doubled terminal, adding exactly1 per whole A2/B9 component pair. B9 can no longer be silently replaced by the old rank9 representative in a finite budget; its new interaction must be counted explicitly. The local theorem by itself supplies neither that maximum nor a new dimension bound.

Verification

Generation5661ec and a separate fresh replay494d67 both terminated exit0/PASS. A separate full mathematical/source review passed the complete principal extension domains, all-native cost bounds, R9 identity, Cauchy and isolated-A1 sealing, physical gauge interpretation, all whole-component payment roles and simultaneous actual-corner/triangle interfaces. The reviewer did not repeat the matrix calculations or perform a finite-forest optimization. Finite/affine classification, the actual factor-fiber setup, and the previously retained weights remain inherited mathematical inputs. No geometric core census is required for this universal proof, and no external review or proof-assistant formalization is asserted. Source, certificate and report are final/frozen after these checks.

Source 199

Source path: `results/fresh_order12_bc_near_terminal_half_transfer_20261018.md`

SHA256: 08c042d7282dbbd6f43ee2ef4c675dedf1dfbbc0b1a13951250eec344eaacb9f

Universal order12 costs B62 and C62

The affine order12 family maximum falls from63 to **62**: B63→62, C63→62, with A60 and D62 unchanged. The B change halves two distance9 pairs; the C change deletes one distance9 pair. The only new finite payment is a half between a whole A1 and the penultimate node adjacent to the doubled terminal of a separate whole B11. There is exactly one eligible marked pair, costing **1/2 per A1/B11 component pair**. This fits the already counted A1/D11 representative allowance5/2. Every triangle still receives at least2 and every quadrilateral at least1. No new ambient dimension or finite forest maximum is asserted.

The producer and certificate prove universal profile identities and whole-component alternatives; they do not use a bounded overlap census as a surrogate for arbitrary supporting roots. The starting rules are the B12 center-zero theorem and B/C12 unit-coupon theorem. The BC11 zero theorem, coupled order11 theorem and D13 center/leaf proof supply the established profile and independent-priority interfaces.

Marked pairs and actual quadrilateral

For B11/order12 use spine0–…–9, doubled bond8–9 and fork leaves10,11 at0. Halve(8,10),(8,11), joining the node just before the doubled terminal to either opposite fork leaf. For C11/order12 use path0–…–11 with doubled end bonds0–1 and10–11. Delete(1,10), the pair just inside the doubled terminals. All three pairs have distance9 and the changed sets are invariant under every native diagram automorphism. All short8 weights are unchanged, including previously zero weights.

At an affected ideal corner let F be the whole varying affine factor containing consecutive face marks a,b. Write the quadrilateral star as W+a+b+c+d with opposite pairs a,c and b,d. The rest of that corner is a positive-definite spectator diagram Z orthogonal to F. All physical root squares are2. Supporting-white and white-plus-one-black principals are strictly positive definite; corner principals are positive semidefinite. These are the inherited actual factor-fiber hypotheses, including nonsimple ideal vertices.

A nonzero adjacent black product gives a separate distance1 unit. If c,d share a nonzero contact coordinate in Z, their corner gives a distance2 unit. Otherwise adjacent products vanish and the nonnegative Z-contact vectors x,y have disjoint supports. Their opposite products with a,b remain arbitrary native magnitudes, including parallel or dotted products. A nonzero magnitude is at least1, and above1 the next possibility is sqrt2. Heights h,k against the positive cusp kernel are strictly positive.

Complete native profiles

At a node just before a doubled terminal, deleting the mark leaves an isolated A1 and a large finite component. For C this large component is B10; for B it is D10. At the B fork-leaf mark, deletion leaves B11. For each large component the inverse diagonal is1 at its relevant long endpoint,2 at the next node and strictly greater than2 elsewhere. These complete lists are checked exactly.

A positive-semidefinite one-root extension of that connected component has no contact, a simple or doubled endpoint contact, or a simple next-node contact. Two contacts would create a cycle in a diagram already containing a fork or a doubled bond. Finite diagrams are trees; the only cyclic affine type is A, which cannot contain this old fork or doubled bond. Exceptional finite/affine types are too small. The single contact Schur condition $q^2 J_{ii} \le 2$ gives the stated all-label list, including equality; the next native magnitude after sqrt2 already has square>2. Additional spectators cannot repair a non-PSD principal extension.

For a near-terminal mark m, outer terminal o, first inward node e and second inward node f, the complete profiles are:

Profile	Contacts	Height h	Moment l	c0	Cost upper bound
isolated	m:t	t	0	2	2
tail only	m:t,o:q	t+q/sqrt2	0	2−q²/2	3
end3	m:t,e:1	t+1+q/sqrt2	1	1−q²/2	3
end4	m:t,e:sqrt2	t+sqrt2	sqrt2	0	2
next3	m:t,f:1	t+1	2	0	4

In the end3 row the optional additional outer contact is o:q. The isolated profile has t≥1. Tail only has1≤q<2 and t≥0, by the strict edge principal on o and the new root. End3 has q=0,1 or sqrt2: the extended large block has remaining Schur norm1, so q²/2≤1. End4 and next3 contain an affine principal with positive kernel, forcing q=0 and sealing it against every other supporting white in that corner. Their opposite t-contact remains allowed because the omitted mark is not in that corner.

For the B fork end use b=10, mate11 and center0. The complete four profiles are:

Profile	Contacts	Height k	Moment m	d0	Cost upper bound
isolated	b:t	t/2	0	2	8
mate3	b:t,11:1	(t+1)/2	1	1	8
mate4	b:t,11:sqrt2	(t+sqrt2)/2	sqrt2	0	4
center3	b:t,0:1	1+t/2	2	0	4

The first two attain8 only at t=1 and t=0 respectively, with k=1/2. All other native instances have cost at most4. Polynomial certificates and the native gap establish these bounds for unbounded t; no numerical label cutoff is imposed.

Give the B spine cusp coefficient1, its doubled terminal1/sqrt2 and fork leaves1/2. Give the C inner spine coefficient1 and both doubled terminals1/sqrt2. Exact elimination of F minus the two marks yields resistance R=10 for B and R=9 for C. With end costs

$$g=2l/h+c0/h^2, \quad f=2m/k+d0/k^2,$$

eliminating the finite support gives the checked identity

$$\det(S)/(h^2k^2) = 1-(g+f)/R + E/R,$$
$$E=(x/h-y/k)^T\, G_Z^{-1}(x/h-y/k) \geq 0.$$

The actual four-black Schur determinant is zero. The producer verifies all19 end identities and45 cross identities, both principal inverse lists per family and the full formal determinant including spectator terms and unequal heights.

C: a uniform strict gap gives an independent unit

For C use a=1,b=10. Both sides have the five near-terminal profiles. Every cost is at most4, so the normalized determinant is at least 1−8/9=1/9. It cannot be zero. Therefore the no-adjacent-contact, disjoint-spectator branch is impossible: the actual face already has a distinct distance1 or2 unit. No original C weight or finite coupon is needed, so its selected unit can be deleted completely.

This is an independent actual-corner theorem. Apply it before the new B half argument if both changed types occur on one face.

B: immediate affine alternatives

For B set a=8,b=10. If c has near-end profile end4, the adjacent corner contains a whole affine B10 of order11 with its doubled-terminal and opposite short-leaf marks. Its current literal weight is zero. Apply the independent BC11 zero theorem first whenever such an actual corner occurs; it supplies a full face payment through its own retained paths and distinct finite alternatives. It does not consume a new B12 half.

For near-end next3, the adjacent whole factor is affine D10 of order11, marked at opposite short leaves. Its current weight is1/2. That corner is distinct from the selected B12 corner; their two retained halves sum to1 directly. A positive kernel seals these affine components against all additional supporting roots in their actual corners. The source reconstructs both kernels, every canonical marked image and the exact current B/D weights. It does not assume an old all-pairs unit cutoff for these order11 factors.

B: arbitrary spectators force a whole finite alternative

After those immediate profiles, the near-end cost satisfies g≤3. If the fork side is nonextreme, f≤4, giving a strict determinant gap at least3/10. Hence the fork side is extreme: a single simple contact to b or its mate, with k=1/2 and f=8. The determinant gives

$E=g-2, \quad 0\le E\le 1.$

The complete near-end possibilities with g≥2 are:

Profile	Parameters	Height	Cost
isolated	t=1	1	2
tail only	t=0,q=1	1/sqrt2	3
end3	t=0,q=0	1	3

Indeed isolation has cost2/t². Tail only at t=0 has cost4/q²−1, so the next native magnitude gives1; if t≥1 and q≥1, monotonicity bounds its cost by the value at t=q=1, below2. End3 decreases in both t and q on0≤q≤sqrt2; either t≥1 or q≥1 gives a value below2. Thus **h≤1** in every remaining case, without bounding the spectator rank.

The entire fork-side vector y vanishes. If y≠0 and x=0, coordinate Cauchy gives E≥(1/k)²/2=2. If both supports are nonempty, choose i in the support of y and j in the disjoint support of x. Then

$E \ge (y_i/k+x_j/h)^2/(4-2(G_Z)_{ij}) > 9/8 > 1.$

The numerator is at least(2+1)²; strict positive definiteness of the two-node spectator principal gives (G_Z)_ij>−2 and the strict denominator bound<8. Both cases contradict E≤1. No restriction on x is needed.

Choose the actual corner containing a,d and omitting b,c. Since y=0, all Z roots are orthogonal to its displayed components; c and all its arbitrary contacts are absent. If d attached to omitted b, d is a whole A1 and the far mark a belongs to whole finite B11, exactly at the node next to its unique doubled terminal. Their new half plus the selected B12 half gives1. If d attached to the mate11, both marks lie in one whole finite B12 path, at marked distance10; its unchanged unit already pays. The source checks positive pivots, actual components, marked degree and doubled-neighbor role in both finite alternatives. These whole-component assertions hold for arbitrary additional spectators.

Cost and simultaneous-rule compatibility

The A1 mark and the penultimate B11 mark are intrinsically unique. There is one new marked half, hence total finite increment1/2. The current actual A1/B11 charge is0. Its representative pair A1/D11 already has allowance5/2; all33,489 simultaneous finite-type comparisons through rank60 continue to dominate the actual rule with this one addition. No representative weight or residual item changes. This is a pointwise cost check, not a fresh finite-forest optimization or a global bound. Other new coupons, such as A4/B9 or D5/D11, concern distinct type pairs and require their own explicit simultaneous binder; this theorem does not silently omit their charges.

The proof is applied to actual changed corners, with the independent BC11 and new C12 unit alternatives before the B12 half branch. The existing B12 center-zero theorem has different marks and outputs only its retained D11 center/leaf unit, whole finite D12 unit or typed A2/B9 and A2/D11 halves. The D13 refinements likewise use smaller B/D11 roles and D10/D12 coupons. None uses the present B12 or C12 changed distance9 pair. The B/C13 and order14/15/16/17 refinements use independent short-path alternatives or their stated finite components; they do not receive a new unpaid output here.

The new B proof itself uses only a retained D11 half, whole finite B12 unit or the new whole A1/B11 half. No changed same-component rank15–17 payment, long affine order13–17 pair, or generic short8 unit is consumed. All new short1/2

alternatives are literal units under the current rule. Thus current deep8→short7 and deep10 finite7→short6 replacements remain independent, and no older long-pair exclusion is called recursively. If no new changed corner occurs, the previous whole-face proof applies pointwise; otherwise the universal actual-corner proof above pays it. This avoids the need to reinterpret any preferred selected-map witness.

Triangles retain their finite7/ideal5, protected ideal6/7 and whole small-affine-order-at-most9 outputs. The three changed pairs have order12 and distance9. Therefore triangle payment2 and quadrilateral payment1 persist. The factor-fiber divisor12 gives new rank11 envelope 62/12=31/6. A new ambient comparison remains a separate obligation.

Verification scope

Generation chunk `deda55` and separate fresh replay `d170eb` both terminated exit0/PASS. Root and construction_audit independently read the complete producer and report and the stated inherited physical profile, determinant and Cauchy interfaces. Both mathematical reviews passed: they checked complete native contact reduction, whole-component alternatives, the distinct retained half, exact new coupon cost and noncircular current priorities. Neither reviewer repeated the formula, map, typed comparison, arithmetic or historical census computations. Exact symbolic end/cross and determinant identities, native bounds, immediate whole-affine maps, finite whole roles, complete66-pair weight tables, automorphisms and finite pointwise allowances are checked. Historical factor-fiber and finite/affine classification inputs are inherited; no original geometric census or forest optimization is rerun.

```
python3 scripts/fresh_order12_bc_near_terminal_half_transfer_20261018.py --verify
```

Source 200

Source path: `results/fresh_order12_bc_unit_coupon_transfer_20261017.md`

SHA256: 079467a45fdd8d2715899de2adce183c7fd9cccf67aaed269c2586024d3b691c

Order12 B64 and C63 with whole A2/D11 unit coupons

The complete face-payment rule may remove the two remaining half-weights of affine B11, and the two remaining nonbasic unit weights of affine C11. Their order12 costs become64 and63 respectively. Upgrade the existing half-coupon between a marked root of a whole A2 and a short fork leaf of a separate whole finite D11 to a unit. The existing full A1/D11 short-leaf coupon is retained. Every triangle still receives at least2 and every quadrilateral at least1.

The order12 A/B/C/D costs are now60/64/63/62, so the normalized affine rank11 envelope is16/3. The added finite cost is **2 per A2/D11 component pair**, or4*a*2d11 in doubled units. This local theorem gives no finite forest maximum or ambient dimension bound; the new coupon must be included in any joint budget.

The source and certificate reconstruct the complete selected support and exact local interfaces. They retain the inherited all-label core classification and arbitrary supporting-wall proofs stated below.

Pair rules and complete support

For affine B11, number the spine0–…–9 with doubled bond8–9 and short leaves10,11 attached to0. Remove its two half-weights at(9,10) and(9,11), both distance10. The remaining64 pairs are exactly the basic ideal5/feature5 set, all with weight1. This is the full leaf-swap orbit.

For affine C11, use the path0–…–11 with doubled bonds0–1 and10–11. The endpoint pair(0,11) was already absent. Remove the reversal orbit {(0,10),(1,11)}, both distance10. The remaining63 pairs are exactly the basic ideal5/feature5 set, all with weight1. All short8 and elementary triangle units remain for both types. No distance9 payment changes.

The checker re-enumerates all758 eligible connected high patterns into both targets, including every pattern with no map, and compares every successful map to the complete frozen support. The B search is exhaustive injective induced-map recursion. The separate C algorithm tries every contiguous path segment in both orientations, checking all labels and nonedges. The affected support is:

Type	Affected contexts	Affected induced maps
B	14	28
C	2	4

The B contexts are

`98,100,119,121,164,167,171,1452,1728,1734,1745,2121,2127,2138.`

The C contexts are164 and1452. Their abstract target maps are proper extensions of selected finite B11 components. The next two sections verify the actual geometry, not just a list of abstract marked images.

B: exact whole finite alternatives replace the ideal halves

The frozen B-half theorem already proves complete actual supporting-wall control for these14 contexts. The new checker reruns its literal inverse, inertia, contact and corner reconstruction and compares every output to the frozen certificate. Its

arbitrary-support reasoning is essential here.

In eight radical contexts, the full indefinite core has signature(13,1,1). Its radical is an actual linear relation: a null vector orthogonal to a timelike vector in the Lorentz space must vanish. Every possible new contact is in the checked nonzero support of that relation. Acuteness forces a nonzero new contact vector to meet both signs, giving an unchanged marked path of length at most3. In an otherwise unpaid face every extra white is therefore orthogonal to the entire old core. The displayed finite components remain whole after arbitrary supporting whites are restored. There is no first contact from which an indirect tail could start.

The other six cores are nonsingular Lorentzian of order14. With norm2 physical roots and saved gauge squares s_i, a native contact q at old node i has projection square q^2*s_iJ*_ii. The complete contact domain splits into two ends; meeting both gives a retained short8 path. The positive inverse blocks give a lower bound exceeding2 for every two-node support on one end, including arbitrary larger contact magnitudes. Exactly one singleton profile survives: a unit contact at an omitted black root outside the selected component, with normalized projection2 and residual0. Labels at least4 are excluded.

In contexts164 and1452 no surviving contact can enter the selected B11. It is a whole finite component and its marked pair retains a full unit. These contexts cannot complete to an actual whole affine B11 or C11.

In the remaining four contexts there is either no such extra root or exactly one. Its zero residual fixes its entire normal in the old Lorentz span, so a second such normal is impossible under strict white positive definiteness. Every core-orthogonal white is also orthogonal to this forced root. Thus arbitrary further supporting whites cannot attach through it or bridge to the displayed finite components. The source reconstructs both actual possibilities, using the correct norm2 interpretation before applying the saved positive diagonal gauge.

There are16 finite-payment configurations: eight sealed radical cores and both variants of four nonsingular cores. Their finite alternatives are whole B12 units, whole A1/D11 short-leaf coupons, or whole A2/D11 short-leaf coupons. A1/D11 is already full by the universal D11 theorem. Making A2/D11 full allows the preferred ideal B11 contribution to vanish. The complete finite totals are:

New finite total	Configurations
1	8
2	8

The checker uses the actual connected components and actual marked roles, not a proper D11 subdiagram. Twelve configurations use the new A2 upgrade and four use the existing A1 upgrade. All same-component units have order at most13. The two already-finite B11 contexts separately retain their full unit.

C: the missing terminal is impossible in both contexts

Only contexts164 and1452 map a selected component to one of the removed C pairs. The preceding argument already proves that their selected B11 component is whole and finite. A direct first-entry exclusion confirms the four abstract C maps.

Each map omits one terminal of the12-node C path. The whole target would require a new norm2 root with a label4 contact to an old marked node i. The root is genuinely new: omitted old black roots are absent from this corner, while every other old corner component is orthogonal to the selected component. Its native squared contact magnitude is2. In every map the correctly scaled bound is

```
2*s_i*J_ii = 16 > 2.
```

Within one end, entrywise positivity means further contacts can only increase this bound. Contacts on the other end give an unchanged short8 path. Thus no additional contacts, labels or indirect spectator branches can repair the missing-terminal obstruction. No unspecified contact has been assigned zero and no norm4 gauge vector has been mistaken for a physical norm2 root.

Every nonselected and later interface

The complete low-core closure has66 chosen cases and69 actual variants. Its only affine B order12 images are basic retained pairs; it has no C order12 output. These are closed actual components, not arbitrary extensions of finite patterns. The disjoint high-core branch remains excluded. Protected ideal outcomes have distance at most9 and remain paid, while all finite protected outcomes are unchanged by this theorem.

The sole additional B provenance is the universal affine-D13 `inner4` immediate alternative. The frozen B-half proof already excludes it for all native opposite contact magnitudes and all spectator forests: its end cost is at most2, while the opposite end costs at most6 or8, giving strict determinant gaps1/9 or1/11. The nonnegative spectator correction cannot cancel them. The checker reruns all three exact gap formulas. No C immediate alternative exists. Universal D14 immediate factors have order13 and are whole by their positive affine kernels.

The frozen C endpoint report audits every conclusion of the earlier universal and local transfers. Its reasoning still applies because the two newly removed pairs have distance10: cycle alternatives are finite or short ideal paths; classical-tree alternatives have distance at most9 with persistent nearby features; all equality, partial-weight and singleton alternatives have their stated finite components, separate coupons or other whole affine factors. The cycle13/14 and longer-cycle proofs therefore cannot introduce either distance10 C pair.

The later cycle12 and D11 transfers return finite alternatives or ideal paths of length at most4 and5 respectively. The B14 projection refinement introduces no replacement payment. The finite rank17 refinement changes no ideal pair, and its finite deletions do not touch the components here. Consequently there is no later nonselected B/C order12 obligation beyond the support and immediate cases just treated.

The new large-D/D10 transfer changes only whole D_n with n>=20 and separate small-component/D10 coupons. None of those halves or coupons is used by the present B/C proof. Conversely, the large-D proof uses short8/basic ideal paths and unchanged protected ideal units, so these distance10 deletions do not debit it. Both local theorems compose, but their finite coupons must be counted together.

Finite cost and verification scope

A whole A2 has two marked positions and a whole D11 has two short fork leaves. Four pair weights increase from1/2 to1, adding2 per component pair. The fork-center coupons and every other old coupon are unchanged; including its two fork-center halves, the full A2/D11 coupon total is5. No coupon has been counted by a rank-only surrogate in the local proof.

Generationefbe5b and separate fresh replayd26928/b55e8a both terminated exit0/PASS. The checker re-enumerates the complete758 pattern domain for both targets, reruns all14 literal B geometry certificates and16 finite variants, checks the four C forced-terminal projections, all basic/short8 sets and automorphisms, the low-domain interfaces and three universal immediate gaps. The complete all-label core census, finite/affine classification, factor-fiber theorem and arbitrary-support sealing remain inherited frozen inputs.

A separate full source/report mathematical and interface read passed. That review checked the inherited B-half geometry reconstruction, all16 finite variants with whole A2/D11 roles, both whole-finite-B11 C contexts and scaled missing-terminal bounds, complete selected/low/immediate/later interfaces, and compatibility with the large-D/D10 transfer without consuming its coupons. It did not repeat the core census or numerical geometry calculations.

Root completed a full source/report read and reread the frozen B-half source/report and C endpoint report. That review passed all758 successful and failed pattern cases, the16 finite variants with actual whole A1/A2 and D11 roles, scaled projection16, arbitrary radical and zero-residual sealing without indirect bridges, all retained short/basic/low/immediate and later interfaces, and independent composition with the large-D rule. Root did not duplicate the geometry or census arithmetic. All37 direct bindings, Python AST and four local report links pass. Source, certificate and report are final/frozen.

No exact joint finite-forest maximum, ambient bound, polytope construction, external refereeing or formal verification is asserted here.

```
python3 scripts/fresh_order12_bc_unit_coupon_transfer_20261017.py
python3 scripts/fresh_order12_bc_unit_coupon_transfer_20261017.py --verify
```

Source 201

Source path: `results/fresh_order12_order16_depth8_ambient60_frontier_20261018.md`

SHA256: 9ba5373780bcbf2cc907b99e12447144108011ab7806b59c037262ec292a9e3a

Ambient60 ideal closure and the remaining finite maximum409

The combined rule has **positive surplus at every ideal vertex**, and its exact formal finite-forest maximum is **409**, against curvature405. Every finite-sided, finite-volume real hyperbolic Coxeter polytope of dimension60 therefore satisfies

$4\,v_f \geq 4500 + 54\,v_i.$

Here v_f and v_i count finite and ideal vertices. Consequently all-ideal polytopes in this class have dimension at most59, using the completed unrestricted dimension60 theorem for the higher-dimensional tail. The unrestricted bound remains60. The formal maximum409 is attained by 4A1+4D5+3D12; this is not a geometric realization of a vertex or polytope.

The producer and certificate combine the frozen order13 complete rule, order12 B/C theorem, order16 B/D theorem and depth8 finite7 theorem. All previous same-component rules and whole-component coupons remain present except for the explicitly stated new deletions below.

The one simultaneous payment rule

At ideal order12 the family totals are A60/B62/C62/D62. The B11 affine diagram halves its two fork-leaf/near-heavy-terminal pairs (8,10),(8,11); the C11 affine path deletes (1,10). All three have distance9. The new finite coupon is a half on the unique pair consisting of a whole A1 and the penultimate node adjacent to the doubled terminal in a whole B11. Thus its cost is exactly1/2 per A1/B11 component pair.

At ideal order16 the totals are A80/B102/C103/D103. The B15 affine diagram deletes the three units (0,10),(10,14),(10,15); the D15 affine diagram deletes the four halves (2,14),(2,15),(9,12),(9,13). Their distances are at least10. The finite compensation upgrades the whole D5 unique long endpoint against either short fork leaf of a whole D11 from a half to a unit. The two eligible marked pairs add exactly1 per D5/D11 pair, raising its total from15/2 to17/2. This does not change the separate whole D5/A11 coupon or any finite D16/D17 same-component weight.

Finite distance7 pairs in whole A/B/D components receive zero whenever both marks have feature depth at least8. Features are endpoints, branches and endpoints of nonsimple bonds, as in the bound local theorem. The previous threshold was10. The total numbers of deleted distance7 pairs at rank n are

Finite family	New depth8 count	Previous depth10 count
A	max(0,n−23)	max(0,n−27)
B	max(0,n−24)	max(0,n−28)
D	max(0,n−25)	max(0,n−29)

The binder subtracts only the difference of these counts from the previous same-component costs. No coupon or ideal weight is changed by this finite rule. In particular it affects none of the finite components of ranks15–17 used by preceding compensation arguments.

Exhaustive actual-corner composition

First consider an actual newly changed C12 corner. Its independent universal near-terminal proof gives a distinct actual short1/2 unit. Those units are unchanged by every present reduction, so this pays the face without using its now-zero C12 pair.

Next consider an actual newly changed B12 corner. The independent B/C11 zero theorem has priority in its immediate whole B11 branch. Otherwise the new B12 half is supplemented by a distinct retained D11 half, a whole finite B12 unit, or the new whole A1/B11 half. The finite B12 unit has marked distance10 and lies in a rank12 component, where none of the current finite long-pair/depth rules changes it. The whole-component proof allows arbitrary supporting roots: its Cauchy argument eliminates the entire fork-side spectator vector, while the other root and its arbitrary contacts are absent from the paying corner. The other new ideal changes have orders16 and distances at least10, so none affects these B12 outputs. The C12-first priority avoids recursion between the two order12 changes.

After excluding every actual changed order12 corner, examine any actual newly deleted finite distance7 corner. Its depth8 theorem proves either an actual short6 alternative or a distinct retained whole A8 unit. The latter has marked leaves. The former uses the independently established current short6 interface: literal short units, the independent cycle11 half transfer, or the universal cycle/tree replacement with protected finite leaf/branch or whole small-component outputs. The depth8 theorem explicitly checks these outputs, including the triangle exception with its fixed marked leaf. None is a newly changed order12 distance9 or order16 distance10/11 pair; no new depth8 finite7 deletion can remove a protected marked leaf or a whole A8 unit. This branch does not call the selected large-affine exclusion back to establish its short-path payment.

In the remaining case there is neither an actual new order12 corner nor an actual new finite7 corner. On this whole face those two changes agree pointwise with the preceding rule; their extra finite coupon can only increase payment. Apply the order16 local theorem to the complete preceding reduction, already including the order13 improvements. Its independent short8 branch is established before its selected-factor exclusions. The existing finite depth8 distance8 rule supplies short7 when necessary; the current finite7 rule supplies short6 or its retained whole A8 unit. The earlier B/C11, cycle11, coupled order11, B12 center, D13 and B/C13 priorities use only their stated smaller ideal roles or separate finite components and remain independent.

In the no-short8 branch, the inherited low-core, high-disjoint, protected/basic and adaptive routes are preserved. The order16 theorem reconstructs all394 eligible patterns per family, including failed native embeddings. Its changed B images are impossible by arbitrary-contact physical projection or a saturated residual with no possible next white. Its changed D images are excluded before imposing completion zeros or seal a whole D5/D11 unit at a distinct corner. That new unit replaces the deleted ideal half without consuming any finite same-component unit. The complete routing and later interfaces are part of that local theorem, not an inference from the current216 literal examples alone.

These cases exhaust actual faces. The order13 A4/B9 half and all older compensations remain included. Triangle payment2 persists: the new ideal pairs have distance at least9; every newly affected finite distance7 triangle output was checked in the depth8 theorem against its persistent feature or fixed marked-leaf witness. Quadrilateral payment1 therefore holds under one simultaneous rule with both new finite coupon costs.

Complete current literal check

The binder reconstructs all216 inherited literal configurations from their native Grams and starts with the final order13 cornerwise lower payments. It maps every shared affine component of order12 or16 to the new canonical targets and checks every finite corner for the D5/D11 upgrade. This complete literal finite domain is simply laced and its largest whole finite component has order17. It therefore contains no new whole B11 coupon and no newly deleted deep finite7 pair.

Exactly four configurations change: 1979,2372,2536,2540 in the inherited D16 partial-weight package. Each loses one ideal half and gains the distinct whole D5/D11 coupon half; each total stays1. There are no new order12 debits and no B16 literal debit. All216 totals remain at least1, and the four previously recorded D16 eighth credits are retained. This exact saved-interface reconstruction supplements the universal geometric composition above; it is not a replacement for it.

Complete finite upper bound and actual attainment

Finite quantities in the producer use eighths. It checks simultaneous same-component and cross-component dominance for all33,489 ordered type pairs through rank60, including the exact new actual charges:

Actual component pair	New actual total	Retained representative allowance
D5/D11	17/2	A5/D11:19/2
A1/B11	1/2	A1/D11:5/2
A4/B9	1	A4/B9:1

The last row is the preceding order13 charge, retained explicitly. The first two new charges require no increase to the representative interaction table. All earlier actual charges, including D5/A11 and D5/D12, are still counted separately. Representative allowances are upper bounds, not claims that their displayed representative forest attains the actual value.

All1,044 fixed-component multiplicity vectors and792 interaction keys are included. Fixed ranks are9 through17; the separate A17 representative and residual D17 treatment are inherited unchanged. Residual ranks are 1 through8 and17 through60. Every actual forest is dominated by this complete split, with both fixed-fixed and residual-residual cross costs zero. The54,288 exact item identities verify every fixed interaction.

New residual tables are required because the large-rank same-component items changed. Generation computes them by the item-first recurrence. Replay checks all1,142,064 last-item upper inequalities and47,520 attaining predecessors. Induction on residual rank gives both an upper bound and attainment for each representative table, with no optimizer called in replay. Twelve fixed-vector alternatives remain above405.

The largest representative value is409. The table displays 4A1+4A5+3D12 at this upper value, whose actual value is only358. This does not make the bound unattainable: replacing the four A5 components by D5 gives the actual formal forest

```
4A1 + 4D5 + 3D12,
```

of total rank60 and exact cost409, checked directly with the full actual cross rule. Thus409 is the exact maximum, not merely a conservative upper estimate. A separate actual witness2D17+D26 now costs817/2=408.5; it does not attain the maximum. No such formal forest is asserted to occur as a vertex of a finite-volume hyperbolic Coxeter polytope.

Exact ideal closure and its global consequence

The producer reconstructs the preceding exact integer convolutions and scalar rows and verifies equality with the frozen order13 certificate. The complete terminal domain has total rank59, at most one part from1 through8, and every other part at least9. Its2,113 profiles use the exact family envelopes at every rank, with no constant high-rank substitution. Only the established low9 concentration is used; arbitrary multiplicities of ranks9 and above are retained.

For each row the new normalized cost saving is exactly

```
count(rank11)/12 + count(rank15)/8.
```

These are total-cost savings1 at affine order12 and2 at affine order16, divided by their true factor-fiber multiplicities12 and16. All ten remaining negative profiles become positive; none is zero. The exact normalized minimum is31/6000 at (9,9,9,9,9,14), and the exact absolute minimum is54 at the singleton profile (59).

For an arbitrary actual cusp, inherited low9 concentration decreases the normalized margin and does not increase the link-vertex product V. Since every terminal normalized margin is now positive, multiplying preserves the comparison: the actual absolute surplus is at least the terminal absolute surplus, hence at least54. This positive-only product argument was not available for earlier negative terminal rows. The factor-fiber counting theorem converts these charges and surpluses to the actual ideal vertices, rather than treating each factor as an independent geometric vertex.

Write C(v) for each ideal vertex's charge and kappa(v) for its shape curvature term. The simultaneous local payments and exact forest bound give the global upper estimate

```
D5 ≤ 409 v_f + sum_i C(i).
```

The inherited shape-curvature inequality in ambient60 gives

```
D5 ≥ 405 v_f + sum_i kappa(i) + 4500.
```

Here the two extrema together contribute 5 floor(60²/4)=4500, counted once. Since kappa(i)−C(i)≥54, subtraction gives4v_f≥4500+54v_i. In particular dimension60 requires at least1125 finite vertices, and at least1139 if an ideal vertex exists. An all-ideal dimension60 polytope is impossible, and the completed unrestricted60 theorem excludes higher dimensions. Therefore the all-ideal class has dimension at most59. The finite maximum409 exceeds405, so this argument does not exclude dimension60 in general.

Restoring the older ideal17 center halves while reducing their finite coupon is a separate possible successor. It is not applied in this certificate; its finite count and simultaneous literal changes are not silently assumed here.

Verification scope

Generation session97211/chunkfa1358 and separate fresh replay 74572/chunk2b03bf both terminated exit0/PASS. Direct metadata checks pass49 source bindings,6 local links and Python syntax. Root's full producer/report mathematical composition read passed, including the exhaustive actual-corner priority, exact representative domination and actual attaining forest, simultaneous literal accounting and positive-only absolute-surplus extension. Root independently checked the actual409 cost breakdown and a saved-margin update, but did not rerun the DP, convolutions, literal computations or historical geometry. A second independent full producer/report mathematical composition read passed: it checked the same exhaustive priorities and protected outputs, both new finite coupon allowances, the complete fixed/residual certificate, actual attaining forest, exact ideal savings and positive-only product comparison. That reviewer did not repeat geometric censuses, DP, convolutions, literal or other arithmetic. Source, data and report are final/frozen after these reviews and the direct metadata checks. All bound local reports were frozen before this generation. The producer checks source hashes, all finite type comparisons, new complete Bellman certificates, reconstructed ideal convolutions and all216 literal interfaces. The local arbitrary-support, all-label, classification, concentration, factor-fiber and shape-curvature proofs remain explicit inherited inputs. No historical geometric census is rerun, and no external refereeing or proof-assistant validation is claimed.

```
python3 scripts/fresh_order12_order16_depth8_ambient60_frontier_20261018.py --verify
```

Source 202

Source path: `results/fresh_order12_payment_support_20261017.md`

SHA256: 18e7168471b18b5221e06b3abc537639e9bf7fa85c41d10d7f431b7a66ed9df7

Complete order12 selected-payment support diagnostic

The saved interfaces require **66 B pairs** and **65 C pairs**, while their basic ideal5/feature-radius5 sets have64 and63 pairs respectively. The A selected interfaces use only its60 basic pairs, but all six antipodal distance-six pairs remain a separate universal short8 obligation. This diagnostic therefore does not establish a rank11 cost below66 or a new dimension bound.

The source and saved data adapt the order13 diagnostic to order12, explicitly adding the complete low-core corner domain and short8 provenance. They use the canonical target numbering of the frozen affine graph generators.

Affine family, order12	Basic pairs	High selected pairs	Actual low-core pairs	Immediate D13 alternatives	Union of basic and selected interfaces	Short8 outside basic
A	60	24	0	0	60	6
B	64	5	2	3	66	0
C	63	5	0	0	65	0
D	66	4	4	8	66	0

The columns are set sizes and are not added: their overlaps are retained exactly in the saved pair sets. The complete domain includes758 eligible high patterns from all2,614 frozen overlap cases, using component bound at least12 and marked pattern size at most12. Their bound histogram is563/116/52/17/10 for bounds12/13/14/15/16. All induced weighted embeddings, including nonedges and all target automorphisms, are enumerated. This is a new support scan of the frozen classification, not a repeated classification of possible stars.

B and C nonbasic support

For B the two nonbasic pairs form one automorphism orbit:

`{(9,10),(9,11)}, distance10.`

They occur in high cases98,100,119,121,164,167,171,1452,1728,1734,1745,2121,2127,2138. These are the L9/L25 single-node and L10/L25 two-node contexts, including their reversed presentations. Exact core IDs, black marks, identifications, preferred methods and all embeddings are saved. Removing either pair orbit requires a new transfer for this support; absence from any smaller sublist is insufficient.

The same B orbit is also used by the frozen D13 long-pair transfer. Its `inner4` alternative is a whole12-node affine B factor: when the other old mark is a far short leaf, its new marked pair maps to(9,10) or(9,11). The far-center alternative maps to the already-basic pair(0,9), at distance9. Thus resolving only the selected high contexts would leave an immediate-payment dependency.

For C the two nonbasic pairs are

`{(0,10),(1,11)}, distance10.`

Only high cases164 and1452 use this orbit. The endpoint pair(0,11), at distance11, is absent from every selected high, actual low and immediate interface considered here. Keeping the basic63 and these two pairs gives65 for those interfaces. Every short8 pair is already basic. This is a support statement, not a promoted new payment rule.

All66 low-core chosen corners, including genuine affine extensions

The source checks every chosen corner in the complete low-core closure, not only those whose printed component bound equals12. It reconstructs the exact marked Gram block, verifies its inertia and retained marks, and matches any whole affine block against the canonical family. For the three `unique_sealed_label3_root` closures it also constructs and checks the optional extra simple-root component. The frozen closure supplies the exhaustiveness and absence of further relevant supporting roots; this diagnostic checks every resulting literal component.

The69 actual component variants comprise10 finite components of order10,21 finite of order11,14 finite of order12,12 affine of order11 and12 affine of order12. Only22 distinct exact Gram calculations are needed. Whole affine components are certified by a strictly positive kernel; additional acute contacts to such a component vanish.

At order12 the low domain contributes:

- Two B variants, low cases28/29, with images(8,10),(8,11). These are basic and add no B nonbasic pair.
- Ten D variants with cross-short-leaf images(8,10),(8,11),(9,10),(9,11). Eight are existing affine chosen components; two are the optional sealed simple-root completions in low cases60/62.
- No A or C component.

The two optional D completions are genuinely affine. Treating all optional extensions as finite would omit real low-core payment obligations. Their low/high IDs and actual matrices are recorded separately. The third optional extension remains finite.

The D13 `inner_next3` immediate alternative also yields a whole order12 D factor, with far-center images at distance8 and far-short-leaf images at distance9. All are basic, but a proposed D pair reduction must still account for these literal immediate alternatives and the ten actual low variants.

Universal short8 provenance remains separate

The finite8/ideal5 composition first handles an existing marked path of length at most8 before invoking the no-short8 core census. At order12 every B/C/D short8 pair is in the basic set. A is a12-cycle: its six antipodal pairs have distance6 and are not basic.

The frozen cycle13 transfer concerns order13, and the general ideal5 local cycle transfer starts at order15. Neither statement supplies an order12 antipodal replacement. Therefore the A union of60 selected/basic pairs cannot be used as a complete payment set merely because the antipodal pairs do not occur in the selected no-short8 patterns. Until a separate universal cycle12 argument is supplied, these six short8 obligations remain explicit.

Generation25610 and separate fresh saved replay99984 both terminated with exit0/PASS. The source binds the authoritative high and low classifications, immediate-transfer data and relevant composition statements. No pair is removed from a frozen rule; no new ideal envelope, feasibility result or global theorem is asserted.

A separate root read of the complete source and report passed the758-pattern high domain, all66 low closures and three optional variants, their exact signatures and positive kernels, all six immediate alternatives, the induced maps and nonbasic orbits, and the explicit unreplaced cycle12 short8 obligation. That read did not repeat the inherited all-label classifications. All26 direct bindings, eight report links and Python syntax were checked. Source, data and report are final/frozen.

Source 203

Source path: `results/fresh_order13_affine_b_fork_projection_transfer_20261017.md`

SHA256: 33144882bd974c15902d7c734c33f9a4f1586344e71a2cfbf91f8036ed3f13d0

Removing the final two nonbasic order13 affine B pairs

Starting with the completed ceiling65 payment rule, the two remaining nonbasic affine B12 pairs may be removed. Its order13 cost falls from75 to73. No finite weight or coupon changes. The full whole-A2/whole-D11 short-leaf coupon already proved in the order12 B/C theorem is essential to two of the supporting cases.

The order13 A/B/C/D costs become65/73/73/74, so the normalized rank12 envelope falls from75/13 to74/13. This is a local payment refinement; it gives no new ambient bound. Other scalar and finite obstructions remain.

The source and certificate re-enumerate the complete selected-map domain and check every affected actual-component interface.

Intrinsic pair rule and complete domain

Number the affine B12 spine0–1–…–10, with doubled bond9–10 and short leaves11,12 at0. Delete(9,11) and(9,12), both at distance10. They form one orbit under the diagram's leaf swap. The earlier singleton (0,10) and terminal/short-leaf pairs(10,11),(10,12) stay absent under their already completed proofs. The retained73 pairs are exactly the basic ideal5/feature5 set, all with weight1.

Every short8 and elementary-triangle pair is retained, and no distance9 payment changes. The source checks the intrinsic basic set, all automorphisms, shortest paths and the full triangle pair set inherited from the order13 singleton theorem.

The frozen complete order13 diagnostic contains801 eligible connected selected patterns, including those whose actual-component upper bound is larger than13. The new source reruns exhaustive injective induced embeddings of every pattern into the13-node B target, checking every edge and nonedge and every failed pattern. Its output agrees with the complete saved maps. Exactly23 contexts, with52 affected maps, contain one of the changed pairs:

149,151,625,630,638,715,721,1356,1364,1556,1642,1648,1668,
1742,1849,1857,1922,1935,2135,2242,2250,2315,2328.

Four contexts already contain the whole affine factor. Nineteen describe proper selected subdiagrams and require additional roots to complete the target. These are different geometric situations and have separate proofs below.

Nineteen proper-completion contexts: impossible doubled terminal

Every affected proper core H is nonsingular Lorentzian, all its saved roots have physical norm2, and all its nonzero off-diagonal entries are−1. The checker recomputes its exact inverse J and actual preferred corner. The complete necessary contact domain of any supporting white is partitioned into two ends. Their inverse submatrices are entrywise positive. A root meeting both ends gives a retained marked path of length at most8.

In each of the44 maps, target node9 is an old selected marked root i, while the doubled terminal10 is absent. An actual whole affine B12 factor would require a norm2 root e with label4 contact to i. That root is outside H: old omitted black roots do not belong to this corner, and every other old corner component is orthogonal to the selected component. It therefore cannot supply the required nonzero first bond. Other target nodes may be supplied later by arbitrary old or new components; the proof uses only this forced first terminal contact.

The native squared magnitude of the label4 contact is2. Since the orthogonal complement of H is positive definite,

```
projection_H(e)^2 <= 2.
```

If e meets both ends the short-path branch already pays. Otherwise its contacts lie on one end, where positivity gives the lower bound2*J_ii. Every extra contact on that end only increases the bound. The exact table over all44 maps is:

Forced norm2 projection lower bound	Maps
5/2	12
3	12
7/2	8
7	4
15	8

Every entry exceeds2, excluding all such actual completions. No unspecified contact is set to zero. This argument covers arbitrary native labels, multiplicities and indirect spectator chains. It does not require the more complicated relevant-component tail bound merely because some saved contexts were originally classified by that method.

Four complete affine factors: radical sealing and an existing unit

Contexts149,151,1742,2135 have15-root cores of signature(13,1,1). Their selected13-node component F is the whole affine B12 factor. The checker reconstructs its positive kernel, the full-core radical, the strictly positive-definite white and edge diagrams, all four corners and their connected finite components with native labels.

For any extra supporting white e, positive semidefiniteness of the actual corner containing F and the positive affine kernel force every contact with F to vanish. The full-core radical is an actual zero linear combination: the core contains a timelike vector, and a null vector orthogonal to it must vanish. Its two coefficients at the omitted black roots are nonzero with opposite signs. Pairing e with this relation says its two remaining core contacts either both vanish or both are nonzero.

Two nonzero contacts give an unchanged adjacent marked path of length2. Otherwise every supporting white is orthogonal to the whole old core. Each finite corner is then the displayed positive-definite corner plus an orthogonal positive-definite white diagram. In particular its marked components are whole actual components: neither a direct attachment nor an indirect bridge can enlarge them.

The reconstructed finite payments are:

Contexts	Whole finite alternatives	Total after deleting the ideal unit
149,1742	B13 unit and A2/D11 short-leaf unit	2
151,2135	A2/D11 short-leaf unit	1

In every A2/D11 corner the small component is exactly A2 and the large mark is a short fork leaf of exactly D11. The source verifies these roles from the complete connected components and the positively scaled native Gram. It uses the already completed A2/D11 unit upgrade, whose cost is already included in the ceiling65 finite budget. No new coupon is added here. The B13 units are also unchanged.

Nonselected and later provenance

The complete short8-first reduction keeps all its original units and universal transfers. Their ideal alternatives have distance at most9 with the retained nearby features, or are direct/shared-neighbour paths. Their finite outputs remain unchanged. The low-core closure has whole marked-component bound at most12 and cannot yield an order13 factor. The disjoint high-core branch is excluded by its inherited all-label argument. Protected high outcomes have retained finite payments or basic ideal payments at distance at most9. The selected cases whose weights can change are therefore exactly the23 cases above.

Universal affine-D14 immediate B alternatives have marked images (0,10),(10,11),(10,12), none of the newly removed pair orbit. Their existing strict-gap replacement remains unchanged. Its other immediate alternatives are affine D, not B. Universal D13 immediate factors have order12. The complete saved immediate map interface is checked directly.

The later universal D12 and D11 transfers end in finite alternatives or ideal paths of length at most4 and5. Cycle12/13/14 and longer-cycle alternatives end in their stated finite units/coupons or retained short ideal paths. The older same-order and higher-order partial-weight replacements retain their stated affine D weights; their other outputs are finite. The B14/B15 projection exclusions introduce no replacement payment, while their finite-completion predecessors use the stated small whole finite components.

The newest order12 B/C theorem ends in finite alternatives and consumes no order13 B unit. The order14 B refinement is a projection exclusion. The large-D/D10 theorem uses retained short8/basic ideal paths and protected distance9 outcomes, so its proof does not lose a payment here. Its finite changes affect whole D_n of order at least20 and D10 coupons; they leave the whole B13 and A2/D11 alternatives above unchanged. The finite rank17 deletions likewise do not affect these components.

Thus all triangle payments remain at least2 and all quadrilateral payments at least1. Actual cusp-fiber multiplicities stay unchanged; only the order13 B factor weight sum changes, from75 to73.

Verification and scope

Generationbfb9cf/8dee57 and separate fresh replaybd3c11/a7cb3c both terminated exit0/PASS on all801 pattern searches, all52 affected maps,19 complete necessary contact domains,44 exact projection bounds, four radical/sealed corner configurations, and the retained basic, short8, triangle and immediate interfaces. The complete all-label geometric core census, no-extra-cross-edge proof, finite/affine classification and arbitrary-support contact-domain theorems remain explicitly inherited frozen inputs. They are not replaced by an abstract finite embedding list.

Two separate full source/report mathematical reads passed. Both reviewers reread the complete contact-domain/projection helper and the radical, kernel and finite-component helper. They checked the physical norm2 interpretation, positive inverse end blocks with arbitrary extra contacts, the genuinely new forced terminal, whole-component sealing and existing A2/D11 unit, and all retained short/basic/low/immediate/later interfaces. Root also checked all four finite totals and the exact existing coupon budget dependency. Neither reviewer repeated the801-pattern census, 44 projection calculations or inherited all-label geometric census. All34 direct bindings, Python AST and three local report links pass. Source, certificate and report are final/frozen.

No finite-forest recomputation is required by this unchanged finite rule, but any new ambient claim still needs its complete ideal comparison and remaining finite obligations. No new bound, construction, external refereeing or formal verification is claimed here.

```
python3 scripts/fresh_order13_affine_b_fork_projection_transfer_20261017.py
python3 scripts/fresh_order13_affine_b_fork_projection_transfer_20261017.py --verify
```

Source 204

Source path: `results/fresh_order13_affine_b_singleton_transfer_20261017.md`

SHA256: 84262d376910422fe21b252af4269509b3dd319db5b4ae54da2273c64ccca2f0

Removing the doubled-terminal / fork-center pair from affine B12

Completed local argument; verification is recorded below. An affine B12 factor has13 nodes. In the canonical numbering used by the order13 support diagnostic, its fork center is0, its short leaves are11,12, and the spine is0–1–…–10 with doubled bond9–10. This package removes only the singleton marked pair `(0,10)`, of graph distance10, from the otherwise all-pairs rule. The factor cost changes from78 to77.

The replacement uses one additional finite half-payment: the two marks lie in a whole A2 component and the unique doubled terminal of a whole B10 component. Either A2 endpoint is eligible. The existing finite half-payment for whole A1 versus the fork center of whole D12 is retained. A separate finite-vertex budget must account for the new rule. This report asserts no ambient dimension bound.

There are two distinct reasons the removed pair was previously required. Both are discharged here: the two selected L10/L25 face-star contexts, and the universal `inner4` alternative inside the affine-D14 transfer. Discharging only the selected contexts would be insufficient. The source and certificate record both interfaces.

Complete support and retained short pairs

The frozen order13 diagnostic enumerated all801 selected patterns that can require a factor of order13, including patterns with larger component bounds, and their induced labelled embeddings. The removed singleton occurs in exactly contexts160 and1206 of the complete finite8 overlap certificate. These are the two orientations of L10 with marks `{3,4}` and L25 with marks `{0,3}`, identifying L10 node5 with L25 node9. The source checks the saved marked-image support, rather than inferring completeness from the two examples.

Every affine-B12 diagram automorphism fixes the fork center and the doubled terminal, so the singleton is invariant. Its distance10 exceeds every short8 payment and every elementary triangle pair in the retained ideal5 triangle rule. The new source checks the complete basic and triangle pair inclusions. Thus the short8-first reduction and triangle theorem are unaffected. The original complete overlap, no-extra-cross-edge and all-support classification remains an explicit frozen input; it is not rerun here.

The immediate D14 interface is also checked literally. Of its six saved profile/mark records, only `inner4` with the opposite fork-center mark maps to the singleton. In fact the argument below excludes all three `inner4` far-mark alternatives under the unpaid-face hypothesis, including the two still-retained distance11 B pairs. The `inner_next3` alternatives have affine D type and are not changed by this package. Whole factors of order at most12 and every other order13 pair remain paid.

The actual radical core seals all supporting whites

Use context160 numbering. Its15-root core consists of the path

`4–3–2–1–0–5–14–13–12–11–10–6`,

with simple bonds except3–2 of label4, and branches6–7 and6–8–9. Roots3 and4 have squared gauge2 in the saved rational congruence; all other squared gauges are1. The black pairs are `{3,4}` and `{6,9}`. The white set is the remaining11 roots.

The complete core has signature `(13,1,1)`. Its rational-gauge radical is

`(-1,-2,-3,-2,-1,0,6,3,4,2,5,4,3,2,1)`.

The corner retaining3,6 omits4,9. Its13-node component F is the entire affine B12 factor, with a positive kernel. The checker verifies the full signature, the white positive definiteness, all four strict edge diagrams, the positive affine kernel, and each of the four corner signatures using exact rational arithmetic.

Let e be any additional supporting white root of the actual quadrilateral. The corner containing F is positive semidefinite. The positive kernel, together with nonpositive native off-diagonal products, therefore forces e to be orthogonal to every root of F. This is a statement about the actual corner; it does not assume simplicity of the ideal vertex.

The displayed indefinite core contains a timelike vector. Its Gram radical represents an actual zero linear combination of its roots: the represented vector is orthogonal to the core and has square zero, while the orthogonal complement of a timelike vector is positive definite. Pairing the radical with e consequently gives

`-<e,scaled_root4> + 2<e,root9> = 0`.

The only remaining possible core contacts are4 and9. They either both vanish or are both nonzero. In the latter case the adjacent marks4,9 have the actual path4–e–9 of length2, which is already paid. Hence, under the unpaid-face hypothesis, every additional supporting white is orthogonal to the entire15-root core. The reversed context1206 has the same proof, with its own saved radical and scaled omitted-root coefficients.

The actual corner diagrams therefore split as the displayed core corners plus a positive-definite supporting-white diagram. In particular each displayed finite component remains a whole component, with no indirect supporting tail or bridge. Each of the three other actual corners is finite, since its full diagram is a direct sum of positive-definite blocks.

The two finite halves

For context160, the finite corners have these marked roles:

Marks	Whole marked components	Payment
3,9	B10 at doubled terminal3; A2 at endpoint9	new1/2
4,6	A1 at4; D12 at fork center6	existing1/2
4,9	A1 at4; A2 at endpoint9	0

The unmarked extra A1 in the first corner and unmarked path component in the last corner do not alter these roles. The reversed context gives the same two halves. The source reconstructs each connected component, its finite type, all bond labels and the marked endpoint/fork roles from the rational-gauge Gram. Thus the two halves replace the removed ideal unit in both selected contexts.

The new rule contributes exactly1 per whole A2/B10 component pair at a finite vertex: there are two A2 endpoints and one doubled terminal. If a is the total rank of the eligible small components and v the number of B10 components, this contribution is at most `a*v/2`. This elementary count does not itself establish the joint finite-vertex budget.

The universal D14 immediate alternative has a strict gap

The affine-D14 long-pair theorem originally paid a connected affine B12 alternative when a marked fork center a is deleted and its replacement c attaches by label4 to the long endpoint of the remaining D12 block. Its positive affine kernel forces all other roots of that adjacent corner to be orthogonal to the block, so no additional outer contact is missing from this profile.

Here that immediate unit is unnecessary. Keep the original D14 factor F of order15, with marks a=0 and b in `{10,13,14}`, and use the same cusp coefficient normalization and Schur coordinates as the frozen proof. Put `x=-a.c>=0`. Direct elimination gives

`c0=0, l=sqrt(2), h=x+sqrt(2), g_c=2sqrt(2)/(x+sqrt(2))<=2`.

This bound covers every nonnegative opposite product, including all finite labels, parallel and dotted cases. The source independently reconstructs the original15-node Gram and these three exact eliminations; it does not merely copy the endpoint maximum.

At the opposite end, the complete inherited profile lists give `g_d<=6` for a fork center and `g_d<=8` for a short leaf. These bounds include the former immediate center profiles themselves: `inner4` has cost at most2 and `inner_next3` at most4. The arbitrary outer-label estimates from the original proof remain in force. Thus the argument does not depend on paying another immediate order13 alternative first.

If any adjacent native product or short spectator path already pays, there is nothing to prove. Otherwise the weighted-tree identity gives

`det(S)/(h^2*k^2) = 1-(g_c+g_d)/R`,

where R is10 for the opposite fork center and12 for a short leaf. Consequently this determinant is at least1/5 or1/6, respectively. Arbitrary positive-definite spectators add

`(1/R)*(x_Z/h-y_Z/k)^T G_Z^(-1)*(x_Z/h-y_Z/k) >= 0`.

The actual quadrilateral requires the determinant to vanish, a contradiction. Reflection gives the same conclusion when the deleted center is at the other end. Therefore the D14 theorem remains valid after removing the B12 singleton; this branch supplies an actual alternative payment rather than an unpaid new assumption.

The distance11 orbit is recorded but retained

The same sealed core also underlies contexts166,170,1744,2137, which support the two canonical pairs `(10,11),(10,12)`. Contexts166 and1744 have a whole finite B13 corner and already receive a unit. Contexts170 and2137 instead have a whole A1 versus a short fork leaf of whole D12; the currently retained rule gives only a half there. A separately justified upgrade of that particular coupon to1 would pay these cases. The checker saves all four literal outcomes, but this package does not supply that upgrade or remove either distance11 pair.

Verification scope

Generation43252 and a separate fresh saved replay89076 both terminated exit0/PASS for both singleton cases, all four conditional distance11 cases, their24 exact corner decompositions and strict edge tests, the three D14 immediate eliminations, and the complete saved-map and pair-set interfaces. Final checks passed for syntax, all16 direct source bindings and all4 report links. Input statuses and every recorded direct source binding are verified before use.

Two separate agents completed full source and written-mathematics reviews, finding no gap in the affine-kernel/radical sealing, whole-component coupon roles, independent immediate-D14 elimination, or the distinction between the removed singleton and conditional distance11 orbit. Neither review repeated the24-corner arithmetic or the frozen801-pattern census. The complete801-pattern embedding search,82,197-overlap construction and inherited all-label profile classifications are explicit frozen dependencies, not repeated finite censuses here. Source, data and report are final and frozen. No global budget, dimension improvement, external refereeing, proof-assistant formalization or geometric realization of a new chamber is asserted.

Source 205

Source path: `results/fresh_order13_b73_scalar_frontier_20261017.md`

SHA256: 868cb27fd435b84598124d57140bde267c45bddc4ea276d888ffec0f112aa21e

The order13 improvement leaves seven ambient65 cusp obstructions

The completed order13 B theorem changes the A/B/C/D factor costs to65/73/73/74 without changing any finite weight. Its normalized rank12 envelope is therefore74/13. On a complete domain of1,650 cusp profiles at ambient dimension65, this reduces the number of negative margins from16 to7. The smallest new margin is −35835/454597 at(10,10,12,16,16).

The source and certificate record every profile and exact rational comparison. This is a scalar diagnostic, not a new dimension theorem. The finite maximum remains487 against curvature480, independently of the remaining ideal deficits.

A complete domain that survives further cost changes

Use the inherited convex-order reduction: spreading two rank summands decreases normalized curvature. Two ranks below10 can be concentrated until one vanishes or the other reaches10, preserving their linear cost. Two ranks at least18 can be concentrated until one reaches17, preserving their constant cost. Every move increases the bounded sum of squared ranks, so it terminates.

The resulting complete domain has arbitrary repetitions of every rank 10 through17, at most one rank1 through9 and at most one rank at least18. It has1,650 profiles of total rank64. It deliberately retains repeated rank13 parts: the previous duplicate13 reduction becomes invalid in one of the hypothetical improvements below. No such reduction is used to justify this domain or its results.

For H(t)=product(1+t+…+t^r), put V=H(1) and G=sum_j[t^j]H(t)*min(j,64−j). Each recorded margin is

`(5G−64V)/(2V) − sum_i c(r_i).`

The source constructs every integer polynomial directly and uses exact rational costs, with the proved new c(12)=74/13. The seven negative profiles are:

Ranks	Margin
(1,10,10,11,16,16)	−3563/209814
(2,10,10,10,16,16)	−1108/67881
(10,10,10,10,10,14)	−14986/219615
(10,10,12,16,16)	−35835/454597
(10,10,14,14,16)	−1376/30855
(10,11,11,16,16)	−3869/114444
(16,16,16,16)	−96/4913

These are comparison obstructions, not geometrically realized cusps. Six have no rank12 part, so further order13 improvements alone cannot resolve the ideal comparison.

Exact simultaneous targets

Starting with the proved order13 maximum74, suppose one could also prove the maximum order15 cost at most94 instead of96, and the maximum order17 cost at most119 instead of120. Then all1,650 margins would be positive, with minimum61/1936 at(9,10,10,10,10,15).

These are maxima over every applicable affine family. Improving only the B family at order15 is insufficient when C or D still has a larger cost. With the order17 reduction and only a one-unit order15 reduction, the single profile(10,10,10,10,10,14) remains negative at−23/14641.

The hypothetical changes leave the low linear and high constant costs used in this domain proof unchanged. Their geometric validity and any additional finite coupons remain obligations. Even if these ideal targets were proved without new finite expense, the finite deficit7 at ambient65 would still prevent a new global bound by this comparison.

Verification scope

Generationfb4876 and a separate fresh saved replay026822 terminated exit0/PASS on all1,650 profiles and all conditional scenario rows. The source imports no project arithmetic. Its actual local input is the frozen order13 B theorem; hypothetical order15/17 drops are clearly separated. A separate full source/report read passed the complete domain, direction of concentration, all-family targets and finite-deficit scope.

An independent standard-library reconstruction also passed all1,650 actual rows and all three hypothetical scenario tables. It enumerates optional low/high parts before multiplicities of10 through17, uses sliding-window convolution, and independently computes G through an inclusion-exclusion cumulative-composition identity. It imports no producer arithmetic. Every domain element, polynomial, V, G and margin agrees, including the old16/current7 negatives. That separate run terminatedfcd4f6 with exit0/PASS.

All11 direct source bindings, three local report links and Python syntax pass. Source, certificate and report are final/frozen. No new ambient bound, external review, formalization or example is asserted.

Source 206

Source path: `results/fresh_order13_bc_inner_terminal_transfer_20261018.md`

SHA256: 4354857b1e4ce2b5b2a67e58d31ccd9cbe1cecf944e1f94502f00007483b6ea0

Universal order13 B71.5/C71, with an absorbed finite half

The ideal order13 family costs become **A65/B143/2/C71/D71**, lowering the all-family maximum from145/2 to143/2. The only new finite payment is1/2 between a whole A1 and the third node from the doubled terminal of a separate whole B12. This unique marked role adds1/2 per component pair, from0 to1/2, within the existing A1/D12 allowance5/2.

Under the resulting simultaneous rule, triangles still receive at least2 and quadrilaterals at least1. The inherited complete finite maximum remains793/2=396.5. The saved complete1,890-profile ambient59 ideal comparison has three negative margins instead of six. The completed unrestricted dimension bound stays59; no bound58 follows.

The producer and certificate bind the current combined rule, the previous B/C13 weights, the order12 profile argument, the independent BC11 theorem and its inherited coupled-D11 half interface. No selected-map census is used to substitute for arbitrary supporting roots.

The actual face and intrinsic pair changes

For affine B12/order13 use spine0–…–10, doubled bond9–10 and fork leaves11,12 at0. Halve the two distance9 pairs(8,11),(8,12). The old sparse half(1,10) and all previous zeros stay unchanged. This saves1.

For affine C12/order13 use path0–…–12 with doubled end bonds0–1 and 11–12. Delete the distance8 pair(2,10), saving1. The old zero(1,11) and all other weights stay unchanged. Both changed sets are invariant under the complete native automorphism groups. The unchanged A65 and D71 totals are the preceding order13 theorem's values.

At an affected ideal quadrilateral corner, let F be the whole varying affine factor containing its adjacent face marks a,b. The full star is W+a+b+c+d, with opposite pairs a,c and b,d. Other supporting roots form a positive-definite diagram Z orthogonal to F at this corner. Every physical root has square2. White and white-plus-one-black principals are strictly positive definite; corner principals are positive semidefinite. These are the established actual factor-fiber hypotheses, including nonsimple ideal vertices.

An adjacent black contact gives an independent distance1 unit. A shared contact coordinate of c,d in Z gives an independent distance2 unit. Otherwise adjacent black products vanish and the nonnegative spectator contact vectors x,y have disjoint supports. Opposite products with a,b remain arbitrary native magnitudes:0 or at least1, with the next value above1 equal to sqrt2. Heights against the positive cusp kernel are strictly positive.

Complete profiles, including the outer B2 block

In B set a=8,b=11. Deleting a leaves the outer B2 on9,10 and an inner D10 on0–7,11,12. Deleting b leaves B12. In C set a=2,b=10; deleting either mark leaves an outer B2 and an inner B10.

The large-component inverse has diagonal1 at its relevant endpoint, 2 at the next node and values strictly greater than2 elsewhere. The source checks all three exact principal types. Their complete PSD one-root extensions therefore permit no contact, a simple or doubled endpoint contact, or a simple next-node contact. Two contacts create a cycle in a diagram already containing a fork or doubled bond; finite diagrams are trees and the only cyclic affine type is A. Exceptional finite/affine diagrams are too small. This invokes the same published classification input as the bound profile theorems, not a native-label cutoff.

The outer B2 inverse has both diagonals1 and positive off-diagonal 1/sqrt2. It lies entirely in the supporting whites, so its extension by the new root is a strict edge principal. Consequently a single outer contact has q²<2 and must be simple. Two contacts have energy at least2+sqrt2>2 and are impossible. Thus there are exactly three outer choices: none, the inner outer-node, or the doubled terminal. Their cusp coefficients are respectively0,1,1/sqrt2; a nonzero choice has energy1.

Let t be the opposite product. Write h for height, l for the inward moment and c0 for the residual constant. The complete near-side list is:

Profile	Height h	Moment l	c0	Cost bound
isolated, t≥1	t	0	2	2
outer inner-node	t+1	0	1	1
outer doubled-terminal	t+1/sqrt2	0	1	2
inward endpoint3, no outer contact	t+1	1	1	3
endpoint3 plus outer inner-node	t+2	1	0	3
endpoint3 plus outer terminal	t+1+1/sqrt2	1	0	3
inward endpoint4	t+sqrt2	sqrt2	0	2
inward next-node3	t+1	2	0	4

The last two profiles contain an affine principal with positive kernel, so every outer and additional supporting contact to that principal vanishes. For endpoint3 the large-component residual norm is1, allowing the listed outer energy0 or1. Enlarging this necessary domain by any subsequently impossible combination only weakens the obstruction.

On the B fork side, the complete four profiles are those of the preceding order12 proof: isolated contact to b, simple or doubled contact to its mate12, or simple contact to center0. Their maximal costs are8,8,4,4. Cost8 occurs only at the isolated t=1 or mate3 t=0 profiles, both of height k=1/2. Every other native instance has cost at most4. The source verifies these bounds using the native gap and polynomials valid on the full unbounded t-rays.

Give B spine coefficient1, doubled terminal1/sqrt2 and fork leaves1/2. Give C inner spine coefficient1 and doubled terminals1/sqrt2. Exact elimination gives resistance R10 for B and R8 for C. Defining

```
g=2l/h+c0/h²,  f=2m/k+d0/k²,
E=(x/h−y/k)^T G_Z^(−1)(x/h−y/k) ≥0,
```

the actual four-black Schur determinant identity is

```
0 = det(S)/(h²k²) = 1−(g+f)/R+E/R.
```

The producer reconstructs both old Schur blocks, all28 end identities, 96 cross identities and the formal determinant with arbitrary unequal heights and spectators. Polynomial coefficient certificates establish the bounds; no opposite native magnitude is silently bounded above.

C first: an independent distinct unit

If either C side has the next-node3 profile, its adjacent corner has a whole affine B10/order11 factor. The marked pair has canonical images (6,9),(6,10), both at distance7 and currently weight1. Its positive kernel seals it against every additional supporting white. This is a distinct actual corner, so the original C weight is unnecessary.

If neither side has next-node3, both costs are at most3. The determinant is then at least1−6/8=1/4, contradicting zero. Together with the earlier distance1/2 escapes, this proves a full face payment independently of the new B halves or any finite coupon.

The source also records a direct triangle exclusion. A third triangle side whose extensions of F minus a and F minus b are PSD can meet F only in{0,1,2,3,4} and only in{8,9,10,11,12}, respectively. These sets are disjoint, so it is orthogonal to F, contradicting its positive height at that ideal triangle vertex. Independently, both newly changed distances8 and9 lie outside the ideal distances used by the bound finite7/ideal5 triangle theorem. Thus no triangle payment is removed.

B: immediate profiles and vanishing far spectator support

The near endpoint4 profile gives a whole affine B10/order11 corner at a currently zero terminal/fork pair. Apply the independent BC11 zero theorem first whenever such an actual corner occurs. Its retained short paths and finite outputs are unchanged here. The near next-node3 profile gives a whole affine D10/order11 opposite-leaf half. That distinct half plus the retained selected B13 half sums to1.

After these immediate profiles, g≤3. A nonextreme fork side has f≤4, giving a strict determinant gap3/10. The fork side is therefore one of the two extremes, with f=8 and k=1/2. The identity becomes

$E=g-2,\ \ 0\le E\le 1.$

The complete near instances with g≥2 are isolated t=1 (h=1,g=2), outer-terminal t=0 (h=1/sqrt2,g=2), and endpoint3 without outer contact at t=0 (h=1,g=3). All other native instances have g<2. In particular h≤1. Monotonicity and the gaps t=0 or t≥1 and t≥sqrt2 after1 prove this for all native products.

The entire fork-side spectator vector y vanishes. If y is nonzero and x=0, coordinate Cauchy gives $E\ge(y_i/k)^2/2\ge 2$. If both supports are nonempty, choose a y-coordinate i and a disjoint x-coordinate j:

$E \ge (y_i/k+x_j/h)^2/(4-2(G_Z)_{ij}) > 9/8 > 1.$

The numerator is at least9; strict positive definiteness gives $(G_Z)_{ij}>-2$, so the denominator is less than8. These inequalities exclude y≠0 without restricting x or the rank/type of Z.

Use the actual corner W+a+d, which omits b,c. Since y=0, all Z is orthogonal to its displayed finite components; c and its arbitrary contacts are absent. For the isolated fork extreme, d is a whole A1 and a lies at the third node from the doubled terminal in a whole B12. The new half plus the selected half pays1. For mate3, both marks lie in a whole B13 path at distance10, with its unchanged unit. The source checks strict positive pivots, complete components and intrinsic roles for both corners. No same-component D16/A17/D17 payment is consumed.

Simultaneous current rule and exact arithmetic consequences

Apply the new C theorem first to any actual face with a changed C13 corner. Otherwise apply the independent BC11 priority when needed, then the new B theorem at any changed B13 corner. These proofs use only the retained B11 units/D11 half, finite B13 unit, short1/2 units or the new typed half. They cannot recurse through another newly removed pair. If no changed B/C13 corner occurs, all old actual ideal weights agree pointwise and finite weights only increase, so the current proved whole-face payment is inherited. This also protects older short8→7→6/whole-A8 and selected/nonselected proofs whose earlier preferred output might have been the changed C distance8 pair.

The current D16, A17 and D17 transfers consume different whole finite components/marked roles. No finite weight decreases here. The producer reconstructs all216 current literal configurations from their Grams, including every new ideal debit and A1/B12 credit; no literal changes occur and every lower total remains at least1. This is an interface check, not a complete geometry census.

The A1/B12 marked role is intrinsically unique (canonical node9), so its exact new actual component cost is1/2. All32,400 ordered actual type pairs still fit the *same* simultaneous representative costs. In particular0+1/2≤5/2 for A1/B12 versus A1/D12. All residual weights, contextual charges and fixed costs therefore remain those of the completed finite certificate. Its upper793/2 remains attained by the unchanged actual2A16+D27 forest. No forest DP is rerun and no geometric realizability is inferred from that formal attainment.

On the bound complete1,890 terminal rank58 ideal rows, the normalized cost decreases exactly by count(rank12)/13. All three negative rows containing rank12 become positive; no zero appears. The remaining negative normalized margins are

Factor ranks	Margin
9,9,9,15,16	−389/5440
8,9,9,16,16	−1087/52020
6,9,9,9,9,16	−19/59500

This is an exact update of the frozen complete domain, not a new convolution or an extension of negative terminal margins to arbitrary cusps. It changes no low-rank concentration hypothesis. The finite curvature remains783/2 and the extremum constant4350 is counted once in the inherited global method. The still-negative ideal margins and finite excess prevent a dimension58 conclusion.

Verification

Final generation20307/chunk0648e5 and separate fresh replay30080/chunkb4bb29 both exited0/PASS, including the explicit old-family/envelope and short-pair retention guards. Root completed a full final producer/report mathematical read and reread the inherited fork profiles, positive-polynomial and determinant helpers: PASS. It checked the complete strict outer-B2 domain,28/96 physical identities, independent C-first whole-B11 payment, entire far-vector Cauchy, whole finite alternatives and unique B12 role, triangle/current-corner priorities, simultaneous cost domination, unchanged finite attainer and exact six-to-three ideal update. That review did not execute field arithmetic, literals, censuses, DP or convolutions. Exact_hub_counts independently completed the full producer/report read and the same inherited fork/polynomial/determinant helper reads: PASS. It checked the complete strict/PSD profiles, C-first unit and B immediate interfaces, unequal-height full-vector Cauchy, whole marked finite roles, triangle/current priorities, unchanged representative bound/attainer and the complete saved-row delta. It did not execute symbolic, literal, type-pair, DP or convolution arithmetic. Final integrity checks pass all340 direct bindings, seven report links and the Python AST. The source, certificate and report are final/frozen. The historical one-root classifications, factor-fiber identities and completed current proof remain explicitly bound inputs; no external refereeing or proof-assistant claim is made.

Source 207

Source path: `results/fresh_order13_bc_sparse_terminal_transfer_20261018.md`

SHA256: 1d1c590ef04f417cf4c72ca6e68868e0da2c07cc962f438461285b0accf4c2f2

Order13 B72.5 and C72 with one sparse B9 coupon

In the varying whole affine factor of order13, lower the B12 pair(1,10) from1 to1/2 and delete the C12 pair(1,11). Their factor totals become **B=145/2** and **C=72**. Add a half-payment between either endpoint of a whole finite A4 and the unique doubled terminal of a separate whole finite B9. There are two eligible marked pairs, so this new actual component-pair cost is **1**, formerly0. All other finite weights and coupons remain unchanged.

The canonical B diagram has spine0 through10, doubled bond9–10, and short leaves11,12 at0. Its changed pair has distance9 and is invariant under the fork swap. The canonical C diagram is the path0 through12 with doubled end bonds0–1 and11–12. Its changed pair has distance10 and is fixed by reflection. Triangles retain payment2 and quadrilaterals payment1 under the complete simultaneous rule below.

The source and certificate bind the current fractional rule, the preceding B13 theorem, and the C13-authorizing composition. Together with the separately completed D13 center/leaf theorem, the family totals are A65/B145/2/C72/D71, so the all-family maximum is 145/2. The new A4/B9 coupon must be included in any finite forest count. This packet proves no forest maximum or new ambient dimension bound.

Complete selected and unbounded protected domains

The complete2614 high-overlap contexts partition as follows:

Route	Count
Whole-component bound at most12	822
Eligible order13 native patterns	801
Displayed component larger than13	908
Protected marked-leaf/feature route	74
Adaptive finite route	9

For each B/C target the source repeats all801 native induced embedding searches, including failures, and compares every successful map and marked image with the complete frozen order13 support record. For B, there are107 successful patterns and240 maps; only contexts157,174,952,2474 are affected, with eight maps. For C there are13 successful patterns and30 maps; only625,1556,1642 are affected, with six maps. These are complete saved necessary domains, not an enumeration of all ambient chambers.

The B pair is basic at distance9 and includes a leaf. Therefore the selected domain alone would not justify changing it. The producer separately reconstructs **all74 protected native bases**, preserving the actual old marked component and both marks. Each potential completion to the order13 target requires an injective induced native embedding of this displayed connected component. A base with more than13 vertices cannot embed; a native label other than3/4 cannot embed into these targets. In every remaining case the source exhausts all native embeddings. None maps the protected marked pair to either changed pair. The complete maps and reasons for rejection are saved.

This excludes arbitrary larger supporting tails: an actual whole order13 factor would still restrict to one of those induced embeddings, regardless of how omitted supporting roots are attached. It is not an assumption that the displayed component

is already whole. The nine adaptive cases have finite outputs whose weights do not decrease. Low-core factors are too small, and the high-disjoint branch remains excluded by its bound all-label theorem.

Proper completions in the physical normalization

For every proper case the source reconstructs the full native Gram, the actual selected corner and necessary extra-white contact domain, the strict common-white and edge principals, and the nonsingular Lorentz inverse J. Allowed contacts split into two ends; every inverse end block is entrywise positive. A root contacting both ends gives an independently paid full-star opposite-black path of length at most8.

Two B cores have old diagonal gauges1 and2. Write an old stored normal as sqrt(g_i) times its physical square2 normal. A new physical square2 root with native contact magnitude q to it has stored contact magnitude q sqrt(g_i). Consequently a forced first entry gives

```
projection square >= q² g_i J_ii.
```

For a simple bond q²=1; for a doubled bond q²=2. No old diagonal gauge is silently replaced by1. Positivity of the entire end block proves this bound even with arbitrary other same-end contacts; both-end contacts already have the independent short-path alternative. The positive-definite complement of the Lorentz core requires the projection square to be at most2.

In B contexts157 and952, the displayed component has12 nodes and is missing one simple short fork leaf. Every affected map forces that leaf to contact the old fork center, with physical projection lower bound14. Thus no such proper completion is possible.

In C contexts625,1556,1642, the displayed A11 is target nodes1 through11. Both doubled terminals are missing. Their forced projection bounds are3 and7 in625/1556, and5/2 at both ends in1642. Every bound exceeds2. These contexts have the inherited relevant-component routing tag; the argument instead checks their actual Lorentz inverses and complete contact domains directly. No optional exterior-root or indirect-tail assertion is substituted for this first-entry contradiction.

Every required first root is genuinely outside the old core. The corner marks are already displayed, so the missing vertex is a supporting white. Omitted old black roots are absent from this corner; other old corner components are orthogonal to the displayed marked component, whereas the missing vertex has a prescribed nonzero first bond to it. Contacts to arbitrary spectator roots impose further conditions and cannot repair a failed necessary projection inequality. No unspecified Gram entry is set to zero.

The two whole B factors and the new half

Contexts174 and2474 have the whole order13 affine B factor inside a 15-root core of signature(13,1,1). The source reruns the full inherited `inspect_case`, checking its physical gauges, positive affine kernel, strict white/edge supports, complete possible contact domain, full radical and all corner components.

The positive affine kernel seals the whole factor against extra supporting whites in its actual ideal corner. The only possible remaining contacts are the two omitted opposite black roots. Their coefficients in the full-core radical have opposite signs. That radical represents zero because the actual span contains a timelike vector. Hence an extra white contacts both omitted roots, giving the independent length2 payment, or neither. In the latter branch every extra supporting white is orthogonal to the whole core. A bridge through an arbitrary spectator would have a first contacting root and is excluded by the same argument.

All displayed finite components are therefore whole actual components. At a distinct corner the marks are an endpoint of whole A4 and the doubled terminal of whole B9. The new coupon contributes1/2; the selected ideal B pair retains1/2. These distinct corners give1. No finite same-component unit is consumed, and the original ideal half is not discarded. The other finite corners need not pay.

Current interfaces and simultaneous priorities

All216 current literal configurations are reconstructed from their native Grams and the final fractional rule's cornerwise lower payments. Every actual affine B/C component of order13 is mapped to its complete canonical target, and every finite

corner is checked for the new whole coupon. No changed ideal image or new coupon occurs in this list; all current literal totals are unchanged and at least1. This supplements the complete actual-face reduction rather than replacing it.

The immediate order13 interface from the older D14 universal theorem is also checked: none of its native marked pairs is a newly changed B/C pair. Its B heavy-terminal/fork-center and heavy-terminal/fork-leaf roles differ from the present heavy-terminal/degree2 role. The universal radius-five tree replacements use immediate ideal paths of length at most7 and sealed finite alternatives; the cycle replacements have ideal paths of length1/2 and longer finite or typed-coupon outputs. They do not consume the present distance9/10 ideal units. All short8 pairs and elementary triangle pairs of these B/C targets remain units.

Current independent changed-ideal-corner priorities are used first. They give retained short paths or their specified finite/ideal alternatives. The separate new D13 center/leaf theorem can also be applied first whenever its changed corner occurs: its only immediate ideal outputs have order11, and its finite outputs are whole D10/D12 coupons or finite D13. None is a changed B/C13 pair or the new B9 coupon. Conversely, the short8 escape here cannot be a D13 distance9 pair.

If a short8 escape uses a deep finite8 zero, the independent depth8 theorem supplies short7. If that uses a deep finite7 zero, the independent depth10 theorem supplies short6. The current small-factor/tree/cycle interfaces pay those paths without either proposed long B/C pair. No circular appeal to the new theorem is needed.

After these priorities the complete selected/nonselected reduction applies. Every affected selected image is excluded by its physical projection or paid by the distinct whole A4/B9 half. The exhaustive 74 protected-base check covers the only unbounded component branch. All other selected images retain their old weights. On a face with no changed actual B/C corner, the old weights agree pointwise and the new finite coupon can only add payment. This gives the claimed whole-face inequality, rather than assuming an old preferred witness remains paid. Triangle payments are unchanged because the changed distances9/10 exceed its protected ideal distances and their complete elementary triangle sets are retained.

The actual A4/B9 component-pair increment is exactly2·(1/2)=1. It is separate from the already existing A2/B9 coupon and from all D10, D11 or D12 coupons. Root's finite preflight is not used as a theorem in this packet; a final joint finite certificate is a separate obligation.

Verification and scope

Final generation4d5976 and separate fresh saved replay9ca3bc both terminated exit0/PASS. This packet reconstructs both801 native target domains, all74 protected bases for each target, proper physical bounds, two radical full-corner proofs and216 current literal interfaces. The inherited all-label geometric census and universal short-path arguments remain bound inputs.

Two independent full source/report mathematical reviews passed. The root review reread the complete affine-core helper and physical native-label conversion, checking the full protected induced-embedding domain, gauge-aware projection, whole finite factors, current literals, immediate and short-path priorities, and family-cost provenance. The construction review additionally reread finite-component typing, actual-corner contact extraction and the full order13 diagnostic/immediate interface; it checked that arbitrary tails cannot bypass the protected-base obstruction and that the selected half and distinct new coupon have their stated roles. Neither review repeated embeddings, projections, inertia, literals, arithmetic or the historical census. Final direct source bindings, report links and Python syntax passed. The source, certificate and report are final/frozen.

No historical census, finite forest optimization, external review, formal verification or new ambient dimension theorem is claimed.

```
python3 scripts/fresh_order13_bc_sparse_terminal_transfer_20261018.py --verify
```

Source 208

Source path: `results/fresh_order13_d_center_leaf_half_transfer_20261018.md`

SHA256: 16a4b399bd0f7d957d43123c3f4ca16e4a7ae84a12a35979f57cb1efb1afd1d7

Order13 D71 using existing whole-component halves

In a whole affine D12 factor of order13, lower the four center-to-opposite- short-leaf pairs from1 to1/2. Number the spine0 through8, with leaves9,10 at0 and11,12 at8. The changed orbit is

`(0,11), (0,12), (8,9), (8,10).`

Every pair has distance9. The factor total drops73 to **71**. No finite weight or coupon changes. The earlier center pair(0,8) remains zero. Every triangle still receives at least2 and every quadrilateral at least1 under the simultaneous rule described below. The B and C families at this order still cost73, so this theorem alone does not improve the all-family envelope or the completed dimension bound60.

The producer and certificate bind the current fractional rule, the earlier center-zero theorem, the complete center/leaf profile argument, and the arbitrary-rank energy-one theorem. The argument is universal in the supporting rank and native labels; it does not infer coverage from the selected overlap maps.

Actual face and complete profiles

Write an actual quadrilateral star as W+a+b+c+d with opposite pairs a,c and b,d. Normalize physical roots to square2. The supporting-white diagram and each white-plus-one-black edge diagram are strictly positive definite; adjacent corners are positive semidefinite. At the selected ideal corner the varying whole affine factor F contains a,b, and its orthogonal complement in that corner is a finite spectator diagram Z. These are the inherited factor-fiber hypotheses, including nonsimple ideal vertices.

By diagram automorphisms it suffices to take a=0 and b=11. Adjacent black contacts give a retained distance1 unit; a common spectator contact of c,d gives a distance2 unit. Otherwise their adjacent products vanish, their nonnegative Z-contact vectors x,y have disjoint supports, and the opposite products remain arbitrary native products, including parallel and dotted products. Each nonzero magnitude is at least1; above1 the next possibility is sqrt2. Both heights relative to the selected cusp are strictly positive.

Give the spine cusp coefficient1 and each short leaf coefficient1/2. The resistance between a,b is R=8+2=10. The finite principal diagram F minus the center splits as two A1 roots and D10. The finite diagram F minus b is D12. For a further root, either large D component permits only isolation, a simple or doubled bond to its long endpoint, or a simple bond to its second node. The inverse diagonals at those nodes are1 and2; all other diagonals exceed2. A second contact creates a cycle in a component with a fork. The finite/affine classification excludes such a cycle and excludes exceptional types containing D10 or D12. The strict white-plus-one-black edge condition is used where the corner alone could have equality. This is an all-label classification, not a maximum-label test.

For the center end, after allowing its two old isolated leaves, the complete symbolic profiles are:

Profile	Contacts	c0	Moment 1	Height h
outer	a:t,9:u,10:v	$2-(u^2+v^2)/2$	0	t+(u+v)/2
inner3	a:t,1:1,9:u,10:v	$1-(u^2+v^2)/2$	1	t+1+(u+v)/2
inner4	a:t,1:sqrt2	0	sqrt2	t+sqrt2
next3	a:t,2:1	0	2	t+1

Profile	Contacts	c0	Moment l	Height h

The inner3 outer contacts are exactly none, one by3, one by4, or both by3. These follow from the same finite/affine classification after the large block has extended to D11. The outer profile admits every native u,v allowed by the strict edge inequality u²+v²<4. The inner4 and next3 profiles are handled by the immediate actual-corner priority below.

For the leaf end, let t be its opposite product and12 its mate:

Profile	Contacts	d0	Moment m	Height k
isolated	b:t	2	0	t/2
mate3	b:t,12:1	1	1	(t+1)/2
mate4	b:t,12:sqrt2	0	sqrt2	(t+sqrt2)/2
center3	b:t,8:1	0	2	1+t/2

The producer checks the two principal inverse lists, all eight end identities and all16 separated cross identities directly at R=10. For costs g=2l/h+c0/h² and f=2m/k+d0/k², eliminating F and Z gives

```
det(S)/(h²k²) = 1−(g+f)/10 + E/10,
E = (x/h−y/k)^T G_Z^(−1)(x/h−y/k) ≥0.
```

The actual four-black Schur determinant is zero. The formal identity is checked symbolically, including the spectator terms and unequal heights.

The two immediate affine cases use current weights

At center inner4, the adjacent corner contains a whole affine B10 factor of order11 with marks at its doubled terminal and opposite short leaf, distance9. Its current literal weight is zero. We do **not** use an old whole-order cutoff here. Apply the completed independent BC11 zero theorem first to any actual changed B/C11 corner; that theorem supplies a full payment using its own retained short paths and distinct finite/ideal alternatives. Its outputs do not involve an order13 center/leaf pair.

At center next3, the adjacent whole affine D10 factor has its two marks at opposite short leaves, distance8. Its literal weight is still1/2 in the coupled order11 rule. It is a distinct corner from the selected D13 half, so the two halves give1 directly. A positive kernel seals each immediate affine component against every other acute root in that corner. The checker reconstructs both whole components, their kernels, all canonical marked images and their exact current weights.

Cost bounds and the unequal-height branch

After the immediate cases, the center cost is at most6. Its unique extreme is a simple new leaf c on either9 or10, with t=0 and h=1/2. Every other center profile has g≤3. More precisely, a nonextreme center with g≥2 has only these possibilities, up to exchanging9 and10:

Profile	Height	Cost
outer isolated, t=1	1	2
outer one doubled contact, t=0	1/sqrt2	2
inner3 with no outer contact, t=0	1	3

To justify this discrete list, the outer isolated cost is2/t². A single outer contact with t=0 has cost8/u²−2; the native gap gives the extreme6 at u=1, cost2 at sqrt2, and smaller values for all larger labels. With t≥1 and a nonzero outer contact its cost is at most2/3. Two outer contacts give at most1. For inner3, any nonzero t gives at most5/4; one outer simple contact gives at most14/9, a doubled contact less, and two simple contacts at most1. Thus the displayed list covers all nonextreme profiles with g≥2, and in particular **h≤1** there.

The leaf cost is at most8. The two extremes are a simple d attached to the selected old leaf11 or to its mate12; both have k=1/2. Every other leaf profile has f≤4, by the monotone table expressions and native gap.

If neither side is extreme, g+f≤7 and the normalized determinant is at least3/10, impossible. If the center is extreme but the leaf is not, g+f≤10. Equality is necessary and E=0. Positive definiteness and disjoint supports then force x=y=0. Choose the adjacent corner that contains c,b and omits a,d. It has whole A2 marked at c and whole D10 marked at the short leaf b. Their existing half plus the selected half pays1. Orthogonal Z cannot enlarge or join these components.

Suppose instead that the leaf is extreme and the center is not. The determinant equation gives E=g+8−10=g−2, so g≥2, h≤1 and E≤1. The entire leaf-side vector y must vanish. If y is nonzero and x=0, coordinate Cauchy gives E≥(1/k)²/2=2. If both supports are nonempty, choose i in the support of y and j in the disjoint support of x. Cauchy with e_i−e_j gives

$$E \geq (y_i/k+x_j/h)^2/(4-2(G_Z)_{ij}) > 9/8 > 1.$$

Here the numerator is at least(2+1)², and strict positive definiteness of the two-node spectator principal gives (G_Z)_ij>−2 and hence the strict denominator bound<8. Both alternatives contradict E≤1.

With y=0, choose the adjacent corner containing a,d and omitting b,c. It is independent of all contacts of c. If d attaches to b, it is a whole A1 against the fork center of whole D12, with the existing half. If d attaches to the mate, both marks lie in whole finite D13 and have its unchanged unit. In either case the selected half gives total≥1. Every remaining supporting white is in Z and orthogonal to F and d, so the stated components are whole without a spectator-rank bound.

Both extreme and the complete energy-one boundary

Now h=k=1/2 and g+f=14. The determinant identity yields

$$D=(x-y)^T\ G_Z^{-1}(x-y)=1.$$

If one vector vanishes, choose the corresponding extreme root's corner as above: center side gives whole A2/D10-shortleaf half; leaf side gives whole A1/D12-center half or finite D13 unit. The other vector can have arbitrary contacts and rank because its root is absent from this corner.

If both vectors are nonzero, the inherited complete energy-one theorem applies: Z+c and Z+d are strict positive-definite principal subdiagrams of the actual edges. With disjoint supports, either the contacts are unit contacts on two distinct whole A1 components, or they are at the two endpoints of A3 or the two short leaves of D4 through D7. The latter cases give an actual c-to-d path of length4 and an unchanged unit. All other spectator components have zero contacts.

For the two separate A1 case, choose the center-root corner c,b. Its whole small component is A3 with c its middle, and the large component is whole D10 with b a short leaf. The pre-existing small-component short-leaf coupon pays1/2 for every mark of A3, so it pays this middle mark. Together with the original half the total is1. No new A3-middle coupon is introduced: that new role in the earlier center-zero theorem concerned the **D10 fork center**, whereas the present large mark is a **short leaf** already covered by the old whole-small rule.

The checker constructs the four center-side finite representatives (two old leaves, with or without the isolated spectator), the two leaf-side representatives, and the five mixed energy-one diagrams. These corroborate the exact component roles. Arbitrary-rank completeness comes from the inherited energy-one theorem and the analytic one-sided argument, not from enumerating these representatives.

Simultaneous payment interfaces

The only finite payments used are the existing whole A2/A3 against whole D10 short-leaf halves, whole A1 against whole D12 fork-center half, and the whole finite D13 unit. No newly reduced finite A15/B15/D15, A16/D16, A17 or D17 pair is used. The separate new D16/A16 local work therefore does not alter these outputs. No finite cost needs a new optimization for this theorem alone.

Apply BC11's independent actual-corner priority first. If such a corner does not occur, the inner4 branch is absent. For any changed actual D13 center/leaf corner the complete argument above then obtains1 from its retained half and another existing half/unit, or directly from a short distance1,2 or4 unit. No result invokes the proposed D13 theorem to pay its own output. On faces with no changed actual corner the prior rule agrees pointwise. This also covers protected-overlap and immediate uses of the four pairs without a new finite selected-map census.

The old D13 center-zero theorem uses only order11 center/terminal units, short4 or whole D10-center coupons. The current B/C completion proofs, order15/17 transfers and finite zero rules retain their independent short8→7→6 routes. Their universal tree/cycle replacements use smaller ideal paths or sealed finite roles, while this theorem handles any newly encountered D13 center/leaf corner directly. All its short outputs have length at most4, so no deep finite7/8 zero is consumed.

Triangle payments are unchanged: all altered pairs have distance9, beyond the elementary finite7/ideal5 triangle interface. The earlier protected small-factor and fixed-leaf triangle exceptions do not use these pairs. All short8 literal D13 weights are unchanged, including the already-zero center pair at distance8.

Verification and scope

Final generationdb9c64 and separate fresh saved replay83e0ba both terminated exit0/PASS. The producer checks exact inverse/profile/cross identities, the spectator determinant, native boundary values, current immediate roles and finite components. It binds the complete inherited profile and energy-one theorems.

Two independent full source/report mathematical reviews passed. The root review included the inverse-diagonal guards and inherited energy-one and below-one arguments, the complete $g \geq 2$ profile list, unequal-height Cauchy, current immediate roles and six whole finite representatives. The construction review additionally reread the large-D one-root profile proof and energy-one checker; it checked that vanishing of the entire extreme-side vector suffices despite arbitrary contacts of the absent opposite root, and distinguished the existing short-leaf coupon from the separate fork-center coupon. Both checked current priorities and triangle protection. Neither repeated the formula, profile, inertia, classification or historical census arithmetic. Direct immutable bindings, report links and Python syntax passed after review. The source, certificate and report are final/frozen.

No historical all-label census, finite forest optimization or global dimension proof is rerun. No external or formal verification is claimed.

```
python3 scripts/fresh_order13_d_center_leaf_half_transfer_20261018.py --verify
```

Source 209

Source path: `results/fresh_order13_d_center_pair_half_transfer_20261018.md`

SHA256: 1c2fc50f2d406d1965e73307d80eed5ff81efae3e7937b8dc0bd37ed4e702e1a

Halving the affine D12 fork-center pair

An actual ideal varying factor of type affine D12 has order13. Its two fork centers are distance8 apart. Their pair may be reduced from1 to1/2 after adding the following finite payments between separate whole components:

- either endpoint of a whole A2 against the fork center of a whole D10: weight1/2, total increment1 per component pair;
- the unique middle of a whole A3 against the fork center of a whole D10: weight1/2, total increment1/2 per component pair.

All other weights are retained. The affine D12 total falls from74 to147/2, with normalized cost147/26. Every affected quadrilateral has a distinct unit-paid corner, or a distinct new half-coupon corner which combines with the retained original half. This is a universal local theorem, allowing arbitrary native labels and arbitrary supporting spectators. Its finite budget and any ambient dimension conclusion are separate obligations.

The source and certificate verify the exact formulas, affine alternatives and finite component roles.

Actual star and complete center profiles

Write the actual quadrilateral star as W+a+b+c+d, with opposite pairs {a,c},{b,d}. All roots have square2; W and each actual edge W plus one black root are positive definite, adjacent corners are positive semidefinite, and the four-root residual Gram after eliminating W has determinant zero. At the selected ideal corner W+a+b, the factor-fiber theorem gives the whole varying affine factor F containing a,b and a positive definite spectator diagram Z orthogonal to F. This formulation applies also to nonsimple ambient cusps.

Use spine0–1–…–8, leaves9,10 at0, and leaves11,12 at8; a=0,b=8. The positive kernel has coefficient1 on the spine and1/2 on the four leaves. Its products with the opposite roots give positive heights h,k. If b.c, a.d or c.d is negative, a distinct actual corner has a retained distance1 unit. A common nonzero spectator contact of c,d gives a retained distance2 unit. Hence assume these three black products vanish and the nonnegative spectator vectors x,y have disjoint supports. Opposite products a.c and b.d remain unrestricted: their magnitudes are zero or at least1; if larger than1 they are at least sqrt2. This includes parallel and dotted opposite sides.

Deleting a leaves a finite D10 block B on1,…,8,11,12, and the two isolated old leaves9,10. In B inverse diagonals at nodes1,2 are1,2; every other diagonal exceeds2 (the fork leaves have5/2). A single nonzero contact therefore can only be simple or doubled at1, or simple at2. There cannot be two contacts to B: they create a cycle in the augmented connected diagram already containing the remote fork. Finite and affine Coxeter classification permits no such diagram. Exceptional types are too small to contain D10. Thus the complete positive-semidefinite principal extensions B+c have precisely these four profiles, including the isolated one. This is a classification argument, not a bounded label search. In the doubled-at1 and simple-at2 cases, a positive affine kernel forces every other root of the actual corner to be orthogonal to that component. Consequently c has no contact with the old isolated leaves or with Z in those cases.

Put t=-a.c and let u,v be the contact magnitudes to9,10. Eliminating F minus a,b gives the following exact data, with R=8:

Profile	c0	1	h
outer only	2-(u²+v²)/2	0	t+(u+v)/2
simple contact at1	1-(u²+v²)/2	1	t+1+(u+v)/2
doubled contact at1	0	sqrt2	t+sqrt2

Profile	c0	l	h
simple contact at2	0	2	t+1

Here the residual c square before Z elimination is c0+l²/8, and its cross term with the far scaled mark is -l/8. Reflection gives the right end data. The supports are separated, and every left/right projection cross product is lm/8. The source checks all four profiles and all16 ordered cross identities directly in the physical norm-two Gram.

The last two profiles already give distinct retained **unit** corners. They are whole affine B10 and D10 factors of order11, respectively, with the old far mark a fork center and c a terminal. Their marked distances are8 and7. The current B/C/D11 rule only halves B11 terminal/fork-leaf distance9 and D11 opposite fork-leaf distance8 pairs; these center pairs remain units. The source checks the four reflected literal affine components, positive kernels, marked roles and the exact retained weights. The C11 endpoint-zero change does not affect either output.

We may therefore discard the last two profiles after paying a distinct unit. For the simple-at1 profile, the principal corner B+c plus its two old leaf contacts is positive semidefinite, so0<=c0<=1. Since h>=1, its cost g=2l/h+c0/h² is at most3. No discrete outer-label list is needed. For the outer-only profile the strict actual edge gives u²+v²<4. The following cases exhaust every native contact and opposite product:

- u=v=0: positive height forces t>=1, so g<=2;
- exactly one outer contact, t>=1: c0<=3/2 and h>=3/2, so g<=2/3;
- exactly one outer contact, t=0: g=8/u²-2, equal to6 only for a simple contact u=1, and at most2 otherwise because u>=sqrt2;
- two nonzero outer contacts: u,v>=1, hence c0<=1,h>=1 and g<=1.

Thus every unpaid end has g<=6; the unique extreme is a simple new leaf on one old outer fork leaf, with t=0 and height1/2. Every nonextreme has g<=3. These statements hold with all additional spectator contacts: those only strengthen the positive-semidefinite restrictions already used.

Determinant and arbitrary spectators

Write J=G_Z^(-1). Eliminating F minus a,b and then Z gives the exact four-root determinant identity

```
0 = 1-(g_c+g_d)/8 + E/8,
E=(x/h-y/k)^T J (x/h-y/k) >= 0.
```

The source verifies this identity symbolically, including the signs of the off-diagonal spectator contribution. Its use precedes any unscaled native-contact estimate. Two nonextreme ends would give g_c+g_d<=6 and a positive gap at least1/4, which is impossible.

Suppose exactly one end, say c, is extreme. Then h=1/2 and E=g_d-2<=1. If some x_i is nonzero, disjoint supports give y_i=0. Gram-metric coordinate Cauchy yields

```
E >= (x_i/h-y_i/k)²/G_ii = 2*x_i² >= 2,
```

a contradiction. Hence x vanishes on the entire Z, including any connected tails. In the actual adjacent corner W+b+c, deleting a separates the old fork leaf and c as a **whole A2**, while the old far component is a **whole D10** marked at its fork center b. The old other fork leaf is an isolated A1. Every remaining spectator is orthogonal to these components, since it was orthogonal to F and x=0. The new A2/D10 half plus the original ideal half gives1. No restriction on y or on components incident only to the opposite root d is necessary: d is absent from this chosen corner.

If both ends are extreme, h=k=1/2 and g_c+g_d=12, so

```
(x-y)^T J (x-y)=1.
```

Both separate spectator extensions Z+c and Z+d are strictly positive definite, being principal subdiagrams of actual edges. Thus the complete energy-one spectator theorem applies with its native magnitudes and disjoint support hypotheses.

If either support is empty, choose that extreme root and obtain the same whole A2/D10 half as above. This does not require classification of the other support. If both are nonempty, the theorem gives exactly:

- one simple contact on each of two distinct whole A1 components;
- simple contacts at the two endpoints of one whole A3;
- simple contacts at the twin short leaves of one whole D_r,4<=r<=7.

In the last two cases the c-to-d path has length4, giving a distinct retained unit. The theorem's whole-component and no-other-contact conclusions are essential: they follow from equality in Gram-metric Cauchy, the native contact gap, and the strict separate edge inequalities, not from enumerating spectator ranks. Its written arbitrary-rank proof is inherited here; the source corroborates the five mixed finite types.

In the two-whole-A1 case, c is simply attached to one old fork leaf and to its isolated spectator. In W+b+c these three roots form a **whole A3**, with c its unique middle. The far old block remains whole D10 with mark b at the fork center. Every other spectator has zero c contact and is orthogonal to this component; the other contacted A1 is isolated in this corner because d is absent. Thus the narrower A3-middle/D10-center half is sufficient. Combined with the retained ideal half it pays1. We do not charge the two A3 endpoints and do not require both symmetric coupon corners. Eight reflected literal corners in the certificate verify the whole A2/A3 and D10 roles and finite signatures.

Simultaneous rule and interfaces

The pair of fork centers is an intrinsic singleton orbit. The completed dimension63 rule inherits affine D12 cost74 from the whole-leaf transfer. The dimension64 composition records the exact paid set; the source binds these completed theorems, checks the center unit, and changes only that weight to1/2. All ideal distance at most7 pairs remain literal units.

For every actual quadrilateral, first ask whether an actual corner has the changed D13 center pair. If so, apply this theorem to that corner. It uses its retained half and a distinct coupon half, or a distinct retained unit. If no such actual corner exists, every actual weight is the inherited weight, apart from finite coupon increases, and the prior whole-face proof applies unchanged. This priority is independent of which corner an earlier proof had selected; no provenance of a selected witness is substituted for an actual-corner statement.

The theorem is compatible with the independent C11 endpoint-zero rule: its only ideal outputs are the retained B/D11 center units or a path of length at most4. The B12 center/terminal deletion likewise cannot change these outputs. Short finite paths at most4 are unaffected by deep finite8 zeros, the D16/D17 exchanges, or the proposed sparse finite refinements. The coupons concern whole D10, not whole D16 or D17. No existing finite same-component distance8 unit is consumed. These observations concern local compatibility, not an assertion that future finite refinements have already been globally budgeted.

The finite7/ideal5 triangle theorem uses finite distances at most7, ideal distances at most5, protected ideal leaf distances6/7, or whole ideal factors of order at most9. It therefore does not use this order13 distance8 center pair. Triangle minimum2 is unchanged; quadrilateral minimum1 is proved above.

The only additional finite costs are exactly1 per whole A2/D10 component pair and1/2 per whole A3/D10 component pair, with the stated middle and fork-center roles. A separate complete finite-forest bound must include them. No new global dimension bound follows from this packet alone.

Verification scope

Generation04dc72 and a separate fresh saved replay6a0f6d terminated exit0/PASS, checking two finite inverse diagonal lists, four exact end profiles,16 cross identities, the formal determinant identity and scalar endpoint values, four retained whole affine unit corners, eight whole finite coupon corners, and five mixed energy-one finite types. The profile classification and arbitrary-rank spectator theorem are written mathematical inputs, not consequences of the finite checks. Direct audit ed80c9 passes26 source bindings,10 local links and Python syntax.

A separate full producer/report mathematical read passes the complete D10 attachment classification, positive-kernel sealing of the affine B10/D10 unit alternatives, all-native extreme/nonextreme bounds, coordinate-Cauchy whole A2 conclusion without restricting the opposite spectator vector, energy-one whole A3-middle alternative, determinant signs, finite roles and actual-corner/triangle/current-zero interfaces. That reviewer did not rerun formulas, inertia, profiles or spectator classification; the arbitrary-rank energy-one theorem remains an explicit inherited input. Parent separately completed a full source/report mathematical read of the complete D10 attachment cases, positive-kernel sealing, cost bounds, determinant

sign, coordinate Cauchy, whole A2 and A3-middle outputs and changed-corner compatibility; no formula or inertia replay was duplicated. Source, data and report are final/frozen. No external refereeing or proof-assistant formalization is claimed.

```
python3 scripts/fresh_order13_d_center_pair_half_transfer_20261018.py
python3 scripts/fresh_order13_d_center_pair_half_transfer_20261018.py --verify
```

Source 210

Source path: `results/fresh_order13_d_center_pair_zero_transfer_20261018.md`

SHA256: c6d617635e8375f5566770e9702e66b514d0a33b1248ab7afee3b7acdce762a1

Deleting the affine D12 center half with full finite coupons

Relative to the completed dimension62 rule, the unique fork-center pair in a whole affine D12 factor of order13 may be reduced from1/2 to0, provided precisely these whole finite coupons are upgraded:

Separate whole components and marked roles	Old weight	New weight	Old pair total	New pair total	Increment
A2 either endpoint / D10 fork center	1/2	1	1	2	1
A3 unique middle / D10 fork center	1/2	1	1/2	1	1/2

Thus the ideal D12 total falls from147/2 to73. The four order13 family totals become65,73,73,73, with all-family normalized envelope73/13. For a finite forest with multiplicities a2,a3,d10, the added charge is

```
a2*d10 +(a3*d10)/2.
```

This local theorem does not bound that enlarged finite charge globally, lower an ambient dimension, or assert that the ideal improvement alone resolves the current frontier.

The source and certificate compose the frozen universal D13 half theorem with the completed dimension62 rule. The arbitrary-label and arbitrary-spectator proof is inherited in full, not replaced by the finite list of literal corner roles.

Every branch has a distinct full unit

Use the preceding theorem's actual quadrilateral star W+a+b+c+d, with the selected ideal factor F containing a,b. In its canonical numbering, F has spine0 through8 and two short leaves at each end; the changed marks are a=0,b=8. All roots have physical norm2. The actual varying factor is whole, its other supporting components form the positive definite spectator Z, and both opposite-root heights are positive. All retained weights below are evaluated in the simultaneous new rule.

If an adjacent black product is negative, or c,d have a common spectator contact, the preceding proof gives a **distinct** actual corner a path of length1 or2. It still carries a unit. No original ideal payment is used.

For the immediate end4 or next3 profiles, the old positive-kernel argument seals a whole order11 affine B10 or D10 factor at another corner. Its marks are respectively center/terminal distance8 or center/leaf distance7. These are retained units under the completed62 rule, not any of the order11 terminal/fork-leaf halves or the deleted C11 endpoint pair. The source compares all four reflected immediate records with their inherited unit payments.

After these cases the exact R=8 determinant and native profile bounds leave only the following possibilities. These are precisely the full analytic branch partition from the half theorem; no additional rank or label cutoff is introduced.

- Two nonextreme ends are impossible: their cost sum is at most6, giving a strictly positive determinant gap at least1/4.
- One extreme and one nonextreme end force the extreme side's entire spectator contact vector to vanish. The weighted spectator energy is at most1, while any nonzero extreme coordinate would cost at least2 by Gram-metric Cauchy. In the adjacent corner the extreme root and its old fork leaf form a whole A2, separate from the whole D10 containing the far mark at its fork center. The upgraded coupon now gives a **distinct unit**, replacing the old coupon half plus original half. No condition on the other spectator vector is needed.
- If both ends are extreme, the unscaled spectator energy equals1. If one support is empty, the same whole A2/D10 corner gives a unit. If both are nonempty, the bound inherited energy-one classification gives a mixed whole A3 or D4–D7, yielding a distinct path4 unit, or one contact on each of two separate whole A1 components. In the latter case the selected adjacent corner has a whole A3 with its new mark at the unique middle, separate from whole D10 marked at its fork center. The upgraded A3-middle coupon gives a distinct full unit.

Whole-component scope is unchanged by upgrading weights. In the one-sided branch the chosen opposite root has no Z contact at all; the other opposite root is absent from that corner. In the two-A1 branch the equality classification forces all other spectator contacts to vanish and identifies those A1s as entire components. Consequently neither small A2/A3 nor far D10 can acquire an undisclosed supporting tail or bridge. Only the middle of A3 is charged; its endpoints never need an upgrade.

The four immediate affine records, eight finite coupon records and five mixed energy-one records are checked as17 literal interfaces. Each now has a distinct payment1 and the original ideal contribution0. Those records corroborate the new roles; completeness comes from the preceding universal proof and the analytic branch partition just given. In particular this theorem is not the invalid inference that an old half-transfer automatically permits deleting its original half.

Simultaneous coverage and finite cost

For every actual quadrilateral, first test whether any actual corner contains the newly zeroed D13 center pair. If one does, the proof above supplies a distinct full unit under the new rule. If none does, every actual ideal weight agrees with the completed62 rule, and finite weights only increased. The preceding whole-face payment at least1 therefore transfers directly. This also covers older proofs which may have selected a D13 corner: the case distinction is about all actual corners, not the provenance of an earlier selected witness.

The new outputs are short paths of length at most4, the retained order11 center units, and the exact whole A2/A3-middle/D10-center coupons. They avoid sparse A16/D16/D17 long deletions, deep11 finite8 zeros, and the already deleted C11/B12 pairs. No changed branch invokes the unchanged-corner case recursively. The order13 distance8 center pair is absent from the inherited finite7/ideal5 triangle proof, so triangle payment2 remains unchanged. Every other pair weight and every other finite coupon is retained.

The finite increments are literal counts of invariant marked roles: A2 has two interchangeable endpoints and D10 one fork center, giving two upgraded halves; A3 has one middle and D10 one fork center, giving one upgraded half. They are distinct from the old D10 short-leaf coupons, which keep their old values. The preceding finite budget counted center totals1 and1/2 only. It cannot be reused as an enlarged budget without adding the displayed increments and recomputing a complete forest bound. That is a separate task.

Verification scope

Final generation61a495 and a separate fresh replay892cd1 terminate exit0/PASS. Root completed a full source/report mathematical read: every upgraded whole-component unit genuinely dispenses with the original ideal half; the17 literal interfaces corroborate the complete inherited analytic branch partition; changed-actual-D13-first priority is noncircular; and the exact increments1 and1/2 concern center roles distinct from the old short-leaf coupons. That review did not rerun matrices, arithmetic or classification. The source imports only standard-library arithmetic and checks frozen bindings, the sole ideal weight change, all four family totals, exact marked-role multiplicities, every inherited immediate/coupon/mixed interface, and the new finite increment. It does not rerun the preceding profile matrices, spectator classification, historical censuses or any finite-forest optimization. Source, data and report are final/frozen. No global dimension claim, external review or proof-assistant formalization is made.

```
python3 scripts/fresh_order13_d_center_pair_zero_transfer_20261018.py
python3 scripts/fresh_order13_d_center_pair_zero_transfer_20261018.py --verify
```

Source 211

Source path: `results/fresh_order13_payment_support_20261017.md`

SHA256: a420e61bce95acf0c79e4019584bb765bbaacebd3cae166692e5ab5690367d61

Exact order-13 payment support and the remaining B/D obstruction

The current rank-12 envelope remains **6**, corresponding to all78 pairs in a factor of order13. The complete selected-pattern diagnostic plus the immediate affine alternatives of the existing D14 transfer require all78 pairs for both the affine B and affine D targets. The analogous unions contain only65 pairs for affine A and73 for affine C. Lowering A or C alone therefore does not lower the worst rank-12 cost.

This package is a diagnostic, not a new payment theorem or dimension bound. In particular, an absent preferred-map demand is not by itself a proof that a pair can be removed from the complete face-payment rule.

The source and exact saved data reconstruct the complete new target-order domain from the frozen 2,614-context classification. They reuse the same exhaustive induced-map algorithm as the established small-factor census, checking both labelled edges and nonedges. No temporary exploratory JSON is an input.

Complete selected-pattern domain

A selected component can embed in an affine factor of order13 only if its proved full-component bound is at least13 and its displayed marked component has at most13 nodes. Exactly801 of the current2,614 contexts satisfy these conditions:

Proved full-component bound	Eligible patterns
13	556
14	108
15	87
16	50

These include606 displayed components of order13,160 of order12,33 of order11 and two of order10. The larger-bound entries cannot be discarded merely because the target is smaller. All injective induced labelled maps of every eligible pattern into each of the four order13 classical affine types are enumerated. The standard affine rank is12 in each case.

Target	Basic ideal5/R5 pairs	Selected marked-pair union	Basic plus selected plus immediate union	Contexts with a map	Emitted maps
A	65	39	65	67	1742
B	73	9	78	107	240
C	72	5	73	13	30
D	74	8	78	305	2344

The basic rule pays distance at most5, or distance6 through9 with a mark within5 of a branch or nonsimple-bond feature. The selected union is not assumed disjoint from that basic set. The saved certificate contains all maps, marked images and whole marked-core keys.

For affine A, the13 pairs absent from this union are exactly its distance-six pairs. Their removal would still require the universal cycle13 replacement in the short8-first stage. The existing cycle theorem starts at larger order, so this note does not silently extend it. For affine C the five absent pairs are `(0,10),(0,11),(0,12),(1,12),(2,12)` in the saved canonical numbering. No new reduced C rule is promoted here either.

The extra B and D orbits

For the B target, the fork center is0 with short leaves11,12; the spine is0–1–…–10 and the bond9–10 is doubled. Its five extra pairs form three symmetry orbits:

Representative	Distance	Orbit size	Selected supporting contexts
0,10	10	1	2
9,11	10	2	23
10,11	11	2	4

The singleton0,10 is required by contexts160 and1206. They are the two orientations of L10 with marks{3,4} joined to L25 with marks{0,3}, sharing L10 node5 with L25 node9. Their preferred13-node component is already affine B12. Unlike the completed order15 reduction, it is not an abstract affine completion of an untouched finite component. Removing this singleton needs an actual new local transfer.

The distance11 orbit has contexts166,170,1744,2137, also orientations and marked choices of this same radical L10/L25 core. The remaining B orbit has23 contexts across several core families. The source saves every literal key rather than classifying these cases by counts alone.

For the D target, the fork centers are0 and8, with leaves9,10 at0 and11,12 at8. The only extra orbit consists of the four cross-fork leaf pairs, each at distance10. It is required by127 selected contexts. Thus the B obstruction is small in one orbit, while the D obstruction already has a substantial complete support domain.

Immediate alternatives in the existing D14 transfer

The universal affine D14 long-pair transfer has an original factor of order15. Deleting a fork center leaves finite D12 plus two isolated leaves. Its replacement profiles `inner4` and `inner_next3` create connected affine factors of order13, already paid in that proof by the whole-small-factor rule.

This package reconstructs both literal13-node labelled diagrams, matches their frozen positive-kernel records, and enumerates all marked images for each of the three possible opposite marks. They give:

Profile	New affine target	Marked-pair roles
inner4	B12, order13	doubled terminal against the far fork center or either short leaf
inner_next3	D12, order13	one new short leaf against the far fork center or either short leaf

The B alternatives therefore include both the singleton0,10 and the two distance11 pairs. The D alternatives include all four cross-fork leaf pairs, as well as already-basic distance-nine pairs. All these immediate images are already in the unions in the preceding table, so the numerical counts do not rise. Their provenance is nevertheless different: proving a new replacement only for the selected L10/L25 or other core contexts would not automatically discharge these universal D14 alternatives.

The lower D13 transfer has immediate factors of order12, so it does not introduce another order13 demand. The previously sealed order16/17 transfers end in finite payments or short paths. The diagnostic does not reprove those frozen conclusions.

A precise next local obligation

For a possible universal affine D12 long-pair theorem, the old profile formulas suggest a useful extreme case. The center/center, center/leaf and leaf/leaf resistances would be8,10,12. Their extreme costs sum to12,14,16, respectively. Conditional on forcing both heights to1/2, the zero-determinant equation requires unscaled spectator energy1. The frozen energy-one classification is therefore directly relevant to this extreme branch.

It does **not** close the entire theorem. Merely reusing the earlier nonextreme bounds, center at most3 and leaf at most4, gives normalized gaps `-1/8`, `-1/10` and `0` in these three end types. Thus there are additional nonextreme profiles and equality cases to analyze; neither the old positive-gap proof nor a height1/2 reduction may be imported without that work. For reducing the four extra D pairs alone, only the leaf/leaf branch is required, so its zero-gap cases are the narrower relevant obligation. The negative center cases matter only to a stronger all-nine-pair theorem. The certificate records these exact conditional comparisons to prevent treating the extreme spectral match as complete coverage.

The formal dimension74 boundary profile `(1,12,12,12,12,12,12)` explains why this is a meaningful obstruction: it uses six rank-12 factors. Nevertheless this report changes no factor cost and draws no new scalar or dimension conclusion.

Verification scope

Generation and a separate fresh saved replay both completed with terminal PASS for all801 eligible patterns,4356 emitted maps, six literal immediate-profile/mark records and the exact orbit counts. Final checks passed for Python syntax, all17 direct source bindings and all5 report links. The checker validates completed input statuses and every recorded direct source digest before using them.

The root agent completed a separate full source and written-mathematics review: the eligible domain includes the larger component bounds, the basic/selected/immediate unions remain distinct, the six literal D14 interfaces are accounted for, and the B/D orbit and conditional D12 gap statements have the stated scope. That review did not repeat the4356-map enumeration. Neither this package nor that review repeats the82,197 overlap construction or its all-label geometric proof. The map enumeration is complete within the explicitly defined selected-component domain; it is not a classification of all actual chambers or all possible face stars. No new local payment rule or dimension bound is asserted. The source, data and report are final and frozen; no frozen predecessor was edited.

Source 212

Source path: `results/fresh_order13_sparse_ambient60_frontier_20261018.md`

SHA256: 0766d99165d8b5774e52453272bd22f584d17e8947abee2a747f3385858e2116

Order13 counted: finite409.5 and ten remaining negative ideal profiles

The complete ambient60 finite maximum remains **819/2=409.5**, including the new whole A4/B9 cost1. The complete2,113-profile ideal calculation now has **ten negative profiles and none zero**, down from17. The finite curvature is405 and the single extremum correction4500. These remaining obstructions prevent a dimension59 theorem; the completed bound is60.

The producer and certificate combine the preceding balanced rule with the universal D13 transfer and B/C13 transfer. All preceding finite costs and ideal order14/17 improvements remain present.

Simultaneous local coverage

Order13 family costs are A65/B145/2/C72/D71, with maximum145/2. The D transfer lowers four center-to-opposite-leaf units to halves and uses existing distinct whole-component halves or a unit. Its arbitrary- spectator proof first applies the independent B/C11 theorem; its other ideal output is a retained D11 half. Its finite outputs are whole D10/D12 coupons or a finite D13 unit, all unchanged by the current rank15–17 rules.

The B transfer lowers one distance9 unit to a half, and the C transfer deletes one distance10 unit. Complete801-pattern native searches per target, including failures, and all74 unbounded protected bases are covered by the local theorem. Proper completions are impossible by physical projection bounds. The two whole B contexts have arbitrary- white radical seals and a distinct whole A4-end/B9-doubled-terminal half. Two eligible finite marked pairs add exactly1 per A4/B9 component pair.

Apply the independent B/C11 and D13 actual-corner priorities before the B/C13 selected/nonselected argument. Neither earlier priority uses a changed B/C13 pair. The B/C13 short8 escape cannot itself be a changed D13 distance9 pair. Its independent deep8→short7→short6 replacements retain their smaller ideal paths and protected finite outputs. Existing order14/15/17 priorities likewise use their specified exclusions, short paths and separate coupons. The new finite charge only increases weights. The frozen local proofs therefore preserve triangle payment2 and quadrilateral payment1 under one rule, with no circular replacement.

All216 inherited literal configurations are reconstructed anew from their native Grams, starting at the balanced rule's final cornerwise lower payments. The binder maps every shared affine order13 block to B/C/D, checks all three changed pair sets, and tests every finite corner for the new whole A4/B9 coupon. No new debit or credit occurs; all lower totals remain at least1 and the four prior D16 eighth credits are present. This check supplements, rather than replaces, the universal geometric proof.

Complete finite budget with the added charge

The new A4/B9 actual and representative pair costs are both1, formerly0. The A4 and B9 representatives are unchanged. All33,489 simultaneous type comparisons verify dominance including the new actual cost. No new fixed rank or interaction key is needed: B9 was already fixed, and A4 is a residual item. In eighth-units its added cost is8 times the B9 multiplicity. The source explicitly identifies that multiplicity as key[5].

All1,044 fixed-component vectors and792 interaction keys are included. Every residual table is newly generated because its A4 item may change. The54,288 item identities include all fixed interactions; fixed-fixed and residual-residual cross costs

remain zero. Replay checks all1,142,064 Bellman upper inequalities and47,520 attaining predecessors without an optimizer. The previous costs A16=110,D16=441/4,A17=117,D17=469/4 and all other coupons and deep7 deletions remain included.

The maximum is still409.5, attained under the actual rule by2D16+D28, D16+D17+D27 and2D17+D26. None of these contains A4/B9. Fifteen representative alternatives remain above405. Formal attainment is an optimization statement, not a geometric realization of a hyperbolic vertex.

Exact ideal comparison

The producer reconstructs the complete preceding integer convolutions and all scalar rows, checks their equality with the frozen balanced certificate, then subtracts the normalized cost

`count(rank12)/26.`

This uses the true order13 maximum73→145/2 and the factor-fiber divisor13. Seven negative terminal profiles become positive. Ten remain negative, none is zero, and the normalized minimum stays−1117/7680 at (9,9,11,15,15). The complete domain has total rank59, at most one part1 through8, and every other part at least9. Only the inherited low9 concentration theorem is used. Negative terminal absolute margins are not asserted to bound arbitrary unreduced cusps.

A separate exact conditional query shows that additional all-family cost savings1/2 at order12 and5/4 at order16 would make every terminal margin positive, with normalized minimum73/104000. Those two improvements are research targets here, not hypotheses already proved, and their possible finite compensation is not assumed free. No new dimension conclusion follows from the conditional query.

Verification

Generation82852/chunk91db7e and separate fresh replay57169/chunk4bcd1e terminated exit0/PASS. Two peers completed full producer/report mathematical composition reads: PASS. They checked the exact actual/representative new coupon, fixed/residual split and complete certificate induction, current literal interfaces and independent local priorities, exact ideal saving, low9 domain and conditional-only scope. Neither repeated DP, convolution, literal, native-map or historical census arithmetic. The producer checks bound sources, simultaneous finite costs, the complete Bellman certificates, reconstructed exact ideal convolutions and every current literal interface. Historical universal geometric classifications and local proofs remain explicit bound inputs; their all-label censuses are not rerun. There is no external refereeing or proof-assistant verification. Source, data and report are final/frozen; all predecessor packets remain frozen.

```
python3 scripts/fresh_order13_sparse_ambient60_frontier_20261018.py --verify
```

Source 213

Source path: `results/fresh_order14_affine_b_fork_projection_transfer_20261017.md`

SHA256: 0d6a5137b86dafe55fb3048d751cd275eec070bddbd7c6554e945a214bab8fe0

Two more redundant pairs in the order14 affine B factor

In affine B13, whose diagram has14 nodes, delete the two pairs(10,12) and(10,13). Its cost falls from86 to84. No finite weight or coupon changes, and no other affine type changes. The order14 maximum is now84, shared by B and C, so its normalized rank13 coefficient becomes6.

The source and certificate check all23 affected selected contexts and all52 affected induced maps. This is a complete local payment refinement, with no ambient dimension claim. Further ideal and finite obstructions remain in ambient66.

The intrinsic pair rule

Number the spine0–1–…–11 with doubled bond10–11 and attach the two short leaves12,13 to0. The changed pairs have distance11 and form one orbit under the diagram's automorphisms. Start from the completed singleton deletion, which already gives cost86. Keep its five previously absent pairs absent; remove only the additional two pairs above.

All82 basic ideal5/feature5 pairs remain at unit weight. In particular, every elementary triangle payment and every marked path of length at most8 remains a unit. No fractional replacement or new finite allowance is being used.

Complete support and necessary contact domains

The complete distance8 small-factor certificate enumerates every induced map of every eligible selected connected marked component. Exactly23 contexts contain one of the two changed marked images:

```
624,629,637,674,713,719,736,778,842,
1555,1559,1567,1572,1584,1641,1647,1667,
1848,1855,1934,2241,2248,2327.
```

For each context the new checker reconstructs the native labelled pattern from its saved full core Gram, reruns the exhaustive induced-map routine, and compares its complete output with the frozen map certificate. There are52 maps with a changed marked image. Unaffected maps in the same context retain their previous proof.

Every affected core H is nonsingular Lorentzian, all old roots have square2, and all its nonzero off-diagonal entries are-1. The imported constraint checker is the full one from the preceding singleton proof: it recomputes the exact inverse J, the actual preferred corner and its complete necessary contact domain, its partition into two end sets, and the entrywise-positive inverse blocks on those ends. It also checks that a root meeting both ends yields a retained marked path of length at most8. These are the complete contact-domain statements of the frozen overlap proof, not bounds inferred only from the selected pattern.

Thus, unless a retained short path already pays the face, any additional supporting white has all its old-core contacts on one end. If a is its nonnegative vector of native contact magnitudes, then

```
projection_H(e)^2 = a^T J a <= e^2 = 2.
```

The complement of the Lorentzian span of H is positive definite. Within one end, positivity of every entry of J means that a forced contact q at node i implies

```
a^T J a >= q^2 J_ii,
```

regardless of any further contacts on that end. Contacts on the other end are already resolved by the short-path alternative. No unspecified contact is assigned zero, and no bound on spectator rank, multiplicity or length of indirect chains is imposed.

Every actual factor forces an impossible new doubled terminal

In every one of the52 maps, target node10 belongs to the old selected component and is marked, while target node11 is missing. To be the actual whole affine B13 factor, the corner must supply a root at node11 joined by label4 to node10.

That root is genuinely outside H. Old roots omitted from the preferred corner are the two other black roots and cannot belong to that corner's factor. Every old corner root outside the selected connected component is orthogonal to it, so none can supply the required nonzero bond to the old node10. This argument remains valid if other target nodes lie in old components later joined by indirect paths: it uses only the forced first terminal contact, not a completed Gram matrix.

With roots normalized to square2, a label4 bond has squared contact magnitude2. The complete exact obstruction table is:

Old inverse diagonal J_ii	Forced projection2*J_ii	Maps
2	4	12
7/2	7	12
8	16	20
26	52	8

Every projection lower bound exceeds2. Therefore no such actual affine completion exists in an otherwise unpaid face. The checker verifies the forced terminal, old marked neighbor, native label and exact inequality for each individual map. This excludes the affected ideal occurrence; it does not reinterpret its proper finite subdiagram as a whole finite component.

Coverage beyond the selected maps

The original complete theorem first pays triangles and marked paths of length at most8; these units are retained. Its low-core domain has whole marked-component bound at most12. Its disjoint high-core branch is excluded geometrically. Protected/basic ideal outcomes have distance at most9. Finite whole-component, adaptive and protected-leaf payments do not change. The remaining affected selected cases are precisely the 23 contexts checked above.

The universal tree and D13/D14 immediate affine outputs have order at most13, so they do not introduce an order14 B obligation. The later cycle12 proof returns finite payments or ideal paths of length at most4; the D11 proof returns finite payments or paths of length at most5. The order12 B half-transfer and C deletion have their stated order12 outputs and finite alternatives. They never call for either changed B13 pair. The finite rank17 deletions used in the completed ceiling66 change no ideal weight. The older same-order partial-weight replacements use affine D factors, not these B pairs.

Consequently every triangle still receives at least2 and every quadrilateral at least1. The only change to cusp costs is B13:86 to84. The order14 A/B/C/D costs are70,84,84,82. Actual cusp-fiber multiplicities remain those of the preceding theorem.

Scope and verification

Generationdca80c and separate fresh replay68c6b2 both terminated exit0/PASS on all52 maps,23 necessary contact domains, exact projection bounds, retained elementary/basic pairs and automorphism invariance. A separate full source/report read, including the inherited constraint and map-witness helpers, passed the norm-two Lorentz projection semantics, complete end-contact partition, positive inverse blocks, retained cross-end short path, genuinely new doubled-terminal witness, arbitrary additional same-end contacts, complete saved-map comparison and later smaller-factor/finite interfaces. That reviewer did not duplicate the 52-map arithmetic or the upstream geometric census.

Root also completed a full source/report read, including the inherited constraint and map-witness helpers. That review passed the norm-two new-terminal argument, exhaustive affected-pattern map reconstruction, end positivity with arbitrary

additional contacts, and all preserved interfaces. Root did not repeat the map arithmetic or geometric census. All31 direct source bindings, Python AST and three local report links pass. Source, certificate and report are final/frozen.

The complete all-label core census, no-extra-cross-edge proof and arbitrary-support contact-domain geometry remain inherited frozen inputs; their original searches are not rerun here.

The previous duplicate-rank12 concentration is invalid after this cost change. A separate complete scalar diagnostic is required; the improved five-rank13 profile alone does not give an ambient bound. No new polytope, external refereeing or formalization is asserted.

```
python3 scripts/fresh_order14_affine_b_fork_projection_transfer_20261017.py
python3 scripts/fresh_order14_affine_b_fork_projection_transfer_20261017.py --verify
```

Source 214

Source path: `results/fresh_order14_ambient62_ideal_margin_closure_20261018.md`

SHA256: 50f8336ade6489b2330c6076a3b9af5b2b12995d069a483ec5eec9ecc3df1b4b

All ambient62 ideal margins are positive

The completed order14 B/C changes give family totals A70/B82/C83/D82 and normalized rank13 envelope83/14. Combined with the preceding three local transfers, **every actual cusp in dimension62 has absolute payment surplus at least73**. All2,641 terminal profiles have positive normalized margin; the minimum is2741/371712 at(10,10,11,15,15).

The finite rule is unchanged and still has maximum441 against curvature434. Thus the unrestricted dimension bound remains62. The available global consequence at dimension62 is the necessary inequality

`7*v_f >=4805+73*v_i.`

In particular, an all-ideal finite-volume polytope cannot have dimension62; together with the completed unrestricted higher tail, the all-ideal class has dimension at most61. This does not exclude a dimension62 polytope with sufficiently many finite vertices.

The source and certificate bind the B14 terminal theorem, the C14 singleton theorem, and the complete three-transfer frontier. This packet converts that frontier's conditional order14 saving into a proved local rule with a stronger saving and no additional finite cost.

A single simultaneous local rule

Keep all payments of the three-transfer rule, including D13 center zero, B16 singleton deletions, D17 distance10 zeros, and the upgraded whole finite coupons. Make precisely two further ideal changes:

- In whole affine B13 of order14 delete(9,12),(9,13), reducing84 to82.
- In whole affine C13 of order14 delete(1,12), reducing84 to83.

The B pairs have distance10 and the C pair distance11. Both local proofs retain every basic, short8 and triangle pair. Neither changes any finite weight or coupon. The two changes concern different factor types and cannot be mistaken for the same literal payment.

For a quadrilateral, first apply the inherited actual-corner priorities: C11 endpoint zero, B12 center/terminal zero, D13 center zero, cycle11 distance5 half, then the coupled order11 B/C/D half cases. Each theorem has an independent retained unit or distinct-half alternative. Its outputs are short ideal paths, whole order11/12 center units, protected whole finite components, or the explicitly budgeted typed coupons. None is a newly deleted order14 pair, B16 singleton or D17 distance10 pair.

Next, if a changed order17 D corner occurs, use its complete zero theorem. The short8 branch is the independent current route already established by the preceding composition, including the cycle8 complementary-arc interfaces and deep11 finite8-zero to short7 replacement. The new B/C14 pairs have distance10/11 and are not used by that route. In the no-short8 branch, the proper completions are impossible and every whole-radical case has its distinct whole E6/D11 unit. This branch needs no original D17 half and no finite D16/D17 same-component unit.

If no changed D17 corner occurs, apply the preceding complete B16 whole-face proof with the two additional selected-domain exclusions proved by the new B14/C14 packets. Their complete necessary domains are the same600 eligible selected patterns. Every possible removed B14 image is among108 checked maps:92 impossible first entries or16 doubled-terminal Cauchy contradictions. Every removed C14 image is among six checked maps, all requiring an impossible physical norm-two terminal. Their cross-end or negative-coupling alternatives give the already established short8 route. There is no circular request that a removed long pair itself pay.

All other selected marked images retain their preceding payment. Low cores have whole-component order at most12; the high-disjoint branch is excluded; protected/basic ideal outputs have distance at most9. Finite payments are unchanged. The complete nonselected and later interfaces are explicitly retained in both local theorems. In particular neither proof outputs the other's removed pair, and neither introduces an unexamined later order14 obligation. This patches the complete preceding whole-face proof, rather than claiming that the selected600 patterns enumerate every quadrilateral. On a face with no new actual changed corner, the prior weights also agree pointwise.

Consequently every quadrilateral still receives at least1. Every triangle receives at least2 by the retained finite7/ideal5, protected6/7 and small whole-factor interfaces; the removed distance10/11 pairs are absent from those routes. Nonnegative payments continue to suffice for faces with at least five sides. The resulting upper bound on the global defect uses the new ideal envelopes and exactly the preceding finite rule.

Complete exact scalar comparison

The preceding frontier independently reconstructed every scalar row for the exact rank-specific ideal costs. It used the complete low9-only concentration domain: total affine rank61, at most one part1 through8, and arbitrary repeated parts at least9. There is no concentration of the high ranks into one part; that would not be justified for the exact nonconstant tail cost. The domain has2,641 profiles.

The present source independently enumerates that whole domain with the optional low part separate and compares it bijectively with the frozen rows. It does not reconvolve the unchanged link polynomials. For each profile r, it verifies the link vertex product V=product(r_i+1) and its saved normalized/absolute margin relation. Let k be its number of rank13 factors. Their former total84/14 is now83/14, so

```
C_new/V = C_old/V −k/14,
margin_new = margin_old +k/14.
```

Every row is updated exactly with rational arithmetic. All2,641 updated margins are strictly positive. The normalized minimum is2741/371712, uniquely at(10,10,11,15,15); this profile contains no rank13 and its margin was already positive. The minimum absolute margin is73, uniquely at the single rank61 factor. The three previous negative rows are now positive. No prospective local or finite coupon hypothesis remains in these costs.

The inherited low9 concentration theorem preserves the linear low-rank cost, decreases normalized curvature, and decreases V. The present change is at rank13 and does not alter the low-rank move. Therefore the normalized margin of an unreduced actual cusp is at least its terminal margin. Since the terminal margins are now positive and V also decreases, the absolute margin V times normalized margin is at least its terminal value. Thus the complete scalar certificate proves for every actual cusp

```
kappa(L_v)−C_v >=73.
```

The actual factor-fiber multiplicity V/(r+1) is unchanged and is the reason the factor-total saving1 enters as1/14 per rank13 factor. No arithmetic, simple-cusp or facet-count assumption is inserted.

Finite obstruction and the limited global consequence

No new finite payment is introduced. The preceding complete finite certificate, including both D13 upgrades and actual E6/D11 total4, still gives exact rank62 maximum441. Its formal witness D17+D17+D28 remains at125+125+191=441. Curvature at a finite vertex is434, leaving deficit7. No Bellman table is rerun or improved by the present ideal change.

The inherited shape-curvature inequality has the same two-extrema constant4805=5*floor(62²/4). With the newly proved ideal surplus, it gives

```
D5 <=441*v_f +sum_(ideal v) C_v,
D5 >=434*v_f +sum_(ideal v) kappa(L_v)+4805.
```

Subtracting yields the necessary vertex inequality at the start. It is not a contradiction for arbitrary finite-vertex count. If v_f=0 it is impossible, proving the stated all-ideal exclusion. The completed unrestricted ceiling62 supplies every dimension above62, so that class ceiling61 has its full higher-dimensional tail. The unrestricted ceiling remains62 until a separate finite comparison or other global argument is completed.

Verification scope

Final source-bound generation3e7486 and separate fresh replay0ca6d1 both terminated exit0/PASS. Auditbaeadf passes13 direct bindings, six local links and Python syntax. A peer completed a full source/report composition read, also reading both frozen B/C local reports. It passed the independent actual-corner and short8 priorities, shared selected-domain exclusions, B16/D17 compatibility, complete low9-only domain, exact cost update, positive-terminal absolute extension, unchanged finite rule and limited global consequences. It did not rerun scalar, DP, map, inverse or census arithmetic and did not claim an additional full B/C producer audit. Root and a second peer also completed full source/report mathematical composition reads. They passed the complete bijective domain comparison, exact rank13 saving, both minima, use of only low9 concentration with decreasing V, independently established short8 route before the B/C selected patch, unchanged finite comparison and single global constant, and the all-ideal higher-tail conclusion without an unrestricted bound. Neither repeated convolution, scalar arithmetic, finite optimization or geometric censuses. All three reviews found no gap in the stated scope. Source, data and report are final/frozen. This binder verifies the two local interfaces, exact family maximum, retained basic/short/triangle sets, no finite change, complete2,641-profile domain, exact cost updates and both minima. The previous scalar convolution, finite optimization, geometric censuses, factor fibers, all-rank concentration and shape theorem are completed bound inputs. They are not independently reproved by this binder. No new unrestricted dimension bound, chamber, external refereeing or formal proof-assistant verification is claimed.

```
python3 scripts/fresh_order14_ambient62_ideal_margin_closure_20261018.py
python3 scripts/fresh_order14_ambient62_ideal_margin_closure_20261018.py --verify
```

Source 215

Source path: `results/fresh_order14_b84_scalar_frontier_20261017.md`

SHA256: 982cbc6e5df0c9301b556a7af90199b071f31f9d119d788b7d79c84b7c052cf9

Exact scalar frontier after the order14 B reduction

The new order14 B transfer changes the A/B/C/D costs from70/86/84/82 to70/84/84/82. Thus the normalized affine **rank13** coefficient falls from43/7 to6. This resolves the former worst formal cusp profile in ambient66, but ten negative ideal margins remain. There is no new ambient dimension bound.

The source and certificate evaluate the complete old, corrected new and common expanded concentration domains with exact rational arithmetic. All other ideal coefficients are those of the completed ceiling66.

Correct rank convention and required improvement

Five rank13 factors have total link rank65. Each factor has14 nodes. Their old limiting type was B with cost86; the order13 costs65/75/73/74 instead belong to affine rank12 and are not the coefficient in this profile.

The old margin of(13,13,13,13,13) is-1565/10976. A total pair-cost decrease delta in each worst order14 factor adds5*delta/14. Resolving this single profile requires delta>313/784. The proved two-unit reduction gives new margin6275/10976. B and C now tie at cost84; A70 and D82 remain below them.

The concentration domain must change

The inherited convex-order theorem decreases normalized curvature under concentration(a,b) to(a-1,b+1), for1<=a<=b. A valid margin reduction also requires that total cost be preserved or increased.

Previously two rank12 factors could become11 and13, with cost increase 23/1092. With the new coefficient this increment is-19/156. The old move is therefore invalid. Indeed the move already fails for an order14 cost drop greater than23/78, which is smaller than the313/784 needed to resolve the former worst profile. Reusing the old terminal list would miss a real proof obligation.

The new valid middle move is13+13 to12+14, whose cost increase is11/65. Low parts still concentrate only up to10, preserving their linear cost; two high parts at least18 still concentrate until one is17, preserving their constant cost. Every move strictly increases the bounded integer sum of squared ranks and therefore terminates.

The old terminal domain retains arbitrary copies of10,11,13,14,15,16,17 and at most one12. The corrected new domain retains arbitrary copies of 10,11,12,14,15,16,17 and at most one13. Both have at most one low part1–9 and at most one high part at least18. These are complete reductions for their respective costs. A third, common expanded domain permits arbitrary copies of every rank10–17, applying only the low/high moves. It provides a common exact comparison without relying on either middle move.

Complete old and new comparisons at ambient66

Domain and cost rule	Profiles	Negative margins	Minimum margin	Minimizing ranks
Old terminal domain, old costs	1,625	23	-1565/10976	(13,13,13,13,13)
New terminal domain, new costs	1,659	10	-31895/248832	(10,11,11,11,11,11)
Common expanded domain, old costs	1,796	28	-1565/10976	(13,13,13,13,13)
Common expanded domain, new costs	1,796	10	-31895/248832	(10,11,11,11,11,11)

There are no zero margins. All ten remaining negative profiles have no rank13 factor, so their numerical margins were already negative before the improvement. Eight were explicitly present in the old terminal list:

Ranks	Unchanged negative margin
(10,11,11,11,11,11)	-31895/248832
(11,11,11,16,16)	-1055/10404
(10,10,11,11,11,12)	-3535/41184
(10,11,12,16,16)	-2330/41327
(9,11,11,11,11,12)	-12799/539136
(1,10,10,11,11,11,11)	-449/50688
(1,10,11,11,16,16)	-103/28611
(11,11,12,15,16)	-863/509184

Two have repeated rank12 and are newly admitted to the terminal list:

Ranks	Unchanged margin	Their old concentration image	Old image margin
(10,10,10,11,12,12)	-205/4719	(10,10,10,11,11,13)	-481/5544
(11,12,12,14,16)	-11/1690	(11,11,13,14,16)	-24277/514080

The checker verifies those old images and the required margin directions. These are newly necessary terminal checks, not negative margins created by lowering a cost. For every profile in the common domain the new margin equals the old margin plus its number of rank13 factors divided by7.

A quantitative next obligation, without a local proof

The order12 maximum remains65, shared by B and C; A and D cost60 and62. Changing that maximum would change c(11). The new worst profile alone would require an order12 cost drop greater than6379/20736. That is not the strongest remaining requirement.

Across the complete corrected domain, the exact required uniform drop in the order12 maximum is greater than27960/41327, controlled by (10,11,12,16,16). Every remaining negative profile contains rank11. A hypothetical decrease65 to64 makes all1,659 corrected-domain margins positive, with minimum13367/495924 at that controlling profile. The concentration proof remains valid because its low and high moves and the duplicate13 increment are unaffected by c(11).

This is only a conditional ideal comparison. Both B and C would need geometrically valid reductions to at most64, with full payment provenance and any new finite coupons counted. This package provides no such second local theorem. The completed finite rule also still has the separate D66 cost496 versus curvature495 obstruction. Neither obstruction can be discarded on the basis of these scalar calculations.

Verification scope

Final generation042067 and separate fresh replay668fe2 both terminated exit0/PASS on all three complete domain enumerations, exact profile curvatures, old/new margins and concentration increments, all ten remaining negative profiles and their old-domain provenance, and the conditional next-order12 comparison. A separate implementation check uses the frozen global66 reviewer's optional-first multiplicity enumeration and its sliding-window/CDF curvature arithmetic: all1,796 common-domain profiles and both terminal subdomains agree exactly. This implementation check was run by the producer; it is not a separate mathematical reviewer.

Root completed a full source/report mathematical read. That review passed the corrected repeated-rank12 and at-most-one-rank13 concentration domain, the common1,796-profile comparison, all ten remaining negative profiles and their provenance, and the conditional order12 threshold27960/41327 and positive maximum64 comparison. Root did not repeat the arithmetic. All eight direct bindings, Python AST and three local report links pass. The local B14 report is final/frozen and its final hash is bound here. Source, certificate and report are final/frozen. The all-rank convex-order theorem and existing ideal cost envelopes remain frozen inputs.

No hyperbolic realization of a formal profile, new ambient bound, external refereeing or formal verification is asserted.

```
python3 scripts/fresh_order14_b84_scalar_frontier_20261017.py
python3 scripts/fresh_order14_b84_scalar_frontier_20261017.py --verify
```

Source 216

Source path: `results/fresh_order14_b_basic_distance9_transfer_20261018.md`

SHA256: 0001c9d4f03dee59e8e332199bdd4c54a2a6bd17a672665d67c61730814f8ab2

Two coupon-free basic-pair deletions in affine B13

The current affine B13 factor of order14 may set the two singleton pairs (1,10) and(2,11) from1 to0. Number its spine0–1–…–11, with doubled bond10–11 and simple fork leaves12,13 at0. Both pairs have distance9 and are invariant under the fork-leaf interchange. The B total decreases from82 to80, with no finite charge change and no added coupon.

With the completed C14 deletion, the order14 family totals become A70/B80/C80/D82. The remaining D family still controls the maximum82. This packet preserves triangle payment2 and quadrilateral payment1; it does not prove a new forest maximum or dimension bound.

The producer and certificate bind the completed dimension59 rule, the preceding B82 theorem and the C80 theorem. All predecessors remain unchanged.

Complete selected and protected domains

The whole-order14 reduction partitions its2614 high-overlap contexts into1378 bounded at order at most13,74 protected bases,9 adaptive finite certificates,553 displayed components larger than14, and600 eligible native patterns. The source recomputes every induced embedding of those 600 patterns into the canonical B13 target, including failed searches. It obtains exactly the frozen207 successful patterns and506 maps.

There is no selected map with marked image(2,11). Exactly28 contexts, with54 maps, have image(1,10). Their displayed components are proper; every map requires a missing doubled terminal11 adjacent to displayed node10. Its complete physical projection is impossible as proved below. The source saves every context, map, forced entry and strict witness.

Neither singleton can be removed from this selected search alone: both were basic distance9 payments. Independently, the source converts all74 protected full component Grams to their native labels and searches all induced embeddings into the target. Again there is no occurrence of (2,11). The only protected occurrences of(1,10) are ten maps in:

`367,382,394,1504,1507,1510.`

In each, target node10 is a protected marked leaf of the old displayed component. It would need a new doubled-terminal neighbor to acquire its required degree2. The exact obstruction below excludes that extension; a stored `forced_finite` tag is corroborating data, not the proof.

No arbitrary supporting tail repairs a missing protected embedding: the old displayed roots remain an induced native subdiagram of any actual whole B13 factor. Thus this is a complete check of the unbounded protected branch, including all failed embeddings, rather than just a check of the selected native patterns.

Arbitrary-contact first-entry bounds

For each affected selected or protected context, let G be the full old physical norm-two Gram and J its inverse. The source reevaluates the complete inherited contact helper, verifies the nonsingular Lorentz signature and selected corner, and explicitly checks positive definiteness of the supporting-white Gram and all four white-plus-one-black edge Grams. It does not infer these properties merely from a method tag.

The full set of possible old-core contacts of a further supporting white splits into two end sets. Each same-end principal block of J is entrywise positive. A contact to both ends supplies an actual path of length at most8 between adjacent black sides, one from each opposite pair. That independent short-path payment is preserved as discussed below.

In the remaining branch, the nonnegative contact vector a of a new square2 root e lies at one end. The actual core span is nonsingular Lorentzian, so its orthogonal complement is positive definite. Hence

```
a^T J a <= 2.
```

A prescribed native label4 contact at i has squared magnitude2. Same-end positivity therefore gives a^T J a>=2 J_ii, regardless of additional contacts to any other old roots at that end. Contacts to arbitrary outside spectators add conditions and cannot repair this failed necessary inequality. No unprescribed product is forced to vanish.

In every one of the54 selected maps, target11 is missing and target10 is already displayed. The required doubled-terminal first entry has 2 J_ii>2. It is genuinely new: omitted old blacks are absent from the actual corner, and every other old corner component is orthogonal to the displayed marked component. Such an old root cannot have the required nonzero contact to target10. Both marks are already displayed, so the new terminal is a supporting white, not another omitted black. This first-entry argument does not exclude possible later joins to spectator blocks by assumption.

The same reasoning applies to the ten protected maps. In contexts367 and1504, the protected marked node has J_ii=7/2, so its required doubled contact has projection at least7. In the other four contexts the lower bound is76/7. Both exceed2 strictly. Thus the protected leaf cannot acquire the doubled neighbor required by the proposed B13 factor.

All64 affected selected or protected maps are therefore excluded in the no-short8 branch. The unsupported singleton(2,11) has no occurrence in either complete domain. No extra finite payment is needed.

Current whole-face composition

Every B14 pair of distance at most8 remains a unit. The source checks the complete91-pair table, both singleton invariances, all short8 pairs and the inherited elementary triangle pair set. It separately reconstructs all216 current literal configurations from their Grams and the frozen sparse-D16/ideal17 lower bounds. Neither changed B14 pair occurs there; all literal totals remain unchanged and at least1. Their finite factors have order at most17, so the later depth6 finite7 deletion adds no debit to those examples. This finite audit supplements the universal argument.

First apply the current independent changed-ideal-corner priorities. Small order11–13 transfers give their specified smaller ideal factors, short paths or typed finite coupons. Later B/D15–17 partial transfers use their own selected exclusions and sealed finite alternatives; their short8 escapes remain independent. The original universal radius-five tree route gives ideal paths of length at most7, while its remaining payments are finite. Universal cycle replacements use ideal paths of length1/2 and longer finite paths or whole-component coupons. The inherited immediate whole-affine interfaces have order at most13. They do not require either new B14 degree2/doubled-terminal singleton.

The preceding B14 long-pair proof gives an independent short8 escape or excludes its own completion. The completed C14 basic singleton proof also supplies only its independent short8 escape or a C completion obstruction. Their routes do not consume a newly deleted B distance9 unit, so they can be applied before the present patch.

If a short8 finite output is one of the already-zero deep finite8 payments, the independent finite8 theorem supplies short7. If a resulting finite7 output is zero under the depth6 rule, that theorem supplies short6 or a retained whole small-component payment. The current short6 cycle/tree and whole-A8 or protected-leaf alternatives avoid the new B14 distance9 pairs. Thus this proof's both-end escape is paid without recursing through the conclusion being proved.

The remaining low-core branch uses whole order at most12, and the high-disjoint branch is excluded. In the high-overlap domain, the bounded and too-large components cannot equal the changed whole factor. The nine adaptive finite certificates retain their original payments. The complete 600 selected and74 protected domains are handled above, and every other selected or protected payment is unchanged. This patches the whole-face proof rather than assuming the old preferred witness remains paid. On a face with no changed actual B14 corner, all weights agree pointwise with the preceding rule. On any affected face, all possible uses of a removed payment are covered by the exhaustive domain split.

The separate finite B14/D14 reductions change only finite distances at least10 and use whole D15/D16 or typed-coupon alternatives. The present proof has no such finite output, and those finite transfers retain the short8-to-short6 routes used here. Their local rules therefore compose with this coupon-free ideal change.

The triangle theorem uses unchanged finite/ideal short paths and protected small factors; its complete elementary B14 target set is retained. Triangle total2 and quadrilateral total1 persist. The cusp factor-fiber charge loses2V/14 for a B14 factor, but the all-family maximum remains82/14 until an independent D14 improvement is proved.

Verification and limits

Generation session24002/chunkf35857 and separate fresh replay session97011/chunka23c26 terminated exit0/PASS. The new checks cover the complete600/506 native domain, all74 protected bases,64 strict first-entry obstructions, current pair invariance and triangle sets, and216 unchanged literal configurations. The historical all-label census and universal actual-face reductions remain inherited inputs; they are not reconstructed here.

Parent completed a full source/report and X.constraints mathematical read: the complete600/506 selected and74 protected domains,64 affected first-entry maps, genuine missing terminals, positive end blocks, independent short8 through short6 escape, elementary triangles,216 literal configurations and finite14/C14 compatibility pass. A second reviewer independently read the full producer/report, X.constraints, the universal finite8 cycle/tree output interfaces and the final C14 white/four-edge PD delta. That read also passes the unsupported singleton and all64 strict witnesses. Neither reviewer reran maps, inertia, census or numerical checks. Source, certificate and report are final/frozen after direct binding, local-link and syntax checks.

No finite compensation, forest optimization, dimension58 theorem, external refereeing or proof-assistant formalization is claimed.

```
python3 scripts/fresh_order14_b_basic_distance9_transfer_20261018.py --verify
```

Source 217

Source path: `results/fresh_order14_b_terminal_residual_transfer_20261018.md`

SHA256: fe70262bedc1dd220aa4e91a703b6eba89e8053a3042d25e108833daee7aa435

Removing the last two extra order14 affine B payments

In a whole affine B13 factor of order14, delete the pairs(9,12) and(9,13). The total falls from84 to82, exactly its basic ideal5/feature5 pair count. No finite weight, finite coupon or other ideal payment changes. The target has spine0 through11, doubled bond10–11, and leaves12/13 at0. The two deleted pairs form one automorphism orbit of distance10. Every short8 and elementary triangle pair remains a unit.

Together with the separately proved C14 singleton deletion, the four order14 totals are A70/B82/C83/D82, with maximum83. This is a local payment improvement. The ambient62 finite obstruction is separate; no new ambient dimension bound follows here.

The source and certificate start from the previous B84 theorem and the completed62 rule. They reconstruct all600 eligible high-overlap selected patterns, including failed searches, compare207 successful patterns and506 induced maps with the frozen complete certificate, and check all108 changed maps in53 contexts. The affected maps are not inferred to be every quadrilateral; the separate nonselected interfaces are treated below.

Complete old-core contact control

Every affected old core H is nonsingular Lorentzian, simply laced and physically normalized to square2. The inherited full contact checker recomputes its inverse J, signature, actual selected corner, and entire necessary contact domain for any additional supporting white. It also checks that old corner components outside the selected component are orthogonal to it and that the omitted blacks are absent from this corner.

The contact domain splits into two end sets with entrywise-positive inverse blocks. Any extra white meeting both end sets gives an actual opposite-black path of length at most8. The present simultaneous payment route for that case is supplied below. In the remaining branch, every extra white's contacts with H lie in one end. If a is its nonnegative physical contact vector, then

```
projection_H(e)^2 =a^T J a <= e^2=2.
```

The positive complement of the Lorentz span proves this inequality. Positivity of the end block gives a^T J a>=q²J_ii for every specified contact q at i, regardless of arbitrary other same-end contacts. In particular a label4 contact has q²=2 in this normalization.

In92 affected maps, the actual target forces a missing root adjacent to the displayed component with q²J_ii>2. This root must be outside H: omitted blacks are not in the corner, and other old corner blocks are orthogonal to the displayed component. These are immediate projection obstructions even when the old classification calls the context a relevant-component case. They do not require a whole completion or a finite replacement coupon.

The sixteen two-root obstructions

The remaining16 maps occur in the eight contexts

```
630,638,1648,1668,1849,1935,2242,2328.
```

Each displayed image contains every target node except10 and11. The actual whole B13 factor therefore needs a root e at target10, simply joined to the old target9 at core node i, followed by a root f at target11, joined to e by label4. Both actual roots have square2. The source checks these exact incidences for every map.

The first root e is outside H by the immediate-neighbor argument above. For these contexts, the full end blocks also satisfy

```
J_aa+J_bb+2J_ab>2
```

for every distinct pair a,b in one end. Native nonzero contact magnitudes are at least1; block positivity therefore excludes two contacts on the same end. Contacts on both ends already give the short8 route. Thus e has exactly the one old-core contact at i. Moreover J_ii is3/2 or5/4, so a label at least4 there would have projection square at least2J_ii>2. Its contact is exactly the simple magnitude1 fixed by the target. Its residual in the positive complement has square

```
rho=2−J_ii =1/2 or3/4.
```

The terminal f is genuinely new as well. It cannot be an omitted black. If it were an old root in another displayed corner component, e would have an additional old-core contact to f through the required doubled bond, contradicting the just-proved uniqueness. Old roots of the selected component already have their fixed distinct images and cannot supply a missing target node. Thus the complete extra-white contact domain applies to f; it is not restricted to a guessed subset of spectator roots.

If f contacts both old end sets, its own contacts give the earlier short8 alternative. Otherwise its contact vector b lies on one positive inverse block, giving b^T J b>=0 and therefore

```
||f_perp||²=2−b^T J b <=2.
```

This bound includes equality when f is orthogonal to H. We do not need to assume that f has one old contact, or that it has no old contacts. For every possible contact node j, the source checks the sign of J_ij. If J_ij<0, the actual path from an opposite black through i,e,f,j to the other black has length at most8; in all eight contexts its saved bound is3. This path uses the actual doubled e–f edge and the actual nonzero f–j contact, independently of their magnitudes. That case pays by the short-path route. After excluding it, every actual term in

```
<projection_H(e),projection_H(f)>=sum_j b_j J_ij
```

is nonnegative. Zero coefficients and an empty support are permitted. Since the physical e–f bond is−sqrt(2), their residual product satisfies

```
<e_perp,f_perp> =−sqrt(2)−sum_j b_j J_ij <=−sqrt(2).
```

Its square is at least2. Cauchy in the positive complement instead gives

```
<e_perp,f_perp>² <=rho∗||f_perp||² <=2rho <2.
```

This is a strict contradiction, with gap1 when rho=1/2 and gap1/2 when rho=3/4. Arbitrary additional contacts of f only retain this necessary contradiction or supply a short path. No unspecified Gram entry has been set to zero. No residual-rank bound, optional exterior-root enumeration, or assumption excluding indirect spectator bridges is required.

Thus all108 changed selected occurrences are impossible unless an independently retained short-path payment already resolves the face. There is no surviving affine completion needing a new typed coupon.

Complete domain and simultaneous present weights

The low-core theorem has whole marked-component bound at most12, so cannot require either changed order14 pair. The all-label high-disjoint branch is excluded. Protected/basic ideal choices have distance at most9 and stay paid; every finite whole-component, adaptive or protected-leaf payment is unchanged. The remaining high-overlap selected cases are precisely the complete600-pattern domain reconstructed here. Both removed pairs are outside the basic set; all82 basic pairs remain units.

The universal tree and older D13/D14 immediate affine outputs have orders at most13. The cycle12, D11 and order12 B/C successors use finite payments or smaller/shorter ideal factors. Later order15/16/17 D transfers use their own D-factor roles and typed finite coupons. The newer C11/B12/D13 and coupled order11 theorems do not output these order14 B distance10 pairs. Neither the separate C14 singleton proof nor B16 first-entry proof uses them. Hence the selected domain has no unaccounted later long-pair use.

To invoke the short8 alternative under the present reduced rule, use actual-corner priority. First apply independent C11-zero, B12-zero and D13-half or D13-zero theorems whenever those changed actual corners occur. Their outputs are short units, retained order11/12 center units, whole D12 leaf9, or the stated whole finite coupons. Next apply the independent cycle11 and coupled B/C/D11 half theorems to any affected actual corners. They give distinct retained halves or finite protected units without calling for a long order14 B pair. These routes do not recursively spend an ideal weight that is being removed here.

Otherwise the explicit current short8 route from the completed63 composition remains: unaffected small-tree units; the established cycle6/7/8 and radius-five tree replacements; and, whenever a finite distance8 output is deep, the independent depth11 corridor supplies its protected short7 replacement. The sparse finite deletions are at distance at least10. The order17 zero refinement only removes its own distance10 orbit and uses this same independent short8 route or a distinct whole E6/D11 unit. Thus neither proof requires the other's changed orbit. The short7 route uses retained finite paths or protected roles, rather than another deep8 zero, so it is not circular.

After these priorities, an old selected payment at either new B14 pair is excluded by the complete local obstructions above. Every other old selected payment is retained. On a face with no actual changed B14 corner, weights agree pointwise with the prior rule. The separate C14 deletion has exactly the same independent short-path scope and removes a different factor type. Its affected occurrences are likewise excluded by their own first-entry proof. The two deletions therefore compose without new coupons or any assumption about which old witness would be selected.

Triangle payment2 is unchanged: all elementary retained pairs are checked, and the removed distance10 orbit is absent from the finite7/ideal5, protected6/7 and whole small-factor triangle routes. Quadrilateral payment1 follows from the complete cases above. The current finite forest rule and its maximum are unchanged. Cusp factor-fiber multiplicity remains V/14, and deleting two unit pairs changes the factor total by exactly2. The separate C14 theorem gives the stated all-family maximum83; a global scalar/budget successor must bind both local theorems explicitly.

Verification scope

Generation486d85 and separate fresh replay3c04d0 both terminated exit0/PASS. Audit080f4c passes35 direct bindings, six local links and Python syntax. Root and a separate peer completed full source/report mathematical reads, including the inherited complete contact checker at their recorded scopes. They passed the92 direct all-contact bounds and16 residual cases: e's unique old contact, f's genuine newness, its complete one-end projection bound, every negative cross coefficient's actual shortcut, and strict Cauchy including the core-orthogonal terminal case. Both also checked the exhaustive nonselected/current short8/triangle interfaces and absence of new coupons. Neither repeated maps, inverses, Cauchy arithmetic or the upstream census. Source, data and report are final/frozen.

The source re-enumerates all600 selected patterns and506 successful maps, checks53 full necessary contact domains, and verifies all 92 first-entry inequalities and16 two-root Cauchy contradictions. It binds the inherited all-label overlap reduction, basic/triangle rules and current simultaneous short-path composition. It does not repeat the entire original core census or prove those inherited classifications anew. No finite optimization, external refereeing, formal verification or global dimension bound is asserted.

```
python3 scripts/fresh_order14_b_terminal_residual_transfer_20261018.py
python3 scripts/fresh_order14_b_terminal_residual_transfer_20261018.py --verify
```

Source 218

Source path: `results/fresh_order14_c_basic_distance9_transfer_20261018.md`

SHA256: 8f90d05d37dc56732bd7140ef71cfb93bb5fd33ea8af50c2274aa96f71cec75f

Removing the C13 basic distance-nine singleton

For the current completed dimension59 payment rule, the affine C13 factor of order14 may assign weight0 to the pair(2,11), previously1. Use the canonical path0–1–…–13 with doubled terminal bonds0–1 and12–13. This is a reflection-invariant singleton of marked distance9. The C-family total decreases from81 to80, with no finite-weight change and no new cross-component coupon. All triangles still receive total2 and all quadrilaterals total1 under the simultaneous rule.

This is a local payment theorem. The order14 family totals become A70/B82/C80/D82, so B and D still determine the maximum82. No improved finite forest maximum or ambient dimension bound is claimed.

The producer and certificate bind the completed dimension59 theorem and the preceding C81 theorem. They reconstruct the complete selected native domain, every unbounded protected native base and the216 current literal interfaces.

Complete domains and the newly affected basic pair

The frozen high-overlap classification has2614 contexts. Its exhaustive partition for a possible whole order14 factor is:

Branch	Contexts
Whole selected component bounded by13	1378
Protected finite leaf or ideal feature	74
Adaptive finite certificate	9
Displayed selected component larger than14	553
Remaining eligible native patterns	600

For each of the600 patterns, the source recomputes every induced native embedding into the canonical affine C13 diagram, including failed searches. There are43 successful patterns and92 maps, exactly matching the frozen predecessor. Exactly11 contexts, with22 maps, send the two selected marks to(2,11):

`361,378,390,525,1095,1096,1097,1108,1145,1160,1620.`

The displayed components have order11 or12. Every such embedding has an immediately missing doubled terminal, which is excluded below.

Because the pair is basic and has distance9, selected-map exclusion alone is insufficient. The producer separately takes **all74 protected bases**, converts their full physical Grams to native labels, and recomputes every induced embedding into the target. This includes their full displayed component, rather than merely the path between the marks. Only contexts367 and1504 can send their marks to(2,11), with two maps apiece. Both display an11-node path with a protected marked simple leaf. In the proposed C13 completion that mark would have degree2. We prove the required new neighbor impossible from the complete contact domain; we do not use the saved `forced_finite` tag as a substitute for this proof.

No arbitrary enlargement of a protected base can evade this test: its old roots remain an induced subdiagram of the actual whole affine factor. The other72 protected bases have no relevant induced embedding. The nine adaptive branches retain their independent finite payments.

Physical first-entry obstruction

For each of the13 affected contexts, let H be the full old core, G its Gram and J=G inverse. The inherited necessary-contact helper is re-evaluated, checking that all old roots have physical square2, G is nonsingular with one negative direction, the actual white and one-black edge subdiagrams are positive definite, and the selected corner data agree with the frozen classification.

The helper supplies the complete possible old-core contacts of any additional supporting white. They split into two disjoint end sets, and every entry of each corresponding principal block of J is strictly positive. A root contacting both ends gives an actual path between adjacent black sides of length at most8. That is the independent retained short-path branch described below; it is not a failed Gram test.

In the remaining branch, the contact vector a of a supporting white e lies at one end. The orthogonal complement of the actual Lorentz span of H is positive definite, so

```
a^T J a <= e^2 = 2.
```

All contact magnitudes are nonnegative. A forced contact q at an old node i therefore yields the lower bound

```
a^T J a >= q^2 J_ii,
```

regardless of any additional contacts at that end. Contacts to roots outside H cannot repair this necessary inequality. No unspecified old or new contact is artificially assigned zero.

The first required missing root is genuinely outside H. An omitted old black is absent from this actual corner. Every other old corner block is orthogonal to the displayed selected component, whereas the proposed first root has a prescribed nonzero product with it. Thus no old spectator can supply that immediate neighbor. Both marks are already in the displayed component, so the missing root is a supporting white. The argument makes no assertion that later tail roots cannot join an old spectator block.

For the22 selected maps, at least one required first root is a doubled terminal0 or13. Its physical native label4 has q^2=2. The exact lower bounds2 J_ii used as witnesses are strictly greater than2, and their minimum is5/2. In the18 maps with a12-node displayed path both doubled terminals are immediately missing. In the four11-node maps, the adjacent doubled terminal suffices even though the other first entry can have smaller projection. The certificate records every forced entry and the chosen strict witness.

For each of the four protected maps, a new simple neighbor must attach to the protected marked leaf. In context367 this is old node9; in context1504 it is old node1. Its full-core inverse diagonal is7/2. The prescribed simple contact therefore gives q^2 J_ii=7/2>2. Additional same-end contacts only increase the bound; both-end contacts already give the independent short8 alternative. Consequently that marked leaf cannot acquire the degree2 required by target mark2 or11. This excludes the entire protected C13 completion with arbitrary supporting roots, not only its displayed path.

Hence all26 affected selected or protected maps are impossible in the no-short8 branch. No label cutoff, bounded spectator census, whole-path assumption or completed Gram with forced zeros is used.

Simultaneous current rule and whole-face coverage

All ideal C14 pairs of distance at most8 remain literal units. The certificate checks the complete91-pair old/new table, reflection invariance, the retained short8 set and the inherited elementary triangle pair set. The other basic pairs remain unchanged.

Use the existing independent changed-ideal-corner priorities first. The current BC11, B/C12, B/C/D13, B/D15, B/D16 and order17 transfers give their own retained short paths, specified smaller whole factors or separate finite coupons. None outputs an order14 C pair of distance9. The earlier B14 residual exclusion supplies its short8 alternative or excludes a B completion; it does not call this C pair. The universal radius-five tree replacements give immediate ideal paths of length at most7 and sealed finite alternatives. Universal cycle replacements give ideal paths of length1/2, finite paths with persistent features, or whole-component coupons. Their longer outputs are finite. The older D-family immediate affine alternatives have order at most13.

A short8 finite output that has become zero invokes its independent finite8 theorem, obtaining short7. A newly zero finite7 output invokes the current depth6 theorem, obtaining short6 or a retained whole small-component payment. The current

small-factor, tree and cycle replacements for those paths do not consume the present distance9 C14 unit. The central depth6 whole-A8/leaf alternatives are also unchanged. Thus the short8 escape in the projection proof is independently paid; no call back to this new deletion is needed.

After these priorities, apply the inherited complete local reduction. The low-core and high-disjoint interfaces are unchanged; the former uses whole ideal order at most12, and the latter is excluded in the remaining no-short8 branch. The2614 high-overlap cases partition as above. Whole order at most13 cannot equal this target; a displayed component larger than14 cannot lie in it. All affected selected maps and all protected occurrences have now been excluded. Every other selected or protected payment is unchanged. The adaptive nine finite certificates keep their original payments, including their whole bounded factors and fixed leaves.

This is an exhaustive whole-face patch, rather than an assertion that the old proof's selected witness happens to survive. On a face with no changed actual C14 corner, the new weights agree pointwise with the old ones. If a changed corner exists but is not used by the old proof, its deletion does not alter that proof's payment; if it is used, it is covered by the complete selected/protected analysis and independent short-path alternatives. Finite charges are identical throughout.

As a supplementary interface check, all216 current literal examples are rebuilt from their original Grams and current sparse-D16/ideal17 payment totals. Every whole affine order14 component is typed through all native embeddings. There are no occurrences of the deleted pair, so every current literal total is unchanged and at least1. Their finite components have order at most17, so the subsequent depth6 distance7 zeros introduce no extra change. This finite list supplements the arbitrary-support proof; absence from216 examples is not its basis.

The triangle theorem uses ideal distances at most5, protected smaller ideal distances and whole factors of order at most9; its finite outputs are unchanged. Its complete elementary C14 target set is retained. Therefore triangle total2 and quadrilateral total1 survive simultaneously. The cusp factor-fiber charge loses exactly V/14 for a C14 factor. Since B and D remain82, the present all-family envelope is still82/14.

Verification and scope

Final generation session32376/chunk9f521c and separate fresh replay session61837/chunk314342 both terminated exit0/PASS. They reconstruct all600 selected searches,92 successful maps,22 selected obstructions, all74 protected bases and four protected obstructions, complete routing and216 unchanged current literal totals. The source explicitly verifies 13 white and52 white-plus-one-black positive-definite Grams in addition to the full Lorentz and necessary-contact checks. Initial generation and replay also passed; the final source added these explicit edge checks to match the verification scope precisely.

Two independent full producer/report mathematical reads passed. The parent additionally reread the complete necessary-contact and selected- corner helpers, checking genuine-newness, the physical projection bounds, protected marked-leaf extension, complete routing and independent short8 escape. The construction reviewer also reread physical native-label conversion and the relevant universal tree/cycle interfaces. The parent read the final white/edge-signature checks and adjacent-black path wording as a separate small delta. The construction full read preceded that check-only source addition; no second execution or final-delta read by that reviewer is claimed. Neither reviewer reran maps, projections, inertia, literals, arithmetic or the historical census.

The simultaneous finite B14/D14 local reductions are compatible: they preserve the independent short8-to-short6 routes, and their distinct D15, D16 or whole-component coupon alternatives are unchanged here. This C14 proof supplies no long finite rank14 output. No new finite charge is needed for their joint local composition.

Final direct source bindings, report links and Python syntax were checked. Source, data and report are final/frozen.

The original all-label catalogue, complete local reduction and universal short-path theorems remain explicit inherited inputs. No finite forest optimization, new global bound, external refereeing or proof-assistant formalization is claimed.

```
python3 scripts/fresh_order14_c_basic_distance9_transfer_20261018.py --verify
```

Source 219

Source path: `results/fresh_order14_c_distance10_terminal_projection_20261018.md`

SHA256: b534ccc252c927a5c3ab435d753081d377ca41174c48f94ea4ab7d54258ecf20

Order14 C81 by two free distance10 terminal exclusions

The whole affine C13 factor of order14 may delete the two unit pairs (1,11) and(2,12), reducing its cost83 to81 without any new finite payment. The canonical diagram is the path0 through13 with doubled end bonds0–1 and12–13. Both pairs have distance10 and form one invariant orbit. The current order14 family costs become A70/B82/C81/D82, so the all-family maximum falls from83 to82 and its envelope from83/14 to41/7.

Every triangle still receives at least2 and every quadrilateral at least1. Every finite weight and cross-component coupon is unchanged; the current finite411 bound is therefore unaffected by this local change. This packet does not recompute the ideal scalar frontier or claim an ambient dimension bound. The completed bound remains60.

The source and certificate bind the current simultaneous rule, the preceding C14 singleton deletion, and the completed B/C14 family composition. The full necessary-contact helper is inherited from the bound B14/B15 completion theorem and is re-evaluated here on each actual core.

Complete support

The preceding C14 rule has83 unit pairs. Its81 basic pairs are retained and are exactly the new unit set; the two deleted pairs are its only remaining nonbasic units. The checker verifies every native diagram automorphism, all short8 pairs and the elementary triangle requirements. The already deleted inner-terminal-neighbor pair(1,12) remains zero.

The complete high-overlap domain has2614 contexts. The new source checks the partition into1378 with whole-component bound at most13, 83 protected/adaptive cases,553 with displayed marked component larger than14, and600 eligible order14 patterns. The protected/adaptive group splits into74 fixed leaf/feature and9 adaptive finite cases.

All600 native embedding searches, including557 failures, are repeated. The43 successful contexts and92 maps agree exactly with both the complete frozen small-factor record and the preceding C14 certificate. The changed orbit occurs in precisely13 contexts:

```
366,380,392,531,614,1115,1159,
1503,1505,1508,1541,1619,1642.
```

The first12 have two reflected maps each and display A12 at target nodes1 through12. Context1642 has four maps and displays A11 at1 through11 or2 through12. Thus there are28 affected maps. All are proper displayed subcomponents of a putative whole affine factor; none is assumed to be a whole finite component.

Ten contexts carry the original `lorentz_residual` routing tag and 366,1503,1642 carry `relevant_component`. The present argument does not infer a completion theorem from those names. In every context the full Gram directly passes the same norm2, nonsingular-Lorentz, complete-contact and positive-end checks below. No optional exterior root or indirect-tail exclusion is needed to obtain this first-entry principal obstruction.

Physical first-entry proof with arbitrary contacts

Let H be the displayed full old core, with physical diagonal2 and simple native bonds, and let J be its inverse. The helper verifies signature(|H|−1,1,0), reconstructs the complete actual selected corner and its possible extra-white contact domain,

and partitions that domain into two end sets. Each inverse end block is entrywise positive. Any root meeting both ends gives an opposite-black path of length at most8, whose independent current payment is described below.

Otherwise all contacts of a further supporting white e with H lie at one end. Write its physical contact magnitudes as the nonnegative vector a. The actual span of H is nondegenerate Lorentzian, so its orthogonal complement is positive definite and

`a^T J a <= e^2 = 2.`

Because every entry in that end block is positive, a forced contact q at i gives `a^T J a >= q² J_ii`, even if e contacts arbitrarily many additional old roots at that end. Contacts into arbitrary spectator components impose further conditions but cannot repair this failed necessary inequality. No unspecified contact is assigned zero.

Every affected map has an old endpoint adjacent to a missing doubled terminal0 or13. In each A12 image both doubled terminals are missing; in each A11 image at least one is immediately adjacent to its displayed path. The required root e is a common supporting white: both corner marks are already displayed. It is genuinely outside H. An omitted old black is absent from this actual corner, and every other displayed old corner component is orthogonal to the selected path. Neither can supply its required nonzero terminal bond. This argument concerns the immediate neighbor and remains valid even if later new roots could join old spectator components.

A native label4 bond between physical norm2 roots has contact squared magnitude2. The forced terminal therefore gives the following exact projection lower bounds across all affected maps:

Lower bound2 J_ii	Terminal entries
5/2	4
8/3	4
4	16
7	8
16	12
43	8

There are52 doubled entries:48 from the24 A12 maps, and4 from the four A11 maps. Every bound is strictly greater than2. Either doubled terminal obstructs an A12 completion; the adjacent doubled terminal obstructs each A11 completion. Context1642's other first entry is simple, with lower bound5/4, and is not used as an obstruction. Thus every affected actual whole C13 completion is impossible unless the independent short-path alternative has already paid.

This proof uses physical norm2 throughout. It does not rescale a contact before comparing it to2, assume that a proper A11/A12 path is whole finite, construct a completed Gram by forcing absent contacts, or place an upper bound on spectator rank.

Complete current payment interfaces

First use the current independent short8 route and its actual-corner priorities. The current small ideal changes C11/B11, cycle11 and coupled halves, B12/D13, D105, B/C14 predecessors, B/D15 and the E6/D11 and D5/D12 order17 transfers retain their own short, protected or whole-component alternatives. None uses this order14 C distance10 orbit as an output. The new projection proof adds no finite debit, so every such finite alternative is unchanged.

If a short8 finite output is a deep8 zero, its independent theorem gives short7. If that finite7 output is a deep10-feature zero, the depth10 theorem gives the independent short6 alternative. Current small-factor and cycle/tree replacements then use their retained short/basic paths, fixed leaves/forks or specified finite coupons. They do not call back to an order14 distance10 pair. The current finite D17 zeros and fractional ideal order17 payments retain exactly these same short routes and their separate typed coupons.

In the remaining no-short8 branch, the inherited low-core bound is whole order at most12, and high-disjoint cases are excluded. Protected basic ideal choices have marked distance at most9. The74 fixed and9 adaptive finite certificates are unchanged, including their whole bounded components and fixed-leaf/distance9 roles. A selected displayed component larger than14 cannot be this actual whole affine factor. Every remaining changed selected image is one of the28 maps above. All unchanged selected payments retain their old proof.

The older D13/D14 immediate affine outputs have orders at most13. Later universal tree/cycle routes use protected short/basic paths, smaller whole factors or separate coupons. The partial D transfers use their own D-family roles and distinct finite payments. The current B14 residual argument gives its independent short8 escape or excludes its own B completion; it requires no C14 long pair. Consequently no nonselected, immediate or later interface introduces an unanalysed occurrence of the deleted orbit. This uses the complete inherited interface arguments, not just absence from selected maps.

Equivalently, perform the independent actual-corner priorities first, then patch every affected selected C occurrence by the obstruction above. On a face with no changed actual C14 corner, its weights agree pointwise with the preceding whole-face rule. This preserves total1 on every quadrilateral without assuming an old preferred witness remains paid. The elementary triangle sets remain units, and the finite7/ideal5 triangle theorem and its protected small-factor or fixed-leaf exceptions use no distance10 C14 pair. Thus total2 on triangles is retained as well.

The cusp factor-fiber multiplicity stays V/14 per marked pair. The C sum loses2; since B and D both remain82, the all-family sum improves by1 to82. There is no finite compensation and no finite optimization to preflight. Any simultaneous finite improvements being developed separately remain separate theorems and are not assumed here.

Verification and scope

Generationfcfa84 and separate fresh saved replaydc40ca both terminated exit0/PASS. All600 native searches,92 successful maps,28 affected maps, 13 full necessary-contact domains,52 doubled projection bounds, retained pair sets, current family provenance, complete2614 routing and direct immutable bindings are checked. The original all-label census and the complete nonselected/universal short-path classifications remain inherited inputs; they are not rerun.

Two independent full source/report mathematical reviews passed. The root reviewer also read the predecessor C14 source and the full necessary-contact helper, checking physical doubled magnitude, genuine new-root status, all contact and routing interfaces and current priorities. The construction reviewer independently reread that helper and checked the three relevant-component contexts through their actual Lorentz inverses, the arbitrary-contact bounds and all preserved interfaces. Neither review repeated the map, projection or original census arithmetic. Direct source bindings, report links and Python syntax were checked after these reviews. The source, certificate and report are final/frozen. No external review, formal verification, new finite forest maximum or ambient dimension theorem is claimed.

Source 220

Source path: `results/fresh_order14_c_inner_endpoint_projection_transfer_20261018.md`

SHA256: 5ff7df37109ea31447f318a49a590bc1ab298925337558d15ec4bfe72668ec2e

A free unit removed from the order14 affine C factor

In the whole affine C13 factor of order14, delete the pair(1,12) from the current unit rule. Its total falls from84 to83 without any finite coupon or other payment change. The canonical diagram is the path0 through13, with doubled terminal bonds0–1 and12–13. The removed pair is the singleton invariant pair of both inner terminal neighbors, at distance11. Every81 basic pair and every short8/triangle pair remains a unit. The order14 totals become A70/B84/C83/D82, so the all-family maximum is still84, now attained only by B. No ambient bound improves from this theorem alone.

The source and certificate use the complete saved high-overlap and small-factor domains from the current62 rule. The contact-domain helper is the one from the B14/B15 singleton theorem; it is applied to the actual old core and corner, independently of the name of the proposed affine target.

Complete support and exact physical projections

There are600 eligible selected patterns with displayed order at most14 and whole-component bound at least14. The checker reconstructs every induced labelled embedding into the canonical C13 target, including all 557 failed pattern searches. The43 successful patterns and92 maps agree exactly with the complete frozen support certificate. The removed marked image occurs only in contexts624,1555,1641, each with two reflected maps.

For each of these three actual cores H the helper rechecks physical norm2, simple old bonds, nonsingular Lorentz signature, the exact inverse J, and the actual preferred corner. It reconstructs the entire necessary contact domain of an extra supporting white, its two end sets, and the entrywise-positive inverse blocks on those ends. A root contacting both ends gives an opposite-black path of length at most8. Its complete payment route under the present weights is treated below.

In a face without that route, all contacts of a new root e with H lie on one end. With a the nonnegative physical contact vector, the positive orthogonal complement of the Lorentz span gives

```
a^T J a <= e^2 =2.
```

Every entry of the corresponding end block of J is positive. A forced contact q at i therefore gives a^T J a >=q^2 J_ii even when e makes arbitrarily many further contacts on that same end. No unspecified old contact is assigned zero. Contacts with additional spectator roots are unrestricted by this necessary inequality.

Every one of the six affected maps has displayed image exactly1 through12. Both missing target roots0 and13 must be supporting whites joined by label4 to the marked old roots1 and12. Either one supplies the obstruction. They are genuinely outside H: the two old blacks omitted from this corner cannot belong to its actual factor, while every other old corner component is orthogonal to the selected component. Thus none of those old roots can supply the required nonzero terminal bond. Possible later bridges between old components do not alter this immediate-neighbor argument.

In physical norm2 a label4 contact has squared magnitude2, not1. The12 forced terminal entries have exact lower bounds

Projection square2 J_ii	Entries
4	4
7	4
16	4

All exceed2. Hence the actual whole affine completion cannot occur in any of these selected cases unless the retained short-path route already pays. This excludes the affine occurrence; it does not assume its displayed finite A12 is a whole finite component. There is no completed Gram with artificial zero contacts, no bound on external spectator rank, and no finite replacement payment.

Other domains and current-rule composition

The selected maps are only one branch of the complete inherited theorem. The low-core bound is whole marked-component order at most12. The high disjoint branch is excluded by its all-label proof. Protected/basic ideal choices have marked distance at most9 and are untouched here; all finite whole-component, adaptive and protected-leaf payments are unchanged. Every remaining high selected occurrence of the deleted pair belongs to the three contexts above. Elementary triangle requirements were checked separately and remain paid.

The universal tree and older D13/D14 immediate affine outputs have orders at most13. The later cycle12, D11, B12 and C12 refinements use finite payments or smaller/shorter ideal outputs. The order15/16/17 partial D transfers use distinct D-factor roles and their typed finite coupons. No later proof introduces this order14 C distance11 pair as an output. The earlier same-order B projections alter only B pairs; the present proof does not change their alternatives.

For simultaneous use with the completed62 rule, apply its independent actual-corner priorities before requiring an old short8 unit. A changed C11 endpoint yields a path1/2 unit; a changed B12 center/terminal yields its retained affine D11 unit, finite D12 leaf9 unit, or two whole-component halves; a changed D13 center uses its independent half transfer, or its new zero theorem with upgraded full finite coupons. None outputs the deleted C14 pair. Likewise handle cycle11 distance5 and coupled order11 B/C/D half corners by their complete independent theorems. Their outputs are short ideal pairs, protected whole finite components or typed coupons, not a long order14 C pair.

After those branches, the complete short8 route of the completed63 composition applies: unaffected small-tree short8 units, retained cycle6/7/8 rules, radius-five tree replacements, and the independent depth11 finite8-zero to short7 replacement. Sparse finite deletions have distance at least10 and do not remove those outputs. In particular no proof here silently counts a current order11 half or zero as an unchanged unit. The short7 replacement cannot call back to this distance11 pair.

Now, if a selected proof requires the newly removed order14 C occurrence, the complete necessary projections above exclude it unless that route has paid already. Otherwise its former payment is unchanged. Equivalently, on a face having no actual changed C14 corner, all weights agree pointwise with the preceding complete rule. This preserves quadrilateral payment1. The newer B16 distance10 singleton deletions are compatible for the same reason: neither proof uses the other's changed long pair as an output. The completed order17 zero refinement also uses only this protected short8 route or a separate whole E6/D11 unit; its changed distance10 orbit is never an output needed here. Its independent completion theorem remains a separate input when the two rules are composed.

Triangle payment2 follows from the retained finite7/ideal5 and protected small-factor routes. The removed distance11 pair is absent from all of them. Cusp factor-fiber multiplicity remains V/14. Every finite weight, coupon and every other ideal weight is unchanged. The current scalar frontier still requires a separate reduction of the B14 maximum, as well as a finite-vertex improvement; this packet makes neither claim.

Verification scope

Generation7aff88 and separate fresh replay1ebc2b both terminated exit0/PASS; direct frozen-input bindings were rechecked successfully in0d60b1. Root and a separate peer completed full producer/report mathematical reads, including the complete contact-domain helper at their recorded scopes. Both passed the reconstructed support, genuinely new norm-two doubled terminals, arbitrary same-end contacts, opposite-end short8 route, and the retained nonselected and current local interfaces. The peer also checked compatibility with B16 and the separate order17 zero theorem. Neither reviewer repeated a census, inverse calculation or arithmetic run. Source, data and report are final/frozen. The source reconstructs the600 eligible pattern searches and compares every successful map with the frozen certificate. It then rechecks three necessary contact domains and all12 forced terminal bounds across six affected maps, plus invariance, basic/short8/triangle inclusion and current factor totals. The upstream all-label geometric census and the written completeness of its domain remain inherited theorem inputs. No external review, formal verification or new global bound is claimed.

```
python3 scripts/fresh_order14_c_inner_endpoint_projection_transfer_20261018.py
```

```
python3 scripts/fresh_order14_c_inner_endpoint_projection_transfer_20261018.py --verify
```

Source 221

Source path: `results/fresh_order14_contextual_ambient59_frontier_20261018.md`

SHA256: eb9c165229666038f9f5f9624bb73b197fecb5a7edd76d5b8973eb5e481ef1fa

The proved order14 envelope80 and the ambient59 frontier

The simultaneous local rule now has ideal order14 totals A70, B80, C80, D80, so its all-family envelope is40/7. The new D14 finite coupon is fully counted together with the contextual rank13 rebalance. The complete rank59 finite maximum remains795/2=397.5, with14 representative alternatives above curvature783/2=391.5. The complete terminal ideal domain has seven negative profiles, down from ten. The proved unrestricted dimension bound remains59; this packet does not establish dimension58.

The source and certificate compose frozen local proofs and the complete finite certificate. They independently enumerate all1,890 terminal rank58 profiles, verify their saved polynomial and curvature identities, and update every exact margin. No forest optimization, fresh convolution or historical geometric census is repeated.

The compatible current local rule

The coupon-free B14 theorem deletes the two intrinsic distance9 singletons(1,10),(2,11), giving B80. The C14 theorem deletes its remaining basic distance9 singleton(2,11), giving C80. Both include the complete selected and protected native domains and keep their independent short8-to-short7-to-short6 escapes. They do not create any finite coupon.

The universal D14 theorem halves the four near-fork/far-leaf distance9 pairs, giving D80. It adds only a whole A1/whole D13 long-arm-neighbor half. The large mark is the unique neighbor9 of fork center10, not the center or either short leaf. The actual A1/D13 component-pair cost rises3/2 to2. The old fork-center ideal pair(0,9) was already zero and is unchanged. A70 is inherited from the exact family table bound by the B14 theorem.

The contextual rank13 theorem reduces A2/A13-middle and A2/D13-center half weights to7/16, raises the A1/A13-middle half to9/16, and adds1/8 between marked distinct whole A2 components only when the same finite diagram has an unmarked whole A10 and at least two unmarked whole A1 components. The trigger is counted once per marked pair, independently of the number of possible witnesses. It keeps all ideal weights unchanged. Its three repaired geometric interfaces are the cycle14 equality, original order16 case384, and the two later finite D15 cases1147/1269.

Here is the noncircular simultaneous actual-face priority. If an actual changed BC11 corner occurs, first apply its independent zero-weight theorem. Its short1/2, smaller finite/ideal and retained D11-half outputs are unchanged by the present rules. Next, if an actual changed D14 near-fork corner occurs, apply the new universal D14 theorem. Its outputs are a distinct retained D11 half, the whole A4/D10-shortleaf half, the new A1/D13-neighbor half, the retained finite D14(0,10) unit, or a short1/2/3 unit. These are all distinct from A2/D13-center and A2/A13-middle, and from the A1/A13-middle role. Thus the contextual rebalance cannot remove any of its payments.

If neither changed actual corner occurs, the ideal weights agree pointwise with the preceding B80/C80/contextual rule, while the new finite coupon only increases. That whole-face theorem applies directly. Its cycle14, order16 and D15 repairs use only their stated finite roles and independent short paths, so do not depend on a removed D14 distance9 unit. The independently retained short8-to-short6 and whole-A8/protected- leaf alternatives also make the B14/C14 selected-domain exclusions safe under the current finite and contextual changes.

All triangle elementary weights are retained: the new ideal pairs have distance9, while the inherited triangle proof uses the unchanged short and protected small-factor outputs. Thus triangle total2 and quadrilateral total1 hold for the simultaneous rule.

The source also matches all216 literal configurations by source and case, with the same ordered actual marks. The B14, C14 and D14 packets have no affected literal corners. The contextual packet's two center debits and one contextual credit in case384 are therefore the complete combined literal changes; every corner lower bound stays nonnegative and every face total is at least1. This is supplementary to the universal actual-corner argument, not a finite enumeration of all polytopes.

The complete finite certificate is now unconditional locally

The frozen contextual packet included a separate complete certificate named `conditional_neighbor_preflight`. It already counts exactly the A1/D13 increment1/2 proved by the new D14 theorem. That was its only geometric condition, now discharged. No arithmetic of its ordinary or no-rank2 residual tables changes, and the contextual proof and D14 proof use disjoint paying roles as just shown.

The bound certificate covers all961 fixed multiplicity vectors,960 interaction keys,1,920 residual tables,32,400 ordered actual-type comparisons,49,011 item identities,2,613,120 Bellman upper inequalities, 113,280 attaining predecessors and2,691 contextual options. Fixed rank17 means A17; residual D17 remains in the complete residual domain. The ordinary/no-rank2 decomposition explicitly enumerates every A2 count and reserves two A1 components whenever the contextual trigger is active. Any rank10 component may serve as an upper-bound witness, whereas actual costs require a whole A10. This only increases the representative bound.

The current component-pair costs checked by this binder are

Component types	Actual total
A2/A13	39/8
A2/D13	39/8
A1/A13	25/16
A1/D13	2

The contextual A2/A2 total is k(k-1)/4 when a whole A10 and at least two whole A1 components are present, and zero otherwise. All other coupons and bases stay as in the bound finite certificate. In particular, actual D15 is201/2 while its rank15 representative allowance is101. The finite B14/D14 refinements are included: B14=87, D14=181/2, A14=91.

The exact maximum397.5 has actual typed witnesses

- D16+D17+D26;
- A17+D16+D26;
- 2D16+D27.

Fourteen representative alternatives remain above391.5. The maximum among those containing a rank13 component is1563/4=390.75, and among those containing rank10 it is386. The actual forests10A2+3A13 and 10A2+3D13 now cost1561/4=390.25, below curvature. These are formal weighted forests and no geometric realization is asserted.

This binder checks the exact certificate interfaces, all saved fixed-row maxima and the listed attainments; it does not re-execute the already generated Bellman certificate. The full frozen source/data are bound by hash and retain the complete proof of the finite upper bound and actual formal attainment.

All1,890 terminal ideal comparisons

The low9 terminal domain consists of arbitrary repeated ranks at least9, and at most one rank1 through8, totaling58. A new independent recursion enumerates it and proves equality with the frozen1890-row domain. For each row the binder checks the saved positive, symmetric degree58 h-polynomial, its coefficient sum V=product(r+1), G, curvature, the exact absolute-deviation identity and the old envelope cost/margin. No high-rank concentration is used; the exact all-family tail formulas are unchanged.

The only envelope change is at affine rank13/order14:

```
82/14 - 80/14 = 1/7.
```

Consequently the new normalized margin is exactly the old margin plus count(rank13)/7. Every row is checked both by this update and by summing the new envelopes. The prior ten negative rows become the following seven; there are no zero rows. All tuples list affine ranks, not orders.

Ranks	Normalized margin	Terminal absolute margin
(9,9,12,12,16)	-5241/57460	-26205
(9,9,9,9,10,12)	-5227/71500	-104540
(9,9,9,15,16)	-389/5440	-19450
(1,9,9,9,9,9,12)	-6893/130000	-137860
(8,9,9,16,16)	-1087/52020	-5435
(8,9,9,9,9,14)	-43/4500	-12900
(6,9,9,9,9,16)	-19/59500	-380

The normalized minimum is -5241/57460 at(9,9,12,12,16); the terminal absolute minimum is -137860 at(1,9,9,9,9,9,12). The former worst profile (9,9,9,9,9,13) is now positive. Negative absolute margins are not extended to arbitrary profiles by the positive-margin/product monotonicity lemma. The negative terminal comparisons themselves are genuine scalar obstructions to this payment estimate.

The ambient59 curvature coefficient remains783/2 and the global extremum constant remains4350, counted once and bound to the prior frontier. The finite deficit is6, and the ideal terminal domain still has negative rows. Neither a new dimension58 theorem nor a new all-ideal ceiling follows.

Exact next scalar targets, without new geometry

All seven negative tuples contain rank9. On this terminal domain alone, an all-family order10 total saving strictly greater than5241/11492 is sufficient; a half-unit saving gives minimum101/11492. This would mean all-family cost45 falling to89/2. It is only an arithmetic target: changing rank9 also requires revalidating the low9 concentration argument, as well as proving the local geometry and counting any added finite coupons. The certificate flags that extra concentration obligation.

Two alternatives leave the low-rank concentration ingredient unchanged:

Additional all-family total savings by affine order	Terminal normalized minimum
Order13:1; order15:1/4; order16:1/2; order17:3/4	21/5500
Order13:1; order15:1/4; order17:5/4	11/5440

Both make all1890 terminal rows strictly positive and give terminal absolute minimum91/2 at singleton rank58. These are conditional scalar calculations, not local weight theorems or a finite-budget improvement. Every cost refers to the maximum across all types of that order.

For the first coordinated alternative, with the other listed savings fixed, the exact strict thresholds are5227/5500 at order13,43/300 at order15,149/340 at order16 and219/320 at order17. In particular, reducing order13 alone leaves four negative tuples unaffected; reducing order17 alone leaves three unaffected. The complete single-order and coordinated threshold records are saved to guide concrete compatible local proofs.

Verification

Final generation f88d55 and a separate fresh saved replay aea70c terminated exit0/PASS. They check frozen dependency hashes, local rule and coupon matching, the complete finite-certificate interfaces,216 literal lower totals, all1890 saved polynomial/margin identities and every conditional scalar scenario.

Parent completed a full producer/report mathematical composition read: the exact optional-neighbor hypothesis match, independent actual-corner priority, contextual interfaces,216-row alignment, inherited finite certificate scope, independent1890-row domain and saved moment identities, seven negative profiles, exact conditional thresholds and rank9 concentration

flags pass. A second reviewer independently read the full producer/report, contextual whole-corner interfaces and inherited low9 statement, after previously reading the complete D14 local proof. That read also passes the disjoint new role, pointwise unchanged-face branch, finite hypothesis discharge, exact1/7 update and scope flags. Neither reviewer executed DP, convolution, scalar or geometric calculations. Source, certificate and report are final/frozen after direct binding, local-link and syntax checks. No DP, convolution, historical geometry, external refereeing or proof-assistant formalization is claimed here.

```
python3 scripts/fresh_order14_contextual_ambient59_frontier_20261018.py --verify
```

Source 222

Source path: `results/fresh_order14_d_near_fork_half_transfer_20261018.md`

SHA256: 26dc0b458e6d84b7553572c0e051f172f63279bc315781fdb80c936b0b313e61

Four universal affine D13 near-fork halves

For an actual ideal factor of type affine D13, of order14, the four near-fork/far-leaf pairs may be reduced from1 to1/2. In the labeling with spine0–1–…–9, leaves10,11 at0 and leaves12,13 at9, they are (1,12),(1,13),(8,10),(8,11). They form one intrinsic distance9 orbit. The total falls from82 to80. The fork-center pair(0,9) was already zero and is unchanged.

There is exactly one new finite coupon: weight1/2 between a whole A1 and the unique long-arm neighbor of the fork center of a separate whole D13. For finite D13 with spine0–…–10 and short leaves11,12, the large mark is9. There is one eligible marked pair, so the increment is exactly1/2 per component pair, raising its current actual total from3/2 to2. This coupon is distinct from the D13 fork-center and short-leaf roles.

Together with the frozen B14 and C14 reductions, the ideal order14 family totals are A70, B80, C80, D80, with envelope80/14. The theorem covers arbitrary native labels, arbitrary supporting spectators and every actual occurrence of the changed ideal orbit. A complete finite-forest count and any new ambient bound are separate obligations. The producer and certificate check the exact physical Gram identities and marked component roles.

The actual star and complete principal profiles

Write the quadrilateral star as W+a+b+c+d, with opposite pairs{a,c},{b,d}. The actual edge diagrams W plus one black root are strictly positive definite; adjacent corners are positive semidefinite. Roots have square2. At W+a+b, the inherited factor-fiber theorem supplies the whole varying affine factor F and a positive-definite spectator diagram Z orthogonal to F. This applies without a simple-cusp assumption. Use a=1 and b=12; automorphisms give the other three changed pairs. The positive kernel of F has coefficient1 on the spine and1/2 at each leaf.

If b.c, a.d or c.d is negative, a distinct actual corner has a retained distance1 unit. If c and d share a nonzero spectator contact, a retained distance2 unit pays. Thus assume those black products vanish and the nonnegative spectator contact vectors x,y have disjoint supports. The opposite products a.c and b.d remain unrestricted. Every nonzero native magnitude is at least1; a magnitude greater than1 is at least sqrt2, including the parallel and dotted possibilities. Kernel heights are positive.

Deleting a from F leaves A3 on10–0–11 and finite D10 on2,…,9,12,13. Deleting b leaves finite D13, with long endpoint13. In either large finite block, inverse diagonals at its first two long-arm vertices are1,2, and all other diagonals exceed2. A single nonzero contact in a positive- semidefinite one-root extension can therefore only be simple or doubled at the first vertex, or simple at the second. Two contacts would create a cycle in a connected diagram still containing the old fork; finite and affine Coxeter classification excludes this, and exceptional diagrams are too small to contain D10. This proves the complete profile list for all native labels, rather than assuming a finite label cutoff.

The doubled-first and simple-second profiles create an affine component with positive kernel. In the actual positive-semidefinite corner that kernel forces every other corner root to be orthogonal to that whole component. In particular the opposite new root's old A3 and spectator contacts vanish when such a near profile occurs. Its product with the omitted opposite black root remains unrestricted.

Put t=-a.c, and u=-10.c, v=-0.c, w=-11.c. Define

```
K = [[3/4,1/2,1/4],[1/2,1,1/2],[1/4,1/2,3/4]],
e=(u,v,w) K (u,v,w)^T,     h0=t+(u+w)/2+v.
```

Eliminating F minus a,b gives resistance R=10. The near profiles have height h, residual constant c0 and moment l as follows:

Near profile	h	c0	l
Only outer A3/opposite contacts	h0	2-e	0
Simple contact at2	h0+1	1-e	1
Doubled contact at2	t+sqrt2	0	sqrt2
Simple contact at3	t+1	0	2

For the far root put t=-b.d. Its complete profiles are:

Far profile	k	d0	m
No contact to F minus b	t/2	2	0
Simple contact at13	(t+1)/2	1	1
Doubled contact at13	(t+sqrt2)/2	0	sqrt2
Simple contact at9	t/2+1	0	2

The residual new-root squares before eliminating Z are c0+l²/10 and d0+m²/10. Every cross projection is lm/10. These statements, the two complete inverse-diagonal lists and all16 cross identities are checked directly in the norm-two Gram, not after an unrecorded gauge change.

The last two **near** profiles already have independent alternatives. A doubled contact at2 gives a distinct whole affine B10 factor of order11, with doubled-terminal/fork-leaf marks. Its weight is currently zero, so apply the independent BC11-zero theorem at that actual corner first. It supplies a full retained payment. A simple contact at3 gives a whole affine D10 factor of order11 with opposite fork-leaf marks, of current weight1/2. That distinct half plus the selected D14 half is1. The source checks the positive kernels, native canonical embeddings and current weights of both outputs. The far doubled/second profiles need not be spent as alternative payments; the inequalities below exclude them.

Complete native cost bounds

For an end define g=2l/h+c0/h², with the corresponding far variables. After the two immediate near alternatives have been handled, every near profile has g<=5, with a unique extreme type: a simple contact to one of the old outer leaves10,11, no opposite contact, height1/2 and g=5. Every other near profile has g<=3. Here is a complete argument.

For the outer-only profile, A3+c is a principal subdiagram of the strict actual edge. Thus e<2. Any two nonzero contacts give e>=Kii+Kjj+2Kij>=2, so at most one A3 contact is possible. With none, positive height implies t>=1 and g<=2. A contact at the middle0 is simple and gives g<=1. A single leaf contact q with t=0 gives g=8/q²-3: this is5 for q=1 and at most1 for q>=sqrt2. If t>=1 instead, c0<=5/4 and h>=3/2, giving g<=5/9. These bounds include all allowed native opposite products, not just simple ones. For the simple-at2 profile, positive semidefiniteness of D10+c plus A3 gives0<=c0<=1 and h>=1, hence g<=3. Additional spectators only strengthen these restrictions.

At the far end the isolated profile with t=1 and the simple-at13 profile with t=0 both have k=1/2 and g=8. Every other far profile has g<=4: the isolated nonunit case has t>=sqrt2; simple-at13 with t>=1 has g<=3; and the doubled-at13 or simple-at9 cases have g<=4 for every t>=0. Thus these are the complete two far extreme types.

Eliminating Z, with J=G_Z^(-1), gives the exact determinant identity

```
0 = 1-(g_c+g_d)/10 + E/10,
E=(x/h-y/k)^T J (x/h-y/k) >= 0.
```

The source verifies this identity formally, including the spectator off-diagonal sign. If the far end is not extreme, g_c+g_d<=5+4=9, giving a strict positive gap at least1/10. Hence the far end is extreme.

Whole-factor payments with arbitrary spectators

First suppose the near end is not extreme. Then

```
E=g_c+8-10=g_c-2<=1,     k=1/2.
```

If any y_i is nonzero, disjoint supports give x_i=0. Gram-metric coordinate Cauchy, using G_ii=2, implies

$$E \geq (x_i/h-y_i/k)^2/G_{ii} = 2*y_i^2 \geq 2,$$

a contradiction. Thus the entire far vector y vanishes, without any restriction on x or on arbitrary connected spectator tails. In the chosen paying corner W+a+d, the opposite root c is absent.

- For the isolated far profile, d contacts only the omitted b. In the chosen corner it is a whole A1; F minus b is a whole finite D13, with a at the unique long-arm neighbor of its fork center. The new half coupon plus the selected ideal half gives1.
- For the simple-at13 far profile, d extends that D13 to a whole finite D14. The marked canonical pair is(0,10), of retained weight1. The current finite D14 refinement changes(0,11), so this unit survives.

The whole-factor assertions follow because Z was orthogonal to F and y=0. Contacts belonging only to c do not enlarge this paying corner. The source reconstructs both actual finite Grams and marked roles.

If both ends are extreme, h=k=1/2 and the determinant gives

$$(x-y)^T J (x-y) = (5+8-10)/4 = 3/4.$$

Both separate spectator extensions are strictly positive definite, as principal subdiagrams of actual edges. The complete below-one spectator theorem therefore applies. At rational energy3/4 its only possibilities are a single simple contact at an endpoint of one whole A3, with the other support empty, or two simple contacts at adjacent nodes of one whole A3. Every other component has zero contacts. In the two-nonempty case there is an actual path of length3 between c and d, giving a distinct unit.

If the far support is empty, use the previous far-corner payment. If the near support is empty, c extends the old A3 to a whole A4 with c an endpoint. The old far component is a whole D10 with b a short leaf. The existing small-component/D10 short-leaf coupon has weight1/2 for every mark of a whole component of order at most5. It applies to this whole A4 endpoint and combines with the selected half. The other root d is absent from the chosen corner, so its arbitrary spectator contacts cannot spoil either whole factor. The source checks the literal A4/D10 roles; no A7 or additional cross-component coupon is being assumed.

Current rule, priorities and complete scope

The canonical current D14 table is bound to the completed dimension64 interface and its frozen successors: total82, center pair(0,9) zero, and all four proposed pairs literal units. The native automorphism orbit is checked exactly. The current B14 reduction and C14 reduction both have total80; their recorded family provenance supplies the unchanged A70. Thus all four current family totals are 70,80,80,80 after this theorem.

For any actual quadrilateral, first apply the independent BC11-zero theorem if a changed BC11 corner occurs. Its short1/2, small finite and retained D11-half or typed-coupon outputs avoid every new D14 role. Otherwise apply this theorem whenever a changed D14 corner occurs. The two immediate near profiles, the far-empty cases and the complete energy3/4 alternatives above give an independent full payment. If neither occurs, the actual ideal weights agree pointwise with the preceding proved rule and the finite coupon only increases, so the whole-face proof is inherited. This priority concerns actual corners and does not assume that an earlier selected witness avoids the new orbit.

The proof is universal and consequently covers occurrences in every selected, protected, adaptive or low/disjoint catalogue branch, including all74 protected bases. No absence from saved embeddings is substituted for this coverage. In contrast to the B/C basic-distance9 deletions, there is no new selected-map or protected-base census in this packet.

All ideal distances at most8 are unchanged literal units for this D14 factor. The independent current short8-to-short7-to-short6/whole-A8 route is therefore available to the other B14/C14 and finite14 proofs. The new theorem itself uses only short1/2/3, retained affine D11 halves, whole finite D14(0,10), A4/D10 short-leaf halves and the new A1/D13-neighbor half. It uses no reduced finite D15–D17 same-component payment, no A2/D13-center coupon, and no A1/A13-middle coupon. Its outputs therefore do not depend on the separate proposed rebalances of those roles.

The elementary D14 triangle target set is explicitly unchanged. The inherited finite7/ideal5 triangle theorem uses short or protected small-factor outputs, and does not spend these distance9 near-fork pairs. Triangle minimum2 and quadrilateral minimum1 persist. A fresh audit of all216 current literal configurations, starting from the frozen finite B14/D14 table,

finds zero occurrences of either new ideal debit or new coupon. This is an interface check; universal coverage rests on the arbitrary-contact proof above, not on the216-row observation.

The only finite increment is exactly1/2 for the one whole A1/D13 role. The existing A4/D10 coupon is already budgeted. No finite maximum is asserted here, and the family envelope80/14 alone is not a dimension58 proof.

Verification

Final generation session29119/chunka50bf9 and separate fresh replay session47272/chunk0e7165 terminated exit0/PASS. The checks verify the two inverse lists, four near and four far profiles, all16 cross identities, determinant formula and native-bound endpoint values, two immediate affine corners, three whole finite corner roles, native orbit/triangle retention, current family provenance and216 literal configurations. The all-native principal-extension classification and arbitrary-rank below-one theorem are written mathematical inputs; these are not inferred from finite checks.

Parent completed a full producer/report mathematical read and checked the bound below-one theorem at3/4. The all-native principal profiles, R10 physical identities, strict cost gap, entire-vector Cauchy argument, whole finite roles, independent BC11 priority, retained D11 half and triangle/current216 interfaces pass. A second reviewer independently read the full producer/report, complete below-one report, analogous D13 center-profile and sealing proof, factor-fiber/determinant interfaces and BC11-zero outputs. That read also passes the native bounds, complete energy3/4 alternatives and unique new D13-node9 coupon. Neither reviewer executed inverse, profile, literal, census or finite-count calculations. Source, certificate and report are final/frozen after direct binding, local-link and syntax checks. No external refereeing or proof-assistant formalization is claimed.

```
python3 scripts/fresh_order14_d_near_fork_half_transfer_20261018.py --verify
```

Source 223

Source path: `results/fresh_order15_affine_b_pair_reduction_20261017.md`

SHA256: f4cf09bb4d6dc2e7d051d7b3a7a0069155d5ea2882fb6eef3fdb5fe8cc76b2d6

Three removable payments in the order-15 affine B factor

The order-15 affine B factor can use **97 paid pairs instead of 100**, with no new finite coupon. The three omitted pairs were required only by seven selected overlap contexts. In six contexts, the marked component is a whole finite D14 that no supporting white can touch. In the seventh it remains a whole finite A12 or grows to a whole finite A13. Every case therefore already has a finite unit payment.

This is a local improvement to the complete finite8/ideal5 face-payment theorem. It is not a new ambient dimension bound. Standard type names are used below: the affine target is **affine B14**, of rank14 and order15; the finite D14 has14 roots.

The checker and certificate bind the complete high-overlap domain, its complete selected-pattern map census, and the finite8/ideal5 composition. The all-label supporting-white classification is an explicit frozen input, not inferred from a bounded search in this new checker.

Exact support diagnostic

The cost-maximizing family at order15 is affine B14: its cost100 exceeds A75, C95 and D96. In its canonical numbering the fork center is0 with short leaves13,14; the spine runs0–1–…–12, with the last bond11–12 doubled. There are two automorphisms, exchanging13 and14.

The existing100-pair set consists of91 basic pairs and these six extra orbits. A context is counted once in a row if at least one of its saved marked images meets that orbit.

Representative	Distance	Orbit size	Supporting contexts
0,10	10	1	13
0,11	11	1	3
1,11	10	1	11
9,13	10	2	53
10,13	11	2	17
11,13	12	2	4

The certificate saves every supporting context's whole marked-core key. It also records the analogous orbit diagnostics for B/C/D at orders14 and16; those are diagnostics only, with no pair deletion claimed there.

Remove the invariant union

`{(0,11), (11,13), (11,14)}`.

The exact complete saved-map support is:

Removed orbit	Affected current context indices
0,11	1284, 1623, 1641
11,13 and11,14	1646, 1666, 1933, 2326

All three pairs have distance greater than9. The full basic set, elementary triangle payments, and the short8-first replacement interface remain unchanged. The old small-factor transfers use their own existing finite outcomes or retained short paths; they do not require a new basic pair here.

The previous map search deliberately allowed every induced embedding of the bounded marked component into a target. Such embeddings need not all occur as actual affine completions. The following literal supporting-white checks prove that the seven particular preferred components are always finite. No conclusion is drawn from an abstract target embedding alone.

Six reversed-three-node contexts: the marked D14 is untouched

Contexts1284,1623,1646,1666,1933,2326 all have two L25 cores identified along the reversed path

`left7=right9, left8=right8, left9=right7`.

Use the saved numbering with right nodes0,…,6 becoming10,…,16. The17-root Gram has signature(16,1,0). The complete frozen no-short8 supporting-white classification says that any extra supporting white with a nonzero core contact has exactly one unit contact, at node3 or node13. Both projection squares are2; every other nonzero support or larger native contact is already excluded by the exact corner and Lorentz tests. Additional core-orthogonal whites are unrestricted subject to the actual positive-definite face support.

For all six contexts, the preferred marked component is a finite D14 and contains neither3 nor13. The new checker reconstructs its literal graph and positive definiteness. Consequently every extra supporting white is orthogonal to every root of this D14. No chain of extra whites can connect to it, since any such chain would need a first direct contact. The component is therefore whole and finite, with14 roots.

More explicitly, the complement of this component in the preferred corner consists entirely of supporting whites: both black marks are in D14. That complement is positive definite as a principal subdiagram of the actual face support. The complete preferred-corner Gram is thus an orthogonal sum of positive-definite blocks, and this is an actual finite corner. The whole-component-at-most17 rule gives a unit. This argument does not presume a simple ideal vertex or treat a finite subdiagram at an ideal corner as sufficient by itself.

The absence of extra cross-core edges is supplied by the frozen complete overlap certificate for the literal whole context. It is essential to the stated contact domain and is not assumed from the preferred component alone.

Context1641: a residual-zero endpoint can only give a whole A13

The remaining context has two L25 cores identified along the reversed path

`left6=right9, left7=right8, left8=right7, left9=right6`,

with opposite black pairs{1,2} and{11,12}. Its16-root Gram is nondegenerate Lorentzian, of signature(15,1,0). The preferred marks1,11 are endpoints of the displayed A12 path. The complete supporting-white domain consists of no core contact, or one unit contact at one of1,3,11,13. In that order, the corresponding inverse block is

```
 2      1      −3      −2
 1      4/3    −2      −4/3
−3     −2       2       1
−2     −4/3     1       4/3
```

Only1 and11 belong to the marked A12. A white root attached at either of these nodes has projection square2, hence residual square0. The core spans a Lorentzian subspace, whose perpendicular is positive definite. The residual vector therefore vanishes: this root lies in the core span and is orthogonal to every core-orthogonal white.

It also cannot coexist with a second core-contact white. Its actual product with such a root equals the displayed projection product. For the same endpoint or the other same-end contact, this product is positive, contrary to acuteness. At the opposite end the product is−3 or−2. Two actual supporting whites have a positive-definite principal Gram, so their norm2 product must lie strictly between−2 and2. Those opposite-end possibilities are excluded as well. This argument includes distinct roots with the same contact; no multiplicity assumption is made.

If neither endpoint-contact root exists, every extra white is orthogonal to A12, so it remains whole. If one exists, it has a single simple bond at an endpoint and gives the literal whole A13. It cannot connect to any other extra white by the preceding two arguments. The checker verifies both A13 extension matrices and every mixed product used above. Both marks remain in this component; its orthogonal complement in the corner is contained in the actual positive-definite white support. The corner is finite and the existing whole-component rule pays a unit in either case.

Local consequence and quantitative scope

All seven newly omitted selected images have a retained finite payment under arbitrary supporting-white extensions. Every other selected image is unchanged. The complete predecessor theorem may therefore use the saved97-pair affine B14 set while preserving every triangle and quadrilateral payment. No extra finite weight is introduced and the finite8 vertex budget remains unchanged. The other order15 costs remain A75, C95 and D96, so the worst order15 cost becomes97.

The previous dimension75 boundary profile `(12,12,12,12,14)` has normalized margin `−16873/171366`. Reducing the order15 cost by one pair changes this margin to `−9081/285610`; two pairs change it to `29879/856830`; all three change it to `87001/856830`. These are comparisons for this one formal profile only. They neither certify all profiles nor prove that the formal profile is geometrically realizable. A new global bound requires a fresh complete scalar comparison and valid concentration rules for the changed envelope.

Verification status

Final generation and separate fresh saved replay both terminated exit0/PASS. The source checks every input's completed status and direct recorded hashes. It reconstructs each actual corner from the four black roles and verifies that the saved marked component is its literal connected component, contains both marks, and has only supporting whites in its complement. All17 direct bindings, syntax and report links pass.

Two separate full source/mathematical reads found no gap in the six untouched D14 cases, residual-zero A12/A13 argument, strict white-PD exclusions, complete saved-map affected support or basic-pair retention. Root also read the inherited ordinary-extension classification and requested the explicit component and input-hash guards now present. Neither review duplicated the seven-case arithmetic or the inherited overlap and map searches.

The new arithmetic reconstructs graph automorphisms and exact support orbits from the saved complete map corpus, and verifies seven literal finite outcomes using rational Gram inverses and signatures. It does not repeat the82,197-overlap or1,153-pattern searches. Published diagram classifications and the complete predecessor's all-label support proof remain explicit inputs. Source, data and report are final/frozen. No frozen predecessor was edited, and no new dimension theorem is asserted here.

Source 224

Source path: `results/fresh_order15_affine_b_residual_terminal_transfer_20261017.md`

SHA256: 8bebe8578f85d390d1be52368470b1f627101d1a56283f68cafdf36ecbbb27f9

Two more order15 affine B deletions by a sealed terminal

The order15 affine B14 cost can fall from96 to94 without changing any finite weight or coupon. Delete the two pairs(10,13),(10,14) in the canonical spine0–…–12, doubled bond11–12 and short leaves13,14 at0. These pairs have distance11 and form one automorphism orbit.

The other order15 costs remain A75/C95/D96. Therefore this B-only result does **not** yet lower the order15 maximum96 or its normalized rank14 envelope32/5. In particular it does not by itself supply the scalar target of a maximum94 over all families.

The source and certificate give a complete local refinement of the finished ceiling65 rule, also compatible with the subsequent order13 B refinement. No new ambient bound or finite-budget claim is made.

Complete selected support and retained payments

Start from the96-pair set after the earlier three-pair B14 reduction and the B14/B15 singleton projection theorem. All their absent pairs stay absent. Removing only the new two-element orbit leaves94 units and retains all91 basic ideal5/feature5 pairs, every short8 pair and every elementary-triangle payment. The source checks these sets and the full diagram automorphism group.

The complete frozen mapping corpus has536 patterns with selected order at most15 and whole-component bound at least15. These include every selected occurrence that could require this actual whole affine target. The checker re-enumerates every induced labelled map of all536 patterns, including failed patterns, and compares all successful maps to the frozen complete certificate. Exactly17 contexts have changed marked images, with36 maps:

```
629,637,674,778,1163,1176,1559,1572,1641,1647,1667,1848,
1911,1934,2241,2304,2327.
```

Every affected full core is nonsingular Lorentzian and simply laced, with all saved roots of physical norm2. The existing complete corner contact-domain helper recomputes its inverse J, its actual preferred corner and the complete necessary contact domain. This domain splits into two ends, each with entrywise-positive inverse block. A root meeting both ends supplies a retained marked path of length at most8.

Twenty-eight maps fail a forced first contact

An actual target completion must supply every missing target node and its bond to each already present target node. Any root supplying such a first bond is genuinely outside the old core: omitted black roots are absent from the corner, and other old corner components are orthogonal to the selected component. The checker records the full list of forced first entries for every affected map.

For a norm2 new root with native contact magnitude q at old node i, projection on the Lorentzian old core has square at most2. Within one end positivity gives the necessary lower bound q^2*J_ii; additional contacts on that end cannot lower it. Cross-end contacts have already been paid by the short-path branch. Native labels3 and4 have q^2=1 and2 respectively. No other contacts are arbitrarily assigned zero.

In13 contexts, all28 maps have a forced first entry with bound greater than2:

Forced projection lower bound	Maps
7/2	8
6	4
7	8
8	8

These exclude the proposed actual whole affine factor, irrespective of how other target nodes or spectator components might be supplied.

Eight saturated maps cannot obtain the next terminal

Contexts1647,1667,1934,2327 have two affected maps each. Their selected component has13 nodes and the two missing target nodes are11 and12. The old marked root i maps to target10. Target11 requires a simple contact to i, and J_ii=2.

Let e be the required norm2 root at target11. Its unit i-contact already has projection square2. Same-end inverse positivity forces all its other core contacts to vanish, while any opposite-end contact would have given the retained short path. The perpendicular complement of the old Lorentzian span is positive definite. Therefore e has residual0: its entire vector lies in that span, and its exact old-core contact vector has only the unit contact at i.

For each literal map the source verifies the following full inverse-row data. At every allowed contact on e's end, J_ij is positive. The opposite end consists of a single node j, with J_ij=−2. Hence any other supporting white z falls into one of these exhaustive alternatives:

- If z is orthogonal to the old core, then e.z=0.
- If z meets only the same end, then e.z>0, contrary to acuteness.
- If z meets only the opposite singleton, then e.z=−2q<=−2, since a native nonzero contact has magnitude q>=1. Two distinct supporting norm2 whites have a strictly positive-definite principal Gram and therefore inner product strictly greater than−2. This is impossible.
- If z meets both ends, a retained short8 marked path already pays.

Thus in an otherwise unpaid face e is orthogonal to every other new supporting white. It also has no old-core neighbour except i. The required target12 cannot be an old corner root, since the specified old-core products of e vanish there; omitted black roots are absent from the corner. It must be another supporting white, yet its required label4 bond to e is now impossible.

This seals the entire first root, not only a bounded list of possible attachments. Arbitrary labels, contact multiplicity, orthogonal-white forests and indirect paths are covered. No finite corner replacement or new coupon is needed.

Complete nonselected and later interfaces

All basic and protected ideal payments of distance at most9 remain. The short8-first universal cycle/tree replacements therefore keep their direct and shared-neighbour alternatives and all of their finite outputs. The low-core actual closure has whole-component bound at most12, so it cannot produce an order15 factor. The high-disjoint branch remains excluded. The remaining selected interfaces are exactly the17 contexts above.

The universal D13/D14 immediate affine outputs have orders12 and13, not15. Later cycle12/13/14 and universal D11/D12 alternatives return their stated shorter ideal paths or finite components/coupons. The older partial-weight replacements retain halves on the specified affine D factors. None introduces a new long order15 B payment.

The earlier B15 finite-completion theorem and singleton exclusions keep their original finite or contradiction conclusions. The newer order12 B/C and order13 B proofs use finite B13-or-smaller components and whole A1/A2 versus D11 units; the order14 B proof is a projection exclusion. The large-D/D10 theorem uses retained short/basic and distance9 ideal alternatives. No finite rule changes here, so all those finite outputs and their budgets are untouched.

Consequently the complete rule still gives every triangle at least2 and every quadrilateral at least1. The only change is the affine B14 factor sum96 to94. The actual cusp-factor multiplicity remains the preceding theorem's V/15 copies of each marked pair.

Verification scope

Generation246cc4/ddc542 and separate fresh replay8c0085/e63e40 both terminated exit0/PASS. The checker reconstructs all536 eligible pattern searches, all36 affected native maps,17 necessary contact domains,28 direct projection obstructions and eight saturated first-root certificates, as well as the retained elementary/basic sets and symmetry orbit. The full all-label core census, no-extra-cross edge proof, actual edge positivity and arbitrary-support contact-domain theorems remain inherited frozen inputs.

Root completed a full source/report mathematical read. That review checked that both missing roots are outside the old corner, the physical norm2 and native contact magnitudes, saturated projection2 forcing the single contact and zero residual, and the complete0/positive/at-most−2 product exclusion using strict white positive definiteness. Root also checked the complete536-pattern and nonselected/later interfaces and the unchanged C95/D96 bottlenecks. Root did not repeat the36-map arithmetic. A second full source/report read passed the first-entry bound with complete positive end blocks, the eight saturated maps and all supporting-white product alternatives, the impossibility of using an old corner root as the next terminal, and the retained short/basic/triangle/later interfaces. That reviewer did not repeat the536 searches,36-map arithmetic or inherited all-label contact-domain census. All34 direct bindings, Python AST and two local report links pass. Source, certificate and report are final/frozen. C95 and D96 still obstruct the order15 maximum94 target. No new finite cost, ideal envelope improvement, ambient bound, chamber construction, external refereeing or formal verification is asserted.

```
python3 scripts/fresh_order15_affine_b_residual_terminal_transfer_20261017.py
python3 scripts/fresh_order15_affine_b_residual_terminal_transfer_20261017.py --verify
```

Source 225

Source path: `results/fresh_order15_affine_c_terminal_projection_transfer_20261017.md`

SHA256: e07de9c7f5ae7f63512e2bf83196442b405da32e85bc65513bfe87bd720b073f

Four affine C14 payments excluded by terminal projection

The whole affine C14 factor, of diagram order15, can use91 units instead of95. Remove the four pairs(1,11),(3,13),(1,12),(2,13) from the canonical path0–…–14, whose two terminal bonds have label4. The first two pairs have distance10 and the other two distance11. Each two-element set is a reflection orbit. No finite payment or coupon changes.

Together with the frozen B15 refinement, the order15 costs are now A75/B94/C91/D96. The maximum remains96: D still prevents the desired all-family maximum94. This local theorem makes no ambient dimension claim.

The source and certificate refine the complete distance8/ideal5 payment rule. The existing all-label geometric classification and supporting-root domain theorem are inherited.

Complete selected support

The frozen small-factor corpus contains536 eligible patterns whose selected component has at most15 nodes and whose whole-component bound is at least15. Every induced labelled injection of every one of these patterns into the 15-node C target is reconstructed, including the patterns with no map, and all successful maps are compared exactly with the saved complete corpus. The changed pairs occur in exactly ten contexts and24 maps:

`366,611,986,1031,1159,1503,1614,1619,1641,1642.`

All ten full old cores are simply laced, have physical root norms2 and are nonsingular Lorentzian. The necessary-contact helper reconstructs the preferred corner, its marked component and other old corner components, and the complete permissible old-core contacts of any new supporting white. It checks the inverse Gram J and partitions the contact domain into two ends. The inverse block on each end is entrywise positive. A root meeting both ends gives a retained marked path of length at most8.

The ninety basic ideal5/feature5 pairs remain units, as do all short8 pairs and all elementary-triangle requirements. Of the five previously paid nonbasic pairs, only(2,12) remains. Its proof obligations are unchanged.

A required missing terminal cannot exist

Suppose an affected selected component lies in the actual whole affine C14 factor. Fix its induced map. Every missing target node adjacent to an old target node requires a supporting root with the prescribed native bond. Such a root is genuinely outside the old core: omitted black roots are absent from this corner, and every old corner component outside the selected one is orthogonal to it. A new nonzero adjacency cannot be supplied by any such old corner root.

For a norm2 new root with contact vector of nonnegative magnitudes t, its projection on the old Lorentzian span has square t^T J t at most2. This holds because the orthogonal complement of that span is positive definite. If the root has contacts on both ends, the face already has a retained short8 payment. Otherwise all contacts lie on one end. Positivity of the entire inverse end block then gives

```
t^T J t >= q^2 J_ii
```

for each required contact of magnitude q at i, regardless of additional contacts on that end. No unspecified contact is set to zero.

Each of the24 affected maps requires a missing native label4 terminal entry with q^2=2 and projection lower bound strictly greater than2. The exact chosen obstruction bounds are:

Bound	Maps
5/2	4
8/3	2
4	10
7	2
8	2
16	4

This excludes every such actual factor completion in an otherwise unpaid face. The assertion concerns physical norm2 roots before any diagonal congruence. It allows arbitrary extra labels, multiple contacts, other supporting components and indirect tails: none can supply the missing first terminal entry while lowering its already excessive projection. The relevant-component cases use the same necessary-contact positivity; they do not require uniqueness of an attaching root or a completed Gram with artificial zeros.

Nonselected and later interfaces

Removing only distances10 and11 preserves all previously used basic and protected ideal payments of distance at most9. Thus the universal cycle and tree replacements keep their short and shared-neighbour alternatives. The complete low-core closure has whole-factor bound at most12. The high-disjoint branch remains excluded. The remaining selected branch is exactly the536-pattern domain just checked.

The D13/D14 universal transfers have immediate affine alternatives of orders12 and13, respectively; these do not produce an order15 C factor. Their other nontrivial outputs are finite components and coupons. Cycle transfers keep their stated shorter ideal alternatives and finite outputs. The D16/D17 partial-weight replacements use the original affine D factor and finite replacement corners. The later B terminal exclusions add no ideal payment obligation. The order12 B/C transfer and order13 B transfer use finite components and existing whole A1/A2 versus D11 coupons. The large-finite-D/D10 transfer uses retained short/basic or distance9 ideal alternatives. Consequently none of these interfaces introduces an additional long order15 C payment outside the selected domain.

Finite weights remain unchanged, including the later finite A/B/D17 reductions. Existing proofs of deleted pairs and fractional D payments keep their own finite and ideal contributions. The triangle minimum2 and quadrilateral minimum1 therefore persist. The cusp-factor multiplicity is still V/15 per marked pair; this changes the C factor sum only.

Verification scope

Generation179942 and separate fresh replay54045d/854249 both terminated exit0/PASS. The checker reconstructs536 eligible pattern searches, all24 affected maps, ten complete necessary-contact domains and their exact projection obstructions. It also checks target symmetry, all retained basic/short8/triangle sets and every direct input binding. Root read the complete source and report and reread the inherited necessary-contact helper. That mathematical review passed the complete contact domain, the genuinely new terminal-root argument, positive inverse end blocks, the retained both-end short8 branch and all24 terminal projection exclusions with arbitrary extra contacts. Root also checked the unchanged finite rule and the nonselected/later interfaces. Root did not independently repeat the536-pattern or24-map arithmetic. The inherited all-label core census and supporting-root domain proof are not independently re-enumerated here. No new finite budget, ideal envelope, ambient bound, construction, external refereeing or formal verification is asserted. Source, certificate and report are final/frozen after the completed replay, mathematical review and direct metadata checks.

```
python3 scripts/fresh_order15_affine_c_terminal_projection_transfer_20261017.py
python3 scripts/fresh_order15_affine_c_terminal_projection_transfer_20261017.py --verify
```

Source 226

Source path: `results/fresh_order15_affine_d_half_coupon_transfer_20261017.md`

SHA256: 9d4c47a1c89fc26f75891c499010b760cc5b708d1364e0d2b09a38ae30494443

Four order15 affine D payments become halves

The whole affine D14 factor, of diagram order15, can use total weight94 instead of96. In the canonical spine0–…–10, with leaves11,12 at0 and13,14 at10, change the four pairs(1,13),(1,14),(9,11),(9,12) from weight1 to1/2. They form one automorphism orbit of distance10. The older nine deleted pairs keep their existing proofs and weight0.

No finite weight or coupon changes. Every affected actual quadrilateral either already has a retained short payment, cannot have the proposed whole affine factor, or has a sealed finite corner carrying the existing half coupon between whole A4 or A5 and whole D11 marked at a short fork leaf. That half combines with the retained ideal half to give1.

With the frozen B and C improvements, the order15 costs are A75/B94/C91/D94, so the all-family envelope becomes94/15. The source and certificate prove this local refinement. The separate order17 and finite-budget obstructions remain; no ambient dimension bound follows here.

Complete map domain and unchanged interfaces

The inherited complete distance8 overlap classification has2614 contexts. Exactly536 selected patterns have at most15 displayed nodes and a proved whole-component bound at least15. The checker reconstructs every induced labelled injection of all536 patterns into the15-node affine D target, including every failed search, and compares all successful maps to the frozen complete mapping certificate.

The changed orbit occurs in67 contexts with408 maps. Fifty-five contexts have proper displayed subcomponents, producing312 maps. Twelve contexts already contain the entire affine factor, with eight maps each. The source checks the orbit symmetry, retained ninety-two basic payments, all short8 units and every elementary-triangle requirement. All four changed pairs have distance10, so no triangle payment changes.

The complete low-core closure has whole-factor bound at most12; the high-disjoint branch remains excluded. The protected/basic and universal short-path interfaces retain their weights. The D13/D14 universal immediate alternatives have orders12/13, and their remaining outputs are finite components/coupons or shorter ideal paths. Later cycle and D11/D12 transfers likewise retain their specified shorter ideal and finite alternatives. The D16/D17 partial-weight transfers consume halves on their original D factors and finite coupons, not a new order15 long ideal pair. B/C projection refinements add no ideal payment obligation. Thus the selected536-pattern domain covers the additional affected obligations; it is not being presented as the whole quadrilateral domain.

This theorem still uses the inherited short8 payment interface. A later finite7 theorem must preserve it or prove its replacement separately.

Every proper completion is excluded

All proper affected cores H are nonsingular Lorentzian with physical root norms2 and simple bonds. Write J for their inverse Gram. The necessary-contact helper reconstructs the actual preferred corner and the full permissible old-core contact set for a new supporting white. The contact set splits into two ends with entrywise-positive inverse blocks. A root meeting both ends already gives a retained short8 path.

Every missing target node adjacent to the displayed component must be a genuinely new root: omitted black roots are absent from the corner, and other old corner components are orthogonal to the displayed one. The native target bonds to old nodes have magnitude1. Extra contacts on the same end cannot lower projection square; the orthogonal complement of H is positive definite, so that projection square is at most2.

The exact outcomes are:

Affected proper maps	Reason
232	A forced unit contact has J_ii greater than2
64	Two distinct required entry roots would have positive inner product
16	The uniquely forced completion has two negative eigenvalues

For the second row the checker first verifies, for every two distinct possible contacts i,j on one end,

`J_ii + J_jj + 2J_ij > 2.`

Thus an unpaid new root has at most one old-core contact. All required entries in this row have J_ii>1, so their native simple target contacts are also their full old-core contact vectors. For two required roots at i,j, Cauchy bounds their residual product in magnitude by sqrt((2-J_ii)(2-J_jj)). Every chosen literal witness has positive J_ij strictly larger than this bound. The recorded witness values are2 with residual-product bound0, or4/3 with bound4/9. The resulting positive actual inner product contradicts acuteness. The helper also permits the inherited negative-product/short-path alternative; none of the64 chosen witnesses needs it.

The last row consists of four maps each in contexts1648,1668,1935,2328. Their displayed component has twelve nodes and one forced attaching root G of projection square5/4 and residual square3/4. The inherited complete residual proof allows only one possible exterior core-contact root C when G is present. Their residual Gram is

`[[3/4, −1/2], [−1/2, 1]].`

Its inverse GG entry is2. The inherited saturated-tail/no-bridge argument, with all numeric hypotheses rechecked here, gives at most one further root in the marked component if C exists. Consequently that whole component has order at most12+1+1=14, whereas the hypothesized factor has order15. C is impossible in this completion. This argument includes indirect supporting tails and bridges through other old corner blocks.

With C excluded, G is the sole core-contact root. Every other missing factor node is orthogonal to the old core. The actual target fixes all bonds among these nodes and to the displayed component. Only now does the checker form the completed Gram: its zero entries are forced by the contact and no-bridge conclusions, not chosen to simplify a test. Each of the16 maps gives signature(16,2,0), impossible in a Lorentzian ambient space. Arbitrary further roots cannot repair this principal submatrix. The required affine submatrix itself is separately checked to have signature(14,0,1).

These proper-map arguments exhaust arbitrary native labels and contact multiplicity through the contact-domain and projection inequalities. They do not assume a bounded spectator forest or a simple ambient cusp.

Twelve whole-affine contexts have twenty sealed variants

The entire affine factor is already displayed in contexts

`691,795,825,827,1874,1976,2267,2369,2481,2485,2506,2513.`

Its positive kernel identifies it as the whole actual varying factor. Eight of these cores have a one-dimensional radical and signature (16,1,1). The checker reconstructs the positive affine kernel, actual corner contact domain and radical vector. Every allowed contact has a nonzero radical coefficient. Since the radical represents the zero vector in the actual Lorentz span, a further root contacting H must contact both a positive-coefficient node and a negative-coefficient node. Every such combination gives a retained marked path of length at most3. In an otherwise unpaid face all further whites are therefore orthogonal to the entire old core, including its extra finite component. This seals all three finite corners.

The remaining four contexts1976,2369,2506,2513 have nonsingular17-root cores. Their only allowed contacts are the two omitted black roots. Both inverse diagonals are2. A further contacting white must have a single simple contact, projection square2 and residual0. The complete pair tests exclude two such whites by acuteness or a retained short path. All later whites

orthogonal to H are also orthogonal to the zero-residual root. Thus each context has exactly three exhaustive possibilities: no contacting root, one at the first omitted black, or one at the second. No longer indirect tail is possible.

These eight radical cases and twelve nonsingular variants give twenty literal configurations. Each source Gram, white positive-definite block, strict edge block, preferred affine corner and the three finite corners is checked exactly. Remaining supporting components are orthogonal to these configurations. Every typed component used below is therefore a whole actual finite component.

Only existing cross-component halves are consumed

In each of the twenty variants, exactly one of the other corners has its marks in separate whole components A4 or A5 and D11, with the D11 mark a short fork leaf. The current finite rule pays1/2 at this corner; the small-component mark can be any root. The source records every literal component, marked root and role. The ideal half and this finite half total1 in every variant.

The certificate also records any displayed same-component finite payments, with component type, literal marks and marked distance, but explicitly does not consume them. Some are whole D15 or D16 units. None is needed for this proof, and there is no dependency on a finite D16 or D17 same-component unit. Future reductions of those units will therefore not invalidate this transfer, provided the stated A4/A5-D11 half coupons and short8 interface remain available.

No new finite coupon is added. The actual cusp-factor multiplicity is unchanged at V/15 per marked pair. The factor cost is the sum of weights, so four unit-to-half changes subtract2 from96, giving94. This is a weighted rule; its four nonzero halves are not treated as deleted pairs.

Verification scope

Generation6cc56f/0d4275 and separate fresh replay48c379/dc14c7 both terminated exit0/PASS, reconstructing all536 eligible pattern searches,408 affected maps,312 proper exclusions and twenty whole-component finite variants. Direct bindings and all retained pair sets are checked. Root read the full source/report and relevant dependencies and passed the forced-entry/contact distinction, the inverse-GG=2 saturated no-bridge argument excluding the exterior root before constructing the index-two Gram, and the whole-component coupon composition. A second reviewer read the complete source/report and the contact-control and component helpers, passing the same proper-map logic, radical and zero-residual sealing, exact A4/A5-D11 roles and unchanged nonselected/later interfaces. Neither reviewer independently repeated the536-pattern,408-map or inertia/corner arithmetic. All34 direct bindings, two local report links and source syntax pass. The all-label core census, no-extra-cross-edge classification and full residual/no-bridge theorem remain inherited frozen inputs; the source rechecks their local hypotheses rather than claiming an independent repetition of the entire geometric reduction.

No finite-budget improvement, new ambient theorem, chamber construction, external refereeing or formal verification is asserted. Source, certificate and report are final/frozen after the completed generation, fresh replay and two mathematical/source reviews.

```
python3 scripts/fresh_order15_affine_d_half_coupon_transfer_20261017.py
python3 scripts/fresh_order15_affine_d_half_coupon_transfer_20261017.py --verify
```

Source 227

Source path: `results/fresh_order15_bd_basic_fractional_transfer_20261018.md`

SHA256: a8e10b9a35bd5485eb26593f8c1fedbb3c6375809246067cc59b35a157dda4ed

An economical order15 B/D transfer

The current ideal order15 family totals A75/B92/C91/D92 may be changed simultaneously to **A75/B91/C91/D367/4**. Thus the all-family envelope falls from92/15 to367/60, a total saving1/4. The B change is a unit deletion. The D change reduces four units to15/16 and adds exactly one finite coupon of weight1/16. This is a local payment theorem; it does not assert a new finite maximum or dimension58 bound.

The new coupon joins a whole A1 to a separate whole finite D14, marked at its unique long-arm vertex two edges from the fork center. In the canonical finite D14 with spine0–…–11 and short leaves12,13, the large mark is9. There is one eligible marked pair per component pair. Its actual component-pair cost therefore rises from2 to33/16. No other finite weight changes. The whole-component condition is essential.

The producer and certificate bind the current order14/contextual rule, the order15 B92/D92 predecessor, and the universal D14 near-fork theorem. The D argument here uses a different, strict D4 end and energy1/2; the old A3-end/energy3/4 argument is not assumed to apply unchanged.

Canonical changes and the complete B domain

For affine B14 of order15, use spine0–1–…–12, doubled bond11–12, and simple leaves13,14 at0. Delete the retained singleton(3,12). It has distance9 and is invariant under the fork-leaf interchange. All earlier zeros remain zero, and the resulting total is91.

For affine D14 of order15, use spine0–1–…–10 with leaves11,12 at0 and13,14 at10. Change precisely

```
(2,13), (2,14), (8,11), (8,12)
```

from1 to15/16. They form one native automorphism orbit of distance9. Its total debit is1/4, giving367/4. The earlier distance10 deletions, including(1,13),(1,14), remain zero. The source reconstructs the complete weighted tables, checks all native automorphisms, and retains every recorded short8 and elementary-triangle unit.

The B singleton is basic, so a selected-map absence alone is insufficient. The source reconstructs all536 eligible native embedding searches from the complete high-overlap domain, including failures. There are exactly 215 successful patterns and548 maps, identical to the frozen predecessor; none sends the marked pair to(3,12). It also freshly embeds each of the 74 complete protected native bases into the whole B15 target, not merely their previously selected maps. None uses the changed pair. This covers arbitrary supporting extensions of a protected base because its displayed native diagram remains an induced subdiagram of any actual whole factor.

The inherited2614-context partition is1691 whole-order-at-most14 cases, 83 protected/adaptive cases,304 displayed-too-large cases and536 eligible patterns. The83 consist of the74 protected bases and nine independently paid adaptive finite cases. Low-core closures have order at most12; the high-disjoint obstruction, complete low/protected routing and the universal short-path routes are inherited inputs. The nine adaptive outputs are finite and unchanged. Consequently neither a low closure, an eligible selected payment nor a protected ideal payment can use this new B singleton; the too-large branch cannot be a whole order15 factor. The nonselected/current interfaces are detailed below. No new finite coupon is needed for B.

The actual D15 star and its complete profiles

Let an actual quadrilateral star be W+a+b+c+d, with opposite pairs {a,c},{b,d}. Actual edge diagrams W plus one black root are strictly positive definite; actual corner diagrams are positive semidefinite. Use physical roots of square2. The inherited factor-fiber argument at W+a+b supplies the whole varying affine factor F and a positive-definite spectator diagram Z orthogonal to it, without a simple-cusp hypothesis. It suffices to take a=2 and b=13 in F. Its positive kernel has coefficient1 on the spine and1/2 on each leaf.

If b.c, a.d or c.d is negative, a distinct actual corner has a retained distance1 unit. If c,d share a nonzero spectator contact, a distance2 unit pays. Hence assume those products vanish and the nonnegative spectator contact vectors x,y have disjoint supports. The opposite products a.c and b.d remain unrestricted. Every nonzero native magnitude is at least1; one greater than1 is at least sqrt2. This includes parallel and dotted opposite products. Kernel heights are positive.

Deleting a leaves finite D4 on[1,0,11,12] and finite D10 on [3,4,…,10,13,14]. Deleting b leaves finite D14, with long endpoint14. In both large finite blocks, the inverse diagonals at the first two long-arm vertices are1,2, and every other diagonal exceeds2. Thus a positive-semidefinite one-root extension permits only no contact, simple or doubled contact at the first vertex, or simple contact at the second. Two contacts create a cycle in a connected diagram still containing the fork, excluded by finite/affine classification; the exceptional types are too small to contain D10. This is an all-native classification, not a finite label cutoff.

A doubled-first or simple-second extension is affine with positive kernel. In the actual PSD corner that kernel makes the entire component orthogonal to every other corner root. Thus the corresponding near profiles have no additional D4 or spectator contact. Their product with the omitted opposite black a is still unrestricted.

The D4 inverse, in the stated order, is

```
K = [[1,1,1/2,1/2], [1,2,1,1],
     [1/2,1,1,1/2], [1/2,1,1/2,1]].
```

Put t=-a.c, u=-1.c, v=-0.c, w=-11.c, z=-12.c and

```
e=(u,v,w,z) K (u,v,w,z)^T,
h0=t+u+v+(w+z)/2.
```

The strict actual edge contains D4+c, so e<2. A center contact already has inverse diagonal2 and is impossible. Any two nonzero D4 contacts have projection square at least3, also impossible. Each outer inverse diagonal is1; therefore the only possible D4 contact is one simple outer contact. In particular e is0 or1. This strict argument covers arbitrary additional old and spectator contacts, whose principal restrictions cannot weaken it.

Eliminating F minus a,b gives resistance R=10. The four near profiles have height h, residual constant c0 and moment l as follows:

Near profile	h	c0	l
Only D4/opposite contacts	h0	2-e	0
Simple contact at3	h0+1	1-e	1
Doubled contact at3	t+sqrt2	0	sqrt2
Simple contact at4	t+1	0	2

For the far root put t=-b.d. Its complete profiles are:

Far profile	k	d0	m
No contact to F minus b	t/2	2	0
Simple contact at14	(t+1)/2	1	1
Doubled contact at14	(t+sqrt2)/2	0	sqrt2
Simple contact at10	t/2+1	0	2

Before eliminating Z, the new-root residual squares are c0+l²/10 and d0+m²/10, and their cross projection is lm/10. The producer checks the two complete inverse lists, all eight end identities, all16 cross identities and the formal four-by-four determinant directly in the physical norm-two Gram.

Independent affine alternatives and native cost bounds

The last two near profiles have immediate distinct actual alternatives. A doubled contact at3 gives whole affine B10 of order11, with marked doubled-terminal/fork-leaf pair. That pair currently has weight zero; apply the independently proved BC11-zero theorem at this actual corner first. It supplies a full retained payment. A simple contact at4 gives whole affine D10 of order11 with opposite fork-leaf marks, of current weight1/2. That distinct half plus the selected15/16 is23/16. The source checks both kernels, canonical native embeddings and current weights. An old zero is never spent as a unit.

For each end set g=2l/h+c0/h². After those immediate near alternatives, the complete near bound is g<=4. The unique extreme type is a simple contact to old leaf11 or12, no opposite contact, h=1/2 and g=4. Every other near profile has g<=3. Indeed, in the outer-only profile:

- no D4 contact forces t>=1 and gives g<=2;
- a simple contact at1 has h=t+1 and g<=1;
- a simple contact at11 or12 has h=t+1/2 and g=4 at t=0, whereas t>=1 gives g<=4/9.

For the simple-at3 profile, e is0 or1, hence0<=c0<=1 and h>=1, so g<=3. These cover every allowed native opposite magnitude and every strict D4 contact, not only endpoint substitutions in a polynomial.

At the far end, the isolated profile with t=1 and the simple-at14 profile with t=0 both have k=1/2 and g=8. Every other far profile has g<=4: the isolated nonunit case has t>=sqrt2; the simple-at14 case with t>=1 has g<=3; and the last two profiles have g<=4 for every t>=0.

Let J=G_Z^(-1). Elimination and the exact determinant identity give

```
0 = 1−(g_c+g_d)/10 + E/10,
E=(x/h−y/k)^T J (x/h−y/k) >= 0.
```

The spectator cross sign is part of the checked formal identity. If the far end is nonextreme, g_c+g_d<=4+4=8, so the right side is at least1/5. Thus the far end must be one of its two extreme types.

Entire-support Cauchy and the energy1/2 boundary

Suppose first the near end is not extreme. Then k=1/2 and

```
E=g_c+8−10=g_c−2<=1.
```

If y_i is nonzero, disjoint supports give x_i=0. Gram-metric coordinate Cauchy and G_ii=2 imply

```
E >= (x_i/h−y_i/k)²/G_ii = 2∗y_i² >= 2,
```

a contradiction. Thus the **entire** far spectator vector is zero. There is no assumption on the near vector or on the rank and shape of Z. In the paying corner W+a+d the opposite root c is absent.

For the isolated far profile, d contacts only omitted b. It is a whole A1 in this paying corner; F minus b is whole finite D14, with a at canonical node9. The new coupon supplies1/16, in addition to the selected15/16. For the simple-at14 profile, d extends the old D14 to whole finite D15. Its marked pair is canonically(0,10), of retained weight1. The current finite D15 reductions change other long pairs, including(0,11),(0,12),(1,12), and do not remove this unit. The producer checks the two literal whole-component roles and that exact current unit.

If both ends are extreme, h=k=1/2 and the determinant instead gives

```
(x−y)^T J (x−y) = (4+8−10)/4 = 1/2.
```

Both separate spectator extensions are strictly positive definite, being principal subdiagrams of actual edges. The complete arbitrary-rank below-one spectrum theorem therefore applies. At energy1/2 its rational rank is r=1: there is exactly one simple contact on one whole A1 component, and the other support is empty. The two-adjacent-contact alternative cannot occur in A1, and the irrational I2 alternatives have different energies. Every other spectator component has zero contacts.

If the far support is empty, use the preceding far-corner payment. If the near support is empty, c extends the old D4 by a simple contact to leaf11 or12, giving whole D5 with c its unique long endpoint. The other whole factor in W+b+c is D10, with b a short fork leaf. The existing small-component/D10 coupon pays1/2 for any mark of a whole component of order at most5 against a whole D10 short leaf. This gives at least23/16 together with the selected corner. The other root d is absent here, so its arbitrary contacts cannot enlarge either paying factor. The exact D5-long/D10-shortleaf roles are reconstructed; no new D5 coupon is being assumed.

More generally this proof allows every0<delta<=1/2: use selected weight 1-delta and the one new A1/D14 weight delta. The immediate D11 and old D5/D10 halves still give at least1, the isolated far branch gives (1-delta)+delta=1, and the remaining branches give independent full units. The chosen delta=1/16 is the claimed rule. Other parameter values do not inherit any finite-budget assertion from this packet.

Simultaneous current payments and finite accounting

Apply the independent BC11-zero theorem first whenever such an actual corner occurs. Its short1/2, small finite and retained D11-half outputs avoid the new order15 roles and the new A1/D14 coupon. Next apply the universal D15 proof whenever an actual newly reduced D15 corner occurs. All of its payments above are independent of the B15 deletion. No new ideal B15 pair is used in a D branch.

Otherwise the D ideal weights agree pointwise with the preceding rule. Use the independently established current short8 interface, followed by the complete inherited B selected/protected/low/disjoint routing. The new B singleton cannot occur in any of its complete eligible or protected maps. The universal cycle/radius-five tree replacements use their protected short/basic paths, small whole factors and fixed-leaf or fork outputs, not this degree2/doubled-terminal distance9 singleton. The earlier D13/D14 immediate affine outputs have orders at most13; the new D14 theorem uses order11, whole finite D14 or A4/D10 payments. The current C/B projection patches have no new long order15 output. The independent actual-corner small-ideal priorities and the later sealed-factor transfers use their explicit retained finite roles. Thus no earlier preferred witness is silently assumed to survive.

If a short8 finite alternative is changed by depth8 zero, its independent short7 route applies; a changed finite7 alternative is covered by the depth6 theorem's independent short6 or whole-A8/protected-leaf alternative. These routes keep the new ideal distances9 unchanged as unused outputs. The short6 cycle11 exception uses its independently proved second half or finite short alternative. No new D15 proof branch needs a reduced finite7/8 payment. The contextual A2/A13-middle, A2/D13-center and conditioned A2/A2 changes are disjoint from every new output. The new D5/D10 role is an existing broad small-component coupon, distinct from all D5/D12 changes. Triangles retain their elementary short/protected targets, checked in the complete weighted tables. Quadrilateral total1 and triangle total2 therefore persist for the simultaneous local rule.

The producer also reconstructs all216 current literal configurations starting from the frozen contextual rank13 rule, whose D14 successor has no literal change. It tests complete whole-factor native embeddings and exact new coupon roles, and finds zero new ideal debits and zero new coupon credits. All recorded corner lower bounds are nonnegative and each face total remains at least1. This is supplementary interface validation; the arbitrary-support D proof and full B routing give the universal coverage.

The old actual A1/D14 cost is2, and the new unique node9 mark adds1/16. Its total is33/16, explicitly including the other pre-existing roles. No A2–A5/D16 coupon is restored, no same-component finite weight is increased, and no finite budget is reused without this extra charge. The ideal factor-fiber count is V/15 per pair, giving envelope367/60. The completed unrestricted bound remains59. A complete finite count and a new terminal-ideal comparison belong to a separate composition.

Verification and scope

Generation session41241/chunk0c3a92 and separate fresh saved replay session29179/chunke9816b both terminated exit0/PASS. The source checks complete B native-map and protected-base domains, exact D4/large inverse lists, eight end and 16 cross identities, the formal determinant, native-bound endpoint values, two immediate affine and three whole finite corners, current weight/orbit/triangle interfaces and all216 literals. The complete all-native extension classification, factor-fiber/actual-star framework and arbitrary-rank spectator theorem are bound mathematical inputs; they are not inferred solely from these finite checks.

Parent completed a full producer/report mathematical read. The strict D4 contact reduction, complete near/far cost bounds, physical resistance and spectator identity, entire-vector Cauchy argument, energy1/2 use, whole finite roles, BC11 priority, complete B selected/protected domains, current literal/short6 interfaces and exact1/16 cost pass. That review did not execute calculations or repeat the historical census or the inherited spectrum proof.

A second reviewer completed a full producer/report mathematical read and reread the complete below-one spectrum source/report. The strict D4 contact reduction, complete large-block/native-height bounds, entire-vector Cauchy argument, energy1/2 whole-A1 classification, distinct finite payments, BC11 priority, complete536+74 B support and current short6/216 interfaces pass. The valid delta range and chosen1/16 cost are correct, and the D5/D12 alternate outputs are disjoint. This review did not execute arithmetic, maps, the spectrum checks or the census.

Source, certificate and report are final/frozen after both completed mathematical reads, terminal generation/fresh replay and direct metadata validation of120 immutable bindings, six local links and Python syntax. No historical geometric census, global finite optimization, external refereeing or proof-assistant formalization is asserted.

```
python3 scripts/fresh_order15_bd_basic_fractional_transfer_20261018.py --verify
```

Source 228

Source path: `results/fresh_order15_bd_terminal_coupon_transfer_20261018.md`

SHA256: 8a8b2667fd05e6d318192918655e06c7b4663c605f956862a111ddfd428a16a3

Order15 B/D costs92 with two typed finite coupon upgrades

The simultaneous order15 affine factor costs can be changed from A75/B94/C91/D94 to **A75/B92/C91/D92**, giving the normalized envelope 92/15. This is a local weighted-payment theorem compatible with the completed dimension60 rule. It does not improve the ambient dimension bound by itself.

The B change is coupon-free. The D change upgrades exactly two finite roles, each from weight1/2 to1:

- whole A4, marked at either endpoint, versus whole D11 marked at either short fork leaf;
- whole A5, marked at either second-from-end node, versus whole D11 marked at either short fork leaf.

Each component pair has2×2 affected marked pairs, so each upgrade adds exactly2 to its finite cost. Other marks in these A4/A5 components keep their old half weights. These are distinct from the D11 fork-center coupons. No finite same-component weight changes, and no finite forest upper bound is asserted here. In particular the previous finite420 or ambient60 finite413 bound cannot simply be reused without counting these increments.

The source and certificate bind the completed dimension60 proof, the B94 predecessor, the D94 predecessor, and the C91 theorem. The C family is already91; it needs no change for this envelope.

Changed invariant pairs and complete domain

In canonical affine B14 of order15 the spine is0–1–…–12, with doubled bond11–12 and the two simple leaves13,14 at0. Delete the two units (9,13),(9,14). They form an invariant distance10 orbit. All earlier B deletions stay zero. The resulting B sum is92.

In canonical affine D14 of order15 the chain is0–1–…–10, with leaves11,12 at0 and13,14 at10. Delete the four retained halves (1,13),(1,14),(9,11),(9,12). Their sum was2, and their deletion changes D94 to92. They too form an invariant distance10 orbit. The checker enumerates automorphisms, verifies all current weights, and retains every basic, short8 and elementary-triangle unit for both families.

The inherited complete no-short8 high-overlap domain has2614 contexts. Its exact partition is1691 with whole selected order at most14, 83 protected/adaptive contexts,304 with more than15 displayed selected nodes, and536 eligible selected patterns. The protected partition is 74 finite-leaf9/ideal-feature5 contexts and nine adaptive finite contexts. The source asserts this partition. The536-pattern domain is the selected part of the proof, not a claim that all quadrilaterals have this form.

For B the source reconstructs every induced native-label map of all536 patterns, including failed searches, and compares every successful map and marked image to the complete frozen map certificate. The new orbit occurs in53 contexts, with108 maps. Exactly68 maps fail a first-entry projection bound; the other40 fail a forced completed Gram test. There is no surviving B completion requiring a finite coupon.

For D the frozen complete predecessor supplies408 affected maps in67 contexts:312 proper completions are excluded or paid by an independent short-path alternative, and96 maps in12 contexts display the whole affine factor. This packet rechecks all20 sealed whole-factor variants from those12 contexts. It does not rerun the312 unchanged proper-map exclusions or pretend to repeat the full all-label geometric census.

B: physical contacts before the terminal congruence

Every affected old core H is nonsingular Lorentzian, with physical root norms2 and simple bonds. Let J be its inverse Gram. The complete necessary-contact helper reconstructs the actual preferred corner, possible contacts and two ends. Each inverse end block is entrywise positive. A new supporting white meeting both ends gives a retained short8 path; otherwise all of its contacts lie on one end.

A missing target root adjacent to the displayed component must be outside the entire old core. The omitted black roots do not belong to that corner, and all other old corner components are orthogonal to the displayed one. For a native contact of magnitude q at old node i, the projection square is at least q²J_ii. Additional same-end contacts cannot reduce it. The complement of H is positive definite, so this square is at most the physical norm2. Native labels3 and4 have q²=1 and2. These exact bounds exclude68 maps without assuming any other contact vanishes.

For each remaining map the source reruns the complete inherited residual contact control, including same-end multiple-contact exclusions, the native-label restriction and pairwise residual Cauchy tests. There is one required simple first entry G, followed by missing target nodes [11,12] or[10,11,12]. In the ordinary residual cases G is the only possible root contacting H. In the relevant-component cases only one exterior core-contact root C could coexist with G.

That exterior possibility is excluded before constructing any zero Gram entries. The inherited residual/no-bridge theorem, with every numeric hypothesis rechecked, gives actual marked-component order at most14 when C occurs. The hypothesized whole affine factor has order15. This bound controls arbitrary indirect supporting tails and possible bridges to other displayed components. The proof uses the positive residual relation or saturated inverse-GG=2 alternative, rather than merely counting first neighbors. The complete argument is a bound input in the residual/no-bridge report.

Thus G exhausts all core contacts in every remaining completion. Every subsequent required target node is also genuinely outside the old core. A first such node inside another old corner block would give its predecessor an additional old-core contact, contrary to the established contact control. The same reasoning applies successively along the forced target tail. Other missing roots are orthogonal to H; the target diagram fixes all mutual products. Arbitrary spectators cannot change these necessary principal-submatrix entries.

Only at this point is the completed Gram built. All physical norms remain2 in the preceding argument. The missing terminal12 has native label4 to11 and is not an old-core root. Scaling just that new terminal by$1/\sqrt{2}$ changes its diagonal to1 and its doubled bond to-1. Every other diagonal is2 and every other target bond is-1. This is a positive diagonal congruence preserving inertia, not a change to the physical projection bound. The source separately checks that the15-node target submatrix has inertia(14,0,1) and a strictly positive kernel.

The full completed Gram has inertia(16,2,0) in24 maps and(17,2,0) in16 maps. Two negative directions are impossible in the actual Lorentz space. Adding further supporting roots cannot repair this principal obstruction. All108 affected B maps are therefore covered by a retained short-path payment or an impossible actual completion.

D: a distinct whole finite unit replaces the original half

The predecessor's whole-affine contexts are

`691,795,825,827,1874,1976,2267,2369,2481,2485,2506,2513.`

Eight cores have a one-dimensional radical. Every permissible contact has a nonzero radical coefficient, and every mixed-sign contact pair gives a marked path of length at most3. The actual radical represents zero in the Lorentz span. Hence any extra supporting white either gives that independent short payment or is orthogonal to the whole core. The three finite corners are sealed, including their whole components.

In the other four cores the only allowable contacts are the omitted black roots, each with inverse diagonal2. A contacting white has one simple contact and zero residual. The complete pair tests permit at most one such white in an otherwise unpaid face. Later core-orthogonal whites are also orthogonal to this zero-residual root. There are exactly three exhaustive variants per such core: no added root, or one at either allowed contact. No indirect tail remains.

The source recomputes this full contact control and compares every literal Gram, corner and role to the frozen predecessor. In all20 variants, the original ideal half is at one corner and the existing finite half at a distinct corner. The finite role is always one of:

Whole small component and mark	Variants	Distances to its two endpoints
A4 endpoint	12	0,3
A5 second-from-end node	8	1,3

The other whole component is D11, marked at one of its two short fork leaves. The new checker derives these intrinsic roles from the literal path/fork graphs, verifies that the two components are disjoint and orthogonal, and records both distinct corner marks. Upgrading precisely these halves to full units supplies total1 without using any weight from the original affine corner. The old ideal half can therefore be zero. No same-component finite payment, including any D15/D16/D17 long-pair unit, is consumed.

Simultaneous payment and later interfaces

Use the independently established current short8 interface first. The new order15 changes have distance10, so every direct short8 ideal unit in these families is retained. The current small exceptional ideal routes have their own earlier actual-corner priority: C11/B11 zeros, cycle11 and coupled D11 halves, B12 and D13 changes, and the newer D105 and D17 rules retain their stated short/coupon alternatives. Their finite alternatives are unchanged or increased by the two new coupons. They do not require a deleted order15 distance10 unit.

In particular a short8 finite alternative affected by depth8 zero is replaced by its independent short7 route. If a finite distance7 pair is affected by the newer depth10 zero theorem, that theorem supplies the independent short6 route. The retained small ideal units and cycle11 alternative, and the universal short6 cycle/tree outputs, do not use these new order15 long pairs. This packet is therefore also compatible with the depth10 finite7 theorem; it does not reopen the recursive use of a removed generic payment.

On the remaining no-short8 branch, apply the inherited complete low/disjoint/protected/selected routing. Low-core closures have whole order at most12; high-disjoint cases are excluded. Protected/basic payments keep their weights. The nine adaptive finite cases keep their leaf/distance9 or bounded-component outputs, as certified in the completed dimension60 composition; no finite weight decreases here. The displayed-too-large branch cannot be a whole order15 factor. If the selected old payment uses one of the changed B/D pairs, the complete map proofs above replace or exclude it. Otherwise its payment remains available under the inherited rule.

The later partial affine transfers consume their specified original halves or distinct whole finite coupons. The current B/C projection patches have no long order15 output. Universal cycle and radius-five tree replacements use their protected short/basic paths, small whole components or fixed-leaf/fork outputs. The old D13/D14 immediate affine alternatives have orders12/13. Thus none supplies a new unanalysed order15 distance10 obligation. This is complete inherited interface coverage, not an inference from the absence of a saved map.

Equivalently, actual changed-corner branches are settled with these same independent short-path and selected-completion arguments before pointwise inheritance on a face with no changed ideal corner. A chosen old witness is never assumed to remain paid merely because another corner was preferred. All new non-short replacements are distinct full finite units. The simultaneous rule gives every triangle at least2 and every quadrilateral at least1. Triangle protection is immediate also from the explicit retained elementary pair sets and distance10 of all changes.

The actual cusp factor-fiber multiplicity remains V/15 per marked pair. Reducing each B/D sum by2 gives the new maximum92 and envelope 92/15. The finite additions are exactly2 for each of the two stated typed component pairs. No claim is made that the previous finite bound already includes them.

Verification and scope

Final generation27732/2941ed and the separate fresh saved replay 99652/9431c5 both terminated exit0/PASS after adding the CLI guard that refuses to overwrite an existing certificate. The checker reconstructs all536 B pattern searches and108 affected maps, recomputes the forced completed Grams and their exact inertias, and rechecks all20 D whole-component variants with their typed marked roles. It checks the complete2614 routing partition, retained payment sets, automorphism invariance and direct immutable input bindings. The full all-label geometric census, inherited arbitrary-support/no-bridge theorems and the unchanged312 D proper exclusions remain explicitly bound inputs.

Root completed a full source/report mathematical review, including X.constraints, D.core_constraints, the complete residual/no-bridge report, the B94 predecessor and the D full_case implementation. That review passed arbitrary contacts before completed zeros, genuine new tail roots and the first-old-root bridge exclusion, exterior order14 versus actual15, physical norm2 bounds before the terminal congruence, the principal index-two obstruction, exact20 sealed D role refinements, full short8/depth10 and selected/nonselected priorities, and explicit +2/+2 coupon costs. Root did not repeat map, inertia, census or scalar arithmetic.

A second reviewer completed a full producer/report mathematical read, plus the inherited H.full_case, X.constraints, D.core_constraints and residual/no-bridge report. That review passed the physical first-entry bounds and arbitrary same-end contacts before zeros; exterior order14 exclusion and absence of a first-old-root tail bridge; terminal-only positive congruence and principal index-two exclusion; all20 distinct whole A4-end/A5-second–D11-shortleaf units and exact added costs; and complete536/2614 routing, short8/depth10, nonselected/later and triangle interfaces. It found no issue and did not repeat maps, inertia, the census or arithmetic.

The direct metadata audit ed3520 passes55 immutable source/data/report bindings, eight local links and Python syntax. Source, certificate and report are final/frozen after both completed mathematical/source reads and the final generation/fresh replay. These review scopes do not claim an independent repetition of the historical geometric census.

No new chamber, ambient dimension improvement, complete finite budget, external review or formal proof-assistant verification is asserted.

```
python3 scripts/fresh_order15_bd_terminal_coupon_transfer_20261018.py
python3 scripts/fresh_order15_bd_terminal_coupon_transfer_20261018.py --verify
```

Source 229

Source path: `results/fresh_order15_d16_d17_ambient59_frontier_20261018.md`

SHA256: d68b82f3a99d7eece1d9504be3eaac02a59cccc372c8c094ea6c6e98c78e6c13

Simultaneous order15 and finite16–17 payments at ambient59

The simultaneous proved payment rule has exact formal finite maximum 793/2=396.5 at ambient59, attained by2A16+D27. Nine fixed-multiplicity representative alternatives remain above the finite curvature783/2. The complete1,890 terminal ideal profiles have six negative margins, down from seven. The completed unrestricted dimension bound remains59; this is not a dimension58 proof.

The source combines the frozen order15 fractional ideal theorem with the D5/D12 alternate-corner rule and the new finite D16/A17/D17 local transfers. Every local report is frozen and bound by its final hash. All additional actual coupon costs are included simultaneously, even when an existing representative allowance absorbs them.

The local changes and their simultaneous scope

The order15 theorem gives ideal totals A75, B91, C91, D367/4. Its unique new finite role is a whole A1 against canonical node9 of a whole D14, of weight1/16. The actual A1/D14 total rises2 to33/16. The B singleton deletion is covered by the complete536 selected and74 protected native domains; the D change is universal, with strict D4 contact classification, entire-support Cauchy and the complete energy1/2 spectator alternative.

The D5/D12 rebalance is the finite baseline. Its D5-long/D12-shortleaf weights are1/2, with distinct D5-long/A12-endpoint and D5-long/A1 alternatives of3/8. It keeps the ideal order17 weights unchanged. Its actual D5/D12 total is21/2 and its actual D5/A12 total is11/4 before the A17 additions below. All contextual rank13 payments remain in force.

The D17 node2 transfer changes(2,12) from1 to5/8 and(2,14) from1/2 to0, saving7/8. Its four whole-core seals use the existing D5-long/A1 weight3/8, or the two distinct A1/D14-center and A2/A14-third-node halves. No coupon is added.

The D17 leaf transfer deletes the halves at(3,15),(3,16), saving1. Its four whole-core seals use either the retained whole A17 units(0,13)/(3,16), or an existing A4/D13-shortleaf half plus a new whole A1/A16 fourth-node half. The latter has canonical A16 marks3,12 and adds1 per component pair. Its actual total1 fits the unchanged A1/D16 representative allowance3/2. The two D17 orbit changes are disjoint and give actual D17 cost919/8.

The D16 transfer deletes the two halves(2,14),(2,15). Its eight whole-core seals retain the A1/A4–D13-shortleaf half and use either whole A16 halves(0,13)/(2,15) or new whole A1/A15 third-node halves at2,12. The latter add1, raising actual A1/A15 from1/2 to3/2, exactly its old representative allowance. Actual D16 falls443/4 to439/4. Actual A16 stays441/4, so the rank16 representative decreases only to441/4. It is not identified with the smaller actual D16 value.

The A17 transfer deletes(2,12),(4,14),(2,14) and halves(2,13),(3,14), saving4. Its five whole-core seals use whole D5-shortleaf/A12-third-node units, or whole A1/A14 third/fourth-node halves at2,3,10,11. The precise actual increments are4 for D5/A12 and2 for A1/A14, giving totals27/4 and2. Both fit their existing representative allowances. Actual and fixed representative A17 fall467/4 to451/4.

The outputs are mutually protected. A17's changed pairs avoid the whole A17 units required by the D17 leaf transfer and the older restored or rebalanced A17 pairs. D16's changed pairs avoid the D16 halves (1,12),(1,13),(1,14),(1,15),(2,13) used by the finite14/15 theorems. The D17 node2 and leaf changes avoid the D17 node1 orbit used by the fractional fork exchange. All new small-component roles are distinct: D13 short leaves are not its contextually reduced center, D5 short leaves are not its long endpoint, and the new A1/D14 node9 role is not the D14 center11 used by the D17 node2 proof.

Here is the noncircular actual-face composition. Preserve the established independent BC11 and other small-ideal priorities. A changed ideal D15 corner has the universal order15 replacement, whose whole finite D15 (0,10) unit, D5/D10 half and A1/D14-node9 coupon are unchanged here. Its immediate D11 half and BC11-zero outputs also remain available. The changed B15 singleton is excluded in the complete selected/protected domains after the independent current short8 route. All subsequent finite replacements have the whole-corner alternatives just listed, so none requires a newly removed ideal order15 pair.

For changed finite D16/A17/D17 occurrences, use their exhaustive inherited selected/nonselected routing and reconstructed whole-core seals. Their nonzero-contact escapes invoke the independently paid short8 route, which proceeds through short7 to short6 or a whole-A8/protected-leaf alternative when necessary. Those outputs avoid the present long finite pairs. The whole-factor alternatives are distinct actual corners, not an assumed survival of a formerly preferred witness. Faces with no changed proof output inherit the prior whole-face payment. The elementary triangle/protected outputs remain unchanged. Thus the simultaneous triangle and quadrilateral minima are2 and1.

The producer reconstructs every current literal corner from its Gram, starting from the D5 baseline, and applies all finite and ideal debits and every new typed coupon together. Each local change is also matched to its bound local certificate. This216-row check supplements the universal and complete routing proofs; it is not a census of all actual polytopes.

Complete finite counting

All costs use integer sixteenths. Compared with the D5 baseline, the actual base deltas are D16:−16, D17:−30 and A17:−64. The representative deltas are D16:−8, D17:−30 and A17:−64. Actual A16 remains1764 in these units and dominates actual D16=1756. A17 is separately fixed; D17 is a residual type. The distinction is maintained in every item and forest.

The additional actual component costs are

Types	Increment	New actual total
A1/D14	1/16	33/16
A1/A15	1	3/2
A1/A16	1	1
A1/A14	2	2
D5/A12	4	27/4

Only A1/D14 increases a representative cross cost, also by1/16. Every ordered actual-type pair is checked after all additions using one simultaneous representative map. Thus separate compatible-looking allowances are never added without checking their combined cost.

The complete inherited fixed domain contains961 multiplicity vectors at ranks9 through17, with fixed rank17 meaning A17. Every possible rank5 multiplicity is separately fixed, giving2574 expanded vectors. The residual ranks are1–4,6–8 and17–59, with D17 included. The rank5 separation is needed because the D5/A1 coupon creates a rank1/rank5 interaction. All remaining residual–residual cross costs are zero.

For each interaction key and fixed rank5 count, the changed residual item is exactly

```
new_item(r,key,k5) = old_D5_item(r,key,k5)
                     −30·[r=17] + key[4]·[r=1].
```

Here key[4] is the fixed rank14 count. The latter term is the new A1/D14 charge; the former is the simultaneous D17 saving. The source checks each item directly against the actual representative base plus all fixed-component interactions. D16 and A17 base changes enter the fixed cost, not this residual formula.

The contextual A2/A2 payment is counted once per marked pair when a whole unmarked A10 and at least two unmarked A1 components are present. For an upper bound any fixed rank10 type can witness that condition. The count enumerates the A2 multiplicity, reserves two A1 components, and uses a no-rank2 residual table for all remaining components. This covers every triggered actual forest without multiplying the cost by the number of witnesses. The ordinary branch covers nontriggered forests. All ordinary/no-rank2 tables are regenerated for the changed items, then checked by every last-item inequality and an equality predecessor for each rank. The fresh verifier checks these certificates without invoking the generation recurrence.

The complete certificate has2,572 expanded interaction keys and5,144 ordinary/no-rank2 tables. It checks32,400 ordered actual-type pairs, 128,700 residual-item identities,6,718,064 last-item inequalities and 303,496 attaining predecessors, with6,455 contextual alternatives. Its maximum is793/2, five above783/2. The actual formal witness is

```
2A16+D27: 2·(441/4)+176 = 793/2.
```

The stored maximizing representative row is2D16+D27. Its actual cost791/2 is smaller, because actual D16 costs439/4 while its allowance is441/4. Substituting two actual A16 components realizes the bound; this is not an identification of actual D16 with its representative.

The nine fixed-multiplicity alternatives above783/2 are as follows. These are representative upper costs, not a list of all actual typed forests or a geometric realization claim.

Representative forest	Upper cost
2D17+D25	1571/4
D16+D17+D26	3161/8
A17+D16+D26	393
2D16+D27	793/2
D15+D17+D27	3135/8
D15+D16+D28	1573/4
3D15+2A5+A4	392
D14+3D15	394
2D14+D15+D16	1573/4

A formal weighted forest need not be geometrically realizable. The representative bound nevertheless covers every actual finite Coxeter forest under the stated rank and whole-component rule.

Complete ideal comparison and remaining scope

The prior order14/contextual packet supplies the complete1,890 terminal rank58 low9 profiles, their positive symmetric h-polynomials, exact curvature and current all-family envelopes. The source reconstructs the complete domain and rechecks every saved polynomial/moment/curvature identity before updating the costs. No fresh convolution is required.

The only new ideal envelope change is at rank14/order15:

```
92/15 − (367/4)/15 = 1/60.
```

Each normalized margin therefore increases by count(rank14)/60. The previous negative profile(8,9,9,9,9,9,14) becomes8/1125; six other negative profiles remain, unchanged. No zero profile appears. The finite changes do not alter any ideal envelope or the low-rank concentration ingredient. Negative terminal absolute margins are not extended by the positive- margin/product monotonicity argument.

Terminal factor ranks	Normalized margin
9,9,12,12,16	−5241/57460
9,9,9,9,10,12	−5227/71500
9,9,9,15,16	−389/5440
1,9,9,9,9,9,12	−6893/130000
8,9,9,16,16	−1087/52020
6,9,9,9,9,16	−19/59500

As conditional terminal arithmetic, further all-family total savings of1 at order13,1/2 at order16 and3/4 at order17 make every saved margin positive, with minimum21/5500. Holding the other two savings fixed, the respective strict saving thresholds are5227/5500,149/340 and219/320. Another sufficient terminal scenario saves1 at order13 and5/4 at order17, with minimum11/5440. These are additional savings from the present rule. Their local geometry and any extra finite

compensation are unproved; the source marks both limitations explicitly. They do not change the low-rank concentration ingredient.

The ambient59 finite curvature is783/2 and the strict global extremum constant is4350, counted once. The certificate retains the conditional next-saving targets as terminal arithmetic only, with geometry and any new finite compensation explicitly unproved. None is promoted to an ambient bound. The internally checked completed dimension59 proof stays the unrestricted result; no dimension58 proof, new finite-volume example, external refereeing or formal proof-assistant claim is made.

Verification

Generation session98230/chunk63ceeb and separate fresh replay session84439/chunkb967dd both exited0/PASS. The literal reconstruction checks all216 configurations and4,160 local-certificate corner deltas; it finds16 D17 debits and8 A1/A16 credits, with every total at least1. All other new local deltas are zero on this literal domain.

Root completed a full source/report mathematical composition read, including the inherited finite item/table and ideal/conditional helpers: PASS. The review checked the simultaneous actual/upper costs and coupon absorption, fixed/residual17 distinction, complete rank5/contextual enumeration, Bellman logic, five local literal matches and the saved ideal-polynomial scope. It did not execute DP, arithmetic, geometry, convolution or historical censuses. A second independent full review by exact_hub_counts also passed, including rereads of the inherited item/table and ideal/conditional/threshold helpers. It checked all five local deltas and retained outputs, the literal matches and full coupon costs, the actual/representative distinctions, exhaustive rank5 and contextual A2 separation, residual zero-cross and last-item certificate logic, and the1/60 ideal update with conditional-only future targets. That reviewer did not execute DP, maps, literals, scalar arithmetic or convolutions. Root's final completed-results/report read also passed; its saved headline inspection was not a DP replay. Final metadata validation checks338 direct source bindings, seven links and the producer AST; that check is integrity-only. The packet is final/frozen.

Source 230

Source path: `results/fresh_order16_affine_b_singleton_projection_transfer_20261018.md`

SHA256: 9a3295f5c5394bc76a2d885f65e7d3f7a8f540d782cfb36a379e365c5ad3c29f

Two order16 affine B singleton units are unnecessary

The whole affine B15 factor of order16 can have cost105 instead of107, with no finite weight or coupon change. Delete the two singleton pairs (1,11),(2,12) in the canonical spine0–…–13, doubled bond12–13, and short fork leaves14,15 at0. Both pairs have distance10 and are fixed by every labelled diagram automorphism.

The current four order16 costs80/107/103/106 become80/105/103/106. Thus the all-family rank15 envelope decreases from107/16 to106/16=53/8; the remaining maximum is the D family. No new ambient dimension claim is made. The source and certificate reconstruct the complete eligible map domain and check all forced physical norm-two projection obstructions.

Complete map support and invariant retained payments

The starting107-pair B16 set is the completed finite8/ideal5 rule, with its complete distance8 small-factor maps. It is still the B16 rule in the completed dimension62 proof. The checker compares this exact pair set to the current factor table.

All100 basic ideal5/feature-radius5 pairs, every distance-at-most8 pair, and every elementary triangle requirement remain paid. The two deletions are outside that basic set. To check every selected occurrence, the source re-enumerates all induced labelled injections into the16-node target for all394 patterns having displayed order at most16 and proved whole order bound at least16. Failed patterns are included. The125 successful patterns and296 maps agree exactly with the complete saved certificate.

The changed marked support is:

Deleted singleton	Contexts	Affected maps
(1,11)	1031,1614	4
(2,12)	504,1031,1537,1614	6

Thus four distinct contexts and six target/context occurrences require proof. Maps to other pairs in these same contexts remain unchanged.

Actual first-entry obstruction with arbitrary supporting roots

Each affected full core H is nonsingular Lorentzian, with all roots of physical square2 and all nonzero off-diagonal products−1. The inherited complete contact helper reconstructs its inverse J, actual preferred corner, complete possible contact positions and two end sets. Every inverse block on a single end is entrywise strictly positive. A new supporting white contacting both ends gives a marked path of length at most8, handled by the retained short-path branch discussed below.

Suppose the selected connected marked component K lies in an actual whole affine B15 factor. Its induced labelled embedding is among the complete maps just reconstructed. For each map the target requires a missing node e adjacent to an old node i of K. That e is genuinely outside the old core: the two omitted black roots are absent from the actual corner, and all

other old corner components are orthogonal to K. They cannot supply a node with the required nonzero i-contact. Possible indirect bridges or old roots occurring farther along the completed factor do not change this first-entry necessity.

Let q be the forced native contact magnitude and a the full nonnegative old-core contact vector of e. The Lorentzian H has positive definite orthogonal complement in the ambient Lorentz space, so

a^T J a <= e²=2.

In the otherwise unpaid branch all contacts lie in one end. Entrywise positivity there gives a^T J a>=q² J_ii, even with arbitrary extra contacts and native labels. No unspecified product is set to zero. A label3 has q²=1; a label4 has q²=2. These are physical norm-two quantities, not a changed diagonal gauge.

Every changed embedding forces a strict violation:

Deleted pair	Forced new target node	Forced old target node	Native label	Projection lower bound	Maps
(1,11)	one short fork leaf14 or15	0	3	10/3	4
(2,12)	doubled terminal13	12	4	16	4
(2,12)	doubled terminal13	12	4	4	2

All exceed2. Thus no such actual affine completion exists unless a retained short-path branch already pays the face. The first row is a forced **simple fork leaf**, not a doubled terminal; its obstruction uses the full inverse diagonal10/3. The source records every forced entry, then the precise witness used in each map. It does not assume that every missing target node has a single old contact.

Complete nonselected coverage

This ten-map exclusion is not presented as a census of quadrilaterals. It patches precisely the selected-affine part of the complete finite8/ideal5 reduction and its current successors. Before the selected domain, the short8-first branch uses retained elementary/basic ideal pairs or the established universal tree/cycle alternatives. All their ideal B16 pairs have distance at most9; the changed pairs have distance10. Their finite outputs and coupons are unchanged by this B-only theorem.

In the no-short8 branch, the complete low-core theorem bounds the actual marked component by12 and cannot yield an order16 factor. The all-label high-disjoint theorem excludes its domain. The full high-overlap classification has2,614 contexts with supporting-root certificates. Its finite whole-component, protected-leaf and adaptive outcomes keep their current finite proofs; protected/basic ideal outcomes retain their distance-at-most9 payments. Every other possible order16 affine selected image lies in the394-pattern domain checked here. An unaffected image retains its unit; an affected image has the impossible first entry just proved. These cases exhaust the original geometric reduction.

The old universal D13/D14 immediate affine alternatives have orders12/13, not16. Later cycle12/13/14 and D11/D12 transfers use shorter ideal paths, finite components or their typed coupons. The order15, order16 and order17 D partial-weight proofs retain their original ideal D half or other fixed weight, plus sealed finite payments, or invoke short paths; none introduces a long B16 payment. B/C projection refinements are exclusions or keep their explicit smaller finite alternatives. Thus no additional nonselected long B16 role is silently omitted.

Compatibility with the full dimension62 rule

Current short8 payment is a whole-face argument, not the assertion that every ideal short8 pair is a literal unit. Preserve the independent current ideal priority proofs: C11 zero, B12 zero, D13 half, cycle11 half, coupled B/C/D11 halves and order17 D half. Their changed-corner branches use their specified smaller affine roles, finite/coupon outputs or shorter paths; none requires either deleted B16 distance10 pair. In particular a short8 route meeting a D13 center half or an order11 cycle/tree half is resolved by its independent actual-corner theorem, rather than asking the half to pay a unit alone.

The current sparse finite theorem uses its complete selected/proper, radical, literal and universal interfaces. Its proper obstructions and radical branches may invoke short8, whose current independent payment just described survives. Its actual finite outputs and every coupon keep their values; its six distance-at-least10 finite deletions are not altered. If a short8 finite path has zero weight because it is deep, the independent depth11 corridor supplies the constructive short7 alternative. That current route uses basic small-tree, cycle and protected-feature payments, not a long B16 pair. Hence none of the present selected exclusions relies recursively on this new rule's unaffected-face case.

Equivalently, repeat the completed62 proof with the two affected selected-affine entries replaced by the necessary Gram obstructions above. All independent changed-corner branches and all other selected, low/disjoint, protected, literal and universal clauses remain valid. The required short8 entry alternative is already established before the no-short8 selected domain. The full rule still pays every quadrilateral at least1.

The preceding D13 center-zero coupon upgrade, if later composed, is also independent of these B16 pairs: its only ideal outputs are order11 B/D center units or paths of length at most4, and its finite outputs are whole D10 coupons. Its enlarged finite budget remains a separate obligation and is not assumed here.

Triangles retain at least2: all their finite7/ideal5, protected ideal6/7 and whole-ideal-order-at-most9 outputs are unchanged. All weights remain automorphism invariant. The factor-fiber multiplicity gives new B16 charge105*V/16, while D16 still contributes at most106*V/16. There is no new finite charge, but an ambient dimension comparison must use its own complete ideal profiles; this packet supplies none.

Verification scope

Generation428bb4 and separate fresh replay809c4f terminate exit0/PASS on all394 induced searches, 125 successful patterns and296 saved maps, all10 affected projections, the four complete physical contact domains, retained basic/short8/triangle sets, automorphisms and exact family totals. The all-label core census, no-extra-cross-edge theorem, supporting-root contact completeness and published classifications remain inherited inputs. Direct auditc942cd passes47 bindings,6 local links and Python syntax.

Root and a separate reviewer completed full source/report mathematical reads, including the inherited necessary-contact helper. Both checked the complete map domain, genuinely new first-entry witness, physical norm-two bounds with arbitrary same-end contacts and the cross-end short8 alternative, the distinct simple-fork10/3 and doubled-terminal 4/16 witnesses, retained basic/triangle sets, and the full nonselected, current62 priority, sparse and depth11 interfaces. Neither reran maps, inverses or the historical census. Source, data and report are final/frozen. No global bound, new finite budget, external refereeing or formal verification is claimed.

```
python3 scripts/fresh_order16_affine_b_singleton_projection_transfer_20261018.py
python3 scripts/fresh_order16_affine_b_singleton_projection_transfer_20261018.py --verify
```

Source 231

Source path: `results/fresh_order16_bd_sparse_unit_transfer_20261018.md`

SHA256: c8fb3e418599cc00f6724163522d612f1efe01b5b602a5caca3fa507187aae63

Order16 costs B102 and D103 with a sparse D5/D11 unit

The whole affine order16 family costs become **A80/B102/C103/D103**. The maximum falls from105 to103, giving normalized rank15 charge103/16. Three B units are deleted, and four D halves are deleted. The D deletion requires exactly **+1 per whole D5/D11 component pair**, on two specified marked roles. This increase fits the already counted representative allowance19/2; it does not require a new residual item or forest optimization. Every triangle still receives at least2 and every quadrilateral at least1. This is a local payment theorem, not a dimension59 theorem.

The producer and certificate start with the B105 rule, D105 rule, and the balanced finite rule. The already completed B/C13 sparse transfer and D13 center/leaf transfer are compatible inputs. Their separate A4/B9 cost is not silently included in this packet's finite comparison.

Exact changes and complete selected domains

For B15/order16 use spine0–…–13, double bond12–13, and fork leaves14,15 at0. Delete the unit at(0,10) and the two units(10,14),(10,15). Their marked distances are10,11,11. They form invariant orbits of the labelled diagram. The old100 basic pairs, every short8 pair and every elementary triangle requirement remain unchanged. The resulting B total is102.

For D15/order16 use spine0–…–11, leaves12,13 at0 and14,15 at11. Delete the four current halves(2,14),(2,15),(9,12),(9,13), all at distance10. All other D weights remain exactly as in the D105 theorem, including its two distance9 halves and all prior zeros. The basic pairs, short8 units and triangle payments are unchanged. The resulting D total is103.

For each family the producer reconstructs every induced native-labelled map for all394 eligible patterns, including failures, and compares the successful maps with the complete saved small-factor domain. B has125 successful patterns and296 maps; D has151 and928. The changed B set occurs in14 contexts/26 maps. The changed D set occurs in36 contexts/200 maps. This is a complete reconstruction of these selected target domains, not a fresh enumeration of arbitrary Coxeter diagrams.

The full2614-context overlap routing is also checked:2045 whole-component bounds below16,92 displayed components too large,74 protected cases, 9 adaptive finite cases and394 eligible patterns. The nonselected interfaces are explained below; the394 maps alone do not constitute all quadrilaterals.

B: first entry or a saturated root with no possible successor

The necessary contact helper reconstructs each old Gram H in physical norm-two gauge, its nonsingular Lorentz signature, inverse J, the actual corner, all possible extra-white contacts and its two end sets. Each same-end inverse block is strictly positive. A supporting white meeting both ends already creates an actual opposite-black path of length at most8. The independent current short8 payment is applied first.

Let K be the old connected marked component in a proposed whole B15 factor. A missing target node e adjacent to K is outside the entire old core: other old components of that corner are orthogonal to K, and the two omitted black roots are absent from this corner. Thus a first-entry projection test applies even if later target nodes might otherwise have been old spectator roots.

For any physical norm-two new root with nonnegative old contact magnitudes x, positivity of the orthogonal complement of H gives x^T J x≤2. Within a single end all additional contacts only increase this expression. The forced label4 contact

has square2. Fourteen changed maps force projection4 or8/3, both greater than2, and are excluded.

The remaining twelve maps occur in contexts1285,1624,1647,1667,1934,2327. The first missing node is target11, adjacent simply to target10. Its old contact coordinate i has J_ii=2. Saturation forces the exact single unit contact at i and zero orthogonal residual; arbitrary extra contacts on its end or a larger native magnitude would increase the projection. The other end is a singleton j with J_ij=−2. Every J_ik on its own end is positive.

Target12 must be a further supporting white f simply adjacent to e. It cannot be an old core root: e's sole old contact is i, already target10, and no other old root has the required e-contact. The omitted blacks remain absent. This is the second genuine-newness step; it is not inferred merely from a missing target label.

Because e has zero residual, its product with f is fixed by f's old-core contact vector. In the no-short8 branch f has no contacts, only same-end contacts, or only contacts to j. These give respectively0, a positive number, or a number at most−2. None equals the required simple product−1. The last alternative also violates strict positive definiteness of the supporting-white principal. This covers arbitrary native labels, arbitrary extra contacts and old spectator components, without setting unspecified products to zero. Thus every changed B map is impossible unless the independent short8 branch already pays.

This is the same saturation mechanism as the earlier B15 residual-terminal theorem, with the present order16 target and its required next simple bond checked explicitly.

D: complete proper exclusions and four whole-factor units

The producer reruns the complete necessary-contact and completion helpers of the D16 partial-weight theorem on all200 changed maps. It compares each result with the predecessor's saved certificate; only target orderings merged by its identical-Gram deduplication are ignored in that comparison.

The exact outcomes are64 first-entry projections above norm2,56 cases requiring multiple core-contact roots,48 completed principal Grams with two negative directions, and32 whole-factor maps. Arbitrary-support control precedes every completed zero. In ordinary cases positive same-end blocks force a single simple old-core contact, and the exact residual inequalities permit at most one core-contact root. In relevant cases a possible exterior entry would bound the entire marked component by14, including indirect tails and bridges. It is incompatible with an actual order16 factor. The complete no-bridge argument is an inherited proof input, rechecked numerically by the contact helper. Only after these statements are established can the missing affine bonds and forced zeros specify a principal completion. Its index-two signature then excludes the proper completion.

The32 surviving maps are the whole-factor contexts1979,2372,2536,2540. Each old core has18 roots, signature(17,1,0), and no feasible additional core-contact node in the no-short8 branch. Hence every further supporting white is orthogonal to the entire core. All displayed finite components at the other corners are whole actual components, despite arbitrary additional spectators.

A distinct finite corner has a whole D5 marked at its unique long-arm endpoint and a whole D11 marked at a short fork leaf. The producer reconstructs all finite components, recognizes the unique D5 endpoint intrinsically and checks the two-component separation. Upgrade this marked half to a unit. The selected ideal weight may then be zero and the distinct finite corner alone contributes1. No same-component finite unit is consumed, so none of the rank15–17 deletions or restorations is needed to support this replacement.

In canonical finite D5 the long endpoint is0; in finite D11 the short leaves are9 and10. Exactly1×2 marked pairs rise from1/2 to1, for actual component-pair increment1. The full D5/D11 charge becomes17/2 from15/2. All other marks retain their current half weights.

Finite allowance and simultaneous literal interfaces

The balanced finite rule represents rank5 by A5 and rank11 by D11. Its existing A5/D11 allowance is19/2, whereas the new actual D5/D11 charge is17/2. The source checks all33,489 ordered finite-type pairs through rank60, adding this exact increment only to D5/D11. Simultaneous representative domination continues to hold. In particular the separate whole D5/A11 cost2 remains a distinct actual type pair and is retained. All same-component costs, representative cross costs and residual items are unchanged. This pointwise comparison transports the existing upper certificate for this increment; no dynamic program is rerun here.

The separate recently introduced A4/B9 cost1 belongs to the order13 transfer and must be counted by its own composition. It changes a different representative pair, so it does not consume the D5/D11 allowance established here. This packet does not claim a newly combined finite maximum or geometric attainment of a formal maximizing forest.

All216 current literal configurations are reconstructed starting from the balanced rule's exact cornerwise lower payments. Four rows change, namely the four whole D contexts above. Each loses the selected ideal half and gains the distinct finite coupon half, leaving total1. The other212 rows are unchanged. There is no B debit in this literal domain. The bound order13 packet has zero literal changes, so it composes with these four cancellations. The earlier D16 eighth credits remain in the baseline lower payments.

Whole-face coverage and current priorities

The independent current short-path branch is preserved before any of the new no-short8 exclusions. The B/C11 zero theorem, cycle11 and coupled order11 transfers, B12 center-zero rule and both D13 transfers supply their own whole-face payments when their exceptional actual corners occur. Their retained smaller affine roles, short paths and typed finite coupons do not use the present B/D16 distance10/11 pairs. The new B/C13 rule has the same independent priority, with its distinct whole A4/B9 half. The C14 projection, order15 B/D and order17 zero or fractional transfers use their explicit short-path exclusions and whole finite coupons, not a removed B/D16 unit or half.

If an actual finite short8 pair is zero at the current depth8 threshold, its independent corridor theorem supplies short7; if that finite short7 pair is zero at depth10, the independent finite7 theorem supplies short6. The resulting smaller ideal paths and protected finite leaf/branch outputs remain unchanged. This establishes the current short8 escape without recursion through the new selected-affine theorem.

In the no-short8 branch the low-core theorem bounds whole marked factors by12, and the complete high-disjoint theorem excludes its domain. Protected/basic ideal outputs have marked distance at most9; all those weights are unchanged here. The74 protected finite outcomes and9 adaptive outcomes retain their current finite proofs. All remaining possible order16 selected factors lie in the complete394-pattern reconstruction. Unchanged images keep their prior proof; changed images are excluded or receive the distinct whole D5/D11 unit above. Later literal and universal interfaces are unchanged for these same reasons. Equivalently, patch those specific selected entries in the complete current reduction, after establishing the independent short-path and earlier actual-corner priorities. This is an exhaustive whole-face argument, not an inference that a partial map census covers every face.

Triangles retain their finite7/ideal5, protected ideal6/7 and bounded small-factor outputs; the new changed pairs have distance at least10. The added finite coupon is nonnegative. Thus triangle payment2 and quadrilateral payment1 remain valid. Factor-fiber multiplicities give order16 maximum103 and normalized contribution103V/16; a further ambient comparison must be supplied separately.

Verification scope

Generation53253/chunk9d89be and separate fresh replay22252/chunk46931a terminate exit0/PASS. Direct audit00ccfa passes49 bindings,10 local links and Python syntax. The producer reconstructs394 native searches per family, all26 B projection/saturation cases,200 D completions,32 whole-factor role maps, 216 literal configurations and33,489 finite-type comparisons. The historical all-label core classification and universal no-bridge, short-path, triangle and concentration proofs remain explicit inputs. No historical census, global optimization or ambient proof is rerun.

Root completed a full producer/report mathematical composition read, checking the genuinely new B successor, saturation signs, complete native domains including failures, comparison with the inherited D completions, intrinsic whole-component roles, literal cancellations and simultaneous finite comparisons. A separate reviewer completed the full producer/report read and reread the contact/completion helpers, D partial-weight proof and residual/no-bridge proof. That review checked the B contact products, arbitrary-support exclusions before D zeros, whole D5/D11 units, complete routing and current priorities, and the exact finite allowance. Neither reviewer replayed maps, inertia, literals, type comparisons or the historical census. Source, data and report are final/frozen after the direct metadata audit. No external or formal verification is claimed.

```
python3 scripts/fresh_order16_bd_sparse_unit_transfer_20261018.py --verify
```

Source 232

Source path: `results/fresh_order16_cd_sparse_sixteenth_transfer_20261018.md`

SHA256: 2b690d6a7a4393af171276168f469d39771d5f13b6b4a8fa152869e331fc6675

Order16 costs C101 and D411/4 with one sixteenth coupon

For finite-sided, finite-volume real hyperbolic Coxeter polytopes, the whole-factor payment rule below preserves triangle payment at least2 and quadrilateral payment at least1. Its order16 family costs are **A80/B102/C101/D411/4**, with maximum411/4 and normalized rank15 envelope411/64. The only new finite charge is **1/16 per whole A1/D15 component pair**, at one intrinsic D15 role. This is a local theorem and an exact ideal-frontier update, not a new unrestricted dimension bound.

The producer and certificate start from the current order16 rule and the frozen contextual order17 rule, which already binds the new B/C13 theorem. The separate completed finite6 theorem and its394 forest count are not assumed in this packet's finite accounting. Its new coupon must be included in any subsequent combined forest certificate.

Exact intrinsic changes

The canonical affine C15 diagram has path0–1–…–15, with double bonds0–1 and14–15. Change the two units(2,12),(3,13) to zero. They have distance10 and form a reflection-invariant orbit. The C total falls103→101.

The canonical affine D15 diagram has spine0–…–11, leaves12,13 at0 and leaves14,15 at11. Change the four current halves (1,14),(1,15),(10,12),(10,13) to7/16. Their distance is11, and their automorphism orbit is complete. The D total falls103→411/4. Every other D weight is unchanged, including the previously halved basic pairs and all existing zeros. The source checks the complete old/new weighted tables, automorphism invariance, all basic pairs, every short8 unit and every elementary triangle pair. There are99 C basic pairs and101 D basic pairs; none is changed.

Add a weight1/16 at a finite corner whose two marked roots belong to different **whole** components A1 and D15, where the D15 root is the unique long-arm neighbor of the fork center. In canonical finite D15, spine0–…–12 with short leaves13,14 at12, this is node11. Its prior weight is zero. This ordinary coupon has no additional witness condition and is counted once per actual marked pair, not once per realization of a diagram map. The previous fork-center/short-leaf halves are unchanged.

Complete selected support and physical contact control

The source reconstructs all394 eligible native-labelled patterns for each target, including failed embeddings, and compares every successful map with the complete saved small-factor domain. C has15 successful patterns and46 maps; D has151 and928. Exactly four C maps and80 D maps meet the new changed orbits. This is a complete selected target-domain check, not a new census of all possible supporting diagrams.

For each affected old core H, the necessary-contact helper checks its physical norm-two Gram, nonsingular Lorentz signature, inverse J, actual selected corner and complete allowed contact domain. The producer also checks the positive-definite white principal and each of the four white-plus-one-black principals explicitly. The two permitted end sets have entrywise positive inverse blocks. A supporting white meeting both ends gives a path of length at most8 between adjacent black sides. The independent current short-path payment is applied before proceeding in the no-short8 branch.

For a further norm-two white root with nonnegative old contact magnitudes x, the orthogonal complement of the timelike old-core span is positive definite, so x^T J x≤2. Additional contacts within one end can only increase the projection. Thus a single forced contact gives a lower bound without setting any unspecified products to zero.

C: the doubled terminal cannot enter

The four affected maps occur in contexts1031 and1614, two maps each. Every map requires a missing terminal joined by a double bond to an old node i, with J_ii=4/3. The physical contact has square2, so its projection is at least8/3>2. Each certificate records the exact target terminal, old target neighbor, old core coordinate and this bound.

The missing terminal is genuinely outside the entire old core. It is a white of the actual selected corner, so it is not either omitted black root. An old white outside the selected connected component is orthogonal to that component and cannot have the required terminal bond. This argument uses actual corner membership, not merely a missing target label. Consequently the projection test excludes every changed map even with arbitrary further same-end contacts or additional spectator components. The both-end case was already paid independently.

D: forced zeros precede the completion test

The twelve affected contexts are1036,1050,1647,1667,1906,1934,1975,2299, 2327,2368,2505,2512. All use the ordinary Lorentz-residual branch. The source reruns the full contact and completion helpers of the D16 partial-weight theorem, and compares every result with its saved completion. A predecessor may group maps having the same completed Gram but different target fork-leaf orderings; the comparison ignores only this target ordering and compares any affine kernel as an actual node-to-coefficient mapping.

Here the complete necessary-contact domain and positive same-end blocks allow at most one simple old-core contact per new root. Every candidate inverse diagonal is greater than1, so a nonsimple native magnitude already has projection greater than2. For two possible core-contact roots, the checked inequality

```
J_ij^2 > (2-J_ii)(2-J_jj)
```

either forces a positive actual product, impossible for distinct Coxeter walls, or forces the independent short8 escape. Thus at most one further white can contact H in the unpaid branch. The inherited residual/no-bridge proof controls indirect tails and prohibits an old component from being reached through a chain of core-orthogonal whites. All missing target nodes are therefore genuinely new. Only after these facts are established may the one forced entry and the other forced zeros define a principal completion.

The80 changed maps split into32 forced first-entry projections greater than2,16 completed principal Grams with two negative directions, and32 whole-factor maps. The first two classes cannot occur in the ambient Lorentz space. Their exact matrices, inequalities and signatures are saved; the arbitrary-support argument is not replaced by an inertia test on an arbitrarily zero-filled drawing.

The32 surviving maps are eight each in contexts1975,2368,2505,2512. Their displayed affine factor is already whole. Each full old core has18 roots and signature(17,1,0), and its complete contact test has no feasible additional core-contact coordinate. Every further supporting white is therefore orthogonal to the entire core. The displayed finite components at the other corners are whole actual components, even with arbitrary additional spectators. No simple-cusp or spectator-rank restriction is used.

Distinct finite payments in the four whole cases

The source reconstructs every finite corner and its full components, then recognizes the marked roles by all canonical native-labelled maps.

Contexts	Distinct retained payment	Additional new payment	Selected weight	Certified total
1975,2505	whole A4 endpoint/A12 endpoint unit, and the old A4 endpoint/D12 short-leaf half	none	7/16	31/16
2368,2512	whole A4 endpoint/D12 short-leaf half	whole A1/D15 node11 sixteenth	7/16	1

The whole A4/A12 endpoint unit is the already proved sparse endpoint coupon, not a newly inferred unit at another A12 role. The old whole A4/D12 short-leaf half is retained. In the second row all three payments occur at different actual corners:7/16+1/2+1/16=1. No finite same-component D16 unit is spent. The current D16, D17, A16 and A17 long-pair reductions and restorations therefore do not support or obstruct this replacement.

There is only one D15 role in the new coupon. Its actual A1/D15 component-pair charge increases3/2→25/16. The representative A1/D15 allowance must increase by the same1/16: this increment is **not** claimed to fit the old3/2 allowance. The source verifies all32,400 ordered actual type pairs through rank59 against the new representative allowance, including the already proved A1/B12 half from the B/C13 rule. At scale16, the rank1 residual item increases by the fixed rank15 multiplicity. All contextual order17 costs remain unchanged. A complete changed forest budget is deliberately left separate.

Complete current local composition

The selected-domain check is used only after the inherited full reduction. Its2614-context routing consists of2045 whole-component bounds below16, 92 displayed components too large,74 protected cases,9 adaptive finite cases and394 eligible patterns. The low-core and high-disjoint domains retain their proved complete coverage. The protected/basic ideal outputs have marked distance at most9, so their existing weights are unchanged; the adaptive and protected finite outputs retain their current payments. There is no need to reinterpret absence from the394 selected domain as absence from all actual faces.

The independent actual-corner priorities for BC11, cycle11, the current B12/C12 and B/C/D13 rules, B/C/D14, order15 and the contextual order17 rule apply before the new no-short8 exclusions. Their outputs are their explicit smaller affine pairs, protected short paths or whole typed finite coupons, not these C16 distance10 or D16 distance11 pairs. The order17 contextual A2/D5 halves and their whole A1/A9 witnesses are unchanged. The finite rank14–17 exchanges retain all of their explicitly required same-component alternatives and typed roles.

When a finite short8 pair is already zero, the independent corridor theorem gives short7; a zero finite short7 pair gives short6 or the protected whole A8 alternative. The later finite6 theorem likewise has independent short5 or protected whole finite outputs. These escapes do not use a new C/D16 long pair. Thus there is no circular invocation of a short8 theorem whose only payment was just reduced. In the remaining branch, unchanged selected images keep their previous proofs, changed C images are impossible, and changed D images are impossible or have the distinct whole payments above. This patches the complete whole-face reduction. It is not merely a literal census argument.

Triangles retain their short finite and ideal outputs, the protected leaf/branch outputs and their bounded exceptional whole-factor outputs. The changed ideal distances are10 and11; every elementary required pair is explicitly unchanged. The new finite coupon is nonnegative.

Separately, all216 current literal configurations are reconstructed from the frozen contextual17 cornerwise lower payments. Exactly four selected D corners lose1/16, and two distinct finite corners gain1/16. The four totals are31/16,1,31/16,1; every other row is unchanged. This is an exact simultaneous literal check, supplementary to the universal support proof.

Ideal frontier and scope of the improvement

The frozen contextual17 packet supplies the complete1890 terminal low9 rank58 profiles. This packet preserves its domain and saved polynomial data; it changes each normalized margin by count(rank15)/64, exactly the factor-fiber saving from103→411/4 at order16. All1890 margins are now strictly positive. The minimum normalized margin is1/340 at (9,9,9,15,16), and the minimum terminal absolute surplus is91/2 at(58). This is exact saved-row arithmetic, not a new convolution or global solve. All lower-rank concentration ingredients are unchanged.

The new representative A1/D15 cost is not included in the frozen394 forest certificate. This report neither asserts its new combined finite maximum nor promotes the positive ideal terminal rows to an unrestricted dimension58 result. The completed unrestricted ceiling remains59.

Verification

Generation23602/chunkadec0b and separate fresh saved replay11344/chunke9d3ab both terminate exit0/PASS. Final direct validation checks353 source/data/report bindings, eight local links and Python syntax.

The producer checks both complete394-pattern target domains, four C first-entry exclusions,80 D completions, all32 whole-factor role maps, the affected white/four-edge strict principals, complete weighted-pair invariance and protected sets,216 current literal rows,32,400 simultaneous typed comparisons and1890 exact terminal margin updates. The historical all-label core classification, universal residual/no-bridge, short-path, triangle and concentration proofs remain explicit inherited inputs. No old census, global optimization or complete forest DP is replayed.

Root completed a full source/report mathematical read and reread the inherited necessary-contact and completion helpers. That review checked the doubled-contact8/3 bound, complete twelve-context/80-map partition, arbitrary-support no-bridge and forced-zero scope, all32 whole maps and their distinct retained/new marked payments, the increased25/16 allowance, and the complete ideal-margin update. A separate reviewer completed the full producer/report mathematical read, reread the same helpers and the predecessor routing, and checked current actual-corner/nonselected/triangle interfaces and the216 literal changes. Both reviews passed with no mathematical correction. Neither reviewer executed the producer or replayed maps, inertia, historical classification, literal arithmetic, forest optimization or ideal scalar calculations.

Source, data and report are final/frozen after direct validation. No external review or proof-assistant verification is claimed. Any subsequent forest preflight is separate from this frozen local certificate.

```
python3 scripts/fresh_order16_cd_sparse_sixteenth_transfer_20261018.py --verify
```

Source 233

Source path: `results/fresh_order16_d105_ambient61_frontier_20261018.md`

SHA256: ad48da23138bd03e6787cf663c3390f745d0712797c458a6bf8e18342b64cb49

Four ideal obstructions remain after the proved order16 saving

The proved order16 envelope105/16 leaves exactly four negative terminal ideal profiles at ambient dimension61, with no zero profiles. This is a bounded exact update, not a new dimension bound or a local order11 proof.

The source and certificate bind the complete ambient61 frontier and the completed D105 local theorem. The latter lowers the actual all-family order16 maximum from106 to105, with no added finite cost. Its baseline is the completed dimension61 rule, not a new simultaneous forest count with the later finite changes.

Exact saved-row update

Independently enumerate every partition of link rank60 with at most one part1 through8 and all other parts at least9. This gives2,365 profiles and agrees bijectively with the frozen complete domain. Each occurrence of affine rank15 saves1/16 in normalized cost. Consequently each saved normalized margin increases by count(rank15)/16. Multiplying by its unchanged link-vertex count gives the exact absolute margin.

Ranks	New normalized margin	Absolute margin
9,10,10,15,16	−299/32912	−2990
10,10,10,10,10,10	−3189/161051	−35079
10,10,10,14,16	−108/6655	−5508
10,10,10,15,15	−1159/42592	−9272

The normalized minimum is−1159/42592; the absolute terminal minimum is−35079. Negative terminal absolute margins are not promoted to a uniform lower bound over every uncompressed link. The inherited low9 concentration and exact all-family tails supply the domain reduction; this packet does not repeat their proof or recompute a convolution.

The remaining type and local obligations

Current affine order11 costs are A99/2, B54, C54 and D53. The source reads and asserts all four explicit order11 records of the frozen dimension62 composition, rather than deriving types from its maximum alone. Each of the four negative profiles contains affine rank10. Replacing even one of its B/C-envelope factors by a typeD factor improves the normalized margin by1/11, which exceeds the entire deficit in every row. A typeA factor saves still more. Thus only products in which every rank10 factor has typeB orC can attain these four envelope obstructions. No geometric argument here excludes those products or says their worst costs cannot coexist.

For a uniform order11 total saving delta, each row gains count(rank10)*delta/11. The exact strict threshold over every row is

```
delta > 1159/11616,
```

attained by(10,10,10,15,15). A conditional saving1/4 makes all2,365 terminal margins positive, with normalized minimum3/250 and absolute minimum64. Those numerical comparisons are checked exactly. Such an all-family order11 decrease still requires a compatible local theorem; no fractional saving is authorized by this scalar calculation.

Finite payments are unchanged by the D105 theorem. This packet neither recounts a finite forest nor removes the separate finite obstruction. The completed unrestricted ceiling remains61.

Verification scope

Final generation `cc70c7` and separate fresh replay `530d73` both exited 0/PASS. The parent completed a full source/report mathematical review, including the independent sorted-part domain, exact rank15 saving, negative and absolute minima, strict order11 threshold and explicitly bound type-specific refinement. That review did not repeat the scalar, convolution or historical census arithmetic. The source uses only the standard library and reconstructs the sorted-part domain, all2,365 row updates, all four negative profiles and both exact thresholds. It performs no geometric census, new scalar convolution, local LP or global optimization. Source, certificate and report are final/frozen.

```
python3 scripts/fresh_order16_d105_ambient61_frontier_20261018.py
python3 scripts/fresh_order16_d105_ambient61_frontier_20261018.py --verify
```

Source 234

Source path: `results/fresh_order16_d_distance9_half_transfer_20261018.md`

SHA256: eeab2a357cdd9db85302f7c20304f79865442153289aea22b907a51b8a5d922d

Two affine-D15 distance-nine units become halves

In the completed dimension61 payment rule, reduce the two affine-D15 pairs(0,9),(2,11) from1 to1/2. The canonical16-node diagram has spine 0–1–…–11 and leaves12,13 at0 and14,15 at11. Its total becomes105 instead of106. All finite payments and coupons remain unchanged. Every triangle still receives at least2 and every quadrilateral at least1.

The order16 family totals are now A80/B105/C103/D105, so the rank15 ideal envelope becomes105/16. This is a local theorem, not a new dimension bound or a complete ambient61 comparison. It is independent of the newer depth8 finite theorem; the completed depth11 finite rule suffices.

The source and certificate extend the ten-half D16 theorem under the completed dimension61 rule. The changed set is invariant under every labelled diagram automorphism. One mark has degree3 and the other degree2; neither is a leaf. All earlier ten halves and nine zeros, every distance-at-most8 pair and every triangle pair retain their exact weights.

Complete selected support

The frozen high-overlap domain has2614 contexts. For this order16 target, 2045 have a whole marked-component bound below16,92 have a displayed component too large to embed,74 have a protected marked leaf and9 have an adaptive finite payment. The remaining394 are precisely the eligible selected component patterns. The source checks this partition explicitly.

For every one of the394 patterns, including failed embeddings, it re-enumerates all induced labelled injections into the affine-D15 target and compares every map and marked image with the complete saved small-factor certificate. There are151 successful patterns and928 complete target maps. Exactly24 contexts and128 maps hit the two changed pairs. The source records all128 outcomes, without replacing an actual whole factor by a displayed finite subdiagram.

Every affected core is nonsingular Lorentzian and simply laced in the physical norm-two gauge. The inherited complete contact helper is rerun: the actual preferred corner, all finite blocks or affine kernels, possible additional-white contacts, two end sets, inverse entries and residual tests agree with the bound complete overlap certificate.

Actual completions with arbitrary contacts

Write J for the inverse old-core Gram. An extra root has nonnegative contact magnitudes x and satisfies x^T J x$\leq$2, because the orthogonal complement of the Lorentz span is positive definite. Contacts at both ends already give an actual short8 path. Within one end the inverse is entrywise positive, so arbitrary additional contacts cannot lower any forced single-coordinate projection bound.

The two newly encountered relevant-component contexts359 and966 require missing target nodes directly adjacent to the old marked component. Each such first-entry root is genuinely outside the entire old core: other old corner components are orthogonal to the marked component, and the omitted black roots do not occur at this corner. A required simple bond gives projection lower bound10/3 or4, exceeding physical norm2. All eight corresponding maps are excluded. This first-entry argument does not assume that later target nodes are new or assign them artificial zero contacts.

The other proper completions use the already proved residual mechanism of the ten-half theorem. It is checked again at every new context. In the ordinary cases every possible core-contact root has a single simple contact; exact residual inequalities

exclude two such roots unless a short8 path already pays. In relevant-component cases an optional outside entry root would bound the entire marked component by14, including indirect tails and bridges. An actual order16 factor excludes that root. Thus its single relevant entry is the only root contacting the core.

Only after this arbitrary-support argument are the remaining completion zeros forced. The target supplies every affine bond; the sole entry exhausts allowed core contacts, and no old other component can enter through an indirect bridge. The completed principal Gram is therefore determined. Every remaining proper completion has two negative directions, contradicting the ambient Lorentz signature.

The exhaustive map outcomes are:

Outcome	Maps
New first-entry projection obstruction in359/966	8
Multiple required core-contact roots	48
Inherited single-entry projection above2	32
Forced completion with two negative directions	24
Displayed whole affine factor with sealed finite alternatives	16
Total	128

These are all-label arguments: nonzero native magnitudes are at least1, the next possible magnitude is sqrt2 where used, and positive inverse blocks or exact residual inequalities control larger labels and extra contacts. No unit-label sample is promoted to an arbitrary-contact claim.

The two surviving whole-factor cases

The16 surviving maps are the automorphic presentations of contexts1315 and2530. Each core has18 roots and signature(17,1,0); its preferred component is already the entire16-node affine factor. The necessary additional-white contact domain has no feasible core-contact candidate. Consequently every additional supporting white is orthogonal to the entire core in the no-short8 branch. All displayed finite components at the other corners are therefore whole actual components, including in the presence of arbitrary further spectators.

At a distinct finite corner, the marks lie in whole D5 and whole D11, with the D11 mark at its fork center. The existing whole-small-component coupon allows every marked node of a whole component of rank at most5 against that role, and contributes1/2. The selected ideal corner now contributes1/2. These distinct corners give total1.

The source reconstructs all four corner Grams and every marked component and role, as well as the strict white and white-plus-one-black principals. No same-component finite unit is spent. In particular the recent rank15, rank16 and D17 deletions do not affect this replacement.

This is an existing coupon, not an uncounted new cost. The current exact finite-rule function confirms the whole D5/D11 component-pair total15/2: five possible D5 marks times three D11 fork roles times1/2. The theorem uses only the fork-center part of that already charged rule. It adds nothing to any finite forest.

Nonselected, literal and current-rule compatibility

The new pairs belong to the old basic distance9 set, so selected-map coverage alone would be insufficient. The following extra interfaces are essential.

Every one of the74 protected overlap cases has a marked node that remains a leaf of the whole actual component. At an ideal corner it is a leaf of the actual varying affine factor. Such a pair cannot be either new pair, whose two target degrees are3 and2. All9 adaptive cases have finite outputs, so their payments are unchanged. The low-core theorem bounds whole marked components by12, and the all-label high-disjoint theorem excludes its complete branch. The remaining order-bounded selected cases are exactly the394 domains above.

The universal radius-five tree replacement uses immediate ideal paths of length at most7; at marked separation9 its remaining determinant gap is already strict. Its finite alternatives are short paths or a sealed A7. The universal cycle replacements

use only ideal paths of length1 or2, and their longer outputs are finite or typed finite coupons. Thus neither universal route consumes a new distance9 ideal half as a unit. The inherited depth11 finite-zero to short7 replacement also stays independent of this change.

The source reconstructs all216 literal configurations from the completed dimension61 finite rank15/rank16/D17 composition, whose finite D17 cost is120. It checks every actual affine D15 factor of order16 in them, and verifies that none has a changed marked image. There are19 such literal affine interfaces. Their complete current totals therefore remain unchanged and at least1. These216 configurations are later literal obligations, not a substitute for the2614-context domain split. This packet does not separately audit the newer finite D17=118/depth8 composition.

Apply the completed independent actual-corner priorities first: C11 endpoint zero, B12 center/terminal zero, D13 center zero, cycle11 halves and the coupled order11 transfers. Their ideal outputs have smaller whole factors or short paths; their finite outputs and coupons are unchanged. The order17 zero theorem then uses its distinct E6/D11 unit or the independent current short8 route. The B16 and B14/C14 projection exclusions likewise use only that short8 route. No new distance9 pair is required to pay fully in those arguments.

After these priorities, the complete selected/nonselected reduction and the literal audit above apply. The affected selected images are impossible or receive the new ideal half plus its distinct existing finite half. All other selected payments retain their values. This is a patch of the complete current whole-face proof, not an inference that the24 selected contexts enumerate every quadrilateral. On a face with no newly changed actual corner the old weights also agree pointwise.

The finite7/ideal5 triangle theorem, its protected6/7 and small-factor interfaces, and the cycle11 triangle exclusion use no changed distance9 pair. Hence triangles retain2 and quadrilaterals1 under one simultaneous rule, with every finite budget and coupon unchanged.

Verification and limits

Final generation60522 (`0be232`) and a separate fresh saved replay85510 (`635003`) both terminated exit0/PASS. All188 direct input/self bindings, five report links and source syntax passed a separate metadata check. The parent completed a full source/report mathematical read, including the complete inherited core-contact and completion implementations, the relevant-component completion proof, the residual/no-bridge proof and the universal radius-five tree/cycle interfaces. A second agent independently completed a full source/report mathematical read and read both inherited core-contact and completion helpers. Both checked the genuinely new first entries, arbitrary-contact positivity, forced zeros only after the no-bridge argument, distinct whole D5/D11 coupon, protected-leaf and current short-path interfaces, and the stated completed-dimension61 literal baseline. Neither reviewer repeated the embedding, contact, inertia, literal arithmetic or historical census. No gap was identified in this stated local theorem. The source re-enumerates only the394 target embedding patterns, not the historical all-label minimal-core or overlap census. Exact contact, completion and corner checks are new bounded computations; the complete actual-face domains, arbitrary-support residual/no-bridge proof and universal tree/cycle classification remain explicit inherited inputs.

No global scalar comparison, finite-forest optimization, new chamber, external refereeing or proof-assistant verification is asserted. Source, certificate and report are final/frozen.

```
python3 scripts/fresh_order16_d_distance9_half_transfer_20261018.py
python3 scripts/fresh_order16_d_distance9_half_transfer_20261018.py --verify
```

Source 235

Source path: `results/fresh_order16_finite_cross_payments_20261017.md`

SHA256: 50a91515477062d833f0e22f0df9d9727bc4f3769281f181046583f2d933d8c1

Reduced order16 ideal payments with finite cross-component half-payments

Every triangle receives at least2 and every quadrilateral at least1 under the rule below. Finite-vertex cost is at most9n for n<=108. The normalized ideal cost per affine factor of rank r is bounded by

```
b(r) = r/2       for 1<=r<=14,
       7        for r=15,
       127/17   for r=16,
       217/29   for r>=17.
```

This is a complete local payment and cost statement; dimension transfer is a separate argument. The source and certificate bind the completed preceding payment theorem, the order16 embedding census and the universal antipodal cycle replacement.

The rule and reduction to preferred components

Retain the preceding finite rule: unit payment for connected marked pairs at distance at most9 or in whole components of order at most17, and half-payment between A1/D16 or A2/D14 at the stated D fork marks. Add these finite half-payments:

- Between whole A1 and D15 components, at the D fork center or either short leaf.
- Between whole A2 and D13 components, at the D fork center or either short leaf.
- Between whole A1 and A15 components, at the unique middle node of A15.

Either A2 mark is allowed. No payment is assigned merely from a subdiagram inside a larger component, and these disconnected rules are disjoint from the connected unit rule and from one another.

Retain the preceding ideal rules at every order except16. At order16, the A, B and C paid sets are the complete basic-plus-preferred-image unions in the new census, of sizes112,107,103. For D, start with all120 pairs and remove the eight pairs at distance at least12 and the pair of fork centers, which has distance11. The remaining set has111 pairs. All rules are intrinsic under automorphisms of the whole affine factor.

Every triangle retains its two distance7 unit payments. Suppose a quadrilateral has total payment less than1. It can have no adjacent white-interior path of length at most9. At a finite corner such a path pays directly. The inherited ideal replacement handles all larger factors. In smaller factors all pairs are retained. At order16, all B/C/D distances at most9 are retained. The sole A exception is the antipodal distance8 pair; the new cycle lemma supplies another unit payment or two sealed finite A1/A15 half-payments. Thus this exception cannot occur in an unpaid face either.

The complete inherited no-short9 classification now applies unchanged, including its cross-bond exclusions and arbitrary-support extension bounds. Its preferred connected marked component is either finite, or in an actual affine factor of order at most17. Finite corners and orders at most15 pay directly; order17 retains the completed preceding transfer. For order16, every preferred component must occur among the 260 induced embedding domains. All its images are paid except for the nine D pairs handled next.

The nine D15 exceptions and arbitrary additional support

The nine deleted D pairs occur only in inherited contexts 384,394,401,648,663,676,814,829,842. Each literal core consists of two L25 diagrams with the common edge matched by8->9 and9->8. It has18 roots and Gram signature(17,1,0). The corner omitting roots3 and13 contains the entire16-node affine D15 factor F.

Every extra supporting white e is orthogonal to every root of F. Indeed, at that actual ideal corner the positive affine kernel pairs to zero with e and every individual Coxeter product is nonpositive. The only remaining possible contacts with the core are at3 and13.

If both contacts are nonzero, the actual corner marked by3 and13 has the white-interior path3-e-13 of length2 and is already paid. If exactly one contact is nonzero, write its magnitude as t>=1 in the norm-two normalization. The inverse of the18-root core Gram has contact block

```
[ 4 -4 ]
[-4  4 ]
```

on3,13. Therefore the orthogonal projection of e into the Lorentzian core span has square4t^2>=4. The orthogonal complement of that span is positive definite, so e itself would have square at least4, contradicting its square2. This rules out a single contact without a label cutoff.

Consequently, in an otherwise unpaid face every additional supporting white is orthogonal to the entire18-root core. The remaining white diagram is positive definite, as it supports an actual hyperbolic face. It cannot enlarge or join any of the finite core components. All three other corners remain finite with their displayed whole components. This proves support control in every ambient dimension; it is not an observation restricted to a particular small chamber.

Exact rational corner reconstruction gives the same finite-payment pattern as for the preceding order17 exceptions. Five cases already have a connected finite unit payment. Cases384,401,814,842 instead have two half-payments of the new A1/D15 or A2/D13 kinds. Every affected quadrilateral therefore receives at least1. The certificate checks all nine cases, the positive affine kernel, the Lorentz signature and the inverse contact block exactly.

The separate cycle lemma treats all possible antipodal order16 A stars. Its complete finite/affine extension analysis and determinant equality force the two simple-leaf equality core, unless another unit corner is already paid. Spectator equality then either yields a path2 payment or seals the A1/A15 components. This universal step is essential to the new112-pair A rule.

Combined finite budget

Let a,b be the numbers of whole A1,A2 components; let t count all D15,D16,A15 components and v count all D13,D14 components. Their ranks are at least1,2,15,13 respectively. Compared with9 times their ranks, the retained unit costs save at least9,17,24,35 per component. All finite half-payments together cost at most(3/2)at+3bv; this overestimates the A1/A15-center contribution, whose coefficient is1/2. Thus

```
cost <=9n+a(3t/2-9)+b(3v-17)-24t-35v,
a+2b+15t+13v <=n.
```

For n<=108, t<=7 and v<=8. If t<=6 and v<=5, every term after9n is nonpositive. If t=7, then v=0 and a<=3; the possible positive term is at most9/2, while24t=168. If v=6,7,8, respectively, b<=15,8,2; the b contribution is at most15,32,14, while35v is210,245,280. In these cases t<=2, so the a term is nonpositive. This exhausts the integer domain and proves the finite9n bound. Arbitrary other finite components retain cost at most9 times their ranks.

The source additionally checks26,817 endpoints for ambient dimensions 87 through108. Its expression is affine in a, so checking the two rank-constrained a endpoints for every b,t,v covers all integer counts. These checks corroborate the analytic bound; no bound on ambient dimension is inferred from them alone.

Every-rank ideal envelope

Factors of order at most15 retain every pair, giving rank cost r/2. At order16 the maximum of112,107,103,111 is112, giving112/16=7. At order17 the unchanged maximum is127, giving127/17.

For order m>=18, the retained all-rank A cost is7m. The B/C/D formulas are the previously proved piecewise linear counts, with successive slopes9,8,7. Dividing by m, their largest values occur at their finite breakpoints. The maximum is217/29, attained by D at m=29; B is bounded by7+11/28, C by7+8/27, and A by7. The source checks every linear branch endpoint and the increasing slack of each infinite final branch against217m/29. This is an all-rank argument, not extrapolation from the earlier finite BFS range.

The actual cusp factor-fiber theorem multiplies each factor's pair count by V/(r+1), where V is the number of link vertices. Summing gives the normalized ideal budget sum_i b(r_i), including at nonsimple cusps.

Verification scope

The source uses exact rational linear algebra for the exceptional cores, complete inherited embeddings for the preferred components, the separate universal cycle theorem, finite-budget endpoints and all-rank linear branches. Primary generation and a separate fresh saved replay both terminate exit0 with the identical certificate. A separate full mathematical/source read confirms the260-pattern domain, no-short9 reduction, nine D15 exceptions and inverse4 projection obstruction, whole-component support argument, all new coupon types, combined finite budget and every-rank envelope. That review does not duplicate the 260 embeddings, nine-core arithmetic or26,817 endpoint calculations. The independent cycle package has its own generation and fresh replay. The inherited complete classification remains an explicit input; this package does not claim an independent rerun of that entire census or external refereeing.

Source 236

Source path: `results/fresh_order16_piecewise_dimension86_20261017.md`

SHA256: 32c4728f03e91c119ecfed7b3ef1a5176b766a7a0cbbeb9e077ffd12a3e8dd1f

Piecewise ideal payments: dimension at most86

Complete internally checked computer-assisted proof. This argument gives dimension at most **86** for every finite-sided, finite-volume real hyperbolic Coxeter polytope, with no arithmetic, compactness, simple-cusp, cusp-rank or facet-count hypothesis. The compact-facet ambient ceiling79, compact-arithmetic ceiling21 and arbitrary compact ceiling29 are unchanged.

The source and certificate combine the new local payment theorem with the completed dimension91 proof. The separate exact review records independent scalar reconstruction and its written composition scope. Verification status is stated below; no construction or sharpness is claimed.

Actual face payments and costs

The local theorem retains the preceding unit finite payments and adds finite half-payments between whole A1/D15, A2/D13 and A1/A15 components, at specified fork marks or the unique middle of A15. The preceding A1/D16 and A2/D14 half-payments remain. Every triangle receives at least2 and every quadrilateral at least1. The total finite-vertex cost remains at most9n for n<=108.

These payments reduce the order16 ideal costs to112,107,103,111 for affine A,B,C,D. Complete induced embeddings cover all260 potentially relevant preferred components from the inherited classification. The cycle replacement and nine exceptional D stars are proved universally, including arbitrary additional supporting whites; they do not depend on seeing the same component types in one known small chamber.

For an affine factor of rank r, the pair count divided by r+1 is at most

```
b(r) = r/2       if 1<=r<=14,
       7        if r=15,
       127/17   if r=16,
       217/29   if r>=17.
```

The large-rank envelope follows from the explicit all-rank affine spine/fork formulas. It is not inferred from a finite sample. At a cusp let d=n-1, V=product_i(r_i+1), and h(t)=product_i(1+t+…+t^(r_i)). The inherited factor-fiber theorem gives total ideal cost at most V sum_i b(r_i), including at nonsimple cusps.

For G=sum_j h_j min(j,d-j), the normalized curvature from the shape inequality is (5G-dV)/(2V). Its finite coefficient is

```
beta_n=n(n-6)/8              for even n,
beta_n=(n-1)(n-5)/8          for odd n.
```

The shape inequality bounds D5, the sum of5 minus the number of sides over all two-faces, below by these vertex coefficients plus a strictly positive global slack. Actual corner payments bound D5 above by the finite and ideal costs. Comparing coefficients therefore suffices.

Complete concentration for the piecewise budget

The inherited all-rank convex-order theorem states that for1<=a<=b, replacing(a,b) by(a-1,b+1) increases the spread of the centered uniform sum and can only decrease normalized curvature. It applies with the remaining factors fixed. A zero rank is omitted.

Use only these transfers:

1. For two ranks below14, concentrate them until one is0 or the other14. The budget r/2 is linear throughout, so total cost is preserved.
2. Replace two ranks15 by14 and16. Their budget increases by 7+127/17-14=8/17, while curvature decreases. Thus curvature minus cost can only decrease; cost preservation is not required.
3. For two ranks greater than17, concentrate until one is17. Their budget is the constant217/29, so total cost is preserved.

Each nontrivial transfer strictly increases the integer sum of squared ranks at fixed total d, so iteration terminates. The resulting shapes have any number of14,16,17 parts, at most one smaller part in1,…,13, at most one15, and at most one larger part of size at least18. These shapes form a covering reduction of every positive integer partition; it is harmless if some shapes are not themselves final outcomes of a particular transfer sequence.

At every step curvature-minus-cost does not increase. Positive margin for every reduced shape therefore proves positive margin for every original cusp-rank partition. This argument explicitly handles the nonlinear ranks15 and16; applying the previous single-cap reduction without this adjustment would not be justified.

Exact comparison and conclusion

The source checks all3355 reduced profiles in link dimensions86 through90:

Ambient dimension	Profiles	Minimum curvature minus cost	Finite slack beta_n-9n
87	621	146417/911250	197/2
88	646	203099/607500	110
89	671	136309/253125	123
90	696	34528/50625	135
91	721	606679/708750	297/2

The smallest ideal margin is146417/911250, at ranks(2,14,14,14,14,14,14), ambient87. All ideal margins and finite slacks are strictly positive. The resulting D5 lower bound exceeds its payment upper bound, excluding dimensions87 through91. The completed91 proof excludes all larger dimensions. Hence the ceiling is86.

At ambient86, the profile(1,14,14,14,14,14,14) has margin -112409/1518750. This comparison therefore cannot yield a ceiling85. The formal link profile is not a realized finite-volume polytope.

Verification scope

The primary calculation uses exact integer convolution and a separate inclusion-exclusion moment identity, compares all3355 profiles, checks the finite coefficients and records the boundary obstruction. The separate checker uses a different enumeration, sliding-window convolution and a cumulative-tail identity without project arithmetic imports. Primary generation and a separate fresh saved replay both terminate exit0/PASS for all3355 profiles. The independent comparison agrees on every polynomial, G, curvature, cost and margin, all five dimension minima and finite slacks, and the boundary. Its linked report records the final saved bindings and replay status. Separate full mathematical/source reviews confirm the piecewise concentration, actual-payment composition and completed91 tail, with no identified gap.

The local theorem has a separate full mathematical/source review, including its complete embedding domain, universal support arguments, finite budget and all-rank ideal envelope. The independent scalar checker does not duplicate that embedding census or the inherited complete geometric classification. External refereeing, publication priority and proof-assistant formalization are not established here.

Source 237

Source path: `results/fresh_order16_piecewise_dimension86_20261017_review.md`

SHA256: b3c5f7111d2c3ab1bfc9911d9b7173c918e5b5cd5ffc730f902a13e91d86a29b

Independent scalar and composition review of the dimension86 argument

FINAL/FROZEN: independent hash-bound generation and fresh saved replay both PASS. The independent checker uses only standard-library integer/Fraction arithmetic. It imports neither the producer nor any project arithmetic module. A complete written/source composition read of the dimension86 argument finds no identified gap, with its inherited classification and local payment proofs retained as explicit inputs.

This review independently reconstructs the piecewise scalar comparison. It does not claim another260-case induced-embedding census or another reconstruction of the nine exceptional glued cores.

Independent enumeration and exact curvature

The envelope under review is

```
b(r)=r/2 for1<=r<=14; b(15)=7; b(16)=127/17;
b(r)=217/29 forr>=17.
```

For each link dimension d=86,…,90, the checker chooses the optional high part in{0,18,…,d}, the optional low part in{0,…,13}, and whether a15 is present. It then chooses the multiplicities of17 and16; the remaining rank must be a nonnegative multiple of14, forcing its multiplicity. Zero parts are omitted and sorted shapes are deduplicated. This is a different enumeration from the producer's loops over the three repeated-rank counts, followed by low/middle choices and a forced high part. Both give exactly the same shapes.

Sliding-window convolution reconstructs

```
H(t)=product_i(1+t+...+t^r_i)=sum_j h_j t^j,
V=product_i(r_i+1),  G=sum_j h_j min(j,d-j).
```

The checker verifies polynomial length, symmetry and H(1)=V. It independently expands product_i(1−t^(r_i+1))=sum_l a_l t^l, combining repeated shifts exactly. For s factors and q=floor(d/2), a cumulative-tail/hockey-stick identity gives

```
G=qV-2 sum_l a_l binom(q-l+s,s+1).
```

The coefficient and cumulative computations agree in all3,355 profiles and the boundary profile. It also checks the absolute-deviation form

```
kappa/V=(5G-dV)/(2V)
       =3d/4-(5/2)E|sum_i U_i-d/2|.
```

All polynomial coefficients, G, curvature, budget, margin, dimension minima and boundary values match the primary saved JSON exactly.

Ambient dimension	Profiles	Minimum curvature minus cost	Finite slack beta_n−9n
87	621	146417/911250	197/2
88	646	203099/607500	110
89	671	136309/253125	123

Ambient dimension	Profiles	Minimum curvature minus cost	Finite slack beta_n−9n
90	696	34528/50625	135
91	721	606679/708750	297/2

The minimum is146417/911250 at ranks(2,14,14,14,14,14,14). The ambient86 boundary profile(1,14,14,14,14,14,14) has margin−112409/1518750. This formal link shape is a limitation of the particular coefficient comparison, not a realized finite-volume chamber or a sharpness assertion.

Why the reduced shapes cover every partition

The inherited convex-order theorem applies to centered independent uniform factors: for1<=a<=b, transfer(a,b)->(a−1,b+1) increases dispersion and can only decrease normalized curvature, with the other factors fixed.

Only three types of transfer are used. Two ranks below14 are concentrated until one vanishes or the other becomes14; their linear budget is preserved. Two ranks15 are replaced by14 and16; their budget increases by8/17. Two ranks greater than17 are concentrated until one becomes17; their constant budget is preserved. In all three cases curvature minus budget cannot increase. In particular, the second step requires no false assumption of cost preservation: the increase in cost strengthens the desired reduction.

Every nontrivial transfer increases the integer sum of squared ranks at fixed total d, so the process terminates. It leaves arbitrary copies of14,16,17 and at most one part in each exceptional class:1,…,13;15; and18 or larger. Transfers do not create additional troublesome classes: low-rank moves stop at14, high-rank moves stop at17, and the doubled15 move produces only14 and16. Thus every integer partition reduces to a checked shape. Positivity at the reduced shape implies positivity at the original one.

Local payment composition and budgets

The order16 payment theorem binds the complete260-domain preferred-component embeddings and the universal antipodal-cycle half-coupon theorem. Its finite coupons refer to whole components. They do not infer an A1/A15 or A1/D15 payment merely from those subdiagrams occurring within larger components.

The no-short9-path reduction remains valid: finite corners pay a short path directly; at ideal corners the preceding replacement theorem handles larger factors, and the new cycle lemma handles the sole unpaid antipodal order16 A pair. The complete inherited preferred-component classification can therefore be applied with unchanged hypotheses. All induced images in order16 factors are included, and the nine dropped D cases are treated separately.

For those D cases, the actual affine kernel forces every extra supporting white orthogonal to the full16-root affine factor. Contacts to both remaining black roots3,13 give a paid path2. A single nonzero contact has squared Lorentz projection at least4 by the inverse block4[[1,−1],[−1,1]], contradicting norm2 because the orthogonal complement is positive definite. Thus an otherwise unpaid star is sealed, preserving its whole finite components and the required coupons. The cycle lemma uses the arbitrary-spectator determinant equality to obtain the analogous sealing for A1/A15-center coupons. These are universal support arguments, rather than observations about one small example.

For finite cost, let a,b count A1,A2, let t count D15,D16,A15, and v count D13,D14. The source's upper bound is

```
9n+a(3t/2−9)+b(3v−17)−24t−35v,
a+2b+15t+13v<=n.
```

For n<=108, t<=7 and v<=8. If t<=6 and v<=5, all surplus terms are nonpositive. If t=7, then v=0 and a<=3, so its possible9/2 surplus is dominated by168. For v=6,7,8 the respective bounds b<=15,8,2 give positive contributions at most15,32,14, dominated by210,245,280; the a term is nonpositive. This exhausts the domain. The A15 coupon was safely overestimated in the displayed expression. Other finite components retain their inherited9-rank budget.

The order16 factor costs112,107,103,111 give normalized bound7. Order17 retains127/17. For larger orders the inherited all-rank spine/fork formulas have finite breakpoints and a final slope7 branch. Their maximal normalized cost is217/29,

attained by D at order29. Smaller factors use all pairs and give r/2. This justifies the piecewise envelope for every rank; the finite BFS sample is not the tail proof.

At nonsimple cusps the actual factor-fiber theorem multiplies each factor's pair count by V/(r+1), giving total ideal cost V sum_i b(r_i). No minimum-rank or factor-count restriction has entered.

Dimension conclusion and verification scope

The inherited shape inequality gives

```
D5 >= beta_n*v_f + sum_cusps kappa(L) + 5(n-1).
```

Actual payments give the corresponding upper bound, with finite cost9n and the stated ideal envelope. The independent margins and finite slacks are strictly positive for ambient87,…,91, so the bounds contradict one another. The completed dimension91 theorem excludes every larger dimension. The resulting unrestricted ceiling is86; the separate compact-facet ambient ceiling79 is unchanged.

Independent reconstruction and complete comparison with all3,355 primary rows pass. Final hash-bound generation and a separate fresh saved replay both terminated exit0/PASS, including all input hashes and every primary comparison. A separate frontier mathematical/source review of the full primary composition also passed; it does not duplicate this scalar reconstruction. The root independently read this checker and found no issue in its enumeration or identities. This written/source review covers the full composition, concentration, finite budget, ideal envelope, actual cusp multiplicity and inherited tail. The260-domain embedding census, local core computations and complete historical classification remain explicit inputs rather than duplicated computations. External refereeing, publication priority and proof-assistant formalization are not asserted.

Source 238

Source path: `results/fresh_order16_preferred_embeddings_20261017.md`

SHA256: 007b64cdc08555605624ed3b35dd36e113681ac7db6791fbd9157ad7d6d075d2

Complete preferred-component embeddings into order16 affine factors

This exact local census covers every inherited preferred component that could lie in an actual ideal factor of order16. The source and certificate retain all260 cases with extension bound at least16 from the completed distance9 classification. There are180 cases with bound16 and80 with bound17. All other high-core cases have smaller bounds. The complete small-core supplement has bound at most12 and cannot produce an order16 preferred factor.

For each retained case, reconstruct the literal glued core, its preferred connected component and its two marked nodes. Enumerate every injective induced labelled embedding into each order16 affine A, B, C and D diagram. The inherited explicit backtracking tries every unused target vertex consistent with every earlier mapped edge and nonedge. Its degree test is only a necessary condition; patterns larger than16 cannot embed. Induction on the mapping length proves completeness. No graph-isomorphism library or numerical matrix calculation is used.

The basic paid pairs have distance at most7, or distance8/9 with an endpoint within7 steps of a branch or nonsimple-bond feature. In these order16 B/C/D targets the latter rule includes every pair of distance at most9. The A target has no features and pays only distances at most7.

Target on16 nodes	Basic pairs	Extra preferred images	Total	Cases with an embedding
A15 cycle	112	0	112	7
B15	100	7	107	44
C15	99	4	103	4
D15	101	19	120	63

The certificate saves every embedding and marked image, not just totals. The unions are intrinsic under target automorphisms because all embeddings are included. The D union is the full set of120 pairs. Nine of them can be removed by the separate finite cross-payment argument; this census does not silently discard their originating contexts.

This is a conditional local census under the inherited no-short9-path hypothesis. To use the112-pair cycle rule in a universal face-payment theorem, the antipodal ideal corner requires its separate replacement lemma. A complete dimension bound requires the total finite and ideal budgets and curvature comparison as well.

Primary generation and a separate fresh saved replay both terminate successfully for all260 domains and four targets. A separate full mathematical/source read confirms the domain filter, complete induced backtracking and conditional scope. It does not duplicate the260-domain arithmetic enumeration. The integrating payment theorem records the additional universal replacement and budget obligations.

Source 239

Source path: `results/fresh_order17_affine_d_three_quarter_transfer_20261017.md`

SHA256: 04401edda71cff7b5f5f4ef9885945dc1695c360ae3caa212bc9040107937ebb

One further unit saved on the order17 affine D factor

The whole affine D16 factor, of diagram order17, can use total weight119 instead of120 after adding one precisely typed finite coupon. In its canonical spine0–…–12, with short leaves13,14 at0 and15,16 at12, change pairs(3,15),(3,16),(9,13),(9,14) from1 to3/4. These four pairs form one automorphism orbit of distance10. All previously assigned zero and half weights remain unchanged.

The new finite coupon has weight1/4 when the marks lie in separate whole components E6 and D11, the E6 mark is an endpoint of either arm of length2 from its unique degree3 node, and the D11 mark is a short fork leaf. There are two eligible roots in each component, hence four marked pairs and added cost1 per E6-D11 component pair. It is not a coupon on every root of E6 or on an arbitrary rank6 component.

The source and certificate prove the local transfer with this enlarged finite rule. The new all-family order17 costs are A85/B111/C108/D119, with envelope119/17=7. Bounding the total finite cost of the new coupon is a separate obligation. No ambient dimension bound or finite-budget improvement is claimed here.

Complete support and preserved payments

The inherited complete distance8 overlap corpus has140 selected patterns with whole-component bound at least17 and displayed component order at most17. Every induced labelled injection of all140 patterns into the order17 D target is reconstructed, including failed searches, and all successful maps are compared exactly with the complete saved census. The changed orbit occurs in sixteen contexts, with96 maps:

`506,509,512,515,1037,1051,1827,1830,`
`1907,1981,2220,2223,2300,2374,2549,2551.`

Twelve contexts have proper displayed subcomponents and64 maps. The remaining four contexts1981,2374,2549,2551 already contain the whole affine factor, with eight maps each. All basic payments and elementary triangle requirements remain units, as does every short8 pair.

The low-core whole-component bound is at most12, the high-disjoint branch is excluded, and the protected/basic alternatives are unchanged. The universal D13/D14 immediate affine outputs have orders12/13. The remaining universal and later transfers use their specified shorter ideal paths, finite components or finite coupons. Existing D16/D17 partial-weight replacements retain their original distinct pair orbits and their own finite payments. No later B/C projection refinement introduces a long order17 D obligation. Thus there is no additional nonselected provenance of the four changed pairs.

The short8 interface remains an explicit dependency. A later finite7 rule must preserve its payments or prove their replacements separately.

All sixty-four proper completions are impossible

The full old cores in these twelve contexts are nonsingular Lorentzian, simply laced, with physical norm2 roots. The inherited residual helper reconstructs the actual corner and complete necessary contact domain, then rechecks its inverse Gram, positive end blocks, two-contact bounds, all candidate residuals and complete pair tests.

If an additional supporting white meets both ends, the inherited short8 branch already pays. Otherwise the positive inverse blocks and exact two-contact lower bounds force a single old-core contact. Every possible contact has inverse diagonal greater than1, so a native contact of magnitude at least sqrt(2) would have projection square greater than2. The surviving contact must therefore be simple. The pair tests prove that two distinct core-contact roots would have either positive actual product or a negative product supplying the retained short path.

Consequently at most one extra white can contact the old core in an otherwise unpaid face. Every missing target node adjacent to the old displayed component is genuinely new: other old corner blocks are orthogonal to that component, and omitted black roots are absent. The complete outcomes are:

Affected proper maps	Reason
32	The forced unit contact has projection square greater than2
16	The target requires at least two distinct core-contact roots
16	The forced completion has Lorentz index greater than1

In the last row the sole required entry root exhausts the allowed core-contact roots. All other missing target nodes are orthogonal to H. No old corner component can join through an indirect bridge: this would require another core contact, or a second contact by the unique entry root, both already excluded. The actual affine target fixes all remaining bonds. The source constructs the completed Gram only after these facts force its zeros, and checks its affine submatrix to have signature (16,0,1). The full principal Gram has more than one negative direction, which is impossible in the ambient Lorentz space. Additional supporting roots cannot repair that principal obstruction.

These necessary-contact arguments cover arbitrary extra labels and supporting forests; the mapped affine component is not assumed whole without checking its actual completion.

Four whole-affine cases seal the coupon components

All four remaining contexts use the same19-root simple-bond core with signature(17,1,1). Their marks differ. The checker verifies the positive 15-root white block, all four strict16-root edge blocks, the positive kernel of the whole17-node affine component and the exact core radical.

Every extra supporting white is orthogonal to the affine component by its positive kernel at the actual ideal corner. The only possible old core contacts are the two omitted black roots3 and13. Their coefficients in the full core radical have opposite signs. The radical is an actual zero vector: it is orthogonal to an indefinite span containing a timelike vector, whose orthogonal complement is positive definite.

Pairing this zero relation with any further white forces both omitted contacts to vanish or both to be nonzero. The second possibility gives the retained marked path3–white–13 of length2. In an otherwise unpaid face every further white is orthogonal to the entire19-root core. This conclusion seals every displayed finite component against both direct attachments and indirect supporting tails.

Each of the four configurations has a finite corner whose marked components are whole E6 and whole D11. The D11 mark is a short fork leaf. In E6, the three arms have lengths1,2,2, and the mark is an endpoint of one of the length2 arms. The checker derives these arm lengths and checks that the two eligible E6 endpoints form one automorphism orbit. It records the exact literal components, marks and roles, not merely their ranks. The two D11 short leaves are also explicitly identified.

At that finite corner the new coupon gives1/4. The original ideal corner still gives3/4. Their sum is1 in each configuration. Triangles are unaffected because the changed ideal pairs have distance10.

Simultaneous finite scope

No same-component finite unit is consumed by this proof. The two configurations1981 and2549 also display a whole D17 corner, but its payment is recorded as unused. The other two have separate A1 and D16 components at the corresponding corner. Thus future reductions of finite D16/D17 same-component payments do not invalidate this transfer. The only new finite obligation is the typed E6-D11 quarter coupon above.

The total added cost is exactly

```
(2 eligible E6 roots)∗(2 eligible D11 roots)∗(1/4) = 1
```

per pair of whole components. Since the old small-component coupons have rank cutoff5, there is no old E6-D11 coupon being double-counted or silently upgraded. A finite-budget successor must account for the term e6*d11. This package does not assert a resulting forest maximum.

At an actual cusp the factor-fiber multiplicity remains V/17 per marked pair. Four changes of1 to3/4 save exactly1, giving119. This local rule is compatible with the separate order15 improvements: it changes only its own order17 orbit and adds a nonnegative finite payment. The unresolved finite comparison is not bypassed by the improved ideal cost.

Verification scope

Generationе1edf4/dddfd7 and separate fresh replay5f1299/7d7bfd both terminated exit0/PASS. The checker reconstructs all140 eligible pattern searches and96 affected maps, rechecks the64 proper completion obstructions and the four sealed whole-component cases, and verifies the exact coupon roles, orbit size and added cost. Root read the full source/report and passed the ordinary contact control before forming completion zeros, all four radical sealing arguments, the exact two-by-two coupon roles and factor total119. A second reviewer completed a full mathematical/source/interface read, passing the same contact and no-indirect-bridge logic, whole-component sealing, E6 arm lengths and endpoint orbit, exact quarter-coupon cost, and the preserved prior/later payment interfaces. Neither reviewer repeated the140-pattern, 96-map, inertia or coupon arithmetic. All28 direct bindings, two local report links and source syntax pass.

The complete all-label overlap classification, contact-domain proof and residual/no-bridge theorem remain explicit frozen dependencies. This source rechecks their numeric hypotheses and uses their stated arbitrary-support scope. It does not independently repeat the entire geometric census or prove a finite-budget bound. No ambient theorem, new chamber, external refereeing or formal verification is asserted. Source, certificate and report are final/frozen after generation, fresh replay and the two mathematical/source reviews. The finite-budget obligation remains separate from this local theorem.

```
python3 scripts/fresh_order17_affine_d_three_quarter_transfer_20261017.py
python3 scripts/fresh_order17_affine_d_three_quarter_transfer_20261017.py --verify
```

Source 240

Source path: `results/fresh_order17_center_contextual_zero_transfer_20261018.md`

SHA256: 7d4e96b2d7be94925208016f00f16c3d8f00d3b2f2c918ea168cf32f5a2bf021

Two order17 center halves replaced by distinct contextual finite halves

The current affine D16 pairs `(0,10)` and `(2,12)` may both be reduced from one half to zero. Add a half payment between a marked endpoint of a whole A2 and the unique long-arm endpoint of a whole D5 **only when that finite corner also contains whole unmarked A1 and A9 components**. Each affected selected configuration has this new half and a distinct existing whole D5-long-end/D12-center half. Their sum is one without the original ideal half.

The order17 family totals become A85, B111, C108 and D227/2, so the all-family envelope decreases from229/34 to227/34. The new contextual cost is one per A2/D5 component pair when its witnesses exist; multiple witnesses do not multiply the payment. The complete current ambient59 finite maximum remains793/2, attained formally by2A16+D27. With the separately proved order13 improvement, the complete1890 terminal ideal profiles have one negative row, `(9,9,9,15,16)`, of margin−69/5440. The completed unrestricted dimension bound remains59. No bound58 is claimed.

The producer and certificate bind the current finite and ideal composition, the order13 theorem, the original center-orbit seals and the current order17 weight table. The historical quarter/three-quarter rule supplies its geometry here; its superseded weights are not imported into the current rule.

Exact local rule and complete inherited support

Use the canonical affine D16 numbering: spine0 through12, short leaves13/14 at0 and15/16 at12. The changed pairs form one full automorphism orbit at distance10. All136 current weights are reconstructed. Their sum changes from229/2 to227/2; the110 basic pairs, every short8 pair, and the bound elementary triangle payment set stay at unit weight. In particular the four distance11 pairs already at1/8 are unchanged.

The bound center-orbit certificate performed all140 eligible native embedding searches, including failures, and found71 successful contexts and488 maps. This producer checks its complete successful-map identity against the original small-factor certificate, checks the140 eligible pattern count, and filters the changed orbit. Exactly two contexts remain, 1314 and2529, with eight maps each. Their displayed affine factors already have17 roots; there is no unproved proper-completion exclusion. The source does not repeat the historical embedding search. It reconstructs both full-core proofs and compares them with the frozen exact records.

The inherited high-overlap domain has2614 contexts:2391 whole-factor order bounds at most16,83 protected or adaptive cases, and140 eligible order17 patterns. The83 split into74 fixed feature/leaf cases and nine adaptive finite cases. Low-core, disjoint, protected and later universal routes remain inherited local theorems, not consequences of the16-map filter. Their required short/basic outputs remain unchanged here.

Arbitrary supporting roots and the two paying corners

Both selected configurations use the same19-root norm2 Gram core, of signature(17,1,1). Its affine component consists of every core root except 3 and13 and has a strictly positive kernel. The inherited exact check verifies the full radical, the positive affine kernel, the positive definiteness of the supporting whites and each white-plus-one-black edge, and every actual displayed corner.

An additional actual supporting white has nonpositive products with the affine roots. The actual ideal corner is positive semidefinite, so its row is orthogonal to the affine Gram kernel. Positivity of that kernel forces all these products to vanish. The full-core radical maps to zero in the ambient Lorentz space: the core span contains a timelike vector, so a vector in that span orthogonal to the whole span is zero. The radical coefficients at3 and13 are nonzero with opposite signs. Consequently an additional white contacts both of these roots or neither. Contact with both yields a path of length2 between adjacent black sides, paid by the independent current short-path theorem. Otherwise the white is orthogonal to the entire core. The same first-contact argument excludes a bridge through arbitrarily many additional supporting whites.

Thus, away from that independent short-path branch, all displayed finite components remain whole components in the actual corner. Their positive definite Gram blocks and the extra orthogonal spectator forest give an actual finite corner. This is the reason the new witness condition holds universally; its validity is not inferred from absence in a finite list of saved extensions.

The three relevant corners are as follows. All labels in this table are literal core labels, not canonical component indices.

Context	Newly zero ideal corner	Existing half corner	New contextual half corner
1314	0,15	0,13: D12 center0 and D5 long endpoint13	3,13: A2 endpoint3 and D5 long endpoint13
2529	5,10	3,10: D5 long endpoint3 and D12 center10	3,13: D5 long endpoint3 and A2 endpoint13

In context1314, the new corner has components A1 on`[1]`, A2 on`[2,3]`, A9 on`[4,5,6,7,8,9,16,17,18]`, and D5 on`[10,11,12,13,14]`. In context2529 they are D5 on`[0,1,2,3,4]`, A9 on `[6,7,8,9,14,15,16,17,18]`, A1 on`[11]`, and A2 on`[12,13]`. The A1 and A9 are unmarked in both. The new producer reconstructs all displayed finite component decompositions, checks the A2 endpoint orbit and the unique D5 long endpoint, and records the witness components.

At each face the two finite corners in the table are distinct. The old D5/D12-center half is unchanged. The new contextual half gives `0 + 1/2 + 1/2 = 1`. No same-component finite unit, original ideal half, or second use of one corner is needed.

Compatibility with the current rule

The new coupon changes only A2/D5 pairs satisfying its whole-component witness condition. It changes none of the D5/D12 center, short-leaf or finite-D17 coupon roles; none of the D5/A12 or D5/A1 alternate payments; and none of the finite D16, A17 or D17 same-component weights. In particular it does not consume the current retained A17 units, A16 halves, or D16(1,14)/(1,15) halves used by later finite transfers.

First apply the independent current small-ideal and short-path payment routes. The order13 C-first/B-second theorem just bound here uses only whole B11/D11 alternatives, protected finite B13, or its explicit A1/B12 half. Its proof does not use the changed order17 orbit. The other current C11/B11, cycle11, B12, D13, B/C14, B/D15 and B/D16 routes similarly use retained short pairs or their precise finite coupons. The current finite-distance8 theorem gives short7; the finite-distance7 theorem at depth6 gives short6 or its protected whole small-factor alternatives. Those routes do not return either distance10 pair changed here.

If none of those routes already pays the face, apply the complete bound selected/nonselected reduction. A changed selected order17 pair has one of the two full cores above and receives the two distinct halves. Unchanged selected pairs retain their own proofs. The other order17 sealed replacements use their own finite coupons or independent short paths, not these two ideal halves. The low/disjoint/protected/adaptive routes retain their required basic or short outputs. On a face with no changed ideal corner, every ideal weight agrees pointwise with the preceding theorem and the finite weights only increase. Thus the prior whole-face inequality transfers there without choosing an old witness that might lose its payment.

Triangles retain total payment at least2. The changed orbit lies outside the retained elementary ideal triangle set and has distance10, while the current small-ideal triangle payments and fixed-leaf finite exceptions are unchanged. Quadrilaterals retain total payment at least1.

The source also freshly reconstructs all216 current literal records starting from the order13 certificate. It identifies two ideal half debits and two contextual half credits, and checks every resulting corner lower bound and every face total. This is an interface check; the arbitrary-support argument and complete inherited routing above establish the universal local statement.

Finite accounting with complete contextual coverage

Let `a1,a2,a9,d5` count whole components of those exact types in a finite forest. Its new actual cost is

`a2*d5, if a1>0 and a9>0; otherwise0.`

There are two marked endpoints of A2 and one eligible D5 endpoint, each worth one half. Each marked component pair is counted once, irrespective of how many witness A1 or A9 components are available. The existing contextual A2/A2 payment is disjoint: it has two marked A2 components, requires an unmarked A10 and at least two unmarked A1 components, and contributes `a2*(a2-1)/4` when triggered.

For the upper certificate, a whole A9 witness is relaxed to any rank9 component, and a D5 is relaxed to the rank5 A5 representative. This only increases the possible new payment. Ordinary base/cross costs retain their preceding simultaneous representative bounds, including the new order13 A1/B12 half. All32400 actual type-pair inequalities are checked. Fixed rank17 means A17; D17 remains a residual item. Actual D16 and its A16-dominating representative upper remain distinct.

The complete961 fixed rank9-through17 vectors are expanded by every rank5 multiplicity, giving2574 possibilities. The frozen5144 residual tables are unchanged: this new interaction is handled by explicit A2 multiplicity rather than an altered additive item. The producer checks all6,718,064 last-item upper inequalities,303,496 attaining predecessors, and128,700 residual item identities before using the tables. It does not rerun a dynamic-programming optimizer.

The ordinary and previously contextual branches already have their complete saved upper bounds. A new-trigger branch must have at least one fixed rank9 factor, one rank5 factor and one A1. For every such fixed vector, the source enumerates all A2 multiplicities and reserves one A1. If a fixed rank10 factor is also present, it separately reserves two A1 and adds the complete old A2/A2 interaction. The remaining table excludes rank2, so every A2 is explicitly counted. There are3405 new context choices over585 eligible fixed vectors.

For the one-A1 branch when rank10 is present, the actual class has exactly one A1 and hence no old A2/A2 payment. Allowing extra A1 items in its residual table is a relaxation used only to bound that class. It is not an assertion that the displayed table witness has that exact contextual cost. Actual forests with two or more A1 use the separate two-A1 branch, where both interactions are included. If rank10 is absent, the old interaction never occurs. These classes exhaust every newly charged forest; repeated witnesses give no further terms.

The maximum upper remains6344 sixteenths, or793/2. The actual formal forest2A16+D27 has that cost and none of the new contextual components, so it proves attainment. Nine fixed-vector alternatives remain above the ambient59 finite curvature783/2. The largest new-trigger upper is 5871/16, well below the maximum. These are exact forest costs, not a claim that the maximizing forest occurs in a geometric example.

This count uses the current completed depth6 finite-distance7 rule. The separate new finite-distance6 theorem under development is not assumed or counted here; composing it requires a new finite certificate.

Complete ideal frontier and remaining obstruction

The producer checks the complete saved1890 rank58 low9 terminal domain, its moment/curvature identities and the current envelope chain. It first matches the new order13 saving `count(rank12)/13` exactly, then adds `count(rank16)/17` from the two deleted order17 halves. The factor-fiber normalization is unchanged. No convolution or historical geometric census is repeated.

Only `(9,9,9,15,16)` remains negative. Its normalized margin is−69/5440 and absolute terminal margin is−3450; there are no zero rows. Negative terminal absolute margins are not extended to arbitrary actual cusp profiles by concentration. There is no new global inequality closing dimension58.

If a further all-family order16 cost saving is available without losing the other rules, the exact strict threshold is69/340. A saving of1/4 would make all1890 terminal margins positive: normalized minimum1/340 and absolute terminal minimum91/2. This is a conditional target only; the required local geometry and any additional finite costs are not proved by that scalar calculation. It does not change the low-rank ingredient of the inherited low9 concentration theorem.

Verification

Source-bound generation6908/chunk47e0ca and separate fresh replay73245/ chunk2c938b both terminated0/PASS. Both runs reconstruct the two full seals, the216 current literal records, the unchanged-table finite certificate and the saved exact ideal update. Historical native search and original classification completeness remain bound inputs.

Root completed a full producer/report mathematical composition read, including the immediate inherited `Z.inspect` helper. It checked the complete saved140/488 domain and16 affected maps; positive-kernel and full-radical arbitrary-support seal and short2 branch; distinct existing and contextual halves with whole unmarked witnesses; finite actual versus representative accounting and exhaustive one-/two-A1 residual classes; rank17 type separation; complete1890 ideal update, triangle and literal composition. No execution or historical map, geometry, Bellman or convolution replay was performed by that reviewer.

A second reviewer completed a full producer/report mathematical read, including `Z.inspect` and its underlying `O.inspect`. It checked the arbitrary-white seal, distinct whole paying corners and witness roles, complete inherited support versus universal-routing scope, all-rank5 and A2 contextual coverage with the one-A1 relaxation correctly distinguished from attainment, typed domination, actual finite witness, ideal update and conditional next-quarter target. That reviewer did not execute maps, seals, literals, Bellman or convolution computations.

Final metadata validation checks all344 direct immutable bindings, six local report links and source syntax. Source, certificate and report are final and frozen. No external review or proof-assistant verification is claimed.

Source 241

Source path: `results/fresh_order17_d5_d12_center_quarter_transfer_20261018.md`

SHA256: ec94fd0c6f5ae5ba87d289bc6621b4361b308b5c1d887d8de129a7c8b194a5bb

Two order17 ideal quarters with a whole D5/D12 three-quarter coupon

Starting from the frozen D114 rule, reduce the affine D16 pairs(0,10) and(2,12) from1/2 to1/4. Upgrade the separate whole-component coupon between the unique long-arm endpoint of D5 and the fork center of D12 from1/2 to3/4. Each affected face retains its original ideal quarter and has a distinct finite three-quarter payment, totaling1.

The ideal D total falls from114 to227/2. All other order17 families remain A85/B111/C108, so the all-family envelope becomes227/34. Exactly one marked pair per D5/D12 component pair changes, adding1/4 to the finite cost. This packet proves the local payment theorem, including compatibility with the separate finite D17 transfer; it asserts no finite forest maximum or new dimension bound. The completed unrestricted bound remains60.

The source and certificate bind the preceding D114 theorem, the original complete half transfer, the complete selected-map/overlap data, and the separately completed finite D17 transfer.

Complete native support and exact retained weights

Use the affine D16 numbering with spine0 through12, leaves13/14 at0, and leaves15/16 at12. The two changed pairs form a full automorphism orbit, both at distance10. Each previous half becomes exactly1/4; all other weights, including the four preceding distance11 zeros, are retained. The checker compares all136 pair weights and checks that all110 basic pairs, all short8 pairs and every recorded elementary triangle pair still have weight1.

The producer repeats all140 eligible native embedding searches, including failures. The71 successful contexts and488 maps agree exactly with the complete preceding certificate. Exactly two contexts, 1314 and2529, meet this new orbit, each in8 maps. Both displayed components already have17 roots and are the entire actual affine factor. No proper completion is omitted or excluded by assumption.

The complete high-overlap routing remains the bound2614-context partition:2391 whole-order bounds at most16,83 protected/adaptive cases and140 eligible order17 patterns, with no displayed-too-large case. The83 split into74 fixed leaf/feature and9 adaptive finite cases. Low-core, high-disjoint and later universal interfaces remain those of the preceding theorem. The two new contexts are the entire affected selected support, not a claimed enumeration of every face.

Actual supporting roots and the distinct three-quarter payment

Both contexts use the same19-root norm2 core of signature(17,1,1). The full affine component consists of all roots except3 and13 and has a strictly positive kernel. The producer rechecks the complete inherited O.inspect certificate, then independently reconstructs the actual contact domain, finite components and the more precise marked roles needed here.

Every extra actual supporting white is orthogonal to the positive kernel of the affine corner. Since its individual products with those roots are nonpositive, it is orthogonal to the whole affine component. The full-core radical represents zero in the actual Lorentz space: it is orthogonal to a span with a timelike vector. Its coefficients at3 and13 are nonzero and have opposite signs. Therefore an extra white either contacts both3 and13 or neither. The first possibility supplies the independent opposite-black path2 unit. In the second, every extra white is orthogonal to the entire core. A spectator bridge would have a first root contacting the core and is excluded by the same argument.

The common supporting-white diagram and every displayed finite corner are positive definite. Thus the extra orthogonal spectator forest cannot change the type or marked role of a displayed whole component, or make its actual corner ideal. This establishes the whole-component coupon for arbitrary extra supporting roots and native labels, rather than only for the saved core.

Context	Original ideal marks	Distinct finite coupon marks	Whole D5 role	Whole D12 role
1314	0,15	0,13	long endpoint13	fork center0
2529	5,10	3,10	long endpoint3	fork center10

The marks in the table are literal core labels. The D5 fork arms have lengths(1,1,2); its mark is the unique endpoint of the length2 arm. The D12 fork arms have lengths(1,1,9); its mark is the unique degree3 vertex. The source derives the trees and roles from the actual finite Gram matrices, checks that the two components are disjoint and orthogonal, and verifies the distinct corner identities. In standard finite numbering these roles are D5 node0 and D12 node9.

At the original ideal corner the new payment is1/4. At the distinct finite corner the upgraded coupon is3/4. Their sum is1 in each configuration. No same-component finite payment is used. This is a fractional transfer; deleting the original quarter is not justified by the present theorem.

Exact costs and simultaneous finite compatibility

The D114 theorem already raised the two long-end/short-leaf coupons by1/2 each, from the older D5/D12 total15/2 to17/2. The present change has one eligible mark in each component and adds1/4, yielding35/4 before the separate finite D17 transfer.

The finite D17 theorem upgrades a disjoint set: the two D5 short leaves3/4 against the D12 center9 and short leaves10/11. Those six halves become units, adding3. The source explicitly compares all three sets of canonical marked pairs:

- preceding D114: `(0,10),(0,11)`;
- this packet: `(0,9)`;
- finite D17: `{3,4} × {9,10,11}`.

They are pairwise disjoint. Their simultaneous increment is `1 + 1/4 + 3 = 17/4`, and the whole D5/D12 component-pair total is `15/2 + 17/4 = 47/4`. These are actual costs, not a claim that an older representative allowance already contains them.

The finite D17 theorem's six affected configurations are wholly finite and pay by their own distinct upgraded unit. They do not use any ideal quarter. The two present configurations pay only by the original ideal quarter and the whole D5-long-end/D12-center coupon; they use none of the three newly zero finite D17 pairs. The prior D114 configurations likewise keep their separate full units. Hence the three local transfers compose directly. Their common short-path escape is the independently established current short8 route.

Complete face and triangle coverage

As in the D114 predecessor, first use the current independent small ideal and short-path routes. All direct short8 ideal weights in this order remain units; this new orbit has distance10. Current C11/B11 zeros, cycle11 and coupled halves, B12/D13, D105, B/C14, B/D15 and prior order17 transfers retain their specified short paths or whole coupons. Their outputs do not use either of these two long ideal pairs. The new finite weights only increase.

If a short8 finite alternative is a deep8 zero, its independent corridor gives short7. If that finite7 alternative is a depth10 zero, its independent theorem gives short6. The current small ideal, cycle/tree and triangle protections supply the corresponding paid routes without these order17 long pairs. This prevents recursion through the changed weights.

On the remaining branch, apply the complete inherited selected and nonselected reduction. If its selected order17 pair is changed here, the complete two-context argument above supplies either a length2 unit or the explicit quarter-plus-three-quarter total. Unchanged selected pairs retain their previous proof. All other original order17 half/zero configurations retain their own separate whole coupons. The low, disjoint, protected, adaptive, immediate and later interfaces are inherited

with the same retained short/basic or bounded-component outputs. None introduces a new unsupported order17 distance10 obligation.

For an actual face with no changed ideal corner, its ideal weights agree pointwise with the preceding theorem and its finite weights only increase. The predecessor's whole-face inequality transfers there. For the exceptional selected contexts the distinct payments are explicit above. The separately changed finite D17 cases use their own independent full units. This is the same actual-corner priority and full-interface composition as the bound predecessor; it does not assume that a historically selected witness remains paid.

Every triangle still receives2: the elementary retained ideal sets are checked, the changed orbit is too long for their small protected outputs, and the current finite7/ideal5 triangle certificates retain their fixed-leaf exceptions. Every quadrilateral still receives1. The ideal factor-fiber multiplicity remains V/17 per marked pair, so the lost two quarters subtract exactly1/2 from114 and give227/34.

Verification and scope

Generationfcf696 and separate fresh saved replay926d0e both terminated exit0/PASS. They reconstruct all140 native searches and488 maps, isolate the16 affected maps, recompute the two full radical/corner certificates, check intrinsic whole-component roles, retain all required pair sets and verify direct immutable bindings. The original all-label geometric census and complete universal interface proofs remain inherited inputs. No finite forest optimization is run.

Root completed a full source/report mathematical review, including a reread of the full O.inspect implementation and preceding D114 role and inspect definitions. That review passed the complete140/488 support and16 affected maps, intrinsic marked roles, arbitrary-white kernel/radical sealing and independent length2 escape, distinct quarter-plus-three-quarter payment without same-component finite units, retained short/basic/triangle and depth8-to7/depth10-to6 priorities, and the three disjoint coupon sets and exact costs. Root did not execute maps, inertia or the inherited census.

A second reviewer completed a full producer/report mathematical review. That review passed the same arbitrary-support sealing, distinct whole-component payment, complete two-context support and absence of proper-completion assumptions; exact disjoint cost accounting; current small-ideal and short8-to7-to6 routes; protected leaves/basic pairs, separate finite D17 units and other order17 replacements; pointwise unchanged-face inheritance and triangle protection. It did not replay arithmetic, maps, inertia, the census or a forest count.

Final metadata checks cover all direct immutable bindings, report links and source syntax. Source, data and report are final and frozen. No external review, proof-assistant verification or new dimension theorem is claimed.

Source 242

Source path: `results/fresh_order17_d5_d12_unit_coupon_zero_transfer_20261018.md`

SHA256: 9f5e8855daa90aaabd7a05dff86da4eff0ef70108513addf2da24344e80b560a

Four order17 ideal halves replaced by whole D5/D12 units

The current affine D16 factor of order17 may reduce four distance11 pair weights from1/2 to0. Its total falls from116 to114. The other order17 families retain A85/B111/C108, so the new all-family envelope is114/17. Upgrade precisely the finite coupon between a whole D5 marked at its unique long-arm endpoint and a separate whole D12 marked at a short fork leaf from1/2 to1. The eligible marks number1 and2, respectively, so this adds exactly1 per such component pair.

This is a local triangle/quadrilateral payment theorem. It preserves triangle total at least2 and quadrilateral total at least1. It does not prove a new finite forest budget or ambient dimension bound. In particular the added finite cost must be included in any subsequent count. The completed unrestricted bound remains60.

The source and certificate bind the current simultaneous payment rule, the current order17 weights, and the complete original half transfer. The new proof uses the latter's complete geometric support, with fresh native-map reconstruction and full rechecking of the four required radical cores. It never treats the old ideal half as a residual payment.

The exact changed orbit and complete support

Number the affine D16 diagram with spine0–1–…–12, leaves13/14 at0, and leaves15/16 at12. The changed orbit is

`(2,15), (2,16), (10,13), (10,14).`

All four pairs have distance11. Their old weights are1/2 and their new weights0. The checker verifies invariance under every native diagram automorphism and retains all110 basic pairs, all pairs of distance at most8, and every elementary triangle pair at weight1. All other pair weights remain those of the bound current rule.

The complete high-overlap domain contains2614 contexts, partitioned as 2391 with whole-component bound at most16,83 protected/adaptive cases, zero cases whose displayed marked component is larger than17, and140 eligible order17 patterns. The protected cases split into74 fixed leaf/feature certificates and9 adaptive finite certificates. These counts and the entire partition are checked in the new source.

The checker redoes all140 induced native embedding searches, including failures, and compares every successful map and marked image to the frozen complete certificate. There are71 successful contexts with488 maps. Exactly the four contexts1978,2371,2535,2539 meet the changed orbit; each has8 maps. In all32 changed maps the displayed component already has17 roots and is the entire actual affine factor. There are no proper-component completions to exclude in this orbit.

Each of these four contexts has the same19-root native core: two L25 minimal diagrams sharing node9, with core key `(25,25,((9,9),))`. The inherited all-label overlap reduction provides that core on the branch with no independently paying short-path alternative. All core norms are2 and all gauges1. The new source rechecks its inertia, radical, positive affine kernel, strict actual-edge principals, complete possible-contact domain and four literal corner decompositions. The 140 patterns are the selected portion of the reduction, not a claimed list of all quadrilaterals.

Arbitrary supporting roots and whole-component sealing

Let F be the displayed affine component. It consists of every core root except3 and13, has signature(16,0,1), and has a strictly positive kernel. The full19-root core has signature(17,1,1). Its radical is an actual zero relation: its span contains

a negative direction, whose orthogonal complement in the Lorentz space is positive definite. The radical coefficients at3 and13 are nonzero and opposite in sign.

For any further actual supporting white root e, positive semidefiniteness of the actual corner containing F makes e orthogonal to its positive kernel. All root products with distinct walls are nonpositive, so e is orthogonal to every root of F separately. Pairing e with the full-core zero relation then forces its contacts with3 and13 either both to vanish or both to be nonzero. The latter case gives the opposite-black path3–e–13 of length2, with its retained full unit.

On the remaining branch every additional supporting white is orthogonal to the entire19-root core. This excludes indirect tails as well as direct contacts: the first root of any bridge would have a nonzero core contact. The remaining supporting-white diagram is positive definite. Consequently each displayed finite core corner is a whole finite corner after adjoining an orthogonal positive-definite spectator forest, and each displayed connected component stays a whole component. No simple-cusp or bounded number of spectator roots is assumed.

The distinct finite unit and exact cost

The finite corner supplying the new unit is explicitly distinct from the original ideal corner:

Context	Original ideal marks	Finite coupon marks	Whole small role	Whole large role
1978	1,15	1,13	D5 long endpoint13	D12 short leaf1
2371	2,15	2,13	D5 long endpoint13	D12 short leaf2
2535	5,11	3,11	D5 long endpoint3	D12 short leaf11
2539	5,12	3,12	D5 long endpoint3	D12 short leaf12

The labels in this table are literal core labels. Intrinsically the D5 fork arms have lengths(1,1,2), and its mark is the unique endpoint of the length2 arm. The D12 arms have lengths(1,1,9), and its mark is an endpoint of a length1 arm. The source derives these properties from the actual finite graphs, checks the components are disjoint and orthogonal, and verifies all four new unit totals. In the usual finite canonical numbering the eligible marks are D5 node0 and D12 nodes10/11.

The existing small-component rule already pays1/2 for each of the five D5 marks against each of the D12 fork center and two short leaves: 15 marked pairs, total15/2. The frozen current counting function is checked in both component orders to give exactly that total. Only the two long-end/short-leaf pairs change, each by1/2. The new total is17/2, with increment1. These roles are invariant under both component isomorphism groups.

The four original ideal halves are now zero, and each configuration has a distinct whole finite unit. No same-component finite payment is used. In particular this argument does not rely on old whole D17 units appearing at other corners of two cores; those old units have since been removed. It also does not use any D5 short-leaf coupon upgrade. All finite same-component weights and every other cross-component coupon are unchanged by this packet.

Complete current-rule composition

Use the current independently proved short-path routes before the selected long-pair reduction. The changed order17 pairs have distance11, so no direct current short8 ideal unit changes. The current C11/B11 zeros, cycle11 and coupled D11 halves, B12 and D13 changes, D105, B/C14 and B/D15 transfers, and the earlier E6/D11 order17 zero theorem retain their own stated short/coupon alternatives. None of those alternatives uses this distance11 orbit. Their finite outputs are unchanged or increased by the new coupon.

A finite short8 alternative affected by the depth8 zero rule supplies its independent short7 route. A distance7 finite alternative affected by the depth10 zero theorem supplies its independent short6 route. Current small ideal units and the cycle11 exception, and the universal short6 cycle/tree transfers, use their protected short paths, fixed leaves/forks, small whole components or specified finite coupons. Thus this short-path priority does not recur through a removed order17 long pair. The separate prospective finite D5-short-leaf/D12 upgrade being explored in parallel is not an input here; this theorem is independent of its completion or budget.

On the remaining no-short8 branch, the inherited complete low-core closures have whole order at most12; the high-disjoint cases are excluded; protected/basic choices remain paid; and the nine adaptive cases retain their bounded-component or

fixed-leaf/distance9 finite outputs. In the selected high-overlap branch, the complete native maps identify precisely the four changed occurrences above. Their arbitrary support proof gives either the independent length2 unit or the distinct whole D5/D12 unit. Unchanged selected weights retain their previous proof. The other order17 half or zero orbits retain their own independent finite replacements; the current rule does not use this orbit as a replacement for a different removed long pair.

Later immediate affine alternatives have smaller order and retain their specified center/leaf or short-pair weights. Universal cycle and radius-five tree interfaces retain their protected short/basic outputs or separate whole-component coupons. The new orbit therefore creates no unanalysed low, disjoint, immediate or later obligation. This uses the complete bound interface proofs, rather than inferring coverage from the four saved contexts alone.

Equivalently, resolve all independently treated actual-corner changes first; use the complete selected replacement when its selected payment belongs to this orbit; and inherit a prior whole-face theorem pointwise when none of its actual ideal weights changes. The new finite rule only increases weights. This supplies the same noncircular actual-face priority as the bound current composition, not an assumption that a historically preferred witness stays paid.

Triangles retain total2: all changed pairs have distance11, the retained elementary ideal pair sets are checked, and the current finite7/ideal5 triangle routes and their fixed-leaf exceptions survive. The added finite units cannot decrease a triangle payment. Factor-fiber multiplicity remains V/17 per marked pair. The four lost halves lower D116 toD114 and give envelope114/17, with no scalar concentration or finite-budget assertion in this packet.

Verification and limits

Generation76b709 and a separate fresh saved replayc6fd44 both terminated exit0/PASS. They recheck the complete140-pattern native domain, all488 successful maps,32 affected maps, four full radical/corner configurations, marked whole-component roles, retained weights and complete2614 routing. Every direct immutable input binding is checked. The original all-label geometric census and the universal nonselected and short-path classifications remain explicit inherited inputs.

Root completed a full source/report mathematical read, including the inherited O.inspect implementation. That review passed the complete 140-pattern/488-map domain and four whole-affine32-map support; the actual positive-kernel and zero-radical argument for arbitrary supporting roots; the length2 alternative and distinct whole D5-long-end/D12-short-leaf unit; exact added cost1; absence of any same-component finite payment; and retained basic, triangle and current-priority interfaces. Root did not execute maps, inertia, scalar or census arithmetic.

A second reviewer completed a full producer/report mathematical read, including the inherited O.inspect implementation and current depth10 and E6-priority interfaces. That review passed the positive affine kernel and full-core radical control of arbitrary supporting whites, the independent length2 route and whole-component sealing, complete selected-domain scope, distinct unit with no same-component finite payment, exact1×2 half upgrades, and all retained basic, short8, triangle, current-priority and nonselected/later interfaces. It did not replay arithmetic, maps, inertia or the all-label census.

Final direct metadata validation passes40 bindings,6 report links and source syntax. Source, data and report are final and frozen. No external review, formal verification, new finite forest maximum or new dimension bound is claimed.

Source 243

Source path: `results/fresh_order17_e6_half_coupon_transfer_20261018.md`

SHA256: 8f3bc79f375679f3ecd26a0f7a4a16cc02ccafb9117ba7fa5ca8c8b1ff62d863

Order17 affine D cost118 with a typed finite half coupon

The whole affine D16 factor (17 roots) can use total weight **118**, with all earlier weights preserved except the four distance10 pairs (3,15), (3,16), (9,13), (9,14). Their weights fall from 3/4 to 1/2. The required finite change is equally specific: raise the coupon from 1/4 to 1/2 between a whole E6, marked at an endpoint of either length 2 arm, and a whole D11, marked at a short fork leaf. There are four eligible marked pairs per pair of components, so the additional finite cost is **one per E6–D11 component pair**.

The source and certificate refine the frozen quarter-coupon theorem. This is a local face-payment theorem. It does not by itself prove a finite forest budget or an ambient dimension bound.

Why the same complete geometric proof applies

The frozen theorem reconstructs all 140 eligible selected patterns and 96 affected maps. Its 64 proper completions are impossible, using complete necessary contact control, projection obstructions and Lorentz index. These obstructions do not depend on the amount of the finite coupon. Only four whole-affine contexts remain: 1981,2374,2549,2551, each with 8 maps.

Each has a 19-root core of signature (17,1,1). The positive affine kernel forces every further supporting white to be orthogonal to the 17-root ideal component. The two remaining possible core contacts have opposite coefficients in the actual zero radical relation. Consequently both contacts vanish, or the retained length 2 path pays. In an otherwise unpaid face every extra white is orthogonal to the entire core. The finite E6 and D11 components at the alternative corner are therefore whole components, with no indirect supporting tail.

The present certificate checks the saved radical and positive affine kernel exactly, and reconstructs the two finite component trees directly from the saved Gram matrix. Their fork-arm lengths are respectively (1,2,2) and (1,1,8). The actual marks are an E6 length 2 endpoint and a D11 short leaf. In each of the four configurations, the old proof used exactly one ideal 3/4 payment and one finite 1/4 coupon, at different corners. The new proof uses 1/2+1/2=1 at the same corners. No other finite payment is consumed. In particular, the displayed finite D16 or D17 units remain explicitly unused.

The four changed ideal pairs form a complete automorphism orbit and all have distance10. Every basic pair, short8 pair and retained triangle pair stays a unit. Completeness of the old selected, low-core, disjoint, immediate and later interfaces is inherited from the frozen theorem; this small refinement does not replace that all-label classification. Its geometric alternatives are independent of the numerical split of 1 between the two specified corners.

Simultaneous scope

All other ideal weights and finite payments are retained. The new finite half coupon dominates its old quarter value, so faces not using the changed ideal orbit cannot lose payment. When a changed ideal occurrence is needed, the complete preceding argument gives either a retained short alternative or the sealed finite half, retaining the quadrilateral total 1. The changed pairs have distance10, and the prior triangle argument remains unchanged with total at least 2.

The finite coupon lies in components of ranks 6 and 11. It is therefore unaffected by the depth11 finite-distance8 zero rule, or by the separate finite D16/D17 two-unit exchange. No existing same-component D16/D17 payment is used. The only

interaction with newer order11 rules is through a short8 alternative in an inherited branch. Among the proposed B/C/D changes, only the opposite-fork pairs of order11 D have distance8; they retain weight 1/2. At a distinct alternative corner that half already combines with the original retained order17 half to give 1. More generally the independent complete order11 transfer supplies its own other half using another order11 half, a retained finite payment or a typed finite coupon; it never uses the changed order17 distance10 orbit. A simultaneous binder must use that independent transfer, rather than assume an old short8 unit is unchanged. The frozen cycle11 short5 halves can also occur in a short8 branch: a distinct such half and the original order17 half suffice. A deep finite8 zero alternative must instead be handled through its independent short7 replacement theorem. Complete simultaneous branch bookkeeping belongs to the global binder; this packet alone does not certify that final composition or the pending order11 theorem.

The order17 family totals become A85/B111/C108/D118, giving envelope118/17. The usual cusp factor-fiber multiplicity is unchanged. The added finite component-pair cost must be included in a separate complete forest budget; no global conclusion is asserted in this report.

Verification scope

Generation a4fd87 and separate fresh replay f68e65 both terminated with exit 0/PASS. The parent completed a full producer/report mathematical read, including the frozen predecessor's complete written argument. A separate reviewer completed the full new source/report and predecessor-proof read, passing the role counts, inherited proper exclusions, weight refinement and explicit global-composition caveat. Neither reviewer repeated the geometric census or arithmetic execution. The implementation uses only the standard library and imports no producer arithmetic. It checks all 136 saved pair weights, the changed orbit, exact radical/kernel identities, four literal coupon-role trees and four payment totals. It binds the frozen complete geometric theorem and current dimension64 package. It does not rerun the 140-pattern/96-map census or independently prove its arbitrary-support sealing and completeness arguments.

All seven direct source bindings, the Python syntax and all three local links pass. Source, data and report are final and frozen.

Source 244

Source path: `results/fresh_order17_e6_unit_coupon_zero_transfer_20261018.md`

SHA256: d315bd51a5ba55323c5565fc1a8ceeb358e527e2c438879dafcfc4aca4f0ac62

Four zero ideal weights with whole E6/D11 unit coupons

Relative to the completed dimension62 payment rule, the four pairs (3,15), (3,16), (9,13), (9,14) in a whole affine D16 factor of order17 may fall from1/2 to0. Its total falls from118 to116. Upgrade exactly the finite coupon between a whole E6 marked at an endpoint of either length2 arm and a separate whole D11 marked at a short fork leaf from1/2 to1. There are two eligible marks on each component, hence four pairs; the component-pair cost rises from2 to4, an increment of2.

This increment fits pointwise inside the existing finite budget's conservative E6/D11 allowance4. No new forest optimization is necessary for this upgrade alone. The order17 family totals become A85/B111/C108/D116, with normalized envelope116/17. This is a local payment theorem and an absorption statement for its specific finite increment, not a new ambient dimension bound. The independent finite obstruction at ambient62 remains.

The source and certificate compose the frozen half-coupon theorem, its complete geometric predecessor, and the completed62 rule. The complete inherited geometric arguments are used, not just the four literal coupon configurations. The original ideal half is not used in any new branch.

Proper completions and sealed whole components

The canonical target has spine0 through12, leaves13/14 at0 and leaves15/16 at12. The four changed pairs form an invariant orbit, all at distance10. The original selected-support theorem reconstructs all140 eligible patterns, including failed maps; its96 affected maps belong to16 contexts. Twelve proper displayed-component contexts give64 maps. The other four contexts1981,2374,2549,2551 already contain the whole affine factor, with eight maps each.

In an actual face without a paying short8 alternative, the proper cases are excluded before choosing any coupon. The original complete contact domain uses physical norm-two roots and positive inverse end blocks. Cross-end contacts produce the short8 route. Otherwise at most one new root can contact the old core, and it has exactly one simple contact. All further target contacts are then forced:32 maps require projection square greater than2,16 require two distinct core-contact roots, and16 give a completed principal Gram matrix with two negative directions. The no-bridge argument excludes an indirect connection through another old corner component before zero entries are imposed. Extra supporting roots or labels cannot repair these necessary principal obstructions. These are inherited geometric statements, unchanged by the size of the finite coupon; their short8 premise is supplied below under the new rule.

The four full cases have a19-root core of signature(17,1,1). The positive kernel of its whole affine17-root component makes every further white orthogonal to that component. The remaining possible old-core contacts are the two omitted blacks3,13. Their coefficients in the zero radical relation have opposite signs, so a further white either meets both or neither. Meeting both gives a distinct opposite-black path2 and its unit. Otherwise every further white is orthogonal to the entire core. This seals its displayed finite components against both direct attachments and indirect spectator tails.

At a distinct actual corner, the two marked components are therefore whole E6 and whole D11. Their fork-arm lengths are(1,2,2) and(1,1,8), and the marks have exactly the stated endpoint/short-leaf roles. The new finite coupon is a full unit there. The original ideal contribution is zero, so the quadrilateral still receives1 without using a second half. The source rechecks the exact radical and positive kernel identities, reconstructs these two finite trees from the saved Gram matrices, and checks distinct corner marks and unit totals in all four configurations. It does not repeat the geometric census or signature computation.

No same-component finite payment is used. In particular the displayed whole D17 corner in two configurations, and the other configurations' A1/D16 corner, remain unused. The sparse finite D16/D17 deletions thus do not invalidate this argument.

Complete simultaneous short-path and actual-corner scope

The preceding proper-completion argument must not assume that every old short8 unit is unchanged. Use the following priority on all actual corners of a quadrilateral, before applying that argument.

1. If there is a changed C11 endpoint corner, its independent zero theorem gives a distinct path1 or2 unit. This does not use any original C11 half or any order17 distance10 pair.
2. Otherwise, if there is a changed B12 center/terminal corner, its independent theorem gives a whole affine D11 center/leaf unit, a whole finite D12 leaf9 unit, or the two typed whole A2/B9-terminal and A2/D11-center halves. All are retained.
3. Otherwise handle a changed D13 center corner. Under the completed62 rule its original half plus its distinct typed half pays; alternatively the independent D13 zero theorem supplies a distinct full unit with the upgraded A2/D10-center or A3-middle/D10-center coupon. Its other outputs are path at most4 or whole order11 B/D center units. Neither version uses this order17 orbit or any sparse finite long pair.
4. Otherwise handle any cycle11 distance5 corner by its standalone half theorem: it gives its original half plus a distinct retained half, or a finite short unit. Finite outputs have distance at most6 and ideal outputs at most5. The affected cycle11 corner is excluded on triangles by its independent triangle theorem.
5. Otherwise handle any changed order11 B/C/D tree corner by the complete coupled half theorem. Its immediate affine alternatives supply distinct halves directly, rather than recursively asking for a full unit from another changed pair. Any actual C11 zero pair was already handled in step1. Other outputs are finite whole components of order at most14, whole A9/A18 endpoint8, D18 leaf9, E8 path6, or the retained typed finite coupons. None is a changed order17 pair.
6. If a changed order17 corner now occurs, apply the complete geometric theorem above. Any short8 alternative pays under the routes described in the next paragraph. If no such alternative occurs, the64 proper completions are impossible and the four sealed cases each have their distinct full finite unit.
7. If no changed order17 corner occurs, all ideal weights agree pointwise with the preceding complete rule on this actual face, and the only new finite change is nonnegative. Its whole-face payment therefore transfers directly.

The short8 route in step6 is the explicit route already proved in the completed63 composition, with the three newer short-pair changes handled first above. All18 small tree targets of orders12 through17 retain every pair of distance at most8 in that preceding rule. The sole subsequently altered such pair is the D13 center, already excluded from step6 by step3. The B12 zero pair has distance9 and the B16 deletions distance10. Small exceptional whole factors retain their prescribed units. Larger radius-five tree outputs use retained finite short paths, protected whole-leaf9 roles or separate coupons. The cycle6/7 interfaces are retained. For cycle arc8, complementary arcs8 through11 have protected nondeep outputs, including the sealed cycle16 case; complementary arcs at least12 have the strict depth11 cycle-interface gap7/48. A finite short8 output retains its unit unless it is one of the deep distance8 zeros; its independent depth11 corridor then supplies the protected short7 route. The cycle and large-tree short7 outputs use finite paths at most7 or their established protected roles, so this replacement does not recurse through another deep8 zero. Sparse finite deletions all have distance at least10. Thus this complete short-path route never needs the original order17 half, or any removed order17 distance10 unit.

The cases use actual corners, not the provenance of an older selected witness, and are exhaustive. The inherited low-core whole-component bound12, excluded high-disjoint case, unchanged protected/basic choices, and immediate/later finite and shorter-ideal interfaces remain those of the complete order17 predecessor. The140-pattern selected list is not asserted to enumerate every quadrilateral. Its nonselected domain is supplied by those bound inherited theorems.

The same reasoning is compatible with the new B16 singleton deletions: all their changed pairs have distance10, none is an output used above. If no order17 change occurs, apply their complete whole-face theorem with the old order17 rule, which agrees pointwise on that actual face. The optional D13 zero theorem is handled explicitly in step3; its extra finite counting obligation is kept separate below.

Triangles still receive2. The changed orbit has distance10; the inherited finite7/ideal5 triangle routes, protected ideal6/7 pairs, and whole small ideal factors remain unchanged. New coupons can only increase payment. Every other ideal and finite weight is retained.

Exact absorption of the finite increment

The sparse finite budget already uses a representative E6/D11 allowance4, although its actual coupon had total2. This was needed to dominate the actual E6/A11 total4 after replacing A11 by D11. Now the actual E6/D11 coupon also has total4; the representative rule and every one of its Bellman tables stay fixed.

For all192 finite types through rank63 and all36,864 ordered type pairs, the source takes the saved actual and representative doubled costs and adds4 to the actual doubled cost exactly for(E6,D11) or(D11,E6). The new actual value is then8, equal to the old upper value8. All other pointwise inequalities are unchanged. The same-component domination is unchanged. Summing over components and unordered component pairs gives the same upper bound for every forest; simultaneous two-end domination is used, not an invalid sequential replacement.

As an additional saved-data check, all1,329 formal upper-attainer forests are recounted with the increment4*e6*d11 in doubled units and remain at most their saved representative upper values. The unchanged actual witness2D17+D29 has no E6/D11 interaction and still costs449. Hence for the completed62 finite rule plus **this upgrade alone**, the exact rank63 maximum is still449. The inherited complete Bellman certificate provides the upper bound; the present script does not rerun that optimization.

The separate D13 zero theorem adds a2*d10*+*(a3*d10)/2. Those increments are not the E6 allowance and are not included in the absorption claim. A simultaneous successor must use their separate enlarged forest bound. Likewise this rank63 statement does not settle the ambient62 comparison or assert a new global dimension theorem. The factor-fiber multiplicity at a cusp remains V/17, so four decreases of1/2 save exactly2 in the order17 factor total, giving116.

Verification scope

Generation0254a5 and separate fresh replayee44e3 both terminated exit0/PASS. Audit314751 checks all46 direct bindings, nine local report links and Python syntax. Root completed a full source/report mathematical read, passing the distinct sealed whole E6/D11 unit without the original ideal half, the actual-corner priority before the proper-completion short8 branch, and compatibility with the B16 distance10 deletions. The review also passed the pointwise absorption under allowance4, unchanged rank63 attainment449 for this upgrade alone, and the explicit separation of D13 increments. Root did not rerun inherited censuses, inertia or Bellman arithmetic. A second peer completed a full producer/report mathematical read, also reading the D13-zero report and the frozen E6-half/current63 interfaces. It passed the independent actual-corner priority, pointwise inheritance, all four distinct whole-component units, absence of a same-component D16/D17 debit, four-role coupon increment, simultaneous two-end budget absorption and unchanged actual attainment. It did not rerun a census, inertia, matrix, scalar calculation or Bellman table. Neither review identified a mathematical gap in the stated local and counting scope. Source, data and report are final/frozen.

The producer imports only the standard library. It rechecks136 weights, their invariant changed orbit, all basic and short8/triangle inclusions, four literal radical/kernel/whole-role interfaces, all36,864 pointwise finite inequalities and1,329 saved formal attainers. It binds the complete geometric and simultaneous short-path proofs. It does not repeat the140-pattern/96-map census, proper-completion inertia, arbitrary-support classification or Bellman tables. No external refereeing, formal verification, new chamber or global bound is asserted.

```
python3 scripts/fresh_order17_e6_unit_coupon_zero_transfer_20261018.py
python3 scripts/fresh_order17_e6_unit_coupon_zero_transfer_20261018.py --verify
```

Source 245

Source path: `results/fresh_rank13_contextual_coupon_rebalance_20261018.md`

SHA256: 2b5d7ac343174bc4e6df4ad1c74c7c0edbb03a92c322867d38e7a5cc3e4b6de7

A contextual payment removes the rank13 finite obstruction

Two changes lower the actual costs of both A2/A13 and A2/D13 component pairs from5 to39/8. Their replacements preserve every triangle payment2 and quadrilateral payment1. The complete rank59 finite count includes a new payment conditioned on other whole components at the same vertex. Both formal forests10A2+3A13 and10A2+3D13 fall394 to1561/4=390.25, below curvature783/2=391.5. Every representative forest containing a rank13 component costs at most1563/4=390.75.

The overall exact finite maximum remains795/2=397.5, with the same three actual typed attainers D16+D17+D26, A17+D16+D26 and2D16+D27. There is no new dimension bound or constructed finite-volume polytope.

The producer and certificate start from the completed59 rule with the finite B14/D14 refinements. The earlier frozen packages remain unchanged.

Exact new rule

Put delta=1/16. All conditions below refer to whole finite components at the actual corner, not embedded subdiagrams.

- Between A2 and D13, reduce the fork-center weight1/2 to7/16. Either A2 mark is allowed; the D13 center is canonical node10. The four A2/short-fork-leaf unit roles remain units.
- Between A2 and A13, reduce the unique-middle weight1/2 to7/16. Either A2 mark is allowed; the A13 middle is canonical node6. The four A2/A13-endpoint unit roles remain units.
- Between A1 and A13, increase the unique-middle weight1/2 to9/16. Its other roles are unchanged.
- Between distinct A2 components, add1/8 for the marked pair if the same finite diagram contains a whole A10 component and at least two whole A1 components. All these additional components are unmarked. This is a single payment determined by their presence; it is not multiplied by the number of possible A10 or A1 witnesses.

All finite same-component and ideal weights stay unchanged. In particular, actual D15 remains201/2. The rule is intrinsic under all component and diagram automorphisms. It does not require choosing an orientation or distinguishing one of several identical witness components.

The three geometric interfaces that use a decreased coupon

In the universal cycle14 transfer, the only remaining unpaid equality is the sealed14-cycle with two opposite simple leaves and an isolated spectator attached to one new leaf. Its two adjacent finite corners have whole A2/A13-middle and A1/A13-middle roles. They now pay7/16+9/16=1. The proof's all-label profile analysis, strict spectator-energy hypotheses, height-first equality and arbitrary-support orthogonality remain unchanged. The source reconstructs the equality Gram, signature(15,1,1), positive-definite white block and four white-plus-black edge diagrams, and checks the exact whole marked roles. It does not repeat the predecessor's energy-spectrum classification.

In the original order16 transfer, only literal case384 uses the A2/D13-center role. Its two adjacent finite corners each lose1/16. The opposite finite corner has exactly the displayed whole components2A1+2A2+A10, with the marks in the two A2 components. It gains1/8, so the four-corner total remains1.

The full18-root core has signature(17,1). Its actual affine16-node factor has a strictly positive kernel, forcing every extra supporting white orthogonal to that factor. The only possible remaining core contacts are at roots3 and13. Both contacts

supply an independently paid path of length2. A single nonzero contact of magnitude t>=1 has projection square4t^2>=4, exceeding norm2, since the inverse contact block is(4,-4;-4,4). Thus in the remaining branch every additional supporting white is orthogonal to the whole core. Neither the marked A2 factors nor the unmarked A1/A10 factors can be enlarged or joined by a hidden spectator tail. The context condition therefore holds in the full actual diagram. The source freshly reconstructs the complete old nine-case order16 seal and checks the new corner roles.

Finally, the three-singleton D15 theorem uses the center coupon in cases1147 and1269, paired with the selected D15(1,12) half. Both cases also have the same opposite whole2A1+2A2+A10 corner. The old whole-D15 proof seals every other core component: allowed remaining inverse diagonals are4,4,14, with strictly positive same-end inverse blocks; a contact at both ends supplies the independent short8 route. The new coupon therefore applies. Each face now receives 1/2+7/16+1/8=17/16. No restoration of D15(1,12) is necessary. The source rechecks the Lorentz signature, white positivity, remaining inverse diagonals and all marked roles against both frozen seals.

Complete output inventory and simultaneous composition

The current coupon-output audit divides the complete face-payment proof into universal short-path routes, original sealed literals, later selected transfers and finite exchanges. We refine that inventory by exact marked role, since the broad budget allowances alone do not specify which geometric coupons are needed.

- The universal cycle/feature-free tree routes use A1/A15, A1/A17, A1/A11 or A2/A11, apart from the cycle14 middle pair just repaired. Whole connected alternatives and independently retained short paths are unaffected.
- The universal nine-pair affine-D13 theorem has exactly A5/D11, A2/D11, A4/D13 and A1/D13 cross outputs. Its description's separate lists of possible small and large factors must not be combined into an additional A2/D13 output. The affine-D14 theorem uses D12/D14. Neither consumes either decreased role.
- Original order16/17 sealed literal outputs are covered by the complete current216-configuration reconstruction. The only changed record is order16 case384, with two center debits and the contextual credit.
- Later ideal transfers use the separately sealed A4/D13, D5/D12, A4/D12, A5/D11, E6/D11, D5/D11 and A4/D11 roles, or smaller B/D factors. None is an A2/D13-center or A2/A13-middle role.
- The finite A15 transfer upgrades the A2/A13 **endpoint**, and the sparse rank15 exchange upgrades the A2/D13 **short fork leaves**. Those units remain unchanged. The earlier finite D15 alternative also uses a short fork leaf, not its center. The later three-singleton D15 center uses are the two seals repaired above. The finite14 and rank16/17 transfers use A4/D13 or their named same-component weights.

This is an audit of all outputs of the inherited exhaustive proof, not a claim that every actual quadrilateral occurs in the216 saved cores. Replay that proof with the same actual-corner priorities. Every unchanged output retains its payment; the three changed interfaces now have the replacement totals proved above. Short8-to-short7-to-short6 routes retain their complete protected alternatives, including the repaired cycle14 case, and do not rely on a decreased coupon without this replacement. Triangle payments use unchanged same-component or ideal elementary weights, so retain2. Quadrilateral totals retain1.

The subsequent coupon-free B14 and C14 ideal deletions are compatible: their first-entry projection exclusions use independent short8 escapes and no new finite coupon. Conversely the present three seals use finite roles or an independent short2 escape. Neither depends on their removed ideal distance9 pairs.

As an exact supplementary interface check, all216 current literal configurations are reconstructed from their Grams and actual marks, starting from the finite14 packet's cornerwise lower payments. Exactly two A2/D13-center debits and one contextual credit occur, all in case384. Every corner lower bound is nonnegative and every total remains at least1. The universal cycle equality and the later D15 cases are checked separately as above; they are not silently assumed part of those216 literals.

Complete finite count with the contextual charge

The two reduced component-pair costs are

$$4*1 +2*(7/16) =39/8.$$

A1/A13 becomes25/16, while A1/D13 stays3/2. If a finite diagram has k whole A2 components, a whole A10 and at least two A1, its new contextual total is

```
4 * binomial(k,2) * (1/8) = k*(k-1)/4.
```

There are four choices of the two marks for each unordered A2-component pair. The charge is counted once for each marked pair regardless of how many witness components exist. Without the required witnesses it is zero.

The rank59 certificate retains all961 fixed multiplicity vectors for ranks9–17, with A17 fixed separately and D17 among the residual ranks. There are960 interaction keys. All32,400 ordered actual type pairs satisfy simultaneous representative domination, including both A2/A13 and A2/D13 reductions. Same-component bases and49,011 residual item identities are checked exactly. The representative treats any rank10 component as an A10 witness; this can overcount the contextual charge, so gives a safe upper bound. It maps other rank2 types to A2 and preserves A1 counts.

For each key generate two exact tables: the ordinary residual table and a table excluding rank2. The latter is necessary because the contextual charge is quadratic in the number of A2 components. When rank10 is present, enumerate the total A2 count k, reserve two A1 components, and use the no-rank2 table for the remaining rank. Additional A1 components are allowed in that table. This covers every diagram where the charge is active. Compare all such options with the ordinary table, which also bounds every diagram where the trigger is absent.

Fresh replay checks2,613,120 last-item upper inequalities and113,280 attaining predecessors across the1,920 tables, and all2,691 context options. Each chosen row reconstructs its forest and exact contextual charge. The maximum397.5 has actual typed attainers, so is exact rather than a loose representative upper bound. Fourteen representative rows remain above391.5, down from fifteen before this transfer. The maximum among rows with rank13 is390.75; among rows with rank10 it is386. Actual D15 remains100.5 while its representative allowance is101.

The certificate also includes a separate complete counting preflight: increase the whole A1/D13 component-pair cost by1/2, as would result from a new half at its unique long-arm neighbor of the center. Both the overall397.5 maximum and the390.75/386 subdomain bounds remain unchanged. This optional field asserts no geometric theorem for that extra role. Any ideal improvement using it needs its own full proof.

All ideal weights are unchanged by the main rule, so the previous ten negative terminal ideal comparisons are not repaired here. The other finite obstructions also remain. No dimension58 conclusion follows.

Verification scope

Final generation session97323/chunke0c336 and a separate fresh saved replay session52827/chunk471153 both terminated exit0/PASS. The earlier development replay revealed that both D15 seals also receive the new context payment; the final rule therefore needs no D15 restoration. The final source explicitly checks actual attaining forests and binds the frozen B14/C14 interfaces.

A separate full source/report mathematical review reread the order16 projection seal, cycle14 equality, both D15 seals, complete coupon-output inventory, exact affine-D13 output table and finite counting helpers. It found no gap in arbitrary-support whole-component protection, the once-per-marked-pair charge, noncircular payment composition, simultaneous representative domination, the reserve-two-A1/no-A2-table enumeration, or actual attainment. That review did not repeat arithmetic, native maps, Bellman checks or convolutions. The root supplied the composition and full source/report read; no external referee review is implied.

The producer rechecks the stated local Grams/seals, current literal interfaces and all complete new forest certificates. It does not rerun the inherited all-label census, old spectator-energy spectrum, cusp convolutions, or construction searches. No external refereeing or proof-assistant formalization is asserted.

Source 246

Source path: `results/fresh_rank15_order15_ambient60_frontier_20261018.md`

SHA256: 8c8a82a7ee7c719edc5cba868b0f90fb674b63d78e8d2376e414f82df0272800

Combined rank15/order15 frontier: finite413 and49 negative cusp profiles

The new finite-rank15 and affine-order15 local theorems compose. With all compensating finite payments included, the complete ambient60 finite maximum remains **413**, against curvature405. The number of negative terminal cusp margins falls from77 to **49**, among all2,113 profiles; none has zero margin. These improvements do not yet exclude dimension60. The completed unrestricted dimension bound remains60, and no new finite-volume example is established.

The source and certificate bind the frozen finite exchange, order15 B/D theorem, depth10 local theorem and preceding complete frontier. They leave the completed dimension60 proof unchanged.

One simultaneous local rule

The finite exchange gives A15=101, B15=97, D15=102 and A17=116. The last cost includes the restored whole A17 pair(1,15) at weight1. All other same-component finite costs follow the current rule, including the depth10 distance7 deletions. The affine-order15 costs become A75/B92/C91/D92, with maximum92 and normalized envelope92/15.

Exactly three actual finite component-pair costs increase:

Whole component pair and changed marks	Added cost	New total actual cost
A2, either node / D13, short fork leaf	2	5
A4, endpoint / D11, short fork leaf	2	8
A5, second-from-end node / D11, short fork leaf	2	19/2

Each increase upgrades four half payments to units. Every other mark keeps its previous weight, including fork-center marks. These costs are counted together with the A17 increase.

For a changed order15 D corner, the universal local proof supplies a distinct whole A4/D11 or A5/D11 unit, or its independent short-path escape. The changed B corner has an impossible forced completion or the same independent short-path alternative. Neither proof consumes a same-component finite long-pair unit. Thus the new rank15 finite decreases cannot invalidate the new ideal transfers.

Conversely, the finite exchange uses unchanged whole A16 halves, the restored whole A17 unit and the upgraded whole A2/D13 unit, together with the independently established short8 route. Its depth8-to-short7 and depth10-to-short6 replacements retain the necessary protected outputs. The new ideal deletions have distance10 and do not remove those short-path outputs. Current earlier ideal-corner priorities and the complete selected/nonselected and later literal interfaces retain their established scope. Hence the two local theorems apply under a single rule, with every triangle paid by at least2 and quadrilateral by at least1. This composition uses their universal proofs, not merely their finite lists of literal configurations.

Complete finite count with new residual tables

The preceding representative rule already allowed5 for A2/D13 in order to dominate A2/A13. The new actual A2/D13 cost fits that allowance. The two D11 additions genuinely raise the upper rule: they increase the rank4 and rank5 residual item values by4 times the number of fixed D11 factors in doubled units. The source checks both ends of every typed component-pair domination simultaneously.

D15 at102 dominates the new A15/B15 values101/97. A17 remains a separate fixed species at116. The complete fixed species are B9,D10,…,D16,A17. There are1,044 multiplicity vectors and792 residual interaction keys. The residual ranks are1..8 and17..60. All fixed-fixed and residual-residual upper cross costs are zero, and every residual item is checked directly against its base plus all fixed interactions.

Generation constructs fresh item-first unbounded-knapsack tables for all792 keys. Replay does not repeat that optimization. It checks every possible last-item upper inequality and an attaining predecessor at each rank, proving the tables by induction. This covers1,142,064 upper inequalities,47,520 attaining predecessors and54,288 item identities. Every fixed forest is then combined with its entire residual optimum.

The complete maximum is413. The formal forest2D17+D26 attains it and contains neither a newly charged component pair nor a deep distance7 pair. The actual witness costs recorded for other reconstructed forests subtract the depth10 deletions; the representative upper is allowed to ignore those nonnegative savings. Thus attainment is for the current actual payment rule, not just for a conservative upper formula. No geometric realization of these formal forests is asserted.

There are15 representative alternatives above405. In particular, four rank15 factors now have costs404 for all-A,388 for all-B and408 for all-D. The finite obstruction has narrowed, but the worst deficit 413−405=8 remains.

Complete ideal comparison and remaining targets

The low9-only concentration theorem still applies: costs through rank9 remain r/2, and no high-rank factor is concentrated. Independently enumerating every terminal partition of rank59 gives2,113 profiles, with at most one part1 through8 and all other parts at least9.

For each profile, generation and replay reconstruct the exact integer convolution

```
H(t) = product_i (1+t+...+t^(r_i)).
```

The coefficient sum, symmetry, weighted curvature moment and independent absolute-deviation/CDF identities are checked against the prior exact row. The only normalized cost change is

```
saving = 2 * number of factors of rank14 / 15.
```

This includes the actual factor-fiber multiplicity. All other envelopes and family tail formulas are unchanged. Exact recomputation resolves28 previously negative profiles and leaves49 negative, with none zero. The minimum normalized margin is still−1823/11560 at(9,9,9,16,16). The minimum terminal absolute margin is now−123905 at(9,9,9,9,11,12). Negative terminal absolute margins are not universal absolute bounds for unreduced cusps.

The remaining part of the preceding sufficient envelope scenario is:

Affine order	Current maximum	Conditional target	Required saving
12	63	125/2	1/2
13	73	145/2	1/2
14	83	331/4	1/4
16	105	415/4	5/4
17	116	229/2	3/2

Order15 already meets its former target92. Applying these further conditional savings to all2,113 profiles gives normalized minimum 73/104000 and terminal absolute minimum54. These are exact sufficient scalar targets; the corresponding local theorems and any added finite costs remain unproved here. They are not an optimality claim.

The global two-extrema correction remains4500 and is counted once. The finite deficit and49 negative ideal profiles prevent the usual pointwise curvature contradiction, so no bound59 is claimed.

Verification scope

Generation session92271, final chunk71fb95, terminated exit0/PASS. Fresh replay session4000, final chunk589cdf, terminated exit0/PASS. An independent internal agent completed a full mathematical source/report composition review, including the inherited local interfaces, typed costs, complete finite domain and ideal comparison: PASS. That review did not execute a second geometric census, Bellman calculation or convolution. The binder validates the frozen local inputs and exact typed increments, all finite domination and Bellman certificates, all ideal convolutions and scalar comparisons, and the single global constant. It does not rerun a geometric census or prove the local theorems anew. The packet is now frozen. No external refereeing or formal verification is claimed.

```
python3 scripts/fresh_rank15_order15_ambient60_frontier_20261018.py
python3 scripts/fresh_rank15_order15_ambient60_frontier_20261018.py --verify
```

Source 247

Source path: `results/fresh_rank15_rank16_dimension61_20261018.md`

SHA256: ce1a00a2eccd0c913aa14bd70bd60f80bc89ec0f6eeeba0e406c69a21a4b6587

Unrestricted finite-volume dimension bound 61

The simultaneous local rule and complete counts below exclude dimension 62. Together with the completed higher-dimensional exclusion, this gives **dimension at most 61 for every finite-sided, finite-volume real hyperbolic Coxeter polytope**. Arithmeticity, compactness, simple cusps and a bound on the number of facets are not additional hypotheses.

The final contradiction is

```
0 >= 0*v_f + 73*v_i + 4805 > 0,
```

where v_f and v_i are the numbers of finite and ideal vertices. The finite comparison is equality, which suffices because the extremum constant is strictly positive. No bound on the number of vertices is needed.

The source and certificate bind the completed local, finite and ideal packets. This is a complete internally checked computer-assisted proof, with no identified unresolved mathematical step. Generation, fresh replay and internal mathematical composition reviews pass at the scopes recorded below.

One simultaneous local payment rule

Start with the completed order14 ideal closure and finite D17 cost-120 rule. The former includes all current ideal transfers, with order-14 family costs A70/B82/C83/D82. The latter changes no ideal weight or cross-component coupon. Both explicitly preserve the independent short8 route and its replacement when the finite endpoint is a deep distance8 zero.

Make the following additional changes, whose numbering is defined in their local reports:

Whole finite component	Changed marked pairs	New weight	New component cost
A15	(0,13),(1,14)	1/2	104
B15	(0,14)	0	104
D15	(0,13),(0,14)	1/2	104
A16	(0,11),(4,15)	1/2	113
D16	(2,13),(2,14),(2,15)	1/2	225/2
A17	(1,12),(4,15)	1/2, restored from zero	115

All other finite weights and all cross-component coupons retain their current values. In particular D17 remains 120, both D13-zero coupon increments are present, and the actual E6/D11 component-pair cost is 4. The A17 restoration is a genuine cost increase and is counted below.

The rank15 theorem validates the complete 536-pattern selected domains for all three types. The two A15 cases have a radical that forces every additional supporting root either to give an independently paid short path or to be orthogonal to the entire core. This makes the alternative A2 and A13 whole components; their existing endpoint half complements the new A15 half.

In the four D15 cases, the hypothesis that the selected component is a whole D15 removes all contacts with its nodes. Every remaining possible old-core contact has inverse diagonal 4 or 14. Contact with both ends gives a short path; within one

end, positive inverse entries force a projection larger than the new root's norm 2. Hence, in the remaining branch, the entire core is orthogonal to all extra supporting whites. A distinct whole D16 corner retains its half at (1,14)/(1,15), giving total 1 with the new D15 half. These are not the D16 pairs changed above. The unsupported B15 endpoint is removed only after all nonselected and literal interfaces have also been discharged.

The rank16 exchange validates both complete 394-pattern domains, with 238 A16 and 578 D16 maps. Its fourteen affected contexts have either whole-D16 projection sealing or radical sealing. The twelve D16 cases use an existing distinct whole-small/D13 fork half. The two A16 cases use the restored whole A17 half. Every such replacement totals 1 at two distinct actual corners; none relies on a displayed component remaining a proper subdiagram.

These local arguments compose without spending a removed payment:

- The rank15 alternatives use A2/A13 endpoint coupons and the unchanged D16 (1,14)/(1,15) halves. The rank16 alternatives use small/D13 coupons and restored A17 halves. Neither changes the other's needed role.
- The changed rank15 and rank16 selected-context lists are disjoint. The earlier D17 cases use unchanged small/D12 coupons. All these replacements can therefore patch the same complete selected reduction.
- The rank15 certificate reconstructs all 216 inherited literal records after D17=120 and applies rank15 and rank16/A17 changes simultaneously. Exactly 16 D15 corners are debited; every record still totals at least 1. The rank16/A17 exchange affects no corner of those records. This finite list covers inherited literal obligations, not every quadrilateral.

First apply the independently established actual ideal-corner priorities: C11, B12, D13, cycle11, the coupled order11 B/C/D cases, and ideal D17 zero. Their outputs are retained short paths, protected whole components or the budgeted typed coupons. The B16 and B14/C14 projection exclusions use the same independently available short8 alternative. Every new finite decrease above has distance at least 10 and is absent from these routes.

In the remaining branch use the complete selected/nonselected reduction with the explicit replacements above. The low-core bound, high-disjoint exclusion, protected/adaptive cases, universal cycle/tree routes and later interfaces remain as proved in the local reports. A deep finite8 zero has its independent constructive short7 replacement; this route does not invoke the theorem being proved. The retained triangle theorem uses finite distances at most 7 and unchanged ideal interfaces. Thus every triangle receives at least 2 and every quadrilateral at least 1 under this single simultaneous rule.

Complete finite-vertex bound 434

The new finite certificate counts every current coupon and all the changes above. It independently recounts actual base and component-pair costs, checks all 35,721 simultaneous type comparisons, and reconstructs all 1,227 fixed large-component vectors. The representative D16 is conservatively allowed 113, although its actual cost is 225/2, so it still dominates actual A16 of cost 113.

The changes occur only among fixed components: rank15, rank16 and the separately retained A17. Every one of the 66,258 residual item comparisons is unchanged. The 921 complete certified residual tables therefore remain valid. If c15, c16 and a17 are the fixed multiplicities, the exact change in doubled upper cost is

$$-2*c15 - 2*c16 + 2*a17.$$

Applying this shift to every fixed vector gives upper maximum 434. The actual formal forest D17+D45 attains 120+314=434 with no cross coupon, so this is the exact maximum of the actual payment rule. Its use as an attainer asserts no geometric realization. No new Bellman optimization is needed; the existing complete upper certificates are used with identical residual item weights. All actual finite-vertex diagrams are covered.

For a finite link at ambient dimension 62, put d=61. Its H-polynomial has all coefficients 1, so V=62 and

$$G = \sum_{j=0}^{61} \min(j,61-j) = 930,$$
$$\kappa = (5*G - 61*V)/2 = 434.$$

Finite vertices consequently have nonnegative surplus, with lower bound 0.

Every ideal vertex has surplus at least 73

The completed ideal theorem uses the same current ideal weights, unchanged by the new finite transfers. Its complete terminal domain consists of 2,641 rank-61 partitions with at most one part from 1 through 8 and arbitrary repeated parts at

least 9. Every terminal margin is positive. The normalized minimum is 2741/371712, and the absolute minimum is 73.

The established concentration move only combines low ranks; it preserves their linear cost, decreases normalized curvature and decreases the link vertex count V. Positive terminal margins therefore extend to absolute margins for every unreduced actual cusp. The factor-fiber multiplicities are included in that theorem. Hence the actual ideal payment C_v satisfies

```
kappa(L_v) - C_v >= 73.
```

No assumption of simple cusps or arithmeticity enters this argument. The final binder checks all saved exact margin relations and their minima; domain completeness, scalar convolution, concentration and factor fibers are the completed ideal theorem's explicit inputs.

Global contradiction and the higher-dimensional tail

Let D5 be the sum of 5 minus the number of sides over all two-faces. The local payments give

```
D5 <= 434*v_f + sum_(ideal v) C_v.
```

The inherited shape-curvature theorem gives its lower bound with the two-extrema correction. Every coefficient of the relevant simplex or product-of-simplices H-polynomial is at least 1. At d=61 each of the two extreme vertices therefore has

```
E-G >= sum_(2j>61) (2j-61) = 961.
```

Multiplication by 5/2 at each extreme gives the single global correction 4805. This is the same correction already present in the ambient62 frontier, not a second copy of a vertex payment. Thus

```
D5 >= 434*v_f + sum_(ideal v) kappa(L_v) + 4805.
```

Subtracting the two bounds, and using every ideal surplus at least 73, gives the impossible inequality at the start. This excludes dimension 62, including the case v_i=0. The completed bound 62 already excludes every dimension at least 63. The combined conclusion is dimension at most 61.

Compact arithmetic 21 and arbitrary compact 29 are unchanged. No new finite-volume example, external refereeing, publication priority or proof-assistant formalization is supplied by this argument.

Verification record

Final source-bound generation cd6064 and separate fresh replay caf2c7 both terminated exit0/PASS, giving bound 61 and the displayed contradiction. The final binder checks source integrity, the precise actual local costs including A17's increase, simultaneous literal and selected interfaces, the finite witness, all saved ideal margin relations, finite curvature and the single strict global constant. It does not rerun a historical geometric census, finite-forest optimization or scalar convolution. Those completed inputs retain their own recorded verification scopes.

The author read the contributing local, ideal and finite proofs at their recorded scopes. Three peer agents completed full source/report mathematical composition reads. They checked the actual-corner priorities, independent short8/deep-zero route, mutually retained rank15/rank16 alternatives, simultaneous literal debits, counted A17 restoration, conservative D16 allowance and actual D17+D45 attainment, unchanged ideal envelopes and absolute-surplus extension, finite curvature, single extremum correction and completed higher tail. Their additional contributing-source reads are recorded in the respective local and counting reports. None reran a historical census, Bellman calculation or scalar convolution as part of this composition review. No mathematical correction was requested.

Final source/data/report hashes and the completed execution and review scopes are recorded in the checkpoint audit. This proof is not externally refereed or formalized in Lean or another proof assistant. Internal agent reviews are not external mathematician review. No publication-priority or current-best literature claim is made.

```
python3 scripts/fresh_rank15_rank16_dimension61_20261018.py
python3 scripts/fresh_rank15_rank16_dimension61_20261018.py --verify
```

Source 248

Source path: `results/fresh_rank15_sparse_payment_exchange_20261018.md`

SHA256: ae8f89387c6e34ee9a2dec54c6cb47e0c78625d4c340a61d07e20e33d703cc02

Rank15 costs A101, B97 and D102 with all compensating payments counted

This simultaneous local exchange changes the current finite costs from A15/B15/D15=103 to **101/97/102**. It restores the whole A17 pair(1,15) from zero to a unit, raising A17 from115 to116, and upgrades the whole A2 versus D13-short-leaf coupon from half to unit. That upgrade adds2 per A2/D13 component pair; its total actual cost becomes5. These costs are part of the rule, not omitted compensations.

Every triangle still receives at least2 and every quadrilateral at least1. Every ideal payment is unchanged. The complete ambient60 finite maximum remains413 against curvature405, including the new A17 and coupon costs. Consequently no dimension59 theorem follows. The completed unrestricted bound60 remains the current global result.

The producer and certificate bind the current ambient60 frontier, D16/D17 rule, A15 rule, B15/D15 rule and depth10 theorem.

Intrinsic changes and complete selected support

Use the path0 through14 for A15 and B15, with B15's double bond13–14. For D15 use spine0 through12 and short leaves13/14. Apply precisely:

Whole finite type	Changed pairs	Old to new	Saving
A15	(0,12),(2,14)	1 to1/2	1
A15	(0,14)	1 to0	1
B15	(0,12),(1,13),(1,14),(2,14),(3,14),(4,14)	1 to0	6
D15	(1,13),(1,14)	1/2 to0	1
A17	(1,15)	0 to1	−1

Each changed set is invariant under every labelled automorphism. All decreased pairs have distance at least10; every finite short8 and protected leaf9 unit is retained. The existing A15 zeros and D15 halves at(0,13)/(0,14) remain as before.

The inherited selected domain has536 eligible patterns per rank15 type, including patterns with no embedding. The new producer validates every saved native induced map, its complete target labels and marked images: 482 maps for A,240 for B and870 for D. These are the complete saved map domains, not newly executed exhaustive searches.

The A changes meet exactly five contexts:1029,1612,1638,1671,2416. The six B pairs meet none. The D changes meet the same four proper and four whole contexts as the preceding B15/D15 theorem. Selected absence for B is used only with the full inherited routing and later interfaces described below.

Five new A15 whole-component transfers

All five selected marked components already contain15 nodes. When the actual selected factor is A15, these displayed components are whole. The white and white-plus-one-black principal Grams are positive definite, and all four displayed corners are finite.

In1029 and1612 the18-root norm2 core is nonsingular Lorentzian. A further supporting white is orthogonal to the stipulated whole A15. Its remaining possible old-core contact positions have inverse diagonals4,8,14. The inherited complete contact theorem splits these positions into two ends: contacting both ends supplies a full-star short8 path, while each single-end inverse block is entrywise positive. A nonzero native contact has magnitude at least1, so its projection square is at least4, exceeding norm2. Additional contacts cannot decrease that lower bound. Thus, in the no-short8 branch, every extra white is orthogonal to the whole core.

In1638,1671 and2416 the19-root core has inertia(17,1,1). Its exact radical represents zero in an actual Lorentz realization: the span contains a timelike vector, whose orthogonal complement is positive definite. Every additional supporting root annihilates this radical. All allowed contact positions have nonzero radical coefficient. A nonzero nonnegative contact vector must therefore meet both signs; every such pair gives a short8 path. In the remaining branch every additional white is core-orthogonal. These arguments cover arbitrary contact magnitudes and numbers of extra roots. An indirect bridge would have a first root meeting the core and is excluded by the same argument.

Hence the other displayed factors are whole actual factors. The exact marked roles give the following independent payments:

Contexts	Changed selected payment	Distinct retained/restored payment
1029,1612,1671,2416	A15(0,12)/(2,14), half	Whole A16(1,13)/(2,14), retained half
1638	A15(0,14), zero	Whole A17(1,15), restored unit

The first row uses the existing A16 half theorem. Both payments occur at distinct corners. The second row needs just the restored A17 unit. The A17 pair is fixed by path reflection, so this adds exactly one unit per whole A17 factor. The producer checks that it was absent from both the original retained set and the later restored halves.

D15 zeros use an explicitly upgraded coupon

The eight affected D contexts are reconstructed and compared exactly with the preceding B15/D15 certificate. Four proper displayed D14 components require a new root whose forced contact projects to7/2>2; their arbitrary-contact exclusions are unchanged. In the four whole D15 contexts, the remaining inverse diagonals are4,4,14. The same whole-factor and positive-end argument seals every displayed component unless there is a short8 escape.

The preceding rule paid a half at the selected D15 corner and a half at a distinct whole A2/D13 corner. Set the selected half to zero and upgrade the distinct coupon to a unit. Its exact roles are either node of A2 and either short fork leaf of D13, canonical nodes11/12. There are four marked pairs per component pair, so four half-increments cost2. The old broader A2/D13 rule cost3; the new actual total is5. The fork-center half is unchanged. No other small-component or D13 role is upgraded.

This is an actual whole-component coupon justified by sealing, not a coupon on an arbitrary embedded subdiagram. The proper exclusions consume no alternate payment. The whole cases each retain total1.

Simultaneous complete-face interfaces

The producer reconstructs all216 current literal configurations at the D16=111/D17=119 baseline. It applies all new A15/B15/D15 debits together. No literal corner is debited; every lower total remains at least1 even before counting the nonnegative additions. It also checks the positive replacement corners of the earlier rank15, rank16, D17, A15, A16 and D16 transfers. None consumes a newly decreased pair. The changed selected D15 halves are covered explicitly by the upgraded proof above.

The inherited all-label actual-face routing has2614 high-overlap contexts: 1691 have marked whole-component bound at most14;83 use protected/adaptive outputs;304 display more than15 nodes; and536 form the selected rank15 domain. The first group cannot use the new whole rank15 weights, the third cannot embed in a rank15 factor, and the protected group keeps its short/leaf9 outputs. The low-core whole-component bound is12; the high-disjoint branch retains its existing exclusion. Thus the new five A proofs, unchanged four D proper exclusions, four upgraded D whole proofs and absence of B selected images cover every newly changed selected obligation.

Apply the current independent changed-ideal-corner routes before this finite selected reduction. The latest B11/C11 routes output short paths, whole B11/B12, small components or typed coupons. The D105 adaptive outputs are protected distance9 leaves or marked distance9 pairs; none is one of the decreased distance-at-least10 pairs. Other finite/ideal cycle and tree routes retain their independently proved short paths, sealed small factors and separate coupons. All those coupons are unchanged or increased.

If a short8 output is a deep finite8 zero, its established corridor gives a short7 alternative. If that is a newly deep-zero finite7 pair, the completed depth10 theorem independently gives its short6 route. The protected cycle/tree outputs of those routes have distance at most9 or their separate coupons; they consume none of the new long rank15 decreases. This ordering does not invoke the present finite theorem to justify its own short-path premise. If no newly changed actual finite corner occurs, all old actual weights are unchanged or increased. Triangles use the retained finite7/ideal5 interfaces. Hence triangles remain paid by2 and quadrilaterals by1 under one simultaneous rule.

Complete finite comparison, including both increases

The actual A2/D13 cost5 is within the existing conservative representative allowance5, already needed to dominate the separate actual A2/A13 cost5. The producer checks simultaneous domination on both ends of every typed component pair. The updated D15 representative at102 dominates A15=101 and B15=97. A17 remains its own fixed species at116. All other actual base and coupon costs remain unchanged.

The complete ambient60 domain still has1,044 fixed vectors and792 residual keys. No residual base cost or upper interaction changes, so the frozen residual Bellman tables remain applicable. The producer checks every upper inequality and attaining predecessor again, along with all residual item identities and zero within-group upper interactions. For each complete fixed vector, its new doubled upper is exactly

```
old doubled upper + 2*(number of A17 - number of D15).
```

This explicitly includes the A17 increase. The maximum remains826/2=413, attained by the unchanged formal forest2D17+D26. The depth10 deletions cannot increase this upper and do not affect that attainer. There are15 representative alternatives above405, down from17 in the prior frontier. Four A15, four B15 and four D15 now cost404,388 and408 respectively; the rank15 obstruction has narrowed to D15 and mixtures with enough D15. The overall eight-unit finite deficit remains, and the ideal weights still have the prior77 negative terminal profiles. No geometric realization of these formal forests or new global dimension bound is claimed.

Verification record

Generation session93020, final chunk0ca025, and separate fresh saved replay session38100, final chunk0a64d9, both terminated exit0/PASS. Two peer agents completed full source/report mathematical reads, including the inherited end-projection, D15 proper/whole and A16-half interfaces. Both checked the five arbitrary-support A15 seals, exact A17 restoration, distinct upgraded D15 coupon and its cost, complete selected/literal and positive replacement interfaces, current ideal/depth10 priorities, and the complete finite fixed-vector shift with unchanged attainment. Neither reviewer repeated map, inertia, census, literal, Bellman or scalar arithmetic. No mathematical correction was identified. These are internal agent reviews, not external mathematician refereeing or formal verification.

The producer checks exact Fraction core geometry, all saved rank15 maps,216 simultaneous literal records, previous positive finite interfaces, typed budget domination and the complete saved finite Bellman certificates. It shares the predecessor geometry/classification helpers and does not repeat the historical all-label census, native-map search or a new optimization. Written universal routing remains an explicit inherited proof input. Source, certificate and report are frozen after generation, fresh replay and the two completed reviews.

```
python3 scripts/fresh_rank15_sparse_payment_exchange_20261018.py
python3 scripts/fresh_rank15_sparse_payment_exchange_20261018.py --verify
```

Source 249

Source path: `results/fresh_rank16_existing_payment_exchange_20261018.md`

SHA256: 2d5d5dc422017c6602d5b0fdc0e986e78f3c37e4ef6b3a53c3b47ba9881ce7f5

Rank16 savings by an existing-payment exchange

The current finite rule admits the following simultaneous change, with no new cross-component coupon:

Whole component	Changed pairs	Change	New component cost
A16	(0,11),(4,15)	unit to half	113
D16	(2,13),(2,14),(2,15)	unit to half	225/2
A17	(1,12),(4,15)	zero to half	115

A paths use nodes0 through n−1. D_n uses spine0 through n−3 and short leaves n−2,n−1 at n−3. The changes are invariant under diagram automorphisms. Every decrease has distance at least10; finite short8 and actual whole-leaf9 payments are untouched. All ideal weights and all cross-component coupons retain their values, including the latest D17=120 finite rule and its existing small/D12 coupons.

Every triangle still receives at least2 and every quadrilateral at least1. This is a local payment exchange, not a new finite-forest maximum or ambient bound. In particular, the A16 saving is accompanied by an actual A17 cost increase. A later finite count must include that increase.

The source and certificate bind the sparse finite rule, the D17=120 refinement, the current ideal-transfer interfaces, and the existing whole-small/D13 fork coupon. The saved complete rank16 map domains are consumed and validated, not re-enumerated.

Bounded support diagnostic

The frozen complete domains have394 eligible selected patterns each, with238 induced maps into A16 and578 into D16. The new checker validates every saved map's injectivity, every edge and nonedge label, and its marked images against the original literal core. Completeness of those domains remains an inherited input. No new82,197-identification or2,614-context census is run.

The certificate inventories every automorphism orbit of a remaining positive rank16 payment at distance at least10, together with its selected supporting contexts. This selects a small constructive exchange rather than treating absence from an incomplete search as permission to delete a payment.

The proposed changes affect precisely fourteen selected contexts:

Changed component	Contexts	Mechanism
A16	1657,2067	full radical core; restore the opposite A17 half
D16	1018,1035,1049,1265,1905,2298	whole selected D16 and a Lorentz projection bound
D16	1305,1964,2357,2409,2421,2429	full radical core; existing small/D13 half

All are full16-node selected components. No proper extension of a shorter displayed component into a new target must be guessed or classified.

The six Lorentz cores seal the other finite corner

For each of the six nonradical D16 contexts, the old18-root physical Gram has signature(17,1,0), with all root squares2. The selected actual corner is precisely the displayed whole D16. Any further supporting white belongs to that corner, so it must be orthogonal to every root of that whole component. A contact would enlarge the component and contradict the hypothesis that the marked whole component is D16.

The inherited complete necessary contact domain then leaves only the two omitted black roots as possible old-core contacts. One lies in each of the two contact end sets. Their inverse-Gram diagonal entries are4 and8. The source checks the complete contact domain, both positive inverse end blocks, the selected whole component and this exact remaining set in every context.

If an extra white contacts both ends, the inherited distance check gives an actual opposite-black path of length at most8. The independent current short-path payment described below resolves that case. Otherwise its nonzero contacts lie in a single end. Every nonzero native norm-two contact has magnitude at least1, and the inverse block is entrywise positive. Its projection to the old Lorentz span would therefore have square at least4, exceeding its own square2. The orthogonal complement of a nonsingular Lorentz span is positive definite, so this is impossible. Larger bonds or extra same-end contacts only strengthen the bound.

Thus, in the no-short8 branch, every further supporting white is orthogonal to the entire old core. This permits arbitrary extra roots and arbitrary native labels; it does not set unspecified entries to zero by convention. An indirect spectator bridge would require a first root contacting the core and is excluded by the same argument.

All four old corner supports are checked positive definite, as are the white support and every white-plus-one-black edge support. Their displayed components therefore remain whole in the actual supporting diagram. At a corner distinct from the selected D16, the two marks lie in whole A1 and whole D13, the latter marked at its fork center for1018/1265 and at a short leaf for the other four contexts. The existing whole-small/D13 coupon pays one half there. Together with the retained selected D16 half, the face receives1.

The eight radical cases

The remaining eight contexts use the same19-root L25/L25 core as the frozen sparse theorem. It has signature(17,1,1), positive-definite15-root white support, positive-definite16-root edge supports and four positive-definite17-root corners. The source checks its radical and the full necessary contact domain again.

The represented radical is zero in an actual Lorentz realization: it is null and orthogonal to a span containing a timelike vector. Every extra supporting white must annihilate it. Each possible contact position has nonzero radical coefficient. A nonzero acute contact vector annihilating this radical must touch both coefficient signs; every such pair gives a checked opposite-black path of length at most8. Consequently either the independent short-path route pays, or all further supporting whites are core-orthogonal and every displayed corner component is whole.

In the six D16 radical cases, a distinct corner has whole A4 and whole D13. The A4 mark may be an internal node; the existing coupon allows any marked node in any whole small component of rank at most5. The D13 mark is its fork center or one of its short leaves. This existing half again complements the retained selected D16 half. No A4-endpoint restriction or new coupon is introduced.

For A16 contexts1657 and2067, the opposite finite corner is a whole A17 with marked images(1,12) or(4,15). Those pairs were previously zero. Restore them to one half. The selected A16 pair now pays one half, so these two distinct corners give total1. The old radical proof seals the A17 component as well; a displayed A17 subdiagram inside a larger component would not suffice. The source reconstructs the literal components and both canonical marked images, rather than inferring their role from the component order alone.

For every one of the fourteen contexts the saved proof displays exactly the two required half-payments. Other nonnegative payments may exist but are not needed.

Complete literal and universal interfaces

The latest D17=120 certificate supplies216 complete literal configurations, including three- and four-corner replacements. The new source reconstructs their actual corner components and transports the proposed A16/D16 decreases and A17 increases simultaneously. None of the changed marked roles occurs in those216 configurations. Every recorded payment and every total is therefore unchanged, with minimum total1. This includes the four retained ideal D17 halves explicitly checked by the preceding refinement. The216 records are inherited literal obligations, not a census of all quadrilaterals.

The original complete finite16/17 theorem supplies the remaining coverage. The low-core bound is at most12. Protected and adaptive branches use basic ideal payments, finite short paths, actual whole-leaf9 pairs, or their two explicitly protected whole-order16 distance9 cases. None uses a new distance-at-least10 decrease. The all-label high-disjoint branch is excluded. Every other selected high-overlap component that can lie in whole A16 or D16 is in the394-pattern domains consumed here. The fourteen affected contexts are treated above; all other selected payments retain their values.

The later universal tree/cycle interfaces give finite short paths, protected whole-leaf9 roles, their specified small whole components or cross-component coupons. Their exceptional later A16 uses are among the216 literal configurations. The more recent order11 outputs are whole finite components of order at most14, protected A9/A18 endpoint8, D18 leaf9, E8 path6, or typed coupons. The current D13-center theorem uses whole D10-center coupons, short paths or retained order11 center units. B12 uses whole D12 leaf9 or its coupons; ideal D17 zero uses a distinct whole E6/D11 unit. These outputs avoid the new A16/D16 long-pair decreases.

Apply the independent actual ideal-corner priorities first: C11, B12, D13, cycle11 and the coupled order11 B/C/D transfers, followed by ideal D17 zero. Each paying branch remains valid under this finite exchange by the explicit outputs just listed. The B16 and new B14/C14 projection or residual exclusions use the same independent current short8 route and introduce no new finite long-pair requirement. After those priorities, use the complete current selected/nonselected reduction with the fourteen replacements above and the unchanged literal obligations.

If a short8 alternative reaches a deep finite8 zero, the independent depth11 corridor supplies its constructive short7 replacement. The completed small-tree, cycle-tail and protected-feature interfaces pay it without either new distance-at-least10 decrease. Thus the short8 branch used to seal a core does not invoke the unchanged-corner case of this theorem recursively. The A17 restoration only increases payments, so it cannot invalidate any earlier A17 endpoint deletion, sparse transfer, ideal priority or literal replacement. It also leaves the old cross-component A17 positions5/11 coupons untouched.

Triangles retain their completed finite7/ideal5, protected6/7 and whole-small-factor routes. No new decrease occurs in those routes. This proves simultaneous local face payment under the current ideal rule. The finite count is a separate obligation and must use A16=113, D16=225/2 and A17=115 together with every unchanged coupon.

Validation and limits

The initial bounded generation43721 exited0/PASS (terminal f41358). Final source-bound generation5404 exited0/PASS (terminal9b0cdf), and separate fresh replay86162 exited0/PASS (terminal72c267), including the explicit current B14/C14 bindings and A17 reflection guard. The source checks816 saved induced maps, fourteen complete local cores, all strict Lorentz projection inequalities and radical short-path interfaces, and216 simultaneous literal records. Metadata check eb67ae passed173 source bindings, four local links and the Python syntax tree.

Root and an independent sibling completed full producer/report mathematical reads: PASS. Both also reread the inherited endpoint-contact constraints and sparse/literal interfaces. The reads checked whole-selected-D16 orthogonality, the complete remaining omitted contacts and inverse4/8 projection bounds, radical mixed-sign sealing, whole-factor alternate halves, A17 reflection and restoration, all saved-map/current literal obligations, and current ideal/deep11/B14/C14 and nonselected priorities. Neither review repeated map enumeration, inertia or literal arithmetic, the historical census, or the fresh numerical execution. Saved-map completeness remains explicitly inherited. Source, data and report are final and frozen; no owned process remains active.

There is no new global forest optimization, scalar computation, historical overlap census, external review or proof-assistant verification in this packet. The separate frozen rank62 budget still states its previously proved maximum438 until a successor incorporates these component changes. No ambient61 claim or new geometric construction is made here.

Source 250

Source path: `results/fresh_rank17_reduced_forest_frontier_20261017.md`

SHA256: 29cca978768eb0ab3dceec4c9cb9a7a046ae770e8b15b3203f6484b13e9914df

What reduced finite rank17 costs would suffice to prove

Reducing whole finite A17/B17/D17 costs to114/105/127, while preserving all current coupons and face coverage, would make the existing comparison exclude ambient dimensions67 and68. With the completed ceiling68, that would give ceiling66. This package proves the exact finite counting under explicit assumed rank17 costs. Geometric coverage and the final global composition remain separate obligations; it makes no new dimension claim.

The source and certificate evaluate two abstract rules. All costs other than the specified rank17 bases, and all finite cross-component coupons, are unchanged from the completed68 proof.

Rule	A17 base	D17 base	Required B17 upper bound
Mild reduction	131	131	131
Full proposed reduction	114	127	127

The proposed geometric B17 cost105 would satisfy the second upper bound. Neither rule is inferred merely from an absence in a partial selected-map list. The geometry must independently establish full face coverage.

Complete abstract finite-forest count

Keep the inherited finite-type domination in ranks other than17. At rank17, keep A17 as a separate coupon-bearing type and D17 as the residual type. B17 has no such coupons and is dominated by D17 under the explicit base-cost hypothesis. Do not identify A17 with D17 merely because they have the same rank.

The finite forest is partitioned into small residual components of ranks 1 through9, residual D components of ranks17 through75, and the large coupon-bearing types B10,D11,…,D16,A17. Enumerate every feasible vector of counts of the latter types:1,608 vectors through rank75. Their coupon environments give675 distinct additive Bellman tables, each with76 states including rank0. Every state maximizes over every residual component rank that fits; the saved last-rank predecessor supplies an attaining forest. Combining every large vector with every residual rank proves the upper bound and attainment at each total rank.

Writing a for total rank in components of rank at most5 and t for the number of D11 through D16 components, the doubled cross-cost is

```
a*(3*t + 2*b_A17)
+ 2*a1*(b_D11+b_D12) + 2*a2*b_B10 + 2*a4*b_D13.
```

The full D11 and D12 upgrades replace their old halves rather than count them twice. The two A11 cycle coupons are also included in the bound: the frozen A11-to-D11 substitution still preserves rank/base and dominates every interaction, since no rank11 cost or coupon was changed. Every saved maximizing forest omits A11, so the maxima remain attained.

Ambient rank	Mild maximum	Mild finite slack	Full maximum	Full finite slack
66	502	−7	496	−1
67	513	−3/2	504	15/2

Ambient rank	Mild maximum	Mild finite slack	Full maximum	Full finite slack
68	524	3	512	15
69	528	16	520	24

The full-rule maxima in these four ranks are attained by the single finite components D66,D67,D68,D69. Their unchanged cost is$8n-32$. Hence removing the rank17 obstruction exposes a different finite obstruction at ambient66: D66 costs496 against curvature495.

The mild rule would suffice only to exclude68, giving ceiling67 once coverage were proved. It fails the finite comparison at67 by3/2. The stronger rule fits both67 and68; these statements must not be conflated.

The unchanged ideal comparison

The source freshly enumerates the completed68 theorem's concentration domain and imports its exact polynomial arithmetic. Arbitrarily repeated ranks are10,11,13,14,15,16,17; at most one part lies in1 through9, at most one is12 and at most one is at least18.

At ambient67 all1,761 terminal profiles have positive margin. The minimum is851/165888, attained by six rank11 factors. At68 all1,906 are positive, with minimum411499/3234816. At69 all2,060 remain positive. The exact finite slacks above and the positive global constants330/335 would therefore exclude67/68 under the full rule. Ceiling68 supplies n>=69.

Ambient66 still has a negative ideal margin$-1565/10976$ at five rank13 factors, in addition to its finite deficit1. No ceiling65 follows from these costs. The ideal comparisons are inherited-formula calculations, not a new independent implementation of that arithmetic.

Verification scope

Generation22631 and fresh separate-process replay38315 terminated exit0/PASS after explicitly distinguishing the mild and full rules. A separate standard-library certificate verifier independently reconstructs both1,608-vector domains and675 coupon tables per variant. It checks every Bellman upper alternative and attaining predecessor, every global upper bound and every typed attaining witness. Its report records final bound generation and replay. A full mathematical source/report read passed the rank17 A/D separation, coupon counts, unchanged A11 domination and the distinct mild/full conclusions. The ideal profiles are not claimed as independently reconstructed by that finite-budget verifier. Source, data and report are final/frozen. The finite-type classification, unchanged type domination, A11 coupon replacement and convex-order lemma remain frozen proof inputs. No external refereeing or formalization is claimed, and this abstract budget alone is not an ambient dimension proof.

Source 251

Source path: `results/fresh_rank17_reduced_forest_frontier_20261017_review.md`

SHA256: 83b6d6421859f39d215e3193e7fe874b6d562e4cea987a4e43183e8fd8523dd5

Independent conditional rank17 finite-budget certificate

Both abstract finite-forest budgets in the primary report pass an independent standard-library certificate verifier. Under assumed whole-component costs A17=114 and B17≤127, D17=127, the exact finite maxima at ranks67 and68 are504 and512, attained by D67 and D68. Their finite curvature margins are15/2 and15. The weaker A17=B17=D17=131 envelope has maxima513 and524, with margins−3/2 and3.

This verifies the finite counting conditional on the stated rank17 costs. Geometric face coverage and a final global dimension argument are separate obligations. This certificate does not by itself assert a new dimension bound.

Independent source · saved certificate

Independent domain and coupon reconstruction

The checker imports no project or producer arithmetic. It enumerates large components by their number and a nondecreasing tuple of ranks, independently of the producer's successive-rank count recursion. All1,608 multisets of B10,D11,…,D16,A17 through total rank75 agree with the saved domain. Their six coupon counts give exactly675 distinct environments for each variant.

The independent component-pair coupon function includes all inherited broad small-component halves, the A17 coupons, the full A1/D11 and A1/D12 upgrades, the A2/B10 and A4/D13 additions, and the A1/A11 and A2/A11 cycle coupons. Checking every partner type verifies rank-preserving type domination for228 finite type representatives, including an I2(*) representative for all labels at least4. The finite Coxeter classification and unchanged same-component graph costs remain inherited inputs. The checker independently counts the coupon interactions; it does not reproduce the geometric coupon-transfer proofs.

Rank17 requires two representatives. A17 has its separate interaction with small components, whereas B17 is dominated by residual D17 under the stated B17 upper bound. The algorithm keeps A17 among the explicit large components and D17 among the residual Bellman choices; it never merges them merely because their ranks coincide.

The A11 coupons are included explicitly before domination. Replacing A11 by D11 preserves rank and same-component base cost and increases every relevant coupon interaction. Thus their omission from the maximizing representative list does not omit a payment obligation.

Complete Bellman certificates

Residual choices are A1 through A9 and D17 through D75. Once a large-component multiset is fixed, all their cross-costs with those large components are additive. Residual components have no mutual coupon interaction. A dummy D14 is used only to reconstruct the common interaction profile of the residual number of D14/15/16 large components; it is not asserted to preserve their total rank.

For each environment, the saved value V(n) is checked against every fitting residual rank r:

$$V(n) \geq V(n-r) + c(r).$$

Every positive n also has a saved fitting predecessor attaining equality. Induction from V(0)=0 therefore proves both an upper bound for every residual forest and attainment by a forest; no optimizer or trust in the producer's maximization is needed.

For **each** of the two variants the checker verifies:

Certificate part	Count
Complete large count vectors	1,608
Distinct Bellman tables	675
Bellman upper-bound alternatives	1,626,075
Attaining Bellman predecessors	50,625
Global upper-bound alternatives	22,504
Positive-rank global maxima	75

At every total rank0 through75, the saved global value dominates every fitting large-vector/residual combination. Its explicit typed forest is independently recounted, including all coupon interactions, and attains that value. Exact rational finite curvature coefficients and slacks are also checked.

Rank	Mild maximum	Mild slack	Full maximum	Full slack
67	513	−3/2	504	15/2
68	524	3	512	15

All saved maxima for the full variant use actual representatives whose costs attain the asserted bound. In particular the witnesses D67/D68 have no rank17 component or cross coupon. A stricter geometric B17 cost105 therefore leaves these displayed maxima unchanged.

Scope and validation

Final generation and a separate fresh saved-certificate replay both terminated directly with exit0/PASS. Both process logs record the counts above. Sixteen direct source bindings, Python syntax, and all local report links pass. The primary source/data/report hashes are bound in the saved review.

The reviewer read the complete primary source and mathematical report, including the two-type rank17 distinction, unchanged A11 domination, and conditional versus global scope. Root separately read the entire independent verifier: complete multiset domain, dummy-interaction construction, type domination, every Bellman alternative and attaining predecessor, global attainment, and the B17≤D17 assumption all passed. Root did not duplicate the1.6million alternative checks per variant.

The primary ideal-profile comparison uses the frozen dimension68 arithmetic. This review does **not** independently reconstruct those ideal profiles, rerun a physical Gram census, or prove the proposed finite rank17 deletions. Those remain separately identified inputs to any global composition. Source, data, and this review are final/frozen.

Source 252

Source path: `results/fresh_rank61_complete_finite420_budget_20261018.md`

SHA256: b9f1d86d93e978a3fcb36f875a092f9d8e9a43900e678a31a730d46ddfbaab3c

Complete finite rank61 maximum420

The simultaneous proved finite rule has exact rank61 maximum420, equal to the finite curvature coefficient. All added coupons and the D17 restoration are counted. This closes the finite counting deficit at ambient61; it does not by itself supply the missing positive ideal comparison or a dimension60 theorem.

The source and certificate start from the frozen depth8 and D17 budget and explicitly incorporate all new local costs and coupon roles. The producer computes new residual tables; it does not reuse tables with obsolete D17 or A2 interaction costs.

Simultaneous rule and actual charges

The simultaneous same-component costs are:

Rank	A	B	D
15	103	103	103
16	111	104	111
17	115	105	119

Every other same-component cost remains the frozen depth8 cost. In particular the D17 exchange restores two halves to the preceding118, giving119; it is not silently counted as118. The earlier A17 restoration remains115. All four new cross-component charges are explicit increments beyond the frozen425 rule:

Whole component pair and marked roles	Increment per component pair
A2 either node / A13 endpoint, half upgraded to unit	2
A1 / A13 endpoint, new half	1
A2 either node / A14 third node, new half	2
A4 endpoint / A12 third node, new half	2

The A15 theorem, B15/D15 theorem, A16 theorem and D16 exchange are final frozen inputs. Their actual marked choices, weights, costs and retained alternatives are checked explicitly. These are all the extra finite roles in this packet. In particular no prospective ideal order11 change is assumed.

The finite transfers compose using the independent current ideal-corner priorities and generic short8 escape. The depth8 corridor replaces any new deep finite8 zero by the completed short7 cycle/tree route. The long-pair theorems check their complete selected domains and simultaneous latest216 literal obligations separately. Added coupons and restored D17 halves can only increase other payments. Their precise whole-face coverage is supplied by the bound local proofs, not by this counting argument.

Type domination and exact cost allowance

The predecessor's finite classification gives the complete finite types used here. Whole I2 families with nonstandard labels have the same counting data as the retained I2 representative. Use paths through rank8 except that E6 is retained separately,

B9 at rank9, D-types from rank10, and a separate A17. Every actual type has the same rank as its chosen representative. The source checks its base cost is no larger and checks every ordered pair of the186 types through rank61 simultaneously:

```
actual_cross(type1,type2)
    <= upper_cross(representative1,representative2).
```

This is simultaneous two-end replacement, not an unjustified sequential replacement of only one component. It retains the distinctions needed for B9, E6 and A17.

Only one new conservative allowance is necessary: add2 to the A2/D13 representative cross cost. This absorbs the A15 upgrade, whose actual A2/A13 total is now5. It is not an actual new A2/D13 geometric coupon. The other new actual totals are absorbed by unchanged allowances: A1/A13 becomes3/2, A2/A14 becomes4 and A4/A12 becomes6, matching A1/D13, A2/D14 and A4/D12 respectively. Every pair is checked rather than relying solely on these examples. Actual and conservative costs are recounted separately in the saved alternatives.

The source also records a narrow optional inequality: upgrading the existing A1/B10 half to a unit would still be dominated by the A1/D10 allowance1. This is an accounting observation only; neither its local geometry nor the ideal reduction it might support is assumed here.

Complete forest domain and new residual certificates

Separate the fixed types B9,D10,…,D16,A17 from the residual types at ranks1 through8 and D-ranks at least17. Their complete multiplicity domain of total rank at most61 has1,132 vectors. The unchanged seven interaction coordinates distinguish854 keys. All coupon costs cross from the fixed block to the small residual block; there are no fixed–fixed or residual–residual coupon interactions.

For each key and residual rank r, the item weight is its current base plus its cross cost with every fixed component. The producer checks this literal identity for every fixed vector and every allowed residual rank. Relative to the425 item formula, rank2 gains4*d13 in doubled units, and residual D17 gains2. Both alter the optimizing recurrence, so every residual table is generated anew.

Generation uses an item-first unbounded integer dynamic program. Fresh replay takes the saved table U and checks all last-item upper inequalities

```
U(s) >= U(s−r) + item(r)
```

and a saved equality predecessor at every positive s, starting with U(0)=0. Induction proves both that U bounds every residual forest and that its value is attained. This verification does not rerun the optimizing recurrence. Every fixed vector is then combined with the appropriate residual certificate. Thus neither a proposed maximizer nor a restricted sample stands in for the complete domain.

Attainment and remaining scope

The complete upper is420, attained by actual formal forests D61, 2D17+D27,3D15+A16 and3D15+D16. Their literal current costs are recounted directly. The two saved upper-attaining fixed-vector alternatives have actual costs420 as well; no conservative allowance is needed for their attainment. An actual formal-forest attainer proves sharpness of the counting rule, not existence of a hyperbolic polytope with that link.

At ambient61 the finite link has dimension60 and61 vertices, with G=sum(min(j,60-j),j=0..60)=900. Its curvature coefficient is420. Hence the finite slack is0. An ambient60 theorem still requires an independently proved positive ideal comparison and the global incidence composition.

Final generation `37b845` and separate fresh replay `01904f` both exited successfully with PASS. They check34,596 simultaneous type pairs, 59,996 residual item identities, all1,132 fixed vectors and854 tables. Fresh replay verifies1,276,730 last-item upper inequalities and52,094 attaining predecessors, without rerunning the item-first optimization. Historical classification, native contact and all-label geometric censuses remain the bound local inputs. No ideal convolution or global model optimization is performed. The root and both peer agents completed full source/report mathematical reads, including the bound counting-engine interfaces. They checked all actual versus conservative coupon costs, the complete fixed/residual decomposition, changed item identities, last-item induction and literal attainment. None repeated the DP, typed-pair arithmetic

or geometric census. The local geometric theorems retain their separate completed review scopes. No new ambient bound follows here.

```
python3 scripts/fresh_rank61_complete_finite420_budget_20261018.py
python3 scripts/fresh_rank61_complete_finite420_budget_20261018.py --verify
```

Source 253

Source path: `results/fresh_selective_distance10_dimension116_20261017.md`

SHA256: 4bbbff927ab2e6fc34fffa39471905017d2300a5aa953077c4a94b9f068d3c38

Selective distance 10 and controlled supporting extensions

Complete internally checked computer-assisted proof. Primary generation and a fresh saved-output replay pass. A separate mathematical/source review and independent exact-arithmetic reconstruction also pass; the linked review records its final source bindings and saved replay. No unresolved mathematical step is identified in the argument below.

The argument below gives dimension at most **116** for every finite-sided, finite-volume real hyperbolic Coxeter polytope. It requires no arithmeticity, compactness, simple-cusp, affine-factor-rank, or facet-count assumption. Its compact-facet consequence is ambient dimension at most **87**.

The primary source and exact certificate record the complete finite split. A separate review reconstructs the arithmetic and checks the geometric argument. The preceding dimension-126 proof supplies the inherited incidence and all-rank estimates and the excluded higher-dimensional tail.

Actual face setup

For an actual quadrilateral, write W for its supporting white roots and U,V for the two opposite pairs of black side roots. Normalize every root to square 2. W and W together with any one black root are positive definite; W with an adjacent black pair is positive semidefinite. The opposite sides do not meet, even in the closed quadrilateral, so W+U and W+V are indefinite. Choose minimal indefinite cores L and M. Each contains its two black roots, and deleting either black root gives a positive-definite diagram.

Assume there is no path of length at most 10 from U to V with white internal nodes. The complete order-6 through order-10 catalogue has 384 marked contexts, with maximum marked radius 7. Cores of order at most 4 have radius at most 2. Order-5 cores have radius at most 3 and require a separate treatment in the disjoint case below.

An external supporting white contacting node i of a black-deletion Gram A has squared projection at least (A inverse)ii. This uses the entrywise nonnegative spherical inverse, nonpositive contacts, and minimum nonzero Coxeter magnitude 1. If either black-deletion inverse diagonal is at least 2, positive definiteness of W plus the remaining black root is impossible. Equality 2 is excluded, since this is an edge-support test. Of the 527 high-order ports at marked distance at least 3, exactly 161 pass.

Whenever a required positive-definite diagram fails, the producer records a nonnegative nonzero vector with nonpositive square. Symmetric elimination of a Z-matrix yields such a vector at its first nonpositive pivot; the saved vector and square are checked exactly. Its sign makes the obstruction valid after arbitrary additional nonpositive contacts or larger bond magnitudes are restored. Label-4 diagrams are rationalized by a positive diagonal congruence with squared scales 1 or 2. Inverse diagonals are returned to the norm-two gauge before comparison with 2.

Disjoint high-order cores, including multiple cross edges

Disjoint indefinite cores cannot be orthogonal in an ambient Lorentzian space. Each cross bond joins white ports with marked distances summing to at least 10. Each distance is therefore at least 3.

The 161 allowed high-order ports give 1,478 ordered candidate joins. In 363, the simple cross bond already gives a nonnegative vector of nonpositive square in W. The other 1,115 possible edges occur in 1,007 ordered marked-context pairs: 905

admit one edge, 96 admit two, and six admit three. Thus one cannot assume uniqueness of the cross bond at this distance threshold.

Every possible cross bond is nevertheless simple. Each endpoint has a three-node white path. Coefficients (1,2/3,1/3) on that path have square at most 4/3, since native bond magnitudes are at least 1. The two paths together have square at most 8/3-2 beta for cross magnitude beta; additional acute contacts only decrease this value. Positive definiteness forces beta<4/3, leaving only the simple Coxeter magnitude 1. This excludes all larger finite labels, parallel contacts, and dotted contacts without a label cutoff.

All 35 underlying single-edge Grams have exactly two negative directions. For the context pairs admitting multiple edges, the source tests every subset of size at least two. All 120 such choices have a nonnegative nonpositive-square witness in W plus one black root. Thus every nonempty subset is impossible, and no unspecified cross bond can repair the single-edge inertia obstruction.

Complete small-order supplement

The radius inequality alone does not exclude an order-5 core here: radius 3 plus radius 7 permits a connecting edge at distance 11. This supplement is essential. Orders at most 4 are excluded because 2+7+1<=10, and two small cores are also excluded.

Number the black roots 0,1 and a white root at marked distance 3 by 4. The other whites must form the path 4-3-2. Neither black can contact 3 or 4. Connectivity leaves exactly three tree attachments:

```
0–1–2–3–4;
1–0–2–3–4;
0–2, 1–2, 2–3, 3–4.
```

The only additional possibility would contain the triangle 0,1,2. Three nonright Coxeter contacts give a nonpositive square on the positive all-ones vector, with equality only for the affine simple triangle. In the equality case its attached white 3 makes the proper four-node diagram indefinite. Thus the triangle is impossible.

All edges in these five-node trees have finite labels at most 5. Indeed, each edge belongs to a connected proper four-node principal diagram, which must be positive semidefinite. An edge of magnitude at least sqrt(3), together with an adjacent nonright edge, has a three-node Gram determinant at most zero. Strict inequality contradicts semidefiniteness; equality is the affine [6,3] path with positive kernel. Its connected fourth-node extension again contradicts semidefiniteness. Parallel and dotted contacts are excluded by the same argument. Hence enumerating labels 3,4,5 is exhaustive, rather than a search cutoff.

The 3 times 3^4=243 marked tree recipes split exactly as follows:

Outcome	Recipes
Indefinite proper principal diagram	209
Full diagram positive semidefinite	14
A black deletion is not positive definite	4
Far-port inverse test fails	12
Surviving far ports	4

For the semidefinite tests, the one- and two-node minors are positive from the allowed bond alphabet; every larger proper principal minor and the full determinant are checked exactly. The four survivors are the two black orientations of the path [5,3,3,3], and the fork with respective edge labels (3,4,3,3) or (4,3,3,3). Their far-port inverse diagonals are (4/5,3/4), (3/4,4/5), (1,4/5), and (4/5,1).

After the high-order inverse test, the only radius-7 ports are L25/node 9 with black pairs {1,2}, {1,3}, {2,3}. In each small/high pair the only possible cross edge is 4-to-9. The same white-tail argument makes it simple. Each of the four resulting 15-root Grams has exact inertia (13,2,0), excluding all 24 ordered marked joins. The two labels 5 in this small supplement are handled exactly over the real algebraic field, not by rounding their cosines.

Complete overlap split and fixed core Grams

Every shared white root has marked distances summing to at least 11, so both distances are at least 4. Small-order cores therefore cannot occur in an overlap. Each marked high-order context has at most four eligible nodes. Enumerating injective partial identifications gives 22,881 candidates and 4,862 Gram-consistent overlaps.

External-neighbor inverse tests exclude 4,296, and explicit nonnegative vectors in W plus one black root exclude 254. The remaining 312 contexts are exactly the following; mixed counts include both orientations.

Core intersection	Contexts	Core order	Extension mechanism
L10/node 5 = L25/node 9	6	15	Actual radical relation
L18/node 7 = L25/node 9	6	17	Actual radical relation
L22/node 4 = L25/node 9	18	18	At most one zero-residual root
L22/node 8 = L25/node 9	54	18	Actual radical relation
L22 terminal edge = reversed L25 terminal edge	18	17	Residual 1/2; at most one attached A1
L25/L25 shared node 9	114	19	Actual radical relation
L25/L25 reversed terminal edge	60	18	No additional core contact
L25/L25 reversed terminal three-node path	27	17	At most one zero-residual root
L25/L25 reversed terminal four-node path	9	16	Residual 0 or 2/3; at most an A2 tail

Every unspecified bond between the exclusive core parts is tested. Exactly 21,393 create a path of length at most 10. The other 168 have nonnegative nonpositive-square witnesses in W plus one black root. Stronger or additional nonpositive bonds cannot remove these obstructions. Thus the actual Gram is fixed in all surviving cases before the extension arguments are applied.

Preferred corners and actual radical relations

For each surviving context, the certificate checks all four adjacent corners and chooses one whose marked roots lie in a connected component of order at most 17. Every corner is decomposed into finite or affine components. A new white can contact a finite component only at normalized inverse-diagonal entries at most 2; it is orthogonal to an affine component by the positive kernel. The two omitted black roots are included separately in the possible contact domain.

In the radical cases the Gram has one negative direction and one null direction. The radical gives an actual root relation: its linear combination is orthogonal to a timelike vector in the core span and has square zero, so it vanishes in the positive-definite perpendicular space. For rationalized Grams, the positive diagonal scaling changes coefficients but not their signs or zeros.

The selected contact domains contain no zero-radical coordinate. The certificate assigns the two radical signs to the two marked pairs and checks that any pair of opposite-sign contacts gives a marked path of length at most 10 through the new root. Acuteness and the actual relation force both signs whenever there is any contact. Hence every additional supporting white is orthogonal to these cores. Their marked components are isolated with order at most 17.

Nonsingular cores and the residual norm budget

All other surviving cores are nonsingular Lorentzian. In each preferred corner, the allowed contact domain splits into two end sets, each within distance 4 of its own marked pair. A root contacting both ends gives a path of length at most 10. Within either end, the full inverse-Gram block is strictly positive.

For a norm-two new root with core contact vector b, write its orthogonal decomposition as e=e_core+w. The perpendicular space is positive definite, so

```
b^T G^{-1} b <= 2,     w^2 = 2-b^T G^{-1}b.
```

Exact inverse entries show that two nonzero contacts at one end exceed the budget. Every permitted diagonal is greater than 1, so the next Coxeter magnitude sqrt(2) also exceeds the budget. Consequently a nonorthogonal supporting white must have a sole contact -1 at a listed node i. Its projection square h_i=(G inverse)ii is one of

```
2, 3/2, 4/3,
```

with residual square rho_i=0,1/2,2/3, respectively.

There is at most one such core-contacting white in total. For each pair of candidate nodes i,j, the exact certificate checks

```
((G inverse)ij)^2 > rho_i rho_j.
```

If the inverse entry is positive, Cauchy's inequality makes the actual product positive, violating acuteness. If negative, the actual product is strictly negative and therefore creates an edge between the two new whites. The certificate checks that this edge gives a marked path of length at most 10. The same test with i=j excludes two distinct roots at a single candidate node. In the zero-residual case the root is uniquely determined in the core span as well.

It remains to control whites attached indirectly through that one root. All other supporting whites are core-orthogonal. Their Gram E is a positive-definite principal submatrix of W. If c is the extra root's contact vector with these whites, projection gives

```
c^T E^{-1} c <= rho.
```

Each contacted connected component costs at least 1/2. A connected component of order at least two costs at least 2/3, and one of order at least three costs at least 3/4. These bounds follow by projecting onto a connected two- or three-node principal subdiagram through a contacted node; the explicit inverse inequalities are proved in the supporting-extension lemma. Larger diagrams and additional contacts cannot lower projection cost.

Thus rho=0 permits no attachment, rho=1/2 permits only one A1, and rho=2/3 permits only one A1 or A2. This bounds the entire connected attachment, with no ambient-rank cutoff. The core-contacting root plus this tail adds at most one, two, or three nodes, respectively. Every selected marked component still has order at most 17, as checked context by context. This explicitly handles supporting extensions with positive Euclidean residual; unchanged isolation is not assumed.

At an ideal corner, the inherited factor-fiber theorem places the entire varying affine factor inside W plus the two side roots. The bounded marked component is therefore the actual component relevant to payment. Omitted spectator cusp walls cannot enlarge it. The argument includes nonsimple cusps.

Face payments and dimension inequalities

Pay 1 on a corner pair if its distance is at most 10 or its connected component has order at most 17, and 0 otherwise. The classification above pays every quadrilateral at least 1. The inherited triangle argument gives two paths of length at most 8, so every triangle receives at least 2. Nonnegative payments suffice for larger polygons. Every k-gonal two-face therefore receives at least 5-k.

The inherited all-rank graph counts give finite vertex cost at most 10n. At an ideal vertex with affine factor ranks r_i and link vertex count V=product(r_i+1), factor-fiber multiplicity gives

```
C(v)/V <= sum min(10,r_i/2).
```

The shape inequality has normalized ideal curvature

```
kappa/V = 3d/4 - (5/2) E|sum X_i|,    d=n-1,
```

where X_i is centered uniform on r_i+1 consecutive integers. The proved convex-order transfer applies below and above cap 20, preserving cost and not increasing curvature. It terminates by increasing the sum of squared ranks. Reduced profiles have copies of 20, at most one part 1 through 19, and at most one part at least 21.

For d=116 through 125, all 1,015 reduced profiles have positive curvature-minus-cost. Integer convolution and the separate moment formula agree exactly. The smallest margin is **145234/4084101**, at ranks (20,20,20,20,20,16). The finite slack beta_n-10n is positive for n=117 through 126, and the global shape inequality has its strict positive term 5(n-1). These dimensions are excluded. The completed 126 theorem excludes all higher dimensions, giving 116.

At ambient dimension 116, the profile (20,20,20,20,20,15) has margin -673055/16336404. The same scalar weights do not prove 115; that local profile is not a finite-volume realization.

For a compact facet of dimension d=n-1, finite payments give beta_d<10(d+1). For d=87+2t and 88+2t, eight times the opposite difference has coefficient lists (12,176,4) and (96,180,4), positive for all t>=0. Hence d<=86 and the ambient dimension is at most 87.

Published classifications, actual-face and cusp incidence, the all-rank distance counts, convex-order transfer, and shape inequality are explicit inherited mathematical inputs. The arithmetic certificates do not replace the written geometric proof. This package makes no finite-volume construction, external-refereeing, proof-assistant, or publication-priority claim.

Primary final generation (session 75668) and fresh verification (session 7019) both exited zero with status PASS_SEL ECTIVE_DISTANCE10_CONTROLLED_EXTENSIONS_DIMENSION116. The verifier reconstructs the complete classification and exact inequalities:

```
python3 scripts/fresh_selective_distance10_dimension116_20261017.py --verify
```

Source 254

Source path: `results/fresh_selective_distance10_dimension116_20261017_review.md`

SHA256: ab5debac302b866bc39880b2e965a74727fe5c63f08df12874fe4595fcb185ce

Independent review of the selective distance-10 proof

FINAL / PASS. The mathematical/source review, independent exact generation, and fresh saved replay all pass. No gap remains identified in the stated unrestricted dimension-116 and compact-facet dimension-87 conclusions.

The independent checker and its certificate review the primary proof, source, and data. The checker imports no producer arithmetic, SymPy, numerical linear algebra, or optimizer. Its rational calculations use Python `Fraction`; the small-order supplement uses separately implemented exact arithmetic in Q(sqrt(5)).

Independent finite reconstruction

Starting from the frozen 26-type high-rank catalogue, the checker independently reconstructs all 384 marked contexts and the complete relevant ports, joins, partial identifications, and corner decompositions:

Check	Exact result
High-rank candidate / inverse-admissible ports	527 / 161
Ordered disjoint joins / white obstructions	1,478 / 363
Multiple-edge subsets, all rejected	120
Distinct single-join inertia types, all with two negative directions	35
Candidate / consistent overlaps	22,881 / 4,862
Inverse / actual-edge rejections	4,296 / 254
Surviving overlap contexts	312
Adjacent corner decompositions	1,248
Extra cross-edge shortcut / actual-edge rejections	21,393 / 168
Saved nonnegative producer witnesses independently evaluated	905
Reduced cusp profiles	1,015

Each rational Gram is constructed afresh from the literal diagrams. For diagrams with label 4, the checker recomputes a positive diagonal sqrt(2) congruence on the whole joined graph, rather than joining independently chosen root-length gauges. A separate rational elimination supplies signatures, inverse diagonals, null vectors, and positive or nonnegative-square tests. Every saved nonnegative witness is evaluated against the independently reconstructed principal Gram, with its gauge and square checked exactly.

The overlap enumeration is injective and checks all prescribed contacts on the common nodes. The independent preferred-corner records agree with the producer, including contact domains, finite/affine components, residual norm budgets, candidate-pair Cauchy inequalities, and all extra cross-edge checks. The four adjacent corners are tested even when their two marked roots lie in different components; only a connected preferred corner is used for payment.

Essential small-order supplement

Distance 10 requires an order-5/radius-3 disjoint supplement that was unnecessary at distance 11. A small core can join a radius-7 port at distance 11, so discarding all cores of order at most 5 would leave a gap.

The final primary proof supplies the complete all-label reduction: the three white nodes form a chain, the two black nodes have three possible tree attachments, and the only extra cycle is a black-end triangle ruled out in a proper four-node subdiagram. Every bond lies in a connected proper four-node positive-semidefinite diagram. A label at least 6 produces an indefinite connected triple, or an affine equality whose connected fourth-node extension is indefinite. Thus labels 3, 4, 5 exhaust this domain. This is a classification argument, not an assumed numerical cutoff.

The independent checker computes every principal tree determinant by the matching-polynomial recurrence, which depends only on squared bond magnitudes 1, 2, and (3+sqrt(5))/2. All 243 recipes agree with the producer: 209 have an indefinite proper principal subdiagram, 14 are positive semidefinite, four have a nonspherical black deletion, 12 fail the port test, and four survive. Their normalized far-port inverse pairs are (4/5,3/4), (3/4,4/5), (1,4/5), and (4/5,1).

Each surviving low core has one possible joining edge to L25/node 9, in three high marked contexts and either orientation. A second, independent leaf-Schur algorithm computes inertia (13,2,0) for all four joined trees, excluding all 24 ordered cases. Its zero-leaf rule uses a two-node indefinite pivot and requires no irrational matrix entries. Exact signs in Q(sqrt(5)) are decided by rational square comparison.

Geometric and completeness audit

The actual-edge tests require positive definiteness of W plus a single black root, including edges with ideal endpoints. This makes the inverse cutoff strict at 2. Cross-core endpoints are white under the assumed absence of a short marked path. The three-white-node test forces every possible disjoint cross bond to be simple. All subsets of the remaining possible edges are then exhausted: a single edge has two negative directions, and every multiple-edge subset has a nonnegative actual-edge obstruction. Consequently additional acute bonds cannot repair the excluded cases.

For overlaps, every unspecified bond between exclusive core parts either gives the required short path or has a nonnegative actual-edge obstruction. The latter remains valid when further nonpositive contacts are added. This fixes the core Gram before its extension arguments are used.

A radical of an indefinite realized core is an actual linear relation: the corresponding zero-square vector lies perpendicular to a timelike vector and therefore vanishes. The independently checked preferred contact domain excludes zero-radical coordinates. Any nonzero acute contact vector must meet both signs of the relation, yielding the checked short path.

For nonsingular cores, supporting whites lie in a Euclidean orthogonal complement after projection. A new root's core projection and residual satisfy the exact norm budget. The positive inverse end blocks exclude multiple contacts and every larger Coxeter label. Candidate-pair residual Cauchy inequalities either contradict acuteness or force a short marked path, so at most one white can contact the core.

The separate supporting-extension lemma controls the entire indirectly attached component. The other supporting whites form a positive-definite principal block E. Projection cost is at least 1/2 per contacted component, at least 2/3 for a connected component of order at least two, and at least 3/4 for order at least three. Residual squares 0, 1/2, and 2/3 therefore allow at most zero, one, and two additional roots. Arbitrarily long indirect tails are excluded analytically; no rank cutoff is used. All 312 selected marked components remain of order at most 17.

At ideal corners, the inherited factor-fiber theorem places the whole varying affine factor in W plus the adjacent pair. Extra cusp walls outside that face star cannot enlarge the paid marked component. This is essential for nonsimple cusps and does not assume that an induced face is itself a Coxeter polytope.

Curvature and scope

The weight pays distance at most 10 or connected component order at most 17. The inherited triangle argument gives two short paid corners; the new classification gives one paid quadrilateral corner. The finite cost is at most 10n, and the normalized ideal cost is at most sum min(10,r_i/2).

The checker independently evaluates every cap-20 reduced profile for link dimensions 116 through 125, using both integer convolution and the inclusion-exclusion absolute-moment formula from the shape-curvature framework. All 1,015 margins are positive. The minimum is 145234/4084101; the tested boundary profile at ambient dimension 116 has margin -673055/16336404. The latter is a limitation of this scalar payment, not a realized polytope. The compact-facet parity polynomials have coefficients (12,176,4) and (96,180,4).

The completed dimension-126 proof supplies the excluded higher-dimensional tail. With the written geometric and all-rank curvature inputs, these checks establish the stated dimension-116 and compact-facet dimension-87 consequences. This is a complete internal mathematical/source and independent exact-arithmetic review of the new step. It does not independently reconstruct published finite classifications or every older theorem. It is not external refereeing, proof-assistant formalization, a new construction, or a priority claim.

Final generation 21363 and fresh saved replay 80114 both terminated successfully. The review binds the frozen primary source, certificate, and proof, and preserves the completed 126 predecessor. All 11 direct review bindings, all 23 producer bindings, source AST, and nine local report links pass. The review source and data are frozen; no producer files were edited. The independent low-order implementation and all 905 saved-witness evaluations are included in the fresh replay.

Source 255

Source path: `results/fresh_selective_distance11_dimension126_20261017.md`

SHA256: bf8e11a8d6762dae6a38cf2790048448ccf15c5fbd546ad7e56b9c18f020d3db

Selective distance 11 and the unrestricted dimension 126 bound

Every finite-sided, finite-volume real hyperbolic Coxeter polytope has dimension at most **126**. No arithmeticity, compactness, simple-cusp, cusp-factor-rank, or facet-count restriction is imposed. The compact-facet consequence gives ambient dimension at most **95**.

The change from the completed distance-12 proof is to pay pairs at distance at most 11, while still paying every pair in a connected corner component of order at most 17. Two new overlap types appear. One has an actual radical relation that isolates the component. The other admits a supporting-root extension, but the extension lies in the original core span and can add at most one node.

The primary source and certificate record the finite classification and scalar inequalities. The separate review checks the geometry and reconstructs the arithmetic independently.

Actual face geometry and exact obstruction certificates

Use the actual-face and cusp-factor incidence statements of the preceding proof. For a quadrilateral, let W be its supporting white roots, and let U and V be its two opposite pairs of black side roots. The diagram W, and W together with any one black root, is positive definite. The diagram W with an adjacent black pair is positive semidefinite. Each of W+U and W+V is indefinite. Choose minimal indefinite cores L and M in these two diagrams. Each contains its two marked black roots, and deleting either black root leaves a positive-definite diagram.

The inherited complete classification has 26 diagrams of order at least 6, giving 384 marked contexts. Their maximum distance from a marked pair is 7. Cores of order at most 5 have marked radius at most 3, with no restriction on their finite labels. Assume there is no path from U to V of length at most 11 with white internal nodes.

For a positive-definite black-deletion Gram A, an additional supporting white contacting its node i has squared projection at least (A inverse)ii. The inverse is entrywise nonnegative, all contacts are nonpositive, and every nonzero norm-two Coxeter contact has magnitude at least 1. If either black deletion has inverse diagonal at least 2, its extension contradicts positive definiteness of W with the remaining black. Equality 2 is excluded as well. The complete test of 200 ports at marked distance at least 4 leaves 55 allowed ports.

Some rejections use an explicitly nonnegative vector x with x^T A x <= 0 in a diagram required to be positive definite. These are exact certificates with a monotonicity property: additional nonpositive contacts or larger bond magnitudes can only decrease x^T A x. Thus the certificate remains valid when unspecified contacts are restored.

The certificates are obtained by symmetric elimination of a Z-matrix (a symmetric matrix with nonpositive off-diagonal entries). At every positive pivot q, the remaining basis vectors change by subtracting nonpositive multiples of the pivot vector. Starting from coordinate vectors, all residual basis vectors stay nonnegative. The first nonpositive pivot therefore supplies the required vector. The source checks its entries and its quadratic value exactly. Labels 4 are rationalized by a positive diagonal rescaling with entries 1 and sqrt(2); this preserves inertia and nonnegativity of the witness. No approximate eigenvalue is used for an exclusion.

Disjoint cores

Two disjoint indefinite cores cannot be orthogonal in an ambient space with negative index one. A cross bond must connect white ports whose marked distances sum to at least 11. Both distances are at least 4, because each core radius is at most 7. This also excludes cores of order at most 5 from the present disjoint case.

The 55 allowed ports give 425 ordered marked joins. In 119 joins, the minimum Gram with a simple cross bond already has a nonnegative nonpositive-square vector in the wholly white block. Those joins are impossible for every allowed magnitude of that bond. Among the remaining 306 joins, each ordered marked-context pair admits exactly one cross edge. Every other potential cross edge is already excluded by a short path, an inverse test, or the preceding positive-vector certificate.

The remaining cross bond must be simple, including when the native diagrams contain labels 4. Each endpoint has a three-node wholly white path starting at that endpoint. Give the consecutive nodes coefficients 1, 2/3, 1/3. Its square is at most 4/3, because each native bond magnitude is at least 1. Combining the two paths gives square at most

```
8/3 − 2 beta,
```

where beta is the cross-bond magnitude. White positive definiteness implies beta < 4/3. The norm-two Coxeter alphabet has no nonright bond between the simple magnitude 1 and sqrt(2), so beta=1. Parallel and dotted bonds are also excluded. This argument has no finite-angle cutoff.

The exact full Grams for all 306 remaining joins have two negative directions. Rational congruence certificates verify this inertia. They exhaust 13 underlying unmarked joins, all with one endpoint L25/node 9:

Other diagram and port	Ordered marked occurrences
L10/node 5	6
L13/node 6	6
L14/node 6	6
L16/node 7	6
L18/node 7	42
L20/node 8	6
L21/node 8	6
L22/node 8	54
L22/node 4	42
L23/node 9	6
L24/node 9	6
L25/node 9	114
L25/node 3	6

Since no additional cross bond remains possible, these are the actual core Grams. Negative index two contradicts the ambient Lorentzian space.

Exhaustive overlap classification

A shared white root has its two marked distances sum to at least 12. Each distance is therefore at least 5. In the complete marked catalogue each context has at most three such nodes. Enumerating injective partial identifications of these nodes exhausts all possible overlaps.

There are 2,137 candidate identifications before injectivity and common Gram-entry consistency, and 1,010 after those checks. In 856 cases a shared node has an external white neighbor from the other core and a black-deletion inverse diagonal at least 2. The inverse test excludes them. Of the remaining 154 cases, 40 have a nonnegative nonpositive-square vector in W with one black root, which must be positive definite. Restoring other nonpositive contacts cannot repair that obstruction.

The remaining 114 ordered marked contexts are precisely:

Overlap	Contexts
Two L25 cores sharing their far node 9	60
Two L25 cores sharing the terminal edge in reverse	27
Two L25 cores sharing their terminal three-node path in reverse	9
L22/node 8 identified with L25/node 9, in either orientation	18

Every unspecified bond between the exclusive parts is checked: 8,280 create a marked path of length at most 11, and the other 45 have exact nonnegative nonpositive-square witnesses in W with one black root. All 45 occur in shared-vertex contexts. This determines the actual core Gram in all 114 surviving contexts. The following extension arguments therefore apply to the actual normals, rather than to a freely chosen zero-completion.

Shared vertex: the larger marked domain

The 19-root L25/L25 shared-vertex Gram is the same as in the completed distance-12 proof. Number the first copy 0 through 9 and map the second copy's 0 through 8 to 10 through 18, with node 9 shared. Its actual radical relation is

```
z=(-6,-3,-4,-2,-5,-4,-3,-2,-1,0,
    6,3,4,2,5,4,3,2,1).
```

To see that this is an actual root relation, its Gram-null combination is orthogonal to the core span, which contains a timelike vector, and has square zero. The orthogonal complement of that timelike vector is positive definite, so the combination vanishes.

In each copy choose the preferred marked root as follows: choose 4 if it belongs to the marked pair, then 0 if it belongs; otherwise choose 2 from {1,2} or {2,3}, and 1 from {1,3}. The complete preferred-corner calculation gives a marked component of order at most 17 in all 60 contexts.

For each finite corner component, its inverse diagonal restricts a new supporting white's possible contacts to entries at most 2. For an affine component, its positive kernel forces all contacts to that component to vanish. Include the two omitted black roots separately. Every allowed contact has nonzero z-coordinate; all negative-z contacts are within distance 3 of U, and all positive-z contacts within distance 3 of V.

Pairing an additional supporting white with z forces contacts of both signs if there is any contact. That creates a white-interior marked path of length at most 8. Hence every additional supporting white is orthogonal to the core, and the preferred marked component is isolated.

Shared edge: the larger end sets

The 18-root shared-edge core maps the second L25 copy's 0 through 7 to 10 through 17, its 8 to 9, and its 9 to 8. It is nonsingular Lorentzian, with determinant -1. The two-center Schur complement is

```
[[-5/66,-1/11],[-1/11,-5/66]].
```

Use the same preferred-root choices. All 27 preferred marked components have order at most 16. The finite inverse and affine kernel restrictions confine contacts of an additional supporting white to

```
E_left={0,1,2,3}, E_right={10,11,12,13}.
```

Every allowed end node is within distance 2 of its own marked pair. Contacts at both ends give a marked path of length at most 6. If the contacts are supported at just one end, the corresponding principal block of the inverse of the full core Gram is

```
[[30,15,20,10],
 [15, 8,10, 5],
 [20,10,14, 7],
 [10, 5, 7, 4]].
```

It has strictly positive entries and diagonal minimum 4. Thus a nonzero contact vector b has b^T G^{-1}b >= 4.

The core span contains the unique negative direction, so its ambient orthogonal complement is positive definite. A norm-two additional root would need squared orthogonal residual 2-b^T G^{-1}b >= 0, a contradiction. All supporting whites outside the core are orthogonal to it.

Three-node overlap: the extension is controlled

The new 17-root core maps the second L25 copy by

`(0,1,2,3,4,5,6,7,8,9) -> (10,11,12,13,14,15,16,9,8,7).`

Its determinant is -2 and its signature is (16,1). Both black pairs are one of {1,2}, {1,3}, {2,3}. Choose the preferred node 2, 1, 2 respectively. The separate exact extension lemma proves that each preferred marked component initially has order 15, and that a new supporting white's contacts can lie only in

`{1,2,3} or {11,12,13}.`

A contact at both ends creates a marked path of length at most 4. The full inverse Gram on either one-end set is

`[[7/2,4,2],[4,6,3],[2,3,2]].`

Its strictly positive entries and the norm-two residual inequality imply that the only nonzero possibilities are a sole contact -1 at node 3, or a sole contact -1 at node 13. Each attains squared projection 2, so its orthogonal residual vanishes. The actual roots are uniquely r=-G^{-1}e_3 and s=-G^{-1}e_13, in the core span. Their mutual product is -2. They cannot both belong to the positive-definite supporting diagram W. Duplicate normalized roots do not define distinct facets.

Consequently at most one extra root occurs. Every other supporting white orthogonal to the core is also orthogonal to that extension, because it lies in the core span. The preferred marked component remains isolated and has order at most 16. The argument explicitly permits this extension; it does not assume that every surviving core must be isolated unchanged.

Mixed L22/L25 overlap

Number L22 by 0 through 8, and map L25 by

`(0,1,2,3,4,5,6,7,8,9) -> (9,10,11,12,13,14,15,16,17,8).`

The mixed extension lemma gives inertia (16,1,1) and the actual radical relation

`z=(-4,-2,-3,-2,-1,-3,-2,-1,0,6,3,4,2,5,4,3,2,1).`

Each black pair is one of {1,2}, {1,3}, {2,3} in its original copy. Use the preferred choices 2, 1, 2. The nine contexts, and their nine reversed orientations, have marked corner component order at most 16. The full corner restrictions exclude the zero-radical node from the contact domain. Allowed negative-z contacts lie within distance 2 of the left black pair; positive-z contacts lie within distance 1 of the right pair.

The actual radical relation forces both signs for any nonzero additional supporting-white contact, giving a marked path of length at most 5. Every additional supporting white is therefore orthogonal to the core.

From local isolation to actual corner payment

At finite corners, the resulting isolated marked component is a component of the actual ambient vertex diagram. At ideal corners, the inherited factor-fiber incidence theorem places the entire varying affine factor inside W plus the two side roots. An isolated finite component cannot be that varying factor; an isolated affine component is already the whole factor. Omitted spectator cusp walls do not enlarge the marked component relevant to the face payment. Thus the proof includes nonsimple cusps.

Pay weight 1 on a marked pair if its distance is at most 11 or its connected corner component has order at most 17; otherwise pay 0. The complete classification above pays every quadrilateral at least 1. The inherited minimal-core argument gives every triangle two paths of length at most 8, hence payment at least 2. Nonnegative weights suffice for polygons with at least five sides. Every k-gonal two-face therefore receives total weight at least 5-k.

Exact dimension inequalities

Components of order at most 17 have at most 8 times their order in pairs. For larger finite and affine components, the inherited all-rank path/cycle/fork counts give at most the component order at each distance. The finite vertex cost is at most 11n. At an ideal vertex with affine factor ranks r_i and link vertex count V=product(r_i+1), the factor-fiber multiplicity gives

```
C(v)/V <= sum_i min(11,r_i/2).
```

The inherited shape inequality has normalized ideal curvature

```
kappa/V = 3d/4 - (5/2) E|sum_i X_i|,   d=n-1,
```

where X_i is centered uniform on r_i+1 consecutive integers. The proved all-rank convex-order transfer applies with cap 22. Concentrating two ranks below 22, or two above 22, preserves the cost and cannot increase curvature. Increasing the sum of squared ranks proves termination. The reduced profiles consist of copies of 22, at most one part from 1 through 21, and at most one part at least 23.

For d=126 through 135, all 1,095 reduced profiles have positive curvature-minus-cost. Integer convolution and a separate moment formula give the same exact values. The minimum is

```
40603/559682,
```

at ranks (22,22,22,22,22,16). The finite slack beta_n-11n is positive for n=127 through 136, and the global shape inequality has its strict positive term 5(n-1). These dimensions are therefore impossible. The completed 136 theorem excludes all higher dimensions, proving 126.

At ambient n=126 the reduced profile (22,22,22,22,22,15) instead has margin -238535/12872686. These same scalar weights do not prove 125. This is a local link profile, not a finite-volume construction.

For a compact facet of dimension d=n-1, the finite payment gives beta_d < 11(d+1). For d=95+2t and d=96+2t, eight times the opposite difference has coefficient lists (12,192,4) and (104,196,4), respectively. Both are positive for t>=0. Thus d<=94 and the ambient dimension is at most 95.

The finite/quasi-Lanner classification, actual incidence statements, distance counts, convex-order theorem, and shape inequality remain explicit inherited inputs. Exact finite arithmetic does not by itself formally verify these geometric arguments. No finite-volume realization, external refereeing, proof-assistant formalization, or publication-priority claim follows from this package.

Primary generation 85083 and fresh saved replay 10135 both terminated with exit code zero and PASS. The separately implemented Fraction reconstruction agrees on all 200 tested ports, 425 joins, 1,010 consistent overlaps, 114 preferred-corner cases, 8,325 extra-contact exclusions, and 1,095 profiles. Its written mathematical review finds no gap. These are internal agent-workflow checks; the report does not claim external mathematician refereeing or formal verification.

The primary source and certificate are frozen at SHA256 8fcbea5f00ae2b63cd3e0b7fe3fa8616d77b7ed559c321af0648ec cfc31bfb66 and f6340fcf99d3884b4a22113fc446cef5ac8bd6880f0b1354f525f2bcaab47646, respectively. All 25 direct bindings and source syntax passed. The separate review records its own saved replay and final bindings.

Source 256

Source path: `results/fresh_selective_distance11_dimension126_20261017_review.md`

SHA256: 7d3c6b390f3ee14fc82ade69d619aad7f7f2e8296749dc9823f87339b9045459

Independent review of selective distance 11 and dimension 126

Mathematical and exact-arithmetic PASS. No unresolved geometric case or arithmetic discrepancy was identified in the producer proof. Its conclusion is dimension at most **126** for every finite-sided, finite-volume real hyperbolic Coxeter polytope, without arithmeticity, compactness or cusp restrictions. The separate compact-facet conclusion is ambient dimension at most **95**.

The independent checker and certificate use only standard-library `Fraction` arithmetic and literal graphs. They import no producer arithmetic, SymPy, numerical linear algebra or optimizer. The finite reconstruction and the mathematical review below are distinct checks. This is internal review, not external refereeing or proof-assistant formalization; no priority claim is made.

Complete local domain

The inherited actual-face argument gives a positive-definite supporting white diagram W, positive-definite W plus any one side root, and positive-semidefinite W plus an adjacent pair. The two opposite closed sides are disjoint even when the quadrilateral has ideal vertices. Hence each opposite-side diagram is indefinite, and its minimal indefinite core contains both black roots with both black deletions positive definite.

The reviewer reconstructs all 384 marked contexts in the frozen order-6 through order-10 catalogue, including their graph distances and all relevant inverse diagonals. The maximum marked radius is exactly 7. A core of order at most 5 has radius at most 3 by its size; it cannot participate in a disjoint join requiring radius sum at least 11 or an overlap requiring radius sum at least 12. Thus its potentially unbounded finite labels introduce no omitted case here.

At distance at least 4 there are 200 tested ports, of which 55 survive the inverse test. For a label-4 edge, the checker independently constructs a positive diagonal square-root-of-two congruence to a rational Gram, then returns inverse diagonals to the norm-two gauge. An external supporting white has nonpositive contacts with nonzero magnitude at least 1. Since the spherical inverse is entrywise nonnegative, inverse diagonal at least 2 contradicts the required positive definiteness. Equality 2 is correctly excluded: this test concerns an actual edge support, not merely a semidefinite ideal corner.

Positive witnesses and all disjoint joins

The reviewer independently generates a nonnegative vector whenever a required positive-definite Z-matrix first has a nonpositive elimination pivot. It does so by solving against the positive leading block, rather than importing the producer's accumulated basis transformations. The vector and its quadratic value are checked exactly.

This sign is essential to the proof. Restoring additional nonpositive contacts, or increasing a tested cross-bond magnitude, cannot increase the vector's quadratic value. The rejection therefore applies to the actual Gram matrix, not just its minimal-contact completion. The checker also independently verifies all **204** saved producer witnesses: 119 disjoint-white obstructions, 40 overlapping edge-support obstructions, and 45 additional-cross-edge obstructions.

The 55 allowed ports give 425 ordered marked disjoint joins. Of these, 119 have a nonnegative nonpositive-square witness already in W. The remaining 306 correspond to 306 different ordered marked-context pairs; thus each such pair has exactly one possible cross edge after the short-path, inverse and white-witness exclusions.

For every remaining join the reviewer checks the producer's two wholly white three-node tails directly in the literal native diagrams. With positive tail coefficients (1,2/3,1/3), each tail has square at most 4/3, also when a native edge is labelled 4. A cross bond of magnitude at least sqrt(2) makes their combined square negative. Consequently the only possible nonright contact is simple; parallel and dotted contacts are excluded too. No label cutoff is used.

As a separate check, the reviewer eliminates the full white block on each side and computes the exact upper bound on the squared cross contact; every bound is at most 2 and the positive-definite inequality is strict. Both arguments independently force the same simple bond.

The reviewer reconstructs every resulting Gram and its inertia by rational one-by-one or two-by-two symmetric pivots. All 306 joins have at least two negative directions. It also compares all 13 underlying saved Gram/inertia types. Since every other cross edge is already excluded, these are the actual core Grams, contradicting ambient negative index one.

Exhaustive overlaps and unknown cross contacts

Every shared white root has its two marked distances summing to at least 12. Both distances are at least 5, and each marked core has at most three such roots. The reviewer enumerates subsets of the left and right node sets and all bijections between them, independently of the producer's filtering of combinations of candidate pairs. It obtains 2,137 candidates and 1,010 injective, Gram-consistent overlaps:

Outcome	Count
External-neighbor inverse obstruction	856
Positive witness in W plus one black root	40
L25/L25 shared far vertex	60
L25/L25 reversed terminal edge	27
L25/L25 reversed terminal three-node path	9
L22/L25 shared far vertex, both orientations	18

The matching is the complete intersection. Each external neighbor used by an inverse rejection is therefore genuinely outside the other core, and its large marked distance ensures it is white. The independently recovered categories agree with every saved producer record.

For each of the 114 surviving contexts, the reviewer tests every unspecified edge between the exclusive parts. Exactly 8,280 tests create a path of length at most 11; the other 45 have explicit nonnegative nonpositive-square witnesses in W with one black root. These witnesses remain valid for stronger or additional acute contacts. Thus all surviving core Grams are fixed before their isolation arguments are applied.

Supporting-white extensions

The reviewer reconstructs all 114 preferred corners from scratch, including all finite inverse diagonals, affine positive kernels, possible contacts and marked components. It additionally checks positive semidefiniteness of all four adjacent corners in every surviving context.

For the 60 shared-vertex cases, the 19-root Gram has inertia (17,1,1). Its radical is an actual linear relation: its Gram-null root combination has square zero and is orthogonal to a timelike vector in the core span. Positive definiteness perpendicular to that timelike vector makes the combination vanish. Every allowed new contact has a nonzero radical coordinate. The two signs are within distance at most 3 of their respective marked pairs. Acuteness and radical balance force both signs whenever a new contact exists, yielding a path of length at most 8. Thus all further supporting whites are orthogonal, and the marked component has order at most 17.

For the 27 shared-edge cases, the 18-root core is nonsingular Lorentzian. Preferred-corner restrictions allow contacts only in the two end sets of four nodes. The full inverse blocks on those end sets have strictly positive entries and diagonal minimum 4. A contact at both ends gives a path of length at most 6; a nonzero contact vector supported at one end has squared projection at least 4. The orthogonal complement of the core span is positive definite, so a square-two root cannot have the required negative residual. Every further supporting white is orthogonal, and the marked component has order at most 16.

The nine three-node overlaps require a different argument: they do admit an extension. The reviewer independently recomputes the separate extension lemma. The one-end inverse block is

`[[7/2,4,2],[4,6,3],[2,3,2]].`

Only a sole contact -1 at node 3 or its opposite-copy counterpart is possible. The projection has square 2, forcing zero orthogonal residual and a unique actual root in the core span. The two possible roots have mutual product -2 and cannot both lie in positive-definite W. Every other white orthogonal to the core is automatically orthogonal to the extension. All 18 one-extension corner variants are independently checked; the marked component remains isolated and has order at most 16.

The 18 mixed cases are handled by the mixed radical lemma. The reviewer reconstructs both orientations, including their literal isomorphisms to the frozen lemma. The core has inertia (16,1,1). Its only zero-radical node is excluded by the preferred-corner contact tests, and any nonzero contact forces a short path between opposite signs. Consequently every supporting white is orthogonal and the marked component has order at most 16.

At finite corners, the isolated component is an actual ambient vertex component. At ideal corners, the inherited actual factor-fiber statement puts the entire varying affine factor inside W plus the two side roots. A finite marked component cannot be that factor, and an isolated affine marked component is already the whole factor. Additional incident cusp walls outside this star cannot enlarge the component relevant to the payment. This is why the proof permits nonsimple cusps.

Payment and the dimension inequalities

The selective rule pays a pair when its distance is at most 11 or it belongs to a connected corner component of order at most 17. The preceding argument pays every quadrilateral. The inherited triangle argument provides two paths of length at most 8. Hence every k-gonal two-face receives at least 5-k.

Small components contribute at most 8 times their order in pairs; the inherited all-rank distance counts give the required bound on larger components. Finite vertices cost at most 11n, and an ideal vertex with factor ranks r_i and link vertex count V costs at most `V sum min(11,r_i/2)`.

The proved convex-order transfer is applied separately below and above cap 22. These transfers preserve the charge, do not increase curvature, and terminate by increasing the sum of squared ranks. No transfer across the cap is presumed. The resulting profiles have copies of 22, at most one smaller part and at most one larger part.

The reviewer independently enumerates all 1,095 reduced profiles for link dimension 126 through 135. Integer convolution and a separate inclusion-exclusion moment formula agree exactly. Every margin is positive; the least is **40603/559682**, at ranks (22,22,22,22,22,16). The finite-vertex slack is positive in ambient dimensions 127 through 136, and the global shape inequality retains the strictly positive term 5(n-1). The completed dimension-136 theorem supplies the higher-dimensional tail. This proves 126 with no new restriction.

At ambient dimension 126, the profile (22,22,22,22,22,15) has margin -238535/12872686; the reviewer confirms this boundary. The same scalar argument does not establish 125. For the compact-facet branch, the independent parity polynomials have coefficients (12,192,4) and (104,196,4), giving ambient dimension at most 95.

Final validation scope

Final independent generation and fresh saved replay 53606 both passed. The primary source, data and report are bound at their final frozen hashes. Python syntax, all ten direct source bindings, all 25 producer source bindings and seven report links passed. This review source, certificate and report are final/frozen. The published classification, actual incidence statements, all-rank graph budgets, convex-order theorem and shape inequality remain explicit inherited mathematical inputs. They are reviewed geometric arguments, not consequences of the finite verifier alone. The completed package records a complete internally checked computer-assisted proof; it does not assert formal verification or external refereeing.

Source 257

Source path: `results/fresh_selective_distance12_dimension136_20261017.md`

SHA256: 498aeeb06b3eedaf8754804d24a1f7f97afd62f72148846b94d58d10a356a76b

Selective distance12 and the unrestricted dimension136 bound

Every finite-sided, finite-volume real hyperbolic Coxeter polytope has dimension at most136. The argument has no arithmeticity, compactness, simple-cusp, factor-rank or facet-count restriction. For a chamber having a compact facet, the ambient dimension is at most103.

This strengthens the dimension146 proof by replacing distance13 with distance12. As before, pairs in the same connected corner component of order at most17 are also paid. The new exception consists of two minimal cores sharing a terminal edge. A Lorentzian Schur-complement argument isolates that core, replacing the radical argument used for the previously known shared-vertex exception.

The producer and certificate give the exact finite case split and the remaining scalar inequalities. The separate review independently reconstructs the critical arithmetic and reviews the written geometry. No new finite-volume construction or publication-priority claim is made.

Geometric setup and inverse tests

Use the inherited actual-face and cusp-factor incidence statements from the146 proof. For a quadrilateral, let W be its supporting white roots and U,V the opposite pairs of side roots. W and W plus one black root are positive definite. W plus an adjacent black pair is positive semidefinite. Each of W+U and W+V is indefinite, since the corresponding closed opposite sides do not meet. Minimal indefinite cores L,M have order at most10 and both black deletions positive definite.

The complete high-rank catalogue has384 marked contexts and maximum distance7 from the marked pair. Cores of order at most5 have radius at most3, independently of their edge labels. Assume there is no path of length at most12 between U and V with white internal nodes.

For a black-deletion Gram A, every external supporting white contacting node i must have Schur residual at most2−(A inverse)ii. The normalized contact magnitude is at least1 and A inverse is entrywise nonnegative. If either black deletion has inverse diagonal at least2, this contradicts positive definiteness of W plus the remaining black. Thus equality2 is excluded here too. The producer recomputes all marked distances and all inverse tests at distance at least5. Exactly18 marked ports survive.

Disjoint cores

The cores cannot be orthogonal in a space of negative index one. Every cross bond must join white ports whose distances from their black pairs sum to at least12; both distances are therefore at least5. The18-port list leaves115 ordered marked joins, with six underlying possibilities:

Ports, ignoring marked-context multiplicity	Ordered occurrences
L25 node9 to L25 node9	60
L25 node9 to L25 node8	24
L25 node9 to L22 node8	18
L25 node9 to L18 node7	6

Ports, ignoring marked-context multiplicity	Ordered occurrences
L25 node9 to L22 node4	6
L25 node8 to L25 node8	1

L25 has simple edges0–1,0–2,2–3 and0–4–5–6–7–8–9. The node9-to-node8 join contains the same wholly white affine E8 block as in the146 proof. The node8-to-node8 join contains wholly white affine D5: its two trivalent nodes are the joined nodes8,18 and their four leaves are7,9,17,19. Positive kernels certify both obstructions. Larger bond magnitudes or additional nonpositive contacts only decrease the quadratic form on those kernels, so neither actual white block can be positive definite.

After excluding those types, each marked context pair has only one possible cross bond. Each endpoint has a wholly white simple A3 tail. Their Schur endpoint residuals are4/3, so positive definiteness forces the squared cross-bond magnitude below16/9. The only nonright Coxeter bond in this range is simple. This reasoning covers arbitrary finite labels as well as parallel or dotted candidates; there is no cutoff.

Delete the two fork centers, numbered0 and10, from each resulting joined Gram. The retained block is positive definite. Its two-center Schur matrices are as follows:

Right core/port, with left L25/node9	Schur matrix	Determinant
L18/node7	[[-1/15,-1/10],[-1/10,-7/30]]	1/180
L22/node8	[[-5/66,-1/11],[-1/11,-7/44]]	1/264
L22/node4	[[-1/15,-1/10],[-1/10,-1/5]]	1/300
L25/node9	[[-7/78,-1/13],[-1/13,-7/78]]	1/468

Every matrix is negative definite. Thus every disjoint case has two negative directions and is impossible.

Complete overlap split

Every shared root must have its two distances sum to at least13. Consequently both distances are at least6. Exactly16 marked contexts have such roots, and each has at most two of them. Enumerating injective partial identifications of these roots therefore exhausts the overlap; no intersection of three or more nodes is possible.

There are208 candidate identifications before injectivity and common Gram-entry consistency, and162 after those tests. In120, a shared node has an external neighbor from the other core and its black-deletion inverse diagonal is at least2. That neighbor is white, since the shared root's distance from the other black pair is at least6. The inverse test excludes these identifications.

The42 remaining identifications all use two L25 cores:

- 27 share only node9 with node9. Their marked pairs belong to {0,1},{0,2},{0,3},{1,2},{1,3},{2,3}, with at least one pair from the last three (radius7).
- Six share just node9 with node8, or node8 with node9. The shared node has wholly white arms of lengths1,2,5, so the positive affine-E8 kernel excludes these cases, including any extra contacts.
- Nine share the terminal edge in reverse: left8=right9 and left9=right8. Both marked pairs are from {1,2},{1,3},{2,3}.

For the shared-vertex cases, the only additional cross bond not already creating a path of length at most12 would connect the two neighbors of the shared node. These three whites would be an affine triangle; its positive kernel excludes the bond. In the shared-edge cases every additional cross bond immediately gives a short marked path. Therefore the two surviving core Grams are completely fixed.

Isolation of the shared-vertex core

This is the19-root Gram of the146 proof. Number the first copy0..9 and map the second's0..8 to10..18, with9 shared. Its Schur matrix after deleting centers0,10 is the constant2×2 matrix with entries−1/12. Its radical is the actual root relation

```
z=(-6,-3,-4,-2,-5,-4,-3,-2,-1,0,
    6,3,4,2,5,4,3,2,1).
```

The relation is actual because its Gram-null linear combination is orthogonal to a timelike vector in the core span, and hence must vanish.

Choose side0 for a marked pair containing0. For the remaining pairs choose2 from {1,2},1 from {1,3}, and2 from {2,3}; make the analogous choice in the second copy. In all27 contexts the two preferred sides lie in one corner component of order at most17. Their distance in that component is13 or14.

The producer decomposes each full preferred corner into connected finite or affine blocks. An additional supporting white e has no contact with an affine block, by its positive kernel. In a finite block, positive semidefiniteness permits contact only at inverse diagonals at most2. Together with the two omitted blacks, every allowed contact has nonzero z-coordinate. Every negative-z contact is within distance2 of U; every positive-z contact is within distance2 of V.

Pairing e with z gives zero. Acuteness therefore forces contacts of both signs whenever there is any contact. This gives a white-interior marked path of length at most6, contrary to the assumption. Every additional supporting white is orthogonal to the core.

Isolation of the shared-edge core

The new core has18 roots. Number the first copy0..9; map the second's 0..7 to10..17, its8 to9, and its9 to8. Deleting centers0,10 leaves a positive-definite block. The two-center Schur matrix is

```
[[-5/66,-1/11],[-1/11,-5/66]],
```

with negative determinant. Thus the full core is nonsingular Lorentzian; its determinant is-1. Use the same preferred side choices for the nine radius7 marked contexts. Their16-node preferred corners are one A16, four D16 and four affine D15, and the preferred marked distance is13.

The preferred-corner inverse or positive-kernel tests allow additional supporting-white contacts only in the end sets

```
E_left={1,2,3}, E_right={11,12,13}.
```

Each available end node is within distance1 of its marked pair. If e contacts both ends, it gives a marked path of length at most4. Otherwise all contacts lie at one end. In the inverse of the FULL Lorentzian Gram, the principal block on either end set is

```
[[8,10,5],
 [10,14,7],
 [5,7,4]].
```

All entries are positive and the diagonal minimum is4. For a nonzero contact vector b supported at that end, acuteness and the minimum nonzero magnitude1 imply b^T G^{-1} b>=4. The core span already has the unique negative direction, so its orthogonal complement in the ambient space is positive definite. The additional root's orthogonal residual must therefore satisfy

```
0 <= 2-b^T G^{-1}b <= -2,
```

a contradiction. Every additional supporting white is again orthogonal to the whole core. This step applies to arbitrarily large allowed contact magnitudes and needs no radical relation.

At finite preferred corners the isolated marked component is the actual ambient vertex component. At ideal preferred corners, the inherited factor-fiber incidence theorem puts the entire varying affine factor inside W plus the two sides. An isolated finite component cannot be such a factor; an isolated affine component is already the full factor. Additional omitted spectator cusp walls do not enlarge it. This covers nonsimple cusps and completes the selective quadrilateral payment.

Dimension inequalities

Pay weight1 on a marked pair if its graph distance is at most12 OR its connected component has order at most17; otherwise pay0. Every triangle has two paths of length at most8 by the inherited minimal-core argument. The preceding proof pays

every quadrilateral at least1. Nonnegative weights suffice for polygons with five or more sides. Thus every k-gonal two-face receives total weight at least5−k.

Components of order at most17 have at most8 times their order in pairs. For larger finite or affine components, the all-rank path/cycle/fork counts give at most the component order at each distance. The finite vertex cost is consequently at most12n. At an ideal vertex of factor ranks r_i and link vertex count V=product(r_i+1), the factor-fiber multiplicity gives

```
C(v)/V <= sum_i min(12,r_i/2).
```

The inherited shape inequality gives ideal normalized curvature

```
kappa/V = 3d/4 - (5/2) E|sum_i X_i|,   d=n-1,
```

where each X_i is centered uniform on r_i+1 consecutive integers. The already proved all-rank convex-order transfer applies with cap24: concentrating two ranks below24, or two above24, preserves this cost and cannot increase curvature. Increasing the sum of squared ranks proves termination. The reduced partitions are copies of24, at most one part1..23, and at most one part at least25.

For d=136,…,145, the1175 reduced profiles all have positive curvature-minus-cost. Integer convolution and the independent moment formula agree. The minimum is172096/1953125, at ranks(24,24,24,24,24,16). The finite slack beta_n−12n is positive throughout n=137,…,146; the global shape inequality also has its strict positive5(n−1) term. Hence these dimensions are impossible. The completed146 theorem excludes all larger dimensions, proving the bound136.

At n=136 the profile(24,24,24,24,24,15) instead has margin−59157/3906250. This same scalar comparison does not prove135. It is a local link profile, not a finite-volume chamber construction.

For a compact facet of dimension d=n−1 the finite payment gives beta_d<12(d+1). For d=103+2t and104+2t, eight times the opposite difference has coefficient lists(12,208,4) and(112,212,4), respectively. Both are positive for t>=0. Thus d<=102 and ambient n<=103.

The finite/quasi-Lanner classification, actual incidence results, all-rank distance counts, convex-order theorem and shape inequality remain explicit inherited inputs. The finite arithmetic does not formally verify these written arguments or replace their hypotheses.

Primary generation and a fresh saved replay both terminate with PASS. The separate Fraction-only implementation also passes all384 marked contexts,75 ports,162 consistent overlaps,115 disjoint joins,36 corner domains and1175 profiles. Its written mathematical/source review finds no gap. These are internal checks, not external refereeing or formal proof-assistant verification. The primary source, data and report are final/frozen; the separate review records its own replay and final bindings.

Source 258

Source path: `results/fresh_selective_distance12_dimension136_20261017_review.md`

SHA256: ddad9601719b55b984b7c7e4c9fe09cf1e7bdeec6e8eb040cee5f540c07a8fe3

Independent review of selective distance12 and dimension136

PASS. The producer proof gives a complete internally checked computer-assisted proof of dimension at most136 for finite-sided, finite-volume real hyperbolic Coxeter polytopes, without arithmeticity or cusp restrictions. Its separate compact-facet conclusion is ambient dimension at most103. No unresolved geometric case or arithmetic discrepancy was identified in this review.

The independent checker and saved certificate use standard-library Fraction arithmetic, literal graphs and an independently organized matching enumeration. They import no producer arithmetic, SymPy, NumPy or optimizer. The mathematical review below is separate from those calculations. This is internal review, not external refereeing or proof-assistant formalization; no publication-priority claim is made.

Coverage of the minimal-core split

The inherited actual-edge argument makes both marked deletions of each minimal indefinite opposite-side core positive definite, even when some quadrilateral vertices are ideal. The opposite closed sides remain disjoint. The complete order6–10 quasi-Lanner catalogue and the order-at-most5 elementary radius bound therefore have the hypotheses required by the producer.

The checker reconstructs all384 finite-black-pair contexts directly from the frozen literal diagrams. It recomputes distances and75 ports at distance at least5, obtaining18 allowed ports. For diagrams with a4 bond, a diagonal sqrt2 congruence gives a rational Gram; the inverse diagonal is rescaled back to the norm-two gauge. Every tested black deletion is independently checked positive definite. The bound excludes inverse diagonal equal to2 as well as larger values: the relevant block including the external white is contained in an actual edge star, which is positive definite, not merely semidefinite.

Under the assumption of no marked white path of length at most12, a disjoint joining bond must have endpoint radii summing to at least12, hence both at least5. The18 allowed ports yield exactly115 ordered joins of six underlying types. The checker verifies that both rejected affine blocks are wholly white in every relevant marked context, not just in an unmarked representative.

For overlapping cores, every shared root has its two radii summing to at least13. Thus each radius is at least6. The16 relevant marked contexts each have at most two such roots; intersections of three or more roots are impossible. Rather than using the producer's filtered combinations of candidate identifications, the reviewer enumerates left and right subsets and all bijections between them. It recovers208 candidate identifications and162 consistent ones:

Outcome	Count
External-neighbor inverse obstruction	120
Wholly white affine E8	6
L25 copies sharing their far node	27
L25 copies sharing the reversed far edge	9

An external neighbor used by an inverse rejection is a white root: its distance from the other core's black pair is too large for it to be black. The enumeration treats the matching as the complete intersection, so that this neighbor really is outside

the first core. The independent checks reproduce every rejection category and surviving identification.

Disjoint joins and unspecified cross contacts

The far-near and near-near white blocks have strictly positive affine kernels. Increasing a bond magnitude or adding another acute contact decreases their quadratic form on that positive kernel. Thus their obstruction applies beyond the minimal simple-bond representative.

After those types are removed, every marked context pair permits only one remaining cross edge. The checker verifies this uniqueness and a wholly white A3 tail at each endpoint. Eliminating those tails gives endpoint residuals4/3, forcing squared cross contact below16/9<2; consequently the only possible nonright Coxeter contact is simple. No finite-label cutoff is used, and parallel or dotted contacts are excluded by the same bound.

Independent rational Gaussian elimination reproduces the four two-center Schur matrices and positive determinants:

Right core/port	Schur matrix	Determinant
L18/7	[[-1/15,-1/10],[-1/10,-7/30]]	1/180
L22/8	[[-5/66,-1/11],[-1/11,-7/44]]	1/264
L22/4	[[-1/15,-1/10],[-1/10,-1/5]]	1/300
L25/9	[[-7/78,-1/13],[-1/13,-7/78]]	1/468

Each eliminated block is independently positive definite. The residual matrices are negative definite, proving negative index at least2. This excludes all disjoint possibilities in the actual Lorentzian ambient space.

For each surviving overlap context the reviewer also tests every unspecified cross edge. It either creates a marked path of length at most12 or, in the shared-vertex case only, creates the wholly white affine triangle across the shared node. Hence the two surviving core Grams are fixed; their isolation proofs do not silently assume other cross contacts vanish.

All36 supporting-white isolation cases

The19-root shared-vertex core has the stated one-dimensional Gram radical and a timelike vector in its span. Its Gram-null combination is therefore an actual linear relation: it is null and orthogonal to a timelike vector, whose orthogonal complement is positive definite. This justifies pairing an arbitrary additional supporting white with that relation.

The independent checker reconstructs all27 preferred corners, including every finite/affine block, inverse diagonal, positive affine kernel and allowed extra contact. Their marked components have order16 or17. Every allowed contact has a nonzero radical coordinate; the negative and positive coordinates lie respectively within distance2 of the two opposite marked pairs. Acuteness and the actual radical relation force contacts on both sides whenever any contact exists, producing a white path of length at most6. Thus every additional supporting white is orthogonal to the core.

The18-root shared-edge core is nonsingular. Its eliminated positive block and residual

```
[[-5/66,-1/11],[-1/11,-5/66]]
```

give signature(17,1); its independently recomputed determinant is−1. The nine preferred corners have order16 and marked distance13. Their PSD tests leave possible contacts only in the two end sets{1,2,3} and{11,12,13}. A contact at both ends gives a marked path of length at most4.

The independently calculated full inverse block at either end is

```
[[8,10,5],[10,14,7],[5,7,4]].
```

For a nonzero one-end contact vector b, all entries have the same nonpositive sign and every nonzero magnitude is at least1. Hence b^T G^(-1)b>=4. Because the core already contains the unique negative direction, the root's orthogonal residual must have nonnegative square; instead its square is2−b^T G^(-1)b<=−2. This contradiction holds in every ambient dimension and for arbitrary contact magnitudes. It is not a positive-definite Schur argument applied incorrectly to a Lorentzian block.

At a finite preferred corner, simplicity identifies its isolated component with the actual ambient corner component. At an ideal preferred corner, the inherited actual factor-fiber lemma places the entire varying affine factor in W plus the two side roots. Thus hidden incident cusp walls cannot enlarge an isolated finite or affine marked component. This is the essential justification for the small-component payment in a nonsimple cusp.

Global payment and exact dimension calculation

The selective rule pays a marked pair if its distance is at most12 or its connected component has order at most17. The core analysis pays every quadrilateral. The inherited triangle argument supplies two paths of length at most8. Components of order at most17 have at most8 times their order in pairs, while larger classical finite or affine diagrams satisfy the inherited all-rank distance counts. Therefore finite vertices cost at most12n and a cusp costs at most

```
V sum_i min(12,r_i/2).
```

The convex-order transfer applies with cap24 exactly as with the earlier caps: transfers between two ranks below24, or two above24, preserve the charge and do not increase the curvature. The two rank regions must remain distinct; a transfer across the cap is not being assumed. The resulting normal forms contain copies of24, at most one smaller part and at most one larger part.

The reviewer independently enumerates these normal forms for d=136,…,145. Integer convolution and a separate inclusion-exclusion moment calculation agree on all1,175 profiles. Every margin is positive; the least is172096/1953125. The finite-vertex slack beta_n−12n is positive for every n=137,…,146, and the shape inequality retains its strict positive5(n−1) term. Together with the completed146 upper tail, this proves136 without a new hypothesis on cusp rank or factor count.

At n=136, the profile(24,24,24,24,24,15) has margin−59157/3906250. The checker confirms this boundary; no135 conclusion follows from the same scalar argument. For the compact-facet branch it independently obtains the parity polynomials with coefficients(12,208,4) and(112,212,4), proving ambient dimension at most103.

Validation and limits

Final generation and a fresh saved replay both pass. Syntax, all ten direct bindings, all19 producer input bindings and local report links pass. The producer source/data/report are bound at their final frozen hashes. The computations are independent exact rechecks of the displayed finite cases; the completeness of the26-type quasi-Lanner catalogue, the actual-face incidence theorem, all-rank graph budgets, convex-order theorem and shape inequality remain explicit inherited mathematical inputs. They are not replaced by bounded numerical testing. Source, data and report are final/frozen.

Source 259

Source path: `results/fresh_selective_distance13_dimension146_20261017.md`

SHA256: a006670090d8dcec3494947e929e0a8764ea0fb03146cb9afaea0f19f431c3dd

Selective distance13 and the unrestricted dimension146 bound

Every finite-sided, finite-volume real hyperbolic Coxeter polytope has dimension **at most146**. There is no arithmetic, compactness, simplicity, cusp-rank, factor-count or facet-count restriction. This improves the internally checked156 proof. A compact-facet application also improves the ambient bound127 to111.

The argument uses a selective corner payment: weight1 if the two marked nodes have graph distance at most13, **or if they lie in one connected component of order at most17**; weight0 otherwise. Distance13 alone is false. Its exceptional configuration is the known finite-volume H17 pyramid. The crucial new step is proving that the relevant small corner component cannot acquire additional supporting white attachments without creating a short paid path. No new construction dimension is claimed.

The producer and exact certificate check the geometric matrices and1255 reduced cusp profiles. The separate review records the independent checking and its scope. External refereeing and proof-assistant formalization are not asserted.

Actual quadrilateral stars

Use the actual-face incidence and closed-opposite-side arguments from the distance theorem. For a quadrilateral, W is the positive-definite set of supporting whites, and U,V are its opposite black pairs. W plus one black is positive definite; W plus adjacent blacks is positive semidefinite. Each W+U and W+V is indefinite. Choose inclusion-minimal indefinite cores L,M in them. Both black deletions of each core are positive definite. Minimal cores have order at most10, and the complete high-rank census supplies384 marked contexts, all of radius at most7. Low-order cores have radius at most3, independently of their labels.

Suppose no path of length at most13 joins U to V with only white internal nodes. If the cores overlap, every shared node must have distance7 from both marked sets: any smaller sum gives the forbidden short path. The only radius7 contexts are L23 with blacks{0,1}, far node8, and L25 with blacks{1,2},{1,3},{2,3}, far node9. Each has only one farthest node.

For a positive-definite black deletion A, its inverse is entrywise nonnegative. A new white contacting its node i nontrivially has contact magnitude at least1 and squared norm2. Its Schur residual is at most 2−(A inverse)ii. If that inverse entry is at least2, the enlarged block cannot be positive definite, contradicting its containment in W plus one black. This excludes equality as well as strict negativity.

The producer recomputes every marked distance and all21 distance-at-least6 ports in16 contexts. The only ports surviving both black-deletion tests are L25 node9 for the six pairs

```
{0,1}, {0,2}, {0,3}, {1,2}, {1,3}, {2,3},
```

and L25 node8 for{1,2}. In particular L23's far node8 cannot contact an external white. If L23 shared that node with M, a neighbor of the shared node in M would be an external white: another shared node would already give a shorter marked path. The same inverse test excludes this overlap.

Disjoint cores cannot be exceptional

Disjoint cores cannot be orthogonal, since the ambient Gram has negative index at most one. Any cross bond must have endpoint distances summing to at least13. Both endpoints are white. The preceding inverse tests leave only two underlying possibilities, with both cores of type L25.

For a far9-to-far9 bond, the white A3 tail on each side has residual4/3. Positive definiteness of their union bounds the squared bond magnitude by16/9, below2. Thus the nonright bond is simple, including when arbitrary finite labels, parallel or dotted pairs were initially allowed. Removing the two fork centers from the resulting20-root Gram leaves the positive block A12+2A2+2A1. Its two-center Schur complement is

```
[[-7/78, -1/13], [-1/13, -7/78]],
```

with two negative eigenvalues. This is the obstruction already used in the156 proof. If there were additional far-to-near cross bonds, the next obstruction applies instead.

For a far9-to-near8 bond, both marked contexts have radius7. There is a wholly white T-shaped subgraph with arm lengths1,2,5, namely affine E8. For the bond9–18 in two copies numbered0..9 and10..19, its nodes are

```
[18,19,17,16,9,8,7,6,5].
```

The certificate supplies its positive kernel. Every native nonright bond has magnitude at least1. Increasing that bond or adding other nonpositive contacts makes the quadratic form on the positive kernel no larger. Thus the actual white block cannot be positive definite. Inducedness of the displayed graph is unnecessary. This excludes every far-to-near case, including simultaneous cross bonds.

Consequently the only no-short-path case is two L25 cores sharing node9. They share no other node and have no additional cross bond, since either would supply a path of length at most13.

The shared core and its isolation

Number its first copy0..9 and its second copy10..18, using9 for the shared far node. Each L25 has edges

```
0--1, 0--2, 2--3, 0--4--5--6--7--8--9,
```

with all bonds simple; the second copy uses the analogous labeling. The19-root Gram has signature(17,1,1) and radical vector

```
z=(-6,-3,-4,-2,-5,-4,-3,-2,-1,0,
    6, 3, 4, 2, 5, 4, 3, 2, 1).
```

This is an actual linear relation among the roots, not just a relation in an abstract Gram matrix. Their span contains a timelike vector. The Gram-radical combination is null and orthogonal to that vector, hence lies in a positive-definite space and is zero.

For the left marked pair choose a preferred side as follows:

Marked pair	Preferred side	End type
{1,2}	2	Unbranched
{1,3}	1	Fork
{2,3}	2	Fork

Make the analogous choice on the right. Deleting the two unchosen black nodes gives a connected17-node corner: A17 for two unbranched ends, D17 for one fork, or affine D16 for two forks. There are respectively one, four and four marked contexts. Its preferred sides are distance14.

Let e be any further SUPPORTING white of the actual quadrilateral. In an A17 or D17 preferred corner, positive semidefiniteness after adding e forces e to be orthogonal to every node whose inverse diagonal exceeds2. For A17 the possible

contacts are only the two nodes nearest either end; for D17 only the long-tail endpoint and its neighbor. The inverse is entry-wise nonnegative, so multiple contacts cannot evade these tests. For affine D16, its positive kernel forces e to be orthogonal to all17 corner roots.

Including the two omitted black roots, all remaining possible contacts are at distance at most1 from U on the left or V on the right. Every left candidate has z-coordinate strictly negative and every right candidate strictly positive; the central zero-coordinate node is excluded. Pairing e with the actual radical relation gives

```
sum_i z_i (e,e_i)=0.
```

All products are nonpositive. A nonzero contact therefore requires at least one contact on each side. These give a path from U to V through e of length at most4, with only white internal nodes. This contradicts the assumption. Every additional supporting white is consequently orthogonal to the whole core.

At a finite preferred corner, all incident roots are W plus its two sides, so its marked connected component has exactly17 nodes. At an ideal corner, use the actual cusp-factor incidence lemma: the varying affine factor containing both side marks lies WHOLLY in W plus those two sides. Thus an isolated finite A17 or D17 component cannot be the varying component of an ideal corner. The affine D16 case is already an entire irreducible cusp factor. Missing spectator walls at a nonsimple cusp cannot enlarge the paid marked component.

This proves the selective payment for every quadrilateral. Triangles already have two paths of length at most8 by the inherited minimal-core argument, so receive at least2. Larger polygons require only nonnegative payment. Every k-gonal two-face therefore receives at least5−k.

Vertex costs and the dimension contradiction

A component of order m<=17 has at most m(m−1)/2<=8m pairs in total. For a larger component the payment counts only distances at most13. The inherited all-rank path, cycle and single-/double-fork formulas give at most m pairs at each distance; exceptional finite/affine diagrams have order at most9. Thus finite vertex cost is at most13n.

At an ideal vertex let the irreducible factor ranks be r_i and let V=product(r_i+1). The same pair bound and the trivial total-pair bound, multiplied by the exact factor-fiber multiplicity V/(r_i+1), give

```
C(v)/V <= sum_i min(13,r_i/2).
```

The shape-curvature theorem gives normalized ideal curvature

```
kappa/V = 3d/4 - (5/2) E|sum_i X_i|,
```

where d=n−1 and X_i is centered uniform on r_i+1 consecutive integers. Use the all-rank convex-order transfer already proved in the156 package, now with cap26. Concentrating two ranks below26 until one is0 or26, or two above26 until one is26, preserves the cost and cannot increase the curvature. The strictly increasing sum of squared ranks proves termination. Every partition is therefore dominated by copies of26, at most one part1..25, and at most one part>=27.

For d=146,…,155 there are121,…,130 such profiles,1255 in total. Integer convolution and a separate bounded-composition moment formula agree exactly. All curvature-minus-cost margins are positive; the minimum is830407/9565938 at ranks(26,26,26,26,26,16). The finite slack beta_n−13n is positive throughout n=147,…,156. The global shape lower bound also has the positive5(n−1) term. These inequalities contradict the corner-payment upper bound. The inherited 156 theorem excludes n>=157, completing the unrestricted bound146.

At n=146, the valid local link profile(26,26,26,26,26,15) instead has margin−515185/19131876. Thus this scalar comparison alone gives no145 bound; that profile is not an ambient finite-volume example.

For a compact facet of dimension d=n−1, all vertices are finite and its strict shape inequality gives beta_d<13(d+1). For d=111+2t and 112+2t, eight times the difference has coefficients respectively (12,224,4) and(120,228,4) in increasing powers of t. Both are positive for t>=0. Hence d<=110 and ambient n<=111 in this restricted class.

Known geometric sharpness and verification scope

The shared core is a finite-volume H17 pyramid with19 facets, already within Tumarkin's classification, Theorem2 and Table13. The producer independently reconstructs its entire chamber cone in Gram-dual coordinates p_i=(x,e_i). Its image

is z-perpendicular and its inequalities are p_i<=0. The extreme rays are the single zero-coordinate ray and the81 two-coordinate rays pairing opposite signs of z. Exact inverse-Gram calculations give80 finite vertices and2 ideal vertices, all in one time cone. Their sum is a strict timelike interior point; the tight rays of each wall span a facet. Thus the complete projective polytope lies in the closed Klein ball and has finite volume. This is an exact reconstruction of a known example, not a novelty claim.

It realizes the distance14 obstruction. The unrestricted improvement comes from its forced small component, not from asserting distance13 for every quadrilateral. The producer verifies384 marked contexts, 21 inverse-budget ports, nine preferred-corner contact domains, the physical Gram obstructions,82 chamber rays and1255 profile margins. Published classifications, inherited actual incidence arguments and the written all-rank reductions remain explicit mathematical inputs; finite matrix checks do not replace those arguments.

Producer generation and a fresh saved replay both terminate with PASS. Source syntax and all19 direct source bindings pass. A separate internal mathematical/source review and an independent Fraction-only implementation also pass, including the nine preferred-corner domains and all1255 profiles. The separate review binds this final producer package and records its own saved replay. No mathematical gap was identified by these internal reviews; external refereeing remains unperformed.

Source 260

Source path: `results/fresh_selective_distance13_dimension146_20261017_review.md`

SHA256: 0007ee1c8ddac2af0523b0f7b1122602f460cb44db5241d4484db964c369566b

Independent review of selective distance13 and dimension146

PASS. A separate mathematical/source read finds no gap in the stated unrestricted finite-volume dimension146 proof or the compact-facet ambient111 consequence. The accompanying standalone Fraction checker independently reconstructs the critical exact calculations. No arithmeticity, cusp simplicity, factor-rank, factor-count or facet-count hypothesis is added.

This review binds the final producer, certificate and proof. Its own checker imports no producer arithmetic, SymPy, numerical linear algebra or optimizer. The review certificate records exact inputs and outputs.

Independent arithmetic

Starting from the frozen literal minimal-core catalogue, the checker recomputes all384 finite-black-pair graph distances. There are16 contexts with21 white ports at distance at least6. Rational congruence replaces every square-root-two bond in these tested trees by rational Gram entries, tracking the diagonal scales explicitly; Gaussian elimination then recomputes all42 native inverse-diagonal values. The seven surviving ports agree exactly with the producer, including the radius6 L25 far-node cases with black pairs{0,1},{0,2},{0,3}.

For the disjoint exceptions it independently calculates the two-negative Schur matrix for the far-to-far join, the positive affine-E8 kernel in the wholly white far-to-near subgraph, and the A3 tail residual4/3. For the shared core it reconstructs the19-root Gram, its one-dimensional radical, the positive17-root block and its two-center Schur matrix with every entry$-1/12$. All nine preferred corners are independently eliminated. Their types are one A17, four D17 and four affine D16; all permitted additional contact positions, radical signs and short marked distances agree.

The1255 reduced link profiles of total ranks146 through155 are generated by a separate normal-form loop. Integer convolution and independent bounded-composition inclusion-exclusion agree on every curvature value. All margins are positive, with minimum830407/9565938; the boundary negative value at ambient146 is also reproduced. The compact-facet parity slacks are independently recovered as coefficients(12,224,4) and(120,228,4).

Geometric points checked

The opposite-block minimal cores have both black deletions positive definite because each deletion lies in the supporting diagram of an actual edge, which has a hyperbolic interior even when its endpoints are ideal. The inverse test therefore rejects equality2 as well as larger values. No ideal-corner semidefinite relaxation is substituted for that edge test.

Under absence of a white-interior path of length at most13, any overlapping cores share only their unique radius7 far node. An L23 overlap would have an external white neighbor at its forbidden inverse-budget node, so only shared L25 remains. Disjoint cores must have a cross bond by negative index one. Their endpoint-distance and inverse tests leave exactly the two stated L25 patterns. The far-near positive-kernel argument remains valid with stronger or additional negative contacts: it proves non-positive-definiteness of a wholly supporting-white principal block, without assuming the displayed affine graph stays induced.

The shared Gram radical is a true relation among the ambient normals. Its span contains a timelike vector; a null vector orthogonal to that vector must vanish. Thus pairing any additional supporting white with the radical gives the exact balance used in the proof. The preferred-corner inverse constraints exclude all zero-radical positions and leave negative-radical

contacts only near the left marks and positive-radical contacts only near the right marks. Acuteness and balance force both signs whenever there is a contact, producing a white-interior path of length at most4. This proves orthogonality of every additional supporting white, with no label bound.

The ideal-corner conclusion uses the full factor-fiber statement, not an assumption of simple cusps. The varying affine factor containing both marked sides lies wholly in W plus those sides. An isolated finite A17/D17 component therefore cannot be enlarged by omitted cusp walls into that varying factor. The affine-D16 alternative is itself complete. This is the essential reason the small-component payment is valid for arbitrary nonsimple cusps.

For vertex costs, components of order at most17 have at most8 times their order in pairs, below the allowed13 times their order. Larger finite and affine components are classical, and the inherited all-rank distance counts apply. Exact cusp-fiber multiplicities give the stated minimum of13 and r/2. The previously proved concentration argument applies with cap26; its all-rank convex-order justification is inherited and source-read, rather than inferred from the1255 profiles. The strict shape slack and earlier unrestricted156 theorem complete the bound146.

For a compact facet the proof uses its simple combinatorics and finite ambient vertices. The facet need not be an induced Coxeter polytope, and ideal vertices elsewhere need not be simple. At each facet vertex its paid corners form a subset of the ambient vertex's nonnegative weights, so the13n cost and the stated strict threshold apply.

The plain-distance13 counterexample is actual geometry

The review also independently checks the entire shared-core H17 chamber. In Gram-dual coordinates its cone is p<=0 with the single equation z.p=0. Its extreme rays are exactly the zero-radical coordinate axis and the81 pairs with opposite radical signs. Rational inverse-Gram calculations give80 finite and2 ideal vertices in a common time cone; their sum is a strict timelike center. Every wall's tight rays span an actual facet.

The two complete ideal links are separately certified: their active diagrams have positive affine kernels and total affine rank16, with component orders(9,9) and(17). Thus they are the bounded E8×E8 and D16 alcove links. This supplies an explicit proper-cusp check in addition to the complete causal-ray census.

For each of the nine marked-pair contexts, the15 supporting whites are positive definite and the face cone has exactly four rays spanning a three-dimensional cone. These are actual quadrilateral faces. Every corner's two marks have distance at least14 or belong to different components. Consequently the plain distance13 rule fails on actual finite-volume geometry; the selective small-component clause is necessary for this proof. The producer's identification with the published known pyramid is retained without a fresh independent catalogue-matching or novelty claim.

Verification status and dependencies

The complete finite/quasi-Lanner classification, original actual-face incidence facts, all-rank graph distance counts, convex-order theorem, exact shape inequality and earlier156 tail remain explicit frozen inputs. This review does not reclassify every diagram or repeat the published historical tables. It independently redoes the listed rational calculations and reads the complete new geometry and composition.

FINAL/FROZEN. Generation and a fresh saved replay both terminate with PASS. Syntax, nine direct source/data/report bindings, all19 inherited producer bindings and five local report links pass. The final producer report hash is a006670090d8dcec3494947e929e0a8764ea0fb03146cb9afaea0f19f431c3dd. This is an independent internal implementation and proof/source review, not external refereeing or proof-assistant formalization.

Source 261

Source path: `results/fresh_selective_distance9_dimension108_20261017.md`

SHA256: 05d7d24199f528d6154d1eec0c32ce61bcd759e16f72b1fa5bfe6aa6427e7e57

Selective distance nine and controlled marked components

Complete internally checked computer-assisted proof. Primary generation and a fresh saved replay pass. A separate mathematical/source review and independent exact reconstruction agree with the complete high-order census, all preferred corners, all 4,371 saved obstruction witnesses and the scalar bounds. The linked review records its own final binding and replay status. No gap has been identified. External refereeing and Lean formalization have not been performed.

The argument below gives dimension at most **108** for every finite-sided, finite-volume real hyperbolic Coxeter polytope. It has no arithmetic, compactness, simple-cusp, affine-factor-rank or facet-count hypothesis. Its compact-facet consequence is ambient dimension at most **79**.

The primary source and certificate record the high-order finite split and scalar inequalities. The complete small-core lemma supplies the additional low-order cases. The new supporting-component argument has a separate local review. The independent full review checks the new step. The completed dimension-116 proof supplies the inherited actual-face and cusp incidence, all-rank graph counts, shape inequality and convex-order transfer, and the excluded higher-dimensional tail.

Actual-face setup and complete core domain

For an actual quadrilateral, let W be its supporting white roots and U,V its opposite black pairs. Normalize roots to square two. W and W plus any one black root are positive definite; W plus an adjacent black pair is positive semidefinite. Each opposite pair gives an indefinite diagram. Choose minimal indefinite cores L in W+U and M in W+V. Each contains both its black roots; deleting either black root is positive definite. Their proper principal subdiagrams are positive semidefinite.

Assume no U-to-V path of length at most nine has white internal nodes. The complete high-order catalogue, of orders six through ten, has 384 marked contexts and maximum marked radius seven. Orders at most three cannot participate in a disjoint unpaid join or overlap. Two cores of order at most five cannot do so either. The remaining small/high cases require a separate classification; they are not discarded by the old distance-ten radius argument.

The small-core lemma proves that all order-four labels are at most six, and all order-five labels at most five. It exhausts all weighted graphs, obtaining 32 and 14 canonical minimal-indefinite types, respectively. Black-deletion and port tests leave 104 low-first joining-label cases. Eleven fail white positive definiteness; the other 93 have two negative directions. Six multiple-edge choices have a white affine cycle. Of 30 low-first overlaps, 24 fail the inverse test and six have an isolated marked corner of order at most 12. Reversing the cores covers the other orientation. Both fork/radical and label-five/nonsingular overlap cores are handled over all further supporting whites.

High-order disjoint cores

An external supporting white contacting a node of a positive-definite black deletion has squared projection at least that normalized inverse diagonal. Thus a simple contact requires both deletion inverse diagonals strictly below two. Among 1,262 high-order ports at marked distance at least two, exactly 378 pass. Each distant cross edge has endpoint distances summing to at least nine, giving 4,442 ordered possibilities.

For a simple edge between white ports x,y, eliminating the two white blocks gives the necessary condition

```
(G_WL inverse)xx (G_WM inverse)yy < 1.
```

It excludes 1,103 joins. For every remaining join, either one endpoint black-deletion inverse diagonal is at least one, or twice the displayed product is at least one. Therefore every magnitude at least sqrt(2) violates an actual-edge or white positive-definiteness condition. All labels at least four, parallel contacts and dotted contacts are excluded. Every remaining cross edge is simple, without a label cutoff assumption.

The 3,339 remaining joins have 85 underlying single-edge Gram types, each with exactly two negative directions. Some context pairs admit several candidate edges: 2,007 admit one, 348 admit two, 180 admit three and 24 admit four. All 1,332 subsets with at least two edges have an explicit nonnegative nonpositive-square witness in W plus one black. Extra negative bonds only strengthen these witnesses. Every nonempty cross-edge set is therefore impossible.

The producer rationalizes each required positive-definite principal diagram separately. A full joined indefinite graph may have an odd label-four cycle even though such a principal diagram is a forest. If a principal graph itself has no rational sqrt(2) gauge, replacing its nonright bonds by simple bonds gives an entrywise larger Z-matrix; a nonnegative obstruction for that relaxation remains valid for the native Gram. The certificate records the gauge or relaxation used for each witness. No irrational Gram is silently treated as rational.

High-order overlaps and fixed core Grams

Every shared white has marked-distance sum at least ten. Both distances are at least three. The source exhausts all injective partial identifications of eligible nodes, checking every contact on shared nodes. Of 312,316 candidates, 20,718 are consistent; inverse tests exclude 18,721 and actual-edge witnesses exclude 1,048 more.

The remaining 949 marked contexts have 54 oriented literal core types. All four adjacent corner decompositions are checked as finite/affine. An extra bond between exclusive core parts either creates a path of length at most nine (60,831 tests), or contradicts actual-edge positive definiteness (888 tests). Thus the full Gram of each surviving core is fixed before the extension argument is used.

At a selected corner, a further supporting white is orthogonal to every affine component by its positive kernel. On each finite component, a contact is possible only at normalized inverse diagonal at most two. Omitted black roots are also retained in the possible contact domain. This accounts for contacts anywhere in the core.

The ordinary argument handles 880 contexts. In a singular indefinite core, its radical is an actual root relation in a Lorentzian realization. The chosen contact domain avoids zero radical coordinates; balancing its two signs forces a forbidden short path. In nonsingular cores, projection onto the core leaves a positive-definite residual space. Positive inverse end blocks and short-path tests force a single simple contact. Pairwise residual Cauchy tests allow at most one core-contacting white; the entire indirect tail is bounded by the lemma below.

A rank bound for all indirectly attached roots

Let E be a connected positive-definite Coxeter Gram of order m, normalized to diagonal two. Every diagonal entry of E inverse is at least m/(m+1). To prove this, write E=2I-A with A entrywise nonnegative. Positive definiteness implies spectral radius(A)<2. Removing bonds to a spanning tree and replacing its bond magnitudes by one decreases A and therefore decreases the inverse entrywise, by the convergent Neumann series.

At a chosen root of that unit tree, let its branches have orders m_i. Induction and Schur complementation give

```
(E inverse)root,root >= 1 / (2 - sum m_i/(m_i+1)).
```

Since sum m_i=m-1 and sum m_i/(m_i+1)>=(m-1)/m, the required lower bound follows. The base case is the one-node Gram [2].

If a root of residual square rho contacts several orthogonal supporting components, with total order m, its projection cost is at least the sum of their order/(order+1) bounds, hence at least m/(m+1). Nonpositive multiple contacts cannot decrease this cost because inverse entries are nonnegative. Therefore, for rho<1,

```
m <= floor(rho/(1-rho)).
```

This bounds the entire contacted components, including every indirect tail. It is not a fixed-rank enumeration. Equality can occur for a simple A_m path contacted at an endpoint; rho=1 must be treated separately.

The new relevant-component argument

The remaining 69 contexts have a preferred marked component of order at most 16 after all extensions. Some roots outside that component can have residual square one, so bounding every external root by the preceding lemma would be invalid. Only attachments that can enlarge the marked component need to be controlled.

The exact inverse tests split each contact domain into two positive inverse end blocks. Any contact at both ends gives a short marked path; two contacts at one end already have projection square greater than two. Thus each external core-contacting white has one contact. All candidate projection squares are at least one. A projection-one node can have a simple or label-four contact; every larger label exceeds the norm budget. Every node that can directly enlarge the selected marked component has projection square strictly greater than one, hence only a simple contact.

If there is no such relevant contact, the marked component is isolated, regardless of external attachments elsewhere. Otherwise call its new white G, with residual vector g of square rho<1. Cauchy inequalities and the Coxeter gap between -1 and zero exclude every other relevant root. They also exclude all label-four external contacts in the presence of G. At most one possible outside simple-contact node remains. Call its root C and residual vector c. It lies outside the marked component and must be orthogonal to G: the allowed inner-product interval has lower endpoint greater than -1, while acuteness requires a nonpositive Coxeter product.

Write P for the positive product of the two core projections. Then g.c=-P. If c has square less than one, two roots of its type cannot coexist by the same positive-product test. If c has square one, two such roots would have to be orthogonal with opposite unit residuals. Their common product -P with g makes this impossible. Thus at most one C coexists with G.

The exact surviving residual cases are:

G projection square	C projection square	P	det S	(S inverse)GG
4/3	4/3	2/3	0	—
3/2	1	1/2	1/4	4
7/4	1	1/2	0	—
5/4	1	1/2	1/2	2

Here S is the Gram of g,c, with negative off-diagonal entry -P. If C is absent, every remaining white is core-orthogonal and the rank lemma bounds G’s entire tail. If det S=0, g and c have a positive linear relation; nonpositive products force every core-orthogonal white to be orthogonal to both, so no tail can join them.

If det S>0, its inverse is entrywise positive. A core-orthogonal norm-two white contacting G has projection cost at least (S inverse)GG. The value four excludes it. At the value two, equality forces contacts (-1,0) to G,C and puts that white entirely in their residual span, with negative coefficients. Its product with every other core-orthogonal supporting white is then nonnegative; acuteness forces zero. It is also orthogonal to C. Thus this one possible tail root cannot lead to a longer tail or to C’s other attachments. This explicitly excludes an indirect bridge through an otherwise unbounded projection-one attachment.

All preferred marked components in the 949 high-order cases have order at most 17, and the small-core lemma gives at most 12. At ideal corners, the inherited factor-fiber theorem places the entire varying affine factor inside W plus the two marked sides. Omitted spectator cusp walls cannot enlarge this component. No simple-cusp assumption is made.

Payments and the dimension conclusion

Pay one on a corner pair when its distance is at most nine, or when both marks lie in a connected component of order at most 17. The complete classification pays every quadrilateral at least one. The inherited triangle argument supplies two paths of length at most eight, hence pays each triangle at least two. Nonnegative weights suffice for larger polygons, giving total payment at least 5-k on a k-gonal two-face.

The inherited all-rank finite/affine path, cycle and fork counts give finite cost at most 9n. Small components of order at most 17 have total pair cost at most eight times their order, so the extra payment does not break that bound. At a cusp with affine ranks r_i and link vertex count V=product(r_i+1), cost divided by V is at most sum min(9,r_i/2).

The shape inequality and convex-order transfer apply with cap 18. For link dimensions d=108 through 115, the reduced profiles consist of copies of 18, at most one smaller positive part and at most one larger part. All 756 profiles have positive curvature-minus-cost, with minimum 45873/2476099 at six copies of 18. The finite slack is also positive in ambient dimensions 109 through 116. The completed 116 theorem excludes the higher tail, giving dimension at most 108.

At ambient 108, the profile (18,18,18,18,18,17) has margin -4659/260642. The same scalar weights do not prove 107; that profile is not a finite-volume realization.

For a compact facet of dimension d=n-1, finite payments imply beta_d<9(d+1). At d=79+2t and 80+2t, eight times the opposite difference has coefficient lists (12,160,4) and (88,164,4), respectively. Both are positive for all t>=0. Therefore d<=78 and ambient dimension is at most 79.

The finite classifications and numerical inequalities are exact. The written geometric and all-rank arguments remain explicit mathematical inputs. This package makes no construction, external-refereeing, proof-assistant or publication-priority claim.

Primary generation (session 41240) and the fresh `--verify` replay (session 2981) both terminated with exit code zero and PASS_SELECTIVE_DISTANCE9_CONTROLLED_EXTENSIONS_DIMENSION108. The replay recomputes the complete certificate and compares it with the saved JSON. The low-order package has its own completed generation and fresh replay; its separate mathematical/source review does not claim a second independent implementation of that entire enumeration.

Source 262

Source path: `results/fresh_selective_distance9_dimension108_20261017_review.md`

SHA256: 6b451725fca055fec1d6311bba67552925b55562e6cb6002f6f5e566ed045250

Independent review of selective distance nine

FINAL / FROZEN. Independent exact generation and fresh saved replay PASS; mathematical/source review PASS. No gap was identified in the unrestricted dimension108 or compact-facet ambient79 conclusions under the stated inherited inputs.

The review source reconstructs the new high-order calculation with `Fraction`, using elementary Gaussian, Schur and graph routines from the frozen preceding independent review. It imports no producer arithmetic, SymPy, numerical linear algebra or optimizer. Its certificate is compared with the primary certificate and binds the primary proof. The separately completed small-core classification and 69-context supporting-extension review remain explicit, independently checked inputs; this source does not rerun their entire computations.

Exact reconstruction

The new calculation rebuilds all384 high-rank marked contexts and their1,262 eligible ports, including378 that pass the two black-deletion inverse tests. It exhausts4,442 ordered single joins, all1,332 multiple-cross-edge subsets and all injective overlap identifications. The full raw overlap domain has312,316 candidates and20,718 consistent identifications:18,721 inverse rejections,1,048 actual-edge rejections and949 surviving contexts on54 oriented cores.

Every required positive-definite principal matrix is rationalized separately. The source does not require a rational square-root-two gauge for an entire indefinite diagram. If a required principal diagram has an odd label-four cycle, a unit-bond comparison supplies a nonnegative witness; the original native Gram can only decrease its square. This permits exact checking of the formerly exceptional cyclic cases without silently replacing the native geometry.

The review recalculates all85 underlying single-join inertias, all four corner decompositions of every surviving overlap, and every additional exclusive-core bond. A short path excludes60,831 such bonds;888 further cases have an actual-edge positive-definiteness obstruction. The independent code also evaluates every saved producer nonnegative witness directly in its exact principal Gram or stated comparison matrix. In total this is4,371 saved witnesses, beyond the independently generated rejection checks.

The corner calculation independently obtains all finite inverse diagonals, affine positive kernels, full-core radicals or inverses, complete contact domains and admissible end partitions. It reproduces the selected corner and component bound for every surviving context. The new69 cases are checked by the relevant-component residual inequalities, including the doubled-contact alternatives, multiplicity restriction and two-vector residual Schur test. The remaining880 contexts use radical balance or the ordinary single-root residual argument. All resulting marked components have order at most17.

The scalar calculation independently convolves each product-of-simplices link polynomial, checks its symmetry and total number of vertices, and compares the curvature formula with a separate inclusion-exclusion moment formula. It exhausts756 cap18 reduced profiles in link dimensions108 through115, with minimum positive margin45873/2476099. The ambient108 boundary profile has margin$-4659/260642$. The compact-facet parity polynomials have coefficients(12,160,4) and(88,164,4).

Mathematical review

No gap was found in the stated geometric composition. An actual edge supplies the strict positive-definiteness needed in the port tests, including when its endpoints are ideal. A nonnegative nonpositive-square witness remains an obstruction after adding or strengthening acute bonds. Disjoint cores with no cross bond already have two negative directions; the single- and

multiple-bond cases exhaust the other possibilities. Shared-node compatibility and all extra-cross tests fix each surviving core before its inverse or radical is used.

A singular indefinite core has an actual linear root relation: its Gram radical represents a null vector orthogonal to a timelike vector, and that perpendicular space is positive definite. For a nonsingular Lorentzian core the perpendicular residual space is positive definite, so the norm projection and Cauchy inequalities apply. Every nonzero normalized acute Coxeter contact has magnitude at least one. The positive inverse blocks and the short-path alternatives consequently control all contacts, rather than just chosen one-node recipes.

The general inverse-diagonal bound m/(m+1) for a connected positive-definite Coxeter Gram is valid by inverse monotonicity to a unit spanning tree and rooted Schur induction. It bounds an entire indirectly attached component, including nodes not directly contacting the new wall. At residual square one it does not give a finite rank bound; the new argument correctly restricts attention to the marked component instead. The separate69-context review closes the possible bridge through an outside projection-one root: positive residual relations force orthogonality, and the inverse-diagonal-two equality determines a unique sealed tail root. This is essential; a blanket tail-rank bound at residual one would be invalid.

At an ideal corner the actual factor-fiber lemma places the entire varying affine factor in the supporting whites plus the marked pair. Other cusp walls therefore cannot enlarge the paid component. No assumption that ideal vertices are simple is introduced. The complete small-order lemma covers the order-four and order-five cases that become relevant at distance nine, including all joining labels and the surviving label-five overlaps.

The inherited triangle payment, all-rank finite/affine graph counts, cusp-link shape inequality and convex-order concentration then give the stated9n finite budget and sum min(9,r_i/2) cusp budget. The strict scalar contradiction applies in ambient109 through116; the previously completed116 theorem covers the higher tail. The compact-facet calculation excludes facet dimension at least79 and hence gives ambient bound79. No intrinsic Coxeter assumption on that facet is used.

Scope and status

This is an internal independent arithmetic and mathematical/source review. It does not replace the published minimal-indefinite classification or the inherited written face/cusp and all-rank curvature arguments with a proof-assistant formalization. It makes no external-refereeing, novelty or new-construction claim. Bound generation7674 and fresh saved replay90919 both terminated with exit0 and PASS_INDEPENDENT_FRACTION_SELECTIVE_DISTANCE9_DIMENSION108_REVIEW. All16 direct source bindings,23 producer source bindings, Python syntax and six report links pass. The69-context local certificate matches explicitly, including all24 relevant-root cases and maximum component order16. The full independent comparison matches1,262 ports,4,442 joins,1,332 multiple-edge subsets,20,718 consistent overlaps,949 selected corners,54 core Grams,4,371 saved obstruction witnesses and756 reduced scalar profiles. The preceding unbound preparation and direct comparison also passed; the final saved replay recomputed the bound result without modifying any producer or predecessor file.

Source 263

Source path: `results/fresh_simple_cusp_affine_a_quadrilateral_20261017.md`

SHA256: d3f14e0624e18fd7fb2f1cf07644bbc245aaeadf74a944a8547b94c5f95b46a0

An affine-A corner of a quadrilateral is paid in every rank

Let P be a finite-volume hyperbolic Coxeter n-polytope, with every vertex simple and n>=14. If an actual quadrilateral has a corner whose full n-node diagram is affine A_(n-1), then the proposed mixed weight pays that quadrilateral: either the two marked roots at this corner have cycle distance at most3, or the marked roots at the opposite corner are adjacent. This resolves the complete affine-A-cycle branch, with no bound on labels or arc lengths. Other affine corner types remain outside this result; **no new ambient dimension bound is asserted**.

The checker and data use exact Fraction arithmetic in Q(sqrt(2)), a formal determinant expansion, and256 rectangle-corner checks. The argument below explains why these finite checks certify all ranks rather than a bounded search.

Geometry and complete replacement profiles

Use roots of norm2. Write a,b for the two marked roots of the affine corner, c opposite a, and d opposite b in the quadrilateral. If a,b have cycle distance<=3, the ideal weight already pays. Otherwise the two cycle arcs have edge lengths P,Q>=4, P+Q=n. Deleting a,b leaves the white diagram W=A_(P-1)+A_(Q-1). The roots a,b attach to the first and last ends, respectively, of both paths. In particular a.b=0.

The actual adjacent corner W+b+c is either spherical or, if ideal, a connected affine diagram on n nodes. The old diagram W+b is the single long path A_(n-1). Consequently its complete one-root extension list is:

Contacts of c with W+b	Result
none	A_(n-1)+A1
one bond3 at either endpoint	A_n
one bond4 at either endpoint	B_n
one bond3 at either next-to-end node	D_n
bonds3 at both endpoints	affine A_(n-1)

This is a classification argument in every n>=14. A new root touching the connected old path produces an irreducible diagram of order n. Exceptional finite and affine types are too small. Among the classical finite and affine diagrams, deleting one node to leave the entire path A_(n-1) gives precisely the displayed possibilities. In particular the other three large affine types do not have such a deletion. The finite and affine classification is the standard primary input recalled in Vinberg, section2.

Here the endpoints and next-to-end nodes lie near a, not b, because both arc lengths are at least4. Thus c.b=0 in every profile. The analogous list for d uses the last ends near b and gives d.a=0. The isolated A1 profile is explicitly retained; its height comes from its omitted-root contact with a or b.

If c.d<0, the opposite corner has adjacent marked roots, so it is paid by either the finite or ideal rule. It remains to contradict c.d=0. Write x=-c.a>=0 and y=-d.b>=0. Each is an arbitrary Coxeter contact: zero, 2cos(pi/m) for m>=3, parallel2, or dotted>2. No arithmetic field restriction is imposed on x,y.

Small affine diagrams force uniform height bounds

The frozen global exclusion for this all-simple class forbids every connected affine subdiagram of order below n. We use only affine A2,A3,B3,E8, of orders3,4,4,9. The checker supplies a strictly positive kernel and positive-definite one-node deletions for each of these four literal diagrams.

A simple leaf attached anywhere on the old n-cycle creates a forbidden affine E8: take the leaf arm of length1 and the two cycle arms of lengths2 and5. These nine vertices induce precisely the tree with arms(1,2,5). The unused complementary cycle arc has length n-7>=7, so its two selected ends have no extra edge. A positive kernel has center6 and successive arm coefficients (3), (4,2), and (5,4,3,2,1).

Let w be c's vector of nonnegative contact magnitudes with W, and put h=x+sum(w). The complete low-contact exclusions are:

Profile	Excluded x	Reason	Sufficient remaining bound for h
isolated	0,1,sqrt(2)	orthogonality to the full cusp; leaf E8; four-node B3 star	h>sqrt(2)
endpoint3	0,1	leaf E8; A2 triangle	h>=1+sqrt(2)
next-to-end3	0,1	leaf E8; A3 cycle	h>=1+sqrt(2)
endpoint4	0	B3 star at the old endpoint	h>=1+sqrt(2)
both endpoints3	0,1	A3 cycle; A2 triangle	h>=2+sqrt(2)

For isolated x=sqrt(2), the B3 star has center a, two simple cycle neighbors and the double-bond new leaf. For endpoint4 and x=0, its center is the first white root, with neighbors a, the next white root and c. The A2 and A3 diagrams in the other rows are literal triangles or four-cycles on the two new/marked roots and their stated white neighbors. The Coxeter alphabet has no value strictly between0 and1 or strictly between1 and sqrt(2). After excluding isolated x=1,sqrt(2), the next finite value exceeds sqrt(2); parallel and dotted contacts are larger.

For completeness, isolated x=0 cannot be a new spacelike root: it would be orthogonal to every normal at the full simple cusp. Those n normals are linearly independent, span the null hyperplane nu-perp, and have positive null sum nu. Their orthogonal complement is R nu. Thus a root orthogonal to all of them would be null. This also proves h>0. All other profiles have h>0 directly. The same arguments apply to d and its height k=y+sum(z).

Exact Schur identity

Set a0=1/P, b0=1/Q, alpha=a0+b0. The norm2 A_(P-1) inverse is

```
J_ij=min(i,j)*(P-max(i,j))/P,  1<=i,j<=P-1.
```

After eliminating W, the residual a,b Gram is alpha[[1,-1],[-1,1]]. For c define

```
B=v_b^t J w,  C=2-w^t Jw,
```

and for d define A=v_a^t Jz, D=2-z^t Jz. Finally E=w^t Jz. Here v_a,v_b are the nonnegative old endpoint contact vectors and J is the inverse Gram of all W. Since the first and last inverse rows sum entrywise to1, the residual a.c product is B-h and the residual b.d product is A-k. Under the remaining assumption c.d=0, the full residual matrix in order a,b,c,d is

```
S = [[ alpha, -alpha, B-h,   -A  ],
     [-alpha,  alpha, -B,    A-k ],
     [ B-h,    -B,     C,    -E  ],
     [-A,      A-k,   -E,     D  ]].
```

The actual quadrilateral star has n+2 roots in Lorentz dimension n+1, rank n+1, negative index1 and corank1. In particular det S=0. The formal 24-permutation determinant check gives

```
det S=(hk-kB-hA)^2-alpha*(C*k^2+D*h^2+2E*h*k).            (1)
```

Only this necessary determinant condition will be needed. No Gram matrix with the wrong rank or signature is promoted to an actual face.

For each profile introduce its two endpoint moments l=(l1,l2) and c0:

Profile	l	c0
isolated	(0,0)	2
endpoint3 on arc i	unit vector e_i	1
endpoint4 on arc i	sqrt(2)e_i	0
next-to-end3 on arc i	2e_i	0
both endpoints3	(1,1)	0

Then B=a0*l1+b0*l2 and C=c0+a0*l1^2+b0*l2^2. For the d-profile write m,d0; then A=a0*m1+b0*m2, D=d0+a0*m1^2+b0*m2^2 and E=a0*l1*m1+b0*l2*m2. The latter formula includes the boundary P=4: the two next-to-end positions can coincide, and the same inverse formula still gives4/P.

Put u=1/h>0, v=1/k>0 and L=u*l+v*m. Dividing(1) by h^2*k^2 and substituting the displayed expressions yields

```
F=1-2(a0*L1+b0*L2)
    -(a0+b0)*(c0*u^2+d0*v^2)-a0*b0*(L1-L2)^2 = 0.      (2)
```

The all-rank strict inequality

Use the conservative closed bounds

```
u<=1/sqrt(2)                  for isolated,
u<=sqrt(2)-1                  for endpoint3/4 or next-to-end3,
u<=1-sqrt(2)/2                for both endpoints3,
```

and likewise for v. For fixed a0,b0>0, F is jointly concave in(u,v): its quadratic part is the sum of a negative diagonal quadratic and a negative square. Therefore its minimum on each closed height rectangle is attained at a rectangle corner.

At the enlarged reciprocal-arc values a0=b0=1/4, the exact checker tests all8^4=256 corners in Q(sqrt(2)). Every value is nonnegative. Exactly three values are zero: both isolated profiles at their upper bounds, and the two choices of both next-to-end3 profiles on the same arc at their upper bounds. These are the only finite algebraic checks needed; concavity covers every real height in every rectangle.

Finally, for u,v>0 both partial derivatives with respect to a0,b0 are strictly negative. For example

```
dF/da0=-2*L1-(c0*u^2+d0*v^2)-b0*(L1-L2)^2 < 0.
```

All terms are nonpositive. If L1=0 and the middle term vanishes, some nonzero moment remains in L2 and the last term is negative; if both moments vanish, both profiles are isolated and the middle term is negative. The other derivative is identical with the two arcs exchanged.

Actual P,Q>=4 and P+Q>=14 give a0,b0<=1/4 with at least one strict. Hence F(actual a0,b0;u,v)>F(1/4,1/4;u,v)>=0, contradicting(2). Thus c.d cannot vanish. Its negative native product makes the opposite marked pair adjacent, proving the asserted quadrilateral payment.

Verification and scope

Generation and a fresh saved replay pass all256 exact rectangle checks, all four positive-kernel affine witnesses and their one-deletion tests, and the formal determinant identity. Syntax, all eight direct source hashes and all four local report links pass. An independent internal mathematical/source read finds no gap in the full extension list, low-contact exclusions, height bounds, boundary P=4 cross term, concavity or strict reciprocal-arc monotonicity. That read is distinct from an independent replay of the256 arithmetic values. The finite/affine classification and the inherited actual-cusp affine exclusion are explicit proof inputs. No unbounded rank or label is replaced by a numerical cutoff. The original mixed weight interface and all earlier61/127/163 results remain frozen and unchanged.

Source 264

Source path: `results/fresh_simple_cusp_affine_exclusion_20261017.md`

SHA256: 0dc0a705301f2f9c827bf83ff0162a41c12831e54b34f097231b531e2ff1a0cc

Global affine exclusion and the all-simple dimension bound61

Let P be a finite-sided finite-volume hyperbolic Coxeter polytope in H^n. If every ideal vertex of P is simple, then **n<=61**. Finite vertices of an acute-angled polytope are automatically simple. The statement permits both finite and ideal vertices and requires no arithmetic hypothesis.

The key structural fact is stronger than a restriction on black deletions: **every connected affine subdiagram of an all-simple P has exactly n nodes**. Consequently, when n>=10, every minimal indefinite subdiagram of order at most10 is Lanner, and every actual simplex face has dimension at most4. Supporting white roots, including affine subdiagrams containing every black root, are covered by this global argument.

The checker and saved certificate give the small exact radius, graph-cost, and threshold checks. This is a fresh successor to the frozen distance15 proof. Its compact-facet conclusion n<=127 remains unchanged. The unrestricted finite-volume bound163 remains unchanged: arbitrary ideal vertices need not be simple. No novelty or current-best published-bound claim is made.

An affine diagram must be a whole simple cusp

Use outward spacelike normals e_i and a future timelike strict interior point H, so H.e_i<0. Distinct normals have nonpositive products. Let C be a connected affine principal subdiagram. Its Gram matrix has a strictly positive kernel vector (c_i). Define

```
z = sum_{i in C} c_i e_i.
```

Then z^2=0, z.e_i=0 for i in C, and z.e_j<=0 for every other wall j. Moreover H.z<0, so z is nonzero and lies on the future null sheet. Thus [z] is a feasible point at infinity of P.

The projective closure of a finite-volume polytope is a finite convex polytope contained in the closed Klein ball. Every point of this closure on the boundary sphere is extreme: strict convexity of the ball forbids expressing it as a nontrivial convex combination of two distinct points of the closed ball. Hence [z] is an actual ideal vertex, not merely a feasible direction unrelated to the vertex set. Every wall of C contains this vertex.

At an ideal vertex the Euclidean link is a product of simplices. A simple vertex has exactly n incident walls, so its (n-1)-dimensional link is one simplex; its entire Coxeter diagram is connected affine of order n. Every proper principal subdiagram of this connected affine diagram is positive definite. Since it contains C, which is singular, C must be the whole diagram. In particular,

```
order(C) = n.                                    (1)
```

The primary Potyagailo–Vinberg paper, printed pp.64 and69 states the finite/ideal local face structure and the finite-volume projective convex-hull convention used here. The positive-kernel and strict-convexity argument above supplies the particular global conclusion. There is also direct primary confirmation in Felikson–Tumarkin, Section1, PDF pp.3–4: every parabolic subdiagram lies in an ideal-vertex diagram, and a simple ideal vertex has a connected parabolic diagram. Their displayed condition(iii), that parabolic diagrams have order n, uses these simplicity facts; it does not use the all-ideal assumption needed for their separate condition(ii) excluding elliptic diagrams of order n. We use only the former implication in the mixed-vertex class.

Consequences for small indefinite stars

An inclusion-minimal indefinite principal diagram L is Lanner or quasi-Lanner: if a proper principal subdiagram is affine, take one connected affine component C. Any outside node coupled to its positive kernel makes C plus that node indefinite. Minimality forces L to be exactly C plus that node. Thus a quasi-Lanner diagram of order at most10 contains a connected affine diagram of order at most9.

For n>=10 this contradicts (1). Therefore every minimal indefinite diagram of order at most10 is Lanner and has order at most5. These classification inputs are the same ones used in the frozen original-star simplex proof. The checker additionally reads all26 high-rank quasi-Lanner diagrams and confirms that their listed affine deletions are connected and of order at most9; it does not repeat their classification.

In an actual k-simplex face, the original star has n-k supporting white roots and k+1 black roots. Every indefinite principal subdiagram contains all black roots, because deleting any black leaves normals at an actual finite or ideal vertex. The original-star proof gives a minimal L of order at most10. Hence for n>=10,

```
k+1 <= order(L) <= 5,  so k<=4.                    (2)
```

In particular actual simplex faces of dimensions7,8,9 are all excluded, not just the dimension9 case. No assumption that an induced face is a Coxeter simplex is made. Equation (1) also handles the formerly delicate white affine deletion containing every black node; that affine diagram produces its own actual cusp elsewhere in P and is too small.

Triangle and quadrilateral payments at distance7

Retain the original-wall-star and graph-distance conventions of the distance15 proof. Put weight1 at a polygon corner when its two side normals are joined by a path of length at most7 within that corner's incident diagram, and weight0 otherwise.

For a triangle, a minimal indefinite star contains its three black nodes. It is now Lanner of order at most5. The tree joining the three marked nodes has at least two marked-to-marked paths whose interiors avoid the third black node. Each has length at most order(L)-2<=3. These paths lie in actual corner diagrams, so every triangle receives at least2.

For a quadrilateral, each opposite pair together with its white support is indefinite. Otherwise the positive-definite common-face argument or the nonzero feasible affine-kernel argument would put a finite or ideal point in two disjoint CLOSED opposite edges. This remains impossible for quadrilaterals with ideal vertices. Each resulting minimal indefinite diagram L or M contains both opposite black nodes and has order at most5.

In a connected N-node graph with two marked nodes, the distance of any node to the nearer mark is at most N-2: a shortest path to their set has no marked interior nodes. Hence this radius is at most3 for L and M. The two indefinite diagrams cannot be disjoint and orthogonal, because that would give two negative directions in a Lorentz space. They either overlap or have a joining edge. Joining their paths and taking a shortest path between their two black sets gives length at most

```
3+1+3 = 7.                                         (3)
```

Its interior has only white nodes; its endpoints are consecutive sides, so it lies in an actual corner diagram. Every quadrilateral receives at least1. The checker confirms the radius bound for every connected simple graph of orders2 through5 and every marked pair, rather than assuming a particular list of low-order Coxeter diagrams.

The vertex cost is at most7n, in every rank

At a finite vertex the n incident normals form a finite Coxeter forest. At a simple ideal vertex they form a connected affine diagram on n nodes. For paths, cycles, single forks and double forks, the all-rank formulas proved in the preceding distance15 report give at most N unordered pairs at each positive fixed graph distance. Therefore the total number at distance at most7 is at most7N. Summing over finite components gives7n.

The exceptional connected finite/affine diagrams have at most9 nodes, so even all their unordered pairs number at most4N<=7N. This includes the affine D4 four-leaf star, separated from the double-fork formula. No finite rank cutoff is used. The checker reconfirms the new cost for all245 independently counted classical graphs and the seven saved exceptional graphs; the inherited analytic formulas establish all ranks.

Thus the total polygon-corner weight is at most7n f_0(P), while the triangle/quadrilateral payments give at least2f_2$^{\{(3)\}+f_2}${(4)}.

Strict h-vector threshold

The projective closure of P is simple. The same strict simple-polytope h-vector inequality used in the frozen distance proof gives

```
2 f_2^{(3)} + f_2^{(4)} > b_n f_0(P),
b_n = n(n-6)/8                 if n is even,
      (n-1)(n-5)/8             if n is odd.
```

Consequently b_n<7n. For n=62+2t and n=63+2t, respectively,

```
8(b_n-7n) = 4t^2+124t,
            4t^2+128t+68.
```

Both are nonnegative for every integer t>=0. Equality at n=62 still contradicts the STRICT lower inequality. Therefore n<=61. Dimensions n<=9, where the preceding Lanner reduction was not invoked, already satisfy this conclusion.

Verification and limits

Generation and a fresh saved replay pass: 7,521 marked low-order graph checks, all26 high-rank affine-deletion checks, and245 inherited classical graph-cost checks. Syntax, nine direct source hashes, and report links pass. The independent mathematical/source review finds no gap and explicitly does not claim a duplicate arithmetic replay. The checker uses exact integers/Fractions and frozen source manifests. The polynomial identities and graph formulas, not the finite confirmation ranges, justify the unbounded dimension and rank claims. No optimizer, arithmetic field restriction, or induced-face Coxeter assumption enters the proof.

This lemma supplies neither an unrestricted bound61 nor a construction. The mixed-vertex all-simple class differs from the simple ALL-IDEAL class, for which Felikson–Tumarkin prove the stronger dimension bound8. No transfer of that all-ideal result to the mixed case is used here.

Source 265

Source path: `results/fresh_simple_cusp_affine_exclusion_20261017_review.md`

SHA256: 0fef92f303c359a5fe4fa6fb462b8c60188f7fca0cb36d1850f366c232718ccc

Independent proof and source review: simple-cusp bound61

PASS: no mathematical or source-level gap found in the stated scope. The conclusion is that a finite-sided, finite-volume hyperbolic Coxeter polytope whose ideal vertices are all simple has dimension at most61. It permits finite vertices and imposes no arithmetic hypothesis. It does not change the unrestricted163 bound or the compact-facet127 branch.

This review read the successor source, its saved data, the draft proof, and the inherited distance15 proof. It did not independently repeat the producer's 7,521 marked-graph calculations, 245 inherited graph checks or quasi-Lanner census. The producer reports successful generation and fresh saved replay. The review independently checked the geometric implications, classification scope, graph estimates and all-rank threshold algebra.

Bound frozen inputs:

- Source SHA-256: a710c0f90911ec582ba6133838468466eaabebbd1b2f2982335d4e344bcc7c5c.
- Data SHA-256: 01687a2e7e9896d0f077efad28f1633d61f7f14b12c20813358e823f03bbf34a.

The producer report was still receiving verification-status wording at review time, so its hash is deliberately not bound here.

Affine subdiagrams and actual ideal vertices

For a connected affine principal diagram, its positive kernel gives a normal combination z with $z^2 = 0$. It is not the zero vector: its product with a common strict future timelike interior point is strictly negative. This also fixes its future orientation. Its products with all other outward wall normals are nonpositive by acuteness, so it is a feasible null ray.

Finite volume matters at the next step. The projective closure is a finite convex polytope contained in the closed Klein ball. Strict convexity of that ball makes every feasible boundary-sphere point extreme; therefore this ray is an actual ideal vertex. It cannot be merely an unattained direction or an interior point of a larger projective face. All roots in the affine diagram belong to its incident-wall diagram.

At a simple ideal vertex the Euclidean link is one simplex. Its full incident diagram is irreducible affine of order n; every proper principal subdiagram is positive definite. The original singular affine diagram must consequently be the whole incident diagram. This argument includes affine subdiagrams consisting of white roots or containing all marked black roots.

The primary Potyagailo–Vinberg paper, printed pp.64 and69 supplies the local link structure and finite-volume convex-hull convention. There is also direct support in Felikson–Tumarkin, Section1, PDF pp.3–4: parabolic subdiagrams extend into ideal-vertex diagrams, and a simple ideal vertex has connected parabolic diagram. Their condition(iii) follows from simplicity; their different condition(ii), excluding elliptic diagrams of order n, uses the all-ideal hypothesis. The present mixed-vertex argument does not import condition(ii).

Minimal cores and payments

The inherited classification of inclusion-minimal indefinite diagrams gives Lanner or quasi-Lanner order at most10. A non-Lanner minimal core has a connected affine proper subdiagram of order at most9. This contradicts the affine order n result when $n \geq 10$. Thus every minimal indefinite core used in the argument is Lanner of order at most5. Smaller ambient dimensions already satisfy the final bound.

For an actual simplex face, deleting any black side root leaves roots at a finite or ideal vertex, hence a positive-semidefinite block. Every indefinite subdiagram therefore contains all black roots. The usual minimal-core argument then gives simplex dimension at most4, without assuming the induced simplex has Coxeter angles.

For a triangle, a spanning tree of its minimal core has two paths joining pairs of black nodes without the third in their interiors; each has length at most $5-2=3$. For a quadrilateral, opposite-side blocks are indefinite: a positive-definite block would produce a finite common point, and an affine block a feasible ideal common point, contradicting disjointness of the **closed** opposite polygon edges. Each opposite minimal core contains both marked roots and has radius at most3 from that pair. The two cores must overlap or have a joining edge, since two disjoint orthogonal indefinite blocks would give two negative directions. A shortest path between their black sets has only white interior nodes and length at most7. It belongs to an actual corner diagram.

Consequently distance7 corner weights pay at least2 on triangles and at least1 on quadrilaterals, including ideal corners. This does not assume all polygon vertices are finite.

All ranks and strict threshold

The inherited path, cycle, single-fork and double-fork formulas give at most N unordered pairs at each fixed positive graph distance. In particular the affine double-fork family is included in every rank. The small exceptional finite/affine diagrams have order at most9, so all their pairs total at most $4N \le 7N$. The affine D4 four-leaf star is treated as an exception. Thus every finite rank-n forest and every simple affine order-n vertex costs at most $7n$. Finite sampled ranks merely confirm these explicit formulas; they do not stand in for a tail proof.

The strict simple-polytope lower bound is

```
b_n=\begin{cases}n(n-6)/8&n\text{ even},\\(n-1)(n-5)/8&n\text{ odd}.\end{cases}
```

Independent substitution gives

$$8(b_{62+2t}-7(62+2t))=4t^2+124t, \qquad 8(b_{63+2t}-7(63+2t))=4t^2+128t+68.$$

Both are nonnegative for every integer $t \ge 0$. Equality at62 is excluded because the lower payment inequality is strict. Hence $n \le 61$.

The checker uses exact arithmetic, preserves source manifests and implements the stated checks. Its finite threshold loop confirms polynomial identities whose displayed algebra supplies the unbounded tail. This is an internal proof/source review, not external refereeing or formal proof-assistant verification. No novelty or best-published-bound assertion is supported or needed. The published simple **all-ideal** bound8 has a different hypothesis.

Source 266

Source path: `results/fresh_simple_cusp_affine_tree_completion_20261017.md`

SHA256: e13155b010fb64b9157848249a607deb593de255774749f4757681d5ef2038c4

All classical affine-tree quadrilateral corners: proposed completion

Final internally reviewed lemma: generation and fresh saved replay PASS. Scope: an all-simple finite-volume hyperbolic Coxeter polytope of dimension n>=18, with the weights in the frozen interface. The theorem proved below is that every quadrilateral having an affine B/C/D corner receives its required payment. The all-finite-corner transfer is a separate obligation; no dimension bound is asserted here.

Normalize every root to square2. Denote the old ideal corner by W+a+b, and the other corners by W+b+c, W+a+d, W+c+d. Assume the old corner is unpaid. Then a,b have graph distance at least4 and are not together in either prescribed six-node end segment. If any of b.c, a.d, c.d is nonzero, the corresponding corner is paid immediately. We therefore impose all three products zero. All one-root extensions below must be spherical, or must be the entire connected affine diagram of order n: a smaller affine component is forbidden by the global simple-cusp lemma.

Cusp normalization and the common Schur identity

In an affine tree give cusp coefficient1 to every main-path node and fork center, coefficient1/sqrt(2) to a terminal node on a bond4, and coefficient1/2 to a fork leaf. Write the null vector as nu=sum nu_i n_i. Replace the marked roots in the Schur calculation by nu_a*a and nu_b*b, leaving c,d at norm2. The scaled old Gram is a graph Laplacian: conductance1 on a main-path or terminal bond4 edge, and conductance1/2 on a fork edge. Its effective resistance between a,b is R. Eliminating W gives the marked block R^-1[[1,-1],[-1,1]].

For c write h=-c.nu>0, and similarly k for d. Positivity follows directly: all old contacts are nonpositive and every cusp coefficient is positive; equality would make c orthogonal to all n cusp normals, hence proportional to their nonzero null direction, contrary to c^2=2. If its contacts are localized at the a end, its residual square and cross contact to the scaled b have the form C=c0+l^2/R and B=l/R. For d use D=d0+m^2/R and A=m/R. Provided the contact supports do not cross along the surviving path, the projection cross term is E=lm/R. Consequently the formal four-root determinant identity gives

`det Schur/(h^2 k^2) = 1−(g_c+g_d)/R`, where `g_c=2l/h+c0/h^2`.

After eliminating the n−2 independent white roots, all four residual normals lie in a three-dimensional Lorentz space. The residual a,b,c determinant is −R^-1 h^2<0, so their span has rank3 and the four-root Schur matrix has exactly corank1. In particular its determinant is zero. A strict positive lower bound is a contradiction. Here is an explicit all-rank derivation. In the scaled white coordinates the Dirichlet inverse is the Green matrix of the weighted tree grounded at a,b. Put resistance coordinate z along their unique path. For two separated contact supports, with left coordinate z and right coordinate z', the Green entry is z(R-z')/R. Multiplying by the cusp coefficients in the native contact vectors gives E=lm/R. A left diagonal entry equals its one-ended tree value minus z^2/R. A branch off the path adds its common branch resistance to that one-ended value. Thus ordinary first and second path nodes give (c0,l)=(1,1),(0,2); a bond4 to the first gives (0,sqrt(2)). At a fork-leaf mark, its mate lies on a resistance2 branch attached at z=2 and has cusp coefficient1/2, yielding (1,1), or (0,sqrt(2)) for a bond4. These identities give every displayed terminal row. Outer components grounded directly at a have z=0, yielding the arbitrary outer cost q and zero cross moment. The two exceptional E8 supports overlap these local regions and are computed separately. No finite-rank cutoff is used.

There are three exhaustive positions for a marked root: a main-path node, including a fork center; a terminal bond4 node; or a fork leaf. Two terminal marks at the same fork are already paid. The cases below exhaust the remaining possibilities, up to exchanging a,b.

Main-path marks, including fork centers

Let P be their graph distance. The outer components beyond a have type B_r, r>=1, or D_r, r>=2; D2 here means two orthogonal A1 leaves, with a their common fork center. There are P-1 intervening white path nodes and an analogous outer end beyond b. The total order is n=r+s+P+1 and R=P.

The short-end proof applies to B_r and D_r, r>=3. Its outer inverse formulas and arbitrary-contact argument do not require an upper bound on r. Add D2 with J=I/2, old attachment v=(1,1), and t=Jv=(1/2,1/2). For a nonnegative outer contact vector w let q=w^T Jw<2 and t0=v^T Jw. The strict inequality follows because W+c supports an actual edge and is positive definite, so its subdiagram consisting of the outer block and c is positive definite. If the opposite native magnitude x=-a.c is at least1, q>=t0^2 and t0>=1/2 give `(2-q)/(x+t0)^2<=1`. If x=0 and at least two outer contacts are nonzero, t0>=1 gives q+t0^2>=2. A single outer contact of magnitude1 creates a proper affine E8 with arms1,2,5; magnitude sqrt(2) creates a proper affine B4 (one doubled terminal arm at a fork). Thus any remaining single contact has magnitude d>=phi>3/2, and its contribution is `8/d^2-2<14/9`.

When c joins the inner component to D2, classification gives either one leaf by a bond3 (the other A1 remains separate), or both leaves by bonds3. The inner contact is always a bond3 at its long endpoint. The respective outer data (q,t0) are (1/2,1/2) and (1,1), with h>=1+t0. Their contributions `2/h+(1-q)/h^2` are at most14/9 and1. A doubled bond to a single leaf would create a proper affine component beside the unused A1; a bond5 cannot occur in a large finite classical component. All other B/D joins are the complete short-end list and contribute at most13/9. Isolated and inner-only profiles contribute at most1 by the same proper affine B/D, A2 and small-star exclusions as in that proof. Hence every ordinary main-path profile has g<=14/9.

The only additional finite one-root extensions when an inner component is small are D6 to E7 and D7 to E8. B_j with j>=5 and D_j with j>=8 only admit the long-end bond3 finite extension. Here a small inner rank is r+P<=8; since P>=4 and n>=18 the opposite outer component has rank>=9. D6 to E7 is directly paid, since every pair in a finite component of order<=7 is paid. D7 means either r=2,P=5 or r=3,P=4. In the first case the marked a is the E8 branching vertex, so Vinberg's finite special-vertex lemma pays it (original paper, §6.3, printed p81). The second case is treated below. Thus in every nonexceptional case

`det Schur/(h^2 k^2)>=1-28/(9P)>=2/9>0`.

The main-path D3/P4 to E8 exception

Use an old D-type end whose fork center is0, fork leaves are L,L', and a is the next main-path node1. Let b be main-path node5. The new d has its single bond3 to L', so W+a+d has an E8 component. The arbitrary opposite contact is y=-b.d and k=y+1/2. At y=0 the old affine tree plus d contains a proper affine E8, so y>=1 and v=1/k<=2/3. Exact elimination gives alpha=1/4, A=1/2, D=5/4.

An outer-only c makes c,d belong to the same spherical component of W+c+d of order at most5; that corner is paid. The remaining c profiles and normalized determinant bounds are:

c profile	reciprocal bound	normalized determinant	lower bound
isolated	u^2<=1/2	1-v-v$^{2/16-u}$2/2	1/18
inner-only bond3	u<=1/2	1-u/2-v-(2u-v)^2/16	11/144
join through fork center	u<=1/2	1-u/2-v-v^2/16	1/18
join through one fork leaf	u<=2/5	1-u/2-v-(u+v)^2/16	14/225

For the leaf join, x=0 creates an affine A4 cycle, so h=x+3/2>=5/2. The displayed expression uses the larger cross projection E=3/4 when c,d use the same leaf; the other leaf has E=1/4 and improves it. The inner-only and leaf-join quadratics are concave, so their minima on the stated rectangles occur at corners. Every bound is strictly positive.

A terminal bond4 mark

Let a be that terminal node. An unpaid old pair has graph distance P>=6 from a to a main-path b; R=P and nu_a=1/sqrt(2). Deleting a leaves a large finite B or D diagram. Its complete finite-or-connected-affine one-node extensions are: isolated;

bond3 to its long endpoint; bond4 to that endpoint; or bond3 to its next node. The last two are full affine extensions. This list follows by deleting the new node in the classical finite/affine diagrams: finite B/D permits only an added long-end bond3 at rank>=9; affine C/B adds a long-end bond4, and affine B/D adds a second-node bond3. No exceptional finite or affine diagram has the required large order. The corresponding (c0,l,h) are

`(2,0,x/sqrt(2))`, `(1,1,1+x/sqrt(2))`, `(0,sqrt(2),(x+2)/sqrt(2))`, `(0,2,1+x/sqrt(2))`.

In the first case x=0 is impossible by orthogonality to the full cusp, x=1 induces affine F4, and x=sqrt(2) induces affine C2. The conservative bound x>=sqrt(2) gives g<=2. In the second case x=0 induces the affine star(4,3,3), so x>=1 and again g<2. The third gives g=4/(x+2)<=2. In the fourth, x=0 gives a proper affine B4 with the doubled terminal at the end of its length-two arm. Thus x>=1 and g<=8-4sqrt(2)<5/2. The uniform bound is therefore5/2. The opposite main-path profile has g<=14/9: its near-terminal inner component is B_P of rank>=6, with no exceptional finite extension. Hence the determinant is at least `1-(5/2+14/9)/P>=35/108>0`.

A fork-leaf mark

Let a be one leaf, with mate L and center0. An unpaid pair with a main-path b has graph distance P>=5; R=P+1 and nu_a=1/2. Deleting a leaves a large finite B or D diagram. Complete finite-or-connected-affine one-node extensions are: isolated; a bond3 to L (finite); a bond4 to L (full affine); or a bond3 to0 (full affine). The same complete large B/D deletion list applies. Their (c0,l,h) are

`(2,0,x/2)`, `(1,1,(x+1)/2)`, `(0,sqrt(2),(x+sqrt(2))/2)`, `(0,2,1+x/2)`.

For an isolated c, x=1 gives affine E8, x=sqrt(2) affine B4, and x=0 is impossible. Hence x>=phi>sqrt(2), giving g<=4. For the mate-leaf profile, x=0 gives affine E8 and x=1 gives affine A3; thus h>=(1+sqrt(2))/2 and g<=8-4sqrt(2)<4. The mate-leaf bond4 profile gives g=4sqrt(2)/(x+sqrt(2))<=4 even without excluding x=0. The center profile gives g<=4 for all x>=0. The opposite inner component is D_(P+1). Its D6 to E7 extension is paid. Apart from D7 to E8 below, the opposite profile has g<=14/9. Thus `1-(4+14/9)/R>=2/27>0`.

The fork-leaf P6 to E8 exception

Here b is main-path node5, P=6, and d has a bond3 to the other fork leaf L. The same affine E8 exclusion gives k=y+1/2>=3/2. Literal elimination gives alpha=1/7,A=5/14,D=8/7. If c attaches by a bond3 to L, the opposite corner has a finite D8 component with c,d as its two fork leaves (the far outer end is separated by the deleted b). They lie in the same fork-invariant end segment, so the finite weight rule pays it. If c attaches by a bond4 to L, the opposite corner contains the proper affine star(4,3,3), impossible. For the center profile c→0, the opposite component is finite E8 with arms1,2,4; it must not be called affine. Here x=0 supplies a proper affine D4 centered0 with leaves a,c,L and main node1, and x=1 supplies affine A2(a,0,c). Therefore u=1/(1+x/2)<=2-sqrt(2). Its exact normalized determinant is `1-4u/7-5v/7-v^2/28`, at least `(36sqrt(2)-40)/63>0`. For isolated c put u=2/x<4/3 using x>=phi>3/2, and v=1/k<=2/3. Then

`det Schur/(h^2 k^2)=1-5v/7-v^2/28-2u^2/7>0`,

because the expression is zero at u=4/3,v=2/3 and strictly decreases in both positive variables.

Two terminal marks

They must be at opposite ends. The effective resistance is R=n-1, for a pair of doubled terminals, a doubled terminal and a fork leaf, or two fork leaves. Their respective total g bounds are5,13/2,8. Since n>=18, all are strictly smaller than R. The local supports are separated, so the same factorized Schur formula applies without an E8 overlap.

Status

The exact checker and saved data bind the complete proof inputs. Generation passes 1,191 unpaid marked-pair contexts at orders18,19,23, 82,542 generic profile-pair formula probes, 14 literal exceptional E8 calculations, the scalar margins, and the additional affine B4/D4 positive-kernel witnesses. The finite checks verify explicit formulas; the Laplacian/Green derivation and finite/affine classification supply their all-rank scope. Generation and a fresh saved replay both terminate with PASS. AST and all 11 direct source hashes pass. Separate internal mathematical/source reviews by the root and

construction agents pass the corrected four-profile terminal lists, cusp normalization, complete residual classification and exceptional Schur formulas. Neither review claims to duplicate the 82,542 producer probes; the construction agent separately recomputed the terminal and exceptional formulas. The separate terminal and E8 audit also passes generation and a fresh saved replay, including 48 independent physical normalization checks and symbolic exceptional determinants. That audit binds the final producer source/data hashes. This report and its producer source/data are now frozen.

The two additional terminal affine replacements and the finite D8/E8 descriptions in the fork-leaf exception are included explicitly. No dimension conclusion is asserted: the all-finite-corner transfer and final composition remain separate obligations.

Source 267

Source path: `results/fresh_simple_cusp_affine_tree_corridor_20261017.md`

SHA256: 4e455fbc256390283bd755186e31ee66358ca8dc88fcd25c25c6e679b9ccf7e6

Long central corridors at affine tree corners

There is an all-rank paid branch for quadrilaterals with an affine B/C/D corner. It is narrower than the full tree case. Suppose its two marked roots a,b lie on the central simple-bond path, at distance P>=4, and the two end diagrams beyond a,b are finite B or D diagrams of ranks r,s>=9, attached at their long endpoints. Then the opposite corner has adjacent marked roots and pays the quadrilateral. Here n=r+s+P+1>=23. Marked roots in the bounded decorated-end regions remain open. This lemma gives **no new ambient dimension bound**.

The checker and data bind the affine-A proof and its independent review. The ambient assumptions are again finite volume and simplicity of all vertices. Thus the global exclusion forbids every affine subdiagram of order below n.

The complete four-profile list

Deleting a from the old affine corner gives two finite components, each of type B or D and rank>=9. The same holds after deleting b. A new root c replacing a can therefore have only these contacts with the remaining old diagram:

1. no contacts;
2. one bond3 at the exposed long endpoint of the outer component;
3. one bond3 at the exposed long endpoint of the inner component;
4. bonds3 at both of those long endpoints.

To see completeness, every old component remains whole in any connected new component. A connected finite extension of one B_r or D_r, r>=9, can only be B_(r+1) or D_(r+1), extending its long endpoint by a simple bond: exceptional types are too small, and the retained double bond or fork rules out the other classical finite types. A connected affine extension of only one old component would leave the other component disconnected. This is neither a spherical corner nor the irreducible affine diagram of a simple ideal corner, so it is forbidden.

If c touches both old components, the new connected diagram has all n nodes. Two double-bond ends, a double-bond end and a fork, or two forks can occur only in the corresponding affine C, B, or D diagram. In each case the joining root has simple bonds to exactly the two long endpoints. An extra attachment would produce a cycle or additional branch absent from those diagrams. This proves the four-profile list using the full finite/affine classification, in every rank. It does not extrapolate a finite enumeration. In particular there are no free unbounded-label contacts to old white roots in this list.

For c the outer component lies beyond a, and the inner component contains b. The nearest contact of c on that inner component is the first white node after a. Since P>=4, c.b=0. Analogously d.a=0 for the root d opposite b. If c.d<0 the opposite marked pair is adjacent and already paid; assume c.d=0 for a contradiction.

Explicit low-contact exclusions

Normalize all roots to norm2. Put x=-a.c. For the isolated profile, x=0 would make c orthogonal to the entire simple cusp, hence null. The value x=sqrt(2) creates the four-node affine B3 star centered at a. The value x=1 makes c a simple leaf on the central corridor.

Every such simple leaf creates a proper affine E8 with arms(1,2,5). For a leaf at a, take the length5 arm towards the long outer block and the length2 arm towards b. For a leaf at the exposed outer port, take five steps into that block and two through a. For a leaf at the first inner white node, take two steps towards a and the outer port, and five towards b and, if necessary, into the other long block. Ranks r,s>=9 and P>=4 provide these simple-bond arms; the long port of D9 still has

six simple steps before its fork, and B9 has seven before its double bond. The original diagram is a tree, so no extra edge closes these arms. The new leaf has no other old contacts in each case.

Thus isolated x=1 is excluded. A single-bond profile with x=0 is the same forbidden leaf construction; with x=1 it creates an affine A2 triangle. For the both-ends profile, x=0 creates an affine A3 four-cycle, and x=1 creates an affine A2 triangle. All these affine diagrams have order<=9<n. The literal positive kernels are checked in the inherited affine-A package.

The normalized cusp kernel has coefficient1 at a,b and their neighboring long ports. Consequently the height h=-c.nu equals x, x+1, or x+2 for the isolated, single, or both profiles. The same complete Coxeter alphabet argument as before gives conservative reciprocal bounds

```
u=1/h <= 1/sqrt(2), sqrt(2)-1, or 1-sqrt(2)/2,
```

respectively. All heights are positive. The identical bounds hold for v=1/k of d.

Schur reduction and uniform gap

The inverse Gram at the long port of either finite B_r or D_r equals1. This holds in every rank: its inverse column is1 along the main path, with the last B-node coefficient1/sqrt(2), or the two D-fork leaves each1/2. Direct multiplication gives the unit vector at the port. The checker confirms these literal columns in184 examples of ranks9..100; the displayed multiplication identity proves every rank.

Eliminating W leaves the residual a,b block alpha[[1,-1],[-1,1]], now with alpha=1/P. The intervening A_(P-1) path supplies the only cross term between c and d. Introduce moment l and constant c0 for c:

Profile	l	c0	upper bound on 2*l*u+c0*u^2
isolated	0	2	1
outer single3	0	1	3-2sqrt(2)
inner single3	1	1	1
both3	1	0	2-sqrt(2)

Then B=l/P and C=c0+l^2/P in the notation of the affine-A Schur identity. For d use m,d0, giving A=m/P, D=d0+m^2/P and E=l*m/P. Substituting into that formally checked determinant identity cancels the quadratic cross term and gives

```
detS/(h^2*k^2)
  =1-[2*(l*u+m*v)+c0*u^2+d0*v^2]/P.
```

Each single-root expression is increasing on its permitted reciprocal height interval and is at most1 by the exact table. Therefore the right side is at least 1-2/P>=1/2. The actual quadrilateral star has corank1, so detS=0, a contradiction. Hence c.d<0, as claimed.

Status

The four exact profile costs and184 inverse-column checks are generated and freshly replayed by the checker. Syntax, all ten direct source hashes and all five local links pass. A separate internal mathematical/source read confirms the full B/D extension list, the three leaf-E8 embeddings including P=4, the all-rank port columns and the scalar Schur cancellation. That read does not claim a duplicate184-case arithmetic replay. The completeness arguments above are classification and all-rank identities, not finite cutoff claims. Only this long central-corridor branch is closed. The full affine-tree quadrilateral obligation, including all decorated-end positions, remains open. The mixed rule and earlier dimension bounds are unchanged.

Source 268

Source path: `results/fresh_simple_cusp_affine_tree_short_ends_20261017.md`

SHA256: 8400694a494c598964b90001890b76f116c85fbbebb3c057204fafb9b376ebf2

Affine-tree quadrilaterals with short outer end blocks

There is a further all-rank paid branch for the proposed mixed finite/ideal 3n rule. In an all-simple finite-volume Coxeter polytope of dimension n>=14, let an affine B/C/D corner have central-path marked roots a,b at distance P>=4. Write the diagram as

`F_r – a – (P–1 white path nodes) – b – F_s`,

where each end is B_r (r>=1) or D_r (r>=3). Here B1 means A1 joined to a by a bond4; D3 means A3 joined at its middle node. If **r+P>=9 and s+P>=9**, the opposite marked roots c,d are adjacent. The quadrilateral is therefore paid.

This removes the large-outer-end restriction from the earlier corridor lemma. It leaves only 13 central-position types, up to exchanging the two ends, together with fork-leaf and fork-center positions. It is not a full quadrilateral theorem or a dimension bound.

The checker and data check the closed inverse formulas, the five distinct join costs, and literal positive kernels for the two exceptional affine exclusions. All-rank completeness comes from the arguments below, rather than the range of the confirming matrix calculations.

Complete extension classes

Deleting a leaves the short outer F_r and an inner B/D component of rank s+P>=9. Every new connected component contains whole old components. An extension touching only the outer component must be finite: an affine component plus the untouched inner component cannot be the connected affine diagram of a simple ideal vertex. Its contact vector may otherwise be arbitrary, including unbounded labels for an A1 outer component. An extension touching only the inner component is its unique finite long-end extension by a bond3. Extending it affinely would again leave a disconnected corner. The isolated case is also allowed.

If c touches both components, its connected diagram has all n nodes. The complete finite/affine classification now gives a classical diagram: exceptional diagrams have order at most9. The large retained B/D component determines its long port. At the outer end the possibilities are precisely:

- B1: bond3 for a finite classical extension or bond4 for an affine extension;
- B2: bond3 at either endpoint;
- B_r, r>=3: bond3 at its long port;
- D3: bond3 at either path endpoint for a finite extension, or at the middle for an affine extension;
- D4: bond3 at any of its three leaves;
- D_r, r>=5: bond3 at its long port.

The contact to the inner component is always bond3 at its long port. This list follows by deleting a node from classical paths and single/double forks; there is no remaining exceptional or cycle case with the large B/D component. A proposed type incompatible with the other end may be retained in the table as a harmless relaxation. In particular c.b=0 because P>=4. The same reasoning gives d.a=0. It suffices to contradict c.d=0, since a nonzero nonpositive product is an adjacent pair under the rule.

Exact end identities and arbitrary outer-only contacts

Normalize roots to norm2. Let J=G(F_r)^−1 and let v be the positive contact vector from a to F_r. Except for B1 it is the unit long-port vector; for B1 it is sqrt(2) times that vector. In every case v^T J v=1. The cusp coefficients on F_r are t_j=(Jv)_j.

For B_r, r>=2, the diagonal of J is 1,2,…,r−1,r/2, and the t_j are 1,…,1,1/sqrt(2). For B1 the diagonal is1/2 and t_1=1/sqrt(2). For D_r the diagonal is 1,…,r−2,r/4,r/4, and t_j are 1,…,1,1/2,1/2. The checker displays and multiplies the full closed inverse matrices in 78 examples; sparse multiplication of the same formulas proves them for every rank.

For an outer-only contact vector w>=0 put q=w^T Jw and t=v^T Jw. Its finite corner requires q<2. Every nonzero coordinate of w is at least1, and all t_j>=1/2, so t>=1/2. Cauchy–Schwarz in J gives q>=t^2. If x=−a.c>=1, then

`(2-q)/(x+t)^2 <= (2-t^2)/(1+t)^2 <= 1`.

If x=0, the same bound follows from q+t^2>=2. With two or more contacts, t>=1 and Cauchy–Schwarz already proves this. With one contact of magnitude at least sqrt(2), the displayed diagonals prove it as well. The only remaining possibilities are one bond3 at B1, the non-port endpoint of B2, or a fork leaf of D3,D4,D5,D6.

For B1/B2 that extra leaf and the first old path nodes give the five-node affine F4 path with bonds(3,4,3,3). For D3..D6 the added leaf extends one fork arm to length2; the other fork arm has length1, and the old long direction supplies length5. This is the nine-node affine E8 with arms(1,2,5). The full dimension n>=14 supplies the five simple steps before the other decoration. These are induced diagrams: the new root has precisely its one outer contact and x=0. Both are proper affine subdiagrams, forbidden by the all-simple affine exclusion. The checker verifies their positive kernels and positive definite one-deletions. Thus every outer-only profile contributes at most1, without an angle or rank cutoff.

Heights of isolated and inner-only roots

Give the old cusp kernel coefficient1 at a,b and on the intervening path. For an isolated c its height is h=−c.nu=x. The value x=0 would make c orthogonal to the full simple cusp, hence proportional to its null vector. The value x=1 supplies a proper affine B/D diagram: retain the whole outer end, a, and the two simple leaves c and the first inner path node. This diagram has order r+3<n. The value x=sqrt(2) gives the four-node affine star with bonds(4,3,3), unless the outer bond itself is4, when the two bond4 contacts give the three-node affine C2. Hence the conservative bound h>=sqrt(2) is valid and 2/h^2<=1.

For the inner-only bond3 profile, h=x+1. At x=0 the same affine B/D construction uses the outer end extended by a, with the first inner node as its branch point, and c and the second inner node as its leaves. Its order is r+4<n. At x=1, a, the first inner node and c form an affine A2. Therefore h>=1+sqrt(2), giving `2/h+1/h^2<=1`. These exclusions work with the entire outer block and do not require it to have large rank.

Join costs and the corank contradiction

For a join, write q=w^T Jw and t=v^T Jw for its outer contact. Its height is h=x+1+t, so h>=1+t. The complete join list has only the following five distinct scalar data:

Outer profile	q	t	bound on 2/h+(1−q)/h^2
B1, finite bond3	1/2	1/sqrt(2)	less than13/9
original affine port	1	1	1
B2, exchanged endpoint	1	1/sqrt(2)	less than13/9
D3, path endpoint	3/4	1/2	13/9
D4, other leaf	1	1/2	4/3

All entries are checked exactly in Q(sqrt(2)). No restriction on the opposite contact x beyond x>=0 is needed here.

Eliminate all white roots W. The a,b block is `(1/P)[[1,-1],[-1,1]]`, because each eliminated end costs exactly1. For c use moment l=0 for isolated/outer-only and l=1 for inner-only/join. Its residual norm is `C=c0+l^2/P`, with c0 respectively

2,2−q,1,1−q; its cross-coupling to b is l/P. The analogous quantities for d are m,d0. Since the two possible inner contacts lie at opposite ends of the intervening path, their only projection cross term is lm/P. The formally checked determinant identity consequently reduces to

`det(S)/(h^2 k^2) = 1 - [g_c+g_d]/P`,

where `g_c=2l/h+c0/h^2`, and similarly for d. The preceding exhaustive classes give each g<=13/9. Therefore

`det(S)/(h^2 k^2) >= 1-26/(9P) >= 5/18 > 0`.

The actual quadrilateral star has corank1, so det(S)=0. This contradiction proves c.d<0.

Remaining positions and verification scope

Failure of an inner-rank condition means, after exchanging ends, r+P<=8. Since P>=4, B ends give ten pairs (r,P), and D ends give three. The data save all 13 pairs explicitly; the other end may have arbitrary rank. Fork leaves and fork centers are separate unclosed positions. This is an inventory of remaining positions, not an assertion that they fail.

Generation and a fresh saved replay both pass all 78 inverse-formula checks, five exact join costs and the two positive-kernel witnesses. A separate internal mathematical/source review by the root agent passes the complete join list, the small-affine exclusions, arbitrary outer-contact bounds and the Schur estimate; that review does not duplicate the 78 matrix computations. The all-finite-corner transfer and the complete mixed quadrilateral theorem remain separate obligations. No previously frozen proof or dimension bound is changed.

Source 269

Source path: `results/fresh_simple_cusp_distance_bound_20261017.md`

SHA256: 8e60de5e8bf5247ac751b302d6e8d54f88d8c5f03cfbc185edf8484e9cdeacce

Distance15 bounds for simple cusps and compact facets

Let P be a finite-sided, finite-volume hyperbolic Coxeter polytope in H^n. The following restricted conclusions hold:

- If every ideal vertex is simple, then **n<=125**.
- If P has a compact facet, then **n<=127**, with no simplicity assumption on its other ideal vertices.

Finite vertices are automatically simple. Neither hypothesis is asserted for every finite-volume chamber; the full unrestricted bound163 is unchanged. No novelty or current-best claim is made. The first assertion uses a simple projective polytope, not an assumption that the chamber is compact in hyperbolic space. The compact facet in the second assertion need not itself have Coxeter angles.

The exact checker and certificate refine the earlier compact-facet distance17 argument. They use the frozen complete high-rank quasi-Lannér catalogue to reduce the quadrilateral distance from17 to15. The all-rank finite/affine graph bound and the passage to ideal polygon corners are proved below.

Local face stars, including ideal corners

Every positive-dimensional k-face belongs to exactly n−k facets. The facet normals of its supporting walls have positive-definite Gram, because the relative interior contains a hyperbolic point. These incidence facts, and the product-of-simplices description of an ideal link, are stated for acute chambers in Potyagailo–Vinberg, printed p.64.

Let Q be an actual triangular or quadrilateral two-face. Write W for its n−2 supporting facet normals and mark its side facets black. Each side has exactly one original ambient facet beyond W. Its edge interior gives W plus that black node positive definite. At a finite polygon corner, W plus the two adjacent black nodes is positive definite. At an ideal corner it is positive semidefinite. Under the simple-cusp hypothesis these n normals are precisely the complete incident affine vertex diagram. The compact-facet application has only finite corners.

The full face star is indefinite: after projecting the side normals off the positive span W, they bound a nondegenerate finite-area polygon in a Lorentzian three-space, which they span. Every principal block has negative index at most one, since it is a block of the ambient Gram.

We use two acute-incidence facts from the inherited compact-facet proof. A positive-definite set of original wall normals has an actual common face, obtained by projecting a strict timelike interior point; the inverse of a positive-definite matrix with nonpositive offdiagonals is entrywise nonnegative. A singular positive-semidefinite set contains a connected affine component whose positive null combination is a nonzero future null vector. It is orthogonal to the whole PSD block and pairs nonpositively with all other walls, hence gives an actual ideal point. Nonzeroness and orientation follow from its strictly negative pairing with the common interior point.

In a quadrilateral, the two **closed opposite sides are disjoint**. This remains true when some or all four vertices are ideal: the four vertices are distinct, and only consecutive sides have a common endpoint. Consequently, for either opposite black pair U, the block W+U is indefinite. Positive definiteness would give a finite common point of those opposite sides; singular PSD would give an ideal common point. Both contradict the actual closed polygon incidence. Compactness of Q is not used in this step.

Minimal diagrams and the radius7 certificate

Every inclusion-minimal indefinite principal diagram L is connected and has all proper principal blocks PSD. If a proper connected affine block C occurs, its positive kernel pairs nontrivially with an adjacent outside node; C plus that node is already indefinite. Minimality forces that to be all of L. Thus L has the Lannér/quasi-Lannér classification scope, with order at most10. Strict dotted pairs give only an order-two case. The classification input and primary references are recorded in the frozen high-rank census and the inherited compact-facet proof; this package does not repeat the large classification enumeration.

Choose minimal indefinite diagrams L in W+U and M in W+V, where U,V are the opposite black pairs. Each contains both its black nodes. On deleting either black node the remaining subdiagram lies in W plus one side facet, so that deletion is **positive definite**, not merely PSD. Therefore both marked nodes are in the catalogue's `finite_deletions`.

For every such marked pair in all26 high-rank types, the checker performs a multi-source breadth-first search from the pair. There are384 pairs:

Maximum distance to the marked pair	Number of pairs
1	10
2	123
3	143
4	61
5	31
6	12
7	4

The four radius7 cases are L23 with blacks {0,1}, farthest node8; and L25 with blacks {1,2}, {1,3}, or {2,3}, farthest node9. Every literal distance is saved. A connected diagram of order at most5 has radius at most3 from a two-node set: a shortest path cannot pass the other marked node before reaching the set. This treats all smaller types without a label cutoff. Hence the radius7 bound is complete for every possible L and M.

The two minimal indefinite diagrams cannot be disjoint and orthogonal, as that would give two negative directions. They share a white vertex or have an edge between them. In the overlap case the marked sets are joined by a path of length at most14. In the joined case it is at most 7+1+7=15. Taking a shortest path from U to V ensures that all internal vertices are white. Its endpoints are consecutive sides of Q, so this path lies in their actual corner diagram.

For a triangle, every indefinite subdiagram contains all three black nodes, since any black deletion lies in a PSD corner. A minimal one has at most10 nodes. In a spanning tree, take the minimal subtree joining the three marks. At least two marked leaves can be deleted separately while the other two marks stay joined. Each resulting path has length at most8 and belongs to one actual corner diagram. This argument uses PSD corner blocks and is therefore valid at ideal corners as well.

Give a polygon corner weight1 when its two black nodes are at graph distance at most15 in the full incident vertex diagram; otherwise give weight0. Triangles receive at least2 and quadrilaterals at least1. Larger polygons need only nonnegative payment. Thus every two-face with k sides receives at least5−k.

All-rank finite and affine vertex costs

For a connected graph with N vertices let p_k be the number of unordered pairs at graph distance k. The unweighted underlying graphs of the classical finite and affine Coxeter diagrams are paths, cycles, or the single- and double-fork trees below. Edge multiplicities do not change graph distance.

- A path has p_k=max(N−k,0).
- A cycle has p_k=N for 2k<N, p_k=N/2 for 2k=N, and zero otherwise.
- A path on m>=2 vertices with two extra leaves at one end has N=m+2 and p_k=max(m−k,0)+2[1<=k<=m]+[k=2].
- A path on m>=2 vertices with two extra leaves at each end has N=m+4 and p_k=max(m−k,0)+4[1<=k<=m]+2[k=2]+4[k=m+1].

Each formula follows by separating path–path, leaf–path, and leaf–leaf pairs. In every displayed case p_k<=N. The affine D4 four-leaf star is the m=1 exception and is treated separately. Every remaining exceptional finite/affine diagram has

N<=9, so even its total number of pairs is at most4N. The finite dihedral diagrams and affine A1 are paths of order2; the other small F/G/H types are paths or included exceptions. These classification alternatives cover all connected finite and affine diagrams.

It follows in every case that the number of pairs at distance at most15 is at most15N. The same holds for a spherical forest by adding component orders; pairs in different components have infinite distance. The checker independently counts245 classical graphs and seven exceptional graphs by BFS and compares the formulas. The explicit formulas and the finite exceptional classification, not the tested rank range, prove the all-rank statement.

At a finite vertex the full spherical forest has order n. At a simple ideal vertex the Euclidean link is one simplex, so its full incident diagram is connected affine of order n. In both cases all planar-angle weights sum to at most15n. No such order-n assertion is made for a nonsimple ideal vertex.

Strict h-vector thresholds

The projective closure of P is a bounded simple n-polytope when all its ideal vertices are simple. The inherited nonnegative symmetric h-vector argument gives the strict average corner-payment lower bound

$$b_d = \begin{cases} d(d-6)/8, & d \text{ even}, \\ (d-1)(d-5)/8, & d \text{ odd}. \end{cases}$$

Its strictness follows from the positive endpoint h-coefficients. Applied to the whole chamber, the payment bound gives b_n<15n. For n=126+2t and n=127+2t respectively, eight times b_n−15n is

$$4t^2 + 252t, \qquad 4t^2 + 256t + 132.$$

Both are nonnegative for t>=0. Equality in the first at t=0 still contradicts the strict lower bound. Thus n<=125. Without the new radius certificate, distance17 would similarly give the weaker n<=141.

If P instead has a compact facet F, put d=n−1. F is a simple d-polytope whose vertices are finite ambient vertices. Its two-faces are actual two-faces of P, and their weights count only a subset of the pairs in the ambient rank-n spherical vertex forest. Hence b_d<15(d+1). At d=127+2t and d=128+2t, eight times the difference is

$$4t^2 + 256t + 12, \qquad 4t^2 + 260t + 136,$$

strictly positive. Therefore d<=126 and n<=127. This replaces the older distance17 compact-facet bound143 in this restricted branch only; the other unrestricted finite-volume branches are not changed here.

Comparison, verification, and scope

Felikson–Tumarkin prove the much stronger n<=8 result for **simple all-ideal** Coxeter polytopes. Their primary paper, §§1–3 uses the absence of elliptic order-n diagrams, together with restrictions on ideal triangles and quadrilaterals. That absence fails when finite vertices are allowed, so their proof is not imported to the mixed case here. This report makes no survey-based claim that125 is a best published bound for the restricted class.

Generation and a fresh saved replay pass. Syntax, all nine direct source hashes, and report links pass. Two internal mathematical/source reviews found no gap; the separate independent review records its scope and does not claim a duplicate arithmetic replay. The checker binds the frozen complete catalogue and the preceding geometric proofs; its new arithmetic is the384 exact distance calculations, independent classical graph counts, and rational threshold identities. No coefficient field, arithmeticity, optimizer, finite-tail cutoff, or induced-face Coxeter assumption is introduced.

Source 270

Source path: `results/fresh_simple_cusp_distance_bound_20261017_review.md`

SHA256: 8ee0cec0e1ad16d4ff6032fdd9029b7d81b6d8620b5e961f2cf67eaf61e41cc6

Independent geometric and source review of the distance15 bounds

PASS: no mathematical gap identified in the stated restricted conclusions. The argument gives dimension at most125 when every ideal vertex is simple, and ambient dimension at most127 when a compact facet exists. The latter permits nonsimple ideal vertices elsewhere. Neither condition is asserted for all finite-volume Coxeter chambers, so the unrestricted bound163 is unchanged.

This is a separate mathematical/source review by exact_hub_counts. It does not duplicate the producer's 384 distance computations, 245 classical graph checks, saved replay, or the inherited quasi-Lannér classification census.

Reviewed source SHA256: 0af68dae0bc9625bb7fabf1629117bbfb480bcde002de03f73700d1e64f75f62. Reviewed certificate SHA256: 6eceec3e7999b98511f996c1315d22fae767beb1af0e10c9a219eab6c2d842ee. The proof report supplies the geometric argument; its later validation-status wording is not an arithmetic input to this note.

Ideal quadrilaterals and the finite-black radius condition

The crucial extension from compact to ideal polygons is sound. The relative interior of an actual edge contains a hyperbolic point even if its endpoints are ideal. Its supporting normals therefore have a positive-definite Gram. Thus, for a quadrilateral with white support W, W plus any one black side is positive definite.

For two opposite sides, positive definiteness of their combined block would give an actual finite common point by acute projection. A singular PSD block gives a positive affine-kernel combination. Its pairing with a strict timelike center is strictly negative, so this null vector is nonzero and future directed. It is orthogonal to the PSD block and nonpositive against all other walls by acuteness. Hence it gives a feasible ideal point in both closed opposite sides. Closed opposite edges of a genuine quadrilateral are disjoint, including when endpoints are ideal. Both alternatives are impossible, so each opposite block is indefinite.

A minimal indefinite subdiagram in that block contains both black nodes. Deleting either one leaves a principal subdiagram of W plus a single side, hence is **positive definite**. The catalogue test must therefore use pairs of finite deletions, as the source does, rather than arbitrary pairs of affine-or-finite deletions.

The frozen complete order6–10 catalogue covers those cases. Any connected diagram of order at most5 has distance at most3 to a two-node set: a shortest path to the set avoids the other marked node. The low-order argument also covers arbitrary small labels, and a dotted order-two core has radius zero. There is no unexamined rank or label tail in the radius7 conclusion.

Two opposite minimal indefinite cores cannot be disjoint and orthogonal in an index-one ambient space. An overlap gives a path of length at most14 between the two black sets; an inter-core edge gives at most7+1+7=15. A shortest such path has no black internal vertex. Its endpoints are consecutive sides of the quadrilateral, so it is contained in their actual finite or simple-ideal corner diagram.

For triangles, every black deletion lies in a PSD corner. A minimal indefinite core contains all three marks and has at most10 nodes. Its minimal connecting subtree yields at least two mark-to-mark paths avoiding the third mark, each of length at most8. This is valid for ideal corners as well.

All-rank graph costs

The path, cycle, single-fork and double-fork formulas correctly separate path–path, leaf–path and leaf–leaf pairs. In particular, the affine-D double fork with path order m>=2 has

`p_k=max(m-k,0)+4[k<=m]+2[k=2]+4[k=m+1]`,

with total order N=m+4. At k=2 the count is exactly N; the other cases are at most N as well. The exceptional m=1 affine-D4 star is handled separately. The remaining exceptional finite/affine graphs have order at most9 and at most4N pairs in total. Finite dihedral diagrams, affine A1, and the small path-shaped F/G/H diagrams are covered.

These explicit formulas and the finite classification, rather than the checked rank interval, establish the uniform distance15 cost at most15N. Spherical components add. A simple ideal vertex has a simplex Euclidean link, hence one connected affine diagram of order n; a finite vertex has a spherical forest of order n. No order-n assertion is made for a nonsimple cusp.

Strict thresholds and scope

For an all-simple chamber, the projective closure is a bounded simple polytope, so the inherited strict h-vector inequality applies even when the hyperbolic chamber has ideal vertices. At n=126 the scalar lower bound equals15n, but the actual lower inequality is strict. This excludes126, giving at most125. The two displayed parity polynomials are nonnegative for every t>=0, not merely for the sampled t values in the checker.

For a compact facet of dimension d=n−1, its vertices are finite and its two-faces are actual ambient two-faces. Their angle weights form a subset of ambient rank-n pair weights, so the upper budget is15(d+1). The two parity differences become strictly positive starting at d=127 and128, giving d<=126 and ambient n<=127. The proof does not assume that the induced facet is itself a Coxeter polytope.

The comparison with the stronger published simple **all-ideal** bound is correctly separated from the mixed finite/ideal hypothesis here. No novelty, current-best literature status, arithmeticity, or external refereeing claim is made by this review.

Source 271

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SHA256: 9c1aad19b00720722144f59d12df9b16bb426a03f59fe7e78bc8ec2f6a6f1d69

Vinberg's compact scope and a 3n affine weight interface

The checked primary literature does **not** supply a mixed-simple finite-volume dimension bound29 by direct citation. There is a concrete extension candidate with the correct vertex budget: retain Vinberg's finite-vertex weights and use the rule below at simple ideal vertices. Its total vertex cost is at most3n, and the triangle payment transfers. **Quadrilateral payment remains unproved.** Thus this package gives no new ambient dimension bound; the frozen all-simple bound61 is unchanged.

The checker and data verify the graph counts exactly. Their finite ranges are confirmations of the all-rank formulas below, not cutoff-based proofs.

What the primary theorem says

Vinberg's original Theorem1 on printed p.69 explicitly concerns compact type. Its §6.1 begins with a bounded fundamental polytope, and §6.2 calculates costs for elliptic vertex diagrams. The original §4 is the coarse distance7 argument; §§5–6 provide the much finer weights giving the compact bound29. These are distinct proof steps. Original paper.

One modern wording that could suggest a broader scope occurs at the opening of Burcroff's §9. But its following Proposition9.2 assumes a compact Coxeter polytope, and Theorem10.2 again states the bound for the compact class. That introductory wording does not provide a proof for mixed finite/simple-ideal vertices. Primary preprint, §§9–10.

The primary Felikson–Tumarkin introduction likewise attributes29 to the compact theorem and separately proves8 for the simple all-ideal class. Primary paper. This is a bounded source check, not a claim to have established the best published bound for the mixed-simple class.

The proposed ideal-vertex rule has cost exactly3n

Assume n>=14. A simple ideal vertex has a connected affine diagram on n nodes. Exceptional affine types have at most9 nodes, so only the four classical underlying shapes occur: a cycle, a path, a single fork, or a double fork. Normalize every ideal corner weight to0 or1 as follows:

- weight1 if the two marked roots have graph distance at most3;
- also weight1 if they belong to an end segment of order at most6;
- otherwise weight0.

On a fork an end segment must contain both interchangeable fork leaves or neither. The maximal allowed fork-end segment contains the two leaves and the first four vertices of the central path. A cycle has no end segments. Distances ignore bond multiplicity, just as in the preceding distance proofs.

The exact costs are:

Underlying affine graph	Pairs at distance<=3	Additional end pairs	Total
cycle	3n	0	3n
path	3n-6	6	3n

Underlying affine graph	Pairs at distance<=3	Additional end pairs	Total
single fork	3n-5	5	3n
double fork	3n-4	4	3n

For a path, each end contributes two extra distance4 pairs and one distance5 pair. A fork end contributes two distance4 pairs and no distance5 pair. These extra pairs are disjoint for n>=14. The short-pair counts follow directly from the all-rank path/cycle/fork formulas already proved in the frozen distance package. This proves the table in every rank. The checker reconstructs all348 graphs of orders14 through100 and counts their literal pair sets independently.

At finite vertices use Vinberg's published weights, including the four E8 exceptions. Their total is at most3n by the frozen finite-component check and its inherited all-rank proof. The new affine rule only concerns ideal vertices; it does not reassign weights to finite forests.

Triangle payment does transfer

By the global affine exclusion, each triangle star in the all-simple class, n>=14, contains a Lanner diagram L of order at most5 containing its three black roots. Follow the literal cases in Vinberg §6.4, rather than importing the whole compact face theorem.

If L has order3 or is a4-cycle, there are two adjacent black pairs, so two corners receive weight1. If L contains a bond5, the proof chooses a corner whose connected diagram contains that bond; such a corner cannot be a classical affine n-node diagram. It is finite of type H3 or H4 and receives weight2 under the unchanged compact rule.

The remaining case is a5-cycle. Vinberg's two displayed marked patterns give two corner contributions of weight1 by adjacency and his local endpoint/special-edge lemmas. If either contributing corner is ideal, the required marked pair is connected within L minus the omitted black root, on at most4 vertices, so its distance is at most3. Our ideal rule therefore gives at least the same unit payment. The finite corners retain the original local lemma. Hence every triangle receives at least2.

The classification and local finite-corner lemmas here are explicit published inputs. The graph checker does not claim to reclassify Lanner diagrams or independently replay their angle arithmetic.

The precise remaining quadrilateral obligation

The absence of affine diagrams of order below n does not make the whole corner diagram elliptic. A quadrilateral can have a connected affine n-node corner. In particular an affine A cycle has two marked-to- marked arcs. Vinberg's §6.5 description of W_ij as a tree with the two marked roots at its ends does not apply verbatim to that corner. His later classification of each T_ij as a finite classical diagram or E8 also needs new cases. Thus the compact quadrilateral theorem has not been transferred by the cost calculation above.

The exact target is now clear: prove that every actual quadrilateral in the all-simple class has at least one corner paid by this mixed rule, or give a literal realizable counterexample. Any proof must retain the full original star, all finite/affine vertex conditions, negative index1, corank1, and the global exclusion of affine subdiagrams of order below n. No label cutoff or bounded white-tail scan suffices.

Two tempting shortcuts have specific defects:

1. Taking the maximum compact weight over all single-root deletions of an affine A cycle marks exactly every pair at cyclic distance at most5. Its cost is5n for n>=14, not3n. The checker verifies all27 sizes14..40; the proof is that a pair lies in a6-node end segment of a deleted cycle precisely when one of its cyclic connecting arcs has length at most5.
2. Deleting a supporting white root does not preserve a quadrilateral star's corank1. For a symmetric N-by-N Gram matrix of rank N-1 with kernel vector k, its adjugate is a nonzero scalar multiple of kk^t. Deleting a node w with k_w!=0 therefore leaves a nonsingular principal matrix. Vinberg §6.10 explicitly uses det H=0; a spherical deletion cannot silently be substituted for a full quadrilateral star.

These are obstructions to particular transfer arguments, not counterexamples to the proposed mixed rule or to a possible bound29.

Status

Generation and a fresh saved replay pass, covering348 affine graph checks and27 cycle-deletion envelopes. Syntax, all nine direct source hashes, and report links pass. An internal mathematical/source read independently confirms the triangle transfer, including the original5-cycle case. The certificate status explicitly leaves quadrilateral payment open and assigns no new ambient dimension bound. The older61/127/163 theorems and all frozen predecessors are untouched.

Source 272

Source path: `results/fresh_simple_finite_quad_vinberg_transfer_20261017.md`

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Vinberg payment for quadrilaterals with four finite corners

Claim. Let $P \subset \mathbb{H}^n$ be a finite-sided, finite-volume Coxeter polytope which is simple at every finite and ideal vertex, with $n \geq 18$. If a quadrilateral two-face F has four finite vertices, then at least one of its corners has positive weight under Vinberg's original finite-vertex rule. In particular its total corner payment is at least one.

This is a transfer of the published diagram argument, not a new classification of its finite cases. The audit below retains Vinberg's finite Coxeter diagram classifications and local determinant calculations as published inputs and checks precisely the hypotheses that used compactness. It does not apply the compact theorem directly to a noncompact polytope.

The primary source is Vinberg, *Trudy Moskovskogo Matematicheskogo Obshchestva* **47** (1984), 68–102, especially §3 Proposition 8, §6.3, and §6.5–11, printed pp.82–91. Original paper, bibliographic record.

The simple-cusp affine exclusion is the other geometric input: every connected affine subdiagram of P has exactly n nodes. In particular every subdiagram on fewer than n nodes is affine-free. An order-n affine diagram omitting a supporting white wall is **not** excluded by this fact. No step below assumes otherwise.

Geometric properties of the actual compact face star

Write W for the $n-2$ supporting white walls of F, and write the four black walls as opposite pairs $\{u_1, u_2\}, \{v_1, v_2\}$. The full star has $n+2$ roots. Every $W + u_i + v_j$ is positive definite, since it is the wall diagram at an actual finite vertex. Also W is positive definite. These assertions do not require compactness of the ambient P.

Both $W + u_1 + u_2$ and $W + v_1 + v_2$ are indefinite. Here is a direct argument that deals with the potential ideal case. If the first Gram matrix were positive definite, its inverse would be entrywise nonnegative. Project a strict timelike interior point h onto the common orthogonal complement of its normals. The projection adds a nonnegative combination of those normals; its norm becomes more negative and its products with all other walls remain strictly negative. This produces an actual point of F on both opposite sides, which is impossible.

If instead that Gram matrix were positive semidefinite and singular, eliminate W. Each of the two residual diagonals is positive, and the residual off-diagonal is nonpositive. Singularity gives a kernel vector with both black coordinates strictly positive. Lifting it through the entrywise nonnegative inverse of W gives nonnegative white coordinates. The resulting combination z of actual normals is null and nonzero: $(z, h) < 0$ for a strict center h. It is orthogonal to W and both opposite roots, and has nonpositive product with every other wall. Thus it gives an ideal point of F, contradicting compactness of this four-finite-vertex face. Negative index at most one leaves the asserted indefinite alternative.

Every indefinite subdiagram of the star contains an opposite black pair, since otherwise it lies in one of the four positive-definite corner diagrams. Hence all four black roots belong to the unique indefinite connected component H.

The projected side normals of a nondegenerate compact quadrilateral span the three-dimensional Lorentz space $W^{\perp}$. Consequently the full star has rank $n+1$ and corank one. The component H has corank one as well; its other components are positive definite by the orthogonality lemma below. In particular $\det H = 0$, just as required in §6.10.

The unique dependence has nonzero black coefficients, with the two opposite u-coefficients of one sign and the two opposite v-coefficients of the other. One way to see this is to normalize the projected side normals against an interior point of F. Their

spatial parts are the vertices of the polar convex quadrilateral. Its unique affine dependence has alternating signs around the quadrilateral. This also proves that no black coefficient vanishes.

Two replacements for blanket affine exclusion

First, if an actual subdiagram A is orthogonal to an indefinite subdiagram B, then A is positive definite. The span of B contains a timelike vector, whose orthogonal complement is positive definite. Thus the Gram matrix of A is positive semidefinite. If it were singular, a connected singular Coxeter component would have a positive kernel vector. The associated combination of its normals cannot vanish because it pairs strictly negatively with an interior point. This is impossible in the positive-definite orthogonal complement. This argument works for arbitrarily large A, and needs no affine-order cutoff.

Second, an indefinite subdiagram with at most n nodes is affine-free in the present simple finite-volume setting. A proper connected affine subdiagram would have fewer than n nodes; an affine subdiagram of order n would be the whole diagram and could not be indefinite. The same assertion applies componentwise when appropriate.

These two replacements are stronger than merely observing negative index one: an abstract block-diagonal matrix could have both an indefinite and a singular positive-semidefinite block. The strict center of the actual chamber rules out the singular block here.

Preliminary diagram decomposition

Use the decomposition $H = U \cup V \cup W_{\text{shared}}$ from §6.5: U contains the two u-roots and the white components unable to reach a v-root by a white path; V is defined symmetrically. The notation W_{shared} here distinguishes Vinberg's shared partial diagram from the whole supporting diagram W above.

Assume for contradiction that all four corners have weight zero. The shortest white path between the two black pairs has length at least two, so U, V are orthogonal. There is at least one shared white node. Since each of U, V contains two black nodes and $|H| \le n + 2$,

$$|U|, |V| \le n - 1, \qquad |H - U|, |H - V| \le n.$$

The finite corners and the opposite-pair indefiniteness proved above give the connected indefinite diagrams $H - U, H - V$, exactly as in §6.5 Lemma 4. They are therefore affine-free by the second replacement. The diagrams U, V are affine-free by their strict size bounds. Thus §6.6 Lemma 5 still gives an elliptic member of the orthogonal pair U, V, and Lemma 12 still supplies the required cyclic Lannér core whenever the corresponding elliptic member has no high bond.

All the auxiliary T_{ij} are actual finite corner components. Therefore Lemmas 1–3 and 6–11, including the end-segment conditions, the four exceptional E_8 marked positions, the tree property of the shared white paths, and the classical-versus-E_8 alternatives, apply to these T_{ij} literally. None of these finite-corner statements is asserted for an affine corner.

Lemma 7 is needed before bounding the order of U or V: if either contains a high bond, that whole diagram is connected. Thus an **elliptic** U or V with a high bond is a connected finite diagram, not an arbitrarily large forest containing a small exceptional component. Its high bond necessarily has a finite Coxeter label at least five (multiplicity at least three in Vinberg's convention), and the finite classification gives order at most four.

Among connected affine diagrams, a **finite** bond of label at least five occurs only in the three-node affine G_2 diagram, with label six. Hence a connected diagram of order at least five containing such a finite high bond is indefinite, at any order. This statement does not concern the infinite bond of affine A_1, and does not use a global affine prohibition.

Audit of the five remaining cases

The following table records the affine-sensitive parts of §6.7–11. The published local finite-diagram and determinant conclusions are retained. The cardinalities are deliberately generous carrier bounds, rather than counts of the minimal offending affine diagram.

Part of the published proof	Justification available here
§6.7: U elliptic with a high bond, V indefinite	$\lvert U\rvert \le 4$. Connected high-bond extensions either become indefinite within a small extension or belong to the explicitly displayed exceptional configurations. The latter fit inside U, one E_8 corner component and the remaining black node, at most 13 nodes. The displayed affine obstructions have at most nine nodes. Pieces orthogonal to V are elliptic by the first replacement.
§6.8: U elliptic without a high bond, V indefinite	The cyclic Lannér core has at most five nodes. Its local extensions, including the two three-node end pieces, fit in a carrier of at most 13 nodes. The explicit affine B_4/F_4 exclusions are small. Any larger piece shown orthogonal to V is already elliptic by the first replacement.
§6.9: both U, V elliptic with high bonds	$\lvert U\rvert, \lvert V\rvert \le 4$. The exceptional E_8-corner configurations fit inside $U \cup V \cup T_{ij}$, at most 16 nodes. The displayed affine obstructions have at most nine nodes. For the long classical case, see the separate argument immediately below.
§6.10: only one of the two elliptic diagrams has a high bond	The exceptional configurations are contained in $U \cup T_{ij} \cup L$, at most $4+8+5=17$ nodes. Other contradictions use small high-bond extensions orthogonal to the cyclic Lannér core, or the negative-index and determinant tests. The quadratic determinant equation is nontrivial by the argument below.
§6.11: neither elliptic diagram has a high bond	The two cyclic Lannér diagrams and all their overlap/connecting edges use at most ten nodes. The implicit affine exclusions used to determine their permitted overlap or connection are within this carrier. The later larger end-piece contradictions use the already known ellipticity of U, V and the finite T_{ij} classification; they do not apply a blanket affine exclusion to those larger diagrams.

Every finite carrier in this table has fewer than n nodes when $n \ge 18$. Its affine exclusions therefore remain valid.

For precision in \S6.11, the small-carrier assertion concerns restrictions on the induced union of the two cyclic cores and their connecting edges. It is not a separate assertion that arbitrary white branches have been excluded by an affine test on that union. The subsequent determination of the shared paths uses the already positive-definite T_{ij} and Lemma 8: a cross-distance-two shared diagram is linear, and at the other short cross-distances a nonlinear shared diagram requires the finite E_8 alternative. The published end-piece arguments then use Lemmas 10–11 and ellipticity of U, V. Thus extending a displayed core pattern by additional white nodes does not require a new blanket affine exclusion; the pertinent larger diagrams are already finite corner diagrams. This distinguishes the core-overlap affine tests from the later finite classification of complete shared paths.

There is one additional size point in the long classical branch of §6.9. Its (OU) and (OV) attach every white neighbor of the two black roots to U, V. Each black root has at most three white neighbors, because all these neighbors occur in an actual positive-definite corner diagram. Thus

```
|O\(U\)|,|O\(V\)|\le4+2\cdot3=10<n.
```

They are orthogonal when the shortest white cross-path has length at least four, so one is elliptic. After the long corner component is split into its two end pieces, the high-bond pieces of order at least five are indefinite by the finite/affine classification stated above. Their orthogonal complements are elliptic by the first replacement, regardless of their order. This supplies the ellipticity needed for the final H_4-piece and negative-index contradiction in that branch. No order-n affine end piece has been silently discarded.

Nonzero determinant polynomial in §6.10

Let e be the high bond of U. All the other normalized Gram entries belong to $\mathbb{Q}(\sqrt{2})$: the elliptic U has only this one high bond, the elliptic V has none, and every other relevant bond belongs to the finite corner diagrams covered by Lemma 6. Varying the entry on e gives a determinant polynomial of degree at most two. In the original diagonal-one normalization its quadratic coefficient is minus the determinant obtained by deleting the endpoints of e. Its positivity deserves an explicit check in the present setting.

In the unique dependence of H, every vertex of U has a nonzero coefficient of the same sign as its two black u-coefficients. Indeed, a private white component has contacts only with the u-pair and its own white nodes. Solving its dependence equation using its entrywise nonnegative positive-definite inverse gives that common sign, strictly at every node because the component is connected and attached to the black pair.

Deleting an endpoint of e therefore destroys the only linear dependence. The normals remaining after both endpoints are deleted are independent, and their span still contains the indefinite diagram $H - U$. A subspace containing a timelike vector has nondegenerate Lorentz restriction. Its Gram determinant is consequently strictly negative. Thus the quadratic coefficient is strictly positive (four times as large if the variable is $\cos(\pi/m)$ in diagonal-two normalization), including when the high bond has private white endpoints.

The published algebraic angle argument and the subsequent negative-index contradictions can therefore be used unchanged. This check relies on the actual compact quadrilateral and its full star; it would not be justified by deleting an arbitrary white root and pretending the smaller diagram were another face star.

Conclusion and review scope

Under the assumption of four zero-payment finite corners, the preliminary decomposition and all five published cases retain their required hypotheses. Each case produces the same contradiction. This proves the claim for four-finite-vertex quadrilaterals in the stated all-simple finite-volume class, $n \geq 18$.

This note is an independent mathematical/source audit, not a numerical graph census or an independent rederivation of every determinant table in the original paper. Separate source and mathematical reads by the root and frontier agents pass the geometric replacements, Lemma 7 connectedness, carrier-size controls, §6.11 locality, and the nonzero quadratic coefficient. No remaining compactness gap was identified in the stated transfer. These internal reviews do not claim to duplicate the published local tables or constitute external refereeing or formal verification. This proof note is final and frozen.

The other mixed-corner quadrilateral branches require their separate proofs. This note alone gives no new ambient dimension bound, and changes none of the frozen 61/127/163 proofs or arithmetic certificates.

Source 273

Source path: `results/fresh_small_cusp_defects_20261017.md`

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Small-cusp defects and a one-cusp consequence

This package proves a local refinement of the fine-payment theorem and an exact cusp-incidence balance. A deficient triangle or quadrilateral has a small affine factor in its original wall star. After paying small ideal corners sufficiently, every remaining carrier cusp lies **outside** the defective face. Separately, dimension is at most155 if every cusp has some irreducible factor of rank at most8; in particular, every one-cusp chamber has dimension at most155.

This package does not improve the unrestricted156 ceiling. Its155 consequence is retained as a historical refinement of that bound; the separate selective-distance13 proof, once finalized, supersedes it with a stronger unrestricted result. The local nonincident-defect statement remains a distinct ingredient. No bound on carrier multiplicity, new example, or novelty claim is made.

Localizing the fine-payment failure

Let P be a finite-sided finite-volume hyperbolic Coxeter n-polytope. For an actual triangular or quadrilateral two-face F let H(F) consist of its $n-2$ supporting walls and its side walls. This is the original wall star, without artificially adding all walls at nonsimple ideal vertices.

If H(F) contains no connected affine subdiagram of order at most9, then F receives fine payment at least2 for a triangle or at least1 for a quadrilateral. Use the finite Vinberg weights and the distance-three/end-segment rule at ideal corners.

This is the face-local form of the frozen rank9 theorem. The following audits the hypothesis change. The triangle's minimal indefinite core and its proper affine subdiagrams belong to H(F), so its Lanner reduction and finite-corner alternatives are unchanged. At an ideal corner the varying affine factor belongs to H(F), and thus has order at least10. The ideal-corner proof uses only this factor, the other two sides and the supporting spectators. Every excluded low-contact affine, including the new dual-E8 witnesses, is a principal subdiagram of H(F).

For four finite corners, both complements in the private/shared decomposition lie in H(F). Every witness in the carrier proof uses only its carrier, the actual white connecting path and the four sides. The later published cases use the order-at-most9 tests in the refined source audit; their large orthogonal pieces are positive definite by the actual Lorentz realization. No affine prohibition elsewhere is used. The published finite case tables remain inherited inputs, not a new classification by this checker.

A connected affine diagram in H(F) determines an actual cusp and is one of its full irreducible factors. Its positive kernel combination is a nonzero feasible null vector: nonzero follows by pairing with a strict center, and feasibility follows from acuteness. Strict convexity of the Klein ball makes its ideal point an actual vertex. The positive-semidefinite incident Gram then forces every other incident root right to this affine factor. This is the actual-factor lemma, consistent with the primary unique-ideal-vertex statement in Felikson–Tumarkin, §1, PDF p.4, reopened for this package.

A defect demand that charges only nonincident carriers

Use this nonnegative rule:

- At finite corners keep Vinberg's weights, whose vertex total is at most3n.

- At ideal corners whose varying factor has rank r<=8, use weight2.
- At other ideal corners retain the fine distance-three/end-segment rule.

Let s_large(v) count the factors of rank at least9, V_v=product_i(r_vi+1), and

b(v)=3 s_large(v)+sum_(r_vi<=8) r_vi.

The total cusp weight is at most V_v b(v). Each marked pair in factor i occurs in exactly V_v/(r_vi+1) link fibers, so a small factor costs 2 binom(r+1,2)V_v/(r+1)=rV_v. A large factor costs at most3V_v by the all-rank affine cost proof.

Call a triangle or quadrilateral deficient when its weight is below2 or1 respectively. Such a face has no small varying-factor corner, since that corner alone pays2. The local theorem nevertheless forces a small affine factor C inside H(F).

The cusp determined by C is not on F. Otherwise all its walls contain some ideal vertex v of F. The walls of H(F) containing v are exactly its supporting whites and the two incident sides. The factor-fiber description of this corner star has one complete affine varying factor and one-node-deleted finite pieces from every other factor. Thus C must be the complete varying factor, contradicting the absence of a small varying-factor corner. This allows nonsimple ideal vertices.

Writing T_bad,Q_bad for the deficient counts, the weight upper bound and exact link-shape curvature imply

2T_bad+Q_bad >= (beta_n-3n)v_f +sum_v [kappa(L_v)-V_v b(v)]+5(n-1).

Here beta_n=n(n−6)/8 for even n and (n−1)(n−5)/8 for odd n, and v_f counts finite vertices. A deficient triangle is short by at most2 and a deficient quadrilateral by at most1; polygons with at least five sides need no positive payment. This proves the inequality directly, without assuming all deficient payments are equal to their maximum shortfall.

A positive right side forces face-star charges to **nonincident** small cusps. No multiplicity bound for these charges follows from the argument. In particular, counting only corners of faces actually passing through the carrier cusp does not finish a dimension proof.

Exact example distinguishing carriers from incident cusps

In upper half-space H3 consider

−1<=X<=1, −1<=Y<=1, X²+Y²+Z²>=4, Z>0.

This Coxeter square pyramid has four finite base vertices (±1,±1,sqrt2) and one ideal apex at infinity. Adjacent vertical walls are right, opposite ones parallel, and the base hemisphere has angle pi/3 to each. In metric diag(−1,1,1,1), outward normals of square one are

R=(1, 1, 0,1), L=(1,−1, 0,1),
U=(1, 0, 1,1), D=(1, 0,−1,1),
B=(−3/4,0,0,−5/4).

The strict center (5,0,0,4) has square−9. The full cone has exactly the rays (5,±2,±2,3) and (1,0,0,1). The checker exhausts all ten triple frames in the section defined by the timelike sum of the normals. That section is bounded: its recession direction would pair nonpositively with every normal and zero with their sum, hence zero with all five; they span the Lorentz space. Every facet has an explicit relative-interior point from a positive sum of its vertex rays. There are no exterior rays.

The relations R+L=U+D=2(1,0,0,1) certify the two affine A1 factors and the proper rectangular cusp link. The bounded link and complete finite/ideal cone give finite volume.

The compact base's full star contains both affine factors although their cusp is not on that base. This disproves identifying a carrier in a face star with a cusp on the face. **It is not an unpaid-face example:** each base corner is A3 marked at its two end roots, and the published rank3 rule pays that pair weight2, total8. No adjacency-only finite rule is being substituted for Vinberg's actual weights.

Exact cusp-type balance at dimension156

A separate consequence uses the unrestricted distance14 theorem, rather than the unbounded nonincident charge. Its cusp cost per link vertex is

c(r)=sum_i min(14,r_i/2).

At n=156 the total link rank is155. The frozen convex-order transfer (a,b)->(a−1,b+1), applied when both ranks are strictly below28 or both strictly above28, preserves c and cannot increase normalized curvature. Thus a reduced profile has copies of28, at most one positive rank below28 and at most one above28. All128 reduced shapes of total155 give

kappa(L)/V−c(r) >= −4172075/82044596.

If a cusp has a rank at most8, hold one such rank a fixed and transfer only the other factors. For a=1,…,8 this gives988 marked reduced shapes, proving

kappa(L)/V−c(r) >= 1395158/20511149 >0.

This does not assume that unrestricted transfers preserve a small factor. The exact minima occur at (15,28,28,28,28,28) and (1,14,28,28,28,28,28). Standalone rational convolution and independent inclusion-exclusion for the centered absolute moment agree on all1116 profiles. Completeness comes from the proved concentration reduction, not a sampled partition cutoff.

Let V_small sum V_v over cusps having a rank at most8, and V_large sum V_v over all other cusps. The exact curvature lower bound minus the distance14 payment upper bound gives

4172075 V_large >= 60795045636 v_f +5580632 V_small +63584561900.

The coefficients are exact: beta_156−14*156=741 and the global slack is*5155=775, multiplied by82044596. V_v counts the actual edges incident to a cusp; it is not merely the cusp count or the factor count. Thus the multiplicity in this incidence inequality is explicit.

The rank9 theorem forces V_small>0 in dimension156, and the displayed balance forces V_large>0. Every hypothetical156-dimensional chamber must therefore have at least two cusps of these different types. Consequently:

- Every chamber whose each cusp has some factor of rank at most8 has dimension at most155.
- Every one-cusp chamber has dimension at most155: a large-only cusp would instead put it in the rank9 class, bounded by29.

The unrestricted156 theorem excludes all higher dimensions; compact chambers already satisfy29. No unrestricted155 conclusion follows, since mixed cusp types can satisfy this balance.

Verification scope

The checker and certificate use standalone Fraction arithmetic. They check128 general plus988 marked reduced profiles, two moment formulas, exact balance coefficients, eight small-factor cost identities and the complete rational five-wall example. Frozen proofs supply the localization cases, curvature, concentration and the two earlier dimension bounds. Their published tables and prior arithmetic are not reconstructed.

Generation and a fresh saved replay both pass. Syntax, all13 direct source hashes and11 local report links pass. The parent independently read the localization, nonincident-cusp argument, modified weight budget, deficit inequality, square-pyramid caveat and exact156 balance, finding no gap. That conceptual/source review does not duplicate the arithmetic or published finite tables. These are scoped incidence and historical restricted-dimension results, not an unrestricted improvement or external refereeing. Source, data and report are final/frozen.

Source 274

Source path: `results/fresh_small_d16_coupon_and_ideal17_rebalance_20261018.md`

SHA256: c0c10822e322c31689b21a53b2bc64b03522dd7077fbcac1758dfdf19ddacb28

Sparse D16 coupons and a fractional order17 reversal

Starting from the completed407.5 rule, make two simultaneous changes.

- Remove the half-payment between a whole finite component of rank2,3,4 or5 and a whole D16 marked at its fork center or a short fork leaf. The whole A1/D16 halves remain unchanged.
- At an ideal factor of order17, restore weight1/8 to the four affine-D16 pairs `(2,15),(2,16),(10,13),(10,14)`. Lower the distinct whole-D5 long-endpoint/whole-D12 short-leaf coupon from1 to7/8.

Every triangle still receives at least2 and every quadrilateral at least1. The actual order17 family totals become A85, B111, C108, D229/2. The whole D5/D12 component-pair cost falls from23/2 to45/4. No finite same-component weight changes and no new coupon is added. All statements concern finite-sided, finite-volume real hyperbolic Coxeter polytopes under the same actual-corner hypotheses as the completed rule.

The complete formal rank60 finite maximum remains815/2, against curvature 405. However, the two smaller-component obstructions disappear:

Formal forest	Previous cost	New cost
3D16+2A5+A2	1629/4	1413/4
4A1+4D5+3D12	406	403

Only six representative alternatives now exceed405. Each has cost815/2 and consists of two components chosen from A17,D16,D17, together with D26,D27 orD28 to make rank60. These have actual typed attainers. Thus this theorem removes the smaller-component obstructions; it does not exclude dimension60 or establish a new unrestricted bound59.

The producer and certificate give the exact changed rule, inherited proof bindings, four reconstructed radical seals, all216 current literal corners, complete new finite certificates and all2,113 ideal row updates.

Why the discarded D16 coupons are unnecessary

The broad small-component allowance entered through a counting theorem. The first broad budget enlarged the earlier A1/D16, A2/D14, A1/D15 and A2/D13 coupons to all small components of rank at most4 against D12 through D16. The later rank5 budget enlarged this further. Both separate the budget enlargement from the geometric question of whether a whole-component payment is needed. In particular, the enlargement alone is not a geometric requirement to retain every added small/D16 pair.

The necessary proof here is an audit of the actual outputs of the complete face-payment argument. Absence from a finite sample would not suffice. The following is the complete division of its coupon-producing interfaces.

Interface	Cross-component outputs relevant here
Universal cycle/feature-free tree short-path replacements	Sealed A1/A13, A2/A13, A1/A15 or A1/A17 coupons; cycle12 also uses A1/A11 units and A2/A11 halves; other outputs have both marks in one component or a retained short path
Universal nine-pair affine-D13 transfer, factor order14	Whole A1,A2,A4 orA5 against D11 orD13 fork marks
Universal nine-pair affine-D14 transfer, factor order15	Whole small rank at most3 against D12 orD14 fork marks
Literal order16 and order17 original replacements	Whole A1/D15, A2/D13, A1/A15; whole A1/D16 and A2/D14, respectively
Later selected ideal transfers at orders14–17	Complete literal records, or their separately sealed A4/D13, D5/D12, A4/D12, A5/D11 and E6/D11 roles
Order11 and order12 universal spectator transfers	Small whole A components against B9,B10,B11,D10 orD11; the remaining outputs are short paths, whole finite components or retained small affine factors
Order13 center and center/leaf spectator transfers	Whole A2/A3 against D10, whole A1/D12-center, whole finite D12/D13, or independent retained ideal/short-path outputs
Later finite rank15–17 transfers	Whole same-component alternatives, or the explicit sparse A/D roles listed below; no new small/D16 cross coupon
Large finite-D protected-leaf transfer	Whole small rank at most5 against D10 short leaves

For the first row, the finite8 cycle theorem, ideal5 local theorem and cycle14 theorem classify all supporting contacts. Their equality coupons are the stated sealed A1/A2 versus path roles. The cycle12 theorem adds the stated A1/A11 unit and A2/A11 halves; the cycle13 theorem adds no coupon. A finite D16 may occur as a single whole marked component; that is a same-component payment, which is untouched. The shortened finite8-to-short7 and finite7-to-short6 routes inherit these independent outputs. The current whole-A8 alternative is also a same-component unit.

The two universal fork-transfer rows follow directly from the full order14 and order15 statements. Their finite equalities include arbitrary supporting spectators and prove that the named factors remain whole. The large ranks are bounded by14, not by an unspecified D11-through-D16 budget allowance.

The original order17 literal replacement needs only whole A1/D16 and A2/D14; its positive affine kernel and full-core radical seal the entire actual supporting diagram, or supply an independent length-two unit. The corresponding order16 replacement has the lower ranks in the table. These are the literal transfers used by the complete finite8 theorem, whose full low/disjoint/overlap domain is retained here. Low-core, protected/adaptive and ordinary selected routes give same-component payments. Cross-component replacements enter through the explicitly listed universal and sealed literal cases.

The later order16/order17 partial-weight transfers have explicit whole D5/D12, A4/D13, A4/D12 orA5/D11 outputs. The current refinements add E6/D11, D5/D11, D5/D12, A4/D11 and A5/D11 roles. The order13 sparse terminal transfer adds A4/B9. The current universal B/C11 and B/C12 proofs use only the small B/D factors listed above. Their energy-spectrum boundary cases can give larger single whole components, but never turn them into a small/D16 coupon. Those single-component outputs retain the same weights as before this change.

The subsequent finite exchanges use existing small/D13 or small/D14 halves, A2/A13, A1/A13, A2/A14, A4/A12, D5/A11, and D5/D12 roles, or restored same-component A17/D17 weights. The latest D15 theorem uses whole A2/D13-center or same-component D16 halves. Its D16 pairs `(1,12),(1,13)` are unrelated to the discarded cross-component coupons. The complete current proof bindings retain these whole-component statements; none is inferred from a truncated component.

As a second, exact check of this inventory, the producer reconstructs every finite and ideal corner in all216 current literal records from its Gram matrix and supporting whites. There are exactly ten corners with marks in separate components, one of type D16. In all ten the other component is A1. Six are in the original order17 cross-payment package and four in the old order17 partial-weight package. No rank2–5 coupon is debited. Arbitrary extra whites cannot enlarge these A1 components: their inherited radical/equality arguments seal the whole core before claiming the coupon. The other later explicitly typed coupons above are likewise whole-component conclusions.

This is why the removal is universal. Replay the existing full-domain proof with the same actual-corner priorities and the stated output inventory; every payment it uses survives. It is not an argument that a quadrilateral containing a deleted

coupon must itself be one of the 216 saved cores. Such a face is covered by the original exhaustive proof, whose actual paying outputs have just been retained.

Four ideal eighths and two finite seven-eighths

Use the complete frozen D114 theorem. Its 140 eligible native patterns and488 successful induced maps locate all affected images in four whole-factor contexts:1978,2371,2535,2539, with32 affected maps. Each has a complete19-root core, a whole17-root affine-D16 component and a distinct whole finite D5/D12 coupon corner.

The positive affine kernel forces every additional supporting white orthogonal to the affine component. Pairing with the actual full-core radical then either gives the independent opposite-side path of length2 or forces all remaining core contacts to vanish. In the latter branch all displayed finite factors remain whole; an indirect supporting tail cannot join them. The producer recomputes all four old seal certificates and compares them exactly with the frozen proofs. It does not rerun the historical complete embedding census.

The selected ideal pair now contributes1/8 and the distinct D5/D12 corner contributes7/8. Their sum is1 without spending any same-component finite payment. Intrinsically, the D5 mark is its unique long-arm endpoint, and the D12 mark is either short fork leaf. There is one eligible D5 node and two eligible D12 nodes, so the finite component-pair change is exactly

```
1 * 2 * (7/8 - 1) = -1/4.
```

These two marked roles `(0,10),(0,11)` are disjoint from the reverted center role `(0,9)` and from the six finite-D17 compensation roles `{3,4} × {9,10,11}`. The latter still receive full units. Hence the current D5/D12 total becomes23/2−1/4=45/4. The four ideal pairs gain 4/8=1/2, taking D114 to229/2. All other ideal-family costs stay fixed.

The complete selected/nonselected scope of D114 is inherited: the four changed pairs have distance11, so basic, short8 and elementary triangle weights are untouched. Earlier small-ideal actual-corner priorities, current short8→short7→short6 replacements and the whole-A8 branch remain independent of these roles. In the selected order17 branch, the four seals now use1/8+7/8; all other images retain their proof. Together with the preceding coupon-output inventory this gives a noncircular current whole-face composition. Every other finite reduction and the current rank16/rank17 balances keep exactly their old weights.

The new216-corner audit begins with the final407.5 literal totals. Exactly four ideal credits1/8 and four finite coupon debits1/8 occur; they cancel in their four records, whose totals stay1. The D16 coupon removal has zero occurrences. All216 totals remain at least1. Triangles use retained same-component/ideal elementary payments, so retain2.

Complete changed finite count

Keep the actual finite base costs from the407.5 rule, including A15=101, B15=97, D15=201/2. The rank15 representative allowance remains 101; actual D15 does not dominate A15. Every other actual base is unchanged.

The producer checks all33,489 ordered type pairs for simultaneous representative domination after both coupon decreases. Each actual small rank2–5/D16 total loses3r/2; its representative loses the same amount. Other types mapped to D16 have no such cross coupon. For D5/D12 the actual and representative allowances both lose1/4. Whole A1/D16 still costs3/2. No extra cost is concealed in a representative.

The complete fixed-component domain still contains1,044 multiplicity vectors for ranks9–17. The old seven-coordinate interaction key is extended by the number of D16 representatives; there are now1,042 keys. This is necessary because rank2–5 residual costs now distinguish D16 from the other formerly grouped large components.

In eighth units the residual item change is

```
-12*r*c16,       2 <= r <= 5,
-2*c12,          additionally when r=5.
```

All residual–residual and fixed–fixed interactions remain zero. Every residual item is checked against its explicit base plus all fixed cross costs. Generation computes new item-first Bellman tables. Fresh replay checks every last-item upper inequality and an attaining predecessor; it does not optimize again. There are1,502,564 upper inequalities, 62,520 attaining predecessors and54,288 item identities. Every fixed vector reconstructs an attaining representative forest. The unchanged actual forest2D17+D26 attains815/2, matching the complete upper.

For the first targeted obstruction, each of three D16 factors loses 3(25+2)/2=18 cross payment, so1629/4 becomes1413/4. For the second, twelve D5/D12 component pairs each lose1/4, taking406 to403. These are actual typed cost calculations, not representative forests mistaken for geometric vertices. Neither formal forest is asserted realizable.

Complete positive ideal comparison

Independently enumerate the terminal rank59 partitions with at most one part1–8 and all other parts at least9, and compare them bijectively with the frozen2,113 rows. Their exact convolutions are inherited; every new scalar row is computed exactly. The only normalized cost increase is

```
count(rank16)/34.
```

All2,113 margins remain strictly positive. The normalized minimum is 31/6000 at(9,9,9,9,9,14), and the absolute minimum is54 at(59). The exact allowable total increase of the order17 envelope before a row reaches zero is897/1360, attained at(9,9,9,16,16); the chosen1/2 is strictly smaller. This threshold is a scalar observation, not a license to change any additional geometric payment.

The inherited low9 concentration changes only low-rank factors, whose cost remains r/2. It decreases both normalized curvature and V. Since every terminal margin is positive, the absolute bound54 extends to all actual cusps. The finite bound is still405+5/2, so the same global comparison gives

```
(5/2)*v_f >= 4500 + 54*v_i.
```

The completed unrestricted ceiling60 and all-ideal ceiling59 remain unchanged. The new work supplies a smaller obstruction list, not a proof of unrestricted59, a construction, external review or formalization.

Verification scope

Final source-bound generation42974/98e044 and separate fresh replay17587/badde0 both terminated exit0/PASS for the four seals, all216 literal records, all changed finite certificates and all2,113 ideal row updates. Root completed a full producer/report mathematical composition read, including the broad-coupon and original finite8 provenance and D114 reports. That read checked the exposed D16 count, simultaneous domination, complete fixed/residual domain, disjoint fractional roles and ideal row update, without executing arithmetic or repeating the historical census. A second peer completed a full producer/report mathematical read and the inherited broad/finite8, cycle12/cycle14, universal-fork and original order16/order17 provenance checks. It checked the complete actual-output inventory and arbitrary- white sealing, four disjoint fractional seals, literal deltas, typed domination, the new interaction key, Bellman proof and ideal updates. Its cycle12/cycle14 inventory precision corrections are incorporated above. Neither peer reran arithmetic, maps, geometry, DP or convolutions. No mathematical gap was identified within this stated scope.

All176 direct bindings,16 local report links and Python syntax pass. Source, certificate and report are final/frozen. No historical geometric census or convolution is repeated by this producer; their complete frozen certificates remain explicit inputs. These are internal agent mathematical reviews, not external refereeing or proof-assistant checks.

Source 275

Source path: `results/fresh_small_spectator_difference_gap_20261017.md`

SHA256: 223060b187a7dc7b944654781b14e8489156d9f283174a96f689311f90535226

The discrete spectator-energy spectrum through three quarters

FINAL/FROZEN: exact generation, fresh saved replay and a separate full mathematical/source review PASS.

Let G be a positive-definite norm-two acute Coxeter Gram matrix. Let x,y be nonnegative vectors of native contact magnitudes with disjoint supports. Assume both separate augmented Grams

`[[G,-x],[-x^T,2]],  [[G,-y],[-y^T,2]]`

are **positive definite**. This is the strict actual-edge hypothesis, not merely semidefiniteness at a corner. Put D=(x-y)^T G^(-1)(x-y). Then

`D<=3/4 implies`
`D in {0,1/2,2/3,3/4,2-sqrt(2),1-sqrt(5)/5}.`

In particular **D cannot equal4/7**. Equality D=3/4 has only two forms, up to swapping x,y and diagram symmetries: one unit contact at an endpoint of a whole simple A3 component, or one unit contact from each side at adjacent vertices of a whole simple A3 component. Every other spectator contact vanishes.

The source and certificate verify the seven nonzero literal patterns, their separate strict edge extensions, Schur identities and exact values. Completeness follows from the argument below, which permits arbitrary spectator rank and native finite labels. No ambient dimension or complete face-payment theorem is asserted.

Preliminary positive-definite facts

Every nonzero native magnitude is at least1; if greater than1 it is at least sqrt2. A positive-definite acute Coxeter graph is a forest: on any cycle, the vector with entries1 on that cycle has square at most0, since each cycle bond has magnitude at least1 and extra bonds only decrease the square. This applies to G and to each of the two strict augmented Grams, even if their graph is disconnected.

The inverse of G is entrywise nonnegative and has diagonal at least1/2. One proof writes G=2I-A with A nonnegative and expands the convergent Neumann series. Restricting to a principal diagram cannot increase the corresponding inverse quadratic energy: the full inverse adds a positive-semidefinite Schur correction. We also use the inherited rank bound: a connected positive-definite norm-two Coxeter component of rank r has every inverse diagonal at least r/(r+1). That argument covers all finite labels and indirect tails, not a finite diagram sample.

One nonempty support

If y=0, nonnegative inverse entries give D>=sum_i x_i^2/2. Thus D<=3/4 allows only one nonzero coordinate, and its native magnitude must be1. The rank bound forces its whole component to have rank at most3.

Rank1 gives A1 and D=1/2. At rank2, the inverse diagonal is 2/(4-t^2), where t>=1 is its bond magnitude. Being at most3/4 forces t^2<=4/3, hence t=1 and D=2/3. At rank3 the graph is a path. Its middle inverse diagonal is at least1. At a leaf, the minimum is3/4, attained only when both bonds are simple: increasing either bond strictly increases that diagonal. This is the whole simple A3 endpoint case. Swapping the two supports gives the same list. If both supports vanish, D=0.

Two nonempty supports must be singletons

For any i in the x-support and j in the y-support, Cauchy in the Gram metric gives

D >= (x_i+y_j)^2/(4−2G_ij).

If i and j are not adjacent, this is at least1. Hence under D<=3/4, every vertex of one support is adjacent to every vertex of the other. If one support had two vertices i1,i2 and the other contained j, adjoining the root with contacts to i1,i2 would create the cycle i1-j-i2-new-i1 in its strict positive-definite edge diagram. This is impossible. Thus x=p*e_i and y=q*e_j for adjacent i,j.

Put t=-G_ij. Positive definiteness of each separate extension gives $(G^{(-1)})_{ii}<2/p2<=2$ and $(G^{(-1)})_{jj}<2/q2<=2$. The two-node principal inverse is2/(4-t^2), so t<sqrt3. If either p or q exceeded1, the native gap and the Cauchy bound would give

D >= (1+sqrt2)^2/(4+2t) > 96/125 > 3/4.

The displayed rational lower bound uses sqrt2>7/5 and sqrt3<7/4. Consequently p=q=1.

A nonsimple spectator bond is isolated

Suppose t>=sqrt2. If either endpoint of i-j had another neighbor with bond magnitude u>=1, its inverse diagonal in that three-node principal path would be

2/(4−t^2−u^2) >= 2/(3−t^2) >= 2,

contradicting the strict edge-extension inequality. The denominator is positive because that principal path is positive definite. Therefore i-j is a whole rank-two component. The native conditions sqrt2<=t<sqrt3 leave exactly labels4 and5. For the two unit contacts,

D=2/(2+t),

giving2-sqrt2 and1-sqrt5/5 respectively. This excludes arbitrary high labels by a proved inequality, not an imposed cutoff.

A simple spectator bond has at most one additional root

Now t=1. If i-j is the whole component, it is A2 and D=2/3. Otherwise choose a third node adjoining one of its endpoints, with bond magnitude u>=1. It cannot adjoin both endpoints because G is a forest. On A2, G_A2^(-1)(e_i-e_j)=(1/3,-1/3). Schur complementation therefore gives the three-node principal energy

2/3 + u^2/[6(3−u^2)],
energy−3/4=(u^2−1)/[4(3−u^2)].

Positive definiteness ensures u^2<3. Hence the energy is at least3/4, with equality only at u=1. The principal diagram is then simple A3, and the two contacts occupy adjacent vertices. In one orientation its inverse difference vector is

G_A3^(−1)(e_0−e_1)=(1/4,−1/2,−1/4),

which has no zero coordinate. Any additional root attached to this A3 can meet only one of its nodes, since a second attachment would create a cycle. Its product with the displayed inverse difference is nonzero. The positive Schur correction is then strictly positive. Thus D=3/4 forces this A3 to be the whole connected component, including all tails. Every other component has zero contact from both sides.

This proves the stated list and the sealed A3 equality classification. The exact values2-sqrt2 and1-sqrt5/5 lie on opposite sides of4/7; none of the other listed values equals4/7. The latter exclusion is useful for the cycle14 resistance calculation, while the A3 equality description is a possible input to a smaller affine-D transfer. Those face arguments remain separate obligations.

Verification scope

The literal matrices and separate edge extensions are checked exactly over the relevant quadratic fields. The source also verifies the three-node Schur identity, its positive derivative, the nonvanishing A3 inverse difference, the rational nonunit-

contact bound and the4/7 exclusion. Generation and a separate fresh saved replay both terminated with exit0/PASS, checking all seven literal patterns. A second agent read the complete proof and source and found no gap in the one-support rank reduction, strict-edge singleton argument, native-label exclusions or whole-A3 sealing. That review does not claim a second implementation of the exact arithmetic. The root also read the final proof and source. No external refereeing or formalization is claimed.

Source 276

Source path: `results/fresh_sparse_coupon_order11_13_dimension62_20261018.md`

SHA256: b7a9a249bd15f5db1b6937e3a957964d5158fb67ea7f6177ccaad1f1274f1dac

Unrestricted finite-volume dimension bound62

The simultaneous payment rule below excludes dimension63. Together with the completed higher-dimensional exclusion, it gives dimension at most62 for arbitrary finite-sided finite-volume hyperbolic Coxeter polytopes. No arithmetic, compactness or simple-cusp hypothesis is imposed. **Completed internally checked computer-assisted proof; final verification and review scopes are recorded below. Source, data and report are frozen.**

At ambient63 its exact comparison is

```
0 >= (1/2)*v_f + (139/46)*v_i +4960,
```

which is impossible. Here v_f and v_i count finite and ideal vertices. The source and certificate bind the completed local proofs and independently checked finite/ideal arithmetic, and verify their precise simultaneous interfaces. They do not rerun the historical geometric censuses, finite-forest dynamic programs or2,204-profile scalar computation.

The single simultaneous payment rule

Begin with the completed dimension63 rule. Make exactly these further ideal changes:

- whole affine C10 of order11: endpoint pair(0,10), weight1/2 to0, by the C11 zero theorem;
- whole affine B11 of order12: fork-center/doubled-terminal pair(0,9), weight1 to0, by the B12 zero theorem;
- whole affine D12 of order13: fork-center pair(0,8), weight1 to1/2, by the D13 half theorem.

The order is the number of roots, one more than affine rank. The binder reconstructs all four factor-family totals at each changed order:

Order	A	B	C	D	Maximum
11	99/2	54	54	53	54
12	60	63	63	62	63
13	65	73	73	147/2	147/2

All other ideal weights, including cycle11 halves and the order17 D half-coupon improvement, retain their completed values. The new normalized envelopes are54/11,21/4,147/26 at ranks10,11,12. Every other rank envelope is checked against the preceding complete ideal calculation.

For finite corners, adopt the sparse whole-component theorem. It gives A16=114, B16=104, D16=114, D17=125 by deleting six units, all at marked distance at least10, and adding six typed component-pair coupons. Retain every preceding coupon and the deep11 finite8 zeros. Additionally include the B12 theorem's A2/B9 doubled-terminal half and the D13 theorem's A2/D10-center and A3-middle/D10-center halves. No payment is assigned merely because a component contains a named type as a proper subdiagram: all these roles concern separate **whole** finite components.

The nine added component-pair totals, in the order used by the budget, are A4/A12:4, A4/A13:4, D5/A12:2, A2/A13:2, A2/A14:2, E6/A11:4, A2/B9:1, A2/D10:1, A3/D10:1/2. The source compares every total with its local role certificate and the final budget. The A3 coupon charges only its unique middle. Existing coupons on the same type pairs have disjoint

marked roles and are retained. The enlarged E6/D11 total4 used in the budget is only a conservative representative allowance: the actual E6/D11 rule still costs2.

Exhaustive actual-quadrilateral priority

Apply the following cases to the actual corners of each quadrilateral. They do not refer to which corner an older algorithm selected. All payments below use the final simultaneous rule.

1. Some actual corner has the changed C11 endpoint pair. The independent C11 theorem analyzes every norm-two heavy-end profile, including the affine alternatives. Its strict determinant gap at least1/5 forces a distinct actual adjacent-black or common-spectator path of length1 or2. That corner retains a unit under every ideal and finite change here. This branch consumes neither the deleted C11 half nor another potentially changed affine half. It supplies total at least1.

2. Otherwise some actual corner has the changed B12 center/terminal pair. The universal B12 theorem covers all supporting spectators and native labels. Its possible outputs are a whole affine D11 factor of order12 marked at a center and a short leaf, distance8; a whole finite D12 center-to-long-endpoint distance9 unit; or two distinct finite half-coupon corners, whole A2/B9 doubled-terminal and whole A2/D11 fork-center. The affine pair is still a unit: it is not an opposite-leaf pair, and the new D13 change is in a different order. The finite D12 pair is unaffected by sparse rank16/17 distance-at-least10 deletions or deep distance8 zeros. Both whole-component coupons are included in the budget. The source checks the six literal boundary cases and the four canonical affine center/leaf unit pairs. Total payment is at least1.

3. Otherwise some actual corner has the changed D13 center pair. Keep its original half. The universal D13 theorem gives either a distinct whole A2/D10-center half, a distinct whole A3-middle/D10-center half, or a distinct retained unit. Its unit alternatives are whole order11 affine B10 center/terminal distance8, whole order11 affine D10 center/leaf distance7, or a path of length at most4. The two affine center pairs are not the B/D11 terminal-leaf halves, and no C11 endpoint or B12 center/terminal pair is used. All finite outputs are typed whole D10 coupons or short paths; none uses a finite rank16/17 long unit. The binder checks all four immediate affine unit roles and eight finite coupon corners against the final weights. The half plus distinct half, or the independent unit, gives total at least1. The local theorem applies its bound inherited arbitrary-rank energy-one classification and proves the new whole-component/corner consequence. Neither is inferred from these literal records.

4. No actual corner has any of the three changed ideal pairs. Apply the complete sparse finite **whole-face** theorem under the baseline63 ideal weights. Its ideal weights now agree pointwise with the final weights at every actual corner of this face. Its finite rule is the final finite rule except for the B9/D10 coupon additions, which can only increase payment. Therefore its total at least1 is inherited directly. There is no requirement to predict which alternative corner the sparse proof will select, and no circular appeal to a payment that may have been removed elsewhere in the face.

These four cases are exhaustive and all handle the actual whole components and actual corner marks. In particular, the sparse theorem's old short8 alternatives need not be asserted to remain literal units for every new ideal pair: any actual changed ideal corner was handled first by its independent theorem.

Scope of the inherited sparse finite theorem

The sparse theorem is a complete geometric input, not an inference from its numerical forest bound. Its selected-support proof covers the complete2,614-context high-overlap domain:394 eligible A16 patterns with 238 maps,394 D16 patterns with578 maps, and140 D17 patterns with168 maps. Six proper D16 contexts are excluded by exact norm-two projections, two-leaf Cauchy, or a negative fork residual; arbitrary additional contacts are controlled before zeros are imposed. Ten radical contexts give the new coupons, including the four obligations previously covered by restored D17 units. Their zero radical and all-label contact domain seal entire displayed components unless an actual short8 path occurs.

All216 inherited literal configurations are debited simultaneously; the minimum remains1. This finite list is not treated as every quadrilateral. The inherited low-core bound12, excluded high-disjoint case, protected/adaptive leaf9 outputs, immediate and universal cycle/tree branches, and later whole-component or typed-coupon interfaces are retained separately in that theorem. In particular, its two adaptive bound16 outputs have marked distance9, missing every new deletion. Later baseline63 order11 outputs use components of order at most14, protected A9/A18 endpoint8 or D18 leaf9, E8 path6, or typed coupons; order17 uses no rank16/17 same-component unit.

If an inherited short8 path ends at a deep finite8 zero, the independent depth11 zero theorem forces a short7 alternative. Under baseline63 weights its complete small-tree, cycle11, cycle-tail and protected-feature interfaces pay. The sparse deletions have distance at least10 and do not change those outputs. This route is part of the sparse theorem under its stated baseline ideal rule; the actual-corner priority above transfers it to the final rule without reopening or circularly modifying the route. The present author read the complete sparse source/report and its inherited selected/literal exchange routines; the underlying all-label censuses and published classification retain their inherited proof scope.

Triangles and complete local payment

The finite7/ideal5 triangle theorem uses finite distances at most7, ideal distances at most5, protected ideal leaf distances6/7, or whole ideal factors of order at most9. The three new ideal pairs have distances10,9,8 and orders11,12,13, respectively; none is used by that triangle theorem. The finite deletions have distance at least10, so triangle payment at least2 is unchanged. The previously proved cycle11 triangle exclusion and all prior interfaces remain in the baseline. The new coupons only add weight.

Thus every triangle receives at least2 and every quadrilateral at least1. Weights are nonnegative, so faces with at least five sides need no positive payment to bound5 minus their number of sides. Summing all actual corner payments yields an upper bound on

```
D5 = sum_(two-faces Q) (5 - number_of_sides(Q)).
```

Exact finite and ideal vertex budgets

The frozen joint finite budget includes all old and nine new coupons, with the exact marked-role increments above. Its complete1,329 large multiplicity vectors and993 residual tables give maximum449 at rank63. Pointwise simultaneous two-end type domination retains B9, E6 and A17 distinctly; its larger E6/D11 allowance is an upper bound only. Every Bellman upper inequality and an attaining predecessor are checked, and the actual formal forest 2D17+D29 has cost125+125+199=449. This proves the bound for all finite vertex diagrams; no assertion that this witness occurs in a chamber is needed. The curvature at a finite vertex is899/2, leaving surplus1/2.

At an ideal vertex, its Euclidean link is a product of simplices with factor ranks summing to62. The independent ideal theorem uses exactly the family envelopes above, together with all unchanged rank envelopes. Its complete concentration domain has2,204 profiles: arbitrary repeated ranks9 through17, at most one smaller positive rank, and at most one rank at least18. Every normalized margin is positive; the minimum is3505/201344 at(10,10,12,15,15).

The concentration moves preserve cost, decrease normalized curvature and decrease the product V of factor orders. Since terminal margins are positive, the absolute margin V times normalized margin also decreases. Its exact minimum139/46 at the single rank62 factor is therefore valid for every actual cusp profile, not just the terminal list. The factor-fiber formula counts each marked varying-factor pair with its exact multiplicity V/(r+1). Consequently the ideal payment C_v satisfies

```
kappa(L_v) - C_v >=139/46.
```

No cusp simplicity or arithmetic condition has been inserted. The independent scalar theorem was explicitly conditional on the three new envelopes; the local priority proof above now discharges those geometric hypotheses, and the complete finite budget includes their coupons.

Global contradiction and higher dimensions

Let d=62. The shape-curvature theorem gives the local curvature kappa(L)=(5G-dV)/2. Its extremum correction can be strengthened for these links as follows. Every coefficient h_j of a simplex or product-of-simplices H-polynomial is at least1. At each of the two global extreme vertices,

```
E-G = sum_(j>d/2) h_j*(2j-d)
    >= sum_(j>d/2)(2j-d) = floor(63²/4) =992.
```

The two extrema, each multiplied by5/2 in D5, thus contribute4960. This corrects the same curvature lower bound once; it is not an extra vertex payment or a second copy of a previously counted constant.

The local upper bound and global lower bound are therefore

```
D5 <=449*v_f + sum_(ideal v) C_v,
D5 >=(899/2)*v_f + sum_(ideal v) kappa(L_v) +4960.
```

Subtracting gives the impossible inequality at the start. It excludes ambient dimension63 regardless of the numbers of finite or ideal vertices. The completed unrestricted ceiling63 already excludes every dimension at least64. Combining the two exclusions gives **dimension at most62**. Compact and arithmetic-specific bounds are unchanged. No new Coxeter polytope example is constructed by this argument.

Verification and proof status

Source-bound generation4b3b23 and a separate fresh saved replaye48d4f terminate exit0/PASS on12 complete family interfaces, exact changed-pair identities, protected alternative roles, all nine coupon increments, frozen budget/scalar bindings, finite curvature and the strengthened extremum constant. Direct audit318d0f passes42 source bindings,12 local links and Python syntax.

Root and two separate peer agents completed full source/report mathematical composition reads. All three checked the exhaustive actual-corner priority, pointwise agreement in the fourth branch, protected unit and whole-component half outputs, all nine exact coupon increments, conservative versus actual E6 allowance, triangle/deep11 interfaces, family envelopes, absolute ideal surplus, finite curvature, single extremum correction and completed higher tail. Root separately read the contributing local, scalar and budget proofs at their recorded scopes. The first peer also read the frozen ideal, baseline63 and sparse reports; the second completed full C11/B12 and sparse-budget source/report reads in addition to its earlier D13, sparse-local and ideal reviews. None of these composition reviewers reran the historical census, Bellman tables or2,204 scalar rows. The inherited energy-one attribution was clarified in the report; no mathematical or arithmetic change was required. Source, data and report are final/frozen.

The constituent local and counting packets already have completed generation/fresh replay and mathematical reviews at their recorded scopes. This binder is a local-to-global composition check, not a new independent enumeration of every historical geometric core. The underlying finite/affine classification, all-label reduction and all-rank concentration proofs remain explicit inherited mathematical inputs. All review is internal; no external refereeing or formal proof-assistant verification is claimed.

```
python3 scripts/fresh_sparse_coupon_order11_13_dimension62_20261018.py
python3 scripts/fresh_sparse_coupon_order11_13_dimension62_20261018.py --verify
```

Source 277

Source path: `results/fresh_sparse_coupon_rank63_forest_budget_20261018.md`

SHA256: a88d1323636bcaff51aff76e2b865c9148d7d5b4e1b220b0f252ffb4333c70af

All new sparse coupons fit finite rank63 maximum449

For the explicit finite rule below, every rank63 forest has cost at most **449**, and the formal forest2D17+D29 has actual cost449. This is below the finite curvature899/2 by1/2. The count includes every preceding coupon and all nine new typed component-pair increments. It is a finite counting theorem; simultaneous geometric face coverage and the global ideal comparison are separate obligations.

The source and certificate bind the sparse finite transfer, the B12 transfer, the D13 transfer, and the previous complete finite budget. The typed marked roles and exact cost increments are compared directly with those local certificates before the forest count runs.

Actual rule and the conservative representative allowance

Same-component costs are A16/D16=114, B16=104, A17=114, B17=105 and D17=125. All other preceding costs remain, including deep11 finite8 zeros and the large-D fork-leaf halves. Every old cross-component coupon is preserved. The new increments are:

Actual component pair	Added total cost	Marked roles
A4,A12	4	endpoint/endpoint unit
A4,A13	4	endpoint/second-node unit
D5,A12	2	long-endpoint/second-node unit
A2,A13	2	either A2 node/A13 endpoint half
A2,A14	2	either A2 node/A14 second node half
E6,A11	4	E6 length2-arm endpoint/A11 second node unit
A2,B9	1	either A2 node/doubled terminal half
A2,D10	1	either A2 node/fork center half
A3,D10	1/2	unique A3 middle/fork center half

The last two coupons are distinct from the old small-component/D10 short-leaf halves. The new A2/A13 coupon is distinct from its old middle-node coupon. Thus the totals above are genuine increments, without overlap or suppression of an existing payment.

For a uniform upper bound, replace both types of every component pair simultaneously. Use E6 at rank6, B9 at rank9, retain A17 separately, and otherwise use A_r for ranks at most8 and D_r for ranks at least10. Use an **upper allowance4 for E6/D11**, although its actual coupon still costs2. This absorbs the new actual E6/A11 cost4 when A11 becomes D11. It is not an added E6/D11 payment rule.

For every pair of finite types a,b the source verifies

```
upper_cross(R(a),R(b)) >= actual_cross(a,b),
base(R(a)) >= base(a).
```

This is deliberately a simultaneous two-end comparison. A5 alone need not dominate the new D5/A12 coupon against an unchanged A12. After the second component is also replaced by D12, its existing small-component allowance absorbs that

cost. Summing these pointwise inequalities over all components and all unordered component pairs proves a bound for every finite forest, including repeated types. No sequential one-end domination is assumed.

All192 finite type representatives through rank63 and all36,864 ordered type pairs are checked. Arbitrary dihedral labels use the inherited uniform rank2 representative. Finite/affine classification and the old same-component formulas remain explicit inherited inputs.

Complete count domain and interaction key

Enumerate every multiplicity vector of B9,D10,…,D16,A17 whose rank sum is at most63. There are1,329 vectors. All other components have ranks1 through8, represented by A except E6, or ranks at least17 represented by D. In particular A17 and D17 are different choices with different coupons; B9 is not silently folded into the additive small-rank part.

Write b9,d10,…,d16,a17 for the large multiplicities and set

```
b = 2*d10 +3*(d11+...+d16) +2*a17.
```

The exact doubled residual-item interaction key is

```
(b, d11+d12, d13, d11, d14, b9, d10).
```

It has993 distinct values. For residual rank r, start with the doubled same-component cost. Add r*b for r<=5, and additionally

```
r=1: 2*(d11+d12)+d14;
r=2: 4*d11+2*d14+2*b9+2*d10;
r=3: d10;
r=4: 2*d13;
r=6: 8*d11.
```

The rank6 coefficient8*d11 is the doubled conservative E6/D11 allowance4. The rank2 B9 and D10 terms are the two new component-pair increments1; the rank3 D10 term is the new middle-node increment1/2. All existing special A1/A2/A4 and D14 coupons are retained. Large components have no mutual cross coupon, and residual components have no mutual cross coupon, so these additive items cover every interaction. The source independently compares each item with the literal typed upper-cross formula for every large vector.

Exact upper certificates and actual attainment

Item-first dynamic programming produces each residual table through rank63. Saved verification instead checks every possible last-item upper inequality and an attaining predecessor at every rank. Induction on rank therefore proves both the complete upper bound and attainment in the representative upper-cost problem; no sampled cutoff is used.

The993 tables contain1,592,772 upper inequalities and62,559 attaining predecessors. There are73,095 literal interaction identities and1,329 global alternatives. Every alternative reconstructs a typed forest and recounts both its upper cost and its actual cost, asserting actual<=upper. The two costs need not agree on a forest containing E6 and D11, and such a forest is not claimed to attain its conservative allowance physically or even under the actual counting rule.

The conservative maximum is449. The explicit witness2D17+D29 has no cross coupon, and its actual costs are

```
125+125+199 =449.
```

The D29 value is the inherited8*29−33=199; no deep11 distance8 pair exists in D29. Thus actual and conservative costs agree on this witness, proving the exact actual-rule maximum449 as well as its upper bound. Formal attainment makes no claim that this forest is realized by a chamber.

At ambient63, the finite curvature is

```
(5*sum_(j=0)^62 min(j,62-j) -63*62)/2 =899/2.
```

Consequently the complete finite comparison has slack1/2, with every new coupon included. This report supplies no ideal margin, complete simultaneous face-payment theorem, dimension62 conclusion, or example.

Verification status

Final source-bound generation (session81636, terminal chunkb7b486) and separate fresh saved-certificate replay (session88456, terminal chunk6c791d) both exited0/PASS. Both bind the final frozen D13 source, data and report. The source imports only the standard library. It retains separate actual and conservative cost functions and checks the complete coupon-role records from the local inputs. No numerical optimizer is used.

The root reviewer and a separate peer each completed a full source/report mathematical read. Their scopes include the nine typed coupon guards, simultaneous two-end type domination, distinct B9/E6/A17 representatives, the conservative E6/D11 allowance, complete multiplicity/residual split, interaction compression, Bellman upper-bound and attainment argument, and the actual-cost forest449. Neither review independently reran the 1,329-vector enumeration or Bellman arithmetic. The generation and fresh saved-certificate replay above supply that exact computation; the local geometric classifications and transfers remain explicitly bound inputs.

Source 278

Source path: `results/fresh_spectator_energy_below_one_20261017.md`

SHA256: 90ceea392d5201c531fd9952455bd706bda5f80eb5a80a8fb8d4f840364f1f6e

The complete strict spectator spectrum below one

The new all-rank assertion is as follows. Let G be a positive-definite norm-two acute Coxeter Gram matrix, and let x,y be nonnegative native contact vectors with disjoint supports. Require both separate augmented Grams G+x and G+y to be strictly positive definite. For

`D=(x−y)^T G^(−1)(x−y)<1,`

the complete set of possible values is

`0; r/(r+1) for integers r>=1; 2−sqrt2; 1−sqrt5/5.`

At the rational nonzero value r/(r+1), the contacts lie on one whole simple A_r component: either one unit contact at a leaf, or two unit contacts at adjacent vertices. Both separate extensions must still be positive definite; this excludes some interior adjacent positions in large A_r. Every other component has zero contacts. The two irrational values have two unit contacts at the endpoints of a whole I2(4) or I2(5), respectively. Conversely all the listed energies occur: the one-contact A_r leaf realizes every rational value.

The source and certificate check the exact identities below. This strengthens the spectrum through three quarters. It is a local lemma, without a new ambient dimension or construction claim.

Two elementary obstructions in arbitrary rank

All positive-definite acute Coxeter diagrams are forests. A cycle's all-ones vector would otherwise have nonpositive square. Their inverses are entrywise nonnegative, and an inverse diagonal cannot decrease when a principal diagram is enlarged or its bond magnitudes are increased. The principal comparison is a positive Schur correction; bond monotonicity follows from the nonnegative Neumann series, or differentiation of the inverse along the positive-definite line segment.

Two explicit principal subtrees detect any branching or nonsimple bond. On a simple path ending in a bond of magnitude sqrt2 (type B), the inverse diagonal at its remote leaf is1. At the next vertex it is2 if that next vertex precedes the doubled bond. On a simple path ending in a fork with two short simple leaves (type D), the remote leaf and next vertex again have inverse diagonals1 and2. This includes D4 with the fork center as the second vertex.

Here is an all-rank calculation. For B, scale the terminal root by sqrt2. The rational congruent Gram has last diagonal4 and last bond magnitude2. Eliminating the last root makes the preceding diagonal1; eliminating successive roots repeats the diagonal1. The first inverse column is (1,…,1,1/2). The second is(1,2,…,2,1) when the path has at least3 nodes. The first two diagonal entries are unaffected by the terminal congruence. For D, eliminating the two short leaves changes the fork diagonal to1, and the same recurrence applies. The first inverse column is (1,…,1,1/2,1/2); the second is(1,2,…,2,1,1). These vector identities and the positive elimination pivots prove the assertions for all ranks.

From any vertex of a tree that is not a simple A path, follow a path to the first nonsimple bond or branching node. Keeping the terminal doubled bond, or two outgoing fork leaves, gives one of these subtrees; larger native bonds only increase inverse diagonals. Hence every inverse diagonal of a component that is not a simple path is at least1. If the starting vertex itself is a branch, retain two of its neighbors instead: the resulting simple A3 has inverse diagonal1 at that middle vertex, and larger bonds only increase it. Thus this zero-length path case does not require an incoming edge for the fork construction.

There is a stronger statement for an adjacent simple marked edge i-j. If its component has a nonsimple bond or branch anywhere, it lies in one of the two sides of that edge. Retain a path from the opposite marked endpoint to that feature, together with its doubled bond or fork leaves. The opposite endpoint becomes the remote leaf and the other marked endpoint becomes its next vertex. One marked inverse diagonal is therefore at least2. This contradicts strict positive definiteness of that marked root's separate unit-contact extension.

One nonempty contact support

Nonnegative inverse entries and diagonal lower bound1/2 give D>=sum_i x_i^2/2 when y=0. Under D<1, the support has only one coordinate, with native magnitude1. A nonsimple magnitude is at least sqrt2 and would already give D>=1.

The subtree obstruction forces the whole contacted component to be a simple A_r path. Its inverse is, in one-based indices,

$$(G_A_r^{-1})ij=\min(i,j)*(r+1-\max(i,j))/(r+1).$$

Every interior diagonal is at least1. Thus the contact is at a leaf and D=r/(r+1). The same argument applies after interchanging x and y. If both supports vanish then D=0; conversely positive definiteness and disjoint supports force both to vanish when D=0.

Two nonempty supports

For any i in the x-support and j in the y-support, Gram-metric Cauchy gives

$$D >= (x_i+y_j)^2/(4-2G_{ij}).$$

Nonadjacent support nodes would give D>=1. Every opposite-support pair is therefore adjacent. If one support had two vertices, adjoining its new root would create a cycle through a vertex of the other support. Its strictly positive-definite augmented diagram forbids this. Hence x=p*e_i and y=q*e_j for adjacent i,j, with p,q native magnitudes.

Put t=-G_ij. If p>=sqrt2, the separate edge condition and the two-node principal inverse give

$$p^2*2/(4-t^2)<2,$$

so t^2<2 and the native alphabet forces t=1. Both p and q are then below sqrt3<2. On the two-node principal Gram the difference energy is

$$2*(p^2+q^2-p*q)/3.$$

For fixed p<2 this is increasing in q>=1. At q=1 it is increasing for p>=sqrt2. Its minimum is therefore2-(2/3)sqrt2>1, contradicting D<1. Interchanging the contacts gives the same contradiction when q>=sqrt2. Thus p=q=1.

If t is nonsimple, the separate strict edge tests imply t<sqrt3. Any further neighbor of either endpoint with magnitude u>=1 would give inverse diagonal2/(4-t^{2-u}2)>=2 in that three-node principal path, again impossible. The whole component is consequently I2(m), with sqrt2<=t<sqrt3 forcing m=4 or5. Its energy2/(2+t) is exactly one of the two listed irrational values.

If t=1, the marked-edge subtree obstruction excludes every nonsimple bond and branch in the whole component. It is A_r. For adjacent contacts at k,k+1, its inverse difference vector has entries

$$j/(r+1) \text{ for } j<=k;$$
$$-(r+1-j)/(r+1) \text{ for } j>=k+1.$$

Every entry is nonzero and the difference energy is r/(r+1), independent of k. The remaining strict extension conditions are the two explicit diagonal inequalities(G^(-1))kk<2 and(G^(-1))k+1,k+1<2. This completes the all-rank classification; no cutoff on spectator rank or native labels is imposed.

Consequences and verification

For a rational0<D<1, the forced rank is exactly D/(1-D). Thus4/7, 11/14,9/16 and8/17 are impossible;2/3,3/4 and4/5 force whole A2,A3,A4 contact components. The11/14 exclusion is a new input available for shorter cycle transfers. Such transfers require separate native-height and finite-coupon arguments and are not asserted here.

The checker verifies all A inverse columns through rank64, all2,016 adjacent-contact identities, and124 B/D obstruction templates with positive elimination pivots and exact inverse columns. The formulas and subtree arguments above prove arbitrary rank; these finite checks corroborate them. The two irrational literal cases retain their frozen predecessor's exact verification. Generation and a fresh separate-process saved replay both terminated exit0/PASS. An independent full source/mathematical read passes the arbitrary-rank obstruction, support reduction, strict nonunit bound and exceptional rank-two cases. It identified the zero-length path exposition case above, now made explicit. That review did not independently rerun the finite identity checks. Source, data and report are final/frozen after direct binding, syntax and local-link checks. All reviews are internal; external refereeing and proof-assistant formalization are not claimed.

Source 279

Source path: `results/fresh_spectator_energy_one_boundary_20261017.md`

SHA256: d5d21c576220c1351564764061af608c280769c89a17b7a38870bac9009598d3

The strict spectator boundary at energy one

Let G be a positive-definite norm-two acute Coxeter Gram matrix, and let x,y be nonnegative native contact vectors with disjoint supports. Both separate augmented Grams G+x and G+y are required to be strictly positive definite. The equality

```
(x-y)^T G^(-1)(x-y)=1
```

has exactly the following possibilities, up to exchanging x and y and diagram automorphism. Components not mentioned have no contacts.

- If only one support is nonempty: one sqrt2 contact on a whole A1; two unit contacts on two distinct whole A1 components; one unit contact at the middle of a whole A3; one unit contact at the remote leaf of a whole B_r, r>=2; or one unit contact at the long leaf of a whole D_r, r>=4. In B2 either endpoint is allowed, and in D4 any leaf is allowed.
- If both supports are nonempty: one unit contact from each side, on two distinct whole A1 components, on the two endpoints of a whole A3, or on the two short fork leaves of a whole D_r with4<=r<=7.

Together with the strict spectrum below one, this classifies the complete interval0<=D<=1. The source and certificate check exact identities. This local statement does not by itself prove a new corner transfer or lower an ambient dimension bound.

One nonempty support

As in the preceding proof, inverse entries are nonnegative and inverse diagonals are at least1/2. Thus at most two contacts are possible. If there are two, both are unit contacts, both inverse diagonals equal1/2 and their inverse cross entry vanishes. An inverse diagonal equals1/2 only for an isolated root: any incident bond gives a strictly positive Schur correction. This gives the two whole A1 components.

For one contact of magnitude p, the bound p^2/2<=1 and the native alphabet give p=1 or sqrt2. The latter forces a whole A1. For p=1 the contacted inverse diagonal equals1. On A_r the formula i(r+1-i)/(r+1)=1 is equivalent to (i-1)(r-i)=1, so only the middle of A3 occurs.

For a component that is not a simple path, take the principal B or D obstruction from the preceding proof, beginning at the marked vertex. The remote-leaf inverse diagonal is1, and its entire inverse column is strictly positive. Adding any connected vertex therefore increases that diagonal strictly: the Schur correction contains the nonzero product of this positive column with a nonnegative nonzero contact vector. Increasing any bond likewise increases it strictly, by the inverse derivative formula and positivity of that column. Consequently the principal obstruction must already be the whole component, with exactly its minimal bond labels. This gives B_r or D_r at their stated leaves. If the marked vertex is itself a branch, its principal A3 middle already has inverse diagonal1, and the third neighbor makes it strictly greater, excluding this case.

All these one-sided cases have projection norm1<2, so their separate edge extensions are strictly positive definite.

Two nonempty supports

Suppose first that an x-support node i and a y-support node j are nonadjacent. Gram-metric Cauchy gives

```
1 >= (x_i+y_j)^2/4 >= 1.
```

Equality forces x_i=y_j=1 and

`G^(-1)(x-y)=(e_i-e_j)/2.`

Hence every other coordinate of x-y is (G_ki-G_kj)/2. Since G is positive definite, all off-diagonal entries are in(-2,0]. Their half-differences are strictly between-1 and1. A nonzero native contact has magnitude at least1, so every other coordinate vanishes. Disjoint supports now imply x=e_i, y=e_j, and i,j have identical neighbor sets with identical bonds.

If there are no neighbors, they are two whole A1 components. Otherwise the forest condition permits exactly one common neighbor k. The two equal bonds must be simple: a three-node principal Gram with equal magnitude t has determinant8-4t^2>0, hence t<sqrt2. Thus i,j are simple twin leaves.

The remaining connected component is A3 or D_r. Here is an elementary all-rank argument. The common neighbor has at most one further neighbor, since four simple leaves would contain an affine D4 subdiagram. Follow the remaining simple path. A first doubled bond or second fork would contain an affine B or D principal subdiagram, contradicting positive definiteness. Explicitly, give the first two fork leaves coefficient1/2 and every stem vertex coefficient1. At a terminal doubled bond give the terminal vertex coefficient1/sqrt2; at a terminal fork give its two leaves coefficient1/2. The resulting minimal diagram has positive null vector. Larger bond magnitudes make its quadratic form negative. Hence the remaining path ends at a leaf without a nonsimple bond or second branch.

For D_r with short leaves i,j, including D3=A3, the inverse column at i has stem coordinates1/2,2/2,…,(r-2)/2, short-leaf coordinate r/4 at i and (r-2)/4 at j. Multiplying by G verifies this in every rank. Thus the two separate projection norms are r/4. Strict edge positivity requires r<8, giving precisely A3 and D4,…,D7. Their inverse difference is (e_i-e_j)/2, so the energy is1.

It remains to exclude the case in which every cross-support pair is adjacent. Either support having two nodes would make a cycle in its separate augmented Gram. Thus x=p*e_i,y=q*e_j for adjacent i,j. The preceding below-one proof actually shows that any nonunit p or q gives energy strictly greater than1. For p=q=1, a simple marked bond forces the whole component to be A_r by the strict marked-edge obstruction, giving energy r/(r+1)<1. A nonsimple marked bond forces a whole I2(4) or I2(5), also with energy strictly below1. No adjacent case remains.

Verification scope

The written identities and sealing arguments are in arbitrary rank. The checker corroborates125 one-contact B/D identities through rank64, 62 twin-leaf identities through rank64, and the three isolated cases. It verifies positive pivots, entire inverse contact columns, exact energy and the strict rank7/rank8 boundary. The sqrt2/A1 case uses a rational congruence scaling its isolated root by sqrt2.

Generation and a fresh separate-process saved replay both terminate exit0/PASS. A separate full source/mathematical read passes the Cauchy equality, native-gap argument, all-rank twin-leaf classification, strict rank7/rank8 boundary, whole-component sealing and rational congruences. That review does not duplicate the125/62 finite identity runs. Direct bindings, syntax and local report links pass. Source, data and report are final/frozen. No independent arithmetic reconstruction, outside refereeing or proof-assistant formalization is claimed.

```
python3 scripts/fresh_spectator_energy_one_boundary_20261017.py
python3 scripts/fresh_spectator_energy_one_boundary_20261017.py --verify
```

Source 280

Source path: `results/fresh_spectator_five_quarters_transfer_20261017.md`

SHA256: 6bbe570d6437561cc14e0653c97002dc09ed4bd23cf6334a0ac07026c77ca47f

A short-path alternative at spectator energy five quarters

Let G be a positive-definite norm-two acute Coxeter Gram matrix. Let x,y be nonnegative native contact vectors with disjoint nonempty supports, and require both separately augmented diagrams G+x and G+y to be strictly positive definite. If

```
(x-y)^T G^(-1)(x-y) = 5/4,
```

then either an x-contact and a y-contact have distance at most3 in one connected spectator component, or the only contacted components are whole A1 and whole A3, with one unit leaf contact in each and opposite contact sides. The first alternative gives a path of length at most5 between the two new roots. No bound on spectator rank or native labels is imposed.

The source and certificate check the exact identities below. The finite Coxeter classification and the one-sided spectrum below1 are inherited inputs. This is a local spectator lemma, not a complete face-transfer or dimension theorem.

Contacts in the same component

Every finite Coxeter diagram is a forest. Thus a new root can contact at most one node in each connected component: two contacts would make a cycle in its separately positive-definite augmented diagram. Write two opposite contacts in one component as p at i and q at j, with p,q>=1 and i different from j. Put J=G^(-1), a=Jii, b=Jjj, c=Jij. Strict extension positivity forces a<2 and b<2, even before accounting for other components. The contribution to the difference energy is

```
E(p,q)=a*p^2+b*q^2-2*c*p*q.
```

When a-c and b-c are nonnegative, expand around p=q=1. The linear terms are nonnegative and the remaining quadratic form is positive semidefinite, so E(p,q)>=a+b-2c for every p,q>=1. This does not assume coordinatewise monotonicity at arbitrary p,q.

For A_r, put N=r+1 and use one-based indices. The all-rank inverse is

```
Jij=min(i,j)*(N-max(i,j))/N.
```

For i<j and d=j-i, the row sums are a-c=i*d/N and b-c=(N-j)*d/N, both positive. The unit-contact energy is d*(N-d)/N. Since N>=d+2, for d>=4 we have

```
d*(N-d)/N >= 2d/(d+2) >= 4/3.
```

The two differences are respectively d^2*(N-d-2)/(N*(d+2)) and 2*(d-4)/(3*(d+2)). These identities prove the assertion in arbitrary rank. The certificate separately multiplies the proposed inverse by each Gram and checks every far pair in ranks5 through32 as corroborating examples.

For B_r number the simple spine0,…,r-2 and the terminal doubled-bond node r-1. Its native norm-two inverse is

```
Jij=min(i+1,j+1)                    (i,j<r-1),
Ji,r-1=(i+1)/sqrt2                  (i<r-1),
Jr-1,r-1=r/2.
```

Thus a possible unit contact lies only at0, or at r-1 when r<=3. Two distinct allowed positions never have distance at least4. The formula follows by direct multiplication with the tridiagonal Gram, including the two terminal equations, for every r>=2.

For D_r number the long spine0,…,r-3 and the two short leaves r-2,r-1. Its inverse has Jij=min(i+1,j+1) on the spine, Ji,leaf=(i+1)/2, diagonal r/4 at each short leaf and off-diagonal (r-2)/4 between the short leaves. Direct multiplication at the spine and fork proves these identities for every r>=4. The only positions of diagonal below2 are0 and, when r<=7, the two short leaves. Short leaves are distance2 apart. A long/short pair of distance at least4 therefore has r=6 or7; its row sums are1/2 and r/4-1/2, and its unit energy is r/4>=3/2. The certificate verifies both inverse families through rank24, without using that sample as a rank cutoff.

For E6,E7,E8 the entire finite case is checked exactly. E6 has two possible contact nodes, at distance4, with a=b=4/3 and c=2/3; their energy is at least4/3. E7 has only one possible contact node and E8 has none. The complete Grams and inverses are saved. F4,H3,H4 and every I2(m) have diagram diameter at most3 and need no energy estimate.

Every connected component contributes nonnegative energy. Consequently one mixed component with its contact nodes at distance at least4 would already contribute at least4/3>5/4. Every mixed component instead yields the stated short-path alternative.

No component has contacts on both sides

Suppose the short-path alternative does not occur. Every active component has contacts from only one side. Its energy is at least1/2: a nonzero native contact is at least1 and an inverse diagonal is at least1/2. Since both supports are nonempty, there are at least two active components; three would contribute at least3/2. There are exactly two, one per side, and each has energy between1/2 and3/4.

The inherited one-sided spectrum below1 forces a unit leaf contact on a whole A_r, with energy r/(r+1). Here r<=3. The only two such energies summing to5/4 are1/2 and3/4, from A1 and A3. Every other spectator component has zero contacts. This proves the second alternative.

Verification status

Generation and a fresh separate-process saved replay both terminated exit0/PASS on all exact identities and finite exceptional cases. The arbitrary-rank argument is the written inverse calculation and classification above; finite samples are supplementary checks. A separate internal full source and written-proof review passed the support reduction, positive-quadratic remainder argument, all-rank formulas, strict contact position restrictions and final component-energy split. That review did not duplicate the finite arithmetic tables. Source, certificate and report are final/frozen. No external refereeing, formalization or new finite-volume example is claimed.

Source 281

Source path: `results/fresh_universal_deep_finite8_corridor_20261018.md`

SHA256: c59423d9073aef0c03aa4f9fa60967b59ded752f360c24355283cad3a44d8219

A universal short-path alternative for deep finite distance-eight pairs

Let an actual quadrilateral face of a finite-sided finite-volume real hyperbolic Coxeter polytope have a finite corner whose two marked roots lie in one whole finite A/B/D component at graph distance8. Suppose both marks have distance at least17 from every endpoint, branch and endpoint of a nonsimple bond of that whole component. Then the **full actual supporting star contains a path of length at most7 between the two opposite black pairs, with white internal roots**.

This statement covers all low/high, disjoint/overlapping minimal-core cases, all native labels and arbitrary additional supporting roots. It does not assume the old no-short8 condition. It is a conditional short-path theorem, not a claim that every low or disjoint configuration is impossible. Only in the explicit full-star no-short7 branch does it exclude a deep finite-distance8 pair.

The source and certificate check the exact rational inequalities below. No contact-profile or low/high joining census is needed. A complete changed payment rule, its finite-forest budget and any dimension consequence require their separate composition. The unrestricted ceiling stays65 in this report.

Actual-face inputs and the forced path

Write the star as W+a+b+c+d, with opposite black pairs {a,c} and {b,d}. Every root of W is white. The actual white diagram and each W plus one black are positive definite. Each W plus an opposite pair is indefinite; these are the inherited strict-edge and closed-opposite-side facts of the complete face reduction. The corner under consideration is W+a+b. Roots are normalized to squared norm2, so every nonzero native acute contact has magnitude at least1, including arbitrarily large finite labels, parallel or dotted contacts. All distinct-root products are nonpositive.

Assume there is no full-star path of the stated length. In particular c.d=c.b=d.a=0. The products c.a and d.b are unrestricted nonpositive native products; they may vanish or have arbitrarily large magnitude.

The whole finite A/B/D component is a tree. Feature depth17 gives a simple eight-edge path from a to b and two disjoint simple outward arms of17 whites. A feature exactly at outward distance17 is allowed: every bond inside the corresponding17-white arm is still simple. Keep15 outward whites on each side. The retained39 roots form an induced positive definite A39 path with positions0,…,38, and

`a at15,  b at23.`

Its other37 roots are supporting whites. The two unused whites on each outer arm remain available for actual-edge inverse bounds. This is the same forced geometry as the frozen A39 high-domain theorem, but no high-domain enumeration is imported here.

Four possible contact positions on each side

For a norm-two A_n path the positive inverse entries are

`J_n[i,j]=(min(i,j)+1)(n−max(i,j))/(n+1),  0<=i,j<n.`

All its leading principal determinants are k+1, so the path is positive definite. The checker verifies the determinant recurrence and inverse formula exactly at orders17 and39.

The omitted root c and any17-white outer arm belong to an actual edge diagram. If c meets that arm at outward distance j=3,…,15, the projection square is at least

(A17 inverse)[j−1,j−1]=j(18−j)/18 >=5/2>2.

The native contact is at least1; all further contacts have nonnegative magnitudes and the inverse is positive, so they cannot lower this bound. It contradicts strict positive definiteness of that actual edge. Thus c has no retained contact at those distances on either outer arm.

If c meets an inner white at distance j>=2 from a, following the inner path to b gives a c-to-b path of length9-j<=7. Meeting either of the first two whites beyond b gives a path of length2 or3. These contacts are excluded by the full-star hypothesis. The argument for d is the reflection. Therefore, inside the retained A39 path, the only possible nonzero contact positions are

c: 13,14,15,16; d: 22,23,24,25.

No restriction on the magnitudes at these positions is needed. In particular this proof does not impose a finite-label alphabet, rule out simultaneous contacts, or bound the two own-mark products. Contacts to roots outside A39 are allowed throughout.

Why both retained contact vectors are nonzero

This is the step that covers the low-core and disjoint branches without their old no-short8 hypothesis. Choose a minimal indefinite principal diagram K inside W+a+c. Both a and c belong to K: deleting either would leave a principal diagram of a positive-definite actual edge. Minimal indefiniteness makes K connected.

Use the published all-label fact that a minimal indefinite Coxeter diagram has order at most10. The precise classification input and primary source are recorded in the frozen minimal-indefinite catalogue report and its local primary PDF. The current proof uses only this order ceiling. It neither infers the ceiling from a bounded search nor uses the26 high-order diagrams or the finite low-order census.

Here is the precise bridge to that published statement. If K has order2, a dotted pair already satisfies the ceiling. At order at least3, no dotted bond is possible: its two-root indefinite principal would contradict minimality. The remaining finite and parallel products are exactly the Coxeter cosine-matrix alphabet, including the infinite label, used by Chein. Because K with c deleted is strictly positive definite, interlacing and indefiniteness give full signature (|K|-1,1), with no zero eigenvalue. Every other deletion is positive semidefinite and has finite or affine connected components. On printed page4 (PDF page3), Chein explicitly allows such disconnected deletions in his graph criterion, and states the order-ten bound for irreducible hyperbolic matrices, attributing it to Vinberg. Thus the bound applies to these actual minimal cores. No finite bound on order3 black-black labels is assumed. The primary paper on Numdam and the bound local copy contain the same printed-page statement.

Let B be the connected component containing a in K with c deleted. It has at most9 roots. It is contained in W+a, hence also in the actual finite corner W+a+b. Every root of B lies in the same whole finite component as a, and is at distance at most8 from a. Feature depth17 forces B to be a simple subpath of that component. It cannot pass through b, since b does not belong to K. Consequently B fits inside the retained A39: at most8 outward steps from a and at most7 steps toward b, while15 outward whites were retained.

Since K is connected, c has a nonzero contact with B. Otherwise B would already be a separate connected component of K. Thus c has a nonzero contact inside the retained A39. The same argument with a minimal indefinite core in W+b+d proves the analogous assertion for d. No hypothesis about whether the two cores intersect has entered.

Extra supporting roots, old spectator blocks, or additional cross-core contacts do not affect this reasoning. B is an actual connected principal subdiagram of the whole finite tree; its edges and paths cannot be removed by restoring other roots. Only a necessary subset of actual roots is used in the final Gram obstruction.

Uniform correlation and a strictly positive determinant

Let x,y be the nonnegative native contact-magnitude vectors of c,d against A39, and put J=J_39. Their supports lie respectively in I={13,14,15,16} and R={22,23,24,25}, and both vectors are nonzero. Every nonzero coordinate is at least1. Define

q=x^T J x, r=y^T J y, s=x^T J y.

The positive entries of J give

q,r >= min_{i in I union R} J[i,i] =91/10.

Put U=sum_{i in I} x_i(i+1), V=sum_{j in R} y_j(39-j). The cross block of a path inverse has rank one, and here

s=U*V/40.

For i,k in I, assuming i<=k without loss of generality,

J[i,k]/((i+1)(k+1))=(39-k)/(40(k+1)) >=23/680.

The right block has the reflected inequality. Nonnegative contact coordinates therefore imply

q >=(23/680)U^2, r >=(23/680)V^2,
s^2 <=(289/529)q*r.

The source checks every one of the32 within-block inequalities, all16 cross-block factorizations and the eight inverse diagonals. These finite identities prove the bound for arbitrary real nonnegative contact magnitudes, not just the values appearing in a catalogue.

Because c.d=0, eliminating the positive definite A39 block from the actual41-root Gram matrix leaves the two-by-two Schur matrix

[2-q -s]
[-s 2-r].

Both diagonals are at most-71/10. Its determinant has lower bound

(240/529)q*r -2(q+r)+4.

Write q=91/10+u and r=91/10+v, where u,v>=0. This lower bound becomes

13674/2645 +(1126/529)(u+v) +(240/529)u*v >0.

Thus the Schur matrix is negative definite. By congruence, the full Gram matrix has two negative directions. Actual wall roots lie in a Lorentz space with only one negative direction, a contradiction. Additional roots or contacts outside this principal matrix cannot repair it. The argument does not require a nonsingular old core, a rational gauge for that core, or any finite cutoff on supporting components or contact labels.

This proves the universal full-star short7 alternative. It supplies the missing conditional low-core and high-disjoint coverage for a future deep-distance8 payment change. It does not prove a general finite7 quadrilateral theorem, and it does not itself assign a full unit to an ideal short6/7 pair: the existing ideal-cycle/tree transfers must still be composed with it.

Verification and limits

Final generation28326f and a separate fresh saved replay58e035 both terminated exit0/PASS. The source imports no project arithmetic module and uses exact Fraction arithmetic. It checks path positivity/inversion, actual-edge projection gates, support/correlation identities and the strict determinant polynomial. The proof's published order-ten bound and actual-face facts remain explicit inherited inputs.

Two other agents completed full source/report mathematical reads. Both passed the minimal-core deletion/component argument, its placement inside the retained whole finite path, actual-edge support exclusions, all-label nonzero contact bounds, arbitrary spectators, the uniform rank-one correlation inequality and the Lorentz index-two contradiction. They separately checked Chein's printed-page4 statement and the explicit bridge from the actual minimal cores to its hypothesis, including disconnected positive-semidefinite deletions and order2 dotted cores. Neither duplicated the exact arithmetic run or any original census, and neither review is claimed as a review of the later payment/global composition. No mathematical gap was identified in the stated scope.

All nine direct bindings, Python syntax, six local links and the stated primary-paper link were checked. Source, certificate and report are final/frozen.

No original all-label classification or high-overlap census is re-enumerated here. No complete changed payment system, finite budget, new polytope, new ambient dimension theorem, external refereeing or formal verification is asserted.

```
python3 scripts/fresh_universal_deep_finite8_corridor_20261018.py
python3 scripts/fresh_universal_deep_finite8_corridor_20261018.py --verify
```

Source 282

Source path: `results/review_simple_cusp_affine_a_quadrilateral_20261017.md`

SHA256: 74f9e7a7c05b6c8ba853cef3ab89bb092881f8f9c332100c038419089d8339c1

Independent review of the affine-A quadrilateral argument

PASS within the stated all-simple, n >= 14 scope. The proof pays an actual quadrilateral with an affine-A corner. It does not settle the other affine corner types or lower the ambient dimension bound.

Root separately read the proof and producer source. The independent checker uses Fraction pairs for Q(sqrt(2)), imports no producer, and recomputes all 256 height-rectangle values and the four positive kernel equations. Its certificate binds the reviewed source, data and report. Generation and fresh saved replay pass. The formal determinant expansion and positive-definite deletion checks remain the producer's checks rather than independently duplicated calculations.

The following geometric and analytic points were checked separately:

- Deleting one marked root from the full n-cycle leaves A_(n-1). An adjacent actual corner is finite or is a connected n-node affine diagram, because every vertex is simple. In rank n >= 14 the exceptional types cannot occur. The classical finite/affine diagrams give exactly the isolated, endpoint3, endpoint4, next-to-end3 and both-end3 extension profiles. Neither a hidden large label nor a disconnected affine extension is allowed by this classification.
- When both marked arcs have length at least four, the possible contacts lie away from the other old marked root. All stated low-contact exclusions are literal induced affine diagrams of order at most nine. In particular, a simple leaf on the old cycle supplies the affine-E8 arms of lengths 1,2,5 without an extra edge between their ends. The isolated zero-contact case would be orthogonal to the full cusp span and hence null, contradicting the new wall's positive norm.
- The displayed path inverse gives the residual marked Gram, endpoint moments and cross term. At arc length four the two next-to-end contacts may coincide; the inverse still gives the asserted cross term 4/P. Thus this limiting overlap is included. The actual n+2-wall star necessarily has zero determinant in ambient Lorentz dimension n+1.
- Each height bound follows from the displayed forbidden small affine diagram and the Coxeter contact alphabet. Replacing a strict isolated bound by its closed endpoint enlarges the tested rectangle, which is safe. The residual polynomial is jointly concave in the reciprocal heights because its quadratic part is a negative diagonal form plus a negative square. Its minimum on a rectangle therefore occurs at a corner.
- Both reciprocal-arc partial derivatives are strictly negative at positive reciprocal heights. If a moment vanishes, either the other moment or the positive isolated constant supplies strictness. Since P+Q >= 14, at least one reciprocal arc is strictly below 1/4. The nonnegative rectangle calculation therefore becomes a strict positive determinant obstruction at every actual pair of arc lengths.

The inherited global affine exclusion is an explicit input. This proof uses it only for affine diagrams of orders 3,4 and 9, all below n. It does not assume that every affine subdiagram of any order is forbidden. That distinction must be preserved when extending the argument to other corner types or to quadrilaterals whose corners are all finite.